

# **KAM, Smooth Dynamics, and Billiards**

Student Textbook Series — Volumes K01-K20

Hamiltonian perturbation theory, smooth dynamics, billiards, translation  
surfaces, and rigidity

Student textbook sequence — K01-K20

Built on the certified Dynamical Systems Foundation series.

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# Preface to the twenty-volume KAM Part

These twenty volumes follow invariant structures across a sequence of increasingly different dynamical languages. K01–K07 build the symplectic, normal-form and KAM persistence machinery. K08–K09 contrast exponentially long Nekhoroshev confinement with Arnold diffusion. K10 recasts twist dynamics variationally. K11–K12 study smooth and singular billiards. K13–K16 pass from polygonal unfolding to translation surfaces, Teichmüller/KZ renormalization, affine invariant manifolds, counting and wind-tree transport. K17–K18 turn to partially hyperbolic and generic smooth dynamics. K19 develops infinite-dimensional/PDE KAM. K20 closes with modern billiard rigidity and spectral inverse problems.

The series is intentionally not organized as a list of famous theorems. At every transition, the reader should ask what structure has replaced the previous one: invariant tori by exponentially stable action domains; normal forms by minimizing action; curvature by unfolding; a fixed surface by moduli-space renormalization; uniform hyperbolicity by domination and accessibility; finite-dimensional Newton steps by Nash-Moser/reducibility; and forward dynamics by inverse rigidity.

## How to read the deep proofs

Every difficult result is preceded by a proof roadmap. For Teichmüller/KZ arguments, keep separate the dynamics on a fixed surface, the base flow in moduli space and the cocycle on cohomology. For affine-measure rigidity, distinguish measure classification from orbit-closure classification. For partial hyperbolicity, track domination constants and the role of accessibility. For  $C^1$  genericity, distinguish statements about a residual subset from statements about a fixed map. For PDE KAM, keep a ledger of tangential modes, normal frequencies, divisor classes, derivative loss and excluded parameter sets. For billiard rigidity, record exactly which symmetry, closeness, rational-caustic and spectral hypotheses are present.

## Prerequisites and proof policy

These volumes assume the common Foundation F01–F16. When a Foundation theorem is used, its hypotheses are recalled or checked at the point of application. The KAM volumes develop their principal arguments in place rather than organizing the subject as a list of citations to separate specialist texts.

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## **Part I**

# **Hamiltonian Geometry, Normal Forms, and KAM Persistence**

# Symplectic and Poisson Geometry

## Why this chapter matters

Hamiltonian mechanics is often introduced by a pair of differential equations for position and momentum. KAM theory needs a more invariant viewpoint. The decisive object is not a preferred coordinate system but a nondegenerate closed two-form. It tells us how a function produces a vector field, why Hamiltonian time evolution preserves phase-space geometry, how conserved quantities interact through the Poisson bracket, and why invariant tori are naturally Lagrangian. This chapter builds that geometric language before any perturbation theory begins. Darboux's theorem explains why every symplectic manifold looks locally like ordinary phase space. The Lagrangian-neighborhood theorem explains the local geometry near the invariant tori that later KAM arguments will preserve. Generating functions turn exact symplectic maps into scalar objects, and Noether's theorem connects symmetry with conserved quantities. These are not isolated facts: together they define the class of coordinate changes that every later normal-form and KAM iteration is allowed to use.

**Prerequisites.** Smooth manifolds, differential forms, flows of vector fields, the inverse-function theorem, and elementary Lie-group terminology.

## Learning goals

- Translate Hamilton's equations into the identity  $\iota_{X_H}\omega = -dH$  and keep the book's sign convention consistent.
- Explain why Hamiltonian flows and exact symplectic maps preserve the geometry used later in KAM iteration.
- Recognize Lagrangian submanifolds as the natural dimension and isotropy class for invariant tori.
- Use Darboux, Weinstein, generating functions, and Noether as structural tools rather than disconnected theorems.

## Conceptual roadmap

The chapter follows the chain

symplectic form  $\longrightarrow X_H \longrightarrow$  Hamiltonian flow  $\longrightarrow$  Poisson algebra,

while a second chain

Lagrangian submanifold  $\longrightarrow T^*L$

$\longrightarrow$  exact symplectic maps and generating functions.

prepares the local model for invariant tori and canonical transformations. Darboux removes local coordinate accidents; Weinstein removes local accidents near a Lagrangian; Noether explains how group symmetry enters the same formalism.

### Check your understanding

Before proving anything, verify the sign convention in canonical coordinates: with  $\omega = dp \wedge dq$  and  $\iota_{X_H}\omega = -dH$ , compute  $X_H$  for  $H = (p^2 + q^2)/2$ . Keep this calculation as the sign test for the whole KAM series.

## 1.1 Core definitions

### Intuition

A symplectic form converts differentials of functions into vector fields. Nondegeneracy makes the conversion unique; closedness makes the resulting dynamics compatible with a conserved geometric structure. Most later KAM constructions are elaborate ways of changing coordinates without leaving this category.

### Definition K01.1. Symplectic manifold

A *symplectic manifold* is a smooth manifold  $M$  of even dimension  $2n$  equipped with a smooth two-form  $\omega$  such that  $d\omega = 0$  and, for every  $x \in M$ , the bilinear form  $\omega_x$  on  $T_x M$  is nondegenerate.

### Definition K01.2. Hamiltonian vector field

For  $H \in C^1(M)$ , the *Hamiltonian vector field*  $X_H$  is the unique vector field satisfying

$$\iota_{X_H}\omega = -dH.$$

This sign convention is fixed throughout the series.

### Definition K01.3. Poisson bracket

For smooth functions  $F, G$ , define

$$\{F, G\} := \omega(X_F, X_G) = dG(X_F) = -dF(X_G).$$

With this convention,  $[X_F, X_G] = X_{\{F, G\}}$  and the derivative of  $G$  along the  $F$ -flow is  $\{F, G\}$ .

**Definition K01.4. Lagrangian submanifold**

A submanifold  $L \subset (M^{2n}, \omega)$  is *Lagrangian* if  $\dim L = n$  and  $\omega|_{TL} = 0$ . Thus it has the largest possible dimension on which the symplectic form vanishes.

**Definition K01.5. Hamiltonian action and moment map**

Let a Lie group  $G$  act smoothly and symplectically on  $(M, \omega)$ . For  $\xi \in \mathfrak{g}$ , let

$$\xi_M(x) = \left. \frac{d}{dt} \right|_{t=0} \exp(t\xi) \cdot x.$$

A *moment map* is a smooth  $J : M \rightarrow \mathfrak{g}^*$  whose components  $J_\xi = \langle J, \xi \rangle$  satisfy

$$dJ_\xi = -\iota_{\xi_M} \omega,$$

so  $X_{J_\xi} = \xi_M$ . An action equipped with such a moment map is called Hamiltonian.

## 1.2 How the main proofs work

**Intuition**

**Darboux.** Interpolate between the given symplectic form and its constant value at the base point. A radial Poincaré primitive expresses their difference as an exact form, and Moser's method builds a local isotopy whose pullback freezes the interpolation.

**Hamiltonian invariance and Poisson algebra.** Cartan's formula turns  $\iota_{X_H} \omega = -dH$  and  $d\omega = 0$  into  $\mathcal{L}_{X_H} \omega = 0$ . The commutator identity for contraction and Lie derivative then identifies the Hamiltonian of  $[X_F, X_G]$ .

**Weinstein neighborhood theorem.** The Lagrangian condition identifies the normal directions to  $L$  with cotangent directions. After a linear correction along the zero section, a relative Moser argument makes the neighborhood symplectically identical to one in  $T^*L$  while fixing  $L$  pointwise.

**Generating functions and Noether.** The graph of a symplectic map is Lagrangian in the product with form  $-\omega \oplus \omega$ ; exactness turns its primitive into a scalar generating function when the relevant projection is nonsingular. For Noether, differentiate  $J_\xi$  along the  $H$ -flow and use  $G$ -invariance of  $H$ .

## 1.3 Main results

**Theorem K01.1. Darboux theorem**

**Statement.** Let  $(M^{2n}, \omega)$  be a symplectic manifold and let  $p \in M$ . Then there are neighborhoods  $U$  of  $p$  and  $V$  of 0 in  $\mathbb{R}^{2n}$  and a diffeomorphism  $\Phi : U \rightarrow V$  with  $\Phi(p) = 0$

such that

$$\Phi^* \left( \sum_{j=1}^n dp_j \wedge dq_j \right) = \omega|_U.$$

*Proof.* Let  $p \in M$ . We prove that there are local coordinates

$$(q_1, p_1, \dots, q_n, p_n)$$

centered at  $p$  in which

$$\omega = \omega_{\text{st}} := \sum_{j=1}^n dp_j \wedge dq_j.$$

Here  $\omega_{\text{st}} = \sum_j dp_j \wedge dq_j = d(\sum_j p_j dq_j)$  is the book's canonical orientation, chosen so that  $\iota_{X_H} \omega = -dH$  yields the usual Hamilton equations. The proof has four steps.

**Step 1: linear symplectic coordinates at the base point.** Put  $V = T_p M$  and  $\Omega = \omega_p$ . Since  $\Omega$  is nondegenerate,  $V$  admits a symplectic basis. For completeness, choose  $e_1 \neq 0$ . Nondegeneracy gives  $f_1$  with  $\Omega(e_1, f_1) \neq 0$ , and after rescaling  $f_1$  we may assume

$$\Omega(e_1, f_1) = 1.$$

The plane  $E_1 = \text{span}\{e_1, f_1\}$  is symplectic. Its symplectic orthogonal

$$E_1^\Omega := \{v \in V : \Omega(v, w) = 0 \text{ for every } w \in E_1\}$$

satisfies

$$V = E_1 \oplus E_1^\Omega,$$

and the restriction of  $\Omega$  to  $E_1^\Omega$  is again nondegenerate. Induction on  $n$  therefore produces vectors

$$e_1, f_1, \dots, e_n, f_n$$

with

$$\Omega(e_i, e_j) = \Omega(f_i, f_j) = 0, \quad \Omega(e_i, f_j) = \delta_{ij}.$$

Choose a smooth coordinate chart

$$\chi : U_0 \longrightarrow V_0 \subset \mathbb{R}^{2n}, \quad \chi(p) = 0,$$

and postcompose it with a linear isomorphism so that  $d\chi_p(e_j) = \partial_{p_j}$  and  $d\chi_p(f_j) = \partial_{q_j}$ . If

$$\tilde{\omega} := (\chi^{-1})^* \omega,$$

then

$$\tilde{\omega}_0 = \omega_{\text{st}}|_0.$$

After shrinking  $V_0$ , we may assume that it contains a Euclidean ball  $B_{\varepsilon_0}(0)$ .

**Step 2: an exact comparison form with a primitive vanishing at the origin.** On

$B_{\varepsilon_0}(0)$  set

$$\eta := \tilde{\omega} - \omega_{\text{st}}.$$

Both forms are closed, hence  $d\eta = 0$ , and by construction

$$\eta_0 = 0.$$

On a smaller star-shaped ball define the one-form  $\alpha$  by the radial homotopy formula

$$\alpha_x(v) := \int_0^1 t \eta_{tx}(x, v) dt. \quad (\text{D.1})$$

We record why this is the required primitive. Let  $h_t(x) = tx$ . The standard homotopy identity, obtained by differentiating  $h_t^*\eta$  and using Cartan's formula, is

$$h_1^*\eta - h_0^*\eta = d\left(\int_0^1 h_t^*(\iota_{\partial_t h_t} \eta) dt\right) + \int_0^1 h_t^*(\iota_{\partial_t h_t} d\eta) dt.$$

Since  $d\eta = 0$  and the pullback of a two-form by the constant map  $h_0$  is zero, this gives

$$d\alpha = \eta. \quad (\text{D.2})$$

Formula (D.1) also gives  $\alpha_0 = 0$ .

**Step 3: the local Moser path.** For  $t \in [0, 1]$  define

$$\omega_t := \omega_{\text{st}} + t\eta = (1 - t)\omega_{\text{st}} + t\tilde{\omega}. \quad (\text{D.3})$$

Every  $\omega_t$  is closed and

$$(\omega_t)_0 = \omega_{\text{st}}|_0.$$

Nondegeneracy is an open condition. Because  $[0, 1]$  is compact and the map

$$(t, x) \mapsto (\omega_t)_x$$

is continuous, after shrinking to a ball  $B_\varepsilon(0)$  we may assume that  $\omega_t$  is nondegenerate on  $B_\varepsilon(0)$  for every  $t \in [0, 1]$ .

For each  $t$  there is therefore a unique smooth vector field  $X_t$  on  $B_\varepsilon(0)$  satisfying

$$\iota_{X_t} \omega_t = -\alpha. \quad (\text{D.4})$$

Since  $\alpha_0 = 0$  and  $\omega_t|_0$  is nondegenerate,

$$X_t(0) = 0 \quad (0 \leq t \leq 1). \quad (\text{D.5})$$

We next choose one neighborhood on which the time-dependent flow exists for the full interval  $[0, 1]$ . Smoothness and (D.5) imply, after perhaps shrinking  $\varepsilon$ , that there is  $C > 0$  with

$$|X_t(x)| \leq C|x| \quad (t, x) \in [0, 1] \times B_\varepsilon(0). \quad (\text{D.6})$$

Choose  $0 < \delta < \varepsilon e^{-C}$ . If  $\phi_t(x)$  is the solution of

$$\dot{\phi}_t(x) = X_t(\phi_t(x)), \quad \phi_0(x) = x,$$

then Gronwall's inequality and (D.6) give

$$|\phi_t(x)| \leq e^{Ct}|x| < \varepsilon \quad (0 \leq t \leq 1, |x| < \delta).$$

Thus  $\phi_t$  is defined on  $B_\delta(0)$  for the whole interval, and  $\phi_t(0) = 0$ .

Differentiate the pulled-back time-dependent form. Using  $\dot{\omega}_t = \eta = d\alpha$ , Cartan's formula,  $d\omega_t = 0$ , and (D.4),

$$\begin{aligned} \frac{d}{dt}\phi_t^*\omega_t &= \phi_t^*(\dot{\omega}_t + \mathcal{L}_{X_t}\omega_t) \\ &= \phi_t^*(d\alpha + d(\iota_{X_t}\omega_t) + \iota_{X_t}d\omega_t) \\ &= \phi_t^*(d\alpha - d\alpha) = 0. \end{aligned} \tag{D.7}$$

Hence

$$\phi_t^*\omega_t = \omega_0 = \omega_{\text{st}} \quad (0 \leq t \leq 1),$$

and in particular

$$\phi_1^*\tilde{\omega} = \omega_{\text{st}} \quad \text{on } B_\delta(0). \tag{D.8}$$

**Step 4: return to the manifold.** Let

$$U := \chi^{-1}(\phi_1(B_\delta(0)))$$

and define

$$\Psi := \phi_1^{-1} \circ \chi : U \longrightarrow B_\delta(0).$$

From (D.8), equivalently

$$\tilde{\omega} = (\phi_1^{-1})^*\omega_{\text{st}},$$

we obtain

$$\Psi^*\omega_{\text{st}} = \chi^*(\phi_1^{-1})^*\omega_{\text{st}} = \chi^*\tilde{\omega} = \omega|_U.$$

Thus  $\Psi$  is a local symplectomorphism from a neighborhood of  $p$  to an open neighborhood of 0 in  $(\mathbb{R}^{2n}, \omega_{\text{st}})$ . This is Darboux's theorem.  $\square$

### Theorem K01.2. Hamiltonian flows preserve the symplectic form

**Statement.** Let  $(M, \omega)$  be symplectic and let  $H \in C^2(M)$ . If  $\phi_H^t$  denotes the local Hamiltonian flow of  $X_H$ , then

$$(\phi_H^t)^*\omega = \omega$$

for every  $t$  for which the flow is defined.

*Proof.* Let  $X_H$  be defined by the book convention

$$\iota_{X_H}\omega = -dH,$$

and let  $\phi_H^t$  denote its local flow. Wherever the flow is defined, the standard differentiation formula for pullbacks along a flow gives

$$\frac{d}{dt}(\phi_H^t)^*\omega = (\phi_H^t)^*(\mathcal{L}_{X_H}\omega). \quad (\text{H.1})$$

Cartan's formula and the defining properties of a symplectic form yield

$$\mathcal{L}_{X_H}\omega = d(\iota_{X_H}\omega) + \iota_{X_H}d\omega = d(-dH) + 0 = 0. \quad (\text{H.2})$$

Therefore the right-hand side of (H.1) vanishes identically, so  $(\phi_H^t)^*\omega$  is independent of  $t$ . At  $t = 0$  the flow is the identity, hence

$$(\phi_H^t)^*\omega = \omega$$

for every time for which  $\phi_H^t$  is defined. □

### Proposition K01.3. Poisson bracket and commutator identity

**Statement.** With the Poisson-bracket convention of Definition K01.3, for smooth functions  $F, G$  one has

$$[X_F, X_G] = X_{\{F, G\}}.$$

Consequently, a smooth function  $F$  is a first integral of the Hamiltonian flow of  $H$  if and only if  $\{F, H\} = 0$ .

*Proof.* We use throughout

$$\iota_{X_H}\omega = -dH, \quad \{F, G\} = \omega(X_F, X_G) = dG(X_F) = -dF(X_G). \quad (\text{P.1})$$

By the preceding theorem that Hamiltonian flows preserve the symplectic form, every Hamiltonian vector field generates a symplectic local flow, so

$$\mathcal{L}_{X_F}\omega = 0. \quad (\text{P.2})$$

For any vector fields  $X, Y$  and differential form  $\beta$  one has the standard commutator formula

$$\iota_{[X, Y]}\beta = \mathcal{L}_X(\iota_Y\beta) - \iota_Y(\mathcal{L}_X\beta). \quad (\text{P.3})$$

Apply (P.3) with  $X = X_F$ ,  $Y = X_G$ , and  $\beta = \omega$ . Using (P.2), the Hamiltonian convention, and the fact that Lie derivative commutes with exterior differentiation,

$$\begin{aligned} \iota_{[X_F, X_G]}\omega &= \mathcal{L}_{X_F}(\iota_{X_G}\omega) - \iota_{X_G}(\mathcal{L}_{X_F}\omega) \\ &= \mathcal{L}_{X_F}(-dG) \\ &= -d(\mathcal{L}_{X_F}G) \\ &= -d(dG(X_F)) \\ &= -d\{F, G\}. \end{aligned} \quad (\text{P.4})$$



But the Hamiltonian vector field of the function  $\{F, G\}$  is characterized by

$$\iota_{X_{\{F, G\}}} \omega = -d\{F, G\}.$$

Nondegeneracy of  $\omega$  and (P.4) therefore imply

$$[X_F, X_G] = X_{\{F, G\}}. \quad (\text{P.5})$$

Now let  $\phi_H^t$  be the Hamiltonian flow of  $H$ . Along any of its trajectories,

$$\frac{d}{dt} F(\phi_H^t(x)) = dF(X_H)(\phi_H^t(x)) = -\{F, H\}(\phi_H^t(x)) \quad (\text{P.6})$$

by (P.1). Hence  $\{F, H\} = 0$  implies that  $F$  is constant along every  $H$ -trajectory. Conversely, if  $F$  is a first integral of the  $H$ -flow, differentiating at  $t = 0$  in (P.6) gives  $\{F, H\} = 0$  at every point. This proves both assertions.  $\square$

#### Proposition K01.4. Generating functions for exact symplectic maps

**Statement.** In canonical cotangent coordinates, write  $\vartheta = \sum_i p_i dq_i$  and  $\omega = d\vartheta = \sum_i dp_i \wedge dq_i$ . Let  $F$  be a local symplectomorphism and suppose  $F^*\vartheta - \vartheta$  is exact. Then its graph is an exact Lagrangian submanifold of the product with form  $-\omega \oplus \omega$ . If the projection of the graph to the  $(q, Q)$  variables is nonsingular at a point, then near that point there is a first-kind generating function  $S(q, Q)$  satisfying  $p = -\partial_q S$  and  $P = \partial_Q S$ .

*Proof.* Work in canonical coordinates  $(q, p)$  on the source and  $(Q, P)$  on the target. Put

$$\vartheta = \sum_{i=1}^n p_i dq_i, \quad \omega = d\vartheta = \sum_{i=1}^n dp_i \wedge dq_i. \quad (\text{G.1})$$

On the product of source and target let

$$\widehat{\Omega} = -\pi_1^* \omega + \pi_2^* \omega, \quad \widehat{\Theta} = \pi_2^* \vartheta - \pi_1^* \vartheta. \quad (\text{G.2})$$

Then  $\widehat{\Omega} = d\widehat{\Theta}$ .

Let  $F : U \rightarrow V$  be the given local symplectomorphism and use the graph parametrization

$$j : U \longrightarrow M^- \times M, \quad j(z) = (z, F(z)).$$

Its image is the graph  $\Gamma_F$ . Since  $F^*\omega = \omega$ ,

$$j^* \widehat{\Omega} = -\omega + F^* \omega = 0. \quad (\text{G.3})$$

The graph has dimension  $2n$ , exactly half the dimension of the product. Thus (G.3) says that  $\Gamma_F$  is Lagrangian.

Assume now that  $F$  is exact with respect to the canonical primitive, so

$$F^* \vartheta - \vartheta = dA \quad (\text{G.4})$$

for a smooth function  $A$  on the source neighborhood. Pulling back  $\widehat{\Theta}$  to the graph gives

$$j^*\widehat{\Theta} = F^*\vartheta - \vartheta = dA. \quad (\text{G.5})$$

Hence the graph is exact Lagrangian, with primitive  $A \circ \pi_1|_{\Gamma_F}$  under the natural identification of the graph with the source.

Consider the projection

$$\rho : \Gamma_F \longrightarrow \mathbb{R}_q^n \times \mathbb{R}_Q^n, \quad (q, p, Q, P) \longmapsto (q, Q). \quad (\text{G.6})$$

By hypothesis,  $d\rho$  is invertible at the point under consideration. The inverse function theorem therefore makes  $(q, Q)$  local coordinates on  $\Gamma_F$ . Write the primitive in those coordinates as  $S(q, Q)$ . On the graph,

$$\widehat{\Theta} = P dQ - p dq,$$

so (G.5) becomes

$$dS = -p dq + P dQ. \quad (\text{G.7})$$

Comparison of coefficients in (G.7) yields

$$p = -\frac{\partial S}{\partial q}, \quad P = \frac{\partial S}{\partial Q}. \quad (\text{G.8})$$

Thus  $S(q, Q)$  is the required local first-kind generating function.  $\square$

### Theorem K01.5. Weinstein Lagrangian neighborhood theorem

**Statement.** Let  $L$  be a compact Lagrangian submanifold of a symplectic manifold  $(M, \omega)$ . Then there are neighborhoods  $U$  of  $L$  in  $M$  and  $V$  of the zero section in  $T^*L$  and a symplectomorphism

$$\Phi : (U, \omega) \longrightarrow (V, \omega_{\text{can}})$$

whose restriction to  $L$  is the canonical identification of  $L$  with the zero section.

*Proof.* Let  $L \subset (M, \omega)$  be compact and Lagrangian. Denote by

$$i_0 : L \hookrightarrow T^*L$$

the zero section and by  $\omega_0$  the canonical symplectic form on  $T^*L$ . We construct a symplectic embedding from a neighborhood of the zero section into  $M$  that restricts to the given inclusion of  $L$ .

**Step 1: the normal bundle is canonically  $T^*L$ .** At  $x \in L$  let

$$N_x L := T_x M / T_x L.$$

Define

$$B_x : N_x L \longrightarrow T_x^* L, \quad B_x([v])(u) := \omega_x(v, u), \quad u \in T_x L. \quad (\text{W.1})$$

This is well defined because  $\omega_x|_{T_x L} = 0$ . If  $B_x([v]) = 0$ , then  $v$  is symplectically orthogonal

to  $T_x L$ ; since  $T_x L$  is Lagrangian,

$$(T_x L)^{\omega_x} = T_x L,$$

so  $[v] = 0$ . Thus  $B_x$  is injective, hence an isomorphism by dimension. The maps  $B_x$  vary smoothly, giving

$$NL \cong T^*L. \quad (\text{W.2})$$

The tubular-neighborhood theorem, applied through (W.2), therefore gives neighborhoods  $N_0$  of the zero section in  $T^*L$  and  $N$  of  $L$  in  $M$ , and a diffeomorphism

$$\psi : N_0 \longrightarrow N$$

with  $\psi \circ i_0 = i$ , where  $i : L \hookrightarrow M$  is the given inclusion. Put

$$\omega_1 := \psi^* \omega. \quad (\text{W.3})$$

The zero section is Lagrangian for both  $\omega_0$  and  $\omega_1$ .

**Step 2: make the two forms agree as bilinear forms along the zero section.** We use the following elementary linear fact. Suppose  $V$  carries symplectic forms  $\Omega_0, \Omega_1$  and a common Lagrangian subspace  $U$ . Then there is a linear isomorphism

$$A : (V, \Omega_0) \longrightarrow (V, \Omega_1)$$

with  $A|_U = \text{id}_U$ .

Indeed, choose a complement  $W$  of  $U$ . For  $r = 0, 1$ , the pairing

$$b_r : W \longrightarrow U^*, \quad b_r(w)(u) = \Omega_r(w, u)$$

is an isomorphism. Correct  $W$  to an  $\Omega_r$ -Lagrangian complement

$$W_r = \{w + C_r w : w \in W\}$$

by requiring

$$b_r(w_1)(C_r w_2) = -\frac{1}{2} \Omega_r(w_1, w_2). \quad (\text{W.4})$$

For fixed  $w_2$ , the defining equation has a unique solution  $C_r w_2 \in U$  because the transpose pairing  $U \rightarrow W^*$  induced by  $b_r$  is an isomorphism. The solution depends linearly on  $w_2$ , and a direct substitution shows that  $\Omega_r|_{W_r} = 0$ . The cross-pairing  $W_r \times U \rightarrow \mathbb{R}$  is nondegenerate. Hence there is a unique isomorphism  $D : W_0 \rightarrow W_1$  satisfying

$$\Omega_1(Dw, u) = \Omega_0(w, u) \quad (w \in W_0, u \in U). \quad (\text{W.5})$$

Then  $A = \text{id}_U \oplus D$  has  $A^* \Omega_1 = \Omega_0$ , because both forms vanish on  $U \times U$  and on the corresponding Lagrangian complements, while (W.5) matches the mixed terms.

Apply this construction smoothly at every  $x \in L$  with

$$V = T_{(x,0)} T^*L, \quad U = T_x L, \quad \Omega_0 = (\omega_0)_{(x,0)}, \quad \Omega_1 = (\omega_1)_{(x,0)}.$$

Choosing the auxiliary complements smoothly makes the resulting  $A_x$  smooth in  $x$ . We

spell out the standard first-jet realization. Along the zero section there is the canonical splitting

$$T_{(x,0)}T^*L = di_0(T_xL) \oplus V_x, \quad V_x := \ker d\pi_{(x,0)} \cong T_x^*L.$$

Since  $A_x$  fixes  $di_0(T_xL)$  pointwise, there are smooth bundle maps

$$B_x : V_x \rightarrow T_xL, \quad C_x : V_x \rightarrow V_x$$

with  $C_x$  invertible and

$$A_x(u, v) = (u + B_xv, C_xv) \tag{W.6a}$$

in this splitting. Choose a Riemannian metric and a connection on  $T^*L$ . For  $v$  small define

$$h(x, v) := (\exp_x(B_xv), \mathcal{P}_{x \rightarrow \exp_x(B_xv)}(C_xv)), \tag{W.6b}$$

where  $\mathcal{P}$  is parallel transport along the short radial geodesic. At  $v = 0$  this fixes the zero section; differentiating (W.6b) there gives exactly (W.6a). Since  $C_x$  is invertible,  $dh_{(x,0)} = A_x$  is invertible, and compactness of  $L$  plus the inverse function theorem gives, after shrinking to one neighborhood  $N'_0$ , a diffeomorphism

$$h : N'_0 \longrightarrow h(N'_0) \subset N_0$$

such that

$$h|_L = \text{id}_L, \quad dh_{(x,0)} = A_x. \tag{W.6}$$

Set

$$\tilde{\omega}_1 := h^*\omega_1.$$

By construction,

$$(\tilde{\omega}_1)_{(x,0)} = (\omega_0)_{(x,0)} \quad (x \in L) \tag{W.7}$$

as bilinear forms on the full tangent space, not merely after restriction to  $T_xL$ .

**Step 3: relative Moser deformation fixing  $L$ .** Let

$$\eta := \tilde{\omega}_1 - \omega_0.$$

Then  $d\eta = 0$ , and by (W.7) the tensor  $\eta$  vanishes at every point of the zero section. On a fiberwise star-shaped neighborhood of the zero section, the fiber-dilation homotopy

$$r_s(x, \xi) = (x, s\xi), \quad 0 \leq s \leq 1,$$

gives a one-form  $\sigma$  satisfying

$$d\sigma = \eta, \quad \sigma|_L = 0. \tag{W.8}$$

Explicitly,  $\sigma$  is the usual homotopy operator obtained by integrating the contraction of  $\eta$  with the fiber-radial velocity of  $r_s$ . The homotopy identity gives

$$d\sigma = \eta - \pi^*i_0^*\eta = \eta,$$

since  $i_0^*\eta = 0$ .

Define

$$\omega_t := \omega_0 + t\eta, \quad 0 \leq t \leq 1. \quad (\text{W.9})$$

Equation (W.7) gives  $\omega_t = \omega_0$  along the zero section for every  $t$ . Since  $L$  is compact and nondegeneracy is open, there is one neighborhood  $U$  of the zero section on which all  $\omega_t$  are symplectic. Solve uniquely

$$\iota_{X_t}\omega_t = -\sigma. \quad (\text{W.10})$$

Because  $\sigma|_L = 0$ , we have  $X_t|_L = 0$ . After shrinking  $U$  once more, compactness of  $L$  gives a time-dependent flow  $\varphi_t$  on  $U$  for all  $0 \leq t \leq 1$ ; it fixes the zero section pointwise. Cartan's formula, (W.8), and (W.10) give

$$\begin{aligned} \frac{d}{dt}\varphi_t^*\omega_t &= \varphi_t^*(\eta + \mathcal{L}_{X_t}\omega_t) \\ &= \varphi_t^*(d\sigma + d(\iota_{X_t}\omega_t)) = 0. \end{aligned} \quad (\text{W.11})$$

Hence

$$\varphi_1^*\tilde{\omega}_1 = \omega_0. \quad (\text{W.12})$$

Finally define

$$\Phi := \psi \circ h \circ \varphi_1.$$

On a sufficiently small neighborhood  $U$  of the zero section,

$$\Phi^*\omega = \varphi_1^*h^*\psi^*\omega = \varphi_1^*\tilde{\omega}_1 = \omega_0,$$

and every factor restricts to the identity on  $L$ . Thus  $\Phi$  is the required symplectomorphism from a neighborhood of the zero section in  $T^*L$  onto a neighborhood of  $L$  in  $M$ , fixing  $L$  pointwise.  $\square$

### Theorem K01.6. Noether theorem in Hamiltonian form

**Statement.** Let a Lie group  $G$  act Hamiltonianly on  $(M, \omega)$  with moment map  $J : M \rightarrow \mathfrak{g}^*$  in the convention of Definition K01.5, and let  $H \in C^2(M)$  be  $G$ -invariant. Then for every  $\xi \in \mathfrak{g}$  the component  $J_\xi = \langle J, \xi \rangle$  is constant along every trajectory of the Hamiltonian flow of  $H$  on its interval of definition.

*Proof.* Fix  $\xi \in \mathfrak{g}$  and write

$$J_\xi := \langle J, \xi \rangle.$$

By the moment-map definition above, if  $\xi_M$  is the infinitesimal generator of the one-parameter subgroup  $\exp(t\xi)$ , then

$$dJ_\xi = -\iota_{\xi_M}\omega. \quad (\text{N.1})$$

Let  $\phi_H^t$  be a trajectory of the Hamiltonian vector field  $X_H$ , defined on some interval  $I$ . For every  $t \in I$ ,

$$\frac{d}{dt}J_\xi(\phi_H^t(x)) = dJ_\xi(X_H) = -\omega(\xi_M, X_H) = \omega(X_H, \xi_M). \quad (\text{N.2})$$

The book convention  $\iota_{X_H}\omega = -dH$  therefore gives

$$\frac{d}{dt}J_\xi(\phi_H^t(x)) = -dH(\xi_M). \quad (\text{N.3})$$

It remains only to use the symmetry hypothesis. Because  $H$  is  $G$ -invariant,

$$H(\exp(t\xi) \cdot y) = H(y) \quad (y \in M)$$

whenever the action is evaluated. Differentiating at  $t = 0$  yields

$$dH_y(\xi_M(y)) = 0. \quad (\text{N.4})$$

Substituting (N.4) into (N.3) shows

$$\frac{d}{dt}J_\xi(\phi_H^t(x)) = 0 \quad (t \in I).$$

Hence  $J_\xi$  is constant along every Hamiltonian trajectory of  $H$  on its interval of definition. Since  $\xi \in \mathfrak{g}$  was arbitrary, all moment-map components are conserved.  $\square$

## 1.4 Worked examples

### Worked example: the harmonic oscillator as a sign and geometry test

Take  $M = \mathbb{R}^2$  with coordinates  $(q, p)$  and the book's canonical form

$$\omega = dp \wedge dq, \quad H(q, p) = \frac{1}{2}(p^2 + q^2).$$

Write  $X_H = a \partial_q + b \partial_p$ . Since

$$\iota_{X_H}(dp \wedge dq) = b dq - a dp$$

and  $-dH = -q dq - p dp$ , we obtain  $a = p$  and  $b = -q$ . Hence

$$X_H = p \partial_q - q \partial_p, \quad \dot{q} = p, \quad \dot{p} = -q.$$

The solution is a rotation,

$$q(t) = q_0 \cos t + p_0 \sin t, \quad p(t) = p_0 \cos t - q_0 \sin t,$$

so the flow matrix has determinant one and in fact preserves  $dp \wedge dq$ . This concrete calculation is the coordinate shadow of the general theorem  $(\varphi_H^t)^*\omega = \omega$ .

The energy circles  $p^2 + q^2 = 2E$  are one-dimensional Lagrangian submanifolds: in a two-dimensional symplectic manifold every one-dimensional submanifold is automatically isotropic, and its dimension is  $n = 1$ . Later these circles become the simplest model of invariant Lagrangian tori.

**Boundary case: why nondegeneracy cannot be dropped**

On  $\mathbb{R}^2$ , replace  $dp \wedge dq$  by the zero two-form. It is closed but degenerate. The equation

$$\iota_X \omega = -dH$$

then has no solution whenever  $dH \neq 0$ , and infinitely many solutions when  $dH = 0$ . Thus the Hamiltonian vector field is no longer determined by the Hamiltonian. The very first step in Hamiltonian dynamics has failed, so neither the preservation theorem nor the Poisson construction can be used in their symplectic form.

**Common pitfall**

With  $\omega = dp \wedge dq$ , formulas copied from a text using  $dq \wedge dp$  may acquire a sign. Never repair such signs by memory: return to  $\iota_{X_H} \omega = -dH$  and recompute.

## 1.5 Chapter summary

**What to remember**

- A symplectic form is the geometric structure preserved by Hamiltonian evolution.
- Darboux says symplectic geometry has no local scalar curvature-type invariant: locally every symplectic form is canonical.
- Lagrangian submanifolds are the correct geometric class for the invariant tori of integrable and KAM systems.
- Exact symplectic maps admit generating functions under a projection nondegeneracy condition.
- Moment maps convert infinitesimal symmetry into Hamiltonians; invariance of  $H$  then yields conserved quantities.

**Looking ahead**

K02 asks what happens when a Hamiltonian possesses enough commuting first integrals. The answer is the Liouville–Arnold theorem: regular compact common level sets become Lagrangian tori with action–angle coordinates, giving the unperturbed systems to which KAM theory applies.

**Check your understanding**

Explain, without coordinates, why the Hamiltonian flow preserves  $\omega$ . Then explain why this fact alone does not imply that a given invariant submanifold is Lagrangian.

## 1.6 Exercises and problems

### Exercise 1.1

**Difficulty.** Core.

**Related results.** Darboux theorem; Hamiltonian flows preserve the symplectic form; Poisson bracket and commutator identity.

**Problem.** For the harmonic oscillator, compute  $X_H$ , its flow, and  $\Phi_t^*\omega$ . Then compute the coordinate Hamiltonian fields and verify the Poisson-bracket/commutator identity with the sign convention of this volume.

**Solution sketch.** Solve the contraction equations explicitly, integrate the linear system, and derive both sides of the bracket identity from the definitions.

### Exercise 1.2

**Difficulty.** Core.

**Related results.** Generating functions for exact symplectic maps; Weinstein Lagrangian neighborhood theorem; Noether theorem in Hamiltonian form.

**Problem.** For the degenerate two-form in the boundary example, find its degeneracy locus and exhibit a Hamiltonian differential for which  $\iota_X\omega = dH$  loses existence or uniqueness. Pinpoint the failed symplectic input in the linked constructions.

**Solution sketch.** Write the contraction map as a matrix, compute its rank, and compare solvability on and off the degeneracy set.

### Exercise 1.3

**Difficulty.** Advanced.

**Related results.** Poisson bracket and commutator identity.

**Problem.** Prove from Cartan's formula that Hamiltonian flows preserve  $\omega$ , then derive the Poisson-bracket/commutator identity without using the parent result as a black box.

**Solution sketch.** Compute  $L_{X_f}\omega$ , contract the commutator with  $\omega$ , and check the final sign against the book's convention.

### Exercise 1.4

**Difficulty.** Advanced.

**Related results.** Generating functions for exact symplectic maps.

**Problem.** Give an area-preserving annulus map for which  $F^*\lambda - \lambda$  is closed but nonexact. Show why symplecticity alone does not yield the global exact generating function used here.

**Solution sketch.** Compute the period of  $F^*\lambda - \lambda$  around the circle; a nonzero period is the exactness obstruction.

### Exercise 1.5

**Difficulty.** Advanced.

**Related results.** Darboux theorem; Hamiltonian flows preserve the symplectic form.



**Problem.** Use a Weinstein/Darboux neighborhood of a Lagrangian  $L$  to describe the first-order Hamiltonian deformation of  $L$  as a graph of a one-form and show that the deformed submanifold remains Lagrangian.

**Solution sketch.** Pass to a cotangent model, identify the first variation with  $dH|_L$ , and use symplectic invariance of the flow.

# Integrable Hamiltonian Systems and Action-Angle Variables

## Why this chapter matters

An integrable Hamiltonian system is one for which the geometry of motion can be reorganized into invariant tori. On each regular compact torus the motion is linear in angle variables, while the Hamiltonian depends only on action variables. This is the reference picture that KAM theory perturbs.

The subtle point is that “integrable” is not the same as “globally trivial.” Action-angle coordinates are local near a regular torus, and the period lattice can twist as one moves through the base of a torus fibration. Hamiltonian monodromy records that global obstruction. At the same time, the derivative of the frequency map determines whether nearby frequencies can be used as coordinates; Kolmogorov and isoenergetic nondegeneracy are the first versions of the transversality conditions that later drive KAM persistence.

**Prerequisites.** K01, commuting vector fields, regular-value theory, compact Lie-group quotients, and elementary covering-space/monodromy ideas.

## Learning goals

- Derive the torus structure and action-angle coordinates near a compact regular Liouville fiber.
- Compute frequency maps and distinguish Kolmogorov from isoenergetic nondegeneracy.
- Describe resonance modules and their manifolds in action space.
- Understand period lattices and why monodromy can obstruct global action coordinates.

## Conceptual roadmap

The structural chain is

commuting first integrals  $\longrightarrow$  commuting Hamiltonian flows  
 $\longrightarrow$  compact torus  $\longrightarrow (I, \theta)$  coordinates  
 $\longrightarrow \omega(I)$ .

Perturbation theory then splits according to the arithmetic of  $\omega(I)$ : nonresonant frequencies allow full averaging, while integer relations  $k \cdot \omega = 0$  create resonance

manifolds. The period lattice explains both the local construction of the actions and the possibility of global monodromy.

### Check your understanding

For  $H_0(I) = \omega_0 \cdot I$ , compute  $D\omega$ . The system is perfectly integrable, yet Kolmogorov nondegeneracy fails. This distinction between integrability and nondegeneracy is central to everything that follows.

## 2.1 Core definitions

### Intuition

Integrability produces coordinates adapted to the motion. Nondegeneracy is a different property: it says that the frequencies change sufficiently when the actions change. KAM theory needs both ideas but uses them for different purposes.

### Definition K02.1. Liouville integrability

A Hamiltonian system on a  $2n$ -dimensional symplectic manifold is *Liouville integrable* on a region if it possesses  $n$  smooth first integrals whose differentials are independent there and whose pairwise Poisson brackets vanish.

### Definition K02.2. Action-angle coordinates

*Action-angle coordinates* are canonical coordinates  $(I, \theta) \in U \times \mathbb{T}^n$  in which

$$\omega = \sum_{j=1}^n dI_j \wedge d\theta_j$$

and an integrable Hamiltonian has the form  $H_0 = H_0(I)$ . The equations are then  $\dot{I} = 0$  and  $\dot{\theta} = \nabla H_0(I)$ .

### Definition K02.3. Frequency map and Kolmogorov nondegeneracy

For  $H_0(I)$ , the *frequency map* is  $\omega(I) = \nabla H_0(I)$ . Kolmogorov nondegeneracy at  $I_*$  means

$$\det D\omega(I_*) = \det D^2 H_0(I_*) \neq 0.$$

### Definition K02.4. Resonance module

For  $\omega \in \mathbb{R}^n$ , the exact resonance module is

$$\mathcal{R}(\omega) = \{k \in \mathbb{Z}^n : k \cdot \omega = 0\}.$$

A chosen rank- $r$  module  $\Lambda \leq \mathbb{Z}^n$  specifies the integer relations one intends to retain in a resonant normal form.

**Definition K02.5. Period lattice and Hamiltonian monodromy**

For a regular Lagrangian torus fibration  $\pi : M \rightarrow B$ , the *period lattice* at  $b \in B$  consists of covectors whose associated vertical Hamiltonian flows have time one equal to the identity on the fiber. Transport of a lattice basis around a loop in  $B$  can return a different integral basis; the resulting representation into  $\mathrm{GL}(n, \mathbb{Z})$  is the *Hamiltonian monodromy*.

## 2.2 How the main proofs work

**Intuition**

**Liouville-Arnold.** The independent commuting Hamiltonian vector fields span the tangent bundle of the compact regular fiber. Their  $\mathbb{R}^n$ -action is transitive; compactness turns the stabilizer into a full lattice, so the fiber is  $\mathbb{R}^n/\Lambda \cong \mathbb{T}^n$ . Period one-forms then produce the action variables, and a local Lagrangian section supplies the angles.

**Frequency and nondegeneracy.** Once  $H_0 = H_0(I)$ , the dynamical frequency is simply  $\nabla H_0$ . Kolmogorov nondegeneracy is the inverse-function theorem applied to this map. Isoenergetic nondegeneracy instead asks for transversality of the projective frequency along an energy surface, encoded by the bordered Hessian.

**Resonance manifolds.** Put the chosen integer relations into a map  $G_\Lambda(I)$ . If its derivative has full rank, the implicit-function theorem makes  $G_\Lambda^{-1}(0)$  a codimension- $r$  submanifold.

**Monodromy.** Local action coordinates amount to a local integral basis of the period lattice. A nontrivial return of this basis around a loop gives a  $\mathrm{GL}(n, \mathbb{Z})$  obstruction to one globally single-valued action chart.

## 2.3 Main results

**Theorem K02.1. Liouville-Arnold theorem**

**Statement.** Let  $(M^{2n}, \omega, H)$  be Hamiltonian and let  $F = (F_1, \dots, F_n)$  be smooth first integrals of  $H$  which are pairwise in involution and whose differentials are independent near a compact connected regular fiber  $\Lambda = F^{-1}(c)$ . Then  $\Lambda \cong \mathbb{T}^n$ . After restricting to a neighborhood  $N$  of  $\Lambda$ , there are coordinates  $(I, \theta) \in V \times (\mathbb{R}/2\pi\mathbb{Z})^n$  with  $\omega = \sum_{j=1}^n dI_j \wedge d\theta_j$ , each nearby common level of  $F$  given by  $I = \text{const}$ , and  $H = h(I)$  for a smooth  $h$ .

*Proof.* Let

$$F = (F_1, \dots, F_n) : M \longrightarrow \mathbb{R}^n, \quad \Lambda = F^{-1}(c),$$

where the  $dF_i$  are independent on a neighborhood of the compact connected fiber  $\Lambda$ , the  $F_i$  Poisson commute, and each  $F_i$  is a first integral of  $H$ .

**Step 1: the regular fiber is Lagrangian and is an  $\mathbb{R}^n$ -homogeneous space.** Put  $X_i := X_{F_i}$ . If

$$\sum_{i=1}^n a_i X_i(x) = 0,$$

then contraction with  $\omega$  and the convention  $\iota_{X_{F_i}}\omega = -dF_i$  give

$$\sum_{i=1}^n a_i dF_i(x) = 0.$$

Independence of the differentials implies  $a_i = 0$  for all  $i$ . Hence the  $X_i$  are pointwise independent.

For every  $i, j$ , the K01 Poisson convention gives

$$dF_j(X_i) = \{F_i, F_j\} = 0, \quad (\text{LA.1})$$

so each  $X_i$  is tangent to every regular common level. Since the common level has dimension  $2n - n = n$ , the  $X_i$  span its tangent bundle. Moreover the commutator vanishes directly. Indeed,

$$\mathcal{L}_{X_i}\omega = d(\iota_{X_i}\omega) + \iota_{X_i}d\omega = -d^2F_i = 0,$$

and therefore

$$\begin{aligned} \iota_{[X_i, X_j]}\omega &= \mathcal{L}_{X_i}(\iota_{X_j}\omega) - \iota_{X_j}(\mathcal{L}_{X_i}\omega) \\ &= -\mathcal{L}_{X_i}(dF_j) = -d(dF_j(X_i)) = 0 \end{aligned} \quad (\text{LA.2})$$

by (LA.1). Nondegeneracy of  $\omega$  gives  $[X_i, X_j] = 0$ . The restrictions of the  $X_i$  to the compact manifold  $\Lambda$  are complete. Their commuting flows therefore define an action

$$\Phi : \mathbb{R}^n \times \Lambda \longrightarrow \Lambda. \quad (\text{LA.3})$$

At each point, the differential of the orbit map in the  $\mathbb{R}^n$  direction is an isomorphism onto  $T\Lambda$ . Thus every orbit is open. Since  $\Lambda$  is connected and its disjoint orbits are open, there is only one orbit.

Fix  $x_0 \in \Lambda$  and let

$$\Gamma_c := \{t \in \mathbb{R}^n : \Phi^t(x_0) = x_0\}. \quad (\text{LA.4})$$

The orbit-map differential at 0 is injective, so  $\Gamma_c$  is discrete. The orbit map induces a diffeomorphism

$$\mathbb{R}^n / \Gamma_c \cong \Lambda. \quad (\text{LA.5})$$

Because  $\Lambda$  is compact, the discrete subgroup  $\Gamma_c$  is cocompact. A discrete cocompact subgroup of  $\mathbb{R}^n$  is a full lattice, hence isomorphic to  $\mathbb{Z}^n$ . Therefore

$$\Lambda \cong \mathbb{R}^n / \Gamma_c \cong \mathbb{T}^n. \quad (\text{LA.6})$$

Equation (LA.1) also shows that the tangent spaces of the common levels are spanned by the  $X_i$ ; since

$$\omega(X_i, X_j) = \{F_i, F_j\} = 0,$$

these  $n$ -dimensional fibers are Lagrangian.

**Step 2: restrict to a smooth family of compact connected Liouville tori.** The compact-fiber form of the submersion/Ehresmann theorem, applied after shrinking around  $\Lambda$ , gives a ball  $U \ni c$  and a neighborhood  $N$  of  $\Lambda$  such that

$$F : N \longrightarrow U \tag{LA.7}$$

is a smooth locally trivial fibration whose fibers  $\Lambda_b := F^{-1}(b) \cap N$  are compact, connected, regular Lagrangian  $n$ -tori.

We also need a Lagrangian section. At  $x_0$  choose a Lagrangian complement  $E$  to  $T_{x_0}\Lambda$ . By the Darboux theorem, with a preliminary linear symplectic change of coordinates, there is a local Lagrangian submanifold  $S$  through  $x_0$  tangent to  $E$ . Since  $E$  is transverse to  $\ker dF_{x_0} = T_{x_0}\Lambda$ , the inverse-function theorem makes  $F|_S$  a diffeomorphism onto a smaller base ball. Shrinking  $U$  and  $N$  once more, we obtain a smooth Lagrangian section

$$s : U \longrightarrow N, \quad F \circ s = \text{id}_U. \tag{LA.8}$$

For  $b \in U$ , the commuting  $X_i$  give an  $\mathbb{R}^n$ -action on  $\Lambda_b$ . Let  $\Gamma_b$  be its isotropy lattice at  $s(b)$ . Choose a basis

$$\ell_1(c), \dots, \ell_n(c) \in \Gamma_c.$$

Each basis vector continues uniquely and smoothly for  $b$  near  $c$ : in a fiber chart around  $s(b)$ , the equation

$$\Phi^t(s(b)) = s(b) \tag{LA.9}$$

has, near  $(\ell_i(c), c)$ , a unique solution  $t = \ell_i(b)$  by the inverse-function theorem, because the derivative in  $t$  is the isomorphism

$$\mathbb{R}^n \ni a \longmapsto \sum_j a_j X_j(s(c)) \in T_{x_0}\Lambda.$$

After a further shrinking, the  $\ell_i(b)$  remain linearly independent and generate  $\Gamma_b$ : in the Ehresmann trivialization they represent the continuously transported basis of  $H_1(\Lambda_b; \mathbb{Z})$ , while the orbit map identifies  $\Gamma_b$  with this fundamental-group lattice. Thus

$$L(b) := (\ell_1(b) \ \cdots \ \ell_n(b)) \in \text{GL}(n, \mathbb{R}) \tag{LA.10}$$

for every  $b \in U$ .

**Step 3: normalize the lattice flows and construct the angle coordinates.** Write  $\ell_i(b) = (\ell_i^1(b), \dots, \ell_i^n(b))$ . Define the vertical vector field

$$Y_i := \frac{1}{2\pi} \sum_{j=1}^n \ell_i^j(F) X_j. \tag{LA.11}$$

Because the coefficients depend only on the common integrals, they are constant along every fiber. The flow of  $Y_i$  on  $\Lambda_b$  is the original  $\mathbb{R}^n$ -action with parameter  $t\ell_i(b)/(2\pi)$ ; hence it has period  $2\pi$ .

The section and these normalized flows define

$$\Psi : U \times (\mathbb{R}/2\pi\mathbb{Z})^n \longrightarrow N, \quad \Psi(b, \theta) := \Phi^{\frac{1}{2\pi} \sum_i \theta_i \ell_i(b)}(s(b)). \quad (\text{LA.12})$$

On each fiber this is precisely the quotient identification  $\mathbb{R}^n/\Gamma_b \cong \Lambda_b$  written in the chosen lattice basis. Hence, after the preceding shrinkings,  $\Psi$  is a diffeomorphism.

**Step 4: the period one-forms are closed.** From (LA.11) and  $\iota_{X_j}\omega = -dF_j$ ,

$$\iota_{Y_i}\omega = -F^*\beta_i, \quad \beta_i := \frac{1}{2\pi} \sum_{j=1}^n \ell_i^j(b) db_j. \quad (\text{LA.13})$$

Let  $\varphi_i^t$  be the flow of  $Y_i$ . It is vertical, so  $F \circ \varphi_i^t = F$ , and (LA.13) gives

$$\mathcal{L}_{Y_i}\omega = d(\iota_{Y_i}\omega) = -F^*(d\beta_i). \quad (\text{LA.14})$$

Consequently

$$(\varphi_i^{2\pi})^*\omega - \omega = \int_0^{2\pi} (\varphi_i^t)^*\mathcal{L}_{Y_i}\omega dt = -2\pi F^*(d\beta_i). \quad (\text{LA.15})$$

But  $\varphi_i^{2\pi} = \text{id}$ , hence  $F^*(d\beta_i) = 0$ . Since  $F$  is a submersion, pullback on base forms is pointwise injective, so

$$d\beta_i = 0. \quad (\text{LA.16})$$

Because  $U$  is a ball, there are smooth functions  $I_i : U \rightarrow \mathbb{R}$  with

$$dI_i = \beta_i. \quad (\text{LA.17})$$

The Jacobian of  $I = (I_1, \dots, I_n)$  with respect to  $b$  is  $(2\pi)^{-1}L(b)^T$ , which is invertible by (LA.10). Thus, after shrinking  $U$ , the  $I_i$  are coordinates on the base.

Equation (LA.16) and (LA.14) show that every  $Y_i$  is symplectic. Hence the torus translations in the  $\theta$  variables preserve  $\Psi^*\omega$ . Since the fibers are Lagrangian and the section  $\theta = 0$  is Lagrangian, the only possible terms are the mixed ones. By (LA.13),

$$\iota_{\partial_{\theta_i}}\Psi^*\omega = -\beta_i = -dI_i.$$

Therefore

$$\Psi^*\omega = \sum_{i=1}^n dI_i \wedge d\theta_i. \quad (\text{LA.18})$$

This gives the desired action-angle coordinates.

**Step 5: the Hamiltonian depends only on the actions.** Because each  $F_i$  is a first integral of  $H$ ,

$$dH(X_i) = 0. \quad (\text{LA.19})$$

The  $X_i$  span the tangent spaces of the fibers, so  $H$  is constant on each connected  $\Lambda_b$ . Hence  $H = \hat{h} \circ F$  on  $N$  for a smooth  $\hat{h}$  on  $U$ . Since  $I : U \rightarrow V := I(U)$  is a local diffeomorphism,

$$H = h(I), \quad h := \hat{h} \circ I^{-1}. \quad (\text{LA.20})$$

Together with (LA.6) and (LA.18), this proves every assertion.  $\square$

**Concept: Frequency map and Kolmogorov nondegeneracy**

**Statement.** For  $H_0 \in C^2(U)$  the frequency map is  $\omega(I) = \nabla H_0(I)$ . Kolmogorov nondegeneracy at  $I_*$  means  $\det D\omega(I_*) = \det D^2 H_0(I_*) \neq 0$ ; by the inverse-function theorem,  $\omega$  is then locally invertible at  $I_*$ .

**Theorem K02.2. Hamiltonian monodromy obstruction**

**Statement.** Let  $\pi : (M^{2n}, \omega) \rightarrow B^n$  be a connected regular Lagrangian torus fibration with compact fibers and period lattice  $\mathcal{P}$  as in Definition K02.5. Parallel transport of  $\mathcal{P}$  defines a monodromy representation  $\rho : \pi_1(B, b) \rightarrow \text{GL}(n, \mathbb{Z})$ , well defined up to conjugacy. If global action coordinates exist, then  $\mathcal{P}$  has a global integral basis and  $\rho$  is trivial. Hence nontrivial monodromy obstructs global action coordinates; moreover regular Lagrangian torus fibrations with nontrivial monodromy exist.

*Proof.* Let

$$\pi : (M^{2n}, \omega) \longrightarrow B^n$$

be a regular Lagrangian torus fibration with compact connected fibers.

**Step 1: local action coordinates give a lattice local system.** By the Liouville-Arnold theorem proved earlier in this volume, every  $b \in B$  has a neighborhood  $U$  with action-angle coordinates

$$(I_1, \dots, I_n, \theta_1, \dots, \theta_n), \quad \omega = \sum_{j=1}^n dI_j \wedge d\theta_j, \quad \theta_j \in \mathbb{R}/2\pi\mathbb{Z}, \quad (\text{MON.1})$$

and the fibers are  $I = \text{const.}$

For a covector  $a = \sum_j a_j dI_j \in T_b^*B$ , the vertical vector field determined by

$$\iota_{X_a} \omega = -\pi^* a$$

is

$$X_a = \sum_{j=1}^n a_j \partial_{\theta_j}. \quad (\text{MON.2})$$

Its time-one flow is the identity precisely when every  $a_j \in 2\pi\mathbb{Z}$ . Hence on  $U$  the period lattice is

$$\mathcal{P}|_U = \text{span}_{\mathbb{Z}}\{2\pi dI_1, \dots, 2\pi dI_n\} \subset T^*U. \quad (\text{MON.3})$$

On an overlap of two action charts, both displayed bases generate the same intrinsic period lattice. Therefore the change-of-basis matrix lies in  $\text{GL}(n, \mathbb{Z})$  and is locally constant. Thus  $\mathcal{P} \rightarrow B$  is a rank- $n$   $\mathbb{Z}$ -local system.

Parallel transport in this local system along a loop  $\gamma$  based at  $b$  gives an automorphism of  $\mathcal{P}_b$ . Choosing an integral basis identifies it with a matrix in  $\text{GL}(n, \mathbb{Z})$  and



defines

$$\rho : \pi_1(B, b) \longrightarrow \mathrm{GL}(\mathcal{P}_b) \cong \mathrm{GL}(n, \mathbb{Z}). \quad (\text{MON.4})$$

Changing the basis at  $b$  conjugates all matrices simultaneously, so the representation is intrinsic up to conjugacy.

**Step 2: global action coordinates force trivial monodromy.** Suppose global action coordinates  $I_1, \dots, I_n$  exist on  $B$  in the sense defined above. Then (MON.3) holds globally:

$$\mathcal{P} = \mathrm{span}_{\mathbb{Z}}\{2\pi dI_1, \dots, 2\pi dI_n\}. \quad (\text{MON.5})$$

The displayed covectors give a global integral basis of the local system. Parallel transport fixes this basis along every loop, hence

$$\rho([\gamma]) = I_n \quad \text{for every } [\gamma] \in \pi_1(B, b). \quad (\text{MON.6})$$

Therefore nontrivial monodromy is an obstruction to global action coordinates.

**Step 3: an explicit regular fibration with nontrivial monodromy.** It remains to prove that the obstruction can actually occur without importing a singular model. Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}), \quad \tau(x, y) = (x + y, y + 1) = A \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (\text{MON.7})$$

The iterates of  $\tau$  shift the second coordinate by an integer, so the induced  $\mathbb{Z}$ -action on  $\mathbb{R}^2$  is free and proper. Put

$$B := \mathbb{R}^2 / \langle \tau \rangle. \quad (\text{MON.8})$$

On

$$\widetilde{M} := \mathbb{R}_I^2 \times \mathbb{T}_\theta^2, \quad \omega_0 = dI_1 \wedge d\theta_1 + dI_2 \wedge d\theta_2, \quad (\text{MON.9})$$

define

$$\widetilde{\tau}(I, \theta) := (AI + (0, 1)^T, A^{-T}\theta), \quad (\text{MON.10})$$

where  $\theta$  is taken modulo  $2\pi\mathbb{Z}^2$ . Since  $A^{-T} \in \mathrm{SL}(2, \mathbb{Z})$ , this is well defined on the torus. Moreover

$$\widetilde{\tau}^*\omega_0 = \sum_i d(AI)_i \wedge d(A^{-T}\theta)_i = \sum_i dI_i \wedge d\theta_i = \omega_0. \quad (\text{MON.11})$$

The action generated by  $\widetilde{\tau}$  is free and proper because its projection to the base action is. Hence

$$M := \widetilde{M} / \langle \widetilde{\tau} \rangle \quad (\text{MON.12})$$

is a smooth symplectic manifold, and the projection to the first factor descends to

$$\pi : M \longrightarrow B. \quad (\text{MON.13})$$

Each fiber is a copy of  $\mathbb{T}^2$ , and (MON.9) shows that it is Lagrangian. Thus (MON.13) is a regular Lagrangian torus fibration with local action-angle coordinates inherited from  $(I, \theta)$ .

When one goes once around the generator of  $\pi_1(B)$ , the action-coordinate transition

is

$$I \mapsto AI + (0, 1)^T, \quad (\text{MON.14})$$

and therefore the period coframe transforms by

$$(dI_1, dI_2)^T \mapsto A(dI_1, dI_2)^T. \quad (\text{MON.15})$$

Thus the period-lattice monodromy around the generator is represented, in this coframe convention, by the nonidentity matrix  $A$ . (The induced monodromy on a dual basis of fiber one-cycles is  $A^{-T}$ .) In particular the period-lattice monodromy is nontrivial. By Step 2 this fibration admits no global action coordinates.

Thus nontrivial Hamiltonian monodromy exists and obstructs global action coordinates, as claimed.  $\square$

### Theorem K02.3. Isoenergetic nondegeneracy

**Statement.** Let  $H_0 \in C^2(U)$ ,  $\omega(I) = \nabla H_0(I)$ , and  $I_* \in U$ . With  $A = D\omega(I_*) = D^2H_0(I_*)$ , isoenergetic nondegeneracy means

$$\det \begin{pmatrix} A & \omega(I_*) \\ \omega(I_*)^T & 0 \end{pmatrix} \neq 0.$$

It is equivalent to the map  $I \mapsto (H_0(I), [\omega(I)]) \in \mathbb{R} \times \mathbb{RP}^{n-1}$  being a local diffeomorphism at  $I_*$ . Equivalently, on the regular energy hypersurface through  $I_*$ , the projective frequency map is locally nondegenerate.

*Proof.* Let

$$\omega_* := \omega(I_*) = \nabla H_0(I_*), \quad A := D\omega(I_*) = D^2H_0(I_*),$$

and set

$$B_* := \begin{pmatrix} A & \omega_* \\ \omega_*^T & 0 \end{pmatrix}. \quad (\text{IE.1})$$

If  $B_*$  is nonsingular, then necessarily  $\omega_* \neq 0$ ; otherwise the last row and column vanish. Hence the projective class  $[\omega(I)]$  is defined near  $I_*$  after shrinking the domain.

Consider

$$\mathcal{E}(I) := (H_0(I), [\omega(I)]) \in \mathbb{R} \times \mathbb{RP}^{n-1}. \quad (\text{IE.2})$$

For  $v \in \mathbb{R}^n$ ,

$$DH_0(I_*)v = \omega_*^T v. \quad (\text{IE.3})$$

The differential of the projectivization map at a nonzero vector  $\omega_*$  kills precisely the radial direction  $\mathbb{R}\omega_*$ . Therefore

$$D[\omega]_{I_*}v = 0 \iff Av \in \mathbb{R}\omega_*. \quad (\text{IE.4})$$

Consequently  $D\mathcal{E}(I_*)v = 0$  if and only if there is a scalar  $\lambda$  such that

$$\omega_*^T v = 0, \quad Av = \lambda\omega_*. \quad (\text{IE.5})$$

Equivalently,

$$B_* \begin{pmatrix} v \\ -\lambda \end{pmatrix} = 0. \quad (\text{IE.6})$$

Suppose first that  $\det B_* \neq 0$ . Then (IE.6) forces  $v = 0$ , so  $D\mathcal{E}(I_*)$  is injective. Both source and target of  $\mathcal{E}$  have dimension  $n$ , hence  $D\mathcal{E}(I_*)$  is an isomorphism. The inverse-function theorem on manifolds gives that  $\mathcal{E}$  is a local diffeomorphism at  $I_*$ .

Conversely, suppose  $\omega_* \neq 0$  and  $B_*$  is singular. Choose a nonzero  $(v, s)$  in its kernel. If  $v = 0$ , then  $s\omega_* = 0$ , hence  $s = 0$ , a contradiction. Thus  $v \neq 0$ , and

$$\omega_*^T v = 0, \quad Av = -s\omega_*.$$

By (IE.3)–(IE.4),  $D\mathcal{E}(I_*)v = 0$ . Hence  $\mathcal{E}$  cannot be a local diffeomorphism. This proves the equivalence.

Finally let  $E = H_0(I_*)$ . Since  $\omega_* \neq 0$ , the energy level

$$\Sigma_E := \{I : H_0(I) = E\}$$

is a smooth hypersurface near  $I_*$  with

$$T_{I_*}\Sigma_E = \{v : \omega_*^T v = 0\}. \quad (\text{IE.7})$$

By the same kernel calculation, under  $\det B_* \neq 0$  the map

$$[\omega]|_{\Sigma_E} : \Sigma_E \longrightarrow \mathbb{RP}^{n-1}$$

has injective differential at  $I_*$ . The two manifolds have the same dimension  $n-1$ , so this differential is an isomorphism and the restricted projective-frequency map is locally nondegenerate. Conversely, local nondegeneracy on the energy hypersurface gives the same kernel exclusion and therefore nonsingularity of  $B_*$ . This is precisely the isoenergetic condition claimed.  $\square$

#### Theorem K02.4. Resonant modules and resonance manifolds

**Statement.** Let  $\omega : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$  be  $C^1$  and let  $\Lambda \subset \mathbb{Z}^n$  be a rank- $r$  submodule with basis  $k^1, \dots, k^r$ . Define

$$G_\Lambda(I) := (k^1 \cdot \omega(I), \dots, k^r \cdot \omega(I)).$$

If  $G_\Lambda(I_*) = 0$  and  $DG_\Lambda(I_*)$  has rank  $r$ , then near  $I_*$  the exact  $\Lambda$ -resonance set  $G_\Lambda^{-1}(0)$  is a  $C^1$  submanifold of codimension  $r$ . The set and the rank condition are independent of the chosen  $\mathbb{Z}$ -basis of  $\Lambda$ .

*Proof.* Let  $\Lambda \subset \mathbb{Z}^n$  have rank  $r$  and choose a  $\mathbb{Z}$ -basis

$$k^1, \dots, k^r.$$

Write  $K$  for the  $r \times n$  integer matrix whose  $a$ th row is  $(k^a)^T$ . The resonance map is

$$G_\Lambda : U \longrightarrow \mathbb{R}^r, \quad G_\Lambda(I) = K\omega(I). \quad (\text{RM.1})$$

Thus

$$G_\Lambda(I) = 0 \iff k \cdot \omega(I) = 0 \text{ for every } k \in \Lambda, \quad (\text{RM.2})$$

because every element of  $\Lambda$  is an integer linear combination of the basis vectors.

Its differential is

$$DG_\Lambda(I) = K D\omega(I). \quad (\text{RM.3})$$

If  $G_\Lambda(I_*) = 0$  and  $DG_\Lambda(I_*)$  has rank  $r$ , then 0 is a regular value of  $G_\Lambda$  near  $I_*$ . The regular-level-set theorem therefore gives a neighborhood  $V$  of  $I_*$  such that

$$\mathcal{M}_\Lambda := \{I \in V : G_\Lambda(I) = 0\} \quad (\text{RM.4})$$

is a  $C^1$  embedded submanifold of  $V$  of dimension  $n - r$ , hence codimension  $r$ .

It remains to verify that neither the set nor the transversality condition depends on the chosen basis. If  $\tilde{k}^1, \dots, \tilde{k}^r$  is another  $\mathbb{Z}$ -basis of  $\Lambda$ , then there is a matrix

$$A \in \text{GL}(r, \mathbb{Z})$$

with  $\tilde{K} = AK$ . Consequently

$$\tilde{G}_\Lambda = AG_\Lambda, \quad D\tilde{G}_\Lambda = A DG_\Lambda. \quad (\text{RM.5})$$

Since  $A$  is invertible,  $\tilde{G}_\Lambda^{-1}(0) = G_\Lambda^{-1}(0)$  and

$$\text{rank } D\tilde{G}_\Lambda(I_*) = \text{rank } DG_\Lambda(I_*) = r. \quad (\text{RM.6})$$

This proves both claims.  $\square$

## 2.4 Worked examples

### Worked example: the harmonic oscillator in action-angle variables

Consider

$$H(q, p) = \frac{1}{2}(p^2 + q^2), \quad \omega = dp \wedge dq.$$

On the punctured plane set

$$q = \sqrt{2I} \cos \theta, \quad p = -\sqrt{2I} \sin \theta, \quad I > 0.$$

A direct calculation gives  $dp \wedge dq = dI \wedge d\theta$ , so these are canonical coordinates in the convention of this series. Moreover

$$H = I, \quad \dot{I} = 0, \quad \dot{\theta} = 1.$$

Every energy circle is therefore an invariant one-torus and the motion is rigid rotation. This is Liouville–Arnold in its simplest form.

The same example also shows why integrability and Kolmogorov nondegeneracy must not be conflated. Here  $\omega(I) = 1$ , so  $D\omega = 0$ : the system is completely

integrable but Kolmogorov-degenerate. If instead one considers the integrable normal form

$$H_a(I) = I + \frac{a}{2}I^2,$$

then  $\omega_a(I) = 1 + aI$  and  $D\omega_a = a$ ; for  $a \neq 0$  the frequency is locally invertible.

### Boundary case: the isotropic two-frequency oscillator

Let

$$H = I_1 + I_2.$$

The frequency vector is  $(1, 1)$ , so  $(1, -1) \cdot \omega = 0$ . Thus the exact resonance module is nontrivial. The system remains integrable, but a perturbation containing a Fourier mode  $e^{i(\theta_1 - \theta_2)}$  cannot be removed by dividing by  $k \cdot \omega$ : the denominator vanishes. K03 will keep precisely such modes in a resonant normal form.

### Common pitfall

Action-angle coordinates near one regular torus do not imply that a whole integrable fibration admits one global action coordinate system. Monodromy is the standard obstruction.

## 2.5 Chapter summary

### What to remember

- A compact connected regular Liouville fiber is a torus, and nearby dynamics admits action-angle coordinates.
- The frequency map describes linear motion on each invariant torus; its derivative measures the twist used later by KAM theory.
- Kolmogorov and isoenergetic nondegeneracy are transversality hypotheses, not definitions of integrability.
- Integer frequency relations define resonance modules and resonance manifolds.
- The period lattice can have nontrivial monodromy, obstructing global action variables even when local action-angle charts exist everywhere.

### Looking ahead

K03 perturbs the integrable picture. The first task is to solve a homological equation that removes nonresonant oscillations. Its denominators are exactly the quantities  $k \cdot \omega$  isolated in this chapter.

**Check your understanding**

Give one example of a system that is integrable but Kolmogorov-degenerate, and one example of a frequency vector with a nontrivial resonance module. Explain why these are logically different failures.

**2.6 Exercises and problems****Exercise 2.1**

**Difficulty.** Core.

**Related results.** Liouville–Arnold theorem; Frequency map and Kolmogorov non-degeneracy; Isoenergetic nondegeneracy.

**Problem.** Put the planar harmonic oscillator in action-angle variables, compute its frequency map, and show that Liouville–Arnold applies while Kolmogorov twist fails. Repeat the nondegeneracy tests for  $H(I) = \omega I + aI^2$ ,  $a \neq 0$ .

**Solution sketch.** Compute the action integral, differentiate  $H$ , and evaluate both the Hessian and the isoenergetic bordered determinant.

**Exercise 2.2**

**Difficulty.** Core.

**Related results.** Resonant modules and resonance manifolds; Hamiltonian monodromy obstruction.

**Problem.** For the isotropic two-frequency oscillator, compute the resonance module for  $\omega \propto (1, 1)$ . Explain why this resonance is not by itself Hamiltonian monodromy and identify the missing global singular-fibration ingredient.

**Solution sketch.** Solve  $k_1 + k_2 = 0$ , then contrast local resonance with transport of an integral homology basis around singular values.

**Exercise 2.3**

**Difficulty.** Advanced.

**Related results.** Isoenergetic nondegeneracy.

**Problem.** In two degrees of freedom, prove that isoenergetic nondegeneracy is equivalent to nonvanishing variation of the projectivized frequency along the energy curve.

**Solution sketch.** Differentiate the energy constraint, project the frequency derivative modulo its radial direction, and recover the bordered determinant.

**Exercise 2.4**

**Difficulty.** Advanced.

**Related results.** Resonant modules and resonance manifolds.

**Problem.** For fixed  $k \neq 0$ , prove that  $k \cdot \omega(I) = 0$  is a smooth resonance hypersurface when its derivative is nonzero; give a polynomial example where that derivative vanishes and the geometry degenerates.

**Solution sketch.** Apply the regular-value theorem, then compute a multiple-zero

example explicitly.

### Exercise 2.5

**Difficulty.** Advanced.

**Related results.** Liouville–Arnold theorem; Frequency map and Kolmogorov non-degeneracy.

**Problem.** Assume the frequency map is locally invertible in a Liouville–Arnold chart. Show that nearby frequencies label unique nearby Liouville tori and that primitive resonance hyperplanes pull back to smooth action-space hypersurfaces.

**Solution sketch.** Use the inverse-function theorem and compute the pulled-back normal  $D\omega(I)^T k$ .

# Averaging and Hamiltonian Perturbation Theory

## Why this chapter matters

A nearly integrable Hamiltonian contains fast oscillatory terms that may have little long-time effect—unless a resonance prevents them from averaging out. Perturbation theory is therefore a controlled attempt to separate removable oscillations from dynamically persistent terms.

The basic device is a near-identity canonical transformation generated by a Hamiltonian  $S$ . The first-order homological equation chooses  $S$  so that nonresonant Fourier modes disappear. Repeating the idea leads to high-order or exponentially accurate averaging; retaining the zero divisors leads to resonant normal forms. The same perturbative viewpoint also measures separatrix splitting: Melnikov's integral is the first variation of the distance between stable and unstable branches.

This chapter is where arithmetic first becomes dynamical. A small denominator is not merely an estimate to control; it is a warning that a Fourier mode is resonant and must be treated as part of the effective dynamics.

**Prerequisites.** K01–K02, Fourier series on tori, Hamiltonian flows, elementary asymptotic estimates, and the stable/unstable-manifold intuition for a planar hyperbolic equilibrium.

## Learning goals

- Solve the first-order Hamiltonian homological equation mode by mode and identify its divisors.
- Use Lie transforms to derive a controlled transformed Hamiltonian with an explicit remainder.
- Distinguish nonresonant averaging from resonant averaging.
- Derive adiabatic drift bounds directly from Hamilton's equations.
- Interpret exponentially accurate averaging and Melnikov splitting as two different perturbative limits.



**Conceptual roadmap**

For a perturbation  $H_0 + \varepsilon H_1$ , the first-order chain is

$$\begin{aligned}
 H_1 &\xrightarrow{\text{Fourier expansion}} H_{1,k} \\
 &\xrightarrow{k \cdot \omega} \text{homological equation} \\
 &\xrightarrow{\varphi_S^\varepsilon} \text{average or resonant normal form} + O(\varepsilon^2).
 \end{aligned}$$

If the normal form is angle-independent, the action equation immediately yields adiabatic stability. In analytic one-fast-phase systems the normalization can be iterated until the remainder is exponentially small. Near a homoclinic orbit, by contrast, the first-order object of interest is a stable/unstable splitting function, whose leading term is the Melnikov integral.

**Check your understanding**

Suppose  $H_0(I) = \omega I$  and  $H_1(\theta) = \cos(k\theta)$ . Write the divisor appearing in the homological equation. What changes when the divisor vanishes? Your answer should predict the distinction between ordinary and resonant averaging before reading the proofs.

**3.1 Core definitions****Intuition**

Averaging is a coordinate-change statement, not a claim that oscillatory terms are literally zero. The transformation moves them into a smaller remainder. Resonance tells us which terms cannot be moved in this way.

**Definition K03.1. Averaging**

*Averaging* is a near-identity change of variables that replaces an oscillatory perturbation by its orbit or angle average up to a quantitatively controlled remainder on a specified domain and time scale.

**Definition K03.2. Lie transform**

Let  $\varphi_S^t$  be the Hamiltonian flow of  $S$ . With the fixed convention  $\iota_{X_S}\omega = -dS$ , define

$$\mathcal{L}_S F := \{S, F\}.$$

Then  $F \circ \varphi_S^t$  is generated by repeated application of  $\mathcal{L}_S$ , with a finite Taylor formula and integral remainder whenever the required derivatives and flow tube exist.

**Definition K03.3. Resonant normal form**

Given a selected resonance module  $\Lambda$ , a *resonant normal form* retains the Fourier modes whose wave vectors lie in  $\Lambda$  and removes, to the stated order, those modes for which the homological divisor is bounded away from zero.

**Definition K03.4. Adiabatic invariant**

An *adiabatic invariant* is a quantity whose change remains within a stated small bound over a time interval asymptotically longer than the basic dynamical period as the perturbation parameter tends to zero.

**Definition K03.5. Hyperbolic equilibrium and homoclinic orbit**

For  $\dot{z} = f(z)$ , an equilibrium  $p$  is hyperbolic if  $Df(p)$  has no eigenvalue with zero real part. A nonconstant complete orbit  $z_0(t)$  is *homoclinic* to  $p$  if  $z_0(t) \rightarrow p$  as  $t \rightarrow \pm\infty$ .

**Definition K03.6. Selected stable and unstable trajectory branches**

A selected stable trajectory branch converges to the chosen hyperbolic invariant set in forward time, and a selected unstable trajectory branch converges to it in backward time. Whenever a particular branch is used, the invariant set and the branch component are part of the data.

**Definition K03.7. Planar wedge and affine transverse section**

For vectors  $a, b \in \mathbb{R}^2$  put  $a \wedge b = \det(a, b)$ . An affine line segment through  $z_*$  with direction  $n$  is transverse to the vector field  $f$  at  $z_*$  when  $f(z_*) \wedge n \neq 0$ .

## 3.2 How the main proofs work

### Intuition

**First-order averaging.** Expand  $H_1$  into finitely many Fourier modes. For  $k \neq 0$ , solve the homological equation by dividing by  $k \cdot \omega(I)$ . The time- $\varepsilon$  Hamiltonian flow of the resulting real generator cancels the oscillatory first-order part, while an integral Taylor formula controls the  $O(\varepsilon^2)$  remainder.

**Resonant averaging.** Perform the same construction only for  $k \notin \Lambda$ . Modes in the selected resonance module have zero or deliberately untreated divisors, so they remain in the effective Hamiltonian.

**Adiabatic invariance.** Once the transformed Hamiltonian is angle-independent up to  $\varepsilon^2 R$ , Hamilton's equation gives  $\dot{I} = -\varepsilon^2 \partial_\theta R$ . Integrating in time gives the drift estimate directly.

**Exponentially accurate one-fast-phase averaging.** Analyticity permits repeated normalization on slightly shrinking complex domains. Truncation order is chosen proportional to  $1/\varepsilon$  so the factorial/analytic loss and the small perturbation balance at an exponentially small remainder.

**Melnikov precursor.** Differentiate the selected stable and unstable branch

equations with respect to  $\varepsilon$ . Wedge the variational equations with the unperturbed tangent vector; Hamiltonian divergence cancellation reduces the difference to one scalar integral along the homoclinic orbit. A simple zero then persists by the implicit-function theorem.

### 3.3 Main results

#### Theorem K03.1. Exponentially accurate averaging in analytic systems

**Statement.** Let  $f(z, \varphi, \varepsilon)$  be real analytic on a fixed complex neighborhood,  $2\pi$ -periodic in the single fast phase  $\varphi$ , real on the real slice, and uniformly bounded there. Consider

$$\dot{z} = f(z, \varphi, \varepsilon), \quad \dot{\varphi} = \varepsilon^{-1}.$$

After shrinking to a fixed smaller complex domain there exist  $c, C, \varepsilon_1 > 0$  such that, for every  $0 < \varepsilon \leq \varepsilon_1$ , there is an analytic  $2\pi$ -periodic change of variables

$$z = \Psi_\varepsilon(y, \varphi), \quad \|\Psi_\varepsilon - \text{id}\| \leq C\varepsilon,$$

which transforms the system to

$$\dot{y} = F_\varepsilon(y) + R_\varepsilon(y, \varphi), \quad \dot{\varphi} = \varepsilon^{-1},$$

where  $F_\varepsilon$  is independent of the fast phase and the phase average

$$\langle R_\varepsilon \rangle_\varphi(y) := \frac{1}{2\pi} \int_0^{2\pi} R_\varepsilon(y, \theta) d\theta$$

vanishes identically, while

$$\|R_\varepsilon\| \leq C \exp(-c/\varepsilon).$$

#### Scope note

This is the one-fast-phase analytic Neishtadt theorem. Multifrequency exponentially small averaging requires a separate theorem.

*Proof.* We prove the estimate on a fixed smaller complex domain. This is exactly the “after shrinking the complex domain” clause in the statement.

**Complex domains and norms.** Choose a complex polydisc  $D_0 \Subset \mathbb{C}^d$  contained in the given complex  $z$ -neighborhood and choose  $s > 0$  so that the complex phase strip

$$\mathcal{S}_s := \{\varphi \in \mathbb{C}/(2\pi\mathbb{Z}) : |\text{Im } \varphi| < s\} \quad (\text{EA.1})$$

is still contained in the given analyticity neighborhood. After shrinking  $D_0$  once more,

fix a number  $\rho > 0$  for which the nested interiors

$$D_r := \{z \in D_0 : \text{dist}_\infty(z, \partial D_0) > r\}, \quad 0 \leq r \leq \rho, \quad (\text{EA.2})$$

are nonempty. For a bounded analytic vector-valued function  $u$  on  $D_r \times \mathcal{S}_s$ , write

$$\|u\|_{r,s} := \sup_{D_r \times \mathcal{S}_s} |u|. \quad (\text{EA.3})$$

The norm in the theorem may be any fixed equivalent finite-dimensional norm; we use the max norm and its induced operator norm.

By the uniform boundedness hypothesis, after decreasing the allowed  $\varepsilon$ -interval if necessary there is  $M \geq 1$  such that

$$\|f(\cdot, \cdot, \varepsilon)\|_{0,s} \leq M \quad (\text{EA.4})$$

for every  $0 < \varepsilon \leq \varepsilon_0$ . All constants below are independent of such  $\varepsilon$ .

**One analytic averaging step.** Consider, more generally, a system

$$\dot{z} = F(z) + G(z, \varphi), \quad \dot{\varphi} = \varepsilon^{-1}, \quad (\text{EA.5})$$

on  $D_r \times \mathcal{S}_s$ , where  $F$  is independent of  $\varphi$  and  $G$  has zero phase average,

$$\langle G \rangle(z) := \frac{1}{2\pi} \int_0^{2\pi} G(z, \theta) d\theta = 0. \quad (\text{EA.6})$$

Suppose

$$\|F\|_{r,s} \leq B, \quad \|G\|_{r,s} \leq m \leq B. \quad (\text{EA.7})$$

We now establish a quantitative reduction step.

Because of (EA.6),  $G$  has a  $2\pi$ -periodic analytic primitive in the fast phase. One convenient choice is

$$V(z, \varphi) := \int_0^\varphi G(z, \zeta) d\zeta, \quad \partial_\varphi V = G. \quad (\text{EA.8})$$

The integral is taken in the simply connected universal strip and then passes to the quotient because  $\int_0^{2\pi} G d\theta = 0$ . If  $\varphi = x + iy$  with  $x$  represented in  $[-\pi, \pi]$ , integrate first along the real segment from 0 to  $x$  and then vertically. This path has length at most  $\pi + s$ . Thus, with the harmless fixed constant  $L_s := 2\pi + s$ ,

$$\|V\|_{r,s} \leq L_s m. \quad (\text{EA.9})$$

This elementary bound is the point at which the single-fast-phase hypothesis enters: no divisor depending on an integer vector occurs.

Let  $0 < \delta < \rho - r$ . On  $D_{r+\delta} \times \mathcal{S}_s$  make the near-identity substitution

$$z = y + \varepsilon V(y, \varphi). \quad (\text{EA.10})$$

Cauchy's estimate on nested polydiscs supplies a constant  $C_d$ , depending only on the dimension and on the chosen max norm, such that the  $z$ -derivative of any bounded

analytic function loses at most a factor  $C_d/\delta$  when a  $\delta$ -wide buffer is available. Consequently there is a constant  $A = A(d, s) \geq 1$  with the following property: if

$$\delta \geq A\varepsilon B, \quad (\text{EA.11})$$

then (EA.9), Cauchy's estimate, and  $m \leq B$  give simultaneously

$$\varepsilon \|V\|_{r,s} \leq \frac{\delta}{4}, \quad \varepsilon \|D_y V\|_{r+\delta,s} \leq \frac{1}{4}. \quad (\text{EA.12})$$

We enlarge  $A$  once and for all whenever one of these elementary inequalities requires it. The first estimate means that (EA.10) maps the smaller domain into the larger one. The second implies that  $I + \varepsilon D_y V$  is invertible and

$$\|(I + \varepsilon D_y V)^{-1}\| \leq \frac{4}{3}. \quad (\text{EA.13})$$

Moreover, on each convex polydisc the map  $y \mapsto y + \varepsilon V(y, \varphi)$  is injective, since its perturbation has Lipschitz constant at most  $1/4$ . Hence (EA.10) is a legitimate analytic coordinate change onto its image. It preserves the real slice because  $G$ , and therefore  $V$ , is real on real arguments.

Differentiate (EA.10) along the flow. Since  $\varepsilon \partial_\varphi V \dot{\varphi} = G(y, \varphi)$ , equation (EA.5) becomes

$$(I + \varepsilon D_y V) \dot{y} = F(y) + \Delta(y, \varphi), \quad (\text{EA.14})$$

where

$$\Delta := F(y + \varepsilon V) - F(y) + G(y + \varepsilon V, \varphi) - G(y, \varphi). \quad (\text{EA.15})$$

Cauchy's estimate on an intermediate domain, together with the displacement bound in (EA.12), yields

$$\|\Delta\|_{r+\delta,s} \leq A \frac{\varepsilon B m}{\delta} \quad (\text{EA.16})$$

after another harmless enlargement of the same constant  $A$ . The same Cauchy estimate gives

$$\|\varepsilon D_y V F\|_{r+\delta,s} \leq A \frac{\varepsilon B m}{\delta}. \quad (\text{EA.17})$$

Subtracting  $F(y)$  after applying the inverse matrix in (EA.14), we therefore obtain

$$\dot{y} = F(y) + Q(y, \varphi), \quad (\text{EA.18})$$

with

$$Q = (I + \varepsilon D_y V)^{-1} (\Delta - \varepsilon D_y V F), \quad \|Q\|_{r+\delta,s} \leq A \frac{\varepsilon B m}{\delta}. \quad (\text{EA.19})$$

Here and below  $A$  denotes a fixed constant large enough to dominate all the preceding dimension- and strip-dependent constants.

Split  $Q$  once again into its phase average and zero-average part:

$$F_+ := F + \langle Q \rangle, \quad G_+ := Q - \langle Q \rangle. \quad (\text{EA.20})$$

Then

$$\|F_+ - F\|_{r+\delta, s} \leq A \frac{\varepsilon B m}{\delta}, \quad \|G_+\|_{r+\delta, s} \leq 2A \frac{\varepsilon B m}{\delta}. \quad (\text{EA.21})$$

This completes the one-step estimate.

**Iteration.** Return to the vector field in the theorem and set

$$F_0(z, \varepsilon) := \langle f \rangle(z, \varepsilon), \quad G_0(z, \varphi, \varepsilon) := f(z, \varphi, \varepsilon) - F_0(z, \varepsilon). \quad (\text{EA.22})$$

Then

$$\|F_0\|_{0, s} \leq M, \quad m_0 := \|G_0\|_{0, s} \leq 2M. \quad (\text{EA.23})$$

Fix

$$B := 2M, \quad \delta := 8A\varepsilon B = 16AM\varepsilon. \quad (\text{EA.24})$$

Choose  $\varepsilon_1 > 0$  so small that  $0 < \varepsilon \leq \varepsilon_1$  implies  $\delta < \rho/4$  and  $\rho/(2\delta) \geq 2$ . Define

$$N := \left\lfloor \frac{\rho}{2\delta} \right\rfloor. \quad (\text{EA.25})$$

Apply the one-step construction successively on  $D_{j\delta} \times \mathcal{S}_s$ ,  $j = 0, \dots, N-1$ . If  $m_j := \|G_j\|$ , then (EA.21) and (EA.24) give

$$m_{j+1} \leq \frac{1}{4}m_j, \quad \|F_{j+1} - F_j\| \leq \frac{1}{8}m_j. \quad (\text{EA.26})$$

Inductively,

$$m_j \leq 4^{-j}m_0. \quad (\text{EA.27})$$

The phase-independent part remains uniformly bounded by  $B$ , because

$$\|F_j\| \leq M + \frac{1}{8} \sum_{\ell=0}^{j-1} m_\ell \leq M + \frac{1}{8} \frac{2M}{1 - 1/4} = \frac{4}{3}M < 2M = B. \quad (\text{EA.28})$$

Thus all hypotheses needed for the next averaging step remain valid. Also  $j\delta \leq N\delta \leq \rho/2$ , so every step is defined before the fixed inner domain  $D_{\rho/2}$  is reached.

Let  $\Psi_\varepsilon$  be the composition of the  $N$  elementary coordinate changes. Each elementary map is analytic and  $2\pi$ -periodic in the phase, so the same is true of the composition. From (EA.9) and (EA.27), the total coordinate displacement is bounded by a geometric series:

$$\|\Psi_\varepsilon - \text{id}\| \leq \varepsilon L_s \sum_{j=0}^{N-1} m_j \leq \frac{8}{3} L_s M \varepsilon. \quad (\text{EA.29})$$

Hence  $\Psi_\varepsilon$  is near the identity uniformly as  $\varepsilon \rightarrow 0$ . Here “analytic change of variables” refers to analyticity in the phase-space variables for each fixed sufficiently small  $\varepsilon$ ; the theorem does not require the integer  $N$  in (EA.25) to vary analytically with the parameter.

In the final coordinates, still denoted  $(y, \varphi)$ , the system is

$$\dot{y} = F_N(y, \varepsilon) + R(y, \varphi, \varepsilon), \quad \dot{\varphi} = \varepsilon^{-1}, \quad \langle R \rangle = 0, \quad (\text{EA.30})$$

where  $F_N$  is independent of the fast phase and  $R := G_N$ . This is the claimed separation of slow drift and fast oscillation.

It remains only to rewrite the geometric estimate as an exponential one. Since  $\rho/(2\delta) \geq 2$ , the elementary inequality  $\lfloor x \rfloor \geq x/2$  for  $x \geq 2$  gives

$$N \geq \frac{\rho}{4\delta}. \quad (\text{EA.31})$$

Therefore, by (EA.23), (EA.27), and (EA.24),

$$\begin{aligned} \|R\|_{\rho/2,s} &\leq 2M 4^{-N} \\ &\leq 2M \exp\left(-\frac{(\log 4)\rho}{4\delta}\right) \\ &= 2M \exp\left(-\frac{(\log 4)\rho}{64AM} \frac{1}{\varepsilon}\right). \end{aligned} \quad (\text{EA.32})$$

Thus the theorem holds with, for example,

$$C = M \max\left\{2, \frac{8}{3}L_s\right\}, \quad c = \frac{(\log 4)\rho}{64AM} > 0, \quad (\text{EA.33})$$

and with the fixed shrunken domain  $D_{\rho/2} \times \mathcal{S}_s$ . The constants depend only on the original analyticity buffer, the uniform vector-field bound, the dimension, and the chosen smaller domain, and not on  $\varepsilon$ . This proves the asserted one-fast-phase exponentially accurate averaging theorem.  $\square$

### Theorem K03.2. Melnikov precursor

**Statement.** Let

$$\dot{z} = f(z) + \varepsilon g(z, t + t_0), \quad f = X_{H_0}, \quad g(\cdot, t) = X_{H_1(\cdot, t)}$$

be a  $C^3$  planar Hamiltonian system with  $H_1$   $T$ -periodic in  $t$ . Suppose  $f$  has a homoclinic orbit  $z_0(t)$  to a hyperbolic equilibrium  $p$ , with exponential convergence as  $t \rightarrow \pm\infty$ . Choose an affine line segment  $\Sigma = z_0(0) + (-\delta, \delta)n$  with  $f(z_0(0)) \wedge n \neq 0$ . Assume that for small  $\varepsilon$  the selected stable and unstable trajectories meet  $\Sigma$  at  $z_\varepsilon^{s,u}(0; t_0)$ , depend  $C^2$  on  $(\varepsilon, t_0)$ , agree with  $z_0$  at  $\varepsilon = 0$ , and their first  $\varepsilon$ -variations remain bounded in the corresponding forward/backward time directions. If

$$D(\varepsilon, t_0) := f(z_0(0)) \wedge (z_\varepsilon^u(0; t_0) - z_\varepsilon^s(0; t_0)),$$

then, uniformly in  $t_0 \in \mathbb{R}/T\mathbb{Z}$ ,

$$D(\varepsilon, t_0) = \varepsilon M(t_0) + O(\varepsilon^2), \quad M(t_0) = \int_{-\infty}^{\infty} f(z_0(t)) \wedge g(z_0(t), t + t_0) dt.$$

With the Poisson-bracket convention fixed in K01 this is equivalently  $M(t_0) = -\int_{-\infty}^{\infty} \{H_0, H_1(\cdot, t + t_0)\}(z_0(t)) dt$ . Because both section points lie on the same affine line  $\Sigma$ ,  $D(\varepsilon, t_0) = 0$  is equivalent to  $z_\varepsilon^u(0; t_0) = z_\varepsilon^s(0; t_0)$ . If  $M(t_*) = 0$  and  $M'(t_*) \neq 0$ ,

then for every sufficiently small nonzero  $\varepsilon$  the splitting equation  $D(\varepsilon, t_0) = 0$  has a unique solution near  $t_*$ .

*Proof.* Write

$$A(t) := Df(z_0(t)), \quad g_{t_0}(t) := g(z_0(t), t + t_0).$$

For the selected stable and unstable branches let

$$\eta^{s,u}(t; t_0) := \partial_\varepsilon z_\varepsilon^{s,u}(t; t_0)|_{\varepsilon=0}. \quad (\text{M.1})$$

Differentiating the perturbed differential equation at  $\varepsilon = 0$  gives, for both branches,

$$\dot{\eta} = A(t)\eta + g_{t_0}(t). \quad (\text{M.2})$$

The unperturbed tangent vector

$$v(t) := f(z_0(t)) = \dot{z}_0(t)$$

satisfies the homogeneous variational equation

$$\dot{v} = A(t)v. \quad (\text{M.3})$$

For vectors  $a, b \in \mathbb{R}^2$  write  $a \wedge b = \det(a, b)$ . The elementary two-dimensional identity

$$(Aa) \wedge b + a \wedge (Ab) = (\text{tr } A)(a \wedge b) \quad (\text{M.4})$$

holds for every  $2 \times 2$  matrix  $A$ . Set

$$\delta^{s,u}(t; t_0) := v(t) \wedge \eta^{s,u}(t; t_0). \quad (\text{M.5})$$

By (M.2)–(M.4),

$$\frac{d}{dt} \delta^{s,u} = (\text{tr } A(t)) \delta^{s,u} + v(t) \wedge g_{t_0}(t). \quad (\text{M.6})$$

Because  $f = X_{H_0}$  is a planar Hamiltonian vector field,  $\text{div } f = \text{tr } Df = 0$ . Hence

$$\frac{d}{dt} \delta^{s,u}(t; t_0) = f(z_0(t)) \wedge g(z_0(t), t + t_0). \quad (\text{M.7})$$

The homoclinic orbit approaches the hyperbolic equilibrium exponentially, so  $f(z_0(t))$  tends to zero exponentially as  $t \rightarrow \pm\infty$ . The perturbing vector field is bounded on the compact time circle and on a fixed neighborhood of the orbit. Therefore the right-hand side of (M.7) is absolutely integrable, uniformly in  $t_0$ . By hypothesis the first variation of the stable branch remains bounded as  $t \rightarrow +\infty$ , and that of the unstable branch remains bounded as  $t \rightarrow -\infty$ . Consequently

$$\lim_{t \rightarrow +\infty} \delta^s(t; t_0) = 0, \quad \lim_{t \rightarrow -\infty} \delta^u(t; t_0) = 0. \quad (\text{M.8})$$

Integrating (M.7) from 0 to  $+\infty$  for the stable branch and from  $-\infty$  to 0 for the unstable



branch gives

$$\delta^s(0; t_0) = - \int_0^\infty f(z_0(t)) \wedge g(z_0(t), t + t_0) dt, \quad (\text{M.9})$$

$$\delta^u(0; t_0) = \int_{-\infty}^0 f(z_0(t)) \wedge g(z_0(t), t + t_0) dt. \quad (\text{M.10})$$

Subtracting,

$$f(z_0(0)) \wedge (\eta^u(0; t_0) - \eta^s(0; t_0)) = M(t_0), \quad (\text{M.11})$$

where

$$M(t_0) := \int_{-\infty}^\infty f(z_0(t)) \wedge g(z_0(t), t + t_0) dt. \quad (\text{M.12})$$

Now  $D(0, t_0) = 0$ . Differentiating its definition in  $\varepsilon$  at 0 and using (M.1) and (M.11) yields

$$\partial_\varepsilon D(0, t_0) = M(t_0). \quad (\text{M.13})$$

The assumed  $C^2$  dependence on  $(\varepsilon, t_0)$  and compactness of the phase circle give the uniform Taylor expansion

$$D(\varepsilon, t_0) = \varepsilon M(t_0) + O(\varepsilon^2). \quad (\text{M.14})$$

We translate (M.12) into the Poisson convention used in this book. In canonical coordinates  $(q, p)$  with  $\omega = dp \wedge dq = d(p dq)$  and  $\iota_{X_H} \omega = -dH$ ,

$$X_H = (H_p, -H_q).$$

Therefore

$$X_{H_0} \wedge X_{H_1} = H_{0,q} H_{1,p} - H_{0,p} H_{1,q} = -\{H_0, H_1\}, \quad (\text{M.15})$$

because the book's bracket is  $\{F, G\} = dG(X_F) = F_p G_q - F_q G_p$ . Thus the minus sign here is the orientation sign between the Euclidean wedge  $dq \wedge dp$  used in  $D$  and the book's symplectic form  $dp \wedge dq$ . Substitution into (M.12) gives exactly the asserted bracket formula and fixes its sign.

Because  $z_\varepsilon^u(0; t_0), z_\varepsilon^s(0; t_0) \in \Sigma$ , their difference is a scalar multiple of  $n$ . Since  $f(z_0(0)) \wedge n \neq 0$ , we have

$$D(\varepsilon, t_0) = 0 \iff z_\varepsilon^u(0; t_0) = z_\varepsilon^s(0; t_0). \quad (\text{M.16})$$

Thus zeros of  $D$  are precisely intersections of the selected stable and unstable branches on the section. Finally suppose  $M(t_*) = 0$  and  $M'(t_*) \neq 0$ . For  $\varepsilon \neq 0$  put

$$\widehat{D}(\varepsilon, t_0) := \frac{D(\varepsilon, t_0)}{\varepsilon},$$

and extend it at  $\varepsilon = 0$  by  $\widehat{D}(0, t_0) = M(t_0)$ . Taylor's formula and the  $C^2$  hypothesis show that  $\widehat{D}$  is  $C^1$  near  $(0, t_*)$ . Moreover

$$\widehat{D}(0, t_*) = 0, \quad \partial_{t_0} \widehat{D}(0, t_*) = M'(t_*) \neq 0.$$

The implicit-function theorem therefore gives, for every sufficiently small  $\varepsilon$ , a unique  $t_0(\varepsilon)$  near  $t_*$  such that

$$D(\varepsilon, t_0(\varepsilon)) = 0.$$

This is the claimed Melnikov splitting criterion at the precursor scope stated here.  $\square$

### Theorem K03.3. First-order Hamiltonian averaging

**Statement.** Let  $U \subset \mathbb{R}^n$  be open,  $K \Subset U$ , and

$$H_\varepsilon(I, \theta) = H_0(I) + \varepsilon H_1(I, \theta), \quad \theta \in \mathbb{T}^n := \mathbb{R}^n / (2\pi\mathbb{Z})^n,$$

where  $H_0 \in C^\infty(U)$  and  $H_1 = \sum_{k \in \mathcal{K}} H_{1,k}(I) e^{i\langle k, \theta \rangle}$  is a real-valued trigonometric polynomial with finite symmetric Fourier support  $\mathcal{K} \subset \mathbb{Z}^n$ . Put  $\omega(I) = \nabla H_0(I)$  and assume that on a neighborhood of  $K$  there is  $\alpha > 0$  such that  $|\langle k, \omega(I) \rangle| \geq \alpha$  for every  $0 \neq k \in \mathcal{K}$ . If  $\bar{H}_1 := H_{1,0}$ , then there is a real-valued smooth zero-average generator  $S$  and, for all sufficiently small  $|\varepsilon|$ , its time- $\varepsilon$  Hamiltonian map  $\Phi_\varepsilon = \varphi_S^\varepsilon$ , defined near  $K \times \mathbb{T}^n$ , such that

$$H_\varepsilon \circ \Phi_\varepsilon = H_0 + \varepsilon \bar{H}_1 + \varepsilon^2 R_\varepsilon.$$

For every fixed  $r$ , after shrinking to a fixed neighborhood of  $K$ ,  $\|\Phi_\varepsilon - \text{id}\|_{C^r} = O_r(|\varepsilon|)$  and  $\sup_{|\varepsilon| \leq \varepsilon_0} \|R_\varepsilon\|_{C^r} < \infty$ .

*Proof.* We use the action-angle symplectic form

$$\omega_{\text{symp}} = \sum_{j=1}^n dI_j \wedge d\theta_j$$

and the convention used in this book  $\iota_{X_F} \omega_{\text{symp}} = -dF$ . Thus

$$X_{H_0} = \sum_{j=1}^n \omega_j(I) \partial_{\theta_j}, \quad \{H_0, G\} = dG(X_{H_0}) = \omega(I) \cdot \partial_\theta G. \quad (\text{AV.1})$$

Let  $V$  be an open neighborhood of  $K$  with  $\bar{V} \Subset U$  on which all the stated divisor bounds hold. For  $k \neq 0$  in the finite support  $\mathcal{K}$ , define

$$S_k(I) := \frac{H_{1,k}(I)}{i\langle k, \omega(I) \rangle}, \quad S(I, \theta) := \sum_{0 \neq k \in \mathcal{K}} S_k(I) e^{i\langle k, \theta \rangle}. \quad (\text{AV.2})$$

Every denominator is bounded away from zero on  $V$ , so  $S$  is smooth there. Since  $H_1$  is real,  $H_{1,-k} = \overline{H_{1,k}}$ ; because  $\langle -k, \omega \rangle = -\langle k, \omega \rangle$ , (AV.2) gives  $S_{-k} = \overline{S_k}$ . Hence  $S$  is real valued. It has zero average by construction. Using (AV.1),

$$\{H_0, S\} = \sum_{0 \neq k \in \mathcal{K}} i\langle k, \omega(I) \rangle S_k(I) e^{i\langle k, \theta \rangle} = H_1 - \bar{H}_1. \quad (\text{AV.3})$$

Antisymmetry of the Poisson bracket therefore gives

$$\mathcal{L}_S H_0 = \{S, H_0\} = \bar{H}_1 - H_1. \quad (\text{AV.4})$$

Let  $\Phi_t = \varphi_S^t$ . Since  $X_S$  is smooth and  $\bar{K} \times \mathbb{T}^n$  is compact, after replacing  $V$  by a slightly smaller fixed neighborhood of  $K$  there is  $\varepsilon_0 > 0$  such that  $\Phi_t$  is defined on that

neighborhood for  $|t| \leq \varepsilon_0$  and its whole swept tube is compactly contained in  $V \times \mathbb{T}^n$ . For every smooth  $F$ ,

$$\frac{d}{dt}(F \circ \Phi_t) = (\mathcal{L}_S F) \circ \Phi_t. \quad (\text{AV.5})$$

Applying the fundamental theorem of calculus twice to  $H_0$  and once to  $H_1$  yields

$$H_0 \circ \Phi_\varepsilon = H_0 + \varepsilon \mathcal{L}_S H_0 + \varepsilon^2 \int_0^1 (1-s)(\mathcal{L}_S^2 H_0) \circ \Phi_{s\varepsilon} ds, \quad (\text{AV.6})$$

$$H_1 \circ \Phi_\varepsilon = H_1 + \varepsilon \int_0^1 (\mathcal{L}_S H_1) \circ \Phi_{s\varepsilon} ds. \quad (\text{AV.7})$$

Consequently

$$\begin{aligned} H_\varepsilon \circ \Phi_\varepsilon &= H_0 + \varepsilon(H_1 + \mathcal{L}_S H_0) + \varepsilon^2 R_\varepsilon \\ &= H_0 + \varepsilon \bar{H}_1 + \varepsilon^2 R_\varepsilon, \end{aligned} \quad (\text{AV.8})$$

where, explicitly,

$$R_\varepsilon = \int_0^1 (1-s)(\mathcal{L}_S^2 H_0) \circ \Phi_{s\varepsilon} ds + \int_0^1 (\mathcal{L}_S H_1) \circ \Phi_{s\varepsilon} ds. \quad (\text{AV.9})$$

It remains only to justify the uniform regularity assertions. Fix  $r < \infty$ . On the compact swept tube,  $X_S$  and all of its derivatives up to the finite order required for  $C^r$  flow estimates are bounded. Standard differentiation of the flow equation

$$\Phi_t(z) - z = \int_0^t X_S(\Phi_s(z)) ds \quad (\text{AV.10})$$

(and its variational equations up to order  $r$ ) gives

$$\|\Phi_\varepsilon - \text{id}\|_{C^r} = O_r(|\varepsilon|). \quad (\text{AV.11})$$

The same compact-tube bounds applied to the two smooth integrands in (AV.9), together with the  $C^r$  composition chain rule, give

$$\sup_{|\varepsilon| \leq \varepsilon_0} \|R_\varepsilon\|_{C^r} < \infty. \quad (\text{AV.12})$$

This proves the claimed first-order averaging normal form.  $\square$

#### Proposition K03.4. Lie-series perturbation formula

**Statement.** Let  $S, H$  be  $C^{N+1}$  on a symplectic manifold, let  $N \geq 1$ , and suppose the Hamiltonian flow  $\varphi_S^t$  is defined on the orbit segment under consideration for  $0 \leq t \leq T$ . With  $\mathcal{L}_S F := \{S, F\}$ ,

$$H \circ \varphi_S^t = \sum_{j=0}^{N-1} \frac{t^j}{j!} \mathcal{L}_S^j H + \frac{t^N}{(N-1)!} \int_0^1 (1-s)^{N-1} (\mathcal{L}_S^N H) \circ \varphi_S^{st} ds.$$

Thus on any compact set whose swept flow tube is compactly contained in the

domain, the remainder is bounded by  $|t|^N N!^{-1} \sup |\mathcal{L}_S^N H|$  on that tube. In particular the pullback is the finite-order Taylor realization of  $\exp(t\mathcal{L}_S)H$ .

*Proof.* Set

$$f(t) := H \circ \varphi_S^t.$$

By the definition  $\mathcal{L}_S F = \{S, F\} = dF(X_S)$  and the flow equation,

$$f'(t) = (\mathcal{L}_S H) \circ \varphi_S^t. \quad (\text{LS.1})$$

Repeating the same calculation gives, for  $0 \leq j \leq N$ ,

$$f^{(j)}(t) = (\mathcal{L}_S^j H) \circ \varphi_S^t. \quad (\text{LS.2})$$

The assumed  $C^{N+1}$  regularity is more than enough for these derivatives on the stated flow tube.

Taylor's formula at 0 with integral remainder is

$$f(t) = \sum_{j=0}^{N-1} \frac{t^j}{j!} f^{(j)}(0) + \frac{1}{(N-1)!} \int_0^t (t-u)^{N-1} f^{(N)}(u) du. \quad (\text{LS.3})$$

For  $t \geq 0$ , substitute  $u = st$  and use (LS.2):

$$H \circ \varphi_S^t = \sum_{j=0}^{N-1} \frac{t^j}{j!} \mathcal{L}_S^j H + \frac{t^N}{(N-1)!} \int_0^1 (1-s)^{N-1} (\mathcal{L}_S^N H) \circ \varphi_S^{st} ds, \quad (\text{LS.4})$$

which is the asserted formula.

If the swept tube of a compact set  $C$  is compactly contained in the domain, then (LS.4) gives

$$\|\text{Rem}_N(t)\|_{C^0(C)} \leq \frac{|t|^N}{(N-1)!} \left( \int_0^1 (1-s)^{N-1} ds \right) \sup_{\mathcal{T}_C} |\mathcal{L}_S^N H| = \frac{|t|^N}{N!} \sup_{\mathcal{T}_C} |\mathcal{L}_S^N H|. \quad (\text{LS.5})$$

The polynomial in (LS.4) is exactly the order- $(N-1)$  truncation of  $\exp(t\mathcal{L}_S)H$ , completing the proof.  $\square$

### Theorem K03.5. Resonant averaging

**Statement.** In the setting of Theorem K03.3, let  $\Lambda \leq \mathbb{Z}^n$  be a rank- $r$  submodule and set

$$Z_\Lambda(I, \theta) := \sum_{k \in \mathcal{K} \cap \Lambda} H_{1,k}(I) e^{i\langle k, \theta \rangle}.$$

Assume that on a neighborhood of  $K$  there is  $\alpha > 0$  such that  $|\langle k, \omega(I) \rangle| \geq \alpha$  for every  $k \in \mathcal{K} \setminus \Lambda$ . Then a real-valued smooth generator containing only modes  $k \notin \Lambda$  has a time- $\varepsilon$  Hamiltonian map  $\Phi_\varepsilon$  for which

$$H_\varepsilon \circ \Phi_\varepsilon = H_0 + \varepsilon Z_\Lambda + \varepsilon^2 R_\varepsilon$$

near  $K \times \mathbb{T}^n$ , with the same fixed- $C^r$  near-identity and bounded-remainder conclusions as in first-order averaging. Thus precisely the selected resonant-module modes are retained to first order.

*Proof.* Write

$$H_1 = Z_\Lambda + H_1^{\text{nr}}, \quad H_1^{\text{nr}} = \sum_{k \in \mathcal{K} \setminus \Lambda} H_{1,k}(I) e^{i\langle k, \theta \rangle}. \quad (\text{RA.1})$$

Choose a fixed neighborhood  $V$  of  $K$  with compact closure in the domain and on which the stated lower bound  $|\langle k, \omega(I) \rangle| \geq \alpha$  holds for every  $k \in \mathcal{K} \setminus \Lambda$ . Define

$$S(I, \theta) := \sum_{k \in \mathcal{K} \setminus \Lambda} \frac{H_{1,k}(I)}{i\langle k, \omega(I) \rangle} e^{i\langle k, \theta \rangle}. \quad (\text{RA.2})$$

The support is finite and the denominators are uniformly nonzero, so  $S$  is smooth on  $V \times \mathbb{T}^n$ . The same  $k \leftrightarrow -k$  calculation as in first-order averaging shows that  $S$  is real valued whenever  $H_1$  is real valued.

For  $H_0 = H_0(I)$  we have

$$\{H_0, S\} = \omega(I) \cdot \partial_\theta S = H_1^{\text{nr}} = H_1 - Z_\Lambda, \quad (\text{RA.3})$$

hence

$$\mathcal{L}_S H_0 = \{S, H_0\} = Z_\Lambda - H_1. \quad (\text{RA.4})$$

Let  $\Phi_\varepsilon = \varphi_S^\varepsilon$ . For sufficiently small  $|\varepsilon|$  its swept tube over a slightly smaller fixed neighborhood of  $K \times \mathbb{T}^n$  is compactly contained in  $V \times \mathbb{T}^n$ . Apply the Lie-series perturbation formula proved above with  $N = 2$  to  $H_0$  and with  $N = 1$  to  $H_1$  (equivalently use its integral formula before taking the first truncation). We obtain

$$H_0 \circ \Phi_\varepsilon = H_0 + \varepsilon \mathcal{L}_S H_0 + \varepsilon^2 A_\varepsilon, \quad H_1 \circ \Phi_\varepsilon = H_1 + \varepsilon B_\varepsilon, \quad (\text{RA.5})$$

where  $A_\varepsilon$  and  $B_\varepsilon$  are uniformly bounded in every fixed  $C^r$  norm on the smaller compact neighborhood. Therefore

$$\begin{aligned} H_\varepsilon \circ \Phi_\varepsilon &= H_0 + \varepsilon(H_1 + \mathcal{L}_S H_0) + \varepsilon^2(A_\varepsilon + B_\varepsilon) \\ &= H_0 + \varepsilon Z_\Lambda + \varepsilon^2 R_\varepsilon. \end{aligned} \quad (\text{RA.6})$$

The generator (RA.2) has no Fourier mode in  $\Lambda$ , while (RA.6) retains exactly the  $\Lambda$ -projection of the first-order perturbation.

Finally, smooth dependence of the flow on time and initial data on the fixed compact tube gives  $\|\Phi_\varepsilon - \text{id}\|_{C^r} = O_r(|\varepsilon|)$  for every fixed  $r$ , and the integral remainders in that Lie-series formula give the stated uniform  $C^r$  bounds on  $R_\varepsilon$ . This completes the resonant averaging step.  $\square$

### Theorem K03.6. Adiabatic invariance from averaging

**Statement.** Assume a near-integrable Hamiltonian has been transformed on a

domain to

$$\tilde{H}(I, \theta) = H_0(I) + \varepsilon \tilde{H}_1(I) + \varepsilon^2 R(I, \theta), \quad \|\partial_\theta R\| \leq M.$$

If an orbit  $(I(t), \theta(t))$  stays in that domain, then  $|I(t) - I(0)| \leq M\varepsilon^2|t|$ ; in particular  $|I(t) - I(0)| \leq M\varepsilon$  for  $|t| \leq \varepsilon^{-1}$ . More generally, if the remainder is  $\varepsilon^{N+1}R_N$  with  $\|\partial_\theta R_N\| \leq M_N$ , then  $|I(t) - I(0)| \leq M_N|\varepsilon|^{N+1}|t|$  and hence the drift is at most  $M_N|\varepsilon|^N$  on  $|t| \leq |\varepsilon|^{-1}$ .

*Proof.* With

$$\tilde{H}(I, \theta) = H_0(I) + \varepsilon \tilde{H}_1(I) + \varepsilon^2 R(I, \theta),$$

Hamilton's equations in the convention used in this book are

$$\dot{I} = -\partial_\theta \tilde{H} = -\varepsilon^2 \partial_\theta R. \quad (\text{AI.1})$$

As long as the orbit remains in the domain on which  $\|\partial_\theta R\| \leq M$ , integration gives

$$|I(t) - I(0)| \leq \int_0^{|t|} |\dot{I}(s)| ds \leq M\varepsilon^2|t|. \quad (\text{AI.2})$$

For  $|t| \leq \varepsilon^{-1}$  this becomes

$$|I(t) - I(0)| \leq M\varepsilon. \quad (\text{AI.3})$$

If instead

$$\tilde{H} = H_0 + \varepsilon \tilde{H}_1 + \varepsilon^{N+1} R_N, \quad \|\partial_\theta R_N\| \leq M_N,$$

then the same calculation gives

$$|I(t) - I(0)| \leq M_N|\varepsilon|^{N+1}|t|, \quad (\text{AI.4})$$

and hence a bound  $M_N|\varepsilon|^N$  on the time interval  $|t| \leq |\varepsilon|^{-1}$ . No comparison equation or Grönwall argument enters because the angle-independent part of the normal form contributes exactly zero to  $\dot{I}$ .  $\square$

### 3.4 Worked examples

#### Worked example: removing a nonresonant Fourier mode

Take one action-angle pair with

$$H_\varepsilon(I, \theta) = \omega I + \varepsilon \cos \theta, \quad \omega \neq 0.$$

The average of  $\cos \theta$  is zero. With the conventions of this series,

$$\{S, H_0\} = -\omega \partial_\theta S.$$

Choose

$$S(\theta) = \frac{1}{\omega} \sin \theta.$$

Then  $\{S, H_0\} = -\cos\theta$ , so the first-order oscillatory term cancels in  $H_\varepsilon \circ \varphi_S^\varepsilon$ . The transformed Hamiltonian therefore has the form

$$\omega I + \varepsilon^2 R_\varepsilon.$$

If  $\|\partial_\theta R_\varepsilon\| \leq M$  on the region traversed by an orbit, then

$$|I(t) - I(0)| \leq M\varepsilon^2|t|.$$

Thus on times  $|t| \leq \varepsilon^{-1}$  the action drifts by at most  $O(\varepsilon)$ .

The factor  $1/\omega$  in  $S$  is the simplest small divisor. In several degrees of freedom it becomes  $1/(k \cdot \omega)$ .

### Boundary case: a resonant term that must remain

In two angles let

$$H_0 = I_1 + I_2, \quad H_1 = \cos(\theta_1 - \theta_2).$$

For  $k = (1, -1)$  we have  $k \cdot \omega = 0$ . The nonresonant homological equation would require division by zero, so this term cannot be removed by the first-order scheme. Taking  $\Lambda = \mathbb{Z}(1, -1)$ , resonant averaging retains exactly this mode in  $Z_\Lambda$ . The surviving term is therefore not an error in the method; it is the effective resonant dynamics the method is designed to expose.

### Common pitfall

“The average is zero” does not imply “the perturbation has no effect.” It means that after a near-identity transformation the leading oscillation has moved into a smaller remainder; the domain, divisor bounds, and time scale still matter.

## 3.5 Chapter summary

### What to remember

- The homological equation is the algebraic heart of Hamiltonian averaging.
- Its divisor  $k \cdot \omega$  separates removable nonresonant modes from resonant modes that must remain.
- Lie transforms provide an exact canonical realization of the formal cancellation and an explicit remainder formula.
- Angle-independent normal forms immediately yield quantitative action-drift bounds.
- Analyticity can upgrade algebraic averaging accuracy to exponential accuracy in a one-fast-phase problem.
- Melnikov theory uses a different first-order expansion: it measures stable/unstable splitting along a homoclinic orbit.

**Looking ahead**

K04 pushes the same homological-equation mechanism to higher order near elliptic equilibria and periodic orbits. Resonant monomials become the Birkhoff normal form, and a new issue appears: a formal normal form can diverge even though every finite normalization step is legitimate.

**Check your understanding**

Explain why a resonant Fourier mode should be retained rather than estimated as a small error. Then derive the  $O(\varepsilon)$  action drift on times  $O(\varepsilon^{-1})$  from an  $O(\varepsilon^2)$  angle-dependent remainder.

**3.6 Exercises and problems****Exercise 3.1**

**Difficulty.** Core.

**Related results.** First-order Hamiltonian averaging; Lie-series perturbation formula; Resonant averaging.

**Problem.** For the weakly forced oscillator, separate averaged, resonant, and non-resonant Fourier modes. Solve the first homological equation for one nonresonant mode and verify the  $O(\varepsilon^2)$  transformed remainder.

**Solution sketch.** Divide by the nonzero Fourier divisor and expand the time-one Lie transform through second order.

**Exercise 3.2**

**Difficulty.** Core.

**Related results.** Adiabatic invariance from averaging; Exponentially accurate averaging in analytic systems; Melnikov precursor.

**Problem.** For the resonant forcing term, show that the relevant Fourier divisor is zero and hence the periodic homological equation cannot remove that mode. Derive the leading slow resonant dynamics.

**Solution sketch.** Write the coefficient equation, prove its solvability obstruction, and pass to the resonant angle with slow time  $\tau = \varepsilon t$ .

**Exercise 3.3**

**Difficulty.** Advanced.

**Related results.** Resonant averaging.

**Problem.** Prove that a trigonometric polynomial splits into its resonant Fourier projection plus the image of  $L_\omega = \omega \cdot \partial_\theta$ , and write the explicit inverse on nonresonant modes.

**Solution sketch.** Work coefficientwise; the zero divisors are exactly the kernel and all other coefficients are inverted explicitly.



**Exercise 3.4****Difficulty.** Advanced.**Related results.** Adiabatic invariance from averaging.**Problem.** Construct a resonant near-integrable example with order-one action drift on a time of order  $\varepsilon^{-1}$ , showing why the adiabatic-invariance hypothesis cannot be dropped.**Solution sketch.** Choose a resonant angle whose conjugate action has nonzero averaged derivative and integrate the slow equation.**Exercise 3.5****Difficulty.** Advanced.**Related results.** First-order Hamiltonian averaging; Lie-series perturbation formula.**Problem.** Derive the first-order averaged Hamiltonian directly from the Lie-series formula and identify the small-divisor estimate needed to extend the finite Fourier calculation to analytic data.**Solution sketch.** Choose the homological generator, cancel the oscillatory term, and bound the remaining brackets with the divisor estimate.**Exercise 3.6****Difficulty.** Lab.**Related results.** None.**Problem.** Numerically compare the weakly forced oscillator with its first-order averaged system for decreasing  $\varepsilon$ , on fixed and slow  $t \asymp \varepsilon^{-1}$  intervals; repeat near resonance.**Solution sketch.** Refine the integrator, fit the action-error scaling in  $\varepsilon$ , and separate numerical error from resonance effects.

# Hamiltonian Birkhoff and Resonant Normal Forms

## Why this chapter matters

Normal-form theory asks a deceptively simple question: how much of a Hamiltonian can be removed by a canonical change of coordinates? Near an elliptic equilibrium the quadratic part produces a collection of rotations. Terms that oscillate against those rotations can often be eliminated order by order; terms whose phases resonate with the linear motion cannot. Birkhoff normal form makes this separation precise.

The distinction between *finite-order*, *formal*, and *convergent* normalization is essential. A finite-order transformation is an analytic local construction with a controlled remainder. A formal normal form is an infinite algebraic series and need not converge. Later KAM and Nekhoroshev arguments therefore use finite quantitative normal forms, not the unsupported assumption that a formal series defines an analytic conjugacy. The same language also applies to the Poincaré map of an elliptic periodic orbit, connecting equilibria and periodic dynamics.

**Prerequisites.** K01 symplectic geometry, K02 action variables and resonance modules, K03 Lie transforms and homological equations, multivariable Taylor series, and elementary Fourier analysis.

## Learning goals

- Diagonalize the quadratic homological operator on complex monomials and identify its resonant kernel.
- Distinguish finite-order analytic normalization from an infinite formal Birkhoff series and from a genuinely convergent analytic normalization.
- Read the quartic Birkhoff coefficients as a twist matrix for the induced frequency map.
- Transfer the equilibrium normal-form mechanism to elliptic periodic orbits through their symplectic Poincaré maps.
- Understand why Nekhoroshev theory needs a quantitative resonant-block normal form rather than merely a formal expansion at one point.

**Conceptual roadmap**

At a fixed normalization degree the proof engine is

Taylor coefficient  $\longrightarrow$  homological operator  $\{H_2, \cdot\}$   
 $\longrightarrow$  resonant projection + removable complement  
 $\longrightarrow$  Hamiltonian Lie transform  
 $\longrightarrow$  higher-order remainder.

Iterating this only to a prescribed finite order gives the classical Birkhoff theorem. Retaining the resonant kernel gives resonant normal form. At quartic order the action polynomial supplies the twist matrix. On a resonant action block the same idea is combined with Fourier truncation and explicit divisor bounds to produce the quantitative normal form used by Nekhoroshev theory.

**Check your understanding**

For a monomial  $\xi^\alpha \eta^\beta$ , compute  $\{H_2, \xi^\alpha \eta^\beta\}$  when  $H_2 = \sum_j \omega_j \xi_j \eta_j$ . Explain why division is legal exactly when  $(\alpha - \beta) \cdot \omega \neq 0$ .

**4.1 Definitions****Intuition**

The chapter uses the phrase *normal form* in several genuinely different senses. The first layer is local and algebraic: an elliptic equilibrium or periodic orbit is expanded near a fixed point, the quadratic dynamics defines a homological operator, and monomials in its kernel are called resonant. A finite-order Birkhoff normal form removes only the nonresonant terms up to a chosen order; the surviving coefficients are the Birkhoff invariants. For a periodic orbit, the same logic is applied to a symplectic Poincaré return map, with multiplier resonances replacing the continuous-time relations  $(\alpha - \beta) \cdot \omega = 0$ .

The second layer is quantitative and domain-dependent. Nekhoroshev theory does not work with a merely formal germ: it needs an analytic transformation on a complex action-angle block, a weighted Fourier norm, a resonance lattice  $\Lambda$ , and a cutoff  $K$ . “ $\Lambda$ -resonant” means that the retained Fourier vectors lie in  $\Lambda$ ; “nonresonant modulo  $\Lambda$ ” supplies lower bounds for the divisors outside that lattice. The size of the complex action tube, the angle strip, the Hessian bound, and the Fourier cutoff are therefore part of the normal-form data, not bookkeeping that can be discarded after the algebra is understood.

A third group of definitions belongs to the one-dimensional exact-symplectic germ problem. Here the formal map Birkhoff series, its radius of convergence, the twist coefficient, continued-fraction scales, and the invariant-circle deficit measure different aspects of the local dynamics. The  $(PM)$ -prevalence notion is a measure-theoretic statement along affine lines in the perturbation space; it

must not be confused with Baire genericity or with finite-dimensional Lebesgue measure on the whole function space.

Finally, the annular definitions—exact symplectic flux, action translation, monotone twist lifts, rational invariant graphs, and hyperbolic lobes—are geometric tools for comparing formal normalization with actual invariant curves. They become important precisely when one asks what a divergent formal normal form does or does not imply for the real dynamics.

**Key idea.** Whenever the word “normal form” appears, ask four questions: finite or infinite order? formal or analytic? pointwise germ or uniform complex block? nonresonant removal or resonant retention? Those answers determine what estimates and dynamical conclusions are legitimate.

#### Definition K04.1. Birkhoff normal form

A Birkhoff normal form near an elliptic equilibrium is a formal or finite-order symplectic normal form in which the Hamiltonian is expressed in resonant invariants/actions plus a specified remainder.

#### Definition K04.2. elliptic equilibrium

An elliptic equilibrium is a fixed point whose linearized Hamiltonian vector field has purely imaginary nonzero eigenvalues (with semisimplicity/nonresonance stated when used).

#### Definition K04.3. resonant monomial

After  $H_2 = \sum_j \omega_j I_j$  has been fixed and  $\xi_j = (q_j + ip_j)/\sqrt{2}$ ,  $\eta_j = (q_j - ip_j)/\sqrt{2}$ , the monomial  $\xi^\alpha \eta^\beta$  is resonant precisely when  $(\alpha - \beta) \cdot \omega = 0$ , equivalently when it lies in the kernel of  $F \mapsto \{H_2, F\}$ .

#### Definition K04.4. Birkhoff invariant

A Birkhoff invariant is a coefficient of the normalized action expansion that is unchanged by the allowed normalized canonical transformations at the specified order.

#### Definition K04.5. Poincaré section, return map, and multipliers

Let  $\phi^t$  be a smooth flow and let  $x_0$  lie on a periodic orbit. A local Poincaré section through  $x_0$  is a local submanifold  $\Sigma$  transverse to the flow direction. Where a smooth positive return time  $\tau(x)$  is defined, the associated local Poincaré return map is

$$F(x) = \phi^{\tau(x)}(x), \quad x \in \Sigma.$$

For a Hamiltonian flow, a section used below lies in a regular energy hypersurface  $H^{-1}(E)$ . We write  $\omega_\Sigma := \omega|_{T\Sigma}$  for the restricted two-form; the fact that  $\omega_\Sigma$  is symplectic and that the Hamiltonian return map preserves it is proved when first used in the Normal form near a periodic orbit rather than built into the definition. The

eigenvalues of  $DF(x_0)$  are the nontrivial multipliers of the periodic orbit.

#### Definition K04.6. elliptic multipliers and multiplier resonance

Let  $F$  be a local symplectic map fixing the origin and let  $A = DF(0)$ . The fixed point is semisimple elliptic when  $A$  is diagonalizable over  $\mathbb{C}$  and all of its eigenvalues lie on the unit circle. In complex canonical coordinates one may write the reciprocal-conjugate multiplier pairs as  $\lambda_j, \lambda_j^{-1} = \bar{\lambda}_j$ . For a scalar monomial  $\xi^\alpha \eta^\beta$ , its multiplier character is

$$\lambda^{\alpha-\beta} := \prod_j \lambda_j^{\alpha_j-\beta_j}.$$

The monomial is multiplier-resonant when  $\lambda^{\alpha-\beta} = 1$ . At normalization order  $N$ , finite-order nonresonance means that among monomials of degree at most  $N+1$  the relation  $\lambda^{\alpha-\beta} = 1$  implies  $\alpha = \beta$ .

#### Definition K04.7. resonant complex block and weighted Fourier norm

Let  $D_\Lambda \subset \mathbb{R}^n$ , let  $\rho, \sigma > 0$ , and put

$$\mathbb{T}_\sigma^n := \{\theta \in \mathbb{C}^n / (2\pi\mathbb{Z})^n : |\operatorname{Im} \theta|_\infty < \sigma\}, \quad \mathcal{D}_{\Lambda, \rho, \sigma} := \{(I, \theta) : \operatorname{dist}_2(I, D_\Lambda) < \rho, \theta \in \mathbb{T}_\sigma^n\}.$$

If  $F(I, \theta) = \sum_{k \in \mathbb{Z}^n} F_k(I) e^{ik \cdot \theta}$  is analytic on this block, its weighted Fourier norm is

$$\|F\|_{\rho, \sigma} := \sup_{\operatorname{dist}_2(I, D_\Lambda) < \rho} \sum_{k \in \mathbb{Z}^n} |F_k(I)| e^{|k|_1 \sigma}.$$

For a lattice  $\Lambda \subset \mathbb{Z}^n$  and integer  $K \geq 1$ , define

$$\mathcal{P}_\Lambda F := \sum_{k \in \Lambda} F_k(I) e^{ik \cdot \theta}, \quad \mathcal{P}_K F := \sum_{|k|_1 \leq K} F_k(I) e^{ik \cdot \theta}.$$

A function  $G$  is called  $\Lambda$ -resonant when  $\mathcal{P}_\Lambda G = G$ . Equivalently, every Fourier vector of  $G$  lies in  $\Lambda$ ; hence  $\partial_\theta G(I, \theta) \in \langle \Lambda \rangle_{\mathbb{R}}$  at every point. This fast-drift property, not commutation with a nonlinear integrable Hamiltonian, is the resonant property used in the Nekhoroshev block argument.

#### Definition K04.8. nonresonance modulo a lattice and normal-form auxiliary constants

Let  $h = h(I)$  be analytic with frequency map  $\omega(I) = \partial_I h(I)$ . A set  $D_\Lambda \subset \mathbb{R}^n$  is  $(\alpha, K)$ -nonresonant modulo  $\Lambda$  with respect to  $h$  when

$$|k \cdot \omega(I)| \geq \alpha \quad \text{for every } I \in D_\Lambda \text{ and every } k \in \mathbb{Z}^n \setminus \Lambda \text{ with } 0 < |k|_1 \leq K.$$

On a complex action tube,  $M$  denotes a stated uniform bound for the Hermitian operator norm of the Hessian  $D^2 h$ . For a map  $\Psi(I, \theta)$ ,  $\Pi_I \Psi$  and  $\Pi_\theta \Psi$  denote its action and angle components, respectively. These conventions are the ones used in the Nekhoroshev-ready resonant normal form.

**Definition K04.9. fixed-frequency symplectic germ space, map Birkhoff normal form, and  $(PM)$ -prevalence**

Fix  $d \geq 1$  and  $\omega \in \mathbb{R}^d$ . Put  $r_j = (x_j^2 + y_j^2)/2$  and let  $R_\omega := \Phi_{2\pi\langle\omega, r\rangle}^1$  be the linear symplectic rotation. For a real-analytic germ  $F(x, y) = O(\|(x, y)\|^3)$ , let  $f_F$  be the exact-symplectic germ defined near the origin by the generating-function relations

$$\tilde{x} = x + \partial_{\tilde{y}} F(x, \tilde{y}), \quad y = \tilde{y} + \partial_x F(x, \tilde{y}),$$

and let  $\mathcal{S}_\omega$  denote the corresponding fixed-frequency germs  $f = R_\omega \circ f_F$ . The vector  $\omega$  is nonresonant for this map problem when

$$k \cdot \omega \notin \mathbb{Z} \quad \text{for every } 0 \neq k \in \mathbb{Z}^d,$$

equivalently when the linear multipliers have no nontrivial multiplier resonance. For such  $f$ , its formal map Birkhoff normal form is the unique formal series

$$B(r) = 2\pi\langle\omega, r\rangle + O(|r|^2) \in \mathbb{R}[[r_1, \dots, r_d]]$$

for which there exists a formal symplectic transformation  $Z_\infty$ , tangent to the identity, satisfying the formal conjugacy relation

$$Z_\infty \circ f \circ Z_\infty^{-1} = \Phi_B^1.$$

Here  $\Phi_B^1$  denotes the formal time-one Hamiltonian map generated by the integrable Hamiltonian  $B(r)$ . We write this unique series as  $\text{BNF}(f)$ ; its *radius of convergence* is called positive when the series converges absolutely on some polydisc  $\max_j |r_j| < \rho$  with  $\rho > 0$ , and zero when no such  $\rho$  exists.

If  $\mathcal{E}$  is a real affine space, a subset  $\mathcal{A} \subset \mathcal{E}$  is  $(PM)$ -prevalent when there exists  $F_0 \in \mathcal{A}$  such that for every  $F \in \mathcal{E}$  the exceptional parameter set

$$\{t \in \mathbb{R} : tF_0 + (1-t)F \notin \mathcal{A}\}$$

has Lebesgue measure zero. A subset of  $\mathcal{S}_\omega$  is called  $(PM)$ -prevalent when its generating functions form a  $(PM)$ -prevalent subset in the corresponding real-analytic perturbation space.

**Definition K04.10. invariant-circle deficit, twist coefficient, and continued-fraction scales**

In one degree of freedom let  $M_{\mathbb{R}}$  denote either the action-angle cylinder  $\mathbb{T} \times \mathbb{R}$ , with action coordinate  $r$ , or the plane  $\mathbb{R}^2$ , with radial action  $r = (x^2 + y^2)/2$ . For a local real-analytic area-preserving map  $f$ , let  $\mathcal{G}_f$  be the family of  $f$ -invariant curves: essential invariant Jordan curves in the cylinder, and invariant Jordan curves in the punctured plane whose winding number about the elliptic fixed point is  $+1$  or  $-1$

in the planar setting. For  $t > 0$  define

$$L_f(t) := \bigcup_{\substack{\gamma \in \mathcal{G}_f \\ \gamma \subset M_{\mathbb{R}} \cap \{|r| < t\}}} \gamma$$

and the *invariant-circle deficit*

$$m_f(t) := \text{Leb}_{M_{\mathbb{R}}}((M_{\mathbb{R}} \cap \{|r| < t\}) \setminus L_f(2t)).$$

Thus  $m_f(t)$  measures the area in the smaller action window not covered by invariant curves that already fit inside the doubled window.

For a one-dimensional elliptic germ whose formal Birkhoff normal form is written

$$\frac{1}{2\pi} \text{BNF}(f)(r) = \omega r + b_2(f)r^2 + O(r^3),$$

the *twist condition* is  $b_2(f) \neq 0$ . If  $p_n/q_n$  are the continued-fraction convergents of the irrational frequency  $\omega$ , put

$$\tau(\omega) := \limsup_{n \rightarrow \infty} \frac{\log q_{n+1}}{\log q_n} \in [1, \infty]$$

and, whenever  $b_2(f) \neq 0$ , define the source comparison scale

$$t_n(\omega, f) := \frac{5 \min\{|b_2(f)|, |b_2(f)|^{-1}\}}{q_n q_{n+1}}.$$

The case  $\tau(\omega) < \infty$  is called the *finite-arithmetic-exponent regime*;  $\tau(\omega) = \infty$  is the *Liouville/large-jump regime* used in the proof of the Divergence caveat for formal normal forms.

### Definition K04.11. complex action strips, holed action domains, and type-II exact-symplectic generators

For  $h > 0$  put

$$\mathbb{T}_h := \{\theta \in \mathbb{C}/(2\pi\mathbb{Z}) : |\text{Im } \theta| < h\}.$$

If  $U \subset \mathbb{C}$  is open, the associated one-degree complex action-angle domain is  $W_{h,U} := \mathbb{T}_h \times U$ , and  $\|F\|_{h,U}$  denotes the sup norm there. A *finite holed action domain* is a connected set

$$U = D(0, R) \setminus \bigcup_{j=1}^J \overline{D(c_j, r_j)}$$

with pairwise distinct removed disks compactly contained in  $D(0, R)$ ; it is *real symmetric* when  $\overline{U} = U$ .

For a holomorphic function  $F(\theta, R)$  whose first derivatives are sufficiently small on a slightly larger domain, the *type-II exact-symplectic map*  $f_F$  is the local biholo-

morphism determined implicitly by

$$r = R + \partial_\theta F(\theta, R), \quad \Theta = \theta + \partial_R F(\theta, R).$$

Equivalently,  $\theta R + F(\theta, R)$  is its type-II generating function. On the real slice, a real-valued  $F$  gives an exact symplectic map for  $dr \wedge d\theta$ . For a holomorphic action function  $\Omega(r)$  we write

$$\Phi_\Omega(\theta, r) := (\theta + \Omega'(r), r).$$

Thus  $\Phi_\Omega = f_\Omega$ . If  $F(\theta, r) = \sum_{k \in \mathbb{Z}} F_k(r) e^{ik\theta}$ , set

$$M_0 F := F_0, \quad T_N F := \sum_{|k| < N} F_k(r) e^{ik\theta}, \quad R_N F := F - T_N F.$$

The  $(K, N)$  *map-nonresonance condition* for  $\Omega$  on  $U$  means

$$\text{dist}\left(\frac{k\Omega'(r)}{2\pi}, \mathbb{Z}\right) \geq K^{-1} \quad (r \in U, 0 < |k| < N).$$

All later K04 uses of strip loss, Fourier truncation, map cohomology, deleted resonance holes, and type-II generating maps refer to these conventions unless another domain is stated explicitly.

### Definition K04.12. complex Cartesian KAM domains, generating maps, and rotation Fourier projectors

In one complex degree of freedom put

$$r(z, w) := -izw, \quad \omega_{\text{CC}} := dz \wedge dw.$$

If  $U \subset \mathbb{C}$  is connected,  $h > 0$ , and  $\rho_U := \sup_{r \in U} |r| < \infty$ , define the *complex Cartesian domain*

$$W_{h,U}^{\text{CC}} := \{(z, w) \in \mathbb{C}^2 : |z|, |w| < e^h \rho_U^{1/2}, -izw \in U\}.$$

For the central disk  $U = D(0, \rho)$  and  $0 < \delta < h$ , the notation

$$e^{-\delta} W_{h,D(0,\rho)}^{\text{CC}} := W_{h-\delta/2,D(0,e^{-\delta}\rho)}^{\text{CC}}$$

denotes the multiplicatively shrunk domain used in the CC normal-form iteration.

For a holomorphic scalar function  $F(z, W)$  whose first derivatives are sufficiently small on a larger CC domain, its *CC generating map*  $f_F$  is the local exact-symplectic map  $f_F(z, w) = (Z, W)$  determined implicitly by

$$Z = z + \partial_W F(z, W), \quad w = W + \partial_z F(z, W).$$

Thus  $zW + F(z, W)$  is the type-II generating function in the canonical pair  $(z, w)$ . For



a holomorphic action function  $\Omega(r)$  we write

$$\Phi_\Omega(z, w) := (e^{-i\Omega'(r)}z, e^{i\Omega'(r)}w), \quad r = -izw.$$

The linear rotation of frequency  $\omega$  is  $R_\omega = \Phi_{Q_0}$  with  $Q_0(r) = 2\pi\omega r$ . For  $F \in \mathcal{O}(W_{h,U}^{\text{CC}})$  define its rotation-Fourier coefficients by

$$M_n F(z, w) := \frac{1}{2\pi} \int_0^{2\pi} e^{-int} F(e^{-it}z, e^{it}w) dt, \quad n \in \mathbb{Z}.$$

Then  $M_0 F$  is rotation-invariant and therefore has the form  $M_0 F = M(F) \circ r$  for a unique holomorphic action function  $M(F)$  on  $U$ . For  $N \geq 1$  put

$$T_N F := \sum_{|n| < N} M_n F, \quad R_N F := F - T_N F.$$

Finally,  $F = O^p(z, w)$  means that every Taylor monomial  $z^a w^b$  with  $a + b < p$  has zero coefficient. This is a total-degree filtration; it is distinct from the action-order notation  $O(r^m)$ .

#### Definition K04.13. exact Cartesian-action-angle branch charts

Let

$$\lambda_{\text{CC}} := \frac{1}{2}(w dz - z dw), \quad \lambda_{\text{AA}} := r d\theta.$$

Write  $\mathbb{R}_- := (-\infty, 0]$  and  $\mathbb{R}_+ := [0, \infty)$ . Choose the holomorphic square-root branch on either of the two cut planes  $\mathbb{C} \setminus \mathbb{R}_-$  and  $\mathbb{C} \setminus \mathbb{R}_+$ . On the corresponding branch domain define

$$\psi_\pm(\theta, r) := \left( e^{i\pi/4} r^{1/2} e^{-i\theta}, e^{i\pi/4} r^{1/2} e^{i\theta} \right).$$

Then

$$r = -izw, \quad \psi_\pm^* \lambda_{\text{CC}} = \lambda_{\text{AA}}, \quad \psi_\pm^*(dz \wedge dw) = d\theta \wedge dr.$$

Thus each  $\psi_\pm$  is an exact holomorphic symplectic branch chart. Near the elliptic fixed point the two branches are used on overlapping angular sectors whose union covers the punctured action disk; on an off-centre action disk  $D(c, \rho)$  with  $|c| > 4\rho$ , one square-root branch is available on the whole disk. Whenever a CC map is transferred by  $\psi_\pm^{-1} \circ f \circ \psi_\pm$ , the integrable action function is unchanged because  $-izw = r$  in both coordinate systems.

#### Definition K04.14. annular symplectic flux and action translation

Let  $g$  be a holomorphic symplectic map, homotopic to the identity, on a connected action-angle annulus, and let  $\lambda_{\text{AA}} = r d\theta$ . The one-form

$$\alpha_g := g^* \lambda_{\text{AA}} - \lambda_{\text{AA}}$$

is closed. Its period along any positively oriented angle circle is independent of

the chosen action level. We call

$$\text{Flux}(g) := \frac{1}{2\pi} \int_{\mathbb{T} \times \{r\}} \alpha_g$$

the *annular symplectic flux* of  $g$ . The map is exact symplectic precisely when this period vanishes. For the vertical symplectic shift  $G(\theta, r) = (\theta, r - e_0(\theta))$  one has  $\text{Flux}(G) = -\langle e_0 \rangle$ .

When two one-dimensional integrable normal forms are related by a near-identity symplectic, but not necessarily exact, coordinate change, their frequency maps are in general comparable only after a real constant translation  $r \mapsto r + \gamma$  of the action variable. We call this  $\gamma$  the *action translation* (or *flux action shift*) of the comparison. Exactness removes the leading constant translation through the zero-period identity; see Lemma 4.43–Lemma 4.54.

#### Definition K04.15. positive monotone twist lift, rational invariant graph, and hyperbolic lobe

Let  $F$  be a  $C^1$  area-preserving annulus diffeomorphism isotopic to the identity and let  $\tilde{F} = (X, R)$  be a lift to  $\mathbb{R} \times [a, b]$ . On a stated compact strip,  $F$  is a *positive monotone twist map* when

$$\partial_r X(\theta, r) \geq \kappa > 0.$$

(The negative-twist convention is reduced to this one by replacing  $F$  with  $F^{-1}$ .) An *essential invariant graph* is an invariant essential Jordan curve written  $r = u(\theta)$ ; when its rotation number is  $p/q \in \mathbb{Q}$ , it is called a *rational invariant graph*. If a hyperbolic  $q$ -periodic point has local stable and unstable branches  $r = w_-(\theta)$  and  $r = w_+(\theta)$  meeting at the periodic point, a *vertical-cap hyperbolic lobe* is the open region between those branches on a one-sided angular interval, closed at the far end by a vertical segment. The quantitative branch separation and lobe area used below are proved in Lemma 4.65–Lemma 4.67.

## 4.2 How the main proofs work

### Intuition

**Finite-order Birkhoff normal form.** Expand the Hamiltonian homogeneously. On each degree, complex monomials diagonalize the operator  $F \mapsto \{H_2, F\}$ . Project the current term onto its resonant kernel, solve the homological equation on the complement, and use the time-one Hamiltonian flow of the solution. Degree counting shows that earlier normalized terms are untouched and the new error starts one order higher.

**Resonant normal form.** The same calculation is performed without assuming nonresonance. Resonant monomials are not “bad errors”; they are the part that genuinely survives because its divisor is zero. The normalized polynomial therefore records the resonance geometry of the linearized motion.

**Twist matrix.** Under fourth-order nonresonance, the quartic normal form depends only on the actions. Writing it as  $\omega \cdot I + \frac{1}{2} I^T A I$  makes the derivative of the frequency map equal to  $A$ . Kolmogorov nondegeneracy near the elliptic point is therefore a statement about Birkhoff invariants.

**Periodic orbit.** Restrict the Hamiltonian flow to a transverse energy section. The first-return map is symplectic. Normalize its linear elliptic part and repeat the monomial homological calculation using multiplier characters instead of continuous-time divisors.

**Divergence caveat.** The deep fixed-frequency result does not prove that every formal Birkhoff series diverges. It establishes a prevalence statement in the specified symplectic-germ space. Its role in this book is conceptual and defensive: a formal Birkhoff series may not be treated as an analytic coordinate change without an independent convergence theorem.

**Nekhoroshev-ready resonant block.** Fourier-truncate the perturbation and split it into lattice-resonant and nonresonant modes. The assumed divisor bound solves the homological equation for the latter. Cauchy estimates control the generating flow on a smaller complex block. Repeating the averaging step yields a resonant part whose angle derivative lies in the real span of the lattice and an exponentially small high-Fourier remainder. This is the local engine used in K08.

### 4.3 Main results

#### Theorem K04.1. Finite-order Hamiltonian Birkhoff normal form

**Statement.** Let  $H$  be real analytic near  $0 \in \mathbb{R}^{2n}$  and suppose that, after a fixed real linear symplectic change of coordinates for the symplectic form  $\omega = \sum_j dp_j \wedge dq_j$ , one has

$$H(q, p) = H_2(q, p) + P(q, p), \quad H_2 = \sum_{j=1}^n \omega_j I_j, \quad I_j = \frac{q_j^2 + p_j^2}{2}, \quad P = O(\|(q, p)\|^3),$$

with real nonzero frequencies  $\omega_j$ . For every integer  $N \geq 3$  there is a real-analytic local symplectic map  $\Phi_N$ , tangent to the identity in the sense  $\Phi_N(z) = z + O(\|z\|^2)$ , such that

$$H \circ \Phi_N = H_2 + Z_3 + \cdots + Z_N + R_{N+1},$$

where each  $Z_m$  is a real homogeneous polynomial of degree  $m$  satisfying  $\{H_2, Z_m\} = 0$  and  $R_{N+1} = O(\|z\|^{N+1})$ . If  $k \cdot \omega \neq 0$  for every  $0 \neq k \in \mathbb{Z}^n$  with  $|k|_1 \leq N$ , then every  $Z_m$  is a polynomial in the actions  $I_1, \dots, I_n$  (hence  $Z_m = 0$  for odd  $m$ ).

*Proof.* Write  $z = (q, p)$ . Since only a fixed finite order is involved, it is enough to work with the Taylor expansion of  $H$  through degree  $N$  and keep the analytic remainder explicitly. We use

$$\xi_j := \frac{q_j + ip_j}{\sqrt{2}}, \quad \eta_j := \frac{q_j - ip_j}{\sqrt{2}}. \quad (\text{BNF.1})$$

On the real locus  $\eta = \bar{\xi}$ , and  $I_j = \xi_j \eta_j$ . Under the convention used in this book  $\omega = \sum_j dp_j \wedge dq_j$  and  $\{F, G\} = dG(X_F)$ , a direct change of variables gives

$$\{F, G\} = i \sum_{j=1}^n (F_{\xi_j} G_{\eta_j} - F_{\eta_j} G_{\xi_j}). \quad (\text{BNF.2})$$

Thus, with  $H_2 = \sum_j \omega_j \xi_j \eta_j$ ,

$$\{H_2, \xi^\alpha \eta^\beta\} = -i((\alpha - \beta) \cdot \omega) \xi^\alpha \eta^\beta. \quad (\text{BNF.3})$$

We first record the homogeneous homological decomposition. Let  $Q_m = \sum_{|\alpha|+|\beta|=m} q_{\alpha\beta} \xi^\alpha \eta^\beta$  be a real homogeneous polynomial of degree  $m$ . Set

$$Z_m := \sum_{(\alpha-\beta) \cdot \omega=0} q_{\alpha\beta} \xi^\alpha \eta^\beta, \quad (\text{BNF.4})$$

and on nonresonant monomials define

$$(S_m)_{\alpha\beta} := -\frac{q_{\alpha\beta}}{i(\alpha - \beta) \cdot \omega}, \quad (S_m)_{\alpha\beta} := 0 \quad \text{on resonant monomials.} \quad (\text{BNF.5})$$

By (BNF.3),

$$Q_m + \{S_m, H_2\} = Z_m, \quad \{H_2, Z_m\} = 0. \quad (\text{BNF.6})$$

Reality of  $Q_m$  means  $q_{\beta\alpha} = \overline{q_{\alpha\beta}}$ . Since the divisor changes sign when  $(\alpha, \beta)$  is interchanged, (BNF.5) has the same reality symmetry. Hence  $S_m$  and  $Z_m$  are real on the real locus.

We now normalize inductively. Suppose for some  $m \in \{3, \dots, N\}$  that a real-analytic local symplectic map  $\Phi_{m-1}$ , tangent to the identity, has been constructed so that

$$H \circ \Phi_{m-1} = H_2 + Z_3 + \dots + Z_{m-1} + Q_m + R_{m+1}, \quad (\text{BNF.7})$$

where every  $Z_j$  is homogeneous of degree  $j$  and commutes with  $H_2$ ,  $Q_m$  is homogeneous of degree  $m$ , and  $R_{m+1} = O(\|z\|^{m+1})$ . For  $m = 3$  this follows from Taylor's theorem with  $\Phi_2 = \text{id}$ .

Let  $S_m$  be the polynomial obtained from  $Q_m$  by (BNF.5), and let  $\psi_m = \varphi_{S_m}^1$  be its time-one Hamiltonian map. Because  $X_{S_m} = O(\|z\|^{m-1})$ , after restricting to a sufficiently small neighborhood the flow exists for  $0 \leq t \leq 1$  and

$$\psi_m(z) = z + O(\|z\|^{m-1}). \quad (\text{BNF.8})$$

It is symplectic by the Hamiltonian-flow theorem. For homogeneous polynomials  $A_a, B_b$  of degrees  $a, b$ ,

$$\deg\{A_a, B_b\} = a + b - 2. \quad (\text{BNF.9})$$

Taylor expansion along the  $S_m$ -flow therefore shows that the only contribution of degree exactly  $m$  created from  $H_2$  is  $\{S_m, H_2\}$ ; every second or higher bracket with  $H_2$  has degree at least  $2m - 2 \geq m + 1$ , every bracket with  $Z_j$  ( $j \geq 3$ ) has degree at least  $m + 1$ ,

and composition of  $Q_m + R_{m+1}$  changes it only in degree at least  $m + 1$ . Consequently

$$H \circ \Phi_{m-1} \circ \psi_m = H_2 + Z_3 + \cdots + Z_{m-1} + Q_m + \{S_m, H_2\} + \tilde{R}_{m+1}, \quad (\text{BNF.10})$$

with  $\tilde{R}_{m+1} = O(\|z\|^{m+1})$ . By (BNF.6) the degree- $m$  term is  $Z_m$ . Taylor-expand  $\tilde{R}_{m+1}$  into its homogeneous degree- $(m + 1)$  part plus a remainder of order  $m + 2$ ; this gives the induction hypothesis for the next step. Set  $\Phi_m := \Phi_{m-1} \circ \psi_m$ .

After the step  $m = N$  we obtain

$$H \circ \Phi_N = H_2 + \sum_{m=3}^N Z_m + R_{N+1}, \quad R_{N+1} = O(\|z\|^{N+1}), \quad (\text{BNF.11})$$

with every  $Z_m$  in the homological kernel. Since the first generator has degree 3, each factor differs from the identity by order at least 2, and therefore so does their finite composition:

$$\Phi_N(z) = z + O(\|z\|^2). \quad (\text{BNF.12})$$

All generators are polynomial and real; their local time-one flows and the finite composition are real analytic.

Finally suppose  $k \cdot \omega \neq 0$  for every nonzero  $k \in \mathbb{Z}^n$  with  $|k|_1 \leq N$ . If a degree- $m \leq N$  monomial is resonant, then  $k := \alpha - \beta$  satisfies  $|k|_1 \leq m \leq N$ , so resonance forces  $k = 0$ , hence  $\alpha = \beta$ . Such a monomial is

$$\xi^\alpha \eta^\alpha = I_1^{\alpha_1} \cdots I_n^{\alpha_n}.$$

Thus each  $Z_m$  depends only on the actions. Since actions have degree two in  $z$ , no odd-degree action polynomial exists, so  $Z_m = 0$  for odd  $m$ . This proves all assertions.  $\square$

#### Theorem K04.2. Resonant Birkhoff normal form

**Statement.** In the setting of the Finite-order Hamiltonian Birkhoff normal form introduce complex variables  $\xi_j = (q_j + ip_j)/\sqrt{2}$ ,  $\eta_j = (q_j - ip_j)/\sqrt{2}$ . For a monomial  $\xi^\alpha \eta^\beta$  put  $\Delta_{\alpha\beta} := (\alpha - \beta) \cdot \omega$ . The homological operator satisfies

$$\{H_2, \xi^\alpha \eta^\beta\} = -i\Delta_{\alpha\beta} \xi^\alpha \eta^\beta.$$

At every homogeneous normalization step one may choose the generator with zero resonant part so that the new normal-form term is exactly the projection of the current homogeneous term onto the span of monomials with  $\Delta_{\alpha\beta} = 0$ ; every monomial with  $\Delta_{\alpha\beta} \neq 0$  is removed by division by its nonzero divisor. Thus the finite-order normal form retains precisely the resonant monomials at each step.

#### Scope note

The retained coefficients are those of the current transformed homogeneous polynomial; they need not equal the same-degree coefficients of the original untransformed Hamiltonian.

*Proof.* The proof of the Finite-order Hamiltonian Birkhoff normal form established, in

the complex variables  $(\xi, \eta)$ , the diagonal identity

$$\{H_2, \xi^\alpha \eta^\beta\} = -i\Delta_{\alpha\beta} \xi^\alpha \eta^\beta, \quad \Delta_{\alpha\beta} := (\alpha - \beta) \cdot \omega. \quad (\text{RBNF.1})$$

Hence on the finite-dimensional space  $\mathcal{P}_m$  of homogeneous degree- $m$  polynomials the homological operator  $\mathcal{A}_m Q := \{H_2, Q\}$  has

$$\ker \mathcal{A}_m = \text{span}\{\xi^\alpha \eta^\beta : |\alpha| + |\beta| = m, \Delta_{\alpha\beta} = 0\}, \quad (\text{RBNF.2})$$

and the complementary monomial span

$$\mathcal{N}_m := \text{span}\{\xi^\alpha \eta^\beta : |\alpha| + |\beta| = m, \Delta_{\alpha\beta} \neq 0\} \quad (\text{RBNF.3})$$

is invariant and  $\mathcal{A}_m|_{\mathcal{N}_m}$  is invertible. Thus

$$\mathcal{P}_m = \ker \mathcal{A}_m \oplus \mathcal{N}_m. \quad (\text{RBNF.4})$$

Let  $Q_m$  be the degree- $m$  polynomial present after the previous normalization steps. Decompose

$$Q_m = Q_m^{\text{res}} + Q_m^{\text{nr}} \quad (\text{RBNF.5})$$

according to (RBNF.4). In the normalization convention of RES-001 the generator has no resonant component and is chosen uniquely on  $\mathcal{N}_m$  by

$$\{S_m, H_2\} = -Q_m^{\text{nr}}. \quad (\text{RBNF.6})$$

Coefficientwise this is precisely division by the nonzero numbers  $i\Delta_{\alpha\beta}$ , with the reality pairing handled as in (BNF.5). Therefore the degree- $m$  term after the Lie transform is

$$Q_m + \{S_m, H_2\} = Q_m^{\text{res}}. \quad (\text{RBNF.7})$$

Every monomial with nonzero divisor has disappeared, while every zero-divisor monomial is retained. Degree counting from RES-001 shows that lower normal-form degrees are unchanged and all effects of the transformation on higher terms enter only at higher degree.

It is essential that  $Q_m$  in (RBNF.5) is the *current* homogeneous polynomial after the transformations of degrees  $3, \dots, m-1$ . Those earlier transformations can change higher Taylor coefficients, so for  $m > 3$  the coefficient of a retained resonant monomial need not equal its coefficient in the original untransformed Hamiltonian. The theorem claims exact resonant projection at each homological step, and (RBNF.7) proves exactly that claim.  $\square$

### Theorem K04.3. Twist matrix from Birkhoff invariants

**Statement.** Assume the frequencies in the Finite-order Hamiltonian Birkhoff normal form are nonresonant up to order 4. Then the Birkhoff normal form through

degree 4 has the unique tangent-to-identity normalized form

$$H \circ \Phi_4 = \omega \cdot I + \frac{1}{2} I^T A I + O(\|z\|^5),$$

for a real symmetric matrix  $A$ . The quartic coefficients, equivalently the entries of  $A$  after the ordered oscillator coordinates are fixed, are Birkhoff invariants. For the integrable Birkhoff truncation  $K(I) = \omega \cdot I + \frac{1}{2} I^T A I$  the frequency map is  $\Omega(I) = \nabla_I K(I) = \omega + AI$ , so  $D\Omega(0) = A$ . Hence the elliptic Kolmogorov twist condition is exactly  $\det A \neq 0$ .

*Proof.* Apply the Finite-order Hamiltonian Birkhoff normal form with  $N = 4$ . Since the frequency vector is nonresonant up to order 4, the degree-3 normal-form term vanishes and the degree-4 term is a homogeneous polynomial of degree two in the actions. Hence there is a real matrix  $A_0$  such that

$$Z_4(I) = \frac{1}{2} I^T A_0 I. \quad (\text{TW.1})$$

Replacing  $A_0$  by its symmetric part does not change the quadratic form, so we take  $A = A^T$  and obtain

$$H \circ \Phi_4 = \omega \cdot I + \frac{1}{2} I^T A I + O(\|z\|^5). \quad (\text{TW.2})$$

We next justify the asserted Birkhoff invariance rather than merely naming the coefficients. At each homogeneous degree  $m$ , the Resonant Birkhoff normal form gives the direct decomposition

$$\mathcal{P}_m = \ker \mathcal{A}_m \oplus \mathcal{N}_m, \quad \mathcal{A}_m Q := \{H_2, Q\}, \quad (\text{TW.3})$$

where  $\mathcal{A}_m$  is invertible on  $\mathcal{N}_m$ . Consider two tangent-to-identity Birkhoff normalizations through degree 4 that use the standard convention that every generating polynomial has zero resonant component. Let  $m \in \{3, 4\}$  be the first degree at which the resulting normalized homogeneous terms differ. The lower-degree coordinate changes agree through the order relevant to degree  $m$ , and the difference at degree  $m$  must therefore have the form

$$\tilde{Z}_m - Z_m = \{S_m, H_2\} \quad (\text{TW.4})$$

for a degree- $m$  generator  $S_m \in \mathcal{N}_m$ . The left side lies in  $\ker \mathcal{A}_m$ , because both terms are normalized, while the right side lies in  $\mathcal{N}_m$  by invariance and invertibility of the complementary monomial space. The intersection in (TW.3) is zero, so  $\tilde{Z}_m = Z_m$ . This contradiction to the choice of  $m$  proves uniqueness through degree 4 under the fixed tangent-to-identity normalization convention. Consequently the coefficients of  $Z_4$ , equivalently the entries of the symmetric matrix  $A$  once the ordered oscillator coordinates have been fixed, are Birkhoff invariants at this order.

Finally consider the integrable Birkhoff truncation

$$K(I) := \omega \cdot I + \frac{1}{2} I^T A I. \quad (\text{TW.5})$$

By the definition of frequency map above its frequency map is the action gradient:

$$\Omega(I) = \nabla_I K(I) = \omega + AI, \quad D\Omega(0) = A. \quad (\text{TW.6})$$

Thus the Kolmogorov nondegeneracy condition at the elliptic equilibrium is

$$\det D\Omega(0) \neq 0 \iff \det A \neq 0. \quad (\text{TW.7})$$

This is exactly the local twist matrix extracted from the quartic Birkhoff invariants.  $\square$

#### Theorem K04.4. Normal form near a periodic orbit

**Statement.** Let  $\gamma$  be an elliptic periodic orbit of a real-analytic Hamiltonian flow and let  $F : (\Sigma, x_0) \rightarrow (\Sigma, x_0)$  be the local symplectic Poincaré return map on a transverse section. Assume  $DF(x_0)$  is semisimple with unit-circle multipliers and fix a finite normalization order  $N$ . Then a local analytic symplectic change of coordinates, tangent to the identity after linear symplectic normalization, puts the  $N$ -jet of  $F$  into Birkhoff normal form: every nonresonant monomial through order  $N$  is removed and only monomials satisfying the corresponding multiplier-resonance relations remain. In the finite-order nonresonant case the normalized  $N$ -jet commutes with the linear rotation and has the usual action-dependent Birkhoff form.

*Proof.* We first justify the symplectic structure implicit in the theorem. Let the periodic orbit lie on the regular energy hypersurface  $M_E := H^{-1}(E)$  and let  $\Sigma \subset M_E$  be a transverse Poincaré section through  $x_0$  as in the definition of Poincaré section, return map, and multipliers above. Since

$$\ker(\omega|_{T_x M_E}) = \mathbb{R}X_H(x)$$

and  $T_x \Sigma$  is transverse to  $X_H(x)$  inside  $T_x M_E$ , the restriction  $\omega_\Sigma := \omega|_{T\Sigma}$  is nondegenerate. If  $F(x) = \phi^{\tau(x)}(x)$ , then for  $v \in T_x \Sigma$ ,

$$DF_x v = D\phi_x^{\tau(x)} v + d\tau_x(v)X_H(F(x)).$$

The Hamiltonian flow preserves  $\omega$ , and both  $D\phi_x^{\tau(x)} v$  and  $D\phi_x^{\tau(x)} w$  are tangent to  $M_E$ . Hence  $\omega(X_H, \cdot) = -dH(\cdot)$  by the convention used in this book  $\iota_{X_H} \omega = -dH$  makes all return-time cross terms vanish, giving

$$\omega_\Sigma(DF_x v, DF_x w) = \omega_\Sigma(v, w).$$

Thus the local return map is indeed symplectic; no unproved symplecticity assertion is hidden in the definition of Poincaré section, return map, and multipliers above.

Move  $x_0$  to the origin and identify the section with a neighborhood of 0 in a real symplectic vector space  $V$  of dimension  $2d$ . Put

$$A := DF(0). \quad (\text{PM.1})$$

The hypothesis says that  $A$  is a semisimple symplectic linear map and all of its multipliers lie on the unit circle. After the stated fixed real linear symplectic normalization and



complexification, choose complex canonical coordinates  $(\xi, \eta)$  in reciprocal-conjugate pairs so that

$$A(\xi_j, \eta_j) = (\lambda_j \xi_j, \lambda_j^{-1} \eta_j), \quad |\lambda_j| = 1. \quad (\text{PM.2})$$

The proof is finite order, so all coordinate changes below are local analytic Hamiltonian time-one maps after the neighborhood has been made sufficiently small.

We first isolate the elementary fact that makes the map normalization Hamiltonian. Suppose  $G_1$  and  $G_2$  are two symplectic germs fixing the origin, have the same linear part, and agree through degree  $m - 1$ . Then

$$E := G_2^{-1} \circ G_1 = \text{id} + Y_m + O(\|z\|^{m+1}), \quad (\text{PM.3})$$

where  $Y_m$  is homogeneous of degree  $m$ . Since  $E$  is symplectic, expansion of  $E^*\omega = \omega$  at the first nonzero homogeneous order gives

$$\mathcal{L}_{Y_m}\omega = 0. \quad (\text{PM.4})$$

Because  $\omega$  is closed, Cartan's identity yields  $d(\iota_{Y_m}\omega) = 0$ . The one-form  $\iota_{Y_m}\omega$  is polynomial and the neighborhood is star-shaped, so the Poincaré lemma gives a homogeneous polynomial  $Q_{m+1}$  of degree  $m + 1$  with

$$\iota_{Y_m}\omega = -dQ_{m+1}. \quad (\text{PM.5})$$

With the Hamiltonian convention used in this book this says  $Y_m = X_{Q_{m+1}}$ . Thus the first discrepancy between two symplectic jets is generated by a homogeneous Hamiltonian.

We now normalize inductively. Assume that for some  $m \geq 2$  an analytic symplectic change  $C_{m-1}$  tangent to the identity has been found such that

$$F_{m-1} := C_{m-1}^{-1} \circ F \circ C_{m-1} \quad (\text{PM.6})$$

agrees through degree  $m - 1$  with an analytic symplectic germ  $N_{m-1}$  that commutes with  $A$ . Initially take  $C_1 = \text{id}$  and  $N_1 = A$ . By (PM.3)-(PM.5), applied to  $F_{m-1}$  and  $N_{m-1}$ , there is a real homogeneous Hamiltonian  $Q_{m+1}$  such that

$$N_{m-1}^{-1} \circ F_{m-1} = \text{id} + X_{Q_{m+1}} + O(\|z\|^{m+1}). \quad (\text{PM.7})$$

Here the vector field  $X_{Q_{m+1}}$  has degree  $m$ .

Consider the pullback operator on homogeneous Hamiltonians

$$\mathcal{U}Q := Q \circ A. \quad (\text{PM.8})$$

For a monomial  $M_{\alpha\beta} = \xi^\alpha \eta^\beta$ , (PM.2) gives

$$\mathcal{U}M_{\alpha\beta} = \lambda^{\alpha-\beta} M_{\alpha\beta}. \quad (\text{PM.9})$$

Hence the space of degree- $(m + 1)$  Hamiltonians is the direct sum of the resonant eigenspace  $\ker(I - \mathcal{U})$  and the span of monomials for which  $\lambda^{\alpha-\beta} \neq 1$ . Write

$$Q_{m+1} = Z_{m+1} + Q_{m+1}^{\text{nr}}, \quad Z_{m+1} \circ A = Z_{m+1}. \quad (\text{PM.10})$$

On every nonresonant monomial the operator  $I - \mathcal{U}$  is invertible. Choose the homogeneous polynomial  $S_{m+1}$  with zero resonant part and

$$(I - \mathcal{U})S_{m+1} = -Q_{m+1}^{\text{nr}}. \quad (\text{PM.11})$$

Explicitly, if  $q_{\alpha\beta}$  is a nonresonant coefficient of  $Q_{m+1}$ , then

$$(S_{m+1})_{\alpha\beta} = -\frac{q_{\alpha\beta}}{1 - \lambda^{\alpha-\beta}}. \quad (\text{PM.12})$$

The conjugate-pair symmetry of the coefficients and  $|\lambda_j| = 1$  show that  $S_{m+1}$  may be chosen real on the real locus.

Let  $\psi_m = \varphi_{S_{m+1}}^1$ . Since  $X_{S_{m+1}} = O(\|z\|^m)$ ,

$$\psi_m(z) = z + X_{S_{m+1}}(z) + O(\|z\|^{m+1}). \quad (\text{PM.13})$$

For a symplectic linear map  $A$  one has

$$A^{-1}X_S(Az) = X_{S \circ A}(z). \quad (\text{PM.14})$$

Because  $N_{m-1} = A + O(\|z\|^2)$ , replacing  $N_{m-1}$  by its linear part in the commutator with the degree- $m$  vector field changes only degree  $m+1$  and higher. Using (PM.7), (PM.13), and (PM.14), a direct degree- $m$  composition gives

$$\psi_m^{-1} \circ F_{m-1} \circ \psi_m = N_{m-1} \circ (\text{id} + X_{Q_{m+1}+S_{m+1}-S_{m+1} \circ A}) + O(\|z\|^{m+1}). \quad (\text{PM.15})$$

Equation (PM.11) reduces the Hamiltonian in (PM.15) to  $Z_{m+1}$ . Define

$$N_m := N_{m-1} \circ \varphi_{Z_{m+1}}^1, \quad C_m := C_{m-1} \circ \psi_m. \quad (\text{PM.16})$$

Since  $Z_{m+1} \circ A = Z_{m+1}$ , its Hamiltonian flow commutes with  $A$ . By induction  $N_{m-1}$  also commutes with  $A$ , so  $N_m$  commutes with  $A$ . Moreover (PM.15) says that  $C_m^{-1}FC_m$  and  $N_m$  agree through degree  $m$ . This closes the induction.

Iterating for  $m = 2, \dots, N$  gives a local analytic symplectic map  $C_N$ , tangent to the identity, for which the  $N$ -jet of  $C_N^{-1}FC_N$  is generated degree by degree only by the Hamiltonian monomials satisfying

$$\lambda^{\alpha-\beta} = 1. \quad (\text{PM.17})$$

Equivalently every nonresonant multiplier character has been removed. This is the asserted finite-order Birkhoff normal form for the Poincaré map.

Finally assume the finite-order nonresonance condition of the definition of elliptic multipliers and multiplier resonance above. Every resonant scalar monomial of degree at most  $N+1$  then has  $\alpha - \beta = 0$ , hence is a monomial in the actions

$$I_j = \xi_j \eta_j. \quad (\text{PM.18})$$

Therefore each retained homogeneous Hamiltonian  $Z_{m+1}$  is an action polynomial. Its Hamiltonian flow preserves every  $I_j$  and rotates each complex oscillator by a phase depending only on  $I$ . Such flows commute with the linear rotation  $A$ , and their fi-

nite composition has the usual action-dependent Birkhoff form. This proves the final assertion as well.  $\square$

### Theorem K04.5. Divergence caveat for formal normal forms

**Statement.** Fix  $d \geq 1$  and a nonresonant frequency vector  $\omega \in \mathbb{R}^d$  in the sense of Definition K04.9. In the fixed-frequency real-analytic symplectic-germ space  $\mathcal{S}_\omega$ , the subset

$$\{f \in \mathcal{S}_\omega : \text{BNF}(f) \text{ has radius of convergence } 0\}$$

is  $(PM)$ -prevalent. In particular, for every such  $\omega$  there exists a real-analytic symplectic germ with that elliptic linear part and divergent formal Birkhoff normal form.

#### Scope note

The displayed prevalent-divergence assertion is Krikorian's fixed-point result. The Hamiltonian fixed-quadratic-part setting is governed by the related Pérez-Marco dichotomy; no analytic convergence is inferred from formal normal-form existence.

#### Scope note

A formal Birkhoff normal form is not thereby identified with a convergent analytic normalization; convergence requires a separate theorem.

#### Proof strategy

The proof is a contradiction argument organized into four mechanisms. Algebraic and arithmetic lemmas control the Birkhoff normal form coefficients and resonant scales. Analytic continuation on holed domains and Hamilton-Jacobi fillings convert convergence of the formal normal form into strong geometric extension properties. KAM iteration then forces an upper estimate on the size of resonant holes, while a lacunary resonant construction produces incompatible lower holes. The final Remez/prevalence argument converts this contradiction into the stated generic divergence conclusion.

**Lemma 4.1 (affine-parameter polynomiality of the map BNF).** Fix a nonresonant frequency vector  $\omega \in \mathbb{R}^d$  in the sense of the definition of fixed-frequency symplectic germ space, map Birkhoff normal form, and  $(PM)$ -prevalence above. Let  $F_0, F_1$  be real-analytic generating-function germs of order at least three and put

$$F_t := tF_0 + (1-t)F_1, \quad f_t := R_\omega \circ f_{F_t}. \quad (\text{PM.1})$$

Write the unique formal Birkhoff normal form as

$$B_t(r) = 2\pi \omega \cdot r + \sum_{|n|_1 \geq 2} b_n(t)r^n, \quad r^n = r_1^{n_1} \cdots r_d^{n_d}. \quad (\text{PM.2})$$

Then every  $b_n(t)$  is a real polynomial in  $t$  and

$$\deg_t b_n \leq 2(|n|_1 - 1). \quad (\text{PM.3})$$

*Proof.* We give the filtered formal argument because the degree estimate, rather than merely formal existence of the BNF, is the load-bearing point. Complexify the phase variables and use canonical pairs  $(z_j, w_j)$ , so that  $r_j = z_j w_j$  on the real-symmetric locus. Total degree below always means total  $(z, w)$ -degree. The affine family  $F_t$  has homogeneous pieces  $[F_t]_m$  of degree  $m \geq 3$ , each of parameter degree at most one. In particular

$$\deg_t [F_t]_m \leq m - 2. \quad (\text{PM.4})$$

We first record three elementary closure facts for formal maps. Suppose

$$U_t(u) = u + \sum_{q \geq 2} U_q(t; u), \quad \deg_t U_q \leq q - 1, \quad (\text{PM.5})$$

where  $U_q$  is homogeneous of phase degree  $q$ . If  $V_t$  has the same property, then both  $U_t \circ V_t$  and  $U_t^{-1}$  have the same property. For composition, the degree- $q$  term is a finite sum of expressions obtained by substituting homogeneous pieces of  $V_t - u$  into a homogeneous polynomial  $U_m$ . If the Taylor term uses  $s$  substituted displacement pieces of degrees  $q_1, \dots, q_s \geq 2$ , then the  $s$ th derivative of the homogeneous polynomial  $U_m$  has phase degree  $m - s$ . Hence the resulting phase degree is

$$q = (m - s) + q_1 + \dots + q_s.$$

The parameter degree of that term is at most

$$(m - 1) + \sum_{j=1}^s (q_j - 1) = m - 1 + \sum_{j=1}^s q_j - s = q - 1.$$

Thus every degree- $q$  contribution satisfies the same filtration, proving the composition assertion. For inversion, write  $U_t^{-1}(u) = u + W_t(u)$  and use

$$W_t(u) = -(U_t - \text{id})(u + W_t(u)). \quad (\text{PM.6})$$

The degree- $q$  coefficient of  $W_t$  depends polynomially only on the coefficients of  $U_2, \dots, U_q$  and  $W_2, \dots, W_{q-1}$ ; induction with the composition estimate gives  $\deg_t W_q \leq q - 1$ .

Next pass from a generating function to its exact-symplectic map. In the real coordinates of the definition of fixed-frequency symplectic germ space, map Birkhoff normal form, and  $(PM)$ -prevalence above the implicit equations are

$$\tilde{x} = x + \partial_{\tilde{y}} F_t(x, \tilde{y}), \quad y = \tilde{y} + \partial_x F_t(x, \tilde{y}). \quad (\text{PM.7})$$

Because  $F_t = O(3)$ , the displacement starts in degree two. Solving the second equation recursively for  $\tilde{y} - y$  and substituting into the first shows, by the same filtered-composition induction, that the degree- $q$  displacement of  $f_{F_t}$  has parameter degree at

most  $q - 1$ . Multiplication by the fixed linear rotation  $R_\omega$  does not alter this estimate. Thus  $f_t$  satisfies (PM.5).

We now normalize degree by degree, using the scalar generating Hamiltonian of the normalizing exact-symplectic map. Assume that all displacement terms below phase degree  $q$  have already been normalized by a formal exact-symplectic change satisfying the filtration (PM.5). The remaining homogeneous displacement of degree  $q$  is encoded by a scalar homogeneous generating Hamiltonian of degree  $q + 1$ . Indeed, the leading homogeneous term of an exact-symplectic map tangent to the identity is a symplectic vector field; contraction with the constant symplectic form is a closed homogeneous one-form and the homogeneous Poincaré lemma writes it as  $d\chi_{q+1}$ . Thus the leading displacement is  $X_{\chi_{q+1}}$ . Integrating one phase derivative does not change parameter degree, so the coefficients of  $\chi_{q+1}$  still have  $t$ -degree at most  $q - 1$ . On a scalar complex monomial  $z^\alpha w^\beta$  the homological operator induced by conjugation with the fixed rotation is multiplication by

$$e^{2\pi i(\alpha - \beta) \cdot \omega} - 1. \quad (\text{PM.8})$$

These divisors are independent of  $t$ . By the definition of fixed-frequency symplectic germ space, map Birkhoff normal form, and  $(PM)$ -prevalence above nonresonance, (PM.8) vanishes exactly when  $\alpha = \beta$ , namely on scalar action monomials. Projecting the scalar generator onto its action part and dividing every nonaction coefficient by its fixed nonzero divisor are linear operations independent of  $t$ . After taking one derivative to recover the exact-symplectic displacement, both the normal-form contribution and the normalizing displacement of degree  $q$  therefore retain parameter degree at most  $q - 1$ . This closes the induction without invoking an analytic small-divisor estimate. The same diagonal splitting also proves uniqueness of the normalized integrable series. Indeed, suppose two tangent-to-identity formal symplectic conjugacies produced integrable series  $B$  and  $C$ , and let  $m$  be the first action degree at which they differ. After cancelling all lower orders, the first nontrivial homogeneous scalar generator of the transition conjugacy changes the degree- $2m$  scalar normal-form term by an element of the range of the same homological operator. But  $B_m - C_m$  is an action polynomial, hence lies in its kernel. In the monomial basis the kernel is spanned by  $z^\alpha w^\alpha$  and the range by the complementary monomials, so their intersection is zero. Thus  $B_m = C_m$ , a contradiction. Induction gives  $B = C$ .

Finally, an action monomial  $b_n(t)r^n$  in the integrable Hamiltonian  $B_t$  has phase degree  $2|n|_1$ . Its Hamiltonian time-one map first contributes to the displacement after one phase derivative, hence at degree  $q = 2|n|_1 - 1$ . The filtration gives

$$\deg_t b_n \leq q - 1 = 2|n|_1 - 2,$$

which is (PM.3). Real symmetry of the original family makes the coefficients real for real  $t$ .  $\square$

**Lemma 4.2 (measurable-set Remez estimate).** Let  $I = [a, b] \subset \mathbb{R}$  have length  $L = b - a > 0$ , let  $E \subset I$  be measurable with  $|E| = m > 0$ , and let  $p$  be a real or complex

polynomial of degree at most  $N$ . Then

$$\sup_{x \in I} |p(x)| \leq \left( \frac{6eL}{m} \right)^N \sup_{x \in E} |p(x)|. \quad (\text{PM.9})$$

The same conclusion holds if the bound on  $E$  is only essential, after deleting a null subset of  $E$ .

*Proof.* The case  $N = 0$  is immediate, so assume  $N \geq 1$ . Divide  $E$ , in increasing order, into  $2N + 1$  measurable slices of equal measure  $m/(2N + 1)$ . More explicitly, the function

$$M(x) := |E \cap [a, x]|$$

is continuous, nondecreasing and satisfies  $0 \leq M(y) - M(x) \leq y - x$ ; choose cut points at the levels  $jm/(2N + 1)$ . From every even-numbered slice choose one point  $x_j \in E$ ,  $0 \leq j \leq N$ , outside any prescribed null exceptional set. Between  $x_j$  and  $x_k$ ,  $j < k$ , lie at least  $k - j$  full odd slices, whence

$$|x_k - x_j| \geq \frac{(k - j)m}{2N + 1}. \quad (\text{PM.10})$$

Lagrange interpolation gives

$$p(x) = \sum_{j=0}^N p(x_j) \prod_{k \neq j} \frac{x - x_k}{x_j - x_k}. \quad (\text{PM.11})$$

For  $x \in I$ , the numerator in the  $j$ th product is at most  $L^N$ , while (PM.10) gives

$$\prod_{k \neq j} |x_j - x_k| \geq \left( \frac{m}{2N + 1} \right)^N j!(N - j)!. \quad (\text{PM.12})$$

Therefore

$$\begin{aligned} |p(x)| &\leq \sup_E |p| \left( \frac{(2N + 1)L}{m} \right)^N \sum_{j=0}^N \frac{1}{j!(N - j)!} \\ &= \sup_E |p| \left( \frac{(2N + 1)L}{m} \right)^N \frac{2^N}{N!}. \end{aligned} \quad (\text{PM.13})$$

Using  $(2N + 1)/N \leq 3$  and the elementary Stirling lower bound  $N! \geq (N/e)^N$  yields (PM.9).  $\square$

**Lemma 4.3 (real-line dichotomy, prevalence closure, and dimension lift).** In the setting of Lemma 4.1, let

$$\mathcal{C}(F_0, F_1) := \{t \in \mathbb{R} : B_t \text{ has positive radius of convergence}\}. \quad (\text{PM.14})$$

Then either  $\mathcal{C}(F_0, F_1) = \mathbb{R}$  or  $|\mathcal{C}(F_0, F_1)| = 0$ . Consequently, if  $F_0$  has divergent BNF, then for every comparison germ  $F_1$  the exceptional set

$$\{t \in \mathbb{R} : tF_0 + (1 - t)F_1 \text{ has convergent BNF}\} \quad (\text{PM.15})$$

has Lebesgue measure zero. Thus a single divergent germ in  $\mathcal{S}_\omega$  forces the divergent-BNF set to be  $(PM)$ -prevalent.

Moreover, if for some irrational  $\omega_j$  there is a one-degree-of-freedom real-analytic symplectic germ  $f_j$  with frequency  $\omega_j$  and divergent BNF, then for every nonresonant vector  $\omega = (\omega_1, \dots, \omega_d)$  having that component there is a  $d$ -degree-of-freedom seed in  $\mathcal{S}_\omega$  with divergent BNF.

*Proof.* Let  $b_n(t)$  be the coefficient polynomials of Lemma 4.1. For integers  $J, Q \geq 1$  put

$$E_{J,Q} := \left\{ t \in [-J, J] : |b_n(t)| \leq Q^{|n|_1} \text{ for every } |n|_1 \geq 2 \right\}. \quad (\text{PM.16})$$

If  $B_t$  converges, Cauchy's coefficient estimate gives constants  $C, R > 0$  with  $|b_n(t)| \leq CR^{-|n|_1}$ ; choosing  $Q$  large enough and using  $|n|_1 \geq 2$  shows  $t \in E_{J,Q}$  for some  $J, Q$ . Hence

$$\mathcal{C}(F_0, F_1) = \bigcup_{J,Q \geq 1} E_{J,Q}. \quad (\text{PM.17})$$

Suppose  $\mathcal{C}(F_0, F_1)$  has positive Lebesgue measure. Then some  $E := E_{J,Q}$  has measure  $m > 0$ . Fix any target parameter  $t_*$  and choose a compact interval  $I$  containing both  $[-J, J]$  and  $t_*$ . Lemma 4.1 and Lemma 4.2 imply, for  $|n|_1 = s \geq 2$ ,

$$\sup_{t \in I} |b_n(t)| \leq \left( \frac{6e|I|}{m} \right)^{2(s-1)} Q^s \leq A_I^s \quad (\text{PM.18})$$

for a constant  $A_I$  independent of  $n$ . The number of multi-indices  $n \in \mathbb{N}^d$  with  $|n|_1 = s$  is  $\binom{s+d-1}{d-1}$ , which grows only polynomially in  $s$ . Therefore

$$\sum_{|n|_1 \geq 2} b_n(t) r^n$$

converges absolutely and uniformly for  $t \in I$  whenever  $\max_j |r_j| < A_I^{-1}/2$ . In particular  $B_{t_*}$  converges. Since  $t_*$  was arbitrary, positive measure of the convergence set forces  $\mathcal{C}(F_0, F_1) = \mathbb{R}$ . This proves the dichotomy.

If  $F_0$  is a divergent seed, the first alternative is impossible on every comparison line through  $F_0$ . Thus (PM.15) is null for every  $F_1$ , which is exactly  $(PM)$ -prevalence as defined in the definition of fixed-frequency symplectic germ space, map Birkhoff normal form, and  $(PM)$ -prevalence above.

For the dimension lift, let  $f_j$  be the one-dimensional seed and define the product map by applying  $f_j$  in the  $(x_j, y_j)$ -plane and the linear rotations  $z_k \mapsto e^{2\pi i \omega_k} z_k$  in every  $k \neq j$  plane. Write  $f_j = R_{\omega_j} \circ f_{F_j}$  as in the definition of fixed-frequency symplectic germ space, map Birkhoff normal form, and  $(PM)$ -prevalence above and extend  $F_j$  trivially in the other variables. The product map is then exactly  $R_\omega \circ f_F$  for that extended generating function  $F$ , so it belongs to  $\mathcal{S}_\omega$ ; it is real analytic and symplectic. Because the variables are decoupled, the formal conjugacy can be chosen as the product of the one-dimensional BNF conjugacy and the identity maps in the remaining factors;

uniqueness of the normalized BNF then gives

$$\text{BNF}(f)(r) = \text{BNF}(f_j)(r_j) + 2\pi \sum_{k \neq j} \omega_k r_k. \quad (\text{PM.19})$$

The restriction  $r_k = 0$  for  $k \neq j$  would make a convergent left-hand side yield a convergent  $\text{BNF}(f_j)$ , contradiction. Hence the product BNF diverges.  $\square$

**Lemma 4.4 (Hamiltonian fixed-quadratic-part real-line dichotomy).** Let

$$H_t = H_2 + tP_0 + (1-t)P_1, \quad H_2 = 2\pi \sum_{j=1}^d \omega_j I_j, \quad P_0, P_1 = O(3), \quad (\text{PM.20})$$

be an affine real-analytic Hamiltonian family with the same fixed quadratic part and assume  $k \cdot \omega \neq 0$  for every  $0 \neq k \in \mathbb{Z}^d$ . Then the set of real  $t$  for which the formal Hamiltonian Birkhoff normal form converges is either all of  $\mathbb{R}$  or a Lebesgue-null set.

*Proof.* Use the homogeneous Hamiltonian Lie-series normalization over the coefficient ring  $\mathbb{R}[t]$ . At each degree one splits the current homogeneous polynomial into the kernel and range of  $\{H_2, \cdot\}$ , retains the kernel part, and solves for the range part by division by the fixed nonzero divisors. We now track the parameter count in this recursion. At homogeneous phase degree  $q$ , the current Hamiltonian coefficient is obtained by finite sums of iterated Poisson brackets of the original homogeneous pieces and earlier normalizing generators. Give a degree- $q$  coefficient the filtered parameter budget  $q - 2$ . The initial nonlinear pieces satisfy this bound because they are affine in  $t$  and  $q \geq 3$ . A Poisson bracket of homogeneous phase degrees  $p$  and  $q$  has degree  $p + q - 2$ , while the parameter degrees add; hence

$$(p - 2) + (q - 2) = p + q - 4 = (p + q - 2) - 2, \quad (\text{PM.21})$$

so the filtration is preserved. The homological inverse divides by the fixed numbers  $(\alpha - \beta) \cdot \omega$  and therefore does not change the  $t$ -degree. Thus every homogeneous degree- $q$  coefficient of the formal Hamiltonian BNF is a polynomial in  $t$  of degree at most  $q - 2$ .

The normal form contains only action monomials. An action monomial  $I^n$  has phase degree  $2|n|_1$ , so its coefficient  $c_n(t)$  satisfies

$$\deg_t c_n \leq 2|n|_1 - 2. \quad (\text{PM.22})$$

The proof of Lemma 4.3 now applies verbatim with  $c_n$  in place of  $b_n$ : if convergence holds on a positive-measure set of real parameters, Lemma 4.2 produces a uniform exponential coefficient bound on every compact real parameter interval, hence convergence for every real  $t$ . Otherwise the convergence set is null.  $\square$

**Lemma 4.5 (quantitative hole-estimate contradiction).** Let  $m : (0, t_0) \rightarrow [0, \infty)$ .

(i) *Finite arithmetic exponent.* Let  $\tau \in [1, \infty)$  and choose

$$0 < \beta < \frac{1}{2} \left( \frac{1}{\tau} - \frac{1}{1 + \tau} \right). \quad (\text{PM.23})$$



There do not exist constants  $C, c > 0$  and a sequence  $t_k \downarrow 0$  for which both

$$m(t) \leq C \exp\left(-t^{-(1/\tau-\beta)}\right) \quad (0 < t < t_0) \quad (\text{PM.24})$$

and

$$m(t_k) \geq c \exp\left(-t_k^{-(1/(1+\tau)+\beta)}\right) \quad (k \geq 1) \quad (\text{PM.25})$$

hold.

(ii) *Large continued-fraction jumps.* Let  $q_k \rightarrow \infty$  and let  $\mathcal{K} \subset \mathbb{N}$  be infinite. Fix  $0 < \beta < 1/5$ . There do not exist constants  $C, c > 0$  such that, for all sufficiently large  $k \in \mathcal{K}$ ,

$$m_k \leq C e^{-q_{k+1}^{1/5}} \quad (\text{PM.26})$$

and for infinitely many  $k \in \mathcal{K}$ ,

$$m_k \geq c e^{-q_{k+1}^\beta}. \quad (\text{PM.27})$$

Thus, once a Liouville construction supplies the lower estimate on the same large-jump subsequence on which the convergent-BNF upper estimate is valid, the two estimates are incompatible.

*Proof.* For (i), put

$$a := \frac{1}{\tau} - \beta, \quad b := \frac{1}{1+\tau} + \beta.$$

Condition (PM.23) says  $a > b > 0$ . If (PM.24)–(PM.25) held, then

$$\frac{c}{C} \leq \exp\left(-t_k^{-a} + t_k^{-b}\right). \quad (\text{PM.28})$$

Since  $a > b$ ,  $t_k^{-a} - t_k^{-b} = t_k^{-a}(1 - t_k^{a-b}) \rightarrow +\infty$ ; hence the right-hand side of (PM.28) tends to zero, contradiction.

For (ii), (PM.26)–(PM.27) would give along the infinite common subsequence

$$\frac{c}{C} \leq \exp\left(-q_{k+1}^{1/5} + q_{k+1}^\beta\right). \quad (\text{PM.29})$$

Because  $\beta < 1/5$  and  $q_{k+1} \rightarrow \infty$ , the exponent tends to  $-\infty$ , again a contradiction.  $\square$

**Lemma 4.6 (continued-fraction resonant action scales).** Let  $\omega_0 \in \mathbb{R} \setminus \mathbb{Q}$  have continued-fraction convergents  $p_n/q_n$ . Let  $\varpi : (-r_0, r_0) \rightarrow \mathbb{R}$  be  $C^1$  with  $\varpi(0) = \omega_0$ , and assume that for some  $A \geq 1$

$$A^{-1} \leq \varpi'(r) \leq A \quad (|r| < r_0). \quad (\text{PM.30})$$

Then for all sufficiently large  $n$  there is a unique  $c_n \in (-r_0, r_0)$  such that  $\varpi(c_n) = p_n/q_n$ , and

$$\frac{1}{2Aq_nq_{n+1}} \leq |c_n| \leq \frac{A}{q_nq_{n+1}}. \quad (\text{PM.31})$$

Moreover, with  $\tau = \tau(\omega_0)$  as in the definition of invariant-circle deficit, twist coefficient, and continued-fraction scales above:

1. if  $\tau < \infty$ , then for every  $\eta > 0$  there are infinitely many  $n$  with

$$q_{n+1} \geq q_n^{\tau-\eta}; \quad (\text{PM.32})$$

for those indices,

$$q_n \leq A^{1/(1+\tau-\eta)} |c_n|^{-1/(1+\tau-\eta)}; \quad (\text{PM.33})$$

2. if  $\tau = \infty$ , there is a subsequence  $n_j$  such that  $\log q_{n_j+1} / \log q_{n_j} \rightarrow \infty$ . Along a further infinite subsequence the sign of  $p_{n_j}/q_{n_j} - \omega_0$ , and hence the sign of  $c_{n_j}$ , is constant.

*Proof.* Write the continued fraction as  $\omega_0 = [a_0; a_1, a_2, \dots]$  and let  $\alpha_{n+1} = [a_{n+1}; a_{n+2}, \dots] > 1$  be the complete quotient. The standard matrix identity for convergents,  $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$ , gives

$$\omega_0 = \frac{p_n \alpha_{n+1} + p_{n-1}}{q_n \alpha_{n+1} + q_{n-1}}$$

and therefore

$$\left| \omega_0 - \frac{p_n}{q_n} \right| = \frac{1}{q_n (q_n \alpha_{n+1} + q_{n-1})}. \quad (\text{PM.34})$$

Since

$$q_n \alpha_{n+1} + q_{n-1} = q_{n+1} + \frac{q_n}{\alpha_{n+2}},$$

the second factor lies strictly between  $q_{n+1}$  and  $q_{n+1} + q_n$ . As  $q_n \leq q_{n+1}$ ,

$$\frac{1}{2q_n q_{n+1}} \leq \left| \omega_0 - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}}. \quad (\text{PM.35})$$

For large  $n$  the rational  $p_n/q_n$  lies in the range of  $\varpi$  on  $(-r_0, r_0)$ . Strict monotonicity from (PM.30) gives a unique  $c_n$ , and the mean-value theorem gives

$$A^{-1} |c_n| \leq |\varpi(c_n) - \varpi(0)| \leq A |c_n|.$$

Combining this with (PM.35) proves (PM.31).

If  $\tau < \infty$ , the definition of the limsup implies (PM.32) for infinitely many indices for every  $\eta > 0$ . At such an index, the upper bound in (PM.31) gives  $|c_n| \leq A q_n^{-(1+\tau-\eta)}$ , which is exactly (PM.33). If  $\tau = \infty$ , a diagonal choice gives  $n_j$  with the displayed logarithmic ratio tending to infinity. Convergents alternate about  $\omega_0$ ; among the two parities one occurs infinitely often in the selected sequence, giving the final constant-sign subsequence.  $\square$

**Lemma 4.7 (analytic resonant forcing and uniform bad-set activation).** Fix  $0 < h < H$ , an integer  $a_3 \geq 1$ , and a strictly increasing sequence of positive integers  $q_k$ . On the parameter space

$$\mathcal{X} := ([-1, 1]^2)^{\mathbb{N}}, \quad \mu := \lambda^{\otimes \mathbb{N}},$$

where  $\lambda$  is normalized planar Lebesgue measure on  $[-1, 1]^2$ , define

$$G_\zeta(\theta, r) := r^{a_3} \sum_{k \geq 1} e^{-q_k H} (\zeta_{1,k} \cos(q_k \theta) + \zeta_{2,k} \sin(q_k \theta)). \quad (\text{PM.36})$$

Then  $G_\zeta$  is real analytic, uniformly in  $\zeta$ , on every closed strip  $|\operatorname{Im} \theta| \leq h$  and on every bounded complex  $r$ -disc.

Suppose in addition that  $c_n \rightarrow 0$ , and that after the exact coordinate operations of a seed construction the normalized  $q_n$ -Fourier coefficient has the form

$$V_n(\zeta) = a_n + U_n \zeta_n + e_n(\zeta), \quad a_n \in \mathbb{R}^2, \quad U_n \in O(2), \quad \sup_{\zeta \in \mathcal{X}} |e_n(\zeta)| \leq C|c_n|. \quad (\text{PM.37})$$

For all sufficiently large  $n$  there is a Borel set  $J_n \subset [-1, 1]^2$  with

$$|J_n| \leq 4\pi|c_n| \leq 4\pi|c_n|^{1/2} \quad (\text{PM.38})$$

such that

$$\zeta_n \notin J_n \implies |V_n(\zeta)| \geq |c_n|^{1/2} \quad \text{for every choice of the other coordinates.} \quad (\text{PM.39})$$

Consequently, for every infinite  $\mathcal{N} \subset \mathbb{N}$ ,  $\mu$ -almost every  $\zeta$  satisfies (PM.39) for infinitely many  $n \in \mathcal{N}$ .

*Proof.* For  $|\operatorname{Im} \theta| \leq h$ ,  $|\cos(q_k \theta)| + |\sin(q_k \theta)| \leq 2e^{q_k h}$ . After any fixed number  $m$  of  $\theta$ -derivatives the  $k$ th summand is therefore bounded by a constant times  $q_k^m e^{-q_k(H-h)}$ . Since the  $q_k$  are distinct positive integers,

$$\sum_k q_k^m e^{-q_k(H-h)} \leq \sum_{q \geq 1} q^m e^{-q(H-h)} < \infty.$$

Uniform normal convergence, also after derivatives, proves the first assertion. The real symmetry is immediate from (PM.36).

Choose  $n$  so large that  $C|c_n| \leq |c_n|^{1/2}$  and  $|c_n| \leq 1$ . Put

$$J_n := \{u \in [-1, 1]^2 : |a_n + U_n u| \leq 2|c_n|^{1/2}\}.$$

Because  $U_n$  is orthogonal, this is the intersection of the square with a Euclidean disc of radius  $2|c_n|^{1/2}$ , hence (PM.38). If  $\zeta_n \notin J_n$ , then by (PM.37)

$$|V_n(\zeta)| \geq |a_n + U_n \zeta_n| - |e_n(\zeta)| > 2|c_n|^{1/2} - |c_n|^{1/2} = |c_n|^{1/2},$$

uniformly in all other coordinates.

Enumerate the infinite set  $\mathcal{N}$  increasingly. The event that all sufficiently large coordinates in  $\mathcal{N}$  are bad is

$$\bigcup_{N \geq 1} \bigcap_{\substack{n \in \mathcal{N} \\ n \geq N}} \{\zeta : \zeta_n \in J_n\}.$$

For fixed  $N$  the measure of the intersection is at most the measure of each of its coordinate events and therefore at most  $\inf_{n \in \mathcal{N}, n \geq N} \lambda(J_n) = 0$ , because  $c_n \rightarrow 0$ . The countable union is null. Hence almost every parameter is good infinitely often.  $\square$

**Lemma 4.8 (arithmetic conversion of resonant lower bounds).** Assume the resonant actions  $c_n$  satisfy Lemma 4.6. Let  $m : (0, t_0) \rightarrow [0, \infty)$  and suppose that along an infinite index set  $\mathcal{N}$  there are comparison scales  $t_n$  and constants  $C_0, C_1, K, L > 0$  such

that

$$C_0^{-1}|c_n| \leq t_n \leq C_0|c_n|, \quad m(t_n) \geq K^{-1}|c_n|^L e^{-C_1 q_n}. \quad (\text{PM.40})$$

Then the following conversions hold.

(i) *Finite arithmetic exponent.* If  $\tau = \tau(\omega_0) < \infty$  and  $\mathcal{N}$  contains infinitely many indices realizing the limsup in the sense of (PM.32), then for every  $\beta > 0$  there is an infinite  $\mathcal{N}_\beta \subset \mathcal{N}$  for which, for all sufficiently large  $n \in \mathcal{N}_\beta$ ,

$$m(t_n) \geq \exp\left(-t_n^{-(1/(1+\tau)+\beta)}\right). \quad (\text{PM.41})$$

(ii) *Liouville large jumps.* If  $\mathcal{N}$  contains a subsequence on which  $\log q_{n+1}/\log q_n \rightarrow \infty$ , then for every  $\beta > 0$ , after passing to a further infinite subsequence,

$$m(t_n) \geq e^{-q_{n+1}^\beta}. \quad (\text{PM.42})$$

*Proof.* For (i), fix  $\beta > 0$  and choose  $\eta > 0$  so small that

$$a := \frac{1}{1+\tau-\eta} < \frac{1}{1+\tau} + \frac{\beta}{2} =: b_0.$$

On infinitely many indices Lemma 4.6 gives  $q_n \leq C|c_n|^{-a}$ , hence by the comparability in (PM.40),  $q_n \leq C't_n^{-a}$ . Therefore

$$-\log(K^{-1}|c_n|^L e^{-C_1 q_n}) \leq C_2 + L \log(t_n^{-1}) + C_3 t_n^{-a}. \quad (\text{PM.43})$$

Since  $a < b_0 < b := 1/(1+\tau) + \beta$ , the right-hand side of (PM.43) is at most  $t_n^{-b}$  for all sufficiently small  $t_n$ . Exponentiating proves (PM.41).

For (ii), the lower bound in Lemma 4.6 and  $q_n \leq q_{n+1}$  give

$$|c_n| \geq (2A)^{-1} q_{n+1}^{-2}. \quad (\text{PM.44})$$

Along a subsequence with logarithmic ratio tending to infinity, for every fixed  $\beta > 0$  one has  $q_n = o(q_{n+1}^\beta)$ . Moreover  $\log |c_n|^{-1} = O(\log q_{n+1}) = o(q_{n+1}^\beta)$ . Thus

$$\log K + L \log |c_n|^{-1} + C_1 q_n \leq q_{n+1}^\beta$$

for all sufficiently large indices after passing to the same large-jump subsequence. Taking exponentials yields (PM.42).  $\square$

**Lemma 4.9 (explicit no-screening barrier on a holed disk).** Let  $0 < \sigma < \rho < 1$  and

$$U := D(0, \rho) \setminus \bigcup_{j=1}^N \overline{D(z_j, \varepsilon_j)}, \quad \overline{D(0, \sigma)} \subset U, \quad (\text{PM.45})$$

where every closed hole lies in  $D(0, \rho)$ , and assume that  $U$  is connected. Let  $f$  be holomorphic on  $U$  and continuous on  $\overline{U}$ , with

$$\|f\|_U \leq 1, \quad \|f\|_{\partial D(0, \sigma)} \leq m, \quad 0 < m \leq 1. \quad (\text{PM.46})$$

Choose  $d_j$  with  $2\varepsilon_j < d_j$  and define

$$\widehat{U} := D(0, \rho) \setminus \bigcup_{j=1}^N \overline{D(z_j, d_j)}.$$

For every  $z \in \widehat{U} \setminus \overline{D(0, \sigma)}$  one has

$$\log |f(z)| \leq \left[ \frac{\log(|z|/\rho)}{\log(\sigma/\rho)} - \sum_{j=1}^N \frac{\log(d_j/(2\rho))}{\log(\varepsilon_j/\rho)} \right] \log m. \quad (\text{PM.47})$$

If  $\widehat{U}$  is empty the assertion is vacuous.

*Proof.* The point is to build the propagation coefficient explicitly. On  $U \setminus \overline{D(0, \sigma)}$  set

$$A(z) := \frac{\log(|z|/\rho)}{\log(\sigma/\rho)} - \sum_{j=1}^N \frac{\log(|z - z_j|/(2\rho))}{\log(\varepsilon_j/\rho)}. \quad (\text{PM.48})$$

Every singularity of the logarithms lies in a removed component, so  $A$  is harmonic on this domain. Put  $u(z) = \log |f(z)|$ , with the usual value  $-\infty$  at zeros. The function  $u$  is subharmonic and  $u \leq 0$  by (PM.46). Since  $\log m \leq 0$ , we claim that

$$u(z) \leq A(z) \log m. \quad (\text{PM.49})$$

It is enough to check the boundary inequality and apply the subharmonic maximum principle to  $u - A \log m$ .

On  $|z| = \sigma$ , the first term of  $A$  is 1. For every  $j$  one has  $|z - z_j| \leq 2\rho$ , hence

$$-\frac{\log(|z - z_j|/(2\rho))}{\log(\varepsilon_j/\rho)} \leq 0,$$

so  $A \leq 1$ . Therefore  $A \log m \geq \log m \geq u$ . On  $|z| = \rho$  the first term is zero and the same sign calculation gives  $A \leq 0$ , hence  $A \log m \geq 0 \geq u$ .

Finally consider a boundary point belonging to a hole  $|z - z_j| = \varepsilon_j$ . Because the central closed disk is contained in  $U$ ,  $|z| \geq \sigma$ , so the first term of  $A$  lies in  $[0, 1]$ . The  $j$ th hole term equals

$$-\frac{\log(\varepsilon_j/(2\rho))}{\log(\varepsilon_j/\rho)} = -1 + \frac{\log 2}{\log(\varepsilon_j/\rho)} < -1,$$

while every other hole term is nonpositive. Hence  $A \leq 0$  also on the hole boundary, and again  $A \log m \geq 0 \geq u$ . If several hole boundaries meet, choosing any one of them only strengthens this estimate. Thus (PM.49) follows from the maximum principle. For completeness, that principle here is the elementary subharmonic one: a positive interior maximum of  $u - A \log m$  would contradict its submean inequality, while the boundary upper limit is nonpositive by the preceding checks.

Now let  $z \in \widehat{U}$ . Since  $|z - z_j| \geq d_j$  and  $\log(\varepsilon_j/\rho) < 0$ ,

$$\frac{\log(|z - z_j|/(2\rho))}{\log(\varepsilon_j/\rho)} \leq \frac{\log(d_j/(2\rho))}{\log(\varepsilon_j/\rho)}.$$

Hence  $A(z)$  is at least the bracket in (PM.47). Multiplication by the nonpositive number  $\log m$  reverses this last comparison, and (PM.49) yields (PM.47).  $\square$

**Lemma 4.10 (action-order exact-symplectic composition and Fourier interface).** Work on a one-dimensional complex action-angle neighborhood and, for an analytic function  $H(\theta, r)$ , write  $f_H(\theta, r) = (\Theta, R)$  for the canonical exact-symplectic map defined implicitly by

$$r = R + \partial_\theta H(\theta, R), \quad \Theta = \theta + \partial_R H(\theta, R). \quad (\text{PM.50})$$

Fix an integer  $a \geq 4$ . Let  $F, G = O(r^a)$  and let  $g$  be an analytic exact-symplectic map (so  $g^*(r d\theta) - r d\theta$  is exact) with

$$g - \text{id} = O(r^2). \quad (\text{PM.51})$$

Assume all maps below are defined on a common smaller neighborhood, and normalize all generating functions to vanish at  $r = 0$ . The right side below is exact symplectic and tangent to the identity; after shrinking the neighborhood its graph projects diffeomorphically to the  $(\theta, R)$  variables, so it has a unique normalized type-2 canonical generator  $H_*$ . Writing

$$f_{H_*} = f_F \circ g^{-1} \circ f_G \circ g, \quad (\text{PM.52})$$

one has

$$H_* = F + G + O(r^{a+1}). \quad (\text{PM.53})$$

Moreover, if the data vary in a bounded parameter family and the remainder in (PM.53) is uniformly analytic on a complex strip of width  $H_0 > h$ , then for every real  $c$  sufficiently small and every Fourier mode  $q \in \mathbb{Z}$ ,

$$\left| \widehat{H_* - F} - G(q, c) \right| \leq C_h |c|^{a+1} e^{-|q|h}. \quad (\text{PM.54})$$

Consequently, after division by the forcing scale  $|c|^a e^{-|q|h}$ , the Fourier error is  $O(|c|)$  uniformly in the bounded parameters.

*Proof.* We keep the order bookkeeping explicit because (PM.53), rather than a merely formal “higher-order term”, is the load-bearing statement. For  $k \geq 2$  call a vector-valued remainder  $E = (E_\theta, E_r)$  an  $\mathcal{E}_k$  remainder if

$$E_\theta = O(r^{k-1}), \quad E_r = O(r^k). \quad (\text{PM.55})$$

Thus the Hamiltonian vector field  $X_H = (\partial_r H, -\partial_\theta H)$  of a function  $H = O(r^k)$  is of class  $\mathcal{E}_k$ .

We first compare the canonical map (PM.50) with the Hamiltonian vector field. Since  $\partial_\theta H = O(r^k)$ , the first equation gives recursively

$$R = r - \partial_\theta H(\theta, r) + O(r^{2k-1}). \quad (\text{PM.56})$$

Substituting into the second equation and using  $\partial_{rr} H = O(r^{k-2})$  gives

$$\Theta = \theta + \partial_r H(\theta, r) + O(r^{2k-2}). \quad (\text{PM.57})$$

Therefore

$$f_H = \text{id} + X_H + \mathcal{E}_{2k-1}. \quad (\text{PM.58})$$

The time-one Hamiltonian flow  $\Phi_H^1$  satisfies the same expansion. Indeed the second Picard term is  $DX_H X_H$ : its angular component is  $O(r^{2k-2})$  and its action component is  $O(r^{2k-1})$ , and every later term is of still higher action order. Hence

$$f_H = \Phi_H^1 + \mathcal{E}_{2k-1}. \quad (\text{PM.59})$$

Write  $g(\theta, r) = (\theta + u(\theta, r), r + v(\theta, r))$ . By (PM.51),  $u, v = O(r^2)$ , and Taylor's formula yields

$$G \circ g - G = \partial_\theta G u + \partial_r G v + O(r^{a+2}) = O(r^{a+1}). \quad (\text{PM.60})$$

Because  $g$  is symplectic, Hamiltonian flows conjugate exactly:

$$g^{-1} \circ \Phi_G^1 \circ g = \Phi_{G \circ g}^1. \quad (\text{PM.61})$$

Composition on either side by  $g$  and  $g^{-1}$  preserves the anisotropic class  $\mathcal{E}_m$  for  $m \geq 2$ : derivatives of their angular components are  $1 + O(r^2)$  in  $\theta$  and  $O(r)$  in  $r$ , while derivatives of their action components are  $O(r^2)$  in  $\theta$  and  $1 + O(r)$  in  $r$ . Combining (PM.59)–(PM.61) therefore gives

$$g^{-1} \circ f_G \circ g = \text{id} + X_G + \mathcal{E}_{a+1}, \quad (\text{PM.62})$$

because  $2a - 1 \geq a + 1$  and  $X_{G \circ g} - X_G \in \mathcal{E}_{a+1}$  by (PM.60).

Similarly, (PM.58) gives  $f_F = \text{id} + X_F + \mathcal{E}_{2a-1}$ . When this map is composed with (PM.62), the only new first-order cross term is  $DX_F X_G$ . Since

$$X_F^\theta = O(r^{a-1}), \quad X_F^r = O(r^a),$$

its angular component is  $O(r^{2a-2})$  and its action component is  $O(r^{2a-1})$ . We conclude that the right side of (PM.52) is

$$\text{id} + X_{F+G} + \mathcal{E}_{a+1}. \quad (\text{PM.63})$$

It remains to translate the map estimate back into its canonical generator rather than silently identifying maps and Hamiltonians. Equation (PM.63), inserted into the two implicit relations (PM.50) for  $H_*$ , first gives  $\partial_r H_* = O(r^{a-1})$  and  $\partial_\theta H_* = O(r^a)$ ; the normalization  $H_*(\theta, 0) = 0$  therefore implies  $H_* = O(r^a)$ . Applying (PM.58) to  $H_*$  and comparing with (PM.63) yields

$$X_{H_* - F - G} \in \mathcal{E}_{a+1}.$$

In particular  $\partial_r(H_* - F - G) = O(r^a)$ . Since the difference vanishes at  $r = 0$ , integration in the action variable proves (PM.53).

For the Fourier statement, write  $R_* = H_* - F - G$ . Uniformity of all preceding Taylor bounds on a fixed smaller complex neighborhood gives  $\sup_{|\Im \theta| < H_0, |r-c| < \eta|c|} |R_*(\theta, r)| \leq C|c|^{a+1}$  for a fixed  $\eta > 0$  and small  $c$ . Shifting the Fourier contour to  $\Im \theta = \pm h$  according to the sign of  $q$  gives the elementary Cauchy estimate

$$|\widehat{R}_*(q, c)| \leq C_h |c|^{a+1} e^{-|q|h},$$

which is (PM.54). The final normalized  $O(|c|)$  assertion follows immediately.  $\square$

**Lemma 4.11 (analytic  $L^2$ -to-sup bootstrap).** For  $0 < h \leq 1$  write

$$\mathbb{T}_h := \{\theta \in \mathbb{C}/(2\pi\mathbb{Z}) : |\Im \theta| < h\}, \quad \|f\|_h := \sup_{\mathbb{T}_h} |f|, \quad (\text{PM.64})$$

and normalize

$$\|f\|_2^2 := \frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta. \quad (\text{PM.65})$$

There is an absolute constant  $C > 0$  with the following property. If  $f$  is  $2\pi$ -periodic and holomorphic on  $\mathbb{T}_h$ ,  $0 < \delta \leq 1/4$ ,  $\mu \geq 0$ , and

$$\|f\|_2 \leq \delta \|f\|_{C^0(\mathbb{T})} + \mu, \quad (\text{PM.66})$$

then

$$\|f\|_{C^0(\mathbb{T})} \leq C\delta^{-1}\mu + \frac{C}{h} \exp\left(-\frac{h}{48\delta^2}\right) \|f\|_h. \quad (\text{PM.67})$$

*Proof.* Write  $f(\theta) = \sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{ik\theta}$ , with normalized Fourier coefficients. Parseval and Cauchy-Schwarz give, for every integer  $N \geq 1$ ,

$$\sum_{|k| \leq N} |\widehat{f}(k)| \leq (2N+1)^{1/2} \|f\|_2. \quad (\text{PM.68})$$

Holomorphy on the strip permits a contour shift to  $\Im \theta = \pm h$ , hence  $|\widehat{f}(k)| \leq \|f\|_h e^{-|k|h}$ . Therefore

$$\|f\|_{C^0} \leq (2N+1)^{1/2} \|f\|_2 + \frac{C}{h} e^{-hN} \|f\|_h. \quad (\text{PM.69})$$

Choose

$$N = \left\lfloor \frac{1}{12\delta^2} \right\rfloor. \quad (\text{PM.70})$$

Since  $\delta \leq 1/4$ , one has  $N \geq 1$ ,  $N \geq (48\delta^2)^{-1}$ , and

$$(2N+1)^{1/2} \delta \leq \left(\frac{1}{6} + \delta^2\right)^{1/2} < \frac{1}{2}. \quad (\text{PM.71})$$

Insert (PM.66) into (PM.69), absorb the resulting  $(1/2)\|f\|_{C^0}$  on the left, and use the lower bound for  $N$ . This proves (PM.67).  $\square$

**Lemma 4.12 (residue-three-circles extension obstruction).** Fix constants  $A \geq 1$ ,  $C_0 \geq 1$ ,  $P \geq 0$ ,  $0 < h \leq 1$ , and choose  $L \gg_A 1$ . Set once and for all

$$M := \frac{51}{10} > 5. \quad (\text{PM.72})$$

Let  $a, e$  be  $2\pi$ -periodic holomorphic functions on  $\mathbb{T}_h$ , real on  $\mathbb{T}$ . Assume  $a(\theta) > 0$  on  $\mathbb{T}$  and that its positive real square root extends to a nowhere-zero holomorphic branch on  $\mathbb{T}_h$  with  $\|a^{1/2}\|_h + \|a^{-1/2}\|_h \leq A$ . Also assume

$$A^{-2} \leq a(\theta) \leq A^2 \quad (\theta \in \mathbb{T}), \quad \varepsilon := \|e\|_{C^0(\mathbb{T})} \in (0, 1), \quad \|e\|_h \leq \varepsilon^{-P}, \quad (\text{PM.73})$$



and impose the gauge

$$\langle a^{1/2}e \rangle = 0, \quad \langle u \rangle := \frac{1}{2\pi} \int_0^{2\pi} u(\theta) d\theta. \quad (\text{PM.74})$$

Put  $r_0 = L\varepsilon^{1/2}$ . For  $|z| > r_0$  define the model branch

$$g^\sharp(\theta, z) := a(\theta)^{-1/2} z (1 + a(\theta)e(\theta)z^{-2})^{1/2}, \quad \Gamma^\sharp(z) := \langle g^\sharp(\cdot, z) \rangle, \quad (\text{PM.75})$$

where the square root is the branch asymptotic to 1 at infinity. Assume that the inverse branch  $H^\sharp = (\Gamma^\sharp)^{-1}$  asymptotic to  $\gamma^{-1}z$ ,  $\gamma := \langle a^{-1/2} \rangle$ , is holomorphic on the annulus used below.

Let  $R \in (0, 1]$  satisfy  $R/r_0 \geq 4^M$ . Suppose that  $H$  is holomorphic on a neighborhood of the closed annulus  $r_0 \leq |z| \leq R$  and that  $\Xi$  is holomorphic on a neighborhood of the closed disk  $|z| \leq R$ , such that, for some  $0 < \nu < 1$ ,

$$\|\Xi - H^2\|_{|z|=R} \leq \nu, \quad \|\Xi - H^2\|_{|z|=r_0} \leq C_0, \quad (\text{PM.76})$$

and, for every  $4r_0 \leq \lambda \leq R/2$ ,

$$\|H^2 - (H^\sharp)^2\|_{|z|=\lambda} \leq C_0 \lambda^3. \quad (\text{PM.77})$$

Then, after decreasing a threshold  $\varepsilon_0 = \varepsilon_0(A, C_0, P, h, L) > 0$ , the implication

$$0 < \varepsilon < \varepsilon_0 \implies \varepsilon \leq C\nu^{20/123} \quad (\text{PM.78})$$

holds, with  $C$  depending only on the displayed fixed data.

*Proof.* The proof has three independent pieces: a residue detecting the defect  $e$ , a three-circles propagation from the outer boundary, and Lemma 4.11 converting an  $L^2$  defect bound into the required sup norm.

Because  $L$  is fixed large enough,  $|ae/z^2| \leq A/L^2 < 1/2$  on  $|z| > r_0$ . The binomial expansion in (PM.75) is therefore normally convergent there and gives

$$g^\sharp(\theta, z) = a^{-1/2}z + \frac{1}{2}a^{1/2}ez^{-1} - \frac{1}{8}a^{3/2}e^2z^{-3} + O(\varepsilon^3|z|^{-5}). \quad (\text{PM.79})$$

Averaging and using the gauge (PM.74),

$$\Gamma^\sharp(z) = \gamma(z + a_{-3}z^{-3} + O(z^{-5})), \quad a_{-3} = -\frac{1}{8\gamma} \langle a^{3/2}e^2 \rangle. \quad (\text{PM.80})$$

Inverting this Laurent series at infinity yields

$$(H^\sharp(z))^2 = \gamma^{-2}z^2 - 2a_{-3}\gamma^2z^{-2} + O(z^{-3}). \quad (\text{PM.81})$$

Consequently every positively oriented circle  $|z| = \lambda$  lying in the inverse branch annulus satisfies the exact residue identity

$$\frac{1}{2\pi i} \int_{|z|=\lambda} z(H^\sharp(z))^2 dz = \frac{\gamma}{4} \langle a^{3/2}e^2 \rangle. \quad (\text{PM.82})$$

Since  $a$  is positive and uniformly bounded on the real circle, the right-hand side controls  $\|e\|_2^2$  from below by a fixed positive constant.

Set

$$\lambda := R^{1/M} r_0^{1-1/M}. \quad (\text{PM.83})$$

The ratio hypothesis  $R/r_0 \geq 4^M$  gives  $4r_0 \leq \lambda \leq R/2$ . The function  $\Xi - H^2$  is holomorphic on  $r_0 < |z| < R$ . Hadamard's three-circles theorem applied to (PM.76) therefore gives

$$\|\Xi - H^2\|_{|z|=\lambda} \leq C\nu^{1/M}. \quad (\text{PM.84})$$

Combine this with (PM.77). Because  $z\Xi(z)$  is holomorphic in the disk bounded by  $|z| = \lambda$ , its contour integral vanishes. Equations (PM.82)-(PM.84) then imply

$$\|e\|_2^2 \leq C\lambda^2(\nu^{1/M} + \lambda^3), \quad (\text{PM.85})$$

whence

$$\|e\|_2 \leq C\lambda\nu^{1/(2M)} + C\lambda^{5/2}. \quad (\text{PM.86})$$

Since  $R \leq 1$  and  $r_0 = L\varepsilon^{1/2}$ ,

$$\|e\|_2 \leq C_L \varepsilon^{\frac{1}{2}(1-1/M)} \nu^{1/(2M)} + C_L \varepsilon^{\frac{5}{4}(1-1/M)}. \quad (\text{PM.87})$$

For our fixed  $M = 51/10$  one has

$$\frac{1}{2} \left(1 - \frac{1}{M}\right) = \frac{41}{102}, \quad \frac{5}{4} \left(1 - \frac{1}{M}\right) - 1 = \frac{1}{204}. \quad (\text{PM.88})$$

Thus (PM.87) has the Lemma 4.11 form

$$\|e\|_2 \leq \delta\varepsilon + \mu, \quad \delta = C_L \varepsilon^{1/204}, \quad \mu = C_L \varepsilon^{41/102} \nu^{5/51}. \quad (\text{PM.89})$$

For  $\varepsilon$  small,  $\delta \leq 1/4$ . Apply Lemma 4.11. Its analytic-tail term is bounded by

$$Ch^{-1} \exp(-c\varepsilon^{-1/102}) \varepsilon^{-P} \leq \frac{1}{4}\varepsilon \quad (\text{PM.90})$$

after decreasing  $\varepsilon_0$ , because an exponential in a negative power of  $\varepsilon$  dominates every algebraic power. We obtain

$$\varepsilon \leq C\varepsilon^{27/68} \nu^{5/51} + \frac{1}{4}\varepsilon. \quad (\text{PM.91})$$

Absorbing the last term and dividing by  $\varepsilon^{27/68}$  gives

$$\varepsilon^{41/68} \leq C\nu^{5/51}. \quad (\text{PM.92})$$

Since  $(5/51)/(41/68) = 20/123$ , (PM.78) follows.  $\square$

**Lemma 4.13 (safe Extension-Property hole exponent).** Assume the hypotheses and conclusion of Lemma 4.12. Suppose, in addition, that the Hamilton-Jacobi construction producing the hole gives a radius  $\rho_{\text{hole}}$  with

$$\rho_{\text{hole}} \leq C_1 \varepsilon^{1/32}. \quad (\text{PM.93})$$

Then

$$\rho_{\text{hole}} \leq C_2 \nu^{5/984}. \quad (\text{PM.94})$$

In particular, after decreasing the smallness threshold for  $\nu$ ,

$$\rho_{\text{hole}} \leq \nu^{1/200}. \quad (\text{PM.95})$$

*Proof.* Substitution of Lemma 4.12 into (PM.93) gives

$$\rho_{\text{hole}} \leq C_2 \nu^{\frac{1}{32} \frac{20}{123}} = C_2 \nu^{5/984},$$

which is (PM.94). The exponent margin is strict:

$$\frac{5}{984} - \frac{1}{200} = \frac{1}{12300} > 0. \quad (\text{PM.96})$$

Hence  $C_2 \nu^{5/984} \leq \nu^{1/200}$  once  $C_2 \nu^{1/12300} \leq 1$ . This proves (PM.95).  $\square$

**Lemma 4.14 (deficit summation and exponential absorption).** Let

$$W_\rho := \mathbb{T} \times (-\rho, \rho), \quad (\text{PM.97})$$

and let  $f$  be a map defined on  $W_{2\rho}$ . Suppose there are a measurable set  $B \subset W_\rho$  and real intervals  $I_1, \dots, I_N \subset (-\rho, \rho)$  such that

$$\text{Leb}(B) \leq E \quad (\text{PM.98})$$

and every point of

$$W_\rho \setminus \left( B \cup \bigcup_{j=1}^N (\mathbb{T} \times I_j) \right) \quad (\text{PM.99})$$

belongs to an  $f$ -invariant essential circle contained in  $W_{2\rho}$ . Then the deficit of the definition of invariant-circle deficit, twist coefficient, and continued-fraction scales above satisfies

$$m_f(\rho) \leq E + 2\pi \sum_{j=1}^N |I_j|. \quad (\text{PM.100})$$

In particular, if  $N \leq C\rho^{-s}$  and  $|I_j| \leq \eta(\rho)$ , then

$$m_f(\rho) \leq E + C' \rho^{-s} \eta(\rho). \quad (\text{PM.101})$$

Two elementary absorption consequences will be used later. If  $a > 0$  and, for all small  $\rho$ ,

$$E + \eta(\rho) \leq C \exp(-\rho^{-a}), \quad (\text{PM.102})$$

then for every  $0 < b < a$ ,

$$m_f(\rho) \leq \exp(-\rho^{-b}) \quad (\text{PM.103})$$

for all sufficiently small  $\rho$ . Likewise, if a large parameter  $Q \rightarrow \infty$  satisfies  $N \leq CQ^s$  and

$$E + \max_j |I_j| \leq C \exp(-Q^a), \quad (\text{PM.104})$$

then for every  $0 < b < a$ ,

$$m_f(\rho_Q) \leq \exp(-Q^b) \quad (\text{PM.105})$$

for all sufficiently large  $Q$ .

*Proof.* By definition,  $m_f(\rho)$  measures points of  $W_\rho$  that are not covered by invariant circles lying inside  $W_{2\rho}$ . Hypothesis (PM.99) therefore puts the entire deficit inside  $B \cup \bigcup_j (\mathbb{T} \times I_j)$ . Subadditivity of Lebesgue measure and  $\text{Leb}(\mathbb{T} \times I_j) = 2\pi|I_j|$  prove (PM.100), hence (PM.101).

For (PM.103), the right side of (PM.101) is bounded by  $C(1 + \rho^{-s})e^{-\rho^{-a}}$ . Since

$$\rho^{-a} - \rho^{-b} \longrightarrow +\infty \quad (\rho \downarrow 0), \quad (\text{PM.106})$$

this polynomial factor is eventually at most  $\exp(\rho^{-a} - \rho^{-b})$ , proving (PM.103). The same logarithmic comparison, with  $Q^a - Q^b \rightarrow \infty$ , absorbs the factor  $Q^s$  and proves (PM.105).  $\square$

**Lemma 4.15 (convergent-BNF jet matching gives exponential closeness).** Let  $\rho > 0$  and let  $\Xi, \Omega$  be holomorphic on a neighborhood of the closed disk  $D(0, \rho)$ . Suppose

$$\|\Xi\|_{D(0, \rho)} \leq M_0, \quad \|\Omega\|_{D(0, \rho)} \leq M_0, \quad (\text{PM.107})$$

and that their Taylor series at the origin agree through degree  $N$ :

$$\Omega(r) - \Xi(r) = O(r^{N+1}) \quad \text{as a formal Taylor identity.} \quad (\text{PM.108})$$

Then

$$\|\Omega - \Xi\|_{D(0, e^{-1}\rho)} \leq CM_0 e^{-N}. \quad (\text{PM.109})$$

Consequently, if  $N = N(\rho)$  satisfies  $N(\rho) \geq c\rho^{-a}$  for fixed  $c, a > 0$ , then

$$\|\Omega - \Xi\|_{D(0, e^{-1}\rho)} \leq CM_0 \exp(-c\rho^{-a}). \quad (\text{PM.110})$$

*Proof.* Write

$$\Xi(r) = \sum_{k \geq 0} \xi_k r^k, \quad \Omega(r) = \sum_{k \geq 0} \omega_k r^k.$$

By (PM.108),  $\xi_k = \omega_k$  for  $0 \leq k \leq N$ . Cauchy's estimate applied on  $D(0, \rho)$  gives

$$|\xi_k| \rho^k \leq M_0, \quad |\omega_k| \rho^k \leq M_0. \quad (\text{PM.111})$$

Hence for  $|r| \leq e^{-1}\rho$ ,

$$|\Omega(r) - \Xi(r)| \leq 2M_0 \sum_{k \geq N+1} e^{-k} \leq \frac{2}{1 - e^{-1}} M_0 e^{-(N+1)} \leq CM_0 e^{-N}, \quad (\text{PM.112})$$

which proves (PM.109). Substituting  $N(\rho) \geq c\rho^{-a}$  gives (PM.110).  $\square$

**Lemma 4.16 (near-identity transport of punctured action windows).** Let

$$I \subset [-1, 1] \quad \text{be an interval,} \quad A := I \setminus \bigcup_{j \in J} I_j, \quad (\text{PM.113})$$

where the intervals  $I_j \subset I$  are pairwise disjoint and

$$\sum_{j \in J} |I_j|^{1/2} \leq 1. \quad (\text{PM.114})$$

For  $E \subset \mathbb{R}$  write  $W_E := \mathbb{T} \times E$ . Let  $g : \mathbb{T} \times \mathbb{R} \rightarrow \mathbb{T} \times \mathbb{R}$  be an area-preserving  $C^1$  diffeomorphism such that

$$\|g - \text{id}\|_{C^1} \leq \frac{1}{10}, \quad \delta := \|g - \text{id}\|_{C^0} \leq \frac{1}{10}. \quad (\text{PM.115})$$

Then

$$\text{Leb}(W_A \triangle g(W_A)) \leq C\delta^{1/2}, \quad (\text{PM.116})$$

where  $C$  is universal for the normalized cylinder.

*Proof.* For any interval  $J_0 \subset [-1, 1]$ , the action component of both  $g$  and  $g^{-1}$  moves points by at most  $\delta$ . Consequently

$$W_{J_0} \triangle g(W_{J_0}) \subset \mathbb{T} \times \{r : \text{dist}(r, \partial J_0) \leq \delta\}, \quad (\text{PM.117})$$

so

$$\text{Leb}(W_{J_0} \triangle g(W_{J_0})) \leq C\delta. \quad (\text{PM.118})$$

Because  $g$  preserves area, the same symmetric difference is also bounded by  $2\text{Leb}(W_{J_0}) \leq C|J_0|$ . Interpolating these two elementary bounds,

$$\text{Leb}(W_{J_0} \triangle g(W_{J_0})) \leq C \min\{\delta, |J_0|\} \leq C\delta^{1/2}|J_0|^{1/2}. \quad (\text{PM.119})$$

Since the  $I_j$  are disjoint,

$$\mathbf{1}_{W_A} = \mathbf{1}_{W_I} - \sum_{j \in J} \mathbf{1}_{W_{I_j}} \quad \text{a.e.} \quad (\text{PM.120})$$

and the same identity holds after applying  $g$ . Hence the triangle inequality in  $L^1$  gives

$$\begin{aligned} \text{Leb}(W_A \triangle g(W_A)) &\leq \text{Leb}(W_I \triangle g(W_I)) + \sum_{j \in J} \text{Leb}(W_{I_j} \triangle g(W_{I_j})) \\ &\leq C\delta^{1/2} \left( |I|^{1/2} + \sum_{j \in J} |I_j|^{1/2} \right) \leq C\delta^{1/2}, \end{aligned} \quad (\text{PM.121})$$

because  $|I| \leq 2$ . This proves (PM.116).  $\square$

**Lemma 4.17 (integrable-core KAM measure transfer on a punctured action window).** Use the notation and hypotheses of Lemma 4.16. Let  $L \subset A$  be compact, let  $\tilde{A} \subset \mathbb{R}$  be measurable with  $A \subset \tilde{A}$ , and let  $f$  be an area-preserving  $C^1$  diffeomorphism on  $W_{\tilde{A}}$ . Suppose there are a  $C^1$  function  $\Omega$  and an area-preserving  $C^1$  diffeomorphism  $g$  as in Lemma 4.16 such that

$$g^{-1} \circ f \circ g(\theta, r) = (\theta + \partial_r \Omega(r), r) \quad \text{for } (\theta, r) \in W_L, \quad (\text{PM.122})$$

and

$$g(W_L) \subset W_{\tilde{A}}. \quad (\text{PM.123})$$

Let  $\mathcal{L}(f, W_{\tilde{A}})$  denote the union of all  $f$ -invariant essential  $C^1$  circles contained in  $W_{\tilde{A}}$ . Then

$$\text{Leb}(W_A \setminus \mathcal{L}(f, W_{\tilde{A}})) \leq 2\pi \text{Leb}_{\mathbb{R}}(A \setminus L) + C\delta^{1/2}. \quad (\text{PM.124})$$

*Proof.* For each  $c \in L$ , the circle  $\mathbb{T} \times \{c\}$  is invariant under the integrable map in (PM.122). Conjugating by  $g$ , the curve

$$g(\mathbb{T} \times \{c\}) \quad (\text{PM.125})$$

is an essential  $f$ -invariant  $C^1$  circle; by (PM.123) it lies in  $W_{\tilde{A}}$ . Therefore

$$g(W_L) \subset \mathcal{L}(f, W_{\tilde{A}}). \quad (\text{PM.126})$$

Since  $W_A = W_L \sqcup W_{A \setminus L}$  up to a null set and  $g$  preserves area,

$$g(W_A) \subset \mathcal{L}(f, W_{\tilde{A}}) \cup g(W_{A \setminus L}), \quad (\text{PM.127})$$

whence

$$\begin{aligned} \text{Leb}(W_A \setminus \mathcal{L}(f, W_{\tilde{A}})) &\leq \text{Leb}(W_A \setminus g(W_A)) + \text{Leb}(g(W_{A \setminus L})) \\ &\leq \text{Leb}(W_A \triangle g(W_A)) + 2\pi \text{Leb}_{\mathbb{R}}(A \setminus L). \end{aligned} \quad (\text{PM.128})$$

Apply Lemma 4.16 to the first term. This gives (PM.124).  $\square$

**Lemma 4.18 (type-II generating-function composition calculus).** Let  $U \Subset U^+ \subset \mathbb{C}$  be open, let  $0 < h < h^+$ , and put

$$d := \min\{h^+ - h, \text{dist}(U, \partial U^+)\} > 0. \quad (\text{PM.129})$$

There are universal constants  $c, C > 0$  with the following property. If  $F, G$  are holomorphic on  $W_{h^+, U^+}$  and

$$d^{-2}(\|F\|_{h^+, U^+} + \|G\|_{h^+, U^+}) \leq c, \quad (\text{PM.130})$$

then the maps  $f_F, f_G$  of the definition of complex action strips, holed action domains, and type-II exact-symplectic generators above are well defined on  $W_{h, U}$ . Their composition admits the type-II generator  $F \star G$  selected by the stationary formula (PM.134), and

$$\|F \star G - F - G\|_{h, U} \leq Cd^{-2}\|F\|_{h^+, U^+}\|G\|_{h^+, U^+}. \quad (\text{PM.131})$$

The inverse  $f_F^{-1}$  has a type-II generator  $F^{[-1]}$ , chosen by the same stationary normalization, with

$$\|F^{[-1]} + F\|_{h, U} \leq Cd^{-2}\|F\|_{h^+, U^+}^2. \quad (\text{PM.132})$$

If  $S$  is a fixed exact symplectic biholomorphism with bounded first two derivatives on the same nested domains and the mixed projection of  $S^{-1}f_F S$  stays in the type-II graph chart, then its stationarily normalized generator has linear term  $F \circ S$  and a quadratic error bounded by the right-hand side of (PM.132), with  $C$  enlarged only by those derivative bounds.

*Proof.* Write the type-II generating function of  $f_F$  as

$$S_F(\theta, R) = \theta R + F(\theta, R). \quad (\text{PM.133})$$

For the composition  $f_F \circ f_G$ , introduce the intermediate canonical pair  $(\theta_1, r_1)$ . The standard stationary composition formula is

$$\mathcal{S}_{F \star G}(\theta_0, R_2) = \underset{(\theta_1, r_1)}{\text{crit}} [\mathcal{S}_G(\theta_0, r_1) + \mathcal{S}_F(\theta_1, R_2) - \theta_1 r_1]. \quad (\text{PM.134})$$

The critical equations are

$$\theta_1 = \theta_0 + \partial_{r_1} G(\theta_0, r_1), \quad r_1 = R_2 + \partial_{\theta_1} F(\theta_1, R_2). \quad (\text{PM.135})$$

Cauchy's estimate on the  $d$ -collar gives

$$\|DF\| + \|DG\| \leq d^{-1}(\|F\| + \|G\|), \quad \|D^2 F\| + \|D^2 G\| \leq 2d^{-2}(\|F\| + \|G\|). \quad (\text{PM.136})$$

Thus (PM.130) makes the right-hand side of (PM.135) a contraction in a ball of radius  $2d^{-1}(\|F\| + \|G\|)$  around  $(\theta_0, R_2)$ . It has a unique holomorphic solution there, and the solution stays inside the larger domain. Substituting in (PM.134) and subtracting  $\theta_0 R_2$  gives

$$F \star G = G(\theta_0, r_1) + F(\theta_1, R_2) + (\theta_1 - \theta_0)(R_2 - r_1). \quad (\text{PM.137})$$

Taylor's formula, (PM.135), and (PM.136) show that replacing  $(\theta_1, r_1)$  by  $(\theta_0, R_2)$  changes the first two terms by at most  $Cd^{-2}\|F\|\|G\|$ , while the last term is bounded by the same quantity. This proves (PM.131).

For the inverse, solve  $F \star H = 0$ . By (PM.131) the map  $H \mapsto -F - [F \star H - F - H]$  is a contraction in the ball  $\|H + F\| \leq 2Cd^{-2}\|F\|^2$  after decreasing  $c$  if necessary. Its fixed point is the inverse generator selected by this stationary normalization and gives (PM.132).

Finally, conjugation by a fixed exact symplectic biholomorphism preserves exactness. Apply the same stationary formula to  $S^{-1}f_F S$  and differentiate at  $F = 0$ . The derivative is the pullback  $F \mapsto F \circ S$ ; the second derivative is a finite sum of products of first derivatives of the two small generating functions and first/second derivatives of  $S, S^{-1}$ . Cauchy's estimate gives the same quadratic bound, with a constant depending only on the stated bounds for  $S$ .  $\square$

**Lemma 4.19 (truncated map cohomology on a nonresonant holed domain).**

Let  $U \subset \mathbb{C}$  be a finite real-symmetric holed action domain and let  $\Omega$  be holomorphic on  $U$ . Assume the  $(K, N)$  map-nonresonance condition of the definition of complex action strips, holed action domains, and type-II exact-symplectic generators above. If  $F$  is holomorphic on  $W_{h,U}$  and  $0 < \delta < h/3$ , assume also  $\|\text{Im } \Omega'\|_U \leq \delta/2$ . Then there is a unique zero-angle-average trigonometric-polynomial solution  $Y$  of

$$Y(\theta + \Omega'(r), r) - Y(\theta, r) = -(T_N F - M_0 F)(\theta, r) \quad (\text{PM.138})$$

on  $W_{h-2\delta,U}$ , and

$$\|Y\|_{h-\delta,U} \leq CK\delta^{-1}\|F\|_{h,U}, \quad \|DY\|_{h-2\delta,U_\eta} \leq CK\delta^{-1} \min\{\delta, \eta\}^{-1}\|F\|_{h,U} \quad (\text{PM.139})$$

for every  $U_\eta := \{r \in U : \text{dist}(r, \partial U) > \eta\}$ . Moreover

$$\|R_N F\|_{h-\delta,U} \leq C\delta^{-1}e^{-N\delta}\|F\|_{h,U}. \quad (\text{PM.140})$$

*Proof.* Write

$$F(\theta, r) = \sum_{k \in \mathbb{Z}} F_k(r) e^{ik\theta}.$$

For  $0 < |k| < N$  define

$$Y_k(r) := -\frac{F_k(r)}{e^{ik\Omega'(r)} - 1}, \quad Y_0 = 0, \quad (\text{PM.141})$$

and put  $Y = \sum_{0 < |k| < N} Y_k e^{ik\theta}$ . Put  $x = k\Omega'(r)/(2\pi)$  and choose the nearest integer  $m$  so that  $x - m = \xi + i\eta$  with  $|\xi| \leq 1/2$ . Then

$$|\sin \pi x|^2 = \sin^2(\pi \xi) + \sinh^2(\pi \eta) \geq 4(\xi^2 + \eta^2) = 4d(x, \mathbb{Z})^2,$$

where  $|\sin(\pi \xi)| \geq 2|\xi|$  on  $[-1/2, 1/2]$  and  $|\sinh(\pi \eta)| \geq 2|\eta|$ . Therefore  $|e^{2\pi i x} - 1| = 2|\sin \pi x| \geq 4d(x, \mathbb{Z})$ . Hence nonresonance gives

$$|e^{ik\Omega'(r)} - 1|^{-1} \leq K/4. \quad (\text{PM.142})$$

Cauchy's Fourier estimate  $|F_k(r)| \leq \|F\|_{h,U} e^{-|k|h}$  therefore yields

$$\|Y\|_{h-\delta,U} \leq \frac{K}{4} \|F\|_{h,U} \sum_{k \neq 0} e^{-|k|\delta} \leq CK\delta^{-1} \|F\|_{h,U},$$

which is the first part of (PM.139). The derivative estimate is now an ordinary Cauchy estimate on the additional angle loss  $\delta$  and action collar  $\eta$ . Because  $Y$  is a finite trigonometric polynomial, the shifted expression  $Y(\theta + \Omega'(r), r)$  is holomorphic algebraically; the additional bound  $\|\text{Im } \Omega'\| \leq \delta/2$  ensures that the displayed cohomological identity is evaluated inside the stated nested strip. Formula (PM.141) therefore proves (PM.138) on  $W_{h-2\delta,U}$ , and uniqueness follows coefficientwise because the zero Fourier coefficient is prescribed to vanish and every displayed divisor is nonzero. Finally

$$\|R_N F\|_{h-\delta,U} \leq \|F\|_{h,U} \sum_{|k| \geq N} e^{-|k|\delta} \leq C\delta^{-1} e^{-N\delta} \|F\|_{h,U},$$

which is (PM.140). □

**Lemma 4.20 (one exact-symplectic analytic map-KAM step).** Fix  $L \geq 1$ . There are constants  $c_L, C_L > 0$  such that the following holds. Let  $U \Subset U^+$  be finite real-symmetric holed action domains and let  $0 < h < h^+$ . Put

$$\sigma := h^+ - h, \quad \eta := \text{dist}(U, \partial U^+), \quad d := \min\{\sigma, \eta\}. \quad (\text{PM.143})$$

Suppose  $\Omega$  is real symmetric and holomorphic on  $U^+$  with  $\|\Omega\|_{C^3(U^+)} \leq L$ , suppose it is  $(K, N)$  map-nonresonant on  $U^+$ , and assume  $\|\text{Im } \Omega'\|_{U^+} \leq \sigma/16$ . Let  $F$  be real symmetric and holomorphic on  $W_{h^+, U^+}$ , put  $\varepsilon := \|F\|_{h^+, U^+}$ , and assume

$$C_L K \sigma^{-1} d^{-2} \varepsilon \leq 1. \quad (\text{PM.144})$$

Then there is a real-symmetric type-II generator  $Y$  and a real-symmetric remainder  $F_+$



on  $W_{h,U}$  such that

$$f_Y \circ \Phi_\Omega \circ f_F \circ f_Y^{-1} = \Phi_{\Omega_+} \circ f_{F_+}, \quad \Omega_+ := \Omega + M_0 F, \quad (\text{PM.145})$$

and

$$\|Y\|_{h,U} \leq CK\sigma^{-1}\varepsilon, \quad (\text{PM.146})$$

$$\|F_+\|_{h,U} \leq C_L \left( K^2 \sigma^{-2} d^{-2} \varepsilon^2 + \sigma^{-1} e^{-N\sigma/8} \varepsilon \right). \quad (\text{PM.147})$$

If  $m \geq 2$  and  $F$  has zero Taylor coefficients of action order  $< m$ , then  $Y$  has the same property and  $F_+$  has zero coefficients of action order  $< m$ .

*Proof.* Apply Lemma 4.19 on  $W_{h^+,U^+}$  with angle loss  $\delta = \sigma/8$  to obtain  $Y$  solving

$$[\Omega]Y := Y \circ \Phi_\Omega - Y = -(T_N F - M_0 F) \quad (\text{PM.148})$$

and satisfying  $\|Y\| \leq CK\sigma^{-1}\varepsilon$  on an intermediate domain that still contains  $W_{h,U}$ . The action collar is not spent in this Fourier solve; it enters only when derivatives of the implicit type-II equations are estimated. Condition (PM.144), after decreasing  $c_L$ , gives  $d^{-2}\|Y\| \ll 1$ , hence all type-II maps below are well defined by Lemma 4.18.

Let  $\mathcal{C}_\Omega(F, Y)$  denote the stationarily normalized generator, relative to the fixed integrable factor  $\Phi_\Omega$ , of  $f_Y \Phi_\Omega f_F f_Y^{-1}$ . Lemma 4.18 applied three times, together with its fixed-map clause, shows that  $\mathcal{C}_\Omega$  is analytic near  $(0, 0)$ . Its differential at the origin is

$$D\mathcal{C}_\Omega(0, 0)[F, Y] = F + [\Omega]Y. \quad (\text{PM.149})$$

Indeed the  $F$ -variation is the inserted near-identity exact map, whereas the  $Y$ -variation is the difference between the conjugated infinitesimal generator  $Y \circ \Phi_\Omega$  and  $Y$ . The stationary formula in Lemma 4.18 and Cauchy's estimate on the separate angle/action collars give, on the final domain,

$$\|\mathcal{C}_\Omega(F, Y) - F - [\Omega]Y\| \leq C_L d^{-2}(\varepsilon + \|Y\|)^2 \leq C_L K^2 \sigma^{-2} d^{-2} \varepsilon^2. \quad (\text{PM.150})$$

By (PM.148),

$$F + [\Omega]Y = M_0 F + R_N F. \quad (\text{PM.151})$$

Because an action-only type-II generator is exactly an integrable map,  $\Phi_\Omega \circ f_{M_0 F} = \Phi_{\Omega + M_0 F}$ . Factoring this action-only term costs another quadratic quantity covered by (PM.150). Crucially, the Fourier tail pays only the angle loss: Lemma 4.19 with  $\delta = \sigma/8$  gives  $\|R_N F\| \leq C\sigma^{-1}e^{-N\sigma/8}\varepsilon$ . This proves (PM.147).

The symmetry follows coefficientwise from (PM.141) and from uniqueness after fixing the stationary normalization. Finally, the linear cohomological solve preserves action order. In the stationary type-II formulas a nonlinear term contains at least two derivatives of small generators. Each action derivative can lower action order by at most one, so the worst quadratic action order is  $2m - 2 \geq m$  when  $m \geq 2$ ; higher products are better. Hence no nonlinear term creates an action Taylor coefficient below order  $m$ .  $\square$

**Lemma 4.21 (superconvergent schedule for repeated map-KAM steps).** Fix

$p > 0$ ,  $0 < \kappa < 2/(3p)$ , and  $b > 0$ . Put

$$N_j = N_0(4/3)^j, \quad \delta_j = \frac{b}{(j+2)^2}, \quad E_j = \exp\left(-\frac{bN_j}{4(j+2)^2}\right), \quad K_j = E_j^{-\kappa}. \quad (\text{PM.152})$$

Let  $P \geq 1$ . For  $N_0$  sufficiently large, depending only on  $b, \kappa, p, P, C$ , any nonnegative sequence satisfying  $\varepsilon_0 \leq E_0$  and

$$\varepsilon_{j+1} \leq C\left(K_j^p \delta_j^{-P} \varepsilon_j^2 + \delta_j^{-1} e^{-N_j \delta_j} \varepsilon_j\right) \quad (\text{PM.153})$$

obeys

$$\varepsilon_j \leq E_j \quad (j \geq 0). \quad (\text{PM.154})$$

Moreover, for every fixed  $q, Q \geq 0$  with  $q\kappa < 1$ ,

$$\sum_{j \geq 0} K_j^q \delta_j^{-Q} \varepsilon_j < \infty, \quad (\text{PM.155})$$

and the sum tends to zero with  $N_0 \rightarrow \infty$ .

*Proof.* Assume inductively  $\varepsilon_j \leq E_j$ . The first term in (PM.153) is bounded by

$$C \delta_j^{-P} E_j^{2-p\kappa}. \quad (\text{PM.156})$$

The hypothesis  $\kappa < 2/(3p)$  says  $2 - p\kappa > 4/3$ . On the other hand

$$\frac{-\log E_{j+1}}{-\log E_j} = \frac{4(j+2)^2}{3(j+3)^2} < \frac{4}{3}. \quad (\text{PM.157})$$

Thus the strict exponential margin between  $E_j^{2-p\kappa}$  and  $E_{j+1}$  dominates the polynomial factor  $\delta_j^{-P}$  uniformly once  $N_0$  is large. Hence the first term is at most  $E_{j+1}/2$ .

For the tail term, (PM.152) gives

$$e^{-N_j \delta_j} E_j = \exp\left(-\frac{5bN_j}{4(j+2)^2}\right) = E_j^5. \quad (\text{PM.158})$$

Again (PM.157) implies  $C \delta_j^{-1} E_j^5 \leq E_{j+1}/2$  for large  $N_0$ . This proves (PM.154).

Finally

$$K_j^q \delta_j^{-Q} \varepsilon_j \leq \delta_j^{-Q} E_j^{1-q\kappa}. \quad (\text{PM.159})$$

The exponent  $1 - q\kappa$  is positive, and  $N_j/(j+2)^2$  grows geometrically up to a polynomial divisor. Hence the right-hand side is summable. Its first term tends to zero with  $N_0$ , and the tail is uniformly dominated by a geometric series; this proves the last assertion.  $\square$

**Lemma 4.22 (twist-Farey resonance geometry and hole counting).** Let  $I \subset \mathbb{R}$  be an interval and let  $\omega : I \rightarrow \mathbb{R}$  be  $C^1$  with

$$0 < a \leq |\omega'(r)| \leq A. \quad (\text{PM.160})$$

Fix  $N \geq 2$  and  $0 < \gamma < a/(20AN^2)$ . For every reduced rational  $l/k$  with  $1 \leq k < N$  lying in

$\omega(I)$ , let  $c_{k,l}$  be the unique point with  $\omega(c_{k,l}) = l/k$  and define

$$H_{k,l} := \left( c_{k,l} - \frac{2\gamma}{ak}, c_{k,l} + \frac{2\gamma}{ak} \right). \quad (\text{PM.161})$$

Define the inner frequency core

$$I_\gamma := \{r \in I : \text{dist}(\omega(r), \partial\omega(I)) \geq \gamma\}. \quad (\text{PM.161a})$$

Then:

(i) for  $r \in I_\gamma$  outside the union of these intervals,

$$\text{dist}(k\omega(r), \mathbb{Z}) \geq \gamma \quad (0 < k < N); \quad (\text{PM.162})$$

(ii) if  $l/k \neq l'/k'$  are reduced with  $k, k' < N$ , then

$$|c_{k,l} - c_{k',l'}| \geq \frac{1}{AN^2}, \quad (\text{PM.163})$$

so the intervals in (PM.161) are pairwise disjoint;

(iii) if  $J \subset I$  has length  $\rho$ , the number of such resonance centers in  $J$  is at most

$$C_A(\rho N^2 + N); \quad (\text{PM.164})$$

(iv) if  $\tilde{\omega}$  also satisfies the lower twist bound  $a$  and  $\|\tilde{\omega} - \omega\|_{C^0(I)} \leq \Delta$ , then the center carrying the same rational label moves by at most  $\Delta/a$  whenever both centers exist.

*Proof.* Monotonicity from (PM.160) gives uniqueness of each center. Fix  $0 < k < N$  and let  $l \in \mathbb{Z}$  minimize  $|k\omega(r) - l|$ . If the fraction  $l/k$  is not reduced, write it as  $l'/k'$  with  $k' \leq k$ ; it represents the same resonant frequency, and the corresponding deleted interval has half-width  $2\gamma/(ak') \geq 2\gamma/(ak)$ . Thus a point outside all reduced-label holes is outside the interval attached to the nearest rational label. The mean value theorem then yields

$$|k\omega(r) - l| = k|\omega(r) - \omega(c_{k',l'})| \geq ka|r - c_{k',l'}| \geq 2\gamma,$$

whenever the rational lies in  $\omega(I)$ . If  $l/k \notin \omega(I)$  and  $r \in I_\gamma$ , then the frequency-core condition gives  $|\omega(r) - l/k| \geq \gamma$ , hence  $|k\omega(r) - l| \geq k\gamma \geq \gamma$ . This proves (PM.162) on the stated inner core.

For distinct reduced rationals the Farey determinant inequality gives

$$\left| \frac{l}{k} - \frac{l'}{k'} \right| = \frac{|lk' - l'k|}{kk'} \geq \frac{1}{kk'} > \frac{1}{N^2}. \quad (\text{PM.165})$$

Since  $\omega$  is  $A$ -Lipschitz, this implies (PM.163). The sum of the two largest possible half-widths in (PM.161) is at most  $4\gamma/a < 1/(5AN^2)$ , so the intervals are disjoint.

To count centers in  $J$ , the image  $\omega(J)$  has length at most  $A\rho$ . For each  $1 \leq k < N$ , the number of integers  $l$  with  $l/k \in \omega(J)$  is at most  $Ak\rho + 2$ . Summing over  $k$  gives (PM.164).

Finally, if  $\tilde{c}$  and  $c$  carry the same rational label,

$$a|\tilde{c} - c| \leq |\omega(\tilde{c}) - \omega(c)| \leq |\omega(\tilde{c}) - \tilde{\omega}(\tilde{c})| \leq \Delta,$$

which proves (iv).  $\square$

**Lemma 4.23 (stretched-exponential scale conversion).** Let  $a > 0$ ,  $c > 0$ , and suppose  $N(\rho)$  satisfies

$$c^{-1}\rho^{-a} \leq N(\rho) \leq c\rho^{-a} \quad (0 < \rho < \rho_0). \quad (\text{PM.166})$$

Set

$$E(\rho) := \exp\left(-\frac{N(\rho)}{(\log N(\rho))^2}\right). \quad (\text{PM.167})$$

Then for every  $0 < \beta < a$  and every  $M > 0$ , after decreasing  $\rho_0$ ,

$$E(\rho) \leq \exp(-\rho^{-(a-\beta)}), \quad E(\rho)^c \leq \rho^M. \quad (\text{PM.168})$$

Consequently, any KAM remainder controlled by  $CE(\rho)$  is smaller than every algebraic power of  $\rho$  and has any strictly weakened stretched-exponential exponent  $a - \beta$ .

*Proof.* From (PM.166),  $\log N(\rho) = a|\log \rho| + O(1)$ . Hence

$$\frac{N(\rho)}{(\log N(\rho))^2} \geq C \frac{\rho^{-a}}{|\log \rho|^2}. \quad (\text{PM.169})$$

Because  $\rho^{-\beta}/|\log \rho|^2 \rightarrow \infty$ , the right-hand side of (PM.169) is at least  $\rho^{-(a-\beta)}$  for small  $\rho$ , proving the first inequality. Likewise  $\rho^{-a}/|\log \rho|^2$  dominates  $M|\log \rho|/c$ , which is equivalent to the second inequality after exponentiation.  $\square$

**Lemma 4.24 (discrete normal-form Gronwall bootstrap).** Let  $0 < \alpha < 1$ ,  $\theta, a > 0$ ,  $C \geq 1$ , and let  $\varepsilon_0, \dots, \varepsilon_K \geq 0$  satisfy

$$\varepsilon_{j+1} \leq C\alpha^\theta \left(1 + \alpha^{-a} \sum_{i=0}^j \varepsilon_i\right) \varepsilon_j \quad (0 \leq j < K). \quad (\text{PM.170})$$

If

$$2C\alpha^\theta \leq \frac{1}{2}, \quad \varepsilon_0 \leq \frac{\alpha^a}{4}, \quad (\text{PM.171})$$

then

$$\varepsilon_j \leq (2C\alpha^\theta)^j \varepsilon_0, \quad \sum_{i=0}^j \varepsilon_i \leq 2\varepsilon_0 \quad (0 \leq j \leq K). \quad (\text{PM.172})$$

*Proof.* Assume the conclusions hold through index  $j$ . Then

$$\alpha^{-a} \sum_{i=0}^j \varepsilon_i \leq 2\alpha^{-a} \varepsilon_0 \leq \frac{1}{2}. \quad (\text{PM.173})$$

Thus (PM.170) gives

$$\varepsilon_{j+1} \leq \frac{3}{2}C\alpha^\theta \varepsilon_j \leq 2C\alpha^\theta \varepsilon_j,$$

which proves the first inequality at  $j+1$ . Since the ratio is at most  $1/2$  by (PM.171), the resulting geometric series gives  $\sum_{i=0}^{j+1} \varepsilon_i \leq 2\varepsilon_0$ . Induction starts trivially at  $j=0$  and proves (PM.172).  $\square$

**Lemma 4.25 (continued-fraction finite-mode cohomology in elliptic complex coordinates).** Let  $\omega \in \mathbb{R} \setminus \mathbb{Q}$  and let  $p_n/q_n$  be a convergent. Put  $R_\omega(z, w) = (e^{2\pi i \omega} z, e^{-2\pi i \omega} w)$ . Let  $H(z, w) = \sum_{a,b \geq 0} h_{ab} z^a w^b$  be holomorphic on the bidisc  $B_s := \{|z| < s, |w| < s\}$  and let  $\Pi_{< q_n}^{\text{nr}} H$  be the sum of those monomials with  $0 < |a - b| < q_n$ . For  $0 < \delta < 1/2$  the equation

$$Y \circ R_\omega - Y = -\Pi_{< q_n}^{\text{nr}} H \quad (\text{PM.174})$$

has the coefficientwise solution

$$Y(z, w) = - \sum_{0 < |a-b| < q_n} \frac{h_{ab}}{e^{2\pi i(a-b)\omega} - 1} z^a w^b \quad (\text{PM.175})$$

and

$$\|Y\|_{B_{(1-\delta)s}} \leq C q_n \delta^{-2} \|H\|_{B_s}. \quad (\text{PM.176})$$

The omitted monomials with  $a = b$  are exactly the action monomials  $(zw)^a$ .

*Proof.* The continued-fraction best-approximation property gives, for  $0 < |k| < q_n$ ,

$$\text{dist}(k\omega, \mathbb{Z}) \geq \frac{1}{2q_n}. \quad (\text{PM.177})$$

As in Lemma 4.19,  $|e^{2\pi i k \omega} - 1| \geq 4 \text{dist}(k\omega, \mathbb{Z})$ , so every divisor in (PM.175) has inverse at most  $q_n/2$ . Cauchy's coefficient estimate gives  $|h_{ab}| s^{a+b} \leq \|H\|_{B_s}$ . Therefore

$$\begin{aligned} \|Y\|_{B_{(1-\delta)s}} &\leq C q_n \|H\|_{B_s} \sum_{a,b \geq 0} (1-\delta)^{a+b} \\ &\leq C q_n \delta^{-2} \|H\|_{B_s}, \end{aligned}$$

proving (PM.176). Formula (PM.174) follows monomial by monomial, and the equality  $a = b$  is exactly the condition that  $z^a w^b$  is a power of the radial action  $zw$ .  $\square$

**Lemma 4.26 (complex twist exclusion around real resonances).** Let  $I = [u, v] \subset \mathbb{R}$  be a compact interval. Put

$$U_s(I) := \{z \in \mathbb{C} : \text{dist}(z, I) < s\}, \quad S_s(I) := \{x + iy : x \in I, |y| < s\}.$$

Suppose  $\omega$  is real symmetric and holomorphic on  $U_s(I)$  and that on  $I$

$$0 < a \leq |\omega'(x)| \leq A, \quad \|\omega''\|_{U_s(I)} \leq B, \quad Bs \leq a/2. \quad (\text{PM.178})$$

For a real number  $\beta \in \omega(I)$  let  $c_\beta \in I$  be the unique solution  $\omega(c_\beta) = \beta$ . Then, whenever  $z = x + iy \in S_s(I)$  and the real segment from  $x$  to  $c_\beta$  lies in  $I$ ,

$$|\omega(z) - \beta| \geq \frac{a}{2} |z - c_\beta|. \quad (\text{PM.179})$$

Consequently, if  $N \geq 2$ ,  $\gamma > 0$ , and one removes for every reduced  $l/k \in \omega(I)$  with  $1 \leq k < N$  the complex disk

$$D\left(c_{l/k}, \frac{4\gamma}{a}\right), \quad (\text{PM.180})$$

then on the remaining part of the inner frequency core

$$S_s(I_\gamma), \quad I_\gamma = \{x \in I : \text{dist}(\omega(x), \partial\omega(I)) \geq 2\gamma\}, \quad (\text{PM.181})$$

one has

$$\text{dist}(k\omega(z), \mathbb{Z}) \geq \gamma \quad (0 < |k| < N). \quad (\text{PM.182})$$

If in addition  $\gamma < a/(100AN^2)$ , the deleted disks whose centers lie in  $I$  are pairwise disjoint provided  $4\gamma/a < s/4$ .

*Proof.* Because  $\omega$  is real symmetric,  $\omega(x)$  and  $\omega'(x)$  are real on  $I$ . For  $z = x + iy$ , Taylor's formula along the vertical segment gives

$$\omega(z) = \omega(x) + iy\omega'(x) + R(x, y), \quad |R(x, y)| \leq \frac{B}{2}|y|^2. \quad (\text{PM.183})$$

The real mean-value theorem and (PM.178) give  $|\omega(x) - \beta| \geq a|x - c_\beta|$ . Since the first two terms on the right of (PM.183) are respectively real and purely imaginary,

$$|\omega(x) - \beta + iy\omega'(x)| \geq a\sqrt{|x - c_\beta|^2 + |y|^2} = a|z - c_\beta|.$$

Moreover  $B|y|^2/2 \leq a|y|/4 \leq a|z - c_\beta|/4$ . Thus in fact the left side of (PM.179) is at least  $3a|z - c_\beta|/4$ , and the weaker displayed constant follows.

Fix  $0 < k < N$  and choose  $l \in \mathbb{Z}$  minimizing  $|k\omega(z) - l|$ . If  $l/k$  lies in  $\omega(I)$ , reduce the fraction to  $l'/k'$ ; its real center is one of those deleted in (PM.180). Outside that disk, (PM.179) gives

$$|k\omega(z) - l| = k|\omega(z) - l'/k'| \geq \frac{ka}{2} \frac{4\gamma}{a} \geq 2\gamma.$$

If  $l/k \notin \omega(I)$  and  $x \in I_\gamma$ , then  $u := \omega(x) - l/k$  is real and  $|u| \geq 2\gamma$ . Put  $v := y\omega'(x)$ . From (PM.183),

$$|\omega(z) - l/k| \geq \sqrt{u^2 + v^2} - \frac{B}{2}|y|^2. \quad (\text{PM.183a})$$

If  $|v| \leq 4\gamma$ , then  $B|y|^2/2 \leq a|y|/4 \leq |v|/4 \leq \gamma$ , so the right side is at least  $|u| - \gamma \geq \gamma$ . If  $|v| > 4\gamma$ , then the right side is at least  $|v| - |v|/4 > 3\gamma$ . Thus in both cases  $|\omega(z) - l/k| \geq \gamma$ , and a fortiori  $|k\omega(z) - l| \geq \gamma$ . This proves (PM.182).

For distinct reduced labels, Lemma 4.22 gives separation of the real centers by at least  $(AN^2)^{-1}$ . The sum of two radii in (PM.180) is  $8\gamma/a < 1/(12AN^2)$  under the stated stronger smallness, so the disks are disjoint.  $\square$

**Lemma 4.27 (analytic KAM iteration on a controlled nested holed-domain schedule).** Fix  $L \geq 1$ ,  $b > 0$ , and  $0 < \kappa < 1/12$ . Let the sequences  $N_j, \delta_j, E_j, K_j$  be those of Lemma 4.21, with  $p = 4$ . Suppose for every  $j \geq 0$  there are finite real-symmetric holed domains

$$U_j \Subset U_j^+ \subset \mathbb{C}, \quad h_j < h_j^+,$$

and real-symmetric holomorphic functions  $\Omega_j, F_j$  on the corresponding product domains such that

$$h_j^+ - h_j = 8\delta_j, \quad \text{dist}(U_j, \partial U_j^+) \geq \delta_j K_j^{-1}, \quad (\text{PM.184})$$

$$\|\Omega_j\|_{C^3(U_j^+)} \leq L, \quad \|\text{Im } \Omega_j'\|_{U_j^+} \leq \delta_j/2, \quad \Omega_j \text{ is } (K_j, N_j)\text{-nonresonant on } U_j^+, \quad (\text{PM.185})$$

and  $\varepsilon_j := \|F_j\|_{h_j^+, U_j^+} \leq E_j$ . Assume also that each one-step output furnished by Lemma 4.20 extends holomorphically to the next-step input domain  $W_{h_{j+1}^+, U_{j+1}^+}$  without increasing its sup norm, and that the total initial scale is small enough that

$$\sum_{j \geq 0} \delta_j^{-3} K_j^3 E_j \leq c_L. \quad (\text{PM.186})$$

Then the KAM steps may be performed for every  $j$ , with

$$f_{Y_j} \Phi_{\Omega_j} f_{F_j} f_{Y_j}^{-1} = \Phi_{\Omega_{j+1}} f_{F_{j+1}}, \quad \Omega_{j+1} = \Omega_j + M_0 F_j, \quad (\text{PM.187})$$

and, after increasing  $N_0$  if necessary,

$$\varepsilon_j \leq E_j \quad (j \geq 0), \quad (\text{PM.188})$$

$$\sum_j \|f_{Y_j} - \text{id}\|_{C^1(W_{h_j, U_j})} < \infty, \quad \sum_j \|\Omega_{j+1} - \Omega_j\|_{C^3(U_j)} < \infty. \quad (\text{PM.189})$$

Consequently the cumulative exact-symplectic coordinate changes converge in  $C^1$  on every nested real compact core, the limiting frequency map is  $C^3$  there, and the remainders converge to zero super-algebraically. If a uniform real twist margin for  $\Omega_0$  exceeds twice the first sum in (PM.186), the limiting frequency map retains half of that twist margin.

*Proof.* For Lemma 4.20 at stage  $j$ , set  $\sigma_j = 8\delta_j$  and  $d_j = \min\{8\delta_j, \text{dist}(U_j, \partial U_j^+)\}$ . By (PM.184),  $d_j \geq \delta_j K_j^{-1}$ . Hence its nonlinear term is bounded by

$$C_L K_j^2 \sigma_j^{-2} d_j^{-2} \varepsilon_j^2 \leq C_L K_j^4 \delta_j^{-4} \varepsilon_j^2, \quad (\text{PM.190})$$

while the Fourier tail is bounded by

$$C \delta_j^{-1} e^{-N_j \delta_j} \varepsilon_j. \quad (\text{PM.191})$$

The step-smallness quantity satisfies

$$K_j \sigma_j^{-1} d_j^{-2} \varepsilon_j \leq C \delta_j^{-3} K_j^3 E_j, \quad (\text{PM.192})$$

which is uniformly small by (PM.186). Thus every Lemma 4.20 step is legal. Combining (PM.190)–(PM.191) with Lemma 4.21 for  $p = 4$  gives (PM.188).

The coordinate displacement generated by a type-II scalar function  $Y_j$  consists of first derivatives of  $Y_j$ . Hence  $C^1$  control of the coordinate map requires second derivatives of the scalar generator. Starting from (PM.146), two further Cauchy derivatives on the reserved action/angle collars give

$$\|f_{Y_j} - \text{id}\|_{C^1} \leq C \|Y_j\|_{C^2} \leq C \delta_j^{-3} K_j^3 E_j. \quad (\text{PM.193})$$

The series (PM.186) therefore proves the first assertion in (PM.189), and the standard composition estimate for near-identity maps makes the cumulative exact symplectic coordinates Cauchy in  $C^1$  on nested real compact cores.

For the frequency update, Cauchy's estimate in the action variable gives

$$\|M_0 F_j\|_{C^3(U_j)} \leq C(\delta_j K_j^{-1})^{-3} E_j = C\delta_j^{-3} K_j^3 E_j. \quad (\text{PM.194})$$

The first series in (PM.186) proves the second assertion in (PM.189). The same estimate applied to one derivative shows that the twist changes by a summable amount; if the initial twist margin dominates twice the sum, the limiting derivative cannot cross zero. Finally Lemma 4.23 and (PM.188) show that  $F_j \rightarrow 0$  faster than every algebraic power of the initial adapted scale whenever  $N_0$  is chosen as a negative power of that scale.  $\square$

**Lemma 4.28 (collared nesting for moving resonance holes).** For each  $j \geq 0$  let  $\Lambda_j$  be a finite label set with  $\Lambda_j \subset \Lambda_{j+1}$ . Fix numbers  $R_j^+ > 0$ ,  $\eta_j > 0$ , and, for  $\lambda \in \Lambda_j$ , real centers  $c_{\lambda,j}$  and outer hole radii  $a_{\lambda,j}^+ > 0$ . Put

$$R_j := R_j^+ - \eta_j, \quad a_{\lambda,j} := a_{\lambda,j}^+ + \eta_j, \quad (\text{PM.195})$$

and suppose the closed disks  $\overline{D(c_{\lambda,j}, a_{\lambda,j})}$  are pairwise disjoint and compactly contained in  $D(0, R_j)$ . Define

$$U_j^+ := D(0, R_j^+) \setminus \bigcup_{\lambda \in \Lambda_j} \overline{D(c_{\lambda,j}, a_{\lambda,j}^+)}, \quad U_j := D(0, R_j) \setminus \bigcup_{\lambda \in \Lambda_j} \overline{D(c_{\lambda,j}, a_{\lambda,j})}. \quad (\text{PM.196})$$

Then  $U_j \Subset U_j^+$  and

$$\text{dist}(U_j, \partial U_j^+) \geq \eta_j. \quad (\text{PM.197})$$

Assume in addition that

$$R_{j+1}^+ \leq R_j \quad (\text{PM.198})$$

and that every old label persists with

$$a_{\lambda,j+1}^+ \geq a_{\lambda,j} + |c_{\lambda,j+1} - c_{\lambda,j}| \quad (\lambda \in \Lambda_j). \quad (\text{PM.199})$$

Then

$$U_{j+1}^+ \subset U_j. \quad (\text{PM.200})$$

In particular, if a twist lower bound  $a > 0$  and a frequency drift  $\|\tilde{\omega} - \omega\| \leq \Delta_j$  give  $|c_{\lambda,j+1} - c_{\lambda,j}| \leq \Delta_j/a$  as in Lemma 4.22, it is enough to grow an old outer radius by  $\eta_j + \Delta_j/a$  before the next step. If, furthermore, all relevant resonance centers at stage  $j$  are  $s_j$ -separated and

$$\max_{\lambda \in \Lambda_j} a_{\lambda,j} < s_j/3, \quad (\text{PM.201})$$

then the required pairwise disjointness follows automatically.

*Proof.* The outer boundary of  $U_j^+$  consists of the circle  $|z| = R_j^+$  and the circles  $|z - c_{\lambda,j}| = a_{\lambda,j}^+$ . Every point of  $U_j$  has distance at least  $R_j^+ - R_j = \eta_j$  from the first circle and at least  $a_{\lambda,j} - a_{\lambda,j}^+ = \eta_j$  from each hole circle. This proves (PM.197); compact containment follows from the strict compact-containment hypotheses on the expanded holes.



For an old label  $\lambda$ , (PM.199) and the triangle inequality give

$$D(c_{\lambda,j}, a_{\lambda,j}) \subset D(c_{\lambda,j+1}, a_{\lambda,j+1}^+). \quad (\text{PM.202})$$

Thus every point of  $U_{j+1}^+$  avoids every expanded old hole occurring in  $U_j$ . By (PM.198) it also lies in  $D(0, R_j)$ , while newly born holes only remove additional points. Hence (PM.200). The center-drift clause is Lemma 4.22(iv) inserted into (PM.199). Finally, if two expanded radii are each  $< s_j/3$  while their centers are at least  $s_j$  apart, the sum of the radii is  $< 2s_j/3 < s_j$ , so the disks are disjoint.  $\square$

**Lemma 4.29 (real  $C^1$  gap extension from a collared complex holed domain).**

Let  $0 < R < R^+ \leq 1$  and let

$$U^+ = D(0, R^+) \setminus \bigcup_{\lambda \in \Lambda} \overline{D(c_\lambda, r_\lambda^+)}, \quad U = D(0, R) \setminus \bigcup_{\lambda \in \Lambda} \overline{D(c_\lambda, r_\lambda)}, \quad (\text{PM.203})$$

where the centers are real, the larger closed disks  $\overline{D(c_\lambda, r_\lambda)}$  are pairwise disjoint and compactly contained in  $D(0, R)$ , and

$$r_\lambda - r_\lambda^+ \geq \eta > 0, \quad R^+ - R \geq \eta. \quad (\text{PM.204})$$

If  $q$  is real symmetric and holomorphic on  $U^+$  with  $\|q\|_{C^2(U^+)} \leq M$ , then the restriction of  $q$  to  $K := U \cap [-R, R]$  has a real  $C^1$  extension  $\mathcal{E}_U q$  to  $[-R, R]$  such that

$$\mathcal{E}_U q = q \quad \text{on } K, \quad \|\mathcal{E}_U q\|_{C^1([-R, R])} \leq CM, \quad (\text{PM.205})$$

where  $C$  is absolute. The extension can be chosen linearly in  $q$ .

*Proof.* Only the deleted real intervals need to be filled. For one hole put  $a = c - r$ ,  $b = c + r$ ,  $L = b - a = 2r$ . The circle  $|z - c| = r$  lies in  $U^+$ : it is outside the smaller deleted disk  $\overline{D(c, r^+)}$ , inside  $D(0, R)$ , and cannot meet any other smaller deleted disk because those smaller disks lie inside the pairwise-disjoint  $r_\mu$ -disks. Hence integration on the upper semicircle gives

$$|q(b) - q(a)| \leq \pi r \|q'\|_{U^+} \leq \frac{\pi}{2} LM, \quad |q'(b) - q'(a)| \leq \frac{\pi}{2} LM. \quad (\text{PM.206})$$

On  $[a, b]$  take the cubic Hermite interpolant

$$P(x) = q(a)h_{00}(t) + q(b)h_{01}(t) + Lq'(a)h_{10}(t) + Lq'(b)h_{11}(t), \quad t = \frac{x - a}{L}, \quad (\text{PM.207})$$

with the standard basis

$$h_{00} = 2t^3 - 3t^2 + 1, \quad h_{01} = -2t^3 + 3t^2, \quad h_{10} = t^3 - 2t^2 + t, \quad h_{11} = t^3 - t^2.$$

The endpoint values and first derivatives match those of  $q$ . For the derivative, use  $h'_{01} = -h'_{00}$  to write the potentially singular value contribution as

$$\frac{q(a) - q(b)}{L} h'_{00}(t),$$

which is uniformly  $O(M)$  by (PM.206); the remaining derivative terms are bounded multiples of  $q'(a)$  and  $q'(b)$ . The basis functions themselves are uniformly bounded as well, so  $\|P\|_{C^1([a,b])} \leq CM$ . Fill every deleted interval by its corresponding Hermite polynomial and leave  $q$  unchanged on  $K$ . Pairwise disjointness makes the construction independent from gap to gap, and endpoint matching makes the resulting function  $C^1$  on the whole interval. Real symmetry makes all endpoint data real. Linearity is clear from (PM.207).  $\square$

**Lemma 4.30 (twist-preserving continuation across successively deleted gaps).**

Let  $I = [-R_*, R_*]$ . Suppose  $U_j^+ \supset U_j$  are collared holed domains as in Lemma 4.29, with  $I \cap U_{j+1}^+ \subset I \cap U_j$ , and let  $\omega_j$  be real symmetric and holomorphic on  $U_j^+$ . Assume

$$\omega_{j+1} = \omega_j + q_j \quad \text{on } U_j, \quad \|q_j\|_{C^2(U_j^+)} \leq d_j, \quad (\text{PM.208})$$

where the equality is understood on the common next-step domain. Suppose  $\omega_0$  is defined on all of  $I$  and

$$0 < a \leq \omega'_0(x) \leq A \quad (x \in I), \quad C \sum_{j \geq 0} d_j \leq a/2, \quad (\text{PM.209})$$

with  $C$  the constant of Lemma 4.29. Then there are  $C^1$  functions  $\bar{\omega}_j : I \rightarrow \mathbb{R}$  such that

$$\bar{\omega}_j = \omega_j \quad \text{on } I \cap U_j, \quad \frac{a}{2} \leq \bar{\omega}'_j \leq A + \frac{a}{2}, \quad (\text{PM.210})$$

and

$$\|\bar{\omega}_{j+1} - \bar{\omega}_j\|_{C^1(I)} \leq C d_j. \quad (\text{PM.211})$$

Consequently every value in  $\bar{\omega}_j(I)$  has a unique real preimage. If the same value  $\lambda$  belongs to both  $\bar{\omega}_i(I)$  and  $\bar{\omega}_j(I)$ ,  $i < j$ , and  $c_\lambda^{(m)}$  denotes its preimage at stage  $m$ , then

$$|c_\lambda^{(j)} - c_\lambda^{(i)}| \leq \frac{2}{a} \sum_{m=i}^{j-1} C d_m. \quad (\text{PM.212})$$

*Proof.* Set  $\bar{\omega}_0 = \omega_0$ . Given  $\bar{\omega}_j$ , apply Lemma 4.29 to  $q_j$  and put

$$\bar{\omega}_{j+1} := \bar{\omega}_j + \mathcal{E}_{U_j} q_j. \quad (\text{PM.213})$$

Because the next input domain lies in  $U_j$ , the extension agrees there with  $q_j$ ; induction therefore proves the matching assertion in (PM.210). Equation (PM.205) gives (PM.211). Summing the derivative bounds and using (PM.209) yields  $a/2 \leq \bar{\omega}'_j \leq A + a/2$ , so every extension is strictly increasing. For a common value  $\lambda$ ,

$$\frac{a}{2} |c_\lambda^{(j)} - c_\lambda^{(i)}| \leq |\bar{\omega}_i(c_\lambda^{(j)}) - \bar{\omega}_i(c_\lambda^{(i)})| = |\bar{\omega}_i(c_\lambda^{(j)}) - \bar{\omega}_j(c_\lambda^{(j)})|,$$

and (PM.211) gives (PM.212).  $\square$

**Lemma 4.31 (adapted-window scale inequalities).** Fix  $\tau \geq 1$  and

$$0 < \beta < \frac{1}{4(1+\tau)}, \quad \mu := \left(1 + \frac{1}{\tau}\right)(1 - \beta) \in (1, 2). \quad (\text{PM.214})$$

Use the sequences of Lemma 4.21 with  $p = 4$  and  $0 < \kappa < 1/12$ . Let  $0 < \rho \ll 1$  and choose  $N_0$  so that

$$c_0 \rho^{-1/(1+\tau)} \leq N_0 \leq C_0 \rho^{-1/(1+\tau)}. \quad (\text{PM.215})$$

Let  $J$  be the least index with  $N_J \geq N_0^\mu$ . Define

$$S_j := \delta_j^{-3} K_j^3 E_j, \quad \eta_j := \delta_j K_j^{-1}, \quad \hat{r}_j := \exp[-(\log N_j)^2]. \quad (\text{PM.216})$$

After increasing  $N_0$ , uniformly for  $0 \leq i \leq j \leq J$  one has, for every fixed  $M \geq 0$ ,

$$N_J^M \sum_{m \geq i} S_m = o(K_i^{-1}), \quad \sum_{m \geq i} \eta_m = O(K_i^{-1}), \quad (\text{PM.217})$$

$$K_i^{-1} + \sum_{m \geq i} S_m + \sum_{m \geq i} \eta_m \ll \hat{r}_i \ll N_J^{-2}, \quad (\text{PM.218})$$

and

$$N_J \asymp \rho^{-(1-\beta)/\tau}, \quad \rho N_J (\log N_J)^6 + N_0^{-1} (\log N_0)^6 = o(|\log \rho|^{-1}). \quad (\text{PM.219})$$

Moreover, if  $\omega_*(0)$  is  $(\kappa_D, \tau)$ -Diophantine and  $0 < a \leq \omega'_* \leq A$  on a real neighborhood of the origin, then for

$$\sigma_\rho := \frac{\kappa_D}{8A} N_J^{-(1+\tau)} \asymp \rho^\mu \quad (\text{PM.220})$$

every rational  $l/k$  with  $1 \leq k < N_J$  has its  $\omega_*$ -center, when it exists, outside  $D(0, 4\sigma_\rho)$ .

*Proof.* By Lemma 4.21,

$$S_j = \delta_j^{-3} E_j^{1-3\kappa}.$$

Since  $1 - 3\kappa > \kappa$ , the ratio  $S_j/K_j^{-1}$  is a fixed polynomial in  $j$  times  $E_j^{1-4\kappa}$  and tends to zero super-exponentially. Geometric growth of  $N_j$  makes the tail comparable to its first term. Because  $N_J \leq C N_i^2$  in the finite window, multiplying by any prescribed fixed power  $N_J^M$  is only a polynomial cost and is still absorbed by  $E_i^{1-4\kappa}$ . This proves the first part of (PM.217) with its stated polynomial reserve. The second part is identical with  $\eta_j = \delta_j E_j^\kappa$ .

For every fixed  $M > 0$ ,  $E_i^\kappa = o(N_i^{-M})$ . Also  $\hat{r}_i = N_i^{-\log N_i}$  is smaller than every fixed negative power of  $N_i$  but larger than  $E_i^\kappa$  and the quantities in (PM.217). Since  $N_J \leq (4/3)N_0^\mu$  and  $\mu < 2$ , one has  $N_J \leq C N_i^2$  throughout the finite window. Taking a fixed power larger than four proves (PM.218).

Minimality of  $J$  gives  $N_J \asymp N_0^\mu$ . Inserting (PM.215) and the definition of  $\mu$  gives the first relation in (PM.219). The exponent of  $\rho$  in  $\rho N_J$  is

$$1 - \frac{1-\beta}{\tau} \geq \beta > 0,$$

while  $N_0^{-1} \asymp \rho^{1/(1+\tau)}$ ; logarithmic factors are therefore absorbed, which proves the second relation.

Finally the Diophantine condition gives  $|\omega_*(0) - l/k| \geq \kappa_D N_J^{-(1+\tau)}$ . If  $\omega_*(c) = l/k$ , the upper twist bound gives

$$A|c| \geq |\omega_*(c) - \omega_*(0)| \geq \kappa_D N_J^{-(1+\tau)} = 8A\sigma_\rho,$$

so  $|c| \geq 8\sigma_\rho$ ; the weaker displayed exclusion follows.  $\square$

**Lemma 4.32 (laminar evolving resonance family and finite adapted KAM schedule).** Work in the finite adapted window of Lemma 4.31. Suppose the successive holomorphic frequency maps  $\omega_j = (2\pi)^{-1}\Omega'_j$  are produced by the Lemma 4.20 iteration, that their action-frequency increments satisfy

$$\|\omega_{j+1} - \omega_j\|_{C^2(U_j^+)} \leq C_* S_j, \quad (\text{PM.221})$$

and that the initial real twist on a common interval  $I = [-2\rho, 2\rho]$  satisfies  $0 < a \leq \omega'_0 \leq A$ . Assume the total outer-radius shrink is smaller than  $\rho/10$  and the initial angle/action data are small enough for Lemma 4.27.

Then, after increasing  $N_0$ , one can construct inductively finite holed domains  $U_j^+ \supset U_j$  for  $0 \leq j \leq J$  with the following properties. At each stage  $j$  we also write  $V_j$  for the *pre-deletion action domain*: take  $V_0$  to be the initial action disk, and for  $j \geq 1$  take  $V_j$  to be the preceding inner post-deletion domain (with the harmless outer shrink prescribed by Lemma 4.28). New rational disks are first excised from  $V_j$  to form  $U_j^+$ ; all hole boundaries and the outer boundary are then moved inward by the action collar  $\eta_j$  to form  $U_j$ , which is the next pre-deletion input. Thus every newly selected center belongs to  $V_j \cap \mathbb{R}$ , the frequency  $\omega_j$  is holomorphic near that center, and a “birth stage” always refers to this pre-deletion stage.

(a) The real frequencies have global continuations  $\bar{\omega}_j : I \rightarrow \mathbb{R}$  with

$$a/2 \leq \bar{\omega}'_j \leq A + a/2, \quad \|\bar{\omega}_{j+1} - \bar{\omega}_j\|_{C^1} \leq C S_j. \quad (\text{PM.222})$$

(b) At stage  $j$ , for every reduced rational  $\lambda = l/k$  with  $1 \leq k < N_j$  whose center  $c_\lambda^{(j)}$  under  $\bar{\omega}_j$  lies in the surviving pre-deletion real core  $V_j \cap \mathbb{R}$ , remove a disk only if it is not already contained in an older disk. (If the continued center falls inside an older hole, no new disk is born.) A disk born at stage  $i$  has initial radius

$$r_i^+ := C_1 K_i^{-1}, \quad (\text{PM.223})$$

with  $C_1$  fixed sufficiently large; at later stages it is enlarged only by the Cauchy collars  $\eta_m$  from Lemma 4.31. The resulting family is laminar: disks carrying distinct rational labels are disjoint, while every later disk carrying the same label is contained in the disk created at its first birth.

(c) The nested post-deletion domains satisfy the collar and restriction hypotheses of Lemma 4.27, and on  $U_j^+$  one has  $(K_j, N_j)$  map nonresonance. Hence the finite KAM iteration through stage  $J$  is legal and the final remainder is bounded by  $E_J$ .

(d) More generally, if  $\mathcal{H}_{\leq i}(t)$  denotes the minimal final holes whose first birth stage

is at most  $i$  and that meet a real interval of length  $t$ , then

$$\#\mathcal{H}_{\leq i}(t) \leq C_{a,A}(tN_i^2 + 1), \quad 0 \leq i \leq J. \quad (\text{PM.224})$$

In particular  $\#\mathcal{H}_J(t) \leq C_{a,A}(tN_J^2 + 1)$ . If  $\omega_0(0)$  is  $(\kappa_D, \tau)$ -Diophantine, the final domain contains  $D(0, 2\sigma_\rho)$  with  $\sigma_\rho$  from Lemma 4.31.

*Proof.* Apply Lemma 4.29 to each frequency increment in (PM.221) and Lemma 4.30 inductively. The first estimate of Lemma 4.31 makes the total  $C^1$  extension load smaller than  $a/2$ , which proves (a).

We next check the laminar geometry. If a rational label  $\lambda$  is born at stage  $i$ , then by (PM.212) and (PM.217) every later center carrying the same label satisfies

$$|c_\lambda^{(j)} - c_\lambda^{(i)}| \leq Ca^{-1} \sum_{m \geq i} S_m = o(K_i^{-1}). \quad (\text{PM.225})$$

The later nonresonance radius is  $O(K_j^{-1}) = o(K_i^{-1})$ . Choosing  $C_1$  large and then  $N_0$  large therefore makes every later same-label disk lie inside the birth disk after paying all future collars from (PM.217).

For two distinct labels  $\lambda = l/k$  and  $\lambda' = l'/k'$  born at stages  $i \leq j$ , both denominators are  $< N_J$ , so  $|\lambda - \lambda'| \geq N_J^{-2}$ . At  $c_{\lambda'}^{(j)}$  the two global frequency extensions differ by at most  $C \sum_{m=i}^{j-1} S_m$ . Therefore

$$|\bar{\omega}_i(c_\lambda^{(i)}) - \bar{\omega}_i(c_{\lambda'}^{(j)})| \geq N_J^{-2} - C \sum_{m \geq i} S_m \geq \frac{1}{2} N_J^{-2}. \quad (\text{PM.226})$$

Using the upper derivative bound in (PM.222) gives

$$|c_\lambda^{(i)} - c_{\lambda'}^{(j)}| \geq c_{a,A} N_J^{-2}. \quad (\text{PM.227})$$

By (PM.218) the radii, all center drifts, and all collars are  $o(N_J^{-2})$ , so distinct-label disks are disjoint. This proves (b) and, together with Lemma 4.28, gives the nested collar geometry.

For nonresonance, consider first a newly born label. By construction its center lies in the surviving pre-deletion real domain  $V_j \cap \mathbb{R}$ , so  $\bar{\omega}_j = \omega_j$  there and Lemma 4.26 applies before the new disk is removed; the choice of  $C_1$  makes the birth disk contain the complex exclusion disk required for threshold  $K_j^{-1}$ . For an old label born at  $i < j$ , Lemma 4.26 applied at its birth stage gives a  $K_i^{-1}$  divisor margin outside a fixed fraction of the birth disk. On the nested surviving domain,

$$\|\omega_j - \omega_i\|_{C^0} \leq C \sum_{m=i}^{j-1} S_m. \quad (\text{PM.228})$$

Multiplication by  $|k| < N_j \leq N_J$  still leaves  $N_J \sum_{m \geq i} S_m = o(K_i^{-1})$  by the exponential-polynomial separation in Lemma 4.31. Thus at least half of the birth divisor margin survives, and it dominates  $K_j^{-1}$ . Boundary rationals are excluded by working on the fixed inner action core  $[-3\rho/2, 3\rho/2]$ ; the total frequency drift is  $o(\rho)$ , so every rational responsible for a small divisor there has its continued center in  $I = [-2\rho, 2\rho]$ . This

proves the  $(K_j, N_j)$  nonresonance. Lemma 4.20 and Lemma 4.21 may therefore be iterated, while Lemma 4.28 supplies the exact restriction/collar nesting. This proves (c).

For the stagewise count, restrict to labels whose first birth stage is at most  $i$ . Their reduced denominators are  $< N_i$ , hence distinct labels are separated in frequency by at least  $N_i^{-2}$ . Repeating (PM.226)–(PM.227) with  $N_i$  in place of  $N_J$  is legitimate because the post-birth center drift is exponentially smaller than every polynomial scale by (PM.217). Thus their final centers are  $cN_i^{-2}$ -separated, and an interval of length  $t$  meets at most  $C(tN_i^2 + 1)$  of them. This proves (PM.224), including its final-stage specialization. For the central core, let a stage- $j$  rational hole have label  $\lambda = l/k$  with  $k < N_J$  and center  $c = c_\lambda^{(j)}$ . The Diophantine condition at the origin gives

$$|\lambda - \omega_0(0)| \geq \kappa_D N_J^{-(1+\tau)} = 8A\sigma_\rho. \quad (\text{PM.228a})$$

On the other hand, because  $\bar{\omega}_j(c) = \lambda$  and the global continuation increments telescope,

$$|\lambda - \omega_0(0)| \leq |\bar{\omega}_j(c) - \omega_0(c)| + |\omega_0(c) - \omega_0(0)| \leq C \sum_{m < j} S_m + A|c|. \quad (\text{PM.228b})$$

The first term is  $o(\sigma_\rho)$  by the polynomial reserve in (PM.217). Hence no such center can lie in  $D(0, 4\sigma_\rho)$  once  $N_0$  is large. Every final hole radius and later center displacement are also  $o(\sigma_\rho)$ , so no final hole meets  $D(0, 2\sigma_\rho)$ . This argument does not assume that the same rational already belonged to the initial frequency image.  $\square$

**Lemma 4.33 (buffered Hamilton-Jacobi disks and a clean radial boundary).**

Under the hypotheses of Lemma 4.32, let  $D_\lambda = D(c_\lambda, r_\lambda)$  be the minimal final holes that meet  $D(0, \rho)$ . If the hole was born at stage  $i(\lambda)$ , define

$$\hat{r}_\lambda := \exp[-(\log N_{i(\lambda)})^2], \quad \hat{D}_\lambda := D(c_\lambda, \hat{r}_\lambda). \quad (\text{PM.229})$$

For  $\rho$  sufficiently small:

(i) the disks  $\hat{D}_\lambda$  are pairwise disjoint and

$$D_\lambda \Subset \frac{1}{10} \hat{D}_\lambda; \quad (\text{PM.230})$$

(ii) if  $i = i(\lambda)$ , then  $6\hat{D}_\lambda$  is contained in the *pre-deletion* action domain  $V_i$  on which the birth-stage frequency is holomorphic, and

$$\hat{D}_\lambda \setminus \frac{1}{10} \hat{D}_\lambda \subset U_J; \quad (\text{PM.231})$$

(iii) there is a radius

$$\rho' \in \left[ \frac{5}{4}\rho, \frac{4}{3}\rho \right] \quad (\text{PM.232})$$

such that every final hole meeting  $D(0, \rho')$  is actually contained in  $D(0, \rho')$ ;

(iv) the square-root size of the enlarged real windows is summable:

$$\sum_{\lambda} |\widehat{D}_{\lambda} \cap \mathbb{R}|^{1/2} \leq 1. \quad (\text{PM.233})$$

*Proof.* Items (i)–(ii) follow from the two strict scale inequalities in (PM.218), the center separation (PM.227), and the pre-deletion convention in Lemma 4.32. Indeed the actual hole radii, all later center drift, and all accumulated collars are  $o(\widehat{r}_{\lambda})$ , whereas  $\widehat{r}_{\lambda} = o(N_J^{-2})$ . At the instant of birth the center lies in  $V_i \cap \mathbb{R}$  and is separated from every older distinct-label boundary by  $\gg N_J^{-2}$ ; the fixed inner action core also keeps it away from the outer boundary. Hence  $6\widehat{D}_{\lambda} \Subset V_i$ . After the birth disk is removed, later same-label holes remain absorbed inside it while every distinct-label buffered disk stays disjoint. This proves both the birth-stage availability of  $6\widehat{D}_{\lambda}$  and the final annulus inclusion in (PM.231).

For (iii), a circle  $|z| = s$  can cut a final hole only if  $s \in [|c_{\lambda}| - r_{\lambda}, |c_{\lambda}| + r_{\lambda}]$ . Hence the set of forbidden radii in the interval in (PM.232) has total length at most

$$2 \sum_{\lambda: D_{\lambda} \cap D(0, 3\rho/2) \neq \emptyset} r_{\lambda}. \quad (\text{PM.234})$$

By (PM.224) and (PM.218), this is smaller than  $\rho/100$  for small  $\rho$ , while the interval in (PM.232) has length  $\rho/12$ . A permitted radius therefore exists.

Finally group holes by their birth stage. If  $B_i$  denotes the set of minimal holes whose *first* birth occurs at stage  $i$  and that meet  $D(0, \rho)$ , then the cumulative count (PM.224) gives the crude but sufficient bound  $\#B_i \leq C(\rho N_i^2 + 1)$ . Hence each minimal hole is counted exactly once and

$$\sum_{\lambda} |\widehat{D}_{\lambda} \cap \mathbb{R}|^{1/2} \leq C \sum_{i=0}^J (\rho N_i^2 + 1) \exp \left[ -\frac{1}{2} (\log N_i)^2 \right]. \quad (\text{PM.235})$$

The super-polynomial factor dominates every power of  $N_i$ , uniformly over the finite window; the right side tends to zero as  $N_0 \rightarrow \infty$ . This proves (iv).  $\square$

**Lemma 4.34 (quantitative no-screening on the realized adapted domain).** Assume Lemma 4.31–Lemma 4.33 and let  $U' := U_J \cap D(0, \rho')$ , with  $\rho'$  chosen by Lemma 4.33. Let  $D_* = D(c_*, r_*)$  be a final hole meeting  $D(0, \rho)$  and let  $\widehat{D}_* = D(c_*, \widehat{r}_*)$ . Put

$$\sigma := \sigma_{\rho}, \quad d_* := \widehat{r}_*/10, \quad d_{\lambda} := 2\widehat{r}_{\lambda} \quad (\lambda \neq *). \quad (\text{PM.236})$$

Then, for all sufficiently small  $\rho$ , the bracket in Lemma 4.9 satisfies uniformly on

$$\widehat{D}_* \setminus \frac{1}{10} \widehat{D}_* \quad (\text{PM.237})$$

the lower bound

$$\frac{\log(|z|/\rho')}{\log(\sigma/\rho')} - \sum_{\lambda} \frac{\log(d_{\lambda}/(2\rho'))}{\log(r_{\lambda}/\rho')} \geq \frac{c}{|\log \rho|}, \quad (\text{PM.238})$$

where  $c > 0$  depends only on the twist/Diophantine/adapted-window constants. Thus Lemma 4.9 propagates a central logarithmic smallness estimate to the complete

buffered annulus around every selected hole with a loss of only  $O(|\log \rho|)$  in the exponent.

*Proof.* By Lemma 4.33,  $|c_*| \leq \rho + o(\rho)$  and  $\hat{r}_* = o(\rho)$ . Since  $\rho' \geq 5\rho/4$ , every point of (PM.237) satisfies  $|z| \leq (5/6)\rho'$  for small  $\rho$ . Also  $\sigma \asymp \rho^\mu$  with  $\mu > 1$ . Hence

$$\frac{\log(|z|/\rho')}{\log(\sigma/\rho')} \geq \frac{c_0}{|\log \rho|}. \quad (\text{PM.239})$$

It remains to bound the screening correction. A hole born at stage  $i$  has  $r_\lambda \asymp K_i^{-1}$  up to the fixed collar factor, while its chosen enlarged radius is  $e^{-(\log N_i)^2}$ . From (PM.152), uniformly in the adapted window,

$$\left| \frac{\log(d_\lambda/(2\rho'))}{\log(r_\lambda/\rho')} \right| \leq C \frac{(\log N_i)^6}{N_i}. \quad (\text{PM.240})$$

(The extra logarithmic powers harmlessly absorb the polynomial factor  $(i+2)^2$  and  $|\log \rho| \asymp \log N_0$ .) If  $B_i$  is the set of minimal holes whose first birth occurs at stage  $i$  and that meet  $D(0, \rho')$ , the cumulative center-separation count from Lemma 4.32 gives  $\#B_i \leq C(\rho N_i^2 + 1)$ . Grouping the final holes by these disjoint birth sets therefore gives

$$\sum_\lambda \left| \frac{\log(d_\lambda/(2\rho'))}{\log(r_\lambda/\rho')} \right| \leq C \sum_{i=0}^J \left[ \rho N_i (\log N_i)^6 + \frac{(\log N_i)^6}{N_i} \right]. \quad (\text{PM.241})$$

Because  $N_i$  grows geometrically, the first sum is dominated by its last term and the second by its first term. Equation (PM.219) implies that (PM.241) is  $o(|\log \rho|^{-1})$ . Decreasing  $\rho$  makes it at most half the right side of (PM.239), proving (PM.238).

Finally Lemma 4.33 gives  $d_\lambda > 2r_\lambda$  and ensures that the annulus in (PM.237) lies in the shrunken exterior region required by Lemma 4.9; the central disk  $\overline{D}(0, \sigma)$  lies in  $U'$  by Lemma 4.32. All hypotheses of Lemma 4.9 are therefore satisfied.  $\square$

**Lemma 4.35 (continued-fraction CC scale schedule).** Fix a convergent denominator  $q = q_n$  and set

$$\rho_0 = q^{-6}. \quad (\text{PM.242})$$

Let  $h > 0$ , choose  $C_\delta$  so large that

$$\delta_k := C_\delta^{-1} \frac{h}{(k+1)(\log(k+2))^2}, \quad \sum_{k \geq 0} \delta_k \leq \frac{h}{10}, \quad (\text{PM.243})$$

and put

$$\rho_k = \rho_0 \exp \left( - \sum_{j < k} \delta_j \right), \quad W_k = e^{-\sum_{j < k} \delta_j} W_0. \quad (\text{PM.244})$$

Then  $\rho_k \asymp_h \rho_0$  and  $W_k$  retains a fixed positive fraction of the initial complex domain. Moreover, for every  $0 < \theta < 1/6$  and every sufficiently small fixed  $\sigma > 0$ , there is  $\mu > 0$  such that for  $0 \leq k \leq \rho_0^{-\mu}$ ,

$$q\delta_k \geq c\rho_0^{-\theta}, \quad (\rho_k \delta_k)^{-1} \leq C\rho_0^{-7/6+\theta}, \quad q^2 \delta_k^{-1} \rho_k^{1/2} \leq C\rho_0^\theta. \quad (\text{PM.245})$$



Consequently

$$\delta_k^{-1} e^{-q\delta_k/2} \leq C_M \rho_0^M \quad (\text{PM.246})$$

for every prescribed  $M > 0$ , uniformly over that finite iteration window.

*Proof.* The first assertion follows directly from the summability in (PM.243). Since  $q = \rho_0^{-1/6}$  and  $\rho_k \asymp \rho_0$ , while  $\delta_k \geq c_{h,\sigma}(k+1)^{-(1+\sigma)}$ , one has for  $k \leq \rho_0^{-\mu}$

$$q\delta_k \geq c\rho_0^{-1/6+\mu(1+\sigma)}.$$

Choose  $\mu$  so that  $\theta = 1/6 - \mu(1+\sigma) > 0$ . This is the first estimate. The other two are the same substitution, using  $\delta_k^{-1} \leq C\rho_0^{-\mu(1+\sigma)} = C\rho_0^{-1/6+\theta}$ :

$$(\rho_k \delta_k)^{-1} \leq C\rho_0^{-1-1/6+\theta},$$

and

$$q^2 \delta_k^{-1} \rho_k^{1/2} \leq C\rho_0^{-1/3} \rho_0^{-1/6+\theta} \rho_0^{1/2} = C\rho_0^\theta.$$

Finally the first estimate in (PM.245) implies  $e^{-q\delta_k/2} \leq e^{-c\rho_0^{-\theta}}$ , which dominates every algebraic power of  $\rho_0$ ; multiplying by the polynomial factor  $\delta_k^{-1}$  proves (PM.246).  $\square$

**Lemma 4.36 (finite-iteration closure of the corrected CC BNF recurrence).** Assume the CC normalization step at stage  $k$  produces nonnegative quantities

$$\varepsilon_k := \|F_k\|_{W_k}, \quad s_k := \|\Omega_k - \Omega_0\|_{D(0,\rho_k)} \quad (\text{PM.247})$$

satisfying, as long as the step is defined,

$$\begin{aligned} \varepsilon_{k+1} &\leq C_L \left[ \delta_k^{-1} e^{-q\delta_k/2} + q^4 \rho_k^{-2} \delta_k^{-4} \varepsilon_k + q^2 (\rho_k \delta_k)^{-2} s_k \right. \\ &\quad \left. + q^2 \delta_k^{-1} \rho_k^{1/2} \right] \varepsilon_k, \quad s_{k+1} \leq C \sum_{j=0}^k \varepsilon_j. \end{aligned} \quad (\text{PM.248})$$

Suppose moreover that the initial perturbation has total degree  $2m$  and

$$\varepsilon_0 \leq C_0 \rho_0^m, \quad m \geq 30. \quad (\text{PM.249})$$

Then for every sufficiently small  $\beta > 0$  there is  $q_*(\beta)$  such that, for  $q \geq q_*(\beta)$ , the recurrence is defined through

$$\bar{k} := \lfloor q^{1-\beta} \rfloor \quad (\text{PM.250})$$

and satisfies

$$\varepsilon_k \leq (C_\theta \rho_0^\theta)^k \varepsilon_0, \quad s_k \leq 2C\varepsilon_0 \quad (0 \leq k \leq \bar{k}), \quad (\text{PM.251})$$

for some  $\theta > 0$ . In particular,

$$\varepsilon_{\bar{k}} \leq \exp(-q^{1-2\beta}) \quad (\text{PM.252})$$

after decreasing  $\beta$  and increasing  $q_*(\beta)$ .

*Proof.* Use Lemma 4.35 and bootstrap  $s_k \leq 2C\varepsilon_0$ . Since  $q^2 = \rho_0^{-1/3}$ ,  $q^4 = \rho_0^{-2/3}$ , and  $\delta_k^{-1} \leq C\rho_0^{-1/6+\theta}$  on the finite window, the four coefficients in (PM.248) are bounded

respectively by a super-algebraically small quantity,

$$C\rho_0^{-10/3+4\theta}\varepsilon_k, \quad C\rho_0^{-8/3+2\theta}s_k, \quad C\rho_0^\theta. \quad (\text{PM.253})$$

The exponents  $10/3$  and  $8/3$  include the powers of  $q$ ; this is the collar loss that was missing from the earlier provisional recurrence. Since  $m \geq 30$ , (PM.249) makes the middle two quantities  $o(\rho_0^\theta)$  after reducing  $\theta$ . Thus, for small  $\rho_0$ ,

$$\varepsilon_{k+1} \leq C_\theta \rho_0^\theta \varepsilon_k. \quad (\text{PM.253a})$$

The geometric sum then gives  $s_{k+1} \leq C \sum_{j \leq k} \varepsilon_j \leq 2C\varepsilon_0$ , closing the bootstrap. Fix first  $\sigma > 0$  in Lemma 4.35 so small that  $\sigma < \beta/4$ . Then choose  $\theta > 0$  small enough that the resulting  $\mu = (1/6 - \theta)/(1 + \sigma)$  satisfies  $6\mu > 1 - \beta$ . Hence  $\rho_0^{-\mu} \geq q^{1-\beta}$ , while  $\theta$  remains strictly positive. Hence (PM.251) holds through  $\bar{k}$ . Finally

$$\log \frac{\varepsilon_{\bar{k}}}{\varepsilon_0} \leq \bar{k}(\log C_\theta + \theta \log \rho_0) \leq -cq^{1-\beta} \log q,$$

which is stronger than  $-q^{1-2\beta}$  for large  $q$ .  $\square$

**Lemma 4.37 (accumulated  $C^1$  size of the CC normalization).** In the setting of Lemma 4.36, assume that the stage- $k$  CC type-II generator  $Y_k$  satisfies on the shrunk domain

$$\|Y_k\| \leq Cq^2\varepsilon_k, \quad \|D^2Y_k\| \leq Cq^2\delta_k^{-2}\rho_k^{-1}\varepsilon_k. \quad (\text{PM.254})$$

Let  $g_{\bar{k}} := f_{Y_{\bar{k}-1}}^{-1} \circ \dots \circ f_{Y_0}^{-1}$ , where the  $f_{Y_k}$  are the CC generating maps of the definition of complex Cartesian KAM domains, generating maps, and rotation Fourier projectors above. Then, if  $m \geq 30$ ,

$$\|g_{\bar{k}} - \text{id}\|_{C^1} \leq Cq^{-(m-26)}. \quad (\text{PM.255})$$

*Proof.* The implicit equations of the definition of complex Cartesian KAM domains, generating maps, and rotation Fourier projectors above and the inverse-function estimate used in Lemma 4.40 show, whenever the stage smallness condition holds, that

$$\|f_{Y_k}^{\pm 1} - \text{id}\|_{C^1} \leq C(\|DY_k\| + \|D^2Y_k\|) \leq C\|D^2Y_k\|, \quad (\text{PM.255a})$$

after shrinking the fixed initial radius once; the second derivative dominates the first on these small CC domains. From Lemma 4.35,  $\delta_k^{-2} \leq C\rho_0^{-1/3+2\theta}$  and  $\rho_k^{-1} \leq C\rho_0^{-1}$ . Together with  $q^2 = \rho_0^{-1/3}$  and (PM.251),

$$\|D^2Y_k\| \leq C\rho_0^{-5/3+2\theta}(C_\theta\rho_0^\theta)^k\varepsilon_0. \quad (\text{PM.256})$$

Hence

$$\sum_{k < \bar{k}} \|D^2Y_k\| \leq C\rho_0^{m-5/3+2\theta}. \quad (\text{PM.257})$$

For a composition of near-identity  $C^1$  maps, the displacement is bounded by this sum times the product of the derivative norms; that product is at most  $\exp(C \sum_k \|D^2Y_k\|)$  and is therefore uniformly bounded. Since  $\rho_0 = q^{-6}$ , (PM.257) is much smaller than  $q^{-(m-26)}$  for  $m \geq 30$ .  $\square$

**Lemma 4.38 (high-order BNF matching from finite CC normalization).** Assume in addition to Lemma 4.36 that the nonlinear normalization preserves the order filtration

$$F_k = O^{k+2m}(z, w) \quad (\text{PM.258})$$

and that the integrable increments are the radial averages extracted at each stage, so that the formal conjugacy produced by the finite iteration agrees degree by degree with the unique formal Birkhoff normalization. Then, with  $\bar{k}$  from (PM.250),

$$F_{\bar{k}} = O^{\bar{k}+2m}(z, w) = O^{q^{1-2\beta}}(z, w) \quad (\text{PM.259})$$

for large  $q$ , and

$$J^{q^{1-2\beta}} \Omega_{\bar{k}} = J^{q^{1-2\beta}} \text{BNF}(f). \quad (\text{PM.260})$$

Together with Lemma 4.36–Lemma 4.37 this yields, on the  $q^{-6}$  complex domain, a finite exact-symplectic normal form whose remainder is stretched-exponentially small, whose integrable part matches the formal BNF through order  $q^{1-2\beta}$ , and whose normalizing map is  $C^1$ -near the identity by (PM.255).

*Proof.* The first equality in (PM.259) is the assumed filtration gain. Since  $\bar{k} = \lfloor q^{1-\beta} \rfloor$ , one has  $\bar{k} + 2m \geq q^{1-2\beta}$  for all sufficiently large  $q$ , proving the second. At every degree below the first nonzero degree of  $F_{\bar{k}}$ , the finite normalization has removed the non-radial term by the same homological equation used in the formal construction, and has placed its radial average in  $\Omega_k$ . Formal BNF uniqueness, already used in Lemma 4.1, therefore identifies the resulting integrable jet with the formal BNF jet, giving (PM.260). The norm and coordinate conclusions are exactly Lemma 4.36 and Lemma 4.37.  $\square$

**Lemma 4.39 (rotation-Fourier cohomology and CC Cauchy bounds).** Let  $q = q_n$  be a continued-fraction denominator of the irrational frequency  $\omega$ , let  $Q_0(r) = 2\pi\omega r$ , and put  $W = W_{h,D(0,\rho)}^{\text{CC}}$  with  $0 < \rho < 1$ . For  $0 < \delta < \min\{h, 1/4\}$  assume  $q\delta \geq 1$ . If  $F$  is holomorphic and real symmetric on  $W$ , then there is a unique zero-average trigonometric component  $Y$  with  $M_n Y = 0$  for  $|n| \geq q$  satisfying

$$-[Q_0]Y := Y - Y \circ \Phi_{Q_0} = T_q F - M_0 F. \quad (\text{PM.261})$$

On  $e^{-\delta/2}W$  and  $e^{-\delta}W$  respectively,

$$\|Y\| \leq Cq\delta^{-1}\|F\| \leq Cq^2\|F\|, \quad \|R_q F\| \leq C\delta^{-1}e^{-q\delta/2}\|F\|, \quad (\text{PM.262})$$

and

$$\|DY\| \leq Cq^2\rho^{-1/2}\delta^{-1}\|F\|, \quad \|D^2Y\| \leq Cq^2\rho^{-1}\delta^{-2}\|F\|. \quad (\text{PM.263})$$

If  $F = O^p(z, w)$ , then  $Y = O^p(z, w)$ .

*Proof.* Under the rotation flow  $\varphi^t(z, w) = (e^{-it}z, e^{it}w)$ , the definition of complex Cartesian KAM domains, generating maps, and rotation Fourier projectors above gives  $M_n F \circ \varphi^t = e^{int} M_n F$  and  $\Phi_{Q_0} = \varphi^{2\pi\omega}$ . Therefore

$$[Q_0]M_n F = (e^{2\pi in\omega} - 1)M_n F. \quad (\text{PM.264})$$

For  $0 < |n| < q_n$ , continued-fraction best approximation gives  $\text{dist}(n\omega, \mathbb{Z}) \geq (2q_n)^{-1}$ , and hence  $|e^{2\pi i n \omega} - 1|^{-1} \leq Cq$ . Thus

$$Y = - \sum_{0 < |n| < q} \frac{M_n F}{e^{2\pi i n \omega} - 1} \quad (\text{PM.265})$$

solves (PM.261). Along every rotation orbit the function  $t \mapsto F(\varphi^t(z, w))$  extends holomorphically by at least  $\delta/2$  when  $(z, w) \in e^{-\delta/2}W$ . Shifting the  $t$ -contour yields  $\|M_n F\|_{e^{-\delta/2}W} \leq C e^{-|n|\delta/2} \|F\|_W$ . Summing (PM.265) proves the first estimate in (PM.262); the same Fourier estimate for  $|n| \geq q$  proves the tail bound.

It remains to make the derivative scale explicit. There is a universal  $c_h > 0$  such that every point of  $e^{-\delta}W$  is the center of a Euclidean bidisc of radius

$$\Delta = c_h \rho^{1/2} \delta \quad (\text{PM.266})$$

contained in  $e^{-\delta/2}W$ . Indeed the coordinate radii have a margin  $\asymp \rho^{1/2}\delta$ , while changing each coordinate by at most  $c_h \rho^{1/2}\delta$  changes  $|-izw|$  by at most a fixed small multiple of  $\rho\delta$ , which is below the action-radius margin between the two nested domains. Cauchy's estimate therefore gives  $\|D^j Y\| \leq C \Delta^{-j} \|Y\|$  for  $j = 1, 2$ . Using  $q\delta \geq 1$  in the first estimate of (PM.262) gives exactly (PM.263). Finally every  $M_n$  preserves total Taylor degree, so (PM.265) preserves the filtration  $O^p$ .  $\square$

**Lemma 4.40 (nonlinear CC generating-map conjugation calculus).** Fix  $L \geq 1$ . There are  $c_L, C_L > 0$  such that the following holds. Let  $W = W_{h,D(0,\rho)}^{\text{CC}}$ ,  $0 < \rho < 1$ , let  $0 < \delta < \min\{h, 1/4\}$ , and set  $\Delta = c_h \rho^{1/2} \delta$  as in Lemma 4.39. Let  $\Omega$  be real symmetric and holomorphic on  $D(0, \rho)$  with  $\|\Omega\|_{C^3} \leq L$  and assume  $\Phi_\Omega^{\pm 1}$  carries  $e^{-\delta/4}W$  into  $W$ . If  $F, Y$  are real-symmetric holomorphic functions on  $W$  and

$$C_L \Delta^{-2} (\|F\| + \|Y\|) \leq c_L, \quad (\text{PM.267})$$

then on  $e^{-\delta}W$  there are uniquely stationarily normalized generators  $E$  and  $\mathcal{Q}$  such that

$$f_{M_0 F} = \Phi_{M(F)} \circ f_E, \quad \|E\| \leq C_L \Delta^{-4} \|F\|^2, \quad (\text{PM.268})$$

and

$$f_Y \circ \Phi_\Omega \circ f_F \circ f_Y^{-1} = \Phi_{\Omega+M(F)} \circ f_{F-M_0 F + [\Omega+M(F)]Y+\mathcal{Q}}, \quad (\text{PM.269})$$

where  $[A]Y := Y \circ \Phi_A - Y$  and

$$\|\mathcal{Q}\| \leq C_L \Delta^{-4} (\|F\| + \|Y\|)^2. \quad (\text{PM.270})$$

If  $F, Y = O^p(z, w)$  with  $p \geq 3$ , then  $E, \mathcal{Q} = O^{2p-2}(z, w)$ .

*Proof.* The type-II stationary formula used in Lemma 4.18 is coordinate-free for a canonical pair. With  $(\theta, r)$  replaced by  $(z, w)$  it reads

$$\mathcal{S}_{F \star G}(z_0, W_2) = \underset{(z_1, w_1)}{\text{crit}} \{ \mathcal{S}_G(z_0, w_1) + \mathcal{S}_F(z_1, W_2) - z_1 w_1 \}, \quad (\text{PM.271})$$

where  $\mathcal{S}_F(z, W) = zW + F(z, W)$ . The critical equations are  $z_1 = z_0 + \partial_{w_1} G$  and  $w_1 =$

$W_2 + \partial_{z_1} F$ . By Lemma 4.39 the final domain has a Euclidean collar  $\Delta$  inside the initial one. Cauchy's estimates and (PM.267) therefore give the same contraction argument as Lemma 4.18 and, for any finite composition/inversion of the small generators occurring here, a quadratic stationary remainder bounded by  $C_L \Delta^{-2}$  times the product of their norms. We harmlessly weaken this to  $C_L \Delta^{-4}$  since  $\Delta < 1$ .

There is one CC-specific point: an action-only type-II generator is not literally  $\Phi_A$ . Put  $A = M(F)$  and  $s = -izW$ . From the definition of complex Cartesian KAM domains, generating maps, and rotation Fourier projectors above the map  $f_{A \circ r}$  satisfies

$$Z = z(1 - iA'(s)), \quad w = W(1 - iA'(s)), \quad s = \frac{r}{1 - iA'(s)}, \quad r = -izw. \quad (\text{PM.272})$$

Thus  $s - r = O(rA'(s))$  is first order in  $A$ , but Taylor expansion of  $A'$  gives  $A'(s) - A'(r) = O(A''(r)rA'(s))$ , which is quadratic in the size of  $A$  after Cauchy estimates. Together with  $1 - iu - e^{-iu} = O(u^2)$  this yields  $1 - iA'(s) = e^{-iA'(r)} + O(\Delta^{-4}\|A\|^2)$  on the shrunken domain. Consequently  $\Phi_A^{-1}f_{A \circ r}$  is an exact near-identity map whose mixed projection remains a graph and whose  $C^1$  displacement is quadratic. Applying the stationary graph construction once more gives its unique generator  $E$  and (PM.268).

Now split  $F = (F - M_0 F) + M_0 F$ , use the stationary composition formula to move the radial factor next to  $\Phi_\Omega$ , and use (PM.268). Conjugating the remaining near-identity factor by  $f_Y$  is again a finite stationary composition. Its linearization at  $(F, Y) = (0, 0)$  is

$$(F, Y) \mapsto F - M_0 F + [\Omega + M(F)]Y;$$

the  $M(F)$  contribution may be retained in the displayed linear-looking bracket because it is already bilinear in  $(F, Y)$ . Every term not displayed contains at least two derivatives of small generators, hence is bounded by (PM.270). This proves (PM.269).

Finally, a derivative lowers total degree by at most one. Every stationary quadratic term contains at least two small factors and hence has total degree at least  $2p - 2$ . In (PM.272) the difference between the type-II radial map and the integrable exponential is likewise quadratic. This proves the filtration claim.  $\square$

**Lemma 4.41 (one nonlinear exact-symplectic CC BNF step).** Fix  $L \geq 1$ . Let  $q = q_n$ ,  $Q_0(r) = 2\pi\omega r$ , and  $W = W_{h,D(0,\rho)}^{\text{CC}}$  with  $0 < \rho < 1$ . Let  $0 < \delta < \min\{h, 1/4\}$  satisfy  $q\delta \geq 1$ . Suppose

$$\Omega_0 - Q_0 = O(r^2), \quad \|\Omega_0\|_{C^3(D(0,\rho))} \leq L, \quad (\text{PM.273})$$

and let  $\Omega$  obey the same fixed  $C^3$  bound and the domain margin required in Lemma 4.40. Put  $\varepsilon = \|F\|_W$  and  $s = \|\Omega - \Omega_0\|_{D(0,\rho)}$ . If

$$C_L q^2 (\rho^{1/2} \delta)^{-2} \varepsilon \leq c_L, \quad (\text{PM.274})$$

then the  $Y$  of Lemma 4.39 is a valid CC generating function and there are  $\Omega_+ = \Omega + M(F)$  and  $F_+$ , holomorphic and real symmetric on  $e^{-\delta}W$ , such that

$$f_Y \circ \Phi_\Omega \circ f_F \circ f_Y^{-1} = \Phi_{\Omega_+} \circ f_{F_+}, \quad (\text{PM.275})$$

$$\|Y\| \leq C q^2 \varepsilon, \quad \|D^2 Y\| \leq C q^2 \rho^{-1} \delta^{-2} \varepsilon, \quad (\text{PM.276})$$

and

$$\|F_+\| \leq C_L \left[ \delta^{-1} e^{-q\delta/2} + q^4 \rho^{-2} \delta^{-4} \varepsilon + q^2 (\rho\delta)^{-2} s + q^2 \delta^{-1} \rho^{1/2} \right] \varepsilon. \quad (\text{PM.277})$$

Moreover, if  $F = O^p(z, w)$  with  $p \geq 4$ ,  $q \geq p + 1$ , and  $\Omega - Q_0 = O(r^2)$ , then

$$F_+ = O^{p+1}(z, w), \quad M(F) = O(r^{\lceil p/2 \rceil}). \quad (\text{PM.278})$$

*Proof.* Lemma 4.39 solves  $-[Q_0]Y = T_q F - M_0 F$  and gives (PM.276). The smallness (PM.274), after decreasing  $c_L$ , implies the stationary graph condition (PM.267). Apply Lemma 4.40. From (PM.269),

$$F_+ = R_q F + ([\Omega + M(F)] - [Q_0])Y + \mathcal{Q}. \quad (\text{PM.279})$$

The first term is bounded by the tail estimate in Lemma 4.39. The quadratic term is bounded using (PM.270), (PM.262), and  $\Delta \asymp \rho^{1/2} \delta$ :

$$\|\mathcal{Q}\| \leq C_L \rho^{-2} \delta^{-4} (\varepsilon + q^2 \varepsilon)^2 \leq C_L q^4 \rho^{-2} \delta^{-4} \varepsilon^2, \quad (\text{PM.280})$$

where constants are enlarged and  $q \geq 1$ .

For the middle term put  $A = \Omega + M(F) - Q_0$ . The two rotations in the bracket differ at a point of  $e^{-\delta} W$  by at most  $C_L \rho^{1/2} \|A'\|$ . Cauchy's estimate on the action collar gives

$$\|A'\| \leq C_L \rho + C(\rho\delta)^{-1}(s + \varepsilon). \quad (\text{PM.281})$$

Indeed the first term comes from  $\Omega_0 - Q_0 = O(r^2)$ , while the other two come from  $\Omega - \Omega_0$  and  $M(F)$ ; the definition of complex Cartesian KAM domains, generating maps, and rotation Fourier projectors above gives  $\|M(F)\| \leq \varepsilon$ . By the mean-value formula for  $Y$  along the segment joining the two rotation images and Lemma 4.39,

$$\|([\Omega + M(F)] - [Q_0])Y\| \leq C_L q^2 \delta^{-1} [\rho + (\rho\delta)^{-1}(s + \varepsilon)] \varepsilon. \quad (\text{PM.282})$$

The  $s$  term is bounded by the third term in (PM.277), the  $\rho$  term by its fourth term (we weaken  $\rho$  to  $\rho^{1/2}$ ), and the additional  $\varepsilon^2$  term is absorbed by (PM.280). Equations (PM.279)–(PM.282) prove (PM.277).

For the filtration, Lemma 4.39 preserves  $O^p$ . Every monomial in  $R_q F$  has rotation character  $|b - a| \geq q$  and therefore total degree  $a + b \geq q \geq p + 1$ . Lemma 4.40 gives  $\mathcal{Q} = O^{2p-2} \subset O^{p+1}$ . Since  $\Omega - Q_0 = O(r^2)$ , the displacement between  $\Phi_\Omega$  and  $\Phi_{Q_0}$  has total degree at least three; composing  $Y = O^p$  with those rotations changes it only in degree at least  $p + 2$ . Finally the radial part of an  $O^p$  series consists of monomials  $(zw)^j$  with  $2j \geq p$ , so  $M(F) = O(r^{\lceil p/2 \rceil})$ ; its bracket with  $Y$  has degree at least  $2p - 2$ . Hence (PM.278) follows.  $\square$

**Lemma 4.42 (exact  $CC \leftrightarrow AA$  generating transfer).** Let  $\psi_\pm$  be one of the exact branch charts of the definition of exact Cartesian-action-angle branch charts above and let  $V \subseteq \mathbb{C} \setminus \mathbb{R}_\mp$  be a simply connected action domain. Fix a strip  $\mathbb{T}_h$  and a positive collar  $\delta < h/8$ . Let  $F^{\text{CC}}$  be real-symmetric and holomorphic on the corresponding CC image of  $\mathbb{T}_h \times V$ , and suppose the CC type-II map  $f_{F^{\text{CC}}}$  is defined there. Assume that

on the  $4\delta$ -shrunk branch chart

$$\eta := \|f_{F^{\text{CC}}} - \text{id}\|_{C^1} \leq c\delta, \quad (\text{PM.283})$$

where the norm is measured after pulling tangent vectors back by  $\psi_{\pm}$ . Then the conjugated map

$$f_{\pm}^{\text{AA}} := \psi_{\pm}^{-1} \circ f_{F^{\text{CC}}} \circ \psi_{\pm} \quad (\text{PM.284})$$

is exact symplectic and, on a further  $\delta$ -shrink, is represented by a unique stationarily normalized AA type-II generator  $F_{\pm}^{\text{AA}}$ . Moreover

$$\|f_{\pm}^{\text{AA}} - \text{id}\|_{C^1} \leq C\eta \quad (\text{PM.285})$$

and, if  $\Delta$  denotes the Euclidean collar of the final graph inside the initial branch chart,

$$\|F_{\pm}^{\text{AA}} - F^{\text{CC}} \circ \psi_{\pm}\| \leq C\Delta^{-2} \|F^{\text{CC}} \circ \psi_{\pm}\| \eta. \quad (\text{PM.286})$$

In the central case, if  $F^{\text{CC}} = O^3(z, w)$ , the statement applies on both square-root sectors down to every punctured inner radius and the two transferred maps agree on their overlap. If  $V = D(c, \rho)$  and  $|c| > 4\rho$ , one branch chart works on the full disk.

*Proof.* Write

$$z = e^{i\pi/4} r^{1/2} e^{-i\theta}, \quad w = e^{i\pi/4} r^{1/2} e^{i\theta}.$$

A direct differentiation gives

$$\frac{1}{2}(w dz - z dw) = r d\theta, \quad (\text{PM.287})$$

so  $\psi_{\pm}$  preserves not only the symplectic form but the chosen Liouville form exactly. Consequently exact symplecticity is preserved under (PM.284); no flux constant is created by the coordinate change.

The collar hypothesis in (PM.283) guarantees that the image of the final shrunk chart remains in the same square-root branch. One can see the mechanism directly. If  $(Z, W) = f_{F^{\text{CC}}}(z, w)$  and the relative CC displacement is at most  $c\delta$  on the pulled-back chart, then

$$\frac{-iZW}{-izw} = 1 + O(\eta), \quad \frac{W/Z}{w/z} = 1 + O(\eta). \quad (\text{PM.288})$$

The first quotient controls the action displacement and the second, after taking the branch logarithm, controls the angle displacement. Thus  $|R - r| \leq C\eta|r|$  and  $|\Theta - \theta| \leq C\eta$ , proving the  $C^0$  part of (PM.285); differentiating (PM.288) and using the assumed pulled-back  $C^1$  bound gives the derivative estimate. When the central fixed point is in the closure,  $F^{\text{CC}} = O^3$  makes the relative displacement tend to zero with  $|r|^{1/2}$ , so the argument is uniform on punctured sectors as the inner radius tends to zero.

It remains to recover a type-II generator. Write  $f_{\pm}^{\text{AA}}(\theta, r) = (\Theta, R)$ . Since it is  $C^1$ -close to the identity, the mixed projection  $(\theta, r) \mapsto (\theta, R)$  is biholomorphic on the final shrunk chart. Exact symplecticity and (PM.287) imply that on the graph the form

$$r d\theta + \Theta dR$$

is exact. Hence there is a holomorphic function  $S(\theta, R)$  satisfying  $dS = r d\theta + \Theta dR$ .

Subtracting  $\theta R$  gives  $S = \theta R + F_{\pm}^{\text{AA}}$  with

$$r = R + \partial_{\theta} F_{\pm}^{\text{AA}}, \quad \Theta = \theta + \partial_R F_{\pm}^{\text{AA}}. \quad (\text{PM.289})$$

The additive constant is fixed by stationary normalization, proving existence and uniqueness.

For the comparison of generators, introduce a scalar parameter  $t$  and replace  $F^{\text{CC}}$  by  $tF^{\text{CC}}$ . At  $t = 0$  both coordinate systems give the identity. Differentiating the graph equations at  $t = 0$  and using symplectic covariance of the first variation under  $\psi_{\pm}$  shows that the derivative of the AA generator is exactly  $F^{\text{CC}} \circ \psi_{\pm}$ . The remaining stationary-graph terms contain at least two small derivatives. Cauchy's estimate on the collar  $\Delta$  gives the right-hand side of (PM.286). The branch-overlap assertion follows because both expressions in (PM.284) represent the same holomorphic CC map, while the off-centre assertion follows from the existence of one square-root branch on  $D(c, \rho)$  when  $|c| > 4\rho$ .  $\square$

**Lemma 4.43 (exact comparison of one-dimensional normal forms).** Fix  $A, B \geq 1$ . There are constants  $c, C > 0$  depending only on  $A, B$ , the fixed ambient strip, and the normalized ambient action window such that the following holds. Let  $U = A(c_0; \rho_1, \rho_2)$  be an annulus, with the disk case  $\rho_1 = 0$  allowed, and let  $\Omega_j$  be real-symmetric holomorphic action functions satisfying

$$A^{-1} \leq \Omega_j''(r) \leq A \quad (r \in U \cap \mathbb{R}), \quad \|\Omega_j'''\|_U \leq B, \quad (\text{PM.290})$$

for  $j = 1, 2$ . Let  $F_j$  be holomorphic AA type-II generators and let  $g_j$  be exact symplectic holomorphic maps on  $W_{h,U}$  such that

$$\|g_j - \text{id}\|_{C^1} \leq \varepsilon, \quad \|F_j\|_{h,U} \leq \nu, \quad (\text{PM.291})$$

and suppose the two normal-form representations define the same physical map on their common image:

$$g_1 \circ \Phi_{\Omega_1} \circ f_{F_1} \circ g_1^{-1} = g_2 \circ \Phi_{\Omega_2} \circ f_{F_2} \circ g_2^{-1}. \quad (\text{PM.292})$$

If  $0 < \delta < c$  satisfies

$$C\varepsilon < \delta, \quad C\delta^{-20}\nu < 1, \quad 6\delta < \rho_2 - \rho_1 \quad (\text{PM.293})$$

(with the last condition read as  $6\delta < \rho_2$  in the disk case), then

$$\|\Omega_1' - \Omega_2'\|_{A(c_0; \rho_1 + 6\delta, \rho_2 - 6\delta)} \leq C\delta^{-20}\nu, \quad (\text{PM.294})$$

with the obvious disk interpretation.

The same conclusion holds for exact CC normal forms on an off-centre disk  $D(c_0, \rho)$  with  $|c_0| > 4\rho$ . In the central CC fixed-point case, assume the near-identity CC generators vanish to order at least three; then (PM.294) holds on  $D(0, \rho - C\delta)$  after transfer through the two branch sectors of Lemma 4.42.

*Proof.* We first work in AA coordinates. Put  $g = g_2^{-1} \circ g_1$ . Shrinking the strip and



action domain by  $C\varepsilon$ , equation (PM.292) can be rearranged as

$$g \circ \Phi_{\Omega_1} = \Phi_{\Omega_2} \circ g \circ f_E, \quad (\text{PM.295})$$

where  $f_E$  is exact symplectic and near the identity. Indeed,  $f_E$  is obtained by composing  $f_{F_1}^{-1}$ ,  $f_{F_2}$  and their conjugates by  $g$ . The type-II graph formula, inversion, and Cauchy estimates on a  $\delta$  collar give, on a further shrink,

$$\|f_E - \text{id}\|_{C^1} \leq C\delta^{-4}\nu. \quad (\text{PM.296})$$

The exponent 4 is deliberately conservative.

Write

$$g(\theta, r) = (\theta + u(\theta, r), r + v(\theta, r)), \quad \omega_j := \Omega'_j. \quad (\text{PM.297})$$

Comparing the two components of (PM.295) gives

$$\begin{aligned} \omega_2(r + v(\theta, r)) - \omega_1(r) &= u(\theta + \omega_1(r), r) - u(\theta, r) + I(\theta, r), \\ v(\theta + \omega_1(r), r) - v(\theta, r) &= II(\theta, r), \end{aligned} \quad (\text{PM.298})$$

where, after one more collar loss,

$$\|I\| + \|II\| \leq C\delta^{-4}\nu. \quad (\text{PM.299})$$

For each  $k \neq 0$ , remove the components on which

$$\text{dist}\left(\frac{\omega_1(r)}{2\pi}, \frac{1}{k}\mathbb{Z}\right) < \frac{\delta^2}{|k|^3}. \quad (\text{PM.300})$$

The twist lower bound in (PM.290), together with the bound on  $\Omega_1'''$ , implies (after shrinking the normalized action window once) that every connected excluded component is contained in a disk centred on the real axis, and the total sum of these containing radii is  $O(\delta^2)$ : for a fixed  $k$  there are  $O(|k|)$  relevant rationals and each excluded radius is  $O(\delta^2|k|^{-3})$ . Outside their union,  $|e^{ik\omega_1(r)} - 1| \geq c\delta^2|k|^{-2}$ . Fourier inversion of the second line of (PM.298), followed by a strip loss  $\delta$ , therefore gives

$$\|v - \langle v \rangle\| \leq C\delta^{-9}\nu \quad (\text{PM.301})$$

on the nonresonant boundary set. Because the total excluded radial size is  $O(\delta^2)$ , one may choose concentric inner and outer boundaries within a  $2\delta$  collar that avoid every excluded disk. For each complex angle the function  $v - \langle v \rangle$  is holomorphic in  $r$ ; the maximum principle propagates (PM.301) through the deleted resonance disks to the whole inner annulus. Cauchy's estimate then gives  $\|\partial_\theta v\| \leq C\delta^{-10}\nu$ .

Differentiate the first line of (PM.298) in  $\theta$ . The bounded twist and the last estimate reduce the resulting equation for  $\partial_\theta u$  to the same rotation cohomology problem, with forcing at most  $C\delta^{-10}\nu$ . Repeating the resonance-exclusion and maximum-principle argument gives

$$\|\partial_\theta u\| \leq C\delta^{-16}\nu. \quad (\text{PM.302})$$

Since  $g$  is symplectic,

$$(1 + u_\theta)(1 + v_r) - u_r v_\theta = 1.$$

Together with (PM.291), (PM.301)–(PM.302), and  $C\varepsilon < \delta$ , this yields  $\|v_r\| \leq C\delta^{-17}\nu$ . Hence on the connected inner annulus there is a constant  $\gamma$ ,  $|\gamma| \leq C\varepsilon$ , for which

$$\|v - \gamma\| \leq C\delta^{-18}\nu. \quad (\text{PM.303})$$

Choose the reference point defining  $\gamma$  on the real slice. Real symmetry of  $g$  then makes  $\gamma \in \mathbb{R}$ . Averaging the first line of (PM.298) in  $\theta$  eliminates the  $u$ -coboundary; the bounded derivative of  $\omega_2$  and (PM.303) give  $\|\omega_2(\cdot + \gamma) - \omega_1\| \leq C\delta^{-19}\nu$ .

Exactness removes the action translation. Indeed  $g^*(r d\theta) - r d\theta$  is exact, so its integral over every angle circle is zero:

$$0 = \int_0^{2\pi} \{v + (r + v)u_\theta\} d\theta. \quad (\text{PM.304})$$

Because  $\int u_\theta d\theta = 0$ , (PM.302)–(PM.303) imply  $|\gamma| \leq C\delta^{-20}\nu$ . Replacing  $r + \gamma$  by  $r$  in the preceding frequency estimate and using (PM.290) proves (PM.294).

For an off-centre CC disk, Lemma 4.42 supplies one exact AA branch chart on the full disk and leaves the action function  $\Omega_j$  unchanged, so the AA conclusion applies immediately. At the central fixed point use the two branch charts. The resonant disks in (PM.300) are centred on the real axis; on each angular branch sector the same boundary-selection argument applies, while the sector sides are uniformly separated from the opposite real cut. Thus (PM.294) holds on both punctured sectors. Their union covers a punctured inner disk, and  $\Omega'_1 - \Omega'_2$  is holomorphic at  $r = 0$ ; the maximum principle fills the central puncture. This proves the CC assertion.  $\square$

**Lemma 4.44 (model inverse for the Hamilton-Jacobi square-root map).** Assume the hypotheses (PM.73)–(PM.75) of Lemma 4.12, including the gauge  $\langle a^{1/2}e \rangle = 0$ , and put  $r_0 = L\varepsilon^{1/2}$ . There is  $L_0 = L_0(A)$  such that, for  $L \geq L_0$ , the averaged model map  $\Gamma^\sharp$  is univalent on  $\{|z| > r_0\}$  and satisfies

$$|\Gamma^\sharp(z) - \gamma z| \leq C_A \varepsilon^2 |z|^{-3}, \quad |(\Gamma^\sharp)'(z) - \gamma| \leq C_A \varepsilon |z|^{-2}, \quad (\text{PM.305})$$

where  $\gamma = \langle a^{-1/2} \rangle$  and  $A^{-1} \leq \gamma \leq A$ . Its image contains  $\{|\zeta| > C_A r_0\}$ , and on this exterior domain the inverse branch  $H^\sharp = (\Gamma^\sharp)^{-1}$  is holomorphic and obeys

$$|H^\sharp(\zeta) - \gamma^{-1}\zeta| \leq C_A \varepsilon^2 |\zeta|^{-3}, \quad |(H^\sharp)'(\zeta)| \leq C_A. \quad (\text{PM.306})$$

Consequently the model-inverse hypothesis in Lemma 4.12 is automatic after enlarging its fixed constant  $L$  by a factor depending only on  $A$ .

*Proof.* Set  $u(\theta, z) = a(\theta)e(\theta)z^{-2}$ . The model average defining  $\Gamma^\sharp$  is taken over the real angle circle. Hence, for real  $\theta$  and  $|z| \geq r_0$ , the real-slice bounds in (PM.73) give  $|u(\theta, z)| \leq C_A L^{-2}$ . For  $L$  large the chosen square root is therefore holomorphic in  $z$  uniformly for real  $\theta$ , and its binomial series converges normally under the angle integral.

The gauge cancels the term linear in  $e$  after averaging:

$$\Gamma^\sharp(z) - \gamma z = z \left\langle a^{-1/2} \left( (1+u)^{1/2} - 1 - \frac{1}{2}u \right) \right\rangle,$$

which gives the first estimate in (PM.305). More directly,

$$\partial_z \{z(1+u)^{1/2}\} = (1+u)^{-1/2}, \quad (\text{PM.307})$$

so averaging (PM.307) gives the second estimate.

Write  $\Gamma^\sharp(z) = \gamma z + q(z)$ . On the exterior domain,  $|q'(z)| \leq C_A L^{-2}$ . Any two exterior points can be joined outside  $D(0, r_0)$  by a path of length at most  $(\pi/2 + 1)|z_1 - z_2|$ . Taking  $L$  so large that  $(\pi/2 + 1)\|q'\| < \gamma/2$  gives

$$|\Gamma^\sharp(z_1) - \Gamma^\sharp(z_2)| \geq \frac{\gamma}{2}|z_1 - z_2|, \quad (\text{PM.308})$$

so  $\Gamma^\sharp$  is injective.

To locate its image, fix  $\zeta$  with  $|\zeta| > C_A r_0$  and let  $z_0 = \gamma^{-1}\zeta$ . On the circle  $|z - z_0| = |z_0|/3$ , the disk stays outside  $D(0, r_0)$  once  $C_A$  is large enough. The linear term  $\gamma(z - z_0)$  has size  $\gamma|z_0|/3$ , whereas the first estimate in (PM.305) is  $o(|z_0|)$  uniformly for  $L \geq L_0$ . Rouché's theorem therefore gives one zero of  $\Gamma^\sharp(z) - \zeta$  inside this circle. Injectivity makes it unique, and the inverse-function theorem plus (PM.305) makes the resulting inverse holomorphic. Finally

$$H^\sharp(\zeta) - \gamma^{-1}\zeta = -\gamma^{-1}q(H^\sharp(\zeta)), \quad (\text{PM.309})$$

and (PM.308) makes  $|H^\sharp(\zeta)| \asymp_A |\zeta|$ . Equations (PM.305) and (PM.309) yield the first bound in (PM.306), while  $(H^\sharp)' = 1/(\Gamma^\sharp)' \circ H^\sharp$  gives the second.  $\square$

**Lemma 4.45 (actual Hamilton-Jacobi level root and averaged inverse with the cubic term retained).** Fix  $A \geq 1$  and  $h > 0$ . There are constants  $c_A, C_A > 0$  with the following property. Let  $0 < \rho < c_A$  and let

$$\Pi(\theta, r) = a(\theta)(r^2 - e(\theta) + r^3 f(\theta, r)) \quad (\text{PM.310})$$

be holomorphic on  $\mathbb{T}_h \times D(0, \rho)$  and real symmetric. Assume that  $a$  is positive on the real circle, that its positive square root extends holomorphically and nowhere vanishing to  $\mathbb{T}_h$ , and that

$$\|a^{1/2}\|_h + \|a^{-1/2}\|_h + \|f\|_{h,\rho} \leq A, \quad \|e\|_h \leq 1. \quad (\text{PM.311})$$

For  $0 \leq s \leq 1$  put

$$\varepsilon_s := \|e\|_{sh}, \quad \lambda_{s,L} := L\varepsilon_s^{1/2}. \quad (\text{PM.312})$$

If  $L \geq C_A$  and  $\lambda_{s,L} \leq c_A \rho$ , then on  $\mathbb{T}_{sh} \times A(C_A \lambda_{s,L}, c_A \rho)$  there is a unique branch, selected by the condition  $g(\theta, z)/(a(\theta)^{-1/2}z) \rightarrow 1$  as both  $\varepsilon_s/|z|^2$  and  $|z|$  tend to zero, such that

$$\Pi(\theta, g(\theta, z)) = z^2. \quad (\text{PM.313})$$

It has the form

$$g(\theta, z) = a(\theta)^{-1/2}z(1 + u(\theta, z)), \quad \|u\| \leq C_A(L^{-2} + \rho). \quad (\text{PM.314})$$

Branches obtained at two strip widths agree on every connected overlap.

Let  $\varepsilon := \varepsilon_0$ , let  $\gamma := \langle a^{-1/2} \rangle$ , and average the real-angle branch,

$$\Gamma(z) := \langle g(\cdot, z) \rangle. \quad (\text{PM.315})$$

After enlarging the fixed constant in the inner radius,  $\Gamma$  has a holomorphic inverse  $H$  on  $A(C_A L \varepsilon^{1/2}, c_A \rho)$ , and

$$H(\zeta) = \gamma^{-1} \zeta (1 + v(\zeta)), \quad \|v\| \leq C_A (L^{-2} + \rho), \quad \|H'\| \leq C_A. \quad (\text{PM.316})$$

*Proof.* Fix an admissible  $s$  and write  $g = a^{-1/2} z (1 + u)$ . Substitution into (PM.310) gives the fixed-point equation

$$u = \left( 1 + a e z^{-2} - z a^{-1/2} (1 + u)^3 f(\theta, a^{-1/2} z (1 + u)) \right)^{1/2} - 1, \quad (\text{PM.317})$$

where the square root is the branch equal to 1 at the origin of its argument. On the stated annulus  $|a e z^{-2}| \leq C_A L^{-2}$  and  $|z| \leq c_A \rho$ . Choose

$$B := C_A (L^{-2} + \rho). \quad (\text{PM.318})$$

For  $B < 1/10$ , the right-hand side of (PM.317) maps the ball  $\|u\| \leq B$  into itself. To check contraction, restrict the action argument to  $D(0, \rho/2)$  by decreasing  $c_A$ . Cauchy's estimate gives  $\|\partial_r f\| \leq C_A \rho^{-1}$  there. Differentiating the expression inside the square root with respect to  $u$  therefore costs at most  $C_A(|z| + |z|^2/\rho) \leq C_A \rho$ , while the square-root derivative stays bounded. Hence the Lipschitz constant is  $C_A(L^{-2} + \rho) < 1/2$  after the same choice of constants. Banach's fixed-point theorem proves (PM.313)–(PM.314), holomorphic dependence follows from uniform contraction, and uniqueness in the fixed-point ball proves compatibility on overlaps.

On the real-angle branch, (PM.314) gives

$$\Gamma(z) = \gamma z + q(z), \quad |q(z)| \leq C_A B |z|. \quad (\text{PM.319})$$

On a multiplicatively smaller annulus, Cauchy's estimate on a collar comparable to  $|z|$  gives  $|q'(z)| \leq C_A B$ . Thus, for  $B$  small, the map  $z \mapsto \gamma^{-1}(\zeta - q(z))$  is a contraction in a disk of radius  $C_A B |\zeta|$  about  $\gamma^{-1} \zeta$ , whenever  $\zeta$  lies in the stated shrunk target annulus. Its fixed point is the inverse  $H(\zeta)$  and yields the first two bounds of (PM.316). Finally  $H' = 1/(\Gamma' \circ H)$  and  $\Gamma' = \gamma + O_A(B)$  give the derivative bound.  $\square$

**Lemma 4.46 (actual/model cubic comparison for the Hamilton-Jacobi inverse).**

Assume Lemma 4.45 and the square-root model of Lemma 4.12–Lemma 4.44 with the same  $a, e$ . Fix once and for all  $L_0 \gg_A 1$  and put  $r_0 = L_0 \varepsilon^{1/2}$ . After multiplying  $L_0$  by a fixed constant depending only on  $A$ , for every

$$4r_0 \leq \lambda \leq c_A \rho \quad (\text{PM.320})$$

the actual inverse  $H$  of Lemma 4.45 and the model inverse  $H^\sharp$  of Lemma 4.44 are

simultaneously defined on  $|\zeta| = \lambda$  and satisfy

$$\|H - H^\sharp\|_{|\zeta|=\lambda} \leq C_A \lambda^2, \quad \|H^2 - (H^\sharp)^2\|_{|\zeta|=\lambda} \leq C_A \lambda^3. \quad (\text{PM.321})$$

Consequently the comparison hypothesis (PM.77) of Lemma 4.12 is automatic on every annulus lying inside the actual Hamilton-Jacobi level-root domain.

*Proof.* Let  $g$  be the actual branch of Lemma 4.45 and  $g^\sharp$  the model branch (PM.75). They solve

$$a(g^2 - e + g^3 f) = z^2, \quad a((g^\sharp)^2 - e) = z^2. \quad (\text{PM.322})$$

Subtracting gives the exact identity

$$(g - g^\sharp)(g + g^\sharp) = -g^3 f. \quad (\text{PM.323})$$

On  $|z| = \lambda$ , Lemma 4.45 and Lemma 4.44 give  $|g| + |g^\sharp| \asymp_A \lambda$  and  $|g + g^\sharp| \geq c_A \lambda$  once  $L_0$  is fixed large and  $\rho$  is small. Thus

$$\|g - g^\sharp\|_{|z|=\lambda} \leq C_A \lambda^2. \quad (\text{PM.324})$$

Averaging over the real angle circle yields  $|\Gamma(z) - \Gamma^\sharp(z)| \leq C_A \lambda^2$  on the corresponding circle.

Now fix  $|\zeta| = \lambda$  and set  $z = H(\zeta)$ . By Lemma 4.45,  $|z| \asymp_A \lambda$ , and

$$|\Gamma^\sharp(z) - \zeta| = |\Gamma^\sharp(z) - \Gamma(z)| \leq C_A \lambda^2. \quad (\text{PM.325})$$

The two target points in (PM.325) lie in a disk of radius  $O(\lambda^2)$  that stays inside the exterior inverse domain after the fixed enlargement of  $L_0$ . The bound  $\|(H^\sharp)'\| \leq C_A$  from Lemma 4.44 therefore gives

$$|H^\sharp(\zeta) - H(\zeta)| = |H^\sharp(\zeta) - H^\sharp(\Gamma^\sharp(z))| \leq C_A \lambda^2. \quad (\text{PM.326})$$

Both inverse values are  $O_A(\lambda)$ , so multiplying (PM.326) by  $|H + H^\sharp|$  proves the second estimate in (PM.321). Renaming the enlarged fixed  $L_0$  as the fixed  $L$  allowed in Lemma 4.12 gives exactly (PM.77).  $\square$

**Lemma 4.47 (Hamilton-Jacobi generating straightening and the  $1/32$  inner scale).** Fix  $A \geq 1$  and  $h > 0$ . Let

$$P(r) = \alpha r^2 + r^3 b(r), \quad A^{-1} \leq \alpha \leq A, \quad \|b\|_{D(0,\rho)} \leq A, \quad (\text{PM.327})$$

with  $0 < \rho \ll_A 1$ , and let  $\Pi(\theta, r) = P(r) + Q(\theta, r)$  be real symmetric and holomorphic on  $\mathbb{T}_h \times D(0, \rho)$ . Suppose there are numbers  $0 < \varepsilon < 1$  and  $0 \leq \eta < \eta_A$  and defect scales  $\varepsilon_s$  for  $s = 0, 1/2$  such that

$$\varepsilon_0 = \varepsilon, \quad \varepsilon_{1/2} \leq \varepsilon^{1/2}, \quad (\text{PM.328})$$

and, for  $\theta \in \mathbb{T}_{sh}$  and  $|r| \leq \rho/2$ ,

$$|Q(\theta, r)| \leq A(\varepsilon_s + \eta|r|^2). \quad (\text{PM.329})$$

Let  $L \gg_A 1$  and assume  $L\varepsilon_{1/2}^{1/2} \ll_A \rho$ . Then the level equation  $\Pi(\theta, g(\theta, z)) = z^2$  has on the

corresponding annuli a branch for which, if  $g_0$  denotes the branch solving  $P(g_0(z)) = z^2$  with  $g_0(z) \sim \alpha^{-1/2}z$ , one has

$$|g(\theta, z) - g_0(z)| \leq C_A |z| (L^{-2} + \eta) \quad \text{when } |z| \geq C_A L \varepsilon_s^{1/2}. \quad (\text{PM.330})$$

Let  $\Gamma = \langle g \rangle$  and let  $H = \Gamma^{-1}$  on a fixed shrunk annulus. Then

$$p(\theta, R) := g(\theta, H(R)) \quad \text{satisfies} \quad \langle p(\cdot, R) \rangle = R, \quad |p(\theta, R) - R| \leq C_A |R| (L^{-2} + \eta). \quad (\text{PM.331})$$

Consequently

$$S(\theta, R) := \int_0^\theta p(\varphi, R) d\varphi, \quad \Sigma(\theta, R) := S(\theta, R) - \theta R \quad (\text{PM.332})$$

defines a periodic holomorphic type-II generator and, on every collar-shrunk annulus,

$$\|\Sigma\| \leq C_A (1 + h) (L^{-2} + \eta). \quad (\text{PM.333})$$

Assume now that  $\varepsilon$  is small enough that

$$L := \varepsilon^{-7/32}, \quad \delta := \varepsilon^{1/16} \quad (\text{PM.334})$$

satisfy the preceding domain condition and  $\delta \ll \min\{h, \rho\}$ . Then the inner action radius may be chosen

$$\rho_{\text{HJ}} = C_A \varepsilon^{1/32}, \quad (\text{PM.335})$$

and on a fixed further shrink of  $\mathbb{T}_{h/2} \times A(\rho_{\text{HJ}}, \rho/3)$ ,

$$\|\Sigma\|_{C^2} \leq C_A (1 + h) (\varepsilon^{5/16} + \eta \varepsilon^{-1/8}). \quad (\text{PM.336})$$

If the right-hand side is smaller than the universal type-II graph threshold, set  $W := f_\Sigma^{-1}$ . Then  $W$  is a real-symmetric exact symplectic biholomorphism,

$$\Pi \circ W = H^2, \quad W^{-1} \circ \Phi_\Pi^1 \circ W = \Phi_{H^2}^1, \quad (\text{PM.337})$$

and

$$\|W - \text{id}\|_{C^1} \leq C_A (1 + h) (\varepsilon^{5/16} + \eta \varepsilon^{-1/8}). \quad (\text{PM.338})$$

In particular, if  $\eta \leq \varepsilon^{3/8}$ , the right-hand side is  $O_A((1 + h)\varepsilon^{1/4})$ . Equation (PM.335) supplies the  $\rho_{\text{hole}} \lesssim \varepsilon^{1/32}$  input of Lemma 4.13 without a Whitney extension across the deleted core.

*Proof.* For  $\rho$  small,  $P'(r) = 2\alpha r + O_A(r^2)$ , so the selected branch  $g_0$  exists on  $A(0, c_A \rho)$  and satisfies

$$|g_0(z)| \asymp_A |z|, \quad |P'(g_0(z))| \asymp_A |z|. \quad (\text{PM.339})$$

Fix  $s \in \{0, 1/2\}$  and a  $z$  with  $|z| \geq C_A L \varepsilon_s^{1/2}$ . On the circle  $|r - g_0(z)| = K|z|(L^{-2} + \eta)$ , Taylor's theorem and (PM.339) give, for  $K$  large and  $L^{-1}, \eta, \rho$  small,

$$|P(r) - z^2| \geq c_A K |z|^2 (L^{-2} + \eta). \quad (\text{PM.340})$$

On the same circle, (PM.329) gives

$$|Q(\theta, r)| \leq C_A(\varepsilon_s + \eta|z|^2) \leq C_A|z|^2(L^{-2} + \eta). \quad (\text{PM.341})$$

Rouché's theorem therefore selects exactly one zero of  $P(r) + Q(\theta, r) - z^2$  in that circle. It depends holomorphically on  $(\theta, z)$  and proves (PM.330).

Average (PM.330). Writing  $\Gamma_0 := g_0$ , one has  $\Gamma = \Gamma_0 + O_A(z(L^{-2} + \eta))$  on a fixed collar. The inverse of  $\Gamma_0$  is  $H_0(R) = P(R)^{1/2}$  on the selected branch. The same contraction argument used in Lemma 4.45 gives

$$H(R) = H_0(R) + O_A(R(L^{-2} + \eta)). \quad (\text{PM.342})$$

Insert this in (PM.330), use the Lipschitz bound for  $g_0$ , and recall that  $g_0(H_0(R)) = R$ . This gives the second estimate in (PM.331); the first is the definition of  $H$ .

The zero-average identity in (PM.331) implies  $S(\theta + 2\pi, R) - S(\theta, R) = 2\pi R$ , so  $\Sigma$  in (PM.332) is periodic. Integrating the second bound of (PM.331) along a path of length  $O(1 + h)$  proves (PM.333). By (PM.328),

$$L\varepsilon_{1/2}^{1/2} \leq \varepsilon^{-7/32}\varepsilon^{1/4} = \varepsilon^{1/32}, \quad (\text{PM.343})$$

which is (PM.335) after the fixed domain constants are absorbed. Cauchy's estimate on a  $\delta$  collar gives

$$\|\Sigma\|_{C^2} \leq C_A(1 + h)(L^{-2} + \eta)\delta^{-2}. \quad (\text{PM.344})$$

Substituting (PM.344) yields (PM.336).

Finally  $\partial_\theta S = p$ , so the type-II equations for  $f_\Sigma$  read  $r = p(\theta, R)$  and  $\Theta = \partial_R S(\theta, R)$ . Hence  $\Pi(\theta, r) = H(R)^2$ , or equivalently  $\Pi \circ f_\Sigma^{-1} = H^2$ . The small  $C^2$  norm in (PM.336) makes the graph equations biholomorphic by Lemma 4.18, and exactness is built into the type-II generator. Write  $\Phi_\Pi^1$  and  $\Phi_{H^2}^1$  for the time-one Hamiltonian maps generated by  $\Pi$  and  $H^2$ , respectively. A symplectic change satisfying  $\Pi \circ W = H^2$  conjugates the corresponding Hamiltonian vector fields and therefore their time-one maps, proving (PM.337). The implicit graph equations and (PM.336) give (PM.338). If  $\eta \leq \varepsilon^{3/8}$ , then  $\eta\varepsilon^{-1/8} \leq \varepsilon^{1/4}$ , while  $\varepsilon^{5/16} \leq \varepsilon^{1/4}$ .  $\square$

**Lemma 4.48 (affine rescaling of a disk normal form produces the HJ holomorphic filling).** Let  $q \geq 1$ ,  $c \in \mathbb{R}$ ,  $R > 0$ , and let  $\ell(r) = \ell_0 + \ell_1(r - c)$  be affine. Suppose  $H$  is holomorphic on a neighborhood of  $r_0 \leq |z| \leq R$  and define on the corresponding physical annulus

$$\Omega_{\text{HJ}}(r) := \ell(r) + q^{-2}H^2(q(r - c)). \quad (\text{PM.345})$$

Let  $\Omega_{\text{disk}}$  be holomorphic on a neighborhood of  $\overline{D(c, R/q)}$  and put

$$\Xi(z) := q^2(\Omega_{\text{disk}}(c + z/q) - \ell(c + z/q)). \quad (\text{PM.346})$$

Then  $\Xi$  is holomorphic on a neighborhood of  $\overline{D(0, R)}$  and

$$\|\Xi - H^2\|_{|z|=R} = q^2\|\Omega_{\text{disk}} - \Omega_{\text{HJ}}\|_{|r-c|=R/q}. \quad (\text{PM.347})$$

Moreover, if

$$\|\Xi\|_{D(0,R)} \leq C_1, \quad \|H^2\|_{|z|=r_0} \leq C_2, \quad (\text{PM.348})$$

then

$$\|\Xi - H^2\|_{|z|=r_0} \leq C_1 + C_2. \quad (\text{PM.349})$$

Thus, after the exact frequency comparison of Lemma 4.43 and the convergent-jet estimate of Lemma 4.15 have supplied the outer physical discrepancy, (PM.346) is the actual disk-holomorphic function required as  $\Xi$  in Lemma 4.12; no extension of  $H$  through the hole is asserted.

*Proof.* The affine map  $z \mapsto c + z/q$  sends  $D(0, R)$  biholomorphically to  $D(c, R/q)$ , so (PM.346) is holomorphic on the required disk. For  $|z| = R$ , substitute (PM.345) into (PM.346):

$$\Xi(z) - H^2(z) = q^2(\Omega_{\text{disk}}(c + z/q) - \Omega_{\text{HJ}}(c + z/q)), \quad (\text{PM.350})$$

which is (PM.347). The triangle inequality gives (PM.349).  $\square$

**Lemma 4.49 (birth-stage  $q$ -resonant normal form with exponentially small non-resonant correction).** Fix  $A, B \geq 1$  and  $h_0 > 0$ . There are constants  $a_*, c, C > 0$ , depending only on  $A, B, h_0$ , with the following property. Let  $h \geq h_0$ , let  $0 < \varrho < c$ ,  $c_0 \in \mathbb{R}$ , and let  $p/q$  be reduced with  $q \geq 1$ . Suppose  $\Omega$  is real symmetric and holomorphic on  $D(c_0, \varrho)$  and

$$A^{-1} \leq (2\pi)^{-1}\Omega''(r) \leq A \quad (r \in D(c_0, \varrho) \cap \mathbb{R}), \quad \|(2\pi)^{-1}\Omega'''\|_{D(c_0, \varrho)} \leq B, \quad (\text{PM.351})$$

$$(2\pi)^{-1}\Omega'(c_0) = \frac{p}{q}, \quad \varrho < (10Aq)^{-8}. \quad (\text{PM.352})$$

Let  $F$  be real symmetric and holomorphic on  $W_{h, D(c_0, \varrho)}$ , put  $E := \|F\|_{h, D(c_0, \varrho)}$ , and assume

$$E \leq \varrho^{a_*}. \quad (\text{PM.353})$$

Then, on  $W_{h/2, D(c_0, \varrho/2)}$ , there are a real-symmetric exact symplectic biholomorphism  $g_{\text{RNF}}$  and real-symmetric type-II generators  $F^{\text{res}}, F^{\text{cor}}$  such that

$$g_{\text{RNF}}^{-1} \circ \Phi_\Omega \circ f_F \circ g_{\text{RNF}} = \Phi_\Omega \circ f_{F^{\text{res}}} \circ f_{F^{\text{cor}}}, \quad (\text{PM.354})$$

$$F^{\text{res}}(\theta + 2\pi/q, r) = F^{\text{res}}(\theta, r), \quad \|F^{\text{res}}\| \leq 2E, \quad (\text{PM.355})$$

$$\|F^{\text{cor}}\| \leq C \exp(-c\varrho^{-1/4})E, \quad \|g_{\text{RNF}} - \text{id}\|_{C^1} \leq Cq^2\varrho^{-4}E. \quad (\text{PM.356})$$

The constants in (PM.356) are uniform in  $q$  subject to (PM.352). More precisely, at any application scale for which  $E = o(q^{-2}\varrho^{M+4})$ , the final estimate in (PM.356) gives  $\|g_{\text{RNF}} - \text{id}\|_{C^1} = o(\varrho^M)$ .

*Proof.* Let

$$R := \Phi_{2\pi(p/q)r}, \quad \mathcal{P}_q G(\theta, r) := \frac{1}{q} \sum_{j=0}^{q-1} G(\theta + 2\pi jp/q, r), \quad \mathcal{N}_q := I - \mathcal{P}_q. \quad (\text{PM.357})$$

Because  $p$  is invertible modulo  $q$ ,  $\mathcal{P}_q$  is exactly the projector onto Fourier modes divisible by  $q$ ; in particular its range consists of the  $2\pi/q$ -periodic functions. If  $U$  denotes



pullback by  $R$ , then  $U^q = I$  and  $\mathcal{P}_q = q^{-1} \sum_{j=0}^{q-1} U^j$ . Hence, on  $\ker \mathcal{P}_q$ ,

$$(U - I)^{-1}G = \frac{1}{q} \sum_{j=0}^{q-1} jU^j G, \quad \|(U - I)^{-1}G\| \leq q\|G\|. \quad (\text{PM.358})$$

(The displayed finite sum satisfies  $(U - I) \sum jU^j = q(I - \mathcal{P}_q)$ .) Thus for every current generator  $F_k$  the equation

$$[2\pi(p/q)r]Y_k := Y_k \circ R - Y_k = -\mathcal{N}_q F_k, \quad \mathcal{P}_q Y_k = 0, \quad (\text{PM.359})$$

has the unique normalized solution with

$$\|Y_k\| \leq qe_k, \quad e_k := \|\mathcal{N}_q F_k\|. \quad (\text{PM.360})$$

We now iterate this exact rational homological cancellation on nested product domains. Set

$$\delta := c_1 \varrho^{1/4}, \quad K := \lfloor c_2 \varrho^{-1/4} \rfloor, \quad (\text{PM.361})$$

where  $c_1, c_2 > 0$  are fixed so small that the total angle loss  $K\delta$  is  $< h_0/8$  and the total relative action shrink is  $< 1/8$ . At step  $k$  reserve an angle collar  $\delta$  and an action collar comparable to  $\varrho\delta$ . The twist assumptions imply on  $D(c_0, \varrho)$

$$|\Omega'(r) - \Omega'(c_0)| \leq C_A \varrho. \quad (\text{PM.362})$$

By Cauchy's estimate and (PM.360), on the next smaller domain,

$$\|([\Omega] - [2\pi(p/q)r])Y_k\| \leq C_A \varrho \|\partial_\theta Y_k\| \leq C_A q \varrho \delta^{-1} e_k. \quad (\text{PM.363})$$

Apply the stationary type-II calculus of Lemma 4.18 to  $f_{Y_k} \Phi_\Omega f_{F_k} f_{Y_k}^{-1}$ . Since this expression equals  $\Phi_\Omega f_{F_{k+1}}$  for a uniquely normalized small generator, analyticity of the stationary formula at  $(F_k, Y_k) = (0, 0)$  and its differential there give

$$F_{k+1} = F_k + [\Omega]Y_k + Q_k. \quad (\text{PM.364})$$

Every term of  $Q_k$  contains at least one factor  $Y_k$  (when  $Y_k = 0$  there is no conjugation), and Cauchy's estimate on the two collars gives the conservative bound

$$\|Q_k\| \leq C(\varrho\delta)^{-4}(\|F_k\| + \|Y_k\|)\|Y_k\|. \quad (\text{PM.365})$$

Combining (PM.359), (PM.363), and (PM.365), and using  $\|\mathcal{P}_q\| \leq 1$ , yields as long as  $\|F_k\| \leq 2E$

$$e_{k+1} \leq C(q\varrho\delta^{-1} + q^2(\varrho\delta)^{-4}E)e_k. \quad (\text{PM.366})$$

Condition (PM.352) gives  $q \leq C_A \varrho^{-1/8}$ . With (PM.361), the first coefficient in (PM.366) is  $O_A(\varrho^{5/8})$ , while the second is  $O_A(\varrho^{-21/4}E)$ . Choosing  $a_* \geq 20$  and then  $\varrho$  small gives

$$e_{k+1} \leq \varrho^{1/2} e_k. \quad (\text{PM.367})$$

The same decomposition in (PM.364) shows that the resonant part changes by at most

$C\varrho^{1/2}e_k$ . Hence

$$\|F_k\| \leq E + C \sum_{j < k} \varrho^{1/2} e_j \leq 2E, \quad (\text{PM.368})$$

so the bootstrap used in deriving (PM.366) closes for all  $0 \leq k \leq K$ . All implicit type-II maps are legal because their  $C^2$  smallness quantities are a fixed polynomial in  $q, \varrho^{-1}, \delta^{-1}$  times  $E$ , and (PM.353) absorbs that polynomial after increasing  $a_*$ .

From (PM.367),

$$e_K \leq \varrho^{K/2} E \leq \exp(-c\varrho^{-1/4} |\log \varrho|) E \leq \exp(-c\varrho^{-1/4}) E. \quad (\text{PM.369})$$

Let  $g_{\text{RNF}}$  be the accumulated inverse conjugacy selected by the identities leading to (PM.364). The displacement of one  $f_{Y_k}^{\pm 1}$  in  $C^1$  is bounded, by the implicit graph equations and two Cauchy derivatives, by  $Cq^2\varrho^{-4}e_k$  on the final common domain. Summing the geometric sequence in (PM.367) gives the second estimate of (PM.356).

Finally put  $F^{\text{res}} := \mathcal{P}_q F_K$  and define  $F^{\text{cor}}$  by  $f_{F_K} = f_{F^{\text{res}}} f_{F^{\text{cor}}}$ . Lemma 4.18 and (PM.369) give

$$\|F^{\text{cor}}\| \leq C(1 + \varrho^{-C} E) e_K \leq C e^{-c\varrho^{-1/4}} E,$$

after one harmless exponent loss. The first factor is  $2\pi/q$ -periodic by construction, and (PM.368) gives (PM.355). Real symmetry is preserved at every step because the rational projector, the homological inverse (PM.358), and the stationary normalization commute with the real involution. This proves the lemma.  $\square$

**Lemma 4.50 (analytic map-to-autonomous-flow interpolation with a stretched-exponential residual).** Fix  $A \geq 1$  and  $h_0 > 0$ . There are  $c, C > 0$  such that the following holds. Let  $0 < s < c$ ,  $h \geq h_0$ , and let  $\Omega$  be real symmetric and holomorphic on  $D(0, s)$  with

$$\Omega(0) = \Omega'(0) = 0, \quad \|\Omega'\| \leq As, \quad \|\Omega''\| + \|\Omega'''\| \leq A. \quad (\text{PM.370})$$

Let  $F$  be real symmetric and holomorphic on  $W_{h,D(0,s)}$ , put  $E := \|F\|$ , and assume

$$E \leq s^{10}. \quad (\text{PM.371})$$

Then on  $W_{h/2,D(0,s/2)}$  there are a real-symmetric autonomous Hamiltonian  $\Pi$  and a real-symmetric type-II generator  $R$  such that

$$\Phi_\Omega \circ f_F = \Phi_\Pi^1 \circ f_R, \quad (\text{PM.372})$$

$$\|\Pi - \Omega - F\| \leq Cs^{1/4}E, \quad \|R\| \leq C \exp(-cs^{-1/4})E. \quad (\text{PM.373})$$

Here  $\Phi_\Pi^1$  is the time-one Hamiltonian map of the generally angle-dependent Hamiltonian  $\Pi$ ; the notation is deliberately distinguished from the action-only map  $\Phi_\Omega$  of the definition of complex action strips, holed action domains, and type-II exact-symplectic generators above.

*Proof.* We first prove the one-step estimate used in the iteration. Let  $S, G$  be real symmetric and holomorphic on a current domain of angle width  $h_*$  and action radius  $s_*$  with  $s/2 \leq s_* \leq s$ , and let  $0 < \delta < h_*/4$ . Assume

$$C(s_*\delta)^{-4}(\|S\| + \|G\|) < 1, \quad Cs_*\delta^{-3} < 1. \quad (\text{PM.374})$$

Then, after losing an angle collar  $\delta$  and an action collar comparable to  $s_*\delta$ , there is a normalized type-II generator  $\mathcal{A}(S, G)$  such that

$$\Phi_{\Omega+S}^1 \circ f_G = \Phi_{\Omega+S+G \circ \Phi_{-\Omega/2}}^1 \circ f_{\mathcal{A}(S, G)}, \quad (\text{PM.375})$$

$$\|\mathcal{A}(S, G)\| \leq C [(s_*\delta)^{-4}(\|S\| + \|G\|) + s_*\delta^{-3}] \|G\|. \quad (\text{PM.376})$$

We prove this estimate at the map level rather than assuming a logarithm for a canonical map. Write  $\omega(r) = \Omega'(r)$ . Cauchy's estimate on the reserved collars gives, for  $a + b \leq 3$ ,

$$\|\partial_\theta^a \partial_r^b H\| \leq C_{a,b} \delta^{-a} (s_*\delta)^{-b} \|H\|, \quad H \in \{S, G\}. \quad (\text{PM.377})$$

The smallness conditions (PM.374) imply that, for  $0 \leq t \leq 1$ , the Hamiltonian flows of  $\Omega$ ,  $\Omega + S$ , and  $\Omega + S + G$  issued from the twice-shrunk domain remain in the once-shrunk domain. For such an initial point  $x = (\theta, r)$  set

$$\Delta(t, x) := \Phi_{\Omega+S+G}^t(x) - \Phi_{\Omega+S}^t(x), \quad \sigma := \max\{\|DS\|, \|DG\|\}. \quad (\text{PM.378})$$

The variation equation, written componentwise and using  $\omega(r) = O(s_*)$ ,  $\omega'(r) = O(1)$ , gives

$$\dot{\Delta} = J\nabla\Omega(\Phi_{\Omega+S+G}^t) - J\nabla\Omega(\Phi_{\Omega+S}^t) + J\nabla S(\Phi_{\Omega+S+G}^t) - J\nabla S(\Phi_{\Omega+S}^t) + J\nabla G(\Phi_{\Omega+S+G}^t). \quad (\text{PM.379})$$

Taylor's formula and Gronwall first yield  $\|\Delta\| \leq C\sigma$  and  $\|\Phi_{\Omega+S}^t - \Phi_{\Omega}^t\| \leq C\sigma$ . Substituting these bounds once more into (PM.379), and then using the second component of the same equation to eliminate the first-order action displacement, gives at  $t = 1$

$$\Phi_{\Omega+S+G}^1 = \Phi_{\Omega+S}^1 \circ \Phi_{G \circ \Phi_{-\Omega/2}}^1 \circ f_B, \quad \|B\| \leq C [(s_*\delta)^{-4}(\|S\| + \|G\|) + s_*\delta^{-3}] \|G\|. \quad (\text{PM.380})$$

For completeness, the midpoint term in (PM.380) is obtained as follows. Along the integrable orbit  $(\theta + t\omega(r), r)$ , the first-order contribution of  $G$  to (PM.379) is  $J\nabla G(\theta + t\omega(r), r)$ . Integrating the action component and feeding it back into the angle component produces the symmetric half-step  $G \circ \Phi_{-\Omega/2}$ ; all omitted terms contain either one factor  $D^2S\Delta$ , one factor  $D^2G\Delta$ , or an additional factor  $\omega = O(s_*)$ . Formula (PM.377) bounds these respectively by the two terms on the right of (PM.380). The residual map in (PM.380) is exact symplectic and  $C^1$ -close to the identity; the stationary graph theorem in Lemma 4.18 therefore produces the normalized type-II generator  $B$  with the same bound.

Apply (PM.380) with  $G \circ \Phi_{-\Omega/2}$  in place of  $G$ . Since the two half-step pullbacks cancel, exact rearrangement gives

$$\Phi_{\Omega+S}^1 \circ \Phi_G^1 = \Phi_{\Omega+S+G \circ \Phi_{-\Omega/2}}^1 \circ f_{B_1},$$

with  $B_1$  satisfying the bound in (PM.376). Finally, the implicit equations of the definition of complex action strips, holed action domains, and type-II exact-symplectic generators above and the time-one flow equation give on one further collar

$$f_G = \Phi_G^1 \circ f_C, \quad \|C\| \leq C(s_*\delta)^{-3} \|G\|^2. \quad (\text{PM.381})$$

Indeed both  $f_G$  and  $\Phi_G^1$  have first jet  $\text{id} + X_G$  and their difference is quadratic in  $G$ ; Lemma 4.18 converts the exact residual map to  $f_C$ . Composing  $f_{B_1}$  and  $f_C$  with Lemma 4.18 and using (PM.374) absorbs the quadratic term into the first bracket of (PM.376). This proves (PM.375)–(PM.376).

We now iterate. Put

$$\delta := c_0 s^{1/4}, \quad K := \lfloor c_1 s^{-1/4} \rfloor, \quad (\text{PM.382})$$

with  $c_0, c_1$  chosen so that  $K\delta$  spends less than one quarter of the angle strip and less than one quarter of the action radius in relative logarithmic scale. Set  $S_0 = 0$ ,  $G_0 = F$  and, using (PM.375), define

$$S_{k+1} = S_k + G_k \circ \Phi_{-\Omega/2}, \quad G_{k+1} = \mathcal{A}(S_k, G_k). \quad (\text{PM.383})$$

The exact identities telescope:

$$\Phi_\Omega \circ f_F = \Phi_{\Omega+S_K}^1 \circ f_{G_K} \quad (\text{PM.384})$$

on the  $k$ th nested domain. If  $e_k := \|G_k\|$  and  $\sigma_k := \|S_k\|$ , (PM.376) gives

$$e_{k+1} \leq C [s\delta^{-3} + (s\delta)^{-4}(\sigma_k + e_k)] e_k, \quad \sigma_{k+1} \leq \sigma_k + e_k. \quad (\text{PM.385})$$

Assume inductively  $\sum_{j \leq k} e_j \leq 2E$ . Since  $s\delta^{-3} = O(s^{1/4})$  and  $(s\delta)^{-4}E = O(s^{-5}E) = O(s^5)$  by (PM.371), (PM.385) gives, for  $s$  small,

$$e_{k+1} \leq C_0 s^{1/4} e_k \leq \frac{1}{2} e_k. \quad (\text{PM.386})$$

This proves the induction and the legality of (PM.374) at every step. Therefore

$$e_K \leq (C_0 s^{1/4})^K E \leq C \exp(-cs^{-1/4}) E. \quad (\text{PM.387})$$

Set  $\Pi := \Omega + S_K$  and  $R := G_K$ ; (PM.384) is (PM.372). Finally,

$$S_K - F = F \circ \Phi_{-\Omega/2} - F + \sum_{k=1}^{K-1} G_k \circ \Phi_{-\Omega/2}. \quad (\text{PM.388})$$

On a fixed reserve strip, the first difference is  $O(sE)$  because  $\|\Omega'\| = O(s)$ , while (PM.386) makes the sum  $O(s^{1/4}E)$ . This proves the first estimate in (PM.373), and (PM.387) proves the second. Real symmetry is preserved by the flow, pullback, and stationary normalization at every step.  $\square$

**Lemma 4.51 (birth-stage Hamilton-Jacobi package and propagation of the exponentially small residual).** Work in the adapted window of Lemma 4.31–Lemma 4.33. Let  $D_\lambda = D(c, r_\lambda)$  be a final minimal hole first born at stage  $i$ , let  $q$  be the denominator of its reduced rational label  $p/q$ , and put

$$\hat{\rho} := \hat{r}_\lambda = \exp[-(\log N_i)^2], \quad s := q\hat{\rho}. \quad (\text{PM.389})$$

In the stage- $i$  adapted coordinates furnished by Lemma 4.20 and Lemma 4.32, write

the birth-stage map on the pre-deletion neighborhood  $6\hat{D}_\lambda$  as

$$M_i = \Phi_{\Omega_i} \circ f_{F_i}, \quad \|F_i\| \leq E_i, \quad (2\pi)^{-1}\Omega'_i(c) = p/q. \quad (\text{PM.390})$$

Then, after increasing  $N_0$ , there is an annular Hamilton–Jacobi normal form on a punctured subdisk of  $\hat{D}_\lambda$ :

$$g_{\text{HJ},\lambda}^{-1} \circ M_i \circ g_{\text{HJ},\lambda} = \Phi_{\Omega_{\text{HJ},\lambda}} \circ f_{F_{\text{HJ},\lambda}}, \quad (\text{PM.391})$$

where

$$\Omega_{\text{HJ},\lambda}(r) = \ell_\lambda(r) + q^{-2}H_\lambda^2(q(r-c)) \quad (\text{PM.392})$$

for an affine  $\ell_\lambda$ , and the deleted physical inner radius obeys

$$\rho_{\text{HJ},\lambda} \leq Cq^{-1}(q^2E_i)^{1/32} = o(\hat{\rho}). \quad (\text{PM.393})$$

Moreover

$$\|F_{\text{HJ},\lambda}\| \leq C \exp[-c(q\hat{\rho})^{-1/4}]E_i, \quad (\text{PM.394})$$

$$\|g_{\text{HJ},\lambda} - \text{id}\|_{C^1} \leq C(q^2E_i)^{1/4} + o(\hat{\rho}^{10}) = o(1), \quad (\text{PM.395})$$

and the complete coordinate change and its inverse carry the real punctured HJ annulus into  $6\hat{D}_\lambda \subset V_i$ . On the part outside  $\hat{D}_\lambda/10$ , this annulus is contained in the final adapted domain  $U_J$  by Lemma 4.33. The lift through the  $q$ -cover has zero deck-translation integer.

*Proof.* The adapted scales provide more than the high-power reserve required in Lemma 4.49 and Lemma 4.50. Indeed  $q < N_i$  at the birth stage, while  $E_i = \exp[-cN_i/(i+2)^2]$  and  $\hat{\rho} = \exp[-(\log N_i)^2]$ . Hence for every fixed  $M$ ,

$$q^M E_i = o(\hat{\rho}^M), \quad q^2 E_i = o(s^M), \quad E_i^{-M} e^{-cs^{-1/4}} \leq e^{-c_M s^{-1/4}}. \quad (\text{PM.396})$$

The first two statements are the same exponential-versus-superpolynomial separation used in Lemma 4.31, written at the individual birth scale. For the third, put  $L_i = \log N_i$ . Since  $q \leq N_i$ ,

$$s^{-1/4} = q^{-1/4} e^{L_i^2/4} \geq \exp\left(\frac{L_i^2 - L_i}{4}\right).$$

On the other hand  $N_i = N_0(4/3)^i$  implies  $i+2 \asymp L_i$ , and therefore  $\log E_i^{-M} = O_M(e^{L_i}/L_i^2)$ . Thus  $\log E_i^{-M} = o(s^{-1/4})$ , so after decreasing  $c$  to a positive  $c_M$  the last inequality in (PM.396) follows. The same comparison gives  $\hat{\rho} < (10Aq)^{-8}$  and all the smallness hypotheses of Lemma 4.49 after  $N_0$  is increased. Lemma 4.33(ii) supplies the required holomorphic neighborhood before the hole is deleted. Applying Lemma 4.49 (with a harmless fixed-radius rescaling from  $6\hat{\rho}$  to  $\hat{\rho}$ ) yields

$$g_{\text{RNF}}^{-1} M_i g_{\text{RNF}} = \Phi_{\Omega_i} \circ f_{F^{\text{res}}} \circ f_{F^{\text{cor}}}, \quad (\text{PM.397})$$

where  $F^{\text{res}}$  is  $2\pi/q$ -periodic,  $\|F^{\text{res}}\| \leq CE_i$ , and

$$\|F^{\text{cor}}\| \leq C e^{-c\hat{\rho}^{-1/4}} E_i. \quad (\text{PM.398})$$

Pass to the  $q$ -resonant quotient by the affine conformally symplectic map

$$\Lambda_q(\theta, r) := (x, R) = (q\theta, q(r - c)). \quad (\text{PM.399})$$

Because  $F^{\text{res}}$  is  $2\pi/q$ -periodic it descends to a  $2\pi$ -periodic function of  $x$ . Multiplying Hamiltonians/generators by  $q^2$  compensates for  $\Lambda_q^*(dR \wedge dx) = q^2 dr \wedge d\theta$ . After removing the rigid rational rotation, define

$$\tilde{\Omega}(R) := q^2 [\Omega_i(c + R/q) - \Omega_i(c) - \Omega'_i(c)R/q], \quad \tilde{F}^{\text{res}}(x, R) := q^2 F^{\text{res}}(x/q, c + R/q). \quad (\text{PM.400})$$

Then

$$\tilde{\Omega}(0) = \tilde{\Omega}'(0) = 0, \quad \tilde{\Omega}''(R) = \Omega''_i(c + R/q), \quad \tilde{\Omega}'''(R) = q^{-1}\Omega'''_i(c + R/q), \quad (\text{PM.401})$$

so the twist/ $C^3$  constants remain uniform. On  $D(0, cs)$ ,

$$\tilde{\Omega}(R) = \alpha R^2 + R^3 b(R), \quad A^{-1} \leq 2\alpha/(2\pi) \leq A, \quad \|b\| \leq C_{A,B}. \quad (\text{PM.402})$$

Put  $\varepsilon := Cq^2 E_i$ . By (PM.396),  $\varepsilon \leq s^{10}$  and in fact  $\varepsilon = o(s^M)$  for every fixed  $M$ . Lemma 4.50 therefore gives

$$\Phi_{\tilde{\Omega}} \circ f_{\tilde{F}^{\text{res}}} = \Phi_{\Pi}^1 \circ f_{R_{\text{vf}}}, \quad (\text{PM.403})$$

$$\|\Pi - \tilde{\Omega}\| \leq C\varepsilon, \quad \|R_{\text{vf}}\| \leq Ce^{-cs^{-1/4}}\varepsilon. \quad (\text{PM.404})$$

The rescaled image of  $F^{\text{cor}}$  has norm  $\leq Cq^2 e^{-c\hat{\rho}^{-1/4}} E_i$ ; since  $s = q\hat{\rho} \geq \hat{\rho}$  and every polynomial factor is absorbed by (PM.396), Lemma 4.18 combines it with  $R_{\text{vf}}$  into a single residual  $\tilde{R}$  satisfying

$$\|\tilde{R}\| \leq Ce^{-cs^{-1/4}}\varepsilon. \quad (\text{PM.405})$$

Now apply Lemma 4.47 to  $\Pi = P + Q$  with  $P = \tilde{\Omega}$ ,  $Q = \Pi - \tilde{\Omega}$ , and  $\eta = 0$ . Take its two defect scales to be  $\varepsilon_0 = C\varepsilon$  and  $\varepsilon_{1/2} = C\varepsilon$ ; for  $\varepsilon$  small the required inequality  $\varepsilon_{1/2} \leq \varepsilon_0^{1/2}$  is automatic. The very strong reserve in (PM.396) verifies the domain condition  $L\varepsilon_{1/2}^{1/2} \ll s$  for  $L = \varepsilon_0^{-7/32}$ . Thus there are an exact type-II HJ map  $W$  and a holomorphic  $H$  on the annulus outside

$$R_{\text{HJ}} \leq C\varepsilon^{1/32} \quad (\text{PM.406})$$

such that  $W^{-1}\Phi_{\Pi}^1 W = \Phi_{H^2}^1$  and  $\|W - \text{id}\|_{C^1} \leq C\varepsilon^{1/4}$ . Conjugating the residual exact map by  $W$  is handled by the fixed-map clause of Lemma 4.18 on one further HJ collar. Cauchy estimates for  $W, W^{-1}$  may cost fixed powers of  $s^{-1}$  and of  $\varepsilon^{-1/32}$ . The third estimate in (PM.396), together with (PM.405), absorbs every such fixed power into a harmless decrease of the stretched-exponential constant and gives

$$W^{-1}(\Phi_{\Pi}^1 f_{\tilde{R}})W = \Phi_{H^2}^1 f_{\tilde{R}_{\text{HJ}}}, \quad \|\tilde{R}_{\text{HJ}}\| \leq Ce^{-cs^{-1/4}}\varepsilon. \quad (\text{PM.407})$$

This is the promised propagation of the exponentially small vector-field residual through the HJ straightening.

Undo the quotient. If

$$\ell_\lambda(r) := \Omega_i(c) + \Omega'_i(c)(r - c),$$

then (PM.400) and (PM.407) give exactly the action function (PM.392). The scaled type-II residual loses the factor  $q^2$  on returning to physical coordinates, so (PM.407) becomes (PM.394). Likewise (PM.406) becomes  $q^{-1}R_{\text{HJ}}$ , proving the first inequality in (PM.393). Taking  $M > 40$  in (PM.396) gives  $q^{-1}\varepsilon^{1/32} = o(\hat{\rho})$ , proving the second. The accumulated resonant coordinate has  $C^1$  size  $o(\hat{\rho}^{10})$  by (PM.356) and (PM.396), while the rescaled HJ coordinate contributes  $O(\varepsilon^{1/4})$ ; this proves (PM.395).

It remains only to make the covering/lift convention explicit. Choose all type-II maps on the universal cover with the lift continuously connected to the identity. The angle displacement is  $o(1/q)$  by (PM.396), so two possible lifts cannot differ by the nonzero deck shift  $2\pi m/q$ ; hence  $m = 0$ . The  $C^0$  displacement is  $o(\hat{\rho})$ , and (PM.393) puts the deleted HJ core inside  $\hat{D}_\lambda/10$ . Therefore both the coordinate and its inverse remain in  $6\hat{D}_\lambda \Subset V_i$ . Lemma 4.33(ii) says that the buffered annulus outside  $\hat{D}_\lambda/10$  survives in  $U_J$ , giving the last containment assertion.  $\square$

**Lemma 4.52 (polynomial-input KAM startup and extraction of the finite adapted window).** Let  $\alpha_0 \in \mathbb{R} \setminus \mathbb{Q}$  have finite irrationality exponent  $\tau_0$  in the sense of the definition of invariant-circle deficit, twist coefficient, and continued-fraction scales above, and fix  $\tau_+ > \tau_0$ . Fix also  $0 < \beta < 1/(4(1 + \tau_+))$  and put

$$\mu := \left(1 + \frac{1}{\tau_+}\right)(1 - \beta) \in (1, 2). \quad (\text{PM.408})$$

For every prescribed finite reserve exponent  $M_* > 0$  there is a finite normalization order  $m_* = m_*(\tau_+, \beta, M_*)$  with the following property. Suppose the one-dimensional elliptic germ has nonzero twist and, after the Normal form near a periodic orbit, is transferred by Lemma 4.42 to the two exact action-angle branch charts over the punctured action disk  $0 < |r| < 3\rho$ . Assume the branchwise bounds below are uniform on nested punctured sectors and that the integrable action function extends holomorphically through  $r = 0$ :

$$M_0 = \Phi_{\Omega_0} \circ f_{F_0}, \quad \omega_0 = (2\pi)^{-1}\Omega'_0, \quad \omega_0(0) = \alpha_0, \quad (\text{PM.409})$$

with

$$0 < a \leq \omega'_0 \leq A, \quad \|\Omega_0\|_{C^3} \leq L, \quad \|F_0\| \leq C\rho^m, \quad m \geq m_*. \quad (\text{PM.410})$$

Then, for all sufficiently small  $\rho$ , one may start the Lemma 4.20–Lemma 4.21 KAM scheme at a logarithmic Fourier cutoff and reach indices  $I < J$  such that

$$N_I \asymp \rho^{-1/(1+\tau_+)}, \quad N_J \asymp N_I^\mu \asymp \rho^{-(1-\beta)/\tau_+}, \quad (\text{PM.411})$$

with the following conclusions.

- (a) Up to stage  $I$  the fixed central disk  $D(0, 2\rho)$  may be taken hole-free. All Lemma 4.20 steps are legal there, the cumulative  $C^1$  coordinate motion and the  $C^2$  frequency drift are  $o(\rho^{M_*})$  after  $m_*$  is enlarged, and the stage- $I$  remainder satisfies the inherited Lemma 4.21 envelope.
- (b) On the extracted window  $I \leq i \leq J$ , with the *absolute* Lemma 4.21 indices,

$$S_i := \delta_i^{-3} K_i^3 E_i, \quad \eta_i := \delta_i K_i^{-1}, \quad \hat{r}_i := e^{-(\log N_i)^2}, \quad (\text{PM.412})$$

one has for every fixed  $0 \leq M \leq M_*$

$$N_J^M \sum_{j \geq i} S_j = o(K_i^{-1}), \quad \sum_{j \geq i} \eta_j = O(K_i^{-1}), \quad (\text{PM.413})$$

$$K_i^{-1} + \sum_{j \geq i} (S_j + \eta_j) \ll \hat{r}_i \ll N_J^{-2}. \quad (\text{PM.414})$$

The laminar evolving resonance-family construction of Lemma 4.29–Lemma 4.30 and Lemma 4.32, with these absolute indices, therefore gives a legal finite KAM schedule, buffered birth disks of radius  $\hat{r}_i$  as in Lemma 4.33, and the no-screening geometry of Lemma 4.34 on the target disk.

(c) The final remainder obeys

$$\|F_J\| \leq E_J \leq \exp \left[ -c \frac{\rho^{-(1-\beta)/\tau_+}}{|\log \rho|^2} \right]. \quad (\text{PM.415})$$

Consequently, for every  $\beta_* > \beta/\tau_+$ ,

$$\|F_J\| \leq \exp \left[ -\rho^{-(1/\tau_+ - \beta_*)} \right] \quad (\text{PM.416})$$

for all sufficiently small  $\rho$ .

Thus the finite-arithmetic adapted window used later is an actual descendant of a polynomially small fixed-order Birkhoff remainder; it is not postulated to begin with an exponentially small remainder at  $N \asymp \rho^{-1/(1+\tau_+)}$ .

*Proof.* We first record the arithmetic input. Since  $\tau_+ > \tau_0$ , the definition of the finite continued-fraction exponent implies, after absorbing finitely many denominators into a constant, that

$$|k\alpha_0 - \ell| \geq \kappa_+ |k|^{-\tau_+} \quad (k \neq 0, \ell \in \mathbb{Z}) \quad (\text{PM.417})$$

for some  $\kappa_+ > 0$ . Indeed, if  $q_n \leq |k| < q_{n+1}$ , the best-approximation property gives  $\|k\alpha_0\|_{\mathbb{R}/\mathbb{Z}} \geq \|q_n\alpha_0\|_{\mathbb{R}/\mathbb{Z}} \geq (2q_{n+1})^{-1}$  for large  $n$ , while  $q_{n+1} \leq q_n^{\tau_+}$  eventually. Since  $q_n \leq |k|$ , (PM.417) follows after shrinking  $\kappa_+$ .

Choose a large fixed number  $A_0$ , to be specified below, and start Lemma 4.21 with

$$N_b := \left\lfloor \frac{16A_0}{b} |\log \rho| \right\rfloor, \quad N_j = N_b (4/3)^j. \quad (\text{PM.418})$$

Then

$$E_0 = \exp(-bN_b/16) = \rho^{A_0+o(1)}, \quad K_0^{-1} = \rho^{\kappa A_0+o(1)}. \quad (\text{PM.419})$$

The finite-order theorem the Normal form near a periodic orbit is available at every prescribed fixed order. Taylor's theorem on a fixed analytic CC neighborhood gives a remainder  $O(\rho^m)$ . Lemma 4.42 transfers this estimate to each punctured action-angle branch with constants uniform on nested sectors; the action-only normal-form coefficients extend holomorphically through  $r = 0$ . Thus (PM.410) holds branchwise with  $m$  as large as desired. Take  $m > A_0 + 2$ . For small  $\rho$  the constant in (PM.410) is then absorbed and  $\|F_0\| \leq E_0$ . Next choose  $A_0$  and finally  $m_*$  so large that every



fixed power of  $N_J$  of degree at most  $M_* + 20$ , multiplied by the startup sums in Lemma 4.21, is  $o(\rho^{M_*+10})$ . This is possible because  $N_J$  will be only a fixed negative power of  $\rho$ , whereas (PM.419) permits an arbitrarily large fixed power of  $\rho$  at startup.

Choose a sufficiently small constant  $c_* = c_*(a, A, \kappa_+, \tau_+) > 0$  and let  $I$  be the first index for which

$$N_I \geq c_* \rho^{-1/(1+\tau_+)}. \quad (\text{PM.420})$$

The geometric overshoot is at most  $4/3$ . Hence (PM.417), after division by  $|k|$ , shows that every reduced rational  $\ell/k$  with  $1 \leq k < N_I$  is separated from  $\alpha_0$  in frequency by at least

$$\kappa_+ N_I^{-(1+\tau_+)} \geq 20A\rho \quad (\text{PM.421})$$

after  $c_*$  is chosen small enough. The variation of  $\omega_0$  on  $D(0, 3\rho)$  is at most  $3A\rho$ . Lemma 4.21 together with (PM.419) and the choice of  $A_0$  makes the cumulative frequency drift, action collars, and  $C^1$  coordinate motion before  $I$  all  $o(\rho^{M_*+10})$ , in particular  $o(\rho)$ . Therefore no rational center with denominator below the current cutoff can enter  $D(0, 2\rho)$  before stage  $I$ . The complex divisor estimate is the same vertical-Taylor argument as Lemma 4.26, now with no deletion disks because of the strict real margin (PM.421). Lemma 4.27 is thus legal on each punctured branch core from the logarithmic startup through stage  $I$ . All estimates are uniform as the auxiliary inner radius tends to zero, while the action-frequency quantities are holomorphic at  $r = 0$ ; hence the two branch conclusions patch across the puncture for the scalar bounds used below. This proves (a), including the inherited envelope  $\|F_I\| \leq E_I$ .

Let  $J$  be the first index after  $I$  with  $N_J \geq N_I^\mu$ . Minimality and the  $4/3$  growth give (PM.411). Since  $N_b \asymp |\log \rho|$  while  $N_I$  is a fixed negative power of  $\rho$ , the index  $I$  is  $O(|\log \rho|)$ ; the same is true of  $J$ . Thus, throughout  $I \leq i \leq J$ ,

$$E_i = \exp \left[ -\frac{bN_i}{4(i+2)^2} \right] \leq \exp \left[ -c \frac{N_i}{|\log \rho|^2} \right]. \quad (\text{PM.422})$$

Exactly as in Lemma 4.31,  $S_i = \delta_i^{-3} E_i^{1-3\kappa}$  and  $K_i^{-1} = E_i^\kappa$ . Since  $1 - 3\kappa > \kappa$  and  $N_J \leq CN_i^2$  on this finite window, the strict exponential gap  $E_i^{1-4\kappa}$  absorbs every fixed factor  $N_J^M$  with  $M \leq M_*$ . Geometric growth of  $N_i$  makes each tail comparable to its first term. This proves (PM.413). Likewise  $E_i^\kappa \ll e^{-(\log N_i)^2}$ , whereas  $e^{-(\log N_i)^2} \ll N_J^{-2}$  because  $N_J$  is at most a fixed power of  $N_i$ . This proves (PM.414).

We can now repeat the already certified geometric argument without resetting the iteration index. Lemma 4.29–Lemma 4.30 continue each real twist map across older gaps; Lemma 4.26 creates the current complex resonance disks; the tail estimate (PM.413) controls later center drift and old-divisor erosion; the exact Farey separation at cutoff  $N_J$  and (PM.414) make distinct labels disjoint; and Lemma 4.28 pays the collars. Thus the proof of Lemma 4.32 applies with  $I$  as the first adapted index and with the absolute sequences  $(N_i, \delta_i, E_i, K_i)$ . The proof of Lemma 4.33 uses only the same three inequalities (PM.413)–(PM.414), so its buffered disks and birth-stage holomorphic neighborhoods follow unchanged. Finally Lemma 4.34 depends only on the stagewise  $O(tN_i^2 + 1)$  count, the buffer separation, and the two logarithmic sums; the factors  $(i+2)^q = O(|\log \rho|^q)$  introduced by the absolute indexing are absorbed by the strict power margins in (PM.411). Hence the no-screening conclusion holds as

claimed in (b).

By Lemma 4.27,  $\|F_J\| \leq E_J$ . Equations (PM.411)–(PM.422) give (PM.415). If  $\beta_* > \beta/\tau_+$ , then

$$\frac{1-\beta}{\tau_+} > \frac{1}{\tau_+} - \beta_*.$$

A positive power of  $\rho^{-1}$  absorbs  $|\log \rho|^2$ ; consequently the exponent in (PM.415) eventually dominates  $\rho^{-(1/\tau_+ - \beta_*)}$ , proving (PM.416). This completes the construction.  $\square$

**Lemma 4.53 (critical-graph reduction to the weighted-gauge Hamilton-Jacobi form).** Fix  $A, B \geq 1$  and  $h > 0$ . There are constants  $c, C > 0$ , depending only on  $A, B, h$ , with the following property. Let  $0 < \rho < c$ , let  $0 < \eta < c$ , and let

$$\tilde{\Pi}(\theta, r) = \varpi(\theta)r^2 + f_0(\theta) + f_1(\theta)r + f_2(\theta)r^2 + r^3b(\theta, r) \quad (\text{PM.423})$$

be real-symmetric and holomorphic on  $\mathbb{T}_h \times D(0, 2\rho)$ . Assume

$$A^{-1} \leq \varpi(\theta) \leq A \quad (\theta \in \mathbb{T}), \quad \|\varpi\|_h + \|\varpi^{-1}\|_h + \|b\|_{h, 2\rho} \leq B, \quad (\text{PM.424})$$

and

$$\|f_0\|_h \leq \eta\rho^2, \quad \|f_1\|_h \leq \eta\rho, \quad \|f_2\|_h \leq \eta. \quad (\text{PM.425})$$

After decreasing  $c$ , there are unique holomorphic  $e_0$  in the ball  $\|e_0\|_h \leq C\eta\rho$ , a real constant  $C_*$ , and real-symmetric holomorphic functions  $a, e$  on  $\mathbb{T}_h$  and  $f$  on  $\mathbb{T}_h \times D(0, \rho/2)$  such that the vertical symplectic shift

$$G(\theta, r) := (\theta, r - e_0(\theta)) \quad (\text{PM.426})$$

satisfies

$$(\tilde{\Pi} + C_*) \circ G^{-1}(\theta, r) = a(\theta)(r^2 - e(\theta) + r^3f(\theta, r)), \quad (\text{PM.427})$$

$$\langle a^{1/2}e \rangle = 0. \quad (\text{PM.428})$$

Here the positive real square root of  $a$  extends to a nowhere-zero holomorphic branch on  $\mathbb{T}_h$ , and

$$\|e_0\|_h \leq C\eta\rho, \quad \|a - \varpi\|_h \leq C\eta, \quad |C_*| + \|e\|_h \leq C\eta\rho^2, \quad \|f\|_{h, \rho/2} \leq C. \quad (\text{PM.429})$$

The additive constant  $C_*$  does not alter the autonomous Hamiltonian flow. The map  $G$  is symplectic, but it is exact symplectic if and only if  $\langle e_0 \rangle = 0$ ; no exactness is asserted in general.

*Proof.* Put  $A_0 := \varpi + f_2$ . On  $D(0, \rho)$ , Cauchy's estimate applied to  $b$  on  $D(0, 2\rho)$  gives

$$\partial_r \tilde{\Pi}(\theta, r) = 2A_0(\theta)r + f_1(\theta) + r^2c_1(\theta, r), \quad \|c_1\|_{h, \rho} \leq C_B. \quad (\text{PM.430})$$

For  $c$  small,  $A_0$  is nowhere zero and  $|A_0^{-1}| \leq 2A$  on the strip. The map

$$\mathcal{T}_\theta(u) := -\frac{f_1(\theta) + u^2c_1(\theta, u)}{2A_0(\theta)}$$

maps the ball  $|u| \leq C\eta\rho$  into itself and has derivative at most  $C\eta\rho < 1/2$ . Uniform con-

traction gives a unique holomorphic solution  $e_0(\theta) = \mathcal{T}_\theta(e_0(\theta))$  and hence  $\partial_r \tilde{\Pi}(\theta, e_0(\theta)) = 0$ . Real symmetry makes  $e_0$  real on the real circle.

Set

$$V(\theta) := \tilde{\Pi}(\theta, e_0(\theta)), \quad a(\theta) := \frac{1}{2} \partial_r^2 \tilde{\Pi}(\theta, e_0(\theta)).$$

Taylor's formula with integral remainder on  $|r| < \rho/2$  gives

$$\tilde{\Pi}(\theta, e_0 + r) = V(\theta) + a(\theta)r^2 + r^3 b_1(\theta, r), \quad \|b_1\|_{h, \rho/2} \leq C_B.$$

Equations (PM.424)–(PM.425) and the bound for  $e_0$  give  $\|a - \varpi\|_h \leq C\eta$  and  $\|V\|_h \leq C\eta\rho^2$ . Thus  $a$  remains positive on the real circle. Since  $\varpi$  is nowhere zero on  $\mathbb{T}_h$  and is positive on the real angle circle, its winding number around the fundamental angle cycle is zero; hence  $\varpi$  has a holomorphic logarithm and the positive real square root  $\varpi^{1/2}$  extends holomorphically to the strip. Moreover  $\|(a - \varpi)\varpi^{-1}\|_h < 1/2$  after decreasing  $c$ . Therefore  $a^{1/2} := \varpi^{1/2}(1 + (a - \varpi)/\varpi)^{1/2}$ , with the principal second factor, is holomorphic and nowhere zero.

Adding a constant does not change  $e_0$  or  $a$ . For a real constant  $C_*$  put

$$e_{C_*}(\theta) := -\frac{V(\theta) + C_*}{a(\theta)}, \quad f(\theta, r) := \frac{b_1(\theta, r)}{a(\theta)}.$$

Then (PM.427) holds with  $e = e_{C_*}$ . Moreover

$$\langle a^{1/2} e_{C_*} \rangle = -\langle a^{-1/2} V \rangle - C_* \langle a^{-1/2} \rangle.$$

The last coefficient is positive on the real circle. Hence there is a unique real choice

$$C_* = -\frac{\langle a^{-1/2} V \rangle}{\langle a^{-1/2} \rangle}$$

for which (PM.428) holds. The displayed bounds imply (PM.429). Finally

$$G^*(r d\theta) - r d\theta = -e_0(\theta) d\theta.$$

This one-form is exact on the annulus exactly when its period is zero, i.e. exactly when  $\langle e_0 \rangle = 0$ . This proves the final assertion and the lemma.  $\square$

**Lemma 4.54 (symplectic normal-form comparison with the flux action shift retained).** Under the hypotheses (PM.290)–(PM.293) of Lemma 4.43, replace the requirement that  $g_1, g_2$  be exact symplectic by the weaker requirement that they be symplectic. Then there is a real constant  $\gamma$  with

$$|\gamma| \leq C\varepsilon \tag{PM.431}$$

such that, on the annulus shrunk by  $7\delta$ ,

$$\|\Omega'_2(\cdot + \gamma) - \Omega'_1\| \leq C\delta^{-20}\nu. \tag{PM.432}$$

The same statement holds on either Lemma 4.42 action-angle branch, and the con-

stants are uniform on nested punctured central sectors. If  $g_2^{-1}g_1$  is exact, Lemma 4.43 improves  $\gamma$  to  $O(\delta^{-20}\nu)$  and removes it from the frequency comparison after one further fixed collar loss.

*Proof.* The proof of Lemma 4.43 through (PM.303) uses only symplecticity of  $g_1, g_2$ , not exactness. The auxiliary error maps remain exact: zero flux is preserved under conjugation by a symplectic annulus map, and composition preserves zero flux. Hence the type-II remainder reduction used in (PM.295)–(PM.296) is still legal. After the resonance-exclusion, maximum-principle, and symplectic-Jacobian steps, the argument produces a real constant  $\gamma$  satisfying (PM.431) and

$$\|v - \gamma\| \leq C\delta^{-18}\nu$$

for  $g = g_2^{-1}g_1 = (\theta + u, r + v)$ . Averaging the first line of (PM.298) kills the  $u$ -coboundary. The bounded twist and the preceding estimate give

$$\|\Omega'_2(r + \gamma) - \Omega'_1(r)\| \leq C\delta^{-19}\nu.$$

Because  $|\gamma| \leq C\varepsilon < \delta$ , one additional action collar makes every shifted argument legal and yields the weaker displayed exponent in (PM.432). The branchwise statement follows exactly as in the final paragraph of Lemma 4.43; no use of (PM.304) is made. If  $g$  is exact, (PM.304) supplies  $|\gamma| \leq C\delta^{-20}\nu$ , which is precisely the exact refinement already certified in Lemma 4.43.  $\square$

**Lemma 4.55 (shifted affine disk filling and conversion of frequency discrepancy).** Let  $q \geq 1$ ,  $c, \gamma \in \mathbb{R}$ ,  $0 < r_0 < R$ , and suppose

$$\Omega_{\text{HJ}}(r) = \ell(r) + q^{-2}H^2(q(r - c)) \quad (\text{PM.433})$$

on  $r_0/q \leq |r - c| \leq R/q$ . Let  $\Omega_{\text{disk}}$  be holomorphic on a neighborhood of  $\overline{D(c + \gamma, R/q)}$ . Since an action Hamiltonian is determined by the associated integrable map only up to an additive constant, choose a real constant  $C_\gamma$  and define

$$\Xi_\gamma(z) := q^2(\Omega_{\text{disk}}(c + \gamma + z/q) + C_\gamma - \ell(c + z/q)). \quad (\text{PM.434})$$

Then  $\Xi_\gamma$  is holomorphic on a neighborhood of  $\overline{D(0, R)}$  and

$$\Xi_\gamma(z) - H^2(z) = q^2(\Omega_{\text{disk}}(r + \gamma) + C_\gamma - \Omega_{\text{HJ}}(r)), \quad r = c + z/q. \quad (\text{PM.435})$$

If, on a neighborhood of the outer circle  $|r - c| = R/q$ ,

$$\|\Omega'_{\text{disk}}(\cdot + \gamma) - \Omega'_{\text{HJ}}\| \leq d, \quad (\text{PM.436})$$

then  $C_\gamma$  can be chosen so that the two action functions agree at one real point of that circle and

$$\|\Xi_\gamma - H^2\|_{|z|=R} \leq 2\pi q R d. \quad (\text{PM.437})$$

Thus a flux shift from Lemma 4.54 is harmless for the Lemma 4.12 holomorphic filling, provided the shifted physical disk lies in the domain of the disk normal form.

*Proof.* The affine map  $z \mapsto c + \gamma + z/q$  identifies  $D(0, R)$  with  $D(c + \gamma, R/q)$ , proving

holomorphy. Substitution of (PM.433) gives (PM.435). Choose the additive constant so equality holds at a real basepoint  $r_*$  on the outer circle. For any other  $r$  on that circle, integrate the derivative difference in (PM.436) along the shorter circular arc from  $r_*$  to  $r$ ; its length is at most  $2\pi R/q$ . Hence

$$|\Omega_{\text{disk}}(r + \gamma) + C_\gamma - \Omega_{\text{HJ}}(r)| \leq 2\pi(R/q)d.$$

Multiplication by  $q^2$  yields (PM.437).  $\square$

**Lemma 4.56 (Liouville large-jump adapted domain with one surviving resonance).** Fix  $A \geq 2$ ,  $0 < \beta < 1/10$ , and a finite reserve exponent  $M_* > 0$ . Choose once and for all an auxiliary exponent  $P_A = P(A) > 1000A$ , enlarged if necessary to pay the fixed one-hole screening constants below. Let  $\alpha_0 \in \mathbb{R} \setminus \mathbb{Q}$  have convergents  $p_n/q_n$ . There is a finite normalization order  $m_* = m_*(A, \beta, M_*, P_A)$  such that the following holds for all sufficiently large indices  $n$  satisfying

$$q_{n+1} \geq q_n^{10}. \quad (\text{PM.438})$$

Set

$$\rho_n := \frac{10A}{q_n q_{n+1}}. \quad (\text{PM.439})$$

Suppose the Normal form near a periodic orbit is taken to order  $m \geq m_*$  on the CC disk of action radius  $10\rho_n$  and transferred to the two punctured exact AA branches by Lemma 4.42. Choose the logarithmic Lemma 4.21 startup  $N_0 = \lfloor (16A_0/b) |\log \rho_n| \rfloor$  with  $A_0 = A_0(A, P_A)$  so large that the initial envelope satisfies  $E_0^\kappa \leq \rho_n^{P_A+2}$  and all startup coordinate/frequency tails are  $o(\rho_n^{P_A+1})$ ; then choose  $m_*$  large enough that the fixed-order Birkhoff remainder is below  $E_0$ .

Let  $i_n^-$  be the first absolute KAM index  $i \geq 0$  for which  $N_i > q_n$ ; thus

$$i_n^- = 0 \quad \text{if } N_0 > q_n, \quad i_n^- > 0 \implies N_{i_n^- - 1} \leq q_n < N_{i_n^-}. \quad (\text{PM.440})$$

and let  $i_n^+$  be the first index with

$$N_{i_n^+} \geq \frac{3q_{n+1}}{4(10A)^2}. \quad (\text{PM.441})$$

Then  $N_{i_n^+} < q_{n+1}/(10A)^2$ , and the KAM construction can be run through  $i_n^+$  with the following properties.

- (a) No resonance disk born before  $i_n^-$  meets  $D(0, \rho_n)$ . Between  $i_n^-$  and  $i_n^+$  the only reduced rational label whose continued center can meet  $D(0, \rho_n)$  is  $p_n/q_n$ . If  $c_n$  denotes its birth-stage center, then

$$\frac{\rho_n}{40A^2} \leq |c_n| \leq \frac{\rho_n}{5}. \quad (\text{PM.442})$$

The same-label deletion disks are nested. Thus for a final disk  $D_n = D(c_n, \varepsilon_n)$ ,

$$U^{(n)} := U_{i_n^+} \cap D(0, \rho_n) = D(0, \rho_n) \setminus \overline{D_n}, \quad \varepsilon_n \leq \rho_n^{P_A}. \quad (\text{PM.443})$$

(b) With

$$\widehat{D}_n := D\left(c_n, \frac{|c_n|}{24}\right), \quad (\text{PM.444})$$

one has

$$6\widehat{D}_n \Subset V_{i_n^-}, \quad D(0, q_{n+1}^{-6}) \Subset U^{(n)}, \quad (\text{PM.445})$$

where  $V_{i_n^-}$  is the pre-deletion birth-stage action domain. Hence the HJ construction has a physical birth disk comparable to  $|c_n|$ , while a polynomially smaller central disk survives the complete large-jump schedule.

(c) Put

$$\sigma_n := \frac{1}{2}q_{n+1}^{-6}, \quad d_n := \frac{|c_n|}{240}. \quad (\text{PM.446})$$

For every  $z \in \widehat{D}_n \setminus \frac{1}{10}\widehat{D}_n$ , the Lemma 4.9 barrier coefficient for the one-hole domain  $U^{(n)}$ , central disk  $D(0, \sigma_n)$ , and enlarged hole radius  $d_n$  satisfies

$$\frac{\log(|z|/\rho_n)}{\log(\sigma_n/\rho_n)} - \frac{\log(d_n/(2\rho_n))}{\log(\varepsilon_n/\rho_n)} \geq \frac{c_A}{|\log \rho_n|}. \quad (\text{PM.447})$$

(d) The final KAM remainder obeys

$$\|F_{i_n^+}\| \leq \exp\left[-c \frac{q_{n+1}}{(\log q_{n+1})^2}\right] \leq \exp(-q_{n+1}^{1-\beta}) \quad (\text{PM.448})$$

for all sufficiently large selected  $n$ .

Thus the Liouville large-jump upper geometry is an internally realized KAM descendant of polynomial Birkhoff input and is not obtained by formally setting  $\tau = \infty$  in the finite-arithmetic adapted window.

*Proof.* The continued-fraction estimates give

$$\frac{1}{2q_n q_{n+1}} \leq \left| \alpha_0 - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}}, \quad 0 < k < q_n \implies \left| \alpha_0 - \frac{\ell}{k} \right| > \frac{1}{2k q_n}. \quad (\text{PM.449})$$

Use the same admissible Lemma 4.21 constants as in Lemma 4.52, in particular  $1-3\kappa > \kappa$ . With the displayed choice of  $N_0$  one has  $E_0 = \rho_n^{A_0+o(1)}$  and  $K_0^{-1} = E_0^\kappa \leq \rho_n^{P_A+2}$  for large  $n$ . Because the Lemma 4.21 tails carry a strictly larger power of  $E_0$  than  $K_0^{-1}$ ,  $A_0$  may be enlarged so that the total coordinate and  $C^2$ -frequency drift is  $o(\rho_n^{P_A+1})$ . the Normal form near a periodic orbit is available at every prescribed fixed order, so after choosing  $A_0$  we choose  $m_*$  with  $m_* > A_0 + 10$ ; its  $O(\rho_n^{m_*})$  CC remainder, transferred branchwise by Lemma 4.42, is then at most  $E_0$ . Thus Lemma 4.20–Lemma 4.21 starts legally. Since the envelopes decrease,  $K_j^{-1} \leq K_0^{-1} \leq \rho_n^{P_A+2}$  for every later divisor disk, and every coordinate/frequency tail used below is  $o(\rho_n^{P_A+1})$ . The branchwise AA estimates are uniform as the inner auxiliary radius tends to zero, exactly as in the revised Lemma 4.52 startup.

If  $j < i_n^-$ , then every reduced resonance label has denominator  $k < q_n$ . Equation (PM.449), the twist bound, and the  $o(\rho_n)$  frequency drift put every continued center outside  $4\rho_n$ . Hence the central disk remains hole-free until  $i_n^-$ . Now let  $i_n^- \leq j \leq i_n^+$  and

let  $\ell/k$  be a reduced label with  $k < N_j$  and  $\ell/k \neq p_n/q_n$ . By (PM.441),  $k < q_{n+1}/(10A)^2$ , so

$$\left| \alpha_0 - \frac{\ell}{k} \right| \geq \left| \frac{\ell}{k} - \frac{p_n}{q_n} \right| - \left| \alpha_0 - \frac{p_n}{q_n} \right| \geq \frac{1}{kq_n} - \frac{1}{q_n q_{n+1}} \geq \frac{99A^2}{q_n q_{n+1}} \geq 9A\rho_n. \quad (\text{PM.450})$$

The continued twist maps of Lemma 4.29–Lemma 4.30 differ from the initial one by  $o(\rho_n)$ , so their  $\ell/k$  centers stay outside  $4\rho_n$ . For the special label  $p_n/q_n$ , (PM.449), the two-sided twist bound, and the same negligible drift give (PM.442) after harmlessly weakening constants. This proves the uniqueness of the label that enters the target disk.

On the surviving target domain, Lemma 4.26 supplies the complex divisor margin for every nondeleted label, and Lemma 4.28 pays the collar losses required to continue Lemma 4.20–Lemma 4.21 through  $i_n^+$ . For the special label, the Lemma 4.21 tail bound gives later frequency drift  $o(K_{i_n^-}^{-1})$ ; the uniform lower twist bound converts this into center drift  $o(K_{i_n^-}^{-1})$ , while the later divisor radii satisfy  $K_j^{-1} = o(K_{i_n^-}^{-1})$ . We therefore choose, from its first birth onward, the deleted disk for  $p_n/q_n$  to be the cumulative envelope containing its current Lemma 4.26 exclusion disk and all preceding same-label disks. The preceding estimates make these cumulative disks nested, and their final radius is bounded by a fixed multiple of  $K_{i_n^-}^{-1}$ . Since  $K_{i_n^-}^{-1} \leq K_0^{-1} \leq \rho_n^{P_A+2}$ , the fixed enlargement is at most  $\rho_n^{P_A}$  for large  $n$ . Hence (PM.443) follows. There are no other labels in  $D(0, \rho_n)$ , so the displayed equality for  $U^{(n)}$  follows.

For the birth-stage room, (PM.442) gives  $|c_n| + 6|c_n|/24 \leq (5/4)(\rho_n/5) < \rho_n$ , while every older distinct-label hole is outside  $4\rho_n$  and the current hole has not yet been deleted on  $V_{i_n^-}$ . Hence  $6\widehat{D}_n \subseteq V_{i_n^-}$ . On the other hand,

$$|c_n| - \varepsilon_n \geq \frac{\rho_n}{40A^2} - \rho_n^{P_A}.$$

Using (PM.439) and (PM.438), the right side is much larger than  $q_{n+1}^{-6}$  for large  $n$ . Thus the sole final hole misses  $D(0, q_{n+1}^{-6})$ , proving (PM.445).

It remains to verify the no-screening coefficient directly. If  $z \in \widehat{D}_n \setminus (1/10)\widehat{D}_n$ , then (PM.442)–(PM.444) imply

$$c'_A \leq \frac{|z|}{\rho_n} \leq \frac{1}{4}$$

for a constant  $c'_A > 0$ . Therefore the numerator of the radial term in (PM.447) is negative with absolute value bounded below by a positive  $A$ -dependent constant. Moreover

$$|\log(\sigma_n/\rho_n)| \leq C_A |\log \rho_n|,$$

because  $\rho_n = 10A/(q_n q_{n+1})$  and  $\sigma_n = q_{n+1}^{-6}/2$ . Hence the radial term is at least  $c''_A/|\log \rho_n|$ . For the single screening correction,  $d_n/\rho_n$  stays between two positive  $A$ -dependent constants, whereas (PM.443) gives

$$|\log(\varepsilon_n/\rho_n)| \geq (P_A - 1)|\log \rho_n|.$$

Thus the absolute value of the second term in (PM.447) is at most  $C_A/((P_A - 1)|\log \rho_n|)$ . After the fixed radial constants  $c'_A, C_A$  are determined, our initial choice of  $P_A$  is en-

larged so that  $C_A/(P_A - 1) \leq c_A''/2$ . Thus the screening correction is at most half the radial gain. This proves (PM.447), and Lemma 4.9 is applicable because (PM.445) supplies its central disk and the one-hole domain is connected.

Finally  $N_{i_n^+} \asymp_A q_{n+1}$  and the absolute Lemma 4.21 index satisfies  $i_n^+ = O(\log q_{n+1})$ . Hence its envelope gives the first estimate in (PM.448). For every fixed  $\beta > 0$ ,  $q/(\log q)^2 \gg q^{1-\beta}$ , proving the second. This completes the large-jump construction.  $\square$

**Lemma 4.57 (finite-arithmetic deep-core BNF comparison under a convergent formal BNF).** Assume the finite-arithmetic setting of Lemma 4.52, and suppose that the formal Birkhoff normal form converges to a holomorphic action function  $\Xi$  on a fixed disk. Fix

$$0 < a < \frac{1}{\tau_+}. \quad (\text{PM.451})$$

After enlarging the fixed normalization reserve in Lemma 4.52, there are  $B > 1$  and, for every sufficiently small  $\rho$ , a continued-fraction denominator  $Q = Q(\rho)$  such that

$$\rho^{-B} \leq Q \leq C\rho^{-B\tau_+}, \quad D(0, Q^{-6}) \Subset U_J, \quad (\text{PM.452})$$

and

$$\|\Omega'_J - \Xi'\|_{D(0, Q^{-6/4})} \leq e^{-\rho^{-a}}. \quad (\text{PM.453})$$

Thus the finite-arithmetic branch obtains its quantitative central BNF comparison without extending Lemma 4.35–Lemma 4.41 beyond continued-fraction denominators.

*Proof.* Choose  $0 < \beta_B < 1/20$  and then  $B$  so that

$$6B > \mu, \quad B(1 - 3\beta_B) > a, \quad (\text{PM.454})$$

where  $\mu$  is from (PM.408). Since  $\tau_+ > \tau_0$ , eventually  $q_{j+1} \leq q_j^{\tau_+}$ . Let  $Q = q_j$  be the first convergent denominator with  $q_j \geq \rho^{-B}$ . Its predecessor is  $< \rho^{-B}$ , hence  $Q \leq C\rho^{-B\tau_+}$ . Since Lemma 4.52 leaves a central disk of radius  $\sigma_\rho \asymp \rho^\mu$ , (PM.454) gives  $Q^{-6} \ll \sigma_\rho$ , proving (PM.452).

Take the preliminary finite exact Birkhoff normalization from the Normal form near a periodic orbit to normalization order at least  $2m - 1$ , with  $m$  chosen below, so that the remaining CC generator starts in total degree at least  $2m$ ; use the resulting map  $M_*$  as the common input for both constructions. Applying Lemma 4.35–Lemma 4.41 at the denominator  $Q$  gives

$$g_B^{-1} M_* g_B = \Phi_{\Omega_B} \circ f_{F_B}, \quad (\text{PM.455})$$

with

$$\|F_B\| \leq e^{-Q^{1-2\beta_B}}, \quad J^{Q^{1-2\beta_B}} \Omega_B = J^{Q^{1-2\beta_B}} \Xi, \quad \|g_B - \text{id}\|_{C^1} \leq CQ^{-(m-26)}. \quad (\text{PM.456})$$

The jet identity uses Lemma 4.38 and the fact that  $\Xi$  is the convergent realization of the same formal BNF.

Let  $g_J$  be the cumulative exact KAM normalizer from the same common map  $M_*$ . The startup reserve in Lemma 4.52 is arbitrary, while the adapted-window coordinate tail is superpolynomial by (PM.413). Thus, for every prescribed fixed  $M$ ,

$$\|g_J - \text{id}\|_{C^1} \leq C\rho^M \quad (\text{PM.457})$$



after increasing the preliminary Birkhoff order. Choose  $M > 8B\tau_+ + 10$  and  $m > 40$ , and put  $\delta_Q = Q^{-7}$ . The upper bound in (PM.452) then yields

$$\|g_B - \text{id}\|_{C^1} + \|g_J - \text{id}\|_{C^1} = o(\delta_Q), \quad 6\delta_Q < Q^{-6}/4. \quad (\text{PM.458})$$

On either exact Lemma 4.42 branch, Lemma 4.43 compares the two exact normal forms of  $M_*$ ; holomorphy fills the central puncture. Hence

$$\|\Omega'_J - \Omega'_B\|_{D(0, Q^{-6/3})} \leq CQ^{140}(\|F_J\| + \|F_B\|). \quad (\text{PM.459})$$

The uniform twist/ $C^3$  bounds are inherited from the preliminary normal form and the summable KAM drift.

Lemma 4.15 applied to (PM.456), followed by one Cauchy derivative on a fixed smaller disk, gives

$$\|\Omega'_B - \Xi'\|_{D(0, Q^{-6/4})} \leq CQ^6 e^{-cQ^{1-2\beta_B}} \leq e^{-Q^{1-3\beta_B}}. \quad (\text{PM.460})$$

Choose the loss parameter in Lemma 4.52(c) so that  $a < 1/\tau_+ - \beta_*$ . Then  $\|F_J\| \leq e^{-\rho^{-(1/\tau_+ - \beta_*)}}$ . Since every  $Q$ -polynomial is a fixed power of  $\rho^{-1}$  by (PM.452), it is absorbed by this strict exponent margin. The second inequality in (PM.454) and  $Q \geq \rho^{-B}$  similarly absorb (PM.460). Combining with (PM.459) proves (PM.453).  $\square$

**Lemma 4.58 (Liouville large-jump central BNF comparison).** Assume the large-jump setting of Lemma 4.56 and that the formal BNF converges to a holomorphic action function  $\Xi$  on a fixed disk. Put  $Q := q_{n+1}$ . After enlarging the fixed preliminary normalization order and the reserve exponent  $P_A$ , the final KAM integrable action function satisfies, for every fixed  $0 < \beta_0 < 1/10$  and all sufficiently large selected  $n$ ,

$$\|\Omega'_{i_n^+} - \Xi'\|_{D(0, Q^{-6/4})} \leq \exp(-Q^{1-\beta_0}). \quad (\text{PM.461})$$

*Proof.* Use the same preliminary exact finite Birkhoff normalization as in Lemma 4.56, but take its order at least  $2m - 1$  with  $m > 50$ . Denote the resulting common map by  $M_*$ . Since  $Q = q_{n+1}$  is itself a convergent denominator, Lemma 4.35-Lemma 4.41 apply directly on the action disk  $D(0, Q^{-6})$  and give

$$g_B^{-1} M_* g_B = \Phi_{\Omega_B} \circ f_{F_B}, \quad (\text{PM.462})$$

where, after choosing a small auxiliary  $\beta_B < \beta_0/10$ ,

$$\|F_B\| \leq e^{-Q^{1-2\beta_B}}, \quad JQ^{1-2\beta_B} \Omega_B = JQ^{1-2\beta_B} \Xi, \quad (\text{PM.463})$$

$$\|g_B - \text{id}\|_{C^1} \leq CQ^{-(m-26)}. \quad (\text{PM.464})$$

Lemma 4.56(b) places this entire disk inside the final large-jump KAM domain.

Let  $g_+$  denote the cumulative exact KAM normalizer from  $M_*$  through  $i_n^+$ . The startup choice in Lemma 4.56 makes every coordinate tail  $o(\rho_n^{P_A+1})$ ; increasing  $P_A$  if necessary and using  $\rho_n = 10A/(q_n Q) \leq C_A Q^{-1}$  gives

$$\|g_+ - \text{id}\|_{C^1} = o(Q^{-20}). \quad (\text{PM.465})$$

Take  $\delta_Q = Q^{-7}$ . Equations (PM.464)–(PM.465) imply both normalizers are  $o(\delta_Q)$  and  $6\delta_Q < Q^{-6}/4$ . Their remainders satisfy Lemma 4.56(d) and (PM.463), so Lemma 4.43, applied on either exact Lemma 4.42 branch and then filled across the removable central puncture, yields

$$\|\Omega'_{i_n^+} - \Omega'_B\|_{D(0, Q^{-6/3})} \leq CQ^{140} \left( e^{-Q^{1-\beta_B}} + e^{-Q^{1-2\beta_B}} \right). \quad (\text{PM.466})$$

Here we weakened Lemma 4.56(d) harmlessly from its displayed exponent to the common  $1 - \beta_B$  scale by choosing the large-jump loss parameters consistently.

Finally Lemma 4.15 applied to the jet identity in (PM.463), followed by one Cauchy derivative, gives

$$\|\Omega'_B - \Xi'\|_{D(0, Q^{-6/4})} \leq CQ^6 e^{-cQ^{1-2\beta_B}}. \quad (\text{PM.467})$$

Every fixed power of  $Q$  is absorbed by the strict gap between  $1 - 2\beta_B$  and  $1 - \beta_0$ . Combining (PM.466)–(PM.467) proves (PM.461).  $\square$

**Lemma 4.59 (deep-core no-screening propagation to every Hamilton-Jacobi annulus).** Assume that the formal Birkhoff normal form in the one-dimensional upper branch converges to a holomorphic action function  $\Xi$  on a fixed disk. Then the central comparisons Lemma 4.57–Lemma 4.58 propagate to the complete buffered HJ annuli selected by Lemma 4.33 and Lemma 4.56 as follows.

(a) *Finite arithmetic exponent.* In the setting of Lemma 4.52 and Lemma 4.57, fix

$$0 < a_1 < a < \frac{1}{\tau_+}. \quad (\text{PM.468})$$

For all sufficiently small outer radii  $\rho$ , every selected final hole  $D_*$  meeting  $D(0, \rho)$ , with buffered birth disk  $\widehat{D}_*$ , satisfies

$$\|\Omega'_J - \Xi'\|_{\widehat{D}_* \setminus (\widehat{D}_*/10)} \leq \exp(-\rho^{-a_1}). \quad (\text{PM.469})$$

(b) *Liouville large jump.* In the setting of Lemma 4.56 and Lemma 4.58 put  $Q = q_{n+1}$ . For every

$$0 < \beta_0 < \beta_1 < 1/10 \quad (\text{PM.470})$$

and all sufficiently large selected jumps,

$$\|\Omega'_{i_n^+} - \Xi'\|_{\widehat{D}_n \setminus (\widehat{D}_n/10)} \leq \exp(-Q^{1-\beta_1}), \quad (\text{PM.471})$$

where  $\widehat{D}_n$  is the buffered HJ birth disk of Lemma 4.56.

*Proof.* We prove the finite case first. Take the deep convergent denominator  $Q = Q(\rho)$  of Lemma 4.57 and replace the central radius used in Lemma 4.34 by

$$\sigma_Q := Q^{-6}/5. \quad (\text{PM.472})$$

This circle lies inside the Lemma 4.57 comparison disk. The screening part of the barrier proof Lemma 4.34 is unchanged, because its denominators and enlarged-hole numerators depend only on the actual and buffered holes. Only the radial term changes.

By (PM.452),  $Q \leq C\rho^{-B\tau_+}$ , so

$$|\log(\sigma_Q/\rho')| \leq C|\log \rho|. \quad (\text{PM.473})$$

On every target annulus Lemma 4.33 still gives  $|z| \leq (5/6)\rho'$ . Therefore

$$\frac{\log(|z|/\rho')}{\log(\sigma_Q/\rho')} \geq \frac{c_0}{|\log \rho|}. \quad (\text{PM.474})$$

The screening sum is the same  $o(|\log \rho|^{-1})$  quantity as in (PM.241), hence Lemma 4.9 applies with a bracket at least  $c/|\log \rho|$ .

The derivatives of  $\Omega_J$  are uniformly bounded by the twist/ $C^3$  reserve, and  $\Xi'$  is bounded on the fixed convergent-BNF disk. Divide  $\Omega'_J - \Xi'$  by one fixed constant so that the resulting holomorphic function has sup norm at most 1 on the relevant connected holed domain. Lemma 4.57 gives central boundary size at most  $Ce^{-\rho^{-a}}$ . Thus Lemma 4.9 and (PM.474) imply

$$\log \|\Omega'_J - \Xi'\|_{\widehat{D}_* \setminus (\widehat{D}_*/10)} \leq -\frac{c\rho^{-a}}{|\log \rho|} + O(1). \quad (\text{PM.475})$$

Since  $a_1 < a$ , the logarithmic loss is absorbed for small  $\rho$ , proving (PM.469).

For the Liouville branch take  $\sigma_Q = Q^{-6}/5$ , again inside the Lemma 4.58 disk. The one-hole barrier estimate in Lemma 4.56 is unchanged by this fixed-factor shrink of its central  $Q^{-6}$  core; explicitly, its radial denominator is  $O(|\log \rho_n|)$  and its unique screening correction remains smaller than half the radial contribution. Lemma 4.58 supplies the central size  $e^{-Q^{1-\beta_0}}$ . Hence

$$\log \|\Omega'_{i_n^+} - \Xi'\|_{\widehat{D}_n \setminus (\widehat{D}_n/10)} \leq -\frac{cQ^{1-\beta_0}}{|\log \rho_n|} + O(1). \quad (\text{PM.476})$$

Because  $\rho_n \asymp (q_n Q)^{-1}$  and  $q_n \leq Q$ , one has  $|\log \rho_n| = O(\log Q)$ . Every logarithm is absorbed by the strict exponent gap  $\beta_1 - \beta_0$ , proving (PM.471).  $\square$

**Lemma 4.60 (flux-aware outer filling data for the HJ extension obstruction).**

At any HJ birth disk in either arithmetic regime, assume the convergent-BNF hypothesis and the propagated estimate of Lemma 4.59. Write the special HJ Hamiltonian provided by Lemma 4.53 as

$$\Pi(\theta, r) = a(\theta)(r^2 - e(\theta) + r^3 f(\theta, r)), \quad \langle a^{1/2} e \rangle = 0, \quad (\text{PM.477})$$

and let  $H$  be its actual averaged inverse from Lemma 4.45. After fixed collar shrinks, there is a disk-holomorphic function  $\Xi_\gamma$  of the form Lemma 4.55 and numbers  $R > 0$ ,  $r_0 = L\|e\|_{C^0}^{1/2}$  such that

$$\|\Xi_\gamma - H^2\|_{|z|=R} \leq \nu, \quad \|\Xi_\gamma - H^2\|_{|z|=r_0} \leq C_0, \quad (\text{PM.478})$$

where  $C_0$  is uniform and, after arbitrary fixed exponent weakening,

$$\nu \leq \begin{cases} \exp(-\rho^{-a_2}), & \tau(\omega) < \infty, \quad a_2 < a_1, \\ \exp(-Q^{1-\beta_2}), & \tau(\omega) = \infty, \quad \beta_2 > \beta_1. \end{cases} \quad (\text{PM.479})$$

Moreover Lemma 4.46 gives, on every  $4r_0 \leq \lambda \leq R/2$ ,

$$\|H^2 - (H^\sharp)^2\|_{|z|=\lambda} \leq C_0 \lambda^3. \quad (\text{PM.480})$$

Thus all hypotheses of Lemma 4.12 except the defect-sensitive relation between the actual HJ puncture and  $\|e\|_{C^0}$  are supplied quantitatively.

*Proof.* Compare the final KAM representation with the HJ representation (PM.391) on the fixed outer part of the buffered birth annulus. Lemma 4.51 propagates the HJ residual through the HJ coordinate with a stretched-exponential bound, while Lemma 4.52(c) in the finite regime and Lemma 4.56(d) in the Liouville regime give a bound of the same or better type for the final KAM residual. The cumulative coordinate motions are smaller than every fixed power of the buffered birth radius by Lemma 4.31–Lemma 4.33, Lemma 4.52 and Lemma 4.56. Therefore a collar  $\delta$  equal to a fixed small power of that birth radius satisfies the hypotheses of Lemma 4.54; all factors  $\delta^{-20}$  are absorbed by the strict exponential reserves.

Lemma 4.54 consequently provides a real action translation  $\gamma$ , smaller than the reserved collar, for which

$$\|\Omega'_{\text{KAM}}(\cdot + \gamma) - \Omega'_{\text{HJ}}\| \leq \nu_1 \quad (\text{PM.481})$$

with  $\nu_1$  obeying (PM.479) after a harmless exponent loss. Since the shifted argument remains in the buffered annulus, Lemma 4.59 and the triangle inequality give

$$\|\Xi'(\cdot + \gamma) - \Omega'_{\text{HJ}}\| \leq 2\nu_1 \quad (\text{PM.482})$$

after again weakening constants and exponents. Apply Lemma 4.55 with  $\Omega_{\text{disk}} = \Xi$ . The free additive action-Hamiltonian constant is fixed at one real point of the outer circle, and (PM.437) gives

$$\|\Xi_\gamma - H^2\|_{|z|=R} \leq CqR\nu_1. \quad (\text{PM.483})$$

The factors  $qR$  are polynomial in the finite branch and at most polynomial in  $Q$  in the large-jump branch, hence are absorbed into the strict exponent gaps. This is the first estimate in (PM.478) with  $\nu$  as in (PM.479).

On the inner circle,  $\Xi_\gamma$  is bounded because it is obtained by affine rescaling of the fixed holomorphic convergent BNF on a disk strictly larger than the HJ disk. Lemma 4.45 gives  $|H(z)| \leq C|z|$  on its inverse annulus. Since all variables are in the normalized quotient disk, both bounds are uniform after the fixed quadratic normalization; enlarging  $C_0$  gives the second estimate in (PM.478). Finally Lemma 4.46 is exactly (PM.480).  $\square$

**Lemma 4.61 (defect-sensitive HJ puncture after regular-baseline straightening).** Fix the uniform data of Lemma 4.45 and Lemma 4.12. Let

$$\Pi_e(\theta, r) = a(\theta)(r^2 - e(\theta) + r^3 f(\theta, r)) \quad (\text{PM.484})$$

be real symmetric on  $\mathbb{T}_h \times D(0, \rho)$ , with the same bounds on  $a^{\pm 1/2}$  and  $f$  as in Lemma 4.45, and put  $\varepsilon = \|e\|_{C^0(\mathbb{T})}$ . Assume

$$\|e\|_h \leq \varepsilon^{-P} \quad (\text{PM.485})$$

for a fixed  $P \geq 0$ . Then, after shrinking the angle strip by a fixed factor depending only on  $P$  and the uniform HJ data, there is an exact symplectic holomorphic change of variables on the punctured disk outside

$$|r| \leq C\varepsilon^{1/32} \quad (\text{PM.486})$$

which conjugates the autonomous flow of  $\Pi_e$  to an integrable autonomous flow. The coordinate is the composition of a map holomorphic through  $r = 0$  and a near-identity HJ map whose only puncture scale is (PM.486). In particular, the physical HJ hole occurring in Lemma 4.13 satisfies

$$\rho_{\text{hole}} \leq C\varepsilon^{1/32}. \quad (\text{PM.487})$$

*Proof.* First set  $e = 0$  but keep the same functions  $a$  and  $f$ :

$$\Pi_0(\theta, r) := a(\theta)(r^2 + r^3 f(\theta, r)). \quad (\text{PM.488})$$

Write the level root as  $g_0(\theta, z) = zu(\theta, z)$ . The equation  $\Pi_0(\theta, g_0) = z^2$  becomes

$$a(\theta)u^2(1 + zu f(\theta, zu)) = 1. \quad (\text{PM.489})$$

At  $z = 0$  it has the simple solution  $u = a^{-1/2}$ . The derivative of the left side with respect to  $u$  is  $2a^{1/2}$  there and is uniformly bounded away from zero. The holomorphic implicit-function theorem therefore gives a branch  $u$  on a fixed smaller product disk, uniformly bounded together with its first derivatives. Hence  $g_0$  is holomorphic through  $z = 0$ .

Put  $\Gamma_0(z) = \langle g_0(\cdot, z) \rangle$ . Since

$$\Gamma'_0(0) = \langle a^{-1/2} \rangle > 0, \quad (\text{PM.490})$$

its local inverse  $H_0$  is holomorphic through zero. Define  $p_0(\theta, R) = g_0(\theta, H_0(R))$ . By construction  $\langle p_0(\cdot, R) \rangle = R$ . Thus

$$S_0(\theta, R) := \int_0^\theta (p_0(t, R) - R) dt \quad (\text{PM.491})$$

is  $2\pi$ -periodic. The canonical relation  $r = R + \partial_\theta S_0$ ,  $\Theta = \theta + \partial_R S_0$  therefore defines an exact type-II map  $T_0 : (\theta, r) \mapsto (\Theta, R)$ , holomorphic through the central disk, and the level-root identity gives

$$\Pi_0 \circ T_0^{-1}(\Theta, R) = H_0(R)^2 =: P_0(R). \quad (\text{PM.492})$$

Furthermore  $P_0(R) = \alpha_0 R^2 + O(R^3)$  with  $\alpha_0 > 0$  bounded above and below by constants depending only on the uniform data.

Now use  $T_0$ , which is independent of  $e$ , as the baseline coordinate for the full Hamil-

tonian. Since  $\Pi_e - \Pi_0 = -a(\theta)e(\theta)$ ,

$$\Pi_e \circ T_0^{-1} = P_0(R) + Q_e(\Theta, R), \quad \|Q_e\|_{C^0(\mathbb{T} \times D)} \leq C\varepsilon. \quad (\text{PM.493})$$

We also need the intermediate-strip hypothesis of Lemma 4.47. For  $M(y) = \sup_{|\Im \theta| \leq y} |e(\theta)|$ , the elementary three-lines inequality for a holomorphic periodic strip function says that  $\log M(y)$  is convex in  $y$ ; therefore

$$M(sh) \leq M(0)^{1-s} M(h)^s \leq \varepsilon^{1-s(1+P)}. \quad (\text{PM.494})$$

Choose  $s_* = [4(P+1)]^{-1}$  and then take the Lemma 4.47 ambient strip to have half-strip contained in  $\mathbb{T}_{s_* h}$ . Equation (PM.494) gives  $\|e\| \leq \varepsilon^{3/4} \leq \varepsilon^{1/2}$  there. Since  $T_0$  and its inverse have fixed holomorphic bounds on a further fixed shrink, the transformed perturbation  $Q_e$  satisfies Lemma 4.47 with real defect scale  $C\varepsilon$ , half-strip defect scale  $C\varepsilon^{1/2}$ , and  $\eta = 0$ .

Apply Lemma 4.47 with  $L = (C\varepsilon)^{-7/32}$ . Its deleted radius is

$$CL(C\varepsilon^{1/2})^{1/2} = C\varepsilon^{-7/32} \varepsilon^{1/4} = C\varepsilon^{1/32}, \quad (\text{PM.495})$$

and the corresponding HJ type-II map is near identity on the punctured disk. Composing it with the regular exact map  $T_0$  (with the appropriate inverse according to the displayed conjugacy orientation) gives the asserted integrable straightening. Because  $T_0$  is holomorphic through the center, it creates no additional puncture; the only deleted scale is (PM.495), proving (PM.486)–(PM.487).  $\square$

**Lemma 4.62 (Extension-Property collapse of every selected HJ core).** Under the convergent-BNF hypothesis, every selected HJ birth disk in either arithmetic regime may be replaced, for the purposes of the upper invariant-circle deficit, by a much smaller real action interval. More precisely:

(a) *Finite arithmetic exponent.* Given  $0 < b < a_2 < 1/\tau_+$ , every selected HJ core meeting the target disk has physical action radius at most

$$\exp(-\rho^{-b}) \quad (\text{PM.496})$$

for all sufficiently small  $\rho$ .

(b) *Liouville large jump.* Given  $0 < \beta_2 < \beta_3 < 1/10$ , with  $Q = q_{n+1}$ , every selected HJ core has physical action radius at most

$$\exp(-Q^{1-\beta_3}). \quad (\text{PM.497})$$

In particular the exponent in (PM.497) may be weakened to any fixed exponent strictly below  $1 - \beta_3$ , including  $1/4$ .

*Proof.* Use Lemma 4.60 with the special form and gauge of Lemma 4.53. Put  $\varepsilon = \|e\|_{C^0}$ . Lemma 4.61 verifies precisely the missing hypothesis (PM.93) of Lemma 4.13, while Lemma 4.46 supplies (PM.77); Lemma 4.60 supplies the disk filling and the two boundary bounds (PM.76). The birth-scale reserves in Lemma 4.51, Lemma 4.52 and Lemma 4.56 give  $R/[L\varepsilon^{1/2}] \rightarrow \infty$ , so after increasing the initial scale the ratio condition  $R/r_0 \geq 4^M$  of Lemma 4.12 also holds. The strip bound required in Lemma

4.12 follows from Lemma 4.53 and, if necessary, weakening it to the polynomial form  $\|e\|_h \leq \varepsilon^{-P}$  after a fixed strip shrink; if  $\varepsilon = 0$  there is no HJ puncture and the conclusion is immediate.

Lemma 4.12 and Lemma 4.13 therefore give

$$\rho_{\text{HJ}}^{(q)} \leq \nu^{1/200} \quad (\text{PM.498})$$

in the quotient action coordinate. Returning through the  $q$ -cover divides this radius by  $q$ , so it can only improve the estimate. In the finite branch, (PM.479) gives  $\nu \leq e^{-\rho^{-a_2}}$  after a strict prior weakening; hence

$$q^{-1}\nu^{1/200} \leq C \exp(-c\rho^{-a_2}) \leq \exp(-\rho^{-b}) \quad (\text{PM.499})$$

for every  $b < a_2$ . In the Liouville branch the same argument gives  $C \exp(-cQ^{1-\beta_2})$ , which is bounded by (PM.497) after choosing  $\beta_3 > \beta_2$  and taking  $Q$  large.  $\square$

**Lemma 4.63 (upper invariant-circle abundance and physical measure transfer).** Assume the one-dimensional formal BNF converges and the twist coefficient is nonzero. Then the upper deficit estimates needed in Lemma 4.5 hold.

(a) *Finite arithmetic exponent.* If  $\tau = \tau(\omega) < \infty$ , then for every  $\beta_u > 0$  there is  $\rho_0 > 0$  such that

$$m_f(\rho) \leq \exp\left[-\rho^{-(1/\tau - \beta_u)}\right] \quad (0 < \rho < \rho_0), \quad (\text{PM.500})$$

provided  $1/\tau - \beta_u > 0$ .

(b) *Liouville large jumps.* Along every sufficiently large selected jump of Lemma 4.56, with  $Q = q_{n+1}$  and the associated comparison radius  $\rho_n$ ,

$$m_f(\rho_n) \leq \exp(-Q^{1/5}). \quad (\text{PM.501})$$

*Proof.* We first record a continuation estimate that will be used in every normal-form chart. Suppose on a fixed real action interval an exact area-preserving analytic map is written  $\Phi_\Omega f_F$ , with fixed constants  $0 < a < A$  such that  $a \leq \Omega'' \leq A$  on the real interval and with a uniform  $C^3$  bound for  $\Omega$ , and  $\varepsilon := \|F\| \ll 1$ . Choose the initial cutoff for Lemma 4.21 so that

$$N_0 \asymp |\log \varepsilon|, \quad E_0 \asymp \varepsilon^{1/2}, \quad \varepsilon \leq E_0. \quad (\text{PM.502})$$

At each subsequent stage remove the complex resonance disks of Lemma 4.26 with  $\gamma_j = K_j^{-1}$ . Lemma 4.22 and Lemma 4.26 give, on an interval of bounded length,

$$\text{Leb}_{\mathbb{R}}(\text{new real deletions at stage } j) \leq CK_j^{-1}(N_j^2 + N_j). \quad (\text{PM.503})$$

The same scale controls the total boundary collars. Since  $K_j^{-1} = E_j^\kappa$  and  $E_j$  decays exponentially in  $N_j/(j+2)^2$ , while  $N_j$  grows geometrically, (PM.502) and Lemma 4.21 imply

$$\sum_{j \geq 0} K_j^{-1}(N_j^2 + N_j) + \sum_{j \geq 0} \delta_j K_j^{-1} \leq \varepsilon^{c_0} \quad (\text{PM.504})$$

for one fixed  $c_0 > 0$ . Components shorter than the first collar may simply be discarded;

their total length is bounded by the same right-hand side because the number of components is polynomially bounded by Lemma 4.22. On every remaining component, Lemma 4.28 supplies nested collared domains, Lemma 4.26 supplies the complex finite-mode divisor bound, and the next outer domain is chosen inside the preceding inner domain. Thus the extension hypothesis of Lemma 4.27 is satisfied by restriction. Lemma 4.27 gives convergent exact coordinate maps, a limiting twist function, and remainder tending to zero. Passing to the limit in (PM.187), every action in the compact survivor set  $L$  lies on an invariant essential circle. Moreover

$$\text{Leb}_{\mathbb{R}}(A \setminus L) \leq \varepsilon^{c_0}, \quad \|g_{\infty} - \text{id}\|_{C^0} \leq \varepsilon^{c_1} \quad (\text{PM.505})$$

for a fixed  $c_1 > 0$ , by (PM.193)–(PM.194) and Lemma 4.21. Applying Lemma 4.16–Lemma 4.17 on each normalized chart therefore contributes at most another fixed power of  $\varepsilon$  to the uncovered area.

We now apply this continuation to the finite upper engine. Choose  $\tau_+ > \tau$  so close to  $\tau$  and all strict losses in Lemma 4.52, Lemma 4.57–Lemma 4.62 so small that there exists

$$a_* > \frac{1}{\tau} - \frac{\beta_u}{2} \quad (\text{PM.506})$$

for which the final KAM remainder, every HJ residual, and every collapsed HJ core are bounded by  $\exp(-\rho^{-a_*})$  after a further harmless weakening. Lemma 4.32–Lemma 4.34 cover the target action disk by the final KAM domain together with the buffered HJ birth disks; Lemma 4.62 replaces the latter by HJ charts punctured only by intervals of that same stretched-exponential size. Run the continuation estimate (PM.502)–(PM.505) on the real-centered collared components of the final KAM domain and on the corresponding fixed outer collars of every HJ annulus. This point is important: no complex nonresonance is asserted on an entire disk-minus-holes set; Lemma 4.26 is used only on the thin real-centered corridors on which the Newton steps are actually run.

The number of initial charts/holes is polynomial in  $\rho^{-1}$  by Lemma 4.32 and the adapted scale bounds. Hence Lemma 4.14 absorbs their count and every fixed power loss in (PM.505), yielding

$$m_f(\rho) \leq \exp[-\rho^{-a_{**}}] \quad (\text{PM.507})$$

for every  $a_{**} < a_*$ . Choose  $a_{**} = 1/\tau - \beta_u$  to obtain (PM.500).

For the Liouville branch there is only the selected large-jump HJ hole inside the target disk, together with a polynomial number of continuation resonances. Lemma 4.56, Lemma 4.59–Lemma 4.62 give initial errors and the collapsed core bounded by  $\exp(-Q^{a_L})$  for every fixed  $a_L < 1$  after the displayed losses. Apply (PM.502)–(PM.505) in the surviving central/final KAM region and the HJ annulus. Taking, for example,  $a_L > 1/4$ , the polynomial chart count and all fixed power losses are absorbed by Lemma 4.14, giving  $m_f(\rho_n) \leq e^{-Q^{b_L}}$  for every  $b_L < a_L$ . Choosing  $b_L = 1/5$  proves (PM.501).

Finally we justify that these action-angle estimates are estimates for the original elliptic germ. The two branch charts of the definition of exact Cartesian-action-angle branch charts above are exact symplectic on the punctured disk, and every preliminary normalizer is symplectic. We apply the chart construction on a fixed enlargement of the target action disk; tangency to the identity and the fixed outer margin ensure that



invariant circles constructed over the target preimage lie inside the doubled physical action window in the definition of invariant-circle deficit, twist coefficient, and continued-fraction scales above. Thus area preservation transfers the exceptional sets themselves, rather than paying the polynomial displacement of the preliminary normalizer as a boundary error. The fixed point has zero area. The only near-identity symmetric-difference loss used above is the exponentially small continuation tail controlled by Lemma 4.16. This proves the physical deficit estimates.  $\square$

**Lemma 4.64 (Birkhoff graph theorem for invariant essential curves).** Let

$$\mathbb{A} := \mathbb{T} \times [a, b]$$

and let  $F : \mathbb{A} \rightarrow \mathbb{A}$  be a  $C^1$  area-preserving diffeomorphism, isotopic to the identity and preserving the two ends. Choose a lift  $\tilde{F} = (X, R) : \mathbb{R} \times [a, b] \rightarrow \mathbb{R} \times [a, b]$  and assume that on a compact subannulus containing an invariant essential Jordan curve  $\Gamma$  one has

$$\partial_r X(\theta, r) \geq \kappa > 0. \quad (\text{PM.510})$$

Then  $\Gamma$  is the graph of a Lipschitz function  $u : \mathbb{T} \rightarrow [a, b]$ . If on the same compact subannulus the first derivatives of  $F$  and  $F^{-1}$  are bounded by  $M$ , the Lipschitz constant is bounded by a constant depending only on  $\kappa$  and  $M$ .

*Proof.* Lift  $\Gamma$  to a properly embedded line  $\tilde{\Gamma} \subset \mathbb{R} \times (a, b)$  satisfying  $\tilde{\Gamma} + (2\pi, 0) = \tilde{\Gamma}$ . Its complement has exactly two unbounded components  $U^-$  and  $U^+$ , characterized respectively by containing the lower and upper ends. End preservation and invariance of  $\Gamma$  imply

$$\tilde{F}(U^-) = U^-, \quad \tilde{F}(U^+) = U^+. \quad (\text{PM.511})$$

We prove first that every vertical fiber meets  $\tilde{\Gamma}$  exactly once. Essentiality gives at least one intersection. Suppose a fiber  $\{\theta_0\} \times [a, b]$  had two distinct frontier points  $z_- = (\theta_0, r_-)$  and  $z_+ = (\theta_0, r_+)$  with  $r_- < r_+$ . Choose two consecutive such frontier points, so that the open vertical segment between them lies entirely in one complementary component, say  $U^-$ . By (PM.510),

$$X(z_+) - X(z_-) = \int_{r_-}^{r_+} \partial_r X(\theta_0, s) ds \geq \kappa(r_+ - r_-) > 0. \quad (\text{PM.512})$$

Take a thin closed rectangle whose vertical middle segment is the chosen segment, whose lower horizontal side lies in  $U^-$ , and whose upper horizontal side lies in  $U^+$  except at arbitrarily short neighborhoods of the two frontier points. Under  $\tilde{F}$ , the two end components remain on their respective sides by (PM.511), while (PM.512) reverses the vertical ordering of the two frontier gates when they are read on a common nearby vertical fiber. Closing the two gates by the corresponding short arcs of  $\tilde{\Gamma}$  produces two Jordan arcs with alternating endpoints. The planar Jordan separation theorem forces an intersection between an image arc contained in  $U^-$  and  $\tilde{\Gamma}$ , contrary to (PM.511). Thus no vertical fiber has two intersections. This is the classical Birkhoff fold-area argument; area preservation is what forbids an end component from being folded through the invariant frontier.

Hence the angular projection  $\pi_\theta : \tilde{\Gamma} \rightarrow \mathbb{R}$  is a continuous proper bijection and there

is a continuous  $2\pi$ -periodic function  $u$  with  $\tilde{\Gamma} = \{(\theta, u(\theta))\}$ .

It remains to obtain a quantitative graph bound. The restriction of  $F$  to  $\Gamma$  is an orientation-preserving circle homeomorphism. On a sufficiently short lift interval,  $\theta < \theta'$  therefore implies

$$X(\theta, u(\theta)) < X(\theta', u(\theta')). \quad (\text{PM.513})$$

Writing  $A := \sup |\partial_\theta X|$  and applying the mean-value formula to (PM.513) gives, whenever  $u(\theta') < u(\theta)$ ,

$$u(\theta) - u(\theta') \leq A\kappa^{-1}|\theta' - \theta|. \quad (\text{PM.514})$$

Apply the same argument to  $F^{-1}$  to control the opposite sign. A finite cover of the circle by such lift intervals, followed by the trivial bound for pairs separated by more than the chosen local radius, gives

$$|u(\theta') - u(\theta)| \leq C(\kappa, M)|\theta' - \theta| \quad (|\theta' - \theta| \leq \pi). \quad (\text{PM.515})$$

Thus  $\Gamma$  is a global Lipschitz graph.  $\square$

**Lemma 4.65 (rational separatrix/no-crossing lemma).** Assume the hypotheses of Lemma 4.64. Let  $\Gamma = \{r = u(\theta)\}$  be an invariant essential graph with rational rotation number  $p/q$ , and let  $z_*$  be a hyperbolic  $q$ -periodic point of  $F$  lying on  $\Gamma$ . Then, in a sufficiently small neighborhood of  $z_*$ , the graph  $\Gamma$  can meet the local stable or unstable manifold only by coinciding with the corresponding local branch. In particular it cannot cross either branch transversely.

*Proof.* Choose a lift and write

$$G := T_{-2\pi p} \circ \tilde{F}^q. \quad (\text{PM.516})$$

The restriction of  $G$  to the lifted graph is an orientation-preserving homeomorphism of the line with rotation number zero. Its fixed set is nonempty. If  $I = (\alpha, \beta)$  is a component of the complement of the fixed set, then either  $G(x) > x$  throughout  $I$  or  $G(x) < x$  throughout  $I$ . In the first case

$$G^{-n}(x) \longrightarrow \alpha, \quad G^n(x) \longrightarrow \beta, \quad (\text{PM.517})$$

and in the second case the limits are reversed. This follows simply from monotonicity: the forward orbit is monotone and bounded by the appropriate fixed endpoint, and its limit must itself be fixed.

Suppose an endpoint represents  $z_*$  and consider graph points sufficiently close to it on the adjacent component. Equation (PM.517) says that their forward or backward iterates converge to  $z_*$ . The local stable-manifold theorem the Foundation stable-manifold theorem, together with uniqueness of the local stable and unstable sets, therefore places that portion of  $\Gamma$  in  $W_{\text{loc}}^s(z_*)$  or  $W_{\text{loc}}^u(z_*)$ . A transverse crossing would give points on the same side of the graph with incompatible forward/backward convergence and contradict this uniqueness. Hence crossing is impossible.  $\square$

**Lemma 4.66 (vertical-cap lobe barrier and area lower bound).** Let  $F$  satisfy Lemma 4.64 and suppose a hyperbolic  $q$ -periodic point has local stable and unstable branches that, after translating the point to  $(0, 0)$ , are graphs  $r = w_-(\theta)$  and  $r = w_+(\theta)$

on  $0 \leq \theta \leq \rho$ , with

$$\frac{m}{2} \theta \leq w_+(\theta) - w_-(\theta) \leq 2m\theta. \quad (\text{PM.518})$$

Let

$$V := \{(\theta, r) : 0 < \theta < \rho/2, \quad w_-(\theta) < r < w_+(\theta)\}. \quad (\text{PM.519})$$

Then no invariant essential Jordan curve of  $F$  intersects  $V$ , and

$$\text{area}(V) \geq \frac{m\rho^2}{16}. \quad (\text{PM.520})$$

Consequently, if

$$m \geq c_1 q A^{1/2}, \quad \rho \geq c_2 A, \quad (\text{PM.521})$$

then

$$\text{area}(V) \geq c_3 q A^{5/2}. \quad (\text{PM.522})$$

*Proof.* By Lemma 4.64 every invariant essential curve is a graph. By Lemma 4.65 it cannot cross the stable or unstable boundary of  $V$ . If such a graph met  $V$ , follow it in decreasing  $\theta$ . It cannot leave through either separatrix, so it remains between them and converges to their common endpoint at  $\theta = 0$ . The graph therefore contains the hyperbolic periodic point. Lemma 4.65 then forces it locally to coincide with one of the two separatrices, contradicting that it contains interior points strictly between them. Equivalently, an attempted entrance through the sole vertical cap at  $\theta = \rho/2$  can never be completed by a global single-valued invariant graph. Thus  $V$  is invariant-circle-free.

The area estimate is elementary:

$$\text{area}(V) = \int_0^{\rho/2} (w_+(\theta) - w_-(\theta)) d\theta \geq \frac{m}{2} \int_0^{\rho/2} \theta d\theta = \frac{m\rho^2}{16}. \quad (\text{PM.523})$$

Substitution of (PM.521) gives (PM.522).  $\square$

**Lemma 4.67 (retained-harmonic resonant saddle and lower-hole engine).**

There are constants  $a_*, C, c > 0$  with the following property. Let the one-dimensional exact twist map be written on a complex resonant block as

$$f = \Phi_\Omega \circ f_F, \quad (2\pi)^{-1} \Omega'(c) = p/q, \quad (\text{PM.524})$$

where  $p/q$  is reduced, the twist bounds of Lemma 4.49 hold, and

$$\varrho := \min\{|c|/4, q^{-9}\}, \quad E := \|F\|_{h,D(c,\varrho)} \leq \varrho^{a_*}, \quad e^{-qh} < \varrho^{10}. \quad (\text{PM.525})$$

Assume the retained  $q$ -th Fourier coefficient satisfies

$$|\widehat{F}(q, c)| \geq \eta E e^{-qh}, \quad \eta \geq q^2 \varrho, \quad (\text{PM.526})$$

and put  $A_q := \eta E e^{-qh}$ . For all sufficiently large  $q$ ,  $f$  has a hyperbolic  $q$ -periodic point with stable/unstable branches satisfying (PM.518) on a real interval for which

$$m \geq c q A_q^{1/2}, \quad \rho \geq c A_q. \quad (\text{PM.527})$$

Hence there is an open set, contained in the resonant action window and disjoint from every invariant essential circle, whose area is at least

$$cqA_q^{5/2}. \quad (\text{PM.528})$$

In particular, if  $E \leq C_0|c|^{a_3}$  and  $\eta \geq |c|^{1/2}$ , then for fixed constants  $L, C_1 > 0$  independent of the resonance,

$$m_f(t) \geq C^{-1}|c|^L e^{-C_1 qh} \quad (\text{PM.529})$$

for every comparison radius  $t$  whose smaller window contains the resonant lobe and whose doubled window contains the resonant block.

*Proof.* Apply Lemma 4.49 on  $D(c, \varrho)$ . It gives an exact symplectic near-identity map  $g_{\text{RNF}}$  for which

$$g_{\text{RNF}}^{-1} f g_{\text{RNF}} = \Phi_\Omega \circ f_{F^{\text{res}}} \circ f_{F^{\text{cor}}}, \quad (\text{PM.530})$$

where  $F^{\text{res}}$  is  $2\pi/q$ -periodic and the correction is exponentially small in  $\varrho^{-1/4}$ . Fourier projection and Cauchy estimates split

$$F^{\text{res}}(\theta, r) = u_0(r) + a(r) \cos(q\theta) + b(r) \sin(q\theta) + R_{\geq 2}(\theta, r), \quad (\text{PM.531})$$

with

$$\|R_{\geq 2}\| \leq CEe^{-2qh}, \quad |(a(c), b(c))| \geq \frac{1}{2}A_q, \quad (\text{PM.532})$$

for large  $q$ . Indeed, Lemma 4.49 changes the selected Fourier coefficient by at most  $Cq\varrho Ee^{-qh}$ , which is  $o(A_q)$  by (PM.526), while every resonant multiple  $|kq|$ ,  $|k| \geq 2$ , pays the extra factor  $e^{-qh}$ . Rotate  $\theta$  by a real phase so that  $b(c) = 0$  and  $a(c) > 0$ .

Absorb  $u_0$  into the integrable part and let  $\bar{c} = c + O(E)$  be the unique nearby critical action of the resulting rationally rebased frequency. The twist bounds give

$$\Omega_{\text{eff}}(r) - 2\pi(p/q)r = \text{const} + \pi\varpi(r - \bar{c})^2 + O(|r - \bar{c}|^3), \quad C^{-1} \leq \varpi \leq C. \quad (\text{PM.533})$$

On the  $q$ -fold resonant coordinates  $\vartheta = q(\theta - \alpha)$ ,  $s = q(r - \bar{c})$ , the autonomous part has quadratic model

$$H_Q(\vartheta, s) = \frac{\varpi}{2}s^2 + q^2a(\bar{c}) \cos \vartheta. \quad (\text{PM.534})$$

At the appropriate extremum of the cosine its Hamiltonian linearization has real eigenvalues

$$\pm q\sqrt{\varpi a(\bar{c})}, \quad (\text{PM.535})$$

so the corresponding time-one map is hyperbolic. From (PM.525)–(PM.532), two Cauchy derivatives of the cubic Taylor remainder,  $R_{\geq 2}$ , and  $F^{\text{cor}}$  are  $o(q^2A_q)$  on the real box of size  $cA_q$  used below. The implicit-function theorem therefore moves the fixed point by  $o(A_q)$  and preserves its real hyperbolic splitting. Applying the internally proved Foundation stable-manifold theorem and its graph-transform estimates gives local branches which, after undoing the  $q$ -cover, satisfy

$$\frac{m}{2}|\theta - \theta_*| \leq |w_+(\theta) - w_-(\theta)| \leq 2m|\theta - \theta_*|, \quad m \asymp qA_q^{1/2}, \quad (\text{PM.536})$$

on  $|\theta - \theta_*| < \rho$  with  $\rho \geq cA_q$ . The exponentially small correction in (PM.530) is absorbed

by the same graph-transform margin.

The angular derivative of the full map remains bounded below by a positive twist constant on this block because the perturbation has  $C^2$  norm  $o(1)$ . Thus Lemma 4.64–Lemma 4.66 apply and give (PM.528). The map  $g_{\text{RNF}}$  and all preceding normalizers are exact symplectic, hence preserve area; the lobe estimate is therefore unchanged when pulled back to the original coordinates.

Finally, under  $E \leq C_0|c|^{a_3}$  and  $\eta \geq |c|^{1/2}$ ,

$$qA_q^{5/2} \geq c q |c|^{(5/2)(a_3+1/2)} e^{-(5/2)qh}. \quad (\text{PM.537})$$

After increasing a fixed polynomial exponent  $L$  and a fixed exponential constant  $C_1$ , (PM.537) implies (PM.529). Since the lobe is centered at action  $c$  and has action width  $o(|c|)$ , it lies in every fixed-comparable target action window for all large resonances.  $\square$

**Lemma 4.68 (Cartesian lacunary forcing seed).** Fix an irrational frequency  $\omega_0$  and choose a sign  $\sigma \in \{-1, 1\}$  as follows. In the finite-arithmetic case, choose a side of  $\omega_0$  containing infinitely many convergents that realize the limsup exponent of Lemma 4.6; in the Liouville case choose a side containing infinitely many of the selected large-jump convergents. Put

$$\Omega(r) = 2\pi(\omega_0 r + \sigma r^2), \quad (\text{PM.538})$$

so that the corresponding resonant actions  $c_n > 0$  on the selected infinite set  $\mathcal{N}$ . Let  $a_3$  be larger than the fixed exponent required in Lemma 4.49 and define, in the complex Cartesian coordinates of the definition of exact Cartesian-action-angle branch charts above,

$$G(z, w) = (-izw)^{a_3} \sum_{n \in \mathcal{N}} \frac{1}{2} \left[ (i^{-1/2}z)^{q_n} + (i^{-1/2}w)^{q_n} \right]. \quad (\text{PM.539})$$

Then  $G$  is a real-symmetric analytic germ and the exact-symplectic germ

$$f_* := \Phi_\Omega \circ f_G \quad (\text{PM.540})$$

has, at every sufficiently large selected resonance, an invariant-circle-free lobe which for the comparison scale  $t_n$  of the definition of invariant-circle deficit, twist coefficient, and continued-fraction scales above satisfies

$$m_{f_*}(t_n) \geq C^{-1} c_n^L \exp[-C q_n \log(c_n^{-1})] \quad (\text{PM.541})$$

with fixed constants  $C, L > 0$ .

*Proof.* Because  $q_n \rightarrow \infty$ , for every  $0 < \rho < 1$  and every derivative order  $j$  the series  $\sum q_n^j \rho^{q_n}$  converges. Hence (PM.539) and all of its derivatives converge uniformly on compact bidisks. The involution defining the real Cartesian slice exchanges the two displayed monomials, so  $G$  is real symmetric. Also  $G = O((zw)^{a_3})$ , and therefore (PM.540) has the required fixed elliptic linear part and twist coefficient  $b_2 = \sigma$ .

On either exact action-angle branch of the definition of exact Cartesian-action-angle branch charts above one has

$$-izw = r, \quad i^{-1/2}z = r^{1/2}e^{-i\theta}, \quad i^{-1/2}w = r^{1/2}e^{i\theta}, \quad (\text{PM.542})$$

so

$$G \circ \psi_{\pm}(\theta, r) = r^{a_3} \sum_{n \in \mathcal{N}} r^{q_n/2} \cos(q_n \theta). \quad (\text{PM.543})$$

For a selected  $c_n > 0$  put

$$h_n := -\frac{1}{2} \log(c_n + c_n^2), \quad \varrho_n := \min\{c_n^2, q_n^{-9}\}. \quad (\text{PM.544})$$

The corresponding Cartesian bidisk pulls back over  $\mathbb{T}_{h_n} \times D(c_n, \varrho_n)$ , and (PM.543) gives

$$\|G \circ \psi_{\pm}\| \leq C c_n^{a_3}, \quad |\widehat{G}(q_n, c_n)| = \frac{1}{2} c_n^{a_3 + q_n/2}. \quad (\text{PM.545})$$

Moreover

$$e^{-q_n h_n} = (c_n + c_n^2)^{q_n/2} = c_n^{q_n/2} e^{O(q_n c_n)}, \quad (\text{PM.546})$$

and  $q_n c_n \rightarrow 0$  by Lemma 4.6. Thus the normalized retained coefficient in (PM.526) is bounded below by an absolute constant, in particular by  $c_n^{1/2}$ , for all large  $n$ . Also  $e^{-q_n h_n} < \varrho_n^{10}$  and the high-order smallness required by Lemma 4.67 follows from the choice of  $a_3$ .

Lemma 4.67 therefore produces a lobe near  $r = c_n$  and gives

$$\text{area}(V_n) \geq C^{-1} c_n^{L_0} e^{-C_0 q_n h_n}. \quad (\text{PM.547})$$

Since  $b_2 = \pm 1$ , Lemma 4.6 and the definition of  $t_n$  give  $c_n \asymp t_n$  and, after increasing  $n$ , the lobe lies in  $|r| < t_n$  while the whole resonant block lies in  $|r| < 2t_n$ . No invariant circle meets  $V_n$ , hence  $V_n$  is contained in the deficit set defining  $m_{f_*}(t_n)$ . Finally  $h_n \leq C \log(c_n^{-1})$ , so (PM.547) implies (PM.541).  $\square$

**Lemma 4.69 (logarithmic-loss arithmetic conversion).** Assume the resonant actions  $c_n$  satisfy Lemma 4.6,  $t_n \asymp |c_n|$ , and along an infinite arithmetic subsequence

$$m(t_n) \geq K^{-1} |c_n|^L \exp[-C q_n \log(|c_n|^{-1})]. \quad (\text{PM.548})$$

Then:

(i) if  $\tau(\omega_0) < \infty$  and the subsequence realizes the limsup as in Lemma 4.6, then for every  $\beta > 0$ , after passing to a further infinite subsequence,

$$m(t_n) \geq \exp\left[-t_n^{-(1/(1+\tau)+\beta)}\right]; \quad (\text{PM.549})$$

(ii) if  $\log q_{n+1}/\log q_n \rightarrow \infty$  on the subsequence, then for every  $\beta > 0$ , after passing to a further infinite subsequence,

$$m(t_n) \geq e^{-q_{n+1}^\beta}. \quad (\text{PM.550})$$

*Proof.* For (i), choose  $\eta > 0$  so small that

$$a := \frac{1}{1+\tau-\eta} < \frac{1}{1+\tau} + \frac{\beta}{3}. \quad (\text{PM.551})$$

Lemma 4.6 gives  $q_n \leq C|c_n|^{-a}$  on infinitely many selected indices. Since  $t_n \asymp |c_n|$ ,

$$q_n \log |c_n|^{-1} \leq C t_n^{-a} \log t_n^{-1} \leq t_n^{-(a+\beta/3)} \quad (\text{PM.552})$$

for all sufficiently small  $t_n$ . The fixed polynomial factor in (PM.548) costs only  $O(\log t_n^{-1})$ , which is absorbed by another  $\beta/3$  of exponent. This proves (PM.549).

For (ii), Lemma 4.6 gives  $|c_n|^{-1} \leq C q_n q_{n+1}$  up to a fixed factor, hence

$$\log |c_n|^{-1} = O(\log q_{n+1}). \quad (\text{PM.553})$$

Because  $\log q_{n+1}/\log q_n \rightarrow \infty$ , for every  $\beta > 0$  and all sufficiently large selected indices

$$q_n \leq q_{n+1}^{\beta/3}, \quad \log q_{n+1} \leq q_{n+1}^{\beta/3}. \quad (\text{PM.554})$$

Thus  $q_n \log |c_n|^{-1} = o(q_{n+1}^{\beta})$ , and the polynomial factor in (PM.548) is also  $e^{o(q_{n+1}^{\beta})}$ . After one harmless constant absorption, (PM.550) follows.  $\square$

**Lemma 4.70 (lower/upper contradiction and prevalence seed).** For every irrational one-dimensional frequency  $\omega_0$  there exists a real-analytic exact-symplectic elliptic germ with linear frequency  $\omega_0$  and zero radius of convergence of its formal Birkhoff normal form. Consequently, for every nonresonant  $\omega \in \mathbb{R}^d$ ,  $d \geq 1$ , the divergent-BNF subset of  $S_\omega$  is (PM)-prevalent.

*Proof.* Take the Cartesian seed  $f_*$  of Lemma 4.68. Suppose for contradiction that its formal BNF converges.

If  $\tau(\omega_0) < \infty$ , choose  $\beta > 0$  so small that

$$\frac{1}{1+\tau} + \beta < \frac{1}{\tau} - \beta. \quad (\text{PM.555})$$

Lemma 4.63 with upper loss  $\beta$  gives, for all sufficiently small  $t$ ,

$$m_{f_*}(t) \leq \exp\left[-t^{-(1/\tau-\beta)}\right]. \quad (\text{PM.556})$$

On the one-sided limsup subsequence selected in Lemma 4.68, Lemma 4.69 gives infinitely many  $t_n \downarrow 0$  for which

$$m_{f_*}(t_n) \geq \exp\left[-t_n^{-(1/(1+\tau)+\beta)}\right]. \quad (\text{PM.557})$$

The two bounds contradict Lemma 4.5(i).

If  $\tau(\omega_0) = \infty$ , use the one-sided large-jump subsequence selected in Lemma 4.68. Lemma 4.63 gives, with  $Q = q_{n+1}$ ,

$$m_{f_*}(t_n) \leq e^{-Q^{1/5}} \quad (\text{PM.558})$$

for all sufficiently large selected jumps, while Lemma 4.69 with any  $0 < \beta < 1/5$  gives infinitely many of the same selected indices with

$$m_{f_*}(t_n) \geq e^{-Q^\beta}. \quad (\text{PM.559})$$

This contradicts Lemma 4.5(ii). Thus the seed BNF diverges for every irrational  $\omega_0$ .

Now let  $\omega = (\omega_1, \dots, \omega_d)$  be nonresonant in the sense of the definition of fixed-frequency symplectic germ space, map Birkhoff normal form, and  $(PM)$ -prevalence above. Every component  $\omega_j$  is irrational, since a rational component would yield a nonzero integer vector  $k$  with  $k \cdot \omega \in \mathbb{Z}$ . Apply the one-dimensional construction to one component and use the product lift in Lemma 4.3 to obtain a divergent seed in  $S_\omega$ . The real-line dichotomy of Lemma 4.3 then says that on every affine line through this seed the convergent-BNF parameters form a null set. By the definition of fixed-frequency symplectic germ space, map Birkhoff normal form, and  $(PM)$ -prevalence above this is exactly  $(PM)$ -prevalence of divergence. The Hamiltonian fixed-quadratic-part dichotomy asserted in the theorem is already proved independently in Lemma 4.4.  $\square$

**Proof synthesis.** Lemma 4.1–Lemma 4.4 establish the formal coefficient filtration, the real-line Pérez–Marco dichotomy, the dimension lift, and the Hamiltonian fixed-quadratic-part dichotomy. Lemma 4.5–Lemma 4.63 establish the arithmetic, exact symplectic calculus, convergent-BNF upper invariant-circle engine, both finite-arithmetic and Liouville adapted regimes, and the exact Cartesian/action-angle and physical-area transfers. Lemma 4.64–Lemma 4.67 provide the Birkhoff graph, separatrix, resonant-saddle, and invariant-circle-free lobe machinery. Lemma 4.68–Lemma 4.70 construct a genuine Cartesian analytic seed, absorb its shrinking-strip logarithmic loss in both arithmetic regimes, contradict the convergent-BNF upper bounds, and invoke the already proved affine-line dichotomy. This proves every clause of the Divergence caveat for formal normal forms.

#### Theorem K04.6. Nekhoroshev-ready resonant normal form

**Statement.** Let  $H(I, \theta) = h(I) + f(I, \theta)$  be analytic on a complex resonant block  $\mathcal{D}_{\Lambda, \rho, \sigma}$ , where  $D_\Lambda$  is  $(\alpha, K)$ -nonresonant modulo a lattice  $\Lambda$  with respect to  $h$ , and let  $M$  bound the Hessian of  $h$  there. Fix  $\xi > 1$ . If

$$\|f\|_{\rho, \sigma} \leq \varepsilon \leq \frac{\alpha \rho'}{256 \xi K}, \quad \rho' \leq \min \left\{ \rho, \frac{\alpha}{2 \xi M K} \right\}, \quad K \sigma \geq 6,$$

then there is a real-analytic symplectic map  $\Psi : \mathcal{D}_{\Lambda, \rho'/2, \sigma/6} \rightarrow \mathcal{D}_{\Lambda, \rho, \sigma}$  such that

$$H \circ \Psi = h + g + f^*, \quad \mathcal{P}_\Lambda g = g.$$

Writing  $g_0 = \mathcal{P}_\Lambda \mathcal{P}_K f$ , one has on the reduced block

$$\|g - g_0\|_{\rho'/2, \sigma/6} \leq 64 \frac{K}{\alpha \rho'} \varepsilon^2, \quad \|f^*\|_{\rho'/2, \sigma/6} \leq \varepsilon e^{-K \sigma/6}.$$

Moreover, for every  $(I, \theta) \in \mathcal{D}_{\Lambda, \rho'/2, \sigma/6}$ ,

$$\frac{|\Pi_I \Psi(I, \theta) - I|_2}{\rho'} \leq 8 \frac{K}{\alpha \rho'} \varepsilon \leq \frac{1}{32 \xi}, \quad \frac{|\Pi_\theta \Psi(I, \theta) - \theta|_\infty}{\sigma} \leq \frac{32 K}{3 \alpha \rho'} \varepsilon \leq \frac{1}{24 \xi}.$$

All domains, norms, projectors, and the meaning of  $(\alpha, K)$ -nonresonance modulo  $\Lambda$  are fixed in Definition K04.7–Definition K04.8.



*Proof.* Put  $\omega(I) = \partial_I h(I)$  and write

$$f = f^{\text{res}} + f^{\text{nr}} + f^{>K}, \quad f^{\text{res}} = \mathcal{P}_\Lambda \mathcal{P}_K f, \quad f^{\text{nr}} = (1 - \mathcal{P}_\Lambda) \mathcal{P}_K f, \quad f^{>K} = (1 - \mathcal{P}_K) f.$$

We use the weighted Fourier norm of the definition of resonant complex block and weighted Fourier norm above. It is a Banach algebra and, if  $0 < \delta < \sigma$ , Cauchy's inequalities give

$$\|\partial_\theta F\|_{\rho, \sigma - \delta} \leq \delta^{-1} \|F\|_{\rho, \sigma}, \quad \|\partial_I F\|_{\rho - d, \sigma} \leq d^{-1} \|F\|_{\rho, \sigma}. \quad (4.6.1)$$

For cutoff functions one may replace  $\delta^{-1}$  in the first inequality by  $K$ . All estimates below are componentwise and hence are unchanged, up to the fixed Euclidean/max-norm constants already absorbed in the numerical constants of the statement.

**Lemma 4.71 (weighted Fourier tail).** If  $F$  is analytic on width  $\sigma$ , then for  $0 < \delta < \sigma$ ,

$$\|(1 - \mathcal{P}_K)F\|_{\rho, \sigma - \delta} \leq e^{-K\delta} \|F\|_{\rho, \sigma}. \quad (4.6.2)$$

Indeed, for  $|k|_1 > K$  one has  $e^{|k|_1(\sigma - \delta)} \leq e^{-K\delta} e^{|k|_1\sigma}$ ; summing the Fourier coefficients proves (4.6.2).

**Lemma 4.72 (truncated nonresonant homological equation).** On the tube of action radius  $\rho'$  every cutoff divisor outside  $\Lambda$  has modulus at least  $\alpha/2$ , and

$$\chi(I, \theta) = \sum_{\substack{0 < |k|_1 \leq K \\ k \notin \Lambda}} \frac{F_k(I)}{\mathbf{i} k \cdot \omega(I)} e^{\mathbf{i} k \cdot \theta} \quad (4.6.3)$$

solves  $\{h, \chi\} = F^{\text{nr}}$  with  $\|\chi\| \leq 2\alpha^{-1} \|F^{\text{nr}}\|$ . To see the divisor bound, choose for a complex action  $I$  a real base point  $I_0 \in D_\Lambda$  with  $|I - I_0|_2 < \rho'$ . The Hessian bound and  $\rho' \leq \alpha/(2\xi MK)$  give, for  $0 < |k|_1 \leq K$ ,

$$|k \cdot (\omega(I) - \omega(I_0))| \leq KM\rho' \leq \frac{\alpha}{2\xi} < \frac{\alpha}{2}.$$

Thus  $|k \cdot \omega(I)| \geq \alpha/2$  whenever  $k \notin \Lambda$ , proving the claim. Formula (4.6.3) then follows coefficientwise from  $\{h, e^{\mathbf{i} k \cdot \theta}\} = \mathbf{i}(k \cdot \omega) e^{\mathbf{i} k \cdot \theta}$ .

**Lemma 4.73 (one controlled resonant Lie step).** Let  $H_* = h + Z + F$  with  $\mathcal{P}_\Lambda Z = Z$ ,  $\|F\| = e$ , and assume  $e \leq \alpha d/(32K)$  on an action collar  $d$  and an angle collar  $\delta$  with  $K\delta \geq 1$ . The time-one map  $\Phi = X_\chi^1$  generated by Lemma 4.72 is well defined on the domain shrunk by  $(d, \delta)$  and

$$H_* \circ \Phi = h + Z + \mathcal{P}_\Lambda \mathcal{P}_K F + F_+, \quad \|F_+\| \leq 8 \frac{K}{\alpha d} e^2 + 2e^{-K\delta} e. \quad (4.6.4)$$

Moreover the action displacement is at most  $4Ke/\alpha$  and the angle displacement is bounded by the corresponding Cauchy derivative of  $\chi$ . Use the integral Lie formula

$$H_* \circ \Phi = H_* + \{H_*, \chi\} + \int_0^1 (1-t) \{ \{H_*, \chi\}, \chi \} \circ X_\chi^t dt. \quad (4.6.5)$$

The homological equation cancels  $F^{\text{nr}}$ . The new retained term is  $\mathcal{P}_\Lambda \mathcal{P}_K F$ . Since Fourier supports in  $\Lambda$  are closed under addition, every retained increment is again  $\Lambda$ -resonant. The remaining terms contain either the Fourier tail or at least one factor  $X_\chi$ . Applying (4.6.1), the Banach-algebra inequality, and  $\|\chi\| \leq 2e/\alpha$  gives (4.6.4); the smallness assumption makes the Hamiltonian flow stay inside the unshrunk domain. Since  $\dot{I} = -\partial_\theta \chi$ , the action displacement bound follows from the cutoff estimate  $\|\partial_\theta \chi\| \leq 2Ke/\alpha$  and the factor two used to pass between nested domains. The angle bound follows similarly from  $\dot{\theta} = \partial_I \chi$  and differentiation of both  $F_k$  and the divisor.

**Lemma 4.74 (finite Lie iteration and resonant accumulation).** There are nested domains from  $(\rho', \sigma)$  to  $(\rho'/2, \sigma/6)$  and a finite sequence of Lemma 4.73 transformations for which the active low-mode nonresonant perturbations  $e_j$  satisfy

$$e_{j+1} \leq \frac{1}{2}e_j + 2e^{-K\sigma/6}\varepsilon, \quad \sum_j e_j \leq 2\varepsilon, \quad (4.6.6)$$

and the accumulated resonant correction obeys

$$\|g - g_0\| \leq 64 \frac{K}{\alpha\rho'} \varepsilon^2. \quad (4.6.7)$$

Choose the action losses geometrically with total less than  $\rho'/2$  and the angle losses with total less than  $5\sigma/6$ , while keeping each  $K\delta_j \geq 1$  until the last active low-mode error is below the target tail. The initial inequality  $\varepsilon \leq \alpha\rho'/(256\xi K)$  leaves a factor  $8\xi$  of slack in Lemma 4.73 at every stage. Inserting  $d_j \geq \rho'/8$  in (4.6.4), and subdividing a step whenever its quadratic contribution exceeds  $e_j/4$ , gives the first recurrence in (4.6.6). Summation of the quadratic resonant increments gives

$$\sum_j 8 \frac{K}{\alpha d_j} e_j^2 \leq 64 \frac{K}{\alpha\rho'} \varepsilon^2,$$

which is (4.6.7). This is the numerical bookkeeping used in Poeschel's finite normalization: the constants 256 and 64 are chosen precisely so that the domain and summability inequalities close simultaneously.

**Lemma 4.75 (exponential remainder).** The terminal perturbation of Lemma 4.74 may be chosen so that

$$\|f^*\|_{\rho'/2, \sigma/6} \leq \varepsilon e^{-K\sigma/6}. \quad (4.6.8)$$

At each stage all Fourier modes outside  $\Lambda$  up to  $K$  are removed. Equation (4.6.2) on the final strip contributes at most  $e^{-K\sigma/6}$  times the current perturbation, while the finite iteration is continued until its low-mode remainder is no larger than that same quantity. Because (4.6.6) is geometric and  $K\sigma \geq 6$ , the required number of steps fits inside the allocated strip loss. Absorbing the two equal terminal contributions by performing the last subdivision with half the target gives exactly (4.6.8).

**Lemma 4.76 (near-identity coordinate bounds).** The composition  $\Psi$  of the Lie maps satisfies

$$\frac{|\Pi_I \Psi - I|_2}{\rho'} \leq 8 \frac{K}{\alpha\rho'} \varepsilon, \quad \frac{|\Pi_\theta \Psi - \theta|_\infty}{\sigma} \leq \frac{32K}{3\alpha\rho'} \varepsilon. \quad (4.6.9)$$

For the action component sum the Lemma 4.73 bound over the geometric sequence (4.6.6). For the angle component differentiate (4.6.3) in  $I$ ; the derivative of the divisor contributes  $MK/\alpha$  and is converted, using  $\rho' \leq \alpha/(2\xi MK)$ , to the same  $1/\rho'$  scale as the derivative of  $F_k$ . Summing the allocated action collars and the geometric  $e_j$  gives the second constant in (4.6.9). The initial smallness then gives respectively  $1/(32\xi)$  and  $1/(24\xi)$ , so every intermediate map is defined on its assigned domain and their composition maps the final block into the original one.

Set  $g$  equal to the sum of all retained increments. Fourier support is contained in  $\Lambda$ , so  $\mathcal{P}_{\Lambda}g = g$  and  $\partial_{\theta}g \in \langle \Lambda \rangle_{\mathbb{R}}$ . Lemma 4.74 gives the estimate relative to  $g_0$ , Lemma 4.75 gives the exponentially small remainder, and Lemma 4.76 gives the two coordinate estimates. This proves the corrected Nekhoroshev-ready resonant normal form.  $\square$

## 4.4 Worked examples

### Worked example: an elliptic oscillator already in quartic Birkhoff normal form

Consider one degree of freedom with

$$H(q, p) = \frac{\omega}{2}(q^2 + p^2) + \frac{a}{4}(q^2 + p^2)^2, \quad \omega \neq 0.$$

With  $I = (q^2 + p^2)/2$  this is exactly

$$H = \omega I + aI^2 = \omega I + \frac{1}{2}AI^2, \quad A = 2a.$$

There are no angle-dependent terms to remove, so the identity map is already a Birkhoff normalization through every finite order represented by this polynomial. The frequency of the integrable motion is

$$\Omega(I) = \frac{dH}{dI} = \omega + 2aI,$$

and therefore

$$D\Omega(0) = 2a = A.$$

The elliptic twist condition is  $a \neq 0$ . This example makes the later KAM logic transparent: the Birkhoff coefficient is not decorative data; it measures how the rotation frequency changes when one moves away from the elliptic orbit.

### Boundary case: a resonant coupling that the homological equation cannot remove

Take two equal linear frequencies,

$$H_2 = \omega(I_1 + I_2), \quad \omega \neq 0.$$

For the complex monomial  $\xi_1\eta_2$  we have  $\alpha - \beta = (1, -1)$ , hence

$$(\alpha - \beta) \cdot (\omega, \omega) = 0.$$

Consequently

$$\{H_2, \xi_1\eta_2\} = 0.$$

If this monomial occurs in the perturbation, the equation

$$\{H_2, S\} = -\xi_1\eta_2$$

has no monomial solution obtained by division by a nonzero frequency divisor. The term must therefore remain in the resonant normal form. The failure is not a defect of the method: it is the mathematical signature of an actual resonance.

### Common pitfall

Do not infer analytic convergence from the existence of a Birkhoff normal form to every finite order. “For every  $N$  there exists a finite-order normalization” and “one analytic transformation normalizes to all orders” are different assertions.

## 4.5 Chapter summary

### What to remember

- The quadratic Hamiltonian makes the homological operator diagonal on complex monomials.
- Nonresonant monomials are removed by division; resonant monomials form the unavoidable normal-form part.
- Fourth-order Birkhoff invariants determine the local twist matrix and hence the first KAM nondegeneracy test near an elliptic point.
- Elliptic periodic orbits are treated through their symplectic Poincaré return maps and multiplier resonances.
- Formal Birkhoff series need not converge; finite-order and blockwise analytic normal forms are the reliable quantitative tools.
- The resonant-block theorem is the bridge from local normal forms to Nekhoroshev stability.

### Looking ahead

K05 uses the small-divisor and normal-form machinery to prove persistence rather than merely simplification. A Diophantine invariant torus survives a sufficiently small perturbation when the twist/nondegeneracy mechanism can correct the mean frequency error at every Newton step.

**Check your understanding**

Explain why a zero divisor forces a monomial to remain in resonant normal form, but does not by itself imply instability. Then explain why the quartic twist matrix is the quantity  $K_0$  needs in order to move the frequency back to its target value.

**4.6 Exercises and problems****Exercise 4.1**

**Difficulty.** Core.

**Related results.** Finite-order Hamiltonian Birkhoff normal form; Resonant Birkhoff normal form; Twist matrix from Birkhoff invariants.

**Problem.** Compute the quartic Birkhoff normal form of the elliptic equilibrium in the worked example, rewrite the surviving quartic term in the action, and extract the twist coefficient.

**Solution sketch.** Use complex/action coordinates, remove nonresonant monomials, and differentiate the resulting action polynomial.

**Exercise 4.2**

**Difficulty.** Core.

**Related results.** Normal form near a periodic orbit; Divergence caveat for formal normal forms; Nekhoroshev-ready resonant normal form.

**Problem.** For the resonant two-frequency oscillator, list the lowest-degree resonant monomials and prove from the homological operator that they cannot be eliminated. Explain why finite-order normal form does not imply convergence of the formal series.

**Solution sketch.** Compute the divisor  $(\alpha - \beta) \cdot \omega$  for each monomial and separate the finite algebraic statement from analytic convergence.

**Exercise 4.3**

**Difficulty.** Advanced.

**Related results.** Twist matrix from Birkhoff invariants.

**Problem.** Prove the monomial formula for  $\text{ad}_{H_2}$  and deduce the kernel/range decomposition into resonant and nonresonant monomials.

**Solution sketch.** Calculate one Poisson bracket and invert the scalar divisor whenever it is nonzero.

**Exercise 4.4**

**Difficulty.** Advanced.

**Related results.** Normal form near a periodic orbit.

**Problem.** Let a transverse Floquet multiplier of a periodic orbit be a root of unity. Exhibit the resulting resonance in the Poincaré-map homological equation and a monomial that cannot be removed.

**Solution sketch.** Compute the multiplier acquired by a monomial under conjuga-

tion and impose that it equals one.

#### Exercise 4.5

**Difficulty.** Advanced.

**Related results.** Finite-order Hamiltonian Birkhoff normal form; Resonant Birkhoff normal form.

**Problem.** For one primitive resonance, normalize through the first resonant order and separate the action-only normal form from the resonant-angle part used later in Nekhoroshev theory.

**Solution sketch.** Solve successive homological equations, retain the kernel, and rewrite it in a resonant angle.

# Kolmogorov KAM Theorem

## Why this chapter matters

An integrable Hamiltonian is filled with invariant tori, but a generic perturbation destroys the exact foliation. Kolmogorov's theorem says that a large arithmetic subset survives. The theorem is subtle because two obstructions interact at every scale: Fourier modes create divisors  $k \cdot \omega$  that may be very small, while correcting the torus changes its position and therefore its frequency. Diophantine arithmetic controls the first problem and Kolmogorov nondegeneracy controls the second.

The proof is best understood as a Newton scheme in phase space. One solves cohomological equations to remove the oscillatory defect, uses the averaged Hessian to correct the mean frequency defect, shrinks the complex domain, and repeats. The new defect is quadratic, so convergence beats the accumulated small-divisor losses. The a-posteriori theorem isolates this mechanism in its most reusable form: an approximately invariant torus satisfying torsion and arithmetic hypotheses can be corrected to a true one.

**Prerequisites.** K02 action-angle coordinates and Kolmogorov nondegeneracy, K03 homological equations and Lie transforms, K04 normal forms, and Foundation F05/F07/F08/F12 small-divisor and Newton machinery.

## Learning goals

- State precisely the Diophantine and twist hypotheses that compensate for small divisors.
- Follow one Newton step from torus defect to cohomological correction, mean-frequency correction, and quadratic error reduction.
- Distinguish a single surviving torus from the positive-measure Cantor family obtained by excluding resonances.
- Understand why the surviving family is naturally Whitney smooth rather than analytic across resonance gaps.
- Reduce the variational equation along a KAM torus to constant coefficients by solving a matrix cohomological equation.

### Conceptual roadmap

The core iteration is

approximate torus  $K \rightarrow E = X_H \circ K - \partial_\omega K$   
 $\rightarrow$  solve oscillatory cohomological equations  
 $\rightarrow$  use torsion/twist for the mean equation  
 $\rightarrow$  new torus with defect  $O(E^2)$ .

The classical Kolmogorov theorem constructs the initial approximate torus from a nearly integrable Hamiltonian. Parameter exclusion then assembles many target frequencies into a Cantor family; Whitney extension describes the smooth transverse dependence left after the resonance gaps are removed.

### Check your understanding

Suppose  $f(\theta) = \sum_{k \neq 0} f_k e^{ik \cdot \theta}$  has zero average. Write the Fourier solution of  $\omega \cdot \partial_\theta u = f$  and identify exactly where the Diophantine lower bound is used.

## 5.1 Definitions

### Intuition

KAM terminology is easiest to organize by the equation each object serves. The geometric object is a Lagrangian embedding  $K : \mathbb{T}^n \rightarrow M$ . It is an invariant torus with frequency  $\omega$  when

$$X_H \circ K = DK \omega.$$

For an approximate torus the difference between the two sides is the invariance defect  $E$ . An a-posteriori theorem starts from a small  $E$  and corrects the embedding; the classical nearly-integrable theorem first constructs such an approximate object from  $H = h + f$ .

The arithmetic layer belongs to the oscillatory part of the linearized equation. A Diophantine inequality for  $\omega$  bounds the divisors  $k \cdot \omega$  and therefore controls inversion of the zero-average cohomological equation. It does *not* solve the mean equation. That finite-dimensional obstruction is controlled by nondegeneracy: for an integrable action chart, Kolmogorov nondegeneracy is invertibility of  $D^2 h$ ; for an already parameterized approximate torus, the corresponding object is the averaged a-posteriori torsion matrix. The adapted frame packages tangent and symplectic-normal directions so that these two pieces of the Newton equation can be separated cleanly.

The parameter layer explains why persistence produces a Cantor family rather than an open foliation. A Kolmogorov frequency chart makes the action-frequency correspondence uniform on a compact set. Removing resonant frequencies leaves the KAM Cantor parameter set. The surviving embeddings can



nevertheless vary smoothly in the Whitney sense: one records compatible jets on the closed surviving set, or equivalently an extension when the definition is stated in extension form. The finite-order jet seminorms in the later definitions are the quantitative version needed to keep those parameter derivatives under control during iteration.

Two regularities must therefore never be conflated. Analyticity in the angle variable is measured on a complex strip and is spent when solving the small-divisor equations; Whitney regularity concerns how the surviving tori vary across a real parameter set with gaps. Likewise, reducibility of the variational equation is a statement about the linearized dynamics along an already constructed torus, not another definition of invariance.

**Key idea.** KAM closes two inversions at once: arithmetic controls the infinite-dimensional oscillatory equation, while twist/torsion controls the finite-dimensional mean equation. Parameter exclusion and Whitney jets then organize the collection of all tori for which both inversions remain uniform.

### Definition K05.1. Kolmogorov nondegeneracy

Let an integrable Hamiltonian be written in action variables as  $H_0(\theta, I) = h(I)$ . It is *Kolmogorov nondegenerate at  $I_*$*  if

$$\det D^2 h(I_*) \neq 0.$$

Equivalently, the differential of the frequency map  $I \mapsto Dh(I)$  is invertible at  $I_*$ ; hence, by the inverse-function theorem, the frequency map is a local diffeomorphism near  $I_*$ .

### Definition K05.2. KAM torus

For a Hamiltonian  $H$  on a symplectic phase space, a *KAM torus with frequency  $\omega \in \mathbb{R}^n$*  means in this volume a Lagrangian embedding  $K : \mathbb{T}^n \rightarrow M$  satisfying the invariance equation

$$X_H(K(\theta)) = DK(\theta)\omega.$$

Equivalently,  $K(\mathbb{T}^n)$  is invariant and  $\phi_H^t(K(\theta)) = K(\theta + \omega t)$ . The adjective KAM indicates that such a quasiperiodic torus is obtained or persisted by a KAM argument from a nearly integrable or approximately invariant configuration.

### Definition K05.3. Diophantine torus

A quasiperiodic invariant torus with frequency vector  $\omega \in \mathbb{R}^n$  is  *$(\gamma, \tau)$ -Diophantine* if, in the F05 convention,

$$|k \cdot \omega| \geq \gamma |k|_1^{-\tau} \quad \text{for every } 0 \neq k \in \mathbb{Z}^n.$$

A *Diophantine torus* is an invariant torus whose frequency satisfies such an inequality for specified  $\gamma > 0$  and exponent  $\tau$ .

**Definition K05.4. a-posteriori KAM theorem**

Given a candidate torus embedding  $K$  and a prescribed frequency  $\omega$ , its *invariance defect* is

$$E(\theta) := X_H(K(\theta)) - DK(\theta)\omega.$$

An *a-posteriori KAM theorem* is a persistence theorem whose hypotheses are quantitative properties of this already constructed approximate torus—typically analyticity, a Diophantine condition, geometric nondegeneracy/twist bounds, and smallness of  $E$ —and whose conclusion corrects  $K$  to an exact invariant torus, rather than assuming that the Hamiltonian itself is presented as a small perturbation of an integrable normal form.

**Definition K05.5. variational equation and reducibility along a quasiperiodic torus**

Let  $\dot{z} = X(z)$  be a smooth dynamical system and let  $z_*(t)$  be a solution. Its variational equation is the linearized equation  $\dot{v} = DX(z_*(t))v$ . Along a quasiperiodic torus with angle  $\theta(t) = \theta_0 + \omega t$ , the variational equation is analytically reducible on a strip if there is an invertible analytic matrix  $B(\theta)$  on that strip such that  $v = B(\theta(t))w$  changes it to a constant-coefficient linear equation. In a Hamiltonian variational problem the reducing gauge is called symplectic when each  $B(\theta)$  is a linear symplectic map. For a matrix-valued function  $A = (A_{ij})$  holomorphic on  $\mathbb{T}_s^n$ , we use the entrywise maximum strip norm  $\|A\|_s := \max_{i,j} \sup_{\theta \in \mathbb{T}_s^n} |A_{ij}(\theta)|$ .

**Definition K05.6. complex action-angle domain and Hamiltonian sup norm**

For  $I_0 \in \mathbb{R}^n$  and  $r > 0$ , write

$$D_r(I_0) := \{I \in \mathbb{C}^n : |I - I_0|_\infty < r\}.$$

Together with the analytic strip  $\mathbb{T}_s^n$  of the analytic-strip definition from Foundation F07, the product  $\mathbb{T}_s^n \times D_r(I_0)$  is the complex action-angle domain used in this volume. For a scalar Hamiltonian  $P$  holomorphic there,  $\|P\|_{s,r} := \sup_{(\theta, I) \in \mathbb{T}_s^n \times D_r(I_0)} |P(\theta, I)|$ . When  $I_0 = 0$  we abbreviate  $D_r = D_r(0)$ .

**Definition K05.7. fixed-frequency Kolmogorov normal form along a torus**

A Hamiltonian  $G$  is in fixed-frequency Kolmogorov normal form along the zero section with frequency  $\omega$  if, on a neighborhood of  $\mathbb{T}^n \times \{0\}$ ,

$$G(\theta, 0) = c, \quad \partial_I G(\theta, 0) = \omega$$

for a constant  $c$ . Equivalently, its Taylor expansion has the form  $G(\theta, I) = c + \langle \omega, I \rangle + O(|I|^2)$ . Hamilton's equations then give  $\dot{I} = 0$  and  $\dot{\theta} = \omega$  on  $I = 0$ .

**Definition K05.8. adapted frame for a canonical a-posteriori torus**

On the standard exact symplectic phase space with coordinates  $z = (q, p)$ , write

$$\mathbb{J} := \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}, \quad \Omega := -\mathbb{J},$$

so that  $\omega_{\text{st}}(u, v) = u^\top \Omega v$  and, with the K01 sign convention,  $X_H = \mathbb{J} \nabla H$ . For a holomorphic torus parameterization  $K$  put

$$L := DK, \quad G_K := L^\top L, \quad B_K := G_K^{-1}, \quad N_K := \mathbb{J} L B_K, \quad P_K := (L \ N_K).$$

Whenever  $G_K$  is invertible,  $P_K$  is called the *canonical adapted frame* of  $K$ . It satisfies  $L^\top \Omega N_K = I_n$  exactly. For a frequency  $\omega$  we write  $\partial_\omega F := DF \omega$ .

**Definition K05.9. a-posteriori torsion and torsion nondegeneracy**

For the adapted frame of Definition K05.8 define

$$\mathcal{X}_N := DX_H(K)N_K - \partial_\omega N_K, \quad T_K(\theta) := -N_K(\theta)^\top \Omega \mathcal{X}_N(\theta).$$

The matrix  $T_K$  is the *a-posteriori torsion matrix*. With normalized Haar average  $\langle F \rangle := \int_{\mathbb{T}^n} F(\theta) d\theta$ , the approximate torus is *torsion nondegenerate* when  $\langle T_K \rangle$  is invertible. Quantitative torsion data consist of bounds  $\|T_K\|_h \leq M_{T,0}$  and  $\|\langle T_K \rangle^{-1}\| \leq M_{T,1}$  on the indicated analytic strip.

**Definition K05.10. Kolmogorov frequency chart**

Let  $A \subset \mathbb{R}^n$  be a compact set of positive Lebesgue measure contained in the action domain of an integrable Hamiltonian  $h$ . We call  $A$  a *Kolmogorov frequency chart* when there is an open neighborhood  $U \supset A$  on which the frequency map

$$\nu : U \longrightarrow \nu(U), \quad \nu(I) := Dh(I),$$

is a  $C^1$  diffeomorphism. Quantitative chart data consist of a bound for  $\|D\nu\|$ , a bound for  $\|D\nu^{-1}\|$  on  $\nu(U)$ , and a fixed complex action neighborhood of  $A$  on which  $h$  is holomorphic. Kolmogorov nondegeneracy gives such charts locally; the point of fixing  $A$  is to make all constants uniform.

**Definition K05.11. KAM Cantor parameter set**

Let  $A$  be a Kolmogorov frequency chart with frequency map  $\nu = Dh$ . For  $\gamma > 0$  and  $\tau > n - 1$ , define

$$A_{\gamma,\tau} := \{I \in A : |k \cdot \nu(I)| \geq \gamma |k|_1^{-\tau} \text{ for every } 0 \neq k \in \mathbb{Z}^n\}.$$

We call  $A_{\gamma,\tau}$  the *KAM Cantor parameter set* in the chart  $A$ . Here “Cantor” has the standard KAM meaning of a closed resonance-excluded parameter set; in dimension  $n > 1$  it is not intended to assert that the set is topologically homeomorphic to the middle-thirds Cantor set.

**Definition K05.12. Whitney- $C^\infty$  family on a closed parameter set**

Let  $E \subset \mathbb{R}^m$  be closed and let  $\mathcal{B}$  be a Banach space. A map  $F : E \rightarrow \mathcal{B}$  is called *Whitney- $C^\infty$*  in this volume if there is an open neighborhood  $V \supset E$  and a  $C^\infty$  map  $\tilde{F} : V \rightarrow \mathcal{B}$  with  $\tilde{F}|_E = F$ . For a family of analytic torus embeddings we apply this definition with  $\mathcal{B} = \mathcal{A}_\rho$ , the Banach space of maps holomorphic on  $\mathbb{T}_\rho^n$  and continuous on its closure with the supremum norm, for every fixed reduced strip  $0 < \rho < s$ .

**Definition K05.13. Whitney-smooth KAM Cantor foliation**

Let  $E \subset A$  be a KAM Cantor parameter set. A family of invariant embeddings  $K_I : \mathbb{T}^n \rightarrow M$ ,  $I \in E$ , is a *Whitney-smooth KAM Cantor foliation* when  $I \mapsto K_I$  is Whitney- $C^\infty$  in the sense of Definition K05.12 on every fixed reduced analytic strip and the combined parameterization

$$\mathcal{K}(\theta, I) := K_I(\theta)$$

has a Whitney- $C^1$  extension whose derivative is everywhere invertible along  $\mathbb{T}^n \times E$ . Thus nearby leaves are locally disjoint and vary smoothly in the transverse action parameter, although  $E$  itself has resonance gaps.

**Definition K05.14. first integrals on a KAM Cantor family**

Let  $\mathcal{C} = \bigcup_{I \in E} K_I(\mathbb{T}^n)$  be a Whitney-smooth KAM Cantor foliation. A collection  $F_1, \dots, F_n$  is a system of *first integrals on the KAM Cantor family* if each  $F_j$  is the restriction to  $\mathcal{C}$  of a smooth function defined near  $\mathcal{C}$ , is constant along the Hamiltonian trajectories contained in  $\mathcal{C}$ , and the differentials  $dF_1, \dots, dF_n$  are independent on  $\mathcal{C}$ . The system is *in involution on  $\mathcal{C}$*  when  $\{F_i, F_j\} = 0$  there for every  $i, j$ .

**Definition K05.15. finite-order parameter-jet seminorm along a closed set**

Let  $E \subset U \subset \mathbb{R}^m$  with  $E$  closed, let  $\mathcal{B}$  be a Banach space, and let  $\tilde{F} : U \rightarrow \mathcal{B}$  be  $C^r$ . The *order- $r$  parameter-jet seminorm of the chosen extension  $\tilde{F}$  along  $E$*  is

$$|\tilde{F}|_{r,E;\mathcal{B}} := \max_{|\alpha| \leq r} \sup_{I \in E} \|\partial_I^\alpha \tilde{F}(I)\|_{\mathcal{B}}.$$

For analytic phase-space families on a strip we write  $|\tilde{F}|_{r,E,\rho}$  when  $\mathcal{B} = \mathcal{A}_\rho$ . This seminorm records the derivatives of a *specified smooth extension*; it is not claimed to be intrinsic to the trace  $F = \tilde{F}|_E$ . In the proof of Theorem K05.4, the extension is part of the Newton construction. Compatibility of the limiting jets and the existence of one common  $C^\infty$  extension are separate statements and are not inferred merely from finiteness of these seminorms.

**Definition K05.16. Whitney parameter jet and extension norm**

Let  $E \subset \mathbb{R}^m$  be compact, let  $\mathcal{B}$  be a Banach space, and let  $r \in \mathbb{N}$ . A *Whitney  $r$ -jet on  $E$*  is a collection  $\mathcal{F} = (F^\alpha)_{|\alpha| \leq r}$  of continuous maps  $F^\alpha : E \rightarrow \mathcal{B}$  such that, for every

$$|\alpha| \leq r,$$

$$F^\alpha(x) - \sum_{|\beta| \leq r-|\alpha|} \frac{F^{\alpha+\beta}(y)}{\beta!} (x-y)^\beta = o(|x-y|^{r-|\alpha|})$$

uniformly for  $x, y \in E$  as  $|x-y| \rightarrow 0$  (for  $|\alpha| = r$  this means uniform continuity of the top-order jet). Its *order- $r$  extension norm* is

$$\|\mathcal{F}\|_{\text{wh},r} := \inf_{\tilde{F}} \max_{|\alpha| \leq r} \sup_{x \in \mathbb{R}^m} \|\partial^\alpha \tilde{F}(x)\|_{\mathcal{B}},$$

where the infimum runs over  $C^r$  extensions  $\tilde{F} : \mathbb{R}^m \rightarrow \mathcal{B}$  satisfying  $\partial^\alpha \tilde{F}|_E = F^\alpha$  for every  $|\alpha| \leq r$ ; before the extension theorem is invoked the infimum is allowed to be  $+\infty$ . A compatible family of Whitney  $r$ -jets for every  $r$  is called a *Whitney  $C^\infty$  jet*. For analytic torus maps the Banach space is  $\mathcal{A}_\rho$  on each fixed reduced strip.

## 5.2 How the main proofs work

### Intuition

**A-posteriori KAM.** Start from an embedding  $K$  with small invariance defect. Pull the linearized equation into an adapted tangent/normal frame. The symplectic identities make the system nearly triangular. Solve the zero-average components with Diophantine Fourier estimates and the average component with the torsion matrix. Compose the correction, estimate the nonlinear Taylor remainder, and iterate on shrinking strips until the corrections are summable and the defect tends to zero.

**Kolmogorov analytic theorem.** Translate the target action to the zero section. The defect consists of the nonconstant value of the Hamiltonian on that section and the mismatch between its tangential frequency and the target vector. Two cohomological equations remove the oscillatory parts; the averaged action Hessian corrects the mean. The resulting canonical change produces a quadratic new defect. A Newton recurrence gives convergence to fixed-frequency Kolmogorov normal form.

**Positive-measure Cantor family.** Use the local inverse of the frequency map to parameterize candidate tori by frequency. Remove the resonance slabs where  $|k \cdot \omega|$  is too small. The Foundation measure estimates show that the total excluded measure is  $O(\gamma)$  (with the row's stated constants and geometry). Apply the one-torus theorem uniformly on the surviving set.

**Whitney smooth family.** Differentiate the KAM equations with respect to the surviving parameter and solve the differentiated cohomological equations with strengthened estimates. The jets satisfy Whitney compatibility on the closed Cantor set, so they admit smooth extensions even though no analytic continuation across the deleted resonant gaps is asserted.

**Linearized reducibility.** In Kolmogorov coordinates the variational equation is triangular with an angle-dependent symmetric block  $Q(\theta)$ . Solve  $\omega \cdot \partial_\theta U = Q - \bar{Q}$

entrywise and use the symplectic triangular gauge determined by  $U$  to replace  $Q(\theta)$  by its constant average.

### 5.3 Main results

#### Theorem K05.1. A-posteriori KAM theorem

**Statement.** Let  $n \geq 2$ , and let  $\mathcal{U} \subset \mathbb{C}^{2n}$  be an open complex neighborhood, invariant under complex conjugation, equipped with the standard exact symplectic form of Definition K05.8. Let  $H : \mathcal{U} \rightarrow \mathbb{C}$  be holomorphic and real on the real phase space, and suppose

$$\max_{0 \leq j \leq 2} \|D^j X_H\|_{\mathcal{U}} \leq M_H.$$

Let  $K : \mathbb{T}_h^n \rightarrow \mathcal{U}$  be holomorphic, real on the real torus, with real restriction an embedding. Put  $L := DK$  and assume that the complex bilinear Gram matrix  $G_K := L^\top L$  is invertible throughout  $\mathbb{T}_h^n$ . Define

$$E := X_H \circ K - \partial_\omega K, \quad B_K := G_K^{-1},$$

together with the adapted normal frame  $N_K$  of Definition K05.8 and the torsion  $T_K$  of Definition K05.9. Assume

$$\begin{aligned} |k \cdot \omega| &\geq \gamma |k|_1^{-\tau} \quad (0 \neq k \in \mathbb{Z}^n), \quad \tau \geq n-1, \\ \|L\|_h &\leq M_L, \quad \|B_K\|_h \leq M_B, \quad \|T_K\|_h \leq M_{T,0}, \quad \|\langle T_K \rangle^{-1}\| \leq M_{T,1}. \end{aligned}$$

and assume a positive complex-domain margin

$$d_0 := \text{dist}(K(\mathbb{T}_h^n), \partial\mathcal{U}) > 0.$$

Choose an embedding-stability radius  $r_{\text{emb}} > 0$  such that every real  $C^1$  map within  $r_{\text{emb}}$  of  $K|_{\mathbb{T}^n}$  is an embedding; such a radius exists because the source torus is compact and  $K|_{\mathbb{T}^n}$  is an embedding. Fix  $0 < \delta < h$  and abbreviate the quantitative data by

$$\mathcal{D}_{\text{ap}} := (n, \tau, \gamma, h, \delta, d_0, M_H, M_L, M_B, M_{T,0}, M_{T,1}, 1 + |\omega|).$$

Then there are constants

$$\varepsilon_* = \varepsilon_*(\mathcal{D}_{\text{ap}}, r_{\text{emb}}) > 0, \quad C_* = C_*(\mathcal{D}_{\text{ap}}),$$

such that, if  $\|E\|_h \leq \varepsilon_*$ , there exists a holomorphic map  $K_* : \mathbb{T}_{h-\delta}^n \rightarrow \mathcal{U}$ , real on the real torus and with real restriction an embedding, satisfying

$$X_H(K_*(\theta)) = DK_*(\theta) \omega, \quad \|K_* - K\|_{h-\delta} \leq C_* \|E\|_h.$$

Moreover  $K_*^* \omega_{\text{st}} = 0$ ; hence the real torus  $K_*(\mathbb{T}^n)$  is Lagrangian and is a KAM torus of frequency  $\omega$  in the sense of Definition K05.2.

*Proof.* We prove the theorem by a quasi-Newton iteration on the invariance equation

$$\mathcal{F}(K) := X_H \circ K - \partial_\omega K = 0. \quad (\text{AP.1})$$

Here *quasi-Newton* means that at each step we invert an explicitly solvable leading part of  $D\mathcal{F}(K)$  rather than the exact derivative; the difference between the two operators is itself proportional to the current defect, so the new residual is still quadratic. The important point is that this leading equation is not solved in the ambient coordinates. Instead we use the tangent/normal frame canonically attached to the current approximate torus. In that frame the linearized equation is upper triangular up to an error controlled by the old defect. We call this defect-generated triangularization the *automatic reducibility mechanism*; no separate exact reducibility theorem is being invoked.

Throughout the proof all vector and matrix norms are fixed max norms. Constants  $C$  may increase from line to line but depend only on the data displayed in the theorem (including the fixed number  $1+|\omega|$ ) and on  $n, \tau$ ; they never depend on the current defect. Put

$$\kappa := \tau + n. \quad (\text{AP.2})$$

We first record the only arithmetic estimate used in the proof. If  $f$  has zero average on  $\mathbb{T}_\rho^n$ , then

$$\partial_\omega u = f, \quad \langle u \rangle = 0 \quad (\text{AP.3})$$

has a unique solution on  $\mathbb{T}_{\rho-\sigma}^n$  and

$$\|u\|_{\rho-\sigma} \leq C\gamma^{-1}\sigma^{-\kappa}\|f\|_\rho. \quad (\text{AP.4})$$

For self-containment, write  $f(\theta) = \sum_{k \neq 0} f_k e^{2\pi i k \cdot \theta}$  and set  $u_k = f_k / (2\pi i k \cdot \omega)$ . Analyticity gives  $|f_k| \leq \|f\|_\rho e^{-2\pi\rho|k|_1}$ , while the Diophantine bound gives  $|k \cdot \omega|^{-1} \leq \gamma^{-1}|k|_1^\tau$ . Summing on lattice shells,  $\#\{k : |k|_1 = m\} \leq C_n m^{n-1}$ , yields  $\sum_{k \neq 0} |k|_1^\tau e^{-2\pi\sigma|k|_1} \leq C_{n,\tau} \sigma^{-(\tau+n)}$  and hence (AP.4). The zero-average normalization gives uniqueness.

**lemma Lemma 5.3 (approximate isotropicity and the adapted frame).** Let  $K$  be one approximate torus on  $\mathbb{T}_\rho^n$ , let  $L = DK$ , and put

$$\Omega_K := L^\top \Omega L. \quad (\text{AP.5})$$

If  $0 < 2\sigma < \rho$ , then

$$\langle \Omega_K \rangle = 0, \quad \|\Omega_K\|_{\rho-2\sigma} \leq C\gamma^{-1}\sigma^{-(\kappa+1)}\|E\|_\rho. \quad (\text{AP.6})$$

Consequently, if  $G_K = L^\top L$  is invertible and  $B = G_K^{-1}$ , the frame

$$N = \mathbb{J}LB, \quad P = (L \ N) \quad (\text{AP.7})$$

satisfies

$$L^\top \Omega N = I, \quad N^\top \Omega N = B\Omega_K B, \quad (\text{AP.8})$$

and hence

$$P^\top \Omega P = \begin{pmatrix} \Omega_K & I \\ -I & B\Omega_K B \end{pmatrix}. \quad (\text{AP.9})$$

In particular  $P$  is invertible, with a uniformly bounded inverse, whenever the right-hand side of (AP.6) is sufficiently small.

To prove the lemma, first use exactness of the ambient symplectic form. If  $\omega_{\text{st}} = d\alpha$  and  $K^*\alpha = \sum_j a_j(\theta) d\theta_j$ , then

$$(\Omega_K)_{ij} = \partial_{\theta_i} a_j - \partial_{\theta_j} a_i.$$

The normalized torus average of each derivative of a periodic function is zero, so  $\langle \Omega_K \rangle = 0$ . This is the first assertion of (AP.6). Next put  $A(\theta) = DX_H(K(\theta))$ . Differentiating the defect gives

$$DE = AL - \partial_\omega L. \quad (\text{AP.10})$$

The Hamiltonian identity

$$A^\top \Omega + \Omega A = 0 \quad (\text{AP.11})$$

follows by differentiating  $\Omega X_H = \nabla H$ . Hence

$$\begin{aligned} \partial_\omega \Omega_K &= (\partial_\omega L)^\top \Omega L + L^\top \Omega \partial_\omega L \\ &= -(DE)^\top \Omega L - L^\top \Omega DE. \end{aligned} \quad (\text{AP.12})$$

Cauchy's estimate gives  $\|DE\|_{\rho-\sigma} \leq C\sigma^{-1}\|E\|_\rho$ . Equation (AP.12), the zero-average property and (AP.4) applied entrywise yield (AP.6). Finally, (AP.8) follows directly from  $\Omega = -\mathbb{J}$  and  $\mathbb{J}^2 = -I$ :

$$L^\top \Omega N = L^\top (-\mathbb{J}) \mathbb{J} L B = L^\top L B = I,$$

while  $N^\top \Omega N = B \Omega_K B$ . If

$$\Omega_0 := \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \quad Q := -\Omega_0 P^\top \Omega, \quad (\text{AP.13})$$

then (AP.9) gives

$$QP = I + S, \quad S = \begin{pmatrix} 0 & -B \Omega_K B \\ \Omega_K & 0 \end{pmatrix}. \quad (\text{AP.14})$$

Thus  $\|S\| < 1/2$  implies that  $QP$ , and therefore  $P$ , is invertible; moreover  $P^{-1} = (I + S)^{-1}Q$ . This proves the lemma.

**lemma Lemma 5.4 (approximate automatic reducibility and triangular solve).** Assume the hypotheses of the preceding lemma and define

$$R_L := DX_H(K)L - \partial_\omega L = DE, \quad R_N := DX_H(K)N - \partial_\omega N, \quad (\text{AP.15})$$

and the torsion

$$T := -N^\top \Omega R_N. \quad (\text{AP.16})$$

Then

$$Q(DX_H(K)P - \partial_\omega P) = \Lambda + \mathcal{R}, \quad \Lambda := \begin{pmatrix} 0 & T \\ 0 & 0 \end{pmatrix}, \quad (\text{AP.17})$$

where, after one Cauchy loss,

$$\|\mathcal{R}\|_{\rho-\sigma} \leq C\sigma^{-1}\|E\|_\rho. \quad (\text{AP.18})$$



Moreover, if  $\langle T \rangle$  is invertible,  $\|T\| \leq M_{T,0}$  and  $\|\langle T \rangle^{-1}\| \leq M_{T,1}$ , the triangular equation

$$\Lambda \xi - \partial_\omega \xi = -QE \quad (\text{AP.19})$$

has a solution on  $\mathbb{T}_{\rho-3\sigma}^n$  satisfying

$$\|\xi\|_{\rho-3\sigma} \leq C\gamma^{-2}\sigma^{-2\kappa}\|E\|_\rho, \quad \|D\xi\|_{\rho-4\sigma} \leq C\gamma^{-2}\sigma^{-(2\kappa+1)}\|E\|_\rho. \quad (\text{AP.20})$$

For (AP.17), only the lower-right block is not immediate. Since  $L^\top \Omega N = I$ , differentiation along  $\omega$  gives

$$(\partial_\omega L)^\top \Omega N + L^\top \Omega \partial_\omega N = 0. \quad (\text{AP.21})$$

Using (AP.11) and (AP.10),

$$L^\top \Omega R_N = -(DX_H(K)L)^\top \Omega N - L^\top \Omega \partial_\omega N = -(DE)^\top \Omega N. \quad (\text{AP.22})$$

Together with  $R_L = DE$ , this yields the exact block formula

$$\mathcal{R} = \begin{pmatrix} -N^\top \Omega DE & 0 \\ L^\top \Omega DE & -(DE)^\top \Omega N \end{pmatrix}, \quad (\text{AP.23})$$

and (AP.18) follows by Cauchy.

It remains to solve (AP.19). Write  $\xi = (\xi_L, \xi_N)$  and

$$-QE =: \eta = (\eta_L, \eta_N) = (N^\top \Omega E, -L^\top \Omega E). \quad (\text{AP.24})$$

The compatibility condition  $\langle \eta_N \rangle = 0$  is automatic. Indeed, using  $\Omega X_H = \nabla H$ ,

$$L^\top \Omega E = D(H \circ K)^\top - \Omega_K \omega, \quad (\text{AP.25})$$

and both terms have zero average: the first is a derivative of a periodic function and the second has zero average by (AP.6). Let  $u_N$  be the zero-average solution of

$$\partial_\omega u_N = -\eta_N. \quad (\text{AP.26})$$

Set

$$c_N := \langle T \rangle^{-1} (\langle \eta_L \rangle - \langle T u_N \rangle), \quad \xi_N := c_N + u_N. \quad (\text{AP.27})$$

Then  $-\eta_L + T\xi_N$  has zero average, so there is a unique zero-average  $\xi_L$  solving

$$\partial_\omega \xi_L = -\eta_L + T\xi_N. \quad (\text{AP.28})$$

Equations (AP.4), (AP.26)-(AP.28), the uniform bounds on  $P, Q, T$  and the inverse-torsion bound give the first estimate in (AP.20); one additional Cauchy estimate gives the second. This proves the lemma.

**lemma Lemma 5.5 (one a-posteriori Newton step).** There are constants  $C > 0$  and an integer

$$m := 4\kappa + 6 \quad (\text{AP.29})$$

with the following property. Suppose  $K$  is defined on  $\mathbb{T}_\rho^n$ , the analytic/geometric bounds

of the theorem hold with fixed slack, and  $0 < 5\sigma < \rho$ . If

$$C\gamma^{-2}\sigma^{-(2\kappa+2)}\|E\|_\rho < 1, \quad (\text{AP.30})$$

then the correction

$$\Delta K := P\xi, \quad K_+ := K + \Delta K, \quad (\text{AP.31})$$

constructed in Lemma 5.4 is defined on  $\mathbb{T}_{\rho-5\sigma}^n$ , remains in the phase domain, is real on the real torus, and satisfies

$$\|\Delta K\|_{\rho-4\sigma} \leq C\gamma^{-2}\sigma^{-2\kappa}\|E\|_\rho, \quad (\text{AP.32})$$

$$\|D\Delta K\|_{\rho-5\sigma} \leq C\gamma^{-2}\sigma^{-(2\kappa+1)}\|E\|_\rho, \quad (\text{AP.33})$$

and

$$\|E_+\|_{\rho-5\sigma} \leq C\gamma^{-4}\sigma^{-m}\|E\|_\rho^2, \quad E_+ := X_H \circ K_+ - \partial_\omega K_+. \quad (\text{AP.34})$$

The Gram inverse and averaged torsion remain invertible, with bounds enlarged by at most a fixed factor, if the left side of (AP.30) is made sufficiently small.

Indeed, (AP.32)–(AP.33) follow from (AP.20) and the uniform bounds for  $P$  and  $DP$ . The linearized residual is

$$\mathcal{L}_K \Delta K + E, \quad \mathcal{L}_K \Delta := DX_H(K)\Delta - \partial_\omega \Delta. \quad (\text{AP.35})$$

Multiplying by  $Q$  and using (AP.14), (AP.17) and the core equation (AP.19) gives

$$Q(\mathcal{L}_K \Delta K + E) = \mathcal{R}\xi - S\partial_\omega \xi. \quad (\text{AP.36})$$

The approximate-isotropy estimate (AP.6), (AP.18), and (AP.20) show that the right-hand side is bounded by  $C\gamma^{-2}\sigma^{-(2\kappa+2)}\|E\|_\rho^2$  on the five-times shrunk strip. Because  $QP = I + S$  and  $\|S\| < 1/2$ , the square matrices  $Q$  and  $P$  are invertible and  $Q^{-1} = P(I + S)^{-1}$ . Hence  $Q^{-1}$  is uniformly bounded, so the same quadratic bound holds for (AP.35). Taylor's formula for the vector field gives

$$X_H(K + \Delta K) = X_H(K) + DX_H(K)\Delta K + \mathcal{N}, \quad \|\mathcal{N}\| \leq \frac{1}{2}M_H\|\Delta K\|^2. \quad (\text{AP.37})$$

Combining (AP.32), (AP.35)–(AP.37) and enlarging the harmless exponent to  $m = 4\kappa + 6$  gives (AP.34).

We finally check the geometric data used at the next step. From (AP.33),  $L_+ - L = D\Delta K$  is small. Hence  $G_{K_+} - G_K = L_+^\top L_+ - L^\top L$  is small. More explicitly,

$$G_{K_+} = G_K(I + B_K(G_{K_+} - G_K)).$$

If  $\|B_K(G_{K_+} - G_K)\| < 1/2$ , the factor in parentheses has inverse  $\sum_{m \geq 0} [-B_K(G_{K_+} - G_K)]^m$ ; hence  $G_{K_+}$  remains invertible and  $B_{K_+}$  has a uniformly enlarged bound. The normal-frame correction is explicit:

$$N_{K_+} - N_K = \mathbb{J}((L_+ - L)B_{K_+} + L(B_{K_+} - B_K)), \quad (\text{AP.38})$$

so it has the same first-order bound as  $D\Delta K$ . Put  $A_K := DX_H(K)$ . Since

$$R_{N,+} - R_N = (A_{K_+} - A_K)N_{K_+} + A_K(N_{K_+} - N_K) - \partial_\omega(N_{K_+} - N_K), \quad (\text{AP.39})$$

the bound on  $D^2X_H$ , one additional Cauchy loss, and the fixed factor  $1 + |\omega|$  give

$$\|R_{N,+} - R_N\| \leq C\gamma^{-2}\sigma^{-(2\kappa+2)}\|E\|_\rho. \quad (\text{AP.40})$$

On  $\mathbb{T}_{\rho-5\sigma}^n$  the new normal frame is bounded on the slightly larger strip  $\mathbb{T}_{\rho-4\sigma}^n$ , so Cauchy's estimate and  $R_{N,+} = A_{K_+}N_{K_+} - \partial_\omega N_{K_+}$  give  $\|R_{N,+}\| \leq C\sigma^{-1}$ . Consequently, expanding  $T = -N^\top \Omega R_N$  gives

$$\|T_{K_+} - T_K\| \leq \|N_{K_+} - N_K\| \|\Omega\| \|R_{N,+}\| + \|N_K\| \|\Omega\| \|R_{N,+} - R_N\| \leq C\gamma^{-2}\sigma^{-(2\kappa+2)}\|E\|_\rho. \quad (\text{AP.41})$$

Thus  $\|\langle T_{K_+} \rangle - \langle T_K \rangle\|$  is small. Writing

$$\langle T_{K_+} \rangle = \langle T_K \rangle \left( I + \langle T_K \rangle^{-1} (\langle T_{K_+} \rangle - \langle T_K \rangle) \right),$$

and requiring the second factor to differ from  $I$  by norm less than  $1/2$  shows by the same convergent geometric series that the averaged torsion remains invertible with a uniformly enlarged inverse bound. The Fourier solutions in (AP.26) and (AP.28) preserve the reality condition, so  $K_+$  is real on the real torus. The  $C^0$  estimate (AP.32) also keeps the complex image a positive distance from  $\partial\mathcal{U}$ . This proves the one-step lemma.

**4. Iteration.** Fix the loss  $\delta$  from the theorem and define

$$\sigma_j := \frac{\delta}{10} 2^{-j}, \quad \rho_0 := h, \quad \rho_{j+1} := \rho_j - 5\sigma_j. \quad (\text{AP.42})$$

Since  $\sum_{j \geq 0} 5\sigma_j = \delta$ , one has  $\rho_j \downarrow h - \delta$ .

Let  $K_0 = K$  and  $\varepsilon_j = \|E_j\|_{\rho_j}$ . As long as the one-step hypotheses hold, (AP.34) gives

$$\varepsilon_{j+1} \leq A\gamma^{-4}\sigma_j^{-m}\varepsilon_j^2 = a_0 2^{mj}\varepsilon_j^2, \quad a_0 := A\gamma^{-4}\sigma_0^{-m}. \quad (\text{AP.43})$$

Define

$$z_j := a_0 2^{m(j+1)}\varepsilon_j. \quad (\text{AP.44})$$

A direct substitution in (AP.43) yields

$$z_{j+1} \leq z_j^2. \quad (\text{AP.45})$$

Choose the initial threshold so that  $z_0 \leq 1/4$  and so that all the smallness inequalities in Lemmas 5.3–5.5 have a factor-two margin. Then  $z_j \leq z_0^{2^j}$ , hence  $\varepsilon_j$  tends to zero doubly exponentially. In particular the series of corrections and their first derivatives converge absolutely on the common strip  $\mathbb{T}_{h-\delta}^n$ , because (AP.32)–(AP.33) give

$$\sum_{j \geq 0} \|K_{j+1} - K_j\|_{h-\delta} \leq C\|E_0\|_h, \quad \sum_{j \geq 0} \|D(K_{j+1} - K_j)\|_{h-\delta} < \infty. \quad (\text{AP.46})$$

The constants here contain the fixed factors  $\gamma^{-2}$  and powers of  $\delta^{-1}$ . The same summability, together with (AP.38)–(AP.41), preserves the Gram-inverse, torsion-inverse and phase-domain margins for every step, so the induction is legitimate.

Thus  $K_j$  converges uniformly on  $\mathbb{T}_{h-\delta}^n$  to a holomorphic map  $K_*$  and  $DK_j$  converges uniformly there to  $DK_*$ . Passing to the limit in  $E_j = X_H \circ K_j - \partial_\omega K_j$  gives

$$X_H \circ K_* = DK_* \omega. \quad (\text{AP.47})$$

Equation (AP.46) gives the advertised correction estimate

$$\|K_* - K\|_{h-\delta} \leq C_* \|E\|_h. \quad (\text{AP.48})$$

The real restriction of  $K_*$  is an embedding. Indeed, embeddings of the compact torus into the real phase space form an open subset of the  $C^1$  space of maps. Use the radius  $r_{\text{emb}}$  fixed in the statement. By decreasing  $\varepsilon_*$ , the  $C^1$  version of (AP.46) places  $K_*$  inside this radius. Thus the claimed real embedding property is preserved quantitatively.

**5. The limiting torus is Lagrangian.** Apply Lemma 5.3 to the exact solution  $K_*$ . Its defect is zero, so (AP.12) gives  $\partial_\omega \Omega_{K_*} = 0$ , while exactness still gives  $\langle \Omega_{K_*} \rangle = 0$ . In Fourier coefficients,

$$2\pi i(k \cdot \omega) \widehat{\Omega}_{K_*}(k) = 0. \quad (\text{AP.49})$$

The Diophantine condition implies  $k \cdot \omega \neq 0$  for every  $k \neq 0$ ; therefore all nonzero Fourier coefficients vanish, and the zero Fourier coefficient vanishes by the average condition. Hence

$$K_*^* \omega_{\text{st}} = 0. \quad (\text{AP.50})$$

Since  $K_*$  has dimension  $n$  in a  $2n$ -dimensional symplectic phase space, it is Lagrangian. Finally, (AP.47) is exactly the invariance equation of the definition of KAM torus above, so the Hamiltonian flow restricted to the torus is conjugate by  $K_*$  to the rigid translation with frequency  $\omega$ . Every clause of the statement has now been proved.  $\square$

### Theorem K05.2. Kolmogorov analytic KAM theorem

**Statement.** Let  $D \subset \mathbb{R}^n$  be open, let  $I_0 \in D$ , and choose  $s, r > 0$  so that the complex action domain  $D_r(I_0)$  of Definition K05.6 lies in a complex neighborhood on which  $h$  is holomorphic. Let

$$H(\theta, I) = h(I) + P(\theta, I)$$

be real analytic on  $\mathbb{T}_s^n \times D_r(I_0)$ , with norm  $\|P\|_{s,r}$  as in Definition K05.6. Put  $A_0 = D^2 h(I_0)$  and  $\omega_0 = Dh(I_0)$ . Assume  $A_0$  is invertible and

$$|k \cdot \omega_0| \geq \gamma |k|_1^{-\tau} \quad (0 \neq k \in \mathbb{Z}^n)$$

for some  $\gamma > 0$  and  $\tau > n - 1$ . There exists  $\varepsilon_0 > 0$ , depending only on  $n, \tau, s, r, \gamma$ , an analytic bound for  $h$  on the chosen complex action neighborhood, and a bound for  $\|A_0^{-1}\|$ , such that  $\|P\|_{s,r} < \varepsilon_0$  implies the existence of a real-analytic Lagrangian embedding  $K : \mathbb{T}^n \rightarrow \mathbb{T}^n \times D_r(I_0)$  admitting a holomorphic extension to  $\mathbb{T}_{s/2}^n$  with

values in  $\mathbb{T}_s^n \times D_r(I_0)$  and satisfying

$$X_H(K(\theta)) = DK(\theta)\omega_0$$

(on the real torus, hence throughout the strip by analytic continuation). Consequently  $K(\mathbb{T}^n)$  is an invariant Lagrangian torus and the restricted Hamiltonian flow is analytically conjugate by  $K$  to the rigid translation  $\theta \mapsto \theta + \omega_0 t$ .

*Proof.*

### Proof strategy

Iterate a quantitative analytic KAM step. At each scale solve the homological equation on the nonresonant Fourier modes, adjust the constant/frequency part so that the target torus remains fixed, and use twist nondegeneracy to control the parameter correction. The new perturbation is quadratic up to an exponentially small Fourier tail. A superconvergent choice of cutoffs makes the conjugacies and parameter corrections summable, while the Diophantine exclusions give the Cantor-set measure estimate.

We write the proof for one prescribed Diophantine frequency. This distinction matters: because no frequency parameter has to be continued through the iteration, every homological equation below can be solved by its full Fourier series. No Fourier cutoff or parameter-exclusion argument is hidden in the proof.

**1. Normalization and the torus defect.** The constant action translation  $I \mapsto I + I_0$  is symplectic, so we may assume from now on that  $I_0 = 0$ . Put

$$\omega := \omega_0 = Dh(0), \quad A_0 := D^2h(0). \quad (\text{K.1})$$

For  $0 < s' \leq s$  and  $0 < r' \leq r$  write

$$\mathcal{D}(s', r') := \mathbb{T}_{s'}^n \times D_{r'} \quad (\text{K.2})$$

and use supremum norms there. All vector and matrix norms are fixed finite-dimensional max norms; changing them only changes constants.

For a real-analytic Hamiltonian  $G$  on a neighborhood of  $\mathbb{T}^n \times \{0\}$  define its *fixed-frequency torus defect* by

$$\begin{aligned} c_G &:= \int_{\mathbb{T}^n} G(\theta, 0) d\theta, \\ a_G(\theta) &:= G(\theta, 0) - c_G, \\ b_G(\theta) &:= \partial_I G(\theta, 0) - \omega. \end{aligned} \quad (\text{K.3})$$

Here and below Haar measure on the real torus is normalized to have mass one. Thus  $a_G$  has zero average. The defect vanishes exactly when

$$G(\theta, 0) = c_G, \quad \partial_I G(\theta, 0) = \omega, \quad (\text{K.4})$$

which is precisely the normal form of the definition of fixed-frequency Kolmogorov

normal form along a torus above. In that case Hamilton's equations give

$$\dot{I} = -\partial_\theta G(\theta, 0) = 0, \quad \dot{\theta} = \partial_I G(\theta, 0) = \omega, \quad (\text{K.5})$$

so  $I = 0$  is already the desired invariant torus.

We shall make the norm of  $(a_G, b_G)$  tend to zero by symplectic changes of coordinates. The only nontrivial arithmetic input is the following elementary continuous-time cohomological estimate.

**lemma Lemma 5.1 (continuous-time analytic cohomology).** Let  $f$  be scalar- or vector-valued, holomorphic on  $\mathbb{T}_\rho^n$ , real on the real torus, and of zero average. If

$$|k \cdot \omega| \geq \gamma |k|_1^{-\tau} \quad (k \neq 0), \quad (\text{K.6})$$

then for every  $0 < \sigma < \min\{\rho, 1\}$  there is a unique zero-average solution of

$$\omega \cdot \partial_\theta u = f \quad (\text{K.7})$$

on  $\mathbb{T}_{\rho-\sigma}^n$ , and with  $\kappa := \tau + n$  one has

$$\|u\|_{\rho-\sigma} \leq C_0 \gamma^{-1} \sigma^{-\kappa} \|f\|_\rho, \quad \|\partial_\theta u\|_{\rho-2\sigma} \leq C_0 \gamma^{-1} \sigma^{-(\kappa+1)} \|f\|_\rho. \quad (\text{K.8})$$

The constant  $C_0$  depends only on  $n$  and  $\tau$ .

Indeed, with Fourier convention  $f(\theta) = \sum_k f_k e^{2\pi i k \cdot \theta}$ , set

$$u_0 = 0, \quad u_k = \frac{f_k}{2\pi i k \cdot \omega} \quad (k \neq 0). \quad (\text{K.9})$$

Analytic Fourier decay and (K.6) give

$$|u_k| \leq (2\pi\gamma)^{-1} |k|_1^\tau \|f\|_\rho e^{-2\pi\rho|k|_1}. \quad (\text{K.10})$$

The lattice-shell count  $\#\{k : |k|_1 = m\} \leq C_n m^{n-1}$  then yields

$$\sum_{k \neq 0} |k|_1^\tau e^{-2\pi\sigma|k|_1} \leq C_{n,\tau} \sigma^{-(\tau+n)}. \quad (\text{K.11})$$

This proves the first estimate. Applying Cauchy's estimate on one further  $\sigma$ -loss proves the derivative estimate. The Fourier formula gives reality on the real torus, and uniqueness follows because every nonzero divisor  $k \cdot \omega$  is nonzero. The same argument applies componentwise to vector-valued data. This proves the lemma.

**2. One Newton-Kolmogorov step.** We now prove the specialist step that will be iterated. Suppose that  $G$  is real analytic on  $\mathcal{D}(\rho, R)$  and set

$$a := a_G, \quad b := b_G, \quad A(\theta) := \partial_I^2 G(\theta, 0), \quad \bar{A} := \int_{\mathbb{T}^n} A(\theta) d\theta. \quad (\text{K.12})$$

Assume on this step that

$$\|a\|_\rho + \|b\|_\rho \leq \varepsilon, \quad \|A\|_\rho \leq M, \quad \|\bar{A}^{-1}\| \leq \mu, \quad (\text{K.13})$$

and that the derivatives of  $G$  through order three are bounded by  $M$  on the slightly larger domain on which the Cauchy estimates below are taken. The symbol  $M$  may be enlarged from line to line by a universal factor; in the iteration it will be held fixed once and for all.

Choose positive losses  $\sigma$  in angle and  $\varrho$  in action, with  $5\sigma < \rho$  and  $5\varrho < R$ , and put

$$d := \min\{\sigma, \varrho, 1\}. \quad (\text{K.14})$$

All constants below may depend on  $n, \tau, \gamma, M, \mu$  and on an upper bound for  $R$ , but not on  $\varepsilon, \sigma, \varrho$ .

First solve

$$\omega \cdot \partial_\theta u = a, \quad [u] = 0, \quad (\text{K.15})$$

where brackets denote angle average. By Lemma 5.1,

$$\|u\|_{\rho-\sigma} + d\|\partial_\theta u\|_{\rho-2\sigma} \leq Cd^{-\kappa}\varepsilon. \quad (\text{K.16})$$

The averaged twist matrix is used only once, to remove the mean frequency defect. Define the constant vector

$$\xi := \bar{A}^{-1}([A\partial_\theta u] - [b]). \quad (\text{K.17})$$

Then

$$|\xi| \leq Cd^{-(\kappa+1)}\varepsilon. \quad (\text{K.18})$$

The vector-valued function

$$g := b - A\partial_\theta u + A\xi \quad (\text{K.19})$$

has zero average by (K.17). Hence there is a unique zero-average vector  $v$  satisfying

$$\omega \cdot \partial_\theta v = g. \quad (\text{K.20})$$

Two applications of the cohomological estimate, together with (K.16)–(K.19), give, after harmlessly enlarging  $C$ ,

$$\|v\|_{\rho-3\sigma} + d\|\partial_\theta v\|_{\rho-4\sigma} \leq Cd^{-(2\kappa+2)}\varepsilon. \quad (\text{K.21})$$

Let

$$F(\theta, I) := u(\theta) + v(\theta) \cdot I \quad (\text{K.22})$$

and denote its Hamiltonian flow by  $\Phi_F^t$ . Let  $T_{t\xi}(\theta, I) = (\theta, I + t\xi)$  and consider the symplectic isotopy

$$\Psi_t := T_{t\xi} \circ \Phi_F^t, \quad 0 \leq t \leq 1. \quad (\text{K.23})$$

Both factors are symplectic; they preserve the real phase space because  $u, v, \xi$  are real on real arguments. From (K.16), (K.18), (K.21) and Cauchy's estimate there is an exponent

$$p := 2\kappa + 4 = 2(\tau + n) + 4 \quad (\text{K.24})$$

such that

$$\sup_{0 \leq t \leq 1} (\|\Psi_t - \text{id}\|_{C^0} + d\|D\Psi_t - I\|_{C^0}) \leq Cd^{-p}\varepsilon \quad (\text{K.25})$$

on  $\mathcal{D}(\rho - 5\sigma, R - 5\rho)$ , provided

$$Cd^{-p}\varepsilon \leq \frac{1}{4}d. \quad (\text{K.26})$$

To justify (K.25) rather than merely quote a flow estimate, note that

$$X_F(\theta, I) = (v(\theta), -\partial_\theta u(\theta) - (D_\theta v(\theta))^T I), \quad (\text{K.27})$$

so (K.16), (K.21) bound  $X_F$  and its first derivative on the four-times shrunk strip. The elementary integral equation  $\Phi_F^t(z) = z + \int_0^t X_F(\Phi_F^q(z)) dq$  and Gronwall's inequality then give (K.25). Condition (K.26) keeps every trajectory inside the domain on which those estimates were obtained.

It remains to show that this correction is a Newton correction for the whole torus defect, not merely a formal cancellation. Define the jet map

$$\mathcal{J}(G) := (a_G, b_G). \quad (\text{K.28})$$

Split

$$G = G^0 + a(\theta) + b(\theta) \cdot I, \quad (\text{K.29})$$

so that

$$G^0(\theta, 0) = c_G, \quad \partial_I G^0(\theta, 0) = \omega, \quad \partial_I^2 G^0(\theta, 0) = A(\theta). \quad (\text{K.30})$$

We use the book Poisson-bracket convention

$$\{F, G\} = \partial_\theta G \cdot \partial_I F - \partial_I G \cdot \partial_\theta F, \quad (\text{K.31})$$

which is consistent with  $\dot{\theta} = \partial_I F$ ,  $\dot{I} = -\partial_\theta F$  and with  $\frac{d}{dt}(G \circ \Phi_F^t) = \{F, G\} \circ \Phi_F^t$ . Differentiating  $G^0 \circ \Psi_t$  at  $t = 0$  gives

$$\left. \frac{d}{dt} \right|_0 G^0 \circ \Psi_t = \{F, G^0\} + \partial_I G^0 \cdot \xi. \quad (\text{K.32})$$

Using (K.30), the zero-average constant part of (K.32) is

$$-\omega \cdot \partial_\theta u = -a \quad (\text{K.33})$$

by (K.15); the extra term  $\omega \cdot \xi$  is constant and disappears under the zero-average projection. Differentiating (K.32) once in  $I$  at  $I = 0$  gives the linear-action part

$$-A\partial_\theta u - \omega \cdot \partial_\theta v + A\xi. \quad (\text{K.34})$$

Equations (K.19)–(K.20) turn (K.34) into  $-b$ . Thus the derivative of the jet map at the normalized Hamiltonian  $G^0$  cancels the entire defect:

$$\mathcal{J}(G) + D\mathcal{J}(G^0)[F, \xi] = 0. \quad (\text{K.35})$$

This identity is the algebraic heart of Kolmogorov nondegeneracy: the oscillatory part is removed by cohomology, while invertibility of  $\bar{A}$  removes the mean part of  $b$ .



For completeness we now estimate the nonlinear error in (K.35). Set

$$R(t) := \mathcal{J}(G \circ \Psi_t). \quad (\text{K.36})$$

Taylor's formula gives

$$R(1) = R(0) + R'(0) + \int_0^1 (1-t)R''(t) dt. \quad (\text{K.37})$$

The difference between  $R'(0)$  computed with  $G$  and with  $G^0$  is obtained by differentiating the residual Hamiltonian  $a(\theta) + b(\theta) \cdot I$  in the direction  $X_F + (0, \xi)$ . Cauchy's estimate, (K.13), and (K.25) therefore give

$$\|R'(0) - D\mathcal{J}(G^0)[F, \xi]\| \leq Cd^{-(p+2)}\varepsilon^2. \quad (\text{K.38})$$

The second derivative  $R''(t)$  is a finite sum of products containing a first, second, or third derivative of  $G$ , and two factors chosen among the components or first derivatives of  $X_F + (0, \xi)$ . The derivative bound in (K.13), Cauchy's estimate on the reserved fifth losses, and (K.25) imply

$$\sup_{0 \leq t \leq 1} \|R''(t)\| \leq Cd^{-(2p+4)}\varepsilon^2. \quad (\text{K.39})$$

No unstated cancellation is used here: (K.38) is quadratic because one factor is the old defect and the other is the correction; (K.39) is quadratic because it contains two correction factors. Combining (K.35)–(K.39), and setting

$$q := 2p + 4 = 4(\tau + n) + 12, \quad (\text{K.40})$$

we obtain

$$\|a_{G_+}\|_{\rho-5\sigma} + \|b_{G_+}\|_{\rho-5\sigma} \leq Cd^{-q}\varepsilon^2, \quad G_+ := G \circ \Psi_1. \quad (\text{K.41})$$

This is the required quadratic Newton estimate.

We also need persistence of the twist bound. Twice differentiating  $G_+ = G \circ \Psi_1$  in  $I$  at  $I = 0$ , using (K.25) and one further Cauchy estimate, gives

$$\|A_+ - A\|_{\rho-5\sigma} \leq Cd^{-(p+3)}\varepsilon, \quad A_+(\theta) := \partial_I^2 G_+(\theta, 0). \quad (\text{K.42})$$

Consequently, if the right-hand side is at most  $(2\mu)^{-1}$ , then

$$\|\bar{A}_+^{-1}\| \leq 2\mu \quad (\text{K.43})$$

by the Neumann-series criterion. More explicitly, differentiating the composition formula and using the two still-unused Cauchy buffers gives, for  $m = 0, 1, 2, 3$  on the final step domain,

$$\|D^m G_+ - D^m G\| \leq Cd^{-(p+6)}\varepsilon. \quad (\text{K.44})$$

The exponent  $p + 6$  is intentionally nonoptimal; it dominates every first, second, and third derivative of the affine Hamiltonian flow occurring in the chain rule. Thus a summable sequence of step corrections preserves a fixed uniform analytic derivative bound. This completes the one-step proof.

**lemma Lemma 5.2 (analytic torus Newton step).** The preceding calculation proves the following reusable statement. Under (K.13), on a domain with five reserved angle/action losses, there is a real-analytic symplectic correction  $\Psi$  such that

$$\|\Psi - \text{id}\|_{C^1} \leq C d^{-p-1} \varepsilon, \quad \varepsilon_+ \leq C d^{-q} \varepsilon^2, \quad (\text{K.45})$$

where  $\varepsilon_+ = \|a_{G \circ \Psi}\| + \|b_{G \circ \Psi}\|$  on the smaller domain,  $p, q$  are given by (K.24), (K.40), and the averaged Hessian remains invertible whenever the displayed smallness conditions hold. Everything in this lemma has been proved in (K.15)–(K.44).

**3. Initialization for  $H = h + P$ .** Return to the Hamiltonian in the theorem. Work first on the smaller fixed domain

$$\mathcal{D}(s_0, r_0), \quad s_0 := \frac{3s}{4}, \quad r_0 := \frac{3r}{4}. \quad (\text{K.46})$$

The unused outer quarter supplies the Cauchy buffer needed to estimate action derivatives of  $P$ . At the zero section,

$$\begin{aligned} a_H(\theta) &= P(\theta, 0) - [P(\cdot, 0)], \\ b_H(\theta) &= \partial_I P(\theta, 0). \end{aligned} \quad (\text{K.47})$$

Therefore Cauchy's estimate in the action variables gives

$$\varepsilon_0^{\text{jet}} := \|a_H\|_{s_0} + \|b_H\|_{s_0} \leq C_r \|P\|_{s,r}. \quad (\text{K.48})$$

Likewise

$$A_H(\theta) = A_0 + \partial_I^2 P(\theta, 0), \quad (\text{K.49})$$

so

$$\|\bar{A}_H - A_0\| \leq C_r \|P\|_{s,r}. \quad (\text{K.50})$$

Since  $A_0$  is invertible, choosing  $\|P\|_{s,r}$  sufficiently small gives

$$\|\bar{A}_H^{-1}\| \leq 2\|A_0^{-1}\|. \quad (\text{K.51})$$

The same outer-domain Cauchy bounds give a finite uniform constant  $M_*$  for the derivatives through order three of  $H$  on  $\mathcal{D}(s_0, r_0)$ , depending only on the analytic bound for  $h$ ,  $s, r$ , and an upper bound for  $\|P\|_{s,r}$ . Once the perturbation threshold is chosen below one,  $M_*$  may be fixed independently of the particular  $P$ .

**4. Infinite iteration.** Choose limiting widths

$$s_\infty := \frac{s}{2}, \quad r_\infty := \frac{r}{2}, \quad (\text{K.52})$$

and define

$$s_j = s_\infty + (s_0 - s_\infty)2^{-j}, \quad r_j = r_\infty + (r_0 - r_\infty)2^{-j}. \quad (\text{K.53})$$

At step  $j$ , split the total loss  $s_j - s_{j+1}$  and  $r_j - r_{j+1}$  into the five equal buffers required by the one-step argument. Thus there is a constant  $d_* > 0$ , depending only on  $s, r$ , such

that the step parameter  $d_j$  of (K.14) satisfies

$$d_j = d_* 2^{-j}. \quad (\text{K.54})$$

Let  $H_0 := H$ . Inductively apply Lemma 5.2 to obtain symplectic maps  $\Psi_j$  and

$$H_{j+1} := H_j \circ \Psi_j \quad (\text{K.55})$$

on  $\mathcal{D}(s_{j+1}, r_{j+1})$ . Write

$$\varepsilon_j := \|a_{H_j}\|_{s_j} + \|b_{H_j}\|_{s_j}. \quad (\text{K.56})$$

The one-step estimate has the form

$$\varepsilon_{j+1} \leq A 2^{qj} \varepsilon_j^2, \quad (\text{K.57})$$

for one constant  $A$  depending only on the data listed in the theorem. Set

$$z_j := A 2^{q(j+1)} \varepsilon_j. \quad (\text{K.58})$$

Then (K.57) gives

$$z_{j+1} \leq z_j^2. \quad (\text{K.59})$$

Indeed the harmless fixed powers of two have been absorbed into  $A$  in (K.57). If the initial perturbation is so small that  $z_0 \leq \frac{1}{2}$ , then

$$z_j \leq 2^{-2^j}, \quad (\text{K.60})$$

and hence  $\varepsilon_j$  tends to zero double-exponentially. This elementary normalization is the entire numerical convergence argument; no abstract KAM convergence theorem is being used as a black box.

The coordinate estimate (K.45), (K.54), and (K.60) imply

$$\|\Psi_j - \text{id}\|_{C^1} \leq C 2^{(p+1)j} \varepsilon_j, \quad (\text{K.61})$$

and the series on the right is summable because double-exponential decay beats every exponential weight. After decreasing the initial threshold once more, its sum is smaller than the reserved outer action and angle margins. Thus every composition is defined on the limiting domain.

Equations (K.42) and (K.44) are controlled by the same summable series. Therefore  $\bar{A}_j$  stays within a fixed small neighborhood of  $A_0$  and remains invertible, with a uniform inverse bound, at every step; simultaneously the derivatives through order three stay below one fixed bound. Hence none of the hypotheses used to construct step  $j + 1$  is lost during the iteration.

Let

$$\Phi_N := \Psi_0 \circ \Psi_1 \circ \cdots \circ \Psi_{N-1}. \quad (\text{K.62})$$

The summability in (K.61) first gives uniform convergence of the compositions on every compact subdomain of  $\mathcal{D}(s_\infty, r_\infty)$ . Indeed,  $\Phi_{N+1} = \Phi_N \circ \Psi_N$ , while the derivative products of the preceding factors are bounded by  $\prod_j (1 + \|D\Psi_j - I\|) < \infty$ ; hence the telescoping

difference  $\Phi_{N+1} - \Phi_N$  is bounded by a fixed multiple of  $\|\Psi_N - \text{id}\|$ . The latter series converges by (K.61). Weierstrass' theorem therefore gives a holomorphic limit  $\Phi_\infty$ . On each still smaller compact subdomain, Cauchy's estimate applied to  $\Phi_N - \Phi_\infty$  gives convergence of the first derivatives as well. Each  $\Phi_N$  is symplectic, so

$$D\Phi_N^T J D\Phi_N = J. \quad (\text{K.63})$$

Passing to the  $C^1$  limit yields the same identity for  $\Phi_\infty$ . Moreover  $D\Phi_\infty$  remains close to the identity, so the holomorphic inverse-function theorem makes  $\Phi_\infty$  a local analytic symplectic change of coordinates on the limiting domain. Reality of every step implies that  $\Phi_\infty$  maps the real phase space to itself.

Since  $H_N = H \circ \Phi_N$ , the same convergence gives

$$H_N \longrightarrow H_\infty := H \circ \Phi_\infty \quad (\text{K.64})$$

in  $C^1$  on the zero section. From (K.56), (K.60),

$$a_{H_N} \rightarrow 0, \quad b_{H_N} \rightarrow 0. \quad (\text{K.65})$$

Because  $H_N \rightarrow H_\infty$  uniformly on the zero section by (K.64), their angle averages  $c_{H_N}$  converge automatically to  $c_\infty := [H_\infty(\cdot, 0)]$ . Consequently

$$H_\infty(\theta, 0) = c_\infty, \quad \partial_I H_\infty(\theta, 0) = \omega. \quad (\text{K.66})$$

Thus  $H_\infty$  is in the fixed-frequency Kolmogorov normal form of the definition of fixed-frequency Kolmogorov normal form along a torus above.

**5. Construction of the invariant torus.** Define the holomorphic map

$$K(\theta) := \Phi_\infty(\theta, 0), \quad \theta \in \mathbb{T}_{s/2}^n. \quad (\text{K.67})$$

The zero section is Lagrangian for the standard action-angle symplectic form, and  $\Phi_\infty$  is symplectic, so the restriction  $K(\mathbb{T}^n)$  is Lagrangian. It remains to check the word "embedding" globally rather than only infinitesimally. Each correction is homotopic to the identity (the Hamiltonian flow starts at the identity and the action translation does not change the angle), so the limiting angular map has degree one. Write its real lift to  $\mathbb{R}^n$  as

$$\tilde{K}_\theta(x) = x + u_\infty(x), \quad u_\infty(x + m) = u_\infty(x) \quad (m \in \mathbb{Z}^n). \quad (\text{K.68})$$

By decreasing the initial smallness threshold once more, the summable  $C^1$  estimate (K.61) makes  $\|Du_\infty\| < 1/2$ . If two real points have the same angular image modulo  $\mathbb{Z}^n$ , choose  $m \in \mathbb{Z}^n$  so that  $\tilde{K}_\theta(x) = \tilde{K}_\theta(y + m)$ . Periodicity and the mean-value inequality then give

$$|x - y - m| = |u_\infty(y + m) - u_\infty(x)| \leq \frac{1}{2}|x - y - m|,$$

so  $x = y + m$ . Hence the angular component, and therefore  $K|_{\mathbb{T}^n}$ , is injective. Its derivative is also injective because it is  $C^1$ -close to the zero-section inclusion. Since the real torus is compact, this injective immersion is a real-analytic embedding. Thus (K.67) is the claimed holomorphic extension of a real-analytic Lagrangian embedding.

By (K.5) and (K.66), the Hamiltonian vector field of  $H_\infty$  satisfies

$$X_{H_\infty}(\theta, 0) = (\omega, 0). \quad (\text{K.69})$$

Symplectic changes of coordinates intertwine Hamiltonian vector fields:  $D\Phi_\infty X_{H_\infty} = X_H \circ \Phi_\infty$ . Restricting this identity to the zero section gives

$$X_H(K(\theta)) = D\Phi_\infty(\theta, 0)(\omega, 0) = DK(\theta)\omega. \quad (\text{K.70})$$

This is exactly the invariance equation stated in the theorem. Integrating the invariance equation (K.70) gives

$$\phi_H^t(K(\theta)) = K(\theta + \omega t), \quad (\text{K.71})$$

so the flow on the image torus is analytically conjugate to rigid translation with the prescribed frequency  $\omega = \omega_0$ .

All smallness requirements encountered above are inequalities of the form  $C\|P\|_{s,r} < c$  after the fixed data  $n, \tau, s, r, \gamma$ , an analytic bound for  $h$ , and  $\|A_0^{-1}\|$  have been fixed. Taking  $\varepsilon_0$  smaller than the finite collection of resulting thresholds proves the claimed existence of a positive  $\varepsilon_0$  with precisely the dependency asserted in the statement. Translating the action coordinate back by  $I_0$  completes the proof.  $\square$

### Theorem K05.3. Positive-measure Cantor family of KAM tori

**Statement.** Let  $A$  be a Kolmogorov frequency chart for a real-analytic integrable Hamiltonian  $h$  in the sense of Definition K05.10, and assume that  $h$  is holomorphic on a fixed complex neighborhood of  $A$  large enough to contain  $D_r(I)$  for every  $I \in A$ , for some  $r > 0$ . Fix  $s > 0$  and  $\tau > n - 1$ . Let

$$H(\theta, I) = h(I) + P(\theta, I)$$

be real analytic on the corresponding common complex action-angle neighborhood, and write  $\varepsilon := \|P\|$  there.

There exist constants

$$c_0 > 0, \quad C_0 < \infty, \quad \gamma_0 > 0, \quad N \in \mathbb{N},$$

depending only on  $n, \tau, s, r$ , uniform analytic bounds for  $h$  on that neighborhood, and the quantitative frequency-chart bounds, with the following property. Whenever  $0 < \gamma \leq \gamma_0$  and

$$\varepsilon \leq c_0 \gamma^N,$$

then for every  $I \in A_{\gamma, \tau}$  there is a real-analytic Lagrangian embedding  $K_I : \mathbb{T}^n \rightarrow \mathbb{T}^n \times \mathbb{R}^n$  satisfying

$$X_H(K_I(\theta)) = DK_I(\theta) \nu(I), \quad \nu(I) = Dh(I).$$

Thus the perturbed Hamiltonian possesses a family of invariant Diophantine La-

grangian tori indexed by the KAM Cantor parameter set  $A_{\gamma,\tau}$ . Moreover

$$\text{Leb}(A \setminus A_{\gamma,\tau}) \leq C_0 \gamma.$$

Consequently, for all sufficiently small  $\varepsilon > 0$ , one may choose  $\gamma_\varepsilon = (\varepsilon/c_0)^{1/N} \leq \gamma_0$  and obtain

$$\frac{\text{Leb}(A_{\gamma_\varepsilon,\tau})}{\text{Leb}(A)} \geq 1 - \frac{C_0}{\text{Leb}(A)} \left( \frac{\varepsilon}{c_0} \right)^{1/N} \rightarrow 1 \quad (\varepsilon \rightarrow 0).$$

### Scope note

The measure estimate concerns the action-parameter set in the fixed frequency chart. Whitney regularity and foliation regularity are separate assertions.

*Proof.* Write

$$\nu(I) := Dh(I).$$

By the frequency-chart hypothesis there is an open neighborhood  $U \supset A$  on which  $\nu$  is a  $C^1$  diffeomorphism. Shrinking  $U$  once, if necessary, we may assume that its closure stays inside the real action domain and that the common complex  $r$ -neighborhood appearing in the theorem is available over all of  $A$ . Since  $A$  is compact, all analytic and inverse-Hessian bounds needed in the fixed-frequency theorem are uniform on  $A$ .

### 1. Uniform polynomial dependence of the one-torus threshold on $\gamma$ .

The fixed-frequency theorem proved earlier says that, for each prescribed  $(\gamma, \tau)$ -Diophantine frequency, a sufficiently small perturbation leaves an invariant torus with that frequency. For the present measure argument we need one additional piece of bookkeeping: the smallness threshold must be uniform over  $I \in A$  and its dependence on  $\gamma$  must not be hidden.

**lemma Lemma 5.6 (uniform polynomial KAM threshold on a compact frequency chart).** There exist  $c_* > 0$ ,  $\gamma_* > 0$  and an integer  $N \geq 1$ , depending only on the fixed analytic/chart data and on  $n, \tau$ , such that, for every  $0 < \gamma \leq \gamma_*$ , every  $I_0 \in A$  whose frequency  $\nu(I_0)$  is  $(\gamma, \tau)$ -Diophantine satisfies the conclusion of the fixed-frequency KAM theorem whenever

$$\|P\| \leq c_* \gamma^N.$$

The constants are uniform in  $I_0 \in A$ .

*Proof of the lemma.* We inspect the already written fixed-frequency Newton proof. The only place in that argument where the Diophantine constant enters is the cohomological solve

$$\omega \cdot \partial_\theta u = f, \quad [f] = 0.$$

Its Fourier proof gives, after a strip loss  $\sigma$ ,

$$\|u\| \leq C \gamma^{-1} \sigma^{-(\tau+n)} \|f\|, \quad \|D_\theta u\| \leq C \gamma^{-1} \sigma^{-(\tau+n+1)} \|f\|.$$

Every subsequent quantity in one Newton step is obtained from these solutions by

finitely many additions, products, Cauchy derivatives, multiplication by the action Hessian, and multiplication by an inverse averaged Hessian. On the compact chart the analytic Hessian bounds are uniform, while the inverse averaged Hessian is kept in a fixed bounded neighborhood of the uniformly invertible matrix  $D^2h(I_0)$  by the same Neumann-series estimate used in the fixed-frequency proof. Consequently the constants in the one-step estimates can be bounded by

$$C\gamma^{-m}$$

for some finite integer  $m$  depending only on  $n, \tau$  and on the number of cohomological inversions/derivatives in the step. In particular, after the fixed geometric domain losses have been chosen, the Newton lemma may be written uniformly in the form

$$\varepsilon_+ \leq C\gamma^{-m}d^{-q}\varepsilon^2, \quad \|\Psi - \text{Id}\|_{C^1} \leq C\gamma^{-m}d^{-p}\varepsilon,$$

with fixed integers  $p, q$  and with the same type of polynomial bound for all bootstrap conditions preserving the action domain and the averaged twist inverse.

The infinite iteration uses a prescribed geometric sequence of domain losses  $d_j \asymp 2^{-j}$ . Thus its numerical recurrence has the form

$$\varepsilon_{j+1} \leq C\gamma^{-m}2^{qj}\varepsilon_j^2.$$

After rescaling  $z_j = C\gamma^{-m}2^{q(j+1)}\varepsilon_j$ , exactly the same elementary calculation as in the fixed-frequency proof gives double-exponential decay provided  $z_0$  is smaller than an absolute constant. All the remaining initial smallness conditions have the form

$$C_a\gamma^{-m_a}\varepsilon \leq c_a$$

for finitely many exponents  $m_a$ . Choose an integer  $N$  larger than every  $m_a$  and larger than the exponent required by the numerical recurrence. For  $0 < \gamma \leq 1$ , choosing  $c_* > 0$  sufficiently small makes  $\varepsilon \leq c_*\gamma^N$  imply every one of those inequalities. Compactness of  $A$  makes the constants uniform in  $I_0$ . This proves the lemma.  $\square$

The lemma is deliberately qualitative in the integer  $N$ : the present theorem only needs a positive power of  $\gamma$ . No sharp exponent is claimed.

## 2. Measure of the bad frequency set.

Put

$$\Omega := \nu(A) \subset \mathbb{R}^n.$$

Because  $A$  is compact,  $\Omega$  is compact. Choose  $R > 0$  with  $\Omega \subset [-R, R]^n$ . For  $0 \neq k \in \mathbb{Z}^n$  define the resonant slab

$$S_k(\gamma) := \{\omega \in \Omega : |k \cdot \omega| < \gamma|k|_1^{-\tau}\}.$$

Then

$$\Omega \setminus \Omega_{\gamma, \tau} \subset \bigcup_{0 \neq k \in \mathbb{Z}^n} S_k(\gamma),$$

where  $\Omega_{\gamma, \tau}$  is the set of frequencies satisfying all the Diophantine inequalities.

Choose an index  $j$  such that  $|k_j| = |k|_\infty$ . Holding the other  $n - 1$  coordinates fixed, the inequality

$$|k \cdot \omega| < \gamma |k|_1^{-\tau}$$

restricts  $\omega_j$  to an interval of length at most

$$\frac{2\gamma |k|_1^{-\tau}}{|k_j|} \leq 2n\gamma |k|_1^{-(\tau+1)},$$

because  $|k|_\infty \geq |k|_1/n$ . Fubini's theorem therefore gives

$$\text{Leb}(S_k(\gamma)) \leq 2n(2R)^{n-1}\gamma |k|_1^{-(\tau+1)}.$$

**lemma Lemma 5.7 (summable continuous-time resonance slabs).** If  $\tau > n - 1$ , then

$$\sum_{0 \neq k \in \mathbb{Z}^n} |k|_1^{-(\tau+1)} < \infty,$$

and consequently there is  $C_{\Omega, n, \tau} < \infty$  such that

$$\text{Leb}(\Omega \setminus \Omega_{\gamma, \tau}) \leq C_{\Omega, n, \tau} \gamma.$$

*Proof of the lemma.* The number of lattice points with  $m \leq |k|_1 < m + 1$  is  $O_n(m^{n-1})$ . Hence the displayed series is bounded by a constant multiple of

$$\sum_{m \geq 1} m^{n-1} m^{-(\tau+1)} = \sum_{m \geq 1} m^{n-\tau-2},$$

which converges precisely when  $\tau > n - 1$ . Countable subadditivity together with the slab estimate proves the measure bound.  $\square$

This is the reason for the exponent restriction  $\tau > n - 1$  in the family theorem. The fixed-frequency proof itself did not need that summability.

### 3. Pulling the measure estimate back to actions.

Because  $\nu$  is a  $C^1$  diffeomorphism on a neighborhood of  $A$ , its inverse has bounded Jacobian determinant on the compact set  $\Omega$ . Set

$$J_* := \sup_{\omega \in \Omega} |\det D\nu^{-1}(\omega)| < \infty.$$

By the change-of-variables formula,

$$\begin{aligned} \text{Leb}(A \setminus A_{\gamma, \tau}) &= \int_{\Omega \setminus \Omega_{\gamma, \tau}} |\det D\nu^{-1}(\omega)| d\omega \\ &\leq J_* C_{\Omega, n, \tau} \gamma. \end{aligned}$$

Thus the theorem's measure estimate holds with

$$C_0 := J_* C_{\Omega, n, \tau}.$$



Notice that no assertion about topological total disconnectedness is used here; “Cantor parameter set” means precisely the resonance-excluded set defined in the definition of KAM Cantor parameter set above.

#### 4. Persistence over the whole good set.

Let  $0 < \gamma \leq \min\{1, \gamma_*\}$  and assume

$$\varepsilon = \|P\| \leq c_* \gamma^N.$$

Take  $I \in A_{\gamma, \tau}$ . By definition,  $\omega_I := \nu(I) = Dh(I)$  satisfies

$$|k \cdot \omega_I| \geq \gamma |k|_1^{-\tau} \quad (0 \neq k \in \mathbb{Z}^n).$$

All the remaining analytic and twist constants in the fixed-frequency theorem are bounded uniformly on the compact chart. The uniform-threshold lemma therefore applies at this  $I$  and supplies a real-analytic Lagrangian embedding  $K_I$  such that

$$X_H(K_I(\theta)) = DK_I(\theta) \omega_I.$$

Since  $I$  was arbitrary, such a torus exists for every  $I \in A_{\gamma, \tau}$ . This proves the persistence assertion.

The conclusion at this stage is intentionally only a family indexed by the good action parameters. We have not used any continuity of  $I \mapsto K_I$ , and we do not infer a Whitney foliation from the present theorem. That parameter regularity is the separate content of the next result.

#### 5. Asymptotically full relative measure.

Set  $c_0 := c_*$  and shrink  $\gamma_0$  if needed so that  $C_0 \gamma_0 < \text{Leb}(A)$ . For  $0 < \varepsilon \leq c_0 \gamma_0^N$  define

$$\gamma_\varepsilon := \left( \frac{\varepsilon}{c_0} \right)^{1/N}.$$

Then  $\varepsilon = c_0 \gamma_\varepsilon^N$ , so the persistence conclusion applies and

$$\text{Leb}(A \setminus A_{\gamma_\varepsilon, \tau}) \leq C_0 \left( \frac{\varepsilon}{c_0} \right)^{1/N}.$$

Dividing by the positive number  $\text{Leb}(A)$  gives the displayed relative-measure estimate in the theorem, and the right-hand side tends to zero as  $\varepsilon \rightarrow 0$ . In particular the good action set has positive measure for every sufficiently small perturbation.  $\square$

#### Theorem K05.4. Whitney smooth dependence of KAM tori

**Statement.** Let  $A$  be a Kolmogorov frequency chart, fix  $\tau > n-1$  and  $0 < \gamma \leq \gamma_0$ , and let  $A_{\gamma, \tau}$  be the KAM Cantor parameter set of Definition K05.11. For a sufficiently small real-analytic perturbation  $H = h + P$  (with a smallness threshold allowed to be stronger than the one needed only for the Positive-measure Cantor family of KAM tori), the invariant tori with frequencies  $Dh(I)$ ,  $I \in A_{\gamma, \tau}$ , can be selected so

that, for every fixed reduced strip  $0 < \rho < s/2$ , the map

$$I \mapsto K_I \in \mathcal{A}_\rho$$

is Whitney- $C^\infty$  in the sense of Definition K05.12. The combined map  $\mathcal{K}(\theta, I) = K_I(\theta)$  is Whitney- $C^1$  close to the unperturbed action-angle parameterization and therefore defines a Whitney-smooth KAM Cantor foliation in the sense of Definition K05.13.

Moreover, the smooth transverse parameter coordinates supplied by this foliation restrict to  $n$  independent first integrals  $F_1, \dots, F_n$  on the Cantor family, and these integrals are in involution there in the sense of Definition K05.14.

### Scope note

The transverse parameter dependence is Whitney smooth on the Cantor set; analytic continuation across resonance gaps is not asserted.

### Theorem K05.5. Linearized reducibility along a KAM torus

**Statement.** Let an analytic invariant Lagrangian torus with frequency  $\omega \in \mathbb{R}^n$  be written on  $\mathbb{T}_s^n \times D_r$  in Kolmogorov coordinates as

$$H(\theta, I) = c + \langle \omega, I \rangle + \frac{1}{2} \langle I, Q(\theta) I \rangle + O(|I|^3),$$

where  $Q(\theta) = Q(\theta)^T$  is real analytic and  $|k \cdot \omega| \geq \gamma |k|_1^{-\tau}$  for every  $0 \neq k \in \mathbb{Z}^n$ . Put  $\bar{Q} := \int_{\mathbb{T}^n} Q(\theta) d\theta$  with normalized Haar measure. For every  $0 < \delta < \min\{s, 1\}$  there is a unique zero-average real-analytic symmetric matrix  $U$  on  $\mathbb{T}_{s-\delta}^n$  solving

$$\omega \cdot \partial_\theta U = Q - \bar{Q}, \quad \|U\|_{s-\delta} \leq C_{n,\tau} \gamma^{-1} \delta^{-(\tau+n)} \|Q - \bar{Q}\|_s.$$

Along the torus  $I = 0$ , whose variational equation is

$$\delta \dot{I} = 0, \quad \delta \dot{\theta} = Q(\theta_0 + \omega t) \delta I,$$

the quasiperiodic linear symplectic change

$$\eta = \delta I, \quad \xi = \delta \theta - U(\theta_0 + \omega t) \delta I$$

reduces the system to the constant equation  $\dot{\eta} = 0$ ,  $\dot{\xi} = \bar{Q} \eta$ .

*Proof.* The torus is  $I = 0$ . Differentiating the displayed Kolmogorov expansion gives, near  $I = 0$ ,

$$\partial_I H(\theta, I) = \omega + Q(\theta) I + O(|I|^2), \quad \partial_\theta H(\theta, I) = \frac{1}{2} \partial_\theta \langle I, Q(\theta) I \rangle + O(|I|^3). \quad (\text{LR.1})$$

Hence Hamilton's equations  $\dot{\theta} = \partial_I H$ ,  $\dot{I} = -\partial_\theta H$  restrict on  $I = 0$  to  $\theta(t) = \theta_0 + \omega t$ ,  $I(t) = 0$ .

Their derivative along this orbit is exactly

$$\delta \dot{I} = 0, \quad \delta \dot{\theta} = Q(\theta_0 + \omega t) \delta I. \quad (\text{LR.2})$$

Thus the variational equation stated in the theorem has no omitted  $\delta\theta$  term.

We next solve the continuous-time cohomological equation. Expand

$$Q(\theta) = \sum_{k \in \mathbb{Z}^n} Q_k e^{2\pi i k \cdot \theta}, \quad Q_0 = \bar{Q}. \quad (\text{LR.3})$$

For  $k \neq 0$  define

$$U_k := \frac{Q_k}{2\pi i k \cdot \omega}, \quad U_0 := 0, \quad U(\theta) := \sum_{k \neq 0} U_k e^{2\pi i k \cdot \theta}. \quad (\text{LR.4})$$

The Diophantine hypothesis makes every divisor in (LR.4) nonzero and gives

$$|U_k| \leq \frac{1}{2\pi\gamma} |k|_1^\tau |Q_k|. \quad (\text{LR.5})$$

If  $\|\cdot\|_s$  is the entrywise maximum analytic sup norm, Cauchy's Fourier estimate gives  $|Q_k| \leq \|Q - \bar{Q}\|_s e^{-2\pi s |k|_1}$  for  $k \neq 0$ . Therefore, on the smaller strip,

$$\|U\|_{s-\delta} \leq \frac{\|Q - \bar{Q}\|_s}{2\pi\gamma} \sum_{k \neq 0} |k|_1^\tau e^{-2\pi\delta |k|_1}. \quad (\text{LR.6})$$

Counting lattice points with  $|k|_1 = m$  and comparing the resulting polynomial-exponential series with the corresponding Gamma integral gives, for  $0 < \delta \leq 1$ ,

$$\sum_{k \neq 0} |k|_1^\tau e^{-2\pi\delta |k|_1} \leq C'_{n,\tau} \delta^{-(\tau+n)}. \quad (\text{LR.7})$$

Absorbing  $(2\pi)^{-1}$  into  $C_{n,\tau}$  proves the asserted strip-loss estimate. Termwise differentiation of the normally convergent series on every smaller strip yields

$$\omega \cdot \partial_\theta U = Q - \bar{Q}. \quad (\text{LR.8})$$

The zero Fourier coefficient was fixed to be zero, so  $U$  is the unique zero-average solution. Since  $Q(\theta)$  is real on the real torus,  $Q_{-k} = \overline{Q_k}$ ; since the denominator in (LR.4) is purely imaginary and changes sign under  $k \mapsto -k$ , one gets  $U_{-k} = \overline{U_k}$ . Hence  $U$  is real on the real torus. Moreover  $Q_k^T = Q_k$  for every  $k$ , so  $U_k^T = U_k$  and therefore

$$U(\theta)^T = U(\theta). \quad (\text{LR.9})$$

Set  $\eta = \delta I$  and

$$\xi = \delta\theta - U(\theta_0 + \omega t)\eta. \quad (\text{LR.10})$$

For each fixed angle the corresponding linear change has block matrix

$$S(\theta) = \begin{pmatrix} I & -U(\theta) \\ 0 & I \end{pmatrix}. \quad (\text{LR.11})$$

Because  $U^T = U$ , a direct block multiplication gives  $S(\theta)^T J S(\theta) = J$  for the standard variational symplectic matrix  $J$ . Thus the reducing gauge is symplectic, not merely invertible.

Finally, (LR.2),  $\dot{\eta} = 0$ , and (LR.8) give

$$\begin{aligned} \dot{\xi} &= Q(\theta_0 + \omega t)\eta - (\omega \cdot \partial_\theta U)(\theta_0 + \omega t)\eta \\ &= \overline{Q}\eta, \quad \dot{\eta} = 0. \end{aligned} \quad (\text{LR.12})$$

This is the claimed constant-coefficient variational equation. If  $Q$  is already constant, then  $U = 0$ , so the construction also covers the degenerate edge case without an unnecessary coordinate change.  $\square$

## 5.4 Worked examples

### Worked example: a two-degree nearly integrable torus

Let  $\omega_* \in \mathbb{R}^2$  be Diophantine and consider, near  $I = 0$ ,

$$H(\theta, I) = \omega_* \cdot I + \frac{1}{2}|I|^2 + \varepsilon \cos \theta_1.$$

The integrable part is

$$h(I) = \omega_* \cdot I + \frac{1}{2}|I|^2, \quad Dh(0) = \omega_*, \quad D^2h(0) = \text{Id}.$$

Thus the Kolmogorov matrix is invertible. The perturbation on the unperturbed torus has zero average. The first scalar homological equation has the form

$$\omega_* \cdot \partial_\theta u = -\varepsilon \cos \theta_1.$$

Because the only Fourier vectors are  $k = \pm(1, 0)$  and  $k \cdot \omega_* \neq 0$ , one may take

$$u(\theta) = -\frac{\varepsilon}{\omega_{*,1}} \sin \theta_1.$$

This is only the first correction, not the full KAM conjugacy. Its purpose is to display the two central mechanisms: the Fourier divisor is controlled arithmetically, while the identity Hessian makes the mean frequency correction solvable. The full Newton scheme repeats this on successively smaller complex strips and produces a true invariant torus with frequency  $\omega_*$  for  $|\varepsilon|$  below the theorem's quantitative threshold.

**Boundary case: a Liouville vector and loss of uniform analytic inversion**

For a Liouville vector  $\omega$ , there are integer vectors  $k_j \neq 0$  with  $|k_j \cdot \omega|$  smaller than any prescribed power of  $|k_j|^{-1}$  along a subsequence. If a zero-average analytic forcing has nonnegligible Fourier coefficients on those  $k_j$ , the formal solution

$$u_{k_j} = \frac{f_{k_j}}{i k_j \cdot \omega}$$

may lose more than any fixed analytic strip estimate can tolerate. The classical Diophantine KAM theorem in this chapter therefore cannot be invoked. This does not say that every Liouville torus is destroyed; it says that the theorem's inversion mechanism no longer supplies the required uniform bound.

**Common pitfall**

A positive-measure KAM set is not an open foliation of phase space. Resonant gaps remain. This is why transverse regularity is formulated in the Whitney sense on a closed Cantor parameter set.

## 5.5 Chapter summary

**What to remember**

- KAM is a Newton theorem with arithmetic losses: each step solves small-divisor equations but makes the defect quadratically smaller.
- Diophantine bounds control the oscillatory equations; Kolmogorov/torsion nondegeneracy controls the average obstruction.
- A-posteriori KAM turns a sufficiently accurate approximate invariant torus into an exact one and is the most reusable formulation of the mechanism.
- Uniform parameter exclusion converts one-torus persistence into a positive-measure Cantor family.
- The family is Whitney smooth across the surviving parameter set, not analytic across deleted resonance gaps.
- The linearized dynamics along a KAM torus is reducible to constant coefficients under the stated Diophantine assumptions.

**Looking ahead**

K06 changes the nondegeneracy and regularity regimes. Isoenergetic KAM keeps the energy fixed, Moser's theorem treats exact symplectic twist maps, and finite-differentiability KAM shows how analytic smoothing and parameter book-keeping replace the infinite analytic reserve used here.

**Check your understanding**

Separate the KAM proof into its infinite-dimensional and finite-dimensional versions. Which hypothesis controls each one, and why would removing either hypothesis prevent the stated Newton step from closing?

**5.6 Exercises and problems****Exercise 5.1**

**Difficulty.** Core.

**Related results.** Kolmogorov analytic KAM theorem; Positive-measure Cantor family of KAM tori; Whitney smooth dependence of KAM tori.

**Problem.** For the nearly integrable model, compute the unperturbed frequency and twist matrix and perform the first linearized KAM correction for the displayed Fourier modes, separating the oscillatory cohomological equation from the mean equation.

**Solution sketch.** Invert the twist matrix for the mean and the Diophantine divisors for nonzero Fourier modes.

**Exercise 5.2**

**Difficulty.** Core.

**Related results.** Linearized reducibility along a KAM torus; A-posteriori KAM theorem.

**Problem.** Using the Liouville frequency in the boundary example, construct Fourier modes whose divisors are exceptionally small and show that the formal cohomological solution violates the required analytic bound.

**Solution sketch.** Place analytic Fourier mass on a Liouville approximation sequence and compare exponential decay before and after division.

**Exercise 5.3**

**Difficulty.** Advanced.

**Related results.** Whitney smooth dependence of KAM tori.

**Problem.** Prove that jets of a  $C^r$  parameter family restricted to a closed Cantor set satisfy Whitney compatibility estimates, explaining the meaning of Whitney-smooth KAM dependence.

**Solution sketch.** Apply Taylor's theorem with remainder between two Cantor-set points for every derivative order up to  $r$ .

**Exercise 5.4**

**Difficulty.** Advanced.

**Related results.** Linearized reducibility along a KAM torus.

**Problem.** For a linear normal equation, derive the divisor  $k \cdot \omega + \lambda_i - \lambda_j$  and show that an exact resonance obstructs reducibility for generic forcing in that matrix entry.

**Solution sketch.** Fourier-expand entrywise and exhibit a nonzero forcing coefficient when the divisor vanishes.

### Exercise 5.5

**Difficulty.** Advanced.

**Related results.** Kolmogorov analytic KAM theorem; Positive-measure Cantor family of KAM tori.

**Problem.** From a resonant-strip measure bound  $O(\gamma)$ , prove the positive-measure Cantor-family conclusion and explain why the parameter family is Whitney-smooth rather than smooth on an open set.

**Solution sketch.** Sum excluded strips, pull back through the nondegenerate frequency map, and interpret the gaps as a Whitney-domain issue.

### Exercise 5.6

**Difficulty.** Lab.

**Related results.** None.

**Problem.** Implement one Fourier-truncated Newton/KAM correction for the invariant-torus equation at a Diophantine and a near-resonant frequency; track smallest divisors, correction norms, and defect reduction as cutoff grows.

**Solution sketch.** Double the Fourier cutoff and compare the same quantities; numerical convergence is evidence only, not a proof of a torus.

# Arnold, Moser, Isoenergetic and Finite-Regularity KAM

## Why this chapter matters

The analytic Kolmogorov theorem is the cleanest KAM model, but many dynamical problems do not fit its exact hypotheses. One may want to preserve a torus on a fixed energy surface rather than fix its full frequency vector; one may study a Poincaré or twist map instead of a Hamiltonian flow; or the system may have only finitely many derivatives. This chapter shows how the KAM mechanism changes when those constraints change.

The unifying lesson is that there is no single universal “nondegeneracy condition.” Isoenergetic KAM needs invertibility of a bordered Hessian because only the frequency direction matters on an energy surface. Twist-map KAM needs monotonic variation of the rotation number. Finite-regularity KAM must budget derivatives through smoothing and loses quantitative exponent compared with the analytic setting. Parameterized, reversible, and conformally symplectic variants require their own source conditions rather than being silently identified with the Hamiltonian theorem.

**Prerequisites.** K05 analytic KAM, Foundation F05/F07/F08/F12/F15, exact symplectic maps, basic measure theory, and Hölder spaces.

## Learning goals

- Explain the bordered Hessian in isoenergetic KAM and why it fixes frequency direction rather than full frequency.
- Apply Moser’s invariant-curve theorem to an exact symplectic twist map at a Diophantine rotation number.
- Track the differentiability and measure exponents in the finite-regularity theorem without replacing them by analytic heuristics.
- Distinguish geometric regularity of each torus from Whitney regularity in the Cantor parameter.
- Use the measure-bookkeeping lemma to convert countably many resonance exclusions into a positive-measure surviving set.



**Conceptual roadmap**

The chapter compares three persistence mechanisms:

- fixed energy: bordered Hessian + projective  
Diophantine direction,
- annulus map: exact symplecticity + twist  
+ Diophantine rotation,
- finite regularity: smoothing + analytic KAM at successive scales  
+ an explicit derivative budget.

All three end with a Cantor-type survival statement, and the final bookkeeping lemma explains how row-by-row excluded sets are converted into a global measure estimate.

**Check your understanding**

For  $h(I)$  on an energy surface, compare invertibility of  $D^2h$  with invertibility of the bordered matrix

$$\begin{pmatrix} D^2h & Dh \\ Dh^T & 0 \end{pmatrix}.$$

What extra scalar freedom does the bordered system encode?

**6.1 Definitions****Intuition**

This chapter changes the persistence problem three times, and each change alters the correct nondegeneracy notion. In isoenergetic KAM one does not insist on preserving the entire frequency vector. Motion on a fixed energy surface is unchanged, up to a constant reparametrization of time, when the frequency vector is multiplied by a positive scalar. The natural invariant is therefore the projective frequency direction. The bordered Hessian in the definition of isoenergetic nondegeneracy supplies exactly one additional scalar unknown: it solves simultaneously for an action displacement and for the allowed rescaling of frequency.

For an annulus map, the relevant object is an essential invariant circle rather than a Hamiltonian torus. Exact symplecticity is stronger than area preservation:  $F^*(I d\theta) - I d\theta$  must be exact. This exactness removes the average obstruction in the invariant-circle equation, while the twist derivative changes the rotation number when the action is moved. Thus “area preserving,” “exact symplectic,” and “twist” are three different properties, and Moser’s theorem uses the latter two in distinct places.

Finite differentiability introduces a second distinction. A function in  $C^\ell$  has a finite Hölder derivative budget in the phase variables. A Cantor family of resulting tori is described by a Whitney  $C^r$  jet on the surviving parameter set. The exponent  $r$  governing parameter jets need not equal the regularity  $\nu$  of each individual torus. The proof uses analytic smoothing to convert the finite  $C^\ell$  reserve into a sequence of analytic problems, spending derivatives at every scale. Finally, parameterized KAM and the bad-parameter bookkeeping definitions explain how the nonresonance conditions are made uniform. External parameters are not decorative labels: their derivatives may supply missing frequency directions. Once each resonance family has its own measure estimate, the surviving parameter set is the complement of their countable union. The elementary union bound is the final step only after the dynamical transversality estimates have been established.

**Key idea.** Match the nondegeneracy condition to the obstruction in the actual invariance equation: bordered Hessian for fixed energy, twist for annulus rotation, source-parameter transversality for families, and a derivative budget plus smoothing in finite regularity.

### Definition K06.1. isoenergetic nondegeneracy

Let  $h$  be a  $C^2$  integrable Hamiltonian and write  $\omega(I) = Dh(I)$ . At a regular action  $I_*$ , so  $\omega(I_*) \neq 0$ ,  $h$  is *isoenergetically nondegenerate* if the bordered Hessian

$$\mathcal{B}_h(I_*) := \begin{pmatrix} D^2h(I_*) & \omega(I_*) \\ \omega(I_*)^T & 0 \end{pmatrix}$$

is invertible. Equivalently, on the energy hypersurface through  $I_*$ , the projective frequency direction varies nondegenerately: in any projective chart containing  $[\omega(I_*)]$ , the map  $I \mapsto [\omega(I)]$  restricted to the energy hypersurface has nonsingular derivative.

### Definition K06.2. twist map

Let  $F(\theta, I) = (\Theta(\theta, I), J(\theta, I))$  be a lift of an annulus diffeomorphism. It is a *positive monotone twist map* on a region if  $\partial_I \Theta \geq m > 0$  there, and a *negative monotone twist map* if  $\partial_I \Theta \leq -m < 0$ . When only the integrable map  $F_0(\theta, I) = (\theta + \omega(I), I)$  is under discussion, this is exactly the fixed-sign condition  $|\omega'(I)| \geq m > 0$ .

### Definition K06.3. Whitney smoothness

Let  $E \subset \mathbb{R}^n$  be closed and write  $r = m + \eta$  with  $m = \lfloor r \rfloor$  and  $0 \leq \eta < 1$ . A *Whitney  $C^r$  jet* on  $E$  is a family  $\{F^a : E \rightarrow \mathbb{R}^N : |a| \leq m\}$  such that, for every  $|a| \leq m$ ,

$$F^a(x) - \sum_{|b| \leq m-|a|} \frac{F^{a+b}(y)}{b!} (x-y)^b = \begin{cases} O(|x-y|^{r-|a|}), & \eta > 0, \\ o(|x-y|^{m-|a|}), & \eta = 0, \end{cases} \quad x \rightarrow y \text{ in } E,$$

uniformly on compact subsets of  $E$ . A map  $F : E \rightarrow \mathbb{R}^N$  is Whitney  $C^r$ , written  $F \in C_W^r(E)$ , if it is the zeroth component of such a jet. Whitney  $C^\infty$  means that compatible jets exist to every finite order. On  $E \times \mathbb{T}^d$  the definition is imposed in periodic coordinate charts. For  $0 < r < 1$  this is exactly  $r$ -Hölder continuity on  $E$ ; for  $1 < r < 2$  it means that a first jet  $A$  exists with  $A$  Hölder of exponent  $r - 1$  and  $F(x) - F(y) - A(y)(x - y) = O(|x - y|^r)$ .

#### Definition K06.4. parameterized KAM

Parameterized KAM refers to persistence theorems for families  $X_\mu/H_\mu$  in which external parameters are part of the nondegeneracy mechanism and the surviving tori occur on a Cantor/Whitney parameter subset.

#### Definition K06.5. frequency direction and frequency ratio

For a nonzero frequency vector  $\omega \in \mathbb{R}^n$ , its *frequency direction* is the projective class  $[\omega] \in \mathbb{RP}^{n-1}$ . Two nonzero frequency vectors have the same *frequency ratios* when they differ by multiplication by a positive scalar. Thus frequency direction records the oriented quasi-periodic orbit geometry while forgetting a constant rescaling of time.

#### Definition K06.6. orbital equivalence on invariant tori

Two flows on invariant tori are *orbitally equivalent* if a diffeomorphism sends oriented flow orbits to oriented flow orbits, without requiring equality of their time parametrizations. For constant linear flows this occurs when their frequency vectors differ by a positive scalar. A conjugacy is stronger: it also preserves the time parametrization.

#### Definition K06.7. projective Diophantine direction

Fix  $\tau > n - 1$  and  $\gamma > 0$ . A nonzero frequency vector  $\omega \in \mathbb{R}^n$  has a  $(\gamma, \tau)$  *projective Diophantine direction* if its unit representative  $u = \omega/|\omega|$  satisfies

$$|k \cdot u| \geq \gamma |k|_1^{-\tau} \quad \text{for every } 0 \neq k \in \mathbb{Z}^n.$$

This condition depends only on the frequency direction and is unchanged by multiplication of  $\omega$  by a positive scalar.

#### Definition K06.8. isoenergetic action chart

Let  $h$  be isoenergetically nondegenerate and choose an index  $j$  with  $\partial_{I_j} h \neq 0$ . An *isoenergetic action chart* is an action neighborhood on which

$$I \mapsto \left( h(I), \left( \frac{\omega_i(I)}{\omega_j(I)} \right)_{i \neq j} \right), \quad \omega = Dh,$$

is a diffeomorphism onto its image. On a fixed energy hypersurface its transverse coordinates are therefore precisely the local frequency ratios.

**Definition K06.9. exact symplectic annulus map**

On  $\mathbb{T} \times (a, b)$  let  $\lambda = I d\theta$  and  $\Omega = d\lambda = dI \wedge d\theta$ . A symplectic annulus diffeomorphism  $F$  is *exact symplectic* if there is a single-valued periodic function  $S$  such that

$$F^*\lambda - \lambda = dS.$$

This exactness condition is stronger than preservation of area and is the compatibility mechanism that removes the mean normal obstruction in the invariant-circle Newton equation.

**Definition K06.10. essential invariant circle**

An *essential invariant circle* for an annulus map  $F$  is the image of an embedding  $K : \mathbb{T} \rightarrow \mathbb{T} \times (a, b)$  whose angular component has degree one and for which  $F(K(\mathbb{T})) = K(\mathbb{T})$ . If  $F \circ K = K \circ R_\rho$  with  $R_\rho(\theta) = \theta + \rho \pmod{1}$ , then the dynamics on the circle is analytically conjugate to the rigid rotation of rotation number  $\rho$ .

**Definition K06.11. finite Hölder class  $C^\ell$** 

Let  $\ell = m + \eta > 0$  with  $m = \lfloor \ell \rfloor$  and  $0 \leq \eta < 1$ . A function belongs to  $C^\ell$  if all derivatives of order at most  $m$  are continuous and, when  $\eta > 0$ , every derivative of order  $m$  is  $\eta$ -Hölder continuous. We use the norm

$$\|f\|_{C^\ell} := \max_{|\alpha| \leq m} \|\partial^\alpha f\|_{C^0} + \mathbf{1}_{\{\eta > 0\}} \max_{|\alpha|=m} [\partial^\alpha f]_{C^{0,\eta}}.$$

On  $D \times \mathbb{T}^d$  the same definition is taken in all action-angle variables, using local lifts in the torus factor.

**Definition K06.12. bad-parameter family and surviving KAM parameter set**

Let  $(\Pi, \mu)$  be a finite measure space. For a scale parameter  $\gamma > 0$ , a sequence  $E_j(\gamma) \subset \Pi$  of measurable sets is a *bad-parameter family* if membership in  $E_j(\gamma)$  records failure of the  $j$ th required nonresonance or admissibility condition. The associated *surviving (good) parameter set* is

$$\Pi_\gamma := \Pi \setminus \bigcup_{j \geq 1} E_j(\gamma).$$

Thus the excluded set is exactly  $\Pi \setminus \Pi_\gamma = \bigcup_j E_j(\gamma)$ ; all measure conclusions below refer to this convention.

## 6.2 How the main proofs work

### Intuition

**Isoenergetic KAM.** Use the energy constraint together with the torus invariance equation. The bordered Hessian simultaneously solves for the action dis-

placement and the scalar rescaling of the frequency vector. Diophantine estimates handle the zero-average equations, while the bordered inverse handles the mean equation. Parameter exclusion is performed on frequency directions along the energy surface.

**Moser invariant curves.** Write an invariant circle as an embedding conjugating the perturbed map to the rigid rotation  $R_\rho$ . The linearized difference equation contains divisors  $e^{2\pi i k \rho} - 1$ , controlled by the Diophantine condition. Exact symplecticity removes the average obstruction and the twist derivative solves the remaining scalar correction. Newton iteration then produces the invariant essential circle.

**Finite differentiability.** Replace the original  $C^\ell$  Hamiltonian by analytic approximants on shrinking complex domains. At scale  $j$ , apply an analytic KAM step with a smoothing error small enough to be absorbed into the next correction. Choose the scales so that derivative losses are summable in  $C^\nu$ . The inequalities  $\ell > 2\nu > 2d$  are the derivative budget that makes the construction close. A separate boundary-localization argument yields the stated measure exponent.

**Parameterized variants.** Add external parameters to the normal form and identify the source map whose transverse derivative supplies the missing average directions. Hamiltonian, reversible, volume-preserving, and conformally symplectic classes have different source equations; they are compared here but not conflated.

**Measure bookkeeping.** Once each resonance family has an estimate  $\mu(E_j) \leq a_j$ , countable subadditivity gives the total loss. No dynamical information is hidden in this last step; all difficult estimates are upstream in the resonance-slab bounds.

## 6.3 Main results

### Lemma K06.1. Measure bookkeeping lemma for KAM parameter sets

**Statement.** Let  $(\Pi, \mu)$  be a finite measure space and let  $E_j(\gamma) \subset \Pi$  be measurable bad-parameter sets with  $\mu(E_j(\gamma)) \leq a_j(\gamma)$  and  $\sum_{j \geq 1} a_j(\gamma) < \infty$ . For  $\Pi_\gamma = \Pi \setminus \bigcup_{j \geq 1} E_j(\gamma)$  one has  $\mu(\Pi \setminus \Pi_\gamma) \leq \sum_{j \geq 1} a_j(\gamma)$ . Hence  $\Pi_\gamma$  has positive measure whenever the sum is smaller than  $\mu(\Pi)$ ; if  $\sum_j a_j(\gamma) \rightarrow 0$  as  $\gamma \rightarrow 0$ , then  $\mu(\Pi_\gamma) \rightarrow \mu(\Pi)$ ; and if  $\sum_j a_j(\gamma) = O(\gamma^b)$ , then  $\mu(\Pi \setminus \Pi_\gamma) = O(\gamma^b)$ . In KAM applications the individual bounds  $a_j$  come from resonance-slab estimates and parameter-exclusion lemmas of the type proved in F05 and F15.

*Proof.* By definition,

$$\Pi \setminus \Pi_\gamma = \Pi \cap \bigcup_{j \geq 1} E_j(\gamma).$$

Hence monotonicity and countable subadditivity of  $\mu$  give

$$\mu(\Pi \setminus \Pi_\gamma) \leq \mu\left(\bigcup_{j \geq 1} E_j(\gamma)\right) \leq \sum_{j \geq 1} \mu(E_j(\gamma)) \leq \sum_{j \geq 1} a_j(\gamma). \quad (\text{M.1})$$

This is the asserted basic estimate.

If the sum on the right is strictly smaller than  $\mu(\Pi)$ , then

$$\mu(\Pi_\gamma) = \mu(\Pi) - \mu(\Pi \setminus \Pi_\gamma) > 0,$$

where finiteness of  $\mu(\Pi)$  is used to subtract the two finite numbers. If instead  $\sum_j a_j(\gamma) \rightarrow 0$  as  $\gamma \rightarrow 0$ , then (M.1) implies

$$0 \leq \mu(\Pi) - \mu(\Pi_\gamma) = \mu(\Pi \setminus \Pi_\gamma) \leq \sum_j a_j(\gamma) \rightarrow 0,$$

so  $\mu(\Pi_\gamma) \rightarrow \mu(\Pi)$ . Finally, if  $\sum_j a_j(\gamma) \leq C\gamma^b$  for all sufficiently small  $\gamma$ , then (M.1) immediately yields

$$\mu(\Pi \setminus \Pi_\gamma) \leq C\gamma^b,$$

which is exactly the claimed  $O(\gamma^b)$  estimate.

No independence, disjointness, or Borel–Cantelli hypothesis is required: overlaps between the bad sets can only make their union smaller than the sum of the individual upper bounds. In a concrete KAM application, all dynamical content lies in proving the row-specific estimates  $\mu(E_j(\gamma)) \leq a_j(\gamma)$  and their summability; the present lemma is only the final measure-bookkeeping step.  $\square$

### Theorem K06.2. Isoenergetic KAM theorem

**Statement.** Let  $D \subset \mathbb{R}^n$  be open, let  $h : D \rightarrow \mathbb{R}$  be real analytic with a holomorphic extension to a fixed complex neighborhood, and let  $E \in \mathbb{R}$ . Let  $\Sigma \Subset \{I \in D : h(I) = E\}$  be a compact subset of positive relative  $(n-1)$ -dimensional measure. Assume that  $Dh(I) \neq 0$  and that

$$\mathcal{B}_h(I) = \begin{pmatrix} D^2h(I) & Dh(I) \\ Dh(I)^T & 0 \end{pmatrix}$$

is nonsingular on a neighborhood of  $\Sigma$ . Fix  $\tau > n-1$ .

Then, for every sufficiently small projective Diophantine constant  $\gamma > 0$ , there exists  $\varepsilon_0(\gamma) > 0$  such that every real-analytic Hamiltonian  $H(\theta, I) = h(I) + f(\theta, I)$  with  $\|f\| < \varepsilon_0(\gamma)$  on the fixed complex neighborhood has the following property. On the fixed perturbed energy surface  $H^{-1}(E)$  there is a positive-measure Whitney- $C^\infty$  Cantor family of real-analytic Lagrangian invariant tori. More precisely, after restricting to a positive-measure compact isoenergetic chart  $\Sigma' \subset \Sigma$ , there is a closed subset  $\Sigma'_\gamma \subset \Sigma'$  whose relative-measure complement is  $O(\gamma)$  and analytic embeddings

$$K_I : \mathbb{T}^n \longrightarrow H^{-1}(E), \quad I \in \Sigma'_\gamma,$$

which depend Whitney- $C^\infty$  on  $I$ , are  $C^1$ -close to  $\theta \mapsto (\theta, I)$ , and satisfy

$$X_H \circ K_I = DK_I(\lambda_I Dh(I)) \quad \text{for some } \lambda_I > 0$$

with  $\lambda_I$  uniformly close to 1. Hence the surviving tori preserve the unperturbed frequency ratios, equivalently the frequency directions, but not necessarily the full frequency vectors. Their union has positive relative  $(2n-1)$ -dimensional measure

in the local energy hypersurface.

*Proof.* Write

$$\omega(I) := Dh(I), \quad A(I) := D^2h(I).$$

The proof has four stages. The first three isolate the finite-dimensional geometry that distinguishes the isoenergetic theorem from ordinary KAM. The last stage inserts the Whitney-smooth ordinary KAM family proved in K05.

**Lemma 6.1 (bordered Hessian and projective-frequency transversality).** Fix  $I_0$  with  $\omega(I_0) \neq 0$  and assume that

$$\mathcal{B}_h(I_0) = \begin{pmatrix} A(I_0) & \omega(I_0) \\ \omega(I_0)^T & 0 \end{pmatrix} \quad (\text{I.1})$$

is invertible. Choose an index  $j$  for which  $\omega_j(I_0) \neq 0$  and, after shrinking the action neighborhood, put

$$r(\omega) := \left( \frac{\omega_1}{\omega_j}, \dots, \frac{\widehat{\omega_j}}{\omega_j}, \dots, \frac{\omega_n}{\omega_j} \right) \in \mathbb{R}^{n-1}. \quad (\text{I.2})$$

Then the map

$$\Xi_h(I) := (h(I), r(\omega(I))) \quad (\text{I.3})$$

has invertible derivative at  $I_0$ . Consequently the restriction of the frequency-direction map to  $h^{-1}(E)$  is locally nondegenerate, and frequency rays are transverse to the image of that energy hypersurface in frequency space.

Indeed, let  $v \in \ker D\Xi_h(I_0)$ . The first component gives  $\omega(I_0) \cdot v = 0$ . Vanishing of all derivatives of the ratios in (I.2) means precisely that

$$A(I_0)v = \lambda \omega(I_0) \quad (\text{I.4})$$

for some scalar  $\lambda$ : infinitesimally the frequency may change only along its own ray. Therefore

$$\mathcal{B}_h(I_0) \begin{pmatrix} v \\ -\lambda \end{pmatrix} = 0. \quad (\text{I.5})$$

Invertibility of (I.1) gives  $v = 0$  and  $\lambda = 0$ . Since the domain and codomain of  $D\Xi_h(I_0)$  both have dimension  $n$ , its kernel being zero makes it invertible. The inverse function theorem proves the assertion. Notice also that (I.1) itself forces  $\omega(I_0) \neq 0$ .

**Lemma 6.2 (Douady auxiliary Hamiltonian).** On a sufficiently small neighborhood of a compact isoenergetic action patch there is a real constant  $a$  such that

$$h_a(I) := h(I) + a(h(I) - E)^2 \quad (\text{I.6})$$

is Kolmogorov nondegenerate, while on  $h^{-1}(E)$  it has the same frequency vector as  $h$ . If

$$H_a := H + a(H - E)^2, \quad (\text{I.7})$$

then, in a sufficiently small tube about the selected energy hypersurface,

$$H_a^{-1}(E) = H^{-1}(E), \quad X_{H_a} = X_H \quad \text{on } H^{-1}(E), \quad (\text{I.8})$$

and  $H_a - h_a$  is  $O(\|f\|)$  in the fixed analytic norm.

To prove the nondegeneracy assertion, work first on the compact piece  $\Sigma' \subset \Sigma$  contained in one chart from Lemma 6.1. On  $h = E$ ,

$$D^2 h_a = A + 2a \omega \omega^T. \quad (\text{I.9})$$

Let  $\text{adj}(A)$  denote the classical adjugate, i.e. the transpose of the cofactor matrix. The rank-one determinant identity, valid also for singular  $A$ , gives

$$\det(A + 2a \omega \omega^T) = \det A + 2a \omega^T \text{adj}(A) \omega. \quad (\text{I.10})$$

Expansion of the bordered determinant along its last row and column gives

$$\det \mathcal{B}_h = -\omega^T \text{adj}(A) \omega. \quad (\text{I.11})$$

Thus the second term in (I.10) has a coefficient that is nowhere zero on  $\Sigma'$ . By compactness the continuous ratio

$$\frac{\det A}{\omega^T \text{adj}(A) \omega} \quad (\text{I.12})$$

is bounded. Choose  $a$  so large in absolute value that  $2a + (\det A)/(\omega^T \text{adj}(A) \omega)$  never vanishes on  $\Sigma'$ . Then (I.10) is nonzero there, and continuity gives an action neighborhood  $A_0$  of  $\Sigma'$  on which  $D^2 h_a$  remains invertible.

The identities on the fixed energy are immediate. Since

$$Dh_a = (1 + 2a(h - E))Dh, \quad (\text{I.13})$$

we have  $Dh_a = Dh$  on  $h = E$ . Likewise

$$X_{H_a} = (1 + 2a(H - E))X_H, \quad (\text{I.14})$$

so the vector fields agree on  $H = E$ . Moreover

$$H_a - E = (H - E)(1 + a(H - E)). \quad (\text{I.15})$$

After shrinking the phase-space tube so that  $|a(H - E)| < 1/2$ , equation  $H_a = E$  is equivalent to  $H = E$ . Finally,

$$H_a - h_a = f(1 + 2a(h - E)) + af^2, \quad (\text{I.16})$$

which is bounded by  $C_a \|f\|$  on the fixed tube when  $\|f\|$  is small. Thus ordinary analytic KAM applies to  $(h_a, H_a)$  on  $A_0$ .

**Lemma 6.3 (projective Diophantine directions have small excluded surface measure).** Let  $S \subset h^{-1}(E)$  be a compact subset of one isoenergetic chart from Lemma



6.1, with positive relative surface measure, and put

$$u(I) := \frac{\omega(I)}{|\omega(I)|}. \quad (\text{I.17})$$

For  $\tau > n - 1$  define

$$S_\gamma := \{I \in S : |k \cdot u(I)| \geq \gamma |k|_1^{-\tau} \text{ for every } 0 \neq k \in \mathbb{Z}^n\}. \quad (\text{I.18})$$

Then, for all sufficiently small  $\gamma$ ,

$$\text{vol}_{n-1}(S \setminus S_\gamma) \leq C_S \gamma. \quad (\text{I.19})$$

For  $n = 1$  the projective space has dimension zero and the assertion is immediate; assume  $n \geq 2$ . On the unit sphere the strip

$$Z_k(\gamma) := \{u \in \mathbb{S}^{n-1} : |k \cdot u| < \gamma |k|_1^{-\tau}\} \quad (\text{I.20})$$

has surface measure at most

$$C_n \gamma |k|_1^{-(\tau+1)}. \quad (\text{I.21})$$

Indeed, after an orthogonal rotation taking  $k/|k|_2$  to the first coordinate axis, (I.20) is a spherical zone of width  $O(\gamma |k|_1^{-\tau} / |k|_2)$ , and  $|k|_2$  is comparable with  $|k|_1$  in fixed dimension. The number of integer vectors with  $|k|_1 = m$  is  $O_n(m^{n-1})$ , so

$$\sum_{0 \neq k \in \mathbb{Z}^n} |k|_1^{-(\tau+1)} \leq C_n \sum_{m \geq 1} m^{n-1} m^{-(\tau+1)} < \infty \quad (\text{I.22})$$

exactly because  $\tau > n - 1$ . Countable subadditivity therefore bounds the bad directions on the sphere by  $C\gamma$ . By Lemma 6.1,  $u|_S$  is locally a  $C^1$  diffeomorphism onto a piece of the sphere (equivalently, one may use the projective chart (I.2)); compactness bounds both Jacobians. Pulling the spherical estimate back to  $S$  gives (I.19).

**Application of the ordinary Whitney-smooth KAM family.** Because  $\Sigma$  has positive relative measure, one of the finitely many isoenergetic charts covering it contains a compact subset  $S$  of positive measure. Shrink  $S$  slightly so that it is contained in the interior of the action neighborhood  $A_0$  supplied by Lemma 6.2. On  $A_0$  the auxiliary frequency map

$$F := Dh_a \quad (\text{I.23})$$

has nonsingular derivative, hence after one further shrinking it is a frequency chart in the sense required by the Whitney smooth dependence of KAM tori. On  $S \subset \{h = E\}$  we have  $F = Dh = \omega$ .

Since  $S$  is compact and  $\omega$  never vanishes there, choose constants  $0 < m < M < \infty$  with

$$m \leq |\omega(I)| \leq M \quad (I \in S). \quad (\text{I.24})$$

Fix a sufficiently small projective Diophantine constant  $\gamma$  and put  $\gamma_* := m\gamma/4$ . By the Whitney smooth dependence of KAM tori, if  $\|f\|$  is sufficiently small (depending on  $\gamma_*$  and the fixed analytic data), the auxiliary Hamiltonian  $H_a$  has a Whitney- $C^\infty$  family of

analytic Lagrangian KAM tori

$$J \mapsto K_J, \quad J \in (A_0)_{\gamma_*, \tau}, \quad (\text{I.25})$$

with exact frequency  $F(J)$ , and the combined torus map is Whitney- $C^1$  close to  $(\theta, J) \mapsto (\theta, J)$ .

For  $J$  in the Cantor parameter set,  $H_a$  is constant on  $K_J$ . Indeed,  $H_a$  is a first integral of its own flow and the Diophantine linear flow on the torus is dense; equivalently the Fourier equation  $F(J) \cdot \partial_\theta(H_a \circ K_J) = 0$  has only a constant solution. Denote this constant by

$$e(J) := H_a(K_J(\theta)). \quad (\text{I.26})$$

The Whitney- $C^\infty$  dependence in the Whitney smooth dependence of KAM tori and ordinary smooth composition show that  $e$  is a Whitney- $C^\infty$  scalar function on the Cantor set. Choose one smooth Whitney extension  $\bar{e}$  to a neighborhood of  $S$ . Since the KAM family is  $C^1$  close to the unperturbed family and  $H_a$  is close to  $h_a$ , this extension can be chosen  $C^1$  close to  $h_a$  on a smaller fixed neighborhood.

Use the same projective chart  $r$  as in Lemma 6.1 and define

$$G_0(J) := (h_a(J), r(F(J))), \quad G(J) := (\bar{e}(J), r(F(J))). \quad (\text{I.27})$$

On  $S$ , the first-order isoenergetic geometry of  $h_a$  is the same as that of  $h$ : for a tangent vector  $v$  to  $h^{-1}(E)$  one has  $\omega \cdot v = 0$ , and (I.9) gives  $(D^2 h_a)v = Av$ . Hence Lemma 6.1 applies to  $G_0$  on  $S$ . After restricting to one compact subchart of positive measure,  $G_0$  is a diffeomorphism onto its image. Since  $G$  is  $C^1$  close to  $G_0$ , the inverse function theorem and compactness imply that  $G$  is also a diffeomorphism on a common smaller neighborhood.

For  $I \in S_\gamma$  define  $\Psi(I)$  by

$$G(\Psi(I)) = (E, r(\omega(I))). \quad (\text{I.28})$$

The map  $\Psi$  is smooth and uniformly close to the identity. The equality of projective coordinates in (I.28) means that

$$F(\Psi(I)) = \lambda_I \omega(I) \quad (\text{I.29})$$

for a scalar  $\lambda_I$ . Since  $F(\Psi(I))$  is close to  $F(I) = \omega(I)$ , we may shrink the perturbation size so that

$$\frac{1}{2} \leq \lambda_I \leq 2. \quad (\text{I.30})$$

For  $I \in S_\gamma$ , (I.18), (I.24), and (I.29)–(I.30) imply

$$|k \cdot F(\Psi(I))| \geq \frac{m\gamma}{2} |k|_1^{-\tau} \geq \gamma_* |k|_1^{-\tau}. \quad (\text{I.31})$$

Thus  $\Psi(I)$  belongs to the very Cantor parameter set on which the KAM torus  $K_{\Psi(I)}$  exists. This point is essential: the smooth extension  $\bar{e}$  was used only to locate the candidate parameter; the Diophantine estimate (I.31) puts that parameter back on the genuine invariant KAM set. Consequently  $\bar{e}(\Psi(I)) = e(\Psi(I))$ , and the first component of

(I.28) gives

$$H_a(K_{\Psi(I)}(\theta)) = E. \quad (\text{I.32})$$

By (I.8), the torus therefore lies in  $H^{-1}(E)$  and the vector fields  $X_{H_a}$  and  $X_H$  agree on it. Since the KAM invariance equation for  $H_a$  is

$$X_{H_a} \circ K_{\Psi(I)} = DK_{\Psi(I)} F(\Psi(I)), \quad (\text{I.33})$$

we obtain from (I.29)

$$X_H \circ K_{\Psi(I)} = DK_{\Psi(I)} (\lambda_I Dh(I)). \quad (\text{I.34})$$

This is exactly preservation of the frequency direction, or equivalently of all frequency ratios; the positive scalar  $\lambda_I$  is the permitted constant rescaling of time.

Put

$$K_I^E := K_{\Psi(I)}, \quad I \in S_\gamma. \quad (\text{I.35})$$

Because  $\Psi$  is smooth and the ordinary KAM family is Whitney- $C^\infty$ ,  $I \mapsto K_I^E$  is Whitney- $C^\infty$ . Each torus is analytic in the angle, Lagrangian, and  $C^1$  close to the standard torus  $\theta \mapsto (\theta, I)$ . By Lemma 6.3,  $S_\gamma$  is closed and has positive relative surface measure for all sufficiently small  $\gamma$ , with complement  $O(\gamma)$  inside the chosen chart. Finally the combined map

$$(\theta, I) \mapsto K_I^E(\theta) \quad (\text{I.36})$$

is  $C^1$  close to the standard embedding of  $\mathbb{T}^n \times S$  into the unperturbed energy hypersurface. After one last shrinking it is injective with Jacobian bounded below on the compact Cantor family. Hence the union of the selected tori has positive relative  $(2n-1)$ -dimensional measure in the local perturbed energy hypersurface. This proves every clause of the theorem.  $\square$

### Theorem K06.3. Moser invariant-curve theorem for twist maps

**Statement.** Let  $F_0(\theta, I) = (\theta + \omega(I), I)$  be real analytic and exact symplectic on  $\mathbb{T} \times [a, b]$ , with  $|\omega'(I)| \geq m > 0$ . Let  $I_\rho \in (a, b)$  satisfy  $\omega(I_\rho) = \rho$ , and assume  $\rho$  is Diophantine:  $|k\rho - \ell| \geq \gamma|k|^{-\tau}$  for all  $k \in \mathbb{Z} \setminus \{0\}$  and  $\ell \in \mathbb{Z}$ , for fixed  $\gamma > 0$  and  $\tau > 1$ . On every fixed complex neighborhood of the circle  $I = I_\rho$  there is  $\varepsilon_0 > 0$ , depending on the neighborhood,  $m, \gamma, \tau$  and analytic bounds for  $F_0$ , such that every real-analytic exact symplectic map  $F$  with  $\|F - F_0\| < \varepsilon_0$  has a real-analytic essential invariant circle close to  $I = I_\rho$ ; equivalently, there is an analytic embedding  $K : \mathbb{T} \rightarrow \mathbb{T} \times (a, b)$  with  $F \circ K = K \circ R_\rho$ , where  $R_\rho(\theta) = \theta + \rho \pmod{1}$ .

*Proof.* Work with a lift of the annulus map to  $\mathbb{R} \times (a, b)$  and with period one in the angular variable. Put

$$K_0(\theta) := (\theta, I_\rho), \quad R_\rho(\theta) := \theta + \rho. \quad (\text{T.1})$$

Since  $F_0 \circ K_0 = K_0 \circ R_\rho$ , the initial invariance defect

$$E_0 := F \circ K_0 - K_0 \circ R_\rho \quad (\text{T.2})$$

has analytic norm  $O(\|F - F_0\|)$ . We prove a quantitative Newton step for a general approximate circle and then iterate it.

**Lemma 6.4 (analytic rotation cohomology).** Let  $g$  be holomorphic and one-periodic on  $\mathbb{T}_s$ , with zero average, and suppose

$$|k\rho - \ell| \geq \gamma|k|^{-\tau} \quad (k \in \mathbb{Z} \setminus \{0\}, \ell \in \mathbb{Z}). \quad (\text{T.3})$$

Then for every  $0 < \sigma < s$  the equation

$$u(\theta) - u(\theta + \rho) = g(\theta), \quad [u] = 0, \quad (\text{T.4})$$

has a unique zero-average holomorphic solution on  $\mathbb{T}_{s-\sigma}$  and

$$\|u\|_{s-\sigma} \leq C_\tau \gamma^{-1} \sigma^{-(\tau+1)} \|g\|_s, \quad \|u'\|_{s-2\sigma} \leq C_\tau \gamma^{-1} \sigma^{-(\tau+2)} \|g\|_s. \quad (\text{T.5})$$

Indeed, if  $g = \sum_{k \neq 0} g_k e^{2\pi i k \theta}$ , set

$$u_k := \frac{g_k}{1 - e^{2\pi i k \rho}}. \quad (\text{T.6})$$

For  $x \in \mathbb{R}$ ,  $|1 - e^{2\pi i x}| = 2|\sin \pi x| \geq 4 \operatorname{dist}(x, \mathbb{Z})$  when the distance is at most  $1/2$ . Thus (T.3) gives

$$|1 - e^{2\pi i k \rho}|^{-1} \leq C \gamma^{-1} |k|^\tau. \quad (\text{T.7})$$

Analytic Fourier decay and  $\sum_{k \geq 1} k^\tau e^{-2\pi \sigma k} \leq C_\tau \sigma^{-(\tau+1)}$  prove the first estimate; Cauchy's estimate proves the second. Uniqueness follows because a zero-average solution of the homogeneous equation has no nonzero Fourier coefficient.

**Lemma 6.5 (automatic reducibility and the exact-symplectic mean obstruction).** Let  $K$  be a real analytic degree-one embedding on  $\mathbb{T}_s$  and put

$$E(\theta) := F(K(\theta)) - K(\theta + \rho), \quad L(\theta) := DK(\theta). \quad (\text{T.8})$$

Assume the analytic Gram factor

$$G(\theta) := L(\theta)^T L(\theta)$$

has no zero on the strip and  $G^{-1}$  is uniformly bounded, and set

$$N(\theta) := G(\theta)^{-1} \begin{pmatrix} -L_2(\theta) \\ L_1(\theta) \end{pmatrix}, \quad P(\theta) := (L(\theta), N(\theta)). \quad (\text{T.9})$$

Then  $P$  is holomorphic and  $\det P = 1$ . For embeddings sufficiently close to  $K_0$ , one has  $G = 1 + O(D(K - K_0))$ , so this analytic nonvanishing condition and its inverse bound are automatic. If

$$M(\theta) := P(\theta + \rho)^{-1} DF(K(\theta)) P(\theta), \quad (\text{T.10})$$

then

$$M(\theta) = \begin{pmatrix} 1 & T(\theta) \\ 0 & 1 \end{pmatrix} + R(\theta), \quad \|R\|_{s-\sigma} \leq C \|DE\|_{s-\sigma}, \quad (\text{T.11})$$

provided  $P$  and  $P^{-1}$  are uniformly bounded. Moreover, if

$$\eta(\theta) := P(\theta + \rho)^{-1} E(\theta), \quad (\text{T.12})$$

then exact symplecticity gives the quadratic compatibility estimate

$$|[\eta_2]| \leq C \|E\|_s \|DE\|_s. \quad (\text{T.13})$$

For (T.11), differentiate (T.8):

$$DF(K(\theta))L(\theta) = L(\theta + \rho) + DE(\theta). \quad (\text{T.14})$$

Hence the first column of  $M$  is  $(1, 0)^T + O(DE)$ . Since  $F$  is symplectic and both frames  $P$  have determinant one,  $M$  has determinant one. A  $2 \times 2$  determinant expansion then forces the remaining diagonal entry to be  $1 + O(DE)$ ; the upper-right entry is denoted by  $T$ . This proves (T.11).

To prove (T.13), use the primitive  $\lambda = I d\theta$  from the definition of exact symplectic annulus map above. Exactness means  $F^*\lambda - \lambda = dS$  for a periodic  $S$ . Therefore

$$\int_{\mathbb{T}} (F \circ K)^*\lambda = \int_{\mathbb{T}} K^*\lambda. \quad (\text{T.15})$$

Translation of the parameter does not change the integral, so the right side also equals  $\int (K \circ R_\rho)^*\lambda$ . Write  $K \circ R_\rho = (x_+, y_+)$  and  $E = (e_x, e_y)$ . Expanding (T.15) gives

$$0 = \int_{\mathbb{T}} (e_y x'_+ + y_+ e'_x + e_y e'_x) d\theta. \quad (\text{T.16})$$

Integrating the middle term by parts yields

$$\int_{\mathbb{T}} (x'_+ e_y - y'_+ e_x) d\theta = - \int_{\mathbb{T}} e_y e'_x d\theta. \quad (\text{T.17})$$

Because the second coordinate of  $P_+^{-1}E$  is  $\det(L_+, E) = x'_+ e_y - y'_+ e_x$ , the left side is  $[\eta_2]$ . This proves (T.13). This is the exact-symplectic “vanishing lemma”: the normal mean is not assumed to be zero, but it is already quadratic in the invariance defect.

**Lemma 6.6 (one analytic exact-twist quasi-Newton step).** Fix analytic-domain and frame bounds, and assume

$$|[T]| \geq t_0 > 0. \quad (\text{T.18})$$

There are exponents  $p, q < \infty$  and a constant  $C$  depending only on the fixed data such that, whenever  $E$  is sufficiently small, for every fixed strip loss  $0 < \sigma < s/8$  there is an analytic correction  $\Delta$  on  $\mathbb{T}_{s-4\sigma}$  with

$$\|\Delta\|_{s-4\sigma} + \sigma \|D\Delta\|_{s-4\sigma} \leq C \gamma^{-2} \sigma^{-p} \|E\|_s \quad (\text{T.19})$$

and the corrected embedding  $K_1 = K + \Delta$  satisfies

$$\|F \circ K_1 - K_1 \circ R_\rho\|_{s-4\sigma} \leq C \gamma^{-4} \sigma^{-q} \|E\|_s^2. \quad (\text{T.20})$$

The analytic Gram-inverse/frame bounds and the lower bound  $|[T]| \geq t_0/2$  persist if the right side of (T.19) is small enough.

To prove the step, seek  $\Delta = P\xi$ . Ignore for the moment the remainder  $R$  in (T.11)

and solve the triangular leading equation

$$\begin{pmatrix} 1 & T \\ 0 & 1 \end{pmatrix} \xi - \xi \circ R_\rho = -\eta \quad (\text{T.21})$$

up to the already quadratic mean  $[\eta_2]$ . Write  $\eta_2 = \eta_2^0 + [\eta_2]$  with  $[\eta_2^0] = 0$ . By Lemma 6.4 choose the zero-average solution  $\xi_2^0$  of

$$\xi_2^0 - \xi_2^0 \circ R_\rho = -\eta_2^0. \quad (\text{T.22})$$

The general leading normal correction is  $\xi_2 = \xi_2^0 + c$ . Choose

$$c := -\frac{[\eta_1 + T\xi_2^0]}{[T]}. \quad (\text{T.23})$$

Then the right side of

$$\xi_1 - \xi_1 \circ R_\rho = -\eta_1 - T\xi_2 \quad (\text{T.24})$$

has zero average, and Lemma 6.4 gives a unique zero-average  $\xi_1$ . Two uses of (T.5), together with bounded  $P$ ,  $P^{-1}$ , and  $T$ , give (T.19) with a finite power  $p$ .

The residual in the exact linearized equation consists of two terms. First, Lemma 6.5 leaves the normal mean  $[\eta_2]$ , which is  $O(\|E\|\|DE\|)$ . Second, replacing  $M$  by its triangular part creates  $R\xi = O(\|DE\|\|\xi\|)$ . Both are quadratic after one Cauchy derivative loss. Taylor's formula gives

$$F(K + \Delta) = F(K) + DF(K)\Delta + O(\|\Delta\|^2) \quad (\text{T.25})$$

on a smaller strip, so the nonlinear remainder is also quadratic. Combining these estimates gives (T.20) for some finite  $q$ . Finally  $DK_1$  is close to  $DK$ , so  $G_1 = L_1^T L_1$  remains away from zero by a Neumann/geometric-series estimate and the analytic frames persist. Formula (T.10) depends continuously on  $F, K, DK$ , so its upper-right entry and its average change by  $O(\|D\Delta\|)$ ; this preserves (T.18) with  $t_0/2$  under the same smallness condition.

**Newton iteration.** For  $K_0$  in (T.1),  $L_0 = (1, 0)^T$ ,  $P_0 = I_2$ , and for the integrable map

$$DF_0(K_0) = \begin{pmatrix} 1 & \omega'(I_\rho) \\ 0 & 1 \end{pmatrix}. \quad (\text{T.26})$$

Thus the initial torsion average equals  $\omega'(I_\rho)$  for  $F_0$  and, when  $F$  is sufficiently close to  $F_0$ ,

$$|[T_0]| \geq m/2. \quad (\text{T.27})$$

The initial defect and its derivative are  $O(\varepsilon)$ , where  $\varepsilon = \|F - F_0\|$  on the prescribed complex neighborhood.

Choose strip losses  $\sigma_j = \sigma_0 2^{-j-2}$  with  $\sum_j 4\sigma_j < s/2$ . If  $e_j$  denotes the defect norm at step  $j$ , Lemma 6.6 gives

$$e_{j+1} \leq AB^j e_j^2 \quad (\text{T.28})$$

for fixed  $A, B$  after absorbing the powers of  $\sigma_j^{-1}$  and the fixed Diophantine constants.

For  $e_0$  sufficiently small this standard quadratic recurrence implies double-exponential decay, for example after increasing  $A$ ,

$$e_j \leq A^{-1} B^{-j-2} 2^{-2^j}. \quad (\text{T.29})$$

In particular the correction norms and their first derivatives are summable. Hence  $K_j$  converges uniformly with one derivative on a positive strip to a real-analytic degree-one embedding  $K_\infty$ . The total  $C^1$  displacement can be made arbitrarily small, so its angular lift is  $\theta \mapsto \theta + u(\theta)$  with  $\|u'\| < 1/2$ ; it is therefore injective modulo  $\mathbb{Z}$  and the image is an essential circle. Passing to the limit in the invariance equations gives

$$F \circ K_\infty = K_\infty \circ R_\rho. \quad (\text{T.30})$$

Thus the circle is invariant and the induced dynamics is analytically conjugate to rigid rotation by  $\rho$ . All iterates remain inside the fixed complex neighborhood because the total displacement is smaller than the initial domain margin. This proves the theorem.  $\square$

#### Theorem K06.4. Finite-differentiability KAM

**Statement.** Fix  $d \geq 2$  and real numbers  $\ell, \nu$  with

$$\ell > 2\nu > 2d, \quad \tau := \nu - 1,$$

and let  $D \subset \mathbb{R}^d$  be a nonempty bounded domain. Let  $K \in C^\ell(D)$  and  $P \in C^\ell(D \times \mathbb{T}^d)$  in the sense of the finite-Hölder-class convention of Definition K06.11, put

$$H(y, x) = K(y) + P(y, x), \quad \varepsilon := \|P\|_{C^\ell},$$

and assume that  $D^2K(y)$  is invertible for every  $y \in D$  with  $\sup_D \|(D^2K)^{-1}\| < \infty$ . For every fixed interior buffer  $\Delta > 0$  with

$$D_{3\Delta} := \{y \in D : \text{dist}(y, \mathbb{R}^d \setminus D) > 3\Delta\} \neq \emptyset,$$

and for fixed bounds on  $K, (D^2K)^{-1}$  and the geometry of this buffered region, there is  $c > 0$  such that the following holds. If  $0 < \alpha < 1$  is sufficiently small and

$$\varepsilon \leq c \alpha^{\frac{2\ell}{\ell-2\nu}}, \quad (\text{FR.1})$$

then every action  $y \in D_{3\Delta}$  whose unperturbed frequency  $DK(y)$  is  $(\alpha, \tau)$ -Diophantine belongs to a Cantor family of invariant Lagrangian KAM tori for  $H$ . Each surviving torus is a graph of class  $C^\nu$  over  $\mathbb{T}^d$ , carries the frequency  $DK(y)$ , and is a small deformation of the corresponding unperturbed torus. More precisely, with

$$\beta_0 := \min \left\{ \frac{\ell}{2\nu} - 1 + \frac{1}{\nu}, 2 \right\},$$

the Cantor family has a parameterization of Whitney class  $C_W^\beta$  for every  $\nu^{-1} < \beta < \beta_0$ ; for each fixed surviving Cantor parameter its angular parameterization belongs

to  $C^{\beta\nu}$ . This parameter regularity is distinct from the  $C^\nu$  graph regularity of each individual torus.

If, in addition,  $\partial D$  is smooth and the  $C^\ell$  data extend with the same regularity to  $\overline{D}$  (equivalently, have the standard  $C^\ell$  traces there), the localization can be made uniform as the real boundary buffer shrinks with  $\alpha$ . In particular, for a fixed structural constant  $b > 4$  the construction may be run over

$$\mathcal{E}_\alpha := \left\{ y \in D : \text{dist}(y, \mathbb{R}^d \setminus D) \geq b\alpha, \quad DK(y) \text{ is } (\alpha, \tau)\text{-Diophantine} \right\}. \quad (\text{FR.2})$$

Choosing

$$\alpha \asymp \varepsilon^{\frac{1}{2} - \frac{\nu}{\ell}} \quad (\text{FR.3})$$

then gives a resulting KAM set  $\mathcal{K} \subset D \times \mathbb{T}^d$  whose phase-space complement satisfies

$$\text{Leb}((D \times \mathbb{T}^d) \setminus \mathcal{K}) = O\left(\varepsilon^{\frac{1}{2} - \frac{\nu}{\ell}}\right). \quad (\text{FR.4})$$

The constants in (FR.1)–(FR.4) depend only on the frozen  $C^\ell$  nondegeneracy/domain data and on  $d, \ell, \nu$ .

### Scope note

The theorem gives the stated transverse Whitney regularity; Whitney  $C^\nu$  regularity in the Cantor parameter is not asserted.

### Proof strategy

Approximate the finitely differentiable Hamiltonian by analytic Hamiltonians on a shrinking sequence of complex strips. Apply the analytic KAM step at each scale and balance the smoothing error against the small-divisor loss. Bernstein summation converts convergence on shrinking strips into finite-order regularity of the limiting invariant tori. A Whitney extension argument controls the parameter family, and the same scale choice yields the measure estimate for the surviving Diophantine set.

**Lemma 6.7 (analytic heat-kernel approximation).** Let  $m \geq 1$ , let  $\ell > 0$ , and let  $0 \leq r \leq \ell$ . If  $f \in C_c^\ell(\mathbb{R}^m)$ , then for every  $0 < s \leq 1$  there is an entire function  $S_s f : \mathbb{C}^m \rightarrow \mathbb{C}$  which is real on  $\mathbb{R}^m$  and, for every fixed  $0 < c < 1$ , satisfies

$$\sup_{|\text{Im } z| \leq cs} |\partial^\alpha S_s f(z)| \leq C_{c,m,\ell} s^{-|\alpha|} \|f\|_{C^0} \quad (|\alpha| \leq [\ell] + 1), \quad (\text{FRS.1})$$

and, on the real domain,

$$\|S_s f - f\|_{C^r(\mathbb{R}^m)} \leq C_{m,\ell,r} s^{\ell-r} \|f\|_{C^\ell} \quad (0 \leq r \leq \ell). \quad (\text{FRS.2})$$

For two scales  $0 < s' < s \leq 1$  this gives

$$\|S_{s'} f - S_s f\|_{C^r} \leq C s^{\ell-r} \|f\|_{C^\ell}. \quad (\text{FRS.3})$$



The same conclusions hold for functions on  $\mathbb{R}^{m_1} \times \mathbb{T}^{m_2}$ , with periodic heat convolution in the torus variables. Vector- and matrix-valued versions hold componentwise.

*Proof.* Put  $q = \lfloor \ell \rfloor$  and choose an integer  $N > q/2$ . Fix distinct positive numbers  $a_0, \dots, a_N$ . There are unique real coefficients  $c_0, \dots, c_N$  solving

$$\sum_{j=0}^N c_j = 1, \quad \sum_{j=0}^N c_j a_j^{2r} = 0 \quad (1 \leq r \leq N). \quad (\text{FRS.4})$$

Indeed, this is a Vandermonde system in the distinct numbers  $a_j^2$ . With

$$G_s(x) := (4\pi s^2)^{-m/2} e^{-|x|^2/(4s^2)}, \quad \tilde{G}_s := \sum_{j=0}^N c_j G_{a_j s}, \quad (\text{FRS.5})$$

all moments of  $\tilde{G}_s$  of positive total degree at most  $q$  vanish: odd moments vanish by symmetry, while every even multi-index of total degree  $2r \leq q$  is a fixed constant times  $s^{2r} \sum_j c_j a_j^{2r} = 0$ . The zeroth moment equals one. Define

$$S_s f(z) := \int_{\mathbb{R}^m} \tilde{G}_s(z - y) f(y) dy. \quad (\text{FRS.6})$$

Because  $f$  has compact support, this integral and all of its  $z$ -derivatives converge locally uniformly on  $\mathbb{C}^m$ ; hence  $S_s f$  is entire and is real on the real locus.

For  $z = x + i\eta$  with  $|\eta| \leq cs$ , each Gaussian component satisfies

$$|G_{a_j s}(z - y)| \leq e^{c^2/(4a_j^2)} G_{a_j s}(x - y). \quad (\text{FRS.7})$$

Every derivative of  $G_{a_j s}$  is  $(a_j s)^{-|\alpha|}$  times a polynomial in  $(z - y)/(a_j s)$  multiplied by the Gaussian. Since the  $a_j$  and  $c_j$  are fixed once  $m, \ell$  are fixed, Gaussian moment bounds give (FRS.1).

Now let  $0 \leq k \leq q$  be an integer. Differentiation commutes with convolution, so

$$\partial^k (S_s f - f)(x) = \int \tilde{G}_s(y) (\partial^k f(x - y) - \partial^k f(x)) dy. \quad (\text{FRS.8})$$

Write  $\ell = q + \eta$ ,  $0 \leq \eta < 1$ . If  $\eta > 0$ , expand  $\partial^k f(x - y)$  at  $x$  through order  $q - k$ . Every polynomial term of positive degree integrates to zero by the moment conditions above, while the zeroth term cancels  $\partial^k f(x)$  because  $\int \tilde{G}_s = 1$ . Taylor's formula with Hölder remainder therefore gives

$$|\partial^k (S_s f - f)(x)| \leq C \|f\|_{C^\ell} \int |\tilde{G}_s(y)| |y|^{\ell-k} dy \leq C s^{\ell-k} \|f\|_{C^\ell}. \quad (\text{FRS.9})$$

If  $\eta = 0$  and  $k < q$ , expand only through order  $q - k - 1$  and use the integral Taylor remainder with the bounded  $q$ th derivatives; this gives the same estimate with exponent  $q - k = \ell - k$ . For  $k = q$  the required exponent is zero and the estimate follows directly from the  $L^1$  bound for the kernel. Finally, applying the same difference-quotient argument to the  $\lfloor r \rfloor$ th derivatives (or, equivalently, interpolating the two neighboring integer estimates) gives the required Hölder seminorm for noninteger  $r$ . Hence (FRS.2)

holds for every real  $0 \leq r \leq \ell$ . Then

$$\|S_{s'}f - S_sf\|_{C^r} \leq \|S_{s'}f - f\|_{C^r} + \|S_sf - f\|_{C^r} \leq Cs^{\ell-r}\|f\|_{C^\ell}$$

for  $s' < s$ , proving (FRS.3).

For torus variables periodize every Gaussian component in (FRS.5). The periodized series and all derivatives converge normally on fixed strips by Gaussian decay, the moment cancellation is local on the periodic lift, and the same Taylor/difference-quotient estimates hold. Taking products of the Euclidean and periodic kernels proves the mixed action-angle statement. Vector- and matrix-valued cases are component-wise.  $\square$

**Lemma 6.8 (sharp Rüssmann flow-cohomology estimate).** Fix  $d \geq 1$ ,  $\tau \geq 0$ ,  $\alpha > 0$ , and let  $\omega \in \mathbb{R}^d$  satisfy

$$|k \cdot \omega| \geq \alpha |k|_1^{-\tau} \quad (0 \neq k \in \mathbb{Z}^d). \quad (\text{FRR.1})$$

Let  $f$  be bounded and holomorphic on  $\mathbb{T}_s^d$ , with zero average, and use the convention used in this book  $f(x) = \sum_{k \in \mathbb{Z}^d} \widehat{f}(k) e^{2\pi i k \cdot x}$ . Then  $D_\omega g = f$ ,  $\langle g \rangle = 0$ , has a unique holomorphic solution. For every multi-index  $b \in \mathbb{N}^d$ ,  $q := |b|_1$ , and every  $0 < \sigma < \min\{s, 1\}$ ,

$$\|\partial_x^b g\|_{s-\sigma} \leq C_{d,\tau,q} \alpha^{-1} \sigma^{-(\tau+q)} \|f\|_s. \quad (\text{FRR.2})$$

In particular, with  $\nu = \tau + 1$ ,

$$\|g_x\|_{s-\sigma} \leq C_{d,\tau} \alpha^{-1} \sigma^{-\nu} \|f\|_s. \quad (\text{FRR.3})$$

*Proof.* The Fourier equation gives, for  $k \neq 0$ ,

$$\widehat{g}(k) = \frac{\widehat{f}(k)}{2\pi i k \cdot \omega}, \quad \widehat{g}(0) = 0. \quad (\text{FRR.4})$$

The point is not to estimate the resulting Fourier series term by term; that elementary estimate would introduce an unnecessary dimension loss. We instead use the spacing of the small divisors.

First note the square-Fourier bound

$$\sum_{k \in \mathbb{Z}^d} |\widehat{f}(k)|^2 e^{4\pi s |k|_1} \leq 2^d \|f\|_s^2. \quad (\text{FRR.5})$$

Indeed, fix  $0 < s' < s$ . For each sign vector  $\epsilon \in \{\pm 1\}^d$ , Parseval applied to  $x \mapsto f(x + is'\epsilon)$  gives

$$\sum_k |\widehat{f}(k)|^2 e^{-4\pi s' k \cdot \epsilon} \leq \|f\|_{s'}^2.$$

After summing these  $2^d$  inequalities, the inner sign sum is at least  $e^{4\pi s' |k|_1}$ , because one may choose  $\epsilon_j = -\text{sgn}(k_j)$  (arbitrarily when  $k_j = 0$ ). Consequently

$$\sum_k |\widehat{f}(k)|^2 e^{4\pi s' |k|_1} \leq 2^d \|f\|_{s'}^2.$$

Letting  $s' \uparrow s$  and using monotone convergence proves (FRR.5).

We next establish the Rüssmann divisor-counting estimate. For  $N \geq 1$  put

$$A_q(N) := \sum_{0 < |k|_1 \leq N} \frac{|k|_1^{2q}}{|k \cdot \omega|^2}.$$

Among the positive numbers  $k \cdot \omega$  with  $0 < |k|_1 \leq N$ , list the values increasingly as  $0 < d_1 < d_2 < \dots$ . They are distinct: equality would give  $(k - k') \cdot \omega = 0$ , contradicting (FRR.1). Moreover

$$d_1 \geq \alpha N^{-\tau}, \quad d_j - d_{j-1} \geq \alpha(2N)^{-\tau},$$

because  $|k_j - k_{j-1}|_1 \leq 2N$ . Hence

$$d_j \geq \alpha N^{-\tau} (1 + (j-1)2^{-\tau}).$$

The same argument applies to the absolute values of the negative divisors, so, with the finite constant

$$C_\tau := 2 \sum_{j \geq 1} (1 + (j-1)2^{-\tau})^{-2} < \infty,$$

we obtain

$$\sum_{0 < |k|_1 \leq N} \frac{1}{|k \cdot \omega|^2} \leq C_\tau \alpha^{-2} N^{2\tau}. \quad (\text{FRR.6})$$

Consequently

$$A_q(N) \leq C_\tau \alpha^{-2} N^{2(\tau+q)}. \quad (\text{FRR.7})$$

Set  $r = \tau + q$  and  $c = 4\pi\sigma$ . Abel summation applied to the shells  $|k|_1 = n$  and (FRR.7) yields

$$\begin{aligned} \sum_{k \neq 0} \frac{|k|_1^{2q}}{|k \cdot \omega|^2} e^{-4\pi\sigma|k|_1} &= \sum_{n \geq 1} A_q(n) (e^{-cn} - e^{-c(n+1)}) \\ &\leq C_\tau \alpha^{-2} (1 - e^{-c}) \sum_{n \geq 1} n^{2r} e^{-cn} \\ &\leq C'_{\tau,q} \alpha^{-2} \sigma^{-2r}. \end{aligned} \quad (\text{FRR.8})$$

For the last inequality we used  $1 - e^{-c} \leq c$  and the standard integral majorant  $\sum_{n \geq 1} n^{2r} e^{-cn} \leq C_r c^{-(2r+1)}$  for  $0 < c \leq 4\pi$ .

Finally, (FRR.4), Cauchy-Schwarz, (FRR.5), and (FRR.8) give

$$\begin{aligned} \|\partial_{xg}^b\|_{s-\sigma} &\leq (2\pi)^{q-1} \sum_{k \neq 0} |\widehat{f}(k)| \frac{|k|_1^q}{|k \cdot \omega|} e^{2\pi(s-\sigma)|k|_1} \\ &\leq (2\pi)^{q-1} \left( \sum_k |\widehat{f}(k)|^2 e^{4\pi s|k|_1} \right)^{1/2} \left( \sum_{k \neq 0} \frac{|k|_1^{2q}}{|k \cdot \omega|^2} e^{-4\pi\sigma|k|_1} \right)^{1/2} \\ &\leq C_{d,\tau,q} \alpha^{-1} \sigma^{-(\tau+q)} \|f\|_s. \end{aligned}$$

The same bounds imply normal convergence on the smaller strip. Equation (FRR.4) then verifies  $D_\omega g = f$  term by term, and uniqueness follows because (FRR.1) makes

every nonzero divisor nonvanishing.  $\square$

**Lemma 6.9 (finite-mode Diophantine stability radius).** Let  $K$  be holomorphic on an action neighborhood and suppose  $\|K_{yy}\| \leq K$ . Fix a real action  $y_0$  for which  $\omega_0 = K_y(y_0)$  satisfies  $|k \cdot \omega_0| \geq \alpha |k|_1^{-\tau}$ . Put  $\nu = \tau + 1$ . If  $\kappa \geq 1$  and

$$|y - y_0|_\infty \leq \bar{r}, \quad \bar{r} \leq \frac{\alpha}{2dK\kappa^\nu}, \quad (\text{FRD.1})$$

then every integer mode  $0 < |k|_1 \leq \kappa$  satisfies

$$|K_y(y) \cdot k| \geq \frac{\alpha}{2} |k|_1^{-\tau}. \quad (\text{FRD.2})$$

*Proof.* Along the segment from  $y_0$  to  $y$ , the mean-value formula and the chosen coordinate norms give

$$|(K_y(y) - K_y(y_0)) \cdot k| \leq dK\bar{r} |k|_1.$$

By (FRD.1) and  $|k|_1 \leq \kappa$ ,

$$dK\bar{r}|k|_1 \leq \frac{\alpha}{2} \kappa^{-\nu} |k|_1 \leq \frac{\alpha}{2} |k|_1^{-\tau},$$

since  $\nu = \tau + 1$ . Subtracting this from the Diophantine lower bound at  $y_0$  proves (FRD.2).  $\square$

**Lemma 6.10 (shrinking-strip Bernstein convergence below the approximation order).** Let  $0 < \xi < 1$ ,  $s_j = s_0 \xi^j$ , and let  $F_j$  be scalar-, vector-, or matrix-valued functions holomorphic on  $\mathbb{T}_{s_j}^d$ . Suppose that for some  $\mu > 0$  and  $M > 0$ ,

$$\|F_{j+1} - F_j\|_{s_{j+1}} \leq M s_j^\mu \quad (j \geq 0). \quad (\text{FRB.1})$$

Then for every real  $0 \leq r < \mu$  the real restrictions  $F_j|_{\mathbb{T}^d}$  converge in  $C^r$  to a limit  $F_\infty \in C^r(\mathbb{T}^d)$ , and

$$\|F_\infty - F_J\|_{C^r} \leq C_{d,r,\mu,\xi} M s_J^{\mu-r}. \quad (\text{FRB.2})$$

The same statement holds uniformly with passive parameters whenever (FRB.1) is uniform in those parameters.

*Proof.* Write  $U_j = F_{j+1} - F_j$ . If  $q$  is a nonnegative integer, Cauchy's estimate on the strip  $s_{j+1}$  gives

$$\|D^q U_j\|_{C^0} \leq C_{d,q} s_{j+1}^{-q} \|U_j\|_{s_{j+1}} \leq C_{d,q,\xi} M s_j^{\mu-q}. \quad (\text{FRB.3})$$

Hence the derivative series  $\sum_j D^q U_j$  converges uniformly whenever  $q < \mu$ .

Now let  $r = q + \eta$  with  $q = \lfloor r \rfloor$  and  $0 < \eta < 1$ . For a  $C^1$  function  $v$  on the real torus the elementary interpolation estimate

$$[v]_{C^{0,\eta}} \leq C_\eta \|v\|_{C^0}^{1-\eta} \|Dv\|_{C^0}^\eta \quad (\text{FRB.4})$$

follows by bounding  $|v(x) - v(y)|$  by the minimum of  $2\|v\|_{C^0}$  and  $\|Dv\|_{C^0}|x - y|$  and optimizing at the crossover scale. Applying (FRB.4) to  $v = D^q U_j$  and using (FRB.3) at orders  $q$  and  $q + 1$  yields

$$[D^q U_j]_{C^{0,\eta}} \leq C M s_j^{\mu-q-\eta} = C M s_j^{\mu-r}. \quad (\text{FRB.5})$$

The right-hand side is a summable geometric sequence because  $r < \mu$ . Together with

(FRB.3), this proves convergence in  $C^r$ . Summing from  $j = J$  to infinity gives (FRB.2). The argument is componentwise and is unchanged under a uniform passive parameter.  $\square$

**Lemma 6.11 (invariant irrational torus is Lagrangian).** Let  $M = \mathbb{R}^d \times \mathbb{T}^d$  carry the exact standard symplectic form  $\Omega = d\lambda$ ,  $\lambda = \sum_{j=1}^d y_j dx_j$ . Let  $H \in C^2(M)$  and let  $K : \mathbb{T}^d \rightarrow M$  be a  $C^1$  embedding satisfying

$$\phi_H^t \circ K(x) = K(x + t\omega) \quad (t \in \mathbb{R}), \quad (\text{FRL.1})$$

where  $k \cdot \omega \neq 0$  for every  $0 \neq k \in \mathbb{Z}^d$ . Then  $K^*\Omega = 0$ . Since  $K(\mathbb{T}^d)$  has half the ambient dimension, it is a Lagrangian torus.

*Proof.* Hamiltonian flow preserves  $\Omega$ . If  $R_t(x) = x + t\omega$ , then (FRL.1) gives

$$R_t^*(K^*\Omega) = K^*((\phi_H^t)^*\Omega) = K^*\Omega. \quad (\text{FRL.2})$$

Write the continuous two-form  $\beta := K^*\Omega$  in torus coordinates as  $\beta = \sum_{i < j} b_{ij}(x) dx_i \wedge dx_j$ . Rational independence of  $\omega$  makes every orbit  $x + t\omega$  dense. By (FRL.2), each continuous coefficient  $b_{ij}$  is constant along a dense orbit and hence constant on the whole torus. Thus  $\beta$  is a constant two-form.

On the other hand,

$$\beta = K^*d\lambda = d(K^*\lambda)$$

is exact. Integrating a constant two-form over each coordinate 2-torus shows that every coefficient  $b_{ij}$  must vanish. Therefore  $K^*\Omega = 0$ . An isotropic submanifold of dimension  $d$  in a symplectic manifold of dimension  $2d$  is Lagrangian.  $\square$

**Lemma 6.12 (Rüssmann weighted finite-degree approximation).** Let  $d \geq 1$ ,  $r, s > 0$ , and let  $F$  be bounded and holomorphic on  $D_r \times \mathbb{T}_s^d$ , where the torus has period one and

$$\mathbb{T}_s^d := \{x \in \mathbb{C}^d / \mathbb{Z}^d : |\operatorname{Im} x|_\infty < s\}.$$

Write

$$F(y, x) = \sum_{k \in \mathbb{Z}^d} F_k(y) e^{2\pi i k \cdot x}.$$

For every  $0 < \delta < s$ ,  $0 < \eta \leq 1/4$ , and every real  $\kappa$  satisfying

$$\kappa \geq \frac{d}{\pi\delta} \log \frac{8d}{\pi\eta}, \quad (\text{FRA.1})$$

there are real even multipliers  $\lambda_k = \lambda_{-k} \in \mathbb{R}$ , independent of  $F$  and  $y$ , with

$$\lambda_k = 0 \quad \text{when } |k|_1 > \kappa, \quad (\text{FRA.2})$$

such that

$$\tilde{F}(y, x) := \sum_{|k|_1 \leq \kappa} \lambda_k F_k(y) e^{2\pi i k \cdot x} \quad (\text{FRA.3})$$

satisfies

$$\|F - \tilde{F}\|_{r,s-\delta} \leq 2\eta \|F\|_{r,s}, \quad \|\tilde{F}\|_{r,s-\delta} \leq (1 + 2\eta) \|F\|_{r,s}. \quad (\text{FRA.4})$$

If  $F$  is real on the real domain, so is  $\tilde{F}$ .

*Proof.* We give the approximation argument because the absence of any algebraic factor in  $\delta^{-1}$  is load-bearing. The construction has a one-dimensional kernel and then tensorizes.

*Step 1: the one-dimensional Rüssmann kernel.* For  $0 < q < 1$  define the four Jacobi theta functions by their normally convergent products

$$\begin{aligned}\vartheta_1(z, q) &= 2q^{1/4} \sin(\pi z) \prod_{m \geq 1} (1 - q^{2m})(1 - 2q^{2m} \cos 2\pi z + q^{4m}), \\ \vartheta_2(z, q) &= 2q^{1/4} \cos(\pi z) \prod_{m \geq 1} (1 - q^{2m})(1 + 2q^{2m} \cos 2\pi z + q^{4m}), \\ \vartheta_3(z, q) &= \prod_{m \geq 1} (1 - q^{2m})(1 + 2q^{2m-1} \cos 2\pi z + q^{4m-2}), \\ \vartheta_4(z, q) &= \prod_{m \geq 1} (1 - q^{2m})(1 - 2q^{2m-1} \cos 2\pi z + q^{4m-2}).\end{aligned}\tag{FRA.5}$$

Normal convergence on compact subsets follows from  $\sum_m q^m < \infty$ . Put

$$K = \frac{\pi}{2} \vartheta_3(0, q)^2, \quad k = \frac{\vartheta_2(0, q)^2}{\vartheta_3(0, q)^2}, \quad k' = \frac{\vartheta_4(0, q)^2}{\vartheta_3(0, q)^2},\tag{FRA.6}$$

and define the Jacobi elliptic cosine by

$$\text{cn}(u, q) = \frac{\vartheta_4(0, q) \vartheta_2(u/(2K), q)}{\vartheta_2(0, q) \vartheta_4(u/(2K), q)}.\tag{FRA.7}$$

Write  $q = e^{-\pi t}$ ,  $t > 0$ , and define the complementary nome  $q^* := e^{-\pi/t}$ , so that  $\log(q^{-1}) \log((q^*)^{-1}) = \pi^2$ . We record the elementary series identities needed below rather than importing an elliptic-function table. Expanding the normally convergent products in (FRA.5) gives the Jacobi series

$$\begin{aligned}\vartheta_3(z, q) &= \sum_{n \in \mathbb{Z}} q^{n^2} e^{2\pi i n z}, & \vartheta_4(z, q) &= \sum_{n \in \mathbb{Z}} (-1)^n q^{n^2} e^{2\pi i n z}, \\ \vartheta_2(z, q) &= \sum_{n \in \mathbb{Z}} q^{(n+1/2)^2} e^{2\pi i (n+1/2)z}, & \vartheta_1(z, q) &= -i \sum_{n \in \mathbb{Z}} (-1)^n q^{(n+1/2)^2} e^{2\pi i (n+1/2)z}.\end{aligned}\tag{FRA.7a}$$

The Gaussian Poisson formula

$$\sum_{n \in \mathbb{Z}} e^{-\pi t(n+a)^2} = t^{-1/2} \sum_{m \in \mathbb{Z}} e^{-\pi m^2/t} e^{2\pi i m a}\tag{FRA.7b}$$

(applied also with  $a = 1/2$ ) yields

$$\vartheta_3(0, q^*) = \sqrt{t} \vartheta_3(0, q), \quad \vartheta_2(0, q^*) = \sqrt{t} \vartheta_4(0, q), \quad \vartheta_4(0, q^*) = \sqrt{t} \vartheta_2(0, q).\tag{FRA.7c}$$

Consequently

$$K(q^*) =: K' = tK = \frac{K}{\pi} \log \frac{1}{q}, \quad k(q^*) = k', \quad k'(q^*) = k.\tag{FRA.7d}$$

Finally, shifting the summation index in (FRA.7a) by the real and imaginary half-periods  $1/2$  and  $it/2$ , and inserting the resulting four theta identities into the quotient (FRA.7), gives directly

$$\operatorname{cn}(u + K \pm iK', q) = \mp i \frac{k'}{k} \operatorname{cn}(u, q), \quad \operatorname{cn}(iu + 2K, q) = -\frac{1}{\operatorname{cn}(u, q^*)}. \quad (\text{FRA.8})$$

All interchanges are justified by normal convergence on compact subsets. Thus the transformations used by the kernel reduce to the displayed theta series, Gaussian Poisson summation, and index shifts; no KAM approximation theorem has been hidden in this step.

Set

$$a := \frac{1}{2} \log \frac{1}{q} = \frac{\pi K'}{2K} \quad (\text{FRA.9})$$

and

$$\Phi_q(t) := \frac{2K'k}{\pi^2} \int_0^1 \operatorname{cn}(Ku, q) \cos(tu) du. \quad (\text{FRA.10})$$

Twice integrating by parts shows  $\Phi_q(t) = O(t^{-2})$ , hence  $\Phi_q \in L^1(\mathbb{R})$ . Fourier inversion applied to the even function that equals  $\frac{2K'k}{\pi} \operatorname{cn}(Ku, q)$  on  $|u| \leq 1$  and vanishes outside this interval yields

$$\int_{\mathbb{R}} \Phi_q(t) \cos(ut) dt = \begin{cases} \frac{2K'k}{\pi} \operatorname{cn}(Ku, q), & |u| \leq 1, \\ 0, & |u| > 1. \end{cases} \quad (\text{FRA.11})$$

The endpoint vanishes because  $\operatorname{cn}(K, q) = 0$ . The Fourier series

$$\operatorname{sgn}(\cos t) = \frac{4}{\pi} \sum_{m \geq 0} \frac{(-1)^m}{2m+1} \cos((2m+1)t) \quad (\text{FRA.12})$$

has uniformly bounded symmetric partial sums (write a partial sum as a Fejér mean plus its first difference). Dominated convergence and (FRA.11), whose transform vanishes at every odd integer  $> 1$  and whose value at 1 vanishes, therefore give

$$\int_{\mathbb{R}} \Phi_q(t) \operatorname{sgn}(\cos t) dt = 0. \quad (\text{FRA.13})$$

We also need the strip reproducing formula. If  $H$  is bounded and holomorphic for  $|\operatorname{Im} z| < R$  and  $0 < a < R$ , Cauchy's theorem on the rectangle with horizontal sides  $\operatorname{Im} z = \pm a$ , applied to

$$\frac{H(z) - H(x)}{4ia \sinh(\pi(z-x)/(2a))},$$

and then letting the vertical sides tend to infinity, gives

$$H(x) = \int_{\mathbb{R}} \frac{H(x+t+ia) + H(x+t-ia)}{2} \frac{dt}{2a \cosh(\pi t/(2a))}. \quad (\text{FRA.14})$$

The only remaining identity is the kernel splitting. Apply the residue theorem to  $z \mapsto e^{bz} \operatorname{cn}(z + K, q)$  on the rectangle with vertices  $\pm K \pm iK'$ . The unique interior pole is

at the origin. Substituting (FRA.8) into the four boundary integrals gives the explicit identity

$$\frac{\pi}{2} = k \cos(bK') \int_0^K \text{cn}(u, q) \cosh(bu) du + k' \cosh(bK) \int_0^{K'} \text{cn}(u, q^*) \cos(bu) du. \quad (\text{FRA.14a})$$

Normalize both integrals to  $[0, 1]$ , put  $b = t/K'$ , and use  $a = \pi K'/(2K)$ . With (FRA.10) this is exactly

$$\frac{1}{2a \cosh(\pi t/(2a))} = \Phi_{q^*}(t) + \Psi_q(t) \cos t, \quad \Psi_q(t) > 0. \quad (\text{FRA.15})$$

Here positivity follows from the explicit integral

$$\Psi_q(t) = \frac{Kk}{\pi a \cosh(\pi t/(2a))} \int_0^1 \text{cn}(Ku, q) \cosh\left(\frac{\pi tu}{2a}\right) du; \quad (\text{FRA.16})$$

$\text{cn}(Ku, q) > 0$  for  $0 \leq u < 1$ , which follows directly from (FRA.7) and the positive product factors on that interval.

Insert (FRA.15) in (FRA.14). The  $\Phi_{q^*}$  part defines a convolution operator  $\mathcal{P}$  whose action on  $e^{i\xi x}$  is multiplication by zero for  $|\xi| > 1$ , by (FRA.11). If  $H$  has period  $2\pi/\omega$ , then

$$\mathcal{P}H(x) = \sum_{|j| \leq \omega^{-1}} \lambda_j H_j e^{ij\omega x}, \quad (\text{FRA.17})$$

where  $\lambda_{-j} = \lambda_j \in \mathbb{R}$ . Because  $\Phi_{q^*}$  is real and even, the multipliers in (FRA.17) are real and even. An explicit normally convergent product, useful for checking the dependence on  $q$  but not needed for the KAM estimate, is

$$\lambda(q, u) = \prod_{m \geq 1} \left( \frac{1 - q^{2m}}{1 + q^{2m-1}} \right)^2 \frac{1 - 2q^{2m-1} \cosh(2u) + q^{4m-2}}{1 + 2q^{2m} \cosh(2u) + q^{4m}}, \quad (\text{FRA.18})$$

with  $u = j\omega a$ .

The remainder is the  $\Psi_q \cos t$  part. Because  $\Psi_q > 0$ , (FRA.13) lets us subtract the band-limited component and obtain

$$\|H - \mathcal{P}H\|_0 \leq E(a) \|H\|_a, \quad E(a) := \int_{\mathbb{R}} \frac{\text{sgn}(\cos t)}{2a \cosh(\pi t/(2a))} dt. \quad (\text{FRA.19})$$

Using (FRA.12) and the elementary contour integral

$$\int_{\mathbb{R}} \frac{\cos(bt)}{\cosh(\pi t/(2a))} dt = \frac{2a}{\cosh(ab)} \quad (\text{FRA.20})$$

(the poles are  $i(2m+1)a$ ), we find

$$E(a) = \frac{4}{\pi} \sum_{m \geq 0} \frac{(-1)^m}{(2m+1) \cosh((2m+1)a)}. \quad (\text{FRA.21})$$



The alternating terms decrease in absolute value, so

$$0 < E(a) \leq \frac{4}{\pi \cosh a} \leq \frac{8}{\pi} e^{-a}. \quad (\text{FRA.22})$$

This proves the one-dimensional weighted cutoff, including its finite-support multiplier construction and its pure exponential error.

*Step 2: tensorization.* Let now  $G$  be  $2\pi$ -periodic in each of  $d$  variables and bounded holomorphic on  $|\operatorname{Im} \theta|_\infty < R$ . To obtain a cutoff  $|k_j| \leq T$  in one variable, apply the preceding one-dimensional lemma to  $u \mapsto G(\dots, u/T, \dots)$ : its period is  $2\pi T$ , its frequency spacing is  $T^{-1}$ , and a strip of width  $a$  for  $G$  becomes a strip of width  $Ta$  in  $u$ . Hence the one-variable error factor is  $E(Ta)$ . Apply these scaled operators successively to the  $d$  variables. The resulting multiplier is the product

$$\Lambda_k = \prod_{j=1}^d \lambda_{k_j}, \quad \Lambda_k = 0 \quad \text{if } |k|_\infty > T. \quad (\text{FRA.23})$$

If  $A_m$  denotes the error factor after  $m$  variables, the triangle inequality and the one-dimensional estimate give

$$A_m \leq A_{m-1} + E(Ta)(1 + A_{m-1}), \quad A_0 = 0.$$

Thus by induction

$$\|G - \mathcal{P}_T G\|_0 \leq [(1 + E(Ta))^d - 1] \|G\|_a. \quad (\text{FRA.24})$$

The construction depends only on  $d, T, a$ .

*Step 3: return to the normalization used in this book and a smaller strip.* Fix  $b \in \mathbb{R}^d$  with  $|b|_\infty < s - \delta$  and a passive action  $y \in D_r$ . Define

$$G_{y,b}(\theta) = F\left(y, \frac{\theta}{2\pi} + ib\right).$$

It is  $2\pi$ -periodic and is holomorphic for  $|\operatorname{Im} \theta|_\infty < 2\pi\delta$ . Apply (FRA.24) with

$$a = \pi\delta, \quad T = \frac{\kappa}{d}. \quad (\text{FRA.25})$$

The multipliers do not depend on  $y$  or  $b$ , and  $|k|_\infty \leq T$  implies  $|k|_1 \leq dT = \kappa$ . By (FRA.22) and (FRA.1),

$$E(Ta) \leq \frac{8}{\pi} e^{-\pi\delta\kappa/d} \leq \frac{\eta}{d}. \quad (\text{FRA.26})$$

Since  $0 < \eta \leq 1/4$ ,

$$(1 + \eta/d)^d - 1 \leq e^\eta - 1 \leq 2\eta. \quad (\text{FRA.27})$$

Therefore (FRA.24), uniformly in  $y, b$ , proves the first estimate in (FRA.4). The second follows by the triangle inequality. Even real multipliers preserve the identity  $F_{-k}(y) = \overline{F_k(y)}$  on the real action domain, hence preserve the real structure.  $\square$

**Lemma 6.13 (finite-mode Arnold generator and sharp derivative bounds).** Let

$K$  be real analytic on  $D_r(y_0)$  and  $P$  real analytic on  $D_r(y_0) \times \mathbb{T}_s^d$ . Assume

$$\|K_{yy}\|_r \leq \mathbf{K}, \quad \|P\|_{r,s} \leq \varepsilon, \quad |k \cdot \omega_0| \geq \alpha |k|_1^{-\tau}, \quad \omega_0 := K_y(y_0), \quad (\text{FRG.1})$$

with  $\nu = \tau + 1$ . Let  $0 < \rho \leq 1/4$ ,  $0 < \sigma < s/4$ , and choose

$$\kappa \geq C_d \sigma^{-1} \log \rho^{-1} \quad (\text{FRG.2})$$

large enough for Lemma 6.12 with strip loss  $\sigma/4$  and target  $\rho$ . Suppose

$$\bar{r} \leq \min \left\{ \frac{r}{4}, \frac{\alpha}{2d\mathbf{K}(2\kappa)^\nu} \right\}, \quad r \leq c_0 \frac{\alpha}{\mathbf{K}} \sigma^\nu \quad (\text{FRG.3})$$

for a sufficiently small constant  $c_0 = c_0(d, \tau)$ . Then there is a trigonometric polynomial  $\hat{P}$  with modes  $|k|_1 \leq \kappa$  and

$$\|P - \hat{P}\|_{r,s-\sigma/4} \leq 2\rho\varepsilon, \quad \|\hat{P}\|_{r,s-\sigma/4} \leq (1 + 2\rho)\varepsilon, \quad (\text{FRG.4})$$

and the finite sum

$$g(y, x) := - \sum_{0 < |k|_1 \leq \kappa} \frac{\hat{P}_k(y)}{2\pi i K_y(y) \cdot k} e^{2\pi i k \cdot x} \quad (\text{FRG.5})$$

is real analytic on  $D_{\bar{r}}(y_0) \times \mathbb{T}_{s-\sigma/2}^d$ . It satisfies the exact homological identity

$$K_y(y) \cdot g_x(y, x) + \hat{P}(y, x) = \langle \hat{P} \rangle(y) \quad (\text{FRG.6})$$

and the estimates

$$\|g\|_{\bar{r},s-\sigma/2} \leq C \frac{\varepsilon}{\alpha} \sigma^{-\tau}, \quad \|g_x\|_{\bar{r},s-\sigma/2} \leq C \frac{\varepsilon}{\alpha} \sigma^{-\nu}, \quad (\text{FRG.7})$$

plus, on  $D_{\bar{r}/2}(y_0) \times \mathbb{T}_{s-3\sigma/4}^d$ ,

$$\max \{ \|g_y\|, \sigma \|g_{yx}\|, \sigma^2 \|g_{yxx}\| \} \leq C \frac{\varepsilon}{\alpha r} \sigma^{-\tau}. \quad (\text{FRG.8})$$

Here and below  $C$  depends only on  $d, \tau$  and the fixed numerical strip fractions.

*Proof.* Apply Lemma 6.12 to  $P$  to obtain  $\hat{P}$  and (FRG.4). Apply Lemma 6.9 with cutoff  $2\kappa$ . The first inequality in (FRG.3) gives, throughout  $D_{\bar{r}}(y_0)$ ,

$$|K_y(y) \cdot m| \geq \frac{\alpha}{2} |m|_1^{-\tau} \quad (0 < |m|_1 \leq 2\kappa). \quad (\text{FRG.9})$$

Thus (FRG.5) is well-defined and real analytic. Pairing the  $k$  and  $-k$  terms proves reality on the real domain. Termwise differentiation gives (FRG.6).

The proof of Lemma 6.8 applies verbatim to the finite Fourier support: its divisor-spacing step only compares two modes  $k, k'$  from the support and therefore only requires a lower bound for  $k - k'$ , whose 1-norm is at most  $2\kappa$ . This is exactly why (FRG.9) was imposed through  $2\kappa$ . With a strip loss  $\sigma/4$ , (FRG.4) gives

$$\|\partial_x^q g\|_{\bar{r},s-\sigma/2} \leq C_q \alpha^{-1} \sigma^{-(\tau+q)} \varepsilon \quad (q = 0, 1, 2, 3), \quad (\text{FRG.10})$$

which includes (FRG.7).

To estimate action derivatives without losing a second Diophantine power, differentiate (FRG.6). For the  $j$ th action coordinate,

$$D_{K_y(y)} g_{y_j} = -(\hat{P}_{y_j} - \langle \hat{P}_{y_j} \rangle) - (K_{yy}(y) e_j) \cdot g_x. \quad (\text{FRG.11})$$

Both right-hand terms have modes  $|k|_1 \leq \kappa$ . Cauchy's estimate in the action variable, using  $\bar{r} \leq r/4$ , gives

$$\|\hat{P}_y\| \leq C\varepsilon/r. \quad (\text{FRG.12})$$

Apply the finite-support version of Lemma 6.8 to (FRG.11), first with no angular derivative. Using (FRG.10),

$$\|g_y\| \leq C \left( \frac{\varepsilon}{\alpha r} \sigma^{-\tau} + \frac{\mathbf{K}\varepsilon}{\alpha^2} \sigma^{-(2\tau+1)} \right). \quad (\text{FRG.13})$$

The second inequality in (FRG.3), and  $\nu = \tau + 1$ , imply

$$\frac{\mathbf{K}}{\alpha} \sigma^{-\nu} \leq \frac{c_0}{r},$$

so the second term of (FRG.13) is absorbed into the first. Applying one and two angular derivatives to (FRG.11) gives respectively one and two additional powers of  $\sigma^{-1}$ . Multiplication by  $\sigma$  and  $\sigma^2$  therefore gives (FRG.8).  $\square$

**Lemma 6.14 (quantitative exact-symplectic Arnold step).** Under the hypotheses of Lemma 6.13, let

$$\tilde{K}(y) := \langle \hat{P} \rangle(y), \quad K^+(y) := K(y) + \tilde{K}(y). \quad (\text{FRN.1})$$

Assume in addition that

$$\frac{\varepsilon}{\alpha r} \sigma^{-\nu} \leq c_1 \rho, \quad \frac{\mathbf{K}\varepsilon}{\alpha^2} \sigma^{-2\nu} \leq c_1 \rho, \quad (1 + \mathbf{TK}) \frac{\mathbf{T}\varepsilon}{r^2} \leq c_1, \quad \mathbf{T} := \|K_{yy}^{-1}\|_r, \quad (\text{FRN.2})$$

and that the displacement bound in (FRG.7) is at most  $\bar{r}/8$ . For sufficiently small  $c_1 = c_1(d, \tau)$ , the generating function

$$S(y^+, x) = y^+ \cdot x + g(y^+, x) \quad (\text{FRN.3})$$

defines an exact symplectic analytic map on a fixed smaller domain by

$$y = y^+ + g_x(y^+, x), \quad x^+ = x + g_y(y^+, x). \quad (\text{FRN.4})$$

On that domain

$$H \circ \Phi = K^+ + P^+, \quad \|P^+\| \leq C\rho\varepsilon, \quad (\text{FRN.5})$$

$K_{yy}^+$  is invertible with

$$\|(K_{yy}^+)^{-1}\| \leq 2\mathbf{T}, \quad (\text{FRN.6})$$

and, after shrinking the real action ball by a fixed factor, there is a real-analytic map  $G$  satisfying

$$K_y^+(G(z)) = K_y(z), \quad \|G - \text{id}\| \leq C\mathbf{T} \frac{\varepsilon}{r}. \quad (\text{FRN.7})$$

Consequently the new integrable Hamiltonian has exactly the same frequency image

as the old one after the action reparameterization  $G$ .

*Proof.* We separate the coordinate inversion, the nonlinear remainder, and the frequency restoration.

1. *Inversion and exact symplecticity.* By (FRG.8), on a further strip loss  $c\sigma$ ,

$$\|g_{yx}\| \leq C \frac{\varepsilon}{\alpha r} \sigma^{-\nu}, \quad \|g_y\| \leq C \frac{\varepsilon}{\alpha r} \sigma^{-\tau} \leq C c_1 \rho \sigma. \quad (\text{FRN.7a})$$

Choose  $c_1$  so that the first norm is at most  $1/4$  and the second is at most the reserved angular margin. For fixed  $y^+$  the map  $x \mapsto x + g_y(y^+, x)$  is then injective on the real torus, and on a slightly smaller complex strip its inverse  $x = \varphi(y^+, x^+)$  is obtained by the contraction

$$x = x^+ - g_y(y^+, x). \quad (\text{FRN.8})$$

Cauchy estimates and the contraction theorem show that  $\varphi$  is holomorphic. The assumed bound  $\|g_x\| \leq \bar{r}/8$  keeps  $y^+ + g_x$  inside the action domain. Equations (FRN.4) are precisely the generating-function relations for  $S$ , so

$$dS = y \cdot dx + x^+ \cdot dy^+.$$

Hence the canonical one-forms satisfy the exact identity

$$y \cdot dx - y^+ \cdot dx^+ = d(S - y^+ \cdot x^+). \quad (\text{FRN.8a})$$

Taking an exterior derivative gives  $dy \wedge dx = dy^+ \wedge dx^+$ , while (FRN.8a) proves exactness rather than mere symplecticity.

2. *Exact cancellation and the new perturbation.* Before substituting  $x = \varphi(y^+, x^+)$ , Taylor's formula gives

$$\begin{aligned} H(y^+ + g_x, x) &= K(y^+) + K_y(y^+) \cdot g_x + R_K(y^+, x) \\ &\quad + P(y^+, x) + R_P(y^+, x), \end{aligned} \quad (\text{FRN.9})$$

where

$$\begin{aligned} R_K &= \int_0^1 (1-t) g_x^T K_{yy}(y^+ + t g_x) g_x dt, \\ R_P &= \int_0^1 P_y(y^+ + t g_x, x) \cdot g_x dt. \end{aligned} \quad (\text{FRN.10})$$

Using (FRG.6),

$$K_y \cdot g_x + P = \tilde{K} + (P - \hat{P}).$$

Hence the new perturbation before angular inversion is exactly

$$P_+ = R_K + R_P + (P - \hat{P}). \quad (\text{FRN.11})$$

On the smaller action domain, Cauchy's estimate gives  $\|P_y\| \leq C\varepsilon/r$ . Therefore (FRG.4),

(FRG.7), and (FRN.2) yield

$$\begin{aligned} \|P_+\| &\leq 2\rho\varepsilon + C\mathbf{K}\left(\frac{\varepsilon}{\alpha}\sigma^{-\nu}\right)^2 + C\frac{\varepsilon}{r}\frac{\varepsilon}{\alpha}\sigma^{-\nu} \\ &\leq C\rho\varepsilon. \end{aligned} \quad (\text{FRN.12})$$

Composition with  $x = \varphi(y^+, x^+)$  does not increase the supremum beyond the same smaller domain, so  $P^+ := P_+ \circ (\text{id}, \varphi)$  satisfies (FRN.5).

3. *Twist persistence.* Since  $\tilde{K} = \langle \hat{P} \rangle$ , Cauchy's estimate in the action variable gives

$$\|\tilde{K}_{yy}\| \leq C\varepsilon/r^2. \quad (\text{FRN.13})$$

Thus

$$K_{yy}^+ = K_{yy}(I + K_{yy}^{-1}\tilde{K}_{yy}).$$

For  $c_1$  small, (FRN.2) makes the second factor invertible by the Neumann series with inverse norm at most 2, proving (FRN.6).

4. *Frequency restoration.* For  $z$  in a fixed smaller action ball consider

$$\mathcal{F}(y, z) := K_y^+(y) - K_y(z). \quad (\text{FRN.14})$$

At  $y = z$ ,  $\mathcal{F}(z, z) = \tilde{K}_y(z)$  and  $\|\tilde{K}_y\| \leq C\varepsilon/r$ . Put  $A_z := K_{yy}^+(z)^{-1}$ , so  $\|A_z\| \leq 2\mathbf{T}$  by (FRN.6), and write  $y = z + u$ . The equation  $\mathcal{F}(y, z) = 0$  is equivalent to

$$u = \mathcal{N}_z(u) := -A_z \left[ \tilde{K}_y(z) + K_y^+(z+u) - K_y^+(z) - K_{yy}^+(z)u \right]. \quad (\text{FRN.14a})$$

On a fixed smaller action ball, Cauchy's estimate gives

$$\|K_{yyy}\| \leq C\mathbf{K}/r, \quad \|\tilde{K}_{yyy}\| \leq C\varepsilon/r^3. \quad (\text{FRN.14b})$$

Let  $R_* = C_2\mathbf{T}\varepsilon/r$  with  $C_2$  fixed large enough. For  $|u| \leq R_*$ , Taylor's formula and (FRN.14b) imply

$$\|D_u \mathcal{N}_z(u)\| \leq C\mathbf{T} \left( \frac{\mathbf{K}}{r} + \frac{\varepsilon}{r^3} \right) R_* \leq \frac{1}{2}, \quad (\text{FRN.14c})$$

provided  $c_1$  in (FRN.2) is sufficiently small; indeed the load-bearing factor is  $(1 + \mathbf{T}\mathbf{K})\mathbf{T}\varepsilon/r^2$ . The same estimates and  $|\mathcal{N}_z(0)| \leq 2\mathbf{T}C\varepsilon/r$  show that  $\mathcal{N}_z$  maps the ball  $|u| \leq R_*$  into itself. Banach's fixed point theorem, uniformly in the passive complex parameter  $z$ , therefore gives a unique holomorphic solution  $u(z)$  and

$$|G(z) - z| = |u(z)| \leq C\mathbf{T} \frac{\varepsilon}{r}. \quad (\text{FRN.15})$$

Holomorphy follows also from uniform convergence of the holomorphic fixed point iterates. Reality follows by uniqueness and complex conjugation. Equation  $\mathcal{F}(G(z), z) = 0$  is precisely (FRN.7), so the frequency images agree on the reparameterized domains.  $\square$

**Lemma 6.15 (localized analytic smoothing ladder, twist persistence, and frequency initialization).** Let  $\ell > 4$ , let  $D \subset \mathbb{R}^d$  be a bounded domain, and for  $a > 0$

put

$$D_a := \{y \in D : \text{dist}(y, \mathbb{R}^d \setminus D) > a\}. \quad (\text{FRL.1})$$

Fix  $\Delta > 0$  with  $D_{3\Delta} \neq \emptyset$ , and let  $K \in C^\ell(D)$  and  $P \in C^\ell(D \times \mathbb{T}^d)$ . There are extensions  $K^e \in C_c^\ell(\mathbb{R}^d)$  and  $P^e \in C_c^\ell(\mathbb{R}^d \times \mathbb{T}^d)$ , agreeing with  $K, P$  on  $D_{2\Delta}$ , such that

$$\|K^e\|_{C^\ell} \leq C_\Delta \|K\|_{C^\ell(D)}, \quad \|P^e\|_{C^\ell} \leq C_\Delta \|P\|_{C^\ell(D \times \mathbb{T}^d)}. \quad (\text{FRL.2})$$

Here and below  $C_\Delta$  depends only on  $d, \ell, \Delta$  and on the fixed choice of smoothing kernel, not on the smoothing scale.

Let  $S_s$  be the moment-corrected analytic smoothing operator of Lemma 6.7, fix  $0 < \xi \leq 1/2$ , and put

$$s_j = s_0 \xi^j, \quad K_j = S_{s_j} K^e, \quad P_j = S_{s_j} P^e, \quad (\text{FRL.3})$$

where  $0 < s_0 \leq c \min\{1, \Delta\}$ . For every fixed  $0 < c_0 < 1/8$  the functions  $K_j, P_j$  are real-holomorphic on the complex tubes

$$\mathcal{U}_j := \{(y, x) : \text{Re } y \in D_{2\Delta}, \quad |\text{Im } y|_\infty < c_0 s_j, \quad |\text{Im } x|_\infty < c_0 s_j\}. \quad (\text{FRL.4})$$

If  $a$  is any mixed multi-index with  $|a| = q \leq \lfloor \ell \rfloor$ , then

$$\|\partial^a (K_{j+1} - K_j)\|_{\mathcal{U}_{j+1}} \leq CC_\Delta \|K\|_{C^\ell} s_j^{\ell-q}, \quad (\text{FRL.5})$$

and

$$\|\partial^a (P_{j+1} - P_j)\|_{\mathcal{U}_{j+1}} \leq CC_\Delta \|P\|_{C^\ell} s_j^{\ell-q}. \quad (\text{FRL.6})$$

On the real interior  $D_{2\Delta}$  one also has, for every real  $0 \leq r \leq \ell$ ,

$$\|K_j - K\|_{C^r} + \|P_j - P\|_{C^r} \leq CC_\Delta (\|K\|_{C^\ell} + \|P\|_{C^\ell}) s_j^{\ell-r}, \quad (\text{FRL.7})$$

with the two terms interpreted on their respective domains.

Assume now that

$$\mathsf{T} := \sup_D \|(D^2 K)^{-1}\| < \infty, \quad M_\Delta := C_\Delta \max\{1, \|K\|_{C^\ell(D)}\}. \quad (\text{FRL.8})$$

If  $s_0$  is so small that

$$C\mathsf{T}M_\Delta s_0 \leq \frac{1}{4}, \quad (\text{FRL.9})$$

then every  $D^2 K_j$  is invertible throughout the  $y$ -projection of  $\mathcal{U}_j$  and

$$\|(D^2 K_j)^{-1}\|_{\mathcal{U}_j} \leq 2\mathsf{T}. \quad (\text{FRL.10})$$

If in addition

$$C\mathsf{T}^2 M_\Delta^2 s_0^{\ell-1} \leq \frac{1}{4}, \quad C\mathsf{T}M_\Delta s_0^{\ell-1} \leq \frac{\Delta}{4}, \quad (\text{FRL.11})$$

then there is a real  $C^1$  map  $G_0 : D_{3\Delta} \rightarrow D_{2\Delta}$  such that

$$DK_0(G_0(z)) = DK(z), \quad (\text{FRL.12})$$

with

$$\|G_0 - \text{id}\|_{D_{3\Delta}} \leq CTM_{\Delta}s_0^{\ell-1}, \quad (\text{FRL.13})$$

and

$$\|DG_0 - I\|_{D_{3\Delta}} \leq CTM_{\Delta} \left( s_0^{\ell-2} + TM_{\Delta}s_0^{\ell-1} \right). \quad (\text{FRL.14})$$

Thus the analytic smoothing ladder can be started inside a fixed boundary buffer without losing Kolmogorov nondegeneracy or changing the target unperturbed frequencies.

*Proof.* We prove the localization, the complex-strip estimate, and the two nondegeneracy assertions separately.

*1. Interior cutoff and zero extension.* Let  $\eta$  be a nonnegative  $C_c^\infty$  mollifier of mass one supported in the unit ball, and write  $\eta_h(y) = h^{-d}\eta(y/h)$ . Set

$$\chi := \eta_{\Delta/4} * \mathbf{1}_{D_{3\Delta/2}}. \quad (\text{FRL.15})$$

If  $y \in D_{2\Delta}$ , then the ball of radius  $\Delta/4$  about  $y$  lies in  $D_{3\Delta/2}$ , hence  $\chi(y) = 1$ . If  $\chi(y) \neq 0$ , some point of  $D_{3\Delta/2}$  lies within  $\Delta/4$  of  $y$ , so  $y \in D_{5\Delta/4}$ . Consequently  $\chi$  has compact support a positive distance from  $\mathbb{R}^d \setminus D$ , and

$$\|\partial^b \chi\|_\infty \leq C_b \Delta^{-|b|}. \quad (\text{FRL.16})$$

Define  $K^e = \chi K$  on  $D$  and zero outside  $D$ , and similarly  $P^e(y, x) = \chi(y)P(y, x)$ . Because the cutoff already vanishes in a full neighborhood of  $\partial D$ , these zero extensions are  $C^\ell$  globally. The product rule and (FRL.16) prove (FRL.2), while  $\chi = 1$  on  $D_{2\Delta}$  gives exact agreement there.

*2. Successive analytic estimates on complex strips.* We record the extra complex estimate not contained in the real-axis statement of Lemma 6.7. Let  $f \in C_c^\ell(\mathbb{R}^m)$  and let  $0 < s' \leq s \leq 1$ . If  $z = u + iv$  with  $|v| \leq c_0 s'$ , then for every multi-index  $a$ ,  $|a| = q \leq \lfloor \ell \rfloor$ ,

$$|\partial^a (S_s f - S_{s'} f)(z)| \leq C s^{\ell-q} \|f\|_{C^\ell}. \quad (\text{FRL.17})$$

To prove this, let  $p_u$  be the Taylor polynomial of  $f$  at the real point  $u$  through degree  $\lfloor \ell \rfloor$  when  $\ell$  is noninteger, and through degree  $\ell - 1$  when  $\ell$  is an integer. The moment conditions (FRS.4) imply reproduction of these polynomials also at complex points:

$$S_s p_u(z) = p_u(z). \quad (\text{FRL.18})$$

Indeed, for a polynomial the defining Gaussian integral is entire in  $z$ ; shifting the integration contour from  $\mathbb{R}^m$  to  $\mathbb{R}^m + i \text{Im } z$  is justified by Gaussian decay, after which (FRS.4) is exactly the real moment computation.

For the remainder, Taylor's theorem (with the integral remainder at the integer endpoint) gives

$$|f(y) - p_u(y)| \leq C \|f\|_{C^\ell} |y - u|^\ell. \quad (\text{FRL.19})$$

Derivatives of every Gaussian component in (FRS.5), together with  $|v| \leq c_0 s' \leq c_0 s$ , give

$$|\partial^a \tilde{G}_s(z - y)| \leq C s^{-m-q} \left( 1 + \frac{|y - u|}{s} \right)^q e^{-c|y-u|^2/s^2}. \quad (\text{FRL.20})$$

Using (FRL.18)–(FRL.20) yields

$$|\partial^a S_s f(z) - \partial^a p_u(z)| \leq C s^{\ell-q} \|f\|_{C^\ell},$$

and the same estimate with  $s'$  in place of  $s$ . Subtracting the common polynomial term proves (FRL.17). For torus variables, lift  $f$  periodically to the Euclidean universal cover and write the periodized convolution as the ordinary Gaussian convolution of that lift over  $\mathbb{R}^{m_2}$ . Gaussian decay makes the lifted integral absolutely convergent and the output periodic, while the same local Taylor polynomial and moment-reproduction argument applies without treating a nonperiodic polynomial as a function on the torus. Taking products with the Euclidean action kernels therefore gives the identical mixed Euclidean-torus estimate. Apply (FRL.17) to  $K^e, P^e$  with  $s = s_j$ ,  $s' = s_{j+1}$  to obtain (FRL.5)–(FRL.6). The real estimates (FRL.7) are exactly Lemma 6.7 applied to the extensions.

*3. Persistence of the twist on the complex tubes.* On the real interior, (FRL.7) with  $r = 2$  gives

$$\|D^2 K_j - D^2 K\|_{D_{2\Delta}} \leq C M_\Delta s_j^{\ell-2}. \quad (\text{FRL.21})$$

The same Taylor/Gaussian argument used above with  $q = 3$  gives  $\|D^3 K_j\|_{\mathcal{U}_j} \leq C M_\Delta$ , because  $\ell > 4$ . Hence, for  $y = u + iv$  in the  $y$ -projection of  $\mathcal{U}_j$ ,

$$\|D^2 K_j(y) - D^2 K(u)\| \leq C M_\Delta (s_j^{\ell-2} + s_j) \leq C M_\Delta s_j. \quad (\text{FRL.22})$$

Therefore

$$D^2 K_j(y) = D^2 K(u) [I + (D^2 K(u))^{-1} (D^2 K_j(y) - D^2 K(u))].$$

Condition (FRL.9) and the Neumann series make the bracket invertible with inverse norm at most 2, proving (FRL.10).

*4. Matching the initial analytic frequencies to the original ones.* For  $z \in D_{3\Delta}$  we solve  $DK_0(z + u) = DK(z)$ . Put  $A_z = (D^2 K_0(z))^{-1}$  and

$$\mathcal{N}_z(u) := -A_z [DK_0(z) - DK(z) + DK_0(z + u) - DK_0(z) - D^2 K_0(z)u]. \quad (\text{FRL.23})$$

By (FRL.7),

$$\|DK_0 - DK\| \leq C M_\Delta s_0^{\ell-1}, \quad (\text{FRL.24})$$

while Step 3 gives  $\|A_z\| \leq 2T$  and  $\|D^3 K_0\| \leq C M_\Delta$ . On the ball

$$|u| \leq R_0 := C_0 T M_\Delta s_0^{\ell-1} \quad (\text{FRL.25})$$

with  $C_0$  fixed sufficiently large, (FRL.11) implies both  $\|D_u \mathcal{N}_z\| \leq 1/2$  and  $\mathcal{N}_z(B_{R_0}) \subset B_{R_0}$ . The second inequality in (FRL.11) also gives  $R_0 < \Delta/2$ , so the whole fixed-point ball remains inside  $D_{2\Delta}$ . Banach's theorem yields a unique solution  $u(z)$ . Since the contraction depends  $C^1$  on the real parameter  $z$  and its  $u$ -derivative is uniformly bounded by  $1/2$ , the difference-quotient identity for the fixed point (equivalently, the parameter-dependent contraction theorem) shows that  $u$  is  $C^1$ . Setting  $G_0(z) = z + u(z)$  proves (FRL.12)–(FRL.13).



Differentiate (FRL.12):

$$D^2K_0(G_0(z))DG_0(z) = D^2K(z). \quad (\text{FRL.26})$$

Using (FRL.21),  $\|D^3K_0\| \leq CM_\Delta$ , and (FRL.13),

$$\|D^2K(z) - D^2K_0(G_0(z))\| \leq CM_\Delta \left( s_0^{\ell-2} + \mathsf{T}M_\Delta s_0^{\ell-1} \right). \quad (\text{FRL.27})$$

Multiplication by the inverse Hessian bounded in (FRL.10) gives (FRL.14). This completes the localization, smoothing, twist, and initial-frequency claims.  $\square$

**Lemma 6.16 (shrinking-strip Arnold induction with finite-regularity forcing).**

Fix  $\ell > 2\nu$ ,  $\tau = \nu - 1$ , and the buffered setting of Lemma 6.15. Let

$$\mathcal{H}_j := S_{s_j}(K^e + P^e), \quad s_j = s_0 \xi^j, \quad (\text{FRI.1})$$

where the action and angle smoothing are understood componentwise as in Lemma 6.15. Fix one  $z \in D_{3\Delta}$  for which  $\omega := DK(z)$  is  $(\alpha, \tau)$ -Diophantine, and let  $y_0 = G_0(z)$  be the initial frequency-matched action supplied by Lemma 6.15.

Let

$$\mathsf{M}_0 := \max\{1, \|D^2(S_{s_0}K^e)\|\}, \quad \mathsf{T}_0 := \max\{1, \|(D^2(S_{s_0}K^e))^{-1}\|\}, \quad (\text{FRI.1a})$$

on the initial local complex action tube, and freeze the bootstrap bounds  $\mathsf{M} := 2\mathsf{M}_0$  and  $\mathsf{T} := 2\mathsf{T}_0$ . There are constants  $C_*, c_*, c_r > 0$ , depending only on  $d, \ell, \nu$  and on the localized  $C^\ell$  bound together with  $\mathsf{M}, \mathsf{T}$ , with the following property. Let  $A \geq 1$  be such that the complex successive approximation estimate

$$\|\mathcal{H}_{j+1} - \mathcal{H}_j\| \leq As_j^\ell \quad (\text{FRI.2})$$

holds on the nested tubes from Lemma 6.15. Choose  $0 < \rho < 1/4$  and  $0 < \xi < 1/4$ , put  $L := \max\{1, \log \rho^{-1}\}$ , and assume

$$\rho \leq c_* \xi^\ell, \quad \xi L \leq c_*. \quad (\text{FRI.3})$$

Set

$$r_j := c_r \frac{\alpha}{\mathsf{M}} s_j^\nu, \quad (\text{FRI.4})$$

where  $\mathsf{M}$  is the frozen bootstrap bound above. Write  $E_0 := \|S_{s_0}P^e\|$  on the initial analytic domain. Assume the three numerical gates

$$C_* E_0 \leq \rho \alpha^2 s_0^{2\nu}, \quad C_* \rho E_0 \leq As_0^\ell, \quad C_* As_0^{\ell-2\nu} \leq \rho \alpha^2 \xi^{2\nu}. \quad (\text{FRI.5})$$

Finally assume that the total center displacement budget

$$C_* \mathsf{T} \left( \frac{E_0}{r_0} + \frac{A\mathsf{M}}{\alpha} \xi^{-\ell} \frac{s_1^{\ell-\nu}}{1 - \xi^{\ell-\nu}} \right) \leq \frac{\Delta}{4} \quad (\text{FRI.6})$$

holds, where  $\mathsf{T}$  is the frozen bootstrap inverse-Hessian bound above.

Then, after decreasing the fixed constants if necessary, there are for every  $j \geq 0$  a real-analytic exact symplectic composition  $\Phi^{(j)}$ , an analytic integrable Hamiltonian  $L_j$ , an analytic perturbation  $R_j$ , and an action center  $y_j$ , such that

$$\mathcal{H}_j \circ \Phi^{(j)} = L_j + R_j, \quad DL_j(y_j) = \omega, \quad (\text{FRI.7})$$

with

$$\|D^2L_j\| \leq \mathbf{M}, \quad \|(D^2L_j)^{-1}\| \leq \mathbf{T}. \quad (\text{FRI.8})$$

Let

$$W_j := \text{diag}(r_j^{-1}I_d, s_j^{-1}I_d).$$

On the fixed reduced stage domains the cumulative coordinate maps also obey

$$\|W_0 D\Phi^{(j)} W_j^{-1}\| \leq C_\Phi \xi_j^j, \quad j \geq 0, \quad (\text{FRI.8a})$$

with  $C_\Phi$  depending only on the frozen structural constants. The domains contain  $D_{r_j}(y_j) \times \mathbb{T}_{cs_j}^d$  for a fixed numerical  $c > 0$ , and are nested under the successive coordinate maps. Moreover, for a fixed  $B = B(d, \ell, \nu, \mathbf{M}, \mathbf{T})$ ,

$$E_j := \|R_j\| \leq B A s_{j-1}^\ell \quad (j \geq 1), \quad (\text{FRI.9})$$

so  $E_j \rightarrow 0$  geometrically. The one-step frequency-center displacement satisfies

$$|y_{j+1} - y_j| \leq C\mathbf{T} \frac{E_j}{r_j}, \quad (\text{FRI.10})$$

and the sum of these displacements is bounded by the left side of (FRI.6). Thus all centers stay in the fixed interior buffer and every finite-stage normal form preserves the same Diophantine frequency  $\omega$ .

*Proof.* The point of the lemma is that after Lemma 6.12–Lemma 6.15 the iteration reduces to a transparent scalar recurrence. We nevertheless verify the divisor radius, the analytic-step smallness, the domain nesting, and the forcing term from the next smoothing scale.

*1. Geometry of the next action radius.* At stage  $j$  use a fixed strip loss  $\sigma_j = c_\sigma s_j$  in Lemma 6.13, with  $c_\sigma > 0$  numerical, and choose

$$\kappa_j = C_d \sigma_j^{-1} L. \quad (\text{FRI.11})$$

The output action radius allowed by (FRG.3) is bounded below, up to fixed constants, by

$$\frac{\alpha}{\mathbf{M}\kappa_j^\nu} \asymp \frac{\alpha s_j^\nu}{\mathbf{M}L^\nu}. \quad (\text{FRI.12})$$

On the other hand  $r_{j+1} = c_r \alpha \xi_j^\nu s_j^\nu / \mathbf{M}$ . The second inequality in (FRI.3), with  $c_*$  sufficiently small, therefore allows us to choose the  $\bar{r}$  of Lemma 6.13 so that

$$4r_{j+1} \leq \bar{r}_j \leq r_j/4 \quad (\text{FRI.13})$$

and simultaneously satisfies the finite-mode Diophantine radius in (FRG.3). Taking  $c_r$

below the fixed constant in the second inequality of (FRG.3) also gives  $r_j \leq c_0 \alpha \mathbf{M}^{-1} \sigma_j^\nu$ . Thus the next local action domain fits inside the divisor-stable output domain of the present Arnold step.

*2. The dimensionless KAM load.* All smallness conditions in (FRN.2), including the condition-number term, are implied, after enlarging  $C_*$ , by

$$Q_j := C_* \frac{E_j}{\alpha^2 s_j^{2\nu}} \leq c\rho. \quad (\text{FRI.14})$$

Indeed, substitution of (FRI.4) and  $\sigma_j \asymp s_j$  into the first two inequalities of (FRN.2) gives a fixed structural multiple of the left side of (FRI.14); the third gives the same quantity multiplied by the frozen factor  $(1 + \text{TM})\text{TM}^2$ . The bound for  $g_x$  in (FRG.7) is then a fixed multiple of  $\rho \alpha s_j^\nu$  and hence is smaller than  $\bar{r}_j/8$  once  $\rho \leq c_* \xi^\ell$ ; since  $\ell > \nu$ , this is stronger than the  $\xi^\nu$  radius required in (FRI.13). Likewise (FRG.8) gives an angular displacement  $O(\rho s_j)$ , so a fixed fraction of the angular strip is reserved.

For  $j = 0$ , (FRI.14) is the first gate in (FRI.5). Suppose now  $j \geq 1$  and (FRI.9) holds. Then

$$\frac{E_j}{\alpha^2 s_j^{2\nu}} \leq B A \alpha^{-2} s_0^{\ell-2\nu} \xi^{(\ell-2\nu)(j-1)-2\nu}. \quad (\text{FRI.15})$$

The third gate in (FRI.5) makes the right side at  $j = 1$  at most a fixed small multiple of  $\rho$ , and because  $\ell > 2\nu$  it decreases geometrically thereafter. Hence every Arnold step satisfies the full smallness package of Lemma 6.14.

*3. Inductive normal-form identity.* Set  $\Phi^{(0)} = \text{id}$ ,  $L_0 = S_{s_0} K^e$ , and  $R_0 = S_{s_0} P^e$ . By Lemma 6.15,  $DL_0(y_0) = DK(z) = \omega$  and the twist bounds hold after absorbing the factor 2 of (FRL.10) into  $\mathbf{T}$ .

Assume (FRI.7)–(FRI.8) at stage  $j$ . Apply Lemma 6.14 to  $L_j + R_j$ , centered at  $y_j$ , with the parameters just verified. It yields an exact symplectic map  $\phi_j$ , an integrable  $L_j^+$ , a perturbation  $R_j^+$  and a frequency-restoring map  $G_j$  such that

$$(L_j + R_j) \circ \phi_j = L_j^+ + R_j^+, \quad \|R_j^+\| \leq C\rho E_j, \quad (\text{FRI.16})$$

$$DL_j^+(G_j(y_j)) = DL_j(y_j) = \omega. \quad (\text{FRI.17})$$

Put

$$y_{j+1} := G_j(y_j), \quad \Phi^{(j+1)} := \Phi^{(j)} \circ \phi_j, \quad L_{j+1} := L_j^+. \quad (\text{FRI.18})$$

Using  $\mathcal{H}_{j+1} = \mathcal{H}_j + (\mathcal{H}_{j+1} - \mathcal{H}_j)$  gives the exact identity

$$\mathcal{H}_{j+1} \circ \Phi^{(j+1)} = L_{j+1} + R_j^+ + (\mathcal{H}_{j+1} - \mathcal{H}_j) \circ \Phi^{(j+1)}. \quad (\text{FRI.19})$$

The last two terms define  $R_{j+1}$ .

It remains only to justify that the successive-difference estimate (FRI.2) is being evaluated where it is valid. The weighted estimates (FRG.7)–(FRG.8), divided respectively by  $r_j$  and  $s_j$ , show that each  $\phi_j$  differs from the identity in its stage-scaled norm by  $O(Q_j)$ , and the same is true for its derivative after the fixed extra strip/action loss

already built into Lemma 6.14. By (FRI.14)–(FRI.15),

$$\sum_{j \geq 0} Q_j < \infty. \quad (\text{FRI.20})$$

In fact the same calculation gives the quantitative cumulative derivative estimate needed later. Since  $\|W_j W_{j+1}^{-1}\| = \max\{\xi^\nu, \xi\} = \xi$  and, after the reserved fixed domain loss,  $\|W_j D\phi_j W_j^{-1}\| \leq 1 + CQ_j$ , factorization of the derivative of  $\Phi^{(j)} = \phi_0 \circ \dots \circ \phi_{j-1}$  gives

$$\|W_0 D\Phi^{(j)} W_j^{-1}\| \leq \xi^j \prod_{i=0}^{j-1} (1 + CQ_i) \leq \xi^j \exp\left(C \sum_{i \geq 0} Q_i\right) \leq C_\Phi \xi^j, \quad (\text{FRI.20a})$$

which proves (FRI.8a). The constants in (FRI.5) may be chosen so that  $C_\Phi < 2$ . Because every map is real on the real locus, the imaginary part of the image of a complex point is therefore bounded by a fixed multiple of the current  $r_{j+1}, s_{j+1}$ , not by the accumulated real displacement. The real action centers remain in  $D_{2\Delta}$  by Step 5 below. After fixing the working strip constant  $c$  sufficiently smaller than the  $c_0$  in (FRI.4), this proves inductively that  $\Phi^{(j+1)}$  maps its stage domain into the tube on which (FRI.2) holds. Thus (FRI.19) is legitimate and

$$E_{j+1} \leq C\rho E_j + CAs_j^\ell. \quad (\text{FRI.21})$$

*4. Closing the scalar recurrence.* Choose  $B \geq 4C$ . The second gate in (FRI.5) and (FRI.21) give  $E_1 \leq BAs_0^\ell$ . If (FRI.9) holds at stage  $j \geq 1$ , then

$$E_{j+1} \leq CB\rho As_{j-1}^\ell + CAs_j^\ell = As_{j-1}^\ell (CB\rho + C\xi^\ell). \quad (\text{FRI.22})$$

The first inequality in (FRI.3), with  $c_*$  small, makes the bracket at most  $B\xi^\ell$ . Hence  $E_{j+1} \leq BAs_j^\ell$ , proving (FRI.9) for every  $j$ . In particular  $E_j \rightarrow 0$  since  $s_j$  is geometric. It remains to verify rather than assume the frozen twist bounds. From (FRN.13), the Hessian correction at stage  $j$  is bounded by  $CE_j/r_j^2$ . The first gate in (FRI.5) gives

$$\frac{E_0}{r_0^2} \leq C\rho, \quad (\text{FRI.22a})$$

after structural factors are absorbed into  $C_*$ , while for  $j \geq 1$ , (FRI.4) and (FRI.9) give

$$\frac{E_j}{r_j^2} \leq C \frac{AM^2}{\alpha^2} \xi^{-\ell} s_j^{\ell-2\nu}. \quad (\text{FRI.22b})$$

The geometric sum of (FRI.22b) is, by the third gate in (FRI.5), a fixed small multiple of  $\rho$  as well. Choosing  $C_*$  large and  $\rho_*$  small makes the total Hessian change from  $L_0$  less than both  $M_0$  and  $(2T_0)^{-1}$ . Hence  $\|D^2 L_j\| \leq 2M_0 = M$  for all  $j$ ; the Neumann series applied to  $D^2 L_0$  plus the accumulated Hessian correction gives  $\|(D^2 L_j)^{-1}\| \leq 2T_0 = T$ . Thus (FRI.8) is a consequence of the summable corrections, not an iterated doubling of the one-step inverse bound.

*5. Frequency centers and the boundary buffer.* Equation (FRN.7) gives (FRI.10).

The  $j = 0$  contribution is  $CTE_0/r_0$ . For  $j \geq 1$ , (FRI.4) and (FRI.9) give

$$\frac{E_j}{r_j} \leq C \frac{AM}{\alpha} \xi^{-\ell} s_j^{\ell-\nu}. \quad (\text{FRI.23})$$

Summing the geometric series yields exactly the second term in (FRI.6). Together with the initial displacement  $|y_0 - z| < \Delta/4$  from Lemma 6.15, (FRI.6) keeps every  $y_j$  in the fixed interior buffer. Finally, (FRI.17) proves inductively that  $DL_j(y_j) = \omega$  at every stage. This completes the finite-stage shrinking-strip Arnold induction.  $\square$

**Lemma 6.17 (scale selection and the finite-regularity smallness exponent).** Keep the structural constants of Lemma 6.16 fixed and assume  $A \geq 1$ . Put

$$p := \frac{2\ell}{\ell - 2\nu}, \quad q := \frac{\ell - 2\nu}{\ell + 2\nu} > 0. \quad (\text{FRQ.1})$$

There are constants  $c, \alpha_0 > 0$ , depending only on the frozen structural data and on the boundary buffer  $\Delta$ , such that the following holds. Suppose

$$0 < \alpha \leq \alpha_0, \quad 0 < E_0 \leq C_0 \varepsilon, \quad \varepsilon \leq c\alpha^p. \quad (\text{FRQ.2})$$

Then one can choose  $s_0, \rho, \xi$  so that all numerical hypotheses (FRI.3), (FRI.5), and (FRI.6) of Lemma 6.16 hold, with

$$s_0 \asymp \left( \frac{E_0^2}{A\alpha^2} \right)^{1/(\ell+2\nu)}, \quad \rho \asymp \frac{As_0^\ell}{E_0}, \quad \xi \asymp \frac{1}{\log \rho^{-1}}. \quad (\text{FRQ.3})$$

In particular, the exponent  $p = 2\ell/(\ell - 2\nu)$  in the parent smallness condition is exactly the exponent needed to make the shrinking-strip Arnold induction compatible with the  $C^\ell$  smoothing error. If  $E_0 = 0$ , the perturbation is absent at the initial analytic scale and the conclusion is trivial.

*Proof.* The constants hidden by  $\asymp$  will now be chosen in the order needed by the proof. Fix a small numerical  $c_\rho > 0$  and a large structural constant  $C_s > 1$ , and set

$$s_0 := C_s \left( \frac{E_0^2}{A\alpha^2} \right)^{1/(\ell+2\nu)}, \quad \rho := c_\rho \frac{As_0^\ell}{E_0}. \quad (\text{FRQ.4})$$

Then

$$\rho = CA^{\frac{2\nu}{\ell+2\nu}} E_0^{\frac{\ell-2\nu}{\ell+2\nu}} \alpha^{-\frac{2\ell}{\ell+2\nu}} = CA^{\frac{2\nu}{\ell+2\nu}} (E_0 \alpha^{-p})^q. \quad (\text{FRQ.5})$$

Since  $E_0 \leq C_0 c\alpha^p$ , choosing  $c$  sufficiently small makes  $\rho$  smaller than any prescribed structural constant, uniformly in  $\alpha$ . In particular  $0 < \rho < 1/4$ .

The first two gates in (FRI.5) are precisely the two balances encoded in (FRQ.4). Indeed,

$$\rho \alpha^2 s_0^{2\nu} = c_\rho A \alpha^2 \frac{s_0^{\ell+2\nu}}{E_0} = c_\rho C_s^{\ell+2\nu} E_0, \quad (\text{FRQ.6})$$

so choosing  $C_s$  sufficiently large proves  $C_* E_0 \leq \rho \alpha^2 s_0^{2\nu}$ . On the other hand

$$\rho E_0 = c_\rho A s_0^\ell, \quad (\text{FRQ.7})$$

so choosing  $c_\rho$  sufficiently small proves  $C_*\rho E_0 \leq As_0^\ell$ .

Now put  $L = \max\{1, \log \rho^{-1}\}$  and choose

$$\xi := \frac{c_\xi}{L} \quad (\text{FRQ.8})$$

with  $c_\xi > 0$  small enough for the second inequality in (FRI.3). The elementary limit

$$t(\log t^{-1})^m \longrightarrow 0 \quad (t \downarrow 0) \quad (\text{FRQ.9})$$

for every fixed  $m > 0$  shows, after decreasing the smallness constant in (FRQ.2), that

$$\rho \leq c_* \xi^\ell. \quad (\text{FRQ.10})$$

Thus (FRI.3) holds. Since  $\ell > 2\nu$  and  $0 < \xi < 1$ , (FRQ.10) also implies, after a harmless change of constants,  $\rho \leq c\xi^{2\nu}$ .

To verify the third gate in (FRI.5), use (FRQ.7) and the first gate already proved:

$$As_0^{\ell-2\nu} = \frac{\rho E_0}{c_\rho s_0^{2\nu}} \leq C\rho^2 \alpha^2 \leq c\rho \alpha^2 \xi^{2\nu}. \quad (\text{FRQ.11})$$

This is exactly the required inequality after the structural constant is absorbed.

It remains to check that the smoothing scale fits into the geometric buffer and that (FRI.6) holds. From (FRQ.2),

$$s_0 \leq Cc^{2/(\ell+2\nu)} \alpha^{2/(\ell-2\nu)}. \quad (\text{FRQ.12})$$

Hence  $s_0 < c \min\{1, \Delta\}$  by taking  $c$  and  $\alpha_0$  small, as required in Lemma 6.15. For the first displacement term, the first gate of (FRI.5) and (FRI.4) give

$$\frac{E_0}{r_0} \leq C\rho \alpha s_0^\nu. \quad (\text{FRQ.13})$$

For the second, (FRQ.11) gives

$$\frac{AM}{\alpha} \xi^{-\ell} s_1^{\ell-\nu} = \frac{AM}{\alpha} \xi^{-\nu} s_0^{\ell-\nu} \leq C\rho \alpha M \xi^\nu s_0^\nu. \quad (\text{FRQ.14})$$

The denominator  $1 - \xi^{\ell-\nu}$  in (FRI.6) is uniformly bounded away from zero. Therefore the whole left side of (FRI.6) is at most  $C\rho \alpha s_0^\nu$ , with frozen structural factors absorbed into  $C$ . Equations (FRQ.5) and (FRQ.12) make this quantity arbitrarily small under (FRQ.2), so it is below  $\Delta/4$  after one final reduction of  $c$ . All gates of Lemma 6.16 are now verified.

Finally, the origin of the exponent in (FRQ.1) is visible without any hidden algebra. The first balance says  $E_0 \asymp \rho \alpha^2 s_0^{2\nu}$ , while the second says  $\rho E_0 \asymp As_0^\ell$ . Eliminating  $\rho$  gives

$$s_0^{\ell+2\nu} \asymp \frac{E_0^2}{A\alpha^2}. \quad (\text{FRQ.15})$$

Requiring  $\rho \ll 1$  is then equivalent, up to structural factors, to

$$E_0^{\ell-2\nu} \ll \alpha^{2\ell},$$

that is  $E_0 \ll \alpha^{2\ell/(\ell-2\nu)}$ . This proves the claimed scale selection and the exponent.  $\square$

**Lemma 6.18 (pointwise torus convergence and exact limiting invariance).** Assume Lemma 6.15–Lemma 6.17 and fix one buffered action  $z$  for which  $\omega = DK(z)$  is  $(\alpha, \tau)$ -Diophantine. Let  $y_j$  and  $\Phi^{(j)}$  be the centers and cumulative exact symplectic maps of Lemma 6.16, and define on the real torus

$$\iota_j(x) := \Phi^{(j)}(y_j, x). \quad (\text{FRC.1})$$

After the structural smallness constants are decreased once more, the sequence  $\iota_j$  converges in  $C^1(\mathbb{T}^d)$  to a  $C^1$  embedding  $\iota_\infty$ . More precisely, with

$$\mu := \ell - 2\nu + 1 > 1, \quad (\text{FRC.2})$$

the successive differences admit holomorphic extensions to strips of width  $cs_j$  and satisfy

$$\|W_0(\iota_{j+1} - \iota_j)\|_{cs_{j+1}} \leq C \frac{A\xi^{-\ell}}{\alpha^2 s_0} s_j^\mu \quad (j \geq 1), \quad (\text{FRC.3})$$

where all frozen structural factors are absorbed into  $C$ . The original finite-regularity Hamiltonian  $H = K + P$  then satisfies the exact invariance equation

$$X_H(\iota_\infty(x)) = D\iota_\infty(x)\omega, \quad x \in \mathbb{T}^d. \quad (\text{FRC.4})$$

Consequently  $\phi_H^t \circ \iota_\infty(x) = \iota_\infty(x + t\omega)$ , and the image torus is Lagrangian.

*Proof.* We first prove convergence without assuming a limiting invariant object. At stage  $j$ , compare the new center torus with the preceding one before applying the earlier cumulative map. Equations (FRG.7)–(FRG.8), (FRN.7), and (FRI.10) imply on a fixed reduced stage strip

$$\|W_j(\phi_j(y_{j+1}, \cdot) - (y_j, \text{id}))\| \leq CQ_j, \quad Q_j = C_* \frac{E_j}{\alpha^2 s_j^{2\nu}}. \quad (\text{FRC.5})$$

Indeed, the action component consists of the center displacement plus  $g_{j,x}$ , while the angle component is controlled by  $g_{j,y}$ ; after division by  $r_j$  and  $s_j$ , respectively, both have precisely the load  $Q_j$  up to frozen factors. The straight segment between the two stage points stays in the reduced complex domain by the margin already reserved in Lemma 6.16. Hence the mean-value formula and (FRI.8a) give

$$\|W_0(\iota_{j+1} - \iota_j)\|_{cs_{j+1}} \leq C\xi^j Q_j. \quad (\text{FRC.6})$$

For  $j \geq 1$ , (FRI.4) and (FRI.9) yield

$$Q_j \leq C \frac{A\xi^{-\ell}}{\alpha^2} s_j^{\ell-2\nu}, \quad (\text{FRC.7})$$

and  $\xi^j = s_j/s_0$ . Keeping this scale factor explicit gives exactly (FRC.3). Since  $\mu > 1$ , Lemma 6.10 with target regularity  $r = 1$  shows that  $\iota_j \rightarrow \iota_\infty$  in  $C^1$ .

The same summation, including the finite initial step, makes the  $C^1$ -distance of the angular projection of  $\iota_\infty$  from the identity as small as desired. Its lift is therefore of the form  $\text{id} + u$  with  $\|Du\| < 1/2$ ; the standard lift argument gives a global  $C^1$  torus diffeomorphism. Thus  $\iota_\infty$  is an embedding.

It remains to pass the Hamiltonian equation to the limit. On the stage normal form,

$$\mathcal{H}_j \circ \Phi^{(j)} = L_j + R_j, \quad DL_j(y_j) = \omega.$$

Exact symplecticity of  $\Phi^{(j)}$  gives on the center torus

$$X_{\mathcal{H}_j}(\iota_j(x)) - D\iota_j(x)\omega = D\Phi^{(j)}(y_j, x)X_{R_j}(y_j, x). \quad (\text{FRC.8})$$

Cauchy's estimates on a fixed fraction of the stage domain give

$$\|W_j X_{R_j}\| \leq C \frac{E_j}{r_j s_j}. \quad (\text{FRC.9})$$

Combining (FRC.9) with (FRI.8a), (FRI.4), and (FRI.9) yields

$$\|W_0 D\Phi^{(j)} X_{R_j}\| \leq C \xi^j \frac{E_j}{r_j s_j} \leq C \frac{A}{\alpha} s_j^{\ell-\nu} \rightarrow 0. \quad (\text{FRC.10})$$

By Lemma 6.15 the analytic approximants  $\mathcal{H}_j$  converge to the localized finite-regularity Hamiltonian in  $C^1$  on the real interior buffer. Lemma 6.16 keeps all real torus images in the region where the localized extension equals the original  $H$ . Letting  $j \rightarrow \infty$  in (FRC.8), using the  $C^1$  convergence of  $\iota_j$ , proves (FRC.4). The flow conjugacy follows by uniqueness of solutions of the Hamiltonian ODE. Finally  $\omega$  is rationally independent because it is Diophantine, so Lemma 6.11 applied to  $\iota_\infty$  proves that its image is Lagrangian.  $\square$

**Lemma 6.19 ( $C^\nu$  graph regularity by regraphed Bernstein summation).** Under the hypotheses of Lemma 6.18, the invariant torus  $\iota_\infty(\mathbb{T}^d)$  is the graph of a function

$$Y_\infty \in C^\nu(\mathbb{T}^d; \mathbb{R}^d). \quad (\text{FRB.1})$$

This conclusion is obtained independently of the regularity exponent of the full torus parameterization.

*Proof.* Write the  $j$ th one-step canonical transformation as  $\phi_j = (v_j, u_j)$ . In the generating-function variables of Lemma 6.14,  $u_j(y, \cdot)$  is a biholomorphism on the reserved smaller strip. Define its regraphed action map

$$\gamma_j(y, x) := v_j(y, u_j(y, \cdot)^{-1}(x)). \quad (\text{FRB.2})$$

The two generating-function equations (FRN.4) give the exact identity

$$\gamma_j(y, x) = y + g_{j,x}(y, x). \quad (\text{FRB.3})$$



Thus, on a fixed fraction of the  $j$ th strip,

$$\|\gamma_j - \pi_y\| \leq C \frac{E_j}{\alpha} s_j^{-\nu}, \quad \|D_y \gamma_j - I\| \leq C \frac{E_j}{\alpha r_j} s_j^{-\nu} \leq C Q_j. \quad (\text{FRB.4})$$

The first bound is (FRG.7), and the second is the mixed estimate (FRG.8) with  $\tau + 1 = \nu$ .

For a finite composition, eliminate the intermediate angular variables successively by (FRB.2). Equivalently, define the cumulative graph transform recursively by

$$\Gamma_0(y, x) = y, \quad \Gamma_{j+1}(y, x) = \Gamma_j(\gamma_j(y, x), x). \quad (\text{FRB.5})$$

A direct induction using the defining inverse in (FRB.2) shows that the image under  $\Phi^{(j)}$  of the horizontal stage- $j$  torus  $\{y\} \times \mathbb{T}^d$  is exactly the graph  $Y = \Gamma_j(y, x)$  over the original angle  $x$ . All expressions are holomorphic on a strip of width  $cs_j$  by the domain nesting of Lemma 6.16. Moreover, (FRB.4) and (FRI.20) give

$$\|D_y \Gamma_j\| \leq \prod_{i < j} (1 + C Q_i) \leq C_\Gamma. \quad (\text{FRB.6})$$

Set  $Y_j(x) := \Gamma_j(y_j, x)$ . From (FRB.5), the mean-value theorem, (FRB.6), (FRI.10), and (FRG.7),

$$\begin{aligned} \|Y_{j+1} - Y_j\|_{cs_{j+1}} &\leq C_\Gamma \|\gamma_j(y_{j+1}, \cdot) - y_j\|_{cs_{j+1}} \\ &\leq C \left( \mathsf{T} \frac{E_j}{r_j} + \frac{E_j}{\alpha} s_j^{-\nu} \right) \\ &\leq C \frac{A \xi^{-\ell}}{\alpha} s_j^{\ell-\nu} \quad (j \geq 1). \end{aligned} \quad (\text{FRB.7})$$

Here the frozen factors  $\mathsf{M}, \mathsf{T}, \xi^{-\ell}$  are absorbed into  $C$ . Since

$$\ell - \nu > \nu \quad (\text{FRB.8})$$

by the load-bearing hypothesis  $\ell > 2\nu$ , Lemma 6.10 applied to (FRB.7) with approximation order  $\mu = \ell - \nu$  and target regularity  $r = \nu$  shows that  $Y_j$  converges in  $C^\nu$  to some  $Y_\infty$ .

By construction, the finite-stage center tori are the graphs of  $Y_j$ . Lemma 6.18 gives their  $C^0$  convergence to the invariant torus  $\iota_\infty(\mathbb{T}^d)$ , while (FRB.7) gives uniform convergence of the graphs. Therefore the limiting torus is exactly  $\{(Y_\infty(x), x) : x \in \mathbb{T}^d\}$ , proving (FRB.1).  $\square$

**Lemma 6.20 (finite-order Whitney-Bernstein summation on shrinking neighborhoods).** Let  $E \subset \mathbb{R}^n$  be closed, let  $0 < \vartheta < 1$ , and put  $h_j = h_0 \vartheta^j$ . For  $j \geq 0$  let  $F_j$  be an  $\mathbb{R}^N$ -valued function, real on the real locus and holomorphic on

$$U_j(E) := \{z \in \mathbb{C}^n : \text{dist}(z, E) < h_j\}. \quad (\text{FRW.1})$$

Assume also that  $F_0$  is bounded on  $U_0(E)$ . For  $j \geq 1$  set

$$a_j := \|F_j - F_{j-1}\|_{U_j(E)}. \quad (\text{FRW.2})$$

Fix a noninteger  $0 < \beta < 2$ . If

$$\sum_{j \geq 1} h_j^{-\beta} a_j < \infty, \quad (\text{FRW.3})$$

then  $F_j|_E$  converges uniformly to a Whitney- $C^\beta$  map  $F : E \rightarrow \mathbb{R}^N$  in the intrinsic sense of the definition of Whitney smoothness above. More precisely, if  $0 < \beta < 1$ , then

$$|F(x) - F(y)| \leq C|x - y|^\beta, \quad (\text{FRW.4})$$

whereas if  $1 < \beta < 2$ , the derivatives  $DF_j|_E$  converge uniformly to an intrinsic first jet  $A : E \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^N)$  and

$$\|A(x) - A(y)\| \leq C|x - y|^{\beta-1}, \quad |F(x) - F(y) - A(y)(x - y)| \leq C|x - y|^\beta. \quad (\text{FRW.5})$$

The same conclusion holds on  $E \times \mathbb{T}^q$  after working in periodic local lifts. If (FRW.3) holds for some  $\beta' > 1$ , then the limit is also Whitney- $C^1$ ; thus the integer exponent 1 is obtained from any arbitrarily nearby exponent above it.

*Proof.* Put  $\Delta_j = F_j - F_{j-1}$  and  $b_j = h_j^{-\beta} a_j$ , so  $\sum b_j < \infty$ . For every  $j$  the half-width tube  $U_{h_j/2}(E) := \{z : \text{dist}(z, E) < h_j/2\}$  lies a distance  $h_j/2$  inside  $U_j(E)$ . Cauchy's estimate therefore gives

$$\|D^k \Delta_j\|_{U_{h_j/2}(E)} \leq C_k a_j h_j^{-k}, \quad k = 0, 1, 2. \quad (\text{FRW.6})$$

This formulation is important when  $E$  is disconnected: a segment joining two points of  $E$  is used only when its length is small enough that the whole segment remains inside the controlled tube.

Assume first  $0 < \beta < 1$ . Fix  $x, y \in E$  and write  $\delta = |x - y|$ . If  $\delta \geq h_0/2$ , boundedness of  $F_0$  together with  $\sum_j a_j < \infty$  gives (FRW.4) directly, after enlarging the constant. Thus suppose  $\delta < h_0/2$  and choose  $J \geq 0$  so that

$$h_{J+1} < 2\delta \leq h_J. \quad (\text{FRW.7})$$

For  $j \leq J$  one has  $h_j \geq 2\delta$ , and every point of the segment  $[x, y]$  has distance at most  $\delta/2 < h_j/2$  from  $E$ . Hence the mean-value formula may legitimately be combined with (FRW.6), giving

$$|\Delta_j(x) - \Delta_j(y)| \leq C a_j h_j^{-1} \delta = C b_j \delta^\beta (\delta/h_j)^{1-\beta}. \quad (\text{FRW.8})$$

For  $j > J$  the crude estimate and  $h_j < 2\delta$  give

$$|\Delta_j(x) - \Delta_j(y)| \leq 2a_j = 2b_j h_j^\beta \leq C b_j \delta^\beta. \quad (\text{FRW.9})$$

The finite initial function  $F_0$  is handled by the same scale split: for  $\delta < h_0/2$  the segment lies in  $U_{h_0/2}(E)$  and Cauchy's estimate gives a Lipschitz bound, while the already treated case  $\delta \geq h_0/2$  uses boundedness. Summing (FRW.8)–(FRW.9), using the geometric decay of  $(\delta/h_j)^{1-\beta}$  for  $j \leq J$ , proves (FRW.4) and uniform convergence.

Now assume  $1 < \beta < 2$ . Equation (FRW.6) gives

$$\sum_{j \geq 1} \|D \Delta_j\|_E \leq C \sum_{j \geq 1} b_j h_j^{\beta-1} < \infty, \quad (\text{FRW.10})$$

so the first jets converge uniformly to a continuous map  $A$ . If  $\delta < h_0/2$ , choose  $J$  by (FRW.7). For  $j \leq J$  the segment  $[x, y]$  lies in  $U_{h_j/2}(E)$ , so

$$\|D\Delta_j(x) - D\Delta_j(y)\| \leq Ca_j h_j^{-2} \delta = Cb_j \delta^{\beta-1} (\delta/h_j)^{2-\beta}. \quad (\text{FRW.11})$$

For  $j > J$ ,  $h_j < 2\delta$ , hence

$$\|D\Delta_j(x) - D\Delta_j(y)\| \leq Ca_j h_j^{-1} = Cb_j h_j^{\beta-1} \leq Cb_j \delta^{\beta-1}. \quad (\text{FRW.12})$$

If  $\delta \geq h_0/2$ , boundedness of the limiting jet  $A$  gives the same  $C\delta^{\beta-1}$  estimate directly. Summation proves the first inequality of (FRW.5).

For the Taylor remainder and  $\delta < h_0/2$ , analytic Taylor's formula on the controlled segment gives for  $j \leq J$

$$|\Delta_j(x) - \Delta_j(y) - D\Delta_j(y)(x - y)| \leq Ca_j h_j^{-2} \delta^2 = Cb_j \delta^\beta (\delta/h_j)^{2-\beta}. \quad (\text{FRW.13})$$

For  $j > J$ , use the triangle inequality and the derivative bound on  $E$ :

$$\begin{aligned} |\Delta_j(x) - \Delta_j(y) - D\Delta_j(y)(x - y)| &\leq 2a_j + Ca_j h_j^{-1} \delta \\ &\leq Cb_j (h_j^\beta + \delta h_j^{\beta-1}) \leq Cb_j \delta^\beta, \end{aligned} \quad (\text{FRW.14})$$

because  $h_j < 2\delta$  and  $\beta - 1 > 0$ . The initial term  $F_0$  is treated by ordinary Taylor's formula on  $[x, y] \subset U_{h_0/2}(E)$ . When  $\delta \geq h_0/2$ , boundedness of  $F$ ,  $A$ , and  $F_0$  instead gives the remainder bound directly, since both 1 and  $\delta$  are controlled by a fixed multiple of  $\delta^\beta$ . Summing (FRW.13)–(FRW.14) proves the second inequality in (FRW.5). These are exactly the intrinsic Whitney compatibility inequalities of the definition of Whitney smoothness above.

On  $E \times \mathbb{T}^q$ , cover the torus by finitely many coordinate charts and use periodic lifts on slightly larger charts; the estimates above are uniform and therefore glue. Finally, Whitney- $C^{\beta'}$  with  $\beta' > 1$  contains a continuous first jet satisfying a remainder  $O(|x - y|^{\beta'}) = o(|x - y|)$ , which is the Whitney- $C^1$  condition.  $\square$

**Lemma 6.21 (uniform Cantor-family convergence and the source Whitney range).** Assume Lemma 6.15–Lemma 6.19 and write

$$q := \ell - 2\nu > 0, \quad \beta_0 := \min \left\{ \frac{\ell}{2\nu} - 1 + \frac{1}{\nu}, 2 \right\}. \quad (\text{FRF.1})$$

Let

$$\mathcal{C}_\alpha := \left\{ z \in D : \text{dist}(z, \mathbb{R}^d \setminus D) \geq 3\Delta, DK(z) \text{ is } (\alpha, \tau)\text{-Diophantine} \right\}. \quad (\text{FRF.2})$$

After reducing the structural smallness constant in Lemma 6.17, the pointwise constructions glue to a single family with the following properties. There is a bi-Lipschitz map

$$\mathcal{G}_* : \mathcal{C}_\alpha \longrightarrow \mathcal{C}_* := \mathcal{G}_*(\mathcal{C}_\alpha) \Subset D \quad (\text{FRF.3})$$

and an embedding

$$\Phi_* : \mathcal{C}_* \times \mathbb{T}^d \longrightarrow D \times \mathbb{T}^d \quad (\text{FRF.4})$$

such that, for every

$$\frac{1}{\nu} < \beta < \beta_0, \quad (\text{FRF.5})$$

$\Phi_*$  is Whitney- $C^\beta$  on  $\mathcal{C}_* \times \mathbb{T}^d$  and, for each fixed  $y_* \in \mathcal{C}_*$ ,

$$x \mapsto \Phi_*(y_*, x) \text{ belongs to } C^{\beta\nu}(\mathbb{T}^d). \quad (\text{FRF.6})$$

Moreover, if  $z \in \mathcal{C}_\alpha$  and  $y_* = \mathcal{G}_*(z)$ , then  $\Phi_*(\{y_*\} \times \mathbb{T}^d)$  is exactly the invariant Lagrangian torus constructed pointwise in Lemma 6.18; in particular it has frequency  $DK(z)$  and is a  $C^\nu$  graph by Lemma 6.19.

*Proof.* The missing point is not another KAM step but uniform control of the same steps as the center varies in the Diophantine Cantor set. We make the gluing, fixed-limit-set reduction, and Bernstein exponent explicit.

1. *Uniform stage center maps.* Denote the initial smoothing-frequency map of Lemma 6.15 by  $G^{\text{sm}}$ , so  $y_0(z) = G^{\text{sm}}(z)$ . At the  $j$ th Arnold step denote the frequency-restoring map of Lemma 6.14 by  $A_j$  and put recursively

$$y_{j+1}(z) := A_j(y_j(z)), \quad \mathcal{C}_j := y_j(\mathcal{C}_\alpha). \quad (\text{FRF.7})$$

All constants in Lemma 6.13–Lemma 6.17 are uniform in the center. On two overlapping center balls the normalized homological solution is the same, and the solution of the frequency-restoration equation in (FRN.14) is unique. Hence the local analytic pieces of  $A_j$  agree on overlaps and glue to an analytic map on a complex neighborhood of  $\mathcal{C}_j$ .

Differentiate the frequency identity  $DL_j^+(A_j(y)) = DL_j(y)$ . Using (FRN.13), the third-derivative Cauchy bound from (FRN.14b), and (FRN.15), we obtain on a fixed smaller center tube

$$\|DA_j - I\| \leq C \left( \frac{E_j}{r_j^2} + \frac{\text{TM}E_j}{r_j^2} \right) \leq CQ_j, \quad (\text{FRF.8})$$

where  $Q_j$  is (FRI.14). Likewise (FRI.10) becomes

$$\|A_j - I\| \leq Cr_j Q_j. \quad (\text{FRF.9})$$

For  $j \geq 1$ , (FRI.4), (FRI.9), and (FRI.14) give

$$Q_j \leq Cs_j^q. \quad (\text{FRF.10})$$

The initial map  $G^{\text{sm}}$  satisfies the corresponding fixed finite bounds (FRL.13)–(FRL.14). By decreasing the constant in (FRQ.2), we may assume

$$\sum_{j \geq 0} CQ_j < \frac{1}{4}. \quad (\text{FRF.11})$$

We next justify the global Lipschitz estimate on the possibly disconnected Cantor sets rather than applying a mean-value theorem across gaps. The map  $A_j - I$  is holomorphic on a tube  $D_{cr_j}(\mathcal{C}_j)$ , satisfies (FRF.8) there, and has sup norm  $O(r_j Q_j)$  by (FRF.9). Hence,

for  $a, b \in \mathcal{C}_j$ , if  $|a - b| \leq cr_j/2$ , the segment  $[a, b]$  stays in that tube and

$$|(A_j - I)(a) - (A_j - I)(b)| \leq CQ_j|a - b|. \quad (\text{FRF.11a})$$

If  $|a - b| > cr_j/2$ , the sup-norm bound instead gives

$$|(A_j - I)(a) - (A_j - I)(b)| \leq 2Cr_jQ_j \leq C'Q_j|a - b|. \quad (\text{FRF.11b})$$

Thus  $\text{Lip}_{\mathcal{C}_j}(A_j - I) \leq CQ_j$ . The same near/far argument, now using the fixed interior scale  $\Delta$  together with (FRL.13)–(FRL.14), shows after the initial smallness reduction that  $G^{\text{sm}}$  is bi-Lipschitz on  $\mathcal{C}_\alpha$ . Consequently every finite composition  $z \mapsto y_j(z)$  is bi-Lipschitz with upper and lower constants independent of  $j$ ; more precisely,

$$\prod_{i < j} (1 - CQ_i) \leq \frac{|y_j(z) - y_j(z')|}{|G^{\text{sm}}(z) - G^{\text{sm}}(z')|} \leq \prod_{i < j} (1 + CQ_i). \quad (\text{FRF.12})$$

Moreover, (FRF.9)–(FRF.10) and  $r_j \asymp \alpha s_j^\nu$  give

$$\sum_{k \geq j} |y_{k+1}(z) - y_k(z)| \leq Cr_j s_j^q, \quad j \geq 1, \quad (\text{FRF.13})$$

uniformly in  $z$ . Thus  $y_j$  converges uniformly to a bi-Lipschitz map  $\mathcal{G}_*$ , proving (FRF.3). If  $\mathcal{C}_* := \mathcal{G}_*(\mathcal{C}_\alpha)$ , then (FRF.13) also gives

$$\text{dist}_H(\mathcal{C}_j, \mathcal{C}_*) \leq Cr_j s_j^q = o(r_j). \quad (\text{FRF.14})$$

This is the fixed-limit-set reduction: for all sufficiently large  $j$ ,

$$D_{cr_j}(\mathcal{C}_*) \subset D_{2cr_j}(\mathcal{C}_j) \quad (\text{FRF.15})$$

with one numerical  $c > 0$ .

*2. Uniform increments of the cumulative canonical maps.* Let  $\Phi^{(j)}$  be the cumulative maps of Lemma 6.16. Their domain nesting and (FRF.15) place  $\Phi^{(j)}$  on

$$D_{cr_j}(\mathcal{C}_*) \times \mathbb{T}_{cs_j}^d \quad (\text{FRF.16})$$

for every sufficiently large  $j$ . The proof of Lemma 6.16 gives, on the reserved smaller stage domain,

$$\|W_j(\phi_j - I)\| \leq CQ_j, \quad (\text{FRF.17})$$

while (FRI.8a) gives  $\|W_0 D\Phi^{(j)} W_j^{-1}\| \leq C_\Phi \xi^j$ . The mean-value formula therefore yields on the next fixed-limit tube

$$\|W_0(\Phi^{(j+1)} - \Phi^{(j)})\| \leq C_\Phi \xi^j Q_j \leq Cs_j^{q+1}. \quad (\text{FRF.18})$$

Finite initial indices play no role in a regularity summation.

Fix now  $\beta$  satisfying (FRF.5). Since

$$\beta < \frac{q/2 + 1}{\nu}, \quad (\text{FRF.19})$$

we can choose

$$\nu\beta - 1 < m < \frac{q}{2}. \quad (\text{FRF.20})$$

Because  $s_j < 1$  and  $q > 2m > m$ , (FRF.18) implies the deliberately source-sized bound

$$\|W_0(\Phi^{(j+1)} - \Phi^{(j)})\| \leq C s_j^{m+1}. \quad (\text{FRF.21})$$

The smallest complex width in (FRF.16) is the action width  $h_j := cr_j \asymp s_j^\nu$  (after shrinking  $s_0$  once, since  $\nu > 1$ ). Thus

$$\sum_j h_j^{-\beta} \|W_0(\Phi^{(j+1)} - \Phi^{(j)})\| \leq C \sum_j s_j^{m+1-\nu\beta} < \infty \quad (\text{FRF.22})$$

by (FRF.20). Lemma 6.20, applied in periodic local lifts and then multiplied by the fixed matrix  $W_0^{-1}$ , proves that  $\Phi^{(j)}$  converges on  $\mathcal{C}_* \times \mathbb{T}^d$  to a Whitney- $C^\beta$  map  $\Phi_*$ . Injectivity, hence the embedding assertion, is verified in Step 4 below. If  $\beta = 1$ , which can occur only when  $\beta_0 > 1$ , choose  $1 < \beta' < \beta_0$  and use Lemma 6.20 at  $\beta'$ ; the resulting Whitney- $C^{\beta'}$  jet is in particular Whitney- $C^1$ .

For fixed  $y_* \in \mathcal{C}_*$  the angle strip is  $cs_j$ , not  $cr_j$ . Equation (FRF.21) and (FRF.20) give

$$\sum_j s_j^{-\beta\nu} \|\Phi^{(j+1)}(y_*, \cdot) - \Phi^{(j)}(y_*, \cdot)\|_{cs_{j+1}} < \infty. \quad (\text{FRF.23})$$

The shrinking-strip Bernstein lemma Lemma 6.10 therefore yields (FRF.6). Notice that this is stronger in the angle variable than the joint Whitney exponent and uses the same strict inequality  $\beta\nu < m + 1$ .

**3. Identification with the fixed-frequency tori.** Let  $z \in \mathcal{C}_\alpha$ , set  $y_j = y_j(z)$  and  $y_* = \mathcal{G}_*(z)$ . By (FRF.13),  $|y_* - y_j| \leq Cr_j s_j^q$ . Using (FRI.8a),

$$\|W_0(\Phi^{(j)}(y_*, \cdot) - \Phi^{(j)}(y_j, \cdot))\| \leq C\xi^j \frac{|y_* - y_j|}{r_j} \leq C\xi^j s_j^q \rightarrow 0. \quad (\text{FRF.24})$$

Hence the family limit at  $y_*$  agrees with the pointwise limit in Lemma 6.18. It therefore satisfies the exact invariance equation with frequency  $DK(z)$  and is Lagrangian by Lemma 6.11. Lemma 6.19 identifies the same torus as a  $C^\nu$  graph.

**4. Injectivity of the family map.** It remains to justify the word “embedding” in (FRF.4). We use the regraphed action maps from Lemma 6.19, because they compare different tori at the same *physical* angle. The one-step map  $\gamma_j(y, X) = y + g_{j,x}(y, X)$  satisfies, on the reserved smaller center tube,

$$\|\gamma_j - \pi_y\| \leq Cr_j Q_j, \quad \|D_y \gamma_j - I\| \leq CQ_j \quad (\text{FRF.25})$$

by (FRB.4). As in (FRF.11a)–(FRF.11b), these two bounds imply a global restriction estimate

$$\text{Lip}(\gamma_j(\cdot, X) - \text{id}) \leq CQ_j \quad (\text{FRF.26})$$

on the smaller union of center tubes used by the graph recursion: nearby points are joined by a segment inside the larger tube, whereas for points farther than a fixed multiple of  $r_j$  the sup bound in (FRF.25) is used. The estimate is uniform in the real

angle  $X$ . Domain nesting in Lemma 6.16 ensures that every successive image lies in the smaller tube on which the next restricted estimate applies.

Consequently the cumulative regraphed maps  $\Gamma_j$  of (FRB.5), at each fixed physical angle  $X$ , are uniformly bi-Lipschitz on their relevant stage parameter sets, with

$$c_\Gamma |a - b| \leq |\Gamma_j(a, X) - \Gamma_j(b, X)| \leq C_\Gamma |a - b|, \quad c_\Gamma := \prod_{i \geq 0} (1 - CQ_i) > 0. \quad (\text{FRF.27})$$

To put all stages on the fixed limit Cantor set, define

$$B_j := y_j \circ \mathcal{G}_*^{-1} : \mathcal{C}_* \rightarrow \mathcal{C}_j. \quad (\text{FRF.28})$$

The uniform bi-Lipschitz bounds in (FRF.12) for  $y_j$  and for  $\mathcal{G}_*$  imply that the  $B_j$  are uniformly bi-Lipschitz, and (FRF.13) gives  $B_j \rightarrow \text{id}$  uniformly. Therefore

$$\hat{Y}_j(y_*, X) := \Gamma_j(B_j(y_*), X) \quad (\text{FRF.29})$$

is bi-Lipschitz in  $y_*$  with lower and upper constants independent of  $j$  and  $X$ . The graph convergence proved in Lemma 6.19, together with  $B_j \rightarrow \text{id}$  and the uniform bound (FRB.6), shows that  $\hat{Y}_j$  converges uniformly to the graph action function  $\hat{Y}_*$  of the limit family. Passing to the limit in the lower bound gives

$$|\hat{Y}_*(y_*, X) - \hat{Y}_*(y'_*, X)| \geq c|y_* - y'_*|. \quad (\text{FRF.30})$$

Thus two distinct Cantor parameters define disjoint graph tori. Each individual fiber is already an embedded torus by Lemma 6.18 (its angular projection is a degree-one  $C^1$  diffeomorphism). Hence the family map  $\Phi_*$  is injective. Since  $\mathcal{C}_* \times \mathbb{T}^d$  is compact and the target is Hausdorff, the continuous injective map  $\Phi_*$  is a topological embedding. This proves every assertion of the lemma.  $\square$

**Lemma 6.22 (uniform smooth-domain extension and  $\alpha$ -scale localization).** Let  $\ell = m + \eta > 0$ , where  $m = \lfloor \ell \rfloor$  and  $0 \leq \eta < 1$ , and let  $D \Subset \mathbb{R}^d$  have smooth boundary. There is a bounded linear extension operator

$$\mathfrak{E}_D : C^\ell(D) \longrightarrow C_c^\ell(\mathbb{R}^d) \quad (\text{FRU.1})$$

which restricts to the identity on  $D$  and satisfies

$$\|\mathfrak{E}_D f\|_{C^\ell(\mathbb{R}^d)} \leq C_D \|f\|_{C^\ell(D)}. \quad (\text{FRU.2})$$

The same operator acts in the action variable on  $C^\ell(D \times \mathbb{T}^d)$ , with the identical bound for all mixed action-angle derivatives.

Consequently, under the hypotheses of the Finite-differentiability KAM, the smoothing and iteration of Lemma 6.15–Lemma 6.21 admit a smooth-boundary variant in which the constants are independent of a shrinking real boundary buffer. More precisely, fix  $b > 4$ . After decreasing the structural smallness constant, if

$$\varepsilon \leq c\alpha^{2\ell/(\ell-2\nu)}, \quad (\text{FRU.3})$$

then the construction may be run simultaneously for all

$$z \in D^{[b\alpha]} := \{z \in D : \text{dist}(z, \mathbb{R}^d \setminus D) \geq b\alpha\} \quad (\text{FRU.4})$$

whose frequencies  $DK(z)$  are  $(\alpha, \tau)$ -Diophantine. Every real center and every resulting real invariant graph stays inside  $D$ . The constants in the analytic smoothing estimates, twist bounds, Arnold steps, Whitney convergence, and graph estimates depend on the fixed smooth-domain data, but not on  $\alpha$ .

*Proof.* We first construct the extension, because this is exactly what removes the inverse powers of a shrinking cutoff width.

1. *Half-space extension.* On the upper half-space write a point as  $(x', t)$  with  $t \geq 0$ . Choose real coefficients  $a_1, \dots, a_{m+1}$  solving the Vandermonde system

$$\sum_{j=1}^{m+1} a_j (-j)^r = 1, \quad 0 \leq r \leq m. \quad (\text{FRU.5})$$

For  $t < 0$  sufficiently close to zero define

$$(\mathcal{R}f)(x', t) := \sum_{j=1}^{m+1} a_j f(x', -jt), \quad (\text{FRU.6})$$

and use  $\mathcal{R}f = f$  for  $t \geq 0$ . Equation (FRU.5) says exactly that all normal derivatives through order  $m$  agree at  $t = 0$ ; tangential and mixed derivatives commute with the formula. If  $\eta > 0$ , the order- $m$  derivatives on the two sides are  $\eta$ -Hölder because each reflected point is at distance at most  $(m+1)|t|$  from the boundary and the same is true for two points lying on opposite sides. Thus, after multiplication by a fixed cutoff in the  $t$  variable,

$$\|\mathcal{R}f\|_{C^\ell} \leq C_{m,\eta} \|f\|_{C^\ell} \quad (\text{FRU.7})$$

on a two-sided half-space collar. The integer case  $\eta = 0$  follows from the same matching equations and continuity of the derivatives of order  $m$ .

2. *Passage to a smooth bounded domain.* Choose a finite smooth boundary atlas whose charts flatten  $\partial D$  to  $t = 0$ , together with an interior chart, and choose a smooth partition of unity subordinate to a slightly smaller atlas. Pull each localized piece to a half-space chart, apply (FRU.6), push it back, and sum the extended pieces. Composition with the finitely many fixed smooth charts, multiplication by the partition functions, and (FRU.7) give a uniform  $C^\ell$  bound. Multiplying once by a fixed compactly supported cutoff which equals one on a neighborhood of  $\overline{D}$  yields (FRU.1)-(FRU.2). Because the construction acts only in  $y$ , derivatives in the torus variable pass through it, proving the mixed action-angle assertion.

3. *Uniform smoothing and twist control.* Put

$$K^e := \mathfrak{E}_D K, \quad P^e := \mathfrak{E}_D P, \quad (\text{FRU.8})$$

and apply the smoothing operator of Lemma 6.7 directly to these global extensions. The proof of Lemma 6.15, Steps 2-4, now applies verbatim with  $C_\Delta$  replaced by the



fixed constant  $C_D$  and with no condition  $s_0 \ll \Delta$ : the complex smoothing strips may cross the real boundary because the extended Hamiltonian is defined globally. In particular, all estimates (FRL.5)–(FRL.14) hold uniformly with

$$M_D := C_D \max\{1, \|K\|_{C^\ell(D)}\}. \quad (\text{FRU.9})$$

The smoothed Hamiltonian equals the original one only on the real domain  $D$ , so it remains to check that the real KAM objects never leave  $D$ .

By (FRQ.12),

$$s_0 \leq C\alpha^{2/(\ell-2\nu)}. \quad (\text{FRU.10})$$

Hence the initial frequency-matching displacement satisfies

$$\|G_0 - \text{id}\| \leq Cs_0^{\ell-1} \leq C\alpha^{\frac{2(\ell-1)}{\ell-2\nu}} = o(\alpha), \quad (\text{FRU.11})$$

because  $2(\ell-1) > \ell-2\nu$ . The total later center drift is bounded, by (FRQ.13)–(FRQ.14), by

$$C\rho\alpha s_0' = o(\alpha). \quad (\text{FRU.12})$$

The action displacement produced by the graph transforms is controlled by (FRB.4). Since  $\|\gamma_j - \pi_y\| \leq Cr_j Q_j$ , while the derivative of every earlier cumulative graph transform is bounded by the uniform product  $C_\Gamma$  in (FRB.6), telescoping the cumulative graphs gives

$$\|\Gamma_\infty - \pi_y\| \leq C_\Gamma \sum_{j \geq 0} Cr_j Q_j \leq C\rho\alpha s_0' + C\alpha \sum_{j \geq 1} s_j^{\ell-\nu} = o(\alpha). \quad (\text{FRU.13})$$

Here  $Q_0 \leq C\rho$  and  $Q_j \leq Cs_j^{\ell-2\nu}$  for  $j \geq 1$ . All three estimates are uniform in the Cantor parameter. Thus, for the fixed  $b > 4$ , after reducing  $\alpha_0$  and the smallness constant in (FRU.3), their sum is less than  $(b-1)\alpha/2$ . Every real center and every point of every real invariant graph issued from  $D_{b\alpha}$  therefore stays in  $D$ . On those real objects the global extension coincides with the original Hamiltonian, so Lemma 6.18 proves invariance for  $H$ , not merely for the extension. Since no analytic or twist constant contains the boundary width, Lemma 6.16–Lemma 6.21 run with the same structural constants uniformly in  $\alpha$ .  $\square$

**Lemma 6.23 (Diophantine exclusion and near-identity cross-section measure).**

Assume the smooth-boundary setting of Lemma 6.22 and let

$$E_\alpha := \{z \in D^{[b\alpha]} : DK(z) \in \Delta_\alpha^\tau\}, \quad \tau = \nu - 1, \quad (\text{FRM.1})$$

where  $b > 4$  is fixed. Let  $\mathcal{K}_\alpha$  be the union of the KAM graphs furnished by Lemma 6.21 and Lemma 6.22. Then, for all sufficiently small  $\alpha$ ,

$$\text{Leb}(D \setminus E_\alpha) \leq C\alpha \quad (\text{FRM.2})$$

and

$$\text{Leb}((D \times \mathbb{T}^d) \setminus \mathcal{K}_\alpha) \leq C\alpha. \quad (\text{FRM.3})$$

The constants depend only on the fixed smooth-domain,  $C^\ell$ , twist, and dimension data.

In particular, if  $\varepsilon \leq c\alpha^{2\ell/(\ell-2\nu)}$  and  $\alpha \asymp \varepsilon^{1/2-\nu/\ell}$ , then

$$\text{Leb}((D \times \mathbb{T}^d) \setminus \mathcal{K}_\alpha) = O(\varepsilon^{1/2-\nu/\ell}). \quad (\text{FRM.4})$$

*Proof.* We separate arithmetic exclusion from the geometry of the perturbed graph family.

1. *The bad unperturbed actions have measure  $O(\alpha)$ .* Because  $\partial D$  is smooth, there is a tubular radius  $r_D > 0$  for which the inward normal map

$$\partial D \times (0, r_D) \longrightarrow D, \quad (p, t) \longmapsto p - tn(p), \quad (\text{FRM.5})$$

is a diffeomorphism onto the boundary collar. Its Jacobian is  $\prod_{i=1}^{d-1}(1 - t\kappa_i(p))$  and is uniformly bounded above and below for  $0 < t < r_D$ . Hence

$$\text{Leb}(D \setminus D^{[b\alpha]}) \leq C_D \alpha. \quad (\text{FRM.6})$$

It remains to estimate the resonance set. The extension constructed in Lemma 6.22 makes  $DK$  a  $C^1$  map on a neighborhood of  $\overline{D}$ . Since  $D^2K$  is invertible on  $D$  with uniformly bounded inverse, continuity and compactness give a finite cover by convex balls  $U_1, \dots, U_N$  of  $\overline{D}$  in the extension neighborhood such that  $DK$  is injective and bi-Lipschitz on each  $U_i$ , with constants uniform in  $i$ . The uniform inverse bound on  $D$  passes to  $\overline{D}$  by continuity. For completeness, take a center  $y_i \in \overline{D}$ , put  $A_i = D^2K(y_i)$ , and shrink  $U_i$  until

$$\|A_i^{-1}(D^2K(y) - A_i)\| \leq \frac{1}{2} \quad (y \in U_i). \quad (\text{FRM.7})$$

The integral mean-value identity along the segment in the extension neighborhood then gives

$$|DK(y) - DK(y')| \geq c|y - y'| \quad (y, y' \in U_i), \quad (\text{FRM.8})$$

which proves the asserted local bi-Lipschitz property.

The bounded frequency image  $DK(\overline{D})$  is contained in a fixed ball. For  $0 \neq k \in \mathbb{Z}^d$  put

$$\mathcal{R}_k := \{\omega : |\omega \cdot k| < \alpha|k|_1^{-\tau}\}. \quad (\text{FRM.9})$$

After rotating one coordinate axis to  $k/|k|_2$ , the intersection of  $\mathcal{R}_k$  with that fixed ball is a slab of thickness at most  $C\alpha|k|_1^{-\tau}|k|_2^{-1}$ . Therefore

$$\text{Leb}(\mathcal{R}_k \cap DK(\overline{D})) \leq C\alpha|k|_1^{-(\tau+1)} = C\alpha|k|_1^{-\nu}. \quad (\text{FRM.10})$$

The inverse on each  $U_i$  is uniformly Lipschitz, so covering a measurable subset of frequency space by cubes and pulling those cubes back through the inverse gives

$$\text{Leb}\{y \in U_i \cap D : DK(y) \in \mathcal{R}_k\} \leq C \text{Leb}(\mathcal{R}_k \cap DK(U_i \cap D)). \quad (\text{FRM.11})$$

Summing over the finite cover and then over  $k \neq 0$  yields

$$\text{Leb}\{y \in D : DK(y) \notin \Delta_\alpha^\tau\} \leq C\alpha \sum_{k \neq 0} |k|_1^{-\nu} \leq C\alpha, \quad (\text{FRM.12})$$

because the number of integer vectors with  $|k|_1 = n$  is  $O(n^{d-1})$  and  $\nu > d$ . Equations (FRM.6) and (FRM.12) prove (FRM.2).

*2. A geometric near-identity measure lemma.* We use the following elementary fact. Let  $E \subset D$  be measurable and let  $F : E \rightarrow \mathbb{R}^d$  have the form  $F = \text{id} + u$  with

$$\|u\|_E \leq \delta, \quad \text{Lip}_E(u) \leq \lambda, \quad \sqrt{d}\lambda < 1. \quad (\text{FRM.13})$$

Then

$$\text{Leb}(D \setminus F(E)) \leq C_{D,d}\delta + (1 + \sqrt{d}\lambda)^d \text{Leb}(D \setminus E). \quad (\text{FRM.14})$$

To prove this without an extension theorem, extend each scalar component  $u_i$  by the McShane formula and then clip it to the interval  $[-\delta, \delta]$ . The resulting global component  $\tilde{u}_i$  agrees with  $u_i$  on  $E$ , is bounded by  $\delta$ , and has Lipschitz constant at most  $\lambda$ . Thus

$$|\tilde{u}(x) - \tilde{u}(y)| \leq \sqrt{d}\lambda|x - y|, \quad |\tilde{u}(x)| \leq \sqrt{d}\delta. \quad (\text{FRM.15})$$

The map  $\tilde{F} = \text{id} + \tilde{u}$  is a global bi-Lipschitz homeomorphism: the lower Lipschitz bound is  $(1 - \sqrt{d}\lambda)|x - y|$ , and surjectivity follows because, for each  $q \in \mathbb{R}^d$ , the map  $x \mapsto q - \tilde{u}(x)$  is a contraction. If  $q \in D$  has distance greater than  $\sqrt{d}\delta$  from  $\mathbb{R}^d \setminus D$ , then the unique preimage  $x = \tilde{F}^{-1}(q)$  satisfies  $|x - q| \leq \sqrt{d}\delta$ , so  $x \in D$ . Consequently

$$D \setminus \tilde{F}(D) \subset \{q \in D : \text{dist}(q, \mathbb{R}^d \setminus D) \leq \sqrt{d}\delta\}. \quad (\text{FRM.16})$$

The smooth collar estimate bounds the right side by  $C_{D,d}\delta$ . Furthermore,

$$D \setminus F(E) \subset (D \setminus \tilde{F}(D)) \cup \tilde{F}(D \setminus E). \quad (\text{FRM.17})$$

A Lipschitz map of constant  $L$  sends a measurable set to a set of Lebesgue outer measure at most  $L^d$  times its measure; this follows by covering the set by cubes and covering each image cube by a bounded number of cubes of side  $L$  times the original side. Since  $\text{Lip}(\tilde{F}) \leq 1 + \sqrt{d}\lambda$ , (FRM.14) follows.

*3. The physical-angle cross-sections satisfy the geometric lemma.* Use the regraphed family notation of Lemma 6.19–Lemma 6.21. For each real physical angle  $X \in \mathbb{T}^d$  define

$$F_X(z) := \hat{Y}_*(\mathcal{G}_*(z), X), \quad z \in E_\alpha. \quad (\text{FRM.18})$$

Thus  $F_X(E_\alpha)$  is exactly the action cross-section at angle  $X$  of  $\mathcal{K}_\alpha$ . At finite stage this map is obtained by composing the initial frequency map, the restoring maps  $A_j$ , and the regraphed maps  $\gamma_j$ . Equations (FRL.13)–(FRL.14), (FRF.8)–(FRF.9), and (FRF.25)–(FRF.26) therefore give

$$\sup_X \|F_X - \text{id}\|_{E_\alpha} \leq C \left( s_0^{\ell-1} + \sum_{j \geq 0} r_j Q_j \right), \quad (\text{FRM.19})$$

and, by the same near/far restriction argument used in Lemma 6.21,

$$\sup_X \text{Lip}_{E_\alpha}(F_X - \text{id}) \leq \exp \left( C s_0^{\ell-2} + C \sum_{j \geq 0} Q_j \right) - 1. \quad (\text{FRM.20})$$

These estimates are first uniform for the finite-stage maps and then pass to the uniform limit. The scale relations give

$$\sum_{j \geq 0} r_j Q_j \leq C \rho \alpha s_0^\nu + C \alpha \sum_{j \geq 1} s_j^{\ell-\nu} \leq C \alpha, \quad (\text{FRM.21})$$

while (FRU.11) gives  $s_0^{\ell-1} = o(\alpha)$ . Hence

$$\sup_X \|F_X - \text{id}\|_{E_\alpha} \leq C \alpha. \quad (\text{FRM.22})$$

Also  $s_0^{\ell-2} \rightarrow 0$  and  $\sum Q_j \leq C \rho + C \sum_{j \geq 1} s_j^{\ell-2\nu}$  can be made arbitrarily small by the structural choices already allowed in Lemma 6.17. We therefore reduce the smallness constant once more so that the right side of (FRM.20) is less than  $1/(2\sqrt{d})$ , uniformly in  $X$ .

Apply (FRM.14) to  $E = E_\alpha$  and  $F = F_X$ . Using (FRM.2) and (FRM.22), uniformly in  $X$ ,

$$\text{Leb}(D \setminus F_X(E_\alpha)) \leq C \alpha. \quad (\text{FRM.23})$$

The set  $E_\alpha$  is compact:  $D^{[b\alpha]}$  is closed and bounded, and the Diophantine inequalities define a closed frequency set. Since  $\mathcal{G}_*$  and the regraphed family are continuous,  $\mathcal{K}_\alpha$  is a continuous image of the compact set  $E_\alpha \times \mathbb{T}^d$  and is therefore compact, in particular measurable. Fubini's theorem applied to its physical-angle cross-sections now gives (FRM.3).

Finally set

$$p = \frac{2\ell}{\ell - 2\nu}.$$

Choosing  $\alpha = A\varepsilon^{1/p}$  with a sufficiently large fixed  $A$  makes (FRU.3) hold, and

$$\frac{1}{p} = \frac{\ell - 2\nu}{2\ell} = \frac{1}{2} - \frac{\nu}{\ell}. \quad (\text{FRM.24})$$

Substitution into (FRM.3) proves (FRM.4).  $\square$

**Proof of the Finite-differentiability KAM.** We assemble the preceding lemmas and separate the fixed-buffer persistence statement from the smooth-boundary measure statement.

Fix first an interior buffer  $\Delta > 0$  as in the theorem and a target action  $z \in D_{3\Delta}$  with  $\omega = DK(z) \in \Delta_\alpha^\tau$ . Lemma 6.15 applies the internal analytic smoothing lemma Lemma 6.7 to the finite-Hölder Hamiltonian on a fixed localized action neighborhood, preserves the twist, and produces the initial frequency-matching center. Lemma 6.12–Lemma 6.14 give the analytic Arnold step with the sharp Rüssmann approximation/cancellation loss; Lemma 6.8–Lemma 6.9 are the small-divisor and divisor-stability

inputs used there. The finite-stage induction is therefore legitimate by Lemma 6.16. Lemma 6.17 chooses the scales so that its two recurrence gates close simultaneously precisely under

$$\varepsilon \leq c\alpha^{2\ell/(\ell-2\nu)}. \quad (\text{FRP.1})$$

The accumulated real displacement can be made smaller than the reserved fixed buffer, so every real object lies where the localized Hamiltonian coincides with  $H$ .

For this fixed Diophantine action, Lemma 6.18 proves  $C^1$  convergence to an embedded torus satisfying the exact invariance equation

$$X_H(\iota_z(x)) = D\iota_z(x) DK(z). \quad (\text{FRP.2})$$

Its frequency is therefore exactly the unperturbed frequency  $DK(z)$ . Lemma 6.11 then gives Lagrangianity, while Lemma 6.19 proves independently that the image is a  $C^\nu$  graph over the physical angle. The estimates in Lemma 6.16–Lemma 6.19 are summable and tend to zero with the perturbation load, which gives the stated small-deformation conclusion.

The preceding construction is uniform when  $z$  ranges over the fixed Diophantine Cantor subset of  $D_{3\Delta}$ . Lemma 6.21 first glues the local frequency-restoration maps by uniqueness, moves the analytic iterates to shrinking neighborhoods of a single limiting Cantor set, and then applies the internally proved finite-order Whitney–Bernstein theorem Lemma 6.20. Thus for every

$$\nu^{-1} < \beta < \min \left\{ \frac{\ell}{2\nu} - 1 + \frac{1}{\nu}, 2 \right\} \quad (\text{FRP.3})$$

the family is Whitney- $C^\beta$ , while each fixed Cantor fiber has the stronger  $C^{\beta\nu}$  angular parameterization. Lemma 6.21 also proves family injectivity by the uniformly bi-Lipschitz regraphed action maps. This establishes every assertion in the fixed-buffer part of the theorem.

Assume now that  $\partial D$  is smooth and, as required in the smooth-boundary clause, that the  $C^\ell$  data have their standard  $C^\ell$  traces on  $\overline{D}$ . Lemma 6.22 constructs a bounded global  $C^\ell$  extension in the action variable and reruns the same smoothing and KAM construction with constants independent of a shrinking boundary buffer. The center and graph displacements are  $o(\alpha)$ , so for a fixed sufficiently large structural  $b$  the construction applies uniformly to

$$\mathcal{E}_\alpha = \{z : \text{dist}(z, D^c) \geq b\alpha, DK(z) \in \Delta_\alpha^\tau\} \quad (\text{FRP.4})$$

and all real limiting tori remain inside  $D$ , where the extension equals  $H$ .

Finally Lemma 6.23 proves directly that both the excluded action set and the physical phase-space complement have measure  $O(\alpha)$ . Choosing  $\alpha = A\varepsilon^{(\ell-2\nu)/(2\ell)}$  with a sufficiently large fixed  $A$  makes (FRP.1) hold and gives

$$\text{Leb}((D \times \mathbb{T}^d) \setminus \mathcal{K}) \leq C\alpha = O\left(\varepsilon^{1/2-\nu/\ell}\right). \quad (\text{FRP.5})$$

This is (FR.4) and completes the proof.  $\square$

## 6.4 Worked examples

### Worked example: the standard map near a Diophantine invariant circle

Use the exact symplectic standard map

$$I' = I + \kappa \sin(2\pi\theta), \quad \theta' = \theta + I' \pmod{1}.$$

At  $\kappa = 0$  this is

$$F_0(\theta, I) = (\theta + I, I),$$

so the rotation function is  $\omega(I) = I$  and

$$\omega'(I) = 1.$$

Thus the twist hypothesis holds uniformly. Fix a Diophantine number  $\rho \in (a, b)$  and the unperturbed circle  $I = \rho$ . The perturbation is analytic and has size  $O(|\kappa|)$  on every fixed sufficiently small complex neighborhood of that circle. Moser's theorem therefore supplies an analytic embedding  $K_\kappa$  satisfying

$$F_\kappa \circ K_\kappa = K_\kappa \circ R_\rho$$

when  $|\kappa|$  is below the theorem's threshold.

The first-order difference equation already displays the small divisor. Writing the action correction as  $v(\varphi)$  gives, to first order,

$$v(\varphi + \rho) - v(\varphi) = \kappa \sin(2\pi\varphi).$$

For a Fourier mode  $k$  the divisor is  $e^{2\pi i k \rho} - 1$ . The Diophantine hypothesis prevents these divisors from becoming too small too quickly, exactly as continuous-time KAM controls  $k \cdot \omega$ .

### Boundary case: what is lost when twist vanishes

Consider the integrable exact symplectic map

$$F_0(\theta, I) = (\theta + \rho_0, I).$$

Its rotation function is constant, so  $\omega'(I) = 0$  everywhere. The twist hypothesis in Moser's theorem fails. The cohomological equation may still be solvable for zero-average oscillatory terms, but there is no scalar action displacement that changes the mean rotation number. Hence the theorem in this chapter gives no persistence conclusion for a prescribed nearby rotation number. A degenerate KAM theorem would require additional structure not assumed here.

**Common pitfall**

Do not read the finite-regularity theorem as saying that the Cantor family is Whitney- $C^\nu$  in its transverse parameter. The theorem deliberately separates the  $C^\nu$  regularity of each invariant graph from the smaller stated Whitney exponent of the parameter family.

**6.5 Chapter summary****What to remember**

- Isoenergetic KAM preserves invariant tori on a fixed energy surface by solving a bordered mean equation and preserving frequency direction up to time rescaling.
- Moser's theorem is the map analogue for exact symplectic annulus maps with nonzero twist and Diophantine rotation number.
- Finite-regularity KAM replaces complex analyticity by smoothing and an explicit derivative budget; the regularity and measure exponents must be tracked separately.
- Whitney smoothness concerns the surviving Cantor parameter set and is not the same as the smoothness of an individual invariant torus.
- Parameterized/reversible/conformally symplectic KAM variants require class-specific source/nondegeneracy conditions.
- The final measure lemma is elementary but indispensable: it turns infinitely many local exclusions into one global surviving-set estimate.

**Looking ahead**

K07 studies lower-dimensional and resonant tori, where the normal directions themselves have nontrivial dynamics and extra nondegeneracy conditions enter. K08 then changes the question from eternal persistence to exponentially long stability near resonant networks.

**Check your understanding**

Compare the mean equation in Kolmogorov KAM, isoenergetic KAM, and Moser's twist-map theorem. Identify the matrix or scalar derivative that makes each mean equation solvable.

## 6.6 Exercises and problems

### Exercise 6.1

**Difficulty.** Core.

**Related results.** Isoenergetic KAM theorem; Moser invariant-curve theorem for twist maps; Finite-differentiability KAM.

**Problem.** For the standard map at zero parameter, compute invariant circles, area preservation, and the twist derivative. Turn on a small parameter and derive the linearized invariant-circle/cohomological equation.

**Solution sketch.** Differentiate the explicit map, insert a graph or parameterization ansatz, and separate the Fourier and mean/twist equations.

### Exercise 6.2

**Difficulty.** Core.

**Related results.** Measure bookkeeping lemma for KAM parameter sets.

**Problem.** At the twist-degenerate parameter, model the frequency map by  $\nu(s) = \nu_0 + s^2$ . Show that the preimage of a resonance interval of width  $\gamma$  has size  $O(\sqrt{\gamma})$ , explaining why transversality is needed for KAM measure bookkeeping.

**Solution sketch.** Solve  $|s^2| < \gamma$  and compare with the linear-transversality case.

### Exercise 6.3

**Difficulty.** Advanced.

**Related results.** Finite-differentiability KAM.

**Problem.** Prove a Fourier smoothing estimate of the form  $\|f - S_N f\|_{C^s} \leq C N^{s-r} \|f\|_{C^r}$  in a summable range and explain its role in finite-regularity KAM.

**Solution sketch.** Integrate Fourier coefficients by parts, multiply by  $|k|^s$ , and sum the high-frequency tail.

### Exercise 6.4

**Difficulty.** Advanced.

**Related results.** Measure bookkeeping lemma for KAM parameter sets.

**Problem.** Under a derivative lower bound for each resonance function, estimate the total measure of parameters excluded by  $|k \cdot \omega| < \gamma |k|^{-\tau}$  and determine the summability condition on  $\tau$ .

**Solution sketch.** Bound one resonant preimage, sum over lattice shells, and compare the exponent with the number of lattice points per shell.

### Exercise 6.5

**Difficulty.** Advanced.

**Related results.** Isoenergetic KAM theorem; Moser invariant-curve theorem for twist maps.

**Problem.** Explain how isoenergetic nondegeneracy of a two-degree Hamiltonian induces twist in a Poincaré return map on a fixed energy surface and how an invariant return-map curve suspends to an invariant torus.



**Solution sketch.** Differentiate the rotation ratio along the energy curve, identify the return-map twist, and suspend the invariant curve by the flow.

## **Part II**

# **Resonance, Long-Time Stability, Transport, and Variational Dynamics**

# Lower-Dimensional, Degenerate and Resonant Tori

## Why this chapter matters

Full-dimensional KAM tori fill all available action directions. Lower-dimensional tori are subtler: some directions are tangent to the torus, while the remaining directions carry transverse oscillations or hyperbolic motion. A perturbation can therefore resonate not only with the tangential frequency vector but also with one or two normal frequencies. This is the source of the first and second Melnikov conditions.

The chapter develops a precise answer to a natural question: when a torus has genuine transverse dynamics, which nonresonance and nondegeneracy assumptions are strong enough to preserve it? The answer is not a single condition. One needs a bookkeeping system that separates tangential divisors, first-normal divisors, second-normal divisors, parameter transversality, and the finite-dimensional freedom used to remove averaged defects. The same language also explains why elliptic equilibria can be surrounded by positive-measure KAM tori even when the equilibrium frequency itself is not Diophantine.

**Prerequisites.** K01-K06, especially analytic KAM iteration and resonant normal forms; Foundation F05, F07, F08, F12, and F15; basic linear symplectic algebra and Floquet theory.

## Learning goals

- Separate tangential frequencies from normal frequencies for a lower-dimensional torus.
- Recognize the scalar, first-normal, and second-normal divisors appearing in the homological equation.
- Explain why a moving Cantor set of parameters is needed when the corrected frequencies change during iteration.
- Distinguish Kolmogorov, Rüssmann, parameter, and lower-dimensional nondegeneracy mechanisms.
- Follow the rescaling argument that converts Rüssmann nondegeneracy near an elliptic equilibrium into a positive-measure family of invariant tori.

### Conceptual roadmap

The proof architecture has three layers:

linearized normal form  $\rightarrow$  Melnikov divisor estimates  
 $\rightarrow$  Newton/Floquet iteration on a moving Cantor set.

For a torus with tangential frequency  $\omega$  and normal frequencies  $\Omega_j$ , the new denominators are schematically

$$k \cdot \omega, \quad k \cdot \omega \pm \Omega_i, \quad k \cdot \omega \pm \Omega_i \pm \Omega_j.$$

The first theorem controls these divisors in a standard normally elliptic family. The Li-Yi theorem shows how the mechanism survives multiple normal eigenvalues by replacing scalar denominators with finite-dimensional block operators. The final two results explain how weaker frequency-map nondegeneracy can still yield KAM stability near an elliptic fixed point.

### Check your understanding

Suppose a two-dimensional invariant torus sits in a three-degree-of-freedom Hamiltonian system. How many tangent frequencies and how many normal oscillatory frequencies should you expect? Which new resonances can occur that are absent for a full-dimensional torus?

## 7.1 Definitions

### Intuition

**How to read the lower-dimensional definitions.** A lower-dimensional torus has two different linear problems. Along the torus one solves cohomological equations involving  $k \cdot \omega$ . Transverse to the torus one must invert matrices whose eigenvalues contain  $k \cdot \omega$  shifted by one or two normal eigenvalues. The first and second Melnikov conditions are exactly the quantitative statements that these shifted divisors do not become too small on the retained parameter set.

**Why the weighted degree is asymmetric.** Near a torus, action variables are naturally of size  $r^2$  while normal variables are of size  $r$ . Giving actions weight two and normal variables weight one makes the quadratic normal form and the KAM remainder scale consistently.

**Rüssmann versus Kolmogorov.** Kolmogorov nondegeneracy asks for an invertible Hessian. Rüssmann nondegeneracy is weaker: it only requires that the frequency image not be trapped in a hyperplane through the origin to the relevant finite order. The price is that one no longer prescribes every nearby frequency independently.

**Definition K07.1. lower-dimensional torus**

A lower-dimensional invariant torus has dimension  $m$  strictly smaller than the number of degrees of freedom  $n$ ; its normal bundle carries additional elliptic/hyperbolic dynamics.

**Definition K07.2. normal frequency**

A normal frequency is an eigenfrequency/eigenvalue parameter of the linearized dynamics transverse to an invariant torus, separated from the tangent frequency vector.

**Definition K07.3. Melnikov condition**

A first/second Melnikov condition is a lower bound on divisors  $k \cdot \omega + \lambda_i$  or  $k \cdot \omega + \lambda_i \pm \lambda_j$  for integer  $k$ , with the allowed index/sign cases stated.

**Definition K07.4. Rüssmann nondegeneracy**

For a formal frequency series  $\Omega(I) \in \mathbb{R}[[I]]^n$  at  $I = 0$ , *Rüssmann nondegeneracy* means that for every nonzero  $c \in \mathbb{R}^n$  the scalar formal series  $\langle c, \Omega(I) \rangle$  is not identically zero. Equivalently, compactness of the unit sphere implies that some finite jet detects every unit  $c$ : there are  $q \geq 1$  and  $\beta > 0$  such that for every unit  $c$  some derivative of order at most  $q$  of  $\langle c, \Omega \rangle$  at 0 has modulus at least  $\beta$ . For an actual  $C^q$  frequency map on a parameter domain, the finite-order directional version is the hypothesis stated in Result (*Rüssmann-type nondegeneracy*).

**Definition K07.5. uniform order-two Melnikov transversality**

A bounded convex parameter extension domain has uniform order-two Melnikov transversality if for every nonzero order-two resonance there is a unit direction, possibly depending on the resonance, along which its directional derivative has one fixed sign and magnitude at least  $c \langle k \rangle$  throughout the domain.

**Definition K07.6. weighted Taylor degree**

The weighted degree of  $e^{2\pi i \langle k, \theta \rangle} y^a z^\alpha \bar{z}^\beta$  is  $2|a|_1 + |\alpha|_1 + |\beta|_1$ .

**Definition K07.7. Lipschitz and bi-Lipschitz parameter maps**

For a map  $F$  between metric subsets,  $[F]_{\text{Lip}}$  is the usual difference-quotient supremum. It is bi-Lipschitz onto its image when both  $F$  and the inverse on its image have finite Lipschitz constants.

**Definition K07.8. sup-Lipschitz parameter norm**

$[F]_{\text{Lip}(A)} := \sup_{\xi \neq \eta} |F(\xi) - F(\eta)| / |\xi - \eta|$  and  $\|F\|_{0,1;A} := \sup_A |F| + [F]_{\text{Lip}(A)}$ .

**Definition K07.9. complex lower-dimensional KAM domain and scaled norm**

$D(s, r)$  denotes the complex strip/radial KAM domain with action radius  $r^2$  and normal radius  $r$ ; its scaled vector-field norm uses the corresponding action/normal weights.

**Definition K07.10. weighted sup-Lipschitz parameter norm**

The weighted sup-Lipschitz norm combines the scaled phase vector-field norm with the parameter Lipschitz seminorm using the fixed KAM parameter weight.

**Definition K07.11. weighted fiber-polynomial symplectic-map class**

The class consists of triangular symplectic maps whose angle component depends only on angle, normal component is affine in normal variables, and action component is affine in actions and at most quadratic in normal variables, with the KAM weighted-degree bounds.

**Definition K07.12. Li-Yi general lower-dimensional normal form**

This is the parameter family on  $\mathbb{T}^n \times \mathbb{R}^n \times \mathbb{R}^{2m}$  with quadratic part  $e + \langle \omega, y \rangle + \frac{1}{2} \langle (y, z), M(\omega)(y, z) \rangle$ ,  $M$  symmetric and normal block  $M_{22}$ .

**Definition K07.13. weak Li-Yi second Melnikov condition**

With project Fourier modes  $e^{2\pi i \langle k, x \rangle}$  and eigenvalues  $\lambda_a$  of  $JM_{22}$ , require  $2\pi i \langle k, \omega \rangle - \lambda_a - \lambda_b \neq 0$  for  $k \neq 0$  and every pair  $a, b$  for almost every parameter; multiple eigenvalues are allowed.

**Definition K07.14. Floquet lower-dimensional normal form**

Along the invariant torus, the Hamiltonian through quadratic transverse order has constant tangential frequency and a constant transverse quadratic matrix; the remaining transverse terms have order at least three.

**Definition K07.15. block divisors and finite-scale determinant good set**

$L_0, L_1, L_2$  are the scalar/tangential, first-normal and second-normal Fourier-block homological operators. A finite-scale determinant good set is the parameter set where their required determinants satisfy the current cutoff lower bounds.

## 7.2 How the main proofs work

### Intuition

**Normally elliptic persistence.** At each Fourier cutoff, split the homological equation into tangential, first-normal, and second-normal blocks. Melnikov inequalities give inverses for all nonkernel blocks. The averaged kernel is absorbed

into the new normal form, producing slightly corrected tangential and normal frequencies. Because the frequencies move, the good parameter set must also move. Supergeometric cutoffs make both the transformations and the excluded-measure losses summable.

**The Melnikov estimate.** The analytic inverse estimate is just a Fourier estimate once the divisor lower bound is known. The hard point is not summing the Fourier series; it is proving that the bad parameter strips have summable measure. Uniform transversality converts a small-divisor inequality into a one-dimensional interval estimate along a suitable parameter direction.

**Li-Yi with multiple eigenvalues.** Multiple normal eigenvalues destroy a naive scalar diagonalization. The proof therefore works with block homological operators and their determinants. Finite-scale determinant bounds give block inverses, a resonant translation removes the averaged linear defect, and the Newton iteration produces a limiting Floquet normal form without assuming all normal eigenvalues are simple.

**Rüssmann stability of an elliptic equilibrium.** First put the Hamiltonian into a finite Birkhoff normal form. Rescale a small action ball to a fixed domain. Rüssmann nondegeneracy gives a positive-measure set of rescaled parameters where the KAM theorem applies. Scaling back produces invariant tori whose distance from the equilibrium tends to zero.

## 7.3 Main results

### Theorem K07.1. First and second Melnikov nonresonance conditions

**Statement.** Let  $\Pi$  be a compact parameter set and let  $\omega : \Pi \rightarrow \mathbb{R}^n$  and  $\Omega : \Pi \rightarrow \mathbb{R}^m$  be Lipschitz. For  $\gamma > 0$  and  $\tau > n - 1$ , define  $\Pi_\gamma$  by the order-two Melnikov inequalities in the normalization used here  $|2\pi\langle k, \omega(\xi) \rangle + \langle l, \Omega(\xi) \rangle| \geq \gamma\langle k \rangle^{-\tau}$  for every  $(k, l) \in \mathbb{Z}^n \times \mathbb{Z}^m \setminus \{(0, 0)\}$  with  $|l|_1 \leq 2$ , where  $\langle k \rangle = \max(1, |k|)$ . On  $\Pi_\gamma$ , every nonkernel scalar coefficient equation arising from the tangential, first-normal, and second-normal quadratic homological equations is invertible; the  $(k, l) = (0, 0)$  coefficient is retained as normal-form kernel, and analytic Fourier summation loses at most  $C\gamma^{-1}\sigma^{-(n+\tau)}$  on a strip loss  $\sigma > 0$ . If, in addition, every resonance function above is uniformly transverse to a parameter direction with derivative bounded below by  $c\langle k \rangle$  and the parameter-line sections have uniformly bounded length, then  $\text{Leb}(\Pi \setminus \Pi_\gamma) \leq C_1\gamma$ .

*Proof.* Use complex normal variables  $z_j = (u_j + iv_j)/\sqrt{2}$ . A monomial of weighted degree at most two is a scalar multiple of  $e^{2\pi i\langle k, \theta \rangle} y^a z^\alpha \bar{z}^\beta$  with  $2|a| + |\alpha| + |\beta| \leq 2$ . Commutation with the quadratic normal form multiplies its coefficient by

$$iR_{k,l}(\xi), \quad R_{k,l} = 2\pi\langle k, \omega(\xi) \rangle + \langle l, \Omega(\xi) \rangle, \quad (7.2.1)$$

where  $l = \alpha - \beta$  and  $|l|_1 \leq 2$ . Thus the only coefficient not to be divided is  $(k, l) = (0, 0)$ , which is retained in the new normal form.

**Lemma 7.1 (analytic order-two homological inversion).** On  $\Pi_\gamma$ , if  $F$  has zero kernel coefficient and is analytic on strip  $s$ , the coefficientwise solution of  $\mathcal{L}_N X = F$  satisfies on  $s - \sigma$

$$\|X\|_{s-\sigma} \leq C\gamma^{-1}\sigma^{-(n+\tau)}\|F\|_s. \quad (7.2.2)$$

Indeed, (7.2.1) and the Melnikov inequalities imply  $|R_{k,l}|^{-1} \leq \gamma^{-1}\langle k \rangle^\tau$ . Analyticity gives  $|F_{k,l}| \leq \|F\|_s e^{-2\pi|k|\sigma}$ . Hence

$$\sum_{k \in \mathbb{Z}^n} \langle k \rangle^\tau e^{-2\pi|k|\sigma} \leq C_{n,\tau} \sigma^{-(n+\tau)},$$

which proves (7.2.2). There are only finitely many  $l$  with  $|l|_1 \leq 2$ , so the same bound controls simultaneously the tangential, first-normal and second-normal equations.

**Lemma 7.2 (transverse resonance-slab estimate).** Under the definition of uniform order-two Melnikov transversality above, every bad resonance slab has measure  $O(\gamma\langle k \rangle^{-\tau-1})$ , uniformly in  $l$ , and therefore

$$\text{Leb}(\Pi \setminus \Pi_\gamma) \leq C_1 \gamma \quad (\tau > n - 1). \quad (7.2.3)$$

Fix  $(k, l)$ . Slice the bounded convex extension domain by lines parallel to the transverse unit direction supplied by the definition of uniform order-two Melnikov transversality above. On each line the resonance function is monotone with derivative at least  $c\langle k \rangle$ , so the interval on which  $|R_{k,l}| < \gamma\langle k \rangle^{-\tau}$  has length at most  $2c^{-1}\gamma\langle k \rangle^{-\tau-1}$ . Fubini and the uniformly bounded transverse sections give the same estimate for the full slab. Summing over the finite  $l$ -set and over lattice shells, whose cardinality is  $O(r^{n-1})$ , gives  $\gamma \sum_{r \geq 1} r^{n-1} r^{-\tau-1} < \infty$  exactly when  $\tau > n - 1$ . The  $k = 0$  pure-normal slabs are finite in number and satisfy the same  $O(\gamma)$  bound with  $\langle 0 \rangle = 1$ . Combining Lemma 7.1 and Lemma 7.2 proves both claims of the theorem.  $\square$

### Theorem K07.2. Persistence of normally elliptic lower-dimensional tori

**Statement.** Consider a real-analytic family, on a fixed complex neighborhood of  $\mathbb{T}^n \times \{0\} \times \{0\}$ , of Hamiltonians whose unperturbed normal form is  $N_\xi = \langle \omega(\xi), y \rangle + \frac{1}{2} \sum_{j=1}^m \Omega_j(\xi)(u_j^2 + v_j^2)$ , with  $\xi$  in a compact parameter domain  $\Pi \subset \mathbb{R}^n$ . Assume  $\xi \mapsto \omega(\xi)$  is bi-Lipschitz onto its image and the normal frequencies are simple and bounded away from zero. Fix  $\gamma > 0$  and  $\tau > n$ , and define the moving Melnikov inequalities using the corrected KAM frequencies and the divisor in the normalization used here  $|2\pi\langle k, \omega(\xi) \rangle + \langle l, \Omega(\xi) \rangle| \geq \gamma\langle k \rangle^{-\tau}$  for every  $k \in \mathbb{Z}^n$  and  $l \in \mathbb{Z}^m$  with  $|l|_1 \leq 2$  and  $|k| + |l| \neq 0$ , where  $\langle k \rangle = \max\{1, |k|\}$ . Then, for every sufficiently small uniform real-analytic perturbation  $P_\xi$ , a positive-measure Cantor subset of  $\Pi_\gamma$  carries a parameter-Lipschitz family of nearby analytic invariant  $n$ -tori with elliptic normal linear dynamics. Their tangential and normal frequencies differ from the unperturbed ones by quantities tending uniformly to zero with  $\|P\|$ .



### Scope note

Exact preservation of an arbitrarily prescribed tangential frequency requires an additional frequency-fixing hypothesis.

*Proof.* We first freeze the local phase-space scale. Choose  $s_0 > 0$  and  $r_0 > 0$  so that the complex polydisc

$$D(s_0, r_0) = \{(\theta, y, z, \bar{z}) : |\Im \theta| < s_0, |y| < r_0^2, |z|, |\bar{z}| < r_0\}$$

is contained in the analytic neighborhood occurring in the statement. The integrable terms of weighted degree at least three are then  $O(r_0)$  in the weighted Hamiltonian-vector-field norm. Shrink  $r_0$ , once and for all, so that this fixed tail is smaller than one quarter of the KAM threshold. Only after this choice do we require the external perturbation to be small. This order of quantifiers is important: the fixed nonlinear tail cannot be made small by sending the perturbation parameter to zero.

Write, in complex normal variables,

$$H_\nu = N_\nu + P_\nu, \quad N_\nu = e_\nu + \langle \omega_\nu(\xi), y \rangle + \sum_{j=1}^m \Omega_{\nu,j}(\xi) z_j \bar{z}_j, \quad (7.1.1)$$

on  $D(s_\nu, r_\nu)$  and put  $\varepsilon_\nu = \|X_{P_\nu}\|_{s_\nu, r_\nu}^{\text{Lip}}$ . The initial pair is  $(\omega_0, \Omega_0) = (\omega, \Omega)$ . All Lipschitz seminorms below are taken with respect to the original compact parameter domain and are normalized by the fixed parameter scale specified in the Lipschitz norm defined earlier in this chapter.

**Lemma 7.3 (static corrected-frequency positive measure).** Let  $(\omega', \Omega')$  satisfy

$$\|\omega' - \omega\|_{0,1} + \|\Omega' - \Omega\|_{0,1} \leq c_0 \gamma \quad (7.1.2)$$

with  $c_0$  sufficiently small in terms of the bi-Lipschitz constants of  $\omega$  and the fixed normal-frequency data. Then the corrected order-two Melnikov set with threshold  $\gamma$  has complement of measure at most  $C\gamma$  inside a fixed positive-measure subset of the unperturbed  $2\gamma$ -set. Use  $a = \omega(\xi)$  as a parameter coordinate. By bi-Lipschitzness this changes Lebesgue measure by bounded factors. For  $k \neq 0$  and  $|l|_1 \leq 2$  set

$$R'_{k,l}(a) = 2\pi \langle k, \omega'(\omega^{-1}a) \rangle + \langle l, \Omega'(\omega^{-1}a) \rangle.$$

The derivative of the unperturbed part in the direction  $k/|k|$  equals  $2\pi|k|$ . The Lipschitz perturbation in (7.1.2) changes the corresponding one-dimensional slope by at most  $Cc_0|k|$ ; hence, after fixing  $c_0$ , the slope is at least  $\pi|k|$ . The slicing argument from the preceding first-and-second Melnikov theorem therefore gives

$$\text{Leb}\{|R'_{k,l}| < \gamma \langle k \rangle^{-\tau}\} \leq C\gamma \langle k \rangle^{-\tau-1}. \quad (7.1.3)$$

There are finitely many  $l$  with  $|l|_1 \leq 2$ , so summation over  $k \in \mathbb{Z}^n \setminus \{0\}$  converges for  $\tau > n - 1$ . Pure-normal resonances have  $k = 0$ . Because the normal frequencies are simple and bounded away from zero, the finitely many nonkernel combinations with  $|l|_1 \leq 2$

are separated from zero on the compact parameter set after, if necessary, removing the zero-measure exact-resonance locus already excluded by the hypotheses. The small perturbation in (7.1.2) preserves half of this margin. This proves the claim.

Choose increasing cutoffs  $K_\nu$  and decreasing divisor margins  $\gamma_\nu \searrow \gamma$  with  $\gamma_\nu \geq 3\gamma/2$ . Let  $\Pi_\nu$  be the set on which all corrected divisors with  $|k| \leq K_\nu$  obey

$$|2\pi\langle k, \omega_\nu(\xi) \rangle + \langle l, \Omega_\nu(\xi) \rangle| \geq \gamma_\nu \langle k \rangle^{-\tau}, \quad |l|_1 \leq 2, \quad (7.1.4)$$

with the kernel  $(k, l) = (0, 0)$  omitted.

**Lemma 7.4 (moving-frequency shell estimate).** Assume  $\|\omega_{\nu+1} - \omega_\nu\|_{0,1} + \|\Omega_{\nu+1} - \Omega_\nu\|_{0,1} \leq d_\nu$ . If  $d_\nu \leq c\gamma K_\nu^{-\tau-1}$ , then previously accepted divisors remain accepted with the next margin, and the parameters newly removed at stage  $\nu + 1$  satisfy

$$\text{Leb}(\Pi_\nu \setminus \Pi_{\nu+1}) \leq C\gamma K_\nu^{-(\tau-n)}. \quad (7.1.5)$$

For  $|k| \leq K_\nu$  the divisor drift is at most  $C(|k| + 1)d_\nu$ , which is bounded by the reserved difference  $\gamma_\nu - \gamma_{\nu+1}$  after choosing the numerical constant in the hypothesis. Thus only the shell  $K_\nu < |k| \leq K_{\nu+1}$  can create new bad sets. Applying (7.1.3) to this finite shell and summing lattice points gives

$$C\gamma \sum_{|k| > K_\nu} |k|^{-\tau-1} \leq C'\gamma K_\nu^{n-\tau},$$

which is (7.1.5). This is the reason for the slightly stronger exponent  $\tau > n$  in the present theorem.

**Lemma 7.5 (stage-to-limit divisor persistence).** If for every  $\nu$

$$\sum_{j \geq \nu} (\|\omega_{j+1} - \omega_j\|_{0,1} + \|\Omega_{j+1} - \Omega_j\|_{0,1}) \leq \frac{\gamma}{8C_*} K_\nu^{-\tau-1}, \quad (7.1.6)$$

then every divisor accepted at scale  $K_\nu$  converges to a limiting divisor of size at least  $\gamma \langle k \rangle^{-\tau}$ . Indeed, for  $|k| \leq K_\nu$  and  $|l|_1 \leq 2$ , the total future change is at most

$$C_*(|k| + 1) \frac{\gamma}{8C_*} K_\nu^{-\tau-1} \leq \frac{\gamma}{4} \langle k \rangle^{-\tau}.$$

Since the finite-scale margin is at least  $3\gamma/2$ , the limiting divisor is larger than the claimed  $\gamma$  margin. Every fixed  $k$  is eventually below some cutoff, so all limiting order-two Melnikov inequalities follow.

We now prove the specialist step that was missing from the previous proof architecture. Let  $R_\nu$  be the weighted Taylor polynomial of  $P_\nu$  in  $(y, z, \bar{z})$  through weighted degree two and let  $R_\nu^{K_\nu}$  denote its Fourier truncation to  $|k| \leq K_\nu$ . Split

$$R_\nu^{K_\nu} = [R_\nu] + R_\nu^\perp, \quad (7.1.7)$$

where  $[R_\nu]$  consists exactly of the constant term, the average  $y$ -linear term and the diagonal average  $z_j \bar{z}_j$  terms. These are the kernel coefficients for the adjoint action of  $N_\nu$ ; all other coefficients belong to  $R_\nu^\perp$ .

**Lemma 7.6 (parameter-Lipschitz lower-dimensional KAM step).** There are integers  $A, B$  depending only on  $(n, m, \tau)$  and a constant  $C$  depending only on the fixed local data such that the following holds. Suppose (7.1.4) holds through  $K = K_\nu$  on  $\Pi_\nu$ , let  $0 < \sigma < s_\nu/8$ ,  $0 < \eta < 1/8$ , and assume

$$\delta_\nu := C\gamma_\nu^{-1}\sigma^{-A}\varepsilon_\nu \leq c\eta r_\nu. \quad (7.1.8)$$

Then on  $D(s_\nu - 4\sigma, (1 - 4\eta)r_\nu)$  there is a real-analytic, parameter-Lipschitz symplectic map  $\Phi_{\nu+1}$  in the fiber-polynomial symplectic class defined earlier in this chapter such that

$$H_\nu \circ \Phi_{\nu+1} = N_{\nu+1} + P_{\nu+1}, \quad (7.1.9)$$

where

$$\|\Phi_{\nu+1} - \text{Id}\|^{\text{Lip}} \leq C\gamma_\nu^{-1}\sigma^{-A}\varepsilon_\nu, \quad (7.1.10)$$

$$\|\omega_{\nu+1} - \omega_\nu\|_{0,1} + \|\Omega_{\nu+1} - \Omega_\nu\|_{0,1} \leq C\varepsilon_\nu, \quad (7.1.11)$$

and

$$\varepsilon_{\nu+1} \leq C\left(e^{-2\pi K\sigma} + \eta + \frac{\varepsilon_\nu}{\gamma_\nu^2 \sigma^B \eta^2}\right)\varepsilon_\nu. \quad (7.1.12)$$

The new normal frequencies are real and remain uniformly separated from zero provided  $\sum_j \varepsilon_j$  is smaller than the initial elliptic margin. By the preceding first-and-second Melnikov theorem, coefficientwise division of the nonkernel polynomial  $R_\nu^\perp$  solves

$$\{N_\nu, F_\nu\} + R_\nu^\perp = 0 \quad (7.1.13)$$

and gives, after a strip loss  $\sigma$ ,

$$\|X_{F_\nu}\|_{s_\nu-\sigma, r_\nu}^{\text{Lip}} \leq C\gamma_\nu^{-1}\sigma^{-A}\varepsilon_\nu. \quad (7.1.14)$$

The Lipschitz bound is obtained by applying the resolvent identity to every scalar divisor:

$$\frac{1}{R(\xi)} - \frac{1}{R(\zeta)} = \frac{R(\zeta) - R(\xi)}{R(\xi)R(\zeta)},$$

so that one extra divisor factor is absorbed into the exponent  $A$ ; the finite number of normal monomial types does not affect convergence of the Fourier sum. Under the convention used here  $\iota_{X_F}\omega_{\text{symp}} = -dF$ , cancellation in (7.1.13) is produced by the inverse-time map  $\Phi_{\nu+1} = X_{F_\nu}^{-1}$ , as fixed by the Hamiltonian-flow sign convention established in Volume K01.

The weighted structure of  $F_\nu$  is at most quadratic in the fiber variables. Consequently its Hamiltonian vector field is affine in  $(y, z, \bar{z})$  with analytic angle-dependent coefficients. Cauchy's estimate and (7.1.14) show that (7.1.8) keeps its time interval  $[-1, 0]$  inside  $D(s_\nu - 2\sigma, (1 - 2\eta)r_\nu)$ ; the variational equation gives the same bound for the first derivative of the flow. Hence the inverse-time map is well defined, analytic, symplectic, of the required triangular fiber-polynomial class, and satisfies (7.1.10).

Set

$$N_{\nu+1} = N_\nu + [R_\nu]. \quad (7.1.15)$$

The constant kernel changes  $e_\nu$ , the average  $y$ -linear kernel changes  $\omega_\nu$ , and the real

diagonal  $z_j \bar{z}_j$  kernel changes  $\Omega_{\nu,j}$ . Cauchy's estimate for the Taylor coefficients immediately gives (7.1.11). Reality of the Hamiltonian makes these frequency corrections real. If the initial normal frequencies have lower bound  $\Omega_* > 0$  and  $C \sum_j \varepsilon_j < \Omega_*/2$ , then  $|\Omega_{\nu,j}| \geq \Omega_*/2$  at every step.

For completeness we derive the remainder estimate rather than naming it. The Lie formula and (7.1.13)–(7.1.15) give

$$P_{\nu+1} = (P_\nu - R_\nu^K) \circ \Phi_{\nu+1} + \int_0^1 (1-t) \{ \{N_\nu + R_\nu^K, F_\nu\}, F_\nu \} \circ X_{F_\nu}^{-t} dt + \int_0^1 \{P_\nu - R_\nu^K, F_\nu\} \circ X_{F_\nu}^{-t} dt. \quad (7.1.16)$$

The Fourier part of  $P_\nu - R_\nu^K$  is bounded by  $Ce^{-2\pi K\sigma} \varepsilon_\nu$ . Its fiber Taylor remainder has weighted degree at least three; on the radius  $(1-4\eta)r_\nu$  the standard integral Taylor formula gives  $C\eta\varepsilon_\nu$ . Each Poisson bracket costs one phase-space Cauchy factor and one factor  $\|X_{F_\nu}\|$ ; after collecting all fixed losses into  $B$  and using the two radius losses, the two integrals in (7.1.16) are bounded by  $C\gamma_\nu^{-2}\sigma^{-B}\eta^{-2}\varepsilon_\nu^2$ . This proves (7.1.12).

**Lemma 7.7 (supergeometric iteration and moving survivor).** There exist explicit sequences  $\sigma_\nu, \eta_\nu, K_\nu$  and a threshold  $\varepsilon_*$  such that  $\varepsilon_0 < \varepsilon_*$  implies

$$\varepsilon_{\nu+1} \leq \varepsilon_\nu^{3/2}, \quad \sum_\nu \sigma_\nu < s_0/4, \quad \sum_\nu \eta_\nu < 1/4, \quad (7.1.17)$$

all bounds (7.1.8) and (7.1.6) hold, and the nested survivor  $\Pi_* = \bigcap_\nu \Pi_\nu$  has positive measure. Take, for example,

$$\sigma_\nu = \frac{s_0}{2^{\nu+5}}, \quad \eta_\nu = \frac{1}{2^{\nu+5}}, \quad K_\nu = \left\lceil K_0^{(3/2)^\nu} \right\rceil. \quad (7.1.18)$$

Choose  $K_0$  large enough that  $e^{-2\pi K_\nu \sigma_\nu} \leq \varepsilon_\nu^{1/2}/(4C)$  whenever the inductive estimate in (7.1.17) holds. Since all powers of  $\sigma_\nu^{-1}$  and  $\eta_\nu^{-1}$  grow only exponentially in  $\nu$ , whereas  $\varepsilon_\nu$  decays like  $\varepsilon_0^{(3/2)^\nu}$ , a sufficiently small  $\varepsilon_0$  makes the quadratic term in (7.1.12) at most  $\varepsilon_\nu^{3/2}/2$  and simultaneously enforces (7.1.8). Thus (7.1.17) follows inductively.

At this point the generic numerical convergence implication may be invoked without becoming a black box for the specialist KAM step: the abstract analytic Newton–KAM convergence lemma from Foundation Volume F12 applies precisely because (7.1.10)–(7.1.12) have now been proved. It yields summability of the corrections on the limiting domain. In particular, (7.1.11) gives

$$\sum_{j \geq \nu} (\|\omega_{j+1} - \omega_j\|_{0,1} + \|\Omega_{j+1} - \Omega_j\|_{0,1}) \leq C \sum_{j \geq \nu} \varepsilon_j. \quad (7.1.19)$$

Supergeometric decay beats  $K_\nu^{-\tau-1}$ , so after one further decrease of  $\varepsilon_*$ , (7.1.6) holds for every  $\nu$ . Lemma 7.4 then shows

$$\sum_\nu \text{Leb}(\Pi_\nu \setminus \Pi_{\nu+1}) \leq C\gamma \sum_\nu K_\nu^{-(\tau-n)} < \infty. \quad (7.1.20)$$

Lemma 7.3 supplies an initial reserve whose measure exceeds the right side of (7.1.20) when  $K_0$  is enlarged. Hence  $\text{Leb}(\Pi_*) > 0$ .

**Lemma 7.8 (convergence to an actual local symplectic map and parameter-Lipschitz family).** The compositions  $\Psi_\nu = \Phi_1 \circ \cdots \circ \Phi_\nu$  converge uniformly with first phase derivatives on a fixed smaller complex domain and in the parameter Lipschitz norm to an analytic symplectic map  $\Psi_\infty(\cdot, \xi)$  for every  $\xi \in \Pi_*$ . The coefficient functions admit Lipschitz extensions to the ambient parameter domain. From (7.1.10) and (7.1.17),  $\sum_\nu \|\Phi_{\nu+1} - \text{Id}\|^{\text{Lip}} < \infty$ . The same is true for first phase derivatives after reserving one additional fixed strip loss. Hence the compositions are Cauchy there and converge to an analytic local diffeomorphism. Because every  $\Psi_\nu$  is symplectic,

$$\Psi_\nu^* \omega_{\text{symp}} = \omega_{\text{symp}}.$$

Uniform convergence of the first derivatives passes this identity to the limit, so  $\Psi_\infty$  is symplectic. The parameter coefficient maps are uniformly Lipschitz on the nested closed Cantor sets. Extend each scalar coefficient by the McShane extension with its same Lipschitz constant and use the convergent coefficient series to obtain ambient Lipschitz extensions with uniform bounds. No stronger Whitney  $C^r$  conclusion is claimed from merely Lipschitz parameter data.

**Lemma 7.9 (limiting torus and elliptic normal form).** On  $\Pi_*$ ,

$$H \circ \Psi_\infty = e_\infty + \langle \omega_\infty, y \rangle + \sum_{j=1}^m \Omega_{\infty,j} z_j \bar{z}_j + P_\infty, \quad J^2 P_\infty(\theta, 0, 0, 0) = 0, \quad (7.1.21)$$

where  $J^2$  denotes the weighted Taylor jet through degree two. Therefore the zero section is invariant and its normal linearization has eigenvalues  $\pm i\Omega_{\infty,j}$ . Indeed, at each step all nonkernel coefficients in the truncated weighted 2-jet are removed, while the kernel coefficients are absorbed into  $N_{\nu+1}$ . The unremoved high Fourier tail tends to zero because  $K_\nu \rightarrow \infty$ , and the new quadratic error tends to zero by (7.1.17). Consequently the weighted 2-jet of the limiting remainder vanishes, which is (7.1.21). Hamilton's equations on  $y = z = 0$  are therefore  $\dot{\theta} = \omega_\infty(\xi)$  and  $\dot{y} = \dot{z} = \dot{\bar{z}} = 0$ . The image

$$\mathcal{T}_\xi = \Psi_\infty(\mathbb{T}^n \times \{0\} \times \{0\})$$

is an analytic invariant  $n$ -torus. By (7.1.19),

$$\|\omega_\infty - \omega\|_{0,1} + \|\Omega_\infty - \Omega\|_{0,1} \leq C \sum_\nu \varepsilon_\nu \rightarrow 0$$

with the initial perturbation. The initial normal-frequency lower bound and the summable correction estimate give  $|\Omega_{\infty,j}| \geq \Omega_*/2$ , so the normal linearization is elliptic. Finally Lemma 7.5 proves that every  $\xi \in \Pi_*$  satisfies the full limiting corrected-frequency Melnikov inequalities. We have proved only the required inclusion of the constructed survivor in the final Melnikov set, not equality with that set. This completes the theorem.  $\square$

### Theorem K07.3. Li-Yi Floquet persistence with multiple normal eigenvalues

**Statement.** Consider the lower-dimensional Hamiltonian family in the Li-Yi normal form, with parameter  $\omega \in \mathcal{O}$ , quadratic normal block  $M_{22}(\omega)$ , and eigenvalues  $\lambda_1(\omega), \dots, \lambda_{2m}(\omega)$  of  $JM_{22}(\omega)$ . Assume the weak second Melnikov condition that, for Lebesgue-a.e.  $\omega \in \mathcal{O}$ ,  $2\pi i \langle k, \omega \rangle - \lambda_i(\omega) - \lambda_j(\omega) \neq 0$  for every  $k \in \mathbb{Z}^n \setminus \{0\}$  and  $1 \leq i, j \leq 2m$ . Fix  $\tau > n - 1$ ,  $r, s > 0$ ,  $l_0 \geq 4m^2$ , and  $\sigma_0 \in (0, 1)$ . If either  $M_{22}(\omega)$  or the full quadratic matrix  $M(\omega)$  is nonsingular on  $\mathcal{O}$ , and if the analytic perturbation satisfies  $|\partial_\omega^l \partial_x^i \partial_{(y,z)}^j P| \leq \gamma^{3(1+\sigma_0)a} \mu$  for  $|l| + |i| + |j| \leq l_0$ , where  $a = 4m^2(l_0 + 1)$  and  $\mu$  is sufficiently small, then there are Cantor-like resonance-excluded sets  $\mathcal{O}_\gamma \subset \mathcal{O}$  with  $|\mathcal{O} \setminus \mathcal{O}_\gamma| \rightarrow 0$  as  $\gamma \rightarrow 0$  and a Whitney  $C^{l_0-1}$  family of real-analytic symplectic changes of variables putting the perturbed Hamiltonian into Floquet normal form along the surviving tori. If  $M_{22}$  is nonsingular, the toral frequency is a nearby Diophantine vector  $\Omega^*(\omega)$ ; if the full  $M$  is nonsingular, one may take  $\Omega^*(\omega) = \omega$ . Multiple eigenvalues of  $JM_{22}$  are allowed.

*Proof.*

#### Proof strategy

Put the torus and its normal directions into Floquet form and solve the tangential and normal homological equations simultaneously. Multiple normal eigenvalues are handled blockwise, with the Melnikov conditions imposed only on the divisors that are actually inverted. The KAM step preserves the normal spectral clusters and improves the perturbation quadratically. Cantor exclusion plus convergence of the block conjugacies then yields persistence and the stated Floquet normal form.

We work in the Li-Yi normal form introduced earlier in this chapter,

$$H = N + P, \quad N = e(\omega) + \langle \Omega(\omega), y \rangle + \frac{1}{2} \left\langle \begin{pmatrix} y \\ z \end{pmatrix}, M(\omega) \begin{pmatrix} y \\ z \end{pmatrix} \right\rangle + h(y, z, \omega), \quad (7.3.1)$$

where  $h = O(|(y, z)|^3)$  and  $M$  is symmetric. Initially  $\Omega(\omega) = \omega$ . We use the project Fourier convention  $e^{2\pi i \langle k, x \rangle}$ , so every angular divisor contains the factor  $2\pi$ . Put

$$B = M_{22}J.$$

The matrices  $M_{22}J$  and  $JM_{22}$  have the same characteristic polynomial, so the eigenvalues of  $B$ , with algebraic multiplicity, are the  $\lambda_1, \dots, \lambda_{2m}$  appearing in the theorem.

For  $k \in \mathbb{Z}^n \setminus \{0\}$  define the scalar, first-normal and second-normal homological blocks

$$L_{0k} = 2\pi i \langle k, \Omega \rangle, \quad L_{1k} = 2\pi i \langle k, \Omega \rangle I_{2m} - B, \quad (7.3.2)$$

$$L_{2k} = 2\pi i \langle k, \Omega \rangle I_{4m^2} - (B \otimes I_{2m} + I_{2m} \otimes B). \quad (7.3.3)$$

These are exactly the blocks produced by the quadratic Taylor-Fourier homological system. The point of the Li-Yi scheme is that  $y$  and  $z$  are kept on one common small scale. That choice, together with the modified homological equation below, is what

allows the full- $M$  branch even when the normal block itself is singular.

**Step 1 (Kronecker determinant reduction with multiplicity).** For every  $k \neq 0$ ,

$$\det L_{1k} = \prod_{a=1}^{2m} (2\pi i \langle k, \Omega \rangle - \lambda_a), \quad \det L_{2k} = \prod_{a,b=1}^{2m} (2\pi i \langle k, \Omega \rangle - \lambda_a - \lambda_b). \quad (7.3.4)$$

No diagonalizability or simple-spectrum assumption is required.

Choose a complex Schur triangularization  $T^{-1}BT = E$ . Then  $T^{-1}L_{1k}T$  is triangular with diagonal entries  $2\pi i \langle k, \Omega \rangle - \lambda_a$ . Moreover

$$(T^{-1} \otimes T^{-1})(B \otimes I + I \otimes B)(T \otimes T) = E \otimes I + I \otimes E,$$

which is triangular with diagonal entries  $\lambda_a + \lambda_b$ . Taking determinants proves (7.3.4). Jordan nilpotent parts only alter entries above the diagonal, so repeated eigenvalues cause no difficulty. In particular the weak second Melnikov condition stated above is precisely the almost-everywhere nonvanishing of the initial  $L_{2k}$  determinants. If an initial  $L_{1k}$  determinant vanished, one factor would satisfy  $2\pi i \langle k, \omega \rangle - \lambda_a = 0$ ; using  $2k$  and the pair  $(a, a)$  would contradict the weak second Melnikov condition. Thus  $L_{1k}$  is also invertible for almost every parameter.

Fix a finite cutoff  $K_+$  and a divisor margin  $\gamma_+ \leq \gamma$ . On the finite-scale good set require, for  $0 < |k| \leq K_+$ ,

$$|L_{0k}| > \frac{\gamma}{|k|^\tau}, \quad |\det L_{1k}| > \frac{\gamma^{2m}}{|k|^{2m\tau}}, \quad |\det L_{2k}| > \frac{\gamma^{4m^2}}{|k|^{4m^2\tau}}. \quad (7.3.5)$$

At the end of the step the same inequalities will hold with  $\gamma_+$  and the updated blocks. The determinant powers are the matrix dimensions; this is what makes inversion possible without separating multiple eigenvalue branches.

**Step 2 (finite-scale block inversion and the six coefficient equations).** On the good set, with  $d_0 = 1$ ,  $d_1 = 2m$ ,  $d_2 = 4m^2$ ,

$$\|L_{qk}^{-1}\| \leq C \frac{|k|^{d_q\tau + d_q - 1}}{\gamma^{d_q}}, \quad q = 0, 1, 2, \quad (7.3.6)$$

and, for every parameter multi-index  $\ell$ ,  $|\ell| \leq l_0$ ,

$$\|\partial_\omega^\ell L_{qk}^{-1}\| \leq C_\ell \frac{|k|^{|\ell| + (|\ell|+1)d_q\tau}}{\gamma^{(|\ell|+1)d_q}}. \quad (7.3.7)$$

The quadratic Taylor–Fourier homological system is triangular and therefore solvable in the order  $F_{02}$ ,  $F_{01}$ ,  $F_{11}$ , then the scalar blocks. All parameter jets through order  $l_0$  are controlled by the single safe loss exponent  $a = 4m^2(l_0 + 1)$ .

For  $q = 1, 2$ , Cramer’s formula gives  $L_{qk}^{-1} = \text{adj}(L_{qk}) / \det L_{qk}$ . Every entry of  $L_{qk}$  is  $O(|k| + 1)$  on the compact parameter set, hence every cofactor is  $O(|k|^{d_q - 1})$ , which yields

(7.3.6). Differentiating  $L^{-1}L = I$  gives

$$\partial_\omega^\ell L^{-1} = - \sum_{0 < \ell' \leq \ell} \binom{\ell}{\ell'} (\partial_\omega^{\ell-\ell'} L^{-1}) (\partial_\omega^{\ell'} L) L^{-1}. \quad (7.3.8)$$

Since every nonzero parameter derivative of  $L_{qk}$  is  $O(|k|)$ , induction on  $|\ell|$  proves (7.3.7). The worst block has dimension  $4m^2$ , and at most  $l_0 + 1$  inverse factors occur, explaining the source exponent

$$a = 4m^2(l_0 + 1).$$

To see that no coefficient equation is hidden in this estimate, let  $R$  be the Taylor polynomial of  $P$  through degree two in  $(y, z)$ , truncated to  $|k| \leq K_+$ . Write its nonzero Fourier coefficients as  $P_{k00}, P_{k10}, P_{k01}, P_{k20}, P_{k11}, P_{k02}$  and seek  $F$  with the same quadratic form. Vectorizing  $F_{k02}$ , the six equations are

$$\begin{aligned} L_{0k}F_{k00} &= P_{k00}, & L_{1k}F_{k01} &= P_{k01}, \\ L_{0k}F_{k10} &= P_{k10} - M_{12}JF_{k01}, & L_{2k}\text{vec}(F_{k02}) &= \text{vec}(P_{k02}), \\ L_{1k}F_{k11} &= P_{k11} - (F_{k02}^T + F_{k02})JM_{21}, & L_{0k}F_{k20} &= P_{k20} + \frac{1}{2}(F_{k11}^T JM_{21} - M_{12}JF_{k11}). \end{aligned} \quad (7.3.9)$$

Thus the  $L_2$  equation is solved first, then the two  $L_1$  equations, and finally the scalar equations. Combining (7.3.7) with the Fourier coefficient bound

$$|\partial_\omega^\ell P_{kij}| \leq \gamma^a s^{2-i-j} \mu e^{-|k|r}, \quad 0 \leq i + j \leq 2,$$

shows that all factors  $\gamma^{-(|\ell|+1)4m^2}$  are absorbed by  $\gamma^a$ . After a strip loss  $\rho > 0$ , polynomial powers of  $|k|$  are summed against  $e^{-|k|\rho}$ , producing only a fixed power of  $\rho^{-1}$ .

The angle average  $[R]$  contributes constant, linear and quadratic terms. The constant and quadratic terms are put into the new normal form. The average linear terms are removed by a finite-dimensional translation.

**Step 3 (exact averaged translation and the two frequency branches).** Let  $P_{010}$  and  $P_{001}$  denote the averaged coefficients of  $y$  and  $z$ . If the full matrix  $M$  is nonsingular, the small translation  $(y_0, z_0)$  is the unique solution near zero of

$$(M + \partial_{(y,z)}^2 h(y_0, z_0, \omega)) \begin{pmatrix} y_0 \\ z_0 \end{pmatrix} + \partial_{(y,z)} h(y_0, z_0, \omega) = - \begin{pmatrix} P_{010} \\ P_{001} \end{pmatrix}, \quad (7.3.10a)$$

and the tangential frequency remains exactly  $\Omega_+ = \Omega$ . If only  $M_{22}$  is nonsingular, set  $y_0 = 0$  and solve

$$(\text{diag}(0, M_{22}) + \partial_{(y,z)}^2 h(0, z_0, \omega)) \begin{pmatrix} 0 \\ z_0 \end{pmatrix} + \partial_{(y,z)} h(0, z_0, \omega) = - \begin{pmatrix} 0 \\ P_{001} \end{pmatrix}; \quad (7.3.10b)$$

then  $\Omega_+ = \Omega + P_{010}$ . The solutions and their parameter jets are  $O(\gamma^a s \mu)$ , and the active nonsingularity alternative persists under a summable sequence of such corrections.

Both assertions follow from the parameter-dependent implicit function theorem. The derivative of the left side of (7.3.10a) at zero is  $M$ ; the derivative in the active  $z$  variable of (7.3.10b) is  $M_{22}$ . Their inverses are uniformly bounded on the compact



parameter set. Since  $h = O(|(y, z)|^3)$  and the averaged linear coefficients inherit the perturbation estimate, contraction on a ball of radius  $C\gamma^a s\mu$  gives the unique solutions and their  $C^{l_0-1}$  parameter jets. In the full- $M$  branch the complete average linear term is removed, so no frequency counterterm remains. In the  $M_{22}$  branch only the normal average is removed; the surviving  $y$  coefficient is precisely the frequency correction.

We now establish the quantitative KAM iteration. Fix  $0 < \lambda_0 < 1$  and set, for  $p = |j|$ ,

$$a_p = \begin{cases} a, & p \leq 2, \\ 0, & p \geq 3, \end{cases} \quad \kappa_p = \begin{cases} 2, & p = 0, \\ 1, & p = 1, \\ 0, & p \geq 2, \end{cases} \quad d_p = \begin{cases} 1, & p \leq 2, \\ 1 - \lambda_0, & p \geq 3. \end{cases} \quad (7.3.11)$$

At a generic stage the induction hypothesis is

$$|\partial_\omega^\ell \partial_x^i \partial_{(y,z)}^j P|_{D(r,s)} \leq \gamma^{a_p} s^{\kappa_p} \mu^{d_p}, \quad |\ell| + |i| + |j| \leq l_0. \quad (7.3.12)$$

The theorem's original perturbation estimate starts this induction after the source rescaling

$$s_0 = \gamma^{(1+\sigma_0)a} \mu_*^{1/4}, \quad \mu_0 = \gamma^{\sigma_0} \mu_*^{1/2}, \quad \gamma_0 = \gamma, \quad (7.3.13)$$

where  $\mu_*$  denotes the small parameter in the theorem. Indeed, for  $p = 0, 1, 2$  the right side of (7.3.12), evaluated at  $(s_0, \mu_0)$ , dominates  $\gamma^{3(1+\sigma_0)a} \mu_*$  when  $0 < \gamma, \mu_* \leq 1$  and  $a \geq 1$ ; for  $p \geq 3$  the conclusion is even easier after decreasing  $\mu_*$ . Thus no unstated stronger initial hypothesis is introduced.

The averaging equation is not the ordinary linear KAM equation. Li-Yi retain from  $\{N, F\}$  the terms which are  $O(s^3\mu)$  on the common  $(y, z)$  scale, including all terms of transverse degree at least three, and call their sum  $Q$ . The equation solved is

$$\boxed{\{N, F\} + R - [R] - Q = 0.} \quad (7.3.14)$$

Consequently, if  $\phi_F^t$  is the Hamiltonian flow of  $F$ ,

$$H \circ \phi_F^1 = N + [R] + \int_0^1 \{(1-t)\{N, F\} + R, F\} \circ \phi_F^t dt + (P - R) \circ \phi_F^1 + Q. \quad (7.3.15)$$

This identity is the reason the common-scale remainder closes:  $Q$  is estimated as part of the new perturbation rather than incorrectly discarded from the homological equation.

Choose auxiliary constants  $b > 0$  and  $0 < \sigma < \sigma_0$  so small that

$$\sigma > (b + \sigma)(4b + 5\sigma). \quad (7.3.16a)$$

Choose  $\delta \in (0, 1)$  with

$$\delta(1 + b + \sigma) > 1, \quad (1 + \sigma_0)(1 + b + \sigma)(1 - b - 2\sigma) > 1. \quad (7.3.16b)$$

Finally choose  $\varepsilon > 0$  small enough that

$$(1 + \sigma_0)(1 + b + \sigma)(1 - b - 2\sigma - 2\varepsilon) > 1, \quad (7.3.16c)$$

$$\frac{\sigma}{b + \sigma} > 4b + 5\sigma + 6\varepsilon, \quad \lambda_0 - (1 - \lambda_0)\sigma - 2\varepsilon > 0.$$

Such a choice is possible by first taking  $b, \sigma$  sufficiently small, then  $\delta$  close enough to one, and finally  $\varepsilon$  sufficiently small. These inequalities are not cosmetic: they are the exponent margins used below to dominate every Taylor, Lie, high-derivative and domain-loss term.

For one step write

$$s_+ = s^{1+b+\sigma}, \quad K_+ = (1 + \lfloor \log(1/s) \rfloor)^3, \quad \mu_+ = c_0 s^\sigma \mu, \quad (7.3.17)$$

where  $c_0$  is chosen larger than all fixed constants in the truncation, homological, flow, normal-form and remainder estimates. Let  $\beta$  be an outer transverse safety radius, and decrease  $r, \beta, \gamma$  along the explicit iteration below. If

$$\Gamma(r - r_+) = \sum_{0 < |k| \leq K_+} |k|^{\chi_*} e^{-|k|(r-r_+)/8} \quad (7.3.18)$$

with  $\chi_*$  any fixed exponent dominating all Fourier powers in (7.3.7), then all derivatives of  $F$  needed for the step are bounded by  $C s^{\kappa_p} \mu \Gamma(r - r_+)$ .

**Step 4 (complete modified analytic Li-Yi one-step lemma).** Assume (7.3.12), the determinant good-set inequalities (7.3.5), and the following nine quantitative gates:

- (G1) the active matrix  $M$  or  $M_{22}$  is invertible with inverse norm at most twice its initial bound;
- (G2)  $s_+ \leq s/16$ ;
- (G3) the Fourier tail obeys  $\int_{K_+}^\infty t^{n+l_0} e^{-t(r-r_+)/16} dt \leq s$ ;
- (G4) the  $C^{l_0}$  drift of  $M$  and  $\Omega - \text{id}$  from their initial values is at most  $\mu_*^{1/4}$ ;
- (G5)  $C\mu\Gamma(r - r_+) < (r - r_+)/8$ ;
- (G6)  $Cs\mu\Gamma(r - r_+) < s_+$ ;
- (G7)  $C\mu(\Gamma(r - r_+) + 1) < \beta - \beta_+$ ;
- (G8) every differentiated remainder term in (7.3.15), after the translation (7.3.10), is bounded by  $\gamma_+^{a_p} s_+^{\kappa_p} \mu_+^{d_p}$ ;
- (G9) the normal-form increment is small enough that every divisor for  $0 < |k| \leq K_+$  which satisfied (7.3.5) also satisfies the corresponding updated inequality with  $\gamma_+$ .

Then the time-one map followed by the translation is analytic and symplectic on the smaller domain, the transformed Hamiltonian has the same normal-form structure, and its new perturbation satisfies the induction estimate (7.3.12) with  $(r_+, s_+, \gamma_+, \mu_+)$ . In the full- $M$  branch the frequency is unchanged exactly.

We prove the one-step lemma. Gate (G3) and Taylor's formula give, for  $p = |j|$ ,

$$\|\partial_\omega^\ell \partial_x^i \partial_{(y,z)}^j (P - R)\| \leq C \gamma^{a_p} (s^{\kappa_p+1} + s^{\kappa_p-3} s_+^3) \mu^{d_p}. \quad (7.3.19)$$

The first term is the Fourier tail; the second is the Taylor remainder of transverse degree at least three. The inverse estimates (7.3.6)–(7.3.9) give

$$\|\partial_\omega^\ell \partial_x^i \partial_{(y,z)}^j F\| \leq C s^{\kappa_p} \mu \Gamma(r - r_+) \quad (7.3.20)$$

on the working domain. Gates (G5)–(G7) therefore keep the whole flow and the translation inside the nested phase domains. The Lie identity (7.3.15), the bracket product rule, (7.3.19)–(7.3.20), and the cubic vanishing of  $h$  show that the differentiated new remainder is a sum of four classes:

Lie-quadratic terms,  $Q$ -terms, Fourier/Taylor tails, translated high-order terms. (7.3.21)

After division by the target  $\gamma_+^{a_p} s_+^{\kappa_p} \mu_+^{d_p}$ , the cases  $p = 0, 1, 2$  are bounded by a fixed multiple of a sum of positive powers among

$$s^{2(b+\sigma)}, \quad \gamma_0^{-a} s^{1-b-2\sigma-2\varepsilon}, \quad s^{1-2b-3\sigma}, \quad s^b, \quad (7.3.22a)$$

while the cases  $p \geq 3$  are additionally bounded by positive powers among

$$s^{\lambda_0 - (1-\lambda_0)\sigma - 2\varepsilon}, \quad s^{1-\sigma}, \quad s^{3b+2\sigma}. \quad (7.3.22b)$$

The only term in (7.3.22a) containing  $\gamma_0^{-a}$  is controlled by the initial relation  $s_0 = \gamma^{(1+\sigma_0)a} \mu_*^{1/4}$  and the first inequality in (7.3.16c). Every exponent of  $s$  in (7.3.22) is positive by (7.3.16). Hence, after decreasing  $\mu_*$  once, (G8) follows at every stage. This is the quantitative closure that was missing from the earlier schematic “Newton improvement” argument.

The average coefficients satisfy

$$\|\Omega_+ - \Omega\|_{C^{l_0}} \leq C \gamma^a s \mu, \quad \|M_+ - M\|_{C^{l_0}} \leq C \gamma^a \mu, \quad (7.3.23)$$

with the first quantity identically zero in the full- $M$  branch. A determinant is locally Lipschitz in its matrix entries, with polynomial dependence on the block norm. Therefore a sufficient explicit divisor-persistence condition is

$$3C\mu K_+^{8m^2\tau+8m^2} < \min \left\{ \frac{\gamma - \gamma_+}{\gamma_0}, \frac{\gamma^{2m} - \gamma_+^{2m}}{\gamma_0^{2m}}, \frac{\gamma^{4m^2} - \gamma_+^{4m^2}}{\gamma_0^{4m^2}} \right\}. \quad (7.3.24)$$

Indeed, for  $|k| \leq K_+$  the perturbations of  $L_{0k}$ ,  $L_{1k}$  and  $L_{2k}$  are bounded by (7.3.23) times fixed powers of  $K_+$ ; (7.3.24) is deliberately stronger than the three resulting inequalities. It implies (G9).

We now verify that the nine gates propagate indefinitely, rather than assuming that

the schedule can somehow be chosen. Let  $q = 1 + b + \sigma > 1$  and define

$$\begin{aligned} r_\nu &= r_0 \left( 1 - \frac{1-\delta}{2} \sum_{j=1}^{\nu} \delta^{j+1} \right), \\ \beta_\nu &= \beta_0 \left( 1 - \sum_{j=1}^{\nu} 2^{-(j+1)} \right), \quad \gamma_\nu = \gamma_0 \left( 1 - \sum_{j=1}^{\nu} 2^{-(j+1)} \right), \\ s_\nu &= s_0^{q^\nu}, \quad \mu_{\nu+1} = c_0 s_\nu^\sigma \mu_\nu, \\ K_{\nu+1} &= (1 + \lfloor \log(1/s_\nu) \rfloor)^3. \end{aligned} \tag{7.3.25}$$

Then  $r_\infty > r_0/2$ ,  $\beta_\infty \geq \beta_0/2$  and  $\gamma_\infty \geq \gamma_0/2$ . Moreover

$$\mu_\nu = c_0^\nu \mu_0 s_0^{\frac{\sigma}{b+\sigma}(q^\nu-1)}. \tag{7.3.26}$$

Put

$$E_\nu = \frac{r_\nu - r_{\nu+1}}{16} = \frac{r_0(1-\delta)\delta^{\nu+2}}{32}, \quad \Gamma_\nu = \sum_{0 < |k| \leq K_{\nu+1}} |k|^{\chi_*} e^{-|k|E_\nu/2}. \tag{7.3.27}$$

Since  $K_{\nu+1}$  is a polynomial in  $q^\nu$  and  $E_\nu^{-1}$  grows only geometrically, while  $s_\nu = s_0^{q^\nu}$  decays supergeometrically, after one initial smallness choice we have simultaneously for every  $\nu$

$$s_\nu^\varepsilon \Gamma_\nu^4 \leq 1, \quad \frac{\mu_\nu}{s_\nu^{4b+5\sigma+2\varepsilon}} \leq \mu_*^{1/4}, \quad c_0 \mu_\nu \Gamma_\nu \leq \frac{\mu_*^{1/4}}{2^{\nu+1}}. \tag{7.3.28}$$

The first inequality follows because  $q^\nu$  dominates  $\nu$  and  $\delta^{-\nu}$ ; the second follows from (7.3.26) and  $\sigma/(b+\sigma) > 4b+5\sigma+6\varepsilon$ ; the third follows from the first two after further decreasing  $s_0$ .

### Step 5 (verification of the infinite iteration and nonsingularity inheritance).

The sequences (7.3.25) satisfy gates (G1)–(G9) for every  $\nu$  once the initial theorem parameter  $\mu_*$  is sufficiently small. The coordinate increments obey

$$\|\Phi_{\nu+1} - \text{id}\|_{C^{l_0}} \leq \frac{\mu_*^{1/4}}{2^{\nu+1}}, \tag{7.3.29}$$

and the normal-form increments obey analogous summable bounds, with the extra factors  $\gamma^a s_\nu$  for  $\Omega$  and  $\gamma^a$  for  $M$ . Hence the active nonsingularity alternative persists. If the full  $M$  is active, then  $\Omega_\nu \equiv \omega$  for all  $\nu$ .

Here are the gate checks. Because  $s_{\nu+1}/s_\nu = s_\nu^{b+\sigma} \rightarrow 0$ , (G2) holds. Since  $K_{\nu+1}E_\nu \rightarrow \infty$  supergeometrically relative to logarithmic powers, the elementary incomplete-gamma bound

$$\int_K^\infty t^N e^{-Et} dt \leq C_N E^{-(N+1)} (1 + (KE)^N) e^{-KE}$$

gives (G3). Equations (7.3.20), (7.3.27)–(7.3.28) imply (G5)–(G7). The exponent calculation (7.3.22) gives (G8). Since  $\mu_\nu K_{\nu+1}^{8m^2\tau+8m^2}$  decays faster than  $2^{-\nu}$ , while the successive margins  $\gamma_\nu - \gamma_{\nu+1}$  and their fixed powers are of order  $2^{-\nu}$ , (7.3.24) holds and gives

(G9). The summable bounds (7.3.23) and (7.3.28) give

$$\sum_{\nu \geq 0} \|M_{\nu+1} - M_\nu\|_{C^{l_0}} \leq C\gamma^a \mu_*^{1/4}, \quad \sum_{\nu \geq 0} \|\Omega_{\nu+1} - \Omega_\nu\|_{C^{l_0}} \leq C\gamma^{(1+\sigma_0)a+a} \mu_*^{1/4}. \quad (7.3.30)$$

Choose  $\mu_*$  so that the first sum is less than half the reciprocal of the initial inverse bound of the active matrix. The Neumann-series criterion then proves (G1) at every stage. The second sum, together with the corresponding bound for  $M$ , proves (G4). Thus all nine gates are verified inductively and the iteration is genuinely closed.

For the parameter sets, let  $\mathcal{O}_0 = \mathcal{O}$  and define  $\mathcal{O}_{\nu+1} \subset \mathcal{O}_\nu$  by imposing the three determinant inequalities with  $(\gamma_\nu, L_{qk}^\nu)$  on the new shell  $K_\nu < |k| \leq K_{\nu+1}$ . Gate (G9) guarantees that old inequalities persist, so this shell definition is equivalent to imposing all inequalities through  $K_{\nu+1}$ . Put

$$\mathcal{O}_\gamma = \bigcap_{\nu \geq 0} \mathcal{O}_\nu.$$

It remains to prove that this survivor is asymptotically full measure.

For  $q = 1, 2$  set  $g_{q,k}^\nu(\omega) = \det L_{qk}^\nu(\omega)$  and let  $d_1 = 2m$ ,  $d_2 = 4m^2$ . Choose a coordinate  $j$  for which  $|k_j| = \max_i |k_i|$ . By (7.3.30),  $\Omega_\nu = \text{id} + O(\gamma^{(1+\sigma_0)a+a} \mu_*^{1/4})$  in  $C^{l_0}$  in the drifting branch, and it is exactly the identity in the frequency-preserving branch;  $M_{22}^\nu - M_{22}^0 = O(\gamma^a \mu_*^{1/4})$ . Because the determinant is monic of degree  $d_q$  in the scalar  $2\pi i \langle k, \Omega_\nu \rangle$ , differentiating  $d_q$  times along  $\omega_j$  gives

$$\partial_{\omega_j}^{d_q} g_{q,k}^\nu = d_q! (2\pi i k_j)^{d_q} + O(|k|^{d_q-1} + \mu_*^{1/4} |k|^{d_q}). \quad (7.3.31)$$

Since  $d_q \leq 4m^2 \leq l_0$ , this derivative is available. After decreasing  $\mu_*$  and choosing a fixed  $k_0$ , uniformly in  $\nu$ ,

$$|\partial_{\omega_j}^{d_q} g_{q,k}^\nu| \geq c |k_j|^{d_q}, \quad |k| \geq k_0. \quad (7.3.32)$$

**Step 6 (determinant sublevel summation and asymptotically full survivor).** For every sufficiently large mode  $k$ , every scalar, first-normal and second-normal bad set at any KAM stage has measure

$$\leq C \frac{\gamma}{|k|^{\tau+1}}. \quad (7.3.33)$$

Consequently the union of all large-mode exclusions is  $O(\gamma)$  because  $\tau > n - 1$ . The finitely many modes  $|k| < k_0$  contribute  $o(1)$  as  $\gamma \downarrow 0$  by the initial almost-everywhere nonresonance and the  $\gamma$ -dependent convergence (7.3.30). Hence  $|\mathcal{O} \setminus \mathcal{O}_\gamma| \rightarrow 0$ .

We prove the child statement. If a one-variable  $C^p$  function satisfies  $|f^{(p)}| \geq A$  on an interval, the standard divided-difference/sublevel lemma implies

$$|\{t : |f(t)| \leq h\}| \leq C_p (h/A)^{1/p}. \quad (7.3.34)$$

Apply this on sections parallel to the  $\omega_j$  axis, with  $p = d_q$ , using (7.3.32) and the thresh-

old  $h = \gamma_\nu^{d_q} |k|^{-d_q \tau}$ . Since  $|k_j| \geq |k|/n$  up to the fixed choice of lattice norm,

$$\text{length} \left\{ |g_{q,k}^\nu| \leq \frac{\gamma_\nu^{d_q}}{|k|^{d_q \tau}} \right\} \leq C \frac{\gamma}{|k|^\tau |k_j|} \leq C \frac{\gamma}{|k|^{\tau+1}}. \quad (7.3.35)$$

Fubini gives the same estimate in the full parameter domain. For the scalar block the proof is the same with  $p = 1$ . The number of integer vectors of size about  $R$  is  $O(R^{n-1})$ , so

$$\sum_{k \neq 0} |k|^{-(\tau+1)} < \infty \quad \text{exactly when } \tau > n - 1.$$

This yields an  $O(\gamma)$  bound for all  $|k| \geq k_0$ .

For the finitely many  $0 < |k| < k_0$ , the initial scalar resonance set is a hyperplane and has measure zero. The  $L_2$  zero sets have measure zero by the weak second Melnikov hypothesis, and the  $L_1$  zero sets do as well by the  $2k, (a, a)$  argument following (7.3.4). Given  $\eta > 0$ , choose an open neighborhood  $U$  of this finite union with  $|U| < \eta$ . On the compact complement, all finitely many initial determinants have a positive minimum. By (7.3.30), their iterated counterparts converge uniformly to the initial ones as  $\gamma \downarrow 0$  because the total drift is  $O(\gamma^a \mu_*^{1/4})$ , while the determinant thresholds also tend to zero. Thus, for sufficiently small  $\gamma$ , no finite-mode exclusion occurs outside  $U$ . Hence

$$\limsup_{\gamma \downarrow 0} |\mathcal{O} \setminus \mathcal{O}_\gamma| \leq \eta.$$

Since  $\eta$  is arbitrary, the desired asymptotically full measure follows. Notice that this argument never chooses individual eigenvalue branches; only determinants and their degrees enter, so multiplicities remain allowed.

**Step 7 (Whitney convergence and Floquet reconstruction).** On  $\mathcal{O}_\gamma$ , the compositions of the analytic symplectic KAM maps converge on  $D(r_0/2, s_0/2)$ , together with parameter jets through order  $l_0 - 1$ , to a Whitney  $C^{l_0-1}$  family  $\Psi_\omega$ . The limiting Hamiltonian has no angle-dependent terms through quadratic order in  $(y, z)$ . Thus the zero section is an invariant torus and the transverse variational system has constant coefficients. In the  $M_{22}$  branch its tangential frequency is a nearby Diophantine  $\Omega^*(\omega)$ ; in the full- $M$  branch  $\Omega^*(\omega) = \omega$ .

By (7.3.29), the coordinate corrections form a Cauchy sequence in every phase derivative needed on  $D(r_0/2, s_0/2)$ ; the parameter derivatives through order  $l_0 - 1$  are summable as well. At each finite stage the maps and their jets are smooth on the closed good set  $\mathcal{O}_\nu$ . Extend the finite-stage jets to the original parameter domain with uniformly controlled Whitney norms. Their summable differences produce a compatible limiting Whitney jet, and the Whitney extension theorem gives the claimed  $C^{l_0-1}$  parameter family.

The new perturbation satisfies (7.3.12), while  $s_\nu, \mu_\nu \rightarrow 0$  supergeometrically. Hence every Taylor coefficient of  $P_\nu$  of transverse degree at most two tends uniformly to zero. Passing to the limit gives

$$H \circ \Psi_\omega = e^*(\omega) + \langle \Omega^*(\omega), y \rangle + \frac{1}{2} \left\langle \begin{pmatrix} y \\ z \end{pmatrix}, M^*(\omega) \begin{pmatrix} y \\ z \end{pmatrix} \right\rangle + h^*(y, z, \omega) + P^*(x, y, z, \omega), \quad (7.3.36)$$

where  $h^* = O(|(y, z)|^3)$  and

$$\partial_{(y,z)}^j P^*(x, 0, 0, \omega) = 0, \quad |j| \leq 2.$$

Thus  $y = z = 0$  is invariant and the quadratic transverse matrix is independent of  $x$ , which is precisely the Floquet normal form defined above. In the full- $M$  branch (7.3.10a) preserves  $\Omega_\nu \equiv \omega$  at every stage; in the  $M_{22}$  branch the limit of the summable corrections is the nearby Diophantine vector  $\Omega^*(\omega)$ . Step 6 gives the required asymptotically full Cantor-like parameter set, and Step 1 shows that no simple-spectrum hypothesis entered anywhere. All assertions of the theorem follow.  $\square$

#### Theorem K07.4. Rüssmann-type nondegeneracy

**Statement.** Let  $D \subset \mathbb{R}^m$  be compact and let  $\omega$  be  $C^q$  on a neighborhood of  $D$ , with  $q \geq 1$ . Assume there is  $\beta > 0$  such that for every  $\lambda \in D$  and every unit vector  $c \in \mathbb{R}^n$  there are an integer  $0 \leq j \leq q$  and a unit direction  $v \in \mathbb{R}^m$  for which

$$|\partial_v^j \langle c, \omega \rangle(\lambda)| \geq \beta.$$

Then there is a constant  $C$ , depending only on the fixed domain and uniform  $C^q$  data, such that for every  $\tau > nq - 1$  and all sufficiently small  $\gamma > 0$ ,

$$\text{Leb}\{\lambda \in D : \exists k \in \mathbb{Z}^n \setminus \{0\}, |\langle k, \omega(\lambda) \rangle| < \gamma |k|^{-\tau}\} \leq C\gamma^{1/q}.$$

Thus finite-order Rüssmann transversality gives a quantitative large-measure Diophantine parameter set.

#### Scope note

The finite-order sublevel estimate used here yields the stated threshold  $\tau > nq - 1$ ; the sharper  $\tau > n - 1$  regime requires stronger hypotheses.

*Proof.* **Lemma 7.17 (finite-order one-dimensional sublevel lemma).** Let  $J$  be a compact interval and let  $u \in C^q(J)$ . Suppose that at every  $x \in J$  some derivative  $u^{(j)}(x)$ ,  $0 \leq j \leq q$ , has modulus at least  $\beta$ , while all derivatives through order  $q$  are uniformly bounded. Then, for  $0 < \delta < 1$ ,

$$|\{x \in J : |u(x)| < \delta\}| \leq C\delta^{1/q}. \quad (7.4.1)$$

Cover  $J$  by finitely many neighborhoods on each of which one fixed derivative of order  $j \leq q$  remains at least  $\beta/2$ . The standard induction on  $j$  proves the sublevel estimate: for  $j = 0$  the set is empty for sufficiently small  $\delta$ ; for  $j \geq 1$ , split the interval at the finitely many points where  $|u^{(j-1)}|$  crosses a threshold  $\eta$ , use monotonicity between those points, and optimize  $\eta = \delta^{(j-1)/j}$ . This gives  $C_j \delta^{1/j} \leq C_j \delta^{1/q}$ . Compactness makes the constants uniform.

Apply the lemma on parameter lines in a finite coordinate cover of  $D$ . For  $k \neq 0$  put  $c = k/|k|$  and  $u_k(\lambda) = \langle c, \omega(\lambda) \rangle$ . The theorem's derivative hypothesis is invariant under

this normalization and hence

$$\{|\langle k, \omega \rangle| < \gamma |k|^{-\tau}\} = \{|u_k| < \gamma |k|^{-(\tau+1)}\}.$$

Fubini and Lemma 7.17 therefore give

$$\text{Leb}(E_k) \leq C\gamma^{1/q} |k|^{-(\tau+1)/q}. \quad (7.4.2)$$

**Lemma 7.18 (Rüssmann lattice summation).** Summing (7.4.2) over  $k \neq 0$  is finite when  $(\tau+1)/q > n$ , i.e.  $\tau > nq - 1$ , and yields the claimed  $C\gamma^{1/q}$  bad-set measure. Indeed the number of lattice vectors in the shell  $r \leq |k| < r+1$  is  $O(r^{n-1})$ , so the series is bounded by  $C\gamma^{1/q} \sum_{r \geq 1} r^{n-1-(\tau+1)/q}$ . This proves the theorem.  $\square$

### Theorem K07.5. KAM stability of elliptic fixed points

**Statement.** Let  $H$  be real analytic near an elliptic equilibrium with rationally independent linear frequency vector  $\omega_0$ . Write its formal Birkhoff normal form as  $N(I)$  and its formal frequency series as  $\Omega(I) = \nabla N(I)$ . Assume Rüssmann nondegeneracy in the formal sense of Definition K07.4: for every nonzero  $c \in \mathbb{R}^n$ , the scalar formal series  $\langle c, \Omega(I) \rangle$  is not identically zero. Then every sufficiently small neighborhood of the equilibrium contains a positive-Lebesgue-measure family of real-analytic invariant Lagrangian KAM tori accumulating the equilibrium. This is the Rüssmann-nondegenerate branch of the Eliasson-Fayad-Krikorian theorem and does not require a Diophantine hypothesis on  $\omega_0$ .

*Proof.* We write the elliptic equilibrium as the origin. After the fixed real linear symplectic normalization already used in the finite-order normal-form construction, the quadratic part is

$$H_2 = \sum_{j=1}^n \omega_{0,j} I_j, \quad I_j = \frac{q_j^2 + p_j^2}{2}. \quad (7.5.1)$$

The hypothesis that the coordinates of  $\omega_0$  are rationally independent means  $k \cdot \omega_0 \neq 0$  for every nonzero  $k \in \mathbb{Z}^n$ . Hence for every fixed finite order the nonresonance hypothesis in the finite-order Hamiltonian Birkhoff normal-form theorem is satisfied once the order is fixed. We shall exploit this freedom only after extracting the finite Rüssmann order that has to be protected.

**Lemma 7.19 (uniform finite jet from formal Rüssmann nondegeneracy).** There are an integer  $p \geq 1$ , a constant  $\beta > 0$ , and a finite Birkhoff polynomial  $N_*(I)$  such that, for every unit  $c \in \mathbb{R}^n$ , some multi-index  $\alpha$  with  $|\alpha| \leq p$  satisfies

$$|\partial_I^\alpha \langle c, \nabla N_* \rangle(0)| \geq 2\beta. \quad (7.5.2)$$

Consequently, for the fixed cube  $D = [1, 2]^n$ , all sufficiently small  $0 < \rho < \rho_0$  and every  $J \in D$ , there are  $0 \leq j \leq p$  and a unit direction  $v$  such that

$$|\partial_v^j \langle c, \nabla N_*(\rho J) \rangle| \geq \beta \rho^p. \quad (7.5.3)$$

For each unit  $c$ , formal Rüssmann nondegeneracy gives at least one Taylor coefficient



of  $\langle c, \Omega(I) \rangle$  which is nonzero. Let  $V_m$  be the set of unit  $c$  for which some coefficient of order at most  $m$  has modulus  $> 0$ . The  $V_m$  are open and increase to the compact unit sphere, so some  $V_p$  already covers it. The maximum of the finitely many absolute coefficients of order at most  $p$  is then a positive continuous function of  $c$ ; its minimum is positive. Taking  $2\beta$  below that minimum and taking  $N_*$  to contain all Birkhoff coefficients needed through order  $p+1$  proves (7.5.2).

For fixed  $c$ , choose an index  $\alpha$  realizing (7.5.2), polarize the corresponding symmetric derivative, and choose a unit direction  $v$  for which the directional derivative of the same order has modulus bounded below by a fixed dimension-dependent multiple of  $2\beta$ . Absorb that harmless factor into  $\beta$ . Continuity on the compact set  $D$  gives, after shrinking  $\rho_0$ , the same lower bound  $\beta$  at  $\rho J$ . Under the rescaling  $J \mapsto \rho J$ , a derivative of order  $j \leq p$  acquires the factor  $\rho^j$ ; since  $0 < \rho < 1$ ,  $\rho^j \geq \rho^p$ . This proves (7.5.3).

We next choose an integer  $Q$  much larger than  $p$ ; its exact lower bound will be fixed after the counterterm estimates have produced a finite loss exponent. Apply the finite-order Hamiltonian Birkhoff normal-form theorem through degree  $2Q+1$ . Because  $\omega_0$  is irrational, every resonant monomial through that order is an action monomial. Thus in one fixed analytic symplectic chart

$$H = N_Q(I) + R_Q(q, p), \quad R_Q = O(\|(q, p)\|^{2Q+2}), \quad (7.5.4)$$

where  $N_Q$  agrees with the formal Birkhoff normal form through action degree  $Q$  and in particular contains  $N_*$ . We choose  $N_*$  to contain every Birkhoff action term through degree  $p+1$ . Hence, uniformly for  $J \in D$ ,

$$\|\nabla N_Q(\rho J) - \nabla N_*(\rho J)\|_{C^p(J)} \leq C\rho^{p+1}. \quad (7.5.4a)$$

After reducing  $\rho_0$  once more, (7.5.3) therefore holds for  $\nu_\rho(J) = \nabla N_Q(\rho J)$  with lower constant, say,  $\frac{3}{4}\beta\rho^p$ . No Diophantine lower bound on  $\omega_0$  has been used: only finitely many nonzero divisors are inverted in this finite normalization.

Fix the positive action cube

$$A_\rho := \{I = \rho J : J \in D\}. \quad (7.5.5)$$

Because  $D$  stays a fixed distance from every coordinate hyperplane, the standard oscillator action-angle map is analytic on one complex neighborhood of  $D \times \mathbb{T}^n$  independent of  $\rho$  after the action rescaling  $I = \rho J$ . Writing  $I = \rho J$  and dividing the Hamiltonian by  $\rho$  gives the same Hamilton equations in the variables  $(\theta, J)$  with the canonical form; the rescaled Hamiltonian is

$$K_\rho(\theta, J) = h_\rho(J) + P_\rho(\theta, J), \quad h_\rho(J) := \rho^{-1}N_Q(\rho J), \quad (7.5.6)$$

and Taylor's theorem applied to (7.5.4) gives on a fixed complex neighborhood of  $D$

$$\|P_\rho\| \leq C_Q\rho^Q. \quad (7.5.7)$$

Moreover its integrable frequency map is

$$\nu_\rho(J) := \partial_J h_\rho(J) = \nabla N_Q(\rho J). \quad (7.5.8)$$

The decisive point is that we do not apply the twist-dependent theorem the twist-dependent analytic KAM theorem or the lower-dimensional theorem the lower-dimensional persistence theorem to  $\nu_\rho$ . Rüssmann nondegeneracy is weaker than Kolmogorov nondegeneracy, so such an invocation would be invalid. Instead we prove the counterterm KAM lemma that replaces the twist equation by an external frequency correction.

**Lemma 7.20 (twist-free analytic counterterm KAM lemma).** Fix  $p \geq 1$ ,  $\tau > np - 1$ , a compact real action cube  $D$  with a fixed complex neighborhood, and a compact frequency ball. There are finite exponents  $A_0, A_p$  and constants  $c, C > 0$  with the following property. Let

$$K(\theta, J) = h(J) + P(\theta, J), \quad \|P\| \leq \varepsilon, \quad (7.5.9)$$

be real analytic on the fixed domain. For  $0 < \gamma < 1$ , if  $\varepsilon \leq c\gamma^{A_0}$ , there are a  $C^p$  counterterm  $\lambda(a, \omega)$  and a  $C^p$  family of real-analytic exact symplectic maps  $\Phi_{a, \omega}$ , defined for  $a \in D$  and for all  $\omega$  in the fixed frequency ball, such that

$$\|\lambda(a, \omega) - (\omega - Dh(a))\|_{C^p} \leq C\gamma^{-A_p}\varepsilon. \quad (7.5.10)$$

Whenever  $\omega$  is  $(\gamma, \tau)$ -Diophantine, the Hamiltonian  $K + \langle \lambda(a, \omega), J - a \rangle$  has the invariant torus  $\Phi_{a, \omega}(\mathbb{T}^n \times \{a\})$  with frequency  $\omega$ . The maps satisfy the corresponding first-derivative estimate in phase and parameters,

$$\|\Phi_{a, \omega} - \text{id}\|_{C^1_{(\theta, J, a, \omega)}} \leq C\gamma^{-A_0}\varepsilon. \quad (7.5.11)$$

We prove the lemma, since it is exactly the specialist mechanism which was missing from the previous proof. Translate  $a$  to the zero action and write a current Hamiltonian in the form

$$G(\theta, I) = c_G + a_G(\theta) + \langle \omega + b_G(\theta), I \rangle + O(|I|^2), \quad (7.5.12)$$

where  $[a_G] = 0$  and square brackets denote angle average. The torus defect is  $(a_G, b_G)$ . For a Fourier mode  $k \neq 0$  introduce a smooth cutoff  $\chi_k(\omega)$  which equals 1 when  $|k \cdot \omega| \geq \gamma|k|^{-\tau}$  and equals 0 when  $|k \cdot \omega| \leq \frac{1}{2}\gamma|k|^{-\tau}$ , with derivatives of order  $r$  bounded by  $C_r\gamma^{-r}|k|^{r(\tau+1)}$ . Then

$$(\mathcal{L}_\omega f)_k := \frac{\chi_k(\omega)f_k}{2\pi i k \cdot \omega}, \quad (\mathcal{L}_\omega f)_0 = 0, \quad (7.5.13)$$

is a globally  $C^p$  operator in  $\omega$ . On the Diophantine set the cutoff is identically one in a neighborhood of every frequency, so  $\omega \cdot \partial_\theta \mathcal{L}_\omega f = f - [f]$ . Fourier decay and lattice-shell summation give, after a strip loss  $\sigma$ ,

$$\|\mathcal{L}_\omega f\|_{C^p} \leq C\gamma^{-(p+1)}\sigma^{-B_p}\|f\|_{C^p} \quad (7.5.14)$$

for a finite  $B_p = B_p(n, \tau, p)$ . Off the Diophantine set the unsolved cutoff part is smooth and is flat on the good set; it therefore has no effect when we later restrict to a good frequency.

Let  $A(\theta) = \partial_I^2 G(\theta, 0)$ . Solve

$$u = \mathcal{L}_\omega a_G. \quad (7.5.15)$$

Instead of using  $[A]^{-1}$  to translate the action, define the counterterm increment

$$\delta\lambda := -[b_G - A\partial_\theta u]. \quad (7.5.16)$$

Then  $b_G - A\partial_\theta u + \delta\lambda$  has zero average, so set

$$v = \mathcal{L}_\omega(b_G - A\partial_\theta u + \delta\lambda), \quad F(\theta, I) = u(\theta) + v(\theta) \cdot I. \quad (7.5.17)$$

The Hamiltonian time-one map generated by  $F$ , with the sign chosen according to the Hamiltonian-flow sign convention established earlier, is analytic and symplectic on a slightly smaller domain provided the defect is small. Equations (7.5.14)–(7.5.17) give

$$\|F\|_{C^p} + |\delta\lambda|_{C^p} \leq C\gamma^{-A_p}\sigma^{-B_p}\|(a_G, b_G)\|_{C^p} \quad (7.5.18)$$

for some finite  $A_p$ .

Taylor expansion of the jet map at  $I = 0$  now gives the Newton cancellation. The linear terms in the new scalar defect are  $a_G - \omega \cdot \partial_\theta u$ , and the linear terms in the frequency defect are  $b_G - A\partial_\theta u + \delta\lambda - \omega \cdot \partial_\theta v$ ; both vanish on the Diophantine set by (7.5.15)–(7.5.17). Every remaining contribution contains either two factors from  $(u, v, \delta\lambda)$  or one such factor multiplied by the old defect. Cauchy's estimate therefore yields on the next strip

$$\varepsilon_+ \leq C\gamma^{-A_0}\sigma^{-B_0}\varepsilon^2, \quad \|\Phi_+ - \text{id}\|_{C^1} + |\delta\lambda| \leq C\gamma^{-A_0}\sigma^{-B_0}\varepsilon. \quad (7.5.19)$$

The same calculation after differentiating at most  $p$  times in  $(a, \omega)$  uses the product rule and (7.5.14); it changes only the finite powers of  $\gamma^{-1}$  and  $\sigma^{-1}$ . Thus the  $C^p$  version of (7.5.19) holds with  $A_p, B_p$ .

Choose geometric strip losses and start below the resulting smallness threshold. The specialist estimates (7.5.19) are precisely the hypotheses of the abstract Newton-KAM convergence lemma the abstract analytic Newton-KAM convergence lemma. Hence the symplectic corrections and the counterterm increments are summable and the residual tends to zero supergeometrically. The differentiated recurrences have the same quadratic error multiplied only by fixed polynomial loss factors, so the same double-exponential majorant makes the first derivatives of the coordinate corrections summable; iterating this argument through order  $p$  gives a  $C^p$  parameter family. In particular the combined phase/parameter derivative obeys (7.5.11). Their limits give  $\Phi_{a,\omega}$  and  $\lambda(a, \omega)$ . At  $P = 0$  no iteration is needed and the exact counterterm is  $\omega - Dh(a)$ ; summing the perturbative increments proves (7.5.10)–(7.5.11). On a Diophantine frequency all cutoff remainders vanish, so the limiting zero section is genuinely invariant and has the prescribed frequency. This completes the proof of Lemma 7.20.

We apply Lemma 7.20 to  $K_\rho$ . Because  $D\nu_\rho$  need not be invertible, we recover the frequency by solving the counterterm equation, not by inverting  $\nu_\rho$ .

**Lemma 7.21 (zero-counterterm frequency recovery and preservation of the Rüssmann jet).** Let  $\gamma_\rho := \rho^{2p}$ . If  $Q$  is chosen so large that

$$Q - 2pA_p > p + 1, \quad (7.5.20)$$

then, for all sufficiently small  $\rho$ , there is a  $C^p$  map  $\Omega_\rho : D \rightarrow \mathbb{R}^n$  such that

$$\lambda_\rho(J, \Omega_\rho(J)) = 0, \quad \|\Omega_\rho - \nu_\rho\|_{C^p(D)} \leq C\rho^{Q-2pA_p} = o(\rho^p). \quad (7.5.21)$$

Consequently the finite-order Rüssmann bound inherited by  $\nu_\rho$  after (7.5.4a) persists for  $\Omega_\rho$  with lower constant  $\frac{1}{2}\beta\rho^p$ . Indeed, by (7.5.7), (7.5.10) and  $\gamma_\rho = \rho^{2p}$ ,

$$\lambda_\rho(J, \omega) = \omega - \nu_\rho(J) + E_\rho(J, \omega), \quad \|E_\rho\|_{C^p} \leq C\rho^{Q-2pA_p}. \quad (7.5.22)$$

The same bound for  $D_\omega E_\rho$  is  $< 1/2$  when  $\rho$  is small. Thus  $\omega \mapsto \nu_\rho(J) - E_\rho(J, \omega)$  is a contraction on a fixed small ball about  $\omega_0$ , uniformly in  $J$ . It has a unique fixed point  $\Omega_\rho(J)$ , which is  $C^p$  by differentiating the fixed-point equation. This proves (7.5.21). Condition (7.5.20) makes the  $C^p$  error  $o(\rho^p)$ , so the directional derivatives in (7.5.3) lose at most half of their lower bound.

**Lemma 7.22 (positive-measure survivor and phase-space transfer).** For all sufficiently small  $\rho$ , the set

$$D_\rho^* := \{J \in D : \Omega_\rho(J) \text{ is } (\gamma_\rho, \tau)\text{-Diophantine}\} \quad (7.5.23)$$

has positive Lebesgue measure; in fact  $\text{Leb}(D \setminus D_\rho^*) = O(\rho)$ . The union of the corresponding invariant tori of the original Hamiltonian has positive  $2n$ -dimensional Lebesgue measure in phase space. Apply the proof of the Rüssmann-type nondegeneracy theorem with its transversality constant replaced by  $\frac{1}{2}\beta\rho^p$ . For a resonance  $k \neq 0$ , normalization by  $c = k/|k|$  and the one-dimensional sublevel estimate give

$$\text{Leb}\{J : |k \cdot \Omega_\rho(J)| < \gamma_\rho |k|^{-\tau}\} \leq C \left( \frac{\gamma_\rho}{\rho^p} \right)^{1/p} |k|^{-(\tau+1)/p}. \quad (7.5.24)$$

Since  $\tau > np - 1$ , the lattice sum converges. With  $\gamma_\rho = \rho^{2p}$  this yields

$$\text{Leb}(D \setminus D_\rho^*) \leq C\rho. \quad (7.5.25)$$

Thus  $D_\rho^*$  has positive measure for every sufficiently small  $\rho$ . For  $J \in D_\rho^*$ , Lemma 7.20 and the zero-counterterm identity in Lemma 7.21 give an invariant analytic Lagrangian torus of the *original* Hamiltonian, with frequency  $\Omega_\rho(J)$ .

It remains to justify that a positive-measure set of action parameters really produces positive phase-space measure, rather than merely uncountably many zero-measure leaves. Let

$$\mathcal{W}_\rho(J, \theta) := \Phi_{J, \Omega_\rho(J)}(\theta, J) \quad (7.5.26)$$

in the rescaled action-angle chart. Equations (7.5.7), (7.5.11), and the choice of  $Q$  imply

$$\|D\mathcal{W}_\rho - I\| \leq C\gamma_\rho^{-A_0} \rho^Q = o(1). \quad (7.5.27)$$

Hence for small  $\rho$  the combined parameter-angle map is locally bi-Lipschitz with Jacobian bounded away from zero. Therefore  $\mathcal{W}_\rho(D_\rho^* \times \mathbb{T}^n)$  has positive Lebesgue measure in the rescaled action-angle phase space. The physical action rescaling  $I = \rho J$  multiplies this measure by the positive factor  $\rho^n$ , and the oscillator action-angle map is

canonical on the positive-action chart, so it preserves Lebesgue measure up to the fixed normalization of the angle period. Thus the corresponding union of invariant tori has positive phase-space Lebesgue measure. This proves Lemma 7.22.

Finally, the actions in  $A_\rho$  are  $O(\rho)$  and the Cartesian oscillator amplitudes are  $O(\sqrt{\rho})$ . Hence the entire set constructed above lies in an  $O(\sqrt{\rho})$  neighborhood of the elliptic equilibrium. Given any sufficiently small neighborhood, choose  $\rho$  small enough that the relevant complex-normal-form chart and the real KAM tori lie inside it. Lemma 7.22 then gives a positive-Lebesgue-measure family of real-analytic invariant Lagrangian KAM tori in that neighborhood. Since this can be done for  $\rho \downarrow 0$ , the tori accumulate the equilibrium.

The proof has used no Diophantine hypothesis on  $\omega_0$ : irrationality was needed only to carry the finite Birkhoff normalization to the single large order  $2Q+1$ , while Diophantine inequalities were imposed on the nearby corrected frequencies selected from the positive-measure Rüssmann survivor.  $\square$

## 7.4 Worked examples

### Worked example: one tangent frequency and one normal oscillator

Consider the integrable Hamiltonian

$$H_0(I, u, v) = I + \pi\sqrt{2}(u^2 + v^2),$$

with angle  $\theta \in \mathbb{T}$ , action  $I \in \mathbb{R}$ , and one transverse oscillator  $(u, v)$ . The torus  $\{I = 0, u = v = 0\}$  has tangential frequency  $\omega = 1$  and normal frequency  $\Omega = 2\pi\sqrt{2}$ . With the Fourier convention  $e^{2\pi i k \theta}$ , the order-two Melnikov divisors are

$$2\pi k + l\Omega = 2\pi(k + l\sqrt{2}), \quad k \in \mathbb{Z}, \quad |l| \leq 2.$$

For  $l = 0$  and  $k \neq 0$  the divisor is at least  $2\pi$ . For  $l = \pm 1$  the closest integer to  $\mp\sqrt{2}$  gives a positive gap  $2\pi(\sqrt{2} - 1)$ . For  $l = \pm 2$  the closest integer to  $\mp 2\sqrt{2}$  gives the smaller but still positive gap

$$2\pi(3 - 2\sqrt{2}) > 0.$$

Thus every divisor with  $|l| \leq 2$  is separated from zero in this model. A sufficiently small analytic perturbation therefore illustrates the nonresonant lower-dimensional setting: the tangential motion persists together with an elliptic transverse linear dynamics, after the frequency corrections allowed by the theorem.

#### Check your understanding

Replace the normal frequency by  $\Omega = 2\pi$ . Which pair  $(k, l)$  makes a first-normal Melnikov divisor vanish? Explain why the persistence theorem can no longer be invoked without an additional unfolding parameter or a resonant normal-form analysis.

**Boundary case: exact normal resonance**

For

$$H_0(I, u, v) = I + \pi(u^2 + v^2)$$

the normal frequency is  $2\pi$ , so the divisor corresponding to  $(k, l) = (-1, 1)$  is exactly zero. The failure is not a small quantitative loss: the relevant homological equation has a genuine kernel. One must retain the resonant term in the normal form or introduce a parameter that unfolds the resonance. Treating this example as if it satisfied the nonresonant theorem would erase the main geometric phenomenon of the chapter.

**Common pitfall**

Do not check only that the tangential vector  $\omega$  is Diophantine and then declare the lower-dimensional torus nonresonant. The normal directions introduce new divisors  $k \cdot \omega \pm \Omega_j$  and  $k \cdot \omega \pm \Omega_i \pm \Omega_j$ ; multiplicities may also force block rather than scalar inversion. Tangential arithmetic, Melnikov nonresonance, and parameter transversality play different roles.

## 7.5 Chapter summary

**What to remember**

- Lower-dimensional KAM theory adds a transverse spectral problem to the usual tangential small-divisor problem.
- First and second Melnikov conditions control divisors involving one or two normal frequencies, while parameter transversality controls the measure of the moving bad sets.
- Block formulations are essential when normal eigenvalues are multiple and scalar diagonalization is unavailable.
- Rüssmann nondegeneracy shows that an invertible Hessian is not the only route to a positive-measure KAM family near an elliptic equilibrium.

**Looking ahead**

K08 changes the question. Instead of asking whether a torus survives forever, it asks how far an arbitrary nearby orbit can drift when most invariant tori have been destroyed. Resonance geometry and normal forms then produce exponentially long stability rather than exact persistence.

**Check your understanding**

Separate the three sources of invertibility in lower-dimensional KAM. Which equations are controlled by the tangential Diophantine condition, which by

first/second Melnikov bounds in the normal directions, and where does parameter transversality enter only to estimate the excluded set?

## 7.6 Exercises and problems

### Exercise 7.1

**Difficulty.** Core.

**Related results.** Persistence of normally elliptic lower-dimensional tori; First and second Melnikov nonresonance conditions; Li-Yi Floquet persistence with multiple normal eigenvalues.

**Problem.** For the normally elliptic invariant circle, list tangential and normal small divisors and exhibit a normal resonance not excluded by tangential Diophantine control alone.

**Solution sketch.** Diagonalize the normal oscillator, Fourier-expand the homological equations, and compare  $k \cdot \omega$  with shifted Melnikov divisors.

### Exercise 7.2

**Difficulty.** Core.

**Related results.** Rüssmann-type nondegeneracy; KAM stability of elliptic fixed points.

**Problem.** In the boundary example, choose a vanishing Melnikov divisor while the tangential frequency remains Diophantine, and prove that forcing in that normal mode makes the homological equation unsolvable.

**Solution sketch.** Substitute the resonance into the Fourier coefficient equation and choose a nonzero right-hand side in the obstructed component.

### Exercise 7.3

**Difficulty.** Advanced.

**Related results.** Li-Yi Floquet persistence with multiple normal eigenvalues.

**Problem.** For multiple normal eigenvalues, derive the off-diagonal Floquet divisors  $k \cdot \omega + \lambda_i - \lambda_j$  and solve the truncated reducibility equation under a Melnikov bound.

**Solution sketch.** Compare Fourier matrix entries and divide coefficientwise, recording the small-divisor norm loss.

### Exercise 7.4

**Difficulty.** Advanced.

**Related results.** Rüssmann-type nondegeneracy.

**Problem.** Give a one-parameter frequency curve that is Rüssmann nondegenerate but cannot satisfy square Kolmogorov nondegeneracy, and prove its image is not contained in an affine resonance hyperplane.

**Solution sketch.** Use a curved map such as  $(s, 1 + s^2)$  and show no nonzero affine functional vanishes identically.

**Exercise 7.5**

**Difficulty.** Advanced.

**Related results.** Persistence of normally elliptic lower-dimensional tori; First and second Melnikov nonresonance conditions.

**Problem.** Estimate the parameter measure removed by first- and second-Melnikov resonances up to order  $K$  and explain how the surviving Cantor set feeds the lower-dimensional torus theorem.

**Solution sketch.** Estimate each resonant strip by transversality and sum tangential and normal families separately.



# Nekhoroshev Theory

## Why this chapter matters

KAM theory gives invariant tori on a large Cantor set, but it does not control every initial condition. Between the surviving tori lie resonant regions where slow drift is possible. Nekhoroshev theory addresses the complementary question: even when an orbit is not trapped on a KAM torus, can its actions remain close to their initial values for an astronomically long time?

For steep or quasi-convex integrable Hamiltonians the answer is yes. The mechanism is geometric. Action space is decomposed into resonance blocks. Inside a block one removes all Fourier modes that are nonresonant relative to its resonance lattice, leaving a resonant normal form plus an exponentially small remainder. Steepness prevents the frequency map from remaining too flat along the fast-drift planes, so an orbit cannot pass through a long chain of resonances without being confined.

**Prerequisites.** K03–K07, especially analytic averaging and resonant normal forms; Foundation F05, F07, F08, F12, and F15; elementary lattice geometry and analytic Hamiltonian estimates.

## Learning goals

- Distinguish exact KAM persistence from finite but exponentially long Nekhoroshev stability.
- Explain why action space is organized by resonance lattices of increasing rank.
- Use periodic averaging to remove nonresonant modes near a rational frequency.
- Explain how steepness prevents persistent drift along resonance planes.
- Recover the classical quasi-convex exponents  $a = b = 1/(2n)$  from the common cutoff scale.

### Conceptual roadmap

The proof is a local-to-global confinement argument:

resonance hierarchy  $\rightarrow$  normal form on each block  
 $\rightarrow$  small transverse drift  
 $\rightarrow$  steepness confinement  
 $\rightarrow$  exponentially long stability.

A rank- $j$  resonance lattice determines a fast-drift plane. The normal-form remainder drives only exponentially small motion transverse to the resonant dynamics. Steepness then confines each fast-plane excursion. With the extended-block convention used below, an early exit can only move to a strictly lower-rank layer. Since such descents can occur only finitely many times, the orbit stays in a controlled neighborhood for exponentially long times.

### Check your understanding

Why does Nekhoroshev theory not contradict Arnold diffusion? Compare the size of the allowed drift and the time scale in the stability theorem with the order-one drift sought in a diffusion theorem.

## 8.1 Definitions

### Intuition

**Steepness is a geometric anti-flatness condition.** Convexity is stronger than necessary. What the proof really needs is that, on every affine subspace transverse to the frequency direction, the restricted gradient cannot remain anomalously flat for too long. The steepness indices quantify how quickly a detectable change must occur. Quantitatively, on every  $j$ -plane  $\Lambda \subset \nabla h(I)^\perp$ , the largest radius  $\eta \leq \xi$  at which one tests the sphere in  $\Lambda$  must produce a projected frequency of size at least  $C_j \xi^{\alpha_j}$ .

**Resonance blocks are not arbitrary neighborhoods.** A block records which short integer relations  $k \cdot \omega(I) \approx 0$  are active and which are definitely absent. The extended block thickens the corresponding fast-drift plane just enough to run the normal form while keeping new independent resonances away.

**Quasi-convexity is curvature along an energy surface.** The Hessian need not be positive in the frequency direction. It must be uniformly positive only on vectors tangent to the regular energy surface, i.e. orthogonal to  $\nabla h$ .

### Definition K08.1. steep Hamiltonian

Let  $U \subset \mathbb{R}^n$  be bounded and let  $h \in C^1$  on a neighborhood of  $U$ , with frequency map  $\omega = \nabla h$ . We say that  $h$  is *steep on  $U$  with indices  $\alpha_1, \dots, \alpha_{n-1} \geq 1$ , coefficients*

$C_1, \dots, C_{n-1} > 0$ , and radius  $r > 0$  if

$$\inf_{I \in U} \|\omega(I)\| > 0$$

and, for every  $I \in U$ , every  $j$ -dimensional linear subspace  $\Lambda \subset \omega(I)^\perp$ ,  $1 \leq j \leq n-1$ , and every  $0 < \xi \leq r$  for which the indicated points stay in the neighborhood of  $U$ ,

$$\max_{0 \leq \eta \leq \xi} \min_{\substack{u \in \Lambda \\ \|u\| = \eta}} \|\pi_\Lambda \omega(I + u)\| \geq C_j \xi^{\alpha_j}.$$

Here  $\pi_\Lambda$  denotes Euclidean orthogonal projection onto  $\Lambda$ . The collection  $(\alpha_j, C_j, r)$  together with the positive frequency floor is called the *steepness data*.

### Definition K08.2. resonance block

A resonance block is a region of action space assigned to an integer resonance module where exactly those module resonances are small at the chosen hierarchy of thresholds.

### Definition K08.3. Nekhoroshev stability

Nekhoroshev stability is an estimate  $|I(t) - I(0)| \leq C \epsilon^b$  for  $|t| \leq \exp(C \epsilon^{-a})$  (or a specified variant) for sufficiently small perturbation  $\epsilon$ .

### Definition K08.4. quasi-convexity

Let  $D \subset \mathbb{R}^n$  and  $h \in C^2(D)$ . We call  $h$  *uniformly quasi-convex on  $D$  with constants  $l, m > 0$*  if

$$|\nabla h(I)| \geq l \quad \text{for every } I \in D,$$

and

$$v^T D^2 h(I) v \geq m |v|^2 \quad \text{whenever } I \in D \text{ and } v \cdot \nabla h(I) = 0.$$

Thus the Hessian is uniformly positive on the tangent hyperplane to each regular energy surface; no definiteness in the frequency direction itself is required.

### Definition K08.5. maximal $K$ -lattice and extended resonant block

For  $K \geq 1$  write  $\mathbb{Z}_K^n = \{k \in \mathbb{Z}^n : 0 < |k|_1 \leq K\}$ . A rank- $j$  sublattice  $\Lambda \subset \mathbb{Z}^n$  is a  $K$ -lattice if it has a basis in  $\mathbb{Z}_K^n$ , and it is maximal if it is not properly contained in another rank- $j$  sublattice of  $\mathbb{Z}^n$ . Write  $\langle \Lambda \rangle \subset \mathbb{R}^n$  for its real span,  $\pi_{\langle \Lambda \rangle}$  for orthogonal projection onto that span, and  $|\Lambda|$  for the Euclidean  $j$ -volume of a fundamental parallelepiped. Given a resonant zone  $Z_\Lambda \subset B = B(I_*, R)$ ,  $\rho = R/(2n)$ ,  $0 < \eta \leq \rho$ , and  $I \in Z_\Lambda \cap B_{-\rho}$ , define the resonant disc

$$D_{\Lambda, \eta}^\rho(I) = \left( \left[ \bigcup_{I' \in I + \langle \Lambda \rangle} B(I', \eta) \right] \cap Z_\Lambda \cap B_{-\rho} \right)_I,$$

where  $(\cdot)_I$  denotes the connected component containing  $I$ . If  $B_\Lambda$  is the rank- $j$  res-

onant block and  $r_\Lambda > 0$ , the extended resonant block is  $B_{\Lambda, r_\Lambda}^\rho = \bigcup_{I \in B_\Lambda \cap B_{-\rho}} D_{\Lambda, r_\Lambda}^\rho(I)$ . The affine plane  $I + \langle \Lambda \rangle$  is the fast-drift plane of  $\Lambda$  through  $I$ .

### Definition K08.6. periodic average along a rational linear flow

If  $l(I) = \omega \cdot I$  and  $T\omega \in \mathbb{Z}^n$ , define  $[f](I, \theta) = T^{-1} \int_0^T f(I, \theta + t\omega) dt$ ; this is the kernel projection of the homological operator  $\{\cdot, l\}$ .

### Definition K08.7. rank layer of the resonance hierarchy

If  $Z_j$  is the union of all rank- $j$  cutoff resonance zones, the rank layer  $B_j$  is the rank- $j$  region after higher-rank zones have been removed;  $B_0$  is the nonresonant complement.

### Definition K08.8. $\gamma$ - $K$ nonresonance modulo a lattice

A set is  $\gamma$ - $K$  nonresonant modulo  $\Lambda$  when  $|k \cdot \omega(I)| \geq \gamma$  for every point in the set and every  $0 < |k|_1 \leq K$  with  $k \notin \Lambda$ .

### Definition K08.9. one-periodic/two- $\pi$ Hamiltonian normalization bridge

The change  $\phi = 2\pi\theta$ ,  $J = I/(2\pi)$  preserves the symplectic form and converts the one-periodic K08 convention to the two- $\pi$  convention used by the K04 normal-form engine.

## 8.2 How the main proofs work

### Intuition

**Periodic averaging.** Near a rational linear flow, average the perturbation over one period. The nonaveraged part lies in the range of the homological operator, so one Lie transform removes it to first order. Iterating on a suitable scale gives an exponentially small remainder; the one-step lemma in this volume isolates the reusable analytic estimate.

**Resonance geometry.** Order resonance lattices by rank and choose thresholds that shrink with the Fourier cutoff  $K$ . A point in a rank- $j$  block is close to the span of exactly  $j$  independent short resonances but remains quantitatively nonresonant with every short vector outside that lattice. Extended blocks attached to distinct maximal lattices of the same rank are disjoint from the competing resonance zones.

**Steep Nekhoroshev theorem.** On each block, import the resonant normal-form engine from K04. The action derivative of the resonant normal form lies essentially in the resonance span, while the exponentially small remainder controls transverse leakage. Steepness bounds the diameter of a connected resonant excursion. With the block convention used in this book, an early exit can only enter a strictly lower-rank layer; after finitely many descents the orbit reaches a block

in which it is trapped for the common normal-form time.

**Quasi-convex specialization.** Energy conservation and curvature of the energy surface give a sharper confinement estimate. A single cutoff scale  $K \asymp \varepsilon^{-1/(2n)}$  balances the normal-form remainder and the geometric resonance widths, yielding both the radius exponent and the exponential time exponent  $1/(2n)$ .

### 8.3 Main results

#### Lemma K08.1. Periodic averaging lemma

**Statement.** Let  $l(I) = \omega \cdot I$  with  $T\omega \in \mathbb{Z}^n$  for some period  $T > 0$ , and let  $H = l + f$  be real analytic on a complex action-angle domain  $D_{r,s}$ . Write

$$[f](I, \theta) = \frac{1}{T} \int_0^T f(I, \theta + t\omega) dt.$$

There are dimensional constants  $c, C > 0$  such that, whenever  $\|f\|_{r,s} = \varepsilon$  and  $T\varepsilon/(rs) \leq c$ , there is a real-analytic canonical transformation  $\Phi$ , defined on a fixed smaller domain such as  $D_{r/2, s/2}$ , with  $\Phi - \text{Id} = O(T\varepsilon)$  in the corresponding scaled analytic norm, for which

$$H \circ \Phi = l + [f] + f_+, \quad \|f_+\|_{r/2, s/2} \leq C \frac{T\varepsilon^2}{rs}.$$

The generating function is obtained by integrating  $f - [f]$  along the periodic flow of  $l$ .

#### Scope note

This lemma is a single periodic-averaging step. Its iteration to exponentially small normal forms is a separate argument.

*Proof.* Put  $g = f - [f]$ . Because  $T\omega \in \mathbb{Z}^n$ , the linear flow  $\Phi_l^t(I, \theta) = (I, \theta + t\omega)$  is  $T$ -periodic on the torus and  $\int_0^T g \circ \Phi_l^t dt = 0$ . With the Poisson-bracket convention used in this book, set

$$\chi = \frac{1}{T} \int_0^T t g \circ \Phi_l^t dt. \quad (8.4.1)$$

Integration by parts, using periodicity, gives

$$\{l, \chi\} = g = f - [f]. \quad (8.4.2)$$

(The sign in (8.4.1) is chosen for this convention.) Also  $\|\chi\|_{r,s} \leq T\|g\|_{r,s} \leq 2T\varepsilon$ .

**Lemma 8.1 (one-period analytic averaging step).** If  $T\varepsilon/(rs)$  is sufficiently small,

the time-one map  $\Phi = X_\chi^1$  is canonical from  $D_{r/2, s/2}$  into  $D_{r, s}$  and

$$(l + f) \circ \Phi = l + [f] + f_+, \quad \|f_+\|_{r/2, s/2} \leq C \frac{T\varepsilon^2}{rs}, \quad (8.4.3)$$

while the scaled displacement is  $O(T\varepsilon)$ . Cauchy's inequalities give  $\|X_\chi\|_{r/2, s/2} \leq CT\varepsilon \max(r^{-1}, s^{-1})$  in unscaled coordinates, or  $CT\varepsilon/(rs)$  in the standard scaled vector-field norm. Thus the smallness hypothesis makes  $X_\chi^t$  stay in  $D_{r, s}$  for  $0 \leq t \leq 1$ . Under the definition of Poisson bracket above, differentiation along the  $\chi$ -flow is  $dF(X_\chi) = -\{F, \chi\} = \{\chi, F\}$ . Hence the integral Lie formula and (8.4.2) yield

$$(l + f) \circ \Phi = l + f + \{\chi, l + f\} + \int_0^1 (1 - t) \{\chi, \{\chi, l + f\}\} \circ X_\chi^t dt.$$

Since  $\{\chi, l\} = -\{l, \chi\} = -g$ , the first-order part is  $l + f - g = l + [f]$ . Each remaining bracket contains one factor  $f$  (or  $g$ ) and one factor  $\chi$ . Cauchy's inequalities on the half domain therefore bound their sum by  $CT\varepsilon^2/(rs)$ . The flow integral gives the displacement bound. This proves Lemma 8.1 and the theorem. No exponential iteration is claimed here.  $\square$

### Lemma K08.2. Resonance-block geometry lemma

**Statement.** Let  $n \geq 3$ , let  $B = B(I_*, R)$  be an action ball, and let  $h$  be a steep analytic integrable Hamiltonian on a neighborhood of  $B$ , with steepness indices  $\alpha_1, \dots, \alpha_{n-1}$ , coefficients  $C_1, \dots, C_{n-1}$ , steepness radius  $r_{st} > 0$ , and frequency map  $\omega = \partial_I h$ . Assume  $\omega_* := \inf_B \|\omega\| > 0$ , and let  $M > 0$  be a Lipschitz constant for  $\omega$  on  $B$ . Put  $b_\omega = \omega_*/(2\sqrt{2})$ ,  $p_j = \prod_{i=j}^{n-2} \alpha_i$  for  $1 \leq j \leq n-2$ ,  $p_{n-1} = p_n = 1$ ,  $q_j = np_j - j$ , and  $c_j = q_j - q_{j+1}$ . There are constants  $E > 1$ ,  $K_0 \geq 1$ , and  $\kappa_j > 0$ , depending only on the fixed steepness and frequency bounds, such that with  $A = 6E$ , every  $K \geq K_0$  has the following resonance geometry. For every maximal  $K$ -lattice  $\Lambda$  of rank  $j \in \{1, \dots, n-1\}$  set

$$\lambda_j = \frac{b_\omega}{(AK)^{q_j}}, \quad \delta_\Lambda = \frac{\lambda_j}{|\Lambda|}, \quad r_\Lambda = \frac{\delta_\Lambda}{M}, \quad \gamma_\Lambda = (EK)^{c_j} \delta_\Lambda, \quad r_j = \kappa_j \left( \frac{\lambda_j}{C_j} \right)^{1/\alpha_j}.$$

Define  $Z_\Lambda = \{I \in B : \|\pi_{\langle \Lambda \rangle} \omega(I)\| < \delta_\Lambda\}$ ,  $Z_j = \bigcup_{\text{rank } \Lambda = j} Z_\Lambda$ ,  $B_\Lambda = Z_\Lambda \setminus Z_{j+1}$ , and  $B_0 = B \setminus Z_1$ . With  $\rho = R/(2n)$  and the extended resonant blocks of Definition K08.5, every  $I \in B_\Lambda \cap B_{-\rho}$  and every  $I' \in D_{\Lambda, r_\Lambda}^\rho(I)$  satisfy the quantitative fast-disc confinement estimate

$$\|I' - I\| \leq \kappa_j \left( \frac{\delta_\Lambda}{C_j} \right)^{1/\alpha_j} \leq r_j.$$

Furthermore  $B_0$  is  $\lambda_1$ - $K$  nonresonant, while  $B_{\Lambda, r_\Lambda}^\rho$  is  $\gamma_\Lambda$ - $K$  nonresonant modulo  $\Lambda$  when  $j = n-1$  and  $(\gamma_\Lambda/2)$ - $K$  nonresonant modulo  $\Lambda$  when  $j \leq n-2$ ; explicitly,  $|k \cdot \omega(I)|$  is bounded below by the indicated constant for every  $k \in \mathbb{Z}_K^n \setminus \Lambda$ . Moreover, if  $\Lambda \neq \Lambda'$  are maximal  $K$ -lattices of the same rank  $j \geq 1$ , then  $B_{\Lambda, r_\Lambda}^\rho \cap Z_{\Lambda'} = \emptyset$ .

*Proof.* Write  $V_\Lambda = \langle \Lambda \rangle$ . The arithmetic input from the arithmetic-of-frequencies chapter in Foundation is the following elementary covolume fact: if  $\Lambda$  is a maximal rank- $j$  lattice

with a  $K$ -basis and  $k \in \mathbb{Z}_K^n \setminus \Lambda$ , then the maximal rank- $(j+1)$  lattice  $\Lambda_+$  generated by  $\Lambda$  and  $k$  satisfies

$$|\Lambda_+| \leq C_n K |\Lambda|, \quad (8.3.1)$$

and a component of  $k$  transverse to  $V_\Lambda$  has size bounded from below in terms of the reciprocal covolume of  $\Lambda_+$ . This is the determinant identity behind all divisor estimates below.

**Lemma 8.2 (joint resonance-width compatibility).** There exist  $E > 1$  and  $K_0$  depending only on the steepness constants,  $\omega_*$ ,  $M$ ,  $n$  and  $R$  such that, with  $A = 6E$ , all collar, steepness, divisor-transfer and rank-separation inequalities used below hold for every  $K \geq K_0$  simultaneously. The identities  $q_j = np_j - j$  and  $c_j = q_j - q_{j+1}$  imply

$$K^{c_j} \lambda_j = b_\omega E^{c_j} A^{-q_j} K^{-q_{j+1}}, \quad (8.3.2)$$

so each rank- $j$  enlarged width is a fixed  $E, A$  multiple of the next-rank scale, with no uncanceled dangerous power of  $K$ . The steepness radius condition compares a positive power of  $\delta_\Lambda$  with the next rank width; again the definition of  $p_j$  makes the powers match because  $p_j = \alpha_j p_{j+1}$ . There are only finitely many rank indices. Choose  $E$  large enough for all fixed numerical inequalities and then  $K_0$  large enough for the remaining lower-order/collar terms. This proves Lemma 8.2.

**Lemma 8.3 (steep resonant-disc confinement).** There are fixed  $\kappa_j > 0$  such that, for  $I \in B_\Lambda \cap B_{-\rho}$  and every  $I' \in D_{\Lambda, r_\Lambda}^\rho(I)$ ,

$$|I' - I| \leq \kappa_j \left( \frac{\delta_\Lambda}{C_j} \right)^{1/\alpha_j} \leq r_j := \kappa_j \left( \frac{\lambda_j}{C_j} \right)^{1/\alpha_j}. \quad (8.3.3)$$

At  $I \in Z_\Lambda$  put  $\omega_0 = \omega(I)$  and  $W = \pi_{\omega_0^\perp} V_\Lambda$ . We first prove the estimate on an unthickened connected component

$$\mathcal{C} = ((I + V_\Lambda) \cap Z_\Lambda(\tilde{\delta}) \cap B_{-\tilde{\rho}})_I, \quad Z_\Lambda(\tilde{\delta}) := \{J : \|\pi_{V_\Lambda} \omega(J)\| < \tilde{\delta}\},$$

where  $\tilde{\delta}$  obeys the fixed Lemma 8.2 bounds  $\tilde{\delta} \leq \min\{\omega_* \tilde{\rho}/r_{\text{st}}, \omega_*/\sqrt{2}\}$ . Take  $I' = I + u(1) \in \mathcal{C}$  and a continuous path  $u(t) \in V_\Lambda$  from 0 to  $u(1)$  staying in  $\mathcal{C}$ .

Because  $\|\pi_{V_\Lambda} \omega_0\| < \tilde{\delta} < \omega_*/\sqrt{2}$ , the map  $\pi_{\omega_0^\perp} : V_\Lambda \rightarrow W$  is injective, hence  $\dim W = j$ . Moreover

$$\|\pi_{\mathbb{R}\omega_0} u(t)\| \leq \frac{\tilde{\delta}}{\omega_*} \|u(t)\|. \quad (8.3.4)$$

Both  $\omega_0$  and  $\omega(I + u(t))$  make an angle  $O(\tilde{\delta}/\omega_*)$  with  $V_\Lambda^\perp$ ; projecting the two angle estimates onto  $W$  gives the uniform bound

$$\|\pi_W \omega(I + u(t))\| \leq C_0 \tilde{\delta}. \quad (8.3.5)$$

The collar condition and (8.3.4) ensure  $I + \pi_{\omega_0^\perp} u(t)$  remains in the domain where steepness is available. Put  $\xi = \|\pi_{\omega_0^\perp} u(1)\|$ . If  $\xi = 0$ , injectivity gives  $u(1) = 0$ . Otherwise the continuous projected path in  $W$  joins norm 0 to norm  $\xi$ , so it meets every intermediate sphere. Choose the radius  $\eta_*$  supplied by the max-min definition of steepness. At the

corresponding time  $t_*$ ,

$$C_j \xi^{\alpha_j} \leq \|\pi_W \omega(I + \pi_{\omega_0^\perp} u(t_*))\|.$$

The Lipschitz bound for  $\omega$ , (8.3.4), and (8.3.5) give

$$C_j \xi^{\alpha_j} \leq C_0 \tilde{\delta} + M \|\pi_{\mathbb{R}\omega_0} u(t_*)\| \leq C_1 \tilde{\delta}. \quad (8.3.6)$$

Finally (8.3.3) and  $\tilde{\delta}/\omega_* \leq 1/\sqrt{2}$  imply

$$\|I' - I\| \leq C_2 \left( \frac{\tilde{\delta}}{C_j} \right)^{1/\alpha_j}. \quad (8.3.7)$$

Now let  $I' \in D_{\Lambda, r_\Lambda}^\rho(I)$ . Along a path in the thickened component project each point to its fast-drift plane; the projection moves it by at most  $r_\Lambda$ . The projected path lies in an unthickened component with  $\tilde{\delta} = \delta_\Lambda + Mr_\Lambda = 2\delta_\Lambda$  and  $\tilde{\rho} = \rho - r_\Lambda$ . Lemma 8.2 is precisely the joint parameter choice ensuring the hypotheses used above for these two quantities. Hence

$$\|I' - I\| \leq r_\Lambda + C_2 \left( \frac{2\delta_\Lambda}{C_j} \right)^{1/\alpha_j} \leq \kappa_j \left( \frac{\delta_\Lambda}{C_j} \right)^{1/\alpha_j}, \quad (8.3.8)$$

after fixing  $\kappa_j$  large enough. Since  $\delta_\Lambda \leq \lambda_j$ , this is at most  $r_j = \kappa_j(\lambda_j/C_j)^{1/\alpha_j}$ , proving Lemma 8.3.

**Lemma 8.4 (small divisors on unextended blocks).** If  $I \in B_\Lambda$  and  $k \in \mathbb{Z}_K^n \setminus \Lambda$ , adjoining  $k$  to  $\Lambda$  and completing maximally to rank  $j+1$  gives

$$|k \cdot \omega(I)| \geq C^{-1} \gamma_\Lambda, \quad (8.3.9)$$

with the project constants chosen so that the right side is  $\gamma_\Lambda$  for  $j = n-1$  and at least  $\gamma_\Lambda$  before collar transfer for  $j \leq n-2$ . Suppose instead that the scalar product were below this threshold. Together with the  $j$  small projected components defining  $Z_\Lambda$ , the  $(j+1)$  independent integer vectors of  $\Lambda_+$  would make the orthogonal projection of  $\omega(I)$  onto  $V_{\Lambda_+}$  smaller than the next-rank threshold. Formula (8.3.1) converts the basis-coordinate bounds to Euclidean projection bounds. By (8.3.2) and Lemma 8.2 this puts  $I$  in  $Z_{\Lambda_+} \subset Z_{j+1}$ , contradicting  $I \in B_\Lambda$ . For  $j = n-1$ , adjoining  $k$  spans  $\mathbb{R}^n$ ; the same determinant estimate, together with  $\|\omega\| \geq \omega_*$ , gives the stronger terminal bound.

**Lemma 8.5 (small divisors on extended blocks).** For every  $I'$  in the extended block and every cutoff  $k \notin \Lambda$ ,

$$|k \cdot \omega(I')| \geq \frac{1}{2} \gamma_\Lambda \quad (8.3.10)$$

for  $j \leq n-2$ , and the full  $\gamma_\Lambda$  terminal margin can be reserved at rank  $n-1$  by the terminal choice in Lemma 8.2. Choose a point  $I$  in the corresponding unextended component given by Lemma 8.3. The collar and disc-diameter control imply  $|I' - I|$  is at most the transfer radius selected in Lemma 8.2. Hence  $|k \cdot (\omega(I') - \omega(I))| \leq KM|I' - I|$ . Lemma 8.2 makes this no larger than half the Lemma 8.4 margin (and smaller than the reserved terminal surplus at rank  $n-1$ ), proving (8.3.10) with the exact stated rank convention.



**Lemma 8.6 (same-rank nonoverlap).** If  $\Lambda \neq \Lambda'$  are maximal  $K$ -lattices of the same rank  $j$ , then  $B_{\Lambda, r_\Lambda}^\rho \cap Z_{\Lambda'} = \emptyset$ . Assume an intersection point exists. If  $\Lambda + \Lambda'$  has rank at least  $j + 1$ , choose  $k \in \Lambda' \setminus \Lambda$  from a  $K$ -basis after the standard maximal-lattice reduction. Membership in  $Z_{\Lambda'}$  makes the  $k$ -component small, while Lemma 8.5 says every such component outside  $\Lambda$  is bounded below; Lemma 8.2 makes the two inequalities incompatible. If the sum has rank  $j$ , maximality forces  $\Lambda = \Lambda'$ , contrary to assumption. For  $j = n - 1$  the frequency-floor version of the determinant estimate gives the same contradiction directly. This proves Lemma 8.6.

For  $B_0 = B \setminus Z_1$ , if a cutoff vector  $k$  satisfied  $|k \cdot \omega(I)| < \lambda_1$ , the maximal rank-one lattice generated by its primitive vector would put  $I$  in  $Z_1$ ; hence  $B_0$  is  $\lambda_1$ - $K$  nonresonant. Lemma 8.4–Lemma 8.5 give the stated nonresonance on every extended block, and Lemma 8.6 gives the final disjointness assertion. The theorem follows.  $\square$

### Theorem K08.3. Nekhoroshev theorem for steep Hamiltonians

**Statement.** Let  $n \geq 3$ , let  $U \subset \mathbb{R}^n$  be bounded, and let  $H(I, \varphi) = h(I) + \varepsilon f(I, \varphi)$  extend real-analytically to a fixed complex neighborhood of  $U \times \mathbb{T}^n$ . Assume  $h$  is steep on  $U$  with fixed steepness data and steepness indices  $\alpha_1, \dots, \alpha_{n-1}$ . Put  $p_1 = \alpha_1 \cdots \alpha_{n-2}$ ,  $a = (2np_1)^{-1}$ , and  $b = a/\alpha_{n-1}$ . Then there exist  $\varepsilon_0, R_0, T, c > 0$ , depending only on the fixed analytic and steepness data, such that, for  $0 < \varepsilon < \varepsilon_0$  and  $I_0 \in U_{-2R_0\varepsilon^b}$ , every solution satisfies  $\|I(t) - I_0\| \leq R_0\varepsilon^b$  whenever  $|t| \leq (T/\sqrt{\varepsilon}) \exp(c/\varepsilon^a)$ , where  $U_{-\rho} = \{I \in U : B(I, \rho) \subset U\}$ .

*Proof.* Choose once and for all a smaller complex action radius  $r > 0$  and angle strip width  $s > 0$  inside the fixed analytic neighborhood in the statement, and put

$$F = \|f\|_{r,s}, \quad \omega(I) = \nabla h(I), \quad \omega_* := \inf_U \|\omega(I)\| > 0,$$

where the last inequality is part of steepness. Let  $M$  bound  $D\omega$  on that fixed neighborhood. All constants below may depend on  $n, r, s, F, M$  and the fixed steepness data, but never on  $\varepsilon$  or on the particular orbit.

For  $1 \leq j \leq n - 2$  set

$$p_j = \prod_{i=j}^{n-2} \alpha_i, \quad p_{n-1} = p_n = 1, \quad q_j = np_j - j, \quad c_j = q_j - q_{j+1}. \quad (8.1.1)$$

Then  $c_j \geq 1$ ,  $q_{n-1} = 1$ , and

$$q_1 + 1 = np_1 = \frac{1}{2a}, \quad b = \frac{a}{\alpha_{n-1}}. \quad (8.1.2)$$

Take

$$K = \lfloor K_* \varepsilon^{-a} \rfloor, \quad R = 2R_0 \varepsilon^b, \quad \rho = \frac{R}{2n}, \quad (8.1.3)$$

where  $K_* > 0$  will be chosen small and  $R_0 > 0$  large, in that order. For small  $\varepsilon$  the integer  $K$  is larger than the cutoff  $K_0$  in the resonance-block geometry lemma.

Let  $E, A = 6E$  be the constants from that lemma and write

$$\lambda_j = \frac{b_\omega}{(AK)^{q_j}}, \quad \delta_\Lambda = \frac{\lambda_j}{|\Lambda|}, \quad r_\Lambda = \frac{\delta_\Lambda}{M}, \quad \gamma_\Lambda = (EK)^{c_j} \delta_\Lambda \quad (8.1.4)$$

for a maximal  $K$ -lattice  $\Lambda$  of rank  $j$ . We also use the geometric fast-disc radius

$$r_j = \kappa_j \left( \frac{\lambda_j}{C_j} \right)^{1/\alpha_j} \quad (8.1.5)$$

from the resonance-block geometry lemma and the rank-zero action radius

$$r_0 = \frac{\lambda_1}{2MK}. \quad (8.1.6)$$

The harmless fixed constants  $b_\omega, \kappa_j, C_j$  are those of the frozen steepness/resonance data.

**Lemma 8.7 (constructive parameter package and every normal-form gate).** After choosing  $E$  as in the resonance-block geometry lemma, one can choose  $K_* > 0$ , then  $R_0 > 0$ , and finally  $\varepsilon_0 > 0$  so that, for  $0 < \varepsilon < \varepsilon_0$ ,

$$r_0 \leq \rho, \quad r_\Lambda \leq \rho, \quad r_j \leq \rho \quad (1 \leq j \leq n-1), \quad (8.1.7)$$

and the resonant normal-form theorem from K04 applies on  $B_0$  with divisor  $\lambda_1$  and on every extended  $\Lambda$ -block with the the resonance-block geometry lemma divisor. In every chart the action coordinate error is at most a fixed small fraction of  $r_0$  or  $r_\Lambda$ , and the remainder has norm at most

$$\varepsilon F e^{-Ks/6}. \quad (8.1.8)$$

We prove the child statement rather than hiding it in the phrase “choose the usual Nekhoroshev parameters.” First, for  $j \leq n-2$ ,

$$\frac{q_j}{\alpha_j} = np_{j+1} - \frac{j}{\alpha_j} \geq n - j \geq 2,$$

while  $q_{n-1}/\alpha_{n-1} = 1/\alpha_{n-1}$ . Hence

$$r_j \leq CK^{-1/\alpha_{n-1}}, \quad r_\Lambda \leq CK^{-1/\alpha_{n-1}}, \quad (8.1.9)$$

and the same is true of  $r_0$  because  $q_1 + 1 \geq 1$ . Since  $K^{-1/\alpha_{n-1}} \asymp \varepsilon^b$ , increasing  $R_0$  gives (8.1.7). After reducing  $\varepsilon_0$  we also have  $R < r/2$  and  $Ks \geq 6$ .

Put  $e = \varepsilon F$ , the perturbation norm entering the resonant normal-form theorem from K04. In the nonresonant block take the K04 divisor  $\alpha = \lambda_1$  and an action collar  $\rho'_0 = r_0/2$ . The K04 smallness scale is, up to a fixed numerical factor,

$$\frac{\lambda_1 r_0}{K} = CK^{-2(q_1+1)} = CK^{-1/a}. \quad (8.1.10)$$

By (8.1.3),  $K^{-1/a} = K_*^{-1/a} \varepsilon$  up to the integer-rounding constant. Choosing  $K_*$  sufficiently small therefore makes  $e \leq \lambda_1 \rho'_0 / (256\xi K)$  for one fixed  $\xi > 1$ . The tube condition

$\rho'_0 \leq \lambda_1/(2\xi MK)$  follows after replacing  $r_0/2$  by a fixed smaller fraction if necessary.

Now let  $\Lambda$  have rank  $j$ . Use the divisor  $\alpha_\Lambda = \gamma_\Lambda/2$  for  $j \leq n-2$  and  $\alpha_\Lambda = \gamma_\Lambda$  for  $j = n-1$ , and take  $\rho'_\Lambda = r_\Lambda/2$ . Since  $Mr_\Lambda = \delta_\Lambda$ ,

$$\frac{\gamma_\Lambda}{MKr_\Lambda} = E^{c_j} K^{c_j-1}, \quad (8.1.11)$$

which is bounded below by an arbitrarily large fixed constant once  $E$  is chosen large. Thus the K04 complex-divisor tube condition holds uniformly in all ranks. For the perturbative smallness condition, use the elementary covolume bound  $|\Lambda| \leq C_n K^j$ . Relative to the rank-zero right-hand side, the resonant right-hand side satisfies

$$\frac{\gamma_\Lambda r_\Lambda/K}{\lambda_1 r_0/K} \geq CE^{c_j} K^{2(q_1-q_j-j)+c_j+1}. \quad (8.1.12)$$

The exponent is nonnegative: for  $j = 1$  it is  $c_1 - 1 \geq 0$ , and for  $j > 1$  it is larger because  $p_1 \geq p_j$ . Hence the already imposed rank-zero smallness implies every resonant K04 smallness inequality after one fixed adjustment of constants. the resonant normal-form theorem from K04 then gives (8.1.8), and its displacement bound together with the same smallness inequality makes each action coordinate error at most, say,  $r_0/64$  or  $r_\Lambda/64$ . This closes Lemma 8.7.

Define the local normal-form times

$$T_0 = c_0 \frac{sr_0}{\varepsilon F} e^{Ks/6}, \quad T_\Lambda = c_1 \frac{sr_\Lambda}{\varepsilon F} e^{Ks/6}, \quad T_{\text{exp}} = \min\{T_0, T_\Lambda\}, \quad (8.1.13)$$

where the second minimum ranges over all maximal cutoff lattices and the fixed positive  $c_0, c_1$  are chosen small enough for the Cauchy estimates below.

**Lemma 8.8 (nonresonant trap and resonant trap-or-descend).** Let  $B = B(I_*, R)$  be the ball used to construct the hierarchy.

1. If  $\bar{I}_0 \in B_0 \cap B_{-\rho}$ , then  $|I(t) - \bar{I}_0| \leq r_0$  for  $|t| \leq T_0$ .
2. If  $\bar{I}_0 \in B_\Lambda \cap B_{-(j+1)\rho}$  and  $\tau_e$  is the first exit time from  $B_{\Lambda, r_\Lambda}^\rho$ , then either  $|\tau_e| \geq T_\Lambda$  and  $|I(t) - \bar{I}_0| \leq r_j$  for  $|t| < T_\Lambda$ , or  $0 < |\tau_e| < T_\Lambda$  and

$$I(\tau_e) \in B_i \cap B_{-j\rho} \quad \text{for some } 0 \leq i \leq j-1. \quad (8.1.14)$$

For rank zero, the resonant normal-form theorem from K04 has  $\Lambda = \{0\}$ , so every retained cutoff Fourier mode is angle independent. Consequently the entire action velocity in normal-form coordinates comes from the exponentially small remainder. Cauchy's estimate on the reduced angle strip and (8.1.8) give

$$|J(t) - J(0)| \leq C \frac{\varepsilon F}{s} |t| e^{-Ks/6}. \quad (8.1.15)$$

The definition of  $T_0$ , together with the two  $r_0/64$  coordinate errors, gives the first assertion after decreasing  $c_0$ .

For a rank- $j$  lattice, the resonant normal-form theorem from K04 gives

$$H \circ \Psi = h + g + f^*, \quad \mathcal{P}_\Lambda g = g, \quad (8.1.16)$$

so  $-\partial_{\theta}g \in V_{\Lambda} := \langle \Lambda \rangle$  and only  $f^*$  changes the component of the normalized action perpendicular to  $V_{\Lambda}$ . Thus, for  $|t| < \min\{|\tau_e|, T_{\Lambda}\}$ ,

$$|\pi_{V_{\Lambda}^{\perp}}(J(t) - J(0))| \leq C \frac{\varepsilon F}{s} |t| e^{-Ks/6} \leq \frac{1}{4} r_{\Lambda}. \quad (8.1.17)$$

Adding the coordinate errors at time 0 and  $t$  yields the physical representation

$$I(t) = \bar{I}_0 + v(t) + d(t), \quad v(t) \in V_{\Lambda}, \quad |d(t)| < \frac{3}{4} r_{\Lambda}. \quad (8.1.18)$$

As long as the physical orbit has not left the extended block, it is connected to  $\bar{I}_0$  inside the thickened fast-drift plane occurring in the definition of  $D_{\Lambda, r_{\Lambda}}^{\rho}(\bar{I}_0)$ . The quantitative confinement conclusion of the resonance-block geometry lemma therefore gives

$$|I(t) - \bar{I}_0| \leq \kappa_j \left( \frac{\delta_{\Lambda}}{C_j} \right)^{1/\alpha_j} \leq r_j \leq \rho. \quad (8.1.19)$$

This proves the trapped alternative when  $|\tau_e| \geq T_{\Lambda}$ .

Assume instead  $0 < |\tau_e| < T_{\Lambda}$ . Formula (8.1.19) also holds at the exit by continuity, so the starting condition  $\bar{I}_0 \in B_{-(j+1)\rho}$  implies  $I(\tau_e) \in B_{-j\rho}$ . Because (8.1.18) still keeps the point within the  $r_{\Lambda}$  thickening of the fast plane and inside  $B_{-\rho}$ , the exit can only occur by leaving the resonance zone  $Z_{\Lambda}$ . the resonance-block geometry lemma excludes membership at the exit in any competing same-rank zone  $Z_{\Lambda'}$ . Therefore  $I(\tau_e) \notin Z_j$ . Since the rank layers partition the hierarchy, it belongs to some  $B_i$  with  $i < j$ . This proves (8.1.14) and Lemma 8.8. Notice that the rank direction here is not a convention: with the frozen definition  $B_{\Lambda} = Z_{\Lambda} \setminus Z_{j+1}$ , the chronological transition is a strict *descent*  $j \mapsto i < j$ .

**Lemma 8.9 (finite resonance trap and sharp time extraction).** Starting from the center of  $B(I_0, R)$ , repeated use of Lemma 8.8 produces at most  $n - 1$  strict rank descents. Every completed leg and the final trapped leg move the action by at most  $\rho$ , hence

$$|I(t) - I_0| \leq n\rho = \frac{R}{2} = R_0 \varepsilon^b \quad (8.1.20)$$

for  $|t| \leq T_{\text{exp}}$ . Moreover there are fixed  $T, c > 0$  such that

$$T_{\text{exp}} \geq \frac{T}{\sqrt{\varepsilon}} \exp\left(\frac{c}{\varepsilon^a}\right). \quad (8.1.21)$$

Take  $B = B(I_0, R)$ . Its center belongs to every collar needed for the first application of Lemma 8.8 because  $(j+1)\rho \leq n\rho = R/2$ . If the first block has positive rank and the orbit exits early, (8.1.14) puts the new starting point in a lower-rank block and in  $B_{-j\rho}$ . If its new rank is  $i < j$ , then  $j\rho \geq (i+1)\rho$ , so the collar hypothesis for the next application is automatic. Strict descent can occur at most  $n - 1$  times. Each resonant leg has size at most  $r_j \leq \rho$  and the final rank-zero or trapped resonant leg has size at most  $\rho$ . Summing gives (8.1.20). Early transition times only consume part of the requested interval; the final local trapping estimate is valid for a full time  $T_{\text{exp}}$  from its own starting time, so the concatenation controls the original orbit throughout  $[-T_{\text{exp}}, T_{\text{exp}}]$ .

It remains to prove the claimed time, rather than merely call it “exponential.” From (8.1.6) and (8.1.2),

$$r_0 = CK^{-(q_1+1)} = CK^{-1/(2a)}. \quad (8.1.22)$$

For a rank- $j$  maximal  $K$ -lattice, the  $K$ -basis covolume estimate  $|\Lambda| \leq C_n K^j$  gives

$$r_\Lambda = \frac{b_\omega}{MA^{q_j} K^{q_j} |\Lambda|} \geq CK^{-(q_j+j)} = CK^{-np_j} \geq CK^{-np_1} = CK^{-1/(2a)}. \quad (8.1.23)$$

Hence every algebraic prefactor in (8.1.13) is bounded below by

$$C\varepsilon^{-1} K^{-1/(2a)} \geq C'\varepsilon^{-1/2}. \quad (8.1.24)$$

Also  $Ks/6 \geq c\varepsilon^{-a}$  after decreasing  $c > 0$ . Equations (8.1.13), (8.1.22)–(8.1.24) prove (8.1.21).

Finally,  $I_0 \in U_{-2R_0\varepsilon^b} = U_{-R}$  implies  $B(I_0, R) \subset U$ . The stronger bound (8.1.20), together with the fixed small normal-form collars, therefore keeps every chart inside the analytic neighborhood and closes the bootstrap. Repeating the argument for the reversed flow gives the same estimate for negative time. This proves the theorem.  $\square$

#### Theorem K08.4. Quasi-convex Nekhoroshev theorem

**Statement.** Let  $D \Subset \mathbb{R}^n$  and let  $H(I, \theta) = h(I) + f(I, \theta)$  be real analytic on a fixed complex neighborhood of  $D \times \mathbb{T}^n$ . Assume uniform quasi-convexity: there are  $l, m > 0$  such that  $|\nabla h(I)| \geq l$  and  $v^T D^2 h(I) v \geq m|v|^2$  whenever  $v \perp \nabla h(I)$ . Then there are constants  $\varepsilon_0, C, c > 0$ , depending only on the fixed analytic domain and the quasi-convexity bounds, such that if  $\|f\| \leq \varepsilon \leq \varepsilon_0$  and  $\text{dist}(I(0), \partial D) \geq 2C\varepsilon^{1/(2n)}$ , every solution satisfies  $|I(t) - I(0)| \leq C\varepsilon^{1/(2n)}$  for all  $|t| \leq \exp(c\varepsilon^{-1/(2n)})$ . Thus the classical quasi-convex exponents are  $a = b = 1/(2n)$  after the fixed normalization.

*Proof.* Write  $\omega(I) = \nabla h(I)$  and let  $e = \|f\| \leq \varepsilon$ . Fix a slightly smaller real neighborhood  $D'$  of  $D$  inside the given complex domain. All constants below depend only on that domain, on analytic bounds for  $h$ , and on the quasi-convexity constants  $l, m$ .

**Lemma 8.10 (uniform quasi-convexity implies index-one steepness).** There are  $r_* > 0$  and  $C_* > 0$  such that, for every  $I \in D$ , every  $j$ -plane  $\Lambda \subset \omega(I)^\perp$ ,  $1 \leq j \leq n-1$ , and  $0 < \xi \leq r_*$ ,

$$\max_{0 \leq \eta \leq \xi} \min_{\substack{u \in \Lambda \\ |u| = \eta}} \|\pi_\Lambda \omega(I + u)\| \geq C_* \xi. \quad (8.2.1)$$

Thus  $h$  is steep with all indices  $\alpha_j = 1$  and uniform steepness data. Indeed, because  $\overline{D}$  is compactly contained in  $D'$ , uniform continuity of  $D^2 h$  gives  $r_* > 0$  such that

$$\|D^2 h(I + u) - D^2 h(I)\| \leq \frac{m}{2} \quad (I \in D, |u| \leq r_*). \quad (8.2.2)$$

Fix  $u = \eta v \in \Lambda$  with  $|v| = 1$ . Since  $v \perp \omega(I)$ , quasi-convexity at the base point gives  $v^T D^2 h(I) v \geq m$ . Therefore, for  $0 \leq t \leq \eta \leq r_*$ ,

$$v^T D^2 h(I + tv) v \geq \frac{m}{2}.$$

Using  $v \in \Lambda$  and the fundamental theorem of calculus,

$$\|\pi_\Lambda \omega(I + \eta v)\| \geq v \cdot \pi_\Lambda(\omega(I + \eta v) - \omega(I)) = \int_0^\eta v^T D^2 h(I + tv) v dt \geq \frac{m}{2} \eta. \quad (8.2.3)$$

This holds for every  $u$  on the  $\eta$ -sphere in  $\Lambda$ , so taking the minimum and then  $\eta = \xi$  proves (8.2.1) with  $C_* = m/2$ . In particular the frequency floor  $|\omega| \geq l$  is exactly the other clause in the definition of steepness.

We first assume  $n \geq 3$ . Under  $\alpha_1 = \dots = \alpha_{n-1} = 1$ , the the resonance-block geometry lemma exponents reduce to

$$p_j = 1, \quad q_j = n - j, \quad c_j = 1. \quad (8.2.4)$$

Choose

$$K = \lfloor K_* \varepsilon^{-1/(2n)} \rfloor, \quad R = \frac{2R_*}{K}, \quad \rho = \frac{R}{2n}, \quad (8.2.5)$$

with  $K_* > 0$  small and  $R_* > 0$  large but fixed. The hierarchy of the resonance-block geometry lemma then has

$$\lambda_j \asymp K^{-(n-j)}, \quad r_j \asymp \lambda_j \leq \frac{C}{K}, \quad r_\Lambda = \frac{\lambda_j}{M|\Lambda|} \leq \frac{C}{K}. \quad (8.2.6)$$

Increase  $R_*$  so that all radii in (8.2.6) are at most  $\rho$ .

**Lemma 8.11 (single-scale normal-form and time gates).** For the choice (8.2.5), after fixing  $K_*$  and reducing  $\varepsilon_0$ , the resonant normal-form theorem from K04 applies on the nonresonant block and every extended resonant block. Its action coordinate error is a fixed small fraction of the corresponding collar and its remainder is at most  $ee^{-cK}$ . The local trapping time in every rank is at least

$$c\varepsilon^{-1/2}e^{cK}. \quad (8.2.7)$$

Here the decisive power calculation is elementary. At rank zero,  $q_1 = n - 1$  and  $r_0 \asymp \lambda_1/K \asymp K^{-n}$ , so the K04 smallness right-hand side is

$$\frac{\lambda_1 r_0}{K} \asymp K^{-2n} \asymp \varepsilon. \quad (8.2.8)$$

Choosing  $K_*$  small gives the required numerical slack. For a rank- $j$  lattice,  $|\Lambda| \leq C_n K^j$ , while  $\gamma_\Lambda = (EK)\delta_\Lambda$ . Hence  $\gamma_\Lambda r_\Lambda/K$  is no smaller, up to fixed constants, than the rank-zero scale (8.2.8); this is the specialization of the exponent inequality  $2(q_1 - q_j - j) + c_j + 1 \geq 0$ . The complex-tube condition follows from  $\gamma_\Lambda/(MKr_\Lambda) = EK^0 = E$ , after choosing  $E$  large in the resonance-block geometry lemma. Thus the resonant normal-form theorem from K04 applies uniformly. Its remainder is  $ee^{-Ks/6}$  on a fixed fraction of the original strip.

The smallest action collar occurring in the local time estimate is of order  $K^{-n}$ : indeed  $r_0 \asymp K^{-n}$ , while

$$r_\Lambda \geq cK^{-q_j-j} = cK^{-n}. \quad (8.2.9)$$

Cauchy's estimate therefore allows each chart to be followed for at least  $cK^{-n}e^{-1}e^{cK}$ .

Since  $K^{-n} \asymp \varepsilon^{1/2}$  and  $e \leq \varepsilon$ , this is bounded below by (8.2.7). Lemma 8.11 follows.

**Lemma 8.12 (quasi-convex trap with the correct rank direction).** While an orbit remains in a rank- $j$  extended block, the resonant normal form and the resonance-block geometry lemma confine it to action diameter  $C/K$ . If it exits before the local time (8.2.7), it enters a block of strictly smaller rank. The same-rank nonoverlap in the resonance-block geometry lemma rules out lateral switching between distinct maximal rank- $j$  lattices. The proof is the same short dynamical calculation that is needed in the steep case, and we include it here so K08-002 does not depend on K08-001. In K04 coordinates,

$$H \circ \Psi = h + g + f^*, \quad \mathcal{P}_\Lambda g = g.$$

Hence  $-\partial_\theta g \in \langle \Lambda \rangle$  and the action component orthogonal to  $\langle \Lambda \rangle$  changes only through  $f^*$ . Up to the local time in Lemma 8.11, Cauchy's estimate makes that transverse drift less than  $r_\Lambda/4$ ; the two near-identity coordinate errors can be made less than  $r_\Lambda/8$  in total. Thus

$$I(t) = \bar{I}_0 + v(t) + d(t), \quad v(t) \in \langle \Lambda \rangle, \quad |d(t)| < \frac{3}{4}r_\Lambda. \quad (8.2.10)$$

the resonance-block geometry lemma then confines the connected fast-plane excursion to  $r_j \leq C/K$ . If the extended block is left early, the exit point is still inside the next spatial collar. It has left  $Z_\Lambda$  and, by same-rank nonoverlap, belongs to no competing rank- $j$  zone. With the frozen layer convention  $B_\Lambda = Z_\Lambda \setminus Z_{j+1}$  it therefore lies in a layer of rank  $i < j$ . This proves Lemma 8.12.

**Lemma 8.13 (finite hierarchy and global quasi-convex estimate).** There are at most  $n - 1$  strict rank descents. On the common time  $\exp(cK)$  their total action displacement, including the final trapped leg and all coordinate changes, is at most  $C/K$ . Therefore

$$|I(t) - I(0)| \leq C\varepsilon^{1/(2n)}, \quad |t| \leq \exp(c\varepsilon^{-1/(2n)}). \quad (8.2.11)$$

Center the hierarchy ball at the current initial action and use the collar bookkeeping exactly as in Lemma 8.12. Every early exit lowers an integer rank, so there can be at most  $n - 1$  of them. Each leg has diameter  $C/K$ ; because the number of legs is bounded solely by  $n$ , their sum is still  $C/K$ . Lemma 8.11 gives a common local time larger than  $c\varepsilon^{-1/2}e^{cK}$ , and hence in particular larger than  $e^{c'K}$  after reducing  $c'$ . Formula (8.2.5) now gives (8.2.11).

It remains to close the low-dimensional cases, because the resonance-block geometry lemma was frozen for  $n \geq 3$ . If  $n = 1$ , energy conservation gives, as long as the orbit stays in the fixed neighborhood,

$$|h(I(t)) - h(I(0))| \leq |f(I(t), \theta(t))| + |f(I(0), \theta(0))| \leq 2\varepsilon.$$

Since  $|h'| \geq l$  has constant sign on each connected component, the mean value theorem yields  $|I(t) - I(0)| \leq 2\varepsilon/l$  for all such times. For small  $\varepsilon$  this is stronger than the required  $C\varepsilon^{1/2}$  and closes the bootstrap for all time.

For  $n = 2$  there is only the rank-one resonance layer. Repeat the proof of the resonance-block geometry lemma in this single layer: a primitive cutoff vector generates  $\Lambda$ , adjoining any cutoff  $k \notin \Lambda$  gives a full-rank lattice; the determinant/covolume estimate together with  $|\omega| \geq l$  supplies the nonresonance outside  $\Lambda$ , the index-one

estimate (8.2.1) confines the fast disc to  $C/K$ , and the same determinant argument gives same-rank nonoverlap. With  $K \asymp \varepsilon^{-1/4}$  one has  $\lambda_1 \asymp K^{-1}$ ,  $r_0 \asymp K^{-2}$  and the K04 smallness scale  $\lambda_1 r_0 / K \asymp K^{-4} \asymp \varepsilon$ . The rank-one resonant gate has the same power. Thus the preceding normal-form calculation applies with at most one transition, giving radius  $C/K = C\varepsilon^{1/4}$  and time at least  $\exp(cK) = \exp(c\varepsilon^{-1/4})$ .

Finally choose the theorem's constant  $C$  larger than the constants above and reduce  $\varepsilon_0$  so that  $C\varepsilon^{1/(2n)}$  lies inside the fixed real neighborhood. The assumed boundary distance  $2C\varepsilon^{1/(2n)}$  then keeps the entire bootstrap inside  $D$ . Apply the same argument to the reversed flow for negative time. This proves the theorem in every dimension covered by its statement.  $\square$

## 8.4 Worked examples

### Worked example: a quadratic quasi-convex Hamiltonian

Let

$$h(I) = \frac{1}{2}|I|^2, \quad I \in D := \{I \in \mathbb{R}^n : 1 \leq |I| \leq 2\}.$$

Then

$$\nabla h(I) = I, \quad D^2 h(I) = \text{Id}.$$

On  $D$  we have  $|\nabla h(I)| \geq 1$ . If  $v \cdot \nabla h(I) = 0$ , then

$$v^T D^2 h(I) v = |v|^2.$$

Thus  $h$  is uniformly quasi-convex with  $l = m = 1$  on this annular action domain. For an analytic perturbation  $H = h + f$  with  $\|f\| \leq \varepsilon$ , the quasi-convex Nekhoroshev theorem gives constants  $C, c > 0$  such that, away from the boundary,

$$|I(t) - I(0)| \leq C\varepsilon^{1/(2n)}$$

for all

$$|t| \leq \exp\left(c\varepsilon^{-1/(2n)}\right).$$

The exclusion of a neighborhood of  $I = 0$  is not cosmetic: the definition used in this book of uniform quasi-convexity includes the lower bound  $|\nabla h| \geq l > 0$ , which fails at the origin.

### Boundary case: loss of steepness

Take in two degrees of freedom

$$h(I_1, I_2) = I_1.$$

Its frequency is constant,  $\nabla h = (1, 0)$ , and the Hessian vanishes identically. Along the transverse affine direction  $I_2$  there is no restoring curvature or higher-order steepness. A perturbation depending on the conjugate angle can therefore pro-



duce secular motion in  $I_2$  without the geometric obstruction used by Nekhoroshev theory. The example pinpoints the missing hypothesis: no theorem in this chapter justifies exponential confinement for such a flat integrable part.

#### Check your understanding

For  $n = 3$  and  $\varepsilon = 10^{-12}$ , compute the formal scale  $\varepsilon^{1/(2n)}$ . Compare it with  $\varepsilon$  itself. Why is the Nekhoroshev confinement radius small but much larger than the perturbation amplitude?

#### Common pitfall

An exponentially small normal-form remainder is not, by itself, a Nekhoroshev theorem. One still needs a geometric hypothesis such as steepness or quasi-convexity to prevent long motion inside the resonant part of the normal form. Conversely, steepness without a sufficiently accurate resonant normal form does not yield the exponential time scale.

## 8.5 Chapter summary

#### What to remember

- Nekhoroshev theory replaces exact invariant-torus persistence by a global stability estimate for arbitrary initial data in a protected action region.
- Resonance lattices organize action space and determine the fast-drift planes on which resonant dynamics may occur.
- Resonant normal forms make the transverse remainder exponentially small, while steepness controls the diameter of resonant excursions.
- Quasi-convexity is the cleanest special case and yields the classical exponents  $a = b = 1/(2n)$ .

#### Looking ahead

K09 studies the complementary instability mechanism. Exponential stability does not say that order-one drift is impossible; it says such drift must exploit special resonant channels and usually occurs on very long time scales. Arnold diffusion theory builds those channels explicitly using homoclinic intersections, scattering maps, and shadowing.

#### Check your understanding

Explain the division of labor in Nekhoroshev theory: what does the resonant normal form make small, what does steepness or quasi-convexity control geometrically, and why are both ingredients needed to obtain an exponentially long

confinement estimate?

## 8.6 Exercises and problems

### Exercise 8.1

**Difficulty.** Core.

**Related results.** Nekhoroshev theorem for steep Hamiltonians; Quasi-convex Nekhoroshev theorem; Resonance-block geometry lemma.

**Problem.** For  $h(I) = |I|^2/2$ , verify quasi-convexity/steepness, describe a rank- $r$  exact resonance plane, and explain geometrically why energy conservation restricts transverse drift.

**Solution sketch.** Compute  $\nabla h$  and  $D^2h$ , then intersect resonance planes with an energy sphere.

### Exercise 8.2

**Difficulty.** Core.

**Related results.** Periodic averaging lemma.

**Problem.** For  $h(I_1, I_2) = I_1$  perturbed by  $\varepsilon \cos \theta_2$ , find an exact orbit with order-one  $I_2$ -drift on time  $O(\varepsilon^{-1})$ , demonstrating failure of Nekhoroshev stability without steepness.

**Solution sketch.** Choose  $\theta_2$  with nonzero sine and integrate Hamilton's equations explicitly.

### Exercise 8.3

**Difficulty.** Advanced.

**Related results.** Resonance-block geometry lemma.

**Problem.** Prove that the exact resonance set of a rank- $r$  primitive module is a codimension- $r$  linear subspace and give a quantitative distance estimate using a generator matrix.

**Solution sketch.** Use rank-nullity and the smallest nonzero singular value of the generator matrix.

### Exercise 8.4

**Difficulty.** Advanced.

**Related results.** Periodic averaging lemma.

**Problem.** For a periodic unperturbed flow, construct the periodic average and an explicit zero-mean solution of the homological equation by time integration along one orbit.

**Solution sketch.** Integrate  $f - \langle f \rangle$  over the period and differentiate the proposed primitive along the flow.

**Exercise 8.5****Difficulty.** Advanced.**Related results.** Nekhoroshev theorem for steep Hamiltonians; Quasi-convex Nekhoroshev theorem.**Problem.** Using the resonance-block decomposition and quasi-convexity, explain why a hypothetical large drift must enter higher-rank resonance and why the energy geometry then restricts further drift.**Solution sketch.** Follow the first block transition and intersect allowed resonant directions with the quasi-convex energy hypersurface.**Exercise 8.6****Difficulty.** Lab.**Related results.** None.**Problem.** Compare numerically action drift in the quasi-convex model and in the explicit non-steep model across decreasing  $\varepsilon$ , recording the first time a fixed action threshold is crossed.**Solution sketch.** Use a symplectic integrator, validate the non-steep  $O(\varepsilon^{-1})$  benchmark, and check time-step refinement.

# Arnold Diffusion

## Why this chapter matters

K08 showed that near-integrable Hamiltonian systems can be extraordinarily stable. Arnold diffusion shows that this stability is not absolute. In systems with at least three effective degrees of freedom, carefully organized resonant and homoclinic mechanisms can move an action variable by an order-one amount even when the perturbation is arbitrarily small.

The central idea is to separate two motions. The *inner dynamics* moves along a normally hyperbolic invariant manifold. A transverse homoclinic excursion leaves that manifold and returns to it, producing a *scattering map* between asymptotic base points. Alternating inner motion with scattering steps creates a transition chain. A shadowing theorem then turns this designed pseudo-orbit into a true orbit of the full system.

**Prerequisites.** K08 and Foundation F10–F15; stable/unstable manifolds and normal hyperbolicity; basic symplectic maps; elementary symbolic dynamics and topological degree.

## Learning goals

- Compute a Melnikov function and explain why a simple zero gives a transverse homoclinic intersection.
- Derive symbolic dynamics from one transverse homoclinic point via the Smale–Birkhoff mechanism.
- Define wave maps, homoclinic channels, scattering maps, and full transition maps without conflating them.
- Explain how correctly aligned windows turn a finite pseudo-orbit into a true orbit.
- Trace the explicit a-priori unstable model from Melnikov splitting to an order-one drift in the rotator action.

**Conceptual roadmap**

The diffusion mechanism is

Melnikov splitting  $\rightarrow$  transverse homoclinic channel  
 $\rightarrow$  scattering map  
 $\rightarrow$  transition chain  
 $\rightarrow$  shadowed true orbit.

The Smale–Birkhoff theorem explains why transverse homoclinic geometry already creates symbolic complexity. For diffusion, however, symbolic chaos alone is not enough: the scattering step must have a controlled component in the desired action direction, and the inner dynamics must position the orbit so that successive scattering steps can be concatenated.

**Check your understanding**

Why is a scattering map not itself a finite-time map of the full system? What extra entrance and exit iterates are needed before one obtains the transition map used in a window-shadowing argument?

## 9.1 Definitions

**Intuition**

**Melnikov theory measures first-order splitting.** For an unperturbed homoclinic orbit, the stable and unstable manifolds coincide. The Melnikov integral computes their leading-order signed separation on a transverse section. A simple zero says the separation changes sign with nonzero derivative, which is precisely the first-order signal of a transverse intersection.

**The scattering map is asymptotic data.** A point on a homoclinic channel has a past asymptotic base point and a future asymptotic base point on the normally hyperbolic invariant manifold. The scattering map sends the former to the latter. It does not include the finite inner iterates needed to enter and leave the channel neighborhood.

**Correctly aligned windows are a topological shadowing device.** A window has designated exit and entry directions. Correct alignment says, through a homotopy with nonzero Brouwer degree, that the image crosses the target window in the unstable directions while avoiding its forbidden entry boundary. Chaining such crossings forces an actual orbit through the entire finite itinerary.

**Definition K09.1. Melnikov function**

The Melnikov function is the first-order signed displacement between stable and unstable manifolds along an unperturbed homoclinic/heteroclinic orbit, expressed by the specified convergent integral/sum.

**Definition K09.2. wave maps and homoclinic channel**

Let  $N$  be a normally hyperbolic invariant manifold for a diffeomorphism  $F$ . Each  $z \in W^s(N)$  belongs to a unique stable leaf  $W^s(x)$ , and each  $z \in W^u(N)$  belongs to a unique unstable leaf  $W^u(y)$ . The *wave maps* are

$$\Omega^+(z) = x \quad (z \in W^s(N)), \quad \Omega^-(z) = y \quad (z \in W^u(N)).$$

Equivalently,  $F^n z$  approaches  $F^n \Omega^+(z)$  as  $n \rightarrow +\infty$ , while  $F^{-n} z$  approaches  $F^{-n} \Omega^-(z)$  as  $n \rightarrow +\infty$ . A *homoclinic channel* is a submanifold  $\Gamma \subset W^u(N) \cap W^s(N)$  on which both restricted wave maps are diffeomorphisms onto open subsets of  $N$ .

**Definition K09.3. scattering map**

For a homoclinic channel  $\Gamma$ , put  $U_\Gamma^- = \Omega^-(\Gamma)$  and  $U_\Gamma^+ = \Omega^+(\Gamma)$ . The *scattering map* is the map

$$S_\Gamma : U_\Gamma^- \longrightarrow U_\Gamma^+, \quad S_\Gamma = \Omega^+ \circ (\Omega^-|_\Gamma)^{-1}.$$

It records the change between the asymptotic past and future base points on the normally hyperbolic manifold caused by one homoclinic excursion.

**Definition K09.4. transition chain**

A *transition chain* is a finite or bi-infinite ordered family of invariant subsets of the normally hyperbolic manifold, together with specified inner iterates and homoclinic-channel transition steps from each member to the next. To be a shadowing datum rather than only a formal list, the chosen steps must be represented by windows satisfying the window and correct-alignment conditions defined immediately below.

**Definition K09.5. window and correct alignment**

Let  $m_1 + m_2 = m$ . An  $(m_1, m_2)$ -window in an  $m$ -manifold  $M$  is a compact set  $W = \chi([0, 1]^{m_1} \times [0, 1]^{m_2})$ , where  $\chi$  is a homeomorphism from a neighborhood of the product cube to an open subset of  $M$ . Its exit and entry sets are

$$W^{\text{exit}} = \chi(\partial[0, 1]^{m_1} \times [0, 1]^{m_2}), \quad W^{\text{entry}} = \chi([0, 1]^{m_1} \times \partial[0, 1]^{m_2}).$$

For windows  $W_1, W_2$  with charts  $\chi_1, \chi_2$  and a continuous map  $G$  with  $G(\text{im } \chi_1) \subset \text{im } \chi_2$ , write  $G_\chi = \chi_2^{-1} \circ G \circ \chi_1$ . We say  $W_1$  is correctly aligned with  $W_2$  under  $G$  if there is a homotopy  $h : [0, 1] \times [0, 1]^{m_1} \times [0, 1]^{m_2} \rightarrow \mathbb{R}^{m_1} \times \mathbb{R}^{m_2}$  with  $h_0 = G_\chi$ , such that throughout the homotopy the exit set of  $W_1$  avoids the target cube and the whole source cube avoids the entry set of  $W_2$ , and, for some  $y_0 \in [0, 1]^{m_2}$ , the unstable-coordinate map  $A_{y_0}(x) = \pi_{m_1} h_1(x, y_0)$  sends  $\partial[0, 1]^{m_1}$  outside  $[0, 1]^{m_1}$  and has nonzero Brouwer degree at the chosen interior target point. Here the Brouwer degree is the integer-valued signed covering number for regular target values, extended to all admissible target values by homotopy invariance.

**Definition K09.6. transition map associated with a homoclinic channel**

Let  $F$  have a normally hyperbolic invariant manifold  $N$  and a homoclinic channel  $\Gamma$ . Fix a local return section and integers  $n_-, n_+ \geq 0$  for which an orbit segment entering a neighborhood of  $\Gamma$  and then returning to the section is defined. The corresponding *transition map* is the local map obtained by composing the entrance inner iterate, the channel excursion represented by  $S_\Gamma$ , and the exit inner iterate. In coordinates on  $N$  it has the form

$$T_\Gamma = F^{n_+}|_N \circ S_\Gamma \circ F^{n_-}|_N$$

whenever all three factors are defined; on the full return section it denotes the canonically lifted local map obtained by following the same finite orbit segment. Correct alignment is always checked for this full local transition map, not merely for the formal scattering map.

**Definition K09.7. transverse homoclinic point and full two-sided shift**

Let  $f$  be a  $C^1$  diffeomorphism and let  $\mathcal{O}$  be a hyperbolic periodic orbit. A point  $q \notin \mathcal{O}$  is a *homoclinic point* of  $\mathcal{O}$  if  $q \in W^s(\mathcal{O}) \cap W^u(\mathcal{O})$ ; it is *transverse* if

$$T_q W^s(\mathcal{O}) + T_q W^u(\mathcal{O}) = T_q M.$$

For a finite alphabet  $\mathcal{A}$  with at least two symbols, the *full two-sided shift* is the homeomorphism

$$s : \mathcal{A}^{\mathbb{Z}} \rightarrow \mathcal{A}^{\mathbb{Z}}, \quad (s\omega)_j = \omega_{j+1},$$

with the product topology and with no transition restrictions between symbols. A map on an invariant set is *topologically conjugate* to this shift if a homeomorphism intertwines the two dynamics.

## 9.2 How the main proofs work

### Intuition

**Melnikov simple-zero criterion.** Choose a transverse section through the unperturbed homoclinic orbit and express the stable/unstable splitting by an energy-type scalar function. Differentiate the perturbed flow with respect to the perturbation parameter. The first variation is the Melnikov integral. A simple zero and the implicit-function theorem produce a nearby transverse intersection.

**Smale-Birkhoff.** A transverse homoclinic point is stretched forward along the unstable direction and backward along the stable direction. The inclination lemma produces two crossing strips. Iterating the strip construction and intersecting nested forward/backward cylinders yields one point for every bi-infinite symbolic word, giving a horseshoe and a full shift.

**Scattering map.** The stable and unstable wave maps are smooth on a homoclinic channel. Inverting the past wave map and composing with the future wave

map gives the scattering map. Symplecticity follows by comparing the symplectic form along asymptotic orbits; exactness follows from the corresponding action primitive identity.

**Transition-chain shadowing.** Replace the formal scattering step by the full finite transition map, including inner entrance and exit iterates. Correctly aligned windows have a nonzero topological degree in the expanding directions. A finite intersection argument then supplies one orbit whose successive returns meet every prescribed window.

**Explicit Arnold diffusion.** In the two-harmonic a-priori unstable model, first compute the Melnikov potential and obtain local scattering maps with positive action drift. The delicate point is that the true inner motion on the normally hyperbolic cylinder also changes  $I$ , especially near the two first-order resonances. A resonant normal form shows that inner and outer directions are transverse except on simple tangency curves crossed by the outer motion, so one can build a finite alternating inner/scattering pseudo-orbit across the prescribed action range. Recurrence inside a KAM-bounded annulus then turns each asymptotic scattering arrow into a finite transition map, and the window-shadowing theorem produces one genuine diffusion orbit.

### 9.3 Main results

#### Proposition K09.1. Melnikov simple-zero criterion

**Statement.** Let  $\dot{z} = J\nabla H_0(z) + \varepsilon g(z, t, \varepsilon)$  be a  $C^r$  ( $r \geq 2$ ) planar perturbation, with  $g$   $T$ -periodic in  $t$ . Suppose  $z_*$  is a hyperbolic saddle equilibrium of the unperturbed Hamiltonian vector field and  $z_h(t)$  is a homoclinic solution satisfying  $z_h(t) \rightarrow z_*$  as  $t \rightarrow \pm\infty$ . Assume the Melnikov integral  $M(\tau) = \int_{-\infty}^{\infty} \nabla H_0(z_h(t)) \cdot g(z_h(t), t + \tau, 0) dt$  converges, is differentiable in  $\tau$ , and has a simple zero  $M(\tau_0) = 0$ ,  $M'(\tau_0) \neq 0$ . Then, for every sufficiently small nonzero  $\varepsilon$ , the hyperbolic  $T$ -periodic orbit continuing  $z_*$  has stable and unstable manifolds that intersect transversely near the phase  $\tau_0$ .

*Proof.* Adjoin the phase  $s = t \pmod{T}$  and consider the autonomous system on  $\mathbb{R}^2 \times \mathbb{T}_T$ . At  $\varepsilon = 0$  the orbit  $\{z_*\} \times \mathbb{T}_T$  is hyperbolic in the two transverse directions. Hyperbolic persistence therefore gives, for  $|\varepsilon|$  small, a unique nearby hyperbolic  $T$ -periodic orbit  $\gamma_\varepsilon$  and local stable and unstable manifolds depending  $C^{r-1}$  on  $\varepsilon$ .

Fix a point  $z_h(0)$  of the unperturbed homoclinic orbit away from the saddle and a one-dimensional section  $\Sigma$  transverse to  $z_h$  there. For phases  $\tau$  in a small interval and  $|\varepsilon|$  small, the global unstable and stable branches issuing from  $\gamma_\varepsilon$  meet the translated section  $\Sigma_\tau$  in points  $z_\varepsilon^u(\tau)$  and  $z_\varepsilon^s(\tau)$ . The construction by the graph transform gives these intersection points  $C^1$  in  $(\tau, \varepsilon)$  and

$$z_0^u(\tau) = z_0^s(\tau) = z_h(0). \quad (9.1.1)$$

Choose the signed transverse coordinate on  $\Sigma_\tau$  to be the  $H_0$  coordinate, divided by the



nonzero derivative of  $H_0$  in the transverse direction. It is therefore enough to compute

$$\Delta(\tau, \varepsilon) := H_0(z_\varepsilon^u(\tau)) - H_0(z_\varepsilon^s(\tau)). \quad (9.1.2)$$

Let  $z_\varepsilon^u(t; \tau)$  be the unstable trajectory ending at  $z_\varepsilon^u(\tau)$  at section time zero and asymptotic to  $\gamma_\varepsilon$  as  $t \rightarrow -\infty$ ; define  $z_\varepsilon^s$  analogously for  $t \rightarrow +\infty$ . Along every perturbed trajectory,

$$\frac{d}{dt}H_0(z(t)) = \nabla H_0(z(t)) \cdot (J\nabla H_0(z(t)) + \varepsilon g(z(t), t + \tau, \varepsilon)) = \varepsilon \nabla H_0(z(t)) \cdot g(z(t), t + \tau, \varepsilon). \quad (9.1.3)$$

Integrating (9.1.3) backward on the unstable branch and forward on the stable branch and subtracting gives

$$\Delta(\tau, \varepsilon) = \varepsilon \int_{-\infty}^{\infty} \nabla H_0(z_h(t)) \cdot g(z_h(t), t + \tau, 0) dt + R(\tau, \varepsilon). \quad (9.1.4)$$

The stated convergence hypothesis makes the displayed integral  $M(\tau)$ . We record why the remainder is genuinely  $C^1$  and quadratic. Hyperbolicity implies exponential convergence of the stable/unstable parameterizations to the periodic orbit; on every fixed compact time interval their difference from  $z_h$  is  $O(\varepsilon)$  in  $C^1$ , while on the tails the exponential weights are uniform. Splitting the integral at  $|t| = L$ , applying the mean-value theorem on  $[-L, L]$ , and then letting  $L$  be fixed large enough to control the tails gives, uniformly for  $\tau$  near  $\tau_0$ ,

$$|R(\tau, \varepsilon)| + |\partial_\tau R(\tau, \varepsilon)| \leq C\varepsilon^2. \quad (9.1.5)$$

The contribution from the perturbed periodic orbit cancels in the stable minus unstable difference, so no constant tail term remains.

**Lemma 9.1 (the  $C^1$  Melnikov splitting expansion).** Equations (9.1.4)–(9.1.5) give

$$\Delta(\tau, \varepsilon) = \varepsilon M(\tau) + O_{C^1}(\varepsilon^2). \quad (9.1.6)$$

For  $\varepsilon \neq 0$  put  $G(\tau, \varepsilon) = \Delta(\tau, \varepsilon)/\varepsilon$  and extend it at  $\varepsilon = 0$  by  $G(\tau, 0) = M(\tau)$ . By (9.1.6),  $G$  is  $C^1$  near  $(\tau_0, 0)$  and

$$G(\tau_0, 0) = 0, \quad \partial_\tau G(\tau_0, 0) = M'(\tau_0) \neq 0.$$

The implicit-function theorem produces a unique phase  $\tau(\varepsilon) = \tau_0 + O(\varepsilon)$  for which  $\Delta(\tau(\varepsilon), \varepsilon) = 0$ . Thus the stable and unstable branches meet. Moreover

$$\partial_\tau \Delta(\tau(\varepsilon), \varepsilon) = \varepsilon M'(\tau_0) + O(\varepsilon^2) \neq 0$$

for all sufficiently small nonzero  $\varepsilon$ . Because the signed  $H_0$ -coordinate is transverse to the homoclinic direction, this nonzero derivative is exactly the nonzero angle condition for the two one-dimensional branches in the Poincaré section. Their intersection is therefore transverse, and lifting from the section gives the claimed transverse intersection of the stable and unstable manifolds of  $\gamma_\varepsilon$ .  $\square$

**Theorem K09.2. Smale-Birkhoff homoclinic theorem**

**Statement.** Let  $f \in \text{Diff}^r(M)$ ,  $r \geq 1$ , and let  $q$  be a transverse homoclinic point of a hyperbolic periodic orbit. Then there exist a Cantor set  $\Lambda \subset M$  containing  $q$  and an integer  $m \geq 1$  such that  $f^m(\Lambda) = \Lambda$  and  $f^m|_\Lambda$  is topologically conjugate to a full shift automorphism on a finite alphabet with at least two symbols.

*Proof.* Let the hyperbolic periodic orbit have least period  $p$  and put  $g = f^p$ . One point  $a$  on that orbit is a hyperbolic fixed point of  $g$ . Replacing the given homoclinic point by a suitable  $f$ -iterate, if necessary, we may work with a transverse point  $q_0 \in W_g^s(a) \cap W_g^u(a)$  lying in an arbitrarily small local product neighborhood of  $a$ . At the end we pull the construction back by the same  $f$ -iterate to recover the original  $q$ .

Choose coordinates  $(u, s) \in \mathbb{R}^{d_u} \times \mathbb{R}^{d_s}$  in a rectangle

$$R = D^u \times D^s \quad (9.2.1)$$

centered at  $a$  so that the local unstable and stable manifolds are respectively  $D^u \times \{0\}$  and  $\{0\} \times D^s$ . Shrinking  $R$ , the derivative of  $g$  preserves strictly invariant unstable and stable cone fields. Thus there are  $\lambda > 1$  and  $0 < \vartheta < 1$  such that vectors in the unstable cone are expanded by at least  $\lambda$  under  $Dg$  and vectors in the stable cone are expanded by at least  $\lambda$  under  $Dg^{-1}$ .

The transverse homoclinic point supplies a finite excursion from the local unstable manifold back to the local stable manifold. The inclination ( $\lambda$ -) lemma says that forward iterates of a small disk transverse to  $W_g^s(a)$  become  $C^1$ -close to a local unstable disk, while backward iterates of a disk transverse to  $W_g^u(a)$  become  $C^1$ -close to a local stable disk. Hence, for some sufficiently large  $N$ , the return map

$$G = g^N \quad (9.2.2)$$

has two pairwise disjoint compact vertical strips  $H_0, H_1 \subset R$  such that  $G(H_i) \cap R$  is a horizontal strip crossing all of  $R$ , and on every strip the same stable/unstable cone estimates hold. The strip  $H_0$  is the local return staying near  $a$ ;  $H_1$  follows the chosen homoclinic excursion through  $q_0$ . By taking the entrance and exit pieces of  $H_1$  through  $q_0$ , the itinerary that is 1 at time zero and 0 for sufficiently large positive and negative times has intersection point  $q_0$ .

**Lemma 9.2 (two-strip horseshoe from one transverse homoclinic point).** There exist  $N$  and two disjoint strips  $H_0, H_1$  satisfying the crossing and uniform cone conditions above, with the homoclinic strip anchored on the prescribed transverse branch. This is the direct inclination-lemma consequence used below.

For a bi-infinite word  $\sigma = (\sigma_j)_{j \in \mathbb{Z}} \in \{0, 1\}^{\mathbb{Z}}$  define

$$R_k(\sigma) = \bigcap_{|j| \leq k} G^{-j}(H_{\sigma_j}). \quad (9.2.3)$$

The crossing property implies inductively that  $R_k(\sigma)$  is nonempty and compact. Its width in the unstable direction is at most  $C\lambda^{-k}$  when measured backward, and its

width in the stable direction is at most  $C\lambda^{-k}$  when measured forward. Consequently

$$\text{diam } R_k(\sigma) \leq C'\lambda^{-k}. \quad (9.2.4)$$

The nested intersection therefore consists of exactly one point; write it  $\pi(\sigma)$ . Put

$$\Lambda_0 = \pi(\{0, 1\}^{\mathbb{Z}}).$$

Then  $G\Lambda_0 = \Lambda_0$  and, directly from (9.2.3),

$$G \circ \pi = \pi \circ s, \quad (9.2.5)$$

where  $s$  is the full two-sided shift. Estimate (9.2.4) shows that agreement of two words on  $[-k, k]$  forces their images to be  $O(\lambda^{-k})$  apart, so  $\pi$  is continuous. Distinct words have a first coordinate at which their orbit points lie in the disjoint strips, so  $\pi$  is injective. A continuous bijection from the compact shift space to the Hausdorff set  $\Lambda_0$  is a homeomorphism. In particular  $\Lambda_0$  is a Cantor set and (9.2.5) is a topological conjugacy.

**Lemma 9.3 (nested symbolic coding).** The strip data of Lemma 9.2 produce a unique point for every bi-infinite word; the coding map is a homeomorphism and conjugates  $G$  to the full two-shift.

By the anchoring choice in Lemma 9.2,  $q_0 \in \Lambda_0$ . Suppose  $q = f^{-j}q_0$ . Set  $\Lambda = f^{-j}\Lambda_0$ . Since  $G = f^{pN}$  commutes with  $f^{-j}$ ,

$$G\Lambda = \Lambda, \quad G|_{\Lambda} = f^{-j} \circ (G|_{\Lambda_0}) \circ f^j,$$

so the same full-shift conjugacy holds on  $\Lambda$ , and  $q \in \Lambda$ . Taking  $m = pN$  proves the theorem (with a finite alphabet of two symbols, which is all the statement requires).  $\square$

### Proposition K09.3. Scattering-map construction

**Statement.** Let  $f$  be a  $C^r$  diffeomorphism, let  $\Lambda$  be a normally hyperbolic invariant manifold, and let  $\Gamma \subset W^u(\Lambda) \cap W^s(\Lambda)$  be a homoclinic channel on which the wave maps  $\Omega^- : \Gamma \rightarrow \Lambda$  and  $\Omega^+ : \Gamma \rightarrow \Lambda$  are diffeomorphisms onto open subsets. Then  $S_{\Gamma} = \Omega^+ \circ (\Omega^-)^{-1}$  is a  $C^{r-1}$  diffeomorphism between those open subsets (with the regularity allowed by the wave maps). If  $f$  is symplectic,  $\Lambda$  is symplectic, and the channel satisfies the standard transversality hypotheses, then  $S_{\Gamma}$  is symplectic for the induced form; in the exact-symplectic setting it is exact symplectic.

*Proof.* Write

$$U_{\Gamma}^- = \Omega^-(\Gamma), \quad U_{\Gamma}^+ = \Omega^+(\Gamma).$$

By the definition of a homoclinic channel, the two restricted wave maps are diffeomorphisms onto these open sets. Hence

$$S_{\Gamma} = \Omega^+ \circ (\Omega^-)^{-1} : U_{\Gamma}^- \longrightarrow U_{\Gamma}^+$$

is a diffeomorphism of the common regularity of the wave maps. For a  $C^r$  NHIM and its  $C^{r-1}$  wave maps this is  $C^{r-1}$ , as claimed.

Assume now that  $(M, \omega)$  is symplectic,  $f$  is symplectic, and the restriction  $\omega_\Lambda = \omega|_{T\Lambda}$  is nondegenerate. Let  $i : \Gamma \hookrightarrow M$  be inclusion. We claim

$$i^*\omega = (\Omega^-)^*\omega_\Lambda = (\Omega^+)^*\omega_\Lambda. \quad (9.3.1)$$

We prove the plus identity; the minus identity is the time-reversed argument. For  $z \in \Gamma$  and  $v, w \in T_z\Gamma$ , normal hyperbolicity and the defining property of the stable wave map give

$$\|Df_z^n v - Df_{\Omega^+(z)}^n D\Omega_z^+ v\| + \|Df_z^n w - Df_{\Omega^+(z)}^n D\Omega_z^+ w\| \rightarrow 0 \quad (9.3.2)$$

as  $n \rightarrow +\infty$ , after the standard invariant splitting is used to compare the two tangent vectors along the asymptotic orbits. The compared vectors stay uniformly controlled on the compact local channel under the weighted NHIM norm. Since  $f^*\omega = \omega$ ,

$$\omega_z(v, w) = \omega_{f^n z}(Df^n v, Df^n w).$$

Taking the limit in this identity using (9.3.2), continuity of  $\omega$ , and  $f^n z - f^n \Omega^+(z) \rightarrow 0$  gives

$$\omega_z(v, w) = \omega_{\Omega^+(z)}(D\Omega^+ v, D\Omega^+ w),$$

which is the second equality in (9.3.1). Sending  $n \rightarrow -\infty$  gives the first.

**Lemma 9.4 (wave-map pullback identity).** Under the symplectic NHIM transversality hypotheses, (9.3.1) holds on every homoclinic channel.

Now use  $\Omega^+ = S_\Gamma \circ \Omega^-$ . Pulling back by  $\Omega^-$  and using (9.3.1),

$$(\Omega^-)^*(S_\Gamma^* \omega_\Lambda) = (\Omega^+)^* \omega_\Lambda = (\Omega^-)^* \omega_\Lambda.$$

Because  $\Omega^-$  is a diffeomorphism, it follows that  $S_\Gamma^* \omega_\Lambda = \omega_\Lambda$ . Thus the scattering map is symplectic.

Suppose finally that the setting is exact symplectic. Choose a primitive  $\lambda$  with  $d\lambda = \omega$  and a function  $P$  satisfying

$$f^* \lambda - \lambda = dP. \quad (9.3.3)$$

After adding an exact form on  $\Lambda$  if necessary, write  $\lambda_\Lambda = i_\Lambda^* \lambda$  for the primitive induced on  $\Lambda$ . For  $n \geq 1$  put

$$P_n = \sum_{j=0}^{n-1} P \circ f^j, \quad (f^n)^* \lambda - \lambda = dP_n. \quad (9.3.4)$$

The same asymptotic estimate used in (9.3.2), now applied to the one-form, shows that the following action differences converge in  $C^1$  on compact subchannels:

$$A^+(z) = - \lim_{n \rightarrow \infty} (P_n(z) - P_n(\Omega^+(z))), \quad (9.3.5)$$

$$A^-(z) = \lim_{n \rightarrow \infty} (P_{-n}(z) - P_{-n}(\Omega^-(z))), \quad (9.3.6)$$

where  $P_{-n}$  is determined by  $(f^{-n})^* \lambda - \lambda = dP_{-n}$ . Subtract (9.3.4), pulled back once by  $i$  and once by  $\Omega^+$ , and send  $n \rightarrow +\infty$ ; the asymptotic pullback difference tends to zero and yields

$$i^* \lambda - (\Omega^+)^* \lambda_\Lambda = dA^+. \quad (9.3.7)$$

The backward calculation gives

$$i^*\lambda - (\Omega^-)^*\lambda_\Lambda = dA^-. \quad (9.3.8)$$

Subtracting (9.3.8) from (9.3.7) and using  $\Omega^+ = S_\Gamma \circ \Omega^-$  gives

$$(\Omega^-)^*(S_\Gamma^*\lambda_\Lambda - \lambda_\Lambda) = d(A^- - A^+). \quad (9.3.9)$$

Since  $\Omega^-$  is a diffeomorphism, (9.3.9) says  $S_\Gamma^*\lambda_\Lambda - \lambda_\Lambda$  is exact on  $U_\Gamma^-$ . This is exact symplecticity.

**Lemma 9.5 (exact-action primitive identity).** The convergent action differences (9.3.5)–(9.3.6) provide the primitive in (9.3.9), so exactness is not inferred merely from symplecticity.  $\square$

#### Theorem K09.4. Transition-chain shadowing

**Statement.** Let  $F$  have a normally hyperbolic invariant manifold  $\Lambda$  whose stable and unstable manifolds meet transversely along the homoclinic channels used to define the transition maps. Suppose a finite sequence of windows on a return section in the sense of Definition K09.5 is arranged so that each successive pair is correctly aligned, alternately, under a suitable iterate of the inner map  $F|_\Lambda$  and under the transition map associated with the chosen homoclinic channel. Then there exists a true orbit of the full system whose successive returns pass through those windows in the prescribed order. Thus every finite itinerary built from the specified inner and transition steps and satisfying the correct-alignment conditions is shadowed by an actual orbit.

*Proof.* We first isolate the purely topological statement used by the orbit argument. Let  $W_0, \dots, W_N$  be windows of the same unstable/stable dimensions and let  $G_i$  be a continuous local map for which  $W_i$  is correctly aligned with  $W_{i+1}$ ,  $0 \leq i < N$ . In the window charts write the product cube as  $Q = Q^u \times Q^s$ .

For one alignment, the definition of window and correct alignment above supplies a homotopy  $h_i$  from the coordinate map of  $G_i$  to a terminal map whose unstable component has nonzero Brouwer degree, while throughout the homotopy the source exit boundary avoids the target cube and the whole source avoids the target entry boundary. These two avoidance conditions mean that the homotopy defines the same relative intersection class in

$$H_{m_u}(Q, Q^{\text{exit}}) \cong \mathbb{Z}. \quad (9.4.1)$$

The induced integer is precisely the Brouwer degree  $d_i \neq 0$  of the terminal unstable-coordinate map. For a chain, composition of the induced relative homology maps multiplies these integers; hence the class carried from  $W_0$  to  $W_N$  is

$$d_0 d_1 \cdots d_{N-1} \neq 0. \quad (9.4.2)$$

If no  $x \in W_0$  had all its successive images

$$x, \quad G_0 x, \quad G_1 G_0 x, \dots, \quad G_{N-1} \cdots G_0 x \quad (9.4.3)$$

in  $W_0, W_1, \dots, W_N$ , respectively, then at the first failed stage the chain homotopy could be pushed through an exit or entry boundary and the final relative class would be zero. That contradicts (9.4.2). Therefore a point with itinerary (9.4.3) exists.

For readers preferring a degree rather than homology formulation, one can make the same argument on the product of the  $N$  unstable cubes: introduce the intermediate unstable coordinates as independent variables, write the matching equations between successive windows, and homotope them simultaneously using the  $h_i$ . No zero can cross the boundary because of the exit/entry avoidance conditions. At the terminal homotopy the degree of the product system is  $\prod_i d_i$ , which is nonzero, so the original matching equations have a solution. This is exactly the point in (9.4.3).

**Lemma 9.6 (finite correctly-aligned window chain).** A finite chain of correctly aligned windows has a point whose successive local maps visit all windows in order. The invariant is the nonzero product of the Brouwer degrees; no metric contraction is required.

Return now to the dynamical setting of the theorem. By the Scattering-map construction, every selected homoclinic channel has a well-defined scattering map on open subsets of  $\Lambda$ . the definition of transition map associated with a homoclinic channel above converts it, after the chosen finite entrance and exit times, into a local transition map on the return section. The inner steps are actual iterates of  $F|_\Lambda$ , lifted to the same return section. By hypothesis, the prescribed sequence of windows is correctly aligned alternately under precisely these local maps. Apply Lemma 9.6 to this finite list. It produces a point  $x$  in the first full-system window whose image under each listed inner or transition map belongs to the next window.

Each transition map is not an abstract jump: by the definition of transition map associated with a homoclinic channel above it is the restriction of a finite orbit segment of  $F$  that leaves a neighborhood of  $\Lambda$ , follows the chosen homoclinic channel, and returns. Replace every transition arrow in the itinerary by this finite segment, and every inner arrow by its specified iterate of  $F$ . Concatenation is legitimate because the end point of one local map is the initial point for the next one. The resulting single orbit of  $F$  has successive returns through all of the prescribed windows. This is the asserted shadowing orbit.  $\square$

### Theorem K09.5. Arnold diffusion in an a-priori unstable model

**Statement.** Consider the time-periodic a-priori unstable Hamiltonian  $H_\varepsilon = \pm(\frac{1}{2}p^2 + \cos q - 1) + \frac{1}{2}I^2 + \varepsilon \cos q [a_1 \cos(k_1\varphi + l_1s) + a_2 \cos(k_2\varphi + l_2s)]$ , where  $(p, q)$  are pendulum variables,  $(I, \varphi)$  are rotator action-angle variables,  $s \in \mathbb{T}$  is the time phase with  $\dot{s} = 1$ ,  $a_1 a_2 \neq 0$ , and  $k_1 l_2 - k_2 l_1 \neq 0$ . For every  $I^* > 0$  there is  $\varepsilon^* > 0$  such that, for every  $0 < \varepsilon < \varepsilon^*$ , the system has an orbit and a time  $T > 0$  for which  $I(0) \leq -I^* < I^* \leq I(T)$ . Thus this explicit pendulum-rotator family exhibits an order-one drift of the rotator action.

*Proof.* We prove the assertion for the plus sign in front of the pendulum. Reversing that sign reverses the normal homoclinic orientation and changes none of the argu-

ments below. Throughout the proof fix  $I^* > 0$  and the slightly larger compact interval

$$J = [-I^* - 2, I^* + 2]. \quad (9.5.1)$$

All constants below are uniform on this fixed interval.

### 1. The exact NHIM and the Melnikov potential. The cylinder

$$\Lambda = \{p = q = 0\} \quad (9.5.2)$$

is invariant for every  $\varepsilon$ , not merely persistent: indeed  $\partial_q \cos q = -\sin q$  vanishes at  $q = 0$ . The normal linearization is the hyperbolic pendulum linearization plus an  $O(\varepsilon)$  bounded coefficient, so normal hyperbolicity is uniform on  $J$  for sufficiently small  $\varepsilon$ . At  $\varepsilon = 0$  a separatrix is

$$q_0(t) = 4 \arctan e^t, \quad p_0(t) = \frac{2}{\cosh t}, \quad \cos q_0(t) - 1 = -2 \operatorname{sech}^2 t. \quad (9.5.3)$$

Put

$$\omega_j(I) = k_j I + l_j, \quad \Delta = k_1 l_2 - k_2 l_1 \neq 0. \quad (9.5.4)$$

The convergent Poincaré–Melnikov potential, with the value on  $\Lambda$  subtracted, is

$$\mathcal{L}(I, \varphi, s, \tau) = \sum_{j=1}^2 a_j \int_{-\infty}^{\infty} (\cos q_0(t) - 1) \cos(k_j \varphi + l_j s + \omega_j(I)t - \omega_j(I)\tau) dt. \quad (9.5.5)$$

Using

$$\int_{-\infty}^{\infty} \operatorname{sech}^2 t e^{i\omega t} dt = \frac{\pi\omega}{\sinh(\pi\omega/2)} \quad (9.5.6)$$

with its continuous value 2 at  $\omega = 0$ , we obtain

$$\mathcal{L} = \sum_{j=1}^2 B_j(I) \cos \alpha_j, \quad B_j(I) = -\frac{2\pi a_j \omega_j(I)}{\sinh(\pi\omega_j(I)/2)}, \quad \alpha_j = k_j \varphi + l_j s - \omega_j(I)\tau. \quad (9.5.7)$$

At a zero of  $\omega_j$  the continuous value is  $B_j = -4a_j$ . Since  $a_1 a_2 \neq 0$ , neither  $B_j$  vanishes anywhere on  $J$ .

The determinant hypothesis also says that  $\omega_1$  and  $\omega_2$  never vanish simultaneously. Choose a nonzero constant  $\delta$ , so small that on  $J$

$$|\delta\omega_2| < |B_1|, \quad |\delta\omega_1| < |B_2|, \quad (9.5.8)$$

and choose its sign later. Define

$$u_1 = \delta\omega_2, \quad u_2 = -\delta\omega_1. \quad (9.5.9)$$

Because the sign of each  $B_j$  is constant, we may choose real lifts  $\alpha_j(I)$  continuously on  $J$  so that

$$B_j \sin \alpha_j = u_j, \quad B_j \cos \alpha_j > 0. \quad (9.5.10)$$

Then

$$\partial_\tau \mathcal{L} = \omega_1 u_1 + \omega_2 u_2 = 0, \quad \partial_{\tau\tau} \mathcal{L} = - \sum_{j=1}^2 B_j \omega_j^2 \cos \alpha_j < 0. \quad (9.5.11)$$

Thus the selected homoclinic time is nondegenerate, uniformly after a finite compactness cover.

There is one elementary torus point which is worth recording explicitly. Put  $u = \varphi - I\tau$  and  $v = s - \tau$ . The equations

$$k_j u + l_j v = \alpha_j \pmod{2\pi}, \quad j = 1, 2, \quad (9.5.12)$$

are solvable because the integer matrix with rows  $(k_j, l_j)$  induces a surjective finite covering of  $\mathbb{T}^2$  of degree  $|\Delta|$ . In general there are  $|\Delta|$  solutions, not a unique one. Choosing one lift and solving the real linear system gives a continuous local source curve. This corrects the uniqueness shortcut that appeared in the earlier proof dossier.

By the implicit-function theorem, (9.5.11) determines a local homoclinic channel and a reduced Melnikov potential  $\mathcal{L}^*$ . With the exact-symplectic convention used for the scattering map, its first-order form on a two-dimensional return section is

$$S_\varepsilon(I, \theta) = (I + \varepsilon \partial_\theta \mathcal{L}^*(I, \theta), \theta - \varepsilon \partial_I \mathcal{L}^*(I, \theta)) + R_\varepsilon(I, \theta), \quad \|R_\varepsilon\|_{C^1} \leq C\varepsilon^2. \quad (9.5.13)$$

At the selected critical points,

$$\partial_\theta \mathcal{L}^* = - \sum_{j=1}^2 k_j u_j = -\delta(k_1 \omega_2 - k_2 \omega_1) = -\delta \Delta. \quad (9.5.14)$$

Choose the sign of  $\delta$  so that  $\kappa_0 := -\delta \Delta > 0$ . Shrinking the finitely many channel charts and then shrinking  $\varepsilon$  gives a uniform  $\kappa > 0$  such that in every selected outer chart

$$I(S_\varepsilon z) - I(z) \geq \kappa \varepsilon, \quad \|S_\varepsilon z - z\| \leq C\varepsilon. \quad (9.5.15)$$

Notice carefully what has been proved: (9.5.15) is a statement about the *scattering map*. It is not yet a statement about a full transition map containing arbitrary inner iterates.

**Lemma 9.7 (two-harmonic positive scattering channels).** Equations (9.5.5)–(9.5.15) give, over every fixed compact action interval, a finite family of nondegenerate homoclinic-channel charts whose scattering maps have a common positive action increment at first order. The torus phase map is used as a finite covering, so no false uniqueness assertion is needed.

**2. The inner dynamics and its only first-order resonances.** On  $\Lambda$  the true inner Hamiltonian is

$$K_\varepsilon(I, \varphi, s) = \frac{1}{2} I^2 + \varepsilon [a_1 \cos(k_1 \varphi + l_1 s) + a_2 \cos(k_2 \varphi + l_2 s)]. \quad (9.5.16)$$

Thus  $I$  is *not* constant when  $\varepsilon \neq 0$ . We now show why this fact does not obstruct the positive outer drift.



At least one  $k_j$  is nonzero, because otherwise  $\Delta = 0$ . Relabel the two harmonics and, if necessary, replace a phase by its negative (which leaves its cosine unchanged) so that  $k_1 > 0$  and

$$r := \frac{k_2}{k_1} \in [0, 1]. \quad (9.5.17)$$

The affine harmonic coordinates used in the primary proof,

$$\psi = k_1\varphi + l_1s, \quad r\psi - \sigma = k_2\varphi + l_2s, \quad J = k_1I + l_1, \quad (9.5.18)$$

come from a linear symplectic change in the autonomous extension after adding the action conjugate to  $s$ . The second time frequency is a nonzero constant proportional to  $\Delta/k_1$ ; rescaling that conjugate pair and, if necessary, reversing its orientation reduces the inner Hamiltonian, up to nonzero fixed multipliers in the coefficients, to

$$\overline{K}_\varepsilon = \frac{1}{2}J^2 + B + \varepsilon(A_1 \cos \psi + A_2 \cos(r\psi - \sigma)), \quad \dot{\sigma} = 1, \quad (9.5.19)$$

with  $A_1 A_2 \neq 0$ . An order-one drift of  $J$  is equivalent to an order-one drift of the original  $I$ , because  $J = k_1 I + l_1$  with  $k_1 \neq 0$ . We shall therefore analyze (9.5.19) and then return to  $I$ .

The unperturbed divisors of the two harmonics are

$$\nu_1(J) = J, \quad \nu_2(J) = rJ - 1. \quad (9.5.20)$$

Consequently the only first-order resonances are  $J = 0$  and, when  $r > 0$ ,  $J = 1/r$ . This can be made quantitative by a one-step resonant normal form. Choose smooth cutoffs supported in small disjoint neighborhoods of those two values. Away from a resonance, the generating function

$$G(J, \psi, \sigma) = -\frac{A_1}{J} \sin \psi - \frac{A_2}{rJ - 1} \sin(r\psi - \sigma) \quad (9.5.21)$$

(with the corresponding cutoff factors inserted) solves the homological equation for the nonresonant harmonics because

$$\{\frac{1}{2}J^2 + B, G\} = -A_1 \cos \psi - A_2 \cos(r\psi - \sigma). \quad (9.5.22)$$

Here  $\{F, G\}$  denotes the canonical Poisson bracket. Taylor expansion of the time-one map generated by  $\varepsilon G$  gives, uniformly in  $C^2$  on the fixed compact strip,

$$F^{\text{nr}} = \frac{1}{2}J^2 + O(\varepsilon^2) \quad (9.5.23)$$

in the nonresonant region,

$$F^0 = \frac{1}{2}J^2 + \varepsilon A_1 \cos \psi + O(\varepsilon^2) \quad (9.5.24)$$

near  $J = 0$ , and, in the canonical resonant coordinate  $P = (J - 1/r)/r$  and angle  $\chi =$

$r\psi - \sigma$ ,

$$F^r = \frac{1}{2}c_r P^2 + \varepsilon A_2 \cos \chi + O(\varepsilon^2), \quad c_r > 0, \quad (9.5.25)$$

near  $J = 1/r$ . For  $r = 0$  the second resonance is absent. These formulae are used only as local  $C^2$  normal forms for the *actual* inner vector field; we do not declare the displayed truncated level curves to be exact invariant tori.

**3. No inner barrier survives the positive outer direction.** Work in any one of the positive scattering charts from Step 1 and write its reduced generating function as  $L$ . By shrinking the chart,

$$\partial_\theta L \geq 2\kappa > 0. \quad (9.5.26)$$

The Hamiltonian vector field of  $L$  therefore has a strictly positive action component. We compare it with the true inner field through the normal forms (9.5.23)–(9.5.25).

Outside the resonance neighborhoods, the bracket of the inner reference integral with  $L$  is

$$\{F^{\text{nr}}, L\} = -J \partial_\theta L + O(\varepsilon^2). \quad (9.5.27)$$

Hence the inner and outer directions are transverse there except in the  $J = 0$  neighborhood already reserved for (9.5.24). Near that resonance,

$$\{F^0, L\} = -J \partial_\theta L - \varepsilon A_1 \sin \theta \partial_J L + O(\varepsilon^2). \quad (9.5.28)$$

The derivative of the right-hand side with respect to  $J$  is  $-\partial_\theta L + O(\varepsilon)$ , and therefore is bounded away from zero by (9.5.26). The implicit-function theorem shows that the tangency set in the resonance core is a single  $C^1$  graph

$$J = \varepsilon b_0(\theta) + O(\varepsilon^2). \quad (9.5.29)$$

It cannot be a common barrier: the outer vector field crosses this graph, because its normal component is

$$\partial_\theta L - \varepsilon b'_0(\theta)(-\partial_J L) + O(\varepsilon) \geq \kappa \quad (9.5.30)$$

for small  $\varepsilon$ .

The same calculation in the resonant canonical pair  $(P, \chi)$  of (9.5.25) gives, when  $r > 0$ , a tangency graph

$$P = \varepsilon b_r(\chi) + O(\varepsilon^2) \quad (9.5.31)$$

near  $J = 1/r$ , and the positive outer field crosses it transversely. Thus the two first-order resonances may bend the inner motion by  $O(\sqrt{\varepsilon})$ , but neither produces a curve invariant under both the inner and outer dynamics. This is the precise local content needed from the resonance analysis in the primary proof.

We now turn this local fact into a finite pseudo-orbit. Let  $\mathcal{U}$  be the union of the finitely many positive scattering charts covering the action strip corresponding to (9.5.1). Start with a small open box in  $\mathcal{U}$  below the action  $-I^* - 1$ . In a box where the inner and outer directions are transverse, the flow-box theorem supplies coordinates in which short inner segments and short positive outer segments form a two-directional rectangle; in particular an open set can be moved to the upper face of the box while the outer pieces increase the action. If a path reaches one of the tangency graphs

(9.5.29) or (9.5.31), (9.5.30) and its second-resonance analogue move it across that graph by an outer segment, after which the two directions are again transverse.

For completeness, suppose this continuation stopped before reaching the top of the compact strip. The boundary of the set accessible by finite alternations of the two local dynamics would then contain a first separating arc. At a regular point away from the two tangency graphs, invariance under short inner and outer segments would force that arc to be tangent to two transverse directions, impossible. On a tangency graph, the positive outer direction crosses the graph by (9.5.30), again impossible. Thus no separating arc exists. Compactness now provides a finite chain of flow boxes connecting the lower and upper faces.

Replace every short outer flow segment in this finite geometric chain by iterations of the corresponding scattering map (9.5.13). Since that map is the time- $\varepsilon$  Hamiltonian step of  $L$  up to  $O(\varepsilon^2)$  in  $C^1$ , the Euler polygon remains in the same open boxes for all sufficiently small  $\varepsilon$ . Every outer iterate raises the action by at least  $\kappa\varepsilon$  by (9.5.15). The number of outer iterates is finite for each  $\varepsilon$  and is  $O(\varepsilon^{-1})$  on the fixed strip. The inner arrows are genuine finite pieces of the inner dynamics. We have therefore constructed a finite alternating inner/scattering pseudo-orbit whose first action is below  $-I^* - 1$  and whose last action is above  $I^* + 1$ .

**Lemma 9.8 (inner-resonance transversality and bypass).** The first-order inner normal form has only the two harmonic resonances. Away from them the inner and positive outer directions are transverse. In each resonance core their tangency set is a simple  $O(\varepsilon)$  graph and the positive outer field crosses that graph. Hence no common inner/outer barrier can separate the compact action strip.

**Lemma 9.9 (finite alternating diffusion pseudo-orbit).** A finite flow-box cover, together with Lemma 9.8 and the uniform positive scattering estimate (9.5.15), produces an alternating pseudo-orbit from below  $-I^* - 1$  to above  $I^* + 1$ . Outer steps number  $O(\varepsilon^{-1})$ ; inner steps are the true inner dynamics and are not folded into the Melnikov increment.

**4. From scattering arrows to actual transition maps.** The previous step is still a pseudo-orbit on the NHIM. We now replace every asymptotic scattering arrow by a finite orbit segment of the full system. This is where recurrence and the transition-chain shadowing theorem enter.

Choose two actions  $c_- < -I^* - 3/2$  and  $c_+ > I^* + 3/2$  whose unperturbed rotation numbers for one stroboscopic inner return map are Diophantine. Such choices are possible because Diophantine numbers are dense. On the compact annular neighborhood between them, every time-phase stroboscopic map  $P_{\varepsilon, s_0}$  is real analytic and exact symplectic and, after the harmless normalization of the angular coordinate by  $2\pi$ ,

$$\|P_{\varepsilon, s_0} - P_0\| \leq C\varepsilon, \quad P_0(\theta, I) = (\theta + I, I), \quad (9.5.32)$$

uniformly in  $s_0$ . The estimate follows from the bounded vector field and Gronwall's inequality over one forcing period, while the twist derivative of  $P_0$  is 1. The Moser invariant-curve theorem therefore gives two invariant essential circles on one section close to  $I = c_{\pm}$ ; suspending them by the inner flow gives two invariant tori in  $\Lambda$ . Their slices bound, for every time phase  $s_0$ , a compact annulus  $\mathcal{A}_{\varepsilon, s_0}$  containing the whole

action strip used above.

Each  $P_{\varepsilon, s_0}$  preserves finite area on  $\mathcal{A}_{\varepsilon, s_0}$ . By the Poincaré recurrence theorem its recurrent points form a full-area subset  $\mathcal{R}_{\varepsilon, s_0}$ . A scattering map preserves the time phase and is symplectic on its reduced section by the scattering-map construction theorem; inner flow maps between two time-phase sections are symplectic as well. Hence, if  $\Phi_j$  is the finite composition of the inner/scattering arrows up to the  $j$ th scattering endpoint at phase  $s_j$ , then  $\Phi_j^{-1}(\mathcal{R}_{\varepsilon, s_j})$  has full area in its open source domain. There are only finitely many such conditions for the fixed  $\varepsilon$ . Their intersection with the nonempty open family of pseudo-orbits from Step 3 is therefore nonempty. We choose the pseudo-orbit so that every source and target point of every scattering arrow is recurrent under the stroboscopic return map on its own time-phase section.

Fix one scattering arrow  $x \mapsto y = S_\Gamma(x)$  at phase  $s_0$ . Since  $x$  and  $y$  are recurrent under  $P_{\varepsilon, s_0}$ , for every pair of neighborhoods  $U \ni x$  and  $V \ni y$  there are arbitrarily large positive integers  $n_-$  and  $n_+$  for which

$$P_{\varepsilon, s_0}^{n_-} x \in U, \quad P_{\varepsilon, s_0}^{n_+} y \in V. \quad (9.5.33)$$

Take the return times large enough also for the finite channel excursion to be defined on small full-system windows. In the coordinates of the definition of a transition map above its base map is

$$T_\Gamma = P_{\varepsilon, s_0}^{n_+} \circ S_\Gamma \circ P_{\varepsilon, s_0}^{n_-}. \quad (9.5.34)$$

Normal hyperbolicity and the defining exponential convergence of the wave maps make the lifted finite orbit segment approach this base transition uniformly on compact subwindows as  $n_\pm \rightarrow \infty$ . Hence, after (9.5.33) and then a small-window shrink, the full transition map sends a source window contained in the prescribed pseudo-orbit box across a target window contained in the next box. Since the local transition is a diffeomorphism, choose product coordinates so that the target window is a slightly shrunk image of the source window; the exit map then has Brouwer degree  $\pm 1$ . Thus the two windows are correctly aligned in the precise sense of the definition of correct alignment above. The same construction for an inner arrow uses the corresponding finite iterate of  $P_\varepsilon$  and is simpler.

Perform this operation for the finitely many arrows. We obtain a finite sequence of full-system windows alternately correctly aligned under inner iterates and under genuine homoclinic-channel transition maps. The transition-chain shadowing theorem now gives one true orbit of the full Hamiltonian whose successive returns pass through every window in order.

**Lemma 9.10 (recurrent realization by finite transition maps).** Two outer KAM circles trap the compact inner dynamics in a finite-area annulus. Poincaré recurrence and symplecticity allow all finitely many scattering endpoints to be chosen recurrent. Arbitrarily long recurrent entrance/exit iterates then realize every asymptotic scattering arrow as a genuine local transition map with correctly aligned windows.

**Lemma 9.11 (window shadowing gives Arnold drift).** the transition-chain shadowing theorem shadows the finite correctly aligned chain from Lemma 9.10 by a single full-system orbit. The fixed one-unit action margins absorb the window diameters and all  $O(\varepsilon)$  endpoint errors.

The first window was chosen below  $-I^* - 1$  and the last above  $I^* + 1$ . Shrinking the windows and then  $\varepsilon^*$  if necessary, the shadowing orbit therefore has two times, which we translate so that the first is 0, with

$$I(0) \leq -I^* < I^* \leq I(T)$$

for some  $T > 0$ . Returning through the affine harmonic coordinates (9.5.18) does not change this conclusion: the proof was carried out on the compact interval corresponding to the original prescribed endpoints, and the coordinate map has nonzero action slope. This proves the theorem for every  $0 < \varepsilon < \varepsilon^*$ .  $\square$

## 9.4 Worked examples

### Worked example: a periodically forced pendulum

Consider

$$H(q, p, t) = \frac{1}{2}p^2 + \cos q - 1 + \varepsilon \cos q \cos t.$$

For  $\varepsilon = 0$ , the saddle at  $(q, p) = (0, 0)$  has the standard homoclinic orbit

$$q_h(t) = 4 \arctan(e^t), \quad p_h(t) = \frac{2}{\cosh t},$$

up to the usual  $2\pi$  identification and a time translation. The Melnikov function for a phase shift  $\tau$  has the form

$$M(\tau) = \int_{-\infty}^{\infty} \nabla H_0(z_h(t)) \cdot g(z_h(t), t + \tau, 0) dt.$$

For a single harmonic forcing, parity reduces the integral to a nonzero constant multiple of  $\sin \tau$  (the constant is the Fourier transform of a rapidly decaying homoclinic profile). Hence

$$M(0) = 0, \quad M'(0) \neq 0.$$

The simple-zero criterion therefore gives a transverse homoclinic point for all sufficiently small nonzero  $\varepsilon$ . Smale-Birkhoff then produces a horseshoe for a suitable iterate of the Poincaré map.

The logical chain matters:

$$\begin{aligned} \text{explicit integral} &\Rightarrow \text{simple zero,} \\ &\Rightarrow \text{transverse homoclinic point,} \\ &\Rightarrow \text{symbolic dynamics.} \end{aligned}$$

No numerical plot of the separatrix can replace any of these implications.

**Boundary case: the first Melnikov function vanishes identically**

If symmetry forces  $M(\tau) \equiv 0$ , the first-order splitting calculation is inconclusive. It does *not* prove that the stable and unstable manifolds fail to split; the leading splitting may occur at higher order or may be exponentially small. The correct conclusion is only that the simple-zero theorem is unavailable and a higher-order or different geometric analysis is required.

**Check your understanding**

Suppose the scattering map changes the action by  $+c\varepsilon$  on a chosen chart, with  $c > 0$ . Roughly how many successful scattering steps are needed to obtain an order-one action change? Why does this estimate alone still not prove the existence of a true diffusing orbit?

**Common pitfall**

A transverse homoclinic point and the resulting horseshoe do not by themselves prove Arnold diffusion. They give complicated local recurrence near the homoclinic channel. Order-one drift requires a mechanism that changes the slow action—typically a scattering/transition map—plus a shadowing argument that converts the designed chain into one true orbit.

## 9.5 Chapter summary

**What to remember**

- Arnold diffusion is built from a precise chain of mechanisms rather than from “chaos” in the abstract.
- Melnikov theory creates transverse homoclinic geometry, while Smale-Birkhoff turns such geometry into symbolic dynamics.
- A scattering map records the asymptotic change across a homoclinic excursion; a full transition map also includes the finite inner entrance and exit dynamics.
- Correctly aligned windows turn a designed transition chain into a true orbit, and the explicit a-priori unstable model uses this to produce order-one drift.

**Looking ahead**

K10 returns to two-dimensional exact twist dynamics, where global variational order replaces the high-dimensional diffusion mechanism. Aubry-Mather sets, Peierls barriers, and Birkhoff graph properties provide a complementary route to understanding invariant and minimizing dynamics beyond smooth KAM circles.

**Check your understanding**

Reconstruct the full diffusion chain from a Melnikov simple zero to an order-one drift orbit. At which step do you obtain transverse homoclinic geometry, at which step an action-changing map, and at which step a genuine orbit rather than a pseudo-orbit?

**9.6 Exercises and problems****Exercise 9.1**

**Difficulty.** Core.

**Related results.** Melnikov simple-zero criterion; Smale–Birkhoff homoclinic theorem; Scattering-map construction.

**Problem.** Evaluate the Melnikov integral for the periodically forced pendulum, locate a simple zero, and use it to obtain a transverse homoclinic point. State the Smale–Birkhoff horseshoe conclusion without claiming Arnold diffusion.

**Solution sketch.** Use the explicit separatrix, differentiate the final Melnikov function at a zero, and invoke only the local homoclinic theorem.

**Exercise 9.2**

**Difficulty.** Core.

**Related results.** Transition-chain shadowing; Arnold diffusion in an a-priori unstable model.

**Problem.** When the first Melnikov function vanishes identically, prove that first-order theory is inconclusive rather than proving absence of splitting; write the next possible term in the splitting expansion.

**Solution sketch.** Show the  $O(\varepsilon)$  coefficient cancels and distinguish the cases for the  $O(\varepsilon^2)$  coefficient.

**Exercise 9.3**

**Difficulty.** Advanced.

**Related results.** Transition-chain shadowing.

**Problem.** Assuming symplectic wave maps on a homoclinic channel, prove that the scattering map  $S = \Omega^+ \circ (\Omega^-)^{-1}$  is symplectic.

**Solution sketch.** Show that both wave maps pull back the same reduced symplectic form and compose the identities.

**Exercise 9.4**

**Difficulty.** Advanced.

**Related results.** Arnold diffusion in an a-priori unstable model.

**Problem.** Construct a finite transition pseudo-itinerary that fails the matching/transversality hypothesis and explain why it need not be shadowed by a true orbit.

**Solution sketch.** Make one outgoing window miss the next incoming window or

create a tangency, and identify the failed shadowing step.

### Exercise 9.5

**Difficulty.** Advanced.

**Related results.** Melnikov simple-zero criterion; Smale–Birkhoff homoclinic theorem.

**Problem.** Write the exact logical chain from a Melnikov simple zero to a local horseshoe, then identify the two additional ingredients needed for an  $O(1)$  Arnold-diffusion drift.

**Solution sketch.** Separate transverse splitting, symbolic dynamics, action-changing scattering, and transition-chain shadowing.

### Exercise 9.6

**Difficulty.** Lab.

**Related results.** None.

**Problem.** Measure stable/unstable splitting on a transverse section for several small  $\varepsilon$  and compare it with  $\varepsilon M(t_0)$ , testing whether the residual is  $O(\varepsilon^2)$  away from simple zeros.

**Solution sketch.** Refine integration horizon and step size, and track zero locations separately from normalized residuals.



# Twist Maps and Aubry-Mather Theory

## Why this chapter matters

The preceding chapters were dominated by perturbation theory: start from an integrable Hamiltonian, control small divisors, and decide whether motion persists, remains confined, or drifts. Twist maps offer a different point of entry. They convert two-dimensional symplectic dynamics into a variational problem for a generating function. Once that conversion is made, order and minimization become dynamical tools.

This chapter develops the Aubry-Mather viewpoint from first principles. The Poincaré-Birkhoff theorem shows that opposite boundary twist forces periodic dynamics. Aubry's crossing argument then explains why action-minimizing configurations are ordered. Ordered minimizers exist for every rotation number in the twist interval, and their geometry is controlled by Birkhoff's graph theorem. The Peierls barrier measures exactly when the minimizing set fills an invariant circle rather than leaving gaps. Finally, Mather's  $\beta$ -function reorganizes the same variational mechanism at the level of invariant measures and convex duality.

**Prerequisites.** K01-K06 for exact symplectic maps and generating functions; Foundation F01-F02 for invariant measures and rotation ideas; elementary convex analysis; compactness for minimizing sequences.

## Learning goals

- Derive the orbit equations of an exact twist map from a generating function.
- Explain why strict twist submodularity forbids repeated crossings of minimizing configurations.
- Construct ordered minimizing orbits with prescribed rational or irrational rotation number.
- Interpret the Peierls barrier as the variational obstruction to an invariant circle.
- Use Mather's  $\alpha$ - $\beta$  duality to pass between rotation vectors and cohomological tilts of the action.

### Conceptual roadmap

There are two complementary proof chains:

$$\begin{aligned}
 &\text{twist} + \text{exactness} \longrightarrow \text{generating function} \\
 &\quad \longrightarrow \text{ordered minimizers} \longrightarrow \text{Aubry-Mather sets,} \\
 &\text{Tonelli compactness} \longrightarrow \beta(h) \longleftrightarrow \alpha(c) \\
 &\quad \longrightarrow \text{convex duality.}
 \end{aligned}$$

The first chain is discrete and geometric. The second is measure-theoretic and convex. Their common principle is that minimizing action produces order strong enough to survive the loss of smooth invariant circles.

### Check your understanding

For the integrable twist map  $F(\theta, r) = (\theta + r, r)$ , what is the rotation number of the invariant circle  $r = \rho$ ? Why does the existence of all these circles make the Peierls barrier vanish for every irrational  $\rho$ ?

## 10.1 Definitions

### Intuition

**A twist map is a symplectic recurrence relation.** Exactness gives a generating function  $S(x, X)$ , while the twist condition says that one can solve between angle increments and momenta. The discrete Euler-Lagrange equation for  $\sum S(x_i, x_{i+1})$  is therefore the orbit equation of the map.

**Minimizers are ordered because crossing costs action.** For a positive twist,  $\partial_{12}S < 0$ . This strict submodularity means that exchanging the middle pieces of two twice-crossing minimizing strings lowers their total action. Hence global minimizers cannot braid around each other.

**The Peierls barrier detects gaps, not merely instability.** It is a constrained excess action. A zero barrier at every angular point means that minimizing configurations pass through every angle and assemble into an invariant circle. A positive value marks a genuine gap in the projected minimizing set.

**Mather's  $\beta$ -function is an action-versus-rotation graph.** For each homology/rotation vector  $h$ ,  $\beta(h)$  is the least average action among invariant measures with that rotation. The conjugate  $\alpha(c)$  records the least action after tilting by a cohomology class  $c$ .

### Definition K10.1. exact monotone twist map and generating function

Let  $A = \mathbb{T} \times [a, b]$  with symplectic form  $d\theta \wedge dr$ . A  $C^1$  annulus diffeomorphism  $F$  is an *exact positive monotone twist map* if a lift  $\tilde{F}(x, r) = (X, R)$  satisfies  $\partial_r X(x, r) > 0$  everywhere and if  $F^*(r d\theta) - r d\theta$  is exact. On every lifted twist domain there is then

a generating function  $S(x, X)$ , periodic under  $(x, X) \mapsto (x + 1, X + 1)$ , such that

$$r = -\partial_1 S(x, X), \quad R = \partial_2 S(x, X),$$

and the positive twist is equivalent to  $\partial_{12} S < 0$ . The negative-twist convention is obtained by reversing the action coordinate; all variational statements below use the positive convention after this harmless normalization.

### Definition K10.2. minimizing configuration

For a generating function  $S$ , a bi-infinite sequence  $x = (x_i)_{i \in \mathbb{Z}}$  is a *globally minimizing configuration* if, for every  $m < n$ , the finite string  $(x_m, \dots, x_n)$  minimizes

$$\sum_{i=m}^{n-1} S(z_i, z_{i+1})$$

among all strings with  $z_m = x_m$  and  $z_n = x_n$ . Its associated lifted orbit is obtained from  $r_i = -\partial_1 S(x_i, x_{i+1})$ .

### Definition K10.3. Aubry-Mather set

An *Aubry-Mather set of rotation number  $\rho$*  is the compact invariant subset of the annulus obtained from a translation-ordered family of globally minimizing configurations whose asymptotic displacement satisfies

$$\lim_{n \rightarrow \pm\infty} \frac{x_n - x_0}{n} = \rho.$$

Its lifted points inherit the order of the corresponding minimizing configurations.

### Definition K10.4. Peierls barrier

Fix an irrational rotation number  $\rho$  for an exact monotone twist map with generating function  $S$ . Let  $A_\rho \subset \mathbb{R}$  be the angular projection of its lifted minimizing Aubry set. Set  $P_\rho(\xi) = 0$  for  $\xi \in A_\rho$ . If  $\xi$  lies in a component  $(\xi^-, \xi^+)$  of  $\mathbb{R} \setminus A_\rho$ , let  $x^- < x^+$  be the two extremal  $\rho$ -minimizing configurations with  $x_0^- = \xi^-$  and  $x_0^+ = \xi^+$ . For  $m < 0 < n$ , let  $A_{m,n}(\xi)$  be the minimum of  $\sum_{i=m}^{n-1} S(z_i, z_{i+1})$  over strings with  $z_m = x_m^-$ ,  $z_0 = \xi$ ,  $z_n = x_n^+$ , and  $x_i^- \leq z_i \leq x_i^+$ , and let  $A_{m,n}^0$  be the same minimum without the constraint  $z_0 = \xi$ . The *Peierls barrier* is

$$P_\rho(\xi) = \lim_{m \rightarrow -\infty, n \rightarrow +\infty} (A_{m,n}(\xi) - A_{m,n}^0),$$

where the monotone-twist compactness/crossing estimates guarantee existence of the limit. Thus  $P_\rho(\xi)$  is a nonnegative constrained-action defect normalized to vanish on the projected minimizing set.

**Definition K10.5. average action and rotation vector of an invariant measure**

Let  $L : TM \times \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}$  be a time-periodic Lagrangian and let  $\mu$  be an invariant probability measure of its Euler-Lagrange flow with finite action. Its average action is  $A(\mu) = \int L d\mu$ . Its rotation vector  $\rho(\mu) \in H_1(M; \mathbb{R})$  is characterized by

$$\langle c, \rho(\mu) \rangle = \int_{TM \times \mathbb{R}/\mathbb{Z}} \eta_x(v) d\mu(x, v, t)$$

for every  $c \in H^1(M; \mathbb{R})$  and every smooth closed one-form  $\eta$  representing  $c$ . The right-hand side depends only on the cohomology class of  $\eta$  because  $\mu$  is invariant.

**Definition K10.6. Legendre-Fenchel conjugate and subdifferential**

For a finite-dimensional real vector space  $V$  and a proper convex function  $f : V \rightarrow (-\infty, +\infty]$ , its Legendre-Fenchel conjugate is

$$f^*(c) = \sup_{h \in V} \{ \langle c, h \rangle - f(h) \}.$$

The subdifferential of  $f$  at  $h$  is

$$\partial f(h) = \{ c : f(h') \geq f(h) + \langle c, h' - h \rangle \text{ for every } h' \}.$$

For conjugate convex functions, equality in the Fenchel inequality is equivalent to the mutual subgradient relations.

**Definition K10.7. Tonelli positive-definite time-periodic Lagrangian**

A time-periodic Lagrangian  $L : TM \times \mathbb{T} \rightarrow \mathbb{R}$  is *Tonelli (positive definite in Mather's sense)* if it is sufficiently smooth,  $v \mapsto L(x, v, t)$  has positive-definite fiber Hessian everywhere,  $L(x, v, t)/|v| \rightarrow +\infty$  uniformly in  $(x, t)$  as  $|v| \rightarrow \infty$ , and its Euler-Lagrange flow is complete. These hypotheses imply existence and compactness properties for finite-time action minimizers and make the average-action functional lower semicontinuous on bounded-action families of invariant measures.

**Definition K10.8. strict twist submodularity and crossing of configurations**

For a positive twist generating function  $S$ , *strict submodularity* means

$$S(a, c) + S(b, d) < S(a, d) + S(b, c) \quad \text{whenever } a < b \text{ and } c < d;$$

for  $C^2$  generating functions it follows from  $\partial_{12}S < 0$  by integrating the mixed derivative over the rectangle. Two configurations  $x, y$  *cross between  $i$  and  $i+1$*  if  $(x_i - y_i)(x_{i+1} - y_{i+1}) < 0$ ; equality cases are treated by the usual weak-order limit.

**Definition K10.9. rotation number of an ordered twist orbit**

For a lifted ordered orbit  $(x_i, r_i)$  of an annulus twist map, its rotation number is

$$\rho = \lim_{n \rightarrow \pm\infty} \frac{x_n - x_0}{n}$$

when the limit exists. The rotation number of a boundary component is the Poincaré rotation number of the induced orientation-preserving circle homeomorphism. For Birkhoff/minimizing configurations the order relation implies existence of the displayed limit and the bound  $|x_j - x_i - (j - i)\rho| \leq 1$ .

**Definition K10.10. annulus lift and opposite boundary twist**

For  $A = \mathbb{S}^1 \times [0, 1]$  and its covering map  $\pi : \mathbb{R} \times [0, 1] \rightarrow A$ , a *lift* of a boundary-component-preserving map  $F : A \rightarrow A$  is a map  $\tilde{F} : \mathbb{R} \times [0, 1] \rightarrow \mathbb{R} \times [0, 1]$  satisfying  $\pi \circ \tilde{F} = F \circ \pi$  and  $\tilde{F}(x+1, r) = \tilde{F}(x, r) + (1, 0)$ . Writing  $\tilde{F} = (X, R)$ , it has a *strict opposite boundary twist* if either

$$X(x, 0) - x < 0 < X(x, 1) - x \quad \text{for every } x,$$

or both inequalities are reversed. In the proof of the Poincaré-Birkhoff theorem, the *local fixed-point index* of an isolated fixed point is the Brouwer degree (winding number) of the normalized displacement vector  $\tilde{F}(z) - z$  on a sufficiently small positively oriented circle around that point.

## 10.2 How the main proofs work

### Intuition

**Poincaré-Birkhoff.** Opposite boundary twist forces a nonzero index imbalance for the displacement field of a lift. Area preservation rules out the escape alternatives. The fixed-point index bookkeeping then produces at least two interior fixed points; the proof keeps the one-fixed-point index-zero possibility under explicit control.

**Aubry crossing and the graph theorem.** If two minimizers crossed twice, cut them at the crossings and swap the middle pieces. Strict submodularity makes the swapped pair have strictly smaller total action, contradicting minimality. This one inequality yields global order; the twist estimate then converts angular separation into a Lipschitz bound in the action coordinate.

**Aubry-Mather existence.** For rational rotation  $p/q$ , minimize the  $q$ -step action among strings with  $x_q = x_0 + p$ . Crossing estimates make the minimizer Birkhoff ordered. Approximate an irrational  $\rho$  by rationals and use compactness plus the order bound  $|x_j - x_i - (j - i)\rho| \leq 1$ . The limit is a globally minimizing configuration of rotation  $\rho$ .

**Peierls barrier.** The barrier is nonnegative because it is a constrained minimum minus the unconstrained minimum. An invariant minimizing graph lets

every angular constraint be realized at zero excess cost. Conversely, zero barrier at every angle supplies compatible minimizers through every angle; order and compactness glue them into an invariant circle.

**Mather  $\alpha$ - $\beta$  duality.** Tonelli superlinearity makes bounded-action invariant measures compact enough for minimization. Long minimizers in the homology cover realize every rotation vector. The epigraph of the minimal action is closed and convex, while subtracting a closed one-form tilts the action by  $\langle c, h \rangle$ . Legendre-Fenchel duality then gives the Fenchel inequality and the subgradient/minimizer equivalences.

## 10.3 Main results

### Theorem K10.1. Poincaré-Birkhoff theorem

**Statement.** Let  $F$  be an orientation-preserving area-preserving homeomorphism of the closed annulus  $A = \mathbb{S}^1 \times [0, 1]$  that preserves both boundary components. Suppose a lift  $\tilde{F}(x, r) = (X(x, r), R(x, r))$  satisfies the strict opposite boundary twist  $X(x, 0) - x < 0 < X(x, 1) - x$  for every  $x$  (or the reversed inequalities). Then  $F$  has at least two distinct fixed points in the interior of  $A$ .

*Proof.* It is enough to treat the displayed twist orientation. Let

$$S = \mathbb{R} \times [0, 1], \quad \tilde{F}(x, r) = (X(x, r), R(x, r))$$

be the chosen lift, so  $\tilde{F}(x + 1, r) = \tilde{F}(x, r) + (1, 0)$ . Put

$$V(z) = \tilde{F}(z) - z. \tag{10.1.1}$$

On the lower boundary  $V(x, 0)$  is a nonzero horizontal vector pointing to the left, and on the upper boundary  $V(x, 1)$  is a nonzero horizontal vector pointing to the right.

We use two elementary parts of the Brown-Neumann argument and prove them here in the form needed.

**Displacement-index lemma.** Let  $C : [0, 1] \rightarrow S$  be an arc from the lower to the upper boundary which avoids the fixed points of  $F$ . Choose a continuous argument of  $V(C(t))/|V(C(t))|$  and define

$$\text{ind}_{\tilde{F}}(C) = \frac{1}{2\pi} \Delta_C \arg V. \tag{10.1.2}$$

Because the initial vector points left and the final vector points right, (10.1.2) belongs to  $\mathbb{Z} + \frac{1}{2}$ . If  $C_0, C_1$  are two such arcs, the difference of their indices is the sum of the local fixed-point indices in the annular region swept between them (counted in one fundamental domain). Indeed, concatenate  $C_0$ , a top boundary arc,  $-C_1$ , and a bottom boundary arc. The boundary contributions are constant horizontal directions, hence have zero variation. The winding number of  $V/|V|$  on the resulting Jordan curves is, by the defining degree formula for planar fixed-point index, the sum of the indices of the zeros of  $V$  inside. Approximation by finitely many Jordan components handles

self-intersections and gives

$$\operatorname{ind}_{\tilde{F}}(C_0) - \operatorname{ind}_{\tilde{F}}(C_1) = \sum_{z \in \mathcal{R} \cap \operatorname{Fix}(F)} \operatorname{ind}(F, z), \quad (10.1.3)$$

where  $\mathcal{R}$  is the annular region swept between the two arcs.

Assume, toward a contradiction, that  $F$  has at most one fixed point in the annulus. Boundary twist excludes boundary fixed points. Since  $F$  is orientation preserving and preserves the two ends, it is isotopic to the identity. Its Lefschetz number is therefore

$$L(F) = \operatorname{tr}(F_*|H_0(A)) - \operatorname{tr}(F_*|H_1(A)) = 1 - 1 = 0. \quad (10.1.4)$$

If there is no fixed point, the total index is of course zero. If there is one, it is isolated and the Lefschetz-Hopf formula together with (10.1.4) says its local index is zero. Hence (10.1.3) shows that

$$\operatorname{ind}_{\tilde{F}}(C) \text{ is the same for every lower-to-upper arc avoiding the fixed set.} \quad (10.1.5)$$

**Lemma 10.1 (annular displacement-index invariance).** Under the assumption of at most one fixed point, the total fixed-point index in a fundamental annulus is zero and (10.1.5) holds.

**Area-flow-line lemma.** Under the same assumption there are lower-to-upper arcs  $C_+$  and  $C_-$ , avoiding the fixed set, for which

$$\operatorname{ind}_{\tilde{F}}(C_+) = \frac{1}{2}, \quad \operatorname{ind}_{\tilde{F}}(C_-) = -\frac{1}{2}. \quad (10.1.6)$$

Here is the Brown-Neumann construction. Choose a locally finite periodic family  $W$  of very narrow vertical rectangles whose boundaries avoid the possible fixed point and on which  $|V|$  has a positive lower bound. Join a point  $z$  to  $\tilde{F}(z)$  inside each component of the complement of  $W$  by a short polygonal arc chosen continuously in the component. Across  $W$  connect the polygonal pieces by transverse arcs. The resulting oriented “ $F$ -lines” have the property that the direction of  $V$  along them is homotopic, through nonzero vectors, to their positive tangent direction.

Roll the construction up modulo a large horizontal period  $L$ . If no such positive  $F$ -line joins the lower boundary to the upper boundary, let  $U$  be the union of components reachable from the lower boundary by positive  $F$ -lines. Its frontier is an essential compact curve separating the two boundaries and every point of a thin one-sided collar of that frontier is moved strictly into  $U$ . Because the displacement is uniformly nonzero on  $W$ , the collar can be chosen with positive area. Its successive forward images are pairwise disjoint: the first return would cross the frontier in the forbidden direction. They all lie in the compact rolled-up annulus. This contradicts area preservation. Thus a positive  $F$ -line  $C_+$  crosses the annulus.

Orient  $C_+$  from bottom to top. Its tangent, and hence  $V$  through the nonzero-vector homotopy, starts pointing left and ends pointing right after one counterclockwise half-turn. Extra full turns can be removed by the innermost loop surgery used in the construction: an extra loop bounds a disk on which the  $F$ -line points strictly inward or

outward; the corresponding positive-area collar would again have pairwise disjoint iterates. Therefore the net turn is exactly  $+\pi$  and  $\text{ind}_{\tilde{F}}(C_+) = 1/2$ .

Apply the same construction to  $F^{-1}$ . It is area preserving and its boundary displacement has the opposite direction. Reversing the resulting  $F^{-1}$ -line gives a lower-to-upper  $F$ -line  $C_-$  whose displacement turns through  $-\pi$ . Thus its index is  $-1/2$ . The rectangles and connector arcs can be perturbed off the possible isolated fixed point without changing the index because that point has index zero. This proves (10.1.6).

**Lemma 10.2 (Brown-Neumann area-flow-line construction).** Area preservation plus opposite boundary displacement produces the two arcs in (10.1.6). The rolled-up-area contradiction is the load-bearing place where area preservation is used; the result is false for arbitrary annulus homeomorphisms with the same boundary displacement.

Now (10.1.5) says  $C_+$  and  $C_-$  must have the same displacement index, while (10.1.6) says their indices are  $1/2$  and  $-1/2$ . This contradiction shows that  $F$  cannot have zero or one fixed point. Hence it has at least two distinct fixed points, and boundary twist forces both to lie in the interior. The reversed boundary inequalities are obtained by replacing the angular coordinate by its negative.  $\square$

### Lemma K10.2. Aubry crossing lemma

**Statement.** In the scalar monotone-twist variational setting, let  $x = (x_i)$  and  $y = (y_i)$  be two distinct globally minimizing configurations for the same generating function. If their graphs cross, then they cannot cross a second time: two separated crossings would permit a cut-and-paste of finite segments that, by the strict twist (submodularity) inequality, strictly lowers the sum of the two finite actions, contradicting minimality. Thus two distinct minimizers have the single-crossing property.

*Proof.* We use the positive twist convention of the definition of exact monotone twist map and generating function above, so the generating function  $S$  satisfies  $S_{12} < 0$ . If  $a < b$  and  $c < d$ , then

$$\begin{aligned} S(a, c) + S(b, d) - S(a, d) - S(b, c) \\ = \int_a^b \int_c^d S_{12}(u, v) dv du < 0. \end{aligned} \quad (10.3.1)$$

Thus

$$S(a, c) + S(b, d) < S(a, d) + S(b, c). \quad (10.3.2)$$

This is the strict Monge, or strict submodularity, inequality.

**Lemma 10.3 (strict quadrilateral inequality).** Equation (10.3.2) follows directly from the twist sign and is strict whenever both coordinate inequalities are strict.

Let  $x = (x_i)$  and  $y = (y_i)$  be distinct globally minimizing configurations. Suppose there are two separated crossings. After interchanging  $x$  and  $y$  and shifting indices, choose the first and last crossing positions so that for some  $m < n$

$$x_m \leq y_m, \quad x_{m+1} > y_{m+1},$$



and later

$$x_n \geq y_n, \quad x_{n+1} < y_{n+1}, \quad (10.3.3)$$

with no change of order outside the chosen crossing interval. Define two spliced strings by taking the lower of the two configurations through the middle interval for one string and the upper one for the other; equivalently, cut both configurations immediately after the first crossing and immediately after the last crossing and exchange the two middle pieces. The two new strings have exactly the same four exterior endpoints as the original pair. All action terms away from the two splice sites are merely permuted, so they cancel in the sum of the two action differences.

At a splice site, the old pair of edges connects opposite corners of an ordered rectangle in  $(x_i, x_{i+1})$ -space, whereas the new pair connects equal-order corners. By (10.3.2) the sum of the two new edge actions is strictly smaller than the sum of the two old edge actions. Applying this at both separated crossings gives

$$A_{[m,n+1]}(x^{\text{new}}) + A_{[m,n+1]}(y^{\text{new}}) < A_{[m,n+1]}(x) + A_{[m,n+1]}(y). \quad (10.3.4)$$

On the other hand, global minimality of  $x$  with its fixed endpoints gives  $A(x) \leq A(x^{\text{new}})$ , and global minimality of  $y$  gives  $A(y) \leq A(y^{\text{new}})$  on the same finite interval. Adding these two inequalities contradicts (10.3.4).

If one of the inequalities at a crossing is an equality, perturb the cut one index inward to a strict crossing; if the two configurations share a point, the discrete Euler-Lagrange uniqueness implied by the twist relation either makes the adjacent order strict or makes the two configurations coincide. Thus two distinct minimizers can have at most one crossing. This is the single-crossing property.

**Lemma 10.4 (cut-and-paste crossing exclusion).** Two separated crossings give two strict submodularity gains while preserving the exterior endpoints, contradicting the defining finite-segment minimality of both configurations.  $\square$

### Theorem K10.3. Aubry-Mather existence theorem

**Statement.** Let  $F$  be an exact monotone twist diffeomorphism of the closed annulus, with boundary rotation numbers  $\rho_- \leq \rho_+$ . For every  $\rho \in [\rho_-, \rho_+]$  there exists a nonempty compact  $F$ -invariant ordered action-minimizing Aubry-Mather set  $\mathcal{M}_\rho$  of rotation number  $\rho$ . Its lifted angular projection is order preserving and one-to-one. If  $\rho$  is irrational,  $F|_{\mathcal{M}_\rho}$  is minimal and  $\mathcal{M}_\rho$  is the graph of a Lipschitz function over a compact subset of the circle; if  $\rho = p/q$  is rational in lowest terms, there exists a minimizing Birkhoff  $q$ -periodic orbit of rotation number  $p/q$ .

*Proof.* We use the positive twist convention. Let  $S(x, X)$  be the generating function from the definition of exact monotone twist map and generating function above. First extend it, without changing it on the compact lifted annulus under consideration, to a diagonal-periodic function on  $\mathbb{R}^2$  satisfying

$$S_{12} \leq -\kappa < 0 \quad (10.2.1)$$

and quadratic coercivity as  $|X - x| \rightarrow \infty$ . Such an extension is obtained by extending

$S_{12}$  as a negative periodic function and integrating twice, then adding a periodic one-variable correction to match the original first and second derivatives on the boundary of the twist strip. Because the boundary orbits form ordered barriers, every Birkhoff minimizer whose rotation lies between their rotation numbers stays in the original strip. Thus the extension is only a compactness device and creates no new orbit used below.

**Lemma 10.5 (coercive extension and rational minimizers).** The exact twist generating function may be extended coercively while preserving strict submodularity; minimizers with rotation between the two boundary rotation numbers remain between the boundary configurations by the crossing lemma the Aubry crossing lemma.

**Step 1: rational rotation numbers.** Let  $\rho = p/q \in (\rho_-, \rho_+)$  in lowest terms. On sequences satisfying  $x_{i+q} = x_i + p$ , consider

$$\mathcal{A}_{p,q}(x_0, \dots, x_{q-1}) = \sum_{i=0}^{q-1} S(x_i, x_{i+1}), \quad x_q = x_0 + p. \quad (10.2.2)$$

Modulo simultaneous integer translation, coercivity makes the sublevel sets compact, so (10.2.2) has a minimizer. Its Euler equations are

$$\partial_2 S(x_{i-1}, x_i) + \partial_1 S(x_i, x_{i+1}) = 0. \quad (10.2.3)$$

With  $r_i = -\partial_1 S(x_i, x_{i+1})$ , equations (10.2.3) are exactly the generating function equations for a lifted  $F$ -orbit. It is  $q$ -periodic modulo the deck translation  $(x, r) \mapsto (x + p, r)$ .

Compare this minimizing configuration with every deck-time translate  $T_{a,b}x$ , where  $(T_{a,b}x)_i = x_{i+b} + a$ . Both are minimizing. If they had two crossings, the Aubry crossing lemma would give a contradiction; periodicity rules out a single persistent crossing. Hence every translate is everywhere ordered with  $x$ . This is the Birkhoff order property. A standard counting consequence, included here, is

$$|x_j - x_i - (j - i)p/q| \leq 1. \quad (10.2.4)$$

Indeed, choose the unique integer  $a$  for which  $a \leq (j - i)p/q < a + 1$ ; the two translates differing by  $(a, j - i)$  bracket the configuration by order, and evaluation at index  $i$  gives (10.2.4). We have therefore produced a minimizing Birkhoff  $q$ -periodic orbit of rotation  $p/q$ .

**Step 2: irrational rotation numbers.** Let  $\rho \in (\rho_-, \rho_+)$  be irrational and choose reduced rationals  $p_\nu/q_\nu \rightarrow \rho$ . Take the rational minimizing configurations just constructed and translate them so that  $x_0^{(\nu)} \in [0, 1]$ . Estimate (10.2.4) bounds every fixed coordinate  $x_j^{(\nu)}$  uniformly in  $\nu$ . The twist relation  $r_i = -\partial_1 S(x_i, x_{i+1})$  then bounds the associated action coordinates on every fixed finite index range. A diagonal subsequence converges to a bi-infinite configuration  $x = (x_i)$ .

Every finite segment of  $x$  is minimizing: otherwise a strictly better finite competitor would, by continuity of  $S$ , also improve all sufficiently large  $x^{(\nu)}$ , contradicting their

minimality. Passing to the limit in (10.2.4) gives

$$|x_j - x_i - (j - i)\rho| \leq 1, \quad (10.2.5)$$

so the rotation number is  $\rho$ . The crossing lemma passes to limits and shows that the orbit and all of its integer/time translates are ordered. Let  $\tilde{\mathcal{A}}_\rho$  be the closure of this lifted orbit modulo deck translation and let  $\mathcal{A}_\rho$  be its projection to the annulus. It is nonempty, compact, invariant, ordered, and every point on it lies on a globally minimizing orbit. Choose a nonempty compact invariant subset minimal under inclusion and call it  $\mathcal{M}_\rho$ . Then  $F|_{\mathcal{M}_\rho}$  is minimal by construction. Since  $\rho$  is irrational, no point is periodic.

**Lemma 10.6 (irrational compactness limit).** Rational Birkhoff minimizers with  $p_\nu/q_\nu \rightarrow \rho$  have the uniform displacement estimate (10.2.4); a diagonal limit is a globally minimizing ordered configuration with rotation  $\rho$ .

**Step 3: injective projection and the Lipschitz graph bound.** Let  $(x, r), (x', r')$  be two distinct lifted points of an ordered minimizing set, with  $x < x'$ . Order is preserved by  $F$ , so their first image coordinates obey

$$X(x, r) \leq X(x', r'). \quad (10.2.6)$$

On the compact annulus, the twist has a lower bound  $\partial_r X \geq \kappa > 0$  and  $|\partial_x X| \leq L$ . Hence

$$0 \geq X(x, r) - X(x', r') \geq \kappa(r - r') - L|x - x'|,$$

which gives  $r - r' \leq (L/\kappa)|x - x'|$ . Interchanging the two points and using the inverse order gives the opposite inequality, so

$$|r - r'| \leq \frac{L}{\kappa}|x - x'|. \quad (10.2.7)$$

In particular, if  $x = x'$ , then  $r = r'$ ; the angular projection is one-to-one. Equation (10.2.7) says the set is a Lipschitz graph over its compact angular projection.

**Lemma 10.7 (order-to-Lipschitz graph).** The twist lower bound and preservation of order give the quantitative graph estimate (10.2.7), hence injectivity of angular projection.

For rational  $\rho = p/q$  we take the minimizing periodic orbit from Step 1 as  $\mathcal{M}_\rho$ ; for irrational  $\rho$  we take the minimal set from Step 2. At the endpoints  $\rho_\pm$ , approximate from the interior and repeat the compactness argument; the limiting minimizing set lies on or between the corresponding boundary barriers and has endpoint rotation. Thus every  $\rho \in [\rho_-, \rho_+]$  has the required nonempty compact invariant ordered action-minimizing Aubry-Mather set, with all the rational/irrational properties in the statement.  $\square$

#### Theorem K10.4. Birkhoff graph theorem

**Statement.** Let  $F$  be an orientation-preserving  $C^1$  monotone twist diffeomorphism of the annulus that preserves each end. Every essential (homotopically nontrivial) invariant circle of  $F$  is the graph of a Lipschitz function over the angular circle.

More generally, every minimizing Aubry–Mather set is ordered: after lifting to  $\mathbb{R}^2$ , distinct minimizing orbits satisfy the no-crossing order, its angular projection is injective on the set, and the set is contained in a Lipschitz graph over that projection (hence in a Lipschitz extension over the full circle).

*Proof.* Let  $\Gamma$  be an essential invariant circle and choose a lift  $\tilde{\Gamma} \subset \mathbb{R} \times \mathbb{R}$ . Its complement has a lower and an upper unbounded component. Because  $F$  preserves the two ends and  $F(\Gamma) = \Gamma$ , both components are invariant.

**Lemma 10.14 (Birkhoff vertical-fiber lemma).** Every vertical line meets  $\tilde{\Gamma}$  in exactly one point. Existence follows from essentiality. For uniqueness suppose a vertical line  $x = x_0$  meets the frontier at  $(x_0, r_-) \neq (x_0, r_+)$ . Join the two points by the vertical segment  $V$ . For a lift  $F = (X, R)$  of a positive twist map,  $\partial_r X > 0$ , so  $F(V)$  is transverse to vertical fibers and its angular coordinate runs through a nondegenerate interval. Choose two nearby vertical segments, one based in the lower component and one in the upper component, and close them by short arcs avoiding the frontier. Their images have opposite angular order across  $F(V)$ . The Jordan separation theorem then forces the image of one lower point to cross the invariant frontier. This is impossible because the lower component is invariant. Thus the intersection is unique. The same proof for a negative twist uses  $F^{-1}$ ; the conclusion is unchanged.

Consequently  $\tilde{\Gamma}$  is a graph  $r = u(x)$ , with  $u(x+1) = u(x)$ . Continuity follows because the projection  $\tilde{\Gamma} \rightarrow \mathbb{R}$  is a continuous bijection locally between one-manifolds and is proper on a fundamental strip.

We now obtain the Lipschitz estimate. On a compact annulus containing  $\Gamma$  choose

$$\partial_r X \geq \kappa > 0, \quad |\partial_x X| \leq A, \quad (10.6.1)$$

and analogous bounds for the inverse map. The map induced by  $F$  on  $\Gamma$  is an orientation-preserving circle homeomorphism, so for lifted points with  $x < x'$  sufficiently close we have

$$X(x, u(x)) < X(x', u(x')) \quad (10.6.2)$$

after choosing the same lift branch. The mean-value formula gives

$$0 < X(x', u(x')) - X(x, u(x)) \leq A|x' - x| + \partial_r X(\zeta)[u(x') - u(x)].$$

If  $u(x') < u(x)$  this yields

$$u(x) - u(x') \leq A\kappa^{-1}|x' - x|. \quad (10.6.3)$$

Applying the same argument to  $F^{-1}$  controls the opposite sign. By periodicity, finitely many local intervals cover the circle, and enlarging the constant for pairs separated by more than the local radius gives

$$|u(x') - u(x)| \leq C|x' - x| \quad (10.6.4)$$

for all lifted  $x, x'$  with  $|x' - x| \leq 1/2$ . Thus every essential invariant circle is a global Lipschitz graph.

**Lemma 10.15 (quantitative Birkhoff graph bound).** On every compact twist an-

nulus on which both  $F$  and  $F^{-1}$  have bounded first derivatives and  $|\partial_r X|$  is bounded away from zero, the graph constant in (10.6.4) is uniform.

For the Aubry-Mather assertion let  $\tilde{\mathcal{A}}_\rho$  be any minimizing set. the Aubry crossing lemma says that two lifted minimizing configurations are ordered and cannot cross twice. Hence two points of  $\tilde{\mathcal{A}}_\rho$  with the same angular coordinate must coincide: otherwise their reconstructed configurations would have equality at one index and strict order at an adjacent index, contrary to the strict twist Euler relation. Thus angular projection is injective. Lemma 10.7, proved in the Aubry-Mather existence theorem, gives directly

$$|r - r'| \leq C|x - x'| \quad (10.6.5)$$

on the minimizing set. The projected set is closed, and the Lipschitz function on it extends to the whole circle with the same constant by the McShane extension formula. This extension is only a geometric carrier; no invariance is asserted across the complementary gaps. The required ordered, injective, Lipschitz-graph conclusions follow.  $\square$

### Theorem K10.5. Mather beta function

**Statement.** Let  $L : TM \times \mathbb{R}/\mathbb{Z} \rightarrow \mathbb{R}$  be a time-periodic positive-definite Lagrangian satisfying Mather's smoothness, fiberwise strict-convexity, superlinear-growth, and complete Euler-Lagrange-flow hypotheses. Let  $\mathfrak{M}_L$  be the invariant probability measures of the Euler-Lagrange flow with finite average action and rotation vector as in Definition K10.5, and for  $c \in H^1(M; \mathbb{R})$  put  $A_c(\mu) = A(\mu) - \langle c, \rho(\mu) \rangle$ . Then

$$\beta(h) = \min\{A(\mu) : \mu \in \mathfrak{M}_L, \rho(\mu) = h\}, \quad -\alpha(c) = \min\{A_c(\mu) : \mu \in \mathfrak{M}_L\},$$

the functions  $\alpha$  and  $\beta$  are finite, convex, superlinear, and Legendre-Fenchel conjugate, and

$$\alpha(c) + \beta(h) \geq \langle c, h \rangle.$$

Equality holds exactly when  $c \in \partial\beta(h)$ , equivalently  $h \in \partial\alpha(c)$ . If  $c \in \partial\beta(h)$ , then every  $\beta$ -minimizing invariant measure with rotation vector  $h$  also minimizes  $A_c$ ; conversely, a measure that minimizes  $A_c$  and has rotation vector  $h$  realizes equality in the Fenchel relation.

*Proof.* Put

$$P = TM \times \mathbb{T}$$

and denote the Euler-Lagrange flow by  $\Phi^t$ . Since  $M \times \mathbb{T}$  is compact while the velocity is not, compactify  $P$  by one point  $\infty$  and extend  $L$  by  $L(\infty) = +\infty$ . Tonelli superlinearity implies that this extension is lower semicontinuous and bounded below after adding a constant. The set of probability measures on the compact space  $P^* = P \cup \{\infty\}$  is weak-\* compact; invariance under  $\Phi^t$  is a closed condition.

**Step 1: compactness, lower semicontinuity, and rotation.** If  $\mu_j \rightharpoonup \mu$  weakly, Portmanteau gives

$$A(\mu) = \int L d\mu \leq \liminf_j \int L d\mu_j. \quad (10.4.1)$$

In particular, a finite-action limit has no mass at  $\infty$ . Let  $\eta$  be a smooth closed one-form on  $M$ . Since  $M$  is compact,  $|\eta_x(v)| \leq C_\eta|v|$ . Superlinearity says that for every  $R > 0$  there is  $C_R$  such that

$$|v| \leq R^{-1}L(x, v, t) + C_R. \quad (10.4.2)$$

Thus a uniform action bound makes the family  $\eta_x(v)$  uniformly integrable. After truncating it at  $|v| \leq K$ , weak convergence passes the integral to the limit, while (10.4.2) makes the tails uniformly small. Hence

$$\int \eta_x(v) d\mu_j \longrightarrow \int \eta_x(v) d\mu \quad (10.4.3)$$

on every bounded-action family. By the definition of average action and rotation vector of an invariant measure above this is continuity of the rotation vector there.

**Lemma 10.8 (action-sublevel compactness and rotation continuity).** Invariant probability measures with uniformly bounded average action are precompact in the one-point compactification; (10.4.1) is lower semicontinuity of action and (10.4.3) gives continuity of the rotation vector on such families.

**Step 2: every homology class is realized by an invariant measure.** Identify  $H_1(M; \mathbb{R})$  with the finite-dimensional real vector space associated with the free part of the deck group of the maximal abelian (homology) cover  $\widehat{M} \rightarrow M$ . Fix  $h \in H_1(M; \mathbb{R})$ . Choose deck transformations  $T_n$  so that

$$\frac{[T_n]}{n} \longrightarrow h. \quad (10.4.4)$$

Fix  $\widehat{x}_0 \in \widehat{M}$ . Tonelli's finite-time theorem gives an action-minimizing Euler-Lagrange curve  $\gamma_n : [0, n] \rightarrow \widehat{M}$  joining  $\widehat{x}_0$  to  $T_n\widehat{x}_0$ . The average actions are uniformly bounded: represent a basis of the deck group by fixed smooth loops, concatenate  $O(n)$  of them to reach  $T_n\widehat{x}_0$ , and compare  $\gamma_n$  with that uniformly bounded speed path.

Define the occupational measure

$$\mu_n = \frac{1}{n} \int_0^n \delta_{(\gamma_n(t), \dot{\gamma}_n(t), t \bmod 1)} dt \quad (10.4.5)$$

after projection to  $P$ . For every bounded continuous  $u$  and fixed  $\tau$ ,

$$\left| \int u \circ \Phi^\tau d\mu_n - \int u d\mu_n \right| \leq \frac{2|\tau|}{n} \|u\|_\infty. \quad (10.4.6)$$

A subsequential weak limit  $\mu$  is therefore invariant. If  $\eta$  represents  $c \in H^1(M; \mathbb{R})$ , then

$$\int \eta_x(v) d\mu_n = \frac{1}{n} \int_{\gamma_n} \eta = \left\langle c, \frac{[T_n]}{n} \right\rangle \longrightarrow \langle c, h \rangle. \quad (10.4.7)$$

The uniform action bound and Lemma 10.8 allow passage to the limit, so  $\rho(\mu) = h$ . Hence the constraint set defining  $\beta(h)$  is nonempty for every  $h$ .

**Lemma 10.9 (prescribed-rotation invariant measures).** Long Tonelli minimizers in the homology cover and their occupational measures produce at least one invariant

finite-action measure of every prescribed rotation vector.

**Step 3: existence, convexity, and closedness of  $\beta$ .** Let

$$E = \{(\rho(\mu), z) : \mu \text{ invariant, } A(\mu) \leq z\} \subset H_1(M; \mathbb{R}) \times \mathbb{R}. \quad (10.4.8)$$

It is convex because both  $A$  and  $\rho$  are affine in  $\mu$ . It is closed: if  $(h_j, z_j) \rightarrow (h, z)$  with  $(h_j, z_j) \in E$ , choose  $\mu_j$  realizing the inequality. The  $z_j$  are bounded, so Lemma 10.8 gives a subsequential invariant limit  $\mu$  with  $\rho(\mu) = h$  and  $A(\mu) \leq z$ . Thus  $(h, z) \in E$ . By Lemma 10.9 every vertical fiber is nonempty, and therefore  $E$  is exactly the epigraph of

$$\beta(h) = \min\{A(\mu) : \rho(\mu) = h\}. \quad (10.4.9)$$

The minimum is attained by the same compactness argument. Since its epigraph is closed and convex,  $\beta$  is finite and convex.

**Step 4: the  $c$ -minimization and Legendre-Fenchel duality.** For  $c \in H^1(M; \mathbb{R})$  choose a smooth closed representative  $\eta$ . The modified Lagrangian

$$L_c(x, v, t) = L(x, v, t) - \eta_x(v) \quad (10.4.10)$$

is still Tonelli because subtraction of a fiber-linear term does not change strict convexity and is dominated at infinity by superlinearity. Hence the same compactness proof gives an invariant measure minimizing  $A_c(\mu) = A(\mu) - \langle c, \rho(\mu) \rangle$ . By first minimizing at fixed rotation,

$$\begin{aligned} -\alpha(c) &= \min_{\mu} (A(\mu) - \langle c, \rho(\mu) \rangle) \\ &= \min_h (\beta(h) - \langle c, h \rangle). \end{aligned} \quad (10.4.11)$$

Therefore

$$\alpha(c) = \sup_h \{\langle c, h \rangle - \beta(h)\} = \beta^*(c). \quad (10.4.12)$$

Since  $\beta$  is finite, convex, and lower semicontinuous, the Fenchel-Moreau theorem in finite dimension yields

$$\beta = \alpha^*. \quad (10.4.13)$$

Both functions are finite everywhere: finiteness of  $\beta$  is Lemma 10.9 plus finite action of the constructed measure; finiteness of  $\alpha$  follows from existence of a minimum for (10.4.10). A finite convex function whose conjugate is finite everywhere is superlinear. Indeed, if  $f(x_j)/|x_j|$  stayed bounded along  $|x_j| \rightarrow \infty$ , a linear functional pointing along a limiting direction with sufficiently large coefficient would make  $\sup_x (\langle c, x \rangle - f(x)) = +\infty$ . Apply this observation first to  $\beta$  using finiteness of  $\alpha = \beta^*$  and then to  $\alpha$  using finiteness of  $\beta = \alpha^*$ . Thus both are superlinear.

**Lemma 10.10 (epigraph and duality closure).** The closed convex epigraph (10.4.8) gives the attained function  $\beta$ ; minimization of the cohomologically tilted action gives (10.4.12), and finite-dimensional biconjugacy gives (10.4.13) and superlinearity.

**Step 5: Fenchel equality and minimizing measures.** Equations (10.4.12)–

(10.4.13) immediately give the Fenchel inequality

$$\alpha(c) + \beta(h) \geq \langle c, h \rangle. \quad (10.4.14)$$

By the definition of subdifferential in the definition of Legendre–Fenchel conjugate and subdifferential above, equality in (10.4.14) is equivalent to

$$c \in \partial\beta(h), \quad (10.4.15)$$

and, symmetrically, to  $h \in \partial\alpha(c)$ .

Let  $\mu_h$  minimize  $A$  among measures with  $\rho = h$ . If equality holds, then

$$A_c(\mu_h) = \beta(h) - \langle c, h \rangle = -\alpha(c),$$

so  $\mu_h$  minimizes  $A_c$ . Conversely, if  $\mu$  minimizes  $A_c$  and has rotation  $h$ , then

$$-\alpha(c) = A_c(\mu) = A(\mu) - \langle c, h \rangle \geq \beta(h) - \langle c, h \rangle.$$

Fenchel inequality supplies the reverse inequality, hence equality holds and, in particular,  $A(\mu) = \beta(h)$ . This proves all measure-minimizer statements and completes the theorem.  $\square$

### Proposition K10.6. Peierls barrier criterion

**Statement.** Let  $F$  be an exact monotone twist map with generating function  $h$ , and let  $\rho$  be irrational. The Peierls barrier  $P_\rho(\xi)$  is nonnegative. It vanishes identically in  $\xi$  if and only if  $F$  possesses an essential invariant circle of rotation number  $\rho$ . If  $P_\rho(\xi) > 0$  for some  $\xi$ , no invariant circle of rotation number  $\rho$  exists; the minimizing Aubry–Mather set of that rotation number is then a proper ordered invariant set whose angular projection has a gap.

*Proof.* Work in a lift and fix the irrational rotation number  $\rho$ . Let  $\mathcal{M}_\rho$  be the family of globally minimizing configurations and put

$$A_\rho = \{x_0 : x = (x_i)_{i \in \mathbb{Z}} \in \mathcal{M}_\rho\} \subset \mathbb{R}. \quad (10.5.1)$$

By the Aubry–Mather existence theorem this set is nonempty, closed, invariant under integer translation, and totally ordered. By the Aubry crossing lemma two minimizing configurations cannot have two separated crossings. These facts imply that if  $\xi \notin A_\rho$ , the component  $(\xi^-, \xi^+)$  of  $\mathbb{R} \setminus A_\rho$  containing  $\xi$  has unique extremal bounding minimizers  $x^-, x^+$  satisfying

$$x_0^- = \xi^-, \quad x_0^+ = \xi^+, \quad x_i^- < x_i^+ \quad (i \in \mathbb{Z}). \quad (10.5.2)$$

The inequalities are strict because equality at one index and the twist Euler relations would force equality at every index.

Use the normalization in the definition of Peierls barrier above. Thus  $P_\rho(\xi) = 0$  for  $\xi \in A_\rho$ ; in a gap it is the minimum constrained action defect among strings lying between the two bounding minimizers and satisfying  $x_0 = \xi$ . The infinite action defect is the limit of its finite restrictions. The limit exists: for  $m < n$ , the minimizing property



of  $x^-$  and  $x^+$  bounds the variation of the partial defect by the two endpoint crossing terms, and the Birkhoff displacement estimate from the definition of rotation number of an ordered twist orbit above makes those endpoint terms a Cauchy sequence as  $m \rightarrow -\infty$ ,  $n \rightarrow +\infty$ . Equivalently one may use the standard monotone limit over finite constrained strings; the two constructions give the same number.

**Lemma 10.11 (zero set of the irrational Peierls barrier).** For irrational  $\rho$ ,

$$P_\rho(\xi) \geq 0, \quad P_\rho(\xi) = 0 \iff \xi \in A_\rho. \quad (10.5.3)$$

Indeed, every finite piece of either bounding configuration minimizes the action with its endpoints fixed. Cutting a constrained string at the two ends and comparing it with the corresponding piece of  $x^-$  gives a nonnegative action defect; passage to the limit gives  $P_\rho \geq 0$ . If  $\xi \in A_\rho$ , the minimizing configuration through  $\xi$  gives zero by definition. Conversely suppose  $\xi \in (\xi^-, \xi^+)$  and  $P_\rho(\xi) = 0$ . Finite constrained minimizers, normalized by  $x_0 = \xi$ , lie in the compact coordinate intervals  $[x_i^-, x_i^+]$ . A diagonal limit  $z = (z_i)$  therefore exists. Every finite segment of  $z$  is minimizing: if a competitor lowered its action by  $\delta > 0$ , the same replacement in all sufficiently long finite constrained minimizers would lower their defects by at least  $\delta/2$ , contradicting the convergence to zero. Hence  $z$  is a globally minimizing configuration of rotation  $\rho$  with  $z_0 = \xi$ , so  $\xi \in A_\rho$ , a contradiction. This proves (10.5.3). In particular the barrier is strictly positive at every interior point of every genuine gap.

We next identify the case in which  $A_\rho$  has no gaps. For a minimizing configuration  $x$ , recover the momentum by

$$r_i = -\partial_1 h(x_i, x_{i+1}). \quad (10.5.4)$$

If two minimizing orbits have  $x_0 < x'_0$ , the Aubry crossing theorem gives  $x_i < x'_i$  for every  $i$ . On a compact fundamental annulus the twist has a uniform constant  $\kappa > 0$  with  $-h_{12} \geq \kappa$ , while  $|h_{11}| + |h_{12}|$  is bounded by some  $L$ . Comparing (10.5.4) for the two ordered orbits and using the same estimate as Lemma 10.7 gives

$$|r_0 - r'_0| \leq C_0 |x_0 - x'_0|. \quad (10.5.5)$$

Thus the lifted minimizing set is the graph of a Lipschitz function over  $A_\rho$ . If  $A_\rho = \mathbb{R}$ , integer equivariance gives a periodic Lipschitz function  $r = u(x)$  on the whole line. Its graph is invariant because shifting a minimizing configuration sends it to another minimizing configuration and (10.5.4) is precisely the orbit reconstruction rule. Its projection to the annulus is therefore an essential invariant circle. Its rotation number is  $\rho$  by the definition of rotation number of an ordered twist orbit above.

**Lemma 10.12 (full projected Aubry set gives a circle).** If  $A_\rho = \mathbb{R}$ , the minimizing Aubry set is an invariant periodic Lipschitz graph of rotation number  $\rho$ .

It remains to prove the converse implication from an invariant circle to zero barrier. We record the variational fact needed here because it is the place where exactness is essential.

**Lemma 10.13 (rotational invariant circles are calibrated).** Let  $\Gamma$  be an essential invariant circle of an exact monotone twist map. Then its lift is an ordered graph, and every finite segment of every orbit on  $\Gamma$  minimizes the generating-function action

among strings with the same endpoints. To prove it, let  $U^-$  be the component below a lift of  $\Gamma$ . Since the map preserves the two ends and  $\Gamma$ , both  $U^-$  and its complement are invariant. If a vertical fiber met  $\Gamma$  in two points, the positive twist would send the vertical segment between them to a curve whose angular coordinate is strictly monotone; a small rectangle with two opposite sides in  $U^-$  would then have one image side above and one below the invariant frontier. By orientation and area preservation this forces an image point of  $U^-$  into the other component, contradiction. Thus the lift is a graph  $r = u(x)$ . The same argument applied to every integer translate shows that the graph and all its translates are ordered.

Take an orbit segment  $z_0, \dots, z_n$  on the graph and a competitor with the same endpoints. If the competitor had smaller action, replace it by an action-minimizing competitor. The discrete Euler equation then reconstructs an orbit segment. Translate copies of the invariant graph until this new segment is trapped between two consecutive copies. At the first and last contacts, the new segment and a graph orbit have the same endpoints but are on opposite sides at an interior index. Cutting and exchanging the two middle pieces gives the strict quadrilateral improvement from the Aubry crossing lemma, contradicting minimality of the competitor. Hence every graph segment is minimizing. This proves Lemma 10.13. The angular dynamics on the circle is an orientation-preserving circle homeomorphism; for irrational rotation  $\rho$  every point of the circle belongs to a minimizing orbit of rotation  $\rho$ . Since the graph projects onto every angular phase,  $A_\rho = \mathbb{R}$ .

Combining Lemma 10.11–Lemma 10.13 gives

$$P_\rho \equiv 0 \iff A_\rho = \mathbb{R} \iff \text{there is an essential invariant circle of rotation } \rho. \quad (10.5.6)$$

If  $P_\rho(\xi) > 0$  at one point, (10.5.6) excludes such a circle, while Lemma 10.11 says that  $\xi$  lies in a nonempty complementary interval of  $A_\rho$ . Hence the minimizing Aubry–Mather set is a proper ordered invariant set and its angular projection has a gap, exactly as asserted.  $\square$

## 10.4 Worked examples

### Worked example: the integrable exact twist

Consider the lift

$$\tilde{F}(x, r) = (x + r, r)$$

of the annulus map  $F(\theta, r) = (\theta + r, r)$ . The generating function

$$S(x, X) = \frac{1}{2}(X - x)^2$$

satisfies

$$-\partial_1 S = X - x = r, \quad \partial_2 S = X - x = R,$$

so it generates  $F$ . Moreover

$$\partial_{12} S = -1 < 0,$$

which is the positive-twist convention used in this chapter.

For any real  $\rho$  the configuration

$$x_i = x_0 + i\rho$$

is a discrete Euler-Lagrange orbit. It is globally minimizing because for fixed endpoints the quadratic action

$$\sum_{i=m}^{n-1} \frac{1}{2} (z_{i+1} - z_i)^2$$

is minimized when all increments are equal. The associated orbit lies on the invariant graph  $r = \rho$ , and its rotation number is  $\rho$ . Thus, for irrational  $\rho$ , every angular point lies on a minimizing invariant circle and the Peierls barrier is identically zero.

### Check your understanding

Fix  $m < n$  and endpoints  $z_m = a$ ,  $z_n = b$ . Use Cauchy-Schwarz to show directly that

$$\sum_{i=m}^{n-1} (z_{i+1} - z_i)^2 \geq \frac{(b-a)^2}{n-m},$$

with equality exactly for equal increments.

### Boundary case: loss of twist

The rigid rotation

$$F(\theta, r) = (\theta + \alpha, r)$$

is exact symplectic but has  $\partial_r X = 0$ . The angle increment no longer determines the momentum, and there is no strictly submodular generating function of the twist type. Ordered-orbit conclusions that depend on strict crossing inequalities therefore cannot be imported merely from symplecticity. This example separates the twist hypothesis from area preservation.

### Common pitfall

For an irrational rotation number, an Aubry-Mather set need not be a smooth invariant circle. The Birkhoff graph theorem gives an ordered Lipschitz graph on the minimizing set, while gaps may remain. Confusing an ordered minimizing set with a KAM circle erases the distinction that the Peierls barrier is designed to detect.

## 10.5 Chapter summary

### What to remember

- Exact monotone twist maps admit generating functions, turning orbit equations into discrete Euler-Lagrange equations.
- Strict submodularity gives the single-crossing principle and therefore order, graph structure, and compactness for minimizers.
- Aubry-Mather sets exist for every rotation number in the twist interval, even when a smooth invariant circle has broken.
- The Peierls barrier distinguishes a filled invariant circle from a minimizing Cantor set with gaps.
- Mather's  $\alpha$ - $\beta$  duality packages action minimization into finite-dimensional convex analysis on homology and cohomology.

### Looking ahead

K11 applies this twist-map language to a concrete geometric system: a smooth strictly convex billiard table. The chord-length generating function turns reflection into exact twist dynamics, while Lazutkin coordinates reveal a nearly integrable regime close to grazing collisions.

### Check your understanding

For an irrational rotation number, compare a KAM invariant circle, an Aubry-Mather minimizing set, and the condition that the Peierls barrier vanish. Which implications hold in the twist setting, and where can gaps remain?

## 10.6 Exercises and problems

### Exercise 10.1

**Difficulty.** Core.

**Related results.** Poincaré-Birkhoff theorem; Aubry-Mather existence theorem; Aubry crossing lemma.

**Problem.** Derive the standard twist map from its generating function, compute the twist derivative, and obtain the discrete Euler-Lagrange equation for minimizing configurations.

**Solution sketch.** Differentiate the generating function in both variables and compute its mixed second derivative.

### Exercise 10.2

**Difficulty.** Core.

**Related results.** Mather beta function; Peierls barrier criterion; Birkhoff graph theorem.

**Problem.** In the nontwist boundary example, locate the vanishing/change of sign of the twist derivative and show locally how orbit ordering can fail.

**Solution sketch.** Track two nearby initial actions through the angular update across the critical set and compare their order.

### Exercise 10.3

**Difficulty.** Advanced.

**Related results.** Aubry crossing lemma.

**Problem.** Prove the cut-and-paste form of Aubry's crossing lemma: two strictly minimizing configurations cannot cross twice.

**Solution sketch.** Use the strict twist/submodularity four-point inequality and compare the original and spliced actions.

### Exercise 10.4

**Difficulty.** Advanced.

**Related results.** Mather beta function.

**Problem.** Prove convexity of Mather's  $\beta$ -function first for rational rotation numbers using concatenated minimizing segments, then indicate the approximation step to irrational rotations.

**Solution sketch.** Control the bounded joining cost, divide by length, and pass to a rational-approximation limit.

### Exercise 10.5

**Difficulty.** Advanced.

**Related results.** Poincaré-Birkhoff theorem; Aubry-Mather existence theorem.

**Problem.** Take minimizing periodic configurations with  $p_n/q_n \rightarrow \rho$  and use ordering plus compactness to obtain an Aubry-Mather set of rotation number  $\rho$ , even when no smooth invariant circle exists.

**Solution sketch.** Normalize one coordinate, extract a diagonal subsequence, and pass minimality inequalities to the limit.

## **Part III**

# **Smooth and Hyperbolic Billiards**

# Smooth Convex Billiards, Caustics and Lazutkin Theory

## Why this chapter matters

A billiard table looks geometric rather than Hamiltonian: a particle travels in straight lines and reflects elastically at the boundary. Yet for a smooth strictly convex table the collision map is an exact twist map. The generating function is simply the Euclidean chord length between consecutive impact points. This observation imports the variational and KAM machinery of K10 into billiard geometry.

The most delicate region is near grazing incidence. There the billiard map is singular in ordinary coordinates but, after Lazutkin's curvature-weighted change of variables, it becomes a small perturbation of the integrable shear  $(x, y) \mapsto (x + y, y)$ . KAM theory then produces invariant curves corresponding to caustics accumulating at the boundary. The ellipse provides the exactly integrable model, while the Marvizi–Melrose expansion shows that near-boundary periodic trajectories encode an infinite sequence of curvature invariants.

**Prerequisites.** K05–K06 for finite-regularity KAM; K10 for exact twist maps and invariant graphs; elementary differential geometry of plane curves; Taylor expansion of chord length.

## Learning goals

- Derive the exact symplectic and twist identities for the billiard collision map from chord length.
- Understand why grazing collisions require curvature-adapted Lazutkin coordinates.
- Explain how the Lazutkin normal form leads to a positive-measure family of caustics.
- Verify complete integrability of an ellipse through its confocal conics.
- Interpret the Marvizi–Melrose coefficients as asymptotic geometric data encoded by long near-boundary periodic trajectories.

**Conceptual roadmap**

strict convexity  $\longrightarrow$  chord-length twist map  
 $\longrightarrow$  Lazutkin normal form  
 $\longrightarrow$  KAM caustics  
 $\longrightarrow$  near-boundary invariants.

The ellipse runs alongside this chain as the exact integrable benchmark: its invariant curves are produced by confocal caustics rather than perturbative persistence.

**Check your understanding**

Why is the billiard map naturally defined only on the open collision cylinder  $0 < \varphi < \pi$ ? What geometric event occurs as  $\varphi \rightarrow 0$  or  $\pi$ ?

**11.1 Definitions****Intuition**

**The billiard map remembers collisions, not flight time.** A phase point is a boundary position and an outgoing angle. The next collision defines a map of the collision cylinder. Its singular boundary corresponds to grazing trajectories.

**A caustic is an invariant curve drawn in configuration space.** If every segment of a trajectory is tangent to the same interior convex curve, that curve is a caustic. In phase space the corresponding trajectories lie on an invariant graph of the billiard twist map.

**Lazutkin coordinates flatten the boundary dynamics.** The powers  $\rho^{-2/3}$  and  $\rho^{1/3}$  are chosen so that the first nontrivial curvature dependence is absorbed into the coordinates. The leading map becomes the universal shear  $x_1 = x + y$ ,  $y_1 = y$ .

**Marvizi-Melrose invariants are intrinsic asymptotic length data.** As the period  $q$  grows, every simple winding-one  $q$ -periodic billiard polygon enters the glancing collar at scale  $q^{-1}$ . A uniquely normalized up-to-flat first integral converts the billiard map into an intrinsic action-angle model. The common perimeter expansion then has only even inverse powers, and its coefficients are boundary integral invariants determined by finite curvature jets.

**Definition K11.1. billiard map**

The billiard map sends an oriented boundary impact point/direction to the next impact after straight motion and specular reflection; Birkhoff coordinates and orientation are fixed once for the project.



**Definition K11.2. caustic**

A caustic is a curve inside a convex billiard such that any billiard trajectory tangent to it remains tangent after every subsequent reflection for which the trajectory is defined.

**Definition K11.3. Lazutkin coordinate**

Lazutkin coordinates are curvature-weighted near-boundary canonical coordinates chosen so that the billiard map becomes an integrable twist map to leading order with a high-order remainder.

**Definition K11.4. Marvizi-Melrose invariant**

The *Marvizi-Melrose invariants* are the intrinsic coefficients  $\ell_j$  in the common all-orders perimeter expansion of long simple winding-one billiard polygons. Equivalently, they are the coefficients obtained from the uniquely normalized up-to-flat interpolating Hamiltonian at the glancing boundary. Each coefficient is determined by finitely many terms of the boundary curvature jet and can be written in terms of boundary integrals of the corresponding differential expressions.

**Definition K11.5. simple periodic billiard polygon and extremal perimeter**

A *simple  $q$ -periodic billiard polygon* is a closed billiard orbit with exactly  $q$  boundary impacts in one primitive period whose polygonal trace is non-self-intersecting. Its perimeter is the sum of the  $q$  chord lengths. For a strictly convex table,  $L_q$  and  $l_q$  denote respectively the supremum and infimum of these perimeters over all such simple  $q$ -periodic polygons.

**Definition K11.6. confocal conic of an ellipse**

For the ellipse  $x^2/a^2 + y^2/b^2 = 1$  with  $a > b > 0$ , the *confocal family* is

$$\frac{x^2}{a^2 - \lambda} + \frac{y^2}{b^2 - \lambda} = 1.$$

For  $\lambda < b^2$  this is a confocal ellipse, while for  $b^2 < \lambda < a^2$  it is a confocal hyperbola; every member has the same two foci as the original ellipse. The limiting value  $\lambda = b^2$  corresponds to the focal segment/separatrix.

**Definition K11.7. complete integrability of a two-dimensional billiard map**

A two-dimensional billiard map is *completely integrable on a regular region* if that region is foliated, up to separatrices/singular leaves, by one-dimensional invariant level curves of a nonconstant first integral, so that the dynamics on each regular leaf reduces to an orientation-preserving circle map. In the ellipse-integrability theorem the first integral is the confocal-caustic parameter.

## 11.2 How the main proofs work

### Intuition

**Exact twist identity.** Differentiate the chord length  $L(s, s') = |\Gamma(s) - \Gamma(s')|$ . The endpoint derivatives are the cosines of the incidence angles, giving  $dL = -p ds + p' ds'$ . A second differentiation yields the nonzero mixed derivative and therefore the monotone twist.

**Lazutkin normal form.** Expand the next-impact relation in the small grazing angle. Express the coefficients through curvature and its derivatives, then choose the curvature-weighted coordinate  $x$  and scaled angle  $y$  so that the order- $y^2$  defects cancel. The map becomes  $x_1 = x + y + O(y^3)$ ,  $y_1 = y + O(y^4)$ .

**Lazutkin caustics.** Restrict the normal form to a thin annulus  $0 < y < y_0$  and rescale. The map is a sufficiently small finite-regularity perturbation of an integrable twist. The KAM theorem from K06 supplies a positive-measure Cantor family of invariant graphs, which translate back into smooth convex caustics.

**Ellipse integrability.** The optical property of confocal conics implies that reflection preserves the confocal parameter tangent to each segment. This parameter is a first integral. Elliptic and hyperbolic confocals fill the nonsingular phase cylinder on opposite sides of the focal separatrix.

**Marvizi-Melrose expansion.** Build the normalized up-to-flat first integral of the glancing billiard map, prove that all long simple periodic polygons lie uniformly at Lazutkin scale  $q^{-1}$ , and pass to the intrinsic action given by symplectic area. The model winding condition produces an even  $q^{-2}$  expansion; flat comparison transfers it to every true simple  $q$ -periodic polygon, while the first-integral recursion shows that the coefficients are coordinate-free boundary curvature integrals.

## 11.3 Main results

### Proposition K11.1. Billiard map is an exact monotone twist map

**Statement.** Let  $\Gamma$  be a  $C^r$  ( $r \geq 2$ ) arclength parametrization of a strictly convex planar billiard boundary, let  $\varphi \in (0, \pi)$  be the angle with the positive tangent, and put  $p = \cos \varphi$ . If  $T(s, \varphi) = (s', \varphi')$  and  $L(s, s') = |\Gamma(s) - \Gamma(s')|$ , then  $dL = -p ds + p' ds'$ , hence  $T^*(p ds) - p ds = dL$  and  $T$  preserves  $dp \wedge ds = \sin \varphi ds \wedge d\varphi$ . Moreover  $L_{12} = \sin \varphi \sin \varphi' / L > 0$ , so  $\partial s' / \partial p = -1 / L_{12} < 0$  away from grazing collisions:  $T$  is an exact negative monotone twist map on the open collision cylinder.

*Proof.* Write  $t(s) = \Gamma'(s)$  and

$$e(s, s') = \frac{\Gamma(s') - \Gamma(s)}{L(s, s')}. \quad (11.1.1)$$

At the first endpoint the outgoing chord makes angle  $\varphi$  with the positive tangent, hence  $e \cdot t(s) = \cos \varphi = p$ . At the second endpoint specular reflection preserves the tangential

component, so the incoming chord has  $e \cdot t(s') = \cos \varphi' = p'$ . Differentiating distance gives

$$L_1 = -e \cdot t(s) = -p, \quad L_2 = e \cdot t(s') = p'. \quad (11.1.2)$$

Therefore along the graph of the billiard map

$$dL = -p ds + p' ds' = T^*(p ds) - p ds. \quad (11.1.3)$$

This is exactly the exact-symplectic identity in the convention used in this book. Taking  $d$  and using  $d^2 L = 0$  yields

$$T^*(dp \wedge ds) = dp \wedge ds. \quad (11.1.4)$$

Since  $p = \cos \varphi$ ,  $dp \wedge ds = \sin \varphi ds \wedge d\varphi$ , giving the stated invariant form.

For the twist sign differentiate  $e$  with respect to  $s'$ :

$$\partial_{s'} e = \frac{t(s') - (e \cdot t(s'))e}{L}. \quad (11.1.5)$$

Thus

$$L_{12} = -\partial_{s'}(e \cdot t(s)) = -\frac{t(s) \cdot t(s') - pp'}{L}. \quad (11.1.6)$$

Strict convexity implies that the turning angle from  $t(s)$  to  $t(s')$  along the chord is  $\varphi + \varphi'$ . Hence

$$t(s) \cdot t(s') = \cos(\varphi + \varphi') = pp' - \sin \varphi \sin \varphi',$$

and therefore

$$L_{12} = \frac{\sin \varphi \sin \varphi'}{L} > 0. \quad (11.1.7)$$

Finally  $p = -L_1(s, s')$ , so at fixed  $s$

$$1 = \frac{\partial p}{\partial p} = -L_{12} \frac{\partial s'}{\partial p}, \quad \frac{\partial s'}{\partial p} = -\frac{1}{L_{12}} < 0. \quad (11.1.8)$$

Away from grazing collisions all factors are finite and positive. Thus the billiard map is an exact negative monotone twist map on the open collision cylinder.  $\square$

### Theorem K11.2. Lazutkin normal form near the boundary

**Statement.** Let  $\partial\Omega$  be a  $C^6$  strictly convex planar curve with arclength  $s$ , radius of curvature  $\rho(s) > 0$ , and perimeter  $\ell$ . Put  $C_\Omega = \int_0^\ell \rho(s)^{-2/3} ds$  and define Lazutkin coordinates near grazing collisions by  $x = C_\Omega^{-1} \int_0^s \rho(\sigma)^{-2/3} d\sigma$  and  $y = 4C_\Omega^{-1} \rho(s)^{1/3} \sin(\varphi/2)$ , where  $\varphi$  is the angle from the positive tangent. Then, for  $y > 0$  sufficiently small, the billiard map has the form  $x_1 = x + y + y^3 f(x, y)$  and  $y_1 = y + y^4 g(x, y)$ , with  $f$  and  $g$  uniformly bounded as  $y \rightarrow 0$ . In particular, the leading map is the integrable twist  $(x, y) \mapsto (x + y, y)$ .

*Proof.* Put  $\kappa = \rho^{-1}$ . Fix a collision at arclength  $s$  and write  $h = s_1 - s > 0$  for the next boundary increment. In Frenet coordinates based at  $s$ , the tangent angle accumulated along the boundary is

$$\Theta(t) = \int_0^t \kappa(s + u) du. \quad (11.2.1)$$

The chord vector is

$$\Gamma(s+h) - \Gamma(s) = \int_0^h (\cos \Theta(t), \sin \Theta(t)) dt. \quad (11.2.2)$$

Taylor expansion, using  $C^6$  regularity, gives for the angle  $\varphi$  of the chord with the initial tangent

$$\varphi = \frac{\kappa h}{2} + \frac{\kappa' h^2}{6} + \frac{\kappa'' h^3}{24} + \left( \frac{\kappa'''}{120} - \frac{\kappa^2 \kappa'}{720} \right) h^4 + O(h^5). \quad (11.2.3)$$

At the next endpoint the reflected angle from the positive tangent equals  $\varphi_1 = \Theta(h) - \varphi$ , hence

$$\varphi_1 = \frac{\kappa h}{2} + \frac{\kappa' h^2}{3} + \frac{\kappa'' h^3}{8} + O(h^4). \quad (11.2.4)$$

All curvature quantities without an argument are evaluated at  $s$ .

**Lemma 11.1 (Frenet-Lazutkin cancellation).** With

$$X(h) = \int_s^{s+h} \kappa(\sigma)^{2/3} d\sigma, \quad Y = 4\kappa(s)^{-1/3} \sin(\varphi/2), \quad Y_1 = 4\kappa(s+h)^{-1/3} \sin(\varphi_1/2),$$

one has uniformly in  $s$

$$X(h) - Y = O(h^3), \quad Y_1 - Y = O(h^4), \quad (11.2.5)$$

and  $Y = \kappa^{2/3} h + O(h^2)$ . For completeness, expand the three expressions. From (11.2.3),

$$Y = \kappa^{2/3} h + \frac{1}{3} \kappa^{-1/3} \kappa' h^2 + O(h^3). \quad (11.2.6)$$

On the other hand

$$X(h) = \kappa^{2/3} h + \frac{1}{3} \kappa^{-1/3} \kappa' h^2 + O(h^3), \quad (11.2.7)$$

which proves the first cancellation. To one further order, substituting (11.2.3)–(11.2.4) and  $\kappa(s+h) = \kappa + \kappa' h + \kappa'' h^2/2 + \dots$  gives

$$Y_1 - Y = \frac{-9\kappa^4 \kappa' - 36\kappa^2 \kappa''' + 180\kappa \kappa' \kappa'' - 160(\kappa')^3}{6480 \kappa^{7/3}} h^4 + O(h^5), \quad (11.2.8)$$

so in particular the  $h, h^2, h^3$  coefficients cancel. This proves Lemma 11.1. The displayed coefficient is not used later; it is included to make the fourth-order cancellation checkable.

Now divide by the Lazutkin perimeter  $C_\Omega = \int_0^\ell \kappa^{2/3} ds$ . By definition,

$$x_1 - x = C_\Omega^{-1} X(h), \quad y = C_\Omega^{-1} Y, \quad y_1 = C_\Omega^{-1} Y_1. \quad (11.2.9)$$

Positive curvature on the compact boundary gives  $0 < \kappa_- \leq \kappa \leq \kappa_+$ , and therefore from (11.2.6)  $y \asymp h$  uniformly for small  $h$ . Consequently (11.2.5) becomes

$$x_1 = x + y + O(y^3), \quad y_1 = y + O(y^4). \quad (11.2.10)$$

The Taylor remainders depend continuously on  $s$  and on the normalized small variable

$h$ , and  $C^6$  regularity bounds the required curvature derivatives. Hence the quotients

$$f(x, y) = \frac{x_1 - x - y}{y^3}, \quad g(x, y) = \frac{y_1 - y}{y^4} \quad (11.2.11)$$

extend boundedly to  $y = 0$  after defining their boundary values by the Taylor coefficients. This is exactly the asserted Lazutkin normal form.  $\square$

### Theorem K11.3. Lazutkin caustics theorem

**Statement.** Let  $\Omega \subset \mathbb{R}^2$  be a strictly convex planar billiard table whose boundary is  $C^6$  and has everywhere positive curvature. Then there exists a collection of smooth strictly convex caustics accumulating at  $\partial\Omega$  whose union has positive area. Equivalently, the billiard map has a positive-measure Cantor family of invariant curves sufficiently close to the boundary.

*Proof.* By the Lazutkin normal form near the boundary, in Lazutkin coordinates the billiard map is

$$T(x, y) = (x + y + y^3 f(x, y), y + y^4 g(x, y)), \quad (11.3.1)$$

for  $0 < y < \delta$ , with the finite regularity inherited from a  $C^6$  boundary. The invariant billiard two-form becomes a nonzero constant multiple of  $y dy \wedge dx$ ; hence  $I = y^2/2$  is a canonical action coordinate. Exactness comes from the Billiard map is an exact monotone twist map.

Fix a small scale  $a > 0$  and restrict to the annulus  $a \leq y \leq 2a$ . Put  $y = az$ ,  $1 \leq z \leq 2$ . One iterate has

$$x_1 = x + az + O(a^3), \quad z_1 = z + O(a^3). \quad (11.3.2)$$

Use the slow angular variable only in the Newton estimates, not as a change of the circle topology: the unperturbed rotation function is  $\omega_a(z) = az$  and has twist  $\omega'_a = a$ . Dividing the invariance equation by this common factor shows that the small parameter entering the cohomological/Newton step is  $O(a^2)$  relative to the twist. Thus, after the standard normalization of a twist-map KAM step, the perturbation tends to zero as  $a \rightarrow 0$ .

**Lemma 11.2 (finite-regularity exact-map KAM corollary).** For an exact  $C^5$  annulus map of the form  $F(x, z) = (x + a\omega(z) + R_1, z + R_2)$  on a fixed strip, with  $|\omega'| \geq m > 0$  after normalization and  $\|R\|_{C^5} = o(a)$  in that normalization, all sufficiently small  $a$  admit invariant circles for a positive-measure Cantor set of Diophantine rotation numbers; the graphs have at least  $C^1$  regularity and are uniformly close to constant-action circles. The proof is the one-frequency exact-map version of the finite-regularity engine proved in the Finite-differentiability KAM theorem. Namely, smooth the map at scales  $s_j \searrow 0$  by the analytic approximation K06-Lemma 11.7, solve the circle cohomological equation with the Rüssmann estimate K06-Lemma 11.8, use twist to remove the mean angular defect, and perform the exact symplectic Newton correction. The error recurrence is quadratic plus the next smoothing error. With five derivatives choose the K06 exponents  $2 < \nu < 5/2$  (so the one-frequency divisor exponent  $\tau = \nu - 1 > 1$ ); then the Bernstein sums of first derivatives converge because  $5 > 2\nu$ . The Diophantine

excluded intervals satisfy

$$\sum_{k \neq 0} \frac{2\gamma}{|k|^{\tau+1}} = O(\gamma), \quad (11.3.3)$$

so the surviving rotations have positive measure. Exactness is preserved by every smoothing/Newton correction. This proves Lemma 11.2 without importing a new unproved KAM theorem.

Apply Lemma 11.2 to a sequence of disjoint dyadic Lazutkin annuli  $a_j \leq y \leq 2a_j$ ,  $a_j \downarrow 0$ . On each sufficiently small annulus a positive-measure Cantor set of invariant graphs is obtained. Their union in phase space therefore has positive two-dimensional billiard measure and accumulates at  $y = 0$ , hence at the boundary.

It remains to translate invariant graphs into caustics. For a rotational invariant graph  $y = u(x)$  sufficiently close to zero, associate to each phase point its oriented billiard line. Neighboring lines have a unique envelope because the graph is  $C^1$ -close to a constant grazing graph and the twist is nonzero. The envelope depends  $C^1$  on  $x$  and has positive curvature for small  $u$ . Invariance of the graph says that consecutive reflected lines belong to the same family, so the envelope is tangent before and after every reflection: it is a caustic in the sense of the definition of caustic above. Conversely a smooth strictly convex caustic gives its invariant tangent-line graph, so the two objects are equivalent near the boundary.

**Lemma 11.3 (invariant-graph/caustic envelope correspondence).** Every sufficiently grazing  $C^1$  rotational invariant graph has a unique strictly convex line envelope, and invariance of the graph is equivalent to the caustic property of that envelope. Finally, the envelope map  $(x, u) \mapsto q(x, u)$  has nonzero transverse Jacobian at  $u = 0$ : changing the grazing parameter moves the envelope in the normal direction by a nonzero first-order amount. Fubini/change of variables therefore sends a positive-measure set of invariant graph parameters to a positive-area union of caustics. The caustics accumulate at  $\partial\Omega$ , completing the proof.  $\square$

#### Theorem K11.4. Integrability of the ellipse

**Statement.** Let  $\Omega$  be the interior of a noncircular ellipse with foci  $F_1, F_2$ . Every billiard trajectory that does not pass through a focus is tangent, on every segment of the trajectory, to one fixed conic confocal with  $\partial\Omega$ : it is a confocal ellipse when the segment does not cross the focal interval  $F_1F_2$  and a confocal hyperbola when it does. Consequently the nonsingular part of the billiard phase cylinder is foliated by invariant level curves of the confocal-caustic parameter (with the focal trajectories forming the separatrix), giving complete integrability of the elliptic billiard.

*Proof.* Write the boundary ellipse as

$$E_0 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b > 0, \quad (11.4.1)$$

and its confocal family as

$$E_\lambda : \frac{x^2}{a^2 - \lambda} + \frac{y^2}{b^2 - \lambda} = 1. \quad (11.4.2)$$

For  $0 < \lambda < b^2$  this is an ellipse; for  $b^2 < \lambda < a^2$  it is a hyperbola. The limiting value  $\lambda = b^2$  is the focal segment/separatrix.

Fix  $P \in E_0$ . A line  $P + tv$  is tangent to  $E_\lambda$  precisely when the quadratic equation obtained by substituting it into (11.4.2) has zero discriminant. If  $D_\lambda = \text{diag}((a^2 - \lambda)^{-1}, (b^2 - \lambda)^{-1})$ , this condition is

$$(P^T D_\lambda v)^2 - (v^T D_\lambda v)(P^T D_\lambda P - 1) = 0. \quad (11.4.3)$$

Parameterize  $P = (a \cos t, b \sin t)$ . Substitution and cancellation using  $P \in E_0$  show that (11.4.3), regarded as a quadratic equation for the slope of  $v$ , has two roots whose directions are symmetric with respect to the normal

$$n(P) = \left( \frac{x}{a^2}, \frac{y}{b^2} \right). \quad (11.4.4)$$

Indeed the sum of the two signed angles with the tangent is zero; algebraically the product and sum of the two slope roots are unchanged when the normal component of  $v$  changes sign. Specular reflection is exactly

$$v \mapsto v - 2 \frac{v \cdot n}{|n|^2} n, \quad (11.4.5)$$

which keeps the tangential component and reverses the normal component. Thus it exchanges the two tangent lines from  $P$  to the same  $E_\lambda$ .

**Lemma 11.4 (confocal reflection lemma).** At every point of  $E_0$ , specular reflection exchanges the two lines tangent to any fixed admissible confocal conic  $E_\lambda$ .

Suppose one segment of a billiard orbit is tangent to  $E_\lambda$ . At its next impact with  $E_0$ , Lemma 11.4 says the reflected segment is the other tangent from the impact point to the same  $E_\lambda$ . Induction over impacts proves that every segment is tangent to that one confocal conic. Hence  $\lambda$  is a first integral of the nonsingular billiard map.

A segment not meeting the focal interval has two tangency points with an inner confocal ellipse and therefore corresponds to  $0 < \lambda < b^2$ . A segment crossing the focal interval is tangent to a confocal hyperbola,  $b^2 < \lambda < a^2$ . Lines through a focus are exactly the limiting  $\lambda = b^2$  family and separate the two regions. Away from the focal trajectories the level sets of  $\lambda$  are smooth one-dimensional invariant curves filling the phase cylinder. Together with the invariant energy/speed of the billiard flow, this is complete integrability of the elliptic billiard.  $\square$

### Theorem K11.5. Marvizi-Melrose invariants

**Statement.** Let  $\Omega \subset \mathbb{R}^2$  have  $C^\infty$  strictly convex boundary. For each sufficiently large integer  $q$ , let  $L_q$  and  $l_q$  be respectively the supremum and infimum of the perimeters of simple (non-self-intersecting)  $q$ -periodic billiard polygons. Then  $q^N(L_q - l_q) \rightarrow 0$  for every  $N > 0$ , and  $L_q$  has an asymptotic expansion  $L_q \sim \ell_0 + \sum_{j \geq 1} \ell_j q^{-2j}$  as  $q \rightarrow \infty$ , with  $\ell_0 = \text{length}(\partial\Omega)$ . Each  $\ell_j$  is an integrated invariant determined by the boundary curvature and finitely many of its derivatives; the coefficients  $\{\ell_j\}$  are the Marvizi-Melrose invariants.

*Proof.* We separate the proof into five mechanisms. The point of the first mechanism is important: near the glancing boundary one does not merely have a finite Birkhoff normal form. There is a canonically normalized *formal first integral* whose Hamiltonian flow interpolates the billiard map modulo a remainder flat at the boundary. This is what makes the all-orders length expansion intrinsic.

**Step 1: the normalized Marvizi-Melrose first integral.** Use arclength  $s$  on  $\partial\Omega$  and let  $\varphi \in (0, \pi)$  be the oriented angle from the positive tangent. Put

$$u = 1 - \cos \varphi, \quad z = \sqrt{u}. \quad (11.5.1)$$

Because  $du = \sin \varphi d\varphi$ , the exact twist-map theorem shows that  $(s, u)$  are exact symplectic coordinates (up to the fixed orientation convention). The Frenet expansion used in the Lazutkin normal-form theorem gives a smooth lift of the glancing map in  $(s, z)$  of the form

$$\tilde{T}(s, z) = \left( s + w(s)z + O(z^2), z + \frac{q(s)}{2}z^2 + O(z^3) \right), \quad w(s) > 0. \quad (11.5.2)$$

Area preservation forces

$$q(s) = -\frac{2}{3}w'(s). \quad (11.5.3)$$

Indeed, substituting  $u = z^2$  into  $ds \wedge du$  and comparing the first nontrivial coefficient of  $\tilde{T}^*(ds \wedge 2z dz)$  with  $ds \wedge 2z dz$  yields (11.5.3).

**Lemma 11.5A (normalized up-to-flat first integral).** There is a  $C^\infty$  function  $h(s, u)$  in a glancing collar such that  $h(s, 0) = 0$ ,  $\partial_u h(s, 0) > 0$ ,  $h \circ T - h$  is flat at  $u = 0$ , and, after this normalization,  $T$  agrees modulo a flat map with the time-one map of  $\frac{2}{3}h^{3/2}$ . The full Taylor series of  $h$  at  $u = 0$  is uniquely determined by  $T$ . We prove the child result rather than importing it. First construct an even formal series

$$G(s, z) = \sum_{m \geq 1} g_m(s) z^{2m} \quad (11.5.4)$$

whose image  $G \circ \tilde{T}$  is even. At the first order the coefficient of  $z^3$  vanishes exactly when

$$g'_1 w + qg_1 = 0. \quad (11.5.5)$$

Using (11.5.3), one may take  $g_1 = w^{2/3} > 0$ . Suppose  $G_{m-1} = \sum_{j < m} g_j z^{2j}$  has been chosen so that  $G_{m-1} \circ \tilde{T}$  has no odd terms through degree  $2m - 1$ , and let  $b_m(s) z^{2m+1}$  be its next odd term. Adding  $g_m(s) z^{2m}$  removes that term precisely when

$$g'_m w - \frac{2m}{3} w' g_m = -b_m. \quad (11.5.6)$$

On an interval this scalar equation is solved by the integrating factor  $w^{-2m/3}$ . On the closed boundary the local solutions glue: the normalization obtained below is unique at the level of Taylor jets, hence continuation once around the circle returns the same jet. Thus (11.5.4) is globally defined as a periodic formal series.



The billiard lift factors as the product of two involutions,

$$\tilde{T} = \mathcal{I} \circ \beta, \quad \mathcal{I}(s, z) = (s, -z), \quad (11.5.7)$$

where  $\beta$  exchanges the two endpoints of the oriented chord. Since  $G$  and  $G \circ \tilde{T} = G \circ \beta$  are even, the series

$$t := G + G \circ \beta \quad (11.5.8)$$

is even,  $\beta$ -invariant and therefore  $\tilde{T}$ -invariant. Its leading coefficient is  $2w^{2/3} > 0$ . Replacing  $z^2$  by  $u$  and applying Borel realization gives a smooth function  $t(s, u)$  with the prescribed formal series,  $t = 0$  on  $u = 0$ ,  $t_u > 0$  there, and  $t \circ T - t$  flat.

Let  $X_t$  be its Hamiltonian vector field and choose a time coordinate  $\theta$  with  $d\theta(X_t) = 1$ . Then  $(\theta, t)$  are symplectic. The flat invariance of  $t$  and exact symplecticity imply

$$T(\theta, t) = (\theta + \xi(t), t) + \text{flat}(t). \quad (11.5.9)$$

Lift once more by  $\zeta = \sqrt{t}$ . The involution in (11.5.7) implies that the leading angular displacement is odd in  $\zeta$ ; hence

$$\xi(t) = \sqrt{t}\psi(t), \quad \psi \in C^\infty, \quad \psi(0) > 0, \quad (11.5.10)$$

modulo a flat term. We now remove the remaining smooth factor. Seek  $h = v(t)$  so that the Hamiltonian vector field of  $\frac{2}{3}h^{3/2}$  equals  $\xi(t)\partial_\theta$ . This is the scalar equation

$$v(t)^{1/2}v'(t) = \sqrt{t}\psi(t), \quad v(0) = 0, \quad (11.5.11)$$

with the explicit smooth solution

$$v(t) = \left( \frac{3}{2} \int_0^t \sqrt{p}\psi(p) dp \right)^{2/3}. \quad (11.5.12)$$

Thus  $h = v(t)$  satisfies  $h_u(s, 0) > 0$  and

$$T = \exp\left(X_{\frac{2}{3}h^{3/2}}\right) + \text{flat}(h). \quad (11.5.13)$$

The construction also proves uniqueness of the Taylor jet: (11.5.5)–(11.5.6) fix the invariant jet up to the scalar reparametrization removed by (11.5.11), and (11.5.11) with  $v(0) = 0$ ,  $v'(0) > 0$  fixes that last freedom. This proves Lemma 11.5A.

**Step 2: all long simple periodic polygons are uniformly grazing.** Let  $\varphi_i$  be the positive incidence angle at the  $i$ th vertex of a simple  $q$ -periodic polygon, listed in cyclic boundary order. If  $\beta_i$  is the direction of its  $i$ th edge and  $\vartheta_i$  the tangent direction at the  $i$ th impact, the reflection law gives

$$\beta_i = \vartheta_i + \varphi_i = \vartheta_{i+1} - \varphi_{i+1}. \quad (11.5.14)$$

The polygon is simple and inscribed in a strictly convex curve, hence it is a convex

winding-one polygon. Summing the tangent turns in (11.5.14) gives

$$\sum_{i=0}^{q-1} (\varphi_i + \varphi_{i+1}) = 2\pi, \quad \text{so} \quad \sum_{i=0}^{q-1} \varphi_i = \pi. \quad (11.5.15)$$

Let  $y_i$  be the Lazutkin transverse coordinate of the Lazutkin normal-form theorem. Since curvature is bounded above and below and  $y = C\rho^{1/3} \sin(\varphi/2)$ , there are table-dependent constants  $c_*, C_* > 0$  such that

$$c_*\varphi_i \leq y_i \leq C_*\varphi_i \quad (0 \leq \varphi_i \leq \pi). \quad (11.5.16)$$

Consequently  $\min_i y_i \leq C/q$  and  $\sum_i y_i \leq C$ .

**Lemma 11.5B (uniform grazing and the  $q^{-1}$  scale).** For every sufficiently large  $q$ , every simple  $q$ -periodic billiard polygon satisfies  $c/q \leq y_i \leq C/q$  at every impact, with constants depending only on the table. Choose a glancing collar  $0 < y < \delta$  on which the Lazutkin normal-form theorem and its inverse give

$$y_{i+1} = y_i(1 + O(y_i^3)). \quad (11.5.17)$$

Start at an index where  $y_i$  is minimal. If the orbit left this collar, stop at its first exit. Before that exit, after decreasing  $\delta$  if necessary,

$$\left| \log \frac{y_{i+1}}{y_i} \right| \leq Cy_i^3.$$

Hence along the entire pre-exit segment

$$\left| \log \frac{y_k}{y_{\min}} \right| \leq C \sum_i y_i^3 \leq C\delta^2 \sum_i y_i \leq C_0.$$

But  $y_{\min} \leq C/q$ , so for large  $q$  this bound prevents reaching the fixed level  $\delta$ . Therefore every impact stays in the collar. Repeating the same estimate around the whole orbit gives  $\max_i y_i / \min_i y_i \leq e^{C_0}$ . Together with (11.5.15)–(11.5.16) this yields  $c/q \leq y_i \leq C/q$ . In particular the boundary mesh is  $O(q^{-1})$  and the polygon perimeter tends to  $|\partial\Omega|$ . This proves Lemma 11.5B.

**Step 3: intrinsic action-angle coordinates.** The normalized function  $h$  from Step 1 has regular level circles for small  $h > 0$ . In the symplectic coordinates  $(\tau, h)$  of its Hamiltonian flow, let  $P(h)$  be the  $\tau$ -period of the level  $\{h = \text{const}\}$ . It is smooth and  $P(0) > 0$ . Define

$$J(h) = \int_0^h P(a) da, \quad \theta = \frac{\tau}{P(h)} \pmod{1}. \quad (11.5.18)$$

Then  $d\theta \wedge dJ = d\tau \wedge dh$ , so  $(\theta, J)$  are canonical coordinates with angle period one. The interpolating Hamiltonian becomes

$$K(J) = \frac{2}{3} h(J)^{3/2},$$

and its time-one map is

$$\Phi(\theta, J) = (\theta + \omega(J), J), \quad \omega(J) = K'(J) = \frac{\sqrt{h(J)}}{P(h(J))}. \quad (11.5.19)$$

Since  $J(h) = P(0)h + O(h^2)$ ,

$$\omega(J) = c_0 J^{1/2} (1 + c_1 J + c_2 J^2 + \cdots), \quad c_0 > 0. \quad (11.5.20)$$

**Lemma 11.5C (intrinsic action scale and model periodic action).** For the normalized interpolating Hamiltonian, the winding-one action  $J_q$  determined by  $\omega(J_q) = q^{-1}$  has a full even expansion  $J_q \sim \sum_{r \geq 1} d_r q^{-2r}$ . The jet of this expansion is independent of all auxiliary glancing coordinates. Indeed, write  $r = J^{1/2}$ . Equation (11.5.20) has the form  $q^{-1} = c_0 r (1 + c_1 r^2 + c_2 r^4 + \cdots)$ , which is inverted recursively as an odd series in  $q^{-1}$ . Squaring gives

$$J_q \sim d_1 q^{-2} + d_2 q^{-4} + d_3 q^{-6} + \cdots. \quad (11.5.21)$$

The claimed invariance is not a coordinate slogan. The normalized Taylor jet of  $h$  is unique by Step 1, while  $J(h)$  is the symplectic area of the subannulus between the boundary and the level  $h$ , measured with angle period one. Thus  $J$ ,  $K(J)$ ,  $\omega(J)$  and the coefficients in (11.5.21) are intrinsic under exact symplectic changes of glancing coordinates. This proves Lemma 11.5C.

**Step 4: periodic generating action and the common length expansion.** Let  $\lambda = p ds$  be the billiard primitive from the exact twist-map theorem. In the annular action-angle chart,

$$\lambda = (J + c) d\theta + dG \quad (11.5.22)$$

for a smooth  $G$  and a constant  $c$ : the difference of the two primitives is closed, and the only cohomology of the annulus is the constant multiple  $c d\theta$ . For the model map  $\Phi$ , a one-step generating action is therefore

$$A_0(J) = (J + c)\omega(J) - K(J), \quad (11.5.23)$$

because  $dA_0 = \Phi^* \lambda - \lambda$ . At the winding-one  $q$ -periodic level,

$$qA_0(J_q) = c + J_q - qK(J_q). \quad (11.5.24)$$

By (11.5.21) and  $K(J) = J^{3/2}$  times a smooth series, the right side has a full expansion in even powers  $q^{-2}$ .

**Lemma 11.5D (flat comparison of true and interpolating periodic actions).** For every  $N$  and all sufficiently large  $q$ , every simple  $q$ -periodic billiard polygon has perimeter

$$c + \sum_{r=1}^R \ell_r q^{-2r} + O(q^{-N})$$

with coefficients independent of the polygon;  $R$  may be chosen arbitrarily by increasing the flatness order used in the estimate. To prove this, write the true map in the

intrinsic chart as

$$T(\theta, J) = (\theta + \omega(J), J) + \mathcal{R}(\theta, J), \quad \mathcal{R} = O(J^M) \quad (11.5.25)$$

for every prescribed  $M$ , with the corresponding derivative bounds. Lemma 11.5B gives  $J_i \asymp q^{-2}$  on every simple  $q$ -periodic orbit. Hence

$$J_i - J_0 = O(q^{1-2M}) \quad (11.5.26)$$

uniformly for  $0 \leq i \leq q$ . Lifting the angle and using winding one,

$$1 = \sum_{i=0}^{q-1} (\omega(J_i) + O(J_i^M)) = q\omega(J_0) + O(q^{-A}) \quad (11.5.27)$$

for any prescribed  $A$  once  $M$  is large enough. Since  $\omega'(J) \asymp J^{-1/2} \asymp q$ , (11.5.27) implies

$$J_0 = J_q + O(q^{-A}) \quad (11.5.28)$$

for every  $A$  after increasing  $M$ .

Exact symplecticity gives a generating action  $A_T(\theta, J)$  for the true map. From (11.5.25), normalized by zero action on the boundary,

$$A_T(\theta, J) = A_0(J) + O(J^M) \quad (11.5.29)$$

for arbitrary  $M$ . Summing (11.5.29) over the periodic orbit and using (11.5.26)–(11.5.28) gives

$$\text{Per}(\gamma_q) = qA_0(J_q) + O(q^{-N}) \quad (11.5.30)$$

for every  $N$ , uniformly over all simple  $q$ -periodic billiard polygons. The equality with perimeter uses exactly the generating-function convention of the exact twist-map theorem; exact differentials telescope on a closed orbit.

By Lemma 11.5B the mesh of  $\gamma_q$  tends uniformly to zero, so its perimeter tends to  $|\partial\Omega|$ . Comparing this with (11.5.24) identifies

$$c = |\partial\Omega| =: \ell_0. \quad (11.5.31)$$

Combining (11.5.21), (11.5.24) and (11.5.30) yields, for every  $R$ ,

$$\text{Per}(\gamma_q) = \ell_0 + \sum_{r=1}^R \ell_r q^{-2r} + O(q^{-2R-2}) \quad (11.5.32)$$

with the same coefficients for every simple  $q$ -periodic polygon. This proves Lemma 11.5D.

**Step 5: why the coefficients are Marvizi-Melrose integral invariants.** We still have to prove the part that was implicit in the earlier proof: the coefficients are geometric boundary invariants, not artifacts of the chosen normal-form coordinates.

**Lemma 11.5E (curvature-integral extraction and coordinate invariance).** Ev-

ery coefficient  $\ell_r$  in (11.5.32) is determined by finitely many terms of the boundary curvature jet and is obtained from circle integrals of the corresponding differential expressions. It is unchanged by a change of arclength origin, glancing coordinate, or exact symplectic normal-form chart. The Taylor-Puiseux jet (11.5.2) is obtained from the Frenet expansion of the chord and reflection equations. At every fixed order its coefficients use only finitely many arclength derivatives of the curvature. In the recursion (11.5.6), multiplication by the integrating factor gives

$$(g_m w^{-2m/3})' = -b_m w^{-1-2m/3}. \quad (11.5.33)$$

Thus the only scalar data surviving one circuit of the boundary are circle integrals of differential expressions in the already constructed curvature jet; changes of local primitives contribute total derivatives and vanish upon integration around the circle. The involution symmetrization (11.5.8), the scalar normalization (11.5.11), inversion of power series and composition introduce only algebraic operations on these finitely many coefficients.

Equivalently, the period used in (11.5.18) can be read directly from the normalized first integral. Writing the level as  $u = u(s, h)$ ,

$$P(h) = \int_{\partial\Omega} \frac{ds}{\partial_u h(s, u(s, h))}. \quad (11.5.34)$$

Expanding the integrand at  $h = 0$  shows order by order that every coefficient of  $P(h)$  is a boundary integral built from a finite curvature jet. The coefficients of  $J(h)$ , its inverse  $h(J)$ ,  $K(J)$  and finally  $qA_0(J_q)$  are obtained by finite algebraic recursion from those integrals. The uniqueness of the normalized  $h$  and the symplectic-area definition of  $J$  proved in Lemma 11.5C show simultaneously that these numbers are independent of every auxiliary coordinate choice. They are precisely the Marvizi-Melrose integral invariants. This proves Lemma 11.5E.

Finally, (11.5.32) holds uniformly for every simple  $q$ -periodic polygon. Taking the supremum and infimum gives, for every  $N > 0$ ,

$$q^N (L_q - l_q) \longrightarrow 0. \quad (11.5.35)$$

Taking the supremum in (11.5.32) and then allowing  $R$  to be arbitrary gives

$$L_q \sim \ell_0 + \ell_1 q^{-2} + \ell_2 q^{-4} + \cdots, \quad \ell_0 = |\partial\Omega|, \quad (11.5.36)$$

and Lemma 11.5E supplies exactly the claimed geometric description of every coefficient.  $\square$

## 11.4 Worked examples

### Worked example: the circular billiard

Let  $\Omega$  be the unit disk and let  $s$  denote arclength on the boundary. The incidence angle  $\varphi$  with the positively oriented tangent is preserved by reflection because rotational symmetry preserves angular momentum. Thus every orbit with fixed  $\varphi$  advances by a fixed arclength increment and lies on an invariant horizontal curve in collision coordinates.

The chord joining two boundary points with angular separation  $\Delta$  has length

$$L = 2 \sin \frac{|\Delta|}{2}.$$

Consequently the generating function depends only on the difference of the end-point angles. The invariant configuration-space caustic is the concentric circle at distance

$$d = |\cos \varphi|$$

from the center: every chord making angle  $\varphi$  with the tangent is tangent to that circle. Hence the disk is completely integrable, and its near-boundary caustics are not merely a Cantor family—they fill the regular phase cylinder.

#### Check your understanding

As  $\varphi \rightarrow 0$ , what happens to the caustic radius  $d$  and to the chord length? Relate this to the grazing boundary on which Lazutkin coordinates are built.

### Boundary case: a stadium is not a Lazutkin table

A Bunimovich stadium has straight boundary segments, so its curvature vanishes there and jumps at the joins with the semicircular caps. The hypotheses of the Lazutkin theorem—strictly positive curvature and sufficient smoothness—fail. The dynamics is therefore not controlled by near-boundary KAM caustics. K12 shows that the same table instead exhibits hyperbolic/ergodic behavior through a defocusing mechanism.

### Common pitfall

Lazutkin's theorem is a near-boundary persistence theorem, not a proof of global billiard integrability. It produces a Cantor family of caustics in a thin boundary region under strong smoothness and convexity hypotheses; it does not fill the whole phase cylinder, and it does not imply that the table is an ellipse.

## 11.5 Chapter summary

### What to remember

- The smooth strictly convex billiard collision map is an exact monotone twist map generated by chord length.
- Lazutkin coordinates turn the grazing region into a nearly integrable twist map and make finite-regularity KAM applicable.
- The resulting invariant graphs are geometric caustics accumulating at the boundary.
- The ellipse is exactly integrable through a confocal-conic first integral.
- Long near-boundary periodic polygons carry the Marvizi–Melrose asymptotic invariants, which later feed rigidity questions.

### Looking ahead

K12 changes geometry completely. Convex focusing boundaries can support KAM curves, but dispersing scatterers create invariant cone fields and singular hyperbolicity. The resulting theory concerns ergodicity, Bernoulli behavior, decay of correlations, and limit laws rather than caustic persistence.

### Check your understanding

Starting from a smooth strictly convex billiard table, explain how the collision map becomes an exact twist map near the boundary and how a KAM invariant graph becomes a geometric caustic. Why does the resulting Cantor family fall short of global integrability?

## 11.6 Exercises and problems

### Exercise 11.1

**Difficulty.** Core.

**Related results.** Billiard map is an exact monotone twist map; Lazutkin normal form near the boundary; Lazutkin caustics theorem.

**Problem.** For the ellipse, use the confocal-caustic formula to verify preservation of the caustic parameter by reflection and explain why the one-parameter confocal family yields integrability.

**Solution sketch.** Follow the worked coordinate calculation through one bounce and show the same confocal parameter appears at the next collision.

### Exercise 11.2

**Difficulty.** Core.

**Related results.** Integrability of the ellipse; Marvizi–Melrose invariants.

**Problem.** For a polygon, inspect every curvature-dependent ingredient of the Lazutkin coordinate and show precisely why edges/vertices prevent the smooth near-boundary normal form from being defined.

**Solution sketch.** Evaluate the curvature factors on straight edges and near vertices and record the resulting zero/undefined/discontinuous terms.

### Exercise 11.3

**Difficulty.** Advanced.

**Related results.** Lazutkin caustics theorem.

**Problem.** From the leading Lazutkin normal form, verify twist and show that the remainder becomes small on  $0 < y < \delta$  as  $\delta \rightarrow 0$ , putting the map in KAM form.

**Solution sketch.** Differentiate the leading angular update and compare each stated remainder order with the twist scale.

### Exercise 11.4

**Difficulty.** Advanced.

**Related results.** Integrability of the ellipse.

**Problem.** Insert one smooth Fourier boundary deformation into the first-variation formula for a rational-caustic invariant and determine a mode detected at first order.

**Solution sketch.** Use Fourier orthogonality/selection rules, then distinguish a linearized obstruction from a nonlinear rigidity theorem.

### Exercise 11.5

**Difficulty.** Advanced.

**Related results.** Billiard map is an exact monotone twist map; Lazutkin normal form near the boundary.

**Problem.** Combine exact twist structure, Lazutkin normal form, and an invariant-curve theorem to derive a Cantor family of convex caustics accumulating at the boundary.

**Solution sketch.** Translate invariant graphs of the billiard map into caustic envelopes and track the smoothness/curvature hypotheses at each step.

### Exercise 11.6

**Difficulty.** Lab.

**Related results.** None.

**Problem.** Simulate billiards in an ellipse and a small Fourier deformation; estimate the confocal caustic parameter after each bounce and compare its drift in the two tables.

**Solution sketch.** Calibrate numerical error on the exactly integrable ellipse before interpreting deformation drift.



# Dispersing, Semi-Dispersing and Focusing Billiards

## Why this chapter matters

Dispersing billiards reverse the curvature seen in K11. A particle now moves among convex obstacles, so each collision spreads nearby wave fronts rather than focusing them. The price is singularity: grazing collisions make derivatives blow up and split phase space into infinitely many smooth pieces. Hyperbolicity must therefore be proved together with a careful control of singularity geometry. This chapter develops the Sinai-billiard mechanism from local geometry to statistics. Wave-front curvature gives invariant unstable and stable cones. The finite-horizon assumption prevents arbitrarily long free flights and yields uniform expansion in an adapted norm. Local ergodicity then propagates across the collision space. A stronger symbolic/Markov structure gives the K and Bernoulli properties, an exponential-tail Young tower yields exponential correlation decay, and the same tower supplies a central limit theorem. The Bunimovich stadium shows that dispersing curvature is sufficient but not necessary: focusing arcs can also generate hyperbolicity through defocusing.

**Prerequisites.** Foundation F10–F14 for hyperbolicity, ergodicity, symbolic dynamics and statistical laws; elementary differential geometry of billiard wave fronts; K11 for collision coordinates.

## Learning goals

- Track the curvature of an infinitesimal wave front through free flight and reflection.
- Explain the roles of finite horizon, homogeneity strips, and the adapted  $p$ -norm.
- Distinguish hyperbolicity, ergodicity, K/Bernoulli structure, exponential mixing, and the CLT.
- Understand why singularities require local-ergodicity and growth lemmas beyond the smooth Anosov theory.
- Contrast dispersing hyperbolicity with Bunimovich's defocusing mechanism.

**Conceptual roadmap**

dispersing geometry  $\rightarrow$  cone expansion  
 $\rightarrow$  local/global ergodicity  
 $\rightarrow$  Young/Markov structure,

which then yields Bernoulli/K behavior, exponential correlation decay, and a central limit theorem under the hypotheses stated below. The stadium follows a parallel geometric route: focusing on a cap is compensated by a sufficiently long flight before the next focusing event, so wave fronts emerge divergent.

**Check your understanding**

Why does uniform expansion between nonsingular collisions not by itself prove ergodicity? Identify the extra obstruction created by the singularity curves.

**12.1 Definitions****Intuition**

**Finite horizon is a uniform geometric bound.** It says free-flight times between collisions are bounded above. Without it, long corridors produce qualitatively different correlation behavior.

**Homogeneity strips tame grazing singularities.** Near  $|\varphi| = \pi/2$ , derivatives become large. The phase space is sliced into strips on which the derivative distortion has a controlled scale.

**The  $p$ -norm is adapted to billiard expansion.** Euclidean coordinate size is not the right invariant measure of tangent vectors near grazing. The adapted norm absorbs the angle factor so that wave-front expansion becomes multiplicative and uniform along unstable cones.

**A Young tower is a return-time model, not a synonym for mixing.** The proof constructs a Markov extension with exponential return tails. Those tails imply exponential correlation decay and, with standard probabilistic arguments, a central limit theorem for the stated observables.

**Definition K12.1. dispersing billiard**

A dispersing billiard is a billiard whose collision boundary is convex toward the scatterer so that wave fronts are defocused; the standard Sinai convention uses strictly convex scatterers in a flat domain/torus.

**Definition K12.2. grazing singularity**

A grazing singularity is a phase point whose trajectory is tangent to the boundary at collision; the billiard derivative becomes singular as the incidence angle approaches tangency.

**Definition K12.3. defocusing mechanism**

The defocusing mechanism is the geometric property that a converging beam reflected by a focusing component passes through its focal point and becomes sufficiently diverging before the next relevant focusing collision.

**Definition K12.4. Young tower**

A Young tower is an inducing scheme with a Markov return map on a base, return-time function, and hyperbolic/product structure whose tail estimates imply statistical limit laws.

**Definition K12.5. regular collision space and finite horizon**

For a planar billiard, the *regular collision space* is the collision cross-section with grazing/tangent collisions, corner or cusp hits, and all points whose next collision is not uniquely and smoothly defined removed. A periodic Lorentz gas has *finite horizon* if the free-flight time (equivalently, unit-speed flight length) between successive regular collisions is uniformly bounded above. For pairwise disjoint compact scatterers, regular flights also have a positive lower bound away from zero.

**Definition K12.6. wave-front curvature, billiard cone and adapted  $p$ -norm**

For a one-parameter pencil of nearby billiard rays, let  $J$  be the signed transverse Jacobi separation and put  $B = \dot{J}/J$  when  $J \neq 0$ ;  $B$  is the signed curvature of the infinitesimal wave front, with  $B \geq 0$  for a divergent front. The unstable billiard cone consists of tangent vectors whose post-collision wave fronts are divergent, and the stable cone is its time reversal. The *adapted billiard* or  *$p$ -norm* measures a tangent vector by its transverse separation  $|J|$  at collision (equivalently, in  $(s, \beta)$  coordinates, by the standard norm weighted by  $\cos \beta$ ), so free-flight expansion is exactly the Jacobi-field expansion factor.

**Definition K12.7. homogeneity strips**

With  $\beta \in (-\pi/2, \pi/2)$  the angle from the inward normal, *homogeneity strips* are a fixed central collision region together with near-grazing strips

$$H_k = \left\{ (s, \beta) : \frac{1}{(k+1)^2} < \cos \beta \leq \frac{1}{k^2} \right\}, \quad k \geq k_0,$$

and their reflected copies near the opposite grazing boundary. A *homogeneous* stable/unstable curve is contained in one smooth component of one such strip. The strips do not create hyperbolicity; they make derivative and distortion estimates uniform within each near-grazing component.

**Definition K12.8. billiard suspension and Liouville measure**

If  $T : M \rightarrow M$  is the collision map and  $\tau : M \rightarrow (0, \infty)$  is the free-flight time, the *billiard suspension flow* is the special flow on  $\{(x, t) : x \in M, 0 \leq t < \tau(x)\}$  with  $(x, \tau(x)) \sim (Tx, 0)$ . If  $\mu$  is the normalized collision measure, its normalized Liouville

measure is

$$d\bar{\mu}_{\text{flow}} = \frac{d\mu(x) dt}{\int_M \tau d\mu}.$$

In tangent-angle coordinates the collision measure has density  $d\mu = c \sin \varphi ds d\varphi$ .

### Definition K12.9. Bernoulli, K/Pinsker, and mixing of all orders

An invertible probability-preserving map is *Bernoulli* if it is measure-theoretically isomorphic to a two-sided Bernoulli shift. Its *Pinsker sigma-algebra* is the maximal invariant zero-entropy factor; the map has the *Kolmogorov K-property* when this sigma-algebra is trivial. A flow is Bernoulli when every nonzero time map is Bernoulli. A map is *mixing of all orders* if, for every  $r \geq 2$  and measurable sets  $A_1, \dots, A_r$ , the measure of  $A_1 \cap T^{-n_2} A_2 \cap \dots \cap T^{-n_r} A_r$  tends to  $\prod_j \mu(A_j)$  whenever all pairwise time separations tend to infinity.

### Definition K12.10. dynamically Hölder observable

Fix the singularity/homogeneity partition and let  $s_+(x, y)$  (respectively  $s_-(x, y)$ ) be the first nonnegative time at which two points on a common local unstable (respectively stable) curve acquire different smooth homogeneous itineraries. A bounded observable  $f$  is *dynamically Hölder* with parameter  $0 < \eta < 1$  if

$$\begin{aligned} |f(x) - f(y)| &\leq C_f \eta^{s_+(x,y)} && \text{on local unstable curves,} \\ |f(x) - f(y)| &\leq C_f \eta^{s_-(x,y)} && \text{on local stable curves.} \end{aligned}$$

Write  $\|f\|_{DH} = \|f\|_\infty + C_f$  (using the least admissible  $C_f$  after a fixed project choice of  $\eta$ ).

## 12.2 How the main proofs work

### Intuition

**Uniform hyperbolicity.** Encode a tangent vector by the curvature of its infinitesimal wave front. Reflection adds positive curvature at a dispersing boundary, while free flight evolves curvature by a fractional-linear rule. Finite horizon and curvature bounds keep the unstable cone strictly invariant and give a uniform expansion factor in the adapted norm.

**Ergodicity.** Hyperbolicity alone is not enough because grazing preimages repeatedly cut stable and unstable curves. The proof first establishes a strict inverse-expansion sum after the singularity/homogeneity cuts, and from it a growth estimate saying that short manifold pieces have only linear endpoint mass. Homogeneous distortion gives absolutely continuous holonomy. A late-singularity tail estimate and a divergent-wavefront bridge then cross a simple grazing singularity, after which Hopf's argument gives local ergodicity. Finally a visibility graph of the scatterers merges the local components globally.

**K and Bernoulli properties.** Stable and unstable partitions generate the measurable sigma-algebra. The local product structure and singularity control show that the Pinsker factor is trivial. Very weak Bernoulli approximations constructed from Markov rectangles then upgrade K behavior to Bernoulli behavior on the mixing component.

**Exponential correlations and the CLT.** Induce on a reference set whose return map is Gibbs-Markov. The growth lemma gives exponential tails for return times. Coupling on the resulting Young tower yields exponential decorrelation; martingale/coboundary or spectral arguments on the same tower yield the Gaussian limit law.

**Bunimovich stadium.** A focusing reflection can contract wave fronts, but a long enough free flight after the cap moves the focal point behind the next collision. Repeated cap-to-cap excursions therefore produce net expansion. Singularities and neutral bouncing-ball families require a separate ergodicity argument from the dispersing case.

## 12.3 Main results

### Theorem K12.1. Uniform hyperbolicity of finite-horizon dispersing billiards

**Statement.** Let  $Q$  be a finite-horizon planar periodic dispersing billiard with finitely many pairwise disjoint  $C^3$  scatterers, strictly positive curvature bounded away from zero and infinity, and no cusps. On the regular collision space (singular/grazing collisions removed) the billiard map admits invariant unstable and stable cone fields. There exist  $C > 0$  and  $\Lambda > 1$  such that every regular orbit segment  $x, Tx, \dots, T^n x$  satisfies  $\|DT_x^n v^u\|_* \geq C\Lambda^n \|v^u\|_*$  for unstable-cone vectors and the analogous backward estimate for stable-cone vectors, in the standard adapted billiard norm; homogeneity strips are used only to control distortion near grazing singularities.

*Proof.* Write a regular collision state as  $x = (s, \beta)$ , where  $s$  is arclength on one scatterer and  $\beta \in (-\pi/2, \pi/2)$  is the angle with the inward normal. Thus  $\cos \beta = \sin \varphi$  in the tangent-angle convention of the statement. Let  $\tau(x)$  be the free-flight length and  $\kappa(s) > 0$  the curvature of the scatterer. Pairwise disjoint compact scatterers give a positive minimum free-flight length

$$\tau_{\min} > 0, \quad (12.1.1)$$

while finite horizon gives  $\tau(x) \leq \tau_{\max} < \infty$ .

Represent a tangent vector by a transverse Jacobi field  $J$  along the corresponding pencil of rays. Between collisions  $\ddot{J} = 0$ . Whenever  $J \neq 0$  put

$$B = \frac{\dot{J}}{J}; \quad (12.1.2)$$

$B$  is the curvature of the infinitesimal wave front in the sign convention that  $B \geq 0$

means divergent. During a free flight of length  $\tau$ ,

$$J(\tau) = J(0)(1 + \tau B(0)), \quad B(\tau) = \frac{B(0)}{1 + \tau B(0)}. \quad (12.1.3)$$

At a reflection from a dispersing component, differentiation of the specular reflection law gives the mirror equation

$$B^+ = B^- + \frac{2\kappa}{\cos \beta}. \quad (12.1.4)$$

For completeness, (12.1.4) follows by writing the outgoing angle as  $\theta^+ = 2\alpha - \theta^-$ , where  $\alpha$  is the normal direction. For two neighboring rays,  $d\alpha = \kappa ds$  and the transverse separation at the collision is  $J = \cos \beta ds$ ; hence  $d\theta^+ - d\theta^- = 2\kappa ds = (2\kappa/\cos \beta)J$ , which is exactly the jump in  $\dot{J}/J$ .

**Lemma 12.1 (mirror/free-flight curvature equations).** Equations (12.1.3)–(12.1.4) hold for every regular infinitesimal ray pencil. Consequently the cone  $B^- \geq 0$  is mapped strictly into itself by one complete collision step.

Indeed, if  $B^- \geq 0$ , then after reflection

$$B^+ \geq \frac{2\kappa_{\min}}{\cos \beta} \geq 2\kappa_{\min} > 0. \quad (12.1.5)$$

Combining (12.1.3), (12.1.5), and (12.1.1) yields

$$\frac{|J_1^-|}{|J_0^+|} = 1 + \tau B^+ \geq 1 + 2\kappa_{\min}\tau_{\min} =: \Lambda_0 > 1. \quad (12.1.6)$$

The adapted billiard, or  $p$ -, norm of the definition of wave-front curvature, billiard cone and adapted  $p$ -norm above is precisely the norm in which  $|J|$  measures the transverse component; in collision coordinates it is uniformly equivalent on each regular homogeneity component to  $\cos \beta |ds|$  together with the corresponding wave-front slope. Thus (12.1.6) gives, for every unstable-cone vector,

$$\|DT_x^n v\|_* \geq C_0 \Lambda_0^n \|v\|_*. \quad (12.1.7)$$

The harmless constant  $C_0 > 0$  accounts for the convention of measuring the first vector immediately before rather than immediately after reflection. Because  $\tau \leq \tau_{\max}$  and  $\kappa \leq \kappa_{\max}$ , it is uniform.

**Lemma 12.2 (uniform dispersing cone expansion).** The divergent-wave-front cone is invariant and satisfies (12.1.7) with  $\Lambda_0 = 1 + 2\kappa_{\min}\tau_{\min} > 1$ . Reversing velocity produces the stable cone and the corresponding backward estimate.

The billiard map is singular only when a forward collision becomes tangent or hits a corner endpoint. There are no cusps here. The factor  $1/\cos \beta$  in (12.1.4) explains why the ordinary coordinate derivative is unbounded near grazing even though the adapted expansion is favorable. Put

$$H_k = \left\{ x : \frac{1}{(k+1)^2} < \cos \beta(x) \leq \frac{1}{k^2} \right\}, \quad k \geq k_0, \quad (12.1.8)$$

and include the finitely many complementary central strips. These are the homogeneity strips of the definition of homogeneity strips above. On each  $H_k$  the factors involving  $\cos \beta$  and their first derivatives have bounded relative variation at the scale of that strip; cutting a regular orbit whenever it crosses a strip boundary therefore supplies uniform distortion constants on each smooth component. No strip was used to obtain (12.1.7): it is solely a device for controlling derivatives of the Jacobian close to the singular set.

Taking  $\Lambda = \Lambda_0$  (or any smaller number  $> 1$  after changing  $C$ ) proves the asserted forward unstable and backward stable estimates and the stated role of homogeneity strips.  $\square$

### Theorem K12.2. Ergodicity of dispersing billiards

**Statement.** Let  $Q = \mathbb{T}^2 \setminus \bigcup_{i=1}^m O_i$  be a connected periodic Lorentz gas whose scatterers have pairwise disjoint  $C^3$  boundaries, strictly positive curvature and no cusps, and suppose the horizon is finite, meaning that the free-flight time between successive collisions is uniformly bounded. Let  $T$  be the collision map on the regular collision space, with arclength  $s$  and angle  $\varphi \in (0, \pi)$  measured from the positive tangent. Then  $T$  is ergodic with respect to the normalized invariant measure  $d\mu = c \sin \varphi ds d\varphi$ . The corresponding billiard suspension flow is ergodic with respect to normalized Liouville measure.

*Proof.* Write a collision state as  $x = (s, \beta)$ , where  $\beta \in (-\pi/2, \pi/2)$  is measured from the inward normal. Thus the invariant density is  $c \cos \beta ds d\beta$ , which is the density  $c \sin \varphi ds d\varphi$  in the tangent-angle convention of the statement. Let  $S_0$  be the grazing boundary  $\{|\beta| = \pi/2\}$  and let

$$\mathcal{S}_n = T^{-n} S_0, \quad n \in \mathbb{Z} \setminus \{0\}. \quad (12.2.1)$$

Finite horizon and the finite scatterer family imply that  $\mathcal{S}_{\pm 1}$  is a finite union of  $C^2$  curves; on each smooth branch of  $T^{\pm 1}$ , preimages and images of these curves remain  $C^2$ . Hence every fixed  $\mathcal{S}_n$  is a finite union of smooth curves and has  $\mu$ -measure zero. The full regular set

$$M^\# = M \setminus \bigcup_{n \in \mathbb{Z}} \mathcal{S}_n \quad (12.2.2)$$

therefore has full measure.

The difficulty is not hyperbolicity: the uniform-hyperbolicity theorem already gives invariant stable and unstable cones and uniform expansion in the adapted  $p$ -norm. The difficulty is that the curves (12.2.1) repeatedly cut unstable and stable curves. We now prove the quantitative estimate which prevents those cuts from destroying local product structure.

**Lemma 12.3 (singularity growth and absolute continuity).** There exist  $\delta_0 > 0$ ,  $0 < \theta < 1$ , and  $C < \infty$  with the following properties. If  $W$  is a homogeneous unstable curve of adapted length at most  $\delta_0$  and  $TW = \bigcup_i W_i$  is its decomposition at primary

singularity and homogeneity boundaries, then, writing

$$\Lambda_i = \inf_{x \in W \cap T^{-1}W_i} \frac{\|DT_x v^u\|_*}{\|v^u\|_*},$$

one has

$$\sum_i \Lambda_i^{-1} \leq \theta. \quad (12.2.3)$$

Consequently, if  $W_n(x)$  is the homogeneous component of  $T^n W$  containing  $T^n x$  and  $r_n(T^n x)$  is the adapted distance from  $T^n x$  to the nearer endpoint of  $W_n(x)$ , then

$$|\{x \in W : r_n(T^n x) < \varepsilon\}| \leq C\varepsilon + C\theta^n |W| \quad (12.2.4)$$

for  $0 < \varepsilon < \delta_0$ . On every common homogeneous itinerary, unstable Jacobians have bounded logarithmic distortion, and stable and unstable holonomies between transverse homogeneous curves are absolutely continuous with Jacobians bounded above and below on compact regular rectangles.

We prove all parts of the lemma. First consider (12.2.3). If  $W$  stays a fixed distance from  $\mathcal{S}_1$ , then for  $\delta_0$  small it has one smooth image component and the uniform-hyperbolicity theorem gives  $\Lambda_1^{-1} \leq \Lambda_0^{-1} < 1$ . If  $W$  meets a primary singularity curve, points of either adjacent branch have the next collision close to grazing. The mirror equation from the preceding hyperbolicity proof contains the factor  $1/\cos \beta_1$ , so the one-step unstable expansion tends to infinity as the branch endpoint approaches the singularity. Shrinking  $\delta_0$  makes the sum of the inverse expansions of the finitely many such primary pieces as small as desired.

It remains to sum the pieces created by the countably many homogeneity boundaries. In the strip

$$H_k = \left\{ (s, \beta) : \frac{1}{(k+1)^2} < \cos \beta \leq \frac{1}{k^2} \right\}$$

the angular width is  $O(k^{-3})$ , while the dispersing reflection term in the wave-front curvature is of order  $k^2$ . Hence every descendant whose image lies in  $H_k$  has minimal adapted expansion at least  $c_0 k^2$ . A sufficiently short monotone unstable curve crosses at most a uniformly bounded number of components of a fixed strip: the primary singularity curves have the opposite monotonicity, and the finitely many collision targets allowed by finite horizon give a uniform multiplicity bound. Therefore the tail contributes at most

$$C_0 \sum_{k \geq K} k^{-2}, \quad (12.2.5)$$

which can be made arbitrarily small by choosing  $K$  large. For the finitely many central strips, compactness and the preceding primary-singularity argument allow one common  $\delta_0$ . Choose first  $K$  and then  $\delta_0$  so that the central contribution plus (12.2.5) is  $< \theta < 1$ . This proves (12.2.3). Notice that this is stronger than merely saying that each branch expands: the inverse expansions remain summable even after all singular cuts.

Now iterate. Let  $\mathcal{G}_n(W)$  be the homogeneous components of  $T^n W$ . On a component  $V \in \mathcal{G}_n(W)$ , an endpoint interval of adapted length  $\varepsilon$  has preimage length at most



$C\varepsilon/\Lambda_V^{(n)}$ , where the constant  $C$  is the distortion constant proved below. Split the genealogy into generations during which a descendant remains shorter than  $\delta_0$  and generations at which it first becomes macroscopic. Each completely short generation multiplies the total inverse-Jacobian weight by at most  $\theta$  by (12.2.3). Once a descendant has length at least  $\delta_0$ , at most  $2\varepsilon/\delta_0$  of its image length lies within  $\varepsilon$  of an endpoint. Pulling this portion back and summing over the first long generation gives a geometric series. Thus

$$|\{r_n < \varepsilon\}| \leq C_1 \theta^n |W| + C_2 \varepsilon \sum_{j=0}^{n-1} \theta^j,$$

which is (12.2.4) after changing  $C$ .

We next justify the distortion constant used in that argument. On a smooth homogeneous branch the unstable Jacobian is a product of free-flight wave-front factors,

$$J^u T^n(x) = \prod_{j=0}^{n-1} (1 + \tau(T^j x) B^+(T^j x)). \quad (12.2.6)$$

In  $H_k$ , differentiating the logarithm of a one-step factor costs at most  $Ck^2$ , whereas a homogeneous strip has width  $O(k^{-3})$ . Hence its one-step logarithmic oscillation is  $O(k^{-1})$ . If  $x, y$  have a common homogeneous itinerary for  $n$  steps, inverse unstable contraction gives

$$|T^j x - T^j y| \leq C \Lambda_0^{-(n-j)} |T^n x - T^n y|.$$

If  $T^j W$  lies in  $H_{k_j}$ , homogeneity gives  $|T^j W| \leq C k_j^{-3}$ . Hence for  $x_j = T^j x$ ,  $y_j = T^j y$ ,

$$k_j^2 |x_j - y_j| \leq C |x_j - y_j|^{1/3},$$

because  $|x_j - y_j| \leq |T^j W| \leq C k_j^{-3}$ . On the central strips the same inequality holds after increasing  $C$ . Since the common itinerary gives  $|x_j - y_j| \leq C \Lambda_0^{-(n-j)} |T^n x - T^n y|$ , the logarithmic oscillation of the  $j$ th factor in (12.2.6) is at most  $C \Lambda_0^{-(n-j)/3} |T^n x - T^n y|^{1/3}$ . Summing the geometric series gives

$$\left| \log \frac{J^u T^n(x)}{J^u T^n(y)} \right| \leq C |T^n x - T^n y|^{1/3}. \quad (12.2.7)$$

In particular all pullback densities used above are uniformly comparable.

Finally let  $W_1, W_2$  be two short homogeneous unstable curves crossed by the same family of local stable curves, and let  $h : W_1 \rightarrow W_2$  be stable holonomy. For  $x, hx$  whose forward orbits avoid singularities, stable contraction and (12.2.7) show that

$$J_h(x) = \lim_{n \rightarrow \infty} \frac{J^u T^n(x)}{J^u T^n(hx)} \quad (12.2.8)$$

exists and satisfies  $0 < c_R \leq J_h \leq C_R < \infty$  on every compact regular rectangle  $R$ . Approximate a measurable subset of  $W_1$  by finite unions of intervals, apply the change-of-variables formula to  $T^n$  before the first singular cut, and let  $n \rightarrow \infty$  in (12.2.8). This proves  $|hA| = \int_A J_h d|\cdot|$  and therefore absolute continuity. Time reversal gives the unstable-holonomy statement. This proves Lemma 12.3.

The growth estimate now constructs the actual local manifolds. For  $x \in M^\#$ , intersect the nested smooth past cylinders through  $x$  and take the unstable component through  $x$ ; its endpoint distance is denoted  $r^u(x)$ . Applying (12.2.4) to a finite family of unstable transversals in a regular rectangle and then letting  $n \rightarrow \infty$  gives

$$\mu_R\{x : r^u(x) < \varepsilon\} \leq C_R \varepsilon. \quad (12.2.9)$$

The reversed estimate gives

$$\mu_R\{x : r^s(x) < \varepsilon\} \leq C_R \varepsilon. \quad (12.2.10)$$

Thus local stable and unstable manifolds have positive length almost everywhere, and the set on which either foliation is too short has a linear transverse-measure bound. These are the estimates that the earlier proof architecture had left implicit.

**Lemma 12.4 (local ergodicity across singularities).** Let  $x_0$  be a collision point whose complete orbit has at most one singularity. For the present strictly dispersing table every nonsingular branch is sufficient, because its next regular reflection has strictly positive curvature. There exists a neighborhood  $U(x_0)$  such that every  $T$ -invariant measurable set has either zero or full measure in  $U(x_0)$ . Equivalently,  $U(x_0)$  belongs modulo zero to one ergodic component.

We prove the lemma in the specialized planar setting needed here. Choose a small coordinate rectangle  $R$  around  $x_0$  whose horizontal and vertical directions lie strictly inside the unstable and stable cones. Fix a slowly increasing integer  $F(\delta) \rightarrow \infty$ . Separate manifold cuts into *early* cuts, produced by  $\mathcal{S}_{\pm n}$  with  $|n| \leq F(\delta)$ , and *tail* cuts produced later. Iterating (12.2.3) in the proof of (12.2.4) shows that the conditional weight of a tail cut at depth  $n$  contains an additional factor  $\theta^n$ . Therefore, after summing the geometric tail and using absolute continuity,

$$\mu_R(U_\delta^{\text{tail}}) \leq C\delta\theta^{F(\delta)} + C\theta^{F(\delta)}. \quad (12.2.11)$$

Choose, for example,  $F(\delta)$  so that  $\theta^{F(\delta)} = o(\delta)$ . Then  $\mu_R(U_\delta^{\text{tail}}) = o(\delta)$ . This is the singular “tail bound” needed by the fundamental theorem: late singularities are too sparse to form a positive-measure barrier at scale  $\delta$ .

Only finitely many early singularity curves remain. Away from them, (12.2.9)–(12.2.10) and Fubini imply that, except on total measure  $O(\delta)$ , the stable and unstable manifolds cross every regular  $\delta$ -rectangle completely. We also need to cross a *simple* early singular curve rather than silently treating it as a smooth Anosov chart. Synchronize the flow just before the corresponding grazing event. Two sufficiently close phase points  $(q_1, v_1)$  and  $(q_2, v_2)$  on the two regular sides can be shifted by times  $t_1, t_2 = O(|q_1 - q_2| + |v_1 - v_2|)$  so that

$$\langle (q_1 + t_1 v_1) - (q_2 + t_2 v_2), v_1 - v_2 \rangle \geq 0. \quad (12.2.12)$$

The two shifted points then lie on one divergent orthogonal wave front: if the velocity lines meet at  $O$ , take a sufficiently small circle centered at  $O$  whose outward normals are  $v_1, v_2$ ; in the parallel limiting case use the limiting straight wave front. Forward images of this front remain divergent in a dispersing billiard. Because the table is

strictly dispersing, every regular branch immediately encounters positive curvature, so the resulting front is sufficient and acquires uniform expansion. Pulling it back to the collision section supplies an unstable bridge between the two sides; the time-reversed construction supplies the stable bridge. The only points for which this bridge is cut before reaching the neighboring regular rectangles belong either to the  $o(\delta)$  tail set (12.2.11) or to the  $O(\delta)$  endpoint sets controlled by (12.2.9)–(12.2.10).

Tile the inner half of  $R$  by regular rectangles of side  $\delta$ , inserting the finitely many early singularity arcs into tile boundaries. Adjacent good tiles on the same side are joined by the ordinary stable/unstable crossing; adjacent tiles separated by a simple singularity arc are joined by the bridge just constructed. Thus, after discarding a set whose relative measure tends to zero with  $\delta$ , all good tiles in the inner rectangle lie in one stable/unstable accessibility class. Take a sequence  $\delta_j \downarrow 0$  with  $\sum_j \delta_j < \infty$  and choose  $F(\delta_j)$  as above. Borel-Cantelli shows that almost every point avoids the discarded sets for all sufficiently large  $j$ . Hence almost every pair of points of a smaller neighborhood  $U(x_0)$  can be connected by a finite alternating chain of local stable and unstable leaves.

For completeness we now run Hopf's argument on this accessibility class. Let  $A$  be a  $T$ -invariant measurable set. Approximate  $1_A$  in  $L^1$  by a continuous function  $g$ . If  $y \in W^s(x)$ , then  $d(T^n x, T^n y) \rightarrow 0$  exponentially, so the forward Birkhoff averages of  $g$  at  $x$  and  $y$  agree. The maximal ergodic inequality applied to  $1_A - g$  lets  $g \rightarrow 1_A$  and shows that the forward Birkhoff averages of  $1_A$  agree for almost every stable pair. Because  $A$  is invariant, that average equals  $1_A$  almost everywhere. Applying the same argument backward shows that  $1_A$  is also constant along almost every unstable leaf. The accessibility just proved therefore makes  $1_A$  constant almost everywhere on  $U(x_0)$ . This is the claimed local ergodicity and proves Lemma 12.4.

We pass from local to global ergodicity. Let

$$\mathcal{E} = \bigcup_{m \neq n} \mathcal{S}_m \cap \mathcal{S}_n \quad (12.2.13)$$

be the set of collision states whose complete orbit contains at least two singular events. After splitting singularity curves into their smooth continuation branches, two distinct prescribed grazing conditions cannot coincide on an arc: an arc of coincidence would give a one-parameter family of trajectories satisfying two independent tangency equations, contradicting the strict curvature/transversality of the dispersing reflection. Hence for fixed  $m \neq n$  the intersection is finite on the compact collision surface, so  $\mathcal{E}$  is countable. Removing a countable set from a two-dimensional collision cylinder does not disconnect it (a path can be perturbed locally around its countably many possible intersections). By Lemma 12.4 the ergodic-component label is locally constant modulo zero on the complement of  $\mathcal{E}$ , including across every ordinary simple singularity curve. Consequently each collision cylinder over one scatterer belongs modulo zero to a single ergodic component.

It remains to show that the scatterer cylinders all have the same label. Make a graph whose vertices are the scatterers and join  $O_i$  to  $O_j$  if an open set of regular free flights goes from  $\partial O_i$  to  $\partial O_j$ . This graph is connected. Indeed, for any two scatterers choose a shortest path in the connected free region joining their boundaries. Between

successive contacts with obstacle boundaries the path is a straight visible segment. A generic arbitrarily small perturbation removes tangencies and multiple contacts, producing a finite chain of transverse visible segments. Thus consecutive obstacles in the chain give graph edges. For an edge  $i-j$ , the corresponding open set in the  $i$ th collision cylinder is mapped diffeomorphically by  $T$  to a positive-measure open set in the  $j$ th cylinder. An invariant ergodic component meeting one therefore meets the other, so their labels agree. Connectedness of the graph merges all cylinders into one component. Hence  $T$  is ergodic with respect to  $\mu$ .

Finally let  $\tau(x)$  be the free-flight time. Finite horizon gives  $\tau \leq \tau_{\max}$  and hence  $\tau \in L^1(\mu)$ . The billiard flow is the special flow over  $T$  with invariant probability measure

$$d\bar{\mu}_{\text{flow}} = \frac{d\mu(x) dt}{\int_M \tau d\mu}, \quad 0 \leq t < \tau(x). \quad (12.2.14)$$

If  $B$  is flow invariant, then its section at collision times is  $T$ -invariant and therefore has  $\mu$ -measure zero or one. Flow invariance makes almost every vertical fiber either entirely in or entirely out of  $B$ ; Fubini in (12.2.14) gives  $\bar{\mu}_{\text{flow}}(B) \in \{0, 1\}$ . The suspension flow is ergodic as claimed.  $\square$

### Theorem K12.3. Bernoulli/K property for Sinai billiards

**Statement.** For a planar periodic finite-horizon dispersing billiard (periodic Lorentz gas) with finitely many pairwise disjoint strictly convex scatterers whose boundaries are  $C^3$ , have everywhere strictly positive curvature, and have no cusps, the collision map with its natural invariant probability measure is a Bernoulli automorphism. The associated billiard flow is Bernoulli as well. Consequently the collision map has the K property and, in the mixing case, is mixing of all orders.

*Proof.* We first establish the K property rather than assuming it. Let  $\xi$  be a finite partition whose atoms are regular homogeneity rectangles of diameter at most  $\delta$ , with the singularity curves included in their boundaries. Refine it in backward time and put

$$\xi^u = \bigvee_{j \geq 0} T^j \xi. \quad (12.3.1)$$

For almost every point, the atom of  $\xi^u$  is contained in its local unstable manifold: forward images of an unstable separation expand, so two distinct unstable points are eventually separated by  $\xi$ . Conversely, points on the same sufficiently short unstable leaf share all sufficiently remote past names unless a singularity cuts the leaf. The growth estimate Lemma 12.3 makes the total conditional measure of such exceptional cuts summable over a geometric sequence of scales. Thus  $\xi^u$  is subordinate to unstable manifolds modulo null sets.

If  $A$  lies in the Pinsker sigma-algebra, then its conditional entropy relative to every finite partition is zero. In particular, the indicator  $1_A$  is measurable with respect to the past tail of  $\xi$ , hence is constant almost everywhere on atoms of  $\xi^u$ , i.e. on unstable leaves. Applying the same argument to  $T^{-1}$  makes it constant on stable leaves. Lemma 12.3 and the Hopf connectivity used in the Ergodicity of dispersing billiards then make  $1_A$  locally constant modulo zero, and global ergodicity makes it constant

almost everywhere. Hence the Pinsker sigma-algebra is trivial.

**Lemma 12.5 (K property of finite-horizon Sinai billiards).** The natural invariant collision map is a K-automorphism: its Pinsker sigma-algebra is trivial.

We now prove the K-to-Bernoulli step needed here rather than leaving the name “Chernov-Haskell” as a black box.

**Lemma 12.6 (singular-hyperbolic K-to-Bernoulli criterion).** Let  $(X, S, \nu)$  be an invertible completely hyperbolic K-system with a locally finite singularity partition, absolutely continuous stable/unstable holonomy, and the local-product/growth property Lemma 12.3. Then  $S$  is Bernoulli.

To prove Lemma 12.6, fix a finite regular partition  $\mathcal{P}$  whose boundary has zero measure. Given  $\epsilon > 0$ , choose a compact union  $G$  of product rectangles with  $\nu(G) > 1 - \epsilon$  on which: (i) stable and unstable leaves have length at least  $a > 0$ ; (ii) their holonomy Jacobians differ from 1 by at most  $\epsilon$  on rectangles of diameter  $< \delta$ ; and (iii) the first  $N$  iterates avoid the  $\delta$ -neighborhood of singularities except on a set of measure  $< \epsilon$ . Such  $G$  exists by Lemma 12.3, absolute continuity, and Egorov’s theorem.

The K property gives asymptotic independence of a finite present block from a sufficiently remote past block. More precisely, for every  $m$  there is  $n_0(m)$  such that for  $n \geq n_0(m)$ , outside a family of past atoms of total measure  $< \epsilon$ ,

$$\sum_{A \in \mathcal{P}_{-m}^m} |\nu(A | B) - \nu(A)| < \epsilon, \quad B \in \mathcal{P}_{-n-m}^{-n+m}. \quad (12.3.2)$$

Inside a good past atom  $B$ , cut by unstable leaves. Stable holonomy carries conditional measure from each unstable slice to a reference slice with Jacobian  $1 + O(\epsilon)$ . Hyperbolicity makes the images of the stable diameter smaller than the resolution of the central  $2m+1$  block, except on the already removed singular pieces. Therefore the name distribution of the central block on  $B$  can be coupled with its unconditional distribution so that the unmatched mass is  $O(\epsilon)$ . Equation (12.3.2) controls the remaining macroscopic mismatch. Thus  $\mathcal{P}$  is very weak Bernoulli.

For reference, the last measure-theoretic implication can be proved directly: for successive separated blocks choose maximal couplings of their conditional name distributions. If the unmatched masses are  $\epsilon_j$  with  $\sum_j \epsilon_j < \infty$ , Borel-Cantelli gives an isomorphism, outside a null set, between the resulting name process and an independent block process. Refining the block lengths and a generating sequence of partitions gives a Bernoulli shift with the same entropy. Hence every finite generating regular partition is very weak Bernoulli and  $S$  is Bernoulli. This proves Lemma 12.6.

Applying Lemma 12.6 to  $T$  uses the Uniform hyperbolicity of finite-horizon dispersing billiards for complete hyperbolicity, the Ergodicity of dispersing billiards/Lemma 12.3 for local product connectivity and absolute continuity, and Lemma 12.5 for K. The collision map is therefore Bernoulli.

For the flow, use the suspension representation (12.2.4). On each regular flow box the weak stable and weak unstable holonomies are the lifts of those of the collision map and have absolutely continuous Jacobians. The roof is piecewise  $C^2$  and bounded above by finite horizon. The same Pinsker-tail argument gives the K property of the billiard flow in its standard continuous-time sense, and the preceding very-weak-Bernoulli

construction applies to time- $t$  maps after discarding the null set of boxes whose endpoints cross a collision during the chosen finite block. Hence every nonzero time map is Bernoulli, so the billiard flow is Bernoulli.

A Bernoulli automorphism is K. On a mixing Bernoulli component, independence of sufficiently separated coordinate blocks gives mixing of every finite order first for cylinder functions and then, by density, for bounded measurable functions. This proves the final consequence in the statement.

*The quantitative VWB estimate.* To make the Bernoulli upgrade completely explicit, let  $\mathcal{P}_a^b = \bigvee_{j=a}^b T^{-j}\mathcal{P}$  and measure the distance between two length- $m$  names by normalized Hamming distance. Choose product rectangles covering all but  $O(\epsilon^3)$  of phase space, with holonomy Jacobians  $1 + O(\epsilon^3)$ , and choose a boundary thickness  $\eta$  so that  $\mu(N_\eta(\partial\mathcal{P})) < \epsilon^3$ . The growth lemma implies that, on all but  $O(\epsilon)$  of a remote past atom, the image consists of long unstable crossings; the mass cut by a singularity during the next  $m$  steps is  $O(e^{-cn} + m\eta)$ . Match the crossing masses rectangle by rectangle and pair them by stable holonomy. For paired points the future distance decays exponentially, so their symbols differ only when one orbit visits the  $\eta$ -neighborhood of the partition boundary. Fubini and Markov therefore give a coupling whose mean normalized Hamming distance is  $O(\epsilon)$ . After rescaling  $\epsilon$ , this is precisely the very-weak-Bernoulli condition. Successive maximal couplings and Borel-Cantelli identify the name process with a Bernoulli process.

The flow version is not being inferred from an arbitrary suspension theorem. Use regular flow boxes. The lifted stable holonomy pairs both the base point and the vertical coordinate, and a name can fail to match only near a collision roof or a singular base boundary. Finite horizon bounds the roof, and its piecewise- $C^2$  variation along stable leaves is summable. The same Hamming estimate therefore holds for every nonzero time map, proving the Bernoulli-flow assertion.  $\square$

#### Theorem K12.4. Exponential decay of correlations

**Statement.** Let  $T : M \rightarrow M$  be the collision map of a planar periodic finite-horizon dispersing billiard with finitely many pairwise disjoint  $C^3$  strictly convex scatterers, curvature bounded away from zero and infinity, and no cusps, equipped with its natural invariant probability measure  $\mu$ . On each mixing component there exist  $C > 0$  and  $0 < \vartheta < 1$  such that for every pair of bounded dynamically Hölder observables  $f, g$  with zero mean,

$$\left| \int_M f(g \circ T^n) d\mu \right| \leq C \|f\|_{DH} \|g\|_{DH} \vartheta^n, \quad n \geq 0.$$

#### Scope note

The theorem is stated for the collision map. Exponential mixing of the suspension flow is a separate result. The proof route uses a Young-tower/Markov extension with exponential return-time tails.

*Proof.* We use the homogeneous unstable curves of the definition of homogeneity

strips above. If  $W$  is such a curve and  $T(W) = \bigcup_i W_i$  is its decomposition at singularity and homogeneity boundaries, write

$$\Lambda_i(x) = \frac{\|DT_x v^u\|_*}{\|v^u\|_*}, \quad x \in T^{-1}W_i. \quad (12.4.1)$$

The cone estimate the Uniform hyperbolicity of finite-horizon dispersing billiards gives  $\Lambda_i > 1$ . Near a grazing strip  $H_k$ , the width is  $O(k^{-3})$  while the derivative in the unstable direction is  $\asymp k^2$ . Consequently the sum of inverse minimal expansion over all children created at one iterate is finite. After taking a fixed iterate  $n_0$  (so that the uniform dispersing expansion has acted before the next possible cut), one obtains

$$\sup_W \sum_i \frac{1}{\inf_{T^{-1}W_i} \Lambda_i^{(n_0)}} \leq \theta_0 < 1. \quad (12.4.2)$$

Bounded logarithmic distortion follows from (12.2.2) on each homogeneous itinerary.

**Lemma 12.7 (one-step expansion and growth lemma).** For sufficiently short homogeneous unstable curves, (12.4.2) holds for a fixed iterate  $n_0$ . Hence for some  $C > 0$ ,  $0 < \theta < 1$ ,

$$m_W\{x : r_n(T^n x) < \delta\} \leq C(\theta^n m_W(W) + \delta)m_W(W), \quad (12.4.3)$$

where  $r_n$  is the distance to the nearer endpoint of the homogeneous component of  $T^n W$ .

To derive (12.4.3), split the descendants of  $W$  into those whose image length has reached a fixed macroscopic scale and those that remain short. The total preimage weight of the short descendants contracts by  $\theta_0$  every  $n_0$  iterates by (12.4.2). For a long descendant, only a proportion  $O(\delta)$  of its conditional measure lies within distance  $\delta$  of an endpoint; distortion transports this estimate back with a uniform constant. Summing the resulting geometric series gives (12.4.3).

Choose a small product rectangle  $\Delta$  (a “magnet”) whose stable and unstable sides are regular. When a homogeneous unstable curve has grown to macroscopic size, absolute continuity and transversality imply that a fixed fraction  $a > 0$  of it crosses  $\Delta$  properly within at most  $N_0$  further iterates. Remove that fraction and iterate the remainder. By (12.4.3), after  $n$  blocks the unreturned conditional mass is bounded by  $(1 - a/2)^n$  once the finitely many initial short pieces are absorbed into the constant. Doing the same on the stable foliation yields a Markov return map  $F = T^R : \Delta \rightarrow \Delta$  with

$$\mu_\Delta\{R > n\} \leq C_1 e^{-c_1 n}. \quad (12.4.4)$$

The separation-time distortion estimate is

$$\left| \log \frac{J^u F(x)}{J^u F(y)} \right| \leq C_2 \vartheta_0^{s(Fx, Fy)} \quad (12.4.5)$$

for points on the same return atom.

**Lemma 12.8 (finite-horizon billiard Young tower).** The inducing construction above gives a Young tower with Markov return map, hyperbolic product structure, bounded separation-time distortion, and the exponential tail (12.4.4).

Quotient each stable leaf of the tower to a point. The induced quotient operator  $\mathcal{L}$  on separation-time Hölder functions satisfies a Lasota-Yorke estimate

$$|\mathcal{L}^n h|_\eta \leq C_3 \rho^n |h|_\eta + C_3 \|h\|_1, \quad 0 < \rho < 1, \quad (12.4.6)$$

because each inverse branch contracts separation time and (12.4.5) controls the Jacobian. The exponential return tail makes the tower-level renewal operator analytic in a disk of radius  $> 1$ ; on a mixing component the only unit-circle eigenvalue is 1, which is simple. Therefore for zero-mean  $h$

$$\|\mathcal{L}^n h\|_1 \leq C_4 \rho_1^n \|h\|_\eta. \quad (12.4.7)$$

This is the tower version of the spectral-gap/mixing mechanism in F14-THE-006.

**Lemma 12.9 (exponential tower coupling/correlation estimate).** On every mixing component of the tower from Lemma 12.8, bounded separation-time-Hölder observables have exponentially decaying correlations.

A bounded dynamically Hölder billiard observable lifts to a tower observable whose oscillation on an  $n$ -cylinder is  $O(\eta^n)$  by the definition of dynamically Hölder observable above. Stable leaf contraction allows replacement by its stable-quotient representative with an exponentially small error. Apply Lemma 12.9 to those representatives and absorb the two quotient errors. We obtain constants  $C > 0$  and  $0 < \vartheta < 1$  such that

$$\left| \int f(g \circ T^n) d\mu \right| \leq C \|f\|_{DH} \|g\|_{DH} \vartheta^n \quad (12.4.8)$$

for bounded zero-mean dynamically Hölder  $f, g$  on each mixing component.

The construction Lemma 12.8 is the precise structural assertion intended by the second sentence of the theorem. We do *not* use the logically false converse “exponential correlations imply a Young tower”; the reader-facing word “equivalently” has therefore been repaired to “more precisely, the proof constructs”. Nothing in the argument gives exponential mixing of the continuous-time suspension, and no such claim is made.

*Why the return-time tail is genuinely exponential.* For a standard family of homogeneous unstable curves set  $\mathcal{Z} = \int |W|^{-1} d\lambda(W)$ . The one-step inverse-expansion estimate gives

$$\mathcal{Z}(T^n \mathcal{G}) \leq C \vartheta^n \mathcal{Z}(\mathcal{G}) + C.$$

Thus every surviving family recovers a fixed proportion of macroscopic curves in uniformly bounded time. Fix one regular product “magnet”. Compactness of the macroscopic curve family and local ergodicity provide finitely many regular itineraries to the magnet, hence a uniform fraction  $a > 0$  of every recovered family crosses it properly within bounded time. After each failed trial the remainder recovers again, so

$$\mu\{R > jL\} \leq (1 - a/2)^j.$$

This proves the exponential tower tail rather than assuming it.

The correlation estimate can also be obtained directly by coupling. Each time two



tower densities return to the base, bounded distortion allows a fixed fraction of their mass to be paired by stable holonomy. The exponential return tail makes the first successful coupling time exponentially tailed; after coupling, stable contraction makes future discrepancies exponentially small. Integrating over the coupling time yields exponential decay of correlations on the tower and hence, after the stable-leaf projection, for the dynamically Hölder billiard observables in the theorem.  $\square$

### Theorem K12.5. Central limit theorem

**Statement.** Let  $T$  be the collision map from Theorem K12.4 for a planar finite-horizon dispersing billiard, and let  $f$  be a bounded dynamically Hölder observable with  $\int f d\mu = 0$ . Then  $n^{-1/2} \sum_{j=0}^{n-1} f \circ T^j$  converges in distribution to a centered normal law  $N(0, \sigma_f^2)$ , where the absolutely convergent Green-Kubo variance is  $\sigma_f^2 = \int f^2 d\mu + 2 \sum_{k \geq 1} \int f(f \circ T^k) d\mu$ . In the theorem's tower class,  $\sigma_f^2 = 0$  exactly when  $f = g - g \circ T$  for some  $g \in L^2(\mu)$ .

*Proof.* Let  $S_n f = \sum_{j=0}^{n-1} f \circ T^j$ . By the Exponential decay of correlations,

$$\sum_{k \geq 1} \left| \int f(f \circ T^k) d\mu \right| < \infty, \quad (12.5.1)$$

so the Green-Kubo expression in the statement converges absolutely.

Lift  $f$  to the exponential-tail Young tower of Lemma 12.8 and quotient stable leaves. Let  $P$  be the transfer operator of the quotient map. The spectral estimate (12.4.7) gives

$$\sum_{n \geq 1} \|P^n f\|_2 < \infty. \quad (12.5.2)$$

Thus

$$h = \sum_{n \geq 1} P^n f \in L^2, \quad m = f + h - h \circ T \quad (12.5.3)$$

is well defined on the tower (after the standard stable-leaf correction, which is itself an  $L^2$  coboundary because stable oscillations are exponentially summable). Since  $P(h \circ T) = h$  and  $Ph = h - Pf$ , (12.5.3) gives

$$Pm = Pf + (h - Pf) - h = 0. \quad (12.5.4)$$

Consequently  $m, m \circ T, \dots$  is a reverse martingale-difference sequence with respect to the decreasing future filtration.

**Lemma 12.10 (Gordin decomposition on the billiard tower).** Every bounded centered dynamically Hölder observable admits

$$f = m - h + h \circ T, \quad Pm = 0, \quad m, h \in L^2, \quad (12.5.5)$$

with  $h$  obtained by the convergent transfer sum (12.5.3).

The martingale CLT applies to the stationary square-integrable sequence  $m \circ T^j$ :

$$\frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} m \circ T^j \implies N(0, \|m\|_2^2). \quad (12.5.6)$$

A short proof uses conditional characteristic functions. For fixed  $t$ , expand  $e^{itm/\sqrt{n}}$  to second order; the conditional first-order term vanishes by (12.5.4), the sum of conditional second moments converges by the ergodic theorem to  $\|m\|_2^2$ , and the  $L^2$  truncation error is  $o(1)$ . Iteration yields  $e^{-t^2\|m\|_2^2/2}$ . Since

$$S_n f - S_n m = h \circ T^n - h, \quad (12.5.7)$$

the right side divided by  $\sqrt{n}$  tends to zero in probability, so (12.5.6) also holds for  $f$ .

To identify the variance, stationarity and (12.5.1) give

$$\frac{1}{n} \int (S_n f)^2 d\mu = \int f^2 d\mu + 2 \sum_{k=1}^{n-1} \left(1 - \frac{k}{n}\right) \int f(f \circ T^k) d\mu. \quad (12.5.8)$$

Dominated convergence sends the right side to the Green-Kubo series. From (12.5.5), martingale orthogonality gives the same limit  $\|m\|_2^2$ . Therefore

$$\sigma_f^2 = \|m\|_2^2 = \int f^2 d\mu + 2 \sum_{k \geq 1} \int f(f \circ T^k) d\mu. \quad (12.5.9)$$

Finally  $\sigma_f^2 = 0$  iff  $m = 0$  almost everywhere, and then (12.5.5) says  $f = -h + h \circ T$ . Setting  $g = -h$  yields  $f = g - g \circ T$  with  $g \in L^2$ . Conversely, if  $f = g - g \circ T$  with  $g \in L^2$ , then  $S_n f = g - g \circ T^n$  and

$$\frac{1}{n} \|S_n f\|_2^2 \leq \frac{4\|g\|_2^2}{n} \rightarrow 0. \quad (12.5.10)$$

Comparison with (12.5.8)-(12.5.9) gives  $\sigma_f^2 = 0$ . This proves the CLT, variance formula, and exact coboundary criterion.

*The quotient-to-billiard lifting.* On the Young tower, choose one reference point on each local stable leaf. Exponential stable contraction makes

$$\chi(x) = \sum_{j \geq 0} (f(T^j x) - f(T^j x^*))$$

converge in  $L^2$ , and telescoping gives  $f = \bar{f} \circ \pi + \chi - \chi \circ T$  on the stable quotient. If  $P$  is the quotient transfer operator, exponential mixing gives  $\sum_{n \geq 1} \|P^n \bar{f}\|_2 < \infty$ . Hence  $h = \sum_{n \geq 1} P^n \bar{f}$  and  $m = \bar{f} + h - h \circ T$  satisfy  $Pm = 0$ . On the natural extension,  $m \circ \bar{T}^j$  is therefore a stationary reverse martingale-difference sequence. The usual truncation/Lindeberg argument for square-integrable martingale differences gives the Gaussian limit. Both coboundary endpoint terms are  $o_{L^2}(\sqrt{n})$ , so the limit is the limit for the original collision observable. This also shows that the zero-variance  $\iff L^2$ -coboundary criterion proved above lives on the original billiard map, not merely on the quotient tower.  $\square$

**Theorem K12.6. Bunimovich stadium hyperbolicity and ergodicity**

**Statement.** For the standard Bunimovich stadium obtained by joining two parallel line segments by semicircular caps, the billiard map has nonzero Lyapunov exponents at almost every nonsingular phase point (that is, at almost every point whose orbit avoids tangencies and corner singularities) and is ergodic with respect to its natural invariant probability measure. Its hyperbolicity is produced by the defocusing mechanism: after reflection from a focusing cap, a convergent beam passes its focal point before the next focusing collision and becomes divergent. The stadium map and flow have the Kolmogorov  $K$ -property (trivial Pinsker factor) and, by the Chernov–Haskell theorem for nonuniformly hyperbolic  $K$ -systems with singularities, are Bernoulli.

*Proof.* Let the stadium consist of two semicircles of radius  $R$  joined by parallel segments of length  $L$ . The flat pieces have curvature zero and the caps have focusing curvature  $-1/R$  in the dispersing sign convention. Consider the first-return map  $F$  from a cap collision to the next cap collision. If an orbit bounces on a flat side in between, reflect the stadium successively in that side. In the unfolded strip all those neutral reflections disappear and the orbit is one straight segment joining one cap copy to another. Thus only the two cap reflections alter wave-front curvature.

For a wave front let  $f = 1/B$  be its signed focal distance immediately after a cap reflection ( $f < 0$  means the focal point is behind the outgoing ray,  $f > 0$  ahead). The free-flight rule (12.1.3) becomes simply

$$f_{\text{next}}^- = f^+ + \tau. \quad (12.6.1)$$

At a cap the mirror equation has the focusing sign

$$\frac{1}{f^+} = \frac{1}{f^-} - \frac{2}{R \cos \beta}. \quad (12.6.2)$$

The elementary stadium geometry says that for every regular cap-to-cap segment which is not the diameter separatrix, the outgoing focal point specified by (12.6.2) is reached strictly before the next cap collision. Indeed the normal chord of the semicircle has length  $2R \cos \beta$ ; any intervening unfolded translation by the flat-strip width increases the cap-to-cap flight beyond the focal distance furnished by the right side of (12.6.2). Hence the beam has crossed its focus and is divergent by the time it reaches the next cap. Compact subsets away from tangency and the diameter separatrix have a uniform positive margin in this inequality.

**Lemma 12.11 (stadium defocusing cone for the cap return).** For the cap-to-cap induced map  $F$ , the cone consisting of wave fronts whose focal point lies before the next focusing collision is forward invariant. Every compact regular branch expands transverse Jacobi fields by a factor  $> 1$ , and successive returns multiply these factors. Equality can persist only on the neutral bouncing-ball/separatrix families, which have zero collision measure.

To see the expansion explicitly, after the focus has been crossed one has  $B > 0$  before the next cap. Equation (12.1.3) then gives  $|J_{\text{next}}| = |J|(1 + \tau B) > |J|$ . The cap reflection

maps the resulting cone into the next defocusing cone by (12.6.2). Therefore, for every orbit with infinitely many regular cap returns,

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log \|DT^n v^u\| > 0 \quad (12.6.3)$$

when return frequency is positive. Almost every orbit has positive cap-return frequency by recurrence: the set of trajectories that forever collide only with the two parallel flat sides is the exact vertical bouncing-ball family, a one-parameter subset of the two-dimensional collision section and hence has zero measure. Applying the same argument to  $T^{-1}$  gives a negative stable exponent. This proves nonzero Lyapunov exponents almost everywhere on the nonsingular phase space.

The singularity set for the induced map consists of tangencies, cap/flat junction preimages, and their iterates. On each regular branch the defocusing cone gives the same growth estimate as Lemma 12.3; junction curves are finite in each finite return level and their neighborhoods satisfy the corresponding summable cut estimate. Absolute continuity follows from the convergent unstable-Jacobian product as in (12.2.2). Hopf chains therefore make Birkhoff averages constant on local product components. The stadium is connected and a positive-measure open family of trajectories connects each cap to each flat side, so the component graph is connected. Thus the collision map is ergodic.

**Lemma 12.12 (stadium ergodicity and K property).** The defocusing induced map has the singularity-growth and absolutely continuous local-product structure needed for the Hopf argument. Its global transition graph is connected, hence the stadium map is ergodic. The unstable-tail/Pinsker argument of Lemma 12.5 applies verbatim and gives the K property; the same holds for the billiard flow.

Finally Lemma 12.6 applies to this completely hyperbolic K-system with singularities. It gives the Bernoulli property of the stadium collision map. Its flow version applies to the natural suspension/flow boxes and gives the Bernoulli property of the stadium flow. This proves every assertion of the theorem.

*Two points needed for a noncircular stadium argument.* The defocusing condition is most safely stated with focusing times. If  $f^-$ ,  $f^+$  are the signed backward and forward focusing times of a tangent beam, the mirror formula is

$$\frac{1}{f^+} + \frac{1}{f^-} = \frac{2\kappa(s)}{\sin \theta}.$$

After unfolding the flat-side reflections, the next curved collision occurs on a translated or reflected copy of the same semicircle. For every regular cap-to-cap branch away from the diameter/separatrix limit, the forward focus lies strictly before that next cap; on compact regular branches the gap is uniform. Hence the induced cap return has an invariant expanding cone.

It is also not enough to say merely that cap returns occur infinitely often. The only trajectories that never hit a cap are the exactly perpendicular bouncing-ball family, which is null. Therefore the invariant saturation of the cap section has full measure. Birkhoff's theorem applied to the cap indicator gives a positive cap-return frequency

on almost every ergodic component, since a component with zero frequency would lie outside that full saturation. The induced expansion thus yields nonzero Lyapunov exponents per ordinary collision almost everywhere. Tangency and cap/flat junction preimages satisfy the same inverse-expansion/growth bookkeeping as the induced map, so the Hopf/local-ergodicity argument is legitimate before the K-to-Bernoulli criterion is applied.

□

## 12.4 Worked examples

### Worked example: curvature through flight and collision

Let  $B^-$  be the curvature of an unstable infinitesimal wave front immediately before a collision with a dispersing boundary of curvature  $\kappa > 0$ , and let  $\varphi$  be the collision angle measured from the inward normal convention used in the billiard curvature formula. The reflection law has the schematic form

$$B^+ = B^- + \frac{2\kappa}{\cos \varphi}.$$

Thus a dispersing collision adds a strictly positive amount of curvature. During a free flight of length  $\tau > 0$ , geometric optics gives

$$B_{\text{next}}^- = \frac{B^+}{1 + \tau B^+}.$$

If  $B^+ > 0$ , then  $B_{\text{next}}^- > 0$  as well. Moreover the transverse separation of two neighboring rays is multiplied by

$$1 + \tau B^+ > 1.$$

This is the elementary cone-expansion mechanism behind the full theorem. The actual proof must additionally control grazing angles, singularity cuts, and the conversion to a uniform adapted norm.

#### Check your understanding

Suppose  $B^- = 0$ ,  $\kappa = 1$ ,  $\cos \varphi = 1/2$ , and  $\tau = 1$ . Compute  $B^+$ ,  $B_{\text{next}}^-$ , and the free-flight expansion factor.

### Boundary case: infinite horizon

If a periodic Lorentz gas contains an unobstructed corridor, free-flight lengths are unbounded. The finite-horizon hypothesis in the exponential-mixing theorem fails. Uniform hyperbolicity of individual nonsingular collisions can still coexist with this failure, but the return-time tail is no longer the exponential one used in this chapter. Therefore the stated exponential correlation and CLT package

cannot simply be invoked from the finite-horizon theorem.

### Common pitfall

A dispersing billiard is not an Anosov diffeomorphism on a compact smooth phase space. The collision map has singularity curves and unbounded derivatives near grazing collisions. Hyperbolicity estimates must therefore be combined with singularity partitions, distortion control, and recurrence/tail estimates before statistical conclusions such as decay of correlations are available.

## 12.5 Chapter summary

### What to remember

- Dispersing reflection creates invariant hyperbolic cones; finite horizon makes their expansion quantitatively uniform in an adapted norm.
- Singularities force a local-ergodicity/growth analysis beyond smooth uniformly hyperbolic dynamics.
- Finite-horizon Sinai billiards are ergodic and admit strong K/Bernoulli structure on their mixing components.
- An exponential-tail Young tower produces exponential decay of correlations and a central limit theorem for dynamically Hölder observables.
- The Bunimovich stadium achieves hyperbolicity by defocusing rather than pointwise dispersing curvature.

### Looking ahead

K13 removes smooth curvature altogether. Polygonal billiards have zero curvature on each side and singular vertices. Their natural simplification is not hyperbolic expansion but reflection unfolding: a billiard orbit becomes a straight line on a translation surface, and first returns become interval exchanges.

### Check your understanding

Trace one expansion estimate from collision geometry to statistical dynamics: how do free flight and curvature create unstable expansion, and why must singularity control and horizon assumptions be added before one can deduce mixing or limit theorems?

## 12.6 Exercises and problems

### Exercise 12.1

**Difficulty.** Core.

**Related results.** Uniform hyperbolicity of finite-horizon dispersing billiards; Ergodicity of dispersing billiards; Bernoulli/K property for Sinai billiards.

**Problem.** For a finite-horizon periodic Lorentz gas, use the billiard derivative or wave-front curvature update to verify invariance of an unstable cone and identify where finite horizon gives a uniform hyperbolicity constant.

**Solution sketch.** Track an infinitesimal beam through free flight and convex reflection and bound it uniformly away from the cone boundary.

### Exercise 12.2

**Difficulty.** Core.

**Related results.** Exponential decay of correlations; Central limit theorem; Bunimovich defocusing mechanism.

**Problem.** In an infinite-horizon corridor, construct arbitrarily long free flights and explain which finite-horizon estimates and standard CLT normalization can no longer be imported without tail analysis.

**Solution sketch.** Parameterize nearly corridor-parallel trajectories and relate their lengths to the displacement/free-path tail.

### Exercise 12.3

**Difficulty.** Advanced.

**Related results.** Bernoulli/K property for Sinai billiards.

**Problem.** For a two-sided Bernoulli shift, prove independence of cylinder events on disjoint coordinate sets and deduce mixing for cylinder functions; explain what billiard-specific machinery must replace this elementary product structure.

**Solution sketch.** Compute product probabilities, shift until supports are disjoint, and extend by approximation; then identify singularity/coding control as the hard billiard input.

### Exercise 12.4

**Difficulty.** Advanced.

**Related results.** Exponential decay of correlations.

**Problem.** Assuming exponential covariance decay, prove absolute convergence of the Green-Kubo series and  $n^{-1} \text{Var}(\sum_{j < n} f \circ T^j) \rightarrow \sigma^2$ .

**Solution sketch.** Expand the variance as a covariance double sum, regroup by time lag, and apply dominated convergence.

### Exercise 12.5

**Difficulty.** Advanced.

**Related results.** Uniform hyperbolicity of finite-horizon dispersing billiards; Ergodicity of dispersing billiards.

**Problem.** Build a rigorous implication diagram from dispersing geometry to hyperbolicity, ergodicity, mixing, and the CLT, naming one extra ingredient required at each arrow.

**Solution sketch.** Separate cone estimates, singularity/absolute-continuity control, and quantitative tower/correlation estimates rather than collapsing them into one implication.

### Exercise 12.6

**Difficulty.** Lab.

**Related results.** None.

**Problem.** Simulate free-path lengths for finite- and infinite-horizon Lorentz geometries and compare empirical tails and displacement-variance growth under increasing sample size.

**Solution sketch.** Validate collision detection and report finite-sample uncertainty before discussing tail or variance scaling.



## **Part IV**

# **Polygonal Billiards, Translation Surfaces, and Moduli-Space Dynamics**

# Polygonal Billiards, Unfolding and Interval Exchanges

## Why this chapter matters

Polygonal billiards sit outside both previous mechanisms. There is no smooth focusing boundary on which to build caustics and no dispersing curvature to generate hyperbolicity. Instead, reflection can be eliminated geometrically: reflect the polygon rather than the velocity. A broken billiard path then becomes one straight Euclidean segment through reflected copies.

When every polygon angle is a rational multiple of  $\pi$ , only finitely many direction states occur. The reflected copies can therefore be glued into a compact translation surface. Directional billiard flow becomes straight-line flow on that surface, and a transverse first-return map becomes an interval-exchange transformation (IET). Order-theoretic recurrence for IETs, Teichmüller renormalization, and the Kerckhoff–Masur–Smillie theorem then yield unique ergodicity for almost every direction.

**Prerequisites.** Elementary Euclidean reflection geometry; Foundation F01–F04 for minimality, invariant measures, and Oseledets-type reasoning; K10 for one-dimensional order intuition. K14–K15 will supply the moduli and renormalization framework used by the deepest result.

## Learning goals

- Unfold a polygonal billiard trajectory into a straight Euclidean trajectory.
- Explain why rational angles produce only finitely many direction states and hence a compact translation surface.
- Construct an interval-exchange first return from a directional flow.
- Use Keane’s no-connection condition to understand minimality of an IET.
- State precisely what almost-every-direction unique ergodicity means for a rational polygon after finite unfolding.

**Conceptual roadmap**

polygonal reflection  $\rightarrow$  finite unfolding  
 $\rightarrow$  translation surface  
 $\rightarrow$  IET return map  
 $\rightarrow$  renormalization/recurrence  
 $\rightarrow$  unique ergodicity a.e.

The first four arrows are elementary geometric constructions. The last step is deep: it uses the geometry of the moduli space developed in K14–K15 rather than a local billiard estimate.

**Check your understanding**

A rectangle has all angles  $\pi/2$ . How many direction states does the finite unfolding construction predict? Why does this number stay finite for every rational polygon but not for a polygon with an angle  $\pi\alpha$  where  $\alpha$  is irrational?

**13.1 Definitions****Intuition**

**Unfolding trades reflection for topology.** Each time a trajectory reflects, reflect the next copy of the polygon instead. The velocity stays constant, but the orbit now moves through many copies. Gluing those copies by translations packages the entire reflection rule into the topology of one surface.

**A translation surface is locally Euclidean with finitely many cone points.** Away from the vertices inherited from the polygon, transition maps are translations. Thus a fixed direction is globally meaningful and straight-line flow needs no differential equation beyond constant velocity.

**An interval exchange is a piecewise translation.** Choose a segment transverse to the directional flow. Points with the same return itinerary form intervals, and on each such interval the return map is a translation. Discontinuities come from trajectories that hit a cone singularity before returning.

**Keane's condition excludes saddle connections in the return combinatorics.** If forward discontinuity orbits never meet each other, no finite connection blocks an interval from spreading through the whole exchange; this is the engine behind minimality.

**Definition K13.1. rational polygon**

A rational polygon is a Euclidean polygon all of whose interior angles are rational multiples of  $\pi$ .

**Definition K13.2. unfolding**

Unfolding replaces specular reflections by reflecting copies of the polygon so that a billiard orbit becomes a straight geodesic on the glued flat surface.

**Definition K13.3. interval exchange**

An interval exchange transformation cuts an interval into finitely many subintervals and rearranges them by translations according to a permutation, with discontinuities at cut points.

**Definition K13.4. Keane condition**

Keane's infinite distinct orbit condition (i.d.o.c.) requires the forward orbits of discontinuity points of an IET to be infinite, pairwise disjoint, and never hit a discontinuity, in the standard convention.

**Definition K13.5. translation surface and cone singularity**

A *translation surface* is a compact oriented surface equipped away from finitely many marked points with Euclidean coordinate charts whose transition maps are translations. At a marked *cone singularity* the flat metric has total angle  $2\pi m$  for an integer  $m \geq 2$ ; straight-line geodesics are defined in the regular locus and stop if they hit a cone singularity.

**Definition K13.6. direction state and polygon reflection group**

For a polygon, the *reflection group of side directions* is the subgroup of  $O(2)$  generated by reflections in lines parallel to its sides. A *direction state* is the image of an oriented velocity direction under this finite group. For a rational polygon with angles  $\pi m_i/n_i$  and  $N = \text{lcm}(n_i)$ , the orientation-preserving subgroup is cyclic of order  $N$  and the full direction group is dihedral of order  $2N$ .

**Definition K13.7. transverse segment and first-return map**

For a directional translation flow on a translation surface, a *transverse segment* is a compact straight geodesic segment contained in the regular locus whose tangent is nowhere parallel to the flow direction. For a point  $x$  on such a segment  $I$ , its *first-return time* is  $\tau_I(x) = \inf\{t > 0 : \Phi_t(x) \in I\}$  when this set is nonempty and the orbit segment avoids cone singularities; the *first-return map* is  $R_I(x) = \Phi_{\tau_I(x)}(x)$  on the points for which this time is finite.

**Definition K13.8. directional Teichmüller divergence**

Fix a unit-area translation surface  $X$  and a direction  $\theta$ . Rotate  $\theta$  to the vertical and apply  $g_t = \text{diag}(e^t, e^{-t})$ . We say that  $X$  *diverges in direction  $\theta$  in the flat-systole sense* if for every  $\varepsilon > 0$  there is  $T$  such that for every  $t \geq T$  the deformed surface  $g_t r_{-\theta} X$  has a saddle connection of length at most  $\varepsilon$ . Equivalently, its saddle-connection systole tends to zero along that ray.

## 13.2 How the main proofs work

### Intuition

**Reflection unfolding.** Across a side, the reflected outgoing velocity is the mirror image of the incoming velocity. Reflecting the polygon instead applies the inverse mirror to the geometry, so the two velocity vectors become equal. Induction straightens any finite nonsingular collision word.

**Finite rational unfolding.** If every angle is  $\pi m_i/n_i$ , the group generated by the side reflections has a finite rotational part with  $2N$  direction states,  $N = \text{lcm}(n_i)$ . Glue one copy per state along corresponding parallel sides. Edge pairings are translations, while vertex identifications produce only finitely many cone singularities.

**Directional flow as an IET.** Remove the finite set of points whose forward orbit hits a singularity before returning. Between consecutive deleted points the return itinerary is constant, so the first-return displacement is constant. Hence each continuity interval is translated rigidly.

**Keane minimality.** If an orbit were not dense, an endpoint of a maximal missed interval would generate a forward connection between discontinuities. The infinite distinct orbit condition forbids this, so every orbit must be dense.

**Kerckhoff-Masur-Smillie.** Rotate the chosen direction to vertical and follow its Teichmüller geodesic. If the orbit returns often enough to a thick part, quantitative recurrence prevents two distinct transverse invariant measures. A measure estimate shows that almost every direction has the required recurrence. Thus almost every directional flow is uniquely ergodic.

## 13.3 Main results

### Lemma K13.1. Reflection unfolding lemma

**Statement.** Let a billiard trajectory in a polygon hit the relative interior of a side at a non-tangent angle and avoid vertices. Reflect the next copy of the polygon across the supporting line of that side instead of reflecting the velocity. The incoming and reflected outgoing links then form one straight Euclidean segment. Iterating this operation along any finite collision word with no vertex or grazing collision gives a straight segment through successive reflected copies, and folding the copies back recovers the billiard trajectory.

*Proof.* Consider one collision with a side line  $\ell$  and let  $R_\ell$  be Euclidean reflection in that line. If the incoming unit velocity is  $v^-$ , specular reflection is

$$v^+ = R_{\ell,*} v^-. \quad (13.1.1)$$

Instead of changing the velocity, reflect the copy of the polygon lying after the collision

by  $R_\ell$ . In the reflected copy the same geometric segment has velocity

$$R_{\ell,*}^{-1}v^+ = v^-. \quad (13.1.2)$$

Thus the incoming and outgoing billiard segments become one straight segment across the common side.

For a collision word  $\ell_1, \dots, \ell_m$ , successively reflect the next copy by  $R_{\ell_j}$ . Induction on  $j$  using (13.1.2) shows that the first  $j$  billiard legs become a single straight segment in the union of the first  $j + 1$  reflected copies. Conversely, folding those copies back reverses each reflection and recovers equal incidence/reflection angles at every crossed side. This gives a one-to-one correspondence up to the first exceptional collision.

A trajectory hitting a vertex has no unique side normal and hence no unique ordinary specular continuation. At a grazing collision the segment lies on a side and the collision section itself is singular. Excluding these cases is therefore exactly what is needed for the induction. This proves the unfolding lemma.  $\square$

### Theorem K13.2. Finite unfolding of rational polygons

**Statement.** Let  $P$  be a polygon with angles  $\alpha_i = \pi m_i/n_i$  in lowest terms and put  $N = \text{lcm}(n_1, \dots, n_k)$ . The reflection group generated by the side directions has  $2N$  direction states. Taking one reflected copy of  $P$  for each state and identifying corresponding parallel sides by translations produces a compact translation surface with finitely many cone singularities. Every billiard trajectory in  $P$  that avoids vertices unfolds to a straight geodesic on this surface; conversely, every straight geodesic avoiding the cone singularities folds to a billiard trajectory in  $P$ .

*Proof.* Let the side lines be  $\ell_j$  and let  $r_j$  denote reflection in the line through the origin parallel to  $\ell_j$ . Adjacent side directions differ by the exterior angle

$$\pi - \alpha_i = \pi \frac{n_i - m_i}{n_i}. \quad (13.2.1)$$

The product of the two adjacent reflections is therefore rotation through

$$2(\pi - \alpha_i) = 2\pi \frac{n_i - m_i}{n_i}. \quad (13.2.2)$$

Because  $(m_i, n_i) = 1$ , also  $(n_i - m_i, n_i) = 1$ , so this rotation has exact order  $n_i$ . Hence the rotation subgroup generated by all products of side reflections contains cyclic subgroups of orders  $n_i$ . Conversely every side direction is a rational multiple of  $\pi$  with denominator dividing  $N = \text{lcm}(n_i)$ , so every generated rotation is by an integral multiple of  $2\pi/N$ . The orientation-preserving subgroup therefore has exactly  $N$  elements. Adjoining any one reflection gives the other coset, so the full linear reflection group  $G$  is dihedral of order  $2N$ .

**Lemma 13.1 (finite direction-state group).** The reflection group generated by the side directions of a rational polygon has  $N$  rotations and  $N$  reflections, hence exactly  $2N$  direction states.

Take an abstract copy  $P_g$  of  $P$  for every  $g \in G$ . If side  $j$  of  $P_g$  is crossed, the unfolding

rule sends the direction state  $g$  to  $gr_j$ . Pair that side with the corresponding side of  $P_{gr_j}$ . After the linear placements are forgotten and the copies are regarded abstractly, the two paired sides are parallel and have equal length; identify them by the unique translation respecting the endpoint labels. Every side is paired exactly once because  $r_j^2 = 1$ . Thus the quotient is a compact surface: it is obtained from finitely many compact polygons by complete pairwise edge gluing.

Away from vertices, the two Euclidean charts on paired copies differ by a translation, so they define a translation atlas. Only vertex classes can be singular. Around a vertex of angle  $\alpha_i = \pi m_i / n_i$ , the local reflection subgroup has  $2n_i$  sectors before the direction state returns. The total cone angle is therefore

$$2n_i \alpha_i = 2\pi m_i, \quad (13.2.3)$$

an integral multiple of  $2\pi$ . Hence every vertex class is a cone singularity in the sense of the definition of translation surface and cone singularity above, and the quotient is a compact translation surface with finitely many such singularities.

Finally, the Reflection unfolding lemma says that a billiard trajectory avoiding vertices becomes straight whenever the next polygon copy is reflected rather than the velocity. The present edge gluing performs exactly those reflections in the finite direction-state cover, so the unfolded trajectory is a straight geodesic of the translation atlas. Conversely a straight geodesic avoiding cone points crosses paired edges; folding its successive copies back to the original  $P$  applies the inverse reflections and therefore gives the specular billiard law. This proves all assertions.  $\square$

### Theorem K13.3. Directional flow as interval exchange

**Statement.** Let  $(X, \omega)$  be a compact translation surface and fix a direction. Let  $I$  be a compact geodesic segment contained in the regular locus and transverse to that directional flow. Assume  $I$  is chosen so that the first-return map is defined for Lebesgue-a.e. point of  $I$ . After deleting the finitely many singular/discontinuity points generated by trajectories that meet a zero of  $\omega$  before returning,  $I$  is partitioned into finitely many open intervals; on each interval the return itinerary is constant and the first-return map is a translation. Hence the first-return map is an interval-exchange transformation, with interval lengths and permutation determined by the return rectangles.

*Proof.* Rotate the translation charts so that the chosen flow direction is vertical. Write  $I = [a, b]$  in a translation chart as a horizontal segment; transversality allows this after shortening finitely many flow boxes, and the argument is unchanged for an oblique segment because transverse Euclidean length is then only multiplied by one fixed factor.

Let  $\Sigma$  be the finite cone-singularity set. For each singularity and each vertical separatrix prong pointing backwards toward  $I$ , follow the prong backwards until its first intersection with  $I$ , if such an intersection occurs before another singularity. A zero of order  $k$  has only  $k + 1$  vertical prongs in either oriented direction, so this procedure contributes finitely many points. Add the two endpoints  $a, b$ . Call the resulting finite set  $D$ .

We first show that no other discontinuity can occur. Let  $J$  be a connected component of  $I \setminus D$  and choose  $x_0 \in J$  whose first return is defined. The orbit segment

$$\Gamma_0 = \{\Phi_t(x_0) : 0 \leq t \leq \tau_I(x_0)\} \quad (13.3.1)$$

misses  $\Sigma$  by the definition of the return domain. Compactness of  $\Gamma_0$  gives a narrow horizontal interval  $J_0 \Subset J$  around  $x_0$  such that the vertical flow from  $J_0$  follows the same finite sequence of translation charts before returning. In each overlap the coordinate change is a translation. Hence the swept region is a Euclidean flow strip and there are constants  $h > 0$  and  $c \in \mathbb{R}$  for which, in transverse coordinates,

$$\tau_I(x) = h, \quad R_I(x) = x + c, \quad x \in J_0. \quad (13.3.2)$$

The return time need not remain literally equal after changing the original oblique parametrization, but the return map is still a translation in the natural transverse coordinate; this is the invariant assertion needed below.

**Lemma 13.2 (return-strip continuation).** If one point of a component  $J$  of  $I \setminus D$  has a nonsingular first return, then the return itinerary and translation vector extend over all of  $J$ . Failure of continuation at an interior point would produce a vertical separatrix from a cone singularity whose first backward intersection with  $I$  lies in  $J$ , contrary to the definition of  $D$ .

Indeed, let  $J^*$  be the maximal open subinterval of  $J$  containing  $x_0$  on which the itinerary in (13.3.2) persists. If an endpoint  $y$  of  $J^*$  lies in the interior of  $J$ , take  $x_n \rightarrow y$  from  $J^*$ . The corresponding orbit strips have a limiting boundary orbit. It cannot cease returning by escaping to infinity because  $X$  is compact, and it cannot change chart itinerary at a regular point because translation flow has a flow box there. Thus its first obstruction is a cone singularity. Reversing that boundary orbit from the singularity gives precisely a separatrix whose first meeting with  $I$  is  $y$ , so  $y \in D$ , a contradiction. This proves Lemma 13.2.

By the assumption that the first-return map is defined for Lebesgue-a.e. point of  $I$ , every component  $J$  except possibly a null collection of points has a returning point. Since there are only finitely many components, Lemma 13.2 shows that each such  $J$  is a full continuity interval of the return map. Denote them by  $J_1, \dots, J_d$ , with lengths  $\lambda_1, \dots, \lambda_d$ . On  $J_j$  the return map has the form

$$x \mapsto x + c_j. \quad (13.3.3)$$

Vertical flow preserves transverse Euclidean length, hence the images  $R_I(J_j)$  are open intervals of the same lengths  $\lambda_j$ . Two images cannot overlap on a nonempty open interval: otherwise two distinct points on one transversal would lie on the same oriented orbit before first return, contradicting the definition of first return. The images cover  $I$  modulo the finite cut set and the null exceptional return set; equality of total lengths then forces them to cover  $I$  modulo endpoints. Their left-to-right order is therefore a permutation  $\pi$  of  $\{1, \dots, d\}$ .

Thus  $I$  is cut into finitely many subintervals and rearranged by the translations (13.3.3) according to  $\pi$ . This is exactly the interval exchange of the definition of inter-



val exchange above. The rectangles swept by the  $J_j$  record the return itinerary, their widths are the exchanged lengths, and their vertical gluing records the permutation, proving the stated geometric description.  $\square$

#### Theorem K13.4. Keane i.d.o.c. minimality criterion

**Statement.** Let  $T$  be an interval exchange transformation. Assume Keane's infinite distinct orbit condition: the forward orbits of its discontinuity points are infinite and pairwise disjoint (equivalently there are no forward connections between discontinuities, in the fixed endpoint convention). Then every orbit of  $T$  is dense; that is,  $T$  is minimal. Irreducibility is not an additional hypothesis of this implication.

*Proof.* Let  $T : I \rightarrow I$  be an interval exchange, with the fixed half-open endpoint convention used in the definition of interval exchange above, and let  $D$  be its finite discontinuity set. The i.d.o.c. says in particular that

$$T^m d \neq T^n d' \quad \text{for } d, d' \in D \text{ and } m, n \geq 0, \quad (13.4.1)$$

unless  $d = d'$  and  $m = n$ ; moreover none of these forward images is a (discontinuity-convention) endpoint connection.

Assume for contradiction that  $T$  is not minimal. Choose  $x$  whose orbit closure  $K = \overline{\{T^n x : n \in \mathbb{Z}\}}$  is a proper closed invariant set. The two-sided orbit is legitimate away from the finite endpoint set; if  $x$  lies on an endpoint orbit, replace it by a nearby point in the same nonminimal component. Let  $J = (u, v)$  be a component of  $I \setminus K$  of maximal length. Because  $K$  is invariant, whenever  $T$  is continuous at the endpoints and on  $J$ , the image  $T(J)$  is another complementary interval of the same length. Maximality therefore implies that all such images are again maximal gaps.

**Lemma 13.3 (gap-to-connection lemma).** For an interval exchange, if a proper invariant closed set has a maximal complementary interval, then some forward orbit of a discontinuity meets the forward orbit of a (possibly equal) discontinuity. Equivalently, a proper minimal component forces a connection.

We prove the child. Follow  $J, TJ, T^2 J, \dots$  for as long as no image has a boundary point in  $D$ . During this time every image is obtained by one translation and all have the same length  $|J|$ . If their interiors were pairwise disjoint indefinitely, the unit interval would contain infinitely many disjoint intervals of fixed positive length, impossible. Hence there are  $0 \leq r < s$  for which  $T^r J$  and  $T^s J$  meet.

Two maximal complementary intervals of the same closed set have either disjoint interiors or are identical. Thus  $T^r J = T^s J$ . On this interval  $T^{s-r}$  is a translation from the interval onto itself, hence the translation amount is zero. Every point of the interval is then periodic. A periodic open interval has boundary in the finite union of preimages of  $D$  (otherwise one could enlarge it through a common continuity neighborhood), and tracing one boundary through one period gives a forward connection in  $D$ . This contradicts (13.4.1).

Consequently the preceding continuation must instead stop at a first time  $n$  when a boundary of  $T^n J$  is a discontinuity  $d \in D$ . Apply the same argument backwards

from the opposite boundary of  $J$  (or, equivalently, forward for the inverse IET). Since only finitely many interval labels occur, a first boundary collision produces another discontinuity  $d' \in D$  and integers  $p, q \geq 0$  such that the common endpoint is

$$T^p d = T^q d'. \quad (13.4.2)$$

If  $p = q$  then the two discontinuities coincide and the two boundary branches were the same, contradicting that they bound a positive gap; after cancelling common iterates from (13.4.2), one obtains a nontrivial forward connection. This proves Lemma 13.3.

Applying Lemma 13.3 to the orbit closure  $K$  contradicts the i.d.o.c. Therefore no proper orbit closure exists. Every orbit is dense, so  $T$  is minimal. Notice that irreducibility has not been used separately: if the defining permutation had a proper invariant initial block, its boundary cut point would produce a connection, so i.d.o.c. already excludes reducibility.  $\square$

### Theorem K13.5. Kerckhoff-Masur-Smillie unique ergodicity theorem

**Statement.** For every compact translation surface  $X$ , for Lebesgue-almost every direction  $\theta \in S^1$ , the straight-line flow on  $X$  in direction  $\theta$  is uniquely ergodic: the normalized area measure is the unique invariant probability measure for that directional flow.

*Proof.* Normalize the flat area to one; scaling does not change directional unique ergodicity. For a direction  $\theta$ , rotate it to the vertical and put

$$X_{\theta,t} = g_t r_{-\theta} X, \quad g_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}. \quad (13.5.1)$$

We split the proof into the two geometric assertions announced in the roadmap.

**Lemma 13.4 (bounded-systole compactness for a fixed stratum).** Fix the genus and the cone-angle data. For every  $\varepsilon_0 > 0$ , every unit-area translation surface whose saddle-connection systole is at least  $\varepsilon_0$  has a saddle-connection triangulation all of whose edge lengths lie in  $[\varepsilon_0, C(\varepsilon_0)]$ . Consequently every sequence with that systole bound has a subsequence converging, triangle by triangle and with constant gluing pattern, to a connected translation surface with the same topological data.

To prove Lemma 13.4, start with any saddle-connection triangulation and choose an edge  $AC$  of maximal length  $K$ . The two adjacent triangles have total area at most one. If  $B, B'$  are their opposite vertices, their perpendicular distances to  $AC$  are at most  $2/K$ . Since all saddle connections have length at least  $\varepsilon_0$ , Pythagoras applied to  $AB, BC$  (and similarly on the other side) shows, once  $K$  is large compared with  $\varepsilon_0^{-1}$ , that the feet of the perpendiculars stay a definite distance from  $A, C$ . Flipping the diagonal  $AC$  to the saddle connection  $BB'$  therefore reduces the maximal edge length by a quantity depending only on  $\varepsilon_0$ . Repeating the flip terminates with a uniform upper bound. There are only finitely many combinatorial triangulation patterns for the fixed topology. A diagonal subsequence makes every edge vector converge. The lower systole bound prevents a triangle cycle that separates singularities from collapsing, so the same gluing produces a connected limiting surface. This proves Lemma 13.4.

**Lemma 13.5 (Masur nondivergence criterion).** If the vertical ray  $g_t Y$  of a unit-area translation surface does not diverge in the sense of the definition of directional Teichmüller divergence above, then the vertical flow on  $Y$  is uniquely ergodic.

Choose  $t_n \rightarrow \infty$  and  $\varepsilon_0 > 0$  such that  $\text{sys}_{\text{sc}}(g_{t_n} Y) \geq \varepsilon_0$ . By Lemma 13.4, after a subsequence the surfaces  $Y_n = g_{t_n} Y$  converge to a connected translation surface  $Y_\infty$  through fixed triangulations. Let  $f_n : Y \rightarrow Y_n$  denote the identity on the underlying marked topological surface.

Let  $\mu, \nu$  be ergodic invariant probability measures for the vertical flow on  $Y$ . Choose a countable family of embedded regular rectangles  $U = I \times (0, 1)$  generating the Borel sigma-algebra. By the pointwise ergodic theorem there are generic points  $x \in B(\mu)$  and  $y \in B(\nu)$ . Suppose that along one subsequence  $f_n(x)$  and  $f_n(y)$  converge to opposite corners of one embedded rectangle  $R \subset Y_\infty$  of positive vertical height  $c$ . Pull  $R$  back to  $Y$ . Its two vertical sides through  $x, y$  have height

$$T_n = c_n e^{t_n} \rightarrow \infty, \quad c_n \rightarrow c > 0, \quad (13.5.2)$$

and its horizontal width is  $O(e^{-t_n})$ . Every horizontal crossing of the left side through the test rectangle  $U$  also crosses the right side, except for at most one crossing near each horizontal boundary. Therefore

$$|\#(I \cap C_n(x)) - \#(I \cap C_n(y))| \leq 2. \quad (13.5.3)$$

Genericity and the unit height of  $U$  turn the two counts divided by  $T_n$  into  $\mu(U)$  and  $\nu(U)$ . Letting  $n \rightarrow \infty$  in (13.5.3) gives  $\mu(U) = \nu(U)$  for every generating rectangle, hence  $\mu = \nu$ .

It remains to place arbitrary generic points into such rectangle chains. Every nonempty open set of  $Y_\infty$  contains an accumulation point of  $f_n(z)$  for a generic point  $z$  of some ergodic component: otherwise its pullbacks would have positive area while every ergodic component assigned zero mass to them, contradicting the ergodic decomposition of area. Choose finitely many small open sets whose selected points form a chain across the connected surface  $Y_\infty$ , with consecutive selected points lying at opposite corners of an embedded rectangle. Repeated use of (13.5.3) forces all associated ergodic measures to coincide. A generic point for any other ergodic invariant measure has an accumulation point in the same connected limit and can be joined to this chain. Thus every ergodic invariant probability is the same; by ergodic decomposition every invariant probability is that measure. Since area is invariant and normalized, it is the unique one. Lemma 13.5 is proved.

We next show that the exceptional divergent directions have measure zero. A *saddle-connection complex* will mean a closed triangulable region whose boundary is a finite union of saddle connections and which contains every flat triangle bounded by three of its saddle connections. Its complexity is the number of edges in any triangulation; Euler's formula makes this independent of the chosen triangulation, and a triangulation of the whole surface gives a uniform topological upper bound  $M_X$ .

**Lemma 13.6 (complex-enlargement lemma).** Let  $K$  be a saddle-connection complex and let  $\sigma$  be a saddle connection that crosses  $\partial K$  or is disjoint from  $K$ . There is a

complex  $K' \supset K$  with strictly larger complexity and

$$|\partial K'| \leq 2|\partial K| + |\sigma|, \quad \text{Area}(K') \leq \text{Area}(K) + |\partial K|^2 + |\partial K||\sigma|. \quad (13.5.4)$$

Cut  $\sigma$  at its first encounter with  $K$ . If it is disjoint, add it and fill every newly bounded nonsingular triangle. If one endpoint lies outside, straighten the broken path made of the crossing subsegment and the boundary edge until a cone point is reached; one of the resulting saddle connections is not already in  $K$ , and adjoining it increases complexity. If  $\sigma$  crosses both sides, translate it parallel to itself until it hits a cone point and reduce to the preceding case. Every added edge has length bounded by  $|\partial K| + |\sigma|$  and the added polygon is triangulated by triangles whose base/height are bounded by the same quantities. Summing the triangle areas gives (13.5.4). This proves Lemma 13.6.

**Lemma 13.7 (isolated-short-direction estimate).** Fix  $c > 0$ . Let  $A(t, \varepsilon, c)$  be the set of directions  $\theta$  for which  $X_{\theta,t}$  has a saddle connection of length at most  $\varepsilon$  that meets no other saddle connection of length at most  $c$ . There are  $\bar{\varepsilon}, T, C > 0$  such that

$$\text{Leb } A(t, \varepsilon, c) \leq C\varepsilon \quad (13.5.5)$$

for  $t \geq T$  and  $0 < \varepsilon \leq \bar{\varepsilon}$ .

For a fixed saddle connection  $\alpha$  with holonomy vector  $v$ , the set of angles for which  $|g_t r_{-\theta} v| \leq r$  is the union of two short intervals centered at the direction in which  $v$  is contracted. If  $I_\alpha(r)$  and  $J_\alpha(c)$  denote respectively the  $\varepsilon$ - and  $c$ -windows, elementary trigonometry in the ellipse  $e^{2t}h^2 + e^{-2t}v^2 \leq r^2$  gives

$$|I_\alpha(\varepsilon)| \leq C_0 \varepsilon |J_\alpha(c)| \quad (13.5.6)$$

for large  $t$ , uniformly in  $\alpha$ . Halve  $J_\alpha(c)$ . If the halved windows of two saddle connections overlap and one is in the isolated set, the corresponding geodesic segments must be disjoint; otherwise their intersection persists under the nearby angular deformation and supplies a  $c$ -short crossing. A surface has only a topologically bounded number  $p$  of pairwise interior-disjoint saddle connections. Thus the family of halved  $J_\alpha(c)$  has multiplicity at most  $p$ , and summing (13.5.6) gives (13.5.5).

Now suppose the set  $E$  of divergent directions had positive Lebesgue measure. By the definition of divergence, a diagonal selection gives sets  $E_n \subset E$  with  $\inf_n \text{Leb}(E_n) > 0$ , times  $t_n \rightarrow \infty$ , and  $\varepsilon_n \downarrow 0$  such that every  $X_{\theta,t_n}$ ,  $\theta \in E_n$ , has a saddle connection of length at most  $\varepsilon_n$ . We inductively construct, on positive-measure subsets, complexes  $K_n(\theta)$  whose complexity is at least  $q$ , while

$$|\partial K_n(\theta)| \rightarrow 0, \quad \text{Area } K_n(\theta) \rightarrow 0. \quad (13.5.7)$$

For  $q = 1$ , take the short saddle connection as a one-edge complex.

Assume (13.5.7) at complexity  $q$ . Fix  $c > 0$ . If for a positive-measure subsequence the boundary contains an edge that meets no other  $c$ -short saddle connection, Lemma 13.7 contradicts the positive lower bound on  $\text{Leb}(E_n)$  because the boundary edge length tends to zero. Otherwise, after a diagonal choice  $c \downarrow 0$ , almost every retained complex is met by a saddle connection  $\sigma_n$  with  $|\sigma_n| \rightarrow 0$ . Lemma 13.6 enlarges it to complexity

at least  $q + 1$ , while (13.5.4) preserves both limits in (13.5.7). Thus the induction constructs complexes of every complexity. This contradicts the uniform topological bound  $M_X$ . Hence

$$\text{Leb}(E) = 0. \quad (13.5.8)$$

For every  $\theta \notin E$ , the ray fails to diverge, so Lemma 13.5 gives unique ergodicity of the  $\theta$ -direction flow. Equation (13.5.8) says this holds for Lebesgue-almost every  $\theta$ . The normalized flat area is invariant under every directional translation flow and therefore is the unique invariant probability measure in those directions. This proves the theorem.

*Two quantitative details.* The thick-part compactness used above can be made canonical with a Delaunay triangulation. A unit-area surface with saddle-connection systole  $\geq \varepsilon_0$  cannot contain flat disks of arbitrarily large radius: such a disk would contain a long embedded strip, which either creates a saddle connection shorter than  $\varepsilon_0$  or a flat cylinder whose area exceeds one. The Voronoi cells of the singularities therefore have uniformly bounded diameter, and the dual Delaunay triangulation has all edge lengths between  $\varepsilon_0$  and a uniform upper bound. A fixed combinatorial subsequence and convergent edge holonomies give compactness.

For the exceptional directions, write the divergent set as the intersection of sets for which the saddle-connection systole stays below a prescribed  $\rho$  after a prescribed time. On a positive-measure density subset, build a short saddle-connection complex. At complexity  $q$ , either one of its finitely many boundary edges is isolated among all  $c$ -short saddle connections—whose directions have measure  $O(\rho)$  by the angular estimate—or another short connection meets the complex and the enlargement lemma raises the complexity while keeping perimeter  $O(\rho)$  and area  $O(\rho^2)$ . Since a surface has a topological upper bound on the complexity of such a complex, positive measure would force an impossible complexity. Thus the divergent directions have measure zero. □

### Theorem K13.6. Polygonal billiard unique-ergodicity corollary

**Statement.** Let  $P$  be a rational polygon and let  $X_P$  be the compact translation surface obtained by the standard finite unfolding. By Theorem K13.5, for Lebesgue-almost every direction  $\theta$ , the directional flow on  $X_P$  is uniquely ergodic. Folding the finite directional orbit of  $\theta$  back to  $P$  therefore gives the corresponding billiard flow the unique invariant probability measure induced from normalized area on the unfolding; equivalently, the rational-polygon billiard is uniquely ergodic for almost every direction after the finite directional unfolding is fixed.

*Proof.* Let  $G_P$  be the finite direction-state group from the Finite unfolding of rational polygons and fix an oriented direction  $\theta$ . The directional billiard phase space is

$$\mathcal{B}_\theta = \{(x, g\theta) : x \in P^\circ, g \in G_P\} / \text{the side-reflection identifications}, \quad (13.6.1)$$

with vertices and grazing side states removed. the Finite unfolding of rational polygons constructs the translation surface  $X_P$  by taking exactly the same copies and re-

placing the reflection identification by a translation gluing. Therefore the folding map

$$F : X_P \setminus \Sigma_P \longrightarrow \mathcal{B}_\theta \quad (13.6.2)$$

is one-to-one outside the finite union of glued side segments, and those segments have flat area zero. On every orbit segment avoiding them and the cone points,

$$F \circ \Phi_t^{X_P} = \Phi_t^{\text{bil}} \circ F. \quad (13.6.3)$$

Thus (13.6.2) is a measurable flow isomorphism modulo null sets, not merely a many-to-one factor map.

**Lemma 13.8 (finite-folding measure correspondence).** Pushforward by  $F$  is a bijection between invariant probability measures for the nonsingular directional flow on  $X_P$  and invariant probability measures for the finite-state directional billiard flow. Under this bijection normalized flat area goes to the normalized billiard area/state measure.

Indeed, an invariant measure gives zero mass to the side cross-sections unless it has an atom on a finite singular orbit; such orbits are not part of the regular directional phase space. Hence  $F$  and its measurable inverse are defined almost everywhere for every invariant probability under discussion. Equation (13.6.3) then transports invariance in both directions. Since every copy of  $P$  carries ordinary Euclidean area and all side gluings are translations, the area statement follows as well.

By the Kerckhoff-Masur-Smillie unique ergodicity theorem, for Lebesgue-almost every  $\theta$  the directional translation flow on  $X_P$  is uniquely ergodic, with normalized flat area as its unique invariant probability. Lemma 13.8 transfers this unique measure to  $\mathcal{B}_\theta$ . Forgetting the explicit state label is precisely the usual folding description of billiard motion in  $P$ , so the conclusion is the stated almost-every-direction unique ergodicity after the finite directional unfolding has been fixed.

*Why folding really preserves unique ergodicity.* The side cross-sections and singular leaves have zero mass for every invariant probability on the regular flow: sufficiently small distinct time translates of a side cross-section are disjoint, so invariance would otherwise create infinitely many disjoint sets of the same positive mass. Therefore the state-labelled folding map and its copy-labelled inverse are defined almost everywhere for every invariant probability. Pushforward and pullback are inverse bijections on invariant probabilities. The KMS unique measure on the translation surface consequently corresponds to exactly one invariant probability on the finite-state billiard system.

□

## 13.4 Worked examples

### Worked example: the square billiard and a flat torus

Take the unit square. Every angle is  $\pi/2$ , so  $N = 2$  and the finite unfolding has  $2N = 4$  direction states. Reflecting four copies produces a  $2 \times 2$  square whose opposite sides are identified by translations; after rescaling this is a flat torus. A billiard trajectory with velocity  $(v_x, v_y)$  becomes the straight flow

$$(x(t), y(t)) = (x_0, y_0) + t(v_x, v_y) \pmod{2\mathbb{Z}^2}$$

on the unfolded torus. If  $v_x \neq 0$ , use a vertical transversal. The first return advances the transverse coordinate by a fixed amount proportional to  $v_y/v_x$ , so the return map is a circle rotation, the simplest interval exchange. When  $v_y/v_x$  is irrational, every orbit is dense and the rotation is uniquely ergodic.

#### Check your understanding

For slope  $v_y/v_x = p/q \in \mathbb{Q}$  in lowest terms, what happens to the unfolded straight line? Translate the conclusion back into the billiard table.

### Boundary case: an irrational angle

Suppose one polygon angle is  $\pi\alpha$  with  $\alpha \notin \mathbb{Q}$ . Successive reflections generate rotations by integer multiples of  $2\pi\alpha$ , producing infinitely many distinct direction states. The finite-copy compact unfolding theorem therefore does not apply. One can still unfold any finite orbit segment, but not into the finite compact translation surface used by the rational-polygon theory in this chapter.

### Common pitfall

The finite unfolding construction depends essentially on rationality of the polygonal angles. If the reflection group of directions is infinite, finitely many reflected copies do not close to a compact translation surface. One cannot import the compact-surface straight-line-flow theorems merely by drawing more reflected copies.

## 13.5 Chapter summary

### What to remember

- Reflection unfolding straightens polygonal billiard paths while moving the reflection into the geometry of copied polygons.
- Rational angles make the direction group finite, producing a compact translation surface with finitely many cone singularities.

- Directional first returns on a transversal are interval exchanges.
- Keane's infinite distinct orbit condition implies minimality by excluding finite connections among discontinuities.
- Kerckhoff-Masur-Smillie gives unique ergodicity for almost every direction on every compact translation surface, hence for rational polygonal billiards after unfolding.

### Looking ahead

K14 now studies the translation surfaces themselves rather than only the flows on a fixed surface. Abelian differentials, period coordinates, and the  $SL(2, \mathbb{R})$  action turn the collection of all such surfaces into a finite-dimensional moduli space on which directional dynamics can be renormalized.

### Check your understanding

Explain why rational polygonal angles lead to finitely many reflected direction states and hence to a compact translation surface. What exactly breaks in this argument when one angle is an irrational multiple of  $\pi$ ?

## 13.6 Exercises and problems

### Exercise 13.1

**Difficulty.** Core.

**Related results.** Reflection unfolding lemma; Finite unfolding of rational polygons; Directional flow as interval exchange.

**Problem.** Unfold the square billiard to a straight line on  $\mathbb{R}^2$ , quotient to a flat torus, and prove that a direction is periodic exactly when its slope is rational.

**Solution sketch.** Track reflection parity and solve the lattice return equation for the straightened flow.

### Exercise 13.2

**Difficulty.** Core.

**Related results.** Keane i.d.o.c. minimality criterion; Kerckhoff-Masur-Smillie unique ergodicity theorem; Polygonal billiard unique-ergodicity corollary.

**Problem.** For a polygon with an angle irrational relative to  $\pi$ , prove that the adjacent reflection generators produce an infinite-order rotation and hence infinitely many unfolded orientations.

**Solution sketch.** Use that the product of two reflections is rotation by twice the angle and check its order.



**Exercise 13.3****Difficulty.** Advanced.**Related results.** Directional flow as interval exchange.**Problem.** Choose a transverse segment to a directional flow and prove that the first-return map is an interval exchange: translation on finitely many continuity intervals with length preserved.**Solution sketch.** Partition by first-hit itinerary; within each cell the Euclidean flow gives a constant translation.**Exercise 13.4****Difficulty.** Advanced.**Related results.** Keane i.d.o.c. minimality criterion.**Problem.** For a two-interval exchange, show rational rotation gives periodic orbits and failure of minimality, while irrational rotation is minimal, illustrating the role of a no-connection condition.**Solution sketch.** Write the IET as a circle rotation and analyze rational versus irrational orbit arithmetic.**Exercise 13.5****Difficulty.** Advanced.**Related results.** Reflection unfolding lemma; Finite unfolding of rational polygons.**Problem.** Use finite unfolding plus the Kerckhoff-Masur-Smillie theorem to deduce unique ergodicity of rational-polygon billiards in almost every direction, carefully transferring invariant measures through the finite cover.**Solution sketch.** Apply KMS upstairs, lift/average any billiard invariant measure to the unfolding, and use uniqueness before projecting back.**Exercise 13.6****Difficulty.** Lab.**Related results.** None.**Problem.** Implement a rational-polygon billiard and its unfolded straight-line trajectory; verify collision-by-collision agreement and compare rational versus nearby irrational slopes.**Solution sketch.** Use exact/high-precision reflections and refine tolerances before interpreting finite-time orbit pictures.

# Translation Surfaces, Abelian Differentials and Period Coordinates

## Why this chapter matters

K13 converted a polygonal billiard into straight-line motion on a single translation surface. To understand typical directions and deformations, we must now let the surface itself move. A translation surface can be encoded by a holomorphic one-form  $\omega$  on a compact Riemann surface. Integrating  $\omega$  over relative homology classes gives period coordinates, so each stratum is locally a finite-dimensional complex vector space with an integral affine structure.

This coordinate picture makes the natural  $GL^+(2, \mathbb{R})$  action transparent: apply the same real linear map to every period vector. The diagonal subgroup expands one direction and contracts the transverse direction, producing Teichmüller flow. The integral lattice in period coordinates supplies the Masur-Veech measure, while saddle connections measure degeneration. Masur's compactness criterion then identifies the thick part of a stratum exactly by a positive lower bound on saddle-connection length.

**Prerequisites.** K13 for translation surfaces and directional flow; elementary Riemann-surface theory and holomorphic one-forms; singular homology and relative homology; basic measure theory and linear algebra.

## Learning goals

- Pass between a translation atlas and a holomorphic one-form.
- Use the zero divisor of  $\omega$  to identify the stratum  $\mathcal{H}(\kappa)$ .
- Construct local period coordinates from a basis of relative homology.
- Compute the effect of  $GL^+(2, \mathbb{R})$  on periods and area and identify the Teichmüller diagonal flow.
- Explain how the integral period lattice normalizes Masur-Veech measure and why short saddle connections characterize escape to the boundary of moduli space.

## Conceptual roadmap

translation atlas  $\longleftrightarrow$  Abelian differential  
 $\longrightarrow$  relative periods  $\longrightarrow$  stratum coordinates.

From these coordinates come three structures used below: the  $SL(2, \mathbb{R})$  action,

Masur-Veech measure, and the thick/thin geometry detected by saddle connections. This moduli-space language is the renormalization stage needed by K15: long directional trajectories on one surface become the orbit of that surface under Teichmüller flow.

### Check your understanding

Why are relative periods, rather than only absolute periods in  $H_1(X; \mathbb{Z})$ , needed when the zero set  $\Sigma$  has more than one point?

## 14.1 Definitions

### Intuition

**Zeros of  $\omega$  become cone singularities.** If  $\omega = z^k dz$  in a local coordinate, integrating gives a translation coordinate with total cone angle  $2\pi(k+1)$ . The partition of  $2g - 2$  by zero orders therefore labels the stratum.

**Period coordinates are displacement vectors.** Cut the surface along a relative-homology basis. The complex numbers  $\int_\gamma \omega$  record the edge vectors needed to reconstruct the local polygonal presentation. Small changes of these vectors give nearby translation surfaces with the same zero combinatorics.

**The  $SL(2, \mathbb{R})$  action is linear in period coordinates.** Every complex period is a vector in  $\mathbb{R}^2$ . Applying the same determinant-one matrix to all of them preserves area. The diagonal matrix  $\text{diag}(e^t, e^{-t})$  is the Teichmüller geodesic flow in this flat-geometric model.

**A short saddle connection signals degeneration.** A saddle connection is a straight segment joining zeros with no zero inside. If its length tends to zero at area one, the surface leaves every compact subset of the stratum. A uniform lower bound on all such lengths is exactly the thick-part condition.

### Definition K14.1. translation surface

A translation surface is an oriented surface with an atlas to  $\mathbb{C}$  whose transition maps are translations away from finitely many cone singularities; equivalently a pair  $(X, \omega)$  with  $X$  a Riemann surface and nonzero holomorphic 1-form  $\omega$ .

### Definition K14.2. Abelian differential

An Abelian differential is a holomorphic 1-form  $\omega$  on a compact Riemann surface; its zero orders sum to  $2g-2$  and determine the ambient stratum.

### Definition K14.3. period coordinate

Period coordinates are local coordinates obtained by integrating  $\omega$  over a basis of relative homology  $H_1(X, \Sigma; \mathbb{Z})$ , where  $\Sigma$  is the zero set.

**Definition K14.4. saddle connection**

A saddle connection is a straight geodesic segment for the flat metric whose end-points are cone singularities and whose interior contains no singularity.

**Definition K14.5. stratum, area, and area-one locus**

For a partition  $\kappa = (k_1, \dots, k_m)$  of  $2g-2$ , the *stratum*  $\mathcal{H}(\kappa)$  is the moduli space of pairs  $(X, \omega)$  whose nonzero holomorphic one-form has exactly  $m$  distinct zeros of orders  $k_1, \dots, k_m$ . Its flat area is  $\text{Area}(X, \omega) = \frac{i}{2} \int_X \omega \wedge \bar{\omega}$ , and  $\mathcal{H}_1(\kappa)$  denotes the area-one locus.

**Definition K14.6. zero divisor and order of a holomorphic one-form**

If  $\omega$  is a nonzero holomorphic one-form, then at every point  $p$  there is a local holomorphic coordinate  $z$  in which  $\omega = z^k u(z) dz$  with  $u(0) \neq 0$ . The integer  $k \geq 0$  is the *order*  $\text{ord}_p(\omega)$ ; the *zero divisor* is  $(\omega)_0 = \sum_p \text{ord}_p(\omega)[p]$ , with only the positive-order terms included.

**Definition K14.7. relative marking and integral period lattice**

For the zero set  $\Sigma$  of an Abelian differential, a *local relative marking* identifies the lattices  $H_1(X', \Sigma'; \mathbb{Z})$  of all nearby marked surfaces with one fixed lattice  $H_1(X, \Sigma; \mathbb{Z})$ . Its dual  $H^1(X, \Sigma; \mathbb{Z})$  is the integral relative-cohomology lattice. In complex period coordinates the product lattice is  $H^1(X, \Sigma; \mathbb{Z} \oplus i\mathbb{Z})$ .

**Definition K14.8. Masur-Veech measure normalization**

In a local period chart  $H^1(X, \Sigma; \mathbb{C}) \cong \mathbb{C}^d$ , *Masur-Veech Lebesgue measure* is the Lebesgue measure for which the lattice  $H^1(X, \Sigma; \mathbb{Z} \oplus i\mathbb{Z})$  has covolume one. These local measures are required to agree under integral changes of marking; restriction/disintegration to the area-one hypersurface gives the Masur-Veech measure on  $\mathcal{H}_1(\kappa)$ , subsequently normalized to probability on each finite-volume connected component when convenient.

**Definition K14.9. flat systole and thick part**

The *flat saddle-connection systole* of  $(X, \omega)$  is

$$\text{sys}_{\text{sc}}(X, \omega) = \min\{|\text{hol}(\gamma)| : \gamma \text{ is a saddle connection}\}.$$

For  $\varepsilon > 0$ , the  $\varepsilon$ -*thick part* of an area-one stratum is the locus where  $\text{sys}_{\text{sc}} \geq \varepsilon$ .

## 14.2 How the main proofs work

### Intuition

**Strata.** The divisor theorem for a holomorphic one-form gives total zero order  $2g - 2$ . Grouping surfaces by the unordered list of zero multiplicities partitions the moduli space into strata. Genus one is the zero-free exceptional case  $\mathcal{H}(0)$ .

**Period coordinates.** Choose a marking of  $H_1(X, \Sigma; \mathbb{Z})$  and integrate  $\omega$  along a basis. Relative cohomology deformation theory shows that these integrals give a local holomorphic chart. Changing the integral basis multiplies the coordinate vector by an integral unimodular matrix.

$GL^+(2, \mathbb{R})$  **and Teichmüller flow.** Postcompose all translation charts by the same real linear map. Transition maps remain translations, zero orders are unchanged, and area scales by the determinant. The diagonal determinant-one subgroup expands horizontal periods and contracts vertical periods.

**Masur-Veech measure.** The relative integral lattice normalizes Lebesgue measure in every period chart, and unimodular transition maps make the local forms patch. A zippered-rectangle/Rauzy section supplies a finite return model. Positive return matrices give bounded distortion and summable cylinder volumes; the resulting finite invariant section measure suspends to the finite  $SL(2, \mathbb{R})$ -invariant Masur-Veech measure and yields Teichmüller-flow ergodicity on each connected component.

**Masur compactness.** If a sequence has a saddle connection shrinking to zero, it leaves every compact set. Conversely, a uniform lower bound on the flat systole gives finitely controlled polygonal/triangulation data; after passing to a subsequence the period coordinates converge inside the same stratum.

## 14.3 Main results

### Theorem K14.1. Strata of Abelian differentials

**Statement.** Fix  $g \geq 2$  and let  $\mathcal{H}_g$  be the moduli space of pairs  $(X, \omega)$ , where  $X$  is a compact genus- $g$  Riemann surface and  $\omega$  is a nonzero holomorphic 1-form. The zero divisor of  $\omega$  has degree  $2g - 2$ . Hence, if its distinct zeros have multiplicities  $k_1, \dots, k_m$ , then each  $k_i \geq 1$  and  $\sum_i k_i = 2g - 2$ . For every unordered partition  $\kappa = (k_1, \dots, k_m)$  of  $2g - 2$ , let  $\mathcal{H}(\kappa)$  be the locus with exactly those zero orders. Then  $\mathcal{H}_g = \bigsqcup_{\kappa \vdash (2g-2)} \mathcal{H}(\kappa)$ . For  $g = 1$  the standard convention is the zero-free stratum  $\mathcal{H}(0)$ .

### Scope note

Connected-component classification and period coordinates are treated in separate results.

*Proof.* Let  $X$  be a compact Riemann surface of genus  $g$  and let  $\omega \neq 0$  be a holomorphic

one-form. By the definition of Abelian differential above,  $\omega$  is a holomorphic section of the canonical line bundle  $K_X = T^{*(1,0)}X$ . At every point  $p$ , choose a holomorphic coordinate  $z$  centered at  $p$ . the definition of zero divisor and order of a holomorphic one-form above writes

$$\omega = z^{k_p} u(z) dz, \quad u(0) \neq 0, \quad (14.1.1)$$

so  $k_p = \text{ord}_p \omega \geq 0$  and only finitely many  $k_p$  are positive.

**Lemma 14.1 (degree of the canonical divisor).** For a nonzero meromorphic one-form  $\eta$  on a compact genus- $g$  Riemann surface,

$$\sum_{p \in X} \text{ord}_p(\eta) = 2g - 2. \quad (14.1.2)$$

In particular a holomorphic one-form has a zero divisor of degree  $2g - 2$ .

We recall the short proof. Choose a meromorphic vector field  $V$  not identically zero. The product  $\eta(V)$  is a meromorphic function, hence the total order of its zeros and poles is zero. Therefore

$$\deg(\text{div } \eta) + \deg(\text{div } V) = 0. \quad (14.1.3)$$

The divisor of  $V$  represents the holomorphic tangent bundle  $T_X$ , whose degree is the Euler characteristic  $\chi(X) = 2 - 2g$  (one may compute it from the index sum of a generic smooth vector field and compare local holomorphic orders). Thus  $\deg K_X = -\deg T_X = 2g - 2$ , proving (14.1.2).

For  $g \geq 2$ , let  $p_1, \dots, p_m$  be the distinct zeros and put  $k_i = \text{ord}_{p_i} \omega \geq 1$ . Lemma 14.1 gives

$$k_1 + \dots + k_m = 2g - 2. \quad (14.1.4)$$

Hence the unordered multiplicity list is a partition  $\kappa = (k_1, \dots, k_m)$  of  $2g - 2$ . By the definition of stratum, area, and area-one locus above, the locus with exactly that zero data is  $\mathcal{H}(\kappa)$ . Every nonzero Abelian differential has one and only one such unordered list, so the loci are disjoint and

$$\mathcal{H}_g = \bigsqcup_{\kappa \vdash (2g-2)} \mathcal{H}(\kappa). \quad (14.1.5)$$

No connected-component assertion has been used.

If  $g = 1$ , Lemma 14.1 gives degree zero. A holomorphic one-form has no poles and all zero orders are nonnegative, so it has no zeros. This is the conventional stratum  $\mathcal{H}(0)$ . The genus-zero case has no nonzero holomorphic one-form and is outside the stated moduli space. This proves the result.  $\square$

### Theorem K14.2. Period coordinates

**Statement.** Fix a stratum  $\mathcal{H}(k_1, \dots, k_m)$  of Abelian differentials on genus- $g$  surfaces, let  $\Sigma$  be the labeled zero set, and work on a local marking, i.e. identify the relative homology groups of nearby surfaces with one fixed integral lattice. Since  $\text{rank } H_1(X, \Sigma; \mathbb{Z}) = 2g + m - 1$ , choose a locally constant integral basis  $\gamma_1, \dots, \gamma_{2g+m-1}$ . Then the period map  $(X, \omega) \mapsto (\int_{\gamma_1} \omega, \dots, \int_{\gamma_{2g+m-1}} \omega) \in \mathbb{C}^{2g+m-1}$  is a local holomorphic

coordinate chart on the stratum. Changing the marked integral relative-homology basis changes coordinates by an element of  $\mathrm{GL}(2g + m - 1, \mathbb{Z})$ ; after quotienting the marking, these charts give the usual integral-affine/orbifold structure.

*Proof.* Fix  $(X, \omega) \in \mathcal{H}(k_1, \dots, k_m)$  and label its zero set  $\Sigma = \{p_1, \dots, p_m\}$ . The long exact homology sequence of the pair  $(X, \Sigma)$  contains

$$0 \longrightarrow H_1(X; \mathbb{Z}) \longrightarrow H_1(X, \Sigma; \mathbb{Z}) \xrightarrow{\partial} \tilde{H}_0(\Sigma; \mathbb{Z}) \longrightarrow 0. \quad (14.2.1)$$

The first group has rank  $2g$  and the last rank  $m - 1$ , hence

$$d := \mathrm{rank} H_1(X, \Sigma; \mathbb{Z}) = 2g + m - 1. \quad (14.2.2)$$

Choose an integral basis  $\gamma_1, \dots, \gamma_d$  in the local marking of the definition of relative marking and integral period lattice above and write

$$\mathcal{P}(X, \omega) = \left( \int_{\gamma_1} \omega, \dots, \int_{\gamma_d} \omega \right) \in \mathbb{C}^d. \quad (14.2.3)$$

We prove that (14.2.3) is a local chart. Choose a maximal family of pairwise non-crossing saddle connections whose endpoints lie in  $\Sigma$ . If a component of the complement were not a Euclidean triangle, two nonadjacent vertices of its polygonal boundary could be joined by a straight diagonal whose interior avoids  $\Sigma$ , contradicting maximality. Thus the family triangulates  $X$ . Let  $e_1, \dots, e_N$  be the oriented edges. Their holonomy vectors

$$z_j = \int_{e_j} \omega \in \mathbb{C} \quad (14.2.4)$$

satisfy exactly the linear triangle relations  $z_i + z_j + z_k = 0$  and the edge identification relations. Each  $[e_j]$  is a relative-homology class, so every  $z_j$  is an integral linear combination of the  $d$  coordinates (14.2.3).

**Lemma 14.2 (relative-period reconstruction).** For all  $w \in \mathbb{C}^d$  sufficiently close to  $\mathcal{P}(X, \omega)$ , the edge vectors obtained from the same integral linear combinations form nondegenerate Euclidean triangles with the same gluing pattern. Gluing them by translations gives a unique nearby marked translation surface in the same stratum, and its relative periods equal  $w$ .

The triangle relations continue to hold because they are integral linear relations in relative homology. At the base point every triangle has positive oriented area; this is an open condition, so it persists for  $w$  close enough. Gluing corresponding equal/opposite edge vectors therefore gives a compact translation surface  $X_w$ . Around a vertex corresponding to a zero of order  $k_i$ , the same fixed number  $2(k_i + 1)$  of Euclidean half-planes is cyclically glued; hence the cone angle remains  $2\pi(k_i + 1)$  and the zero order remains  $k_i$ . Integrating  $dz$  along each marked basis chain returns exactly the chosen coordinate  $w_j$ . Conversely, any sufficiently nearby marked surface has the same triangulation by continuous deformation of the edges, so its polygonal pieces are determined by their edge holonomies. This proves both existence and uniqueness in Lemma 14.2.

It follows that  $\mathcal{P}$  is a homeomorphism from a neighborhood onto an open subset of  $\mathbb{C}^d$ . The reconstructed complex coordinate on each triangle depends complex-linearly on  $w$ , and the inverse periods are integrals of the holomorphic one-form over locally constant relative cycles; therefore these are holomorphic coordinates. This proves the local holomorphic chart assertion.

**Lemma 14.3 (integral transition maps).** If  $\gamma'_1, \dots, \gamma'_d$  is another locally constant integral basis, then on the common chart domain

$$\mathcal{P}' = A\mathcal{P}, \quad A \in GL(d, \mathbb{Z}). \quad (14.2.5)$$

Indeed, each  $\gamma'_i$  is an integral linear combination of the  $\gamma_j$ ; the coefficient matrix is unimodular because both families are integral bases, and integration is linear. After forgetting the local marking, the finite mapping-class stabilizers act by such integral matrices, yielding the usual integral-affine orbifold structure. Equations (14.2.2)–(14.2.5) prove the statement.  $\square$

#### Geometric interface: Saddle connections, holonomy, and cylinders

**Statement.** On a translation surface  $(X, \omega)$ , a saddle connection is a straight geodesic segment joining zeros of  $\omega$  with no zero in its interior; its holonomy is  $\text{hol}(\gamma) = \int_\gamma \omega \in \mathbb{C} \cong \mathbb{R}^2$ . A maximal cylinder is an embedded Euclidean cylinder  $(\mathbb{R}/c\mathbb{Z}) \times (0, h)$  foliated by parallel closed geodesics; its circumference is  $c$ , height is  $h$ , and modulus is  $h/c$ .

#### Theorem K14.3. $GL^+(2, \mathbb{R})$ action and Teichmüller flow

**Statement.** Postcomposition of the translation charts of a translation surface  $(X, \omega)$  by  $A \in GL^+(2, \mathbb{R})$  produces another translation surface  $A \cdot (X, \omega)$  in the same stratum. Its flat area is multiplied by  $\det A$ . Therefore the area-one locus is invariant under  $SL(2, \mathbb{R})$ . With the convention that the horizontal direction is expanded and the vertical direction contracted,  $g_t = \text{diag}(e^t, e^{-t})$  acts as the Teichmüller geodesic flow on the stratum (or its marked cover).

*Proof.* Let  $\{(U_i, z_i)\}$  be translation charts for  $(X, \omega)$ , so on overlaps  $z_j = z_i + c_{ij}$ . For  $A \in GL^+(2, \mathbb{R})$  define new real-affine charts  $z_i^A = Az_i$ . Then

$$z_j^A = A(z_i + c_{ij}) = z_i^A + Ac_{ij}, \quad (14.3.1)$$

so all new transition maps are again translations. Because  $A$  is an orientation-preserving linear homeomorphism, the cyclic number of sectors at each cone point is unchanged; hence the zero orders and therefore the stratum are unchanged.

If  $\gamma$  is a relative cycle, its holonomy vector transforms by

$$\text{hol}_{A \cdot (X, \omega)}(\gamma) = A \text{hol}_{(X, \omega)}(\gamma). \quad (14.3.2)$$

Thus in the period charts of the Period coordinates the action is the diagonal real-linear action of  $A$  on every coordinate. Euclidean area of every polygonal piece is multiplied



by  $\det A$ , hence after summing a triangulation,

$$\text{Area}(A \cdot (X, \omega)) = (\det A) \text{Area}(X, \omega). \quad (14.3.3)$$

For  $A \in SL(2, \mathbb{R})$  this preserves the area-one locus.

Take

$$g_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}. \quad (14.3.4)$$

In every flat chart it multiplies horizontal lengths by  $e^t$  and vertical lengths by  $e^{-t}$ . The identity map between the old and new underlying surfaces has maximal and minimal infinitesimal dilatations  $e^t$  and  $e^{-t}$ , so its quasiconformal dilatation is  $e^{2|t|}$ . Its horizontal and vertical measured foliations are exactly the extremal directions. By the definition of the Teichmüller geodesic determined by those transverse measured foliations, this affine representative is the time- $t$  Teichmüller map (with the convention that horizontal is expanded for  $t > 0$ ). Therefore the  $g_t$ -subgroup is the Teichmüller geodesic flow on the marked stratum and descends to the unmarked orbifold. This proves every assertion.  $\square$

#### Theorem K14.4. Masur-Veech finite invariant measure

**Statement.** Fix a connected component  $\mathcal{C}$  of an area-one stratum  $\mathcal{H}_1(\kappa)$  of Abelian differentials. The integral lattice  $H^1(X, \Sigma; \mathbb{Z} \oplus i\mathbb{Z})$  in relative cohomology normalizes Lebesgue measure in period-coordinate charts; these local measures patch to the Masur-Veech measure  $\mu_{\text{MV}}$  on  $\mathcal{C}$ . The measure is invariant under the  $SL(2, \mathbb{R})$  action and has finite total mass. Moreover the Teichmüller geodesic flow (equivalently, the diagonal subgroup action) is ergodic with respect to  $\mu_{\text{MV}}$  on  $\mathcal{C}$ .

*Proof.* Put  $d = 2g + m - 1$ . On a marked period chart identify relative cohomology with  $\mathbb{C}^d \cong \mathbb{R}^{2d}$ . By the Masur-Veech normalization definition choose the Lebesgue form  $dL$  for which  $H^1(X, \Sigma; \mathbb{Z} \oplus i\mathbb{Z})$  has covolume one. The integral transition-map lemma for period coordinates says that every change of marked relative-homology basis is in  $GL(d, \mathbb{Z})$ . Its real determinant has absolute value one on both real and imaginary period coordinates, so the local forms patch to a canonically normalized orbifold measure on the stratum.

The  $SL(2, \mathbb{R})$  action described above applies the same matrix to each complex period regarded as a vector in  $\mathbb{R}^2$ . Its Jacobian on  $\mathbb{R}^{2d}$  is  $(\det A)^d = 1$ . If  $a$  denotes flat area, homogeneity under scalar dilation gives the radial disintegration

$$dL = c_d a^{d-1} da d\mu_1. \quad (14.4.1)$$

Thus the induced area-one measure  $\mu_1$  is  $SL(2, \mathbb{R})$ -invariant. It remains to prove that its mass on each connected component is finite and that the diagonal flow is ergodic.

**Zippered-rectangle coordinates and the fast section.** Fix one Rauzy diagram  $\mathcal{D}$  representing the chosen connected marked component and an irreducible vertex  $\pi$  with alphabet  $\mathcal{A}$ ,  $|\mathcal{A}| = d$ . We use Veech's suspension coordinates  $(\pi, \lambda, \tau)$ :  $\lambda_\alpha > 0$  are the interval lengths and  $\tau \in \Theta_\pi$  satisfies the usual top/bottom partial-sum inequalities.

Writing

$$h = -\Omega_\pi^t \tau, \quad A(\lambda, \tau) = \lambda \cdot h, \quad (14.4.2)$$

the corresponding zippered rectangles have widths  $\lambda_\alpha$ , heights  $h_\alpha \geq 0$ , and area  $A(\lambda, \tau)$ . The coordinate change across a Rauzy arrow  $\gamma$  is

$$(\lambda, \tau) \mapsto (\lambda B_\gamma^{-1}, \tau B_\gamma^{-1}) \quad (14.4.3)$$

for row-vector conventions; it preserves the period-coordinate volume and the area pairing. Notice the convention carefully: if one projectivizes lengths, the inverse projective branch associated with a positive return matrix  $B$  is  $x \mapsto Bx/|Bx|_1$  (for column-vector notation), not  $x \mapsto B^{-1}x/|B^{-1}x|_1$ . Its Jacobian is  $|Bx|_1^{-d}$  because  $\det B = 1$ . We will not use a fictitious finite family of such branches; the actual fast return has countably many branches.

Normalize  $|\lambda|_1 = 1$  and area one. On a top Rauzy half put

$$\Theta_\pi^* = \left\{ \tau \in \Theta_\pi : \sum_{\alpha \in \mathcal{A}} \tau_\alpha > 0 \right\}, \quad (14.4.4)$$

and use the opposite sign on a bottom half. The union  $S^*(\mathcal{D})$  of these finitely many signed pieces is the Veech-Zorich fast transversal. The ordinary Rauzy section has infinite hyperarea in general; the sign cut in (14.4.4) is the point that makes the fast section finite.

**Lemma (rational-simplex integrability).** Let  $\Delta^{(1)} = \{\lambda_\alpha > 0 : \sum \lambda_\alpha = 1\}$  and let

$$R(\lambda) = \frac{L_1(\lambda) \cdots L_p(\lambda)}{M_1(\lambda) \cdots M_q(\lambda)} \quad (14.4.5)$$

with all  $L_i, M_j$  nonnegative linear forms, positive in the interior. For a face  $\Delta_B$  supported on  $B \subset \mathcal{A}$ , let  $m_+(B)$  and  $m_-(B)$  be the numbers of numerator and denominator forms vanishing identically on that face and set  $m(B) = m_+(B) - m_-(B)$ . If

$$d + m(B) > |B| \quad (B \subsetneq \mathcal{A}), \quad (14.4.6)$$

then  $R \in L^1(\Delta^{(1)})$ .

Here is the proof, since it is the analytic core of finiteness. For each multi-index  $n = (n_\alpha) \in \mathbb{N}^{\mathcal{A}}$  with  $\min n_\alpha = 0$ , let  $Q(n)$  be the region where a coordinate with  $n_\alpha = 0$  is bounded below by a fixed dimensional constant and, when  $n_\alpha > 0$ ,  $\lambda_\alpha \asymp 2^{-n_\alpha}$ . These regions cover the simplex with bounded overlap and

$$|Q(n)| \asymp 2^{-\sum_\alpha n_\alpha}. \quad (14.4.7)$$

For any nonnegative linear form  $L$ , let  $B(L)$  be the largest face on which  $L$  vanishes. On  $Q(n)$ ,

$$L(\lambda) \asymp 2^{-\min_{\alpha \notin B(L)} n_\alpha}. \quad (14.4.8)$$

Let  $0 = n_0 < n_1 < \cdots < n_s$  be the distinct coordinate depths and  $B_i = \{\alpha : n_\alpha > n_i\}$ .

Applying (14.4.8) to every factor in (14.4.5) and telescoping the depth levels gives

$$R(\lambda) \asymp 2^{-N(n)}, \quad N(n) = \sum_{i \geq 1} (n_i - n_{i-1})m(B_i^c). \quad (14.4.9)$$

Condition (14.4.6) says  $m(B_i^c) > 1 - |B_i|$ . Combining this with (14.4.7) yields

$$\int_{Q(n)} R d\lambda \leq C 2^{-\max_{\alpha} n_{\alpha}}. \quad (14.4.10)$$

There are  $O(N^{d-2})$  multi-indices with minimum zero and maximum  $N$ , so  $\sum_N N^{d-2} 2^{-N} < \infty$ . This proves the integrability lemma.

We apply it to the suspension fibers. Decompose each polyhedral cone  $\Theta_{\pi}^*$  into finitely many simplicial cones  $\Gamma$  generated by linearly independent vectors  $\tau^{(1)}, \dots, \tau^{(d)}$ . Put  $h^{(i)} = -\Omega_{\pi}^t \tau^{(i)}$  and

$$M_i(\lambda) = \lambda \cdot h^{(i)}. \quad (14.4.11)$$

Writing  $\tau = \sum_i s_i \tau^{(i)}$ ,  $s_i > 0$ , the area constraint is  $\sum_i s_i M_i(\lambda) = 1$ . Direct integration over this simplex gives the fiber hyperarea

$$\frac{c_{\Gamma}}{M_1(\lambda) \cdots M_d(\lambda)}. \quad (14.4.12)$$

Thus only (14.4.6) has to be checked with  $p = 0, q = d$ .

For a proper  $B \subset \mathcal{A}$ , let  $E_B^*$  be the span of suspension vectors in the closure of  $\Theta_{\pi}^*$  for which  $h_{\alpha} = 0$  for every  $\alpha \in B$ . The partial-sum definition of  $\Theta_{\pi}$  gives

$$\text{codim } E_B^* > |B|. \quad (14.4.13)$$

We spell out the combinatorial point. If a height  $h_{\alpha} = h_{\alpha}^t + h_{\alpha}^b$  vanishes, the two nonnegative top and bottom partial-sum terms vanish separately. Reading the letters of  $B$  in top order yields  $|B|$  independent triangular linear conditions unless  $B$  is an initial top block; reading them in bottom order gives the same conclusion unless it is an initial bottom block. Irreducibility forbids a proper set from being both. On the signed fast half (14.4.4), the remaining exceptional case acquires the additional independent sign-boundary relation, giving strict inequality in (14.4.13). Hence fewer than  $d - |B|$  of the independent generators  $\tau^{(i)}$  can lie in  $E_B^*$ . Equivalently, fewer than  $d - |B|$  denominator forms in (14.4.12) vanish identically on  $\Delta_B$ , which is exactly (14.4.6). Therefore every piece (14.4.12) is integrable, and the finitely many  $\pi$  and simplicial cones show

$$\mu_{S^*}(S^*(\mathcal{D})) < \infty. \quad (14.4.14)$$

This proves the finite-section part of the finite-section lemma without any finite-branching assumption.

The same calculation also controls the cusp. In the dyadic proof, imposing  $\min_{\alpha} \lambda_{\alpha} < \varepsilon$  forces  $\max n_{\alpha} \gtrsim |\log_2 \varepsilon|$ , and (14.4.10) gives

$$\mu_{S^*}\{\min_{\alpha} \lambda_{\alpha} < \varepsilon\} \leq C\varepsilon |\log \varepsilon|^{d-2}. \quad (14.4.15)$$

For one return of the Teichmüller flow to the fast section, the elapsed time  $\tau_{\text{roof}}$  is bounded above by  $C + \log(1/\min_{\alpha} \lambda_{\alpha})$ . The layer-cake formula together with (14.4.15) therefore gives

$$\int_{S^*} \tau_{\text{roof}} d\mu_{S^*} < \infty. \quad (14.4.16)$$

The suspension over  $S^*(\mathcal{D})$  consequently has finite volume. Since zippered rectangles give a finite marked covering of the connected stratum component away from a null saddle-connection locus, this suspension volume is precisely the period-coordinate measure  $\mu_1$ , up to the fixed marking multiplicity. This proves finiteness of the Masur-Veech mass.

**Lemma (two-sided Hilbert contraction and ergodicity).** For almost every bi-infinite zippered-rectangle orbit, recurrent positive Rauzy words contract projective length cones in backward time and, after a uniformly bounded number of complete words, contract the projectivized suspension cones in forward time. Hence every Birkhoff limit of a continuous function is independent of both length and suspension coordinates; strong connectivity of the Rauzy diagram then makes it constant. The fast return map and its Teichmüller suspension are ergodic.

We prove this rather than importing ergodicity. Let  $\varphi$  be continuous on  $S^*(\mathcal{D})$  and let  $\bar{\varphi}$  be its two-sided Birkhoff limit, defined on the full-measure recurrent set. A positive Rauzy matrix sends the closed positive cone strictly into its interior, so its projective action contracts the Hilbert metric by a factor  $\kappa < 1$ . Poincaré recurrence to a compact positive cylinder supplies, for almost every past itinerary, one fixed positive word occurring infinitely often. If two points have the same  $(\pi, \tau)$  but length data  $\lambda, \lambda'$ , their backward length coordinates therefore satisfy

$$d_H(\lambda_{-n}, \lambda'_{-n}) \longrightarrow 0. \quad (14.4.17)$$

Continuity of  $\varphi$  on compact pieces and the backward Birkhoff formula imply  $\bar{\varphi}(\pi, \lambda, \tau) = \bar{\varphi}(\pi, \lambda', \tau)$  for almost every pair. Thus  $\bar{\varphi}$  is independent of  $\lambda$ .

For the suspension coordinate one needs the dual positivity statement. If a Rauzy path is a concatenation of  $3d - 4$  complete paths, then

$$\overline{\Theta_{\pi} B_{\gamma}^{-1}} \subset \Theta_{\pi'} \cup \{0\}. \quad (14.4.18)$$

To see it, write the top and bottom partial heights  $h_{\alpha}^t, h_{\alpha}^b \geq 0$ . After  $2d - 3$  complete subpaths every total height  $h_{\alpha}$  is strictly positive because each letter has won often enough to add a positive predecessor height. Let  $\ell^t(j)$  and  $\ell^b(j)$  be the largest initial top and bottom positions on which the corresponding partial heights are strictly positive. Rauzy update formulas show these two integers never decrease. During each further complete path, the letter occupying the first position on the opposite row wins once; the update then increases the other  $\ell$  by at least one. The remaining  $d - 1$  complete paths force  $\ell^t = \ell^b = d$ , which is exactly (14.4.18).

Hence a recurrent block of  $3d - 4$  complete words acts as a strict Hilbert contraction on the projectivized suspension cone. Repeating the previous argument in forward time shows that  $\bar{\varphi}$  is also independent of  $\tau$ . It is therefore a function only of the Rauzy vertex  $\pi$ . Since the Rauzy diagram of one component is strongly connected and  $\bar{\varphi}$  is

return-map invariant, this remaining finite-state function is constant. Thus the fast return map is ergodic. An invariant set for the Teichmüller suspension intersects the section in a return-map invariant set; Fubini along flow lines then makes it null or conull. The diagonal flow is ergodic.

Combining the invariant period-coordinate construction, (14.4.14)–(14.4.16), and the two-sided contraction lemma proves all assertions of the theorem.  $\square$

### Theorem K14.5. Masur compactness criterion

**Statement.** Fix a connected component  $\mathcal{H}_1(\kappa)$  of an area-one stratum of Abelian differentials. A closed subset  $K \subset \mathcal{H}_1(\kappa)$  is compact if and only if there exists  $\varepsilon > 0$  such that every surface in  $K$  has every saddle connection of length at least  $\varepsilon$ . Thus the compact subsets are exactly the closed subsets contained in a uniform thick part of the stratum.

*Proof.* Write  $\text{sys}_{\text{sc}}(X, \omega)$  for the definition of flat systole and thick part above.

Suppose first that  $K \subset \mathcal{H}_1(\kappa)$  is compact. If no positive uniform lower bound existed, choose  $(X_n, \omega_n) \in K$  and saddle connections  $\gamma_n$  with  $|\text{hol}(\gamma_n)| \rightarrow 0$ . Pass to a convergent subsequence in  $K$  and to one local period-coordinate chart around the limit. A saddle connection shorter than a fixed chart scale crosses only a bounded number of triangles of a fixed local triangulation; after a subsequence its relative-homology class is fixed. Its period then converges to zero. The limiting relative class is represented by a collapsed connection between zeros, which leaves the fixed stratum, a contradiction. Therefore compact  $K$  has systole at least some  $\varepsilon > 0$ .

For the converse we need a bounded-geometry triangulation.

**Lemma 14.6 (uniform thick triangulation).** For the fixed topological type  $\kappa$  and every  $\varepsilon > 0$  there is  $C = C(\kappa, \varepsilon)$  such that every area-one surface with  $\text{sys}_{\text{sc}} \geq \varepsilon$  admits a saddle-connection triangulation all of whose edge lengths belong to  $[\varepsilon, C]$ .

Choose a saddle triangulation and repeatedly flip a longest diagonal whenever the opposite diagonal is shorter. At termination the triangulation is Delaunay: the circumdisk of each triangle contains no zero in its interior. Let  $R$  be the largest circumradius. If  $R$  were arbitrarily large while area is one and every saddle connection is at least  $\varepsilon$ , the corresponding empty flat disk would either be embedded or meet one of its translates in the surface. An embedded disk of radius  $R/2$  has area  $\pi R^2/4 \leq 1$ , so  $R \leq 2/\sqrt{\pi}$ . In the self-overlap case, the first pair of identified boundary points gives a simple flat closed geodesic of length at most  $2R$ ; its maximal cylinder has circumference at least  $\varepsilon$  because each boundary contains a saddle connection. Since the cylinder area is at most one, its height is at most  $1/\varepsilon$ . Cutting the overlap across that cylinder and continuing to the nearest singularity bounds  $R$  by  $1/\varepsilon + C_0(\kappa)$ . Thus all Delaunay circumradii are uniformly bounded. Every Delaunay edge is a chord of one of these disks and has length at most twice the radius. The lower bound is the systole hypothesis. This proves Lemma 14.6.

Now let  $K$  be closed and contained in the  $\varepsilon$ -thick part. For any sequence  $(X_n, \omega_n) \in K$ , choose Lemma 14.6 triangulations. The number of triangles/edges is fixed by  $g$  and  $m$ , and there are finitely many gluing patterns, so pass to a subsequence with one pattern.

Every oriented edge holonomy lies in the compact annulus

$$\{z \in \mathbb{C} : \varepsilon \leq |z| \leq C\}. \quad (14.6.1)$$

Pass to a diagonal subsequence on all edge vectors. Triangle relations and edge identifications survive in the limit. No edge collapses by (14.6.1), and a triangle cannot collapse to a segment in a way that changes the stratum: a collapsed triangle would force two of its vertices to be joined by a saddle connection of length tending to zero after cancelling the limiting collinear edges. Thus the limiting polygonal data glue to an area-one translation surface in the same stratum. This is exactly convergence in the period coordinates of the Period coordinates. Hence every sequence in  $K$  has a convergent subsequence in  $\mathcal{H}_1(\kappa)$ . Closedness of  $K$  puts the limit in  $K$ , proving compactness.

We have shown that a closed subset is compact exactly when it lies in a uniform  $\varepsilon$ -thick part, as claimed.  $\square$

## 14.4 Worked examples

### Worked example: the square torus

Let

$$X = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z}), \quad \omega = dz.$$

There are no zeros of  $\omega$ , so this is the genus-one stratum  $\mathcal{H}(0)$ . With the standard homology basis  $a, b$ ,

$$\int_a \omega = 1, \quad \int_b \omega = i.$$

Thus the period vector is  $(1, i)$ . The flat area is

$$\text{Area}(X, \omega) = \text{Im}(\bar{1}i) = 1.$$

Applying

$$g_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}$$

sends the periods to  $e^t$  and  $ie^{-t}$ . Their determinant remains one, so area is preserved while horizontal closed geodesics lengthen and vertical closed geodesics shorten. This is the simplest visible model of Teichmüller flow.

#### Check your understanding

At time  $t$ , what is the length of the shortest nonzero vector in the rectangular lattice generated by  $e^t$  and  $ie^{-t}$ ? What does its behavior as  $|t| \rightarrow \infty$  suggest about escape toward the thin part?

**Boundary case: pinching a slit on a genus-two surface**

Take two flat tori and cut in each a horizontal slit of length  $\varepsilon$ . Glue the upper bank of each slit to the lower bank of the other and vice versa. The result is a genus-two translation surface with two cone points at the slit endpoints. The two copies of the slit give saddle connections of length  $\varepsilon$ . After rescaling to keep total area one, these lengths still tend to zero as  $\varepsilon \rightarrow 0$ . Therefore the family cannot remain in a compact subset of its area-one stratum. This model makes the geometric direction of Masur's compactness criterion concrete.

**Common pitfall**

Keep three dynamics separate: straight-line flow on one translation surface, Teichmüller flow on a moduli/stratum space of surfaces, and the Kontsevich-Zorich cocycle on cohomology over that base flow. Statements about recurrence or Lyapunov exponents of one object do not automatically become statements about the other two.

## 14.5 Chapter summary

**What to remember**

- Translation surfaces are equivalently compact Riemann surfaces equipped with nonzero holomorphic one-forms, with zeros producing cone singularities.
- Zero multiplicities partition the moduli space into strata, and relative periods provide local holomorphic/integral-affine coordinates.
- $GL^+(2, \mathbb{R})$  acts linearly on all periods; its diagonal determinant-one subgroup is Teichmüller flow.
- The integral relative-cohomology lattice normalizes Masur-Veech measure; on each connected area-one component this measure is finite and the Teichmüller flow is ergodic.
- A uniform lower bound on saddle-connection lengths is exactly the compactness criterion for closed subsets of an area-one stratum.

**Looking ahead**

K15 uses this moduli-space dynamics to study directional flows quantitatively. Recurrence of Teichmüller flow gives unique ergodicity, the Kontsevich-Zorich cocycle records cohomological growth, and Forni's theory converts its Lyapunov exponents into deviations of ergodic averages.

**Check your understanding**

Given a translation surface, identify the state space and evolution law for (i) directional flow, (ii) Teichmüller flow, and (iii) the Kontsevich–Zorich cocycle. Which of these objects acts on points of the surface, which on surfaces, and which on cohomology?

**14.6 Exercises and problems****Exercise 14.1**

**Difficulty.** Core.

**Related results.** Strata of Abelian differentials; Period coordinates;  $SL(2, \mathbb{R})$  action and Teichmüller flow.

**Problem.** For the square torus with  $dz$ , compute the two basic periods, apply  $A \in GL^+(2, \mathbb{R})$ , and compute transformed periods and area.

**Solution sketch.** Integrate over a homology basis, regard periods as planar vectors, and use  $\det A$  for area scaling.

**Exercise 14.2**

**Difficulty.** Core.

**Related results.** Masur–Veech finite invariant measure; Saddle connections and cylinders; Masur compactness criterion.

**Problem.** Let a saddle-connection holonomy vector tend to zero at fixed area. Prove via the compactness criterion that the surfaces leave every compact subset of the stratum.

**Solution sketch.** Identify saddle length with the norm of a relative period and apply the uniform lower-bound criterion.

**Exercise 14.3**

**Difficulty.** Advanced.

**Related results.**  $SL(2, \mathbb{R})$  action and Teichmüller flow.

**Problem.** Using a saddle-connection triangulation, explain why compatible edge periods give local affine period coordinates and reconstruct a nearby surface from a small compatible perturbation of those periods.

**Solution sketch.** Use the linear gluing/cocycle relations and keep triangles non-degenerate under a sufficiently small perturbation.

**Exercise 14.4**

**Difficulty.** Advanced.

**Related results.** Masur–Veech finite invariant measure.

**Problem.** Show why area normalization is essential for finite Masur–Veech measure by scaling all periods by  $r$  and computing the radial growth of area and Lebesgue volume.



**Solution sketch.** Area scales as  $r^2$ ; integrate the period-coordinate radial Jacobian over an unbounded cone to see divergence.

#### Exercise 14.5

**Difficulty.** Advanced.

**Related results.** Strata of Abelian differentials; Period coordinates.

**Problem.** For  $g_t = \text{diag}(e^t, e^{-t})$ , compute the time at which a fixed holonomy vector  $(x, y)$  has minimal length and show that Teichmüller flow preserves area.

**Solution sketch.** Minimize  $e^{2t}x^2 + e^{-2t}y^2$  and use  $\det g_t = 1$ .

#### Exercise 14.6

**Difficulty.** Lab.

**Related results.** None.

**Problem.** Track several period/saddle-connection vectors under Teichmüller flow and compare their numerical minimum-length times with the analytic formula from the exercises.

**Solution sketch.** Verify determinant-one area preservation and exact period evolution before using the shortest-vector plot as a compactness diagnostic.

# Teichmüller Flow, Masur-Veech Theory and the KZ/Forni Machinery

## Why this chapter matters

K14 gave local coordinates on strata of translation surfaces. K15 changes the question from *where a surface is* to *what its long straight-line dynamics looks like*. The central device is renormalization: Teichmüller flow exponentially stretches one flat direction and contracts the transverse one, turning long orbit segments on a fixed surface into bounded-scale geometry on a moving surface.

This single change of viewpoint links several deep theories. Recurrence of the Teichmüller orbit constrains directional dynamics through Masur's criterion. Gauss–Manin transport records how cohomology changes under renormalization and produces the Kontsevich–Zorich cocycle. Oseledets exponents of that cocycle control the hierarchy of power-law deviations of ergodic integrals in Forni's theorem. Rauzy–Veech and Zorich induction provide the discrete renormalization model, while the Avila–Viana pinching-and-twisting mechanism explains why the Lyapunov spectrum is simple for Masur–Veech measure.

**Prerequisites.** K13–K14 for interval exchanges, translation surfaces, period coordinates and Masur–Veech measure; Foundation F03–F04 for subadditivity and Oseledets theory; F11 for nonuniform hyperbolicity; basic homology and cohomology of surfaces.

## Learning goals

- Explain why Teichmüller flow is a renormalization of a directional translation flow rather than a second copy of the same dynamics.
- Use recurrence in moduli space to understand Masur's unique-ergodicity criterion and its contrapositive.
- Construct the Kontsevich–Zorich cocycle from Gauss–Manin transport and identify its tautological exponents  $\pm 1$ .
- Interpret Forni's invariant distributions as obstructions that determine successive deviation exponents of ergodic integrals.
- Pass between interval-exchange renormalization, Zorich matrices and the KZ cocycle.
- State precisely what pinching and twisting contribute to simplicity of the Lyapunov spectrum.

### Conceptual roadmap

The chapter has three interacting renormalization chains:

- directional flow  $\rightarrow$  Teichmüller renormalization  $\rightarrow$  recurrence/divergence  $\rightarrow$  unique ergodicity;
- IET return map  $\rightarrow$  Rauzy-Veech induction  $\rightarrow$  Zorich matrices  $\rightarrow$  the KZ cocycle;
- KZ Oseledets splitting  $\rightarrow$  basic currents and invariant distributions  $\rightarrow$  power-law deviation exponents.

Avila-Viana simplicity closes the circle by showing that, for Masur-Veech measure, the positive KZ exponents are all distinct.

### Check your understanding

Why does stretching the horizontal direction by  $e^t$  and contracting the vertical direction by  $e^{-t}$  help analyze a long *vertical* orbit? Identify which geometric feature becomes order one after the renormalization.

## 15.1 Definitions

### Intuition

**Teichmüller flow is a microscope for long trajectories.** A long, thin vertical feature becomes a bounded-scale feature after applying  $g_t = \text{diag}(e^t, e^{-t})$  with the appropriate  $t$ . Compact return says that this renormalized geometry does not degenerate indefinitely.

**The KZ cocycle transports topology while the flat metric changes.** The cohomology group  $H^1(X; \mathbb{R})$  is topological. Gauss-Manin transport canonically identifies nearby fibers, so the nontrivial growth comes from following these identifications along Teichmüller flow. Preservation of algebraic intersection forces the Lyapunov spectrum to be symmetric.

**Forni deviation is finer than Birkhoff's theorem.** Birkhoff gives  $T^{-1}I_T(f, p) \rightarrow 0$  for a zero-average observable. Forni identifies the smaller power  $T^\lambda$  that actually governs the leading deviations, after successive invariant-distribution obstructions vanish.

**Rauzy-Veech induction is Euclid's algorithm for IETs.** At each step the two rightmost exchanged intervals compete; the shorter one is removed and the length/homology data are updated by an integer matrix. Zorich acceleration groups many elementary moves so that the induced dynamics has finite invariant measure and integrable matrix growth.

**Pinching separates growth rates; twisting prevents invariant alignments.** Neither property alone forces a simple Lyapunov spectrum. Together they rule out persistent multiplicities in the symplectic cocycle.

**Definition K15.1. Teichmüller flow**

Teichmüller flow on a translation surface is the action of  $g_t = \text{diag}(e^t, e^{-t})$  on its translation charts/period coordinates.

**Definition K15.2. Kontsevich-Zorich cocycle**

The Kontsevich-Zorich cocycle is the Gauss-Manin parallel-transport cocycle on absolute cohomology  $H^1(X; \mathbb{R})$  over Teichmüller flow.

**Definition K15.3. Rauzy-Veech induction**

Rauzy-Veech induction is the first-return renormalization of an IET obtained by removing the shorter of the two rightmost exchanged intervals and updating the permutation/length data.

**Definition K15.4. Forni deviation**

For the generator  $X$  of a translation flow, an *order-one invariant distribution* is a distribution  $D \in H^{-1}$  satisfying  $D(X\varphi) = 0$  for every smooth test function  $\varphi$  supported away from the zeros. A *Forni deviation block* is one summand in the finite-dimensional splitting of these invariant distributions according to their distinct positive Kontsevich-Zorich growth exponents; annihilating successive blocks controls the power-law limsup exponent of zero-average ergodic integrals as stated in the Forni deviation theorem.

**Definition K15.5. recurrent Teichmüller ray**

A forward Teichmüller geodesic ray  $g_t(X, \omega)$  is *recurrent in moduli space* if there is a compact set  $K \subset \mathcal{M}_g$  and an unbounded sequence  $t_n \rightarrow \infty$  such that the projections of  $g_{t_n}(X, \omega)$  lie in  $K$ . It is *divergent* if it eventually leaves every compact subset.

**Definition K15.6. real Hodge bundle, Gauss-Manin transport, and tautological plane**

The *real Hodge bundle* over a stratum has fiber  $H^1(X; \mathbb{R})$ . Along a path of marked surfaces, the Gauss-Manin connection transports cohomology classes by the canonical topological identification of the fibers. The algebraic intersection form on  $H^1(X; \mathbb{R})$  is the symplectic cup-product pairing. For an Abelian differential  $\omega$ , the *tautological plane* is  $\text{span}_{\mathbb{R}}\{[\text{Re } \omega], [\text{Im } \omega]\}$ .

**Definition K15.7. regular  $H_0^1$  observable and logarithmic deviation exponent**

For a translation surface with singular set  $\Sigma$ ,  $H_0^1(X \setminus \Sigma)$  denotes the  $H^1$ -closure of smooth functions compactly supported in the regular locus and having zero area average. A *regular forward orbit* is an orbit segment  $\{\Phi_t(p) : t \geq 0\}$  that never meets  $\Sigma$ . For an ergodic integral  $I_T(f, p) = \int_0^T f(\Phi_t(p)) dt$ , its logarithmic deviation exponent is measured with  $\log(1 + |I_T|)$ , so the exponent is defined even when some integrals vanish.

**Definition K15.8. Rauzy class, Rauzy matrix, and Zorich acceleration**

A *Rauzy class* is a connected component of the directed graph of irreducible interval-exchange permutations under the top and bottom Rauzy moves. Each move has an elementary unimodular nonnegative *Rauzy matrix* recording how the pre-induction interval lengths/homology basis are obtained from the post-induction data. *Zorich acceleration* is the induced map obtained by grouping each maximal consecutive block of Rauzy moves having the same winner type; its matrix is the ordered product of the elementary Rauzy matrices in that block.

**Definition K15.9. pinching, twisting, and simplicity for a symplectic matrix monoid**

Let  $\mathcal{G} \subset \mathrm{Sp}(H)$  be a matrix monoid on a finite-dimensional symplectic space. It is *pinching* if it contains products whose logarithmic singular-value gaps can be made arbitrarily large; it is *twisting* if for every pair of isotropic subspaces in the dimensions allowed by the symplectic form some element of  $\mathcal{G}$  sends the first transversely to the second. A monoid satisfying both properties is called *simple* in the Avila-Viana criterion. The criterion applies to an integrable locally constant/Markov symplectic cocycle with bounded distortion and a simple supporting monoid.

## 15.2 Main results

**Theorem K15.1. Masur criterion for unique ergodicity**

**Statement.** Let  $(X, \omega)$  be a unit-area translation surface and let  $g_t = \mathrm{diag}(e^t, e^{-t})$ , so the vertical foliation is contracted by  $g_t$ . If the projection of the Teichmüller geodesic ray  $g_t(X, \omega)$ ,  $t \geq 0$ , to the moduli space  $\mathcal{M}_g$  is recurrent (equivalently, it returns infinitely often to some compact subset of  $\mathcal{M}_g$ ), then the vertical foliation, hence the vertical straight-line flow, is uniquely ergodic.

*Proof.* Write  $q = \omega^2$  and let  $X_t$  be the Riemann surface underlying  $g_t q$ . We argue by contraposition. A vertical saddle connection is contracted by the Teichmüller flow; the associated extremal-length estimate forces the conformal surfaces to leave every compact subset of moduli. Thus a recurrent ray has no vertical saddle connection, and the vertical foliation is minimal.

Assume that this minimal foliation is not uniquely ergodic. We first record a finite-dimensional fact that makes Masur's compactness argument possible. An invariant transverse signed measure  $\nu$  defines a cohomology class by integrating  $\nu$  across closed transverse curves. This "flux" map is injective: a measure with zero periods has a cumulative mass function on a complete transversal that is invariant under the minimal first-return IET, so the function is constant. Flux classes of mutually singular ergodic measures have zero algebraic intersection, because long generic orbit pieces may be closed by short arcs whose normalized mutual intersection tends to zero. They therefore span an isotropic subspace of  $H^1(X; \mathbb{R})$ , of dimension at most  $g$ . Distinct ergodic

probabilities give linearly independent flux classes. Hence there are at most  $g$  of them, and flat area has a finite ergodic decomposition

$$dA_q = \sum_{i=1}^m c_i d\nu_i.$$

Choose disjoint full-measure generic sets  $E_i$  for the  $\nu_i$ .

Now take recurrent times  $t_n \rightarrow \infty$ . After markings and a subsequence, the unit-area differentials  $g_{t_n}q$  converge to a differential  $q_\infty$ . Label a regular point of the limit by  $i$  if it is a limit of images of generic points from  $E_i$ . The union of the label sets is dense. Indeed, a small zero-free rectangle in the limit pulls back to rectangles of uniformly positive flat area. One of the finitely many measures must give those pullbacks uniformly positive mass along a subsequence, and a generic point for that measure then accumulates inside the chosen rectangle.

The decisive step is a rectangle comparison. Suppose a zero-free limiting rectangle has one corner labelled  $i$  and the opposite corner labelled  $j$ . If  $\nu_i \neq \nu_j$ , choose a horizontal transversal  $I$  on which their transverse masses differ. Nearby rectangles at times  $t_n$  have equal-length vertical sides. Pulling these sides back to the original surface gives two vertical orbit segments of the same length tending to infinity. Genericity says that their intersection counts with  $I$ , divided by the common length, converge to the two different transverse masses. But after applying  $g_{t_n}$  the transversal is very long and horizontal: apart from bounded end effects, every component that crosses one side of the convergent rectangle also crosses the other. The two intersection counts differ by only  $O(1)$  and hence have ratio tending to 1, a contradiction. Opposite-corner labels must therefore represent the same ergodic measure.

Finally, any two regular points of the limiting surface can be joined by a path avoiding the finitely many zeroes and covered by a finite chain of small rectangles. Density of labels and the rectangle comparison identify every label in the chain. Thus all ergodic measures in the area decomposition are the same, contradicting non-unique ergodicity. Hence a recurrent Teichmüller ray has uniquely ergodic vertical flow.  $\square$

### Theorem K15.2. Kontsevich-Zorich cocycle

**Statement.** Over a connected component (or a finite marked cover) of an area-one stratum of Abelian differentials, let  $\mathcal{H}_{\mathbb{R}}^1$  be the real Hodge bundle with fiber  $H^1(X; \mathbb{R})$ . Gauss-Manin parallel transport along the Teichmüller flow  $g_t$  defines the Kontsevich-Zorich cocycle  $G_t$  on  $\mathcal{H}_{\mathbb{R}}^1$ . The cocycle preserves the algebraic intersection form, hence takes values in  $\mathrm{Sp}(2g, \mathbb{R})$ . For every ergodic  $g_t$ -invariant probability measure for which  $\log^+ \|G_{\pm 1}\|$  is integrable, Oseledets theorem applies and, counting multiplicity, the exponents satisfy  $1 = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_g \geq 0 \geq -\lambda_g \geq \cdots \geq -\lambda_2 \geq -\lambda_1 = -1$ ; the exponents  $\pm 1$  are carried by the tautological plane  $\mathrm{span}\{[\mathrm{Re} \omega], [\mathrm{Im} \omega]\}$ .

*Proof.* Work first on a finite marked cover of the stratum, so that the underlying relative and absolute homology groups are identified with fixed lattices. A path of marked translation surfaces determines a topological isotopy of the fibers, hence an isomorphism on  $H^1(X; \mathbb{R})$ . Parallel transport by these topological identifications is the Gauss-

Manin connection of the definition of real Hodge bundle, Gauss–Manin transport, and tautological plane. Along the Teichmüller orbit this gives linear maps

$$G_t(X, \omega) : H^1(X; \mathbb{R}) \longrightarrow H^1(g_t X; \mathbb{R}), \quad (15.2.1)$$

with  $G_{t+s}(x) = G_t(g_s x)G_s(x)$ . Descending from the marked cover gives the Kontsevich–Zorich cocycle on the Hodge bundle.

Indeed, choose integral homology cycles  $\alpha, \beta$  representing the Poincaré duals of  $a, b$ . Gauss–Manin transport is induced by an orientation-preserving topological identification of the marked surfaces, so it preserves the algebraic intersection number  $\alpha \cdot \beta$ . Since the cup-product pairing on  $H^1$  is Poincaré dual to this intersection number, (15.2.2) follows first for integral classes, then for rational and real classes by bilinearity and continuity. Thus  $G_t \in \mathrm{Sp}(2g, \mathbb{R})$ .

Let  $\mu$  be any ergodic  $g_t$ -invariant probability measure satisfying

$$\log^+ \|G_1\|, \quad \log^+ \|G_{-1}\| \in L^1(\mu). \quad (15.2.3)$$

Apply the invertible splitting form of Oseledec's theorem the Foundation Oseledec's theorem: invertible splitting form to the time-one cocycle. For  $\mu$ -almost every  $x$  this produces an equivariant direct sum

$$H^1(X; \mathbb{R}) = \bigoplus_{\lambda} E_{\lambda}(x), \quad \lim_{n \rightarrow \pm\infty} \frac{1}{n} \log \|G_n v\| = \lambda \quad (0 \neq v \in E_{\lambda}). \quad (15.2.4)$$

By Lemma 15.2 the cocycle is symplectic. The internal symplectic spectral-pairing proposition the Foundation Symplectic cocycle spectral pairing therefore gives

$$\dim E_{\lambda} = \dim E_{-\lambda} \quad (15.2.5)$$

and no pairing between  $E_{\lambda}$  and  $E_{\mu}$  unless  $\lambda + \mu = 0$ . Consequently the spectrum, counted with multiplicity, can be written

$$\lambda_1 \geq \cdots \geq \lambda_g \geq 0 \geq -\lambda_g \geq \cdots \geq -\lambda_1. \quad (15.2.6)$$

It remains to identify the outer pair. In period coordinates the Teichmüller action from the  $GL^+(2, \mathbb{R})$  action and Teichmüller flow is

$$\mathrm{Re} \, \omega \longmapsto e^t \mathrm{Re} \, \omega, \quad \mathrm{Im} \, \omega \longmapsto e^{-t} \mathrm{Im} \, \omega. \quad (15.2.7)$$

The Gauss–Manin trivialization keeps the underlying cohomology lattice fixed, so (15.2.7) says that the two tautological lines have Lyapunov exponents  $+1$  and  $-1$ .

Represent a real cohomology class  $c$  by its harmonic representative for the complex structure of  $(X, \omega)$  and let  $\|c\|_H$  be the Hodge norm. Away from the zero set write the  $(1, 0)$  part as  $m\omega$ . The standard first variation calculation under  $g_t$  gives

$$\frac{d}{dt} \log \|c\|_H = - \frac{\mathrm{Re} \int_X m^2 dA_{\omega}}{\int_X |m|^2 dA_{\omega}}. \quad (15.2.8)$$

Cauchy-Schwarz yields absolute value at most 1. Integrating (15.2.8) along an orbit gives  $e^{-|t|}\|c\|_H \leq \|G_t c\|_H \leq e^{|t|}\|c\|_H$ , so every Oseledets exponent belongs to  $[-1, 1]$ . Equality in Cauchy-Schwarz is attained by the constant meromorphic quotient, exactly on the two tautological directions. Thus  $\lambda_1 = 1$  and  $-\lambda_1 = -1$ , carried by  $\text{span}\{[\text{Re } \omega], [\text{Im } \omega]\}$ . Together with (15.2.6) this is the asserted spectrum.  $\square$

### Theorem K15.3. Forni deviation theorem

**Statement.** Fix a connected component  $\mathcal{H}_1(\kappa)$  of a unit-area stratum of Abelian differentials in genus  $g \geq 2$ , equipped with its Masur-Veech measure  $\mu_{\text{MV}}$ . For  $\mu_{\text{MV}}$ -almost every translation surface  $(X, \omega)$ , let  $\Phi_t$  denote the vertical translation flow. Then there are distinct positive exponents  $1 = \lambda'_1 > \cdots > \lambda'_s > 0$  and a splitting  $\mathcal{I}_X^1 = \bigoplus_{j=1}^s \mathcal{I}_X^1(\lambda'_j)$  of the finite-dimensional space of order-one invariant distributions, with  $m_j = \dim \mathcal{I}_X^1(\lambda'_j)$ ,  $\sum_j m_j = g$  and  $m_1 = 1$ . Let  $f \in H_0^1(X \setminus \Sigma)$  in the sense of Definition K15.7. If  $f$  is annihilated by the first  $i$  distribution blocks,  $i < s$ , then along every regular forward orbit

$$\limsup_{T \rightarrow \infty} \frac{\log \left( 1 + \left| \int_0^T f(\Phi_t(p)) dt \right| \right)}{\log T} \leq \lambda'_{i+1};$$

for almost every  $p$  equality holds whenever some distribution in  $\mathcal{I}_X^1(\lambda'_{i+1})$  does not annihilate  $f$ . If all invariant-distribution blocks annihilate  $f$ , the same logarithmic limsup is 0. The  $\lambda'_j$  are the distinct positive Kontsevich-Zorich Lyapunov exponents for  $\mu_{\text{MV}}$ .

#### Scope note

The theorem uses the distinct positive KZ exponents with their multiplicities; simplicity is a separate result.

*Proof.* The proof has four layers: nonuniform hyperbolicity of the KZ cocycle, basic currents, renormalized orbit currents, and evaluation on the observable.

For a harmonic cohomology class write its  $(1, 0)$  representative as  $m\omega$ . Forni's bilinear form

$$B_q(m_1, m_2) = \int_X m_1 m_2 dA_q$$

controls the derivative of the Hodge norm. Its singular values  $1 = \Lambda_1 \geq \cdots \geq \Lambda_g \geq 0$  also enter the determinant formula for the period map: the last singular value vanishes exactly on the determinant locus. Suppose only  $k < g$  KZ exponents were positive. Equality in the Hodge-trace formula would force  $B_q$  to vanish on the harmonic complement of the unstable  $k$ -plane.

Forni rules this out geometrically. Every stratum component contains a completely periodic differential with  $g$  independent cylinder waists. Backward recurrent returns carry the Poincaré-dual cylinder plane arbitrarily close to the unstable plane. Independently pinching those cylinders while preserving the horizontal measured foliation reaches the boundary model with  $2g$  paired poles. On that model the period-matrix determinant is explicitly nonzero, so the last singular value of  $B_q$  is positive. Continu-



ity contradicts the forced zero block. Hence all  $g$  positive KZ exponents are strictly positive; no simplicity is asserted.

Next consider invariant distributions for the vertical flow. A one-current  $C$  is basic when contraction and Lie derivative along the flow both vanish. Invariant order-one distributions and basic currents correspond by  $D(f) = C(f\eta_V)$ . On a compact Teichmüller return set the weighted Sobolev Poincaré inequality gives a uniform solution operator for the contraction equation. Correcting a harmonic representative at successive backward return times produces a Cauchy sequence because a positive KZ class contracts exponentially backward while the Sobolev constants grow only subexponentially. Thus every unstable KZ class has an order-one basic-current representative. An exact basic current is  $dU$  with  $VU = 0$ ; almost-everywhere ergodicity makes  $U$  constant. The positive basic-current bundle is therefore isomorphic to the unstable KZ bundle, with the same Lyapunov blocks.

It remains to control a long orbit segment. Choose a compact return set on which the flat geometry, Sobolev norms, and Oseledets projections are uniformly controlled. If  $t_k < 0$  are its successive backward returns, a principal return trajectory has length comparable with  $e^{|t_k|}$ . Its projection to a KZ block of exponent  $\lambda$  is bounded by  $e^{(\lambda+\delta)|t_k|}$ , while its exact-current part is bounded by  $e^{\delta|t_k|}$ . Forni's greedy return decomposition writes an arbitrary long orbit segment as principal trajectories of decreasing levels, with level multiplicity at most a constant times  $e^{|t_{k+1}| - |t_k|}$ . Birkhoff recurrence gives  $|t_k|/k \rightarrow 1/\mu(P)$ . Summation yields

$$\|\Pi_\lambda \Gamma_T\|_{-1} \leq C_\varepsilon T^{\lambda+\varepsilon}, \quad \|\Pi^E \Gamma_T\|_{-1} \leq C_\varepsilon T^\varepsilon.$$

Recurrent Lagrangian cylinders of uniformly positive area give the matching lower exponent for every nonzero block on a positive-measure limsup set, and ergodicity spreads this detection to almost every point.

Finally evaluate the orbit current on  $f\eta_V$ :

$$\Gamma_T(f\eta_V) = \int_0^T f(\Phi_t(p)) dt.$$

Blocks whose invariant distributions annihilate  $f$  disappear exactly. The first surviving block gives the stated upper power, and the cylinder subsequence gives equality almost everywhere when that block acts nontrivially. If every positive block annihilates  $f$ , only the subpower exact remainder remains, so the logarithmic exponent is zero.  $\square$

#### Theorem K15.4. Rauzy-Veech/Zorich induction

**Statement.** On every Rauzy class of irreducible interval-exchange permutations, Rauzy-Veech induction is the first-return renormalization obtained by comparing the two rightmost exchanged subintervals, deleting the shorter one, and updating the length vector by the associated unimodular Rauzy matrix. For Lebesgue-almost every length vector the induction is defined for all forward times. Zorich acceleration groups consecutive Rauzy steps of the same winner type; after projectivizing the length simplex it preserves an ergodic finite absolutely continuous

invariant probability measure. If  $B_Z$  denotes the corresponding accelerated Rauzy matrix, then  $\log^+ \|B_Z\| \in L^1$ , so Oseledets theorem applies to the Zorich matrix cocycle. Under the suspension/zippered-rectangle identification this cocycle records return-time and relative-homology data and gives the standard discrete model underlying the Kontsevich-Zorich cocycle.

*Proof.* At one Rauzy step, compare the two rightmost exchanged intervals. If the top letter  $\alpha$  wins over the bottom letter  $\beta$ , the induced length vector satisfies

$$\lambda = (I + E_{\alpha\beta})\lambda'.$$

The symmetric formula holds for a bottom win. Thus Rauzy matrices are nonnegative integral unimodular matrices, and their columns are exactly the visitation vectors of the induced towers.

For a fixed finite Rauzy word, termination after that word means equality of the two competing coordinates of  $B^{-1}\lambda$ . This is one rational hyperplane. Since there are countably many words, almost every irreducible IET can be Rauzy-induced indefinitely.

Zorich acceleration groups one maximal run of Rauzy arrows with the same winner. Its finite invariant measure is most cleanly obtained from the fast zippered-rectangle section constructed in the preceding volume. The return map of that signed section, after forgetting suspension data and finitely many sign/combinatorial labels, is exactly the Zorich map. Disintegrating the finite section measure therefore gives a finite measure on the projective length simplex, equivalent to Lebesgue. An invariant set downstairs lifts to an invariant set upstairs, so ergodicity of the fast section implies ergodicity of the Zorich measure.

The same section also gives the needed logarithmic matrix estimate. If  $B$  is a Zorich return matrix,  $x$  and  $y$  are normalized pre- and post-return length vectors, and  $x = \ell B y$ , then

$$\|B\|_1 \leq d \frac{|By|_1}{\min y_b} \quad \text{and hence} \quad \log^+ \|B\| \leq C_d - \log \ell + \log \frac{1}{\min y_b}.$$

The first term on the right is the Teichmüller return time, up to a bounded finite-state correction, and is integrable by the finite-section theorem. The cusp estimate proved there also makes  $\log(1/\min y_b)$  integrable. Thus  $\log^+ \|B_Z\|$  is integrable and Oseledets theorem applies to the natural extension.

Finally, the same elementary cut-and-stack calculation has a homological meaning. When the losing strip is stacked on the winner, the winner's return path acquires the losing path. Therefore the Rauzy matrix changes the relative-homology basis by exactly the same column addition by which it changes the return-time vector. Iterating gives the matrix for every Rauzy word. In the zippered-rectangle suspension, restricting this relative transport to absolute homology and interpolating between returns by Gauss-Manin transport is precisely the discrete model of the Kontsevich-Zorich cocycle.  $\square$

**Theorem K15.5. Avila-Viana simplicity theorem**

**Statement.** On every connected component of every area-one stratum of Abelian differentials, with its Masur-Veech probability measure, the Kontsevich-Zorich cocycle on absolute cohomology has simple Lyapunov spectrum. In genus  $g$ , with the Teichmüller-time normalization for which the top exponent is 1, the exponents may be ordered as  $1 = \lambda_1 > \lambda_2 > \cdots > \lambda_g > 0 > -\lambda_g > \cdots > -\lambda_2 > -1$ , and every exponent has multiplicity one.

*Proof.* Fix a connected component of an area-one stratum and a Rauzy class coding a finite marked cover of it. The Rauzy-Veech/Zorich theorem gives a finite ergodic Zorich measure, an integrable return-matrix cocycle, and the identification of that matrix cocycle on absolute homology with the discrete Kontsevich-Zorich return cocycle.

The first point is to put the base dynamics into a form in which the pinching-twisting criterion can actually be proved. Choose a finite positive Zorich path  $\gamma$ . The corresponding projective simplex lies compactly inside the length simplex. First return to it has inverse branches

$$h_B(x) = \frac{Bx}{|Bx|_1},$$

with  $B$  positive. Their logarithmic Jacobians have a uniform Lipschitz bound in Hilbert's projective metric, so the symbolic natural extension has bounded distortion, or equivalently approximate product structure. The return matrix is locally constant on each branch. Fixed entrance and exit paths show that the supporting return monoid contains conjugates of arbitrarily large pieces of the Rauzy monoid. Thus it remains to prove two facts: the abstract simplicity criterion and simplicity of the Rauzy monoid.

**The abstract simplicity criterion.** Let  $A$  be an integrable locally constant cocycle over an ergodic Markov map with bounded distortion, and suppose its supporting monoid is pinching and twisting. Fix a dimension  $1 \leq k < d$ . We prove that the  $k$ th and  $(k+1)$ st Lyapunov exponents are different.

Pass to the two-sided symbolic extension. Bounded distortion says that the probability of a concatenated cylinder is comparable, with one uniform constant, to the product of the two cylinder probabilities. Cesaro averaging of a product probability on  $\text{Grass}(k, H)$  therefore produces an invariant  $u$ -state. Its conditional probabilities  $m_x$  depend on the past, and the usual martingale convergence theorem recovers  $m_x$  from longer and longer past cylinders.

Choose a pinching word  $l_p$  with simple, widely separated singular scales. Twisting supplies a connector  $l_q$  and finitely many further words  $l_{q_i}$  with the following uniform property: whatever  $k$ -plane one starts from and whatever complementary repelling plane is created by a later tail word, at least one  $l_{q_i}$  moves the concentrated plane a fixed angle away from that repelling set. Hence, for every probability  $\rho$  on the Grassmannian and every tail word  $l_0$ , words of the form

$$l_p^n l_q l_p^n l_{q_i} l_0$$

send more than  $1 - \varepsilon$  of  $\rho$  into an  $\varepsilon$ -ball when  $n$  is large. Bounded distortion gives

a family of such past cylinders of total measure bounded below independently of  $n$ . Comparing these pushforwards with the martingale limits forces  $m_x$  to be a Dirac mass on a positive-measure set, and ergodicity then gives an equivariant measurable  $k$ -plane  $\xi_k^u(x)$  almost everywhere. The same argument also shows that its distribution is not contained in any Schubert hyperplane section.

Apply the construction to the inverse cocycle. It gives an equivariant  $(d - k)$ -plane  $\xi_{d-k}^s$  depending on the future. Fubini's theorem and the hyperplane-avoidance property imply

$$\xi_k^u(x) \cap \xi_{d-k}^s(y) = \{0\}$$

for almost every past-future pair. Choose transverse support planes  $F^u, F^s$  and return to a positive-measure rectangle where the two measurable sections are both very close to those fixed planes. A measurable near-identity change of coordinates sends the actual sections to  $F^u, F^s$ . The induced return matrix then maps a closed ball about  $F^u$  strictly into its interior. The min-max characterization of singular values gives pointwise

$$\sigma_k(B) > \sigma_{k+1}(B).$$

Integrating the logarithmic singular-value gap over the return set gives

$$\theta_k - \theta_{k+1} > 0.$$

Since this works for every  $k$ , the cocycle has simple Lyapunov spectrum.

**Twisting of the Rauzy monoid.** We prove a stronger statement: the Rauzy monoid acts minimally on Lagrangian flags. The proof is by induction on the alphabet. For two letters the flag space is a projective line and the two Rauzy parabolics give the base case. For a larger irreducible permutation one can delete a letter  $B$  so that the new permutation remains irreducible and the old one is a simple extension. If  $P$  denotes deletion of the  $B$ -coordinate, then

$$P\Omega(\pi)P^* = \Omega(\pi'), \quad P\Theta(\mathcal{E}_*\eta) = \Theta(\eta)P.$$

If genus is unchanged,  $P$  is a symplectic isomorphism on absolute homology and the induction transfers directly. If genus rises by one, the extended old monoid fixes  $e_B$  and its action on the symplectic hyperplane  $\omega(h, e_B) = 0$ , after quotienting the null direction, is precisely the old action. The inductive minimality inside this hyperplane, together with Rauzy transvections moving the hyperplane itself, makes every full Lagrangian flag orbit dense. Extending any pair of isotropic subspaces to Lagrangian flags and hitting the open transversality locus proves twisting.

**Strong pinching of the Rauzy monoid.** Strong pinching is unchanged when the base permutation is moved within its Rauzy class. Again induct on alphabet size. Genus one follows from a hyperbolic product of the two parabolic generators. A non-minimal Rauzy class, with more letters than twice the genus, contains a same-genus simple extension of a smaller class; the smaller symplectic action embeds, so pinching transfers.

It remains to treat a minimal class. Such a class contains a good standard permutation. Delete the two boundary letters in the order used by the Rauzy moves. The

resulting genus- $g - 1$  action occurs inside a submonoid that fixes one boundary vector. By induction that submonoid is strongly pinching and, by the previous paragraph, twisting. Choose there a word  $Q$  with arbitrarily large gaps and with 1 absent from its spectrum. Extend  $Q$  and multiply by the cycle in which the omitted boundary letter  $E$  wins against all the other letters. The latter fixes every  $e_x$ ,  $x \neq E$ , and sends

$$e_E \mapsto \sum_x e_x.$$

The product therefore has eigenvalue 1 with exactly one geometric direction, while all the other positive growth rates retain the arbitrarily large inductive separations. Powers of this symplectic matrix turn the one-dimensional neutral direction into a linearly growing singular scale; that scale can be separated from all the exponential scales. Hence every positive logarithmic singular-value gap can be made arbitrarily large. The Rauzy monoid is strongly pinching.

Twisting and strong pinching make the Rauzy monoid simple. The supporting monoid of the positive-cylinder return is therefore simple, so the abstract criterion gives a simple discrete spectrum. Since the cocycle is symplectic, its spectrum is paired. A zero exponent would have even multiplicity, which is impossible in a simple symplectic spectrum. Thus

$$\chi_1 > \chi_2 > \cdots > \chi_g > 0 > -\chi_g > \cdots > -\chi_1.$$

Finally the Rauzy-Veech/Zorich identification says that these matrices are exactly the discrete KZ return maps on absolute homology. The Teichmüller return time has finite mean. By Birkhoff, the number of returns before time  $T$  divided by  $T$  tends to the reciprocal mean roof, so passing from the induced discrete cocycle to the continuous KZ cocycle multiplies every exponent by one common positive constant. Simplicity and signs are therefore unchanged. The tautological plane has top exponent 1, and we obtain

$$1 = \lambda_1 > \lambda_2 > \cdots > \lambda_g > 0 > -\lambda_g > \cdots > -\lambda_2 > -1$$

with multiplicity one throughout. The finite marked Rauzy models cover every connected stratum component, so the conclusion descends to Masur-Veech probability on each component.  $\square$

## 15.3 Worked examples

### Worked example: genus one and the modular surface

Let  $X = \mathbb{R}^2/\mathbb{Z}^2$  with the flat form  $dz = dx + i dy$ . A straight-line flow in direction  $(1, \alpha)$  meets the section  $x = 0$  in the circle rotation

$$y \mapsto y + \alpha \pmod{1}.$$

If  $\alpha \notin \mathbb{Q}$ , this rotation is uniquely ergodic. Rotating coordinates so that the chosen

direction is vertical and applying  $g_t$  turns the same orbit problem into a geodesic on

$$SL(2, \mathbb{R})/SL(2, \mathbb{Z}).$$

Reduction by  $SL(2, \mathbb{Z})$  is the genus-one prototype of renormalization: continued-fraction steps keep returning the lattice to a fundamental region when the corresponding modular geodesic is recurrent.

On cohomology, use the basis  $[dx], [dy]$ . Along the Teichmüller deformation,

$$g_t^*[dx] = e^t[dx], \quad g_t^*[dy] = e^{-t}[dy].$$

Thus the KZ exponents are exactly  $+1$  and  $-1$ . There are no additional positive exponents in genus one; the higher-genus Forni hierarchy begins only when  $g \geq 2$ .

#### Check your understanding

For  $\alpha = p/q \in \mathbb{Q}$ , compute the return period of the rotation and explain why the resulting directional flow has many invariant probability measures instead of one.

#### Boundary case: a saddle-connection direction

Suppose a direction contains a saddle connection. The flow is then not minimal: a singularity-to-singularity trajectory cuts the surface and may bound periodic cylinders. This is exactly the geometry excluded in the recurrent-minimal branch of Masur's argument. One must not infer unique ergodicity merely from the existence of Teichmüller/KZ data; the dynamical hypothesis is a statement about the particular direction under study.

#### Common pitfall

An Oseledets decomposition for the Kontsevich-Zorich cocycle is not yet a theorem about deviation of ergodic averages on a fixed translation surface. One must specify the invariant measure on the stratum, identify the relevant cohomological class, and use the additional bridge from cocycle growth to deviations. The measure and almost-everywhere quantifiers are part of the theorem.

## 15.4 Chapter summary

#### What to remember

- Teichmüller flow renormalizes directional dynamics by converting long thin geometry into bounded-scale geometry in moduli space.
- Recurrence of the renormalized orbit forces unique ergodicity through Masur's contrapositive divergence mechanism.

- Gauss–Manin transport gives a symplectic KZ cocycle with tautological exponents  $\pm 1$ .
- Forni’s invariant distributions identify the successive power laws governing deviations of ergodic integrals.
- Rauzy–Veech/Zorich induction is the discrete coding of the same renormalization and supplies the matrix dynamics used in KZ theory.
- Pinching and twisting make the Masur–Veech KZ spectrum simple.

### Looking ahead

K16 asks what happens when an entire  $SL(2, \mathbb{R})$  orbit is allowed to move through a stratum. Exceptional lattice surfaces exhibit a sharp Veech dichotomy, while the Eskin–Mirzakhani–Mohammadi theory shows that arbitrary orbit closures are still unexpectedly rigid: they are affine invariant manifolds.

### Check your understanding

State the extra bridges needed to pass from Lyapunov exponents of the Kontsevich–Zorich cocycle to quantitative deviation statements for directional flows. Where do the invariant measure, Oseledets filtration, and cohomological observable enter?

## 15.5 Exercises and problems

### Exercise 15.1

**Difficulty.** Core.

**Related results.** Masur criterion for unique ergodicity; Kontsevich–Zorich cocycle; Forni deviation theorem.

**Problem.** In genus one, identify the KZ cocycle with the natural homology action and integrate one Fourier mode along an irrational linear flow, displaying explicitly how a divisor  $k \cdot \omega$  controls deviation.

**Solution sketch.** Compute the matrix action on homology and integrate the Fourier exponential exactly.

### Exercise 15.2

**Difficulty.** Core.

**Related results.** Rauzy–Veech/Zorich induction; Avila–Viana simplicity theorem.

**Problem.** For a saddle-connection/periodic direction, construct distinct invariant measures and track the shrinking saddle connection under Teichmüller flow, illustrating failure of unique ergodicity and nondivergence.

**Solution sketch.** Use closed/cylinder trajectories for invariant measures and apply  $g_t$  to the holonomy vector.

**Exercise 15.3****Difficulty.** Advanced.**Related results.** Forni deviation theorem.**Problem.** For a zero-mean trigonometric polynomial on the two-torus, compute its ergodic integral mode by mode and prove a divisor bound; explain why higher-genus Forni theory requires more than this Fourier calculation.**Solution sketch.** Integrate each Fourier exponential and sum; then identify the KZ/cohomological and singular-geometric inputs absent in genus one.**Exercise 15.4****Difficulty.** Advanced.**Related results.** Rauzy-Veech/Zorich induction.**Problem.** Carry out one Rauzy-induction step for a concrete three-IET: update lengths, permutation, and Rauzy matrix. Give a connection example where induction terminates or becomes undefined.**Solution sketch.** Compare the two rightmost intervals, subtract the loser, record the elementary matrix, then force an endpoint orbit relation in the failure example.**Exercise 15.5****Difficulty.** Advanced.**Related results.** Masur criterion for unique ergodicity; Kontsevich-Zorich cocycle.**Problem.** In genus one, compute unique ergodicity and cocycle growth separately and explain why they answer different questions; state the extra bridge needed for quantitative deviations in higher genus.**Solution sketch.** Separate time-average uniqueness from linear-cocycle growth and identify Forni's cohomological/invariant-distribution machinery as the bridge.**Exercise 15.6****Difficulty.** Lab.**Related results.** None.**Problem.** Evolve a genus-one period matrix under Teichmüller flow, reduce repeatedly by  $SL(2, \mathbb{Z})$ , and record the accumulated matrices as a toy KZ cocycle.**Solution sketch.** Separate continuous flow from discrete modular reduction and verify determinant/area invariants under time-step refinement.



# Veech Surfaces, Affine Invariant Manifolds, Counting and Wind-Tree Applications

## Why this chapter matters

K15 used Teichmüller renormalization to understand almost-everywhere directional dynamics. K16 studies a different source of rigidity: the geometry of the entire  $GL^+(2, \mathbb{R})$  orbit of a translation surface. At one extreme, a Veech surface has a lattice stabilizer and its directional dynamics obeys a striking dichotomy. At the other, a generic orbit may be much larger, but the Eskin-Mirzakhani and Eskin-Mirzakhani-Mohammadi theorems show that invariant measures and orbit closures are still cut out by linear equations in period coordinates.

The volume then turns this rigidity into concrete geometry and counting. Wright's cylinder deformation theorem explains which cylinder shears remain inside an orbit closure and how arithmetic enters through its field of definition. The Siegel-Veech formula converts  $SL(2, \mathbb{R})$  invariance into quadratic counting laws. Finally, the periodic wind-tree model shows how a Lyapunov exponent of a finite translation-surface cover becomes a physical displacement exponent in an infinite billiard.

**Prerequisites.** K13-K15, especially period coordinates, Teichmüller flow, the KZ cocycle and Oseledets exponents; Foundation F04 for cocycles; F11 and F14 for ergodic/measure tools.

## Learning goals

- Recognize Veech surfaces through their lattice derivative group and explain the periodic/uniquely-ergodic dichotomy.
- State what an affine invariant manifold means in local period coordinates and distinguish measure classification from orbit-closure classification.
- Use cylinder shear/stretch deformations to read tangent directions of an affine invariant manifold.
- Explain why the Siegel-Veech transform turns counting into an invariant-measure integral.
- Trace the wind-tree displacement exponent from a deck cocycle to a KZ Lyapunov exponent.

### Conceptual roadmap

The orbit-closure theory develops in four layers.

- A lattice Veech group gives the Veech dichotomy for directional dynamics.
- Invariant-measure classification upgrades the relevant parabolic invariance to  $SL(2, \mathbb{R})$ -invariant affine measures.
- Orbit-closure classification identifies the geometric supports of those measures as affine invariant manifolds in period coordinates.
- Cylinder deformations and fields of definition then make the affine geometry computable and feed counting laws and billiard applications.

The wind-tree theorem is the model application showing that moduli-space Lyapunov data can dictate a spatial transport exponent.

### Check your understanding

Why is a Veech surface much more special than a surface whose  $SL(2, \mathbb{R})$  orbit closure is merely an affine invariant manifold? Identify the extra lattice condition.

## 16.1 Definitions

### Intuition

**The Veech group is a derivative shadow of affine symmetries.** An affine automorphism of a translation surface has a constant derivative in translation charts. The derivatives form a subgroup of  $SL(2, \mathbb{R})$  on the area-one locus. Finite covolume of this subgroup is the lattice condition behind Veech dichotomy.

**An affine invariant manifold is linear only in the right coordinates.** In period coordinates it is locally given by homogeneous real-linear equations, but globally it may be immersed and have nontrivial topology. “Affine” does not mean affine in an arbitrary ambient Euclidean embedding.

**Cylinder deformations are local probes of the orbit closure.** When a family of parallel cylinders may be sheared without leaving the orbit closure, the shear direction is a tangent vector to the affine invariant manifold. Relations among cylinder circumferences and moduli therefore constrain the tangent space.

**The absolute tangent bundle forgets how zeros move relative to one another.** Period coordinates live in relative cohomology  $H^1(X, \Sigma; \mathbb{C})$ . The absolute projection  $p : H^1(X, \Sigma; \mathbb{C}) \rightarrow H^1(X; \mathbb{C})$  forgets the endpoint data. Because an affine invariant manifold is cut out by fixed linear equations in period coordinates, its tangent spaces parallel-transport by the Gauss–Manin connection; the same is true for their absolute images.

**$\mathcal{M}$ -parallel is stronger than “parallel on this one surface.”** Two cylinders are  $\mathcal{M}$ -parallel when their core periods remain proportional under every sufficiently small deformation inside  $\mathcal{M}$ . A maximal class has a canonical shear direc-

tion: add the intersection functional of each core curve, weighted by that cylinder's height. Wright's theorem detects when this cohomology class is tangent to the orbit closure.

**The field of definition measures arithmetic rigidity of the tangent equations.** It is the smallest real field over which the local homogeneous period-coordinate equations of  $\mathcal{M}$  may be written. A different field, the trace field, is generated by traces of monodromy on the absolute tangent local system. Their equality is not terminology; it is one of the substantive conclusions proved in this volume.

**Siegel-Veech counting is an averaging identity.** A surface contributes a discrete multiset of holonomy vectors. Summing a test function over that set and averaging over an invariant measure must, by  $SL(2, \mathbb{R})$  symmetry, be a scalar multiple of Lebesgue integral on  $\mathbb{R}^2$ .

**The wind-tree exponent is not a random-walk exponent.** The displacement is a deterministic deck cocycle. Renormalization puts that cocycle into an Osledeets subspace whose KZ exponent gives the polynomial growth rate  $2/3$ .

#### Definition K16.1. Veech surface

A Veech surface is a translation surface whose orientation-preserving affine automorphism derivative group is a lattice in  $SL(2, \mathbb{R})$ .

#### Definition K16.2. affine invariant manifold and affine measure

Let  $\mathcal{H}_1(\alpha)$  be a unit-area stratum. An *affine invariant manifold*  $\mathcal{M}_1 \subset \mathcal{H}_1(\alpha)$  is the support of an ergodic  $SL(2, \mathbb{R})$ -invariant probability measure  $\nu_1$  for which there exist a manifold  $N$  and a proper immersion  $f : N \rightarrow \mathcal{H}_1(\alpha)$  with image  $\mathcal{M}_1$  such that the self-intersection locus has  $\nu_1$ -measure zero and, in every sufficiently small period-coordinate neighborhood  $U \subset N$ , the cone  $\mathbb{R}f(U) \subset \mathcal{H}(\alpha)$  is the intersection of the period chart with a complex linear subspace obtained by complexifying a real subspace of relative cohomology. If  $d\nu = d\nu_1 da$  denotes the radial/coned measure on  $\mathbb{R}\mathcal{M}_1$ , then on each such chart  $\nu$  is, up to one positive normalizing constant, Lebesgue measure on that linear subspace. Such  $\nu_1$  is called the *affine measure* on  $\mathcal{M}_1$ .

#### Definition K16.3. Siegel-Veech constant

Fix a measurable configuration rule  $\mathcal{C}$  assigning to a translation surface  $X$  a locally finite multiset  $V_{\mathcal{C}}(X) \subset \mathbb{R}^2 \setminus \{0\}$  of holonomy vectors, equivariantly under  $SL(2, \mathbb{R})$ . For an invariant probability measure  $\mu$ , a number  $c_{\mu, \mathcal{C}} \geq 0$  is the *Siegel-Veech constant* when, for every  $f \in C_c(\mathbb{R}^2)$  for which the transform is integrable,

$$\int \sum_{v \in V_{\mathcal{C}}(X)} f(v) d\mu(X) = c_{\mu, \mathcal{C}} \int_{\mathbb{R}^2} f(u) du.$$

The normalization in this volume therefore makes the quadratic counting asymptotic  $N_{\mathcal{C}}(X, R) \sim c_{\mu, \mathcal{C}} \pi R^2$ .

**Definition K16.4. periodic rectangular wind-tree model**

For  $(a, b) \in (0, 1)^2$ , place at every point of  $\mathbb{Z}^2$  the closed axis-parallel rectangle of side lengths  $a$  and  $b$  centered at that lattice point, and remove the interiors of all these rectangles from  $\mathbb{R}^2$ . The *periodic rectangular wind-tree billiard* is unit-speed straight motion in the complement with specular reflection at regular obstacle sides. Orbits hitting a corner are singular and are not continued through that collision. Unfolding the four reflection states gives a compact translation surface together with a  $\mathbb{Z}^2$ -cover; the deck coordinate records planar displacement.

**Definition K16.5. absolute holonomy module and holonomy field**

For a translation surface  $(X, \omega)$ , identify  $\mathbb{C}$  with  $\mathbb{R}^2$  and define its absolute holonomy module by

$$\Lambda_{\text{abs}}(X, \omega) = \left\{ \int_{\gamma} \omega : \gamma \in H_1(X; \mathbb{Z}) \right\} \subset \mathbb{C}.$$

If  $e_1, e_2 \in \Lambda_{\text{abs}}$  are real-linearly independent, the holonomy field is the smallest subfield  $\mathbf{k} \subset \mathbb{R}$  for which  $\Lambda_{\text{abs}} \subset \mathbf{k}e_1 \oplus \mathbf{k}e_2$ ; this field is independent of the chosen real basis  $e_1, e_2$ .

**Definition K16.6. standard cylinder shear and stretch**

Let  $\mathcal{C}$  be a collection of parallel cylinders on a translation surface, all making angle  $\theta$  with the horizontal. Put

$$u_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad a_s = \begin{pmatrix} 1 & 0 \\ 0 & e^s \end{pmatrix}, \quad r_{\theta} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

The standard cylinder deformation  $a_s^{\mathcal{C}} u_t^{\mathcal{C}}$  is obtained by applying  $r_{-\theta}$ , then applying  $a_s u_t$  only to the cylinders in  $\mathcal{C}$  while leaving the complementary polygons unchanged, and finally applying  $r_{\theta}$ . The  $u_t^{\mathcal{C}}$  part is the cylinder shear and the  $a_s^{\mathcal{C}}$  part is the cylinder stretch.

**Definition K16.7. Veech group and lattice**

For a translation surface  $(X, \omega)$ , the *Veech group*  $\Gamma(X, \omega) \subset SL(2, \mathbb{R})$  is the derivative image of its orientation-preserving affine automorphism group (after the area-one normalization). A discrete subgroup  $\Gamma < SL(2, \mathbb{R})$  is a *lattice* if  $SL(2, \mathbb{R})/\Gamma$  has finite Haar volume, equivalently if the associated hyperbolic orbifold has finite area.

**Definition K16.8. completely periodic direction**

A direction on a compact translation surface is *completely periodic* if every regular trajectory in that direction is closed and the surface is covered by finitely many maximal Euclidean cylinders foliated by those closed geodesics. When the singular set is nonempty, the boundary of each maximal cylinder is a finite union of parallel saddle connections; on an unmarked flat torus the single periodic cylinder has no singular boundary.

**Definition K16.9. upper-triangular subgroup and coning**

Write

$$P = AN = \left\{ \begin{pmatrix} e^t & s \\ 0 & e^{-t} \end{pmatrix} : t, s \in \mathbb{R} \right\} < SL(2, \mathbb{R}),$$

with  $A = \{g_t = \text{diag}(e^t, e^{-t})\}$  and  $N = \{u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}\}$ . If  $\nu_1$  is a probability measure on an area-one locus  $\mathcal{M}_1$ , its *coned measure* is the product-disintegration  $d\nu = d\nu_1 da$  on  $\mathbb{R}\mathcal{M}_1 = \{(X, r\omega) : r > 0\}$ , where  $a = r^2$  is the area coordinate; changing this convention only changes the local normalizing constant.

**Definition K16.10. Siegel-Veech transform and counting function**

For a configuration rule  $\mathcal{C}$  and  $f \in C_c(\mathbb{R}^2)$ , the *Siegel-Veech transform* is

$$\widehat{f}_{\mathcal{C}}(X) = \sum_{v \in V_{\mathcal{C}}(X)} f(v),$$

with multiplicities inherited from the configuration. The associated counting function is  $N_{\mathcal{C}}(X, R) = \#\{v \in V_{\mathcal{C}}(X) : \|v\| \leq R\}$ , again with those multiplicities.

**Definition K16.11. polynomial diffusion exponent**

For a nonsingular wind-tree orbit  $p \mapsto \phi_T^\theta(p)$ , its upper polynomial diffusion exponent is

$$\delta^+(p, \theta) = \limsup_{T \rightarrow \infty} \frac{\log(1 + d(p, \phi_T^\theta(p)))}{\log T}.$$

Under unfolding, the displacement is the value on the corresponding orbit segment of the  $\mathbb{Z}^2$  deck cocycle, equivalently the pairing of that segment with the two cohomology classes dual to the deck generators.

**Definition K16.12. absolute projection and flat tangent local systems**

Let  $\Sigma$  be the zero set of the Abelian differential and let

$$p : H^1(X, \Sigma; \mathbb{C}) \longrightarrow H^1(X; \mathbb{C})$$

be the map induced by forgetting the relative endpoints; it is the *absolute projection*. Over a period-coordinate neighborhood of an affine invariant manifold  $\mathcal{M}$ , the locally constant Gauss-Manin identifications carry the tangent spaces  $T_X \mathcal{M}$  to one fixed complex linear subspace of relative cohomology. These spaces form the *flat tangent local system*  $T\mathcal{M}$ , and their absolute images  $p(T_X \mathcal{M})$  form the *absolute tangent local system*  $p(T\mathcal{M})$ . On a finite cover that labels the zeros,  $\ker p$  is a trivial flat local system.

**Definition K16.13. M-parallel cylinders and cylinder shear class**

Let  $\mathcal{M}$  be an affine invariant manifold and let  $C_1, C_2$  be parallel cylinders on  $X \in \mathcal{M}$  with oriented core curves  $\alpha_1, \alpha_2$ . They are *M-parallel* if, after marking them in a sufficiently small neighborhood of  $X$  in  $\mathcal{M}$ , they remain parallel on every nearby

surface; equivalently the restrictions of the core-period functionals  $\alpha_1, \alpha_2$  to  $T_X \mathcal{M}$  are real-proportional. The maximal such equivalence classes are the  $\mathcal{M}$ -parallel cylinder classes. If  $\mathcal{C} = \{C_i\}$  is a horizontal cylinder family, with height  $h_i$  and core class  $\alpha_i$ , its *standard shear cohomology class* is

$$\eta_{\mathcal{C}} = \sum_i h_i I(\alpha_i) \in H^1(X, \Sigma; \mathbb{R}),$$

where  $I(\alpha_i)$  is the relative-cohomology functional given by algebraic intersection with  $\alpha_i$ .

#### Definition K16.14. field of definition and trace field of a flat bundle

For an affine invariant manifold  $\mathcal{M}$ , its *field of definition*  $\mathbf{k}(\mathcal{M}) \subset \mathbb{R}$  is the smallest subfield over which the homogeneous real-linear equations defining its tangent subspace in every period chart can be written. If  $V$  is a flat complex subbundle of the integral absolute cohomology local system, its *trace field*  $k_{\text{tr}}(V)$  is the subfield of  $\mathbb{C}$  generated over  $\mathbb{Q}$  by the traces of the monodromy representation restricted to  $V$ . Equality between these two fields is a theorem proved below, not part of the definition.

#### Definition K16.15. saddle-connection systole

For a unit-area translation surface  $X$ , the *saddle-connection systole*  $\ell(X)$  is the minimum Euclidean length of a saddle connection on  $X$ . Thus  $\ell(X) > 0$  on every fixed compact translation surface, while  $\ell(X_n) \rightarrow 0$  is the basic cusp degeneration detected by Masur compactness.

#### Definition K16.16. translated circle-average operator

Write  $K = SO(2)$ ,  $g_t = \text{diag}(e^{t/2}, e^{-t/2})$ , and let  $r_\theta \in K$  be rotation through  $\theta$ . For a measurable function  $F$  on an  $SL(2, \mathbb{R})$ -space, its *translated circle average* is

$$(A_t F)(X) = \frac{1}{2\pi} \int_0^{2\pi} F(g_t r_\theta X) d\theta.$$

#### Definition K16.17. K-finite observable

For  $K = SO(2)$  acting on a function space by  $(r_\theta \cdot F)(X) = F(r_{-\theta} X)$ , an observable  $F$  is *K-finite* if the complex linear span of  $\{r_\theta \cdot F : \theta \in \mathbb{R}\}$  is finite dimensional. Equivalently, its  $K$ -orbit is a finite sum of angular characters  $e^{im\theta}$ .

## 16.2 Main results

### Theorem K16.1. Veech dichotomy

**Statement.** Let  $(X, \omega)$  be a compact translation surface whose Veech group (the derivative group of orientation-preserving affine automorphisms) is a lattice in  $\mathrm{SL}(2, \mathbb{R})$ . Then in every direction  $\theta$  the straight-line flow is either uniquely ergodic or completely periodic. If the singular set of  $(X, \omega)$  is nonempty, this sharpens to: a direction containing a saddle connection decomposes  $X$  into finitely many cylinders of closed  $\theta$ -geodesics bounded by saddle connections, whereas a direction containing no saddle connection is uniquely ergodic. In the unmarked genus-one case the same dichotomy holds, with periodic versus uniquely ergodic directions given by the usual rational/irrational torus criterion.

*Proof.* Rotate the chosen direction to the vertical. Rotation conjugates the Veech group and preserves the lattice property and both conclusions of the theorem. Write  $\Gamma = \Gamma(X, \omega)$ .

There is one edge case worth separating at once. If  $(X, \omega)$  has no singularities, then it is a flat torus  $\mathbb{R}^2/\Lambda$ . A direction rational with respect to  $\Lambda$  is periodic, while an irrational direction is uniquely ergodic by the classical Kronecker argument. Thus the dichotomy holds on an unmarked torus. We may now assume that the singular set is nonempty.

Let  $g_t = \mathrm{diag}(e^t, e^{-t})$ , with the vertical foliation contracted. The Teichmüller disk of  $(X, \omega)$  is the hyperbolic plane, and the lattice Veech group identifies its image with a finite-area hyperbolic orbifold.

A first geometric fact concerns divergent geodesics in such an orbifold. Choose disjoint standard cusp neighborhoods whose complement is compact. If a geodesic ray leaves every compact subset, it eventually remains in one cusp and is asymptotic to the ideal center of that cusp. The center is fixed by a nontrivial parabolic element of the lattice. Thus a divergent Teichmüller ray produces a parabolic affine automorphism fixing its limiting direction.

We next translate that parabolic symmetry back into flat geometry. After a linear conjugacy and passage to a positive power, the affine automorphism has derivative

$$\begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix}, \quad \alpha \neq 0,$$

and fixes every singularity and every outgoing vertical separatrix. Consequently it fixes each of those separatrices pointwise. A vertical separatrix that never reached a singularity would be dense in an open minimal component of the vertical foliation. Continuity would then force the affine automorphism to be the identity on that open set, contradicting its nontrivial shear derivative. Hence every vertical separatrix is a saddle connection.

Cut the surface along these finitely many vertical saddle connections. A nonclosed regular vertical leaf in one complementary component would again be dense in a minimal component. The affine map commutes with vertical flow, so the horizontal dis-



placement from a point to its image is constant along that dense orbit. The boundary saddle connections are fixed pointwise, however, so approaching the boundary forces the displacement to vanish. The affine map would again be the identity on an open set, a contradiction. Therefore every regular vertical leaf is closed, and the complement is a finite union of maximal Euclidean cylinders.

The same affine map also explains why the cylinder moduli are commensurable. If one cylinder has transverse width  $h_j$  and vertical circumference  $w_j$ , then the common shear acts by

$$(x, y) \mapsto (x, y + n\alpha x) \pmod{w_j}.$$

Because both boundary components are fixed, the displacement at  $x = h_j$  must be an integral number of circumferences:

$$n\alpha h_j = k_j w_j, \quad k_j \in \mathbb{Z}.$$

Thus one parabolic derivative produces compatible Dehn twists on all cylinders.

Now suppose the vertical flow is not uniquely ergodic. The Masur criterion for unique ergodicity says that a Teichmüller ray returning infinitely often to a compact subset of moduli space would force unique ergodicity. Hence our ray is divergent. Its ray in the finite-area Veech quotient is also divergent, because a compact return there would project to a compact return in moduli space. The cusp argument supplies a parabolic element, and the parabolic-cylinder argument makes the vertical direction completely periodic. This proves the dichotomy.

When singularities are present we can sharpen it. If the direction contains a saddle connection  $\sigma$ , then under  $g_t$  its length decreases like  $e^{-t}$ . A compact subset of a stratum has a positive flat-systole lower bound, so the Teichmüller ray must diverge. The preceding cusp and parabolic arguments therefore give a cylinder decomposition. Conversely, a completely periodic direction on a surface with singularities has cylinder boundaries made of saddle connections. Hence a direction with no saddle connection cannot be periodic and must be uniquely ergodic.

The alternatives are mutually exclusive: a periodic direction has many invariant measures supported on its closed leaves, whereas a uniquely ergodic direction has only normalized area. This proves the theorem.  $\square$

### Theorem K16.2. Eskin-Mirzakhani affine-measure classification

**Statement.** Let  $P < \mathrm{SL}(2, \mathbb{R})$  be the upper-triangular subgroup and let  $\mathcal{H}_1(\alpha)$  be a unit-area stratum of Abelian differentials. If  $\nu_1$  is an ergodic  $P$ -invariant probability measure on  $\mathcal{H}_1(\alpha)$ , then  $\nu_1$  is  $\mathrm{SL}(2, \mathbb{R})$ -invariant and affine. More precisely, for  $\mathcal{M}_1 = \mathrm{supp} \nu_1$ , there is a proper immersion  $f : N \rightarrow \mathcal{H}_1(\alpha)$  with image  $\mathcal{M}_1$  such that, near each point of  $N$ , the cone  $\mathbb{R}f(U) \subset \mathcal{H}(\alpha)$  is a complex linear subspace defined over  $\mathbb{R}$  in period coordinates; if  $d\nu = d\nu_1 da$  denotes the coned measure on  $\mathbb{R}\mathcal{M}_1$ , then locally  $\nu$  is, up to normalization, Lebesgue measure restricted to that subspace.

*Proof.* Write

$$G = \mathrm{SL}(2, \mathbb{R}), \quad A = \{g_t = \mathrm{diag}(e^t, e^{-t})\}, \quad N = \left\{ \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} : s \in \mathbb{R} \right\},$$



and let  $\bar{N}$  be the lower unipotent subgroup. The proof naturally breaks into two rigidity problems. First we show that an ergodic  $P = AN$ -invariant probability is actually  $G$ -invariant and identify its stable and unstable conditional measures. Only after that do we prove that its support is affine in period coordinates.

**1. Stable and unstable period plaques.** It is convenient to pass to a finite level cover of the unit-area stratum on which the relative cohomology bundle is an honest vector bundle. The final conclusion descends through this finite cover. At a translation surface  $x = (M, \omega)$ , let

$$p : H^1(M, \Sigma; \mathbb{R}) \rightarrow H^1(M; \mathbb{R})$$

be the natural projection. The relative period coordinates identify a neighborhood of  $x$  with an open set in relative cohomology. The strong unstable and strong stable directions can then be written as affine subspaces  $W^+[x]$  and  $W^-[x]$ . Under  $g_t$ , the first is expanded and the second contracted. The  $N$ -orbit lies inside  $W^+[x]$ , while the  $\bar{N}$ -orbit lies inside  $W^-[x]$ .

By a conditional measure on one of these plaques we mean the Rokhlin conditional for a measurable partition into sufficiently small plaque pieces. Different subordinate partitions give proportional measures on overlaps, so it makes sense to ask whether the conditional is Lebesgue on a linear subspace of the plaque.

Because the original measure is  $N$ -invariant, its conditional on a local  $N$ -orbit is ordinary Lebesgue measure in the horocycle parameter. The identity

$$g_t u_s g_{-t} = u_{e^{2t}s}$$

therefore gives an entropy contribution  $2t$  to the time- $t$  map. To turn this into information about stable plaques one needs an entropy-support estimate, not merely the symmetry of entropy under inversion. For a partition subordinate to stable plaques, intersect each atom with the smallest real-algebraic set supporting its conditional measure. The conditional entropy is bounded above by the sum of the logarithmic Jacobian rates along the tangent of that support. This is proved by covering the conditional atoms with Lyapunov-adapted boxes; equality is the Jensen equality case and forces affine Lebesgue conditionals. If every stable conditional were a point mass, that tangent would have dimension zero and the upper bound would be zero, contradicting the  $2t$  entropy already supplied by the horocycle. Thus the stable conditional is nontrivial. In particular, inside any compact set of measure very close to one, almost every point has a second point of the same stable plaque at a definite small distance.

**2. Additional invariance.** Suppose we already know that the unstable conditional is Lebesgue on a measurable family of affine subspaces  $U^+[x]$ . Initially these are just the  $N$ -orbits. We call such a family compatible when it is equivariant under the cocycle and local plaque holonomies and when the conditional measure on every  $U^+$ -coset is Haar.

There is a technical complication: after the first enlargement the family need not be locally constant. A nearby affine subspace must therefore be encoded by its graph over a chosen transversal. Changing the transversal is not linear in these graph coor-

dinates. The remedy is finite-dimensional: one embeds the graph and translation data into a direct sum of suitable exterior-power tensor spaces. In this enlarged parameter space, changes of transversal and the Gauss–Manin action are linear. Oseledets’ theorem then provides a finite flag of Lyapunov blocks for this linearized affine data.

Assume that the stable conditional still contains a direction not represented by the reflected  $U^+$ -family. Choose two nearby generic points on one stable plaque that are separated from that reflected family. Move them far forward so that they become exponentially close. Then translate each point inside its  $U^+$ -plaque in a synchronized way. The encoded difference of the two affine subspaces splits into finitely many Lyapunov blocks.

Choose the first time at which the first missing block has a fixed small size. If several exponents are too close to distinguish, group them together. The number of exponents is finite, so after finitely many groupings the normalized cocycle on the active block stays in a compact subset of the corresponding general linear group. At the same time, Birkhoff recurrence and a bounded Jacobian estimate for the resulting time changes allow all comparison points to be chosen inside one large compact set on which the conditionals and the Oseledets data vary uniformly continuously.

Let  $f_x$  and  $f_y$  be the quotient conditional measures at the two stopped endpoints. Equivariance of Rokhlin disintegration expresses them as push-forwards of conditionals at two earlier points. Those earlier points are arbitrarily close, and the two normalized cocycle maps are arbitrarily close because of the synchronized stopping time. Passing to a limit therefore gives an exact relation

$$f_y = (P_{x,y}^+)_* f_x,$$

where  $P_{x,y}^+$  is the unipotent holonomy produced by changing transversal. Repeating the construction at smaller and smaller scales gives nontrivial affine transformations converging to the identity and preserving the conditional measure. The identity component of the closed subgroup they generate has positive dimension. Disintegrating along its orbits shows that the conditional measure is Haar there. Thus the compatible family has been strictly enlarged.

Every strict enlargement increases the tangent dimension, so this process can occur only finitely many times. At the terminal stage the stable support is contained in the reflected unstable family.

**3. Entropy closes the induction.** Let  $I$  be the multiset of Lyapunov exponents occurring in the terminal unstable tangent directions transverse to the  $G$ -orbit, and let  $J \subset I$  be those occurring in the stable support. For any invariant absolute-cohomology subbundle, the sum of its Lyapunov exponents is nonnegative. One proves this by taking the Hodge volume of a transported frame, observing that its logarithm is subharmonic on each Teichmüller disk, and applying Green’s formula together with the multiplicative ergodic theorem.

The unstable Lebesgue conditional therefore gives the lower entropy bound

$$\frac{1}{t} h_\nu(g_t) \geq 2 + \sum_{\lambda \in I} (1 + \lambda),$$

while the fact that the reverse conditional is supported on the stable tangent gives

$$\frac{1}{t}h_\nu(g_{-t}) \leq 2 + \sum_{\lambda \in J} (1 - \lambda).$$

Because  $h_\nu(g_t) = h_\nu(g_{-t})$ , because  $J \subset I$ , and because the relevant exponent sums are nonnegative, equality is forced throughout. Thus the stable and unstable tangent spaces coincide under the natural reflection, and equality in the conditional-entropy estimate forces the stable conditional itself to be Lebesgue rather than merely supported on the correct subspace.

The original unstable family contained the  $N$ -direction, so its reflected stable family contains the  $\bar{N}$ -direction. Lebesgue conditionals on local  $\bar{N}$ -orbits imply  $\bar{N}$ -invariance. Since  $N$  and  $\bar{N}$  generate  $SL(2, \mathbb{R})$ , the measure is now fully  $G$ -invariant.

**4. Why full invariance is not yet affineness.** Let  $U(x)$  be the real relative-cohomology subspace generated by the two  $G$ -orbit directions together with the stable/unstable Lebesgue directions, and let  $U_{\mathbb{C}}(x)$  be its complexification. We already know that all conditional measures along the corresponding local directions are Lebesgue. What remains is to show that the support cannot bend transversely to  $U_{\mathbb{C}}$ .

The absolute Kontsevich–Zorich cocycle has a semisimple algebraic hull for an ergodic  $G$ -invariant measure. The source-aligned proof proceeds through invariant complements. If  $K$  is an invariant isotropic subbundle, the Kontsevich–Forni partial-sum formula and monotonicity of its nonnegative second-variation terms force all Lyapunov exponents on  $K$  to have one sign; conjugacy of the diagonal flow with its inverse then forces them all to be zero. Vanishing of the nonnegative Hodge variation makes the cocycle isometric on  $K$ , and the Hodge orthogonal  $K^\perp$  is invariant. For an arbitrary invariant subbundle  $L$ , apply this to  $K = L \cap L^\dagger$ , where  $L^\dagger$  is the symplectic orthogonal. Then  $L^\dagger \cap K^\perp$  is an invariant complement to  $L$ . Thus invariant subbundles split, which is precisely the semisimplicity needed for the random-walk argument.

Denote the maximal isometric subbundle by  $F$ , the Forni subspace. At almost every density point of the support, every sufficiently small supported period displacement has absolute projection orthogonal to  $F$ . Otherwise a synchronized return would preserve a nonzero isometric component while the complementary components expand or contract, producing an additional translation-invariant direction outside the already maximal Lebesgue subspace  $U$ .

**5. Random walk and semisimple stopped blocks.** Choose a compactly supported rotationally symmetric absolutely continuous probability on  $G$  and form the Bernoulli skew product on sequences of group elements and translation surfaces. Since our measure is already  $G$ -invariant, it is stationary for this walk.

On every strongly irreducible factor of the absolute cocycle, semisimplicity implies that each Lyapunov quotient carries a measurable conformal inner product. A useful way to see this is to look at the symmetric space of unit-determinant inner products. If one quotient had two distinct singular value growth rates, the faster flag would define

a proper invariant projective subset, contradicting strong irreducibility. Equal growth rates make the normalized orbit sublinear in the symmetric space; barycenters of Cesaro averages then yield the conformal structure. Repeating on successive quotients gives a semisimple Lyapunov flag.

For each nonzero Lyapunov block, stop the random walk the first time that a chosen vector reaches a fixed norm. Compact support of the increment law and uniform Oseledets bounds give a bounded overshoot. The distribution at such a stopping time is therefore absolutely continuous with respect to the original Bernoulli-surface measure, and on a compact set of arbitrarily large measure the Radon-Nikodym derivative and its inverse are uniformly bounded. The martingale convergence theorem for the past sigma-algebras then guarantees that, after sufficiently long stopping times, almost every admissible random continuation returns the relevant blocks to the same prescribed good set.

**6. Exponential drift rules out transverse support.** Suppose the support were not locally an open subset of  $x + U_{\mathbb{C}}(x)$ . Disintegrate first along the Lebesgue  $U_{\mathbb{C}}$ -plaques and then along a transverse slice. Non-affineness forces the transverse conditional to be non-atomic on a positive-measure set. Thus we can find arbitrarily close generic support points  $x, y$  whose difference stays a definite angle away from both  $U_{\mathbb{C}}(x)$  and the random-walk stable hyperplane. The Forni-orthogonality statement removes the isometric component of their absolute displacement.

Decompose

$$y - x = w_0 + w_1 + \cdots + w_r,$$

where  $w_0$  lies in the relative-cohomology kernel and each  $w_j$  lies in a nonzero semisimple Lyapunov block. For each  $w_j$ , run the walk backward to the first time at which its norm is a fixed small number  $\varepsilon$ . The conformal structure on the block prevents angular collapse, and the martingale/stopping-time estimate lets all stopped points be chosen in one common compact good set.

Select the first block carrying a definite fraction of the original transverse displacement. At its stopping time, faster blocks are already negligible and slower blocks are still negligible, while the selected block has size comparable to  $\varepsilon$ . Move both trajectories forward to the first time their total separation again has size comparable to  $\varepsilon$ . Stable contraction now forces the limiting displacement into the local unstable plaque  $W^+$ , but semisimple control preserves a nonzero component transverse to  $U_{\mathbb{C}}$ .

This is impossible. The first half of the proof showed that, on a full-measure set, the conditional support inside every sufficiently small unstable plaque is precisely the Lebesgue  $U^+$ -plaque, which lies inside  $U_{\mathbb{C}}$ . We have therefore produced a support point that must lie in  $U_{\mathbb{C}}$  and simultaneously stays a definite distance from it. The contradiction shows that, locally,

$$\text{supp } \nu = x + U_{\mathbb{C}}(x).$$

Overlapping positive-measure period balls force these linear tangent spaces to agree, so they are locally constant on every connected support component.

**7. Gluing and the affine measure.** Changes of period coordinates are integral real-linear maps. Hence the local complex linear pieces glue to an immersed submanifold. Distinct local sheets can meet only in lower-dimensional affine subsets, which have zero Lebesgue conditional measure. The immersion is proper: if infinitely many separated sheets crossed one compact period chart, recurrence and the local lower bound on Lebesgue conditional density would make each sheet contribute a fixed positive amount of measure, contradicting that the total measure is one.

Finally choose a real basis of one local tangent space and disintegrate successively along the associated translation lines. Every one-dimensional conditional is Haar, so Fubini gives ordinary Lebesgue measure on the whole local tangent space. Positive radial scaling makes the local affine equations homogeneous. With the area coordinate  $a = r^2$ , the coned measure  $d\nu da$  is therefore, up to normalization, Lebesgue measure on a complex linear subspace defined over  $\mathbb{R}$ . This is exactly the affine measure required in the statement.  $\square$

**Theorem K16.3. Eskin-Mirzakhani-Mohammadi orbit-closure theorem**

**Statement.** For every  $x \in \mathcal{H}_1(\alpha)$ , the orbit closures satisfy  $\overline{Px} = \overline{SL(2, \mathbb{R})x}$ , and this common orbit closure is an affine invariant submanifold of  $\mathcal{H}_1(\alpha)$ .

**Scope note**

Only the orbit-closure statement is asserted here; equidistribution and isolation are separate results.

*Proof.* Let

$$G = SL(2, \mathbb{R}), \quad a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \quad u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}.$$

Thus the upper-triangular group  $P$  is generated by  $a_t$  and  $u_s$ . Certainly  $P \subset G$ , so

$$\overline{Px} \subset \overline{Gx}.$$

The difficult direction is to show that the smaller group  $P$  already fills an affine  $G$ -invariant manifold. The proof has one long quantitative part and one short topological part.

**Step 1: a height that detects escape to the cusp.** A saddle connection is a straight geodesic segment joining zeroes of the Abelian differential and containing no zero in its interior. Its period is a vector in  $\mathbb{R}^2$ . For each possible number  $j$  of independent relative-homology classes, let  $\ell_j(y)$  be the smallest scale on which one can find  $j$  independent saddle connections on the surface  $y$ . Taking small positive exponents  $\delta_j$ , decreasing rapidly with  $j$ , one forms

$$u(y) = 2 + \sum_j \ell_j(y)^{-\delta_j}. \tag{16.3.R1}$$

After an inessential smoothing,  $u$  is continuous and rotation invariant. It is proper: bounding  $u$  bounds every saddle connection away from zero, and the compactness

criterion from the translation-surface chapter then gives a compact subset of the stratum.

The key estimate comes from the elementary formula

$$|a_t r_\theta v|^2 = |v|^2 (e^{2t} \cos^2 \theta + e^{-2t} \sin^2 \theta). \quad (16.3.R2)$$

For every  $0 < \delta < 1$ , splitting the circle into a small neighborhood of the contracting direction and its complement gives

$$\frac{1}{2\pi} \int_0^{2\pi} |a_t r_\theta v|^{-\delta} d\theta \leq C_\delta e^{-\delta t/2} |v|^{-\delta} + C_\delta. \quad (16.3.R3)$$

Apply the same estimate to exterior products of independent short saddle connections. If the minimizing family changes, either its scale changes only by a bounded factor or one more independent short class appears. Choosing the exponents in (16.3.R1) successively absorbs this latter possibility. Hence there are constants  $\eta, b > 0$  such that, for all sufficiently large  $t$ ,

$$\frac{1}{2\pi} \int_0^{2\pi} u(a_t r_\theta y) d\theta \leq e^{-\eta t} u(y) + b. \quad (16.3.R4)$$

Thus most directions send a surface of controlled height back to a controlled part of the stratum.

**Step 2: measure distance from a surface to an affine manifold.** Work temporarily on a finite marked cover so that relative cohomology is a vector bundle and period coordinates are single-valued. Near a point, an affine invariant manifold  $\mathcal{N}$  is the complexification of a real affine subspace  $L$ . Split a relative-cohomology displacement into three pieces:

$$c = h + c_0 + v,$$

where  $v$  is tangent to  $L$ ,  $c_0$  is purely relative, and the absolute class of  $h$  lies in a fixed invariant complement to the absolute tangent space. The affine-measure theorem proved just above supplies the invariant splitting needed for this decomposition.

Give the absolute part the Hodge norm and the pure-relative part its ordinary finite-dimensional norm. Near  $L$  define the modified transverse distance by

$$d_0(y, L) = \max\{\|c_0\|, \|p(h)\|_H^{1/2}\}, \quad (16.3.R5)$$

cut off at 1. The square root is important: it keeps the error caused by mixing absolute and purely relative cohomology of lower order.

The Hodge norm satisfies a strict first-variation inequality away from the tautological two-plane. More explicitly, if  $c$  is transverse to that plane, then

$$\frac{d}{dt} \log \|c\|_{H, a_t y} \leq \Lambda(a_t y), \quad 0 < \Lambda < 1. \quad (16.3.R6)$$

On a compact set,  $\Lambda$  is uniformly smaller than 1. Combining this with the recurrence estimate (16.3.R4), and treating the pure-relative term with (16.3.R3), gives a small

exponent  $\delta > 0$  for which

$$\frac{1}{2\pi} \int_0^{2\pi} d_0(a_t r_\theta y, a_t r_\theta L)^{-\delta} d\theta \leq \kappa(y, t) d_0(y, L)^{-\delta} + B(t). \quad (16.3.R7)$$

When  $u(y)$  is not exponentially large compared with  $t$ , the coefficient  $\kappa(y, t)$  decays exponentially. This is the transverse contraction that replaces the informal idea that “different affine planes cannot accumulate too much.”

**Step 3: only finitely many local lifts can be dangerously close.** Choose a large compact recurrence set  $K_0$ . The complexity of an affine manifold  $\mathcal{N}$  is the least number of marked affine period-coordinate pieces needed to cover  $\mathcal{N} \cap K_0$ ; it is finite because the immersion defining  $\mathcal{N}$  is proper.

A surface  $y$  can be moved into  $K_0$  in time  $O(\log u(y))$ . Under this motion, the distance (16.3.R5) changes by at most an exponential factor. If we declare a lift  $L$  to be dangerously close when

$$d_0(y, L) \leq u(y)^{-k}$$

with  $k$  sufficiently large, all such lifts move into one embedded period ball in  $K_0$ . Distinct lifts remain distinct there. Consequently their number is at most the complexity of  $\mathcal{N}$ .

**Step 4: construct the affine-avoidance height.** Let  $J_{\mathcal{N}}(y)$  be the finite set of dangerously close affine lifts. For small  $\varepsilon > 0$  define

$$s_{\mathcal{N}, \varepsilon}(y) = \sum_{L \in J_{\mathcal{N}}(y)} d_0(y, L)^{-\varepsilon \delta}$$

and then

$$f_{\mathcal{N}}(y) = s_{\mathcal{N}, \varepsilon}(y) u(y)^{1/2} + \lambda u(y), \quad (16.3.R8)$$

where  $\lambda$  is large. If  $\mathcal{N}$  is empty, only the last term is needed.

The function  $f_{\mathcal{N}}$  is infinite exactly on  $\mathcal{N}$ , while its finite sublevel sets are compact away from  $\mathcal{N}$ . More importantly, for any prescribed  $0 < c < 1$ , the parameters can be chosen so that

$$\frac{1}{2\pi} \int_0^{2\pi} f_{\mathcal{N}}(a_t r_\theta y) d\theta \leq c f_{\mathcal{N}}(y) + b \quad (16.3.R9)$$

for every sufficiently large  $t$ .

Here is the estimate behind (16.3.R9). A nearby affine lift may remain nearby, leave the nearby list, or enter it. For the first class, Cauchy-Schwarz combines the transverse contraction (16.3.R7) with the cusp contraction (16.3.R4). A lift entering or leaving the list has distance at least  $u^{-k}$  at one endpoint, so its contribution is bounded by a fixed multiple of  $u$ . There are only finitely many such lifts. The large term  $\lambda u$  absorbs those errors. Choosing  $\varepsilon$  small makes the remaining transverse coefficient contract. This proves (16.3.R9).

Thus  $f_{\mathcal{N}}$  simultaneously penalizes two bad behaviors: going far into the cusp and approaching the proper affine manifold  $\mathcal{N}$ .

**Step 5: the affine manifolds form a countable family.** The same height gives

the isolation fact needed below. If  $\nu$  is the affine probability on some affine invariant manifold not contained in  $\mathcal{N}$ , integrating (16.3.R9) and using ergodicity of the diagonal flow gives

$$\int f_{\mathcal{N}} d\nu \leq B, \quad (16.3.R10)$$

where  $B$  depends only on the complexity bound. The diagonal-flow ergodicity needed here follows from the elementary Mautner argument: an invariant function for  $a_t$  is invariant under both horocycle groups because  $a_{-t}u_s a_t \rightarrow e$  in one time direction; the two horocycle groups generate  $SL(2, \mathbb{R})$ .

Now fix a dimension and a complexity bound. If infinitely many distinct affine manifolds occurred, compactness of the finite set of local affine planes over  $K_0$  would give two successive manifolds arbitrarily close on  $K_0$ . The height of one relative to the other would then tend uniformly to infinity on at least half the affine measure, contradicting (16.3.R10). Hence only finitely many occur at each bounded complexity, and therefore only countably many affine invariant manifolds occur in a stratum.

**Step 6: the rectangular  $P$ -averages are Følner.** For fixed  $r > 0$  define

$$\lambda_{T,r} * \delta_x = \frac{1}{Tr} \int_0^T \int_0^r \delta_{a_t u_s x} ds dt. \quad (16.3.R11)$$

Left multiplication by  $a_\tau$  merely translates the  $t$ -interval, producing relative boundary  $O(T^{-1})$ . Also

$$u_v a_t u_s = a_t u_{s+ve^{-2t}},$$

so left multiplication by  $u_v$  changes the horizontal interval by only  $O(e^{-2t})$  at time  $t$ ; after averaging in  $t$  the relative error is again  $O(T^{-1})$ . Therefore every probability weak limit of (16.3.R11) is  $P$ -invariant.

The avoidance height makes these weak limits honest probabilities and keeps them away from every chosen proper affine manifold. The uniformity of the threshold is important. If  $F$  is compact and disjoint from  $\mathcal{N}$ , let  $m_F = \sup_F f_{\mathcal{N}}$ . In (16.3.R9) choose the contraction coefficient smaller than  $(2m_F)^{-1}$ . After a time depending on  $F$ , the angular mean of  $f_{\mathcal{N}}$  is at most  $1 + b$ , where  $b$  is the additive drift constant and does not depend on  $F$ ; the bounded initial time interval contributes vanishingly to the long average.

The small horocycle interval in (16.3.R11) is compared with a fixed small angular interval using

$$r_\theta = g_\theta u_{\tan \theta}.$$

The conjugated correction remains in one fixed compact subset of  $SL(2, \mathbb{R})$ . A finite chain of the local multiplicative estimate for  $f_{\mathcal{N}}$  therefore transfers the angular bound to

$$(\lambda_{T,r} * \delta_y)(f_{\mathcal{N}}) \leq C_r(1 + b), \quad T \geq T_0(F), \quad y \in F, \quad (16.3.R12)$$

where the right-hand side is independent of  $F$ . For general fixed  $r$ , partition the horocycle interval into finitely many short pieces and translate the compact starting set by the finitely many corresponding horocycle elements. Thus the same threshold works after increasing  $T_0(F)$ .



Markov's inequality now gives an open height-neighborhood

$$\Omega_{\mathcal{N},\varepsilon} = \{f_{\mathcal{N}} > C_r(1+b)/\varepsilon\}^\circ$$

that is independent of the starting compact set, has compact complement, and receives less than  $\varepsilon$  mass from all sufficiently long averages. Taking  $\mathcal{N} = \emptyset$  gives tightness, hence no escape of mass. **Step 7: equidistribution in the least-dimensional**

**affine manifold.** Choose an affine invariant manifold  $\mathcal{M}$  of least dimension containing  $x$ . Such a manifold exists because the whole stratum is one. Take any subsequence of the measures (16.3.R11). Tightness gives a probability weak limit  $\nu$ , and the Følner calculation makes  $\nu$   $P$ -invariant. By the affine-measure classification, every ergodic component of  $\nu$  is the affine measure of an affine invariant submanifold of  $\mathcal{M}$ .

Because those submanifolds are countable, we may write

$$\nu = \sum_{\mathcal{N} \subseteq \mathcal{M}} a_{\mathcal{N}} \nu_{\mathcal{N}}. \quad (16.3.R13)$$

If  $\mathcal{N}$  is proper in  $\mathcal{M}$ , then  $x \notin \mathcal{N}$  by minimality of the dimension of  $\mathcal{M}$ . Apply the avoidance estimate (16.3.R12) with the starting compact set  $\{x\}$ . Shrinking the height-neighborhood of  $\mathcal{N}$  shows that  $\nu(\mathcal{N}) = 0$ . Therefore every coefficient in (16.3.R13) attached to a proper affine submanifold vanishes. The only possible limit is  $\nu_{\mathcal{M}}$ .

Since every subsequence has the same limit, the whole family converges:

$$\lambda_{T,r} * \delta_x \longrightarrow \nu_{\mathcal{M}}. \quad (16.3.R14)$$

This also proves that the least-dimensional affine manifold containing  $x$  is unique: two choices would give the same limiting probability and hence the same support.

**Step 8: extract the orbit closure from the support.** Every measure in (16.3.R11) is supported on  $Px$ , so its limit is supported on the closed set  $\overline{Px}$ . The affine measure  $\nu_{\mathcal{M}}$  has full support  $\mathcal{M}$ : locally it is Lebesgue measure on the affine period-coordinate subspace. Hence

$$\mathcal{M} \subset \overline{Px}.$$

The opposite inclusions are immediate:

$$\overline{Px} \subset \overline{Gx} \subset \mathcal{M},$$

because  $P \subset G$  and  $\mathcal{M}$  is closed and  $G$ -invariant. Therefore

$$\boxed{\overline{Px} = \overline{Gx} = \mathcal{M}}.$$

By definition,  $\mathcal{M}$  is an affine invariant submanifold. This proves the theorem.  $\square$

#### Theorem K16.4. Wright cylinder and field-of-definition theorems

**Statement.** Let  $(X, \omega)$  be a translation surface and let  $\mathcal{M} = \overline{GL^+(2, \mathbb{R}) \cdot (X, \omega)}$  be its affine invariant orbit closure. (i) If  $\mathcal{C}$  is the collection of all horizontal cylinders on  $(X, \omega)$ , then for every  $s, t \in \mathbb{R}$  the standard cylinder deformation  $a_s^{\mathcal{C}} u_t^{\mathcal{C}}(X, \omega)$  of

Definition K16.6 belongs to  $\mathcal{M}$ . (ii) For an affine invariant submanifold  $\mathcal{M}$ , let  $\mathbf{k}(\mathcal{M})$  be the smallest subfield of  $\mathbb{R}$  over which its local homogeneous period-coordinate equations can be written. Then  $\mathbf{k}(\mathcal{M})$  is the intersection of the holonomy fields of Definition K16.5 for all translation surfaces in  $\mathcal{M}$ ; moreover  $\mathbf{k}(\mathcal{M})$  is a real number field of degree at most the genus.

*Proof.* Let

$$\mathcal{M} = \overline{GL^+(2, \mathbb{R})(X, \omega)}.$$

The orbit-closure theorem proved earlier makes  $\mathcal{M}$  an affine invariant manifold. This is crucial, but it does not by itself prove either of Wright's conclusions. The cylinder statement is a geometric theorem about flat cylinders; the field statement is an arithmetic theorem about the Gauss–Manin monodromy of the tangent bundle. We treat them separately.

**1. What a cylinder shear does to periods.** Suppose  $C_1, \dots, C_r$  are horizontal cylinders, with heights  $h_i$  and core curves  $\alpha_i$ . If  $\gamma$  crosses  $C_i$  once from its lower to its upper boundary, shearing that cylinder by time  $t$  changes the real part of  $\int_\gamma \omega$  by  $th_i$ ; periods of core curves and of cycles disjoint from the cylinders do not change. Hence the infinitesimal shear is

$$\eta_{\mathcal{C}} = \sum_i h_i I(\alpha_i),$$

where  $I(\alpha_i)$  denotes intersection with the core curve. In a period chart,

$$\Phi(u_t^{\mathcal{C}} X) = \Phi(X) + t\eta_{\mathcal{C}}, \quad \Phi(a_s^{\mathcal{C}} X) = \Phi(X) + (e^s - 1)i\eta_{\mathcal{C}}.$$

Thus the entire cylinder theorem reduces to proving that the shear vector lies in the tangent space of  $\mathcal{M}$ .

**2. Partial shears on a horizontally periodic surface.** Assume temporarily that the whole surface is decomposed into horizontal cylinders, of moduli  $m_i = h_i/c_i$ . A shear by  $1/m_i = c_i/h_i$  is a full Dehn twist of the  $i$ -th cylinder. Modulo these twists, the global horocycle flow is therefore the linear flow

$$t \mapsto t(m_1, \dots, m_n) \pmod{\mathbb{Z}^n}$$

on a torus. The closure of this flow is determined by the rational linear relations among the moduli. In particular, if the moduli of one group of cylinders are rationally independent from those of the remaining cylinders, we can approximate a shear of that first group alone by global horocycle shears. Such a partial shear consequently lies in the orbit closure.

**3. Why parallel cylinder classes can be separated.** Two cylinders are called  $\mathcal{M}$ -parallel when they remain parallel on all sufficiently small deformations inside  $\mathcal{M}$ . Equivalently, their core curves define proportional linear functionals on the tangent space. Now make a *real deformation*: change only the real parts of period coordinates. Horizontal cylinder boundaries remain horizontal and cylinder heights remain fixed.

Only their circumferences change, and the modulus has the form

$$m_i(v) = \frac{h_i}{\alpha_i(v)}.$$

If an  $\mathcal{M}$ -parallel class could never be made rationally independent from the other cylinders, there would be a rational relation among these reciprocal linear forms on an open set. Choose one with minimal support. Reciprocals of pairwise nonproportional linear forms are linearly independent: restricting near a point of one kernel but none of the others exposes a pole. So two core-curve functionals in the minimal relation must be proportional, one inside the class and one outside it. This contradicts maximality of the parallel class. Therefore each parallel class has a tangent shear vector.

#### 4. A horocycle orbit always accumulates on a horizontally periodic surface.

This is the Minsky-Weiss/Smillie-Weiss mechanism used by Wright. The first input is quantitative nondivergence. A saddle connection with holonomy  $(x, y)$  changes under the horocycle according to

$$(x, y) \mapsto (x + sy, y).$$

Thus, provided each saddle connection is reasonably long somewhere on a fixed  $s$ -interval, the times at which it is very short occupy only a small subinterval. A finite-complexity triangulation argument makes this estimate simultaneous over all saddle connections.

There is an important extra step before one may invoke Zorn's lemma. In an arbitrary closed horocycle-invariant set, horizontal saddle connections have constant length along each orbit, but their lengths need not have a uniform positive lower bound across the whole set. For a surface  $q$ , form the horizontal critical graph  $\Xi_q$ : the union of horizontal singular leaves. Its regular neighborhood determines the complementary topology of the associated horizontal geodesic lamination. Measure the complexity by the largest number of times that lamination can be strictly enlarged. This is a bounded integer depending only on the topology.

Choose an orbit whose critical-graph complexity is maximal and take its closure  $X_0$ . Along a horocycle orbit the critical graph is unchanged isometrically. At a limit point the comparison maps give a local isometric map of critical graphs; if it failed to be a global isometry, the associated lamination could be strictly enlarged, increasing the complexity. Maximality rules this out. Hence every surface of  $X_0$  has the same horizontal critical graph and the same lengths of its horizontal saddle connections. There is therefore one positive lower bound for every saddle connection whose length is constant under the horocycle.

Now quantitative nondivergence applies on arbitrarily long positive and negative intervals. It gives a single thick compact set met infinitely often in both time directions by every orbit in  $X_0$ . The translation-surface compactness criterion makes this thick set compact. A chain of closed invariant subsets of  $X_0$  has the finite-intersection property inside that compact set, so Zorn's lemma gives a minimal invariant subset. Its compactness is a short topological argument: choose a relatively compact neighborhood of the common return set. If orbit excursions outside that neighborhood had unbounded length, compactness of its boundary would produce a limiting boundary

point whose entire positive semi-orbit never returned, a contradiction. Thus all excursions have uniformly bounded length, and the whole minimal set is the image of a compact boundary over a bounded time interval, together with the compact neighborhood itself.

Finally take a point in this compact minimal set. If its horizontal foliation were not completely periodic, a horizontal ray from a singularity would run forever in a minimal component and accumulate near another singularity. Following the ray for a long time and then joining transversely to that nearby singularity gives a broken path whose vertical displacement is arbitrarily small. It is not legitimate simply to call this path a saddle connection. Instead rotate the initial direction toward the direction of the broken path and take the *first* angle at which a geodesic from the original singularity hits a singularity within the same length bound. This produces a genuine saddle connection with holonomy  $(x', y')$  and arbitrarily small nonzero  $|y'|$ . The horocycle time  $s = -x'/y'$  makes it vertical of length  $|y'|$ , contradicting the positive systole of the compact minimal set. Therefore the minimal set is horizontally completely periodic, and every horocycle orbit closure contains a horizontally periodic surface.

**5. Finish the cylinder theorem.** Let  $Y$  be a horizontally periodic accumulation point of the horocycle orbit of  $X$ . Every horizontal cylinder already present on  $X$  survives to  $Y$ , with the same height and circumference. The limit surface may have extra horizontal cylinders. Near  $Y$ , however, returning points from the horocycle orbit keep the inherited cylinders horizontal while generically tilting the extra ones. Therefore no inherited cylinder is  $\mathcal{M}$ -parallel to an extra cylinder. The inherited family is a union of parallel classes, and Step 3 shows that the sum of their shear vectors is tangent to  $\mathcal{M}$  at  $Y$ . Flat parallel transport in period coordinates carries the same vector back to  $X$ .

The tangent bundle of an affine invariant manifold is complex, so the stretch vector  $i\eta_{\mathcal{C}}$  is tangent as well. The two deformations are straight lines in period coordinates. Continuing them from chart to chart proves

$$a_s^{\mathcal{C}} u_t^{\mathcal{C}}(X, \omega) \in \mathcal{M} \quad \text{for every } s, t \in \mathbb{R}.$$

This proves the first clause.

**6. Closed Teichmüller orbits detect the tangent monodromy.** For the arithmetic clause, let

$$p : H^1(X, \Sigma; \mathbb{C}) \rightarrow H^1(X; \mathbb{C}), \quad V = p(TM).$$

These are flat bundles for Gauss–Manin parallel transport. We first need many closed Teichmüller geodesics. On a compact part of  $\mathcal{M}$ , the stable coordinates contract forward and the unstable coordinates contract backward. A long orbit segment beginning and ending in the same small product box and spending more than half its time in a fixed compact set therefore defines a contraction on the stable–unstable product box. Its fixed point is a nearby closed Teichmüller orbit. Applying Birkhoff to the affine probability measure supplies such recurrent segments in every open set. Thus closed Teichmüller orbits are dense.

On a closed orbit of period  $T$ , the return map is pseudo-Anosov. Its action on cohomology has the simple extremal eigenvalues  $e^T, e^{-T}$ , with  $\Re \omega, \Im \omega$  as eigenvectors. This simplicity is the lever for the field argument. It also supplies the first arithmetic in-

put. The monodromy matrices are integral, so after scaling the corresponding simple eigenvectors have coordinates in a number field. Density lets us choose finitely many closed-orbit period vectors spanning the local affine tangent space. Their finite compositum is a number field over which that tangent space is defined. Thus the minimal field of definition is already a finite extension of  $\mathbb{Q}$  before the Galois dimension count begins.

**7. Simplicity of the absolute tangent bundle.** The absolute Kontsevich-Zorich cocycle is semisimple by the semisimplicity mechanism already established in the affine-measure proof. Suppose  $V = E_1 \oplus E_2$  were a nontrivial flat splitting. Choose a nearby point whose tautological vector has nonzero components in both summands, then choose such a point on a closed Teichmüller orbit. Its real part would split into two independent eigenvectors for the same simple extremal eigenvalue. Impossible. Hence  $V$  is simple. The same argument shows that  $T\mathcal{M}$  has no nontrivial flat direct-sum decomposition; any proper flat subbundle of  $T\mathcal{M}$  must lie in the relative kernel of  $p$ .

**8. The trace field is the field of definition.** Let  $k$  be the field generated by traces of the monodromy representation on  $V$ . Certainly  $k$  is contained in the coefficient field defining  $\mathcal{M}$ . To prove the reverse inclusion, choose a closed-orbit monodromy matrix  $A$  and its simple extremal eigenvalue  $\lambda$ . Every field automorphism fixing  $k$  preserves the character of the simple representation  $V$ , so it sends  $\lambda$  to another simple eigenvalue whose eigenline still belongs to  $V$ .

Let  $g(z)$  be the product of  $z - \sigma(\lambda)$  over the conjugates fixing  $k$ . The roots are simple, so  $g$  is coprime to the complementary factor of the characteristic polynomial. Bezout's identity therefore expresses the projection onto the corresponding sum of eigenspaces as a polynomial in the integral monodromy matrix with coefficients in  $k$ . This eigenspace contains a non-relative vector with coordinates in  $k$ . Its monodromy orbit spans all of  $T\mathcal{M}$ , because any smaller flat span would be a proper flat subbundle not contained in the relative kernel. Thus  $T\mathcal{M}$  is defined over  $k$ . The trace field and the field of definition coincide.

**9. Galois conjugates give the degree bound.** For each embedding  $\rho : k \hookrightarrow \mathbb{C}$ , Galois conjugation produces a simple flat bundle  $V_\rho \subset H^1$ . Distinct embeddings give different characters, hence nonisomorphic bundles. Integral monodromy makes their multiplicities equal, while the simple extremal eigenvalue on a closed orbit shows that the multiplicity of  $V$  itself is one. Therefore

$$\dim_{\mathbb{C}} V[k : \mathbb{Q}] \leq 2g.$$

Since  $V$  contains the two-dimensional tautological plane,  $[k : \mathbb{Q}] \leq g$ . The previous step already established finiteness; this representation-theoretic count supplies the genus bound. As the field of definition lies in  $\mathbb{R}$ , it is a real number field.

**10. Compare with holonomy fields.** If  $\mathcal{M}$  is defined over  $k$ , its homogeneous period equations can be solved over  $k[i]$ , so  $\mathcal{M}$  contains surfaces whose period coordinates lie in  $k[i]$ . Their holonomy fields are contained in  $k$ , which shows that the intersection of all holonomy fields is contained in  $k$ .

Conversely, let a surface  $Y \in \mathcal{M}$  have holonomy field  $k_Y$ . After a linear change of coordinates and scaling, all absolute periods lie in  $k_Y[i]$ . The corresponding tautological vector in  $V$  therefore has coordinates in  $k_Y[i]$ . Simplicity says that its integral-monodromy orbit spans  $V$ , so  $V$  is defined over  $k_Y$ . Since the field of definition of  $V$  is the same as that of  $\mathcal{M}$ , we obtain  $k \subset k_Y$ . This holds for every  $Y$ , and hence

$$k = \bigcap_{Y \in \mathcal{M}} k_{\text{hol}}(Y).$$

Together with the degree estimate, this proves the second clause.  $\square$

### Theorem K16.5. Siegel-Veech formula and quadratic counting

**Statement.** Fix an ergodic  $SL(2, \mathbb{R})$ -invariant probability measure  $\mu$  and a specified saddle-connection or cylinder configuration  $\mathcal{C}$ . If  $V_{\mathcal{C}}(X) \subset \mathbb{R}^2$  is the multiset of holonomy vectors of that configuration, define its Siegel-Veech transform by  $\hat{f}_{\mathcal{C}}(X) = \sum_{v \in V_{\mathcal{C}}(X)} f(v)$  and assume this transform is  $\mu$ -integrable for compactly supported continuous  $f$ . Then there is a constant  $c_{\mu, \mathcal{C}}$  such that  $\int \hat{f}_{\mathcal{C}} d\mu = c_{\mu, \mathcal{C}} \int_{\mathbb{R}^2} f(u) du$ . For Masur-Veech measure on a connected component of a unit-area stratum, in the Eskin-Masur counting scope for  $\mathcal{C}$ , one has  $N_{\mathcal{C}}(X, R)/(\pi R^2) \rightarrow c_{\mu, \mathcal{C}}$  for  $\mu$ -almost every surface  $X$ .

*Proof.* Write  $G = SL(2, \mathbb{R})$  and  $K = SO(2)$ . The proof has two logically separate halves. The first is an exact identity for the average of a Siegel-Veech transform. The second turns that averaged identity into an almost-everywhere asymptotic on an individual surface. Keeping these two halves separate is important: the first uses only  $G$ -invariance and integrability, whereas the second uses ergodicity and quantitative control of the cusp.

#### Step 1: local finiteness of the holonomy multiset and the cusp exponent.

Let  $\ell(X)$  be the saddle-connection systole from the saddle-connection systole defined above. Fix a compact set  $B \subset \mathbb{R}^2$  and choose  $\alpha$  with

$$1 < \alpha < 2.$$

We prove the estimate in the form needed here. Take a saddle-connection triangulation whose first edge has length  $\ell = \ell(X)$ . Cutting along a maximal collection of disjoint saddle connections of length  $O_B(1)$  gives only a topology-dependent number of Euclidean triangles. Any further saddle connection with holonomy in  $B$  crosses a bounded combinatorial sequence of these triangles. In each triangle its transverse displacement changes in multiples of a strip width comparable with a previously chosen edge; the number of admissible crossing integers is therefore bounded by a product of negative powers of the short edge lengths. The elementary induction used by Eskin-Masur chooses at each stage an edge whose length is the smallest still available and trades one full inverse power for an arbitrarily small loss. Consequently, for every  $\alpha > 1$ , the total number of holonomies in  $B$  is  $O(\ell^{-\alpha})$ . A cylinder in the stated configuration scope has a boundary made of saddle connections of comparable bounded length, so the same bound controls cylinder configurations, with only a topology-dependent

multiplicity. This proves the compact-window estimate.

Choose finitely many period-coordinate charts covering the unit-area thick part at a fixed scale. In a chart a saddle connection is represented by a nonzero integral relative class  $\gamma$ , and its holonomy is the complex linear functional  $z_\gamma$ . The locus  $|z_\gamma| < \rho$  has real codimension two and hence Lebesgue volume  $O(\rho^2)$  after restricting to the area-one hypersurface. Only finitely many classes can become shortest inside one such chart before leaving its controlled coordinate neighborhood; a standard first-short-edge/Delaunay decomposition reduces the thin part to finitely many such chart types. Summing gives the stated  $O(\rho^2)$  bound. Finally, by the layer-cake formula,

$$\int_{\{\ell < \rho_0\}} \ell^{-\beta} d\mu = \rho_0^{-\beta} \mu\{\ell < \rho_0\} + \beta \int_{\rho_0^{-1}}^{\infty} t^{\beta-1} \mu\{\ell < t^{-1}\} dt,$$

and the last integral is bounded by a constant times  $\int^{\infty} t^{\beta-3} dt$ , which converges precisely when  $\beta < 2$ . This proves the quadratic cusp-volume estimate. Masur compactness (the corresponding earlier theorem) supplies the properness interpretation: bounded  $\ell^{-1}$  means remaining in a compact subset of the stratum component.

If  $\text{supp } f \subset B$ , the compact-window estimate gives

$$|\widehat{f_C}(X)| \leq \|f\|_{\infty} C_B \ell(X)^{-\alpha}.$$

Choose the same  $\alpha \in (1, 2)$  as above and apply the quadratic cusp-volume estimate. This proves integrability. If  $|f_j| \leq M \mathbf{1}_B$ , the same integrable majorant is independent of  $j$ , so dominated convergence gives uniform integrability. This supplies the precise cusp condition needed in the argument.

**Step 2: the exact Siegel-Veech identity.** For  $f \in C_c(\mathbb{R}^2)$  define

$$L(f) = \int_{\mathcal{H}} \widehat{f_C}(X) d\mu(X).$$

By the hypothesis in the general invariant-measure part of the theorem (and by the transform-integrability estimate for Masur-Veech measure),  $L$  is a positive locally bounded linear functional. Riesz-Markov therefore gives a Radon measure  $\nu$  on  $\mathbb{R}^2 \setminus \{0\}$  such that  $L(f) = \int f d\nu$ . Equivariance of holonomy gives

$$V_C(gX) = gV_C(X), \quad \widehat{f_C}(gX) = \widehat{(f \circ g)_C}(X),$$

so  $G$ -invariance of  $\mu$  implies  $g_*\nu = \nu$ .

The action of  $G$  on  $\mathbb{R}^2 \setminus \{0\}$  is transitive. The stabilizer of  $e_1$  is  $N = \{(\begin{smallmatrix} 1 & s \\ 0 & 1 \end{smallmatrix}) : s \in \mathbb{R}\}$ , which is unimodular. Hence  $G/N$  carries, up to a scalar, a unique  $G$ -invariant Radon measure. Under  $gN \mapsto ge_1$  this is ordinary area, because every  $g \in G$  has determinant one. Thus  $\nu = c_{\mu,C} du$ , and therefore

$$\int \widehat{f_C} d\mu = c_{\mu,C} \int_{\mathbb{R}^2} f(u) du.$$

This proves the first assertion and the invariant-measure argument.

**Step 3: the pointwise circle-average gate.** For the counting assertion,  $\mu$  is the normalized Masur–Veech measure on one connected component, so the corresponding earlier theorem gives ergodicity. Put

$$g_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}, \quad r_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

and let  $A_t$  be the circle-average operator of the translated circle-average operator defined above.

Here is the required proof, specialized to  $G = SL(2, \mathbb{R})$  and written in the form needed for the Siegel–Veech transform. This is the semi-radial pointwise theorem rather than an application of the one-parameter Birkhoff theorem.

Write  $m_K$  for Haar probability on  $K$ . For  $n \in \mathbb{Z}$  let  $\chi_n$  be the complex measure  $e^{in\theta} dm_K(r_\theta)$  and put

$$\sigma_t(n, m) = \chi_n * \delta_{g_t} * \chi_m, \quad \gamma_t^\eta(n, m) = \int_{\mathbb{R}} \eta(t-s) \sigma_s(n, m) ds.$$

The projection  $\pi(\chi_n)$  is the orthogonal projection onto the  $n$ th  $K$ -type. A  $K$ -finite vector is therefore a finite sum of such projections. We need three ingredients.

(a) *Strong maximal inequality.* For every  $p > 1$ ,

$$\left\| \sup_t |\pi(\gamma_t^\eta(n, m))H| \right\|_p \leq C_{p,\eta} \|H\|_p, \quad (2)$$

uniformly in  $m, n$ . Indeed, since  $|e^{in\theta}| = 1$ , positivity gives pointwise

$$|\pi(\sigma_s(n, m))H| \leq \pi(m_K * \delta_{g_s} * m_K)|H|.$$

After convolution in  $s$  with the fixed nonnegative bump  $\eta$ , the right side is a positive fixed-width radial-annulus average. In polar coordinates on the hyperbolic plane  $G/K$ , Haar volume is  $c \sinh(2s) ds dm_K$ . On every interval of the fixed length  $2 \operatorname{diam}(\operatorname{supp} \eta)$  centered at large  $|t|$ ,  $\sinh(2s)$  is comparable to  $\sinh(2|t|)$ ; hence the normalized annulus average is bounded by a constant times the centered hyperbolic Hardy–Littlewood ball maximal operator. The local range  $|t| \leq t_0$  is controlled by the same local maximal operator. The standard covering proof of the Hardy–Littlewood theorem on the doubling-at-unit-scale hyperbolic metric, followed by Calderón transference from left translation on  $G$  to the given measure-preserving action, gives the  $L^p$  bound for the positive radial averages. The displayed domination then proves (2).

(b) *Spectral decay on a dense subspace.* Let  $\pi$  be the Koopman representation on  $L^2(X)$  and remove the invariant subspace. In an irreducible nontrivial representation  $\tau$ , choose unit vectors  $v_j$  in the one-dimensional  $K$ -types and write

$$\Phi_{j,k}^\tau(t) = \langle \tau(g_t)v_j, v_k \rangle.$$

The principal and complementary series models on the circle give, by direct differentiation of the integral formula for these matrix coefficients, constants  $B$  and parameters



$\varepsilon_\tau > 0$ ,  $C(\tau) < \infty$  such that

$$|\Phi_{0,n}^\tau(t)| \leq B(1 + |t|)e^{-\varepsilon_\tau|t|}, \quad |(\Phi_{0,n}^\tau)'(t)| \leq C(\tau)(1 + |t|)e^{-\varepsilon_\tau|t|}. \quad (3)$$

For the principal/discrete series one may take a fixed positive decay exponent; for complementary series the exponent is  $1 - s$  and tends to zero only when the representation approaches the trivial representation. These estimates follow directly from the circle-model integrand

$$(e^{-2t} \cos^2 \theta + e^{2t} \sin^2 \theta)^{(-1+\Re \alpha)/2} e^{-in\theta} :$$

splitting at  $|\sin \theta| = e^{-2|t|}$  gives the first estimate, while one differentiation introduces the bounded quotient

$$\frac{e^{2t} \sin^2 \theta - e^{-2t} \cos^2 \theta}{e^{2t} \sin^2 \theta + e^{-2t} \cos^2 \theta}$$

and at most the factor  $1 + |\alpha|$ , giving the second.

For  $\varepsilon > 0$ , let  $H_\varepsilon$  be the spectral subspace on which  $\varepsilon_\tau \geq \varepsilon$  and  $C(\tau) \leq \varepsilon^{-1}$ . The union of these subspaces is dense in the orthogonal complement of the invariants. Smooth it further by  $C_c^\infty(G)$  convolution and call the resulting dense subspace  $D(H_\varepsilon)$ . Plancherel for the direct integral decomposition and (3) give, for  $h \in D(H_\varepsilon)$ ,

$$\|\pi(\sigma_t(0, n))h\|_2 + \left\| \frac{d}{dt} \pi(\sigma_t(0, n))h \right\|_2 \leq C_{\varepsilon, h}(1 + |t|)e^{-\varepsilon|t|}. \quad (4)$$

Fubini therefore implies that for almost every  $x$  both functions of  $t$  in (4) belong to  $L^2([1, \infty))$ . The orbit map is absolutely continuous for such  $x$  (first for the smoothed vector, by differentiating the compactly supported convolution kernel, and then outside one null set). Thus  $u_x(t) = \pi(\sigma_t(0, n))h(x)$  belongs to  $W^{1,2}([1, \infty)) \cap L^2([1, \infty))$ . The elementary Sobolev identity

$$|u_x(T)|^2 \leq 2 \int_T^\infty |u_x(s)u'_x(s)| ds \leq 2\|u_x\|_{L^2(T, \infty)}\|u'_x\|_{L^2(T, \infty)}$$

shows that  $u_x(T) \rightarrow 0$ . Hence the unmollified character-spherical operator converges pointwise to zero on a dense subspace of  $L_0^2$ ; convolution with  $\eta$  preserves the limit.

(c) *Banach principle and the finite  $K$ -type sum.* The maximal inequality (2) upgrades convergence from the dense subspace to all of  $L_0^2$ . Since bounded functions lie in  $L^2$  and are dense in  $L^p$  on a probability space, the same maximal inequality upgrades the conclusion to every  $L^p$ ,  $p > 1$ . On constants the  $(0, 0)$  operator is the identity and every nonzero  $K$ -type has zero constant projection. Finally, if  $F$  is  $K$ -finite then

$$F = \sum_{n \in E_F} \pi(\chi_n)F$$

for a finite set  $E_F$ . Expanding the translated circle average term by term turns each summand into one of the operators  $\gamma_t^\eta(0, n)$  (with  $t$  possibly replaced by  $-t$ , depending

on the left/right action convention). The preceding convergence therefore gives

$$\int \eta(t-s) A_s F(X) ds \longrightarrow \int F d\mu$$

for almost every  $X$ . This proves the pointwise circle-average lemma. The role of  $p > 1$  is now explicit: it is exactly the range in which the strong radial maximal inequality closes the Banach-principle step.

**Step 4: continuous compactly supported test functions.** Let  $\varphi \in C_c(\mathbb{R}^2)$ . On a closed disk containing its support, Stone-Weierstrass approximates  $\varphi$  uniformly by finite sums of functions  $r^q e^{im\theta}$  after multiplying by a fixed radial cutoff. Their Siegel-Veech transforms are  $K$ -finite. the compact-window estimate and the quadratic cusp-volume estimate allow us to choose  $\alpha < 2$  and then  $\kappa > 0$  so small that  $\alpha(1+\kappa) < 2$ ; hence these transforms lie in  $L^{1+\kappa}$ . Apply the pointwise circle-average lemma to the approximants. The error is bounded by  $\varepsilon \widehat{\mathbf{1}}_B$ , whose  $L^{1+\kappa}$  norm is finite by the same cusp estimate. Letting the uniform approximation error tend to zero therefore gives, for almost every  $X$ ,

$$A_t \widehat{\varphi}_{\mathcal{C}}(X) \longrightarrow c_{\mu, \mathcal{C}} \int_{\mathbb{R}^2} \varphi(u) du,$$

with the harmless mollification removed by comparing neighboring  $t$ 's and using the same maximal bound. We record this as the continuous-test circle-equidistribution consequence of the pointwise circle-average lemma.

**Step 5: the triangle trick converts dynamics into counting.** Let  $\mathcal{T}$  be the triangle with vertices  $0, 1+i, -1+i$ . If  $z = |z|e^{i\theta_z}$  and

$$\Theta_t(z) = \{\theta : g_t r_\theta z \in \mathcal{T}\}, \quad \psi_t = \arctan(e^{-t}),$$

a direct Euclidean calculation gives

$$\frac{\psi_t}{\pi} \mathbf{1}_{B(0, e^{t/2})}(z) \leq A_t \mathbf{1}_{\mathcal{T}}(z) \leq \frac{\psi_t}{\pi} \mathbf{1}_{B(0, \sqrt{e^t + e^{-t}})}(z). \quad (1)$$

Indeed, rotating  $z$  to the vertical axis shows that the admissible angles form an interval of width  $2\psi_t$  whenever  $|z| \leq e^{t/2}$ , while any vector entering  $g_{-t}\mathcal{T}$  has norm at most  $\sqrt{e^t + e^{-t}}$ . Summing (1) over  $v \in V_{\mathcal{C}}(X)$  yields

$$\frac{\psi_t}{\pi} N_{\mathcal{C}}(X, e^{t/2}) \leq A_t \widehat{\mathbf{1}}_{\mathcal{T}\mathcal{C}}(X) \leq \frac{\psi_t}{\pi} N_{\mathcal{C}}(X, \sqrt{e^t + e^{-t}}). \quad (2)$$

Since  $\psi_t \sim e^{-t}$ , (2) is exactly at the quadratic scale.

Choose inner and outer continuous approximations to  $\mathbf{1}_{\mathcal{T}}$  whose difference is supported in a  $\delta$ -tube of the three boundary segments. Away from the cusp, local finiteness makes the transform of that difference tend pointwise to zero as  $\delta \downarrow 0$ . In the cusp, the transform-integrability estimate gives one integrable majorant, uniformly in  $\delta$ . Dominated convergence therefore makes the integral error arbitrarily small. For the pointwise circle averages, apply the maximal estimate in the pointwise circle-average lemma to a  $K$ -finite approximation of the boundary error; the  $L^{1+\kappa}$  norm tends to zero by the compact-window estimate-the transform-integrability estimate. A summable

sequence  $\delta_j \downarrow 0$  and Borel–Cantelli then gives one full-measure set on which the circle-average error is eventually  $< \varepsilon$ . This proves the boundary-removal lemma.

Apply Step 4 to  $f_\varepsilon$  and use the boundary-removal lemma to replace it by  $\mathbf{1}_\mathcal{T}$ . Since  $\text{area}(\mathcal{T}) = 1$ , we get

$$A_t \widehat{\mathbf{1}_{\mathcal{T}\mathcal{C}}}(X) \longrightarrow c_{\mu, \mathcal{C}}$$

for almost every  $X$ . Insert this in (2), divide by the corresponding disk areas, and use

$$\psi_t e^t \rightarrow 1, \quad e^t + e^{-t} = e^t(1 + e^{-2t}).$$

The lower and upper radii have ratio tending to one. Monotonicity of  $N_{\mathcal{C}}(X, R)$  therefore squeezes every intermediate radius and yields

$$\frac{N_{\mathcal{C}}(X, R)}{\pi R^2} \longrightarrow c_{\mu, \mathcal{C}}.$$

This proves the quadratic squeeze and the counting assertion.

**Step 6: normalization and scope check.** The same multiplicity convention is used in the transform and in  $N_{\mathcal{C}}$ , so no factor is lost when passing from the invariant functional to the disk count. The proof also explains the exact scope of the second assertion: it uses the Masur–Veech quadratic thin-part estimate and the Eskin–Masur compact-window count for the selected saddle-connection or cylinder configuration. For a general invariant probability the first Siegel–Veech identity remains valid under the stated transform-integrability hypothesis, but the pointwise quadratic asymptotic requires the corresponding cusp and pointwise-average gates and is therefore not asserted here.  $\square$

### Theorem K16.6. Wind-tree diffusion exponent 2/3

**Statement.** For every obstacle parameter  $(a, b) \in (0, 1)^2$  in the periodic rectangular wind-tree model, for Lebesgue-almost every direction  $\theta$ , and for every point  $p$  whose forward billiard orbit is infinite, if  $\phi_T^\theta(p)$  denotes the billiard position at time  $T$  and  $d$  is Euclidean distance in the plane, then  $\limsup_{T \rightarrow \infty} \frac{\log d(p, \phi_T^\theta(p))}{\log T} = \frac{2}{3}$ .

*Proof.* Fix  $(a, b) \in (0, 1)^2$ . We keep the obstacle parameter fixed throughout; no measure on the parameter square will be used. There are four distinct pieces of the argument: the geometric unfolding, the Hodge-bundle computation, the deviation theorem, and the directional genericity theorem.

**Step 1: the compact genus-five quotient and the deck cocycle.** Reflect a billiard trajectory whenever it meets a horizontal or vertical side of an obstacle. Four reflected copies of a fundamental torus cell fit together to a compact translation surface  $X(a, b)$  in the stratum  $\mathcal{H}(2^4)$ . Unfolding the whole periodic table gives a translation surface  $X_\infty(a, b)$  with a normal covering

$$X_\infty(a, b) \longrightarrow X(a, b)$$

whose deck group is  $\mathbb{Z}^2$ . The genus is five: the four zeros created by the obstacle

corners have order two, so  $2g - 2 = 4 \cdot 2 = 8$ .

Label the four copies of the toric cell by  $\kappa \in \{00, 01, 10, 11\}$  and let  $h_\kappa, v_\kappa$  be the horizontal and vertical boundary cycles. Define the two integral classes

$$h^- = h_{00} - h_{01} + h_{10} - h_{11}, \quad v^- = v_{00} - v_{10} + v_{01} - v_{11}. \quad (1)$$

Equivalently, these are the two characters that record crossing a horizontal or vertical translate of the original fundamental cell. Put  $f = (h^-, v^-) \in H^1(X(a, b); \mathbb{Z}^2)$  after Poincare duality.

Indeed each crossing of a chosen cut changes exactly one deck coordinate by  $\pm 1$ . Summing the signed crossings telescopes to the difference between the initial and final cells. What remains is the difference of two points inside one unit fundamental cell, whose Euclidean norm is at most  $\sqrt{2}$ . This is the precise geometric bridge from billiard displacement to homological pairing.

**Step 2: the Klein-four splitting and the exponent  $2/3$ .** The compact surface has commuting involutions  $\tau_h, \tau_v$  induced by the two reflections. Their Klein four group acts symplectically on homology, so the Hodge bundle over the  $SL(2, \mathbb{R})$ -orbit closure  $\mathcal{G}$  splits  $SL(2, \mathbb{R})$ -equivariantly as

$$E = E^{++} \oplus E^{+-} \oplus E^{-+} \oplus E^{--}. \quad (3)$$

The deck classes in (1) lie respectively in  $E^{+-}$  and  $E^{-+}$ ; these two bundles have real rank two. The invariant part  $E^{++}$  has rank four and is the pullback of the Hodge bundle of the genus-two quotient  $L(a, b) \in \mathcal{H}(2)$ .

The three order-two quotients identify the sums  $E^{++} \oplus E^{--}$ ,  $E^{++} \oplus E^{+-}$ , and  $E^{++} \oplus E^{-+}$  with Hodge bundles over orientation-cover loci of quadratic differentials on the sphere. The relevant strata are

$$\mathcal{Q}(4, -1^8), \quad \mathcal{Q}(1^2, -1^6). \quad (4)$$

Only the singularities of odd order ramify in the orientation double cover. For an invariant ergodic probability on such a hyperelliptic orientation-cover locus, the curvature/degree computation of the Hodge bundle gives

$$\lambda_1 + \cdots + \lambda_g = \frac{1}{4} \sum_{d_j \text{ odd}} \frac{1}{d_j + 2}. \quad (5)$$

For completeness, the special formula (5) is obtained by decomposing the anti-invariant holomorphic one-forms into the character line bundles of the double cover. If a local quadratic differential is  $q = z^{d_j} dz^2$ , the canonical square-root coordinate contributes  $1/(d_j + 2)$  to the parabolic degree of the anti-invariant Hodge line. The Gauss-Manin curvature identity identifies the sum of positive KZ exponents with the Hodge degree divided by the hyperbolic degree of the Teichmüller disk; the latter normalization contributes the factor  $1/4$ . Summing the ramified singularities gives (5). Since the calculation is local in the branch orders, the resulting sum is independent of the invariant ergodic measure supported on the covering locus.

To verify (6), the genus-two quotient contributes exponents 1 and  $1/3$  on  $E^{++}$ . Formula (5) gives total positive sum 2 for the  $\mathcal{Q}(4, -1^8)$  cover and  $5/3$  for the  $\mathcal{Q}(1^2, -1^6)$

cover. Subtracting the common  $E^{++}$  contribution  $1 + 1/3 = 4/3$  leaves respectively  $2/3$  and  $1/3$ . Matching the three quotient bundles gives exactly (6). In particular both deck factors have top exponent  $2/3$ .

**Step 3: from KZ growth to pairing growth for every infinite leaf.** The required statement is stronger than the assertion that a generic vector of a cocycle grows at rate  $2/3$ : it concerns the homology class of a long flow segment and every nonsingular starting point. We therefore use the actual Zorich–Forni deviation mechanism.

Here is the proof at the scope needed. Choose a nested sequence of compact zippered-rectangle neighborhoods of a recurrent base point. A long vertical orbit is cut at successive returns into principal blocks. Each block closes to one of finitely many relative-homology generators while the return matrix is precisely the Rauzy–Veech/KZ matrix. Generic recurrence says that return times have logarithm  $o(n)$  relative to the  $n$ th induction time, so replacing physical time by induction time changes logarithmic exponents by  $o(1)$ . Oseledets gives the upper estimate  $T^{\nu+o(1)}$  on every principal block; summing the geometrically separated scales gives the same upper bound for an arbitrary prefix. For the lower bound, generic recurrence returns infinitely often to a compact neighborhood in which one of the finitely many principal relative cycles has pairing with the unstable line bounded away from zero. Pulling that cycle back by the inverse KZ matrix produces a subsequence of orbit segments with pairing  $T^{\nu-o(1)}$ . The construction uses only the finite return alphabet and hence works for every infinite nonsingular leaf, not merely almost every starting point. This is the homological version of the deviation mechanism already developed in the preceding deviation theorem and proves (7).

Apply the generic-recurrent deviation lemma to the two components of  $f$ . Because each belongs to a rank-two symplectic factor with exponents  $\pm 2/3$ , any nonzero component not on the stable line has pairing exponent  $2/3$ . The two deck coordinates cannot simultaneously lie in their stable lines: they span the integral deck quotient, while the stable lines vary irrationally on a recurrent generic orbit; equivalently the principal-return cycles span the deck quotient and one has nonzero projection on a top line at infinitely many compact returns. Thus

$$\limsup_{T \rightarrow \infty} \frac{\log(1 + \|\langle f, \gamma_T^\theta(p) \rangle\|_2)}{\log T} = \frac{2}{3} \quad (8)$$

for every infinite leaf whenever the rotated surface is recurrent and Oseledets generic.

**Step 4: almost every direction for the fixed obstacle.** The final quantifier is where the later Chaika–Eskin theorem is essential. Classical Oseledets alone would give (8) only for almost every surface with respect to an invariant measure and would not justify the theorem for an arbitrary fixed  $(a, b)$ .

The proof uses the affine-measure/orbit-closure results the affine-measure and orbit-closure theorems to identify the only stationary limits of the random walk on the Teichmüller disk. Comparing geodesic rays with a rotationally symmetric random walk reduces Birkhoff and cocycle deviations to sums of independent bounded increments; the strong law makes those deviations  $o(n)$ . The cocycle statement is first proved on Lusin sets where the Oseledets splitting is continuous and then extended across

the small exceptional set by the integrable KZ norm. This is exactly the point of the Chaika–Eskin theorem: the starting surface is fixed.

Take  $Y = X(a, b)$ . For almost every  $\theta$ , the fixed-surface directional genericity lemma supplies both hypotheses of the generic-recurrent deviation lemma, while the exponent computation supplies the exponent  $2/3$  for its own affine orbit closure. Hence (8) holds for every infinite nonsingular orbit in that direction. Finally (2) changes the displacement by only a bounded vector, so

$$\limsup_{T \rightarrow \infty} \frac{\log d(\tilde{p}, \phi_T^\theta(\tilde{p}))}{\log T} = \frac{2}{3}.$$

A trajectory hitting a corner stops in the unfolded flow and is precisely the finite-forward-orbit case excluded in the statement. Since  $(a, b)$  was arbitrary, the theorem holds for every rectangular obstacle parameter.  $\square$

## 16.3 Worked examples

### Worked example: horizontal cylinders on the regular octagon

Take a regular octagon of side length 1 and identify opposite sides by translation. In the horizontal direction the surface decomposes into two cylinders. One has width  $1 + \sqrt{2}$  and height 1, hence modulus

$$m_1 = 1 + \sqrt{2}.$$

The other has width  $2 + \sqrt{2}$  and height  $1/\sqrt{2}$ , so

$$m_2 = \frac{2 + \sqrt{2}}{1/\sqrt{2}} = 2 + 2\sqrt{2} = 2m_1.$$

Thus the cylinder moduli are commensurable. A standard horizontal shear  $u_t(x, y) = (x + ty, y)$  closes up on both cylinders when  $th_i/c_i$  is integral for each cylinder. Commensurability lets one choose a common nonzero  $t$ , producing a parabolic affine automorphism. This is the local geometric signature behind the horizontal cusp of the regular-octagon Veech group.

#### Check your understanding

If cylinder moduli  $m_1, m_2$  have irrational ratio, explain why there is no single nonzero shear time that performs an integral number of Dehn twists in both cylinders simultaneously.

### Boundary case: a nonlattice surface

A generic translation surface has no lattice Veech group. It may still have an affine invariant orbit closure—indeed the full connected stratum is itself one—but

Veech dichotomy no longer follows. In a direction without saddle connections one must use almost-everywhere theorems such as Masur-Veech/Kerckhoff-Masur-Smillie rather than asserting the all-directions lattice-surface dichotomy.

### Common pitfall

The Veech dichotomy does not say that every direction on a lattice surface is uniquely ergodic: completely periodic directions are the other side of the dichotomy. Likewise, a quadratic counting law comes with a surface/affine-measure dependent Siegel-Veech constant; it is not a universal numerical constant attached only to the stratum dimension.

## 16.4 Chapter summary

### What to remember

- Lattice Veech groups force an all-directions dichotomy between complete periodicity and unique ergodicity.
- Eskin-Mirzakhani classifies ergodic upper-triangular invariant probability measures as affine and  $SL(2, \mathbb{R})$ -invariant.
- Eskin-Mirzakhani-Mohammadi turns measure rigidity into affine orbit-closure rigidity.
- Cylinder deformations and fields of definition expose the internal linear/arithmetic structure of affine invariant manifolds.
- Siegel-Veech transforms convert invariant-measure symmetry into quadratic counting laws.
- The wind-tree model translates a KZ exponent into deterministic planar diffusion with exponent  $2/3$ .

### Looking ahead

K17 leaves flat surfaces and returns to smooth diffeomorphisms. The common theme remains invariant geometry, but the organizing structure is now a dominated splitting with strong stable and unstable directions, accessibility, and normally hyperbolic persistence.

### Check your understanding

For a lattice surface, describe the two alternatives in the Veech dichotomy and explain how cylinder decompositions feed into counting. Then identify which part of a quadratic counting asymptotic is geometric data rather than a universal constant.

## 16.5 Exercises and problems

### Exercise 16.1

**Difficulty.** Core.

**Related results.** Veech dichotomy; Eskin-Mirzakhani affine-measure classification; Eskin-Mirzakhani-Mohammadi orbit-closure theorem.

**Problem.** For a cylinder direction on the regular-octagon Veech surface, compute cylinder data and the associated parabolic affine action; explain the Veech-dichotomy conclusion and the closed Teichmüller-curve orbit in this special case.

**Solution sketch.** List widths/heights, verify commensurable moduli, and identify the lattice stabilizer of the closed orbit.

### Exercise 16.2

**Difficulty.** Core.

**Related results.** Wright structure theorem interface; Siegel-Veech formula and quadratic counting; Wind-tree diffusion exponent  $2/3$ .

**Problem.** For a nonlattice Veech group, explain exactly which all-directions dichotomy argument is lost while almost-every-direction unique ergodicity can still follow from general theory. Keep this separate from Siegel-Veech counting.

**Solution sketch.** Compare hypotheses of the lattice theorem with KMS/Masur-Veech a.e. statements and with the independent counting framework.

### Exercise 16.3

**Difficulty.** Advanced.

**Related results.** Eskin-Mirzakhani-Mohammadi orbit-closure theorem.

**Problem.** Prove in genus one that the  $\mathrm{SL}(2, \mathbb{R})$ -orbit of the square torus is  $\mathrm{SL}(2, \mathbb{R})/\mathrm{SL}(2, \mathbb{Z})$  and is closed, as the simplest orbit-closure model.

**Solution sketch.** Identify marked lattices  $A\mathbb{Z}^2$ , compute the stabilizer, and use the homogeneous-space model of the unit-area stratum.

### Exercise 16.4

**Difficulty.** Advanced.

**Related results.** Wright structure theorem interface.

**Problem.** Compare the tangent dimensions/linear period-coordinate descriptions of a Teichmüller curve and a full stratum component, illustrating why affine invariant manifolds are far more constrained than arbitrary submanifolds.

**Solution sketch.** Write the real-linear tangent equations for the closed orbit and compare them with the full relative-cohomology tangent space.

### Exercise 16.5

**Difficulty.** Advanced.

**Related results.** Veech dichotomy; Eskin-Mirzakhani affine-measure classification.

**Problem.** Separate Veech dichotomy from Siegel-Veech counting: derive the  $R^2$



scaling of cylinder counts from planar scaling and identify the surface-dependent constant.

**Solution sketch.** Apply the Siegel transform to a radially scaled region/test function and note that Euclidean area grows as  $R^2$ .

## **Part V**

# **Smooth Dynamics and Generic Phenomena**

# Smooth Conservative and Partially Hyperbolic Dynamics

## Why this chapter matters

Uniform hyperbolicity is powerful but too rigid for many conservative systems: neutral or weak directions often coexist with strongly contracting and expanding ones. Partial hyperbolicity isolates this situation by requiring domination rather than uniform behavior in the center. The central question is then whether the stable and unstable geometry is strong enough to control the whole system. This volume develops the two principal mechanisms. Geometrically, the strong bundles integrate uniquely and normally hyperbolic invariant manifolds persist under perturbation. Measure-theoretically, stable and unstable leaves can transmit density information. Accessibility asks whether those leaves connect the phase space; center bunching makes their holonomies regular enough for the Hopf/julienne argument. Burns–Wilkinson then upgrades essential accessibility to ergodicity and even the Kolmogorov property.

**Prerequisites.** Foundation F10 for graph transforms, stable manifolds and normal hyperbolicity; F11 for Pesin theory and absolute continuity; F14 for entropy terminology; basic differential topology.

## Learning goals

- Check the domination inequalities defining absolute partial hyperbolicity in concrete examples.
- Distinguish strong stable/unstable integrability from the generally subtler center and center-stable/center-unstable integrability questions.
- Compute accessibility classes generated by  $su$ -paths.
- Explain why  $r$ -normal hyperbolicity gives persistence with  $C^r$  regularity.
- Follow the Burns–Wilkinson chain from center bunching and julienne density to ergodicity and the  $K$ -property.

## Conceptual roadmap

The chapter follows four geometric steps.

- A dominated splitting  $E^s \oplus E^c \oplus E^u$  first produces the strong stable and unstable foliations  $W^s$  and  $W^u$ .
- Normal domination then gives persistence of invariant manifolds under per-

turbation.

- Center bunching improves the regularity of stable/unstable holonomies and makes julienne density usable.
- Stable/unstable saturation plus accessibility finally feeds the Hopf argument and yields ergodicity, and in the Burns–Wilkinson setting the  $K$ -property.

### Check your understanding

Why can  $E^s$  and  $E^u$  be uniquely integrable even when  $E^c$  is not? Which part of the domination inequalities supplies the contraction needed for the graph-transform argument?

## 17.1 Definitions

### Intuition

**Domination compares rates, not just signs.** Partial hyperbolicity does not require the center to contract or expand. It requires stable contraction to dominate center behavior and unstable expansion to dominate it in the opposite direction. **Accessibility is a control condition.** An  $su$ -path moves only along directions whose geometry is well controlled. If such paths connect almost every measurable accessibility-saturated set, stable/unstable information can propagate globally.

**Center bunching protects regularity across the center.** The stable and unstable holonomies must distort center plaques mildly enough that density points survive transport. The center-bunching inequalities quantify this separation of scales.

**Normal hyperbolicity is relative hyperbolicity.** A normally hyperbolic manifold may have rich internal dynamics. What matters is that contraction/expansion transverse to the manifold dominates every derivative order of the tangential dynamics needed for persistence.

**The  $K$ -property is stronger than ergodicity.** Ergodicity eliminates nontrivial invariant sets; the Kolmogorov property eliminates every nontrivial zero-entropy factor, equivalently makes the Pinsker sigma-algebra trivial.

### Definition K17.1. partial hyperbolicity

A diffeomorphism  $f$  is (absolutely) partially hyperbolic on a compact invariant set  $\Lambda$  if there is a continuous  $Df$ -invariant splitting  $T_\Lambda M = E^s \oplus E^c \oplus E^u$  and constants  $0 < \lambda_s < \lambda_c^- \leq \lambda_c^+ < \lambda_u$  with  $\lambda_s < 1 < \lambda_u$  such that for every unit vector  $v^\sigma \in E_x^\sigma$  one has  $\|Df_x v^s\| \leq \lambda_s$ ,  $\lambda_c^- \leq \|Df_x v^c\| \leq \lambda_c^+$ , and  $\|Df_x v^u\| \geq \lambda_u$ . The strict inequalities  $\lambda_s < \lambda_c^-$  and  $\lambda_c^+ < \lambda_u$  are the domination inequalities.

**Definition K17.2. accessibility and essential accessibility**

An *su*-path is a finite concatenation of arcs, each contained in one strong-stable or one strong-unstable leaf. The accessibility class of a point is the set reachable from it by *su*-paths;  $f$  is accessible if there is only one accessibility class. Relative to the smooth invariant volume,  $f$  is essentially accessible if every measurable union of entire accessibility classes has either zero measure or full measure.

**Definition K17.3. center bunching**

For a partially hyperbolic diffeomorphism choose continuous comparison functions  $\nu, \hat{\nu}, \gamma, \hat{\gamma}$  with  $\nu, \hat{\nu} < 1$  and, in an adapted metric,  $\|Df(v^s)\| < \nu < \gamma < \|Df(v^c)\| < \hat{\gamma}^{-1} < \hat{\nu}^{-1} < \|Df(v^u)\|$  for unit vectors in the indicated bundles. The Burns-Wilkinson center-bunching condition is the pair of pointwise inequalities  $\nu < \gamma\hat{\gamma}$  and  $\hat{\nu} < \gamma\hat{\gamma}$ . It is automatic when  $E^c$  is one-dimensional.

**Definition K17.4. stable ergodicity**

Stable ergodicity means ergodicity with respect to the reference smooth volume persists for every sufficiently small perturbation within the specified conservative regularity class.

**Definition K17.5.  $r$ -normal hyperbolicity**

Let  $N$  be a compact  $C^r$  invariant submanifold of a  $C^r$  diffeomorphism  $f$ . It is  *$r$ -normally hyperbolic* if there is a continuous  $Df$ -invariant splitting  $T_N M = E^s \oplus TN \oplus E^u$  and constants  $C \geq 1$ ,  $0 < \lambda < 1$  such that for every  $n \geq 0$  and every integer  $1 \leq j \leq r$ ,

$$\|Df^n|_{E_x^s}\| \|Df^{-n}|_{T_{f^n x} N}\|^j \leq C\lambda^n, \quad \|Df^{-n}|_{E_x^u}\| \|Df^n|_{T_{f^{-n} x} N}\|^j \leq C\lambda^n.$$

The  $j = 1$  inequalities are normal hyperbolicity; the higher inequalities are the spectral gaps needed for  $C^r$  persistence.

**Definition K17.6. Pinsker sigma-algebra and Kolmogorov property**

For a probability-preserving invertible map  $f$ , the *Pinsker sigma-algebra*  $\mathcal{P}(f)$  is the largest  $f$ -invariant sub-sigma-algebra on which the metric entropy is zero. The map has the *Kolmogorov property* (or is a  $K$ -system) when  $\mathcal{P}(f)$  is trivial modulo null sets. In particular a  $K$ -system is ergodic.

**Definition K17.7. julienne density point**

A *julienne* at a point of a partially hyperbolic system is an anisotropic shrinking neighborhood obtained by taking a short center plaque, saturating it by strong-stable plaques on the forward scale and then by strong-unstable plaques on the backward scale, with the scales chosen from the partial-hyperbolicity comparison functions. A point is a *julienne density point* of a measurable set when the relative volume of the set in these juliennes tends to one. Center bunching is precisely what

makes the stable and unstable holonomies quasi-conformal enough for julienne density to be preserved along strong leaves.

## 17.2 Main results

### Definition and interface: Accessibility and stable-accessible classes

**Context.** Accessibility classes are generated by alternating stable/unstable paths; essential accessibility is the measurable analogue used in ergodicity criteria.

### Theorem K17.1. Partial hyperbolicity and invariant foliations

**Statement.** Let  $M$  be a compact smooth manifold without boundary and let  $f \in \text{Diff}^r(M)$ ,  $r \geq 1$ , be partially hyperbolic with a  $Df$ -invariant splitting  $TM = E^s \oplus E^c \oplus E^u$  satisfying the uniform contraction, expansion, and domination inequalities of Definition K17.1. Then  $E^s$  and  $E^u$  are uniquely integrable. Their maximal integral manifolds form  $f$ -invariant strong stable and strong unstable foliations  $W^s$  and  $W^u$ ; every leaf is a  $C^r$  injectively immersed submanifold tangent to  $E^s$  or  $E^u$ , respectively.

### Scope note

No integrability of the center, center-stable, or center-unstable distributions is asserted.

*Proof.* Choose an adapted Riemannian metric so that the constants in the definition of partial hyperbolicity hold with one-step inequalities. In an exponential chart at  $x$ , write a small neighborhood as  $E_x^s \oplus F_x$ ,  $F_x = E_x^c \oplus E_x^u$ .

The derivative at the zero section is block diagonal with stable block norm  $< \lambda_s$  and inverse complementary conorm  $> \lambda_c^-$ ; thus the slope multiplier is at most  $\lambda_s/\lambda_c^- < 1$ . Shrink  $\rho$  so nonlinear terms preserve a strict contraction factor. the Foundation Parameter-dependent contraction theorem gives the parameter-dependent contraction theorem, producing a unique continuous plaque family. Differentiating the graph equation successively up to order  $r$  and using the same domination gives  $C^r$  leaves. This proves Lemma 17.1.

Points on the fixed graph contract by the stable estimate. Conversely, if a point has a nonzero  $F_x$  displacement, domination says that relative to its stable component that displacement grows by at least  $\lambda_c^-/\lambda_s > 1$  until it leaves the chart, contradicting the stated forward contraction. This proves the characterization and hence uniqueness. The identity  $f(W_{\text{loc}}^s(x)) \subset W_{\text{loc}}^s(fx)$  follows from the characterization; therefore  $W^s(x) = \bigcup_{n \geq 0} f^{-n}W_{\text{loc}}^s(f^n x)$  is a  $C^r$  injectively immersed submanifold tangent to  $E^s$ . These leaves partition  $M$  and form the strong stable foliation. Apply the same proof to  $f^{-1}$  and  $E^u$  for  $W^u$ . Nothing in the argument integrates  $E^c, E^{cs}, E^{cu}$ , so no such claim is made.  $\square$

**Theorem K17.2. Persistence of normally hyperbolic invariant manifolds**

**Statement.** Let  $f \in \text{Diff}^r(M)$ ,  $r \geq 1$ , and let  $N \subset M$  be a compact  $C^r$ ,  $f$ -invariant submanifold that is  $r$ -normally hyperbolic. Then every  $C^r$  diffeomorphism  $g$  sufficiently  $C^1$ -close to  $f$  has a unique nearby  $C^r$ ,  $g$ -invariant submanifold  $N_g$  in a fixed small neighborhood of  $N$ ;  $N_g$  is  $C^1$ -close and diffeomorphic to  $N$ . The local strong-stable and strong-unstable laminations persist with the regularity furnished by the corresponding normal-hyperbolicity inequalities.

**Scope note**

Normal-hyperbolicity persistence does not require a symplectic or volume-preserving structure; those are additional restrictions on the perturbation class.

*Proof.* We use the Foundation theorem on persistence of normally hyperbolic invariant manifolds and verify its hypotheses here rather than duplicating its proof. the definition of  $r$ -normal hyperbolicity gives a  $Df$ -invariant splitting along  $N$ ,

$$T_N M = E^s \oplus TN \oplus E^u,$$

with the  $r$ -normal inequalities comparing normal contraction/expansion with all tangent derivatives through order  $r$ . These are precisely the normal-hyperbolicity and  $r$ -bunching hypotheses of the definition of normal hyperbolicity used in the Foundation Normally hyperbolic invariant manifold persistence (after replacing the metric by an adapted equivalent metric). Compactness of  $N$  gives uniform constants.

the Foundation Normally hyperbolic invariant manifold persistence therefore supplies, for every  $C^r$  diffeomorphism  $g$  sufficiently  $C^1$ -close to  $f$ , a unique invariant graph over  $N$  in a fixed tubular neighborhood. Its image  $N_g$  is  $C^r$ , diffeomorphic and  $C^1$ -close to  $N$ . The same theorem's invariant-section construction produces the persisting local strong-stable and strong-unstable laminations with exactly the regularity allowed by the normal inequalities. Uniqueness is uniqueness among nearby invariant graphs in that neighborhood, which is the scope of the K17 statement.

No step of the Foundation Normally hyperbolic invariant manifold persistence assumes a symplectic form or invariant volume; those structures merely define closed perturbation subclasses if desired. Thus the Foundation conclusion matches every clause of the Persistence of normally hyperbolic invariant manifolds.  $\square$

**Theorem K17.3. Burns-Wilkinson ergodicity and K-property theorem**

**Statement.** Let  $f : M \rightarrow M$  be a  $C^2$  volume-preserving partially hyperbolic diffeomorphism of a closed Riemannian manifold. If  $f$  is center bunched and essentially accessible, then  $f$  is ergodic with respect to volume and has the Kolmogorov property. If  $\dim E^c = 1$ , the center-bunching condition is automatic; hence every essentially accessible  $C^2$  volume-preserving partially hyperbolic diffeomorphism with one-dimensional center is ergodic and has the Kolmogorov property.

**Scope note**

Stable ergodicity is not asserted by this theorem; it is a perturbative property requiring additional input.

*Proof.* Write  $m$  for the invariant smooth probability volume. The strong stable and strong unstable foliations have already been constructed. We shall need to distinguish two notions that look similar but are not the same. A measurable set is *essentially stable-saturated* if, after changing it on a null set, it becomes a union of complete stable leaves; define essential unstable saturation similarly. A set may have the two properties using two different null-set modifications. We call such a set *bi-essentially saturated*. It is *essentially bisaturated* when one single modification is simultaneously a union of stable and unstable leaves. The hard part of the Burns–Wilkinson proof is precisely to pass from the first notion to the second.

*Step 1: the Hopf information is bi-essential, not yet bisaturated.* If  $y \in W^s(x)$  and  $\varphi$  is continuous, then  $d(f^n x, f^n y) \rightarrow 0$  exponentially, and therefore

$$\left| \frac{1}{N} \sum_{j=0}^{N-1} \varphi(f^j x) - \frac{1}{N} \sum_{j=0}^{N-1} \varphi(f^j y) \right| \rightarrow 0.$$

Thus the forward Birkhoff limsup is constant on stable leaves. Applying the same argument to  $f^{-1}$  shows that the backward limsup is constant on unstable leaves. Birkhoff's theorem says that the forward and backward averages exist and agree almost everywhere. Hence every sublevel set of the common almost-everywhere value is bi-essentially saturated. We still have to make the stable and unstable representatives compatible.

*Step 2: strong holonomy and fake weak plaques.* Because no dynamical coherence is assumed, there need not be genuine center, center-stable, or center-unstable foliations. Burns and Wilkinson replace them by *fake invariant foliations*. Fix an orbit  $(p_j)$ . In exponential coordinates at  $p_j$ , modify the local formula for  $f$  outside a small ball so that it agrees with the derivative there. For a sufficiently small cutoff the modified maps remain uniformly partially hyperbolic. Graph transform then produces local foliations

$$\widehat{W}^s, \widehat{W}^u, \widehat{W}^c, \widehat{W}^{cs}, \widehat{W}^{cu}$$

tangent to the corresponding cone fields. The stable and center foliations subfoliate the fake center-stable foliation, while the unstable and center foliations subfoliate the fake center-unstable foliation. At the orbit centers the fake strong plaques agree with the genuine strong plaques by uniqueness of the stable and unstable graph transforms.

We also need absolute continuity of the genuine strong foliations. Let  $h : \Sigma_0 \rightarrow \Sigma_1$  be stable holonomy between two smooth transversals. Approximate  $h$  by applying  $f^n$ , projecting between the exponentially close images of the transversals, and pulling back. The logarithm of the Jacobian ratio is a sum of differences

$$\log J^\perp f(f^j x) - \log J^\perp f(f^j h(x)).$$



Since  $f$  is  $C^2$ , the transverse Jacobian is locally Lipschitz, while stable separation is  $O(\nu_*^j)$  for some  $\nu_* < 1$ . The logarithmic distortion is therefore bounded by a convergent geometric series. Passing to the holonomy limit gives a Radon–Nikodym Jacobian bounded above and away from zero. The same argument for  $f^{-1}$  gives unstable absolute continuity.

Center bunching supplies a stronger estimate inside the fake weak plaques. When the derivative of stable holonomy between two fake center transversals is written as a graph map, one graph-transform step multiplies its slope error by a factor controlled by

$$\frac{\nu(x)}{\gamma(x)\widehat{\gamma}(x)}.$$

The center-bunching inequality makes the supremum of this ratio strictly less than one. Iterating the affine graph recursion therefore gives a convergent geometric series and a uniform Lipschitz bound for stable holonomy inside fake center-stable plaques. The inverse-time inequality gives the analogous unstable statement inside fake center-unstable plaques. This is the exact regularity that replaces the missing genuine center foliation.

*Step 3: center-unstable juliennes and thin orbit tubes.* Choose multiplicative scales  $\sigma_n, \tau_n \rightarrow 0$  using the strict partial-hyperbolicity and center-bunching gaps. The center scale decays more slowly than stable separation, while the unstable scale is prescribed at time  $n$ . Define a fake center ball and a pulled-back fake unstable ball by

$$\widehat{W}^c(x, \sigma_n), \quad \widehat{J}_n^u(y) = f^{-n}(\widehat{W}^u(f^n y, \tau_n)),$$

and define the center-unstable julienne

$$\widehat{J}_n^{cu}(x) = \bigcup_{y \in \widehat{W}^c(x, \sigma_n)} \widehat{J}_n^u(y).$$

All intermediate iterates of this set remain in a thin tube about the orbit. After normalizing an error by the relevant center and unstable scales, the remaining factor is bounded by a power of

$$\rho_s = \sup \frac{\nu}{\gamma\widehat{\gamma}} < 1 \quad \text{or} \quad \rho_u = \sup \frac{\widehat{\nu}}{\gamma\widehat{\gamma}} < 1.$$

Consequently every Hölder cocycle has uniformly bounded distortion on the tube: if  $a > 0$  is Hölder with exponent  $\theta$ , then

$$\left| \log \frac{\prod_{j=0}^{n-1} a(f^j z)}{\prod_{j=0}^{n-1} a(f^j z')} \right| \leq C \sum_{j \geq 0} \kappa^{\theta j} < \infty$$

for points  $z, z'$  in the same tube and some  $\kappa < 1$ .

Two elementary density notions will be used repeatedly. A nested sequence  $Y_n \downarrow \{x\}$  is *regular* if  $m(Y_{n+1}) \geq \delta m(Y_n)$  for one  $\delta > 0$ . Two nested sequences  $Y_n, Z_n$  are *internested* if a fixed integer  $k$  satisfies  $Y_{n+k} \subset Z_n$  and  $Z_{n+k} \subset Y_n$  for every  $n$ . For regular internested sequences, density one in one sequence is equivalent to density

one in the other.

*Step 4: stable holonomy quasi-preserves juliennes.* Let  $x, x'$  lie on one short true stable plaque. Stable holonomy from  $\widehat{W}^{cu}(x)$  to  $\widehat{W}^{cu}(x')$  maps the center base into a fixed-rank enlargement of the center base at  $x'$ , because the center holonomy is uniformly Lipschitz. For an unstable fiber, iterate to time  $n$ . The stable displacement is exponentially small; after it is pulled back through the center and unstable directions, center bunching again leaves a factor  $\rho_s^n$ . A fixed loss of rank therefore absorbs the error. Reversing  $x$  and  $x'$  gives the opposite inclusion. Hence the stable-holonomy images of the juliennes at  $x$  and the juliennes at  $x'$  are internested.

The juliennes are also regular in measure. At time  $n$  each unstable julienne has become a fake unstable ball of the prescribed terminal radius. Their leaf volumes are uniformly comparable. Pulling back introduces an unstable Jacobian, but the thin-tube distortion estimate makes this Jacobian uniformly comparable on each level. Consecutive unstable volumes therefore have a uniform positive ratio. The center bases satisfy the same property, and the  $C^1$  fake unstable subfoliation of each fake center-unstable plaque allows a Fubini comparison. The center-unstable juliennes are consequently a regular density sequence.

*Step 5: ordinary density and julienne density are equivalent.* We use a simple version of Cavalieri's principle. Suppose a neighborhood fibers by plaques of an absolutely continuous foliation, all fiber volumes at a given level are uniformly comparable, and  $X$  is saturated by the fibers. Disintegration of volume then shows that density one of  $X$  in the whole neighborhood is equivalent to density one of the corresponding base set. The same remains true for an essentially saturated  $X$  after replacing it by a saturated representative.

Let  $X$  now be genuinely stable-saturated and essentially unstable-saturated. Starting from the Riemannian ball  $B_n(x)$  of radius comparable with  $\sigma_n$ , form a fake center-stable base

$$D_n^{cs}(x) = \bigcup_{x' \in W^s(x, \sigma_n)} \widehat{W}^c(x', \sigma_n).$$

Over this base form successively neighborhoods using short true unstable plaques, pulled-back true unstable juliennes, pulled-back fake unstable juliennes, the local product with short stable plaques, and finally

$$G_n(x) = \bigcup_{q \in \widehat{J}_n^{cu}(x)} W^s(q, \sigma_n).$$

Uniform transversality makes the ball and the first product neighborhoods internested. Since  $X$  is essentially unstable-saturated, Cavalieri's principle permits replacement of short unstable plaques by true unstable juliennes. The thin-tube estimates make the true and fake unstable constructions internested, and local product geometry gives internesting of the remaining intermediate neighborhoods. The last sequence  $G_n$  is regular because it fibers over the regular center-unstable julienne by true stable plaques with uniformly controlled holonomy Jacobians. Finally, stable saturation of  $X$  and

Cavalieri's principle remove these stable fibers. Thus

$$x \text{ is a Lebesgue density point of } X \iff \frac{m_{cu}(X \cap \hat{J}_n^{cu}(x))}{m_{cu}(\hat{J}_n^{cu}(x))} \rightarrow 1.$$

This seven-neighborhood comparison is the density mechanism that an ordinary Hopf proof lacks.

*Step 6: density points become genuinely bisaturated.* Let  $A$  be bi-essentially saturated and choose a genuinely stable-saturated representative  $A^s$ . If  $x$  is a Lebesgue density point of  $A^s$ , the previous step says that it has julienne density one. Let  $x'$  lie on the same short stable plaque. Stable holonomy carries  $A^s$  to itself, has bounded Jacobian, and maps the julienne sequence at  $x$  to a sequence interstested with the julienne sequence at  $x'$ . Regularity therefore transfers julienne density one to  $x'$ , and the density comparison transfers it back to ordinary Lebesgue density. Hence the set of density points is stable-saturated. Apply the same argument to  $f^{-1}$  to obtain unstable saturation. The Lebesgue density theorem changes  $A$  only on a null set, so every bi-essentially saturated set is essentially bisaturated.

The ergodicity conclusion is now short. Each Birkhoff sublevel set from Step 1 is bi-essentially saturated, hence differs modulo zero from a genuinely bisaturated set. Such a set is a measurable union of accessibility classes. Essential accessibility forces its measure to be zero or one. Doing this for all rational sublevels makes every continuous Birkhoff average almost everywhere constant, which is equivalent to ergodicity.

*Step 7: the Pinsker algebra is also bi-essentially saturated.* Recall that the Pinsker sigma-algebra is the maximal zero-entropy factor. The point is not merely that stable points have similar symbolic names most of the time; a tail event is insensitive only to *finitely many* initial symbols. We therefore build the symbolic partition so that stable points have eventually identical names.

Choose finite partitions  $\mathcal{Q}_n$  whose diameters tend to zero and whose boundaries are finite unions of smooth hypersurface pieces. Thus the volume of a  $\rho$ -neighborhood of  $\partial\mathcal{Q}_n$  is  $O_n(\rho)$ . Following the Rokhlin-Sinai construction, choose very large delays  $k_n$  and form

$$\mathcal{P} = \bigvee_{n \geq 1} f^{-k_n} \mathcal{Q}_n.$$

The delays can be chosen so that two requirements hold at once. First, the successive conditional-entropy defects are summable. The resulting partition is *excellent*: the elementary Pinsker entropy identity and martingale convergence identify its remote future tail

$$\bigcap_{N \geq 1} \sigma \left( \bigvee_{j \geq N} f^{-j} \mathcal{P} \right)$$

exactly, modulo null sets, with the Pinsker sigma-algebra. This is the Rokhlin-Sinai part of the argument.

Second, let  $\bar{\nu} < 1$  be a uniform strong-stable contraction rate. We can make the

same delays so large that the total volumes of all boundary tubes

$$N_{q\bar{\nu}^{i+k_n}}(\partial Q_n), \quad i, n \geq 1,$$

are summable, for every integer  $q$ . Borel–Cantelli gives a conull set on which only finitely many of these tubes are visited. This conull set is itself stable-saturated: if  $x$  and  $y$  are on one stable leaf, then  $d(f^t x, f^t y) \leq D\bar{\nu}^t$  after finitely many iterates, so infinitely many bad boundary visits for one point would force infinitely many slightly enlarged bad visits for the other.

Now take two points of this good set on one stable leaf. For all sufficiently large  $i$  and for every  $n$  simultaneously,  $f^{i+k_n}x$  is farther from  $\partial Q_n$  than the distance between  $f^{i+k_n}x$  and  $f^{i+k_n}y$ . Hence those two points lie in the same  $Q_n$ -atom for every  $n$ , so  $f^i x$  and  $f^i y$  lie in the same  $\mathcal{P}$ -atom. Their  $\mathcal{P}$ -names are eventually identical. No remote-tail event can separate them, and therefore every Pinsker set is essentially stable-saturated. Applying the same construction to  $f^{-1}$  gives essential unstable saturation.

Step 6 now turns every Pinsker set into a genuinely bisaturated set modulo zero. Essential accessibility makes it null or conull. Hence the Pinsker sigma-algebra is trivial and  $f$  has the Kolmogorov property.

*Step 8: one-dimensional center.* If  $\dim E^c = 1$ , put  $a(x) = \|Df_x|_{E_x^c}\|$ . The center norm and conorm coincide with this scalar. The comparison functions can therefore be chosen so that  $\gamma(x)$  is arbitrarily close to  $a(x)$  from below and  $\hat{\gamma}(x)^{-1}$  is arbitrarily close to  $a(x)$  from above. Hence  $\gamma\hat{\gamma}$  can be made uniformly as close to 1 as desired. Since the stable and inverse-unstable comparison functions are continuous and strictly less than 1 on the compact manifold, both center-bunching inequalities follow. The theorem just proved applies, giving ergodicity and the Kolmogorov property in the one-dimensional-center case.  $\square$

## 17.3 Worked examples

### Worked example: Anosov base times a center rotation

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \in SL(2, \mathbb{Z})$$

act on  $\mathbb{T}^2$ , and define

$$F(x, \theta) = (Ax, \theta + \alpha)$$

on  $\mathbb{T}^2 \times \mathbb{S}^1$ . The eigenvalues of  $A$  are

$$\lambda_u = \frac{3 + \sqrt{5}}{2} > 1, \quad \lambda_s = \frac{3 - \sqrt{5}}{2} < 1.$$

The derivative on the circle factor has norm 1. Therefore

$$T(\mathbb{T}^2 \times \mathbb{S}^1) = E_A^s \oplus E^c \oplus E_A^u$$

is an absolutely partially hyperbolic splitting with center rate 1 strictly between  $\lambda_s$  and  $\lambda_u$ .

The strong stable and unstable leaves are

$$W_F^{s/u}(x, \theta) = W_A^{s/u}(x) \times \{\theta\}.$$

Hence every  $su$ -path keeps the circle coordinate fixed. Accessibility classes are contained in  $\mathbb{T}^2 \times \{\theta\}$ , so this product is *not accessible*. It is an excellent warning: partial hyperbolicity by itself does not supply the hypothesis needed by Burns–Wilkinson.

#### Check your understanding

What changes if the skew product is modified to  $F(x, \theta) = (Ax, \theta + \alpha + \varphi(x))$ ? Explain qualitatively how the coupling can allow stable and unstable holonomies to change the center coordinate.

#### Boundary case: an invariant center foliation with no accessibility

Take  $\alpha = 0$  in the product above. Every slice  $\mathbb{T}^2 \times \{\theta\}$  is invariant, so volume is not ergodic: any measurable subset of the circle lifts to a nontrivial invariant set. All strong bundles still exist and are smooth. The failure is exactly accessibility, not partial hyperbolicity or stable/unstable regularity.

#### Common pitfall

Partial hyperbolicity alone does not imply ergodicity, accessibility, or even dynamical coherence of the center bundle. Strong stable and unstable foliations are robust, but center behavior is a separate issue. The ergodicity theorem uses accessibility together with regularity hypotheses such as center bunching; none is a synonym for partial hyperbolicity.

## 17.4 Chapter summary

#### What to remember

- Partial hyperbolicity is a dominated three-way splitting with robust strong stable and unstable directions.
- Strong stable and unstable bundles integrate uniquely; center integrability is a separate question and may be pathological.
- Normal hyperbolicity gives persistence of invariant manifolds because normal rates dominate tangential dynamics.
- Accessibility turns strong leaves into a global communication network.
- Center bunching makes stable/unstable holonomies regular enough for juli-

enne density arguments.

- Essential accessibility plus center bunching yields ergodicity and the Kolmogorov property for smooth volume-preserving partially hyperbolic systems.

### Looking ahead

K18 shifts from robust dominated structure to generic perturbative structure. Closing and connecting lemmas ask what a small  $C^1$  perturbation can force, while Kupka-Smale and Bochi-type theorems describe properties that hold on residual subsets rather than for every nearby system.

### Check your understanding

Compare domination, normal hyperbolicity, accessibility, and center bunching. Which are splitting/rate conditions, which is a global connectivity condition, and which regularity input allows stable/unstable holonomies to participate in the ergodicity argument?

## 17.5 Exercises and problems

### Exercise 17.1

**Difficulty.** Core.

**Related results.** Partial hyperbolicity and invariant foliations; Accessibility definition and stable-accessible classes; Persistence of normally hyperbolic invariant manifolds.

**Problem.** For the product of an Anosov toral automorphism with a center rotation, compute  $E^s \oplus E^c \oplus E^u$ , verify domination, determine accessibility classes, and identify a normally hyperbolic center leaf.

**Solution sketch.** Use the toral eigenvalues and the neutral circle rate; stable/unstable paths keep the center coordinate fixed in the direct product.

### Exercise 17.2

**Difficulty.** Core.

**Related results.** Burns-Wilkinson ergodicity and K-property theorem.

**Problem.** Show that this product fails accessibility yet can be ergodic when the center rotation is irrational. Explain why failure of a Burns-Wilkinson hypothesis blocks that theorem but is not a proof of nonergodicity.

**Solution sketch.** Use weak mixing of the Anosov factor and ergodicity of the rotation to analyze the product.

**Exercise 17.3****Difficulty.** Advanced.**Related results.** Persistence of normally hyperbolic invariant manifolds.**Problem.** Prove a finite-dimensional graph-transform lemma under a normal-hyperbolicity inequality, obtaining a unique invariant Lipschitz graph.**Solution sketch.** Write the graph transform explicitly and estimate its contraction factor from normal-versus-tangent rates.**Exercise 17.4****Difficulty.** Advanced.**Related results.** Burns–Wilkinson ergodicity and K-property theorem.**Problem.** Give explicit rates satisfying partial hyperbolicity but violating center bunching, and identify the regularity/holonomy step of the Hopf argument that is then unavailable.**Solution sketch.** Check the domination inequalities numerically and make one bunching product fail; do not infer nonergodicity.**Exercise 17.5****Difficulty.** Advanced.**Related results.** Partial hyperbolicity and invariant foliations; Accessibility definition and stable-accessible classes.**Problem.** Prove that a measurable invariant function constant on stable and unstable leaves is constant on accessibility classes, and explain how essential accessibility turns this into the Hopf ergodicity mechanism.**Solution sketch.** Use forward/backward contraction for leafwise constancy and concatenate su-paths.

# Genericity, Rigidity and Global Smooth-Dynamical Phenomena

## Why this chapter matters

K17 studied robust structures that survive perturbation. K18 studies a complementary question: what behavior is *generic* when arbitrarily small  $C^1$  perturbations are allowed? Genericity is not probability and not openness. A residual property can be dense and topologically typical while individual systems outside the residual set remain dynamically important.

Three perturbative mechanisms organize the chapter. Kupka-Smale transversality removes nongeneric degeneracies of periodic orbits and invariant-manifold intersections. Pugh's closing lemma turns recurrence into nearby periodic dynamics. The connecting lemma turns arbitrarily accurate pseudo-orbits into true orbit connections after perturbation. Bochi's theorem then shows how restrictive the conservative surface setting is: generically an area-preserving surface diffeomorphism is either uniformly hyperbolic or has zero Lyapunov exponents almost everywhere.

**Prerequisites.** Foundation F10 for hyperbolic periodic points and invariant manifolds; F04/F11 for Lyapunov exponents; F15 for genericity/transversality language; K17 for dominated splittings.

## Learning goals

- Distinguish residual, open dense, full-measure and prevalent statements.
- Explain the two clauses of the Kupka-Smale property and why transversality is stable under small perturbation.
- Use nonwandering recurrence as the input to a  $C^1$  closing perturbation.
- Interpret pseudo-orbit chain relations and the conclusion of a connecting lemma.
- State Bochi's two-dimensional conservative dichotomy without importing a false higher-dimensional analogue.

## Conceptual roadmap

The perturbative logic is best read as four distinct mechanisms.

- Local transversality perturbations produce the Kupka-Smale residual set.
- Recurrence is converted by the closing lemma into a nearby periodic orbit.



- Chain relations are converted by the connecting lemma into true orbit connections after a controlled perturbation.
- On an area-preserving surface, domination versus exponent-reduction arguments lead to Bochi's generic Anosov/zero-exponent dichotomy.

### Check your understanding

A property holds on a residual subset of  $\text{Diff}^1(M)$ . What can you conclude about density? What can you *not* conclude about Lebesgue probability on any finite-dimensional parameter family?

## 18.1 Definitions

### Intuition

**$C^1$  perturbations can change derivatives while staying spatially small.** This is why closing/connecting lemmas are naturally  $C^1$  theorems: one can redirect short orbit pieces without losing control of invertibility.

**Nonwandering is weaker than recurrence of one orbit.** A nonwandering point only requires some point of every small neighborhood to return. Pugh's lemma uses that local return geometry to create a periodic point nearby after perturbation.

**Pseudo-orbits encode chain-level possibility.** A chain relation says there are arbitrarily accurate finite itineraries from  $x$  to  $y$ . The connecting lemma turns this topological possibility into a true orbit relation for a nearby map.

**Bochi's dichotomy is specifically two-dimensional and conservative.** Area preservation forces the two Lyapunov exponents to be opposite. The perturbation proof uses this determinant-one geometry twice: localized rotations can be realized conservatively, and after one mixing block turns the expanding line into the contracting line, the two half-orbit growth factors cancel. The measurable tower/castle terminology in this proof is different from the topological perturbation towers used in the connecting lemma.

### Definition K18.1. Kupka-Smale genericity

Kupka-Smale genericity is the property that all periodic points are hyperbolic and all intersections between stable and unstable manifolds of periodic points are transverse.

### Definition K18.2. closing lemma

A closing lemma is a perturbation theorem asserting that an approximate/recurrent return can be converted into a periodic orbit by an arbitrarily small perturbation in a specified topology.

**Definition K18.3. connecting lemma**

A connecting lemma is a perturbation theorem converting suitable pseudo-orbit/chain connections into true orbit/heteroclinic connections by a small perturbation.

**Definition K18.4. C<sup>1</sup> generic dynamics**

$C^1$ -generic means holding on a residual subset of the  $C^1$  space of diffeomorphisms/vector fields under the specified conservative or unrestricted topology.

**Definition K18.5. nonwandering point**

A point  $x$  of a diffeomorphism  $f : M \rightarrow M$  is *nonwandering* if every neighborhood  $U$  of  $x$  satisfies  $f^n(U) \cap U \neq \emptyset$  for some integer  $n > 0$ . The set of nonwandering points is denoted  $\Omega(f)$ .

**Definition K18.6. pseudo-orbit and forward chain relation**

For  $\varepsilon > 0$ , an  $\varepsilon$ -pseudo-orbit from  $x$  to  $y$  is a finite sequence  $x = x_0, x_1, \dots, x_m = y$  with  $d(f(x_i), x_{i+1}) < \varepsilon$  for every  $i < m$ . Write  $x \dashrightarrow y$  when such a pseudo-orbit exists for every  $\varepsilon > 0$ ; this is the forward chain relation used in K18-003.

**Definition K18.7. hyperbolic/irrational-elliptic periodic orbit and Anosov diffeomorphism**

A period- $q$  orbit is *hyperbolic* when  $Df^q$  at one (hence every) point of the orbit has no eigenvalue on the unit circle. On an area-preserving surface it is *irrational elliptic* when the two eigenvalues are  $e^{\pm i\alpha}$  with  $\alpha/2\pi \notin \mathbb{Q}$ . A diffeomorphism is *Anosov* when the whole tangent bundle admits a continuous invariant splitting  $TM = E^s \oplus E^u$  with uniform exponential contraction on  $E^s$  and expansion on  $E^u$ .

**Definition K18.8. top Lyapunov exponent and its integral**

For an area-preserving  $C^1$  surface diffeomorphism, Oseledets' theorem gives for almost every  $x$

$$\lambda^+(f, x) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \|Df_x^n\| \geq 0.$$

Its integrated top exponent is  $LE(f) = \int_M \lambda^+(f, x) d\mu(x)$ . Subadditivity gives

$$LE(f) = \inf_{n \geq 1} \frac{1}{n} \int_M \log \|Df_x^n\| d\mu(x),$$

which is the formula used in the generic dichotomy proof.

**Definition K18.9. tiled perturbation box and tiling-preserving jump**

Fix a  $C^1$  perturbation budget and an integer  $N \geq 1$ . A *tiled perturbation box of order  $N$*  is a sufficiently small embedded cube  $B \subset M$  with a finite cubical tiling such that the closures  $\overline{B}, f(\overline{B}), \dots, f^N(\overline{B})$  are pairwise disjoint and the following

operation is available: a finite ordered family of pseudo-orbit jumps for which the point before each jump and the inverse image under  $f$  of the point after it lie in the same tile can, after deleting only bypassed intermediate requests, be realized by one allowed  $C^1$  perturbation supported in  $\bigcup_{j=0}^{N-1} f^j(B)$ . Such a jump is said to *preserve the tiling* of  $B$ .

### Definition K18.10. separated set and topological tower

For an integer  $N \geq 0$ , a set  $U \subset M$  is  $N$ -*separated* when the closures  $\overline{U}, f(\overline{U}), \dots, f^N(\overline{U})$  are pairwise disjoint. Their union is the associated *topological tower of height  $N$* ; perturbations supported on distinct levels of such a tower do not interact spatially.

### Definition K18.11. entrance and exit cores near a hyperbolic periodic orbit

Let  $\gamma$  be a hyperbolic periodic orbit and let  $W(\gamma) \in V(\gamma)$  be nested neighborhoods. An *entrance core* is a compact subset of a tile in a perturbation box near a local stable fundamental domain which every backward crossing into  $W(\gamma)$  is arranged to meet; an *exit core* is the time-reversed object near a local unstable fundamental domain which every forward crossing out of  $W(\gamma)$  is arranged to meet. A finite entrance/exit family is *uniformly connectible* when there is one integer  $n_0$  such that every entrance core has a true forward  $f$ -orbit segment of length at most  $n_0$  reaching every exit core.

### Definition K18.12. realizable derivative sequence

Fix  $f \in \text{Diff}_\omega^1(M)$ , a basic  $C^1$  neighborhood  $\mathcal{U}$  of  $f$ ,  $0 < k < 1$ , and a nonperiodic point  $x$ . A sequence of area-preserving linear maps

$$T_x M \xrightarrow{L_0} T_{f x} M \xrightarrow{L_1} \dots \xrightarrow{L_{n-1}} T_{f^n x} M$$

is  $(k, \mathcal{U})$ -*realizable* if for every derivative tolerance  $\gamma > 0$  there exists  $r > 0$  such that for every nonempty open  $U \subset B_r(x)$  there are  $g \in \mathcal{U}$  and compact  $K \subset U$  with: the sets  $U, f(U), \dots, f^{n-1}(U)$  lie in pairwise disjoint orbit-chart levels and  $g = f$  outside their union;  $\mu(K) > k\mu(U)$ ; and for every  $y \in K$  and  $0 \leq j < n$ , after the canonical orbit-chart identifications,  $Dg_{g^j y}$  is within  $\gamma$  of  $L_j$ .

### Definition K18.13. Bochi nondomination region

On the positive-exponent Oseledets set of an area-preserving surface diffeomorphism define

$$\Delta(x, m) = \frac{\|Df_x^m|_{E^s(x)}\|}{\|Df_x^m|_{E^u(x)}\|}, \quad \Gamma_m(f) = \{x : \Delta(x, m) \geq \tfrac{1}{2}\}, \quad \Omega_m(f) = \bigcup_{j \in \mathbb{Z}} f^j \Gamma_m(f).$$

The set  $\Gamma_m(f)$  is the *length- $m$  nondomination region*;  $\Omega_m(f)$  is its invariant saturation. The complement of  $\Omega_m(f)$  inside the positive-exponent set is the region where the  $m$ -step stable/unstable ratio is uniformly  $< 1/2$  at every iterate.

**Definition K18.14. measurable tower and Kakutani castle**

For an invertible measure-preserving map  $f$ , a measurable set  $A$  with  $A, f(A), \dots, f^{n-1}(A)$  pairwise disjoint determines a *measurable tower* of height  $n$ , with base  $A$ . A *castle* is a finite or countable disjoint union of towers. If  $A$  has almost-everywhere finite first-return time  $\tau_A$ , the partition of  $A$  by the values of  $\tau_A$  and the corresponding return towers form the *Kakutani castle with base  $A$* .

## 18.2 Main results

**Theorem K18.1. Kupka-Smale theorem**

**Statement.** Let  $M$  be a compact smooth manifold without boundary and let  $r \geq 1$ . In the  $C^r$  topology on  $\text{Diff}^r(M)$ , the set of diffeomorphisms  $f$  such that (i) every periodic point of  $f$  is hyperbolic and (ii) for every pair of periodic points  $p, q$ , the manifolds  $W^s(\mathcal{O}(p))$  and  $W^u(\mathcal{O}(q))$  are transverse at every intersection point is residual, meaning it contains a countable intersection of open dense subsets. Such diffeomorphisms are called Kupka-Smale.

*Proof.* The theorem is a countable genericity statement, but the real work is finite. For each pair of integers  $N, R$  we impose two finite-stage conditions: all periodic points of period at most  $N$  are hyperbolic, and all compact stable/unstable pieces obtained from those orbits using at most  $R$  iterates meet transversely. We prove that each finite-stage condition is open and dense, then take a Baire intersection.

*Steps 1-2: exact-period induction and immediate hyperbolization.* We proceed through the least periods  $k = 1, 2, \dots, N$ . Assume that all orbits of smaller least period have already been made hyperbolic. They are finite and protected by disjoint continuation neighborhoods. Hyperbolicity is important here: if such an orbit has least period  $d \mid k$ , then an eigenvalue of  $Df^k$  cannot equal 1, because every eigenvalue of  $Df^d$  has modulus different from one. Thus lower-period points are already isolated transverse zeros of  $f^k - \text{id}$ .

Outside those protected neighborhoods, every solution of  $g^k(x) = x$  has exact period  $k$ . Its  $k$  orbit points are distinct. A postcomposition bump supported near the final output  $x = g^k(x)$  therefore changes  $g^k(x)$  in an arbitrary tangent direction without changing the preceding outputs. After a smooth approximation inside the prescribed  $C^r$  neighborhood, relative parametric transversality makes the graph of  $g^k$  transverse to the diagonal at all exact-period- $k$  points. Equivalently,  $Dg^k - I$  is invertible there, so those points form a finite set.

We hyperbolize these new orbits *before* passing to period  $k + 1$ . In disjoint coordinate balls along one orbit, a perturbation can fix the orbit points while changing one derivative by any small matrix. Hence the return matrix can be perturbed arbitrarily in  $GL(d, \mathbb{R})$ . Real Schur form shows that hyperbolic matrices are dense: perturb the real  $1 \times 1$  and complex-pair  $2 \times 2$  diagonal blocks slightly so that every eigenvalue has modulus different from one. Because the existing short-period zeros are isolated and the displacement  $d(g^j x, x)$  is uniformly positive away from their continuation neighbor-

hoods, a sufficiently small perturbation creates no new periods  $j \leq k$ . Induction gives an open dense set on which every period- $\leq N$  orbit is hyperbolic and finite in number.

*Step 3: compact pieces of the invariant manifolds.* The finitely many hyperbolic short orbits persist under nearby maps, as do small local stable and unstable disks. Their global manifolds are unions of forward or backward iterates of these local disks. Fixing an iterate bound  $R$  leaves only finitely many *pairs of compact parameterized disks*. A single pair may contain infinitely many intersections, so those intersections cannot be perturbed one at a time.

For one pair write  $S_g(a)$  and  $U_g(b)$  for its stable and unstable parameterizations and consider

$$(g, a, b) \mapsto (S_g(a), U_g(b)) \in M \times M.$$

The two disks meet transversely exactly when this map, for fixed  $g$ , is transverse to the diagonal.

*Step 4: the free-iterate perturbation.* Suppose  $S_f(a) = U_f(b) = z$ . If  $z$  is on the periodic orbit, hyperbolicity already gives complementary stable and unstable tangent spaces. Otherwise  $z$  is nonperiodic. Move far enough backward along the unstable branch to find an orbit leg disjoint from the finitely many pieces used to define the stable disk and from all protected periodic orbits. A bump perturbation on that free leg moves the unstable disk at  $z$  in any prescribed normal direction while leaving the stable disk and periodic jets unchanged. A finite cover of the compact coincidence set gives finitely many such perturbation directions, so the universal disk-intersection map is transverse to the diagonal.

Smooth finite-dimensional parametric transversality then supplies a nearby map for which the entire compact disk product is transverse. Repeat for the finitely many disk pairs. Compactness and  $C^1$  hyperbolic continuation make the resulting condition open.

*Step 5: countable assembly.* Intersect these open dense finite-stage sets over all  $N, R \geq 1$ . Baire's theorem makes the intersection residual. Every periodic orbit has some finite period, and every global stable/unstable intersection lies in some finite iterate piece, so one finite stage detects each object. Hence every periodic point is hyperbolic and every stable/unstable intersection is transverse. This is exactly the Kupka–Smale property.  $\square$

### Lemma K18.2. Pugh closing lemma

**Statement.** Let  $M$  be a compact smooth manifold,  $f \in \text{Diff}^1(M)$ ,  $x \in \Omega(f)$  a non-wandering point,  $\mathcal{U}$  a  $C^1$  neighborhood of  $f$ , and  $V$  a neighborhood of  $x$ . Then there exists  $g \in \mathcal{U}$  having a periodic point  $p \in V$ .

*Proof.* Write  $m = \dim M$  and fix a Riemannian metric. If  $x$  is already periodic, there is nothing to prove: take  $g = f$ . We therefore assume that  $x$  is not periodic. Choose a coordinate neighborhood  $W$  of  $x$  with  $\overline{W} \subset V$ .

The obstacle in the closing lemma is not the lack of a near return. The nonwandering hypothesis gives those immediately. The obstacle is that a bump used to correct

a return may hit some intermediate iterate of the same orbit, so that later corrections undo earlier ones. Pugh's proof solves this by choosing the *shape* of the perturbation box from the derivative growth, then choosing a return whose intermediate orbit avoids that entire anisotropic box.

**Local lifts.** Fix a smaller  $C^1$  neighborhood of  $f$  whose closure lies in  $\mathcal{U}$ . Compactness of  $M$  gives constants  $c_0, \rho_0 > 0$  with the following uniform property. If  $y \in M$ ,  $0 < \rho < \rho_0$ , and  $w \in T_y M$  satisfies  $|w| \leq c_0 \rho$ , then a diffeomorphism supported in  $B(y, \rho)$  can move  $y$  to  $\exp_y(w)$  while remaining as  $C^1$ -close to the identity as we please. In a normal chart this is simply

$$u \mapsto u + \chi(u/\rho)w,$$

with a fixed bump  $\chi$ . The derivative error is bounded by  $C|w|/\rho$ . We choose  $c_0$  so that this error lies inside the prescribed perturbation budget.

If several such local maps have pairwise disjoint supports, their composition has  $C^1$  size controlled by the largest local size rather than by the sum. Thus the proof reduces to finding finitely many useful correction sites with disjoint supports.

**Selecting derivative scales.** For an injective linear map  $T$  let

$$m(T) = \inf_{|v|=1} |Tv|, \quad \text{bol}(T) = \frac{\|T\|}{m(T)}.$$

The second quantity measures eccentricity. Given a subspace  $L$ , project  $T(L)$  orthogonally away from  $T(L^\perp)$ ; the resulting map on  $L$  is called the  $L$ -altitude. Replacing  $T|_L$  by this altitude makes the images of  $L$  and  $L^\perp$  orthogonal. This is the  $L$ -orthogonalization.

Use polar decomposition on the sequence  $D_x f^n$  and pass to a subsequence on which all ratios of singular values converge. The slowest equivalence class of singular values has a moving eigenspace  $E_n$ ; after another subsequence  $E_n$  converges to a fixed plane  $V^1$ . If  $\phi_n$  is the angle from  $E_n$  to  $V^1$ , the slow block has bounded eccentricity, the complementary singular scale dominates it by a factor  $h_n \rightarrow \infty$ , and elementary projection geometry gives

$$\text{bol}(A_n^1) = O(1), \quad \frac{\|A_n^1\|}{m((A_n^1)^\perp)} \rightarrow 0.$$

The second estimate comes from comparing the cone of angle  $\phi_n$  around  $E_n$  before and after the two singular scales are made conformal; its transformed angle satisfies  $\tan \psi_n = h_n \text{bol}(P_n|_{E_n}) \tan \phi_n$ .

One must not simply take limits of all the remaining singular eigenspaces: those limits need not be the correct altitude blocks. Instead apply the same argument recursively to the complementary altitude  $V^{1\perp} \rightarrow (D_x f^n V^1)^\perp$ . Nested orthogonal projections identify the altitudes obtained recursively with the altitudes for the original derivative. This produces an orthogonal splitting

$$T_x M = V^1 \oplus \dots \oplus V^\ell$$

whose block altitudes have one common eccentricity bound and whose adjacent scales separate asymptotically.

A second induction gives the finite-window rescaling used by Pugh. First make the construction on  $V^2 \oplus \cdots \oplus V^\ell$ . Then enlarge the scalar on  $V^1$  enough that the already chosen later-block windows retain their hyperbolicity. Once this scalar is fixed, asymptotic separation makes  $V^1$  itself arbitrarily dominated sufficiently far out, so choose its window beyond the previous ones. Relabeling the reverse-order construction yields a blockwise dilation  $\Delta$  and ordered windows on which each active block is dominated by the complement by an arbitrarily large factor. We choose each window to contain  $N \dim V^i$  iterates and then fix  $N$  sufficiently large in terms of the local lift constant and the common altitude-eccentricity bound. Altogether there are  $mN$  selected times

$$0 < t(1) < \cdots < t(mN) = T,$$

and each coordinate direction occurs exactly  $N$  times.

**The fundamental return box.** Choose an orthonormal basis  $e_1, \dots, e_m$  subordinate to the splitting and write the block dilation as  $\Delta e_j = \delta_j e_j$ . We now construct a return box whose side lengths have the same proportions as the  $\delta_j$ .

For positive weights  $\xi_j$  define

$$d_\xi(u, v)^2 = \sum_{j=1}^m \left( \frac{u^j - v^j}{\xi_j} \right)^2.$$

Nonwandering gives arbitrarily small ordered return pairs  $u < v$  in the exponential chart at  $x$ . Starting from one such pair, if an intermediate orbit point lies within  $\sqrt{3/4}$  times the present weighted separation of either endpoint, replace the farther endpoint by that intermediate point. The weighted separation decreases by a fixed factor, and only finitely many points lie between the original endpoints, so the procedure terminates. At termination no intermediate orbit point lies in either of those two weighted neighborhoods.

Set  $r = (u + v)/2$ , let  $d = d_\xi(u, v)$ , and define

$$\lambda_j = \xi_j \left( \frac{1}{3} \left( \frac{u^j - v^j}{\xi_j} \right)^2 + \frac{d^2}{8m} \right)^{1/2}.$$

A direct calculation gives two facts. First,

$$\frac{\lambda_i}{\lambda_j} < 2\sqrt{m} \frac{\xi_i}{\xi_j}.$$

Second, the two return endpoints lie in the  $\sqrt{3/4}$ -core of the box

$$B = \exp_x(r + \Lambda[-1, 1]^m), \quad \Lambda e_j = \lambda_j e_j,$$

and the whole outer box lies inside the union of the two excluded weighted neighborhoods. Thus no intermediate point of the return orbit lies in  $B$ .

Take  $\xi_j = \delta_j$ . The diagonal map  $\Lambda^{-1}\Delta$  then has uniformly bounded eccentricity. The return can be made arbitrarily small. Since  $x$  is not periodic, its return time can also

be made arbitrarily large; otherwise a bounded subsequence of return times would force  $f^n x = x$  for some fixed  $n$ . We therefore arrange that the return has time  $s > T$ , that both endpoints lie in  $W$ , and that

$$B, fB, \dots, f^T B$$

are pairwise disjoint.

**The trial orbit.** At the  $n$ th selected time let  $J_n = D_x f^{t(n)}$ . Orthogonalize  $J_n$  along the active block and call the result  $S_n$ . A small multiple of  $S_n \Lambda e(n)$  is the physical lift direction at that stage, where  $e(n)$  is the coordinate direction assigned to the selected time. Give each of the  $mN$  lifts a parameter between  $-1$  and  $1$  and follow the perturbed orbit recursively from the second endpoint of the near return.

Normalize every selected neighborhood by pulling it back to the original return box and then applying  $\Lambda^{-1}$ . Strong domination has one precise role: in block form, orthogonalization deletes an off-diagonal term whose inverse effect is bounded by the reciprocal domination ratio. The selection estimate is first obtained for  $J_n \Delta$ . Since the fundamental box has bounded eccentricity relative to  $\Delta$ , conjugating by  $\Lambda^{-1} \Delta$  transfers the same estimate to  $J_n \Lambda$ :

$$(J_n \Lambda)^{-1} S_n \Lambda \rightarrow I$$

uniformly on the active block. Shrinking the return box then makes the nonlinear branches uniformly close to their derivatives. Since there are only  $mN$  stages, the normalized trial position satisfies

$$Q_n = Q_0 + \kappa \sum_{j=1}^n \theta_j e(j) + E_n, \quad \max_n |E_n| \rightarrow 0,$$

uniformly in all parameters.

The same estimates control supports. Bounded altitude eccentricity and the bounded eccentricity of  $\Lambda$  on each selected block imply that the support radius needed for each lift is only a fixed small fraction of the surrounding image box. Hence, whenever the limiting affine walk stays in a fixed inner core, the actual lift is supported inside  $f^{t(n)} B$ . Those supports are pairwise disjoint because the boxes  $B, fB, \dots, f^T B$  are.

**Closing the endpoint by degree.** Now group the  $mN$  parameters: all  $N$  lifts assigned to the  $j$ th coordinate receive the same parameter  $\sigma_j$ . The limiting endpoint becomes

$$\sigma \mapsto Q_0 + D\sigma, \quad D = \text{diag}(d_1, \dots, d_m),$$

and in fact  $D = N\kappa I$ : every coordinate receives  $N$  equal normalized increments. The fixed integer  $N$  was chosen so large that  $N\kappa$  exceeds the width needed to cross the inner return box.

Choose an intermediate cube  $a_2[-1, 1]^m$  strictly containing the core that contains the first endpoint, and let

$$\mathcal{A} = \{\sigma : Q_0 + D\sigma \in a_2[-1, 1]^m\}.$$

The affine endpoint map is a homeomorphism from  $\mathcal{A}$  onto that cube. The true endpoint



map is uniformly close to it. On the boundary of  $\mathcal{A}$ , the straight-line homotopy between the two maps avoids the smaller core. Their topological degrees about every point in the smaller core are therefore equal and nonzero. Consequently the true endpoint map covers the smaller core. The normalized coordinate of the first return endpoint lies there, so some parameter choice makes the trial endpoint exactly equal to  $f^T$  of that first endpoint.

Fix those parameters and compose the local lifts. Their supports are disjoint, so the resulting map  $g$  lies in  $\mathcal{U}$ . By construction the perturbed orbit joins the second return endpoint to  $f^T$  of the first. The original orbit then runs from  $f^T$  of the first endpoint back to the second endpoint. For each later image point  $f^j a$  with  $T < j \leq s$ , an intersection with a support in  $f^{t(n)}B$  would pull back to the intermediate point  $f^{j-t(n)}a \in B$ , where  $0 < j - t(n) < s$ , which the fundamental return box excluded. Since the perturbation is composed after each application of  $f$ , these later image points are precisely the ones relevant to the tail. Hence that tail is unchanged by the perturbation. The two pieces concatenate to a periodic  $g$ -orbit through the second return endpoint. That point lies in  $W \subset V$ , completing the proof.  $\square$

### Lemma K18.3. Bonatti-Crovisier connecting lemma for pseudo-orbits

**Statement.** Let  $M$  be a compact connected Riemannian manifold and let  $f \in \text{Diff}^1(M)$  have only hyperbolic periodic orbits. Let  $\mathcal{U}$  be a  $C^1$  neighborhood of  $f$ . If  $x, y \in M$  are chain-related in the forward sense, meaning that for every  $\varepsilon > 0$  there is a finite  $\varepsilon$ -pseudo-orbit from  $x$  to  $y$ , then there exist  $g \in \mathcal{U}$  and an integer  $n > 0$  such that  $g^n(x) = y$ .

#### Scope note

The conservative variant and the surface elliptic exception require the corresponding conservative hypotheses and are not part of the displayed dissipative statement.

*Proof.* Write  $x \dashrightarrow y$  for the forward chain relation and fix the prescribed  $C^1$  neighborhood  $\mathcal{U}$  of  $f$ . The aim is to turn one sufficiently accurate pseudo-orbit from  $x$  to  $y$  into a true orbit of one nearby map. Unlike the closing lemma, this is not a one-return problem: a pseudo-orbit may contain many jumps, and the main difficulty is to make all local corrections without allowing their supports to interfere.

**1. A uniform local perturbation box.** First choose a very small neighborhood  $\mathcal{V}$  of the identity so that any finite family of maps in  $\mathcal{V}$  with pairwise disjoint supports may be composed with  $f$  while staying in  $\mathcal{U}$ .

There is a uniform local lift. If  $p$  and  $q$  lie in a coordinate ball of radius  $r$  and  $d(p, q)$  is a sufficiently small fixed multiple of  $r$ , a map  $h \in \mathcal{V}$  supported in that ball can send  $p$  to  $q$ . One obtains it as the time-one map of a small compactly supported vector field. In volume coordinates the vector field can be chosen divergence free, so the same local move is available in the conservative category.

Now take a small cube  $B$  whose first  $N$  iterates are disjoint and subdivide  $B$  into

small tiles. A requested jump through one tile can be spread over successive levels

$$B, f(B), \dots, f^{N-1}(B).$$

Uniform bounds on  $Df$  and  $Df^{-1}$  let us make every stage displacement a small fixed fraction of its support radius. Enlarged equal-size tiles have a bounded intersection degree depending only on  $\dim M$ . Color their overlap graph, use different tower levels for different colors, and repeat over a finite nested subdivision. If two requested connections compete for the same support, keep the earlier entrance and later exit and delete the intervening request. This strictly shortens the list. For one sufficiently large  $N$ , uniform over the compact manifold, all retained local moves therefore have disjoint supports. Their composition realizes the requested concatenation inside the tower. This is the perturbation-box lemma.

**2. A topological tower.** The next problem is global: we need finitely many perturbation boxes that every relevant orbit can reach. Fix a tower height  $N_0$  much larger than  $N$ . For each bounded period there are only finitely many periodic points, since the periodic points are hyperbolic and hence isolated.

A finite coloring argument produces an open set  $U$  such that

$$\overline{U}, f(\overline{U}), \dots, f^{N_0}(\overline{U})$$

are pairwise disjoint and every orbit not trapped in the local stable set of a short periodic orbit meets  $U$  forward; the time-reversed statement holds outside the corresponding local unstable sets. Here is the mechanism. Put boundaries of finitely many small sets in general position, so at most  $\dim M$  of them meet at one point. The complementary components can then be colored with a dimension-dependent finite number of colors. Shift components with different colors by different multiples of  $2N_0$ . The forbidden colors are chosen precisely so that no shifted component can meet one of its first  $N_0$  iterates. Iterating this extension over a finite compact cover gives the desired tower.

Tile several widely separated time shifts of a compact core of  $U$ . Put all tile boundaries in general position. Since only finitely many boundaries can pass through one point, among sufficiently many shifted tilings every point lies well inside some tile. A finite subcover gives a finite family of perturbation boxes whose support towers are pairwise disjoint. Denote the union of their smaller compact tile cores by  $D$ .

**3. Protect the short hyperbolic periodic orbits.** A pseudo-orbit may spend a long time near a short periodic orbit and thus avoid  $D$ . Around each such hyperbolic orbit choose local stable and unstable fundamental domains. Tile small neighborhoods of those domains by additional perturbation boxes. They play the roles of entrance and exit boxes.

The stable-manifold theorem gives the local disks. If  $E$  is an entrance core and  $S$  an exit core, the inclination lemma says that forward iterates of a disk transverse to the local stable manifold eventually contain disks close to the local unstable manifold. Hence a true orbit segment starting in  $E$  reaches  $S$ . There are only finitely many pairs of entrance and exit cores, so all such connecting times are bounded by one number. Shrinking the periodic neighborhoods gives the following useful dichotomy: an orbit

which crosses such a neighborhood must meet an entrance core and an exit core, while an orbit staying away from the protected neighborhoods must return to  $D$  in uniformly bounded time.

**4. Decompose a fine pseudo-orbit into bounded blocks.** There are two endpoint nuisances. If  $y$  already belongs to the positive orbit of  $x$ , nothing has to be done. If  $x$  or  $y$  lies on a protected short periodic orbit, take pseudo-orbits with errors tending to zero and examine the first exit or last entrance from a small isolating neighborhood. A limiting point lies on the local unstable or stable fundamental domain and retains the required chain relation. We connect that nearby point and use one final tiny local adjustment, supported away from the constructed orbit segment, to restore the original endpoint. If an endpoint lies in a perturbation support, a bounded iterate leaves all supports; again we work with that iterate and restore the endpoint afterward.

Now shrink all compact cores. Compactness and the tower property give a number  $n_1$  such that a true orbit starting outside a protected periodic neighborhood reaches  $D$  within  $n_1$  iterates unless it enters one of those neighborhoods. Choose the pseudo-orbit error so small that every pseudo-orbit tracks true dynamics for these finitely many iterates. Mark successive visits to the cores. Every block between two marks then has one of two forms:

- it has length at most  $n_1$ ;
- it crosses one protected periodic neighborhood, beginning in an entrance core and ending in an exit core.

**5. Regroup the jumps so that every one lies inside one tile.** For a short block, replace the initial part by the true forward orbit of its first marked point and the final part by the true backward orbit of its last marked point. The whole block now contains at most one jump. Because the length is uniformly bounded and the pseudo-orbit error was chosen very small, the two sides of that jump lie in the same tile.

For a long block crossing a protected periodic neighborhood, use the bounded entrance-to-exit orbit segment obtained above. Replace the entire crossing by that true orbit. Only the two endpoint jumps remain, and they lie in the same entrance and exit tiles respectively. After doing this for every block, we obtain a finite pseudo-orbit from  $x$  to  $y$  with two decisive properties: every jump lies in a perturbation box and every jump preserves that box's tiling.

**6. Eliminate the jumps box by box.** List the finitely many boxes as  $B_1, \dots, B_s$ . Apply the perturbation-box lemma to all jumps through  $B_1$ . If several requests compete, its shortcut operation keeps the required entrance and exit and deletes the bypassed part of the pseudo-orbit. The perturbation is supported only in the tower of  $B_1$  and removes every jump there.

Proceed through  $B_2, \dots, B_s$ . The support towers are pairwise disjoint, so later perturbations cannot undo an earlier connection and cannot change the tiling geometry in any untouched box. After the last box there is no jump. The resulting sequence is therefore a genuine orbit of one map  $g \in \mathcal{U}$ , and for some  $n > 0$ ,

$$g^n(x) = y.$$

This proves the connecting lemma in the unrestricted category.

**7. Conservative maps and the irrational-elliptic surface case.** In the volume-preserving category all global tower and tiling arguments are unchanged. The local lifts are chosen as time-one maps of divergence-free vector fields in volume coordinates, hence their composition remains volume preserving.

On a conservative surface a short periodic orbit may instead be irrational elliptic. Let  $p$  have period  $q$  and put  $F = f^q$ . In an area chart the linear map  $DF_p$  is conjugate to an irrational rotation. Choose a short arc  $J$  on a small adapted circle. Minimality of an irrational rotation and compactness of the circle give a finite time  $m_0$  such that every point hits  $J$  within  $m_0$  forward iterates, and likewise backward. After rescaling a small annulus around  $p$ , the map  $F$  converges in  $C^1$  to this rotation, so the same finite hitting property holds on sufficiently small annuli for  $F$ . The thickened arc becomes a perturbation box which every orbit crossing the annulus must visit. These elliptic boxes can be chosen first and disjointly; on the invariant compact set outside them, all remaining short periodic orbits are hyperbolic and the preceding tower construction applies. Thus the same jump-elimination proof gives the conservative-surface variant stated in the theorem.  $\square$

#### Theorem K18.4. Bochi generic zero-exponent dichotomy

**Statement.** Let  $M$  be a compact surface with a smooth area form. There is a residual subset of  $\text{Diff}_\omega^1(M)$  such that every  $f$  in this residual set satisfies exactly one of the following alternatives:  $f$  is Anosov; or the two Lyapunov exponents of  $f$  vanish for area-almost every point of  $M$ .

*Proof.* Write  $\mu$  for normalized area and

$$LE(f) = \int_M \lambda^+(f, x) d\mu(x).$$

For an area-preserving surface map the two Lyapunov exponents are  $\lambda^+$  and  $-\lambda^+$ . Bochi's proof has two very different scales. At the microscopic scale one perturbs derivatives along long orbit segments so that an expanding direction is turned into a contracting one. At the macroscopic scale one places these orbit segments into a Kakutani tower so that the local perturbations occupy most of the surface without overlapping. The final Baire argument is short only after both mechanisms have been built.

**1. The integrated exponent is upper semicontinuous.** For  $a_n(f) = \int_M \log \|Df_x^n\| d\mu(x)$ , the cocycle inequality gives  $a_{n+m} \leq a_n + a_m$ . Oseledets' theorem and dominated convergence therefore give

$$LE(f) = \inf_{n \geq 1} \frac{1}{n} \int_M \log \|Df_x^n\| d\mu(x). \quad (18.4.1)$$

For a fixed  $n$  the right-hand finite-time integral varies continuously in the  $C^1$  topology. Hence  $LE$  is upper semicontinuous.

**2. A dense class on which perturbation is possible.** The proof first approximates

any  $C^1$  area-preserving map by a smooth one without losing area preservation. Start with a smooth approximation  $h$  and put  $\beta = h^*\omega$ . Since  $\beta$  and  $\omega$  have the same total area,  $\beta - \omega = d\alpha$ . Along  $\beta_t = \omega + t(\beta - \omega)$  solve  $\iota_{X_t}\beta_t = -\alpha$ . The flow  $\psi_t$  satisfies  $\psi_t^*\beta_t = \omega$ , so  $h \circ \psi_1$  is a smooth conservative approximation.

Inside the smooth conservative class one can make all periodic points nondegenerate. At an exact-period- $q$  point, a Hamiltonian bump supported near one point of the orbit changes  $g^q(x)$  in an arbitrary tangent direction while preserving area. Parametric transversality therefore makes the graph of  $g^q$  transverse to the diagonal. There are then only finitely many period- $q$  points, for each  $q$ , and hence only countably many periodic points in total. Such a map is aperiodic in the measure-theoretic sense.

There is one more preparation fact. For a  $C^2$  conservative surface map, a hyperbolic set of positive area forces the whole map to be Anosov. The reason is geometric. Stable and unstable holonomies of a uniformly hyperbolic set have bounded Jacobians: the logarithm of the holonomy Jacobian is a convergent sum of differences of unstable Jacobians along exponentially converging orbits. At a recurrent density point of a positive-area hyperbolic set, absolute continuity and local product structure therefore fill a smaller stable-unstable rectangle. A locally maximal hyperbolic extension has interior; local product structure makes that interior also closed, and connectedness forces it to be the entire surface. Thus a smooth non-Anosov map in the prepared class has only zero-area hyperbolic sets. If the prepared map is Anosov, the Hopf argument gives ergodicity and hence every invariant hyperbolic set has area zero or one.

We now fix a prepared non-Anosov map  $f$ , a small conservative neighborhood  $\mathcal{U}$ , and a target exponent error  $\delta > 0$ .

**3. Realizable derivative sequences.** A realizable sequence is a prescribed finite sequence of area-preserving linear maps along an orbit which an actual  $C^1$  conservative perturbation can match, to arbitrary derivative accuracy, on a fixed large proportion of every sufficiently small base disk. The perturbation is supported only on the corresponding disjoint orbit levels.

The basic local device is a small rotation. In an area chart write

$$h(\rho, \theta) = (\rho, \theta + \alpha F(\rho/r)),$$

with  $F = 1$  on a smaller concentric disk and  $F = 0$  outside the radius- $r$  disk. It preserves area, is an exact rotation on most of the disk, and becomes arbitrarily  $C^1$ -small with  $\alpha$ . To use many rotations without losing a fixed fraction of the base at every step, the supports are nested ellipses: each local map preserves every concentric copy of its ellipse, so one common inner core survives an arbitrarily long string of rotations.

**4. Where domination fails, rotate unstable into stable.** At a positive-exponent regular point define

$$\Delta(x, m) = \frac{\|Df_x^m|_{E^s(x)}\|}{\|Df_x^m|_{E^u(x)}\|}, \quad \Gamma_m = \{x : \Delta(x, m) \geq \tfrac{1}{2}\}.$$

For a sufficiently large fixed  $m$ , every point of  $\Gamma_m$  admits a realizable length- $m$  product sending  $E^u(x)$  exactly to  $E^s(f^m x)$ . There are three cases. If the stable and unstable lines become nearly parallel, one tiny rotation suffices. If some intermediate derivative

makes the stable vector much longer than the unstable vector, two small endpoint rotations use that imbalance to make the transfer. In the remaining case the angles stay bounded below and all stable/unstable norm ratios stay in a compact interval. Area preservation then bounds every partial derivative product. One distributes the total angle from  $E^u$  to  $E^s$  among many tiny conjugated rotations; the nested-ellipse construction realizes the entire string with one fixed loss of area.

The complement of the orbit-saturation of  $\Gamma_m$  inside the positive exponent set is uniformly hyperbolic. Indeed, there  $\Delta(f^j x, m) < 1/2$  at every iterate, so block multiplication gives an exponential stable/unstable ratio. The angle is uniformly bounded from zero, and determinant one converts the ratio estimate into uniform contraction and expansion. The splitting extends continuously to the closure. Our prepared non-Anosov map has no positive-area hyperbolic set, so this complement has area zero.

**5. Put the mixing block near the middle of a long orbit.** Birkhoff's theorem says that almost every orbit in the saturation of  $\Gamma_m$  visits  $\Gamma_m$  with positive asymptotic frequency. Hence, for every sufficiently long orbit segment and every prescribed relative time, there is a visit to  $\Gamma_m$  arbitrarily close to that relative time. Take a visit close to the midpoint.

Before and after that midpoint, Oseledets' theorem says that the unstable and stable directions grow like  $e^{\pm\lambda n/2}$ , up to a small exponential error. Insert the length- $m$  mixing block at the midpoint. Because that block sends the incoming unstable line into the outgoing stable line, the large first-half expansion is followed by second-half contraction instead of another expansion. In Oseledets bases the unstable-to-unstable matrix entry of the mixing block is zero. Multiplying the three matrices therefore cancels the  $\lambda/2$  growth from the two halves, while the midpoint error, angle error, and fixed mixing-block norm contribute only the chosen small exponential error. For almost every  $x$  and all sufficiently large  $n$  one obtains a realizable product with

$$\|L_{n-1} \cdots L_0\| < e^{(4/5)n\delta}. \quad (18.4.2)$$

Choosing the derivative-realization accuracy sufficiently small turns this into an actual perturbation for which  $\|Dg_y^n\| < e^{n\delta}$  on more than a fixed fraction  $k$  of a small base disk.

**6. From local orbit segments to one global perturbation.** The local construction must now be used on most of the surface without support collisions. Choose  $H$  so large that the points whose required orbit length exceeds  $H$  have very small area. Inside the remaining set take a maximal base whose first  $H$  iterates are disjoint, and form its Kakutani return-time castle. Discarding towers much taller than  $H$  costs only a small amount of area; the retained towers have heights between  $H$  and  $3H$ .

Approximate their measurable bases from inside by compact sets and cover those compact sets, up to small area loss, by finitely many disjoint tiny disks. On each disk apply the preceding long-orbit perturbation. Different tower levels are disjoint, so all local perturbations patch to one area-preserving  $g \in \mathcal{U}$ . There is a subcastle  $K$  occupying almost all of the retained castle such that every complete return of length  $n$  based in  $K$  satisfies

$$\|Dg^n\| < e^{\delta n}. \quad (18.4.3)$$

**7. Estimate the integrated exponent in finite time.** Set  $N = \delta^{-1}H$  and let  $G$  consist of points whose first  $N$  iterates stay inside the good subcastle. The castle boundary floors, discarded towers, base-approximation errors, and the small set lost in realizability give

$$\mu(G^c) < 16\delta.$$

For  $x \in G$ , the length- $N$  segment is a concatenation of complete good returns plus initial and terminal scraps of total length below  $6H$ . If  $C$  bounds the derivative in the perturbation neighborhood, then

$$\frac{1}{N} \log \|Dg_x^N\| < (6 \log C + 1)\delta.$$

On  $G^c$  we use only the crude bound  $N^{-1} \log \|Dg^N\| \leq \log C$ . The finite-time formula (18.4.1) yields

$$LE(g) < (22 \log C + 1)\delta. \quad (18.4.4)$$

Thus every prepared non-Anosov map can be perturbed to have arbitrarily small integrated top exponent.

**8. Baire category.** Let  $\mathcal{A}$  be the open set of Anosov maps and define

$$\mathcal{O}_r = \mathcal{A} \cup \{f : LE(f) < 1/r\}.$$

Upper semicontinuity makes every  $\mathcal{O}_r$  open. It is dense because any open set contains a prepared map: if that map is Anosov we are already in  $\mathcal{A}$ , and otherwise the perturbation just proved gives a nearby map with  $LE < 1/r$ . Hence

$$\mathcal{R} = \bigcap_{r \geq 1} \mathcal{O}_r$$

is residual. If  $f \in \mathcal{R}$  is not Anosov, then  $LE(f) = 0$ . Since  $\lambda^+ \geq 0$ , this forces  $\lambda^+ = 0$  almost everywhere, and area preservation gives  $\lambda^- = -\lambda^+ = 0$  almost everywhere. An Anosov map has exponents uniformly bounded away from zero, so the alternatives are mutually exclusive.  $\square$

## 18.3 Worked examples

### Worked example: transversality in a planar saddle connection

Consider the linear saddle

$$L(x, y) = (2x, \tfrac{1}{2}y).$$

Its stable and unstable axes meet transversely at the fixed point because their tangent directions span  $\mathbb{R}^2$ . Now imagine a smooth global diffeomorphism containing two hyperbolic periodic orbits whose stable and unstable curves have a quadratic tangency. In local coordinates the two curves may be written

$$y = 0, \quad y = x^2 + \eta.$$

At  $\eta = 0$  they are tangent. For  $\eta < 0$  there are two transverse intersections and for  $\eta > 0$  none. The one-parameter displacement illustrates why tangency is codimension one and why a small perturbation can restore transversality.

#### Check your understanding

At the intersections for  $\eta < 0$ , compute the slopes of the second curve and verify that they differ from the slope of  $y = 0$ .

#### Boundary case: neutral periodic points

The identity map has every point periodic and every derivative equal to the identity. It is maximally far from Kupka-Smale: no periodic point is hyperbolic. A small perturbation can create hyperbolic or elliptic periodic points, but the theorem does not say the original map itself has generic properties. This separates a residual statement about the ambient function space from a statement about each individual system.

#### Common pitfall

In this chapter, “generic” means residual in the stated  $C^1$  function space, not probability-one for some unstated measure. Also, the closing and connecting lemmas produce the desired orbit structure for a nearby perturbed diffeomorphism; they do not assert that the original map already has the periodic orbit or connection.

## 18.4 Chapter summary

#### What to remember

- Kupka-Smale genericity removes nonhyperbolic periodic points and non-transverse invariant-manifold intersections on a residual set.
- The closing lemma converts local recurrence into a nearby periodic orbit by a small  $C^1$  perturbation.
- The connecting lemma upgrades arbitrarily accurate pseudo-orbit connections to true connections for a nearby diffeomorphism.
- Generic statements are Baire-category statements and must not be confused with probabilistic typicality.
- In the conservative surface category, Bochi’s theorem gives a sharp generic alternative: Anosov or zero Lyapunov exponents almost everywhere.



**Looking ahead**

K19 returns to KAM theory in an infinite-dimensional setting. The perturbative philosophy is familiar, but infinitely many normal modes, unbounded operators, second Melnikov divisors and quasilinear derivative loss require a substantially stronger analytic machine.

**Check your understanding**

For each of the closing lemma, connecting lemma, Kupka-Smale theorem, and Bochi's conservative-surface dichotomy, specify whether the conclusion concerns the original map, a nearby perturbation, or a residual subset of the ambient function space.

**18.5 Exercises and problems****Exercise 18.1**

**Difficulty.** Core.

**Related results.** Kupka-Smale theorem; Pugh closing lemma; Bonatti-Crovisier connecting lemma for pseudo-orbits.

**Problem.** For the Morse-Smale model, compute periodic derivatives, verify hyperbolicity, perturb one stable/unstable intersection to transversality, and distinguish this fixed-map calculation from Pugh closing for a nearby map.

**Solution sketch.** Compute tangent spaces before/after the local perturbation and track the map symbol whenever closing is invoked.

**Exercise 18.2**

**Difficulty.** Core.

**Related results.** Bochi generic zero-exponent dichotomy.

**Problem.** Treat separately a tangency and a neutral periodic point: identify the violated Kupka-Smale condition and give a small local perturbation removing the degeneracy.

**Solution sketch.** Compare stable/unstable tangent spaces for the tangency and derivative eigenvalues for the neutral orbit.

**Exercise 18.3**

**Difficulty.** Advanced.

**Related results.** Bonatti-Crovisier connecting lemma for pseudo-orbits.

**Problem.** In a hyperbolic linear model, prove a finite pseudo-orbit shadowing/connection statement using local product structure, then explain what extra perturbative content the Bonatti-Crovisier connecting lemma adds.

**Solution sketch.** Replace jumps by bracket points and control errors by contraction; the general theorem changes the map when direct shadowing is unavailable.

**Exercise 18.4**

**Difficulty.** Advanced.

**Related results.** Bochi generic zero-exponent dichotomy.

**Problem.** Explain why a residual/generic Lyapunov-exponent dichotomy is not a theorem about every map. Compute exponents for a hyperbolic toral automorphism and for an identity/isometry example.

**Solution sketch.** Calculate derivative growth in both cases and then restore the exact topology and residual-set quantifiers of the theorem.

**Exercise 18.5**

**Difficulty.** Advanced.

**Related results.** Kupka-Smale theorem; Pugh closing lemma.

**Problem.** Combine closing and Kupka-Smale carefully to state an approximation-by-hyperbolic-periodic-data conclusion for nearby maps, and prove why this does not imply periodic points are dense for an arbitrary original map.

**Solution sketch.** Apply closing first, then a transversality/generic perturbation, keeping each perturbed map distinct.

## **Part VI**

# **Infinite-Dimensional KAM and Billiard Rigidity**

# Infinite-Dimensional and PDE KAM

## Why this chapter matters

Finite-dimensional KAM theory preserves quasiperiodic invariant tori by solving homological equations with controlled small divisors. In a Hamiltonian PDE the same idea faces two new difficulties at once: there are infinitely many normal frequencies, and the linearized operator may be unbounded or depend on the solution itself. A successful PDE-KAM theorem must therefore control arithmetic, regularity and operator structure simultaneously.

This volume begins with Pöschel's analytic infinite-dimensional theorem as a model architecture. It then shows how reversibility and reducibility handle derivative nonlinear wave equations, how special tangential-site separation controls resonances for a quintic wave equation, and how modern quasilinear KAM/Nash-Moser machinery reaches gravity-capillary water waves and vortex patches. These examples are deliberately different: the hypotheses reveal which part of the abstract KAM mechanism is doing the real work in each PDE.

**Prerequisites.** K05–K08 for finite-dimensional KAM, parameter exclusion and Melnikov conditions; Foundation F07–F08 for analytic/Sobolev scales and cohomological estimates; F12 for Newton/Nash-Moser and perturbative reducibility.

## Learning goals

- Formulate a PDE invariant torus as an embedding with finitely many tangential and infinitely many normal directions.
- Explain the role of first and second Melnikov divisors and why infinitely many resonance conditions still leave a large Cantor set.
- Distinguish semilinear KAM from quasilinear/reducibility problems where the highest-order operator changes along the iteration.
- Follow the proof architecture for reversible derivative waves, quintic waves, gravity-capillary water waves and vortex patches.
- Interpret reducibility as the source of linear stability for a quasiperiodic PDE torus.

## Conceptual roadmap

Every theorem in this volume fits into the same infinite-dimensional KAM template, but different PDEs make different steps difficult.

- Choose finitely many tangential modes and retain the remaining infinite spec-

trum as normal directions.

- Impose first- and second-Melnikov nonresonance conditions so that the homological equations can be solved.
- Combine tame estimates with a Newton or Nash-Moser scheme to compensate for derivative loss.
- In quasilinear problems, regularize and reduce the linearized normal operator before inversion.
- Exclude the resonant parameters and retain a large Cantor set on which the quasiperiodic invariant tori persist.

### Check your understanding

Why is the divisor  $\omega \cdot k + \Omega_i - \Omega_j$  more difficult than the finite-dimensional divisor  $\omega \cdot k$ ? Identify the two additional indices and the resulting counting problem.

## 19.1 Definitions

### Intuition

**Tangential modes carry the quasiperiodic clock.** One selects finitely many Fourier modes whose actions become parameters and whose angles generate the torus frequency vector. All remaining modes are normal directions to be stabilized and reduced.

**Second Melnikov conditions control normal-mode interactions.** Divisors such as  $\omega \cdot k + \Omega_i - \Omega_j$  arise when eliminating off-diagonal terms in the linearized normal dynamics. In Pöschel's theorem the lower bound is weighted by  $\langle l \rangle_d = \max(1, |\sum_j j^d l_j|)$  and  $A_k = 1 + |k|^\tau$ . The spectral weight is essential: it records separation of high normal modes and is what makes the infinitely many excluded resonance strips countable with summable total measure.

**Pöschel's cutoff is a resonance cutoff, not a perturbation truncation.** At each analytic Newton step one takes the full Taylor polynomial of weighted degree at most two in the normal variables and solves its full Fourier homological equation on a smaller strip. The cutoffs  $K_\nu$  enter only afterward, to decide which divisors must persist and which new resonance zones are removed from parameter space. This distinction explains why the one-step remainder is a Taylor-remainder plus commutator estimate rather than an exponentially small Fourier tail.

**The reversible derivative-wave proof uses momentum and quasi-Toeplitz structure.** For a Fourier-Taylor monomial, the  $a$ -momentum norm penalizes failure of exact spatial-momentum conservation by an exponential weight. At high normal indices the linear coefficients are not exactly Toeplitz, but they split into a Toeplitz part plus a uniformly small defect. This quasi-Toeplitz structure is what survives the reversible homological equation and the Lie transforms, and it is

what keeps the normal-frequency correction asymptotically scalar.

**The missing quartic term is a theorem hypothesis, not a cosmetic simplification.** For the derivative-wave theorem the source assumes a cubic term followed by terms of degree at least five. After the cubic Birkhoff step this gives the remainder size needed by the KAM theorem. A genuine quartic Taylor term would be too large at that stage and would require an additional Birkhoff normal-form step before the same KAM scheme could be applied.

**The quintic-wave proof uses normalized complex actions.** The physical ellipse action in the theorem statement and the complex Hamiltonian action  $|z_j|^2$  differ by the fixed factor  $\lambda_j/2$ . It is therefore harmless to carry out the Birkhoff and twist calculations in normalized variables, but the conversion must be made explicit when the final Cantor family is pulled back to the physical tori.

**A sextic normal form has both tangential and normal-action pieces.** For the quintic equation the Hamiltonian perturbation has degree six. Pairable sextic monomials supported entirely on the selected modes produce the tangential action polynomial and its twist. Pairable monomials containing one normal action must also be retained: their coefficients give the first nonlinear correction to the infinitely many normal frequencies. Discarding these terms would leave the second-Melnikov verification with the wrong normal spectrum.

**Lie degree bookkeeping forces a degree-ten second step.** For Hamiltonian polynomials of degrees  $p$  and  $q$ , the Poisson bracket has degree  $p + q - 2$ . Hence a degree-six generating function applied to a degree-six perturbation first creates degree-ten terms, not degree seven. In this quintic-wave problem the pure-tangential nonnormal part of that generated degree-ten block is removed by a second partial Birkhoff step before the abstract infinite-dimensional KAM theorem is invoked.

**Reducibility is a linear theorem inside a nonlinear theorem.** The nonlinear iteration produces an approximate invariant torus; at each stage one must invert its linearized normal operator. Conjugating that operator to constant or block-diagonal coefficients both makes inversion possible and often proves linear stability of the final torus.

**Water-wave standing symmetry is more than spatial evenness.** The phase space for the gravity-capillary theorem has zero spatial mean and is even in  $x$ . Reversibility additionally requires  $\eta$  to be even and  $\psi$  odd in the quasiperiodic time angles. These parity restrictions are preserved by every conjugation used in the normal-operator reduction.

**Surface tension supplies finite-order transversality, not a twist matrix.** For a divisor  $f(\kappa)$  the relevant statement is that some derivative  $\partial_\kappa^k f$ , with  $k \leq k_0$ , is uniformly nonzero after natural frequency weights are removed. This one-parameter condition is proved from a Vandermonde argument and the capillary  $j^{3/2}$  asymptotics. Herman's counterterm is then inverted at the end to recover the original frequency curve.

**Almost-approximate inversion is the correct Nash-Moser object.** At a finite nonlinear iteration stage the water-wave normal operator is only almost diagonalized on the current Melnikov set. Its main part has a tame inverse, while the

remaining defects are smaller at the current cutoff or smoothing. This is enough for Nash–Moser. Complete reducibility is reserved for the final invariant torus.

**The reduced water-wave normal form has three distinct spectral levels.** After the good-unknown, block-decoupling and Egorov steps, the principal multiplier is  $m_3 T(D)$  of order  $3/2$ , a constant  $m_1 |D|^{1/2}$  correction remains, and a diagonal order-zero remainder is still present. Only the off-diagonal remainder has been made arbitrarily smoothing. Collapsing all three lower levels into a “smoothing remainder” would give the wrong Melnikov spectrum.

**A vortex-patch aspect interval must stay inside one Kirchhoff spectral regime.** For a Kirchhoff ellipse the  $n$ th Fourier block has coefficients

$$\mu_n^\pm = n\Omega_\gamma - \frac{1}{2} \pm \frac{1}{2} q_\gamma^n, \quad q_\gamma = \frac{\gamma - 1}{\gamma + 1},$$

and elliptic frequency  $\Omega_n = \sqrt{\mu_n^+ \mu_n^-}$  whenever the product is positive. The critical aspect ratios at which a product vanishes divide parameter space into spectral regimes. The KAM theorem fixes one compact interval inside one such regime and chooses tangential sites above the finite hyperbolic block, so no normal eigenvalue crosses zero during the iteration.

**The mode-2 obstruction is angular momentum, not an ordinary small divisor.** The second Fourier block is a universal zero/Jordan block. Instead of trying to invert it, one uses angular momentum  $J$  and an impact time  $\bar{t}$  along the Hamiltonian flow of  $J$  as a canonical pair. The resulting symplectic chart removes the mode from the normal inversion. Its explicit transport-flow inverse is essential because differentiating the chart costs one spatial derivative.

**Aspect-ratio transversality replaces vortex action twist.** With  $z = ((\gamma - 1)/(\gamma + 1))^2$ , the Kirchhoff frequencies have a triangular analytic expansion in  $z$  and a controlled large-mode asymptotic. These two facts imply that every nontrivial tangential, one-normal or two-normal Melnikov divisor has some derivative in  $\gamma$ , of uniformly bounded finite order, that is bounded away from zero after the natural frequency weight is removed. This finite-order one-parameter transversality is the nondegeneracy used in the measure argument.

**Vortex normal reduction is shorter than the water-wave pseudodifferential chain.** After the angular-momentum rectification and torus straightening, the normal linearized vortex operator has transport plus periodic Hilbert-transform structure, with smoothing, finite-rank and invariance-defect remainders. One symplectic spatial straightening makes the leading block coefficients constant. Matrix KAM then needs to handle only the remaining smoothing off-diagonal couplings. This is why the vortex theorem still needs Nash–Moser for derivative loss, but not the full sequence of good-unknown and order- $3/2$ , order- $1/2$ , order-0 reductions required for gravity-capillary water waves.

**Analytic Newton and Nash–Moser are different PDE-KAM engines.** Pöschel’s 1996 theorem uses a genuine analytic Newton iteration with shrinking complex strips and radii; the smoothing relation  $p - \bar{p} \leq \delta < d - 1$  is built into the phase-space mapping property. Later quasilinear problems in this volume,

such as gravity-capillary water waves and vortex patches, require Nash-Moser because their approximate inverses lose derivatives. The book keeps these two mechanisms separate.

**Asymptotically full measure is relative to the small-amplitude parameter set.** The Cantor set may have holes at every scale, but the ratio of discarded parameters to the ambient small parameter region tends to zero as the amplitude tends to zero.

### Definition K19.1. PDE KAM torus

A PDE KAM torus is a finite-dimensional invariant torus in an infinite-dimensional phase space carrying quasiperiodic solutions of the PDE with specified basic frequencies.

### Definition K19.2. normal mode

A normal mode is a linear eigenmode transverse to the selected finite set of tangential modes used to parameterize the KAM torus.

### Definition K19.3. second Melnikov condition

A second Melnikov condition is a nonresonance lower bound involving integer combinations of tangential frequencies and sums/differences of two normal frequencies.

### Definition K19.4. tame PDE reducibility

Tame PDE reducibility is conjugation of the linearized PDE to a time-independent diagonal/block form by transformations controlled on a Sobolev/tame scale with explicit derivative loss.

### Definition K19.5. analytic PDE-KAM domain and scaled vector-field norm

For  $s, r > 0$ , write  $D(s, r)$  for the complex neighborhood of the zero torus in which  $|\operatorname{Im} x| < s$ ,  $|y| < r^2$ , and the two normal coordinates have  $\ell^{a,p}$  norm  $< r$ . For a vector field  $X = (X^x, X^y, X^u, X^v)$  its scaled analytic norm is the supremum of

$$|X^x| + r^{-2}|X^y| + r^{-1}(\|X^u\|_{a,\bar{p}} + \|X^v\|_{a,\bar{p}})$$

on  $D(s, r)$ ; the superscript  $L$  denotes the corresponding parameter Lipschitz seminorm. This is the normalization underlying the smallness quantity in K19-001.

### Definition K19.6. order-two Melnikov divisors and Cantor nonresonance set

For tangential frequency  $\omega \in \mathbb{R}^n$  and normal frequencies  $\Omega = (\Omega_j)_{j \geq 1}$ , an *order-two Melnikov divisor* is  $\langle k, \omega \rangle + \langle l, \Omega \rangle$  with  $k \in \mathbb{Z}^n$  and an integer sequence  $l$  of finite support satisfying  $1 \leq |l|_1 \leq 2$ . A KAM Cantor set is obtained by imposing lower bounds on these divisors (and on  $\langle k, \omega \rangle$ ) at the successive Fourier cutoffs.



**Definition K19.7. analytic reducibility and linear stability of a PDE torus**

The linearized equation along a quasiperiodic torus is *analytically reducible to constant coefficients* if an invertible quasiperiodic transformation, analytic in the angle variables and bounded on the stated phase scale, conjugates it to a time-independent diagonal or finite-block system. A torus is *linearly stable* here when the reduced normal flow is uniformly bounded on the phase norm used in the theorem. For the diagonal or semisimple constant normal forms constructed in this volume, purely imaginary reduced normal frequencies imply this uniform boundedness and hence zero normal Lyapunov exponents; vanishing Lyapunov exponents alone is not used as an equivalent definition of linear stability.

**Definition K19.8. tangential action torus for the Dirichlet wave equation**

For the Dirichlet eigenfunctions  $\varphi_j(x) = \sqrt{2/\pi} \sin(jx)$  and frequencies  $\lambda_j = \sqrt{j^2 + m}$ , a finite tangential set  $J = \{n_1, \dots, n_b\}$  determines the invariant linear subspace spanned by those modes. In canonical coordinates  $(q_{n_\ell}, p_{n_\ell})$ , the positive action vector  $I = (I_1, \dots, I_b)$  specifies the rotational torus

$$q_{n_\ell}^2 + \lambda_{n_\ell}^{-2} p_{n_\ell}^2 = I_\ell, \quad \ell = 1, \dots, b.$$

These are the unperturbed tori used in K19-003.

**Definition K19.9. infinite-depth gravity-capillary water-wave variables**

For a  $2\pi$ -periodic free surface  $y = \eta(t, x)$  above an infinitely deep irrotational fluid, let  $\psi(t, x)$  be the trace on the free surface of the harmonic velocity potential. The Dirichlet-Neumann operator  $G(\eta)$  maps  $\psi$  to the outward normal derivative of its harmonic extension, expressed in the horizontal coordinate. In the normalization  $g = 1$  and surface tension  $\kappa > 0$ , the linear frequency of Fourier mode  $j \geq 1$  is  $\Omega_j(\kappa) = \sqrt{j(1 + \kappa j^2)}$ .

**Definition K19.10. standing wave and tangential-site amplitude vector**

A standing water wave in K19-004 is a solution periodic and even in the space variable and quasiperiodic in time. A finite tangential set  $S^+ \subset \mathbb{N}^+$  selects the cosine modes carrying the principal oscillation; the positive vector  $\xi = (\xi_j)_{j \in S^+}$  denotes their small action/amplitude parameters.

**Definition K19.11. vortex patch and contour dynamics**

A planar Euler vortex patch is a vorticity field  $\omega = \mathbf{1}_{D(t)}$  (up to a fixed nonzero constant factor) whose time dependence is carried by the moving boundary of a simply connected domain  $D(t)$ . Contour dynamics is the induced evolution equation for a parametrization of  $\partial D(t)$  obtained from the Biot-Savart velocity, modulo tangential reparametrization.

**Definition K19.12. Kirchhoff ellipse and aspect ratio**

A Kirchhoff ellipse is an elliptical constant-vorticity patch that rotates uniformly under the planar Euler flow. Its aspect ratio  $\gamma > 1$  is the ratio of its major to minor semiaxis;  $\gamma = 1$  is the circular limiting case and is excluded from the compact aspect-ratio intervals used in K19-005.

**Definition K19.13. angular-momentum prime integral and symmetry zero mode**

For the vortex-patch Hamiltonian, angular momentum  $J$  is the conserved moment functional generated by rotations. The infinitesimal rotational symmetry produces a zero normal frequency at a Kirchhoff ellipse. Symplectic reduction of  $J$  means choosing canonical coordinates with  $J$  as one coordinate and restricting the dynamics to a fixed level of  $J$ , thereby removing that neutral symmetry mode from the normal KAM operator.

**Definition K19.14. asymptotically full relative Lebesgue measure**

For a fixed compact interval  $I$  and a family of measurable sets  $G_\varepsilon \subset I$ , “asymptotically full Lebesgue measure” means  $|I \setminus G_\varepsilon| \rightarrow 0$  as the perturbation/amplitude parameter  $\varepsilon \rightarrow 0$ ; equivalently  $|G_\varepsilon|/|I| \rightarrow 1$ .

**Definition K19.15. Pöschel Melnikov weight and divisor weight**

For a finitely supported integer sequence  $l = (l_j)_{j \geq 1}$  with  $|l|_1 \leq 2$ , set

$$\langle l \rangle_d := \max \left( 1, \left| \sum_{j \geq 1} j^d l_j \right| \right), \quad A_k := 1 + |k|^\tau, \quad k \in \mathbb{Z}^n.$$

For frequency vectors  $(\omega, \Omega)$ , the Pöschel order-two Melnikov lower bound with constant  $\alpha > 0$  is

$$|\langle k, \omega \rangle + \langle l, \Omega \rangle| \geq \alpha \frac{\langle l \rangle_d}{A_k}$$

for every  $(k, l) \neq (0, 0)$  with  $|l|_1 \leq 2$ . The factor  $\langle l \rangle_d$  records the leading polynomial separation of the normal frequencies; it is not interchangeable with the unweighted number  $1 + |l|_1$ .

**Definition K19.16. Pöschel combined parameter-Lipschitz norm**

If  $X(\cdot, \xi)$  is a parameter-dependent analytic vector field on  $D(s, r)$ , write

$$|X|_{r, D(s, r)}^\lambda := |X|_{r, D(s, r)} + \lambda |X|_{r, D(s, r)}^L, \quad 0 \leq \lambda \leq \alpha/M,$$

where the first term is the scaled vector-field norm of Definition K19.5 and the second is its parameter Lipschitz seminorm. The restriction  $\lambda \leq \alpha/M$  makes the extra divisor produced by taking a parameter difference absorbable by the same Melnikov constant  $\alpha$ .

**Definition K19.17. Pöschel resonance zones and stage cutoffs**

For Lipschitz frequencies  $(\omega', \Omega')$  and  $(k, l) \neq (0, 0)$  with  $|l|_1 \leq 2$ , define

$$\mathcal{R}_{kl}(\alpha; \omega', \Omega') := \left\{ \xi \in \Pi : |\langle k, \omega'(\xi) \rangle + \langle l, \Omega'(\xi) \rangle| < \alpha \frac{\langle l \rangle_d}{A_k} \right\}.$$

In the Newton iteration a cutoff  $K_\nu$  is used only to decide which newly dangerous resonance zones are removed and which previously verified divisors must persist under the frequency shift. It is *not* a Fourier truncation of the perturbation in the analytic KAM step.

**Definition K19.18. reversible/even phase space and symmetrization**

Let the tangential set be symmetric,  $I = I^+ \cup (-I^+)$ , and write the action-angle/normal variables as  $(x, y, z, \bar{z})$ . The real-even subspace is

$$E := \{x_j = x_{-j}, y_j = y_{-j}, z_j = z_{-j}, \bar{z}_j = \bar{z}_{-j}\},$$

and the reversing involution is

$$S(x_j, y_j, z_j, \bar{z}_j) = (-x_{-j}, y_{-j}, \bar{z}_{-j}, z_{-j}).$$

A vector field is reversible when  $X \circ S = -DSX$ . Its *symmetrization*  $SX$  is the coefficient replacement on the exactly resonant degree- $\leq 0$  monomials that identifies the  $j$  and  $-j$  coefficients forced to coincide on  $E$ ; by construction  $(SX)|_E = X|_E$ .

**Definition K19.19.  $a$ -momentum majorant norm**

For a Fourier-Taylor monomial vector field  $e^{ik \cdot x} y^i z^\alpha \bar{z}^\beta \partial_v$ , let  $\pi(k, \alpha, \beta; v)$  denote its signed spatial Fourier momentum, including the momentum of the differentiated variable. The  $a$ -momentum majorant norm multiplies the absolute value of that monomial coefficient by  $e^{a|\pi|}$  before taking the usual analytic majorant vector-field norm. The exponential factor measures how far a monomial is from exact momentum conservation and is stable under products and commutators because momenta add.

**Definition K19.20. quasi-Toeplitz vector field**

Fix admissible exponents  $0 < b < L < 1$  and high-mode parameters  $N, \theta, \mu$ . A vector field is *quasi-Toeplitz* if, after projecting its high-high linear part to Fourier modes  $|k| < N^b$  with low external momentum  $O(N^L)$  and normal indices larger than  $\theta N$ , its coefficient matrix can be written uniformly in  $N$  as

$$X_{mn} = T(\text{sgn } m, m - n) + N^{-1} R_{mn},$$

where the Toeplitz coefficient  $T$  and the defect  $R$  are bounded in the same  $a$ -momentum majorant norm, uniformly over all sufficiently large  $N$ . The admissible constants are part of the norm data and may be tightened under a KAM step.

**Definition K19.21. reversible Melnikov set**

At a Fourier cutoff  $K$ , the reversible Melnikov set consists of those parameters for which every nontrivial divisor required by the degree- $\leq 0$  homological equation satisfies the corresponding lower bound  $\gamma \langle k \rangle^{-\tau}$  for  $|k| \leq K$ . The four divisor families are

$$\omega \cdot k, \quad \omega \cdot k \pm \Omega_j, \quad \omega \cdot k + \Omega_i + \Omega_j, \quad \omega \cdot k + \Omega_i - \Omega_j,$$

with the exact kernel/resonant monomials omitted rather than divided out. These omitted terms are retained in the normal form.

**Definition K19.22. DNLW action box and asymptotic density one**

For  $\rho > 0$  and  $q = |I^+|$ , set

$$\mathcal{O}_\rho := \{\xi \in \mathbb{R}_+^q : \rho/2 \leq \xi_i \leq \rho \text{ for all } i\}.$$

A family  $\mathcal{C}_\rho \subset \mathcal{O}_\rho$  has *asymptotic density one at the origin* if

$$\frac{|\mathcal{O}_\rho \setminus \mathcal{C}_\rho|}{|\mathcal{O}_\rho|} \rightarrow 0 \quad (\rho \downarrow 0).$$

For the derivative-wave KAM scaling used below, one may take  $\rho = C_* R^2$  and  $\gamma = R^{3+1/5}$ ; the source measure estimate then gives a relative loss  $O(R^{2/15})$ .

**Definition K19.23. normalized Dirichlet-wave coordinates and complex action**

For the Dirichlet wave frequencies  $\lambda_j = \sqrt{j^2 + m}$ , pass from physical modal coordinates  $(Q_j, P_j)$  to

$$q_j = \sqrt{\lambda_j} Q_j, \quad p_j = \lambda_j^{-1/2} P_j, \quad z_j = \frac{q_j + \mathrm{i} p_j}{\sqrt{2}}.$$

The normalized complex action is  $I_j^c := |z_j|^2$ . If  $I_j^{\text{phys}}$  denotes the physical tangential action convention of Definition K19.8, then

$$I_j^c = \frac{\lambda_j}{2} I_j^{\text{phys}}.$$

Thus the two action parametrizations differ by a fixed positive diagonal linear map and have the same local positive-measure property.

**Definition K19.24. sextic pairable and normal-index classes**

Write  $z_{-j} := \bar{z}_j$  and let  $\pm J := J \cup (-J)$ . A signed sextic monomial  $z_{j_1} \cdots z_{j_6}$  is *pairable* if its six signed indices can be partitioned into three pairs  $\{r, -r\}$ ; these are precisely the action monomials retained in the sextic normal form. Its *normal index count* is the number of factors whose absolute index lies outside  $J$ . The degree-six partial Birkhoff step removes every nonpairable monomial with at most two normal indices and retains the pairable tangential terms together with the pairable

two-normal diagonal terms  $I_j$  that generate the leading normal-frequency correction.

**Definition K19.25. quintic-wave gap function**

For  $0 < m \leq \frac{1}{4}$  define

$$f_m(t) := \sqrt{t^2 + m} - t = \frac{m}{\sqrt{t^2 + m} + t}, \quad t > 0.$$

The function is positive, strictly decreasing and strictly convex. After the integer part of a signed wave-frequency combination is cancelled by the sine-product selection rule, the remaining six-frequency small divisor is expressed as a signed combination of values of  $f_m$ . This is the arithmetic quantity controlled by the separated tangential-site hypothesis in K19-003.

**Definition K19.26. scaled quintic-wave action box and KAM radii**

Fix a compact box  $\Theta \Subset (0, \infty)^b$  containing a neighborhood of the frequency-weighted ray  $I_{n_i}/\lambda_{n_i} = \text{constant}$  and write the normalized tangential actions as

$$I = \rho\eta, \quad \eta \in \Theta,$$

with  $0 < \rho \ll 1$ . In the K19-003 verification use normal radius  $r = \rho^{3/2}$  and a Melnikov constant

$$\alpha = \rho^{2+\sigma}, \quad 0 < \sigma < \frac{1}{2}.$$

The tangential frequency derivative is of size  $\rho^2$ , whereas the post-normal-form perturbation vector field is  $O(\rho^3)$  in the scaled KAM norm. This scale separates twist, perturbative smallness and resonance-strip width.

**Definition K19.27. gravity-capillary standing-wave phase space**

For the gravity-capillary problem, the standing-wave phase space is the zero-mean subspace of pairs  $(\eta, \psi)$  that are  $2\pi$ -periodic and even in  $x$ . The reversing involution is  $\rho(\eta, \psi) = (\eta, -\psi)$ . A quasiperiodic embedding  $u(\varphi, x) = (\eta, \psi)(\varphi, x)$  is reversible when

$$\eta(-\varphi, x) = \eta(\varphi, x), \quad \psi(-\varphi, x) = -\psi(\varphi, x).$$

Thus  $\eta$  is even and  $\psi$  odd in the time angles; this is the precise standing-wave symmetry used in K19-004.

**Definition K19.28. finite-order surface-tension transversality**

Let  $f(\kappa)$  be one of the tangential or first/second Melnikov divisor functions in K19-004. A compact interval  $[\kappa_1, \kappa_2] \Subset (0, \infty)$  has *finite-order surface-tension transversality* with index  $k_0$  and amount  $\rho_0$  when, after the natural weight of the integer

tangential vector has been factored out,

$$\max_{0 \leq k \leq k_0} |\partial_\kappa^k f(\kappa)| \geq \rho_0$$

for every nontrivial divisor and every  $\kappa$  in the interval. The integer  $k_0$  is the non-degeneracy index used in the one-dimensional sublevel estimates.

### Definition K19.29. $D^{k_0}$ -tame and modulo-tame operator bookkeeping

A parameter-dependent operator family  $A(\omega, \kappa)$  is  $D^{k_0}$ -tame on the Sobolev scales used in K19-004-005 when every parameter derivative  $\partial_{(\omega, \kappa)}^\alpha A$ ,  $|\alpha| \leq k_0$ , satisfies a one-high-factor estimate

$$\|\partial^\alpha A h\|_s \leq C_s \gamma^{-|\alpha|} (\|h\|_{s+\mu} + M_s(A) \|h\|_{s_0+\mu})$$

with a fixed derivative loss  $\mu$ . It is  $D^{k_0}$ -modulo-tame when the same estimates hold for the majorant matrix of its Fourier coefficients, so that composition, commutators, truncation and the reducibility homological equation preserve the bounds. These terms are bookkeeping properties of the quasilinear K19 normal operators, not assertions of exact diagonalization.

### Definition K19.30. almost-approximate right inverse

For an invariance equation  $\mathcal{F}(U) = 0$ , an *almost-approximate right inverse* at  $U_0$  is an operator  $T_0$  with a fixed-loss tame bound such that

$$D\mathcal{F}(U_0)T_0 - I = \mathcal{P}_0,$$

where  $\mathcal{P}_0$  is quantitatively one smoothing scale smaller than the main inverse, or is a sum of such small/smoothing defects. Nash-Moser uses this weaker object rather than an exact inverse: after projecting to the current smoothing scale the defect is absorbed in the next residual estimate.

### Definition K19.31. admissible Kirchhoff spectral interval and tangential sites

For an integer  $\bar{n} \geq 2$ , let  $(\gamma_n)_{n \geq 3}$  denote the critical Kirchhoff aspect ratios at which the  $n$ th normal frequency crosses zero. An *admissible Kirchhoff spectral interval* is a compact interval  $\mathcal{G} = [\gamma_1, \gamma_2]$  satisfying

$$\mathcal{G} \Subset (1, 3) \quad \text{if } \bar{n} = 2, \quad \mathcal{G} \Subset (\gamma_{\bar{n}}, \gamma_{\bar{n}+1}) \quad \text{if } \bar{n} \geq 3.$$

A tangential set is then any finite set  $S \subset \{\bar{n} + 1, \bar{n} + 2, \dots\}$ . On such an interval the modes  $n \geq \bar{n} + 1$  used as tangential and infinite-dimensional elliptic directions do not cross zero, while the finitely many modes  $3 \leq n \leq \bar{n}$  (when present) remain hyperbolic throughout the interval.

**Definition K19.32. Kirchhoff mode coefficients and spectral blocks**

Set

$$\Omega_\gamma := \frac{\gamma}{(1+\gamma)^2}, \quad q_\gamma := \frac{\gamma-1}{\gamma+1}, \quad \mu_n^\pm(\gamma) := n\Omega_\gamma - \frac{1}{2} \pm \frac{1}{2}q_\gamma^n.$$

For an elliptic mode define

$$\Omega_n(\gamma) := \sqrt{\mu_n^+(\gamma)\mu_n^-(\gamma)} = \sqrt{\left(n\Omega_\gamma - \frac{1}{2}\right)^2 - \frac{1}{4}q_\gamma^{2n}}.$$

The cosine-sine pair of mode  $n$  is an elliptic  $2 \times 2$  block when  $\mu_n^+\mu_n^- > 0$  and a hyperbolic block when  $\mu_n^+\mu_n^- < 0$ . The mode  $n = 2$  is degenerate because  $\mu_2^+(\gamma) = 0$  for every  $\gamma > 1$ ; this is the symmetry zero mode removed by angular-momentum reduction.

**Definition K19.33. impact-time symplectic rectification of angular momentum**

Let  $\Phi_J^t$  be the local Hamiltonian flow generated by the normalized angular momentum  $J$ , and let  $c_2, s_2$  be the cosine/sine basis vectors of the degenerate mode 2. For  $\xi$  close to the Kirchhoff ellipse, the *impact time*  $\bar{t}(\xi)$  is the unique small time such that

$$(\Phi_J^{\bar{t}(\xi)}(\xi), s_2)_{L^2} = 0.$$

The impact-time rectification is

$$\Phi(\xi) := J(\xi)c_2 + \bar{t}(\xi)s_2 + \Pi_2^\perp \Phi_J^{\bar{t}(\xi)}(\xi).$$

It is designed so that  $J$  and  $\bar{t}$  form a canonical pair, the projected coordinates are constant along the  $J$ -flow, and restricting to  $J = 0$  removes the mode-2 Jordan block. Its local inverse is obtained from the same transport flow rather than from a Banach inverse-function theorem, because  $d\Phi - I$  loses one spatial derivative.

**Definition K19.34. finite-order aspect-ratio Melnikov transversality**

Let  $\tilde{\omega}(\gamma) = (\Omega_n(\gamma))_{n \in S}$  on an admissible Kirchhoff spectral interval. We say that the Kirchhoff frequency curve has *finite-order aspect-ratio Melnikov transversality* with index  $k_0$  and amount  $\rho_0$  if every nontrivial divisor needed by the tangential and normal homological equations satisfies

$$\max_{0 \leq k \leq k_0} |\partial_\gamma^k f(\gamma)| \geq \rho_0 \langle \ell \rangle$$

with the natural tangential weight, for the families  $\tilde{\omega} \cdot \ell$ ,  $\tilde{\omega} \cdot \ell + j\Omega_1$ ,  $\tilde{\omega} \cdot \ell + \Omega_n$ , and  $\tilde{\omega} \cdot \ell + \Omega_n \pm \Omega_{n'}$ , with only the exact trivial kernels omitted. This is a one-parameter degenerate-KAM nondegeneracy condition; it is not an action-twist hypothesis.

## 19.2 Main results

### Theorem K19.1. KAM for vortex patches

**Statement.** Fix an integer  $\bar{n} \geq 2$  and an admissible Kirchhoff spectral interval  $\mathcal{G} = [\gamma_1, \gamma_2]$  in the sense of Definition K19.31, and let  $S \subset \{\bar{n} + 1, \bar{n} + 2, \dots\}$  be a finite set of distinct tangential sites. There exist  $s_* > (|S| + 1)/2$  and  $\varepsilon_0 > 0$  such that, for every amplitude vector  $a = (a_n)_{n \in S}$  with  $a_n > 0$  and  $|a| \leq \varepsilon_0$ , there is a Borel set  $\mathcal{G}_a \subset \mathcal{G}$  satisfying

$$|\mathcal{G}_a| \longrightarrow \gamma_2 - \gamma_1 \quad (|a| \rightarrow 0).$$

For every  $\gamma \in \mathcal{G}_a$  there are a scalar  $\mu(\gamma, a) \rightarrow 0$ , a Diophantine vector  $\tilde{\omega}(\gamma, a) = (\tilde{\omega}_n)_{n \in S}$  with  $\tilde{\omega}(\gamma, a) \rightarrow (\Omega_n(\gamma))_{n \in S}$ , and a real  $H^{s_*}$  radial deformation  $\xi(\varphi, \theta)$  that is even under simultaneous reversal  $(\varphi, \theta) \mapsto (-\varphi, -\theta)$  and solves the contour equation at angular velocity  $\Omega_\gamma + \mu$ . Writing

$$M_n(\gamma) := |\mu_n^+(\gamma)|^{1/4} |\mu_n^-(\gamma)|^{-1/4},$$

the solution has the leading form

$$\xi(\tilde{\omega}t, \theta) = \sum_{n \in S} a_n \left( M_n \cos(\tilde{\omega}_n t) \cos(n\theta) + M_n^{-1} \sin(\tilde{\omega}_n t) \sin(n\theta) \right) + r(\tilde{\omega}t, \theta), \quad \frac{\|r\|_{H^{s_*}}}{|a|} \rightarrow 0.$$

Consequently

$$z(t, \theta) = e^{i(\Omega_\gamma + \mu)t} (1 + 2\xi(\tilde{\omega}t, \theta))^{1/2} (\gamma^{1/2} \cos \theta + i\gamma^{-1/2} \sin \theta)$$

parametrizes a genuine time-quasiperiodic vortex patch for the parameter-independent planar Euler equation.

*Proof.*

#### Proof strategy

Write the vortex-patch boundary dynamics in a Hamiltonian contour formulation and isolate the elliptic linear modes around the reference patch. A normal-form reduction separates the finitely many tangential modes from the infinite normal directions. The reducibility and Melnikov estimates control the linearized normal operator, while a Nash-Moser iteration solves the nonlinear invariance equation. Parameter exclusion supplies the Cantor family on which the full scheme converges.

Fix  $\bar{n} \geq 2$ , an admissible compact aspect-ratio interval  $\mathcal{G} = [\gamma_1, \gamma_2]$ , and a finite tangential set  $S \subset \{\bar{n} + 1, \bar{n} + 2, \dots\}$ . We use the Kirchhoff ellipse

$$D_\gamma = \{(x, y) \in \mathbb{R}^2 : x^2/\gamma + \gamma y^2 < 1\}, \quad \Omega_\gamma = \frac{\gamma}{(1 + \gamma)^2}.$$



In a frame rotating with angular velocity  $\Omega$ , write its boundary as

$$w(t, \theta) = \sqrt{1 + 2\xi(t, \theta)}(\gamma^{1/2} \cos \theta + i\gamma^{-1/2} \sin \theta). \quad (19.1)$$

The square-root coordinate is not cosmetic: it is the radial coordinate in which the contour dynamics has the constant symplectic operator  $\partial_\theta$ .

**Step 1. Hamiltonian radial contour equation and reversible phase space.** For  $\xi$  small in a Sobolev space above the algebra threshold, the planar vortex-patch equation is equivalent, modulo tangential reparametrization, to

$$\partial_t \xi = \partial_\theta \nabla H_\Omega(\xi), \quad H_\Omega = -\frac{1}{2}E + \frac{\Omega}{2}J, \quad (19.2)$$

where  $E$  is the patch energy and  $J$  its angular momentum. The equation is Hamiltonian and reversible, preserves the regularity class, and  $\xi = 0$  is an equilibrium precisely when  $\Omega = \Omega_\gamma$ .

Indeed, the boundary velocity is the Biot-Savart velocity restricted to the contour. Only its normal component changes the domain; adding a tangential component changes the parametrization but not the patch. Substituting (19.1) in the normal-velocity identity and differentiating the logarithmic stream-function integral gives a scalar equation of the form

$$\partial_t \xi = \frac{\Omega}{2} \partial_\theta (g_\gamma(\theta)(1 + 2\xi)) + \partial_\theta \mathcal{W}(\xi), \quad g_\gamma(\theta) = \gamma \cos^2 \theta + \gamma^{-1} \sin^2 \theta.$$

The first variation of  $-E/2 + \Omega J/2$  is exactly the quantity in parentheses, so this is (19.2). The reflection  $(t, \theta) \mapsto (-t, -\theta)$  reverses the Hamiltonian vector field, hence the simultaneous-even subspace used in the statement is invariant. Direct substitution at  $\xi = 0$  leaves the factor  $\Omega - \Omega_\gamma$ , proving the equilibrium assertion.

We next diagonalize the quadratic Hamiltonian. With

$$q_\gamma = \frac{\gamma - 1}{\gamma + 1}, \quad \mu_n^\pm = n\Omega_\gamma - \frac{1}{2} \pm \frac{1}{2}q_\gamma^n,$$

write the  $n$ th real Fourier pair as  $(\alpha_n, \beta_n)$ . The linearized equations are

$$\dot{\alpha}_n = -\mu_n^+ \beta_n, \quad \dot{\beta}_n = \mu_n^- \alpha_n. \quad (19.3)$$

**Step 2. Kirchhoff spectral decomposition and admissible intervals.** The eigenvalues of (19.3) satisfy  $\lambda^2 = -\mu_n^+ \mu_n^-$ . Thus the elliptic frequency is

$$\Omega_n(\gamma) = \sqrt{\left(n\Omega_\gamma - \frac{1}{2}\right)^2 - \frac{1}{4}q_\gamma^{2n}}. \quad (19.4)$$

The mode  $n = 2$  is a Jordan block at eigenvalue zero for every  $\gamma > 1$ . For an admissible interval, all modes  $n \geq \bar{n} + 1$  are elliptic, the modes  $3 \leq n \leq \bar{n}$  are hyperbolic when  $\bar{n} \geq 3$ , and no eigenvalue changes type on  $\mathcal{G}$ .

The eigenvalue formula follows immediately by taking the determinant of the matrix

in (19.3). Also

$$\mu_2^+ = 2 \frac{\gamma}{(1+\gamma)^2} - \frac{1}{2} + \frac{1}{2} \left( \frac{\gamma-1}{\gamma+1} \right)^2 = 0,$$

so the zero mode is exact, not perturbative. For  $n \geq 3$  the zeros of  $\mu_n^+ \mu_n^-$  are isolated. Denote the successive critical values by  $\gamma_n$ . The sign is constant between two successive critical values. At  $\gamma \downarrow 1$  all  $n \geq 3$  blocks are elliptic, while crossing  $\gamma_n$  changes precisely the  $n$ th block. Hence the interval convention in the statement fixes the spectral type and proves the asserted splitting. This is why the older claim for an arbitrary compact subinterval of  $(1, \infty)$  was too broad.

The aspect ratio supplies the only external nondegeneracy parameter. Put

$$z = q_\gamma^2 \in (0, 1). \quad (19.5)$$

Since  $4\Omega_1 = 1 - z$ , formula (19.4) becomes, for every elliptic  $n \geq 3$ ,

$$4\Omega_n(z) = \sqrt{(n-2-nz)^2 - 4z^n}. \quad (19.6)$$

Near  $z = 0$  this has the triangular expansion

$$4\Omega_n(z) = (n-2) - nz - \frac{2}{n-2} z^n + O(z^{n+1}). \quad (19.7)$$

**Step 3. Finite-order aspect-ratio transversality without action twist.** There exist  $k_0^* \in \mathbb{N}$  and  $\rho_0 > 0$ , depending only on  $S$  and  $\mathcal{G}$ , such that every nontrivial tangential, one-normal and two-normal Melnikov divisor has a derivative of order at most  $k_0^*$  bounded below by  $\rho_0 \langle \ell \rangle$  on  $\mathcal{G}$ . The proof uses the analytic frequency curve in  $\gamma$  and the large-mode expansion

$$\Omega_n(\gamma) = n\Omega_1(\gamma) - \frac{1}{2} + r(n, \gamma), \quad \sup_{n \geq \bar{n}+1, \gamma \in \mathcal{G}} n |\partial_\gamma^k r(n, \gamma)| \leq C_k. \quad (19.8)$$

No finite-dimensional action-twist determinant is used.

First consider any finite collection of the analytic functions  $1, \Omega_1, \Omega_{n_1}, \dots, \Omega_{n_r}$  with  $3 \leq n_1 < \dots < n_r$ . Suppose a real linear combination vanished identically. In (19.7), the first nonlinear Taylor coefficient  $z^{n_1}$  can only come from  $\Omega_{n_1}$ , so its coefficient is zero. Inductively the coefficients of all  $\Omega_{n_j}$  vanish. The remaining affine combination of 1 and  $\Omega_1 = (1-z)/4$  then has zero constant and linear coefficients. Thus the finite family is linearly independent.

Analytic linear independence and compactness give a finite derivative order: if no  $k_0^*$  worked, there would be normalized coefficient vectors and parameters  $\gamma_j \in \mathcal{G}$  for which successively more derivatives vanish. A convergent subsequence gives a nonzero analytic combination whose all derivatives vanish at one point, hence which vanishes identically, a contradiction.

It remains to make the estimate uniform in the infinitely many normal indices. Factoring the square root in (19.6) gives (19.8); the exponentially small  $q_\gamma^n$  term in fact gives more decay than the stated  $1/n$  bound. For a one-normal divisor, if  $n \gg \langle \ell \rangle$ , the zeroth-order term  $n\Omega_1$  dominates. Otherwise  $n = O(\langle \ell \rangle)$  and normalization by  $\langle \ell \rangle$

reduces a hypothetical failure to the finite analytic independence just proved. For a difference divisor a small value first forces  $|n - n'| = O(\langle \ell \rangle)$  by (19.8); for a sum divisor it forces  $n + n' = O(\langle \ell \rangle)$ . Passing to normalized subsequences again produces one of the forbidden analytic identities. This proves all divisor families in the aspect-ratio transversality definition.

We now remove the mode-2 obstruction before any normal inversion. Normalize angular momentum so that near the ellipse

$$J(\xi) = (\xi, c_2)_{L^2} + O(\|\xi\|^2),$$

where  $c_2, s_2$  are the cosine/sine basis vectors of the second mode. Its Hamiltonian vector field is an affine transport field

$$X_J(\xi) = -s_2 + \mathcal{T}_\gamma \xi, \quad (19.9)$$

where  $\mathcal{T}_\gamma$  is first order and generates a smooth local transport flow  $\Phi_J^t$ .

**Step 4. Impact-time symplectic rectification of angular momentum.** There is a unique small impact time  $\bar{t}(\xi)$  with  $(\Phi_J^{\bar{t}(\xi)} \xi, s_2) = 0$ , satisfying

$$\bar{t}(\Phi_J^\tau \xi) = \bar{t}(\xi) - \tau, \quad (19.10)$$

and the map

$$\Phi(\xi) = J(\xi)c_2 + \bar{t}(\xi)s_2 + \Pi_2^\perp \Phi_J^{\bar{t}(\xi)} \xi \quad (19.11)$$

is symplectic, reversibility preserving, satisfies  $d\Phi(0) = I$ , and is locally invertible by an explicit transport-flow formula despite the one-derivative loss of  $d\Phi - I$ .

At  $(\xi, t) = (0, 0)$  the derivative in  $t$  of  $(\Phi_J^t \xi, s_2)$  equals  $(X_J(0), s_2) \neq 0$ . The scalar implicit-function theorem therefore produces  $\bar{t}$ ; no infinite-dimensional inverse is used. The group property of  $\Phi_J^t$  gives (19.10). Differentiating it along  $X_J$  yields  $d\bar{t}[X_J] = -1$ , equivalently  $\{J, \bar{t}\} = 1$  with the present sign convention. The projected coordinate

$$W(\xi) := \Pi_2^\perp \Phi_J^{\bar{t}(\xi)} \xi$$

is invariant along the  $J$ -flow by (19.10). Since  $\Phi_J^t$  is symplectic,  $J$  and  $\bar{t}$  form a canonical pair, are symplectically orthogonal to  $W$ , and the bracket on  $W$  is the original bracket restricted to the mode-2 complement. This proves that (19.11) is symplectic.

For invertibility, write a target point as  $qc_2 + ps_2 + w$ ,  $w \perp c_2, s_2$ . Seek

$$\xi = \Phi_J^{-p}(vc_2 + w).$$

The scalar equation  $J(\Phi_J^{-p}(vc_2 + w)) = q$  has  $\partial_v = 1$  at the origin, so it has a unique small solution  $v = v(q, p, w)$ . Substitution and (19.10) verify directly that this is  $\Phi^{-1}$ . The transport composition loses one derivative upon differentiation, so this explicit inverse is load-bearing: an ordinary Banach inverse-function theorem would not justify the map. Reflection commutes with the  $J$ -flow and reverses the impact time, hence  $\Phi$  preserves the reversible subspace.

Write the rectified coordinates as  $(\alpha_2, \beta_2, u)$  with  $\alpha_2 = J$ . Since  $J$  is a first integral,

the transformed Hamiltonian is independent of  $\beta_2$ . In the theorem we fix

$$\alpha_2 = J = 0. \quad (19.12)$$

Rescale  $u = \varepsilon \tilde{u}$ . The reduced Hamiltonian is

$$\tilde{K}(0, \tilde{u}) = H_L(\tilde{u}) + \varepsilon P(\tilde{u}), \quad (19.13)$$

where  $H_L$  has precisely the nonzero elliptic/hyperbolic blocks from the spectral decomposition above.

**Step 5. Fixed-momentum reduction, tangential variables and counterterm setup.** On  $J = 0$ , convert the selected elliptic modes  $S$  into action-angle variables  $(\vartheta, I)$ , leave the remaining modes in a normal variable  $z$ , and introduce an auxiliary tangential frequency parameter  $g \in \mathbb{R}^{|S|}$ . The modified invariance equation

$$\mathcal{F}(i, g) := \omega \cdot \partial_\varphi i - X_{K_g}(i) = 0 \quad (19.14)$$

has a freely adjustable finite-dimensional average equation. This is the counterterm mechanism used to recover the parameter-independent Euler problem; no action twist is assumed.

The linear tangential torus is obtained from the normalizations  $M_n = |\mu_n^+|^{1/4} |\mu_n^-|^{-1/4}$ : the corresponding physical radial oscillation is

$$\sum_{n \in S} a_n \left( M_n \cos(\Omega_n t) \cos(n\theta) + M_n^{-1} \sin(\Omega_n t) \sin(n\theta) \right). \quad (19.15)$$

The action-angle map is nonsingular because  $n \in S$  lies in the elliptic range. Replacing the coefficients of  $I$  in the integrable part by  $g$  creates exactly  $|S|$  scalar parameters, enough to solve the averaged tangential equation. The original system is recovered only after the limiting frequency map is inverted below.

Let  $i_0(\varphi)$  be an approximate reversible torus. The pullback of the symplectic form by an exact invariant Diophantine torus is zero: invariance makes it translation invariant on  $\mathbb{T}^{|S|}$ , while exactness of the ambient symplectic form makes its average zero. For an approximate torus the same calculation bounds the pullback by the invariance defect. Correcting the action component by a primitive produces an isotropic torus  $i_\delta$  with a correction of the same defect size.

**Step 6. Isotropic straightening and triangular linearized system.** A symplectic tubular map around  $i_\delta$  sends the torus to  $\{y = z = 0\}$  and conjugates  $D\mathcal{F}(i_0, g)$ , modulo an operator proportional to  $\mathcal{F}(i_0, g)$ , to a triangular system whose tangential cohomological blocks are explicit and whose only infinite-dimensional obstruction is a normal operator  $\mathcal{L}_\omega$ . Therefore a tame almost inverse for  $\mathcal{L}_\omega$  implies an almost-approximate right inverse for the full linearized invariance equation.

In straight coordinates the derivative has the schematic Hamiltonian form

$$\begin{pmatrix} \omega \cdot \partial_\varphi & * & * & -I \\ 0 & \omega \cdot \partial_\varphi & * & 0 \\ 0 & 0 & \mathcal{L}_\omega & 0 \end{pmatrix} + \mathcal{R}_Z,$$

where the last column is the derivative with respect to  $g$  and  $\mathcal{R}_Z$  is linear in the invariance defect. The zero-average cohomological equation is inverted under the tangential Diophantine bound; the average is fixed by  $g$ . Back substitution leaves only the normal inverse. The coordinate changes are tame because they contain one high Sobolev factor at a time; hence composition of their bounds has a fixed rather than accumulating derivative loss.

The special issue is now to identify the normal operator after the nonlinear angular-momentum rectification. Differentiating (19.11) gives

$$\begin{aligned} d\Phi(\xi)[h] &= dJ(\xi)[h]c_2 + d\bar{t}(\xi)[h]s_2 \\ &\quad + \Pi_2^\perp X_J(\Phi_{J\xi}^{\bar{t}})d\bar{t}(\xi)[h] + \Pi_2^\perp d\Phi_{J\xi}^{\bar{t}(\xi)}(\xi)[h]. \end{aligned} \quad (19.16)$$

The first three terms are finite rank; the final term is the differential of a transport flow.

**Step 7. Differentiated rectification preserves the PDE class.** Conjugating the contour linearization by  $d\Phi$  and  $d\Phi^{-1}$  preserves its first-order transport principal part. All terms generated by  $dJ$ ,  $d\bar{t}$  and the variation of the impact time are finite rank or regularizing, while the mismatch between the conjugated linearization at an approximate torus and the linearization of the rectified Hamiltonian is proportional to the invariance defect.

The last term of (19.16) transports  $h$  by a smooth diffeomorphism of the circle, so conjugation of  $a(\theta)\partial_\theta$  again has order one. The other terms consist of scalar functionals of  $h$  multiplied by fixed smooth mode functions or smooth transported functions; they are finite rank and hence smoothing in the scale relevant to the normal operator. Differentiating the identity defining the rectified Hamiltonian shows that the only failure of exact conjugacy contains the first derivative of the invariance defect. Thus it vanishes on a true torus and is one Nash–Moser residual smaller on an approximate one.

**Step 8. Rectified vortex normal operator.** On the symplectic complement of the tangential and mode-2 directions, the normal operator has the form

$$\mathcal{L}_\omega = \Pi^\perp \left[ \omega \cdot \partial_\varphi - \partial_\theta \circ V(\varphi, \theta) + \partial_\theta \circ W_0 + R_\varepsilon \right] \Pi^\perp + R_Z, \quad (19.17)$$

where  $V = -\Omega_\gamma + O(\varepsilon)$ ,

$$\partial_\theta \circ W_0 = \frac{1}{2}\mathcal{H} + Q_\infty, \quad (19.18)$$

$\mathcal{H}$  is the periodic Hilbert transform,  $Q_\infty$  is infinitely smoothing,  $R_\varepsilon$  is small and regularizing, and  $R_Z$  is proportional to the invariance defect.

The logarithmic contour kernel has the same diagonal singularity as the periodic Hilbert kernel. Subtracting that singular part leaves a kernel whose mixed derivatives are bounded; repeated differentiation therefore gains an arbitrary number of Sobolev derivatives, which gives  $Q_\infty$ . The only order-one piece comes from differentiating the moving parametrization and is the transport coefficient  $V$ . The differentiated rectification step above shows that no additional order-one term is hidden in  $R_\varepsilon$  or  $R_Z$ . This establishes (19.17) with tame parameter bounds.

We now exploit a simplification special to this contour equation. Let

$$(\mathcal{B}h)(\varphi, \theta) = h(\varphi, \theta + b(\varphi, \theta)), \quad \mathcal{S} = (1 + b_\theta)\mathcal{B}.$$

The Jacobian factor makes  $\mathcal{S}$  symplectic for the  $\partial_\theta$  Hamiltonian structure.

**Step 9. One-step almost straightening of transport.** For every prescribed smoothing order  $M$ , one can choose a small reversible  $b$  and a constant  $\mathfrak{m} = \Omega_\gamma + O(\varepsilon)$  so that

$$\mathcal{S}^{-1}(\omega \cdot \partial_\varphi - \partial_\theta \circ V + \partial_\theta \circ W_0)\mathcal{S} = \omega \cdot \partial_\varphi + \mathfrak{m}\partial_\theta + \frac{1}{2}\mathcal{H} + \mathcal{Q}_M, \quad (19.19)$$

where  $\mathcal{Q}_M$  gains  $M$  derivatives. The construction is an almost-straightening because the cohomological equation for  $b$  is solved at the current Fourier scale; the residual is absorbed into  $\mathcal{Q}_M$ .

At first order in  $b$ , the coefficient of  $\partial_\theta$  changes by  $\omega \cdot \partial_\varphi b + Vb_\theta + V - \mathfrak{m}$ . Choose  $\mathfrak{m}$  as the space-time average and solve the zero-average equation after the current Fourier truncation. The Diophantine estimate gives the fixed derivative loss and the smallness of  $b$ . Iterating the correction a finite number of times pushes its residual below order  $-M$ .

For the Hilbert term, write its kernel as  $\frac{1}{2\pi} \cot((\theta - \theta')/2)$ . Under the same diffeomorphism, the principal odd singularity is unchanged after multiplication by the Jacobian. Taylor-expanding the difference between the transformed kernel and the original one cancels the diagonal singularity. Repeating the expansion to order  $M$  makes the difference  $C^M$  on the diagonal, hence its integral operator gains  $M$  derivatives. This proves (19.19) without importing the much longer water-wave pseudodifferential reduction.

After removing tangential and mode-2 indices, Fourier decomposition in  $\theta$  converts the constant part into real  $2 \times 2$  blocks. Put  $\kappa_n = q_\gamma^n$ . Up to the small scalar correction  $\mathfrak{m} - \Omega_\gamma$ , the unperturbed block is

$$D_n^{(0)} = \begin{pmatrix} 0 & \mathfrak{m}n - \frac{1}{2} + \frac{1}{2}\kappa_n \\ -\mathfrak{m}n + \frac{1}{2} + \frac{1}{2}\kappa_n & 0 \end{pmatrix}. \quad (19.20)$$

**Step 10. Constant block normal form and spectral split.** There is a bounded reversible block transformation that puts the constant part of (19.20) into its elliptic oscillator form for  $n = 1$  and  $n \geq \bar{n} + 1$ , and into real hyperbolic blocks for  $3 \leq n \leq \bar{n}$ . The transformed operator is

$$\mathcal{L}_0 = \omega \cdot \partial_\varphi I + D_0 + R_0, \quad (19.21)$$

where  $D_0 = \text{diag}(D_{0,n})$  has the same finite hyperbolic/infinite elliptic split as the Kirchhoff spectrum, while  $R_0$  has finite  $s$ -decay norm and can be made arbitrarily smoothing apart from a small order-zero block correction.

Because the admissible interval stays a positive distance from every spectral crossing, the symmetrizers of the elliptic blocks and the bases of the finite hyperbolic blocks are uniformly bounded together with the required  $\gamma$ -derivatives. The large-

mode asymptotics (19.8) imply

$$\Omega_n^{(0)}(\omega, \gamma) = nm - \frac{1}{2} + r_0(n, \omega, \gamma), \quad \sup_n n |\partial_{(\omega, \gamma)}^k r_0| \leq C_k.$$

The smoothing and finite-rank remainders from the previous steps have finite  $s$ -decay matrix norm, so they form  $R_0$ .

Choose a rapidly increasing sequence  $N_{j+1} = N_j^{3/2}$ . At the  $j$ th reducibility step, split  $R_j$  into its diagonal average, its Fourier block with  $|\ell| \leq N_j$ , and a high tail. Solve

$$[\omega \cdot \partial_\varphi + D_j, \Psi_j] + \Pi_{N_j} R_j - \text{diag}\langle R_j \rangle = 0. \quad (19.22)$$

**Step 11. Melnikov matrix inversion and finite-step almost reducibility.** On the finite Cantor set where the tangential and first/second Melnikov bounds with constant  $\delta_M$  hold, equation (19.22) has a reversible solution satisfying

$$|\Psi_j|_s \leq C_s \delta_M^{-1} N_j^{\tau+1} |R_j|_{s+b}. \quad (19.23)$$

The conjugated remainders satisfy a quadratic-plus-tail estimate, so after any finite number of steps

$$U_j^{-1} \mathcal{L}_0 U_j = \omega \cdot \partial_\varphi I + D_j + R_j, \quad |R_j|_{s_0} \rightarrow 0, \quad (19.24)$$

with uniform tame control.

For two elliptic modes, the entries of (19.22) divide by  $\omega \cdot \ell \pm \Omega_n^{(j)} \pm \Omega_{n'}^{(j)}$ . These are exactly the second Melnikov divisors. A finite hyperbolic block has eigenvalues with nonzero real parts uniformly separated from zero on  $\mathcal{G}$ , so its pure hyperbolic denominators are uniformly invertible; mixed hyperbolic-elliptic blocks are treated by the same finite matrix resolvent. The diagonal average is not divided out but is added to  $D_j$ . Hence (19.23) follows from the divisor bounds and finite-dimensional matrix inversion. Exponentiating  $\Psi_j$  gives

$$R_{j+1} = (I - \Pi_{N_j}) R_j + O(\delta_M^{-1} N_j^{\tau+1} |R_j|^2).$$

The smoothing tail decays faster than the growth of the small-divisor factor; choosing  $N_0$  large and the initial remainder small closes the induction. This proves finite-step almost reducibility. We deliberately do not claim an exact diagonalization at an intermediate Nash-Moser iterate.

**Step 12. Normal almost-approximate inverse.** At a Nash-Moser scale  $K$ , choose a reducibility depth for which the remainder in (19.24) is one smoothing scale smaller than the projected residual. Then

$$\mathcal{L}_\omega = \mathcal{L}_\omega^< + R_\omega + R_\omega^K + R_\omega^Z, \quad (19.25)$$

where  $\mathcal{L}_\omega^<$  is invertible on the extended Melnikov set,  $R_\omega$  is small,  $R_\omega^K$  is smoothing at scale  $K$ , and  $R_\omega^Z$  is proportional to the invariance defect. The operator  $T_\omega = (\mathcal{L}_\omega^<)^{-1}$  has a fixed-loss tame bound and  $\mathcal{L}_\omega T_\omega - I$  is an admissible almost-inverse defect.

Indeed, invert the diagonal blocks in the conjugated coordinates and conjugate back

by  $U_j$ . The Melnikov bound gives

$$\|T_\omega h\|_s \leq C_s \delta_M^{-1} (\|h\|_{s+\sigma} + (1 + \|i_0\|_{s+\sigma}) \|h\|_{s_0+\sigma}) \quad (19.26)$$

for a fixed  $\sigma$ . Multiplying (19.25) by  $T_\omega$  gives the error  $(R_\omega + R_\omega^K + R_\omega^Z)T_\omega$ . Smallness absorbs the first piece, the high-frequency gain absorbs the second after the Nash-Moser projection, and the third is quadratic in the current residual. This is precisely the almost-approximate inverse property needed below.

**Step 13. Full almost-approximate right inverse.** Combining the tangential cohomological inverse, the average equation solved by the parameter  $g$ , and the normal inverse above yields an operator  $T_0$  such that

$$\|T_0 h\|_s \leq C_s \delta_M^{-1} (\|h\|_{s+\bar{\sigma}} + (1 + \|i_0\|_{s+d+\bar{\sigma}}) \|h\|_{s_0+\bar{\sigma}}), \quad (19.27)$$

while  $D\mathcal{F}(i_0, g)T_0 - I$  is the sum of a smoothing error and an invariance-defect error. Thus the vortex-patch invariance equation satisfies the abstract Nash-Moser inverse hypothesis with fixed derivative loss.

To prove this, first solve the normal equation with  $T_\omega$ . Insert that solution in the upper triangular tangential equations. Their zero-average parts are solved Fourier by Fourier using  $|\omega \cdot \ell| \geq \delta_M \langle \ell \rangle^{-\tau}$ ; their averages determine the correction to  $g$ . Back substitution gives (19.27). Every transformation used to reach the triangular coordinates is tame, so the losses add to one fixed number  $\bar{\sigma}$ . Composing the previously identified smoothing and defect errors proves the right-inverse identity up to an admissible defect.

We can now run Nash-Moser. Let  $K_j = K_0^{(3/2)^j}$  and let  $\Pi_j$  be the corresponding smoothing projector. Given  $(i_j, g_j)$  with residual  $e_j = \mathcal{F}(i_j, g_j)$ , set

$$(\Delta i_j, \Delta g_j) = -\Pi_j T_j \Pi_j e_j, \quad (i_{j+1}, g_{j+1}) = (i_j, g_j) + (\Delta i_j, \Delta g_j). \quad (19.28)$$

**Step 14. Nash-Moser convergence on nested Cantor sets.** If  $\delta_M = \varepsilon^a$  with  $a > 0$  sufficiently small and  $K_0$  sufficiently large, the corrections in (19.28) are summable in the low norm and satisfy a controlled high-norm growth estimate. The residuals decay superlinearly, the parameter sets are nested, and the sequence converges to a reversible embedding  $i_\infty(\omega, \gamma)$  and a limiting counterterm

$$g_\infty(\omega, \gamma) = \omega + r_\varepsilon(\omega, \gamma), \quad |r_\varepsilon|_{k_0, \delta_M} \leq C\varepsilon \delta_M^{-1}. \quad (19.29)$$

Taylor expansion gives

$$e_{j+1} = (I - D\mathcal{F}_j T_j) \Pi_j e_j + (I - \Pi_j) e_j + O(\|\Delta i_j\|^2).$$

The first term is smaller by the almost-inverse estimate, the second by the smoothing gain, and the third is quadratic. With the standard two-norm induction, choose exponents  $p, q > 0$  so that

$$\|e_j\|_{s_0} \leq C\varepsilon K_j^{-p}, \quad \|i_j\|_S \leq C K_j^q.$$



The growth  $K_{j+1} = K_j^{3/2}$  makes each of the three displayed errors at most  $C\varepsilon K_{j+1}^{-p}$  after increasing  $p$  and the smoothing order. Estimate (19.27) then makes  $\sum_j \|\Delta i_j\|_{s_0} < \infty$ . Differentiating the iteration in  $(\omega, \gamma)$  at most  $k_0$  times introduces one factor  $\delta_M^{-1}$  per weighted derivative, which is exactly the weighted norm in (19.29). This proves convergence and the stated parameter regularity.

**Step 15. Counterterm inversion and recovery of the Euler frequency curve.** For  $\varepsilon \delta_M^{-1}$  small, the map  $\omega \mapsto g_\infty(\omega, \gamma)$  is a  $C^{k_0}$  diffeomorphism near the unperturbed tangential-frequency curve. Therefore

$$\omega_\varepsilon(\gamma) := g_\infty^{-1}(\tilde{\omega}(\gamma), \gamma) = \tilde{\omega}(\gamma) + o(1) \quad (19.30)$$

and the modified equation becomes the original reduced Euler Hamiltonian. The conjugate mode-2 equation determines a scalar angular-velocity shift  $\mu(\gamma, \varepsilon) = o(1)$ .

Indeed  $D_\omega g_\infty = I + D_\omega r_\varepsilon$  and the latter perturbation has norm  $< 1/2$  for the chosen powers. The finite-dimensional inverse-function theorem and differentiation of  $g_\infty(\omega_\varepsilon(\gamma), \gamma) = \tilde{\omega}(\gamma)$  give (19.30) with all weighted derivatives required for the measure argument. Setting the target counterterm to the original frequency curve is the analogue of Herman's parameter trick; again no action twist enters. The cyclic coordinate removed with  $J$  reappears only through its scalar equation, which shifts the rotating speed by  $\mu$ .

It remains to prove that the aspect ratios surviving every iteration have asymptotically full measure. We record the elementary one-dimensional sublevel estimate used here. If  $f \in C^{k_0}(I)$  and  $\max_{0 \leq k \leq k_0} |f^{(k)}(x)| \geq \rho$  on a compact interval, then, after partitioning  $I$  into finitely many subintervals on each of which one fixed derivative has size at least  $\rho/2$ , repeated Rolle monotonicity gives

$$|\{x \in I : |f(x)| < \eta\}| \leq C_{I, k_0} \left(\frac{\eta}{\rho}\right)^{1/k_0}. \quad (19.31)$$

**Step 16. Aspect-ratio Cantor measure closure.** Let  $\nu = |S|$  and choose

$$\delta_M = \varepsilon^a, \quad 0 < a < \frac{1}{k_0(1+6k_0)}, \quad \tau > k_0(1+2k_0)(\nu+1). \quad (19.32)$$

Then the perturbed tangential/normal frequencies retain half of the transversality constant for sufficiently small  $\varepsilon$ . Applying (19.31) to every first/second Melnikov resonance and using the large-mode restriction  $|n-n'| \leq C\langle \ell \rangle$  in the difference family yields a summable series. Consequently the total excluded aspect-ratio measure tends to zero as  $\varepsilon \rightarrow 0$ .

The frequency corrections in (19.29) and the normal eigenvalue corrections inherit  $k_0$  weighted derivative bounds. The power restriction in (19.32) makes them smaller than  $\rho_0/2$ , so the finite-order transversality survives. Each resonance set therefore has the sublevel bound (19.31). The asymptotic formula (19.8) shows that a normal-difference resonance can be nonempty only if  $|n-n'| \leq C\langle \ell \rangle$ ; analogous size bounds hold for the one-normal and sum families. Hence, for each  $\ell$ , only polynomially many normal pairs contribute. The condition on  $\tau$  makes the sum over  $\ell \in \mathbb{Z}^\nu$  convergent

after the  $1/k_0$  power in the sublevel estimate. The remaining prefactor is a positive power of  $\varepsilon$  by the restriction on  $a$ . Thus

$$|\mathcal{G} \setminus \mathcal{G}_\varepsilon| \longrightarrow 0. \quad (19.33)$$

**Step 17. Reconstruction of the physical vortex patch and spectral edge case.**

On the limiting Cantor set, undoing the tangential coordinates, the  $J = 0$  reduction, the symplectic rectification and the amplitude scaling gives the radial expansion in the theorem with remainder  $o(|a|)$  and angular velocity  $\Omega_\gamma + \mu$ . For  $|a|$  small,  $1 + 2\xi$  stays positive, so the reconstructed curve is a smooth embedded boundary of a genuine Euler vortex patch. If  $\gamma > 3$ , the finitely many hyperbolic Kirchhoff blocks persist and the resulting quasiperiodic patch is linearly unstable; therefore no blanket linear-stability conclusion is included in the theorem statement.

To verify the leading expansion, take  $J = 0$  as in (19.12). The limiting torus differs from its linear embedding by  $o(1)$  in the rescaled variable. Multiplication by  $\varepsilon$ , followed by the near-identity rectification inverse, therefore gives

$$\xi(\tilde{\omega}t, \theta) = \sum_{n \in S} a_n \left( M_n \cos(\tilde{\omega}_n t) \cos(n\theta) + M_n^{-1} \sin(\tilde{\omega}_n t) \sin(n\theta) \right) + r, \quad \|r\|/|a| \rightarrow 0.$$

The Sobolev index is above the continuous-embedding threshold, so  $\|\xi\|_{C^0} \rightarrow 0$  and  $1 + 2\xi > 0$ . Formula (19.1), followed by multiplication by  $e^{i(\Omega_\gamma + \mu)t}$ , is therefore a smooth embedded contour. Because every change of variables was obtained from the contour equation by a symplectic conjugacy or an invertible coordinate map, this contour solves the original parameter-independent Euler patch equation. The frequency convergence follows from (19.30), and (19.33) gives the required Borel set of asymptotically full measure.

Finally, when  $\mathcal{G} \in (1, 3)$ , all nonremoved normal blocks are elliptic. When  $\mathcal{G}$  lies beyond the first critical aspect ratio, the finite hyperbolic block in the spectral decomposition remains separated from the imaginary axis under the small KAM perturbation, so real characteristic exponents persist. This explains both the source's instability observation for  $\gamma > 3$  and why the existence theorem is formulated without a global stability claim. The proof is complete.  $\square$

**Theorem K19.2. Pöschel infinite-dimensional analytic KAM theorem**

**Statement.** Let  $\Pi \subset \mathbb{R}^n$  be closed, bounded, and of positive Lebesgue measure. On  $\mathcal{P}^{a,p} = \mathbb{T}^n \times \mathbb{R}^n \times \ell^{a,p} \times \ell^{a,p}$ , with  $\|w\|_{a,p}^2 = \sum_{j \geq 1} |w_j|^2 j^{2p} e^{2aj}$ , consider

$$H = N + P, \quad N_\xi = \langle \omega(\xi), y \rangle + \frac{1}{2} \sum_{j \geq 1} \Omega_j(\xi) (u_j^2 + v_j^2).$$

Assume: (i)  $\omega : \Pi \rightarrow \omega(\Pi)$  is a bi-Lipschitz homeomorphism; for every  $(k, l) \in \mathbb{Z}^n \times \mathbb{Z}^{(\mathbb{N})}$  with  $1 \leq |l|_1 \leq 2$  the exact divisor  $\langle k, \omega(\xi) \rangle + \langle l, \Omega(\xi) \rangle$  has a zero set of Lebesgue measure zero, and  $\langle l, \Omega(\xi) \rangle \neq 0$  on  $\Pi$  whenever  $0 < |l|_1 \leq 2$ ; (ii) for some  $d \geq 1$  and  $\delta < d - 1$ ,  $\Omega_j(\xi) = \bar{\Omega}_j + \tilde{\Omega}_j(\xi)$  with  $\bar{\Omega}_j = j^d + \sum_{\nu=1}^s c_\nu j^{d_\nu}$ , where  $d > d_1 > \dots > d_s > \delta$ , and

$\tilde{\Omega} : \Pi \rightarrow \ell_\infty^{-\delta}$  is Lipschitz; when  $d = 1$ , choose  $\kappa > 0$  so that  $(\tilde{\Omega}_i - \tilde{\Omega}_j)/(i - j) = 1 + O(j^{-\kappa})$  uniformly for  $i > j$  and enlarge  $\delta < 0$ , if necessary, so that  $-\delta \leq \kappa$ ; (iii)  $P$  is real analytic in the phase variables and Lipschitz in  $\xi$ , its Hamiltonian vector field maps  $\mathcal{P}^{a,p}$  analytically to  $\mathcal{P}^{a,\bar{p}}$ , with  $\bar{p} \geq p$  if  $d > 1$  and  $\bar{p} > p$  if  $d = 1$ , and the smoothing indices satisfy  $p - \bar{p} \leq \delta < d - 1$ .

Fix  $0 < \alpha \leq 1$  and  $\tau \geq n + 1$ . If

$$\varepsilon := |X_P|_{r,D(s,r)} + \frac{\alpha}{M} |X_P|_{r,D(s,r)}^L \leq \gamma\alpha,$$

where  $M$  bounds the parameter Lipschitz seminorms of  $\omega$  and  $\tilde{\Omega}$  and  $\gamma = \gamma(n, \tau, s)$  is sufficiently small, then Pöschel's analytic Newton scheme produces a Cantor set  $\Pi_\alpha \subset \Pi$ , a Lipschitz family of nearby real-analytic invariant rotational  $n$ -tori, and shifted Diophantine tangential frequencies. The conjugated Hamiltonian has an elliptic constant quadratic normal part modulo terms of weighted order at least three; consequently every persisted torus is linearly stable.

For the full-measure conclusion assume, in addition,

$$\tau \geq \begin{cases} n + 1 + \frac{2}{d-1}, & d > 1, \\ (n+3)\frac{\delta-1}{\delta}, & d = 1. \end{cases}$$

After replacing  $\gamma$  by the smaller constant required to control the finite low-resonance set, the excluded measure satisfies  $|\Pi \setminus \Pi_\alpha| \rightarrow 0$  as  $\alpha \downarrow 0$ .

*Proof.* The theorem has two rather different halves. The analytic half constructs the invariant tori by a Newton iteration. The geometric half proves that the successive small-divisor exclusions discard asymptotically no parameters. Keeping these halves separate is useful, because the bi-Lipschitz assumption on the tangential frequency map is needed only for the second half.

**1. The order-two homological equation.** Introduce complex normal coordinates

$$z_j = \frac{u_j - iv_j}{\sqrt{2}}, \quad \bar{z}_j = \frac{u_j + iv_j}{\sqrt{2}}.$$

At one Newton step we replace the perturbation by its Taylor polynomial  $R$  of weighted degree at most two in  $(y, z, \bar{z})$ ; here  $y$  has degree two and each normal variable has degree one. Thus the part of the perturbation which can alter the torus or its linear normal dynamics is retained exactly.

A Fourier-Taylor monomial has the form

$$e^{ik \cdot x} y^m z^q \bar{z}^{\bar{q}}, \quad 2|m| + |q + \bar{q}| \leq 2.$$

Poisson bracket with the normal form multiplies this monomial by

$$i(k \cdot \omega + (q - \bar{q}) \cdot \Omega).$$

Consequently the homological equation is solved coefficient by coefficient. The an-

gle averages linear in  $y$  and the diagonal averages  $z_j \bar{z}_j$  are not divided out; they are absorbed into the new tangential and normal frequencies. Every other coefficient is divided by an order-two Melnikov divisor.

The loss in the angle variable is the familiar analytic small-divisor loss. On a strip narrowed by  $\sigma$ , Fourier decay gives a factor  $e^{-|k|\sigma}$ , while the inverse divisor contributes  $1 + |k|^\tau$ . Parameter differences produce a second inverse divisor. Hence one obtains a bound of the form

$$|X_F|_{s-\sigma, r}^\lambda \leq \frac{C}{\alpha \sigma^{2\tau+n+1}} |X_R|_{s, r}^\lambda.$$

The combined norm  $|\cdot|^\lambda$  is precisely designed so that the parameter-Lipschitz term is absorbed by the same Melnikov constant.

There is one genuinely infinite-dimensional point here. The coefficients of terms  $z_i \bar{z}_j$  form an infinite matrix. After the weighted phase-space norms are conjugated away, the worst off-diagonal division reduces to the following elementary operator estimate: if  $A = (A_{ij})$  is bounded on  $\ell^2$  and

$$B_{ij} = \frac{|A_{ij}|}{|i-j|} \quad (i \neq j), \quad B_{ii} = 0,$$

then

$$\|B\| \leq \frac{\pi}{\sqrt{3}} \|A\|.$$

Indeed, Cauchy-Schwarz bounds every row sum and every column sum by  $(\pi/\sqrt{3})\|A\|$ , because  $\sum_{m \neq 0} m^{-2} = \pi^2/3$ ; a second Cauchy-Schwarz estimate then gives the operator norm. The spectral separation  $|i^d - j^d| \gtrsim |i-j| \max(i, j)^{d-1}$  and the smoothing relation  $p - \bar{p} \leq \delta < d-1$  supply the remaining normal-index weights. The one-normal block is easier, since division by one normal frequency gains a power  $j^{-d}$ . This closes the infinite-mode homological equation rather than treating it as a finite-dimensional calculation with infinitely many labels.

## 2. One analytic KAM step. Let

$$t = 2\tau + n + 2.$$

Choose  $0 < \eta < 1/8$  and a strip loss  $0 < \sigma < s$ . If the current perturbation is small enough that

$$|X_P|^\lambda \leq c^{-1} \alpha \sigma^t \eta^2,$$

the homological solution generates a well-defined time-one Hamiltonian flow on a smaller complex domain. Cauchy estimates give both the displacement of the flow and the derivative of the flow.

The crucial point is what is, and what is not, truncated. P"oschel takes the *full* Taylor polynomial of  $P$  in  $(y, z, \bar{z})$ . He does not truncate its Fourier series at this stage. Using

$$\{F, N\} + \hat{N} = R,$$

Lie's formula gives

$$P_+ = (P - R) \circ X_F^1 + \int_0^1 \{(1-t)\widehat{N} + tR, F\} \circ X_F^t dt.$$

The first term is a Taylor remainder and the second is quadratic in the size of the perturbation. On the new radius  $\eta r$  this gives

$$|X_{P_+}|^\lambda \leq C \left( \frac{|X_P|^{\lambda^2}}{\alpha \sigma^t \eta^2} + \eta |X_P|^\lambda \right).$$

There is no exponential Fourier-tail term. The cutoffs introduced later serve a different purpose: they tell us which small divisors have already been verified and which new resonance zones must be removed.

The normal-form correction changes the frequencies by

$$|\widehat{\omega}|^\lambda + |\widehat{\Omega}|_{-\delta}^\lambda \leq C |X_P|^\lambda.$$

Therefore all Melnikov inequalities with  $|k| \leq K$  persist after replacing  $\alpha$  by a slightly smaller number, provided the current perturbation is small compared with the finite cutoff  $K$ . This is where  $K$  enters one KAM step.

**3. The Newton sequence.** P"oschel chooses the parameters so that the two terms in the one-step error are balanced. Put  $\kappa_0 = 4/3$  and choose

$$\alpha_\nu = \frac{\alpha_0}{2}(1 + 2^{-\nu}), \quad M_\nu = M_0(2 - 2^{-\nu}), \quad \sigma_{\nu+1} = \frac{1}{2}\sigma_\nu,$$

$$\varepsilon_{\nu+1} = \frac{c_1 \varepsilon_\nu^{4/3}}{(\alpha_\nu \sigma_\nu^t)^{1/3}}, \quad \eta_\nu^3 = \frac{\varepsilon_\nu}{\alpha_\nu \sigma_\nu^t},$$

and

$$s_{\nu+1} = s_\nu - 5\sigma_\nu, \quad r_{\nu+1} = \eta_\nu r_\nu.$$

If  $\sigma_0 = s_0/40$  and the initial perturbation is sufficiently small, a direct induction gives

$$\varepsilon_\nu \leq C \alpha_\nu \sigma_\nu^t 2^{-\nu}.$$

Thus the coordinate changes and the frequency shifts are summable. A cutoff  $K_\nu = K_0 2^\nu$  is chosen so that the loss in the already verified divisors is smaller than  $\alpha_\nu - \alpha_{\nu+1}$ ; only resonance zones beyond the previous cutoff are newly removed.

The compositions of the symplectic changes therefore converge on the fixed strip  $|\operatorname{Im} x| < s_0/2$  to a Lipschitz family of analytic embeddings  $\Phi_\infty$ . The transformed perturbing vector fields tend to zero, so passing to the limit in the vector-field identity gives

$$X_H \circ \Phi_\infty = D\Phi_\infty \omega_*.$$

Each fiber of  $\Phi_\infty$  is consequently an invariant rotational torus.

The same estimates also give convergence of the quadratic jets. Indeed, two extra derivatives cost only two extra geometric powers of  $\sigma_\nu^{-1}$ , whereas the  $4/3$  Newton re-

currence makes  $\varepsilon_\nu$  decay supergeometrically. In limiting symplectic coordinates the Hamiltonian has therefore the form

$$H_* = \langle \omega_*, y \rangle + \sum_j \Omega_{*,j} z_j \bar{z}_j + O_{\text{wt}}(3).$$

The normal variational equation is  $\dot{z}_j = -i\Omega_{*,j} z_j$ , with real  $\Omega_{*,j}$ , and hence preserves the weighted normal norm. The limiting change of variables and its inverse have bounded derivatives, so the original variational flow is bounded as well. This is the asserted linear stability.

**4. A single resonance strip.** We now prove that the surviving Cantor set becomes full measure as  $\alpha \downarrow 0$ . This part is geometric rather than iterative.

First, pure normal divisors have a uniform lower bound

$$|l \cdot \Omega'| \geq c \langle l \rangle_d, \quad 0 < |l|_1 \leq 2,$$

for all sufficiently small perturbations of the frequencies. For large normal indices this follows from the spectral asymptotics; the finitely many remaining patterns are bounded away from zero by the exact nonresonance assumption. Therefore, if a mixed resonance strip is nonempty, then

$$|k| \gtrsim \langle l \rangle_d.$$

Use the unperturbed tangential frequency  $\eta = \omega(\xi)$  as the parameter. This is legitimate because  $\omega$  is bi-Lipschitz. Along a line in the  $k$ -direction, the tangential part of the divisor changes at speed comparable to  $|k|$ . Once  $|k|$  dominates the weighted size of  $l$ , the Lipschitz normal part can consume at most half of that speed. Fubini's theorem then yields

$$|\mathcal{R}_{kl}| \leq C\alpha \frac{\langle l \rangle_d}{A_k |k|}$$

after excluding only finitely many low pairs. Thus the infinite problem has been reduced to counting how many two-normal-mode vectors  $l$  can occur for a fixed  $k$ .

**5. Counting when  $d > 1$ .** A nonempty strip has  $\langle l \rangle_d \lesssim |k|$ . For  $|l|_1 \leq 2$ , Pöschel's elementary weighted-index inequality

$$\sum_j |l_j| j^{d-1} \leq 2 \langle l \rangle_d$$

forces every normal index occurring in  $l$  to be at most  $C|k|^{1/(d-1)}$ . Consequently there are at most  $C|k|^{2/(d-1)}$  possible pairs. The union of all corresponding strips for a fixed  $k$  has measure

$$\leq C\alpha \frac{|k|^{2/(d-1)}}{1 + |k|^\tau}.$$

Since the number of integer vectors with  $|k| \simeq R$  is  $O(R^{n-1})$ , the sum converges with a  $1/K$  tail once

$$\tau \geq n + 1 + \frac{2}{d-1}.$$

**6. Counting when  $d = 1$ .** This case is subtler because a difference  $\Omega_i - \Omega_j$  may stay of order  $i - j$  even when both indices are very large. Same-sign pairs are still easy and contribute only a quadratic counting factor. For the difficult opposite-sign pair  $l = e_i - e_j$ , write  $m = i - j > 0$ . The fixed spectral expansion gives

$$\frac{\bar{\Omega}_i - \bar{\Omega}_j}{i - j} = 1 + O(j^{-\kappa}),$$

while the parameter-dependent tail is  $O(\alpha j^\delta)$  with  $\delta < 0$ . For a cutoff  $j_0$ , the finitely many  $j < j_0$  are estimated one by one; all  $j \geq j_0$  lie in one enlarged strip around the limiting divisor  $k \cdot \omega' + m$ . Balancing the number of low indices against the two high index errors yields

$$|\mathcal{R}'_k| \leq C \alpha^{\kappa/(\kappa+1)} \frac{|k|^2}{A_k^{\delta/(\delta-1)}}.$$

The convention  $-\delta \leq \kappa$  is exactly what makes this balance uniform. The lattice sum has a  $1/K$  tail provided

$$\tau \geq (n+3) \frac{\delta-1}{\delta}.$$

This is why the measure theorem needs a stronger arithmetic exponent than the existence theorem.

**7. The finitely many remaining resonances.** All strips not covered by the preceding estimates form a finite set of index pairs. Reduce the initial smallness constant so that the first KAM cutoff is past all of them. Their defining functions may then be taken to be the unperturbed exact divisors. For each fixed pair, the resonance strips decrease as  $\alpha \downarrow 0$  to the exact zero set, which has Lebesgue measure zero by hypothesis. Continuity from above of measure and finiteness of this exceptional set give a vanishing contribution.

Combining the finite part with the summable infinite tails proves

$$|\Pi \setminus \Pi_\alpha| \longrightarrow 0.$$

Together with the Newton construction and the quadratic-jet argument, this proves all assertions.  $\square$

### Theorem K19.3. Reversible derivative-wave KAM and reducibility theorem

**Statement.** Fix a finite set  $I^+ \subset \mathbb{N} \setminus \{0\}$  of tangential sites and a compact mass interval  $[m_1, m_2] \Subset (0, \infty)$ . Consider

$$y_{tt} - y_{xx} + my = g(x, y, y_x, y_t), \quad x \in \mathbb{T},$$

where  $g$  is real analytic near the origin, satisfies  $g(x, y, y_x, -v) = g(x, y, y_x, v)$  and  $g(-x, y, -y_x, v) = g(x, y, y_x, v)$ , and has the exact Taylor form

$$g(x, y, y_x, v) = y y_x^2 + g^{(\geq 5)}(x, y, y_x, v),$$

with  $g^{(\geq 5)}$  containing only monomials of total degree at least five in  $(y, y_x, v)$ ; in

particular there are no quadratic or quartic terms. Then there is a finite exceptional set  $\mathcal{E}_{I^+, [m_1, m_2]} \subset [m_1, m_2]$  such that, for every  $m \in [m_1, m_2] \setminus \mathcal{E}_{I^+, [m_1, m_2]}$ , a Cantor family of sufficiently small positive actions has asymptotic density one at the origin and parametrizes small-amplitude solutions analytic in  $(t, x)$ , quasiperiodic and reversible in time, and even in  $x$ . Their leading term is

$$\sum_{j \in I^+} \sqrt{8\xi_j} \lambda_j^{-1} \cos(\omega_j^\infty(\xi)t) \cos(jx) + o(\sqrt{|\xi|}), \quad \lambda_j = \sqrt{j^2 + m}.$$

The associated invariant tori have zero Lyapunov exponents, and the equations linearized along them are analytically reducible to constant coefficients.

*Proof.*

### Proof strategy

The proof separates the PDE regularization from the small-divisor iteration. Reversible changes of variables first reduce the linearized derivative-wave operator to a diagonal principal part plus a sufficiently smoothing remainder. Second Melnikov conditions make the remaining homological equations invertible with quantified losses. A Nash-Moser/KAM iteration alternates these reducibility steps with nonlinear corrections, and the tame estimates close because the smoothing gain dominates the derivative loss.

Put

$$D := \sqrt{-\partial_{xx} + m}, \quad u^+ := \frac{Dy - iy_t}{\sqrt{2}}, \quad u^- := \frac{Dy + iy_t}{\sqrt{2}}.$$

Then

$$u_t^+ = iDu^+ + i\mathfrak{g}(u^+, u^-), \quad u_t^- = -iDu^- - i\mathfrak{g}(u^+, u^-), \quad (19.34)$$

where  $\mathfrak{g}$  is obtained from  $g$  by inserting  $y = D^{-1}(u^+ + u^-)/\sqrt{2}$  and  $y_t = (u^- - u^+)/(i\sqrt{2})$ . In Fourier variables

$$u^+ = \sum_{j \in \mathbb{Z}} u_j^+ e^{ijx}, \quad u^- = \sum_{j \in \mathbb{Z}} u_j^- e^{-ijx}, \quad \lambda_j := \sqrt{j^2 + m},$$

the linear part is diagonal with eigenvalues  $\pm i\lambda_j$ . The real phase space is  $\overline{u_j^+} = u_j^-$ ; the space-reflection hypothesis makes

$$E_0 := \{u_j^\pm = u_{-j}^\pm \text{ for all } j\}$$

invariant; and time reversibility becomes the involution  $S(u_j^+, u_j^-) = (u_{-j}^-, u_{-j}^+)$ . Thus the nonlinear vector field is reversible, real-on-real and even in the reversible/even sense defined above.

The only analytic point in this first conversion is the boundedness of  $D^{-1} : \ell^{a,p} \rightarrow \ell^{a,p+1}$  and of  $\partial_x D^{-1} : \ell^{a,p} \rightarrow \ell^{a,p}$ : indeed  $\lambda_j^{-1} \leq \langle j \rangle^{-1}$  up to a constant depending on the fixed compact mass interval, while  $|j|/\lambda_j \leq 1$ . Hence the analytic Taylor series of  $g$  induces an analytic vector field in the sequence space. Reality and the two symmetry identities are checked monomial by monomial, and the inverse transformation recovers



$y = D^{-1}(u^+ + u^-)/\sqrt{2}$ . This proves the reversible complex formulation.

Let  $I^+ \subset \mathbb{N} \setminus \{0\}$  be fixed,  $I := I^+ \cup (-I^+)$ , and put  $q := |I^+|$ . Before introducing actions we first control the nonlinear vector field in the coordinates  $u^\pm$ . For a monomial vector field

$$e^{ik \cdot x} y^i z^\alpha \bar{z}^\beta \partial_v$$

write  $\pi(k, \alpha, \beta; v)$  for its signed Fourier momentum and weight its majorant coefficient by  $e^{a|\pi|}$ , as in the  $a$ -momentum majorant norm defined above. The triangle inequality

$$|\pi([X, Y])| \leq |\pi(X)| + |\pi(Y)|$$

for every monomial produced by differentiation implies the basic commutator estimate

$$\|[X, Y]\|_{s', r', a'} \leq C \delta^{-1} \|X\|_{s, r, a} \|Y\|_{s, r, a}, \quad \delta := \min\{s - s', r - r', a - a'\}, \quad (19.35)$$

whenever the primed domain is strictly smaller. This is simply the analytic Cauchy estimate applied after taking coefficientwise majorants.

To prove this, expand every coefficient of  $g^{(\geq 5)}$  in its Fourier series. analytic Fourier decay gives exponential decay in the explicit  $x$ -frequency, while each occurrence of  $D^{-1}$  contributes a factor  $\lambda_j^{-1}$  and each occurrence of  $\partial_x D^{-1}$  contributes  $j/\lambda_j$ , uniformly bounded. A vector field homogeneous of degree  $d$  in  $(u^+, u^-)$  has scaled vector-field size  $O(R^{d-1})$ ; hence the cubic field is  $O(R^2)$  and the order-at-least-five field is  $O(R^4)$ .

For the high-mode linear projection, choose scales  $0 < b < L < 1$ , and restrict to monomials with  $|k| < N^b$  and low external momentum  $O(N^L)$ . If the input and output normal indices  $m, n$  satisfy  $|m|, |n| > \theta N$ , the coefficient of the projected cubic field can be written as a function of the sign of  $m$  and the difference  $\sigma m - \sigma' n$ , plus an error of order  $N^{-1}$ . The reason is

$$\lambda_j = |j| + \frac{m}{2|j|} + O(|j|^{-3}),$$

so replacing every high  $\lambda_j$  by  $|j|$  changes the coefficient by  $O(N^{-1})$  after the low factors have been frozen. This gives the Toeplitz approximation and defect required by the quasi-Toeplitz definition. The exponential momentum weight suppresses the complementary high-momentum monomials. Taking the supremum over  $N \geq N_0$  proves the quasi-Toeplitz bound and the quasi-Toeplitz input estimate.

We next remove the nonresonant cubic terms. The linear vector field is

$$\mathcal{N}_0 = \sum_{\sigma=\pm, j \in \mathbb{Z}} \sigma i \lambda_j u_j^\sigma \partial_{u_j^\sigma}.$$

On a cubic monomial  $u_{j_1}^{\sigma_1} u_{j_2}^{\sigma_2} u_{j_3}^{\sigma_3} \partial_{u_j^\sigma}$ ,

$$[\mathcal{N}_0, \cdot] = i(\sigma_1 \lambda_{j_1} + \sigma_2 \lambda_{j_2} + \sigma_3 \lambda_{j_3} - \sigma \lambda_j)(\cdot). \quad (19.36)$$

The only zero divisors compatible with zero total momentum are the pairwise resonances. For every other zero-momentum quartet the elementary convexity estimate for  $r \mapsto \sqrt{r^2 + m}$  yields

$$|\sigma_1 \lambda_{j_1} + \sigma_2 \lambda_{j_2} + \sigma_3 \lambda_{j_3} + \sigma_4 \lambda_{j_4}| \geq \frac{cm}{(n_0^2 + m)^{3/2}}, \quad (19.37)$$

with  $n_0 = \min_h \langle j_h \rangle$ . Indeed, after ordering the absolute indices, twice applying the mean-value theorem writes a nontrivial four-wave combination as a product of a nonzero integer difference and a second derivative of  $\sqrt{r^2 + m}$ ; that second derivative is  $m(r^2 + m)^{-3/2}$ , and the finitely many configurations not covered by one ordering are obtained by permutation. Therefore the cubic homological equation can be solved coefficientwise on every nonresonant monomial.

Let  $F$  be defined by dividing the nonresonant cubic coefficients by (19.36). Estimate (19.37) controls the finitely many low denominators, and the same high-index replacement used in the quasi-Toeplitz input estimate gives a Toeplitz approximation of  $F$  with  $N^{-1}$  defect. Hence  $F$  is quasi-Toeplitz and  $\|F\|^T \leq CR^2$ . The flow exists on the  $R/2$  ball by (19.35). Lie's formula and the cubic homological equation give

$$e^{\text{ad } F}(\mathcal{N}_0 + G) = \mathcal{N}_0 + G_1 + G_2 + G^{(\geq 5)} + O([F, G^{(=3)}]) + O(\text{ad}_F^2 \mathcal{N}_0).$$

The last two terms are order five as vector fields and therefore have scaled size  $O(R^4)$ . The higher part already has size  $O(R^4)$  by the quasi-Toeplitz input estimate. Thus the new remainder is  $O(R^4)$ . If a quartic nonlinearity were present, its vector field would instead be  $O(R^3)$  and this estimate would fail at the later KAM smallness scale; this is exactly why the theorem hypothesis excludes quartic terms. the Birkhoff normal-form estimate follows.

Introduce the action-angle coordinates

$$u_j^+ = \sqrt{\xi_{|j|} + y_j} e^{ix_j}, \quad u_j^- = \sqrt{\xi_{|j|} + y_j} e^{-ix_j}, \quad j \in I,$$

and put  $(u_j^+, u_j^-) = (z_j, \bar{z}_j)$  for  $j \notin I$ . On the symmetric subspace  $x_j = x_{-j}$ ,  $y_j = y_{-j}$ ,  $z_j = z_{-j}$ ,  $\bar{z}_j = \bar{z}_{-j}$ . For the special cubic  $yy_x^2$  the Birkhoff normal form has affine frequencies

$$\omega_i(\xi) = \lambda_i + \lambda_i^{-1} \left( -\frac{1}{4} a_i \xi_i + a \cdot \xi \right), \quad i \in I^+, \quad (19.38)$$

$$\Omega_j(\xi) = \lambda_j + \lambda_j^{-1} a \cdot \xi, \quad j \notin I, \quad (19.39)$$

where

$$a_i = -i^2 \lambda_i^{-2} \neq 0.$$

Writing  $J_q$  for the  $q \times q$  all-ones matrix, the twist matrix is

$$A = \text{diag}(\lambda_i^{-1}) \left( -\frac{1}{4} I_q + J_q \right) \text{diag}(a_i). \quad (19.40)$$

Since  $J_q^2 = qJ_q$ ,

$$A^{-1} = \text{diag}(a_i^{-1}) \left( -4I_q + \frac{16}{4q-1} J_q \right) \text{diag}(\lambda_i), \quad (19.41)$$

so the action-to-frequency map is nondegenerate for every  $m > 0$ .

Indeed (19.41) gives

$$(A^T)^{-1} a = \frac{4}{4q-1} \bar{\omega}, \quad \bar{\omega} = (\lambda_i)_{i \in I^+}. \quad (19.42)$$

Each low-mode resonance is therefore an analytic equation in  $m$  involving finitely many square roots  $\sqrt{h^2 + m}$ . It is not identically zero: for a putative identity, repeated squaring would give an algebraic identity whose largest- $m$  asymptotic contradicts the nonzero integer coefficient, while the one-normal equation is even more immediate from the strict monotonicity of  $\lambda_h$ . A nonzero real-analytic function has isolated zeros. Compactness of  $[m_1, m_2]$  and the finite number of pairs  $|i|, |j| \leq C_0$  therefore leave only finitely many masses. This proves the twist and exceptional-mass argument.

We now prove the two closure mechanisms that are genuinely specific to the reversible derivative-wave problem. Let  $\Pi_{N,\theta,\mu}$  denote the projection onto high-output/high-input normal monomials with low Fourier and low momentum complexity from the quasi-Toeplitz definition. If

$$\Pi_{N,\theta,\mu}X = \tilde{X} + N^{-1}\hat{X}, \quad \Pi_{N,\theta,\mu}Y = \tilde{Y} + N^{-1}\hat{Y},$$

with  $\tilde{X}, \tilde{Y}$  Toeplitz, then on slightly weaker parameters  $(s', r', a', \theta', \mu')$  the projected commutator decomposes as

$$\Pi_{N,\theta',\mu'}[X, Y] = \Pi_{N,\theta',\mu'}[\tilde{X}, \tilde{Y}] + N^{-1}\hat{Z}.$$

The first term is Toeplitz: the high indices can enter only through their sign and the difference  $\sigma m - \sigma' n$ , because every intermediate low index is  $O(N^L)$  and the scale gaps  $b < L < 1$  prevent a high-high interaction from switching sign. Every term in  $\hat{Z}$  either contains one old defect or one discarded high-momentum/high-Fourier monomial. The first class is bounded by (19.35); the second gains  $e^{-N^b \min\{s-s', a-a'\}}$ , which is at most  $N^{-1}$  for  $N_0$  large. Thus

$$\|[X, Y]\|_{p'}^T \leq C\delta^{-1}\|X\|_p^T\|Y\|_p^T. \quad (19.43)$$

Iteration of (19.43) proves Lie-series closure whenever  $C\delta^{-1}\|F\|_p^T < 1$ .

The even subspace creates exact resonances between  $j$  and  $-j$ . Define the symmetrization  $SX$  by replacing, only in the resonant degree-at-most zero monomials,  $e^{ik \cdot x}$  with 1 for odd-symmetric  $k$  and replacing  $z_{-j}$  by  $z_j$ ,  $\bar{z}_{-j}$  by  $\bar{z}_j$ . Every replacement is an identity on  $E$ , so

$$X|_E = (SX)|_E. \quad (19.44)$$

Moreover the replacement never increases the momentum majorant; the omitted high- $k$  or high-momentum terms are exponentially small. Hence, after increasing  $N_0$ ,

$$\|SX\|_{s,r,a} \leq \|X\|_{s,r,a}, \quad \|SX\|^T \leq 9\|X\|^T. \quad (19.45)$$

Reversibility, real-on-real and evenness are preserved by the same monomial check.

This proves the required quasi-Toeplitz closure statement. We turn to the homological equation. On the action-angle phase space the normal form is

$$\mathcal{N} = \omega(\xi) \cdot \partial_x + i \sum_{j \notin I} \Omega_j(\xi) z_j \partial_{z_j} - i \sum_{j \notin I} \Omega_j(\xi) \bar{z}_j \partial_{\bar{z}_j}.$$

For the degree-at-most-zero truncation  $R$  of the perturbation, after symmetrization

solve

$$\mathrm{ad}_{\mathcal{N}} F = R - [R]. \quad (19.46)$$

Every coefficient is divided by one of the four families

$$\omega \cdot k, \quad \omega \cdot k \pm \Omega_j, \quad \omega \cdot k + \Omega_i + \Omega_j, \quad \omega \cdot k + \Omega_i - \Omega_j, \quad (19.47)$$

excluding the diagonal kernel. Let the good set impose the lower bound  $\gamma \langle k \rangle^{-\tau}$  for these divisors at  $|k| \leq K$ , with the usual harmless strengthening for the pure tangential divisor. Analytic Fourier decay and a lattice-shell sum give

$$\|F\|_{s-\sigma, r, a} \leq C \gamma^{-1} K^{2\tau+1} \|R\|_{s, r, a}. \quad (19.48)$$

The parameter-Lipschitz estimate has the same form after one more inverse divisor, which is absorbed by the project combined Lipschitz weight.

The nontrivial point is quasi-Toeplitz preservation. In the high-high linear block, replace the coefficients of  $R$  by their Toeplitz approximants and replace

$$\Omega_m - \Omega_n = (|m| - |n|) + O(|m|^{-1} + |n|^{-1})$$

by its Toeplitz leading divisor. On the projected regime the Melnikov lower bound remains uniform after this replacement, and

$$\frac{1}{D} - \frac{1}{\tilde{D}} = \frac{\tilde{D} - D}{D\tilde{D}} = O(N^{-1})\gamma^{-2}\langle k \rangle^{2\tau}.$$

Thus the quotient of the Toeplitz numerator by  $\tilde{D}$  is Toeplitz and the difference is an  $N^{-1}$  defect. All lower-degree components are finite or one-normal blocks and satisfy the same estimate directly.

Anti-reversibility follows because  $\mathrm{ad}_{\mathcal{N}}$  maps anti-reversible fields to reversible fields and the normalized inverse is unique. The real/even properties follow coefficientwise. Together with the preceding Toeplitz divisor calculation, this proves the homological-equation estimate.

We now perform one KAM step. Split the perturbation by degree as  $R = R^{(-1)} + R^{(0)} + R^{(\geq 1)}$ , solve the homological-equation estimate for the truncated  $R_K^{\leq 0}$ , and set

$$X^+ := \mathcal{S}e^{\mathrm{ad}_F}(\mathcal{N} + R) = \mathcal{N}^+ + R^+.$$

The normal-form averages of  $R^{(0)}$  define the shifts  $\hat{\omega}, \hat{\Omega}$ . If

$$\varepsilon^{(h)} := \gamma^{-1} \|R^{(h)}\|^T, \quad h = -1, 0, \quad \bar{\varepsilon} := \varepsilon^{(-1)} + \varepsilon^{(0)}, \quad \Theta := \gamma^{-1} \|R\|^T,$$

then Cauchy estimates, the quasi-Toeplitz closure argument and the homological-

equation estimate give on the smaller parameters

$$\varepsilon_+^{(-1)} \leq C \left( \delta_+^{-2} K^{4\tau+2} \bar{\varepsilon}^2 + \varepsilon^{(-1)} e^{-K\Delta} \right), \quad (19.49)$$

$$\varepsilon_+^{(0)} \leq C \left( \delta_+^{-2} K^{4\tau+2} (\varepsilon^{(-1)} + \bar{\varepsilon}^2) + \varepsilon^{(0)} e^{-K\Delta} \right), \quad (19.50)$$

$$\Theta_+ \leq \Theta(1 + C\delta_+^{-2} K^{4\tau+2} \bar{\varepsilon}). \quad (19.51)$$

Here  $\Delta$  is the minimum loss of analytic and momentum width. These follow by writing

$$R^+ = \mathcal{S} \left( R - R_K^{\leq 0} + [F, R^{\leq 0}] + [F, R^{\geq 1}] + \sum_{j \geq 2} \frac{1}{j!} \text{ad}_F^j(\mathcal{N} + R) \right)$$

and estimating the commutator terms by (19.43); the Fourier tail supplies the exponential term.

The diagonal correction has an additional high-index structure. Because the diagonal part of a quasi-Toeplitz field is its Toeplitz constant plus an  $O(|j|^{-1})$  defect, there is a scalar  $\hat{a}(\xi)$  such that

$$\hat{\Omega}_j(\xi) = \hat{a}(\xi) + O(\gamma/|j|), \quad |j| \rightarrow \infty, \quad (19.52)$$

with the same Lipschitz estimate. Therefore the updated normal frequencies retain the form

$$\Omega_j^+(\xi) = \lambda_j + c_+(\xi) + O(\gamma/|j|). \quad (19.53)$$

The index-independent constant cancels from differences  $\Omega_i^+ - \Omega_j^+$ , which is the key to second Melnikov measure estimates.

This proves the quantitative one-step estimate. Choose a rapidly increasing sequence  $K_\nu$  and geometrically shrinking losses  $\delta_\nu$ ,  $s_\nu - s_{\nu+1}$ ,  $a_\nu - a_{\nu+1}$  so that  $e^{-K_\nu \Delta_\nu}$  is dominated by a fixed superlinear power of  $\bar{\varepsilon}_\nu$ . Define the next Cantor set by retaining exactly the four Melnikov families (19.47) for  $|k| \leq K_\nu$ . If  $R$  is small enough, the initial estimates established below imply inductively

$$\bar{\varepsilon}_{\nu+1} \leq C_0 \bar{\varepsilon}_\nu^{3/2}, \quad \sum_\nu \|F_\nu\|^T < \infty, \quad \sup_\nu \Theta_\nu < \infty. \quad (19.54)$$

Indeed the polynomial factor in  $K_\nu$  and  $\delta_\nu^{-1}$  grows only geometrically, while the recursion  $x \mapsto C_0 x^{3/2}$  decays supergeometrically once  $x_0$  is below a fixed threshold. The infinite products of the near-identity derivatives therefore converge on a positive analytic strip.

For completeness, let  $\Phi_\nu$  be the time-one map generated by  $F_\nu$  and  $\Psi_\nu = \Phi_0 \circ \dots \circ \Phi_{\nu-1}$ . the homological-equation estimate and (19.54) make  $\sum \|\Phi_\nu - \text{id}\|$  and  $\sum \|D\Phi_\nu - I\|$  convergent. Hence  $\Psi_\nu$  converges uniformly with its first derivative on the limiting strip. The normal-form increments are bounded by  $C\gamma\bar{\varepsilon}_\nu$ , so the frequency sequences converge. At stage  $\nu$ , the degree-at-most-zero perturbation tends to zero; evaluating the conjugated vector-field identity on the zero section and passing to the limit gives the displayed invariance equation. Each  $\Phi_\nu$  is anti-reversible generated and is followed by symmetrization that leaves the flow on  $E$  unchanged, so the limiting torus is real,

even in  $x$ , and reversible. This proves convergence of the Cantor iteration.

The same iteration gives reducibility rather than merely existence. The normal linearization of the transformed vector field along the zero torus is

$$\dot{z}_j = i\Omega_j^\infty(\xi)z_j, \quad \dot{\bar{z}}_j = -i\Omega_j^\infty(\xi)\bar{z}_j,$$

because the limiting degree-at-most-zero perturbation vanishes on  $E$ . The derivative of the convergent coordinate change along the torus is an analytic quasiperiodic linear isomorphism, so it conjugates the original variational equation to this constant diagonal system.

This proves the reducibility and stability assertions. It remains to show that the limiting Cantor set has density one. On an action box

$$\mathcal{O}_\rho := \{\xi \in \mathbb{R}_+^q : \rho/2 \leq \xi_i \leq \rho\},$$

the unperturbed frequencies have the affine form

$$\omega(\xi) = \bar{\omega} + A\xi, \quad \Omega_j(\xi) = \lambda_j + \lambda_j^{-1}a \cdot \xi.$$

Change variables to  $\zeta = \omega(\xi)$ ; by the twist and exceptional-mass argument this is bi-Lipschitz with Jacobian comparable to a nonzero constant. A second-Melnikov difference can then be written

$$\omega^\infty(\xi) \cdot h + \Omega_i^\infty(\xi) - \Omega_j^\infty(\xi) = c_{hij} \cdot \zeta + d_{ij} + r_{ij}(\zeta), \quad (19.55)$$

where

$$c_{hij} = h + (\lambda_i^{-1} - \lambda_j^{-1})A^{-T}a, \quad \|r_{ij}\|_{\text{Lip}} \leq C\varepsilon.$$

If  $\min\{|i|, |j|\}$  is large, then  $|c_{hij} - h| < 1/2$  and hence  $|c_{hij}| \geq 1/2$  for  $h \neq 0$ . If one index is large and the other is bounded, the constant term  $d_{ij}$  keeps the divisor away from zero except in a range already covered by a stronger finite set. If both indices are bounded, the twist and exceptional-mass argument is exactly the condition preventing  $c_{hij} = 0$ . Therefore one coordinate direction has derivative at least a fixed  $c_* > 0$ , once the KAM correction is small. Fubini's theorem gives the strip estimate

$$|\{\xi \in \mathcal{O}_\rho : |\omega^\infty \cdot h + \Omega_i^\infty - \Omega_j^\infty| < \gamma \langle h \rangle^{-\tau}\}| \leq C\gamma\rho^{q-1}\langle h \rangle^{-\tau}. \quad (19.56)$$

The sum and one-normal Melnikov families are easier because the leading normal frequency does not cancel.

We spell out the summability. For  $|i| \sim |j| \gg |h|$ , the condition  $|\lambda_i - \lambda_j + O(|h|)| \lesssim 1$  forces  $|i - j| \leq C|h|$ , leaving  $O(|h|)$  relevant differences rather than all pairs. For noncomparable indices the leading value  $||i| - |j||$  already exceeds the strip width once the larger index is  $C\langle h \rangle$ ; hence only  $O(\langle h \rangle^2)$  pairs remain. The bounded-index region is finite and is controlled by the twist and exceptional-mass argument. Together with (19.56), summing over  $h \in \mathbb{Z}^q \setminus \{0\}$  converges when  $\tau$  exceeds the fixed source threshold. At the very first low- $h$  stage one uses the stronger pure frequency cutoff  $\gamma^{2/3}$ ; this is the origin of the final  $\gamma^{2/3}$  rather than  $\gamma$  in the total estimate. The other Melnikov families have no worse count. This proves the resonance-strip measure estimate.

Finally insert the scale generated by the Birkhoff normal form. Let  $R$  be the small amplitude radius and choose

$$r = R^{1+3/4}, \quad \rho = C_* R^2, \quad \gamma = R^{3+1/5}. \quad (19.57)$$

the Birkhoff normal-form estimate gives the order- $R^4$  Birkhoff remainder. After action-angle scaling, the tangential-drift component is  $O(R^6 r^{-2})$  and the other perturbation components are  $O(r^2 + R^5 r^{-1})$ ; with (19.57) these satisfy the smallness assumptions in the quantitative KAM-step estimate for  $R$  small. Thus the Cantor convergence argument and the reducibility argument apply. the resonance-strip measure estimate gives

$$\frac{|\mathcal{O}_\rho \setminus \mathcal{O}_\infty|}{|\mathcal{O}_\rho|} \leq C \rho^{-1} \gamma^{2/3} \leq C R^{-2} R^{\frac{2}{3}(3+1/5)} = C R^{2/15} \rightarrow 0. \quad (19.58)$$

To verify the displayed leading term, on the zero normal section and before the near-identity corrections we have  $u_j^+ = \sqrt{\xi_j} e^{i\omega_j t}$  and the same amplitude at  $-j$ ; inserting the pair into  $y = D^{-1}(u^+ + u^-)/\sqrt{2}$  gives exactly  $\sqrt{8\xi_j \lambda_j^{-1}} \cos(\omega_j t) \cos(jx)$ . The Birkhoff and KAM maps differ from the identity by higher order terms, hence contribute  $o(\sqrt{\xi})$ . Frequency convergence follows from the Cantor convergence argument. Equation (19.58) proves density one, and the reducibility argument gives the zero Lyapunov exponents and analytic reducibility. All conclusions of the theorem follow.  $\square$

#### Theorem K19.4. Quintic nonlinear-wave invariant-torus theorem

**Statement.** Consider  $u_{tt} - u_{xx} + mu + u^5 = 0$  on  $\mathbb{R} \times [0, \pi]$  with Dirichlet boundary conditions. Assume  $0 < m \leq \frac{1}{4}$ , choose  $b \geq 2$ , and let  $J = \{n_1 < \dots < n_b\} \subset \mathbb{N}^+$  satisfy  $n_{\ell+1} \geq 6n_\ell^3/m$  for  $\ell = 1, \dots, b-1$ . Then there is a positive-Lebesgue-measure set  $\mathcal{C}^*$  of action parameters, a family of  $b$ -tori over  $\mathcal{C}^*$  in the unperturbed tangential subspace, and a Lipschitz embedding  $\Phi$  that is a higher-order perturbation of the inclusion, such that  $\Phi$  restricts on every torus in the family to an embedding of a rotational invariant  $b$ -torus for the nonlinear wave equation.

*Proof.* Let

$$\varphi_j(x) = \sqrt{\frac{2}{\pi}} \sin(jx), \quad \lambda_j = \sqrt{j^2 + m}.$$

If  $Q_j, P_j$  are the physical Fourier coordinates of  $u, u_t$ , put

$$q_j = \sqrt{\lambda_j} Q_j, \quad p_j = P_j / \sqrt{\lambda_j}, \quad z_j = (q_j + ip_j) / \sqrt{2}.$$

Then the equation is the Hamiltonian flow of

$$H = \sum_{j \geq 1} \lambda_j |z_j|^2 + \frac{1}{6} \int_0^\pi (Sq)^6 dx, \quad Sq = \sum_{j \geq 1} \frac{q_j}{\sqrt{\lambda_j}} \varphi_j.$$

The nonlinear Hamiltonian is homogeneous of degree six. Since  $S$  gains half a derivative and  $S^*$  gains another half,

$$\|S^*((Sq)^5)\|_{a,s+1} \leq C \|q\|_{a,s}^5, \quad s > \frac{1}{2},$$

so its vector field gains one derivative. This is the smoothing required by the  $d = 1$  Pöschel theorem proved earlier in this chapter.

The first difficulty is arithmetic. Write  $z_{-j} = \bar{z}_j$  and give the signed complex index  $j \in \mathbb{Z}_*$  the frequency  $\lambda_j = \text{sgn}(j)\sqrt{j^2 + m}$ . For a sextic tuple  $\mathbf{j} = (j_1, \dots, j_6)$  put  $N(\mathbf{j}) = \sum j_r$ . If  $N(\mathbf{j}) \neq 0$ , then, with

$$f(t) = \sqrt{t^2 + m} - t = \frac{m}{\sqrt{t^2 + m} + t},$$

we have

$$\left| \sum_r \lambda_{j_r} \right| \geq 1 - \sum_r f(|j_r|) \geq 1 - 3m \geq \frac{1}{4}.$$

Thus only  $N(\mathbf{j}) = 0$  is delicate. In that case the divisor is a signed sum of the positive, decreasing, convex function  $f$ . If the tuple contains at most two indices outside the chosen tangential set  $J$ , cancel every opposite equal pair. The strong growth

$$n_{\ell+1} \geq \frac{6}{m} n_\ell^3$$

forces every remaining normal index to be tied to one of the finitely many large selected sites. The divisor then reduces to differences of  $f$ -values. The elementary estimates

$$|f(a) - f(n)| > 0 \quad (a \neq n), \quad |f(a) - \ell f(n)| \geq \frac{m^2}{2n^3}, \quad \ell = 2, 3,$$

exclude every nonpaired six-wave resonance. Thus all nonresonant sextic monomials with at most two normal indices can be divided by their linear frequency divisor.

Perform that homological transformation. The sextic Hamiltonian splits into three pieces: a pairable normal form  $Z_6$ , a removable nonpaired part with at most two normal indices, and a remainder whose monomials contain at least three normal factors. The transformed Hamiltonian is

$$H_1 = \Lambda + Z_6 + \widehat{G}_6 + K_{10} + R_{\geq 14}.$$

The degree 10 here is important. A Poisson bracket of two degree-six Hamiltonians has degree  $6 + 6 - 2 = 10$ ; there is no degree-seven Lie remainder.

We next compute the part of  $Z_6$  that changes the tangential frequencies. Let  $I_j = |z_j|^2$  and  $x_i = I_{n_i}/\lambda_{n_i}$ . The separated sites rule out the exceptional trigonometric coincidences, so

$$\int \varphi_n^6 = \frac{5}{2\pi^2}, \quad \int \varphi_i^4 \varphi_j^2 = \frac{3}{2\pi^2}, \quad \int \varphi_i^2 \varphi_j^2 \varphi_k^2 = \frac{1}{\pi^2}$$

for distinct selected indices in the last two formulas. Counting paired monomials gives

$$Z_{\text{tan}} = \frac{1}{\pi^2} \left[ \frac{25}{24} \sum_i x_i^3 + \frac{45}{8} \sum_{i \neq j} x_i^2 x_j + 15 \sum_{i < j < k} x_i x_j x_k \right].$$

On the frequency-weighted action ray  $x_1 = \dots = x_b = t$ , the Hessian of  $\pi^2 Z_{\text{tan}}$  has diagonal

$$d = t \left( \frac{45b}{4} - 5 \right)$$



and off-diagonal

$$o = t \left( 15b - \frac{15}{2} \right).$$

Its two eigenvalues are

$$\frac{5t}{4}(2 - 3b), \quad \frac{5t}{4}(12b^2 - 9b + 2).$$

Both are nonzero for every  $b \geq 2$ . Hence the action-frequency map is locally invertible on a whole positive box around that ray. This is the required twist; no triangular-dominance shortcut is needed.

There is a second load-bearing piece of  $Z_6$ . Pairable monomials with one normal action have the form  $B_j(I)I_j$  and therefore alter the normal frequencies. Explicitly,

$$B_j(I) = \frac{1}{\lambda_j} \left[ \frac{15}{4} \sum_i A_{i,j} \frac{I_{n_i}^2}{\lambda_{n_i}^2} + 15 \sum_{i < k} A_{i,k,j} \frac{I_{n_i} I_{n_k}}{\lambda_{n_i} \lambda_{n_k}} \right],$$

where

$$A_{i,j} = \int \varphi_{n_i}^4 \varphi_j^2, \quad A_{i,k,j} = \int \varphi_{n_i}^2 \varphi_{n_k}^2 \varphi_j^2.$$

Thus on an action box of size  $\rho$ ,

$$\Omega_j = \lambda_j + B_j(I), \quad B_j(I) = O(\rho^2 j^{-1}), \quad \lambda_j = j + \frac{m}{2j} + O(j^{-3}).$$

The normal spectrum therefore has exactly the  $d = 1$ , one-smoothing asymptotics required by the Pöschel theorem.

The first Birkhoff step generated the degree-ten polynomial  $K_{10}$ . Its pure-tangential nonnormal part must be removed before the KAM theorem is applied. This is a second, finite partial Birkhoff step. Because a pure-tangential ten-wave monomial contains at most ten selected frequencies, the super-separated growth of the  $n_i$  prevents cancellation of the largest uncanceled selected index. Solving the degree-ten homological equation leaves a tangential action polynomial  $Z_{10}$  and puts every other degree-ten monomial into a class containing at least one normal factor. The remaining Lie terms begin at degree fourteen. Hence

$$H_2 = \Lambda + Z_6 + Z_{10} + \widehat{G}_6 + \widehat{K}_{10} + T_{\geq 14}.$$

Now choose an action box  $I = \rho\eta$  with  $\eta$  in a fixed compact positive box around the frequency-weighted ray  $I_{n_i}/\lambda_{n_i} = \text{constant}$  on which the twist is invertible. The tangential frequency map satisfies

$$D_\eta \omega = \rho^2 \mathcal{T}(\eta) + O(\rho^4), \quad \mathcal{T}^{-1} = O(1),$$

so it is bi-Lipschitz. Pure normal sums and differences do not vanish for small  $\rho$  because the unperturbed  $\lambda_j$  are strictly increasing. A mixed Pöschel divisor

$$\langle k, \omega(\eta) \rangle + \langle l, \Omega(\eta) \rangle, \quad |l|_1 \leq 2,$$

is a real-analytic function of  $\eta$ . If it vanished identically, differentiation would give a

linear identity between the invertible tangential twist and at most two  $O(j^{-1})$  normal-frequency corrections. Large normal indices are excluded by the  $j^{-1}$  decay, and the finitely many low normal pairs reduce to the six-frequency arithmetic already proved. Therefore every nontrivial exact divisor has a zero set of measure zero.

It remains to verify smallness. Take the KAM normal radius  $r = \rho^{3/2}$ . The sextic remainder has at least three normal factors, so its worst scaled vector-field norm is

$$O(\rho^{3/2}r) = O(\rho^3).$$

After the second normal-form step, the worst degree-ten term has one normal factor and nine tangential factors; in the scaled norm it is again  $O(\rho^3)$ . Higher-degree terms are smaller. Choose

$$\alpha = \rho^{2+\sigma}, \quad 0 < \sigma < \frac{1}{2}.$$

The twist Lipschitz scale is  $M = O(\rho^2)$ , and the combined Pöschel perturbation norm is  $O(\rho^3) = o(\alpha)$ . Thus all hypotheses of the analytic PDE-KAM theorem are satisfied for sufficiently small  $\rho$ .

That theorem gives a Cantor subset of the action box and a Lipschitz family of nearby rotational invariant  $b$ -tori. Since the resonance-strip thickness is measured relative to the frequency-image scale  $\rho^2$ , the relative loss is  $O(\alpha/\rho^2) = O(\rho^\sigma)$ ; hence the surviving action set has positive Lebesgue measure. Finally, the two Birkhoff maps are tangent to the identity and the KAM correction is of higher order than the tangential amplitude  $O(\sqrt{\rho})$ . Pulling the tori back through the normalizations therefore gives the required Lipschitz embedding  $\Phi$ , a higher-order perturbation of the inclusion, and each fiber is a rotational invariant  $b$ -torus of the original quintic wave equation.  $\square$

### Theorem K19.5. Gravity-capillary water-wave KAM theorem

**Statement.** For the two-dimensional infinite-depth gravity-capillary water-wave system with gravity normalized to one, fix an arbitrary finite set  $S^+ \subset \mathbb{N}^+$  of tangential sites and a compact surface-tension interval  $[\kappa_1, \kappa_2] \Subset (0, \infty)$ . Let

$$\omega_j(\kappa) = \sqrt{j(1 + \kappa j^2)}.$$

There exist a Sobolev index  $\bar{s}$  and  $\varepsilon_0 > 0$  such that, for every positive amplitude vector  $\xi = (\xi_j)_{j \in S^+}$  with  $|\xi| \leq \varepsilon_0^2$ , there is a Cantor-like set  $\mathcal{G}_\xi \subset [\kappa_1, \kappa_2]$  satisfying  $|\mathcal{G}_\xi| \rightarrow \kappa_2 - \kappa_1$  as  $\xi \rightarrow 0$ . For every  $\kappa \in \mathcal{G}_\xi$  the equations possess a zero-mean standing wave  $u(\tilde{\omega}t, x) = (\eta, \psi)(\tilde{\omega}t, x)$ , even in  $x$ , with  $\eta$  even and  $\psi$  odd in the time angles, and

$$\eta(\tilde{\omega}t, x) = \sum_{j \in S^+} \sqrt{\xi_j} \cos(\tilde{\omega}_j t) \cos(jx) + o(\sqrt{|\xi|}),$$

$$\psi(\tilde{\omega}t, x) = - \sum_{j \in S^+} \sqrt{\xi_j} j^{-1} \omega_j(\kappa) \sin(\tilde{\omega}_j t) \cos(jx) + o(\sqrt{|\xi|})$$

in the stated Sobolev topology. The vector  $\tilde{\omega}$  is Diophantine and  $\tilde{\omega}_j - \omega_j(\kappa) \rightarrow 0$  for  $j \in S^+$  as  $\xi \rightarrow 0$ . The constructed quasiperiodic solutions are linearly stable.

*Proof.* The proof has two rather different difficulties. First, the linear frequencies are not varied by an action twist: the only external scalar parameter is the surface tension  $\kappa$ . One must therefore prove a finite-order nondegeneracy theorem for the one-dimensional frequency curve. Second, the water-wave equation is quasilinear, so the linearized normal equation loses derivatives. Before Nash–Moser can be used, that normal operator has to be reduced through a sequence of good-unknown, pseudodifferential and reducibility steps.

Write the free surface as  $y = \eta(t, x)$  and the trace of the velocity potential on it as  $\psi(t, x)$ . In the normalization  $g = 1$  the Hamiltonian is

$$H(\eta, \psi) = \frac{1}{2}(\psi, G(\eta)\psi)_{L^2} + \frac{1}{2} \int_{\mathbb{T}} \eta^2 dx + \kappa \int_{\mathbb{T}} \sqrt{1 + \eta_x^2} dx.$$

The zero-mean subspace is invariant. We restrict further to functions even in  $x$  and use the reversing involution  $(\eta, \psi) \mapsto (\eta, -\psi)$ . Thus a standing wave has  $\eta$  even and  $\psi$  odd in time. Since  $G(0) = |D|$ , the linear frequencies are

$$\Omega_j(\kappa) = \sqrt{j(1 + \kappa j^2)}, \quad j \geq 1.$$

For the chosen finite set  $S^+$  the corresponding linear standing wave is

$$\eta = \sum_{j \in S^+} \sqrt{\xi_j} \cos(\Omega_j t) \cos(jx), \quad \psi = - \sum_{j \in S^+} \sqrt{\xi_j} j^{-1} \Omega_j \sin(\Omega_j t) \cos(jx).$$

After the small-amplitude rescaling the nonlinear vector field is  $O(\varepsilon)$  in the tame Sobolev scale.

### 1. Surface-tension transversality. Set

$$\tilde{\omega}(\kappa) = (\Omega_j(\kappa))_{j \in S^+}.$$

The functions  $\sqrt{\kappa}$  and the finitely many tangential frequencies are linearly independent. One convenient proof differentiates them successively. After triangular row operations, the resulting Wronskian is a nonzero multiple of a Vandermonde determinant in

$$x_j(\kappa) = \frac{j^2}{2(1 + \kappa j^2)}, \quad x_0(\kappa) = \frac{1}{2\kappa}.$$

These numbers are pairwise distinct, so the determinant cannot vanish. Compactness then gives an integer  $k_0$  and  $\rho_0 > 0$  for which every nontrivial tangential divisor has some derivative of order at most  $k_0$  of size at least  $\rho_0 \langle \ell \rangle$ .

The same conclusion is needed when one or two normal frequencies are added. Here the capillary asymptotics are decisive:

$$\Omega_j(\kappa) = \sqrt{\kappa} j^{3/2} + \frac{c_j(\kappa)}{\sqrt{\kappa j}},$$

with all fixed-order  $\kappa$  derivatives of  $c_j/\sqrt{\kappa}$  uniformly bounded. A potentially small one-normal divisor must satisfy  $j^{3/2} = O(\langle \ell \rangle)$ ; a difference divisor must satisfy  $|j^{3/2} - j'^{3/2}| = O(\langle \ell \rangle)$ , and a sum divisor satisfies the analogous bound with a plus sign. If finite-

order transversality failed, normalize by  $\langle \ell \rangle$  and pass to a subsequence. The limiting identity would be a nontrivial linear relation between  $\tilde{\omega}(\kappa)$  and  $\sqrt{\kappa}$ , contradicting the Vandermonde step. Thus all first and second Melnikov divisor families have uniform finite-order  $\kappa$ -transversality.

**2. Why a counterterm replaces an action twist.** Introduce action-angle variables on  $S^+$ , normal coordinates on its complement, and a vector counterterm  $\alpha$ :

$$H_\alpha = \alpha \cdot I + \frac{1}{2}(z, \Omega z) + \varepsilon P.$$

For a prescribed torus frequency  $\omega$ , solve

$$\mathcal{F}(i, \alpha) = \omega \cdot \partial_\varphi i - X_{H_\alpha}(i) = 0.$$

The iteration will produce

$$\alpha_\infty(\omega, \kappa) = \omega + r_\varepsilon(\omega, \kappa), \quad |r_\varepsilon|_{k_0, \gamma} \lesssim \varepsilon \gamma^{-(1+k_1)}.$$

For fixed  $\kappa$  this map is a small perturbation of the identity, so it can be inverted. The original water-wave Hamiltonian is recovered by setting the counterterm parameter equal to the unperturbed frequency curve:

$$\omega_\varepsilon(\kappa) = \alpha_\infty^{-1}(\tilde{\omega}(\kappa), \kappa).$$

This is the frequency-recovery mechanism. No nonlinear action-twist matrix is used.

**3. The almost inverse reduces to the normal operator.** An exact invariant Diophantine torus is isotropic. For an approximate torus, the failure of isotropy is controlled by its invariance defect. Correcting that defect and straightening the torus by a symplectic coordinate map turns the linearized invariance equation into a triangular tangential/normal block system. The tangential equations are solved by the Diophantine inverse of  $\omega \cdot \partial_\varphi$  and by the counterterm average. The only load-bearing infinite-dimensional task is therefore inversion of the normal operator.

If that normal inverse is available, the straightening maps give an almost-approximate inverse  $T_0$  satisfying

$$\|T_0 g\|_s \lesssim \gamma^{-1} (\|g\|_{s+\sigma} + \|i_0 - (\varphi, 0, 0)\|_{s+\sigma+\mu} \|g\|_{s_0+\sigma}),$$

and

$$D_{i, \alpha} \mathcal{F}(i_0) T_0 - I = \mathcal{P} + \mathcal{P}_\omega + \mathcal{P}_\omega^\perp,$$

where the three defect terms gain the smoothing powers required by the Nash–Moser iteration.

**4. Good unknown and highest-order reduction.** Along an approximate torus define

$$B = \frac{G(\eta)\psi + \eta_x \psi_x}{1 + \eta_x^2}, \quad V = \psi_x - B\eta_x, \quad c = (1 + \eta_x^2)^{-3/2}.$$

The linearized version of Alinhac's good unknown is the triangular change

$$Z = \begin{pmatrix} 1 & 0 \\ B & 1 \end{pmatrix}.$$

It conjugates the physical linearization to

$$\omega \cdot \partial_\varphi I_2 + \begin{pmatrix} \partial_x V & -G(\eta) \\ a - \kappa \partial_x c \partial_x & V \partial_x \end{pmatrix}, \quad a = 1 + \omega \cdot \partial_\varphi B + V B_x.$$

The derivative loss coming from the shape derivative of  $G(\eta)$  has now cancelled at highest order.

A near-identity spatial diffeomorphism is chosen so that the product of the principal coefficients becomes independent of  $x$ :

$$c(1 + \beta_x)^3 = m(\varphi), \quad m(\varphi) = \left( \frac{1}{2\pi} \int c^{-1/3} dx \right)^{-3}.$$

Symmetrization of the two Hamiltonian components, passage to complex variables, and a quasiperiodic time change then make the order-3/2 coefficient constant. At this point the operator has the schematic form

$$\omega \cdot \partial_\varphi + im_3 T(D) + B_1(\varphi, x) \partial_x + iB_0(\varphi, x) |D|^{1/2} + \text{lower order},$$

where  $T(D) = |D|^{1/2}(1 - \kappa \partial_{xx})^{1/2}$  and  $m_3 = 1 + O(\varepsilon)$ .

**5. Why block decoupling comes before Egorov.** The off-diagonal order-1/2 term is removed iteratively. Commuting a generator with the order-3/2 diagonal multiplier lowers the unwanted off-diagonal order by one half. Repeating this finitely many times makes the off-diagonal block arbitrarily smoothing. This preparatory step is essential: parameter derivatives of the subsequent half-order Egorov flow themselves cost half derivatives, so an untreated off-diagonal order-1/2 block would reappear at a dangerous order.

The transport term is then removed by the flow of the pseudo-PDE

$$\partial_t u = ia(\varphi, x) |D|^{1/2} u.$$

Its commutator with an operator of order  $m$  has order  $m - 1/2$ . Solving the Egorov equation order by order removes the variable order-one transport term; reversibility kills its constant average. A final spatial reduction makes the remaining order-1/2 diagonal coefficient constant. The normal operator has therefore reached the precise form

$$\omega \cdot \partial_\varphi I_2 + im_3 T(D) + im_1 \Sigma |D|^{1/2} + R_0 + Q_{-M},$$

where  $m_1 = O(\varepsilon)$ ,  $R_0$  is diagonal of order zero, and  $Q_{-M}$  is off-diagonal and arbitrarily smoothing. The  $m_1 |D|^{1/2}$  term and  $R_0$  are not smoothing errors; keeping them is necessary for the next divisor calculation.

**6. Finite-step almost diagonalization and inversion.** The leading normal frequen-

cies are

$$\mu_j^{(0)} = m_3|j|^{1/2}(1 + \kappa j^2)^{1/2} + m_1|j|^{1/2}.$$

At reducibility stage  $n$ , solve the homological equation only for  $|\ell| \leq N_n$  under the second Melnikov conditions. The diagonal part of the remainder corrects the frequencies, and the unsolved Fourier tail gains a negative power of  $N_n$ . The resulting transformations converge on the limiting Cantor set and produce real limits  $\mu_j^\infty$ .

For Nash–Moser, however, one needs inversion already at each finite stage. The current operator is therefore split into an invertible diagonal low block, a high-frequency diagonal part and two small remainders. The first Melnikov conditions invert the main block, the high-frequency part is smoothing at the current cutoff, and a Neumann series absorbs the small remainder. This proves the fixed-loss normal inverse required by the almost-approximate inverse above. The distinction matters: complete reducibility is a property of the final torus, whereas finite-stage almost invertibility is what makes the nonlinear iteration possible.

**7. Nash–Moser and the surface-tension Cantor set.** Choose cutoffs  $K_n = K_0^{(3/2)^n}$  and  $\gamma = \varepsilon^a$ . The fixed derivative loss of the almost inverse is compensated by the smoothing projectors; the inverse defect has one better scale, and the nonlinear Taylor remainder is quadratic. The abstract Nash–Moser theorem therefore gives a convergent sequence  $(i_n, \alpha_n) \rightarrow (i_\infty, \alpha_\infty)$  with the counterterm estimate used above.

The final normal frequencies have the form

$$\mu_j^\infty = m_3^\infty \sqrt{|j|(1 + \kappa j^2)} + m_1^\infty |j|^{1/2} + r_j^\infty,$$

with  $m_3^\infty - 1$ ,  $m_1^\infty$  and  $r_j^\infty$  small in the required  $k_0$ -parameter norm. Hence the finite-order transversality proved for the linear frequencies survives for the perturbed divisors.

Take

$$0 < a < \min \left\{ a_0, \frac{1}{1 + k_0 + k_1} \right\}, \quad \tau > k_0(\nu + 4).$$

A one-dimensional finite-order sublevel estimate bounds each resonance strip by a constant times the  $1/k_0$  power of its threshold. Nonempty one-normal, difference and sum strips respectively force

$$j^{3/2} = O(\langle \ell \rangle), \quad |j^{3/2} - j'^{3/2}| = O(\langle \ell \rangle), \quad j^{3/2} + j'^{3/2} = O(\langle \ell \rangle).$$

Thus only polynomially many normal indices occur for a fixed  $\ell$ , and the choice of  $\tau$  makes the sum over  $\ell$  convergent. The resulting good set satisfies the explicit estimate

$$|\mathcal{G}_\varepsilon| \geq (\kappa_2 - \kappa_1) - C\varepsilon^{a/k_0}.$$

It is therefore asymptotically full.

**8. Recovery of the stated solution and stability.** For  $\kappa \in \mathcal{G}_\varepsilon$ , invert the counterterm map at  $\tilde{\omega}(\kappa)$ . Undoing the action-angle and amplitude changes gives

$$\eta(\tilde{\omega}t, x) = \sum_{j \in S^+} \sqrt{\xi_j} \cos(\tilde{\omega}_j t) \cos(jx) + o(\sqrt{|\xi|}),$$

$$\psi(\tilde{\omega}t, x) = - \sum_{j \in S^+} \sqrt{\xi_j} j^{-1} \Omega_j(\kappa) \sin(\tilde{\omega}_j t) \cos(jx) + o(\sqrt{|\xi|}).$$

The recovered vector is Diophantine and  $\tilde{\omega}_j - \Omega_j(\kappa) \rightarrow 0$  as the amplitude tends to zero.

Finally, the limiting reducibility maps conjugate the normal variational equation to constant purely imaginary diagonal frequencies  $-i\mu_j^\infty$ . In the adapted tangential coordinates the action variation is constant and the remaining normal forcing is solved using the first Melnikov inequalities. The conjugacies and their inverses are bounded on the stated Sobolev scale, so the full variational flow is bounded for all time. This is the asserted linear stability.

Every step used the uniform ellipticity coming from  $\kappa \in [\kappa_1, \kappa_2] \Subset (0, \infty)$ ; the argument therefore makes no claim at zero surface tension.  $\square$

## 19.3 Worked examples

### Worked example: a finite tangential set and its first Melnikov divisors

Consider the linear Schrödinger frequencies

$$\Omega_j = j^2 + m, \quad j \in \mathbb{Z},$$

and choose tangential sites  $S = \{1, 2\}$ . The tangential frequency vector is

$$\omega = (1 + m, 4 + m).$$

For  $k = (k_1, k_2) \in \mathbb{Z}^2$ , a first normal divisor has the form

$$D_{k,j} = k \cdot \omega + \Omega_j.$$

A second-order difference divisor is

$$D_{k,i,j} = k \cdot \omega + \Omega_i - \Omega_j = k_1(1 + m) + k_2(4 + m) + i^2 - j^2.$$

The large- $i, j$  structure is visible explicitly through

$$i^2 - j^2 = (i - j)(i + j).$$

Thus large normal indices are not arbitrary extra small divisors: their polynomial separation is part of the summability mechanism in PDE parameter exclusion.

#### Check your understanding

For fixed  $k$  and fixed difference  $i - j = d \neq 0$ , show that  $|i^2 - j^2| = |d| |i + j|$  grows linearly with  $|i + j|$ . Why does this make only finitely many large-index pairs dangerous for a fixed threshold?

**Boundary case: a resonant normal cluster**

If two normal modes satisfy an exact identity

$$\omega \cdot k + \Omega_i - \Omega_j = 0$$

that is not protected by a retained symmetry block, the corresponding homological equation cannot be solved by division. A theorem assuming scalar reducibility or a second Melnikov lower bound therefore cannot be invoked. One must either enlarge the resonant normal block, introduce a parameter that unfolds the resonance, or accept a weaker normal form.

**Common pitfall**

Infinite-dimensional KAM cannot be reduced to checking a finite list of scalar Diophantine inequalities. Normal-frequency clusters may require matrix blocks, quasilinear operators may lose derivatives, and the inverse used at a finite Nash-Moser stage is often only an almost inverse on the current Cantor set. Final reducibility or linear stability must be justified separately from finite-step solvability.

**19.4 Chapter summary****What to remember**

- PDE KAM keeps finitely many tangential frequencies while controlling infinitely many normal modes.
- Melnikov conditions, spectral asymptotics and parameter exclusion make the infinite small-divisor set manageable.
- Reducibility of the linearized normal equation is the key inversion mechanism and often yields linear stability.
- Nash-Moser replaces ordinary Newton iteration when quasilinear terms cause derivative loss.
- Reversible waves, quintic waves, gravity-capillary water waves and vortex patches illustrate distinct ways of supplying twist, transversality and regularization.

**Looking ahead**

K20 closes the KAM Part by returning to billiards at the rigidity frontier. The question is no longer whether invariant curves exist, but whether the existence of sufficiently many caustics—or spectral data encoding them—forces the boundary itself to be an ellipse.



**Check your understanding**

Make a four-column ledger for one PDE-KAM theorem: tangential parameters, normal frequencies/divisors, derivative loss in the linearized inverse, and excluded parameter sets. Which column forces Nash-Moser rather than ordinary analytic Newton iteration?

**19.5 Exercises and problems****Exercise 19.1**

**Difficulty.** Core.

**Related results.** Pöschel infinite-dimensional analytic KAM theorem; Reversible derivative-wave KAM and reducibility theorem; Quintic nonlinear-wave invariant-torus theorem.

**Problem.** For the Fourier-truncated NLS model, split tangential/normal modes, compute first- and second-Melnikov divisors, identify resonant clusters, and state which bounds would need to be uniform as the cutoff grows.

**Solution sketch.** Diagonalize the linear frequencies, write scalar/block homological equations, and record the worst divisor versus cutoff.

**Exercise 19.2**

**Difficulty.** Core.

**Related results.** Gravity-capillary water-wave KAM theorem; KAM for vortex patches.

**Problem.** Choose a resonant normal-mode cluster with a vanishing second-Melnikov divisor and prove the corresponding block homological equation is non-invertible for suitable forcing.

**Solution sketch.** Compute the block determinant/eigenvalue gap at resonance and place forcing in the obstructed component.

**Exercise 19.3**

**Difficulty.** Advanced.

**Related results.** Quintic nonlinear-wave invariant-torus theorem.

**Problem.** Under a Melnikov lower bound, solve a toy weighted-sequence homological equation and prove the associated Fourier/Sobolev norm loss.

**Solution sketch.** Divide coefficientwise by the divisor and absorb the factor  $\langle k \rangle^\tau$  into the stronger input weight.

**Exercise 19.4**

**Difficulty.** Advanced.

**Related results.** Water-wave KAM result interface.

**Problem.** Model a PDE KAM inverse estimate with derivative loss and show why naive Newton iteration in one fixed Sobolev norm cannot close; explain schematically how smoothing changes the bookkeeping.

**Solution sketch.** Iterate the loss estimate to expose the regularity climb, then split low/high frequencies with a smoothing cutoff.

### Exercise 19.5

**Difficulty.** Advanced.

**Related results.** Pöschel infinite-dimensional analytic KAM theorem; Reversible derivative-wave KAM and reducibility theorem.

**Problem.** Assume both invariant-torus existence and reducibility to constant purely imaginary normal coefficients. Derive linear stability and explain why torus existence alone does not imply reducibility.

**Solution sketch.** Solve the constant conjugated variational equation and transfer boundedness back through the bounded conjugacy.

### Exercise 19.6

**Difficulty.** Lab.

**Related results.** None.

**Problem.** For increasing Fourier cutoffs in truncated NLS, compute the smallest Melnikov divisors and truncated homological-inverse norm at nonresonant and near-cluster parameters.

**Solution sketch.** Increase cutoff and precision together; interpret the computation only as finite-mode small-divisor diagnostics.

# Billiard Rigidity, KAM Frontiers and Open Problems

## Why this chapter matters

The Birkhoff conjecture asks for a global inverse theorem: if a smooth strictly convex planar billiard is integrable, must its boundary be an ellipse? The conjecture remains open in full generality. What modern rigidity theory has achieved is a collection of sharp theorems under symmetry, near-ellipse and near-boundary integrability assumptions, together with spectral consequences.

This final KAM volume is therefore organized around *scope discipline*. The centrally symmetric Birkhoff–Poritsky theorem is nonperturbative but uses a strong foliation hypothesis near the boundary. Near-central-symmetry results combine a few rational caustics with closeness assumptions. Koval’s theorem gives local strong Birkhoff rigidity near almost every ellipse, and wave-trace invariants convert that dynamical rigidity into local spectral uniqueness. Symplectic billiards form a separate dynamical model with a different reflection law and area/action spectrum; their recent rigidity results are recorded as a frontier, not merged into the ordinary Birkhoff conjecture.

**Prerequisites.** K10–K11 for twist maps, caustics, Lazutkin coordinates and rational rotation numbers; Foundation F12/F15 for perturbative inversion and parameter exclusion; basic support-function geometry and Laplace spectral terminology.

## Learning goals

- State the global Birkhoff conjecture and distinguish it from each proved local or symmetric rigidity theorem.
- Express a strictly convex curve through its support function and relate Fourier modes to ellipse/isometry directions.
- Understand how rational caustics impose linearized constraints on boundary deformations.
- Follow the analytic-continuation/invertibility mechanism behind Koval’s exceptional eccentricity set.
- Explain how the wave trace recovers periodic-length information and why this can feed a dynamical rigidity theorem.
- Keep ordinary and symplectic billiard rigidity statements in separate models.

### Conceptual roadmap

The rigidity mechanisms form a hierarchy rather than a single theorem.

- A sufficiently strong caustic foliation, together with symmetry, can force a billiard table to be an ellipse.
- Near an ellipse, families of rational caustics constrain the boundary through a linearized rigidity operator; invertibility modulo the ellipse directions yields local rigidity.
- The Laplace spectrum enters through the wave trace, which detects periodic-orbit length information and can recover the rational-caustic data needed by the dynamical rigidity theorem.
- Symplectic billiards form a separate branch, governed by their own reflection law, area spectrum and affine symmetry group.

### Check your understanding

Why would proving that every sufficiently smooth *nearby* integrable table is an ellipse not settle the global Birkhoff conjecture? Identify the missing global hypothesis.

## 20.1 Definitions

### Intuition

**A caustic is an invariant curve seen in configuration space.** A convex caustic is tangent to every segment of a billiard orbit in the corresponding invariant rotational graph. Rational caustics organize periodic families; a near-boundary family supplies especially rigid length/action data.

**The support function linearizes boundary geometry.** With normal angle  $\psi$ , a strictly convex boundary is encoded by  $h(\psi)$ . Curvature is controlled by  $h+h''$ , while translations and ellipse-like deformations occupy explicit low Fourier modes. This makes Fourier analysis natural in local rigidity.

**The Bialy-Mironov coordinates give a second twist structure.** Instead of recording a segment by its impact point and incoming angle, one can record the oriented supporting line by its signed distance  $p$  and normal angle  $\varphi$ . The generating function  $S(\varphi, \varphi_1) = 2h(\psi) \sin \delta$  has positive mixed derivative  $\frac{1}{2}(h+h'') \sin \delta$ . This second twist structure is what makes the Hopf-type argument compatible with the support function.

**The rigidity proof is a sign clash.** The invariant foliation produces a measurable monotone invariant line field. Its Riccati equation gives a weighted integral that is nonpositive. The four-periodic support identities turn the same integral into the sharp period- $\pi$  Wirtinger defect, which is nonnegative. Equality is therefore forced, and equality leaves only the first harmonic of  $\cos(2d)$ , exactly the elliptic support function.

**A loop orbit is not automatically periodic.** A  $(1, 4)$ -loop returns to its starting boundary point after four reflections and one winding, but its outgoing angle need not match its initial angle. Near the centrally symmetric reference table, the rational  $1/4$  leaf supplies a uniform twist argument that makes this loop unique and defines a smooth incidence-angle function  $d(\psi)$  on both the reference and perturbed tables.

**Near-central symmetry first extracts an actual ellipse.** The perturbative Hopf-Wirtinger inequality forces  $\mu = \cos(2d)$  to concentrate in the Fourier modes  $0, \pm 2$ . Those are exactly the modes of a centered elliptic support function. Interpolation turns the resulting  $C^1$  control into the characteristic  $C^2$  error  $O(\varepsilon^{1/8})$ , after which the boundary can be written as a small elliptic-polar graph.

**Weak rational integrability means all  $1/q$  caustics.** A table is  $q_0$ -rationally weakly integrable when it has a  $q$ -rationally integrable caustic for every integer  $q \geq q_0$ . In the near-central theorem, the 3-caustic supplies  $q = 3$ , while the  $C^0$ -integrable annulus (or the stronger 4-rational integrability assumption) supplies every  $q \geq 4$ .

**Dynamical modes replace ordinary Fourier modes near an ellipse.** Each rational caustic has its own action-angle coordinate. Pulling its first sine/cosine modes back to the boundary gives functions  $c_q, s_q$  adapted to elliptic billiard dynamics. Together with the five directions tangent to the family of ellipses, these modes form the basis used to separate genuine shape deformations from translations, rotation, scaling, and eccentricity change.

**The local rigidity engine is a superlinear improvement, not a multiplier inversion.** Rational caustics make the non-elliptic dynamical coefficients quadratic up to a controlled  $q^8$  loss. Complex singularities prove mode independence, the correlation matrix gives a basis, and low/high frequency estimates yield an  $L^2$  gain. Sobolev interpolation and recentering by the best nearby ellipse give the exponent  $703/702 > 1$ ; a minimizing-ellipse argument then forces the deformation to vanish.

**Koval's preservation functions are dynamically adapted Fourier probes.** For a rational  $p/q$  caustic, the billiard map is a rigid translation in its own action-angle coordinate. Koval measures how an elliptic Fourier mode contributes to the  $q$ th sine/cosine harmonic in that coordinate. These coefficients are the entries of the rigidity matrix; ordinary Fourier modes alone do not see the correct elliptic dynamics.

**The finite-rank step has a genuine arithmetic ingredient.** Near the circle the preservation coefficients have explicit powers of the eccentricity. After rescaling, the odd and even blocks reduce to polynomial/Vandermonde matrices at cyclotomic cosine values. Koval selects special prime denominators using the theorem of Heath-Brown that at least one of 3, 5, 7 is a primitive root for infinitely many primes. This arithmetic input is stated explicitly because it is load-bearing in the rank argument.

**Koval's exceptional set is an analytic noninvertibility set.** After the finite block is joined to the high near-Fourier tail, the selected preservation modes define a compact holomorphic operator  $L_e$  on  $L^2$ . A singular eccentricity is one

for which 1 is an eigenvalue of  $L_e$ . Small-eccentricity nonsingularity plus analytic Fredholm theory confines these parameters to a locally finite set rather than an interval.

**The nonlinear closure is a superlinear ellipse improvement.** The exact caustic equations are not solved by a one-step inverse operator. Low dynamical coefficients are quadratic with a  $q^8$  loss, while high coefficients have a  $C^1/j$  bound. Splitting at  $q \asymp \|\mu\|_{C^1}^{-1/9}$  yields the  $L^2$  gain 19/18;  $C^{39}$  interpolation yields the  $C^1$  exponent 703/702. Recenter by the five-dimensional ellipse family and minimize the graph size over nearby ellipses: any positive minimum would be strictly improved, so it must be zero.

**For spectral rigidity, one must hear an entire periodic-length window.** For a type  $(p, q)$ ,  $t_{p,q}$  and  $T_{p,q}$  are the shortest and longest periodic lengths of that type. A rational caustic is equivalent to the collapse  $t_{p,q} = T_{p,q}$ . Thus hearing one periodic length is not enough: the proof must isolate the whole interval and force both endpoints to coincide with the ellipse's single length.

**The near-boundary lengths have a rigid  $q^{-2}$  spacing scale.** For bounded winding number,  $L_{p,q} = p\ell - c_{2,p}q^{-2} + O(q^{-4})$ , with  $c_{2,p} = \frac{p^3}{24}(\int \kappa^{2/3} ds)^3$ . The coefficient  $c_{2,1}$  controls the spacing of the infinite  $(1, q)$  sequence accumulating at the perimeter. Equality of wave traces forces this coefficient to agree before the infinite family can be isolated uniformly.

**The spectral detecting family is finite plus an infinite tail.** Koval's dynamical rigidity proof needs all sufficiently large  $(1, q)$  caustics together with only finitely many extra caustics of winding number  $2 \leq p \leq 7$ . The spectral proof is designed around exactly this family, rather than an unspecified finite collection.

**Non-incidence means no competing orbit can hide inside the chosen window.** An isolation interval contains the complete  $(p, q)$  window but no length from another orbit type and no perimeter multiple. In such an interval the localized wave trace separates one orbit type. Its principal microlocal amplitude is positive, and Soga's oscillatory-integral theorem rules out degenerate cancellation; hence the periodic endpoint lengths are genuine wave-trace singularities.

**Wave trace links spectrum to rational caustics through isolated windows.** For the ellipse, an isolated detecting interval contains exactly one periodic length because its rational caustic makes  $t_{p,q} = T_{p,q}$ . Equality of wave traces forces both endpoints of the comparison table's window to be that same singular time. This is the decisive spectral-to-dynamical step; no inference of loop-function constancy is made from a single stationary-phase coefficient.

**Koval's spectral exceptional set is "small" in a stronger topological sense.** A set is small when its accumulation points form a locally finite set. The incidence analysis classifies elliptic-caustic, hyperbolic-caustic, and axis-bouncing length functions and shows that any nonisolated sequence of incidences can accumulate only at a locally finite set of perimeter incidences. Small sets are therefore countable, measure zero, and nowhere dense.

**Symplectic billiards are a different model.** Their reflection law is symplectic rather than elastic, and the relevant spectral invariant is an area/action spectrum. Similar-looking rigidity conclusions do not transfer automatically to ordinary bil-

liards.

### Definition K20.1. Birkhoff conjecture

The planar Birkhoff conjecture asserts that a sufficiently smooth strictly convex integrable billiard table is an ellipse; “integrable” must be specified by the theorem/conjecture version and is not a single universal definition.

### Definition K20.2. rational caustic and $q_0$ -rational integrability

A smooth convex caustic  $\Gamma$  is an integrable rational caustic if the corresponding noncontractible invariant curve in the billiard phase cylinder consists entirely of periodic points; its rotation number is therefore rational. Following Koval, for  $q_0 > 2$  a billiard domain is  $q_0$ -rationally integrable if it admits an integrable rational caustic with rotation number  $p/q$  for every rational  $0 < p/q \leq 1/q_0$ .

### Definition K20.3. spectral rigidity

Local spectral rigidity/uniqueness means that within a specified neighborhood/class, equality of the relevant Laplace spectrum forces the domain to be congruent/equal to the reference domain according to the theorem normalization.

### Definition K20.4. symplectic billiard

A symplectic billiard is a distinct billiard-type dynamical system defined using a symplectic form/area functional rather than the ordinary optical reflection law; results about it are not transferred to ordinary billiards without a theorem.

### Definition K20.5. support function, normal-angle parametrization, and curvature

For a strictly convex domain  $\Omega$  containing the origin, its support function is  $h_\Omega(\psi) = \sup_{z \in \Omega} z \cdot (\cos \psi, \sin \psi)$ . If the boundary is  $C^2$  and positively curved, the outer normal angle  $\psi$  parametrizes it by  $\gamma(\psi) = h(\psi)n(\psi) + h'(\psi)t(\psi)$ , and the radius of curvature is  $h + h'' > 0$ .

### Definition K20.6. centrally symmetric strictly convex billiard table

A planar billiard table is centrally symmetric if, after translating its center to the origin,  $z \in \Omega$  iff  $-z \in \Omega$ ; in support coordinates this is  $h(\psi + \pi) = h(\psi)$ . Strict convexity means every boundary point has a unique supporting line, and positive curvature means  $h + h'' > 0$  in the  $C^2$  setting.

### Definition K20.7. rotational invariant graph and $C^0$ -integrability near the boundary

A rotational invariant graph of the billiard twist map is an invariant simple graph over the angular coordinate that winds once around the phase cylinder. An annulus is  $C^0$ -integrable if it is foliated by continuous rotational invariant graphs. This is

the foliation notion used in K20-002 and in the near-central-symmetry equivalence of K20-003.

### Definition K20.8. ellipse eccentricity and semi-focal distance

For an ellipse with semiaxes  $a \geq b > 0$ , its eccentricity is  $e = \sqrt{1 - b^2/a^2} \in [0, 1)$  and its semi-focal distance is  $c = \sqrt{a^2 - b^2} = ae$ . The circular limit is  $e = 0$ ; the rigidity theorems in K20-004–005 fix  $e \in (0, 1)$  and  $c > 0$ .

### Definition K20.9. Koval local closeness normalization

In K20-004–005, “ $C^k$ - $K$  close” means that after the source’s Euclidean/elliptic-polar normalization the deformation from the reference ellipse has  $C^k$  norm bounded by  $K$ ; “ $C^r$ - $\varepsilon$  close” means the same normalized deformation has  $C^r$  norm  $< \varepsilon$ . The finite-dimensional ellipse/isometry directions are fixed by the same normalization before the rigidity operator is inverted.

### Definition K20.10. Laplace spectrum and wave trace

Throughout K20-005, fix one of the standard Dirichlet or Neumann boundary conditions and use the same choice for the reference ellipse and the comparison domain. The Laplace spectrum is the corresponding eigenvalue multiset  $\{\lambda_j\}$ . Its even wave trace is the distribution  $t \mapsto \sum_j \cos(t\sqrt{\lambda_j})$ . Equality of Laplace spectra therefore gives equality of wave traces; singular times of the latter are linked by the billiard Poisson relation to lengths of periodic billiard trajectories.

### Definition K20.11. oriented-line coordinates and the Bialy-Mironov generating function

An oriented line meeting the table is written as  $(p, \varphi)$ , where  $\varphi$  is the angle of its right unit normal and  $p$  is its signed distance from the origin. If  $(p_1, \varphi_1) = T(p, \varphi)$ , set  $\psi = (\varphi + \varphi_1)/2$  and  $\delta = (\varphi_1 - \varphi)/2$ . The nonstandard billiard generating function is  $S(\varphi, \varphi_1) = 2h(\psi) \sin \delta$ , with  $p = -S_1$ ,  $p_1 = S_2$ , and  $S_{12} = \frac{1}{2}(h + h'') \sin \delta > 0$ .

### Definition K20.12. measurable monotone invariant subbundle

For a twist-map annulus in coordinates  $(p, q)$ , a measurable line subbundle  $\mathcal{L}$  is monotone with respect to the vertical foliation  $\{q = \text{const}\}$  if every line is nonvertical and is oriented by  $dq > 0$ . It is invariant if  $DT(\mathcal{L}_x) = \mathcal{L}_{Tx}$  almost everywhere. A positive generator is written  $u = \partial_q + \omega \partial_p$ .

### Definition K20.13. quarter-turn angle on a four-periodic invariant curve

If the rotation-1/4 invariant curve is written as  $\alpha = \{\delta = d(\psi)\}$  in normal-angle/incoming-angle coordinates, then  $d(\psi)$  is the incidence-angle function along that leaf. In the centrally symmetric four-periodic setting proved in K20-002 one has  $T_\alpha(\psi) = \psi + \pi/2$  and  $d(\psi + \pi/2) = \pi/2 - d(\psi)$ .



**Definition K20.14. weighted Hopf-Wirtinger defect**

For the centrally symmetric four-periodic geometry, the Bialy–Mironov weighted Hopf calculation produces a one-variable functional  $U(\psi)$  with  $\int_0^\pi U \leq 0$ . After the exact support substitution and quarter-turn symmetrization,  $\mu = \cos(2d)$  satisfies  $\int_0^\pi U = \frac{\pi R^4}{512} \int_0^\pi ((\mu'')^2 - 4(\mu')^2)$ . The latter is the period- $\pi$  Wirtinger defect and is non-negative.

**Definition K20.15.  $(p, q)$ -loop orbit and loop-angle function**

For a lifted billiard map, a  $(p, q)$ -loop orbit is an orbit segment whose base point winds  $p$  times and returns to the same boundary base point after  $q$  reflections, so  $\pi_1 F^q(s, r) = s + p|\partial\Omega|$ ; the final reflection angle need not equal the initial one, so a loop need not be periodic. When the  $(1, 4)$ -loop through every boundary point is unique,  $d_\Omega(\psi) \in (0, \pi)$  denotes its initial incidence angle in normal-angle coordinates.

**Definition K20.16. elliptic-polar graph over a reference ellipse**

For an ellipse of eccentricity  $e \in [0, 1)$  and semi-focal distance  $c$  (with the circular case understood as the trigonometric limit), elliptic coordinates are  $\Phi(\lambda, \theta) = c(\cosh \lambda \cos \theta, \sinh \lambda \sin \theta)$ . A nearby boundary is an elliptic-polar graph if it is written as  $\Phi(\lambda_0 + u(\theta), \theta)$  for a periodic graph function  $u$ ; its  $C^r$  size is measured in this graph coordinate.

**Definition K20.17.  $q_0$ -rational weak integrability**

A billiard is  $q_0$ -rationally weakly integrable if it admits a  $q$ -rationally integrable caustic—that is, a caustic all of whose tangent orbits are  $(1, q)$ -periodic—for every integer  $q \geq q_0$ . This is weaker than  $q_0$ -rational integrability, which also requires the rational caustics for every reduced  $0 < p/q \leq 1/q_0$ .

**Definition K20.18. elliptic dynamical modes and elliptic-motion space**

Fix an ellipse. For each rational caustic of rotation number  $1/q$ , pull the first sine and cosine modes in its action-angle coordinate back to the elliptic boundary angle; the resulting weighted functions are the elliptic dynamical modes  $c_q, s_q$ . The elliptic-motion space  $V_E$  is the five-dimensional tangent space to the family of ellipses in elliptic-polar coordinates, generated by homothety, two translations, rotation, and eccentricity/hyperbolic-rotation variation.

**Definition K20.19. quantitative local ellipse improvement**

A quantitative local ellipse improvement is a superlinear recentering estimate: if a sufficiently smooth rationally weakly integrable table is represented over an ellipse by a small graph  $u$ , then one can choose a nearby ellipse so that the new graph  $\tilde{u}$  satisfies  $\|\tilde{u}\|_{C^1} \leq C\|u\|_{C^1}^{703/702}$ . Iteration is not needed in K20-003: minimizing the  $C^1$  graph size over a compact nearby ellipse family and applying this strict improvement forces the minimum to be zero.

**Definition K20.20. Koval rational-caustic preservation coefficients and functionals**

For a confocal elliptic caustic of rotation number  $p/q$ , let  $\theta_{p/q}$  be its action-angle boundary coordinate. The coefficient  $A_{p,q,j}(e) = \pi^{-1} \int_0^{2\pi} \cos(j\varphi(\theta_{p/q})) \cos(q\theta_{p/q}) d\theta_{p/q}$  measures the  $q$ th action-angle harmonic created by the  $j$ th elliptic Fourier mode; the sine coefficient is analogous. The linear preservation functionals  $A_{p,q}^{\pm}$  are the corresponding sums against the elliptic Fourier coefficients of the deformation.

**Definition K20.21.  $q_0$ -good prime and Koval's arithmetic selector**

A prime  $Q$  is  $q_0$ -good if  $Q > 7q_0$  and at least one of 3, 5, 7 is a primitive root modulo  $Q$ . Koval uses Heath-Brown's theorem that at least one of 3, 5, 7 is a primitive root for infinitely many primes; hence every fixed  $q_0$  admits infinitely many  $q_0$ -good primes. This is a number-theoretic denominator-selection input, not a billiard-dynamical assertion.

**Definition K20.22. Koval preservation basis and compact operator**

After five functions tangent to the ellipse/isometry family, choose finitely many low rational-caustic preservation modes and then the  $(1, j)$  preservation modes for all sufficiently large  $j$ . Denote the resulting family by  $f_j(e)$ . If  $e_j$  is the normalized trigonometric basis, Koval's preservation operator is  $L_e e_j = e_j - f_j(e)/\sqrt{\pi}$ . The tail estimate  $\|f_j/\sqrt{\pi} - e_j\|_{L^2} = O(j^{-1})$  makes  $L_e$  Hilbert-Schmidt;  $f_j$  is a basis exactly when 1 is not an eigenvalue of  $L_e$ .

**Definition K20.23. singular eccentricity and Fredholm exceptional set**

For a fixed  $q_0$ , an eccentricity is singular if 1 is an eigenvalue of the compact preservation operator  $L_e$ , equivalently if Koval's selected preservation functions fail to form a basis. The analytic-Fredholm argument shows that the real singular set  $\mathcal{Z}(q_0) \subset (0, 1)$  is locally finite once nonsingularity is known for small positive eccentricity.

**Definition K20.24. periodic-orbit length window and caustic criterion**

For a strictly convex billiard table and integers  $2p \leq q$ , let  $t_{p,q}$  and  $T_{p,q}$  denote respectively the minimum and maximum lengths among periodic billiard trajectories that make  $p$  windings in  $q$  reflections (the period need not be primitive). The corresponding length spectrum lies in  $[t_{p,q}, T_{p,q}]$ . For  $2p < q$ , equality  $t_{p,q} = T_{p,q}$  is equivalent to the existence of an integrable rational caustic of rotation number  $p/q$ ; thus collapse of the whole window, not merely occurrence of one periodic length, is the spectral-to-caustic criterion used below.

**Definition K20.25. bounded-winding Marvizi-Melrose coefficient**

For bounded  $p$  and large  $q$ , write the near-boundary periodic-length expansion as

$$L_{p,q} = p\ell - c_{2,p}q^{-2} + O(q^{-4}), \quad c_{2,p} = \frac{p^3}{24} \left( \int_{\partial\Omega} \kappa^{2/3} ds \right)^3,$$

where  $\ell = |\partial\Omega|$  and  $\kappa$  is boundary curvature. In particular  $c_{2,1} > 0$ . The absence of odd  $q^{-1}$  and  $q^{-3}$  terms is part of the Marvizi-Melrose invariant expansion used in K20-005.

**Definition K20.26. Koval spectral detecting family**

For the  $q_0$  fixed in the spectral argument, Koval's detecting family is  $\mathcal{F} = \mathcal{F}_1 \cup \mathcal{F}_2$ . The infinite tail  $\mathcal{F}_1$  contains every type  $(1, q)$  for all sufficiently large  $q$ ;  $\mathcal{F}_2$  is a finite family of additional types with  $2 \leq p \leq 7$  supplied by the local strong-Birkhoff rank argument. Preservation of the caustics in  $\mathcal{F}$  is exactly the finite-plus-tail input needed to invoke K20-004.

**Definition K20.27. non-incidence and a  $(p, q, \varepsilon)$  isolation interval**

The reference ellipse satisfies the non-incidence condition for  $\mathcal{F}$  if each length belonging to a type in  $\mathcal{F}$  is realized by no other periodic-orbit type, no such length is a multiple of the perimeter, and the perimeter itself is not a periodic-orbit length. Assuming  $c_{2,1}(\Omega) = c_{2,1}(\mathcal{E})$ , an open interval  $I$  is a  $(p, q, \varepsilon)$  isolation interval if every  $C^{10-\varepsilon}$  deformation in the theorem class has  $t_{p,q}, T_{p,q} \in I$  and  $I$  contains neither a length of another orbit type nor a perimeter multiple.

**Definition K20.28. small exceptional set of eccentricities**

A subset  $Z \subset [0, 1)$  is locally finite if it has no accumulation point in  $[0, 1)$ . Following Koval,  $Z$  is *small* if its set of accumulation points in  $[0, 1)$  is locally finite. Every small set is countable, Lebesgue-null and nowhere dense: on each compact subinterval avoiding the locally finite derived set it is discrete, and a countable exhaustion by compact intervals gives countability and nullity; its closure has empty interior because its only nonisolated points form a locally finite set.

**Definition K20.29. localized fixed-type wave trace**

Let  $w_\Omega(t) = \text{Tr} \cos(t\sqrt{\Delta_\Omega})$ . If a cutoff  $\chi_{p,q}$  equals 1 on  $[t_{p,q}, T_{p,q}]$  and its support contains no length of another orbit type and no perimeter multiple, the localized fixed-type wave trace is  $\sigma_{p,q}(t) = \chi_{p,q}(t)w_\Omega(t)$ . Its singular support is the complement of the largest open set on which it is  $C^\infty$ .

**Definition K20.30. elliptic periodic-length type and incidence eccentricity**

For the ellipse family normalized by fixed semi-major axis, a periodic-length type is one of the holomorphic-in-eccentricity length functions arising from an elliptic caustic  $(p, q)$ , a hyperbolic caustic  $(\tilde{p}, q)$ , a minor-axis bouncing-ball iterate, or a major-axis bouncing-ball iterate; the perimeter is treated as an additional length

function. An incidence eccentricity is a parameter at which a detecting-family length or the perimeter coincides with a different periodic-length type.

## 20.2 Main results

### Conjecture / status: Birkhoff conjecture

**Status.** Conjecture: every sufficiently smooth strictly convex planar integrable billiard table is an ellipse; retain as OPEN in full generality.

### Theorem K20.1. Centrally symmetric Birkhoff-Poritsky theorem

**Statement.** Let  $\gamma \subset \mathbb{R}^2$  be a simple closed centrally symmetric  $C^2$  curve of positive curvature, and let  $T$  be its billiard-ball map on the phase cylinder. Suppose  $T$  has a continuous rotational invariant curve  $\alpha$  of rotation number  $1/4$  consisting of 4-periodic orbits. Let  $A$  be the annulus between  $\alpha$  and the grazing boundary  $\{\delta = 0\}$ . If  $A$  is foliated by continuous rotational invariant curves, then  $\gamma$  is an ellipse. In particular, if a neighborhood of the billiard boundary is  $C^1$ -foliated by convex caustics with rotation numbers in  $(0, 1/4]$ , then  $\gamma$  is an ellipse.

*Proof.* Translate the center of symmetry to the origin. Let  $\psi$  be the outer-normal angle, let

$$n(\psi) = (\cos \psi, \sin \psi), \quad t(\psi) = (-\sin \psi, \cos \psi),$$

and write the boundary in support coordinates as

$$\gamma(\psi) = h(\psi)n(\psi) + h'(\psi)t(\psi), \quad \rho(\psi) := h(\psi) + h''(\psi) > 0. \quad (20.1)$$

Central symmetry is  $h(\psi + \pi) = h(\psi)$ .

**Step 1. oriented-line generating function and second twist structure.** For an oriented billiard segment with oriented-line coordinates  $(p, \varphi)$  and its image  $(p_1, \varphi_1)$ , put

$$\psi = \frac{\varphi + \varphi_1}{2}, \quad \delta = \frac{\varphi_1 - \varphi}{2}.$$

Then the billiard map is generated by

$$S(\varphi, \varphi_1) = 2h(\psi) \sin \delta, \quad (20.2)$$

with

$$p = -S_1 = h \cos \delta - h' \sin \delta, \quad p_1 = S_2 = h \cos \delta + h' \sin \delta, \quad (20.3)$$

and

$$\begin{aligned} S_{11} &= \frac{1}{2}(h'' - h) \sin \delta - h' \cos \delta, \\ S_{22} &= \frac{1}{2}(h'' - h) \sin \delta + h' \cos \delta, \\ S_{12} &= \frac{1}{2}(h'' + h) \sin \delta = \frac{1}{2}\rho \sin \delta > 0. \end{aligned} \tag{20.4}$$

Hence  $T$  is a positive twist map also with respect to the vertical foliation  $\{\varphi = \text{const}\}$ .

Indeed, the oriented line containing the segment has right normal angle  $\varphi$  and signed distance  $p$ . The reflection point has normal angle  $\psi$ , and elementary projection of (20.1) on the right normal to the line gives (20.3). Differentiating (20.2) gives the same formulas and then (20.4). Since  $0 < \delta < \pi$  in the open phase cylinder and  $\rho > 0$ , the cross derivative is positive.

**Step 2. the C0 foliation supplies a measurable invariant monotone subbundle.**

If the annulus  $\mathcal{A}$  between the grazing boundary and  $\alpha$  is foliated by continuous rotational invariant curves, then there is a measurable  $DT$ -invariant line subbundle  $\mathcal{L}$  on  $\mathcal{A}$  that is nonvertical and positively oriented in the  $\varphi$ -direction.

By the Birkhoff graph theorem, applied to the positive twist coordinates of Step 1, every rotational invariant curve is a Lipschitz graph  $p = f(\varphi)$ . Order the leaves by  $p$ . At a point of a leaf define the oriented upper tangent line by the upper Dini derivative

$$D^+f(\varphi) = \limsup_{y \rightarrow \varphi} \frac{f(y) - f(\varphi)}{y - \varphi}.$$

For a Lipschitz function this is finite, and the standard measurable-envelope construction makes  $D^+f$  measurable. Because  $T$  is  $C^1$ , sends each leaf to itself, and preserves their order, the image of the upper tangent cone is the upper tangent cone at the image point. In dimension one that cone has the unique extremal line of slope  $D^+f$ ; hence the resulting line field is  $DT$ -invariant. Its  $d\varphi$  component is positive by construction. This is exactly the measurable monotone subbundle needed below; no differentiable foliation is assumed.

**Step 3. invariant-slope identities and equivalence of the two billiard vertical foliations.** Write a positive generator of  $\mathcal{L}$  in  $(p, \varphi)$  coordinates as  $u = \partial_\varphi + \omega \partial_p$ . There are measurable  $\nu_1, \nu_{-1} > 0$  with

$$DT(u(x)) = \nu_1(x)u(Tx), \quad DT^{-1}(u(x)) = \nu_{-1}(x)u(T^{-1}x), \quad \nu_{-1}(Tx) = \nu_1(x)^{-1},$$

and therefore

$$\begin{aligned} \omega(Tx) &= S_{22}(\varphi, \varphi_1) + S_{12}(\varphi, \varphi_1)\nu_1(x)^{-1}, \\ \omega(x) &= -S_{11}(\varphi, \varphi_1) - S_{12}(\varphi, \varphi_1)\nu_1(x). \end{aligned} \tag{20.5}$$

In particular  $\mathcal{L}$  is monotone also in the standard billiard vertical foliation  $\{s = \text{const}\}$ .

Differentiate the generating equations  $p = -S_1(\varphi, \varphi_1)$  and  $p_1 = S_2(\varphi, \varphi_1)$  on  $u$ . The  $d\varphi_1$  component of  $DT(u)$  is  $\nu_1$ , which gives the second identity in (20.5); shifting the inverse identity gives the first. Positivity of  $S_{12}$  and  $\nu_1$  yields

$$S_{22}(\varphi_{-1}, \varphi) < \omega(x) < -S_{11}(\varphi, \varphi_1). \tag{20.6}$$

For completeness, the coordinate change from standard billiard coordinates  $(\cos \delta, s)$  to  $(p, \varphi)$  is

$$p = h(\psi) \cos \delta + h'(\psi) \sin \delta, \quad \varphi = \psi + \delta, \quad d\psi = k(s)ds.$$

If  $(a, b)$  are the  $(d \cos \delta, ds)$  components of  $u$ , direct inversion of the Jacobian gives

$$b = \frac{\omega + h \sin \delta - h' \cos \delta}{\sin \delta} = \frac{\omega - S_{22} + S_{12}}{\sin \delta} > 0$$

by (20.6). Thus nonvertical positive orientation is preserved in the standard vertical foliation as well.

We now use the special inner boundary  $\alpha$ . Write it as  $\alpha = \{\delta = d(\psi)\}$ .

**Step 4. central symmetry forces parallelogram 4-orbits and a quarter-turn map.** The function  $d$  is  $\pi$ -periodic. If  $T_\alpha$  is the map induced on the normal-angle parameter of  $\alpha$ , then

$$T_\alpha^2(\psi) = \psi + \pi, \quad T_\alpha(\psi) = \psi + \frac{\pi}{2}, \quad (20.7)$$

and

$$0 < d(\psi) < \frac{\pi}{2}, \quad d\left(\psi + \frac{\pi}{2}\right) = \frac{\pi}{2} - d(\psi). \quad (20.8)$$

Every 4-periodic orbit on  $\alpha$  is therefore a parallelogram, and the four tangent lines at its vertices form a rectangle.

The central symmetry map on oriented lines commutes with  $T$  and sends  $\alpha$  to another invariant rotational curve with the same rational rotation number. Every orbit segment on an essential invariant circle of an exact monotone twist map is action minimizing. Hence every orbit segment on  $\alpha$  and on  $S(\alpha)$  is minimizing. Since every orbit on  $\alpha$  has rational rotation  $1/4$  and is periodic, the Aubry ordering/crossing argument shows that the whole graph  $\alpha$  is the full minimizing set of that rotation: a second rotational invariant graph of rotation  $1/4$  would also be calibrated, and its minimizing configurations cannot remain strictly ordered from all four time-translates of the periodic minimizing configurations on  $\alpha$ ; the periodic displacement closes after four steps, so strict order would return with an integer translate and contradict the single-crossing theorem. Thus  $S(\alpha) = \alpha$ , proving  $d(\psi + \pi) = d(\psi)$ . For the increasing lift  $\tilde{T}_\alpha$  we have

$$\tilde{T}_\alpha^4(\psi) = \psi + 2\pi, \quad \tilde{T}_\alpha(\psi + \pi) = \tilde{T}_\alpha(\psi) + \pi.$$

If, for example,  $\tilde{T}_\alpha^2(\psi) < \psi + \pi$ , applying  $\tilde{T}_\alpha^2$  and using the second identity gives  $\tilde{T}_\alpha^4(\psi) < \psi + 2\pi$ , a contradiction; the opposite inequality is identical. Hence the first formula in (20.7) holds, so opposite vertices are centrally symmetric. The exterior-angle sum of the resulting parallelogram and the reflection law give  $2(2d(\psi) + 2d(T_\alpha\psi)) = 2\pi$ . The corresponding tangent lines are therefore perpendicular in consecutive pairs, which yields  $T_\alpha(\psi) = \psi + \pi/2$  and (20.8).

**Step 5. exact support identities on the 4-periodic curve.** There is  $R > 0$  such that

$$\tan d(\psi) = \frac{h(\psi)}{h(\psi + \pi/2)} = -\frac{h'(\psi + \pi/2)}{h'(\psi)}, \quad h(\psi)^2 + h(\psi + \pi/2)^2 = R^2, \quad (20.9)$$

and consequently

$$h(\psi) = R \sin d(\psi), \quad h(\psi + \pi/2) = R \cos d(\psi). \quad (20.10)$$

Write the signed distance of the side  $P_0P_1$  in the two endpoint normal frames, using (20.3); do the same for  $P_1P_2$ , which is centrally symmetric to  $P_3P_0$ . Adding and subtracting the two resulting equalities gives

$$h(\psi + \frac{\pi}{2}) \sin d(\psi) = h(\psi) \cos d(\psi), \quad h'(\psi + \frac{\pi}{2}) \cos d(\psi) = -h'(\psi) \sin d(\psi).$$

These are the ratio identities in (20.9). Multiplying them shows

$$\frac{d}{d\psi} (h(\psi)^2 + h(\psi + \frac{\pi}{2})^2) = 0.$$

The positive constant is  $R^2$ , and the ratio together with positivity gives (20.10). Since  $h \in C^2$  and never vanishes, this also shows  $d \in C^2$  at exactly the regularity used below.

**Step 6. weighted pointwise Hopf inequality.** At  $x = (p, \varphi)$  and  $Tx = (p_1, \varphi_1)$ , the invariant-slope identities imply

$$p_1^2 \omega(Tx) - p^2 \omega(x) \geq p^2 S_{11} + p_1^2 S_{22} + 2pp_1 S_{12}. \quad (20.11)$$

Multiply the first line of (20.5) by  $p_1^2$ , the second by  $p^2$ , and subtract. The only remaining term is

$$S_{12}(p^2 \nu_1 + p_1^2 \nu_1^{-1}).$$

Since  $S_{12} > 0$  and

$$p^2 \nu_1 + p_1^2 \nu_1^{-1} - 2pp_1 = (p\sqrt{\nu_1} - p_1/\sqrt{\nu_1})^2 \geq 0,$$

we obtain (20.11) without any sign assumption on  $p$  or  $p_1$ .

**Step 7. integration over the foliated annulus gives the exact one-variable functional.** Let

$$\begin{aligned} U(\psi) := & -h(h')^2 \rho \left( \frac{1}{2}d - \frac{1}{4} \sin 2d \right) \\ & + (h''h^2 + 3h(h')^2) \rho \left( \frac{1}{8}d - \frac{1}{32} \sin 4d \right). \end{aligned} \quad (20.12)$$

Then

$$\int_0^{2\pi} U(\psi) d\psi \leq 0, \quad \int_0^\pi U(\psi) d\psi \leq 0. \quad (20.13)$$

The slope bound (20.6) and continuity of the second derivatives of  $S$  make  $\omega$  bounded. Integrate (20.11) against the invariant symplectic area  $dp d\varphi = S_{12} d\varphi d\varphi_1$ . The left side integrates to zero by  $T$ -invariance. Substitute (20.3) and (20.4). In the coordinates  $(\psi, \delta)$  the annulus is

$$0 \leq \psi < 2\pi, \quad 0 \leq \delta \leq d(\psi),$$

and, after cancelling the common positive Jacobian factor, the inequality is

$$0 \geq \int_0^{2\pi} d\psi \int_0^{d(\psi)} d\delta \left[ \cos^2 \delta \sin^2 \delta (h'' h^2 + 3h(h')^2) \rho - \sin^2 \delta h(h')^2 \rho \right]. \quad (20.14)$$

Now

$$\int_0^d \sin^2 \delta d\delta = \frac{d}{2} - \frac{\sin 2d}{4}, \quad \int_0^d \cos^2 \delta \sin^2 \delta d\delta = \frac{d}{8} - \frac{\sin 4d}{32},$$

so (20.14) is exactly (20.13) with  $U$  as in (20.12). The second inequality follows from the  $\pi$ -periodicity of  $h$  and  $d$ .

**Step 8. quarter-turn symmetrization of the Hopf functional.** Using (20.10), write  $U = U_1 + U_2 + U_3$  with

$$\begin{aligned} U_1 &= \frac{1}{4} h(h')^2 \rho \sin 2d, \\ U_2 &= -\frac{1}{32} h \rho (3(h')^2 + h h'') \sin 4d, \\ U_3 &= \frac{1}{8} h \rho (h h'' - (h')^2) d. \end{aligned} \quad (20.15)$$

If  $\widehat{U}_j(\psi) = U_j(\psi + \pi/2)$  and  $V_j = (U_j + \widehat{U}_j)/2$ , then  $\int_0^\pi U_j = \int_0^\pi V_j$ , and the  $V_j$  are the explicit expressions obtained from  $d(\psi + \pi/2) = \pi/2 - d(\psi)$  by interchanging  $\sin d$  and  $\cos d$  and changing the signs of  $d''$  and  $\sin 4d$ .

This is a purely algebraic normalization, but it is load-bearing because it removes the terms odd under the quarter-turn involution. Substituting

$$h = R \sin d, \quad h' = R \cos d d', \quad h'' = R \cos d d'' - R \sin d (d')^2$$

into (20.15), averaging with its quarter-turn transform and collecting powers of  $d'$  and  $d''$  gives

$$V_1 = \frac{R^4}{16} (d')^2 \sin^2(2d) \left( (1 - (d')^2) \sin 2d + d'' \cos 2d \right), \quad (20.16)$$

$$V_2 = \frac{R^4}{128} \sin 4d \left( 2(d')^2 ((d')^2 - 1) \cos 2d - d'' (1 + (d')^2) \sin 2d \right), \quad (20.17)$$

and

$$\begin{aligned} V_3 &= \frac{R^4}{32} (\pi \cos^2 d - 2d \cos 2d) ((d')^4 - (d')^2) + \frac{\pi R^4}{128} \sin^2(2d) (d'')^2 \\ &\quad + \frac{R^4}{64} \sin 2d (2d - \pi \cos^2 d) d'' \\ &\quad + \frac{R^4}{128} \sin 2d (\pi(3 + \cos 2d) - 12d) (d')^2 d''. \end{aligned} \quad (20.18)$$

The displayed formulas are obtained by direct expansion of the two terms in each average; no dynamical input is hidden here.

**Step 9. periodic integration by parts collapses the symmetrized functional.** After integrating the  $d''$ -linear terms in (20.16)–(20.18) by parts, with all boundary



terms zero by  $\pi$ -periodicity,

$$\int_0^\pi U d\psi = \int_0^\pi W d\psi, \quad (20.19)$$

where

$$W = -\frac{\pi R^4}{32} \sin^2(2d)(d')^2 + \frac{\pi R^4}{192} (3 - \cos 4d)(d')^4 + \frac{\pi R^4}{128} \sin^2(2d)(d'')^2. \quad (20.20)$$

For example, in  $V_1$  use  $d''(d')^2 = \frac{1}{3}((d')^3)'$ ; in  $V_2$  use both  $d''d' = \frac{1}{2}((d')^2)'$  and  $d''(d')^3 = \frac{1}{4}((d')^4)'$ . The same two primitives remove the  $d''$ -linear terms of  $V_3$ . Differentiating only the trigonometric coefficients and collecting the resulting  $(d')^2$  and  $(d')^4$  terms yields (20.20). Since all coefficients are  $\pi$ -periodic and  $d \in C^2$ , these integrations by parts are valid at the theorem's  $C^2$  regularity.

**Step 10. exact reduction to the period- $\pi$  Wirtinger defect.** For

$$\mu(\psi) = \cos(2d(\psi)), \quad (20.21)$$

one has the exact identity

$$\int_0^\pi U(\psi) d\psi = \frac{\pi R^4}{512} \int_0^\pi ((\mu'')^2 - 4(\mu')^2) d\psi. \quad (20.22)$$

Differentiate (20.21):

$$\mu' = -2 \sin 2d d', \quad \mu'' = -4 \cos 2d (d')^2 - 2 \sin 2d d''.$$

Hence

$$\sin^2(2d)(d'')^2 = \frac{1}{4}(\mu'')^2 - 4 \cos^2(2d)(d')^4 - 2 \sin 4d (d')^2 d''.$$

Substitute this into (20.20). The only mixed term is  $\sin 4d (d')^2 d'' = \frac{1}{3} \sin 4d ((d')^3)'$ ; one final periodic integration by parts gives

$$\int_0^\pi W d\psi = \frac{\pi R^4}{512} \int_0^\pi ((\mu'')^2 + 8(\cos 4d - 1)(d')^2) d\psi.$$

Since  $\cos 4d - 1 = -2 \sin^2 2d$  and  $(\mu')^2 = 4 \sin^2 2d (d')^2$ , this is exactly (20.22).

**Step 11. Hopf sign plus sharp Wirtinger forces the first harmonic.** The two opposite signs in (20.13) and (20.22) force equality, and therefore

$$\mu(\psi) = A \cos(2\psi + \alpha) + c. \quad (20.23)$$

Moreover  $c = 0$  and  $0 \leq A < 1$  after a rotation of  $\psi$ .

Because  $\mu$  is  $\pi$ -periodic,  $\mu'$  has zero mean. The sharp Wirtinger inequality on  $[0, \pi]$  gives

$$\int_0^\pi (\mu'')^2 d\psi \geq 4 \int_0^\pi (\mu')^2 d\psi.$$

Thus the right side of (20.22) is nonnegative, whereas Step 7 makes it nonpositive. Equality holds. Equality in Wirtinger says  $\mu' = a \cos 2\psi + b \sin 2\psi$ , and integration gives

(20.23). On the other hand,

$$R^2\mu(\psi) = h(\psi + \frac{\pi}{2})^2 - h(\psi)^2,$$

whose average on  $[0, \pi]$  is zero, so  $c = 0$ . Rotate  $\psi$  to absorb  $\alpha$ . Since  $0 < d < \pi/2$ ,  $|\mu| < 1$  everywhere, hence  $A < 1$ ; clearly  $A \geq 0$ .

**Step 12. ellipse reconstruction from the equality support function.** After the rotation in Step 11,

$$h(\psi)^2 = \frac{R^2}{2}(1 - A \cos 2\psi) = a^2 \cos^2 \psi + b^2 \sin^2 \psi \quad (20.24)$$

with  $a, b > 0$ . Hence  $\gamma$  is the centered ellipse with semiaxes  $a, b$ .

Indeed,  $\mu = \cos 2d = 1 - 2\sin^2 d = A \cos 2\psi$ , and  $h = R \sin d$  by Step 5. Therefore (20.24) holds with  $a^2 = R^2(1 - A)/2$  and  $b^2 = R^2(1 + A)/2$  (interchanging the names of the axes if necessary). The positive square root of this quadratic trigonometric form is exactly the support function of that ellipse. A  $C^2$  strictly convex curve is uniquely determined by its support function through (20.1), so the boundary is the ellipse. The case  $A = 0$  is the circle, included among ellipses.

**Step 13. the C1 caustic-foliation corollary produces the required 4-periodic leaf.** If a boundary neighborhood is  $C^1$ -foliated by convex caustics with rotation numbers in  $(0, 1/4]$ , then their dual invariant graphs  $C^1$ -foliate the corresponding phase annulus, and the leaf of rotation number  $1/4$  consists entirely of 4-periodic orbits. Hence the hypotheses already proved apply.

The caustic-to-invariant-graph correspondence gives the  $C^1$  foliation of the phase annulus. Choose a  $C^1$  foliation chart  $(r, \theta)$  in which the leaves are  $r = \text{const}$ . Because every leaf is invariant, the billiard map has the form

$$T(r, \theta) = (r, f_r(\theta)).$$

Write the symplectic area in this chart as  $J(r, \theta) dr d\theta$  with  $J > 0$ . Area preservation gives, pointwise,

$$J(r, f_r(\theta)) \partial_\theta f_r(\theta) = J(r, \theta).$$

Hence  $J(r, \theta)d\theta$  is an absolutely continuous invariant probability measure after normalization, with positive continuous density and therefore full support, on every leaf. Let  $F_r$  be its cumulative distribution coordinate. Then  $F_r f_r F_r^{-1}$  is an orientation-preserving circle homeomorphism preserving Lebesgue measure. If  $g$  is such a lift, invariance of interval lengths gives  $g(y) - g(x) = y - x$  for all  $x < y$ , so  $g$  is a rigid rotation. On the leaf with rotation number  $1/4$  it is the rotation by  $1/4$ , hence its fourth iterate is the identity. Thus every point of that leaf is 4-periodic. Step 1–Step 12 apply and prove the corollary as well as the theorem.  $\square$

### Theorem K20.2. Near-centrally-symmetric Birkhoff rigidity

**Statement.** Let  $\Omega_0$  be a centrally symmetric strictly convex planar domain with support function  $h_0 \in C^m(\mathbb{R}/2\pi\mathbb{Z})$ ,  $m \geq 40$ , curvature  $k_{\Omega_0}(\psi) > \kappa_0 > 0$ , and  $\|h_0\|_{C^m} <$

$M$ . There is  $\varepsilon_0 > 0$ , depending only on  $\kappa_0$ ,  $|\partial\Omega_0|$ , and  $M$ , such that if a domain  $\Omega_\varepsilon$  with support function  $h_\varepsilon$  satisfies  $\|h_\varepsilon\|_{C^m} < 2M$  and  $\|h_\varepsilon - h_0\|_{C^5} < \varepsilon_0$ , admits a 3-rationally integrable caustic, and either (i) the phase space between the 4-rationally integrable caustic and the boundary is foliated by continuous invariant graphs over the angular component, or (ii) the billiard is 4-rationally integrable, then  $\Omega_\varepsilon$  is an ellipse.

*Proof.*

### Proof strategy

Use the 4-rational caustic to obtain a uniform loop-orbit parametrization and compare the associated angle functions under a small deformation. Rational integrability for the prescribed rotation numbers turns constancy of perimeter along periodic families into quantitative Fourier constraints on the support function. Central symmetry removes the odd sector, and the remaining even modes are controlled recursively; the resulting rigidity estimate forces the deformation to vanish up to the normalized ellipse parameters.

Set

$$\varepsilon := \|h_\varepsilon - h_0\|_{C^5}.$$

As in the source's quantitative perturbation class, decrease the theorem's threshold once and for all so that every support function used below still defines a strictly convex table, satisfies

$$k_{\Omega_h}(\psi) > \frac{\kappa_0}{2}, \quad \|h\|_{C^m} \leq 2M,$$

and lies in the uniform loop-orbit neighborhood constructed below. Thus the normal-angle parametrization

$$\gamma_h(\psi) = h(\psi)n(\psi) + h'(\psi)t(\psi), \quad n = (\cos \psi, \sin \psi), \quad t = (-\sin \psi, \cos \psi) \quad (20.25)$$

is a  $C^{m-1}$  embedding with uniform norm bounds.

**Step 1. Uniform 4-loop orbit and angle function.** For all tables in a sufficiently small  $C^2$  neighborhood of  $\Omega_0$ , the maximal-perimeter 4-loop orbit starting at any boundary point is unique. Its initial incidence angle  $d_h(\psi)$  is  $C^{m-1}$ , uniformly bounded away from 0 and  $\pi$ , and

$$\|d_\varepsilon - d_0\|_{C^3} \leq C\varepsilon. \quad (20.26)$$

We reproduce the loop-orbit stability argument because the reference table itself is not assumed to be 4-rationally integrable. Work first in normalized arclength coordinates  $(s, r)$ , with  $r = -\cos \theta$ , and generating function

$$H_\Omega(s, s_1) = -|\gamma(s_1) - \gamma(s)|.$$

For a strictly convex curve with the theorem's uniform curvature and  $C^3$ -bounds, chord geometry gives, whenever the arclength separation stays in a compact subinterval of  $(0, 1)$ ,

$$|H_{12}(s, s_1)| \asymp \text{dist}_{\mathbb{T}}(s, s_1), \quad |\partial_r \pi_1 F(s, r)| \asymp \text{dist}_{\mathbb{T}}(s, s_1)^{-1}.$$

A  $(1, 4)$ -loop has four chord separations bounded below by a uniform  $\delta_0 > 0$ : their sum is one turn, and the preceding chord/angle estimates make any vanishing separation incompatible with the remaining three steps. Modify  $H_\Omega$  only in the two grazing strips so that it becomes a global uniform negative-twist generating function; this does not alter any 4-loop.

Now use the actual hypothesis on  $\Omega_\varepsilon$ : its 4-rationally integrable graph is a Lipschitz invariant graph  $\Gamma$ . If an orbit starts strictly below (respectively above)  $\Gamma$ , monotonicity of the twist and invariance of the two complementary components imply

$$\pi_1 F^4(s, r) < s + 1 \quad (\text{respectively } \pi_1 F^4(s, r) > s + 1).$$

On  $\Gamma$  equality holds. Moreover the vertical separation is propagated by the twist: if  $c_\Gamma > 0$  is a uniform inverse-Lipschitz constant of the base map on  $\Gamma$ , then

$$\partial_r \pi_1 F^4(s, r_\Gamma(s)) \geq c_\Gamma^4 \rho > 0,$$

where  $\rho$  is the uniform twist constant of the modified map. Continuity keeps this derivative positive in a fixed vertical tube, while the strict sign outside that tube gives a uniform separation from zero. Therefore every sufficiently  $C^2$ -close generating function has, for each  $s$ , a unique solution of  $\pi_1 F^4(s, r) = s + 1$ , with a uniform lower bound on its  $r$ -derivative and with  $r$  uniformly separated from  $\pm 1$ .

Both  $\Omega_0$  and every table in the chosen perturbation class are  $C^2$ -close to  $\Omega_\varepsilon$ . Hence the preceding argument gives unique 4-loop orbits simultaneously for  $\Omega_0$  and  $\Omega_\varepsilon$ . Passing between arclength and normal angle uses  $ds/d\psi = h + h''$ ; the uniform curvature/regularity bounds make these changes of coordinates uniformly  $C^3$ . The implicit-function theorem applied to  $\pi_1 F^4(s, r) = s + 1$  now yields

$$\|d_\varepsilon - d_0\|_{C^3} \leq C\|F_\varepsilon - F_0\|_{C^3} \leq C\|h_\varepsilon - h_0\|_{C^5} = C\varepsilon,$$

and gives the asserted  $C^{m-1}$  regularity and fixed  $0 < \underline{d} < d_0, d_\varepsilon < \bar{d} < \pi$ .

Let  $\Psi_\varepsilon$  be the map taking the starting normal angle to the next normal angle along this 4-loop family.

**Step 2. Perturbative quarter-turn and support identities.** If  $\Omega_\varepsilon$  has a 4-rationally integrable caustic, then

$$d_\varepsilon(\psi + \pi) = d_\varepsilon(\psi) + O_{C^3}(\varepsilon), \quad (20.27)$$

$$\Psi_\varepsilon^2(\psi) = \psi + \pi + O_{C^2}(\varepsilon), \quad (20.28)$$

$$d_\varepsilon(\psi + \frac{\pi}{2}) = \frac{\pi}{2} - d_\varepsilon(\psi) + O_{C^2}(\varepsilon), \quad (20.29)$$

and every 4-periodic trajectory is  $O_{C^2}(\varepsilon)$ -close to a parallelogram, while the four tangent lines are  $O_{C^2}(\varepsilon)$ -close to a rectangle. There is  $R_0 > 0$ , depending only on the

reference table, such that

$$h_\varepsilon(\psi)^2 + h_\varepsilon(\psi + \frac{\pi}{2})^2 = R_0^2 + O_{C^3}(\varepsilon), \quad (20.30)$$

$$h_\varepsilon(\psi) = R_0 \sin d_\varepsilon(\psi) + O_{C^2}(\varepsilon), \quad (20.31)$$

$$h_\varepsilon(\psi + \frac{\pi}{2}) = R_0 \cos d_\varepsilon(\psi) + O_{C^2}(\varepsilon). \quad (20.32)$$

Because  $\Omega_0$  is centrally symmetric,  $d_0(\psi + \pi) = d_0(\psi)$ . Together with (20.26) this gives (20.27). Write  $\Psi_\varepsilon^2(\psi) = \psi + \pi + f(\psi)$ . The identity  $\Psi_\varepsilon^4(\psi) = \psi + 2\pi$  on the 4-periodic leaf, the  $C^3$  closeness of  $\Psi_\varepsilon$  to the reference map, and one Taylor expansion give  $\|f\|_{C^2} \leq C\varepsilon$ , proving (20.28). The four reflection equations can now be compared with their centrally symmetric counterparts. Opposite chords differ by  $O_{C^2}(\varepsilon)$ , giving the parallelogram statement; the reflection law then makes consecutive tangent lines orthogonal up to  $O_{C^2}(\varepsilon)$  and yields (20.29).

In the Bialy–Mironov oriented-line coordinates the generating function is  $S(\varphi, \varphi_1) = 2h((\varphi + \varphi_1)/2) \sin((\varphi_1 - \varphi)/2)$ . Applying its endpoint equations successively to the four nearly rectangular tangent lines and adding/subtracting opposite relations gives

$$h_\varepsilon(\psi + \frac{\pi}{2}) \sin d_\varepsilon(\psi) = h_\varepsilon(\psi) \cos d_\varepsilon(\psi) + O_{C^2}(\varepsilon),$$

$$h'_\varepsilon(\psi + \frac{\pi}{2}) \cos d_\varepsilon(\psi) = -h'_\varepsilon(\psi) \sin d_\varepsilon(\psi) + O_{C^2}(\varepsilon).$$

Multiply sidewise. Since  $\sin 2d_0$  is uniformly bounded away from zero, the non-grazing estimate from Step 1 gives

$$\frac{d}{d\psi} \left( h_\varepsilon(\psi)^2 + h_\varepsilon(\psi + \frac{\pi}{2})^2 \right) = O_{C^2}(\varepsilon).$$

Integrating and comparing with  $\Omega_0$  fixes the constant  $R_0^2$  and gives (20.30); the preceding two relations then yield (20.31)–(20.32).

**Step 3. Rational integrability is equivalent to  $C^0$ -integrability on the twist annulus.** Let two essential rational invariant graphs of an exact monotone twist map bound an annulus. The map is rationally integrable for every rational rotation number in that interval if and only if the annulus is foliated by continuous rotational invariant graphs. Hence alternative (ii) of the theorem supplies the same  $C^0$  foliation between the 1/4 caustic and the boundary as alternative (i).

We spell out the direction used below. Extend the billiard generating function smoothly to a global monotone twist generating function on a strip; this is possible because the relevant annulus stays uniformly away from grazing. For a cohomology parameter  $c$ , let  $A_c$  be the modified one-step action and let  $\mathcal{A}(c)$  be the Aubry set obtained from the Peierls barrier. By the Peierls-barrier calibration theorem proved earlier, calibrated configurations satisfy the crossing lemma and their lifts have a well-defined rotation number  $\rho(c)$ , monotone in  $c$ . Rational integrability says that for every rational  $\rho \in [\rho_1, \rho_2]$  the minimizing periodic orbit is an invariant graph projecting onto the full base circle. Approximate an arbitrary  $\rho(c)$  by such rationals. Equi-Lipschitz compactness of calibrated graphs implies that  $\mathcal{A}(c)$  also projects onto the full base. The graph theorem the Birkhoff graph theorem proved earlier makes it a Lipschitz invariant graph. Distinct cohomology parameters give disjoint ordered graphs; mono-

tonicity plus density of the rationals makes their intersection with each vertical fibre a continuous homeomorphism of the parameter interval onto the vertical interval. Thus the Aubry graphs foliate the annulus. Conversely, every leaf of a  $C^0$  invariant foliation is a calibrated graph, and on a rational-rotation leaf the full graph is periodic, giving rational integrability.

**Step 4. Perturbative Hopf integral inequality.** For a  $C^0$ -integrable annulus between the  $1/4$  caustic and the boundary define

$$U_\varepsilon = -h_\varepsilon(h'_\varepsilon)^2(h_\varepsilon + h''_\varepsilon)\left(\frac{1}{2}d_\varepsilon - \frac{1}{4}\sin 2d_\varepsilon\right) + (h''_\varepsilon h_\varepsilon^2 + 3h_\varepsilon(h'_\varepsilon)^2)(h_\varepsilon + h''_\varepsilon)\left(\frac{1}{8}d_\varepsilon - \frac{1}{32}\sin 4d_\varepsilon\right). \quad (20.33)$$

Then

$$\int_0^{2\pi} U_\varepsilon(\psi) d\psi \leq 0. \quad (20.34)$$

The proof of the weighted Hopf inequality is local to the twist annulus and does not use central symmetry. In the oriented-line coordinates, the  $C^0$ -foliation supplies a measurable invariant monotone line field exactly as in K20-002. Writing its slope through the two endpoint derivatives of  $S$  and applying  $x + x^{-1} \geq 2$  gives the pointwise Riccati inequality. Multiply by the invariant density  $S_{12} d\varphi d\varphi_1$ , integrate over the annulus, and use invariance to cancel the shifted slope terms. The remaining transverse integrals are elementary powers of  $\sin \delta$  and yield precisely (20.33); hence (20.34).

**Step 5. Perturbative Bialy-Mironov functional reduces to a Fourier defect.** With

$$\mu_\varepsilon(\psi) := \cos(2d_\varepsilon(\psi)),$$

one has

$$\int_0^{2\pi} U_\varepsilon d\psi = \frac{\pi R_0^4}{512} \int_0^{2\pi} \left( (\mu''_\varepsilon)^2 - 4(\mu'_\varepsilon)^2 \right) d\psi + O(\varepsilon). \quad (20.35)$$

Consequently

$$\int_0^{2\pi} \left( (\mu''_\varepsilon)^2 - 4(\mu'_\varepsilon)^2 \right) d\psi \leq C\varepsilon. \quad (20.36)$$

Differentiate (20.31) twice and use (20.29). Uniform  $C^m$  bounds imply

$$\begin{aligned} h_\varepsilon &= R_0 \sin d_\varepsilon + O_{C^2}(\varepsilon), \\ h'_\varepsilon &= R_0 \cos d_\varepsilon d'_\varepsilon + O_{C^1}(\varepsilon), \\ h''_\varepsilon &= R_0 \cos d_\varepsilon d''_\varepsilon - R_0 \sin d_\varepsilon (d'_\varepsilon)^2 + O(\varepsilon). \end{aligned}$$

Substitution into (20.33), followed by the same quarter-turn symmetrization and periodic integrations by parts as in K20-002, changes every identity only by  $O(\varepsilon)$  and gives (20.35). Combining it with (20.34) yields (20.36).

**Step 6. Fourier concentration of the four-loop angle.** If  $\mu_\varepsilon = \sum_{k \in \mathbb{Z}} \hat{\mu}_{\varepsilon,k} e^{ik\psi}$ , then

$$\left\| \mu_\varepsilon - (\hat{\mu}_{0,-2} e^{-2i\psi} + \hat{\mu}_{0,2} e^{2i\psi}) \right\|_{C^1} \leq C\sqrt{\varepsilon}. \quad (20.37)$$

The reference  $\mu_0 = \cos(2d_0)$  is  $\pi$ -periodic, so its odd Fourier modes vanish. From (20.26),  $|\widehat{\mu}_{\varepsilon, \pm 1}| \leq C\varepsilon$ . Parseval and (20.36) give

$$\sum_{k \notin \{0, \pm 2\}} k^4 |\widehat{\mu}_{\varepsilon, k}|^2 \leq C\varepsilon.$$

Sobolev embedding  $H^2(\mathbb{T}) \hookrightarrow C^1(\mathbb{T})$  controls all modes except  $0, \pm 2$ . Furthermore

$$R_0^2 \mu_\varepsilon = h_\varepsilon(\psi + \frac{\pi}{2})^2 - h_\varepsilon(\psi)^2 + O_{C^2}(\varepsilon),$$

so averaging gives  $\widehat{\mu}_{\varepsilon, 0} = O(\varepsilon)$ ; finally (20.26) gives  $\widehat{\mu}_{\varepsilon, \pm 2} = \widehat{\mu}_{0, \pm 2} + O(\varepsilon)$ . This proves (20.37).

**Step 7. Quantitative ellipse extraction and the one-eighth exponent.** Define the centered ellipse  $\check{\mathcal{E}}_0$  by

$$\check{h}_0(\psi) = \frac{R_0}{\sqrt{2}} \sqrt{1 - (\widehat{\mu}_{0, -2} e^{-2i\psi} + \widehat{\mu}_{0, 2} e^{2i\psi})}. \quad (20.38)$$

Then

$$\|h_\varepsilon - \check{h}_0\|_{C^2} \leq C\varepsilon^{1/8}, \quad \|\check{h}_0\|_{C^m} \leq C. \quad (20.39)$$

The square root in (20.38) is positive because the reference support function is strictly positive; it is exactly the support function of a centered ellipse. From (20.31) and (20.37),

$$\|h_\varepsilon - \check{h}_0\|_{C^1} \leq C\sqrt{\varepsilon}.$$

At the same time both functions have uniform  $C^m$  bounds. Apply the one-dimensional interpolation inequality twice, first to the function and then to its derivative, in the source's form

$$\|f\|_{C^2} \leq C\|f\|_{C^1}^{1/4} \|f\|_{C^5}^{3/4}.$$

Since  $\|f\|_{C^1} = O(\varepsilon^{1/2})$ , this gives the explicit  $O(\varepsilon^{1/8})$  bound in (20.39). The higher norm of the ellipse is bounded because its semiaxes remain in a compact set fixed by the curvature, perimeter and  $M$  bounds.

**Step 8. Elliptic-polar transfer with high-regularity control.** Let  $c_0$  and  $e_0$  be the semi-focal distance and eccentricity of  $\check{\mathcal{E}}_0$ , and write its elliptic coordinates as

$$\Phi(\lambda, \theta) = c_0(\cosh \lambda \cos \theta, \sinh \lambda \sin \theta).$$

Then the perturbed boundary can be written uniquely as

$$\partial\Omega_\varepsilon = \{\Phi(\lambda_0 + u_\varepsilon(\theta), \theta) : \theta \in \mathbb{T}\}, \quad (20.40)$$

with

$$\|u_\varepsilon\|_{C^{m-1}} \leq C, \quad \|u_\varepsilon\|_{C^1} \leq C\varepsilon^{1/8}. \quad (20.41)$$

Equation (20.25) and (20.39) imply that the two boundary curves are  $C^1$ -close and uniformly  $C^{m-1}$ . On a fixed tubular neighborhood of the ellipse,  $\Phi^{-1}$  is smooth. Write  $(\bar{\lambda}_\varepsilon(\psi), \bar{\theta}_\varepsilon(\psi)) = \Phi^{-1}\gamma_\varepsilon(\psi)$ . For the ellipse,  $\bar{\lambda}_0 \equiv \lambda_0$  and  $\bar{\theta}_0$  is a smooth circle diffeomor-

phism. The inverse-function theorem gives that  $\bar{\theta}_\varepsilon$  is also a diffeomorphism, with inverse uniformly  $C^{m-1}$  and  $C^1$ -close to the reference inverse. Define  $u_\varepsilon = \bar{\lambda}_\varepsilon \circ \bar{\theta}_\varepsilon^{-1} - \lambda_0$ . Standard composition estimates give (20.41).

**Step 9. The theorem hypotheses give 3-rational weak integrability.** The elliptic-polar table in Step 8 possesses an integrable rational caustic of rotation number  $1/q$  for every integer  $q \geq 3$ .

The assumed 3-rational caustic supplies  $q = 3$ . Under alternative (ii), the definition of 4-rational integrability supplies every  $(1, q)$  caustic with  $q \geq 4$ . Under alternative (i), Step 3 gives rational integrability throughout the annulus between the  $1/4$  leaf and the boundary, hence again every  $1/q$  leaf for  $q \geq 4$ . Thus the table is 3-rationally weakly integrable in the precise sense needed below.

We now prove the local near-ellipse rigidity engine internally. Fix an ellipse  $E = E_{e,c}$  and identify a nearby boundary with an elliptic-polar graph  $E + u(\varphi)$ .

**Step 10. Rational caustics give quantitative dynamical-mode constraints.** For each  $q \geq 3$ , let  $k_q \in (e, 1)$  be the elliptic modulus of the confocal caustic of rotation number  $1/q$ , let  $F(\varphi; k_q)$  and  $K(k_q)$  be the incomplete and complete elliptic integrals, and define

$$c_q(\varphi) = \frac{\cos\left(\frac{2\pi q}{4K(k_q)} F(\varphi; k_q)\right)}{\sqrt{1 - k_q^2 \sin^2 \varphi}}, \quad (20.42)$$

$$s_q(\varphi) = \frac{\sin\left(\frac{2\pi q}{4K(k_q)} F(\varphi; k_q)\right)}{\sqrt{1 - k_q^2 \sin^2 \varphi}}. \quad (20.43)$$

If  $E + u$  has an integrable rational  $1/q$  caustic and  $\|u\|_{C^1}$  is small, then

$$|\langle u, c_q \rangle| + |\langle u, s_q \rangle| \leq C(e, c) q^8 \|u\|_{C^1}^2 \quad (20.44)$$

whenever  $q \leq c(e) \|u\|_{C^1}^{-1/8}$ .

For the ellipse, action-angle coordinates on the  $1/q$  caustic are obtained by  $\theta = (2\pi/(4K(k_q)))F(\varphi; k_q)$ , and the orbit advances by  $2\pi/q$ . The perimeter action of the inscribed  $q$ -gon is constant in the base phase. Vary the boundary by  $u$ . The first variation of each chord is the sum of its endpoint normal variations; summing over the  $q$  vertices and using the rigid action-angle spacing shows that every nonconstant Fourier mode of the phase has zero first variation. Choosing the first sine and cosine modes and changing back from  $\theta$  to  $\varphi$  gives precisely (20.42)–(20.43) and orthogonality at linear order. For a finite perturbation the Taylor remainder of the  $q$ -gon action contains at most eight powers of  $q$  after differentiating the  $q$  impact parameters and inverting the twist Hessian; uniform  $C^5$  control gives (20.44). The restriction  $q \|u\|_{C^1}^{1/8} \leq c(e)$  ensures the orbit stays in the uniform elliptic coordinate chart and absorbs the higher Taylor terms.

**Step 11. Five elliptic-motion directions and quadratic realization by a true**



**ellipse.** There are five analytic functions

$$e_h, e_{\tau_1}, e_{\tau_2}, e_r, e_{hr}$$

obtained by differentiating the ellipse family with respect to homothety, two translations, rotation and hyperbolic rotation/eccentricity. They are linearly independent. If  $v$  lies in their span and is sufficiently small in  $C^1$ , there is a genuine nearby ellipse  $E_v$  whose elliptic-polar graph over  $E$  satisfies

$$\|u_{E_v} - v\|_{C^1} \leq C\|v\|_{C^1}^2. \quad (20.45)$$

Differentiate the five-parameter parametrization  $E(x_0, y_0, c, \mu_0, \vartheta)$  in elliptic coordinates. The derivative with respect to each parameter gives one of the listed modes. Their parity and lowest trigonometric components make them linearly independent. Apply the finite-dimensional inverse-function theorem to the map from ellipse parameters to the corresponding elliptic-polar graph. Its differential at the reference ellipse is exactly the isomorphism onto the five-dimensional span. Taylor's theorem in finite dimension gives (20.45).

**Step 12. Complex singularities separate the dynamical and elliptic-motion modes.** The complex continuations of

$$e_h, e_{\tau_1}, e_{\tau_2}, e_r, e_{hr}, \{c_q, s_q\}_{q \geq 3}$$

are linearly independent. The dynamical mode with index  $q$  has its nearest branch singularities at imaginary height  $\rho_q = \text{arcosh}(1/k_q)$ , with  $\rho_q \neq \rho_j$  for  $q \neq j$ , whereas the elliptic-motion modes have their nearest singularities at the boundary height  $\rho_0 = \text{arcosh}(1/e)$ .

The square-root denominator in (20.42)–(20.43) vanishes exactly at the complex zeros of  $1 - k_q^2 \sin^2 z$ . The elliptic integral continuation does not cancel these branch points. Since  $k_q$  is strictly decreasing to  $e$ , the heights  $\rho_q$  are pairwise distinct and increase to  $\rho_0$ . In a finite linear combination choose the term with the singularity closest to the real axis; all other terms are analytic there, so its coefficient must vanish. Iterate. The five elliptic-motion modes are elementary analytic functions whose leading principal parts at the  $\rho_0$  singularities have distinct parity/order; their coefficients therefore vanish as well. This proves linear independence.

**Step 13. Near-diagonal correlation matrix and basis property.** After normalizing the five elliptic-motion modes as  $e_0, \dots, e_4$  and setting  $e_{2q} = c_q$ ,  $e_{2q+1} = s_q$  for  $q \geq 3$ , the family  $\{e_j\}_{j \geq 0}$  is a (nonorthogonal) basis of  $L^2(\mathbb{T})$ . More precisely, for Fourier modes and  $j \geq 6$  the correlation coefficients satisfy

$$\tilde{a}_{qj} = 2K(k_{[j/2]})\delta_{qj} + O_{e,c}\left(j^{-1}e^{-\rho|q-j|}\right) \quad (20.46)$$

for some  $\rho > 0$ .

Write the action-angle change as  $\xi_j(\varphi) = \varphi + \Delta_j(\varphi)$  in a common analytic coordinate. The confocal-caustic asymptotics give  $\|\Delta_j\|_\rho \leq Cj^{-2}$ . Hence  $e^{ij\Delta_j} = 1 + O(j^{-1})$  in the strip, while all nonzero Fourier coefficients of the analytic error decay like  $e^{-\rho|q-j|}$ . This gives

(20.46) by contour shifting. Choose  $q_0$  so large that the off-diagonal Hilbert-Schmidt norm of the map sending the Fourier basis to the dynamical modes is less than half the inverse norm of its diagonal part. A Neumann-series argument makes the tail map invertible. Step 12 and a finite-dimensional dimension count then show that adjoining the first five elliptic-motion modes produces a basis of the full  $L^2$  space.

**Step 14. Low/high dynamical-mode estimates on the non-elliptic component.**

Let  $V_E$  be the five-dimensional elliptic-motion space and decompose  $u = u_V + u_\perp$  in the basis of Step 13, with  $u_\perp \perp V_E$ . If  $q_* = [\|u\|_{C^1}^{-\alpha}]$  with  $0 < \alpha < 1/8$ , then

$$\sum_{5 \leq j \leq 2q_*} |\widehat{u}_{\perp,j}|^2 \leq C \|u\|_{C^1}^{4-17\alpha}, \quad (20.47)$$

$$\sum_{j > 2q_*} |\widehat{u}_{\perp,j}|^2 \leq C \|u\|_{C^1}^{2+\alpha}. \quad (20.48)$$

With  $\alpha = 1/9$ ,

$$\|u_\perp\|_{L^2} \leq C \|u\|_{C^1}^{19/18}. \quad (20.49)$$

For  $5 \leq q \leq q_*$ , the coefficient of  $u_\perp$  in the  $q$ -th dynamical mode is controlled by (20.44); the bounded basis-coordinate map from Step 13 converts this into  $|\widehat{u}_{\perp,j}| \leq C q^8 \|u\|_{C^1}^2$ . Summation over  $q_*$  indices gives (20.47). For the tail, the integration-by-parts estimate for the oscillatory dynamical modes gives  $|\langle f, c_q \rangle| + |\langle f, s_q \rangle| \leq C \|f\|_{C^1}/q$ ; summing  $q^{-2}$  beyond  $q_*$  gives (20.48). Balance  $4 - 17\alpha = 2 + \alpha$  to obtain  $\alpha = 1/9$  and (20.49).

**Step 15. Quantitative local ellipse improvement.** Assume  $u \in C^{39}$ ,  $\|u\|_{C^{39}} \leq L$  and  $\|u\|_{C^1}$  is small. There exists a nearby ellipse  $E_1$  such that, when the same boundary is written as  $E_1 + \tilde{u}$  in the elliptic coordinates of  $E_1$ ,

$$\|\tilde{u}\|_{C^1} \leq C(e, c, L) \|u\|_{C^1}^{703/702}. \quad (20.50)$$

By Step 14 and one-dimensional Sobolev interpolation, for  $j = 1, 2$  and any cutoff  $\Delta > 1$ ,

$$\|D^j u_\perp\|_{L^2} \leq C \left( \Delta \|u_\perp\|_{C^{39}} + \Delta^{-j/(39-j)} \|u_\perp\|_{L^2} \right).$$

Insert (20.49) and optimize  $\Delta$ . The weakest of the  $j = 1, 2$  exponents gives

$$\|u_\perp\|_{C^1} \leq C(e, c, L) \|u\|_{C^1}^{1+1/702}. \quad (20.51)$$

The five-dimensional part satisfies  $\|u_V\|_{C^{39}} \leq C \|u\|_{C^1}$ . By Step 11 choose a true ellipse  $E_1$  whose graph differs from  $u_V$  by  $O(\|u\|_{C^1}^2)$ . Changing elliptic coordinate frames is smooth, so the new graph  $\tilde{u}$  equals  $u_\perp$  plus this quadratic error and a product of the coordinate-frame change with  $u - u_V$ . These terms are dominated by (20.51), proving (20.50).

**Step 16. Minimizing-ellipse argument closes local rigidity.** Fix a compact sufficiently small family  $\mathcal{E}_\sigma(E)$  of ellipses around  $E$ . If a 3-rationally weakly integrable  $C^{39}$  table has uniformly bounded  $C^{39}$  elliptic-polar graph and is sufficiently  $C^1$ -close to  $E$ , then it is itself an ellipse.

For each  $E_\nu \in \mathcal{E}_\sigma(E)$  write the boundary as  $E_\nu + u_\nu$ . Smooth dependence of elliptic coordinate frames on the five ellipse parameters makes  $\nu \mapsto \|u_\nu\|_{C^1}$  continuous. The compact parameter set therefore contains a minimizer  $\nu_*$ . If  $a := \|u_{\nu_*}\|_{C^1} > 0$ , apply Step 15 at  $E_{\nu_*}$  to obtain a new ellipse  $E_1$  with

$$\|u_{E_1}\|_{C^1} \leq Ca^{703/702} < a$$

for  $a$  below a fixed threshold. The Hausdorff distance between  $E_1$  and the original reference ellipse is at most the distance from the reference ellipse to the table plus  $C\|u_{E_1}\|_{C^0}$ , hence remains inside  $\mathcal{E}_\sigma(E)$  if the original  $C^1$  neighborhood is chosen small. This contradicts minimality. Therefore  $a = 0$  and the table equals an ellipse.

**Step 17. Final compatibility and closure.** The bounds and rational-caustic hypotheses from Steps 8–9 satisfy the local engine Steps 10–16 uniformly for the extracted ellipse. Consequently  $\Omega_\varepsilon$  is an ellipse.

Indeed, set  $\ell = m - 1 \geq 39$ . Step 8 gives a  $C^\ell$  elliptic-polar graph with a uniform  $C^\ell$  bound depending only on the theorem data and  $C^1$  norm at most  $C\varepsilon^{1/8}$ . The extracted ellipses remain in a compact family because curvature, perimeter and  $C^m$  norms are uniformly controlled; therefore all constants in Steps 10–16 are uniform over this family. Step 9 supplies every  $1/q$  caustic for  $q \geq 3$ . Choose the original  $\varepsilon_0$  so small that  $C\varepsilon_0^{1/8}$  lies below the uniform local-rigidity threshold from Step 16. Applying Step 16 proves that  $\partial\Omega_\varepsilon$  is an ellipse, as required.  $\square$

### Theorem K20.3. Koval local strong Birkhoff theorem

**Statement.** Fix  $q_0 > 2$ , an ellipse  $\mathcal{E}_0$  of eccentricity  $e \in (0, 1)$  and semi-focal distance  $c$ , an integer  $k \geq 39$ , and  $K > 0$ . A billiard domain is  $q_0$ -rationally integrable if it admits an integrable rational caustic of rotation number  $p/q$  for every rational  $0 < p/q \leq 1/q_0$ . Then there exist a locally finite set  $\mathcal{Z}(q_0) \subset (0, 1)$  and, for every  $e \notin \mathcal{Z}(q_0)$ , a number  $\varepsilon = \varepsilon(e, c, K, q_0) > 0$  such that the following holds: if  $\Omega$  is a  $q_0$ -rationally integrable  $C^k$ -smooth strictly convex billiard domain whose boundary is  $C^k$ - $K$  close and  $C^1$ - $\varepsilon$  close to  $\mathcal{E}_0$  in Koval's normalization, then  $\Omega$  is an ellipse.

#### Source note

The range  $q_0 > 2$  follows Koval's definition of  $q_0$ -rational integrability. The first line of the printed theorem states a broader range, so the book adopts the source definition as the conservative statement convention.

*Proof.*

#### Proof strategy

Linearize rational-caustic preservation in elliptic action-angle coordinates. For a detecting family of rotation numbers, the resulting Fourier functionals form an almost triangular system whose diagonal coefficients fail only on an explicitly controlled exceptional eccentricity set. The nonlinear preservation equations have quadratic errors, so the linear inversion gives a contraction estimate for the deformation. Normalizing the elliptic parameters removes the finite-dimensional

kernel and yields local rigidity.

Fix the reference ellipse  $\mathcal{E}_{e,c}$  with  $0 < e < 1$  and semi-focal distance  $c$ . In elliptic coordinates

$$\Phi(\lambda, \varphi) = c(\cosh \lambda \cos \varphi, \sinh \lambda \sin \varphi)$$

the reference boundary is the level  $\lambda = \lambda_0$ , while every table in the theorem's sufficiently small neighborhood is written uniquely as

$$\partial\Omega = \{\Phi(\lambda_0 + \mu(\varphi), \varphi) : \varphi \in \mathbb{T}\}. \quad (20.52)$$

The theorem's  $C^k$ - $K$  assumption gives  $\|\mu\|_{C^k} \leq K$  after shrinking the neighborhood and changing  $K$  by a fixed chart constant. We use  $\eta := \|\mu\|_{C^1}$  and assume throughout that  $\eta$  is below a threshold to be chosen at the end.

**Step 1. Elliptic action-angle coordinates and preservation functionals.** For the confocal caustic of rotation number  $\omega = p/q < 1/2$ , let  $k_{p/q}$  be its modulus and let

$$\theta_{p/q}(\varphi) = \frac{\pi}{2K(k_{p/q})} F(\varphi, k_{p/q}). \quad (20.53)$$

The billiard map on that caustic is translation by  $2\pi p/q$  in  $\theta$ . For  $j \geq 0$ , define

$$A_{p,q,j}(e) := \frac{1}{\pi} \int_0^{2\pi} \cos(j\varphi(\theta_{p/q})) \cos(q\theta_{p/q}) d\theta_{p/q}, \quad (20.54)$$

and define the sine coefficients analogously. If  $\mu = a_0 + \sum_{j \geq 1} (a_j^+ \cos j\varphi + a_j^- \sin j\varphi)$ , the linearized preservation constraints are

$$A_{p,q}^\pm(e)[\mu] := \sum_{j \geq 0} A_{p,q,j}(e) a_j^\pm. \quad (20.55)$$

The action-angle coordinate follows from the standard confocal parametrization of the ellipse: if  $\lambda_{p/q}$  is the caustic parameter and  $k_{p/q} = e/\sqrt{1 - \lambda_{p/q}^2}$ , the rotation number is

$$\frac{p}{q} = \frac{F(\arcsin(\lambda_{p/q}/\sqrt{1 - e^2}), k_{p/q})}{2K(k_{p/q})}.$$

Thus (20.53) conjugates the restriction of the billiard map to rigid rotation. A  $p/q$  caustic consists of a one-parameter family of  $q$ -periodic critical polygons for the perimeter functional

$$\mathcal{L}(P_1, \dots, P_q) = \sum_{i=1}^q |P_i P_{i+1}|, \quad P_{q+1} = P_1.$$

Differentiating the perimeter in a normal deformation cancels the vertex motions by the reflection law. The remaining first variation is a weighted sum of the deformation at the  $q$  impact points. Since these points are  $2\pi p/q$ -spaced in  $\theta_{p/q}$ , constancy of the first variation in the starting phase forces the  $q$ th action-angle sine and cosine coefficients to vanish. Expanding the elliptic Fourier modes in that coordinate gives exactly

(20.54)–(20.55). Parity of the change of variables implies

$$A_{p,q,j} = 0 \quad \text{when } q \not\equiv j \pmod{2}. \quad (20.56)$$

**Step 2. Nonlinear preservation equation with its quadratic error.** If  $\Omega$  preserves the  $p/q$  caustic,  $p$  is bounded, and  $q < C(e)\eta^{-1/8}$ , then in elliptic coordinates

$$\left| \int_0^{2\pi} \mu(\varphi) c_{p,q}^\varphi(\varphi) d\varphi \right| \leq C(e, p, K) q^8 \eta^2, \quad (20.57)$$

$$\left| \int_0^{2\pi} \mu(\varphi) s_{p,q}^\varphi(\varphi) d\varphi \right| \leq C(e, p, K) q^8 \eta^2. \quad (20.58)$$

Here  $c_{p,q}^\varphi, s_{p,q}^\varphi$  are the elliptic-coordinate preservation modes obtained from the  $q$ th action-angle harmonics.

For completeness, parameterize the perturbed  $q$ -periodic polygon by the starting phase. The implicit-function theorem applies uniformly while  $q\eta^{1/8}$  is small: the unperturbed  $q$ -gon has a nondegenerate transverse Hessian modulo the one tangential phase direction, and the  $C^5$  bound follows from  $k \geq 39$ . Solve the  $q - 1$  transverse critical-point equations. Inserting that solution in the perimeter gives a one-variable function  $L_{p,q,\mu}(\theta_0)$ . Its nonconstant Fourier coefficients vanish when a rational caustic exists. Taylor expansion at  $\mu = 0$  has first derivative precisely (20.55); the displacement of the critical vertices is  $O(q^3\eta)$  and the second variation has four further powers of  $q$  from differentiating the action-angle chart and the chord Hessian. Consequently the nonlinear error is  $O(q^8\eta^2)$ , uniformly for bounded  $p$ . This is the source Proposition 7.1 and yields (20.57)–(20.58).

**Step 3. Small-eccentricity coefficient asymptotics.** Let  $j, q > 1$  have the same parity and set  $y = |q - j|/2$ . As  $e \rightarrow 0$ ,

$$A_{p,q,j}(e) = \binom{j+y-1}{y} \frac{e^{2y}}{2^{4y} \cos^{2y}(\pi p/q)} + O(e^{2y+2}), \quad j \leq q, \quad (20.59)$$

while

$$A_{p,q,j}(e) = (-1)^y \binom{j}{y} \frac{e^{2y}}{2^{4y} \cos^{2y}(\pi p/q)} + O(e^{2y+2}), \quad j > q, \quad (20.60)$$

and for small  $e$

$$|A_{p,q,j}(e)| \leq C^{3y+j+1} e^{2y}. \quad (20.61)$$

Indeed, the inverse relation for the Jacobi amplitude is

$$\varphi(\theta) = \text{am}\left(\frac{4K(k)}{2\pi}\theta, k\right) = \theta + 2 \sum_{n \geq 1} \frac{\mathbf{q}^n}{n(1 + \mathbf{q}^{2n})} \sin(2n\theta),$$

where  $\mathbf{q} = \exp(-\pi K(k')/K(k))$  is the elliptic nome. Substitution in (20.54), expansion of the exponential, and extraction of the zero Fourier mode gives the absolutely convergent composition formula used in Koval's equation (3.12). The least possible nome degree is  $y$ ; the coefficient at that degree is the Stirling-number/binomial sum  $\binom{j+y-1}{y}$

for  $j \leq q$  and  $(-1)^y \binom{j}{y}$  for  $j > q$ . Since

$$\mathbf{q} = \frac{k^2}{16} + \frac{k^4}{32} + O(k^6), \quad k^2 = \frac{e^2}{1 - \lambda_{p/q}^2}, \quad \lambda_{p/q} = \sqrt{1 - e^2} \sin \frac{\pi p}{q} + O(e^2),$$

the leading nome term becomes  $e^{2y}/(2^{4y} \cos^{2y}(\pi p/q))$ . The majorant of the composition series is geometric after factoring out the least degree, giving (20.61).

**Step 4. Odd-mode finite rank near the circle.** For every  $q_0 > 2$  there is an odd cutoff  $q_{1,o}$  such that the rows  $A_{p,q}^\pm$  with  $q \leq q_{1,o}$  and  $0 < p/q < 1/q_0$  have full rank on the odd elliptic Fourier modes  $3, 5, \dots, q_{1,o}$  for all sufficiently small positive  $e$ .

Write  $q = 2r + 1$  and  $j = 2k + 1$ . Multiplying rows and columns by nonzero factors and then taking  $e \rightarrow 0$  turns the odd block into

$$\hat{A}_{p,r,k} = \binom{r+k}{r-k} \left(1 + \cos \frac{2\pi p}{2r+1}\right)^{k-1}, \quad k \leq r, \quad \hat{A}_{p,r,k} = 0, \quad k > r. \quad (20.62)$$

Let  $K_r$  be the kernel of the rectangular matrix containing all available rows with denominator at most  $2r + 1$ . Adding the next denominator adds one column and at least the  $p = 1$  row, whose new-column entry is nonzero; hence  $\dim K_r$  never increases.

We now state explicitly the single arithmetic theorem used in Koval's rank selection. Heath-Brown proved that among  $3, 5, 7$  at least one is a primitive root modulo infinitely many primes. Therefore, for each fixed  $q_0$ , there are infinitely many primes  $Q > 7q_0$  for which one of  $3, 5, 7$  is primitive modulo  $Q$ ; these are Koval's  $q_0$ -good primes. This deep arithmetic theorem is used only to choose denominators; no dynamical conclusion is imported from it.

Suppose  $Q = 2r + 1$  is  $q_0$ -good and that adding the  $Q$ -rows does not decrease the kernel dimension. Take two admissible numerators  $p_1, p_2$ . Let  $W = \text{lcm}(1, \dots, Q-1)$  and  $F = \mathbb{Q}(\zeta_W)$ . All matrix entries with denominator  $< Q$  lie in  $F$ . The new row is polynomial over  $F$  in  $z = \cos(2\pi p_1/Q)$ , and the second row is obtained by replacing  $z$  by the Chebyshev polynomial corresponding to  $p_2 p_1^{-1} \in \mathbb{F}_Q^*$ . The minimal polynomial of  $z$  over  $F$  is still the full real cyclotomic polynomial of degree  $(Q-1)/2$ : adjoining  $z$  and then the matching sine would otherwise produce a field containing  $\zeta_{QW}$  of degree smaller than  $\varphi(QW) = (Q-1)\varphi(W)$ , a contradiction. Thus every vanishing minor persists after replacing  $z$  by any  $\cos(2\pi s/Q)$ .

Let  $G \subset \mathbb{F}_Q^*$  be the subgroup generated by the admissible numerators and by  $-1$ . Because  $Q > 7q_0$ , the available set contains  $3, 5, 7$ , one of which is primitive; hence  $G = \mathbb{F}_Q^*$ . Consequently all real cyclotomic conjugates may be inserted into the two new rows. After nonzero column scalings the resulting square submatrix is

$$\left[ \left( \frac{1 + \cos(2\pi p/Q)}{2} \right)^s \right]_{p,s},$$

a Vandermonde matrix at distinct nodes, hence nonsingular. A plateau of  $\dim K_r$  is therefore impossible unless  $K_r = 0$ . Since  $q_0$ -good primes occur infinitely often and the kernel dimension is a nonnegative integer, it must eventually become zero. Continuity of the finite determinant in  $e$  then proves the asserted full rank for all small positive

eccentricities. The cosine and sine blocks have the same coefficient matrix.

**Step 5. Even-mode finite rank near the circle.** There is an even cutoff  $q_{1,e}$  such that the rational-caustic preservation rows, including the doubled-harmonic rows belonging to the same caustics, have full rank on the even non-elliptic modes  $4, 6, \dots, q_{1,e}$  for all sufficiently small positive  $e$ .

For the even block set  $j = 2k$  and use rows corresponding to the  $2q$ th harmonic of a preserved  $p/q$  caustic. With denominator  $2r$  the same rescaling yields

$$\hat{A}_{p,r,k} = \binom{r+k-1}{r-k} \left(1 + \cos \frac{\pi p}{r}\right)^{k-1}, \quad k \leq r, \quad \hat{A}_{p,r,k} = 0, \quad k > r. \quad (20.63)$$

Again the kernel dimension cannot increase. At a prime  $r = Q$ , split the new numerators by parity. For even  $p = 2s$  the nodes are  $\cos(2\pi s/Q)$  and the odd-block cyclotomic argument applies verbatim. For odd  $p$  use

$$\cos \frac{\pi p}{Q} = -\cos \frac{\pi(Q+p)}{Q}, \quad s := \frac{Q+p}{2} \in \mathbb{F}_Q^*.$$

After multiplying the indexing set by 2, the subgroup generated by the available odd numerators contains  $1, 3, 5, \dots, 2\lfloor Q/q_0 \rfloor - 1$ . At a  $q_0$ -good prime this again contains a primitive root among  $3, 5, 7$ . Hence both parity classes generate all conjugate nodes. The same Vandermonde contradiction forces a rank drop at every plateau, and the even kernel also reaches zero. Taking

$$q_1 := \max(q_{1,o}, q_{1,e}) \quad (20.64)$$

produces a finite detecting family for every Fourier mode from frequency 3 through  $q_1$ .

**Step 6. Finite detecting family plus near-Fourier tail.** One may select finitely many pairs  $(p_i, q_i)$ ,  $q_i \leq q_1$  and  $0 < p_i/q_i < 1/q_0$ , whose low-frequency matrix is invertible for small  $e$ ; for every  $j > q_1$  append the  $(1, j)$  caustic mode. The resulting functions  $f_j$  consist of five elliptic-motion functions followed by selected caustic cosine/sine modes, and satisfy

$$\left\| \frac{1}{\sqrt{\pi}} f_j - e_j \right\|_{L^2} \leq \frac{C(e)}{j} \quad (j \gg 1), \quad (20.65)$$

with exponentially decaying off-diagonal correlations.

Full rank of the finite rectangular matrix means that some square minor is nonzero; retain exactly those rows. For large  $j$ , choose the mode associated to the preserved  $1/j$  caustic. The action-angle coordinate of a  $1/j$  confocal caustic converges to the Lazutkin coordinate, and its analytic strip has width bounded below on compact eccentricity sets. Fourier contour shifting therefore gives, for  $q, j \geq 3$ ,

$$|\langle \cos(q\vartheta), c_{1,j}(\vartheta) \rangle - \pi \delta_{qj}| \leq C j^{-1} e^{-\delta|q-j|}, \quad (20.66)$$

with the analogous sine and mixed estimates. Parseval gives (20.65) and also square summability of the tail deviations from the trigonometric basis.



**Step 7. Holomorphic continuation of the elliptic data.** Fix  $e_{\max} < 1$  and  $\omega_{\max} < 1/2$ . After complexifying eccentricity in a thin strip away from  $e = 1$  and the focal degeneracy, the functions  $\lambda(e, \omega)$ ,  $k(e, \omega)$ , the action-angle map, and every selected preservation mode  $f_j(e)$  are holomorphic in  $e$ , with uniform analytic-strip and  $L^2$  bounds on compact substrips.

The formula

$$\omega(\lambda, e) = \frac{F(\arcsin(\lambda/\sqrt{1-e^2}), e/\sqrt{1-\lambda^2})}{2K(e/\sqrt{1-\lambda^2})}$$

is holomorphic as long as the two square-root branch loci and  $k = 1$  are avoided. On the real interval of interest,  $\partial_\lambda \omega > 0$ . Compactness therefore supplies a positive lower bound for this derivative; the holomorphic implicit-function theorem gives  $\lambda(e, \omega)$  in a uniform complex strip. The Jacobi amplitude and its derivative are then holomorphic there. The selected low modes are finite algebraic/elliptic combinations of these quantities, while the high  $1/j$  modes satisfy the uniform complex-strip estimate used in (20.66). This proves the asserted holomorphic dependence and uniform bounds.

**Step 8. Compact holomorphic preservation operator.** Let  $(e_j)_{j \geq 1}$  be the normalized trigonometric basis of  $L^2(\mathbb{T})$ , ordered compatibly with the five elliptic directions and the selected cosine/sine rows, and define

$$L_e e_j := e_j - \frac{1}{\sqrt{\pi}} f_j(e). \quad (20.67)$$

Then  $L_e$  is Hilbert-Schmidt, hence compact, and  $e \mapsto L_e$  is a holomorphic bounded-operator family. The family  $(f_j(e))$  is a basis if and only if 1 is not an eigenvalue of  $L_e$ .

By (20.65),

$$\sum_{j \geq j_0} \|L_e e_j\|_2^2 < \infty;$$

the finitely many low columns do not affect Hilbert-Schmidt compactness. For basis vectors  $e_i, e_j$ ,

$$\langle L_e e_i, e_j \rangle = \delta_{ij} - \frac{1}{\sqrt{\pi}} \int_0^{2\pi} f_i(e, \vartheta) e_j(\vartheta) d\vartheta,$$

which is holomorphic by Step 7. Local boundedness plus weak holomorphy on the dense trigonometric span gives operator holomorphy. Finally  $(I - L_e)e_j = f_j/\sqrt{\pi}$ , so  $I - L_e$  is invertible exactly when the selected family is a Schauder/Riesz basis equivalent to the trigonometric basis.

**Step 9. Small-eccentricity nonsingularity of the infinite operator.** For every fixed  $q_0$  the operator  $I - L_e$  is invertible for all sufficiently small positive  $e$ .

The finite rank calculation alone is not enough, because the low columns carry different powers of  $e$ . Let  $q_1$  be the cutoff from Step 6. Use the source's diagonal left/right scaling operators  $S_e^l, S_e^r$  on the first  $2q_1 + 1$  trigonometric modes, and choose a finite-dimensional change of basis  $D_e$  whose columns are the deformed low-mode vectors used in the rank proof. Consider

$$M_e := I + S_e^l (L_e^* - I) D_e S_e^r. \quad (20.68)$$



The rescalings exactly cancel the powers of  $e$  from (20.59)–(20.60). In the decomposition into low and high modes,  $M_e - I$  has four blocks. The low-low block converges to the nonsingular finite matrix from Steps 4–5. The low-high and high-low blocks tend to zero by (20.61), including the negative powers inserted by  $S_e^r$ ; the high-high block tends to zero by (20.65). Therefore

$$\|M_e - M_0\|_{L^2 \rightarrow L^2} \rightarrow 0,$$

and  $I - M_0$  has the required finite-block inverse. Hence 1 is not an eigenvalue of  $M_e$  for all sufficiently small  $e > 0$ . Since the scaling and  $D_e$  are invertible at every positive  $e$ ,  $L_e^* - I$ , and therefore  $L_e - I$ , is invertible. This proves Step 9.

**Step 10. Analytic-Fredholm exceptional set.** For fixed  $q_0$ , the set

$$\mathcal{Z}(q_0) := \{e \in (0, 1) : 1 \in \sigma(L_e)\} \quad (20.69)$$

is locally finite. On every compact set disjoint from  $\mathcal{Z}(q_0)$ ,  $(I - L_e)^{-1}$  is uniformly bounded.

This is exactly the analytic-Fredholm mechanism provided by the analytic-Fredholm theorem in the Foundation. Step 8 gives a holomorphic compact family and Step 9 supplies one nonsingular point in each connected eccentricity strip. Hence the alternative “every parameter is singular” is impossible, so the singular parameters are discrete in the complex strip and therefore finite on compact subsets. The real set (20.69) is locally finite in  $(0, 1)$ . The resolvent is holomorphic away from it, hence locally uniformly bounded. Increasing  $e_{\max} < 1$  exhausts the full interval  $(0, 1)$ .

**Step 11. Transfer from Lazutkin to elliptic coordinates.** For  $e \notin \mathcal{Z}(q_0)$  the elliptic-coordinate family  $f_j^\varphi$  obtained by pulling the preservation modes through the Lazutkin-elliptic coordinate change is a basis of  $L^2(\mathbb{T})$ , and its basis constants are uniformly bounded in a small eccentricity neighborhood.

The Lazutkin and elliptic boundary parameters are related by a smooth positive Jacobian  $J_e$  bounded above and below on every compact eccentricity set. Let  $Y_e u(\vartheta) = u(\varphi_e(\vartheta))$ . Then  $Y_e$  and  $Y_e^{-1}$  are bounded on  $L^2$ , and the dual action transforms the source modes to

$$c_{p,q}^\varphi(\varphi) = \frac{\pi}{2K(k_{p/q})} \frac{\cos(q\theta_{p/q}(\varphi))}{\sqrt{1 - k_{p/q}^2 \sin^2 \varphi}}, \quad (20.70)$$

$$h_i^\varphi(\varphi) = \frac{e_i(\varphi)}{1 - e^2 \cos^2 \varphi}, \quad i = 1, \dots, 5, \quad (20.71)$$

with analogous sine modes. Thus the elliptic family is obtained from the Lazutkin basis by a bounded invertible conjugacy. The local uniform inverse bound from Step 10 and smooth dependence of  $Y_e$  give the uniform basis constant.

**Step 12. High-mode coefficient bound.** There is  $j_0 = j_0(q_0)$  such that for every  $j \geq j_0$  and every  $\mu \in C^1(\mathbb{T})$ ,

$$\left| \int_0^{2\pi} \mu(\varphi) f_j^\varphi(\varphi) d\varphi \right| \leq \frac{C(e)}{j} \|\mu\|_{C^1}. \quad (20.72)$$

Write  $f_j^\varphi = \sqrt{\pi}e_j + r_j$ . The transferred version of (20.66) gives  $\|r_j\|_2 \leq C/j$  and exponential localization of its Fourier coefficients around frequency  $j$ . Integration by parts in  $\int \mu e_j$  gives  $C\|\mu\|_{C^1}/j$ , while Cauchy-Schwarz and the refined localized estimate give the same order for  $\int \mu r_j$ . This is the source Lemma 7.1.

**Step 13. Non-elliptic  $L^2$  gain.** Let  $H_e = \text{span}\{f_1^\varphi, \dots, f_5^\varphi\}$  be the five-dimensional tangent space to the ellipse/isometry family and decompose

$$\mu = \mu_H + \mu_\perp \quad (20.73)$$

using the bounded basis projection from Step 11. If  $\Omega$  is  $q_0$ -rationally integrable and  $\|\mu\|_{C^{39}} \leq K$ , then

$$\|\mu_\perp\|_{L^2} \leq C(e, c, K, q_0)\eta^{19/18}. \quad (20.74)$$

The first five basis coefficients of  $\mu_\perp$  vanish. Let

$$N := \left\lfloor \eta^{-1/9} \right\rfloor,$$

and take  $\eta$  small enough that  $N \geq j_0$  and that every selected low-mode caustic with denominator  $q \leq N$  satisfies the range  $q < C(e)\eta^{-1/8}$  of Step 2. For the basis coefficients with  $j_0 \leq j \leq N$ , the selected numerators are uniformly bounded (indeed Koval needs only 1 and one of 3, 5, 7), so Step 2 gives

$$|\widehat{\mu}_{\perp,j}| \leq Cj^8\eta^2.$$

The finitely many remaining indices are bounded by the same estimate after changing  $C$ . Hence

$$\sum_{6 \leq j \leq N} |\widehat{\mu}_{\perp,j}|^2 \leq C\eta^4 \sum_{j \leq N} j^{16} \leq C\eta^4 N^{17} \leq C\eta^{19/9}. \quad (20.75)$$

For  $j > N$ , Step 12 yields

$$\sum_{j > N} |\widehat{\mu}_{\perp,j}|^2 \leq C\eta^2 \sum_{j > N} j^{-2} \leq C\eta^2 N^{-1} \leq C\eta^{19/9}. \quad (20.76)$$

Because the basis and its dual have bounded  $L^2$  constants, Parseval in this Riesz basis and (20.75)–(20.76) give (20.74) after taking square roots.

**Step 14. Sobolev interpolation gives the 703/702 gain.** Under the same hypotheses,

$$\|\mu_\perp\|_{C^1} \leq C(e, c, K, q_0)\eta^{703/702}. \quad (20.77)$$

The bounded projection onto  $H_e$  satisfies  $\|\mu_H\|_{C^{39}} \leq C\eta$ , hence  $\|\mu_\perp\|_{C^{39}} \leq C(K+1)$ . Apply the one-dimensional Gagliardo-Nirenberg interpolation inequalities proved in the Foundation to  $\mu_\perp$  and its first two derivatives. In the source's convenient cutoff form, for  $j = 1, 2$  and  $\Delta > 1$ ,

$$\|D^j \mu_\perp\|_{L^2} \leq C \left( \Delta \|\mu_\perp\|_{C^{39}} + \Delta^{-j/(39-j)} \|\mu_\perp\|_{L^2} \right).$$

Insert (20.74) and optimize  $\Delta$ . Sobolev embedding  $H^2(\mathbb{T}) \hookrightarrow C^1(\mathbb{T})$  then gives the weak-

est gain

$$1 + \frac{1}{702} = \frac{703}{702},$$

which is exactly (20.77). The unusually small exponent is not cosmetic: it is the regularity cost of converting the  $19/18$   $L^2$  gain to  $C^1$  while retaining only a uniform  $C^{39}$  bound.

**Step 15. Five-dimensional elliptic recentering.** There is a smooth five-parameter local manifold of ellipses/isometric copies through  $\mathcal{E}_{e,c}$  whose tangent space in elliptic-polar coordinates is exactly  $H_e$ . Consequently, for  $\eta$  small there exists a nearby ellipse  $\bar{\mathcal{E}}$  such that the same boundary, written over  $\bar{\mathcal{E}}$  as  $\bar{\mu}$ , satisfies

$$\|\bar{\mu}\|_{C^1} \leq C(e, c, K, q_0) \eta^{703/702}. \quad (20.78)$$

The five directions are homothety, the two translations, rotation, and change of eccentricity/semi-axis ratio. Their coordinate derivatives are the five functions  $h_i^\varphi$  in (20.71), and their parameter Jacobian has full rank. The finite-dimensional inverse-function theorem therefore chooses ellipse parameters whose first-order graph is  $\mu_H$ . The discrepancy between the actual ellipse graph and its tangent approximation is  $O(\eta^2)$  in  $C^1$ . Changing the elliptic coordinate frame is smooth, so

$$\bar{\mu} = \mu_\perp + O_{C^1}(\eta^2).$$

Since  $703/702 < 2$ , (20.77) dominates the quadratic term, proving (20.78). This is Koval's Approximation Lemma 7.2.

**Step 16. Closest-ellipse contradiction.** Fix  $e \notin \mathcal{Z}(q_0)$ . After shrinking the theorem's  $C^1$  radius, there is a compact neighborhood  $\mathcal{E}$  of ellipses around  $\mathcal{E}_{e,c}$  disjoint in eccentricity from  $\mathcal{Z}(q_0)$  and on which every constant in Steps 11–15 is uniform. A  $q_0$ -rationally integrable table satisfying the theorem hypotheses is itself an ellipse.

Because  $\mathcal{Z}(q_0)$  is locally finite and  $e \notin \mathcal{Z}(q_0)$ , choose  $\delta_e > 0$  such that the closed eccentricity interval  $[e - 2\delta_e, e + 2\delta_e] \cap (0, 1)$  contains no exceptional eccentricity. Restrict the five ellipse parameters to a compact box  $\mathcal{E}$  inside this interval and with semi-focal distance in a compact neighborhood of  $c$ . For every  $E' \in \mathcal{E}$ , write the same boundary as an elliptic-polar graph  $\mu_{E'}$  whenever this is possible. The map  $E' \mapsto \|\mu_{E'}\|_{C^1}$  is continuous. Hence it has a minimum at some  $E_* \in \mathcal{E}$ .

Choose the original theorem radius so small that the minimizer lies in the interior of  $\mathcal{E}$ ,  $\|\mu_{E_*}\|_{C^{39}}$  stays below a uniform constant determined by  $K$ , and

$$C_* a^{1/702} < \frac{1}{2} \quad \text{for } a := \|\mu_{E_*}\|_{C^1} \leq \varepsilon.$$

If  $a > 0$ , Step 15 produces another ellipse  $E_1$  in the same compact box with

$$\|\mu_{E_1}\|_{C^1} \leq C_* a^{703/702} < a,$$

contradicting minimality. Thus  $a = 0$  and the boundary equals  $E_*$ . Notice why the old one-step inverse argument was invalid: the nonlinear preservation constraints control only finitely many low modes quadratically and the high modes by a  $1/j$  tail; the source

first converts these two estimates into a superlinear change of reference ellipse, then uses minimization.

**Step 17. Uniformity under scale and the theorem's  $C^k$  normalization.** The preceding constants may be chosen as functions only of  $(e, c, K, q_0)$ , and  $k \geq 39$  supplies every derivative used in the nonlinear estimate and interpolation. Rescaling the semi-major axis from the normalized source calculation restores the stated semi-focal distance  $c$  without changing rational rotation numbers or the conclusion “is an ellipse.”

The source often normalizes the semi-major axis to one while proving the operator estimates. If  $x \mapsto sx$ , chord lengths and the semi-focal distance scale by  $s$ , whereas the billiard map in angular variables and all rotation numbers are unchanged. The elliptic graph function and its  $C^r$  norms change by smooth chart factors bounded on the compact parameter box fixed above. Consequently the smallness threshold can be written as  $\varepsilon(e, c, K, q_0)$ . Proposition 7.1 uses at most five derivatives, while the interpolation step uses the uniform  $C^{39}$  bound; hence every  $k \geq 39$  is sufficient exactly as stated.

**Step 18. Range and exceptional-set closure.** The theorem assumes  $q_0 > 2$ , matching Koval's Definition 1.3. Every caustic used above has  $0 < p/q \leq 1/q_0$ ; only finitely many have  $p > 1$ , and in the rank argument one may take  $p$  from  $\{1, 3, 5, 7\}$ . Thus the locally finite set  $\mathcal{Z}(q_0)$  and the threshold from Steps 16–17 give exactly the claimed theorem.

Indeed, Steps 4–6 choose only admissible rows; the  $q_0$ -good prime condition  $Q > 7q_0$  guarantees that the numerators 3, 5, 7 still satisfy  $p/Q < 1/q_0$ . For large frequencies the proof uses only the  $(1, q)$  rows. All these caustics are present under  $q_0$ -rational integrability. Step 10 constructs the locally finite exceptional set, and Step 16 proves rigidity at every eccentricity outside it. This completes the proof.  $\square$

#### Theorem K20.4. Local spectral uniqueness of almost every ellipse

**Statement.** Fix one of the standard Dirichlet or Neumann boundary conditions, and use the same choice for all domains in the comparison. Fix an ellipse  $\mathcal{E}_0$  of eccentricity  $e \in (0, 1)$  and semi-focal distance  $c$ , an integer  $k \geq 39$ , and  $K > 0$ . Koval proves that there is a small exceptional set  $\mathcal{Z} \subset (0, 1)$ —countable, Lebesgue-null, and nowhere dense—such that, for every  $e \notin \mathcal{Z}$ , there exists  $\varepsilon = \varepsilon(e, c, K) > 0$  with the following property:  $\mathcal{E}_0$  is uniquely determined up to Euclidean isometry by its Laplace spectrum for that fixed boundary condition among  $C^\infty$  planar domains whose boundaries are  $C^k$ - $K$  close and  $C^{10-\varepsilon}$  close to  $\mathcal{E}_0$  in Koval's normalization.

*Proof.*

#### Proof strategy

First recover the perimeter and the first Marvizi-Melrose invariant from the spectrum. For each detecting periodic type, construct an isolated length window and use the localized wave trace to show that both endpoints of the loop-length range are audible. The corresponding ellipse has only one singular time in that window, so isospectrality forces the range to collapse and hence produces the rational caustic. Once the full detecting family of caustics is recovered, the local strong

Birkhoff theorem applies.

Write  $\mathcal{E} = \mathcal{E}_{e,c}$  for the reference ellipse and let  $\Omega$  be a comparison domain satisfying the hypotheses of the theorem and having the same fixed Dirichlet or Neumann spectrum. Shrinking the  $C^{10}$  neighborhood once and for all, we may assume that  $\partial\Omega$  remains strictly convex with positive curvature. We use

$$\ell_D := |\partial D|, \quad w_D(t) := \text{Tr} \cos(t\sqrt{\Delta_D})$$

for a domain  $D$ .

For isospectral  $\Omega$  and  $\mathcal{E}$  with the same Dirichlet or Neumann boundary condition,

$$w_\Omega = w_\mathcal{E} \quad \text{in } \mathcal{D}'(\mathbb{R}), \quad |\Omega| = |\mathcal{E}|, \quad \ell_\Omega = \ell_\mathcal{E}.$$

The wave-trace identity is immediate from the spectral expansion  $w_D(t) = \sum_j \cos(t\sqrt{\lambda_j(D)})$ , with multiplicities. For the heat trace, the planar Dirichlet/Neumann expansion begins

$$\text{Tr}(e^{-t\Delta_D}) = \frac{|D|}{4\pi t} \mp \frac{\ell_D}{8\sqrt{\pi t}} + O(1), \quad t \downarrow 0, \quad (20.79)$$

where the sign is fixed by the boundary condition. Equality of the spectra therefore gives equality of the first two coefficients, proving the three claims. We denote the common perimeter by  $\ell$ .

We next record the uniform billiard information that is used to isolate the spectral windows. All constants below are allowed to depend on the fixed reference eccentricity range, on  $c$ , and on the uniform  $C^k$  bound  $K$ , but not on a sufficiently small  $C^{10}$  deformation.

There is  $q_0$  such that, for every  $1 \leq p \leq 7$  and  $q \geq q_0$ , every sufficiently small comparison table possesses a smooth  $(p, q)$  q-loop generating function  $L_{p,q}(s)$ . Its critical points are the periodic  $(p, q)$  trajectories. Moreover, for every fixed bound  $p < p_0$  and uniformly in the theorem class,

$$L_{p,q} = p\ell - c_{2,p}(D)q^{-2} + O(q^{-4}), \quad T_{p,q} - t_{p,q} = O(q^{-4}), \quad (20.80)$$

where

$$c_{2,p}(D) = \frac{p^3}{24} \left( \int_{\partial D} \kappa_D(s)^{2/3} ds \right)^3. \quad (20.81)$$

Indeed, the Lazutkin caustics theorem supplies a uniform Lazutkin collar near the boundary. In that collar the billiard map is a perturbation of the integrable translation normal form. For bounded  $p$  and large  $q$ , the  $q - 1$  transverse critical equations for a  $p$ -winding polygon have invertible transverse Hessian; the implicit-function theorem therefore solves all vertices except the prescribed starting point and gives the smooth q-loop function. Its derivative is the endpoint angular mismatch, so its critical points are exactly the periodic polygons. Applying the order-six Lazutkin normal form and periodic Euler-Maclaurin summation gives an expansion through  $q^{-4}$ . The odd coefficients vanish by the Marvizi-Melrose invariant expansion established in the

near-boundary billiards volume, while its first nontrivial coefficient is (20.81). The same expansion applies to every periodic orbit of the type, giving the uniform window width in (20.80). In particular  $c_{2,1}(D) > 0$  and depends continuously on  $D$  in the  $C^{10}$  topology.

Choose  $q_0$  larger if necessary so that Koval's additional lower bound

$$t_{p,q}(D) \geq \frac{2p-1}{p} \ell_D, \quad 1 \leq p \leq 7, \quad q \geq q_0, \quad (20.82)$$

holds uniformly. It is an immediate consequence of (20.80), the positivity and uniform boundedness of  $c_{2,p}$ , and enlargement of  $q_0$ .

For this  $q_0$ , the proof of the preceding Koval local strong Birkhoff theorem selects a family  $\mathcal{F} = \mathcal{F}_1 \cup \mathcal{F}_2$  such that  $\mathcal{F}_1$  contains every  $(1, q)$  for all  $q$  above a fixed threshold and  $\mathcal{F}_2$  is finite with  $2 \leq p \leq 7$ . Preserving the rational caustic for every type in  $\mathcal{F}$  is sufficient for the local strong-Birkhoff closure proved in the preceding Koval local strong Birkhoff theorem.

This is precisely the detecting-family reduction in the preceding theorem's proof: the finite family supplies the low odd/even rank, while the  $(1, q)$  tail controls all high dynamical modes. Thus the spectral theorem need not reconstruct every rational caustic; it must reconstruct exactly these finite-plus-tail constraints.

There is  $p_0$  such that all periodic types with winding number  $p \geq p_0$  have length uniformly larger than every detecting-family length. For each fixed  $p < p_0$ , periodic lengths of a sufficiently small deformation converge to the corresponding ellipse lengths as the deformation tends to zero: for  $p/q < 1/2$  they converge to the unique elliptic-caustic length  $T_{p,q}(\mathcal{E})$ , while for  $q = 2p$  they remain near the finite set of ellipse lengths of that type.

Here is the geometric reason for the first assertion. Fix, by the Lazutkin caustics theorem, a rotational KAM curve in a common Lazutkin collar which survives all sufficiently small deformations. A periodic orbit lying on the deeper side of this curve has chord lengths bounded below often enough that a uniform upper bound for its total length bounds its winding number. An orbit lying in the grazing side has total chord length comparable, via Lazutkin coordinates, to its winding number. Consequently there is  $p_0$  for which all  $p \geq p_0$  types lie above a fixed length threshold. In the source normalization with semi-major axis one, Koval chooses the separating bounds  $t_{p,q} > 16\pi$  for  $p \geq p_0$  and  $T_{p,q} < 15\pi$  for  $p \leq 7$ ; scaling preserves the gap.

For the continuity assertion, the billiard maps of the deformed tables converge in  $C^9$  to the ellipse map on every compact phase strip away from grazing. A periodic orbit of a fixed type therefore becomes an almost-periodic ellipse orbit with the same translation number. Compactness gives a limiting periodic orbit of the ellipse. If  $p/q < 1/2$ , the confocal caustic makes all ellipse orbits of that type have the single length  $T_{p,q}(\mathcal{E})$ . If  $q = 2p$ , the possible limiting orbits are the axis bouncing balls and the finitely many hyperbolic-caustic branches of that fixed type. The common KAM curve prevents the fixed-type orbit from escaping through the grazing boundary. This proves the claimed fixed-type control.

We now isolate the only genuinely microlocal part of the proof.

Assume  $1 \leq p \leq 7$ ,  $q \geq q_0$ , and an open neighborhood of  $[t_{p,q}(D), T_{p,q}(D)]$  contains no

periodic length of another orbit type and no multiple of  $\ell_D$ . Then every periodic  $(p, q)$  length in this window belongs to  $\text{singsupp } w_D$ ; in particular

$$t_{p,q}(D), T_{p,q}(D) \in \text{singsupp } w_D.$$

Choose a cutoff  $\chi_{p,q}$  equal to one on the window and supported inside the isolating neighborhood, and set  $\sigma_{p,q} = \chi_{p,q} w_D$ . We use the following *declared microlocal input*, which is Koval's Appendix-B fixed-type parametrix:

$$\sigma_{p,q}(t) = \int_0^\infty \int_{\partial D} \text{Re} \left( e^{i\pi r_{p,q}/4 + i\xi(t - L_{p,q}(s))} \xi^{1/2} a(p, q, t, s, \xi) \right) ds d\xi + R_{p,q}(t), \quad (20.83)$$

where  $R_{p,q}$  is smooth and  $a \sim \sum_{j \geq 0} a_j \xi^{-j}$  is a classical symbol whose principal coefficient  $a_0(p, q, s)$  is independent of  $t$  and strictly positive. The second declared input is Soga's theorem: if  $\phi, a \in C^\infty(\mathbb{R})$ ,  $a$  is compactly supported and nonnegative, and  $a$  is positive at a degenerate critical point of  $\phi$ , then  $\int e^{i\lambda\phi(x)} a(x) dx$  is not  $O(\lambda^{-\infty})$ .

We derive the required noncancellation from these inputs. Let  $t_*$  be a periodic  $(p, q)$  length in the window and suppose for contradiction that  $\sigma_{p,q}$  is smooth near  $t_*$ . Choose  $\rho$  supported in a small such neighborhood. Its Fourier transform must then be rapidly decreasing. Substituting (20.83) and applying stationary phase in  $(t, \xi)$  gives

$$I_{p,q}(\lambda) = \pi \lambda^{1/2} e^{-i\pi r_{p,q}/4} \int_{\partial D} e^{i\lambda L_{p,q}(s)} a_0(p, q, s) \rho(L_{p,q}(s)) ds + O(\lambda^{-1/2}). \quad (20.84)$$

If a critical point over  $t_*$  is degenerate, Soga's theorem contradicts rapid decay because the amplitude is nonnegative and positive there. Otherwise the critical points over  $t_*$  are finitely many and nondegenerate. A final one-dimensional stationary-phase expansion has leading coefficient proportional to

$$\sum_j e^{i\pi \text{sgn } L''_{p,q}(s_j)/4} \frac{a_0(p, q, s_j)}{\sqrt{|L''_{p,q}(s_j)|}}. \quad (20.85)$$

Every weight is positive and every phase in the sum is either  $e^{i\pi/4}$  or  $e^{-i\pi/4}$ . Hence the real part of the sum is  $2^{-1/2}$  times the sum of all positive weights and cannot vanish. Again the Fourier transform is not rapidly decreasing. This contradiction proves that every such  $t_*$  is a wave-trace singularity. The minimax and maximum defining  $t_{p,q}$  and  $T_{p,q}$  are realized by periodic trajectories, so the endpoint assertion follows.

We temporarily impose

$$c_{2,1}(\Omega) = c_{2,1}(\mathcal{E}) =: c_2. \quad (20.86)$$

We will remove this assumption spectrally below.

Assume the reference eccentricity satisfies the non-incidence condition. Under (20.86), every  $(p, q) \in \mathcal{F}_2$  admits a  $(p, q, \varepsilon)$  isolation interval; because  $\mathcal{F}_2$  is finite, one may choose one positive  $\varepsilon$  for all its elements.

Fix  $(p, q) \in \mathcal{F}_2$ . Its ellipse length  $T_{p,q}(\mathcal{E})$  is isolated from the rest of the ellipse length spectrum and from the perimeter multiples. To see this without hiding a compactness issue, observe that high winding numbers  $\bar{p} \geq p_0$  are excluded by the high-winding



and fixed-type estimates above; for each bounded  $\bar{p} < p_0$ , the large- $\bar{q}$  expansion (20.80) tends to the perimeter multiple  $\bar{p}\ell$ ; and only finitely many fixed types remain. Non-incidence separates the selected ellipse length from each of these finitely many values and from every relevant  $\bar{p}\ell$ . Choose a small open interval  $I_{p,q}$  around  $T_{p,q}(\mathcal{E})$  realizing this separation.

For a sufficiently small deformation, all  $(p, q)$  lengths stay inside  $I_{p,q}$  by the high-winding and fixed-type estimates above. A competing type cannot enter: high winding number is still excluded by the uniform gap; bounded winding and large denominator stay in small neighborhoods of their corresponding perimeter multiples by (20.80); and the finitely many remaining fixed types stay close to their reference lengths by the high-winding and fixed-type estimates above. Perimeter multiples themselves are fixed by the spectral heat/wave invariants above. Thus  $I_{p,q}$  is an isolation interval. Finiteness of  $\mathcal{F}_2$  lets us take the minimum of the finitely many deformation thresholds.

Under non-incidence there is a one-sided neighborhood  $J = (\ell - \delta, \ell)$  of the perimeter such that, for the ellipse and all sufficiently small comparison domains, no periodic orbit with winding number  $p > 1$  has length in  $J$ .

For the ellipse, high  $p$  is excluded by the high-winding and fixed-type estimates above. If  $2 \leq p < p_0$  and  $q \rightarrow \infty$ , (20.80) tends to  $p\ell \geq 2\ell$ , so these large-denominator lengths stay uniformly away from  $\ell$ . Only finitely many fixed  $(p, q)$  types remain, including the  $q = 2p$  axis/hyperbolic branches; non-incidence says none has length  $\ell$ . Hence some  $J$  is free of  $p > 1$  ellipse lengths. The uniform asymptotic expansion and fixed-type continuity from the high-winding and fixed-type estimates above allow us to shrink  $J$  and the deformation neighborhood so that the same exclusion holds for every comparison domain.

Assume non-incidence and (20.86). There are  $\hat{q}$  and a single  $\varepsilon > 0$  such that every  $(1, q) \in \mathcal{F}_1$  with  $q \geq \hat{q}$  has the isolation interval

$$I_q := \left( \ell - c_2 q^{-2} - \frac{c_2}{10} q^{-3}, \ell - c_2 q^{-2} + \frac{c_2}{10} q^{-3} \right), \quad (20.87)$$

and the finitely many remaining elements of  $\mathcal{F}_1$  can be isolated after a further uniform decrease of  $\varepsilon$ .

By the uniform  $O(q^{-4})$  remainder in (20.80), enlarge  $\hat{q}$  until every  $(1, q)$  length of every allowed domain lies in  $I_q$ . Enlarge it again until all  $I_q$  lie in the collar  $J$  of the near-perimeter collar constructed above. Thus no  $p > 1$  length can enter them. Adjacent intervals are disjoint because

$$c_2(q^{-2} - (q+1)^{-2}) > \frac{c_2}{10}(q^{-3} + (q+1)^{-3})$$

for all sufficiently large  $q$ ; monotonicity then separates nonadjacent intervals. Thus no other high  $(1, r)$  type enters  $I_q$ . Only finitely many low  $(1, r)$  types remain, and fixed-type continuity separates those after one further decrease of  $\varepsilon$ . This proves the uniform tail assertion. Combining it with the finite part of  $\mathcal{F}_1$  gives a common threshold for all of  $\mathcal{F}_1$ .

We now prove that the auxiliary equality (20.86) is itself spectral information.

If  $\Omega$  and  $\mathcal{E}$  are sufficiently close, satisfy non-incidence for the reference ellipse, and



have equal wave traces, then  $c_{2,1}(\Omega) = c_{2,1}(\mathcal{E})$ .

Put  $c_E = c_{2,1}(\mathcal{E})$  and  $c_\Omega = c_{2,1}(\Omega)$ . By continuity of (20.81),  $c_\Omega/c_E$  is as close to one as desired. The collar argument of the near-perimeter collar constructed above does not use equality of these coefficients, so sufficiently near  $\ell$  only high  $(1, q)$  lengths occur in either domain. For each domain separately, (20.80) and positivity of its own coefficient give disjoint high- $q$  windows of width  $o(q^{-3})$  centered at  $\ell - c_D q^{-2}$ ; the isolated-window audibility argument above supplies at least one wave singularity in each such window.

Use the rescaling

$$g_E(\ell - x) := \sqrt{\frac{c_E}{x}}, \quad x > 0. \quad (20.88)$$

This normalization is chosen so that the ellipse singularities satisfy

$$g_E(\ell - c_E q^{-2} + O(q^{-4})) = q + O(q^{-1}). \quad (20.89)$$

For  $\Omega$ , the corresponding high- $q$  windows map to intervals of arbitrarily small fixed radius about

$$\rho q, \quad \rho := \sqrt{\frac{c_E}{c_\Omega}}, \quad (20.90)$$

once  $q$  is large. (This is the spacing argument of Koval's Lemma 8.12; the normalization (20.88) is written so that (20.89) is exact at leading order.)

Because  $w_\Omega = w_\mathcal{E}$ , every such singularity must also be an ellipse singularity. Hence, after taking the windows small enough and  $q$  large enough, for every large integer  $q$  there is an integer  $n_q$  satisfying

$$|\rho q - n_q| < \frac{1}{10}. \quad (20.91)$$

We may also assume  $|\rho - 1| < 1/10$ . Subtracting the inequalities for consecutive  $q$  shows

$$|(n_{q+1} - n_q) - \rho| < \frac{1}{5}.$$

The left integer difference must therefore equal 1. Thus  $n_q - q$  is constant for all sufficiently large  $q$ . Returning to (20.91) gives  $|(\rho - 1)q - C| < 1/10$  for arbitrarily large  $q$ , which is impossible unless  $\rho = 1$ . Hence  $c_\Omega = c_E$ , proving the claim.

For every  $(p, q) \in \mathcal{F}$ , the isolation intervals from the finite-family isolation argument above and the uniform tail-isolation argument above, combined with equal wave traces, force

$$t_{p,q}(\Omega) = T_{p,q}(\Omega) = T_{p,q}(\mathcal{E}).$$

Consequently  $\Omega$  carries the integrable rational  $p/q$  caustic.

By the recovery of  $c_{2,1}$  above, the coefficient equality needed in the finite-family isolation argument above and the uniform tail-isolation argument above is automatic. Fix  $(p, q) \in \mathcal{F}$  and its isolation interval  $I$ . the isolated-window audibility argument above says that both  $t_{p,q}(\Omega)$  and  $T_{p,q}(\Omega)$  are singular times of  $w_\Omega$ . For the ellipse, the confocal  $p/q$  caustic makes the whole  $(p, q)$  window collapse to the single value

$$t_{p,q}(\mathcal{E}) = T_{p,q}(\mathcal{E}) =: T_{p,q}^E.$$

By construction,  $I$  contains no other ellipse periodic length and no perimeter multiple. The billiard Poisson relation therefore gives

$$\text{singsupp } w_{\mathcal{E}} \cap I \subset \{T_{p,q}^E\}.$$

Since the two wave traces are equal, their singular supports are equal. Hence both endpoints of the comparison window must equal  $T_{p,q}^E$ :

$$t_{p,q}(\Omega) = T_{p,q}(\Omega) = T_{p,q}^E.$$

This is the crucial step missing from the former proof: it is the *entire isolated window* that is compared with the ellipse's singleton singular support; no conclusion of constancy is drawn from a single stationary-phase coefficient. By the variational twist criterion in the periodic-orbit length-window criterion defined above,  $t_{p,q} = T_{p,q}$  is equivalent to the existence of the integrable rational caustic of rotation number  $p/q$ .

Outside the dynamical exceptional set of the preceding Koval local strong Birkhoff theorem, preservation of all caustics in  $\mathcal{F}$  forces  $\Omega$  to be an ellipse.

the isolated-window collapse argument above recovers every caustic in the finite-plus-tail detecting family. The proof of the preceding Koval local strong Birkhoff theorem—specifically its finite rank, high-mode basis, superlinear ellipse-improvement, and closest-ellipse closure—uses exactly these preservation constraints. After decreasing the  $C^{10}$  threshold so that it lies inside the  $C^1$  threshold of the preceding Koval local strong Birkhoff theorem, that argument applies and makes  $\Omega$  an ellipse whenever  $e$  is outside the locally finite dynamical Fredholm exceptional set  $\mathcal{Z}_{\text{dyn}}$ .

It remains to prove that the non-incidence hypothesis discarded above fails only on a small set of eccentricities. Since incidence is invariant under common scaling, we normalize the ellipse family in this paragraph to have semi-major axis one.

The elliptic-caustic, hyperbolic-caustic, minor-axis and major-axis periodic-length types, and the perimeter, depend holomorphically on eccentricity on their natural complex neighborhoods. After enlarging  $q_0$  if necessary, no detecting-family length or perimeter is identically equal, as a function of eccentricity, to a distinct periodic-length type.

The holomorphic dependence follows from the elliptic-integral formulae for the confocal caustics and the implicit-function description of their rotation numbers. We spell out the nonidentity argument because it controls the topology of the exceptional set. A minor-axis bouncing-ball length tends to zero as  $e \rightarrow 1$ , so it cannot be identically a detecting-family length or the perimeter. For a major-axis or elliptic-caustic competitor, the  $e \rightarrow 1$  asymptotics determine the winding number  $p$  (counting the perimeter as  $p = 1$ ); an identity would therefore require the same  $p$ . For fixed  $p$ , the elliptic-caustic length is strictly increasing with the period denominator along the rational rotation ordering, so two distinct such types cannot agree identically.

Only a hyperbolic-caustic competitor remains. The high-winding exclusion bounds its libration numbers whenever its length can meet the bounded detecting lengths. Thus there are only finitely many possible hyperbolic types in the identity test; taking  $q_0$  larger reduces the only possible identity involving the infinite tail to an identity with the perimeter. At the limiting parameter at which a hyperbolic caustic is born, such

an identity would imply

$$q \sin \frac{\pi \tilde{p}}{q} = 2E \left( \cos \frac{\pi \tilde{p}}{q} \right), \quad (20.92)$$

where  $E$  is the complete elliptic integral of the second kind. If  $\tilde{p} \geq 2$ , the left side is at least 4, while the right side is less than  $\pi$ . If  $\tilde{p} = 1$ , the left side is increasing in  $q \geq 4$ , the right side is decreasing, and already at  $q = 4$  the inequality  $E(2^{-1/2}) < \sqrt{2}$  makes the left side strictly larger. Thus (20.92) never holds. No distinct pair can therefore coincide identically.

It follows from holomorphic nonidentity that a fixed pair of length types has only finitely many incidences on every compact eccentricity interval  $[0, e_{\max}]$ .

The eccentricities at which a member of  $\mathcal{F}_2$  is incident with another periodic length form a locally finite set.

Indeed, high winding numbers are too long by the high-winding and fixed-type estimates above. Among bounded winding numbers, the large-denominator expansion approaches the separated perimeter multiples, and (20.82) rules out incidences of the selected large- $q$  types with the forbidden perimeter multiples. Only finitely many competitor types remain on any compact eccentricity interval. the holomorphic non-identity argument above makes the zero set for each pair finite there. Since  $\mathcal{F}_2$  itself is finite, the union is locally finite.

Let  $\mathcal{A}$  be the eccentricities at which the perimeter is realized by a minor-axis bouncing-ball orbit or by a hyperbolic orbit of short-axis libration number  $\tilde{p} = 1$ . Then  $\mathcal{A}$  is locally finite. If  $\mathcal{I}$  denotes all non-incidence eccentricities involving  $\mathcal{F}_1$  or the perimeter, every accumulation point of  $\mathcal{I}$  in  $[0, 1)$  lies in  $\mathcal{A}$  (up to the already locally finite incidences of the finite-family incidence argument above).

The normalized lengths in  $\mathcal{F}_1$  and the perimeter are at most  $2\pi$ . Major-axis and elliptic-caustic competitors with  $p > 1$  are separated by their winding length bounds, while hyperbolic competitors with  $\tilde{p} > 1$  have length at least 8 in Koval's normalization. Hence only minor-axis bounces and hyperbolic branches with  $\tilde{p} = 1$  can approach the relevant near-perimeter lengths. For each compact  $e \leq e_{\max} < 1$ , the orbit formulae and the holomorphic nonidentity argument above leave only finitely many actual perimeter intersections, so  $\mathcal{A}$  is locally finite.

Suppose now that distinct incidences  $e_j \in \mathcal{I}$  converge to  $e_* < 1$  and that we are not in the locally finite cases just listed. A fixed pair of length types cannot occur infinitely often by the holomorphic nonidentity argument above, so the corresponding hyperbolic denominators must tend to infinity. For  $\tilde{p} = 1$ , these hyperbolic lengths converge to the ellipse perimeter. Passing to the limit in the incidence equality therefore makes the competing limiting length equal to the perimeter at  $e_*$ . Thus  $e_* \in \mathcal{A}$ .

The union of the dynamical exceptional set and all incidence eccentricities is a small subset  $\mathcal{Z} \subset (0, 1)$ . Consequently it is countable, Lebesgue-null and nowhere dense.

The dynamical set  $\mathcal{Z}_{\text{dyn}}$  from the preceding Koval local strong Birkhoff theorem is locally finite. By the finite-family incidence argument above, the finite-family incidence set is locally finite. By the tail-incidence accumulation argument above, the derived set of the remaining incidence set is contained in the locally finite set  $\mathcal{A}$  together with those already locally finite pieces. Hence the set of accumulation points of their union is locally finite: this is exactly the definition of a small set in the definition of a

small exceptional set above. A discrete subset of the second-countable interval  $(0, 1)$  is countable; adjoining the locally finite derived set preserves countability. The set and its closure therefore have measure zero and empty interior, proving the three stated consequences.

If the preceding argument shows that  $\Omega$  is an ellipse, the common heat coefficients force it to have the same semiaxes as  $\mathcal{E}$  up to interchange, so it is Euclidean-isometric to  $\mathcal{E}$ .

For completeness, this last ellipse-family uniqueness does not require an additional inverse-spectral theorem. the spectral heat/wave invariants above gives equal area and perimeter. Write the semiaxes of an ellipse as

$$a = \sqrt{m} e^u, \quad b = \sqrt{m} e^{-u}, \quad m = ab = |D|/\pi, \quad u \geq 0.$$

At fixed area its perimeter is

$$P_m(u) = \sqrt{m} \int_0^{2\pi} \sqrt{e^{2u} \sin^2 \theta + e^{-2u} \cos^2 \theta} d\theta. \quad (20.93)$$

For  $A = \sin^2 \theta$ ,  $B = \cos^2 \theta$  and  $h(u) = (Ae^{2u} + Be^{-2u})^{1/2}$ , direct differentiation gives

$$h''(u) = \frac{2(Ae^{2u} + Be^{-2u})^2 - (Ae^{2u} - Be^{-2u})^2}{(Ae^{2u} + Be^{-2u})^{3/2}} > 0.$$

Thus  $P_m$  is strictly convex. The change  $u \mapsto -u$  merely interchanges the axes (equivalently shift  $\theta$  by  $\pi/2$ ), so  $P_m$  is even. Therefore it is strictly increasing for  $u > 0$ . Fixed area determines  $m$  and fixed perimeter then determines the unique  $u \geq 0$ ; hence  $a, b$  agree up to interchange.

For each  $e \notin \mathcal{Z}$ , the finitely many local thresholds and the uniform infinite-tail threshold have a positive common minimum  $\varepsilon(e, c, K)$ ;  $k \geq 39$ , the  $C^k$ - $K$  bound, and  $C^{10}$  smallness supply all derivative and convexity hypotheses used above and in the preceding Koval local strong Birkhoff theorem.

Fix  $e \notin \mathcal{Z}$ . Choose first the  $C^{10}$  neighborhood preserving strict convexity, the common Lazutkin collar, and the q-loop construction. Intersect it with the high-winding/fixed-type thresholds of the high-winding and fixed-type estimates above, the finitely many  $\mathcal{F}_2$  isolation thresholds, the uniform  $\mathcal{F}_1$  threshold, the  $c_{2,1}$  spacing threshold, and the  $C^1$  dynamical-rigidity threshold from the preceding Koval local strong Birkhoff theorem. There are only finitely many nonuniform minima, while the tail was uniform by the uniform tail-isolation argument above; hence their minimum is positive. The stated  $C^k$ - $K$  bound with  $k \geq 39$  supplies the uniform high derivatives used in the Lazutkin and nonlinear-rigidity estimates, and  $C^{10}$  smallness supplies the billiard-map and microlocal comparison.

For any isospectral  $\Omega$  in this final neighborhood, the recovery of  $c_{2,1}$  above recovers  $c_{2,1}$ ; the finite-family isolation argument above and the uniform tail-isolation argument above isolate every detecting window; the isolated-window collapse argument above converts equal wave traces into preservation of every detecting caustic; the detecting-family dynamical closure above makes  $\Omega$  an ellipse; and the ellipse-family heat-invariant argument above makes it Euclidean-isometric to  $\mathcal{E}$ . This proves the

theorem. □

## 20.3 Worked examples

### Worked example: support-function modes near a circle

Write a strictly convex boundary by its support function

$$h(\psi) = R + \varepsilon f(\psi), \quad f(\psi) = \sum_{n \in \mathbb{Z}} \hat{f}_n e^{in\psi}.$$

The radius of curvature is

$$\rho(\psi) = h + h'' = R + \varepsilon \sum_n (1 - n^2) \hat{f}_n e^{in\psi}.$$

For  $|\varepsilon|$  small relative to  $R/\|f + f''\|_\infty$ , the curvature remains positive.

The  $n = \pm 1$  modes correspond to translations of the curve and therefore cannot be recovered by a rigidity statement “up to Euclidean isometry.” The  $n = 0$  mode changes the overall size. Quadratic/ellipse-type deformations first appear in the  $n = \pm 2$  sector. Higher Fourier modes are the genuine non-elliptic directions that local caustic constraints must eliminate.

As a concrete perturbation take

$$h(\psi) = R(1 + \varepsilon \cos 3\psi).$$

Then

$$\rho(\psi) = R(1 - 8\varepsilon \cos 3\psi),$$

so strict convexity holds whenever  $|\varepsilon| < 1/8$ . This domain is a smooth, explicitly non-elliptic perturbation. Local rigidity theorems say that if it also satisfied their full rational-caustic hypotheses while sufficiently close to the reference ellipse, the coefficient  $\varepsilon$  would have to vanish.

#### Check your understanding

Repeat the curvature calculation for a translation mode  $f(\psi) = \cos \psi$ . Why does  $h + h''$  not change? Interpret this geometrically.

### Boundary case: one surviving caustic is not integrability

Lazutkin theory supplies many invariant curves near the boundary of a sufficiently smooth strictly convex table, yet generic tables are not ellipses. Therefore the existence of *some* caustics cannot imply Birkhoff rigidity. The modern theorems use much stronger hypotheses: a foliation of a boundary annulus, prescribed rational caustics, or rational integrability across an entire range. This distinction is essential when reading every statement in the chapter.

**Common pitfall**

Do not collapse the local rigidity implications into the global Birkhoff conjecture. A theorem near an ellipse may require symmetry, closeness, a detecting family of rational caustics, or spectral isolation hypotheses, and Koval's results exclude an explicit small exceptional set of eccentricities. Each qualifier is part of the mathematical conclusion, not editorial decoration.

**20.4 Chapter summary****What to remember**

- The global Birkhoff conjecture remains open; the chapter's theorems are symmetric or local rigidity results with explicit hypotheses.
- Support functions and Fourier modes separate Euclidean/elliptic directions from genuine shape deformations.
- Rational caustics yield linearized constraints whose invertibility can force local ellipse rigidity.
- Koval's analytic operator method gives local strong Birkhoff rigidity near almost every ellipse, with a precisely controlled exceptional eccentricity set.
- Wave-trace singularities connect Laplace spectral data to periodic billiard lengths and hence to local dynamical rigidity.
- Symplectic-billiard rigidity is a parallel frontier with a different reflection law and area spectrum; it is not evidence for the ordinary Birkhoff conjecture unless an explicit bridge is proved.

**Looking ahead**

This closes the present sequence, not the subject. The global Birkhoff conjecture, higher-dimensional rigidity, low-regularity KAM phenomena, and spectral inverse problems remain active frontiers. The methods developed here—normal forms, variational order, hyperbolicity, renormalization, parameter exclusion, and rigidity operators—provide several complementary ways to approach those questions.

**Check your understanding**

Write the implication chain used in local spectral rigidity near an ellipse: spectral equality  $\Rightarrow$  wave-trace information  $\Rightarrow$  isolated periodic-length data  $\Rightarrow$  rational caustics  $\Rightarrow$  local Birkhoff rigidity  $\Rightarrow$  Euclidean congruence. Mark the hypotheses or external microlocal inputs needed at each arrow.

## 20.5 Exercises and problems

### Exercise 20.1

**Difficulty.** Core.

**Related results.** Birkhoff conjecture status; Centrally symmetric Birkhoff–Poritsky theorem; Near-centrally-symmetric Birkhoff rigidity.

**Problem.** For an ellipse plus one small Fourier boundary mode, use the chapter’s caustic/length first-variation formula to determine a rational type that detects the mode and identify symmetry modes invisible at that order.

**Solution sketch.** Insert the Fourier perturbation, apply the stated selection rule, and separate the linearized calculation from any global Birkhoff claim.

### Exercise 20.2

**Difficulty.** Core.

**Related results.** Koval local strong Birkhoff theorem; Local spectral uniqueness of almost every ellipse.

**Problem.** For the smooth nonellipse boundary example, identify the first broken rational caustic and show which hypothesis of Koval’s local strong theorem fails. Explain why this geometric failure alone is not yet spectral non-isospectrality.

**Solution sketch.** Match the nonzero caustic variation to the detecting family, then list the separate wave-trace/isolated-length bridge needed in the spectral theorem.

### Exercise 20.3

**Difficulty.** Advanced.

**Related results.** Near-centrally-symmetric Birkhoff rigidity.

**Problem.** In a finite Fourier truncation of the near-centrally-symmetric linearized problem, build the matrix of detecting rational-caustic functionals and show that its kernel consists only of the permitted normalization modes.

**Solution sketch.** Evaluate the functionals on the symmetry-compatible Fourier basis and row-reduce, keeping geometric normalization modes separate.

### Exercise 20.4

**Difficulty.** Advanced.

**Related results.** Koval local strong Birkhoff theorem.

**Problem.** Delete one detecting caustic equation from the same truncated system and exhibit a newly uncontrolled Fourier direction, showing why the full detecting family is load-bearing at the linearized level.

**Solution sketch.** Remove one row, compute the enlarged kernel, and state explicitly that linearized underdetermination is not a nonlinear counterexample.

### Exercise 20.5

**Difficulty.** Research.

**Related results.** Birkhoff conjecture status.

**Problem.** Build a theorem-status diagram for the classical Birkhoff conjecture,

centrally symmetric rigidity, near-symmetric rigidity, Koval's local strong theorem, and local spectral uniqueness. Record regularity, symmetry/locality, closeness, and exceptional-set qualifiers on every proved arrow.

**Solution sketch.** Read theorem statements, make a hypothesis/conclusion table, and draw no implication that is not licensed by those exact qualifiers.

### Exercise 20.6

**Difficulty.** Lab.

**Related results.** None.

**Problem.** For an ellipse and a small smooth deformation, compute several near-boundary  $(p, q)$ -periodic orbit lengths and plot the fixed-type spread  $T_{p,q} - t_{p,q}$ .

**Solution sketch.** Validate collapse of the spread on the ellipse under tolerance refinement before identifying a stable nonzero spread for the deformation.



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## **Where the series ends—and where the subject continues**

The twenty volumes form a continuous route from symplectic geometry through modern KAM, billiards, translation surfaces, smooth dynamics and selected rigidity frontiers. They do not close the subjects themselves. The global Birkhoff conjecture remains open; many high-dimensional and low-regularity KAM questions remain active; generic smooth dynamics continues far beyond the theorem interfaces developed here; and the interaction between rigidity, spectrum and geometry still offers many open directions. The final lesson is therefore methodological: exact hypotheses and mechanisms matter as much as theorem names, especially at a research frontier.